2026Q4 PCB產業-
需求推動漲價，傳輸靠PTFE解方

福邦投顧 研究部
2026/10

26Q4結論-上游材料加速更新，漲價仍為主旋律

1.PCB上游:供不應求推動漲價動力，新材料應用下帶來新機會

- 玻纖布:E-Glass(高階)/LowDK-2/T-Glass 在2026Q4~2027年呈現全面緊缺，成為推動當前上游材料漲價的主要因素，也是決定結構性缺貨的關鍵原料，**玻布廠迎來產品組合改善+報價上揚雙重催化。**
- 銅箔:HVLP4加速驗證日台以外的供應商，中系擴產積極+中系設備加速採用下，**維持2027年供需缺口放緩預期。**
- 銅箔基板:PTFE樹脂為美系客戶解決互聯的確定性解方，成為銅箔基板材料中新晉路線；傳統的PPO/CH樹脂仍有一片天，26Q4 VR300 設計將定案，PTFE+M8/M9設計呼聲較高。
- 鑽針:高階鍍膜針需求攀升，供應商在2026~2028年加快投入建置產能，新材料(PTFE)仍有一定的鑽針消耗，**產品組合的改變遠比漲價來的重要。**

2. IC載板: 產能全數售罄，缺料與缺產能並行，關注供應商漲價態度。

- 供需缺口:維持2026/2027年 20%+/40%+ 缺口預期。
- 主邏輯為面積放大+層數增加，大廠產能售罄，客戶需要付出更高的價格預定產能+料源。

3. PCB:關注AWS、Google新品放量，以及2027~2028年HDI設計的變化，**聚焦Tier1供應體系和伺服器級HDI稀缺產能。**

4. 建議首選關注個股:

- 高階稀缺HDI產能:2368金像電、3037欣興、4958臻鼎
- 產品組合優化顯著:1303南亞、1815富喬、8021尖點、8039台虹
- 缺貨27~28年不緩解+AI Agent受惠:8046南電、3189景碩、3037欣興、4958 臻鼎

僅供福邦投顧客戶使用

重點載板、CCL公司獲利預估

股票代號	股票名稱	股本(百萬)	25EPS	26EPS(F)	27EPS(F)	28EPS(F)	景氣上行期PE
PCB板-國際同業評價P/E評價20~30X							
2368	金像電	5,306	19.02	41.2	82.9	99.9	20~30
載板-國際同業評價P/E評價20~30X							
8046	南電	6,462	3.01	14.5	55	85	20~30
3037	欣興	16,435	4.4	16.8	33	44.5	20~30
3189	景碩	5,268	3.5	13.7	27.1	42	20~30
4958	臻鼎	10,773	7.1	14.3	26	34	20~30
材料-國際同業P/E評價20~30X							
1303	南亞	79,308	0.51	15	3~4(材料) + 8(南電) + 12~20(南亞科)	N/A	20~25
1815	富喬	5,855	1.5	4.8	7.01	10	15~23
8021	尖點	1,473	2.7	10.2	25	36	16~25
8358	金居	2,526	4.2	12	17.2	27	19~25
2383	台光電	3,583	41	117	260	380	20~25
6213	聯茂	3,636	4.2	9.7	22.3	40	15~25
6274	台燿	2,993	11.8	40.2	85	115	20~25
8039	台虹	2,640	2.04	2.8	3(原業務) +15(PTFE)	3(原業務) +22(PTFE)	20~25
資料來源:福邦投顧整理預估(2026.09)							

僅供福邦投顧客戶使用

目錄:

一	美國政策主導AI為主要核心	4
二	雲端資料中心資本支出概況	5
三	GPU與ASIC需求、規格變化	7
四	PCB鑽針供不應求格局確定	10
五	HVLP4銅箔2026~2027年供需缺口放緩	14
六	T-Glass2026~2028年缺口逐年擴大	19
七	高階玻布&石英布26H2見供不應求	21
八	銅箔基板漲價循環直至2027H1	27
九	NVidia帶起的材料變化-PTFE的方案發展	28
十	ABF載板2027~2028年供不應求	34
十一	800G/1.6T光模組/交換機規格變化	40
十二	相關個股介紹	41

【僅供內部教育訓練使用，嚴禁外流】

全年核心:政策主導AI為基本方針

- AI 成為美國政策上的基本方針，OBBBA法案刺激企業加大投資AI算力，AAAP計畫掃除建置AI算力的瓶頸和主權AI方針的確立。

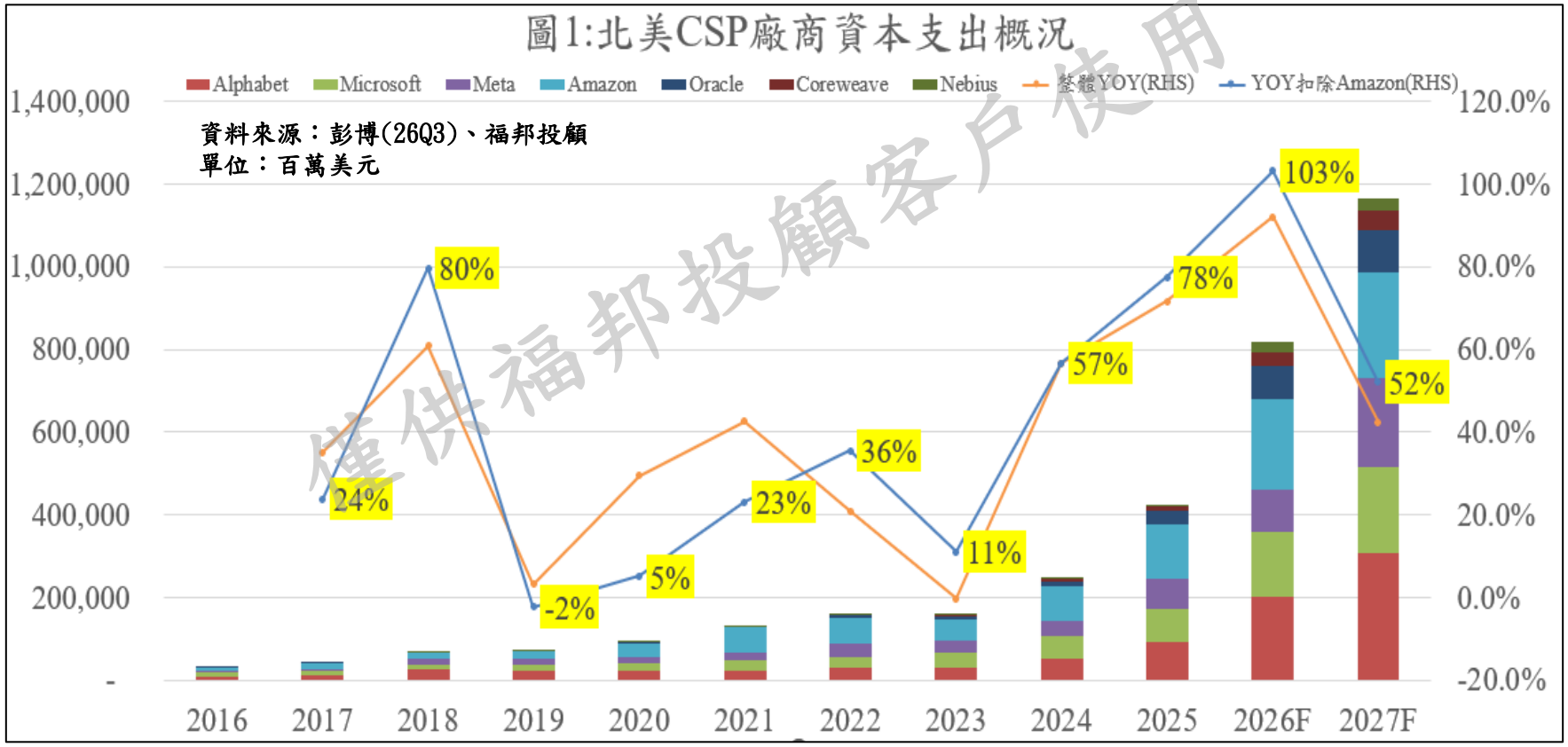
表1:OBBA法案和AI行動計畫　　資料來源:福邦投顧整理

法案	大而美法案(One Big Beautiful Bill Act)	AI行動計劃(America's AI Action Plan)
內容	1. 下調聯邦所得稅:從35%下調至21%，減輕企業稅賦負擔。 2. 恢復2017年減稅與就業法案項目中允許企業**當年資本支出可以全額抵減稅賦(折舊扣除)**，有利重資本支出廠商加快投資回收時間，受惠廠商如CSP大廠。 3. 國內R&D全額費用化:為了獎勵企業在美國進行生產研發，**符合條件的R&D費用可以於當年度全面費用化**，受惠廠商如Open AI、Anthropic等。	1. 加速AI創新:確保美國擁有全球最強大的AI系統，放鬆相關管制與障礙，並鼓勵開源等。 2. 增加AI基礎設施建置:提升美國AI算力基礎設施硬體基礎，以滿足AI推動經濟和安全的需求；加快關鍵項目的審核，**加快建設資料中心**、晶圓代工廠、電力基礎設施，掃除一切AI基礎設施擴充的障礙。 3. 維護國際外交與AI安全:在全球AI競爭中掌握規則制定權，輸出美國AI技術，同時阻止對手利用美國創新超越美國，嚴防高階算力流向競爭國家，向盟友出口美國AI算力技術等，有利**主權AI(Sovereign AI)**的發展。。

僅供福邦投顧客戶使用

雲端巨頭對資本支出態度維持積極

■2026年雲端CSP業者和算力租賃業者正加快擴建算力基礎設施，2026年預估資本支出成長上修至103%(含Amazon成長92%);2027年目前資本支出預估成長52%。



僅供福邦投顧客戶使用

全球伺服器出貨2026~2027年預期雙位數成長

■參考Digitimes最新數據預估，2026年預估出貨量YoY+17%，2027年目前預估11.2%。

圖2:全球伺服器出貨量成長預測

		2023	2024	2025	1Q26	2Q26	3Q26	4Q26	2026(F)	2027(F)
Gartner	出貨量	11,259	11,991	12,544					14,015	15,678
26Q2	YoY	-18.6%	6.5%	4.6%					11.7%	11.9%
	QoQ									
Digitimes	出貨量	14,432	14,888	16,597	4,699	4,877	5,015	4,896	19,487	21,670
26Q3	YoY	-20.4%	3.2%	11.5%	23.5%	26.0%	19.5%	9.0%	17.4%	11.2%
	QoQ				4.6%	3.8%	2.8%	-2.4%		
IDC	出貨量	12,357	14,461	16,779	3,773	3,975	4,261	4,421	16,430	18,083
26Q2	YoY	-17.4%	17.0%	16.0%	2.0%	1.9%	3.0%	0.2%	-2.1%	10.1%
	QoQ				-14.2%	5.4%	7.2%	3.8%		

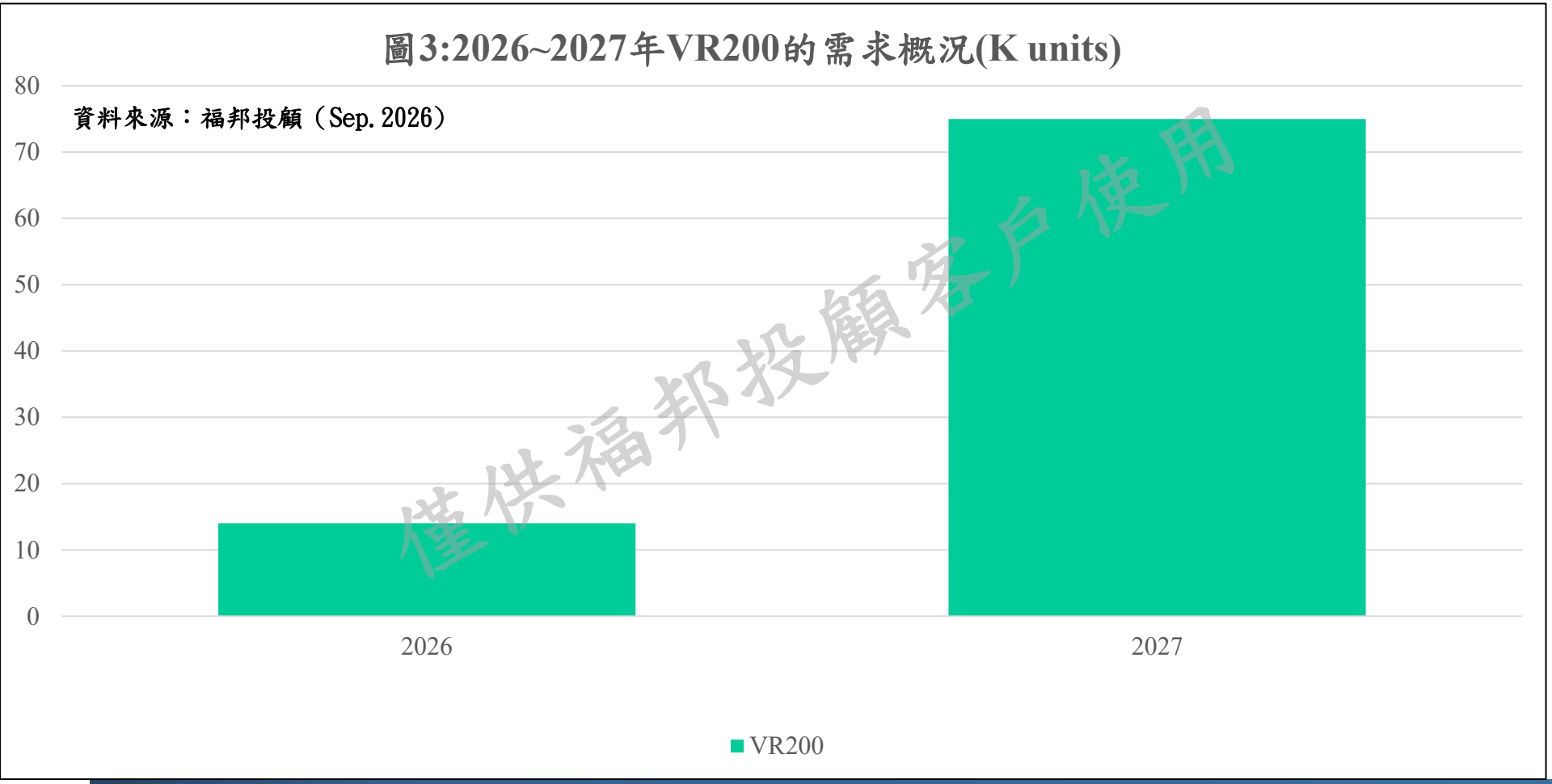
資料來源：福邦投顧整理（26Q2）

僅供福邦投顧客戶使用

GPU:26Q4~2027年VR200合計88K櫃

■根據供應鏈調查統計，2026年Rubin(VR200)預估組裝量約13~14K櫃，2027年預估組裝量約75K櫃。

■Rubin Ultra (VR300)於2027年預估需求量約5~6K組，2028潛在12K組(NVL576)。



僅供福邦投顧客戶使用

GB300/VR200/VR300規格:南亞、台虹為亮點

表2:GB200/300 VS. VR200　　資料來源:產業調研、福邦投顧整理

名稱	GB300 NVL72(72GPU)	VR200 NVL144(72GPU)	VR300 (RubinUltra)(9+9+9)
時間	2026放量	**2027放量**	**2028放量(26Q4 定案相關規格)**
晶圓製程	N4P	N3P	N3P
配合的PCIE版本	6	6	6
CoWoS製程	CoWoS-L 雙GPU封裝	CoWoS-L 雙GPU封裝	CoWoS-L 雙GPU封裝
單晶圓可切割數(顆)	14~15	8(包含I/O chiplet 面積增加)	8(包含I/O chiplet 面積增加)
TDP/散熱方式	1400W	2300W	1,800/2,600
載板供應商	Ibiden80~90% /欣興10~20%	Ibiden90~95% /欣興5~10%	Ibiden90~95% /欣興5~10%
CCL材料供應	計算板:斗山M8(LDK)+M4	計算板:斗山(M8+LDK+HVLP4)+M4	計算板:斗山(M8+LDK2+HVLP4)+M6
	交換板:M8(LDK2)+M2 生益科技(70%)台光電 (30%)	**交換板:M8(LDK2+HVLP4)** 生益科技/台光電/**南亞**	**交換板:M8(K2)/M9+PTFE** **生益科技+台虹/台虹**
		CPX:M8(K2)	CPX:M8(K2)
		中介板:M8(LDK2)	中介板:M9?
CCL上游材料供應	玻布:台玻、Asahi、泰山玻纖	玻布:台玻、Asahi、泰山玻纖、國際複材、富喬、日東紡	玻布:台玻、Asahi、泰山玻纖、國際複材、富喬、日東紡
	樹脂:Sabic、MGC、Asahi	樹脂:Sabic、MGC、Asahi	樹脂:Sabic、MGC、Asahi
	HVLP3銅箔:三井/古河/盧森堡	HVLP4銅箔:三井/古河/盧森堡/金居/福田/德福	HVLP4銅箔:三井/古河/盧森堡/金居/福田/德福
ComputeTray	5壓22L	5壓26L	**5壓30L**
PCB供應商	勝宏>欣興	勝宏>欣興>臻鼎 …定穎?(認證中)	勝宏/欣興/臻鼎/ 定穎(認證中)
NVSwitch Tray	22L HLC	32L(N+N HLC)	**32L(N+N HLC)**
PCB供應商	VGT50%/TTM40%/景旺10%	VGT/WUS/TTM	VGT/WUS /TTM/深南?/景旺?
CPX		Delay(需求不佳)	Delay(需求不佳)
PCB供應商			
小中介板(Midplane)		44L(N+N HLC)	44L(N+N HLC)
PCB供應商		VGT/景旺/滬電/欣興	VGT/景旺/滬電/欣興

僅供福邦證券客戶使用

雲端廠商ASIC相關規格-高階材料/製程加快導入

表3:雲端大廠&LLM廠商方案　　資料來源:產業調研、福邦投顧整理預估

客戶	Design Service/ Code Name	時間	主板 設計規格	L6	L10~L12	CCL Supply	Compute Board PCB Supply
Google	Broadcom - TPU v7 (Ironwood)	26H1	22/24L	Celestica80%/ Jabil 20%	Celestica /Flex/ 鴻海(5%)	M7-Panasonic (80~90%)/EMC	主:ISU/LCS/WUS/TTM/VGT
	Broadcom - TPU v8 Sunfish	**26Q4**	34L			**M8(LK2)-** EMC/松下	主:WUS/TTM/LCS/VGT
	MTK - TPU v8 Zebrafish	**27Q1**	34L			**M8(LK2)-** EMC/Doosan/松下	主:ISU/WUS/TTM/VGT/ **GCE**
	MTK- TPU v9 Humufish	**28Q2**	**6壓28L**			**M9+M8(LK1)** EMC	主:ISU/VGT/WUS/TTM/**欣興** 備用:**臻鼎/LCS**
Amazon	Marvell - AWS Trainium2e (Cayman)	26H1	26L	智邦(氣冷) / 緯穎(水冷)	緯穎 /Jabil/Flex	M8(LK)-EMC/TUC	GCE/SY/FHT
	Annapurna - Trainium3 (Mariana)	26Q2	主: 26L(氣) 40L(水) SW:22L			M8(LK)- EMC(氣+水)/ TUC(氣)	**ComTray:GCE+**SY **(氣+水)** /WUS(水)/**FHT(氣)** **Swtray:GCE/**SY/**ZDT/FHT**
	Annapurna - Trainium4 (Maverick)	27Q4	**主:3/4壓34L SW:30L**			**M9-** EMC/南亞?	**GCE/**SY/**ZDT**/Meiko
Meta	Broadcom - MTIA T V1.0 (Minerva)	25H2	40L	廣達/ Celestica	廣達	M8(LK)- EMC	WUS/TTM/ISU
	Broadcom - MTIA I V1.5 (Iris+ / Santa Barbara)	26H2	36L	廣達 or Celestica	廣達 or Celestica	M8(LK)- EMC	WUS/GCE/ TTM/ISU
	Broadcom - MTIA V2 (Olympus / Santa Cruz)	28E	?	廣達 or Celestica	廣達 or Celestica	**M9-** EMC	WUS/GCE/TTM/ ISU/DY?
MSFT	Marvell-MAIA 300 (Griffin)	27H1	40L+	廣達	廣達	**M9-** EMC/Doosan?	GCE/TTM/WUS/大德/ISU
AMD	MI4系列	27H2	48L	緯創	Celestica、緯穎	M8(LK2)	SCC(90%)/欣興/大德
	MI5系列	28H2	52L	緯創	Celestica、緯穎	M9	

僅供福邦投顧客戶使用

鑽針主邏輯—PCB廠積極擴產

- PCB設計日益複雜，耗用大量產能：過去伺服器板層數約16~2X層，目前主流已經來到20~30層，未來30~50層越顯常見。
- PCB用量增加，耗用大量產能:隨著各家設計陸續參考NVIDIA Rack設計，除了原有的主板(GPU/ASIC 使用)，亦新增更多副板，例如Switchboard、midplane等，終端客戶積極驗證新供應廠商。

表4:全球主要GPU/ASIC陣營PCB供應商擴產概況

億元	產值/產能	2025	2026(F)	YoY	2027(F)	YoY	2028(F)	YoY
勝宏科技	多層板(M^2)	28K	120K	+328%	350K	+190%	590K	+69%
	HDI(M^2)	130K	320K	+146%	640K	+100%	680K	+6%
金像電	sqf	6,800K	10,000K	+47%	11,000K	+10%	12,000K	+9%
ISU Petasy	(M^2)	240K	280K	+16%	320K	+14%	360K	+13%
滬電股份	(M^2)	160K	210K	+30%	340K	+62%	440K	+29%

資料來源:產業調研、福邦投顧整理預估

僅供福邦投顧客戶使用

PCB鑽針需求呈現翻倍成長-CCL材料影響

■ PCB用鑽針受到板材影響，在同樣板厚下，每針可鑽孔次數大幅縮水:

- M6~M7時主針鑽孔壽命約600~800次
- M8材料下可鑽400~500次
- M9材料下僅剩下100~200次
- 材料越高階，鑽針使用量會是過往的6倍以上。

表5:鑽針基於不同板材的使用壽命&各應用對應的鑽針方案

	過去	目前	未來
對應材質	M6~M7	M8/(有布/無布+球型矽粉)	M9~M10
主針壽命	1000/600~800次	400~500次	100~200次

資料來源:產業調研、福邦投顧整理預估

僅供福邦投顧客戶使用

PCB鑽針需求呈現翻倍成長-長徑比帶動報價上揚

- 當前主流GPU用PCB約5mm厚度，隨著AI用PCB板厚的提升，每增加1mm，製造上可能會採用增加一根預鑽針，採用預鑽針預鑽至特定深度，後續用主針完成加工。
- GPU伺服器:GB200採用「1主針配2白針」→GB300採用「1主針配3白針」→VR200採用「1主針配4白針」，

表6:鑽針各應用對應的鑽針方案　　資料來源:產業調研、福邦投顧整理預估

	GPU Server	ASIC Server	交換機
對應鑽針方案	1主針 +1~4根預鑽白針	2主針 +1根預鑽白針	3~4根主針 +1~2根預鑽白針

- AI用的鑽針，長度/直徑比超過30規格的鑽針，報價上會採用**一個方案一個價格**，因為加工精度要求提高，難度變大；目前AI用的鍍膜鑽針報價較傳統白針多20~50%，若是這種報價方式，特別是板厚超過7mm，產品售價將會多40~50%；未來約在2027~2028年時，7.5~8.5mm的鑽針將成為主流，報價將會是一般白針的6倍以上報價。

表7:鑽針不同長徑比的報價倍數　　資料來源:產業調研、福邦投顧整理預估

長徑比	一般(10~20)/鍍膜	30	40	50
售價倍數	1/1.2~1.5	3~4x	6~8x	12~16x

- **毛利率方向:AI用的鍍膜鑽針鑽針，根據產業訪查，產品毛利率約50%+**。

僅供福邦投顧客戶使用

全球主要鑽針廠亦積極擴充產能，供需維持吃緊

- 目前全球主要伺服器PCB廠採用的鑽針廠，高階鑽針以中鎢高新旗下的金洲科技和尖點為主、中階以下以鼎泰高科為主；日本佑能主要用於載板生產用針。
- 鑽針廠的外溢效果:由於前四大廠優先用於AI伺服器相關生產用針，使得中低階針有排擠效應，白針類產品外溢給凱崴等廠商。

表8:全球主要鑽針廠產能/客戶概況　　資料來源:產業調研、福邦投顧整理預估

公司	產能(萬支/月)	高階鑽針占比	主要客戶	規劃
鼎泰高科	1.2億支	>40%	深南、景旺、勝宏、健鼎、生益、方正、廣合、定穎	2026年:1.8~2億支/月
中鎢高新(金洲)	9,000	>15%~30%	勝宏、深南、高技、生益、景旺等((**中系高階主供**)	2026年底至1.2億支/年 2027年新增3600萬支+/年 2028年新增1.3億支/年
佑能	4000	>60%	專注日本、台灣市場、載板廠	2026年底至5,000萬支/月 2027年至6,000萬支/月
尖點	3,500(50%以上高階鍍膜針)	>50%	金像電、ISU、勝宏、TTM、滬電、臻鼎、欣興等	2026年:4,500萬支+/月， 2027年:˙7,000萬支+/月 2028年達到9,000萬支+/月

僅供福邦投顧客戶使用

HVLP4銅箔需求受良率影響，需求驚人

- 根據供應鏈調查，依照良率80~90%的假設下，Nvidia Rubin的需求約300~400噸/、LPU約200~250噸/月，Amazon AWS 800~850噸/月，Meta MTIA 200噸/月，AMD MI450 100~200噸/月，800G交換機 400噸/月，合計**1,200~1,300噸/月/2,700~2,800噸/月。**

- 若依照實際PCB生產狀況，良率約在60~70%不等，總需求可達**1,400噸/月/3,000~3,100/噸/月。**

表9:全球主要HVLP需求估算與供應鏈格局

	HVLP2	HVLP3	HVLP4	HVLP5
加工費	USD15/kg	USD20~25/kg	25~30美金/kg	35+
供應商	三井、古河、福田、金居、盧森堡、銅冠、德福	三井、古河、福田、金居、盧森堡、銅冠、德福	三井、古河、福田、金居、盧森堡、銅冠、德福	三井、古河、福田
客戶	AWS T2 (2024-2025年)	NVIDIA GB200 NVSW	AWS T2MAX(2026年)、Meta M9 896K2/K3、NVIDA Rubin、Google V8、AMD MI450、800G Switch	Panasonic M9
	Google V5/V6	Google V7		SY5300N
需求量	AWS 300-400噸/月	NV 38~40噸/月	AWS800~850噸/月(100萬張/月)	
		Google v7	Nvidia 300~1000噸/月	
			800G 交換機 400噸/月	
			Meta 20噸/月	
	資料來源:產業調研、福邦投顧整理預估		AMD 20噸/月	
			Google Zebra&Sun 80噸/月	

僅供福邦投顧客戶使用

全球主要供應商概況:加快導入新供應體系

表10:全球主要HVLP供應鏈格局

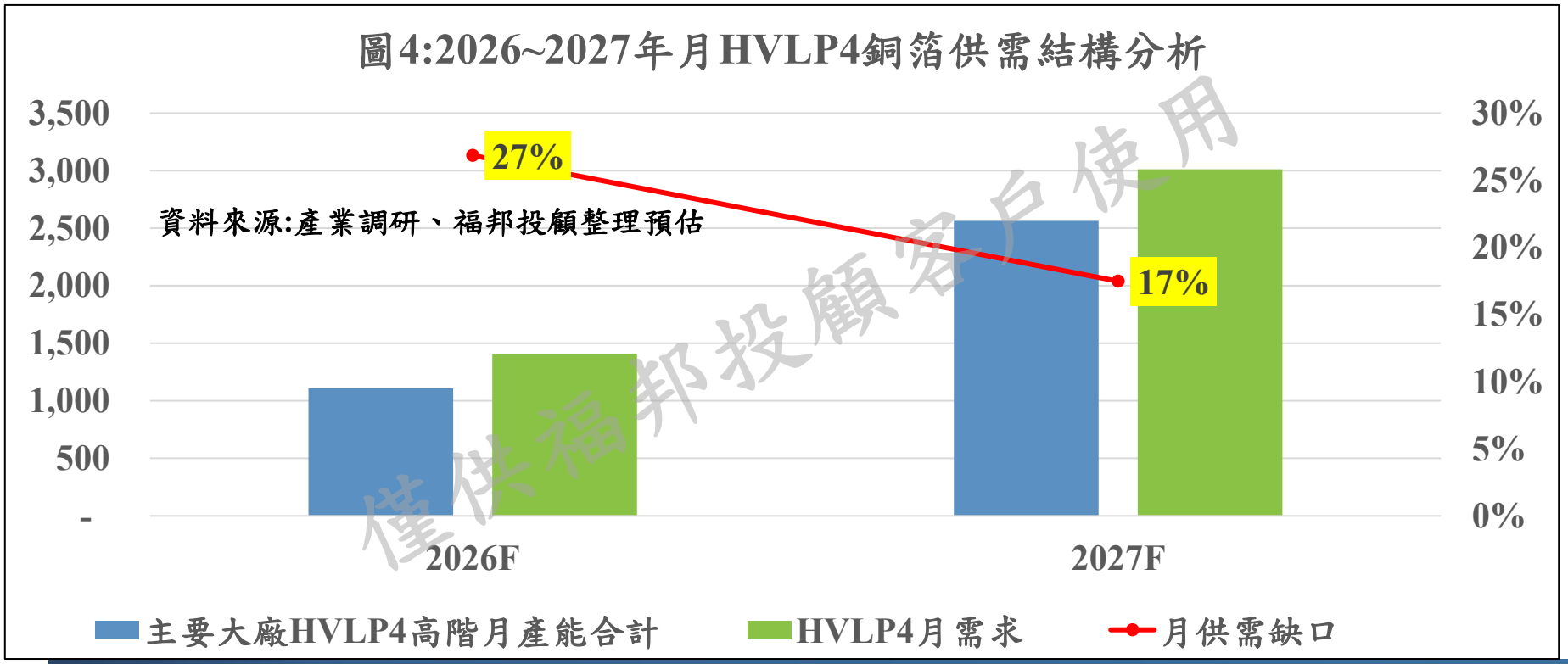
資料來源:產業調研、福邦投顧整理預估

	地區	產品分布	關鍵客戶	關鍵終端客戶	高階產能規劃
三井金屬	日	HVLP1~5	日:松下 台:台光電、台燿 中:生益	Nvidia/Google/AWS等	26Y:550噸/月 27Y:1,000噸/月 28Y:1,200噸/月 30Y:1400噸/月
古河電工	日	HVLP1~4	日:松下 台:台光電	Nvidia/Google/AWS等	26Y:100~200噸/月 27Y:200~300噸/月
福田金屬	日	HVLP1~5	日:松下 台:台光電、台燿、南亞 中:生益	Nvidia/Google	26Y:100~200噸/月
盧森堡	歐	HVLP1~4	韓:斗山 台:台光電、台燿 中:生益	Nvidia/AWS	26Y:100~200噸/月
金居	台	HVLP1~4	韓:斗山 台:台光電 中:南亞新材?	NVidia	26Y:330噸/月 27Y:700噸/月 28Y:800噸/月
德福科技	中	HVLP1~4	日:松下 中:生益 台:聯茂、台光電	NVidia/Google	26Y:200噸/月 27Y:500~600噸/月 (潛在1,200噸)
銅冠銅箔	中	HVLP1~4	中:生益、南亞新材 台:台光電、台燿	NVidia	26Y:50~100噸/月 27Y:200噸/月
長春	台	HVLP4	台:台光電	Nvidia/AWS	27Y:200噸/月
Lotte(樂天)	韓	HVLP2~4	韓:斗山	NVidia	26Y:300~350噸/月 27Y: 900~1,000噸/月

僅供福邦投顧客戶使用

銅箔供需缺口往2027年將縮小，供應廠商增加

- 根據供應鏈調查，由於供需缺口明顯，主要CCL廠商(台光電、生益科技等)積極導入新的供應體系，2026年主要生產廠商擴產後，可達1,200~1,300噸/月，2027年預估達2,300~2,400噸/月。(供應體系:台銅/古河/盧森堡/金居/福田/德福/銅冠/樂天)、台系長春後續有望驗證進入供應體系。
- **根據福邦投顧對於供需格局的假設預估，基於客戶加快認證相關可用產能，2026~2027年的供需缺口更新預估為27%(下修)/17%(上修)。**



僅供福邦投顧客戶使用

玻纖布使用種類

表11:玻纖布使用種類彙整

應用		元件	基板類型		材料等級要求	玻布類型	
						高階	中階
資通訊	AI伺服器/資料中心/交換機/基地台等	處理器	**封裝載板**	**CPU/GPU/ASIC**	**Low CTE**	**T**	**E**
				NAND	**Low CTE**	**T**	**E**
				DRAM	**Low Dk**	**NE/NER**	**E**
		主板	伺服器(AI/一般型)		Low Dk	NE/NER/Q	E
			交換機(核心/AI)		Low Dk	NE/NER/NEZ/Q	E
			網卡(NIC)		Low Dk	NE/NER	E
邊緣端	手機/平板等	處理器	封裝載板	AP/CPU	Low CTE	Ultra-Thin T, T	Ultra-ThinE
		非動態記憶體		NAND	Low CTE	Ultra-Thin T	Super Ultra-Thin E
		動態記憶體		DRAM	Low CTE	Ultra-Thin T(行動)	
				DRAM	Low Dk	NE(PC)	
		主板	主板		Low Dk	Ultra-Thin NE	Ultra-ThinE
		無線通訊	RF 封裝載板		Low Dk	Ultra-Thin NE	Ultra-ThinE
	DT/NB/AI PC	CPU/記憶體	封裝載板	CPU/GPU/NPU	Low CTE	T	E
		主板		DRAM	Low Dk	NE(PC)	E
	AR/VR	先進SoC	封裝載板		Low CTE	T	Super Ultra-Thin E
車用	EV/ADAS	先進SoC	封裝載板		Low CTE	T	E
		mmWave 雷達	模組載板		Low Dk	Ultra-Thin NE	E

資料來源:產業調研、福邦投顧整理預估

僅供福邦投顧客戶使用

高階電子布供應商概況:台光電開始取得日東紡料源

表12:電子玻纖布供應商與其主要客戶

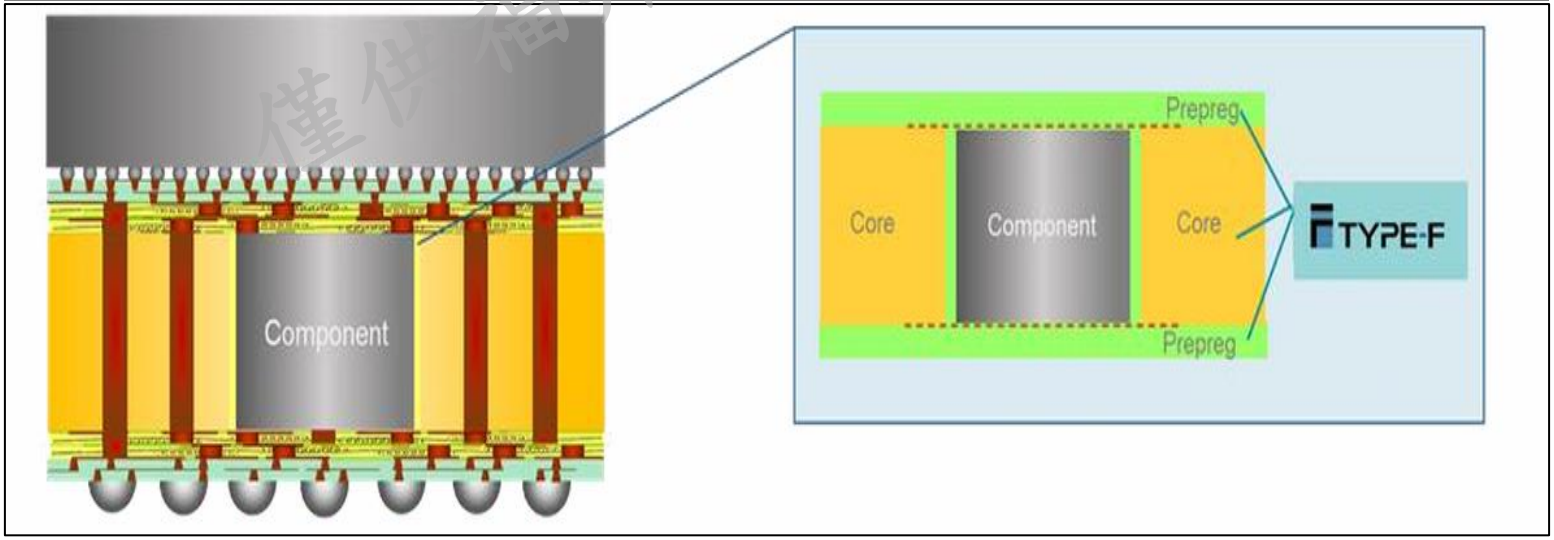
廠商	地區別	供應電子布產品	主要客戶
Nittobo	日	E/lowDK/LowCTE	• LowDK:松下、Doosan、TUC、南亞、EMC • LowCTE:Resonac、MGC、松下
Asahi	日	E/lowDK/LowCTE/Q	• LowDK:EMC、Doosan • Q:EMC、Doosan、松下
台玻	台	E/lowDK/LowCTE	• Low DK: EMC、Doosan、TUC • LowCTE: Resonac、MGC、EMC
德宏	台	E/Q	• Q:EMC、南亞、聯茂
富喬	台	E/lowDK/LowCTE	• LowDK:EMC、Doosan、TUC、Asahi(紗) • LowCTE:Asahi(紗)
南亞塑膠	台	E/lowDK/LowCTE	• LowDK:南亞 • LowCTE:Resonac、MGC、南亞
泰山玻纖	中	E/lowDK/LowCTE/Q	• Low DK: EMC、Doosan、TUC • LowCTE: Resonac、MGC、EMC • Q:EMC、生益科技、Asahi(紗)
光遠新材	中	E/lowDK/LowCTE	• LowDK:EMC、生益科技
宏和科技	中	E/lowDK/LowCTE	• LowDK:EMC、生益科技 • LowCTE:Resonac、MGC、EMC
建滔積層板	中	E/lowDK/T	• LowDK:EMC、生益科技
菲力華	中	Q	• Q:生益科技、EMC、松下、TUC
國際複材CPIC	中	E/lowDK/LowCTE	• LowDK:生益科技

僅供福邦投顧客戶使用

T-Glass:客戶積極導入抗翹曲設計

- 下個世代的半導體封裝需要更大的面積來放置更多元件、支持2D~3D封裝(CoWoS~SOIC)，因此材料的要求需要Low CTE、高彈性變量來達到低翹曲度 (Warpage)和連接可靠度。
- 根據Resonac白皮書，為了解決內埋元件載板對於核心層的厚度精度要求，較薄的核心層或不均勻的厚度，會導致內埋元件的損壞，因此之後許多載板核心的設計會採用厚Core+ low CTE 膠片 (採用T/Q Glass)。
- **2026年的TPU，2027年的Maia300都有見到這類核心Core CCL+PP的設計方案(Multilayer Core)，此也會多消耗T-Glass的供應量。**

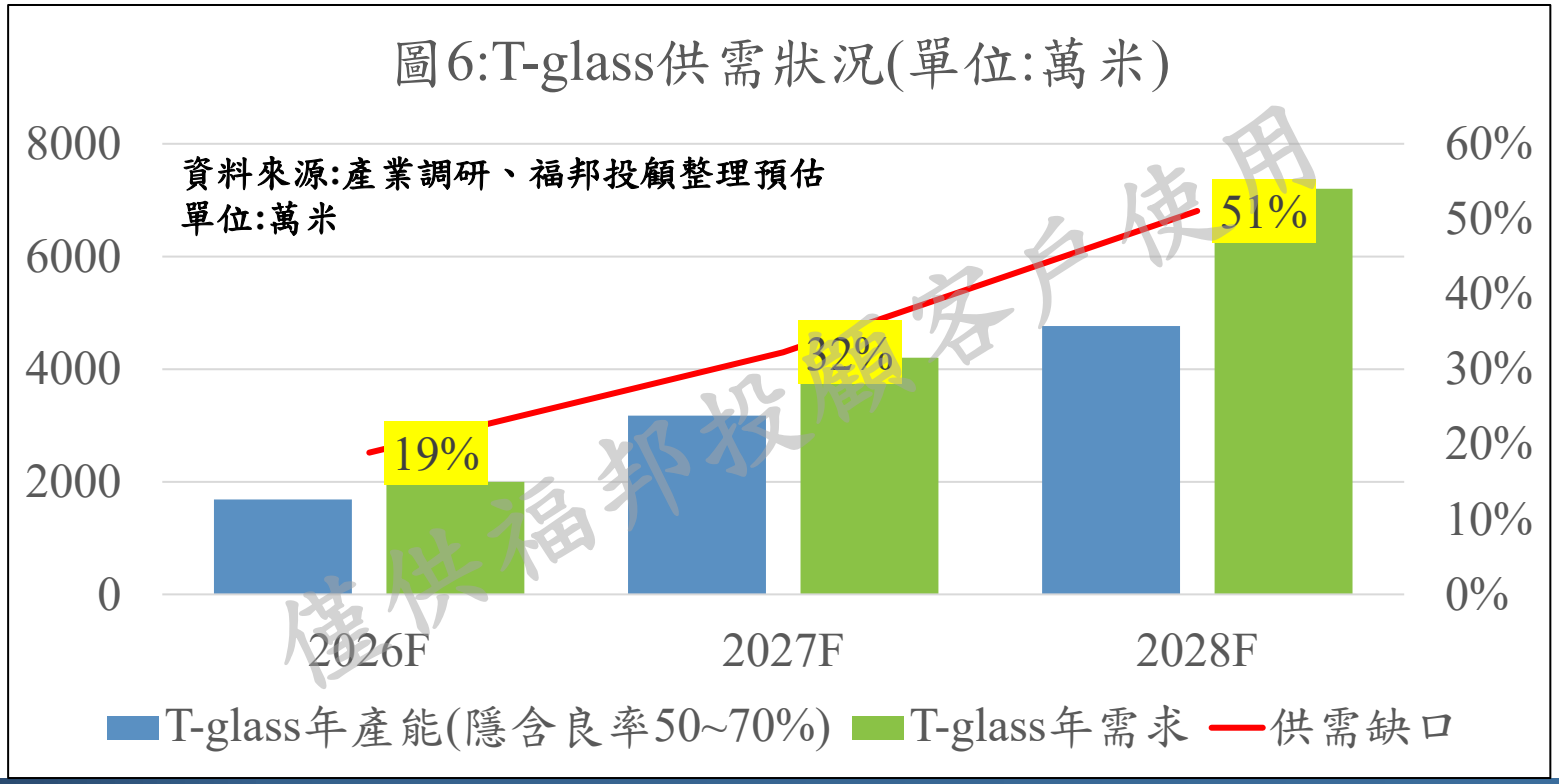
圖5:Resonac Type F產品



僅供福邦投顧客戶使用

T-glass至2027年仍將呈現供不應求格局，缺口30%+

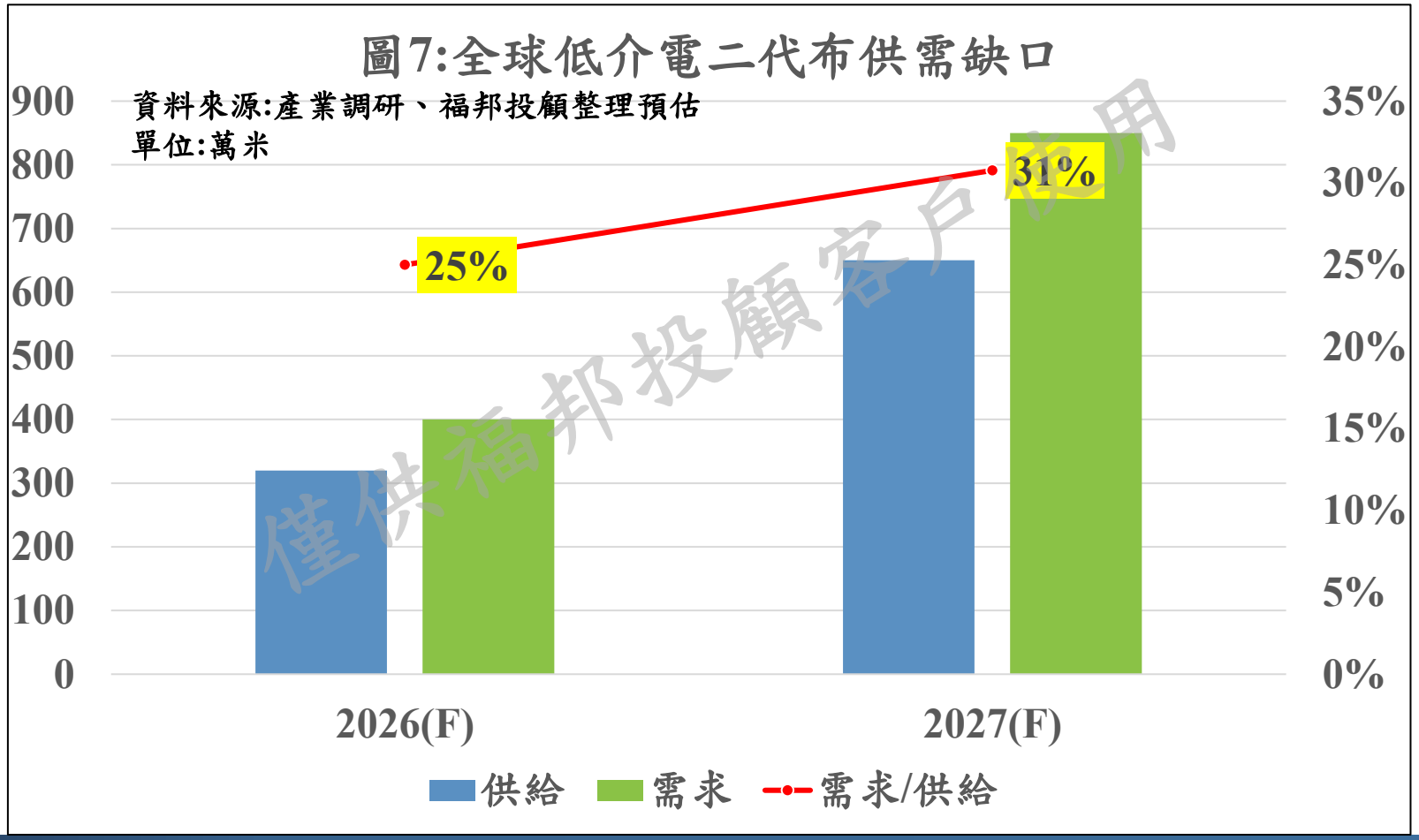
- 目前需求來看，T-glass需求中有70%用於FCBGA用的ABF載板，30%用於手機等FCCSP用的BT載板上。
- T-glass產能劃分來看，日東紡佔有全球主要產能的40%，台玻約15~20%，宏和約20~30%，其他(包含泰山玻纖等)15~20%。
- **福邦投顧預期儘管日東紡於2027年陸續開產能出來，但只緩解了Intel/AMD/Nvidia的用料，對於ASIC陣營的Broadcom/Marvell/MTK/Annapurna而言，仍難以取得足夠料源，維持T-Glass持續供不應求的判斷，缺口仍會逐年往上。**



僅供福邦投顧客戶使用

LowDK Gen2 2027年預期缺口31%

- 根據供應鏈訪查，2026H2開始，低介電二代布需求約400~500萬米/月，其中主要需求來源是Nvidia VR200、Google V8T/V8I。
- 以其銅箔基板廠商供應來看，主要以台玻、泰山玻纖、國際複材、Asahi為主，不足部分將由富喬、光遠、Nittobo補足，2027年供應量預估達650萬米/月以上。



僅供福邦投顧客戶使用

Q-Glass:LPU產品將率先採用

- 目前最新的NVIDIA LPX Rack從原先的全M9Q設計，改採用M8(K2)+M9(Q)的設計，配置為1:1，其中Q Glass主要採用10系列的極薄布，供應廠商少。
- 根據產業調查，預估NVIDIA LPU產品需求約在12~13K(2026~2027Y)，潛在Q glass需求約110萬米/月，隱含良率損耗則需要210~220萬米/月。

表13：LPU 主要供應資訊

	LPU Rack
PCB設計規格	52L(26+26)
Q-Glass供應商	菲利華、泰山玻纖、德宏
HVLP4供應商	三井、古河、盧森堡、金居
M9 CCL供應商	生益>台光電
PCB供應商	勝宏、Wus、TTM(30%)
組裝	鴻海

資料來源:福邦投顧預估

表14：LPU 供應試算

LPU 產品試算(單櫃)	
Tray 數量 per Rack	32
PCB/Tray	1
LPU/PCB	8
LPU總量(顆)	256
CCL 用量/PCB	8
總用量	256
M9Q/張 ASP(TWD)	10,000
CCL總產值(TWD) per Rack	2,560,000
CCL ASP per PCB (USD)	2,667
PCB ASP (USD)	4,444
Q布 ASP per PCB (USD)	1,440
Q布 ASP (USD)	45
Q布用量(米) per PCB	32

僅供福邦投顧客戶使用

Q-Glass：供應者寡，關注非日系供應

- 根據產業調查，統計台系(1家)、日系(2家)、中系(2家)產能規劃，2026年底預估產能超過300萬米/月，2027年預估產能可達600萬米/月以上。

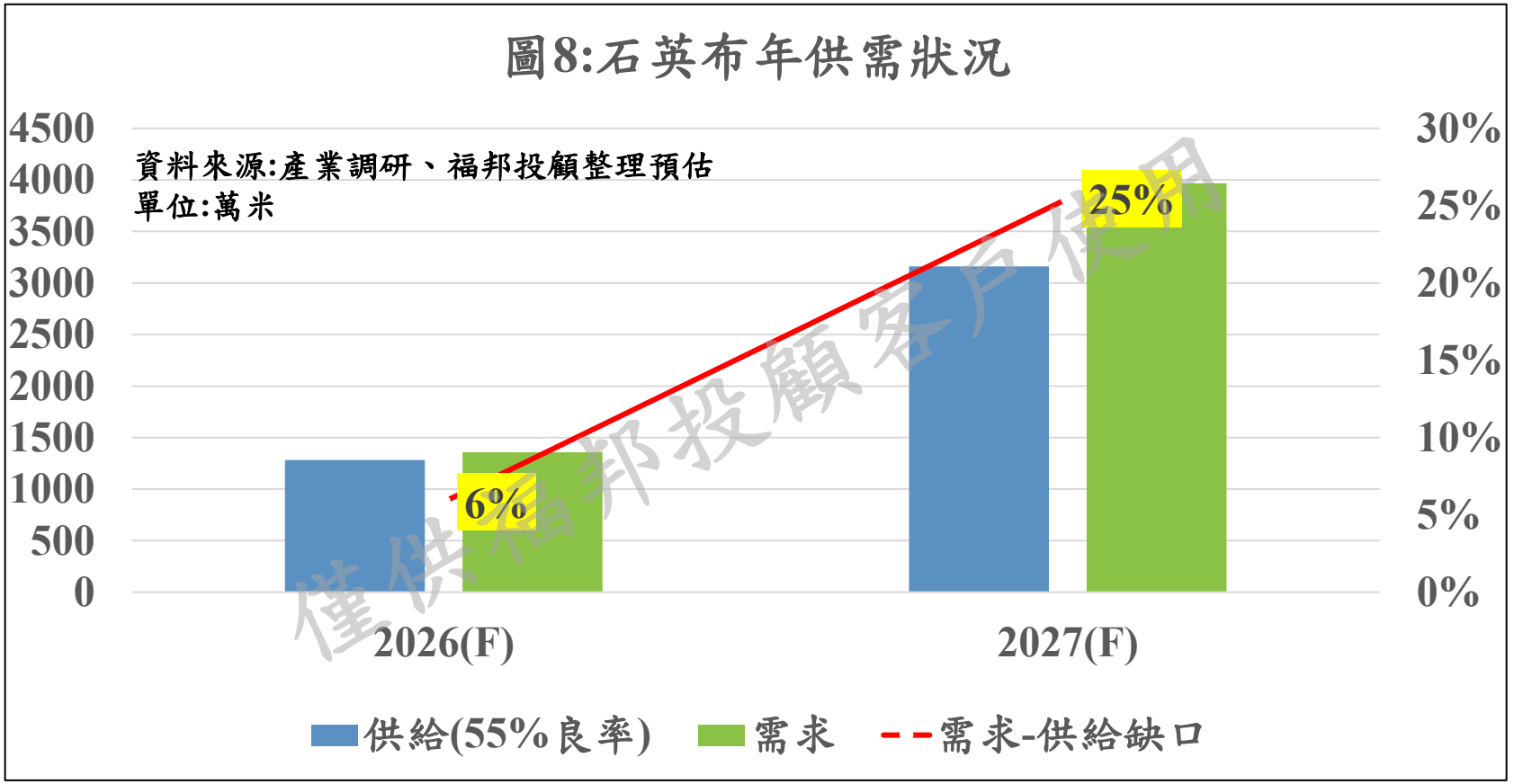
表15:Q-Glass供應商概況　　資料來源:福邦投顧整理

公司	地區	關鍵客戶	產能(K米/月)		說明
			2026F	2027F	
信越化學	日	日:松下、Resonac、MGC 台:台光電	75	240	外觀與同業不同，影響採用意願
Asahi	日	日:松下、Resonac、MGC 台:台光電	50	50	大客戶優先指定選用
泰山玻纖	中	台:台光電 中:生益科技	1,000	1,500~2,000	• 優先供應台光電
菲利華	中	日:松下 中:生益科技 台:台光電、台燿、南亞 韓:斗山	960	1,500~2,000	• 70%+供應生益科技 • 26H1 30W米/月 • →26H2 90~100W米/月
德宏	台	台:台光電、南亞、台燿(?)、聯茂 韓:斗山(?)	1,000	1,500~2,000	• 優先供應台光電 • 26H1 60W米/月→26年底100W米/月

僅供福邦投顧客戶使用

Q-Glass:26客戶進入庫存建立期，缺口下修至25%

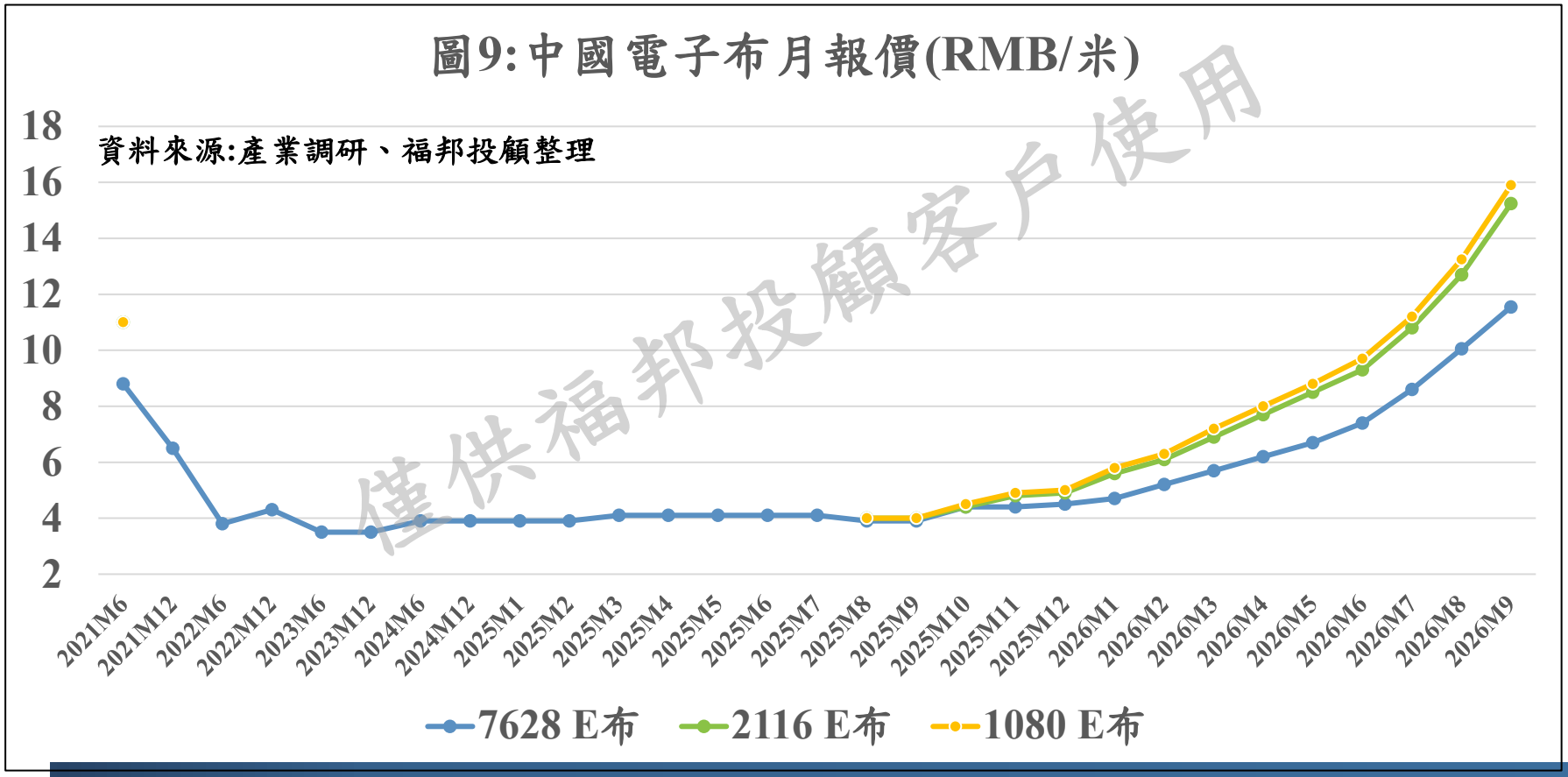
- 目前供應鏈的良率約在55~60%，26H2 進入Nvidia LPU 產品的備貨期，將帶動銅箔基板廠商的備貨需求，其中以**生益科技需求最為迫切，台光電供過於求。**
- ASIC陣營態度:目前Meta選用M9Q進行打樣，Amazon/MSFT選用M9-(K2)進行主板打樣，Google TPU V9至少選用M9以上材料。



僅供福邦投顧客戶使用

價格:E glass漲幅驚人，2H26將續漲至創新高價

- 玻纖布於2021~2022年週期係受到需求端推動帶動漲價，以7628系列為例，報價達到8.8RMB/米，本輪低階起漲，除受原物料之外，也受到主要供應商陸續轉做高階玻布(283，排擠低階布生產(打破供需平衡情況)，進而帶動漲價。
- 7628布累計至26Q3 YTD150%+，2116/1080 布YTD 210%+，均已創下歷史新高，2027年2116/1080的規格將繼續延續漲價趨勢。



僅供福邦投顧客戶使用

價格:高階電子布漲價，漲勢可望持續至2027H1

- 高階布種由於生產效率較傳統布下滑，以及良率因素影響，配合需求持續攀升下，每年均有約20~30%的漲幅，考量缺貨持續至2027年，漲價趨勢將延續至27H1。

表16:玻纖布漲價幅度

產品	目前ASP(USD/米)	2025	2026
LowDK	6~8	+20~30%	維持全年20~30%漲幅預期
LowDK2	18~22	+20~30%	• 26H1 YTD 漲幅+30~50% • **26Q3 漲幅+10%**。
T-Glass	23~30	+20~30%	• 日東紡26H1 YTD漲幅30% • 中系廠商26H1漲幅20% • **26H2預期會有10%+漲幅**。
Q-Glass	40~45	0%	26H1漲幅10~20%

資料來源:產業調研、福邦投顧整理

僅供福邦投顧客戶使用

銅箔基板漲價循環直至2027H2

- 目前AI相關CCL交期延長至4~6個月，上游料況吃緊（庫存1~2周），**福邦投顧預期考量上游材料漲價循環至少漲至27H1，將推動下游CCL客戶漲價續漲至2027年。**

表17:主要CCL廠商漲價幅度　　資料來源:產業調研、福邦投顧整理

公司名稱	地區	過去產品漲價幅度	26H2漲幅預期
Panasonic	日	• 26M5 高速CCL漲20%、FR4漲30%、 • PP漲15%	• 26M8 高速CCL漲10%、FR4漲15%、PP漲15%、載板材料漲5%
Resonac	日	• 26M3 CCL漲30%	
MGC	日	• 26M4 全品項漲40%	
台光電	台	• 26M3全品項漲15%+，M6以下漲20~25% • 26M4 M7以上漲5~10%	• 26Q3 M7以上漲10%。
台燿	台	• 26M4 漲幅20~40%。	• 26Q3 M7以上漲15%，M6以下上漲25%。
南亞	台	• 26H1 YTD+30%。	• 26Q3漲幅30~40%；26Q4會再繼續調漲
聯茂	台	• 26H1 YTD+40%+(4/6月)。	• 26H2每季預計漲20%+
建滔積層板	中	• 26H1累計五次漲價函，YTD 漲幅達50%+	• 26Q3漲幅30%+
生益科技	中	• 26M5 高速CCL /PP漲20%	• 26Q3漲幅30%+

僅供福邦投顧客戶使用

【僅供內部教育訓練使用，嚴禁外流】

Rubin Ultra帶起的材料變化-PTFE的方案發展

- 由於傳輸的需求(337G)，PPO或HC系列樹脂的M9無法滿足NVIDIA要求，NVIDIA 在Rubin Ultra 將導入混壓PTFE材料應用於SW Tray、Feynman 將導入純PTFE板材應用於正交背板。
- 目前銅板基板廠商正在積極送樣測試純無布的方案因應客戶需求。

表18:PTFE 方案組成　　資料來源:產業調研、福邦投顧整理

	PTFE CCL(無布) +CH PP(有布)	PTFE CCL(無布) +CH PP(無布)	PTFECCL +PTFE PP(無布)
樹脂配方	PTFE	PTFE	PTFE
銅箔搭配	HVLP4/5	HVLP4/5	HVLP4/5
供應商	三井金屬SI3-VSP		
電子布	LowDK 4(二代Q)	X	X
膠片配方	CH樹脂	CH樹脂	PTFE樹脂
供應商	生益科技、台光電、聯茂	生益科技、台光電	生益科技、台虹
電性	優於M10(CH樹脂系列)	次之	最佳
加工性	最佳	次之	最差
評論	備案，加工性較佳	供應鏈目前積極測試的方案	最終目標
PCB廠加工良率	30%	10~30%	10~30%
CCL供應商	走FCCL製程，生益科技SG5300N(塗佈+熱壓法)、台虹(熱壓法)		

僅供福邦投顧客戶使用

無布VS.有布方向選擇

- PTFE 由於Dk/Df的性能優於電子布，如果PTFE搭配電子布，會導致其Dk/Df轉差。

表19:PTFE與電子布的性能比較

	PTFE樹脂	LowDK 電子布	LowDK 3~4(1/2代Q-glass)
Dk(10Ghz)	~2.1	4.6~4.8	3.7~3.8
Df(10Ghz)	~0.0002	0.002~0.003	0.0002~0.0003

- 未來走向:

1. 有布方案(M9~M10搭配LowDK2~4電子布):若PCB上須放置CPU/GPU/ASIC等晶片，考量尺寸安定性因素，會需要搭配具有電子布的銅箔基板，例如UBB、ComputeBoard、SwitchBoard、MotherBoard。
2. 無布方案(搭配無布膠片):若僅考連傳輸性能(不放置晶片)，目前採取的無布方案，樹脂填矽粉/球型矽微粉的方案為主流，以改善膨脹性。

表20:PTFE的CTE比較

	PTFE樹脂	球型矽微粉	LowDK 3~4(1/2代Q-glass)
CTE(ppm/°C)	100~200+	0.4~0.6	0.5~0.55
說明		X/Y/Z軸的保護	X/Y軸保護

僅供福邦投顧客戶使用

不同填料對於銅箔基板的影響

- 在樹脂中加入填料能降低熱膨脹係數(CTE)、提高尺寸安定性和降低Dk/Df，近年的填料主要以球型粉料為主。

表21:各類填料的對應用途

填料	用途	相關填料供應商
二氧化矽粉(Si02)	降低CTE	日:電氣化學、Admatechs、龍森、新日鐵 中:聯瑞新材、錦藝新材、國瓷材料
陶瓷粉	調整Dk/Df	
氧化鋁	提高散熱性能	
碳酸鈣/滑石粉	通用材料	
氫氧化鈣/鎂	作為阻燃劑使用	
氧化鋁	改善導熱等	
碳酸鈣	降低複合材料成本	

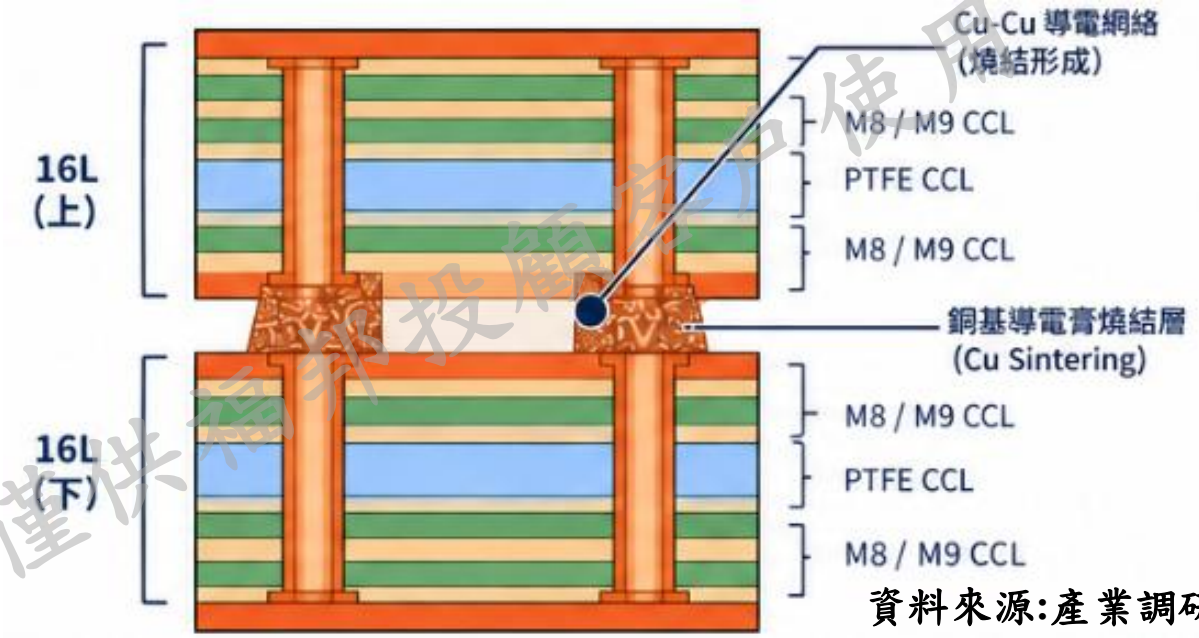
資料來源:產業調研、福邦投顧整理

僅供福邦投顧客戶使用

PCB製造端的製程變化-銅漿燒結

- 銅漿燒結製程源於瞬態液相燒結(TLPS)技術，主要用於實現PCB盲/埋孔電路層間的Z軸互連。
- 過去傳統高多層多採用背鑽的製程，但在NV陣營，開始嘗試使用”銅漿燒結“的製程來處理高多層版的訊號連接，以解決背鑽精度控制的問題。
- 目前具有【燒結製程+大板製造能力】的主要為板廠，例如**深南電路、景旺電子、勝宏科技、滬電股份、生益電子；台系則有金像電具備此製程能力。**

圖10:PCB銅燒結概念示意圖



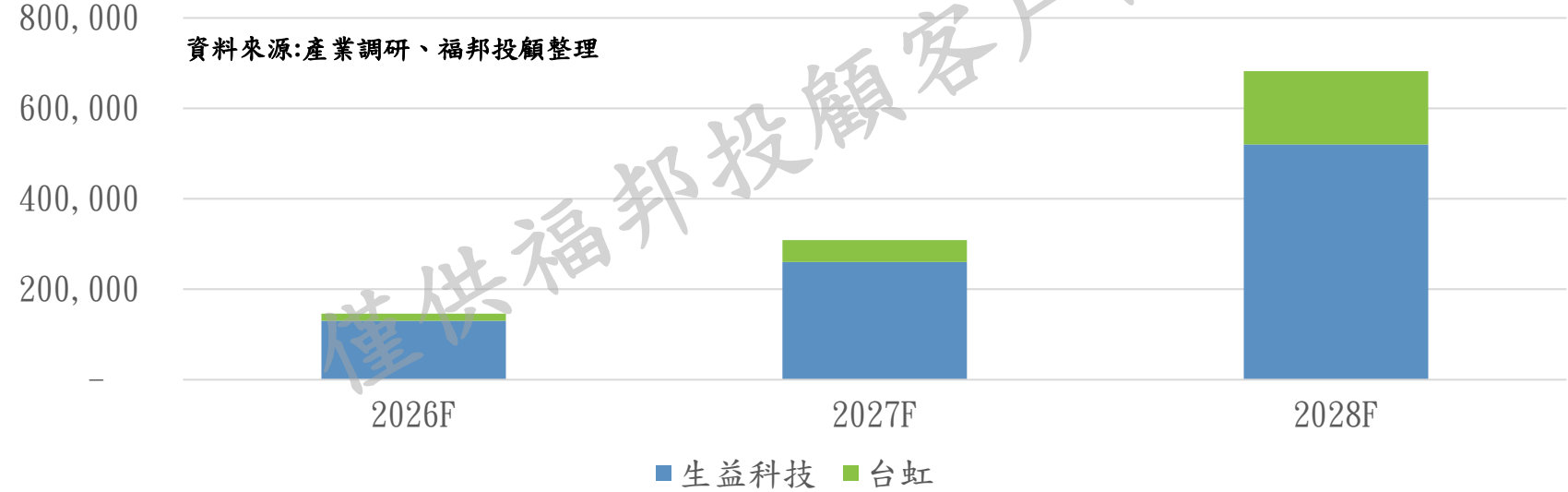
資料來源:產業調研、福邦投顧整理

僅供福邦投顧客戶使用

PTFE供應商產能預期

- 目前Nvidia的供應來看，主要會以生益科技為第一供應商(合作最久)，第二供應商則從非中國供應商中選擇，目前與生益科技與台虹入列的優先供應商，**生益科技具有CCL+PP(PTFE/CH)的製造能力，台虹PTFE CCL (PFA)可搭配生益、台光電等CCL的CH PP方案，亦能搭配自產的PFA PP，可提供客戶彈性的產能需求。**
- 其餘具有PTFE樹脂CCL開發能力的供應商，開發進度較落後，不在考量範圍內。
- PCB段的生產: 以目前客戶的意願上，採用**無布方案**為優先目標，PCB生產良率約在10~30%，主要量產生產廠商預期為深南電路、景旺電子、勝宏科技、滬電股份；欣興僅參與配合RD階段(台郡負責燒結部分,但設備有限制)。

圖:PTFE主要供應商產能(米/月)



僅供福邦投顧客戶使用

PTFE CCL 潛在使用需求預估

- 假設: Rubin Ultra 於2027~2028年的晶片需求量預估400萬、900萬顆，所有機櫃均採用18層SWITCH TRAY設計，PCB板材採用混壓結構下，PTFE於2027~2028年的月均需求26、63萬張。

表22:PTFE CCL的需求試算

	SwitchTray	
產品	Rubin Ultra(VR300)	
層數	16+16=32L	
PTFE per PCB (米)	3	
Yield rate	30%	
實際用量	10	
	2027F	2028F
機櫃需求	35,000	84,000
單櫃SW Tray數	9	9
SW Board 量(張)	315,000	756,000
PTFE per PCB(米)	10	10
PTFE CCL 年總用量(米)	3,150,000	7,560,000
月均量(米)	262,500	630,000
產值預估(USD)	91M/月	220M/月

資料來源:產業調研、福邦投顧整理

僅供福邦投顧客戶使用

OpenClaw:加速從LLM往AI Agent 發展

- OpenClaw:開放式AI助理項目，可以根據使用者指令在應用程式和線上服務之間自行處理複雜任務，並被設計為可以代替使用者執行任務的AI軟體，並非過往的聊天機器人。
- OpenAI推出GPT5.4，此次更新提供Thinking和Pro兩種模型； Pro主要適用於需要進行複雜任務的使用者，Thinking模型可以在產出答案前提出思考計畫，使用者可以從中調整任務方向→強化Agent能力，可以支持智慧裝置執行複雜的工作流程。
- NVIDIA將推出類似OpenClaw的開放式平台-NemoClaw。

表23：LLM 與 Agent 推理比較

	傳統LLM推理	AI Agent 推理
流程	輸入→LLM(GPU)→輸出	感知→規劃→推理→工具與執行(CPU)
運作本質	單向生成	循環決策
互動對象	與使用者進行Q&A	與環境/工具/API互動
對硬體影響	主要消耗GPU算力	CPU需要介入成為主要調度，其中涉及沙盒隔離、儲存、記憶，複雜工具調用，

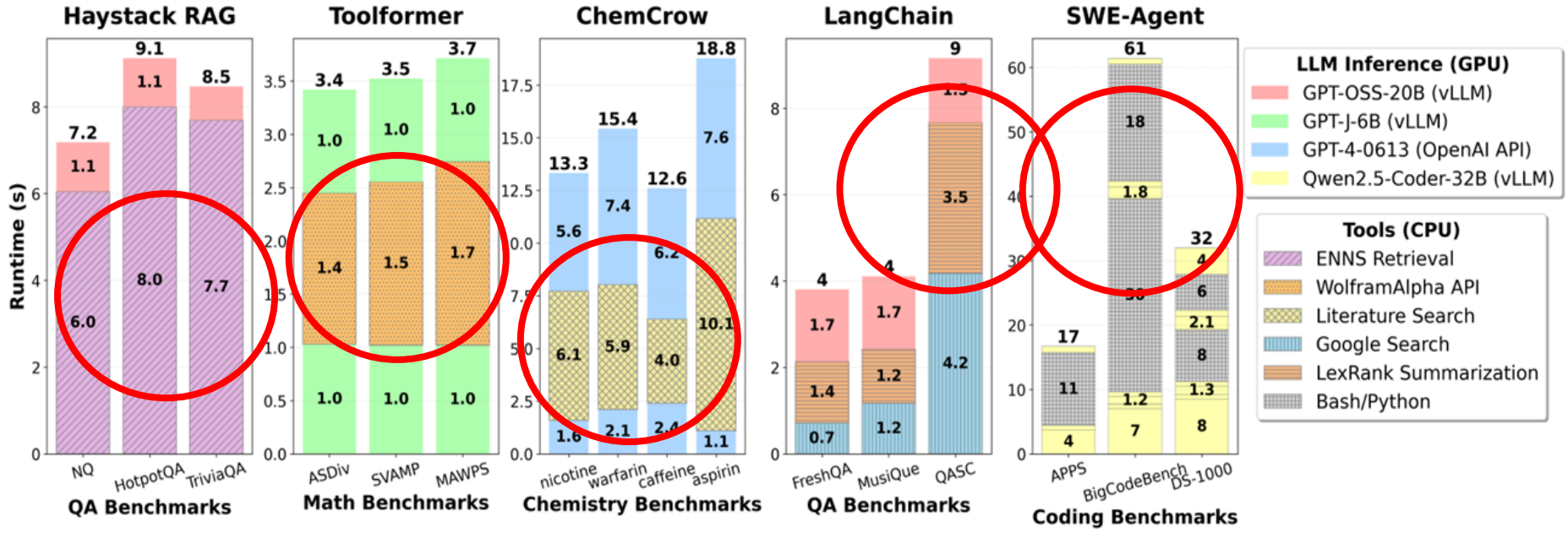
資料來源:產業調研、福邦投顧整理

僅供福邦投顧客戶使用

Agentic AI的瓶頸主要來自CPU

- 根據論文《A CPU-Centric Perspective on Agentic AI》一文指出，可以看出執行時間中CPU延遲占了大比例（紅圈），最高可達到90%以上；吞吐量瓶頸除了來自GPU，亦會來自CPU;能耗方面，大批次處理中(Batch size提升至128)，CPU平行處理效率低於GPU，佔到總能耗超過44%，延遲會從2.8秒提升至6秒以上，超過2倍。

圖11:Agent下，CPU佔據了大部分的延遲



資料來源:《A CPU-Centric Perspective on Agentic AI》 、福邦投顧整理

說明:Haystack RAG（問答檢索）**/Toolformer**（數學運算與 API 呼叫）**/ChemCrow**（化學文獻檢索）**Langchain**（網路搜尋與摘要）**/SWE-Agent**（程式碼與 Bash/Python 執行)

【僅供內部教育訓練使用，嚴禁外流】

CPU朝向多核發展→推升載板面積

- Agent執行:API處理、資料解析、腳本執行，90%由CPU完成，為了降低延遲，CPU往多核心發展是趨勢。
- 記憶體與上下文切換:CPU需要頻繁在不同Agent狀態做切換，多核心架構能讓每個Agent有獨立的處理單元，降低上下文切換帶來的效率損耗。
- 沙盒隔離/安全:Agent 需要大量獨立的一般伺服器建立Sandbox來執行，使用量與使用者和使用量正相關，需要更多核心數、高頻寬。

表24：伺服器CPU核心數

廠商	2023	2024	2025	2026	2027
Intel	Purley/Whitley	EagleStream	BirchStream	Oakstream	?
核心數	28/28	56	128	144~288	500?
AMD	Milan/Rome	Genoa/Bergamo	Turin	VENICE	VERANO
核心數	64/64	96/128	128~192	256	256~320?

資料來源:Intel、AMD、福邦投顧整理預估

【僅供內部教育訓練使用，嚴禁外流】

資料來源:各公司、福邦投顧整理

IC載板:主流晶片使用面積攀升

表25：主流晶片面積　　資料來源:各公司、福邦投顧整理

	產品名稱	推出時間	封裝方式	面積*層數	與前代的增幅
Intel	OAK Stream	2026	LGA	?	
	Birch Stream	2024H1	LGA	6,000*20	**+15%**
	Eagle Stream	2023	LGA	5,000*20	
AMD	**Venice**	**2026**	**FCLGA**	**6,100*26**	**+44%**
	Turin	2024Q4	FCLGA	5,500*20	-
	Genoa	2022Q4	FCLGA	5,500*20	
NV	**Rubin**	**2026~2027**	**CoWoS-L**	**9,000*18**	**+75%**
	Blackwell	2024~2025	CoWoS-L	5,900*14	**+116%**
	Hopper	2023~2024	CoWoS-L	3,200*12	
	Vera	2024~2025	CoWoS-R	5,700*18	**+46%**
	Grace	2023~2024		5,000*14	
Google	**HumuFish**	**2027**	**EMIB-T**	**14,000*24**	**+160%**
	Zebrafish	2026	CoWoS-S	5,300*24	**+22%**
	SunFish	2026	CoWoS-S	7,000*18	**+21%**
	GhostFish	2025	CoWoS-S	6,500*16	
Amazon	**Tranium4**	**2027**	**EMIB-T**	**14,000*24**	**+238%**
	Trainium3	2026	CoWoS-R	6,200*16	

僅供福邦投顧客戶使用

IC載板:產業能見度拉長

- 根據供應鏈訪查，目前主流載板對於供需的意見陸續在26H1看到供不應求，與之對應的生產相關輔材訂單能見度可至2028年。

表26：主流大廠對於供需的想法

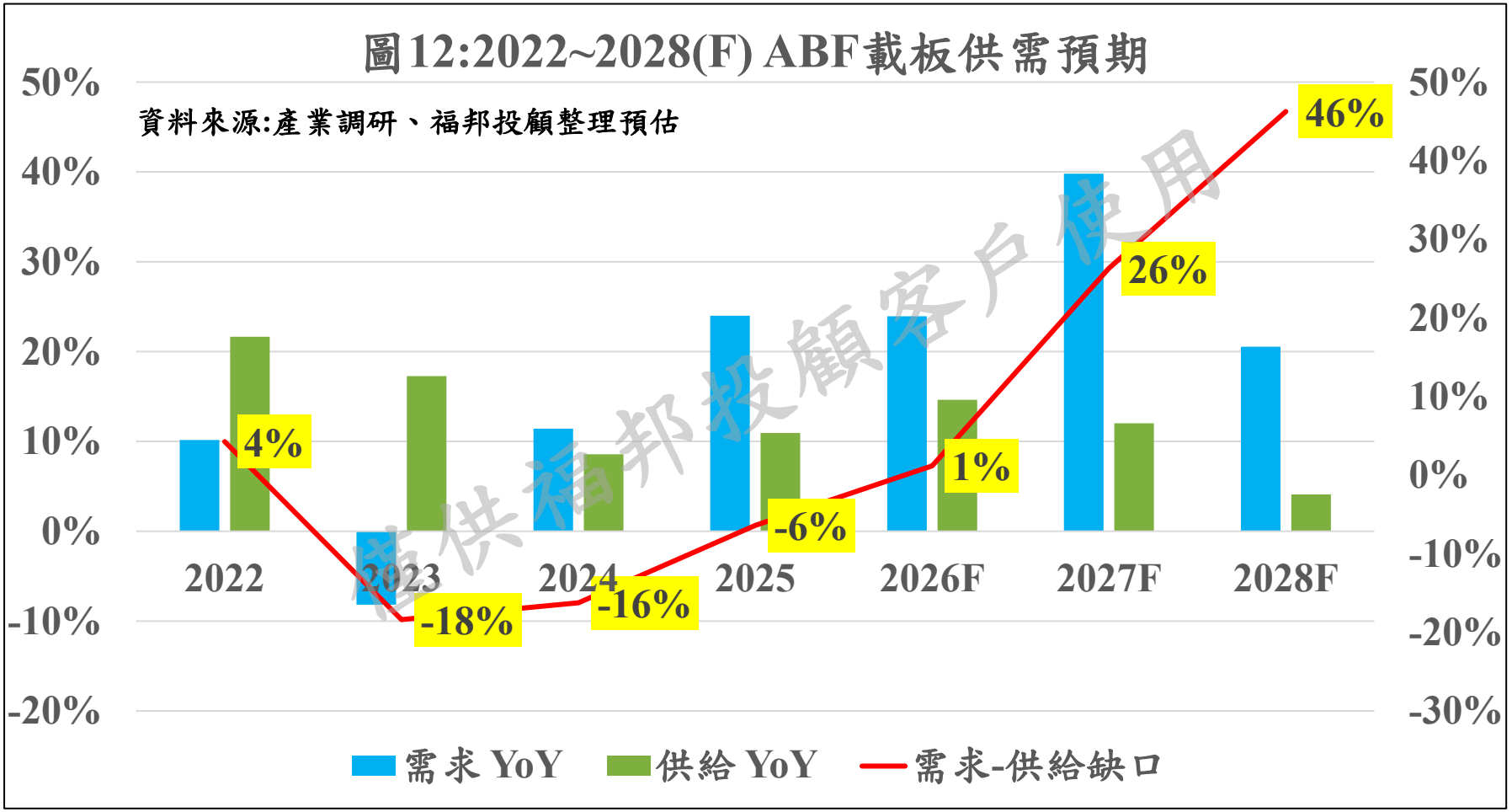
供應	公司	稼動率概況	對供需的想法	對應客戶
載板	Ibiden	AI相關滿載	2027年載板產能供不應求	NV/Intel/AMD
	Semco	26H2滿載	2027年載板產能供不應求	Intel/AMD/NV/AMZN
	AT&S	26Q1滿載	2027年載板產能供不應求	Intel/AMD/Marvell
	欣興	26Q1滿載	2026年玻纖布缺料，2027年供不應求	NV/Intel/AMD/Broadcom/AMZN
	南電	26Q1滿載	2027年載板產能供不應求，LowCTE缺料至2028年。	Broadcom(主要)/AMD
	景碩	26H2滿載	2027年載板產能供不應求，LowCTE缺料至2028年。	NV/AMD
生產輔材	鉅橡	26H2滿載	訂單能見度可至2028年	全球主要載板廠
上游材料	味之素	滿載	若需求持續暢旺，2027年產能嚴重不足：公司於2028年才有70%新產能開出	全球主要載板廠
	日東紡	滿載	2027年開始量產新產能。	全球主要載板廠
	Resonac	滿載	公司供給仍明顯低於需求，2027年有機會再次調價	全球主要載板廠

僅供福邦證券客戶使用

【僅供內部教育訓練使用，嚴禁外流】

IC載板:ABF載板供需將從2026年開始見到供不應求

- 根據福邦投顧預估，2025年ABF載板仍處於供過於求的格局，2026年開始在AI的需求強勢帶動影響下，ABF載板將開始見到供不應求格局，並於2027~28年逐步擴大。（**本圖不含缺T-glass影響**）



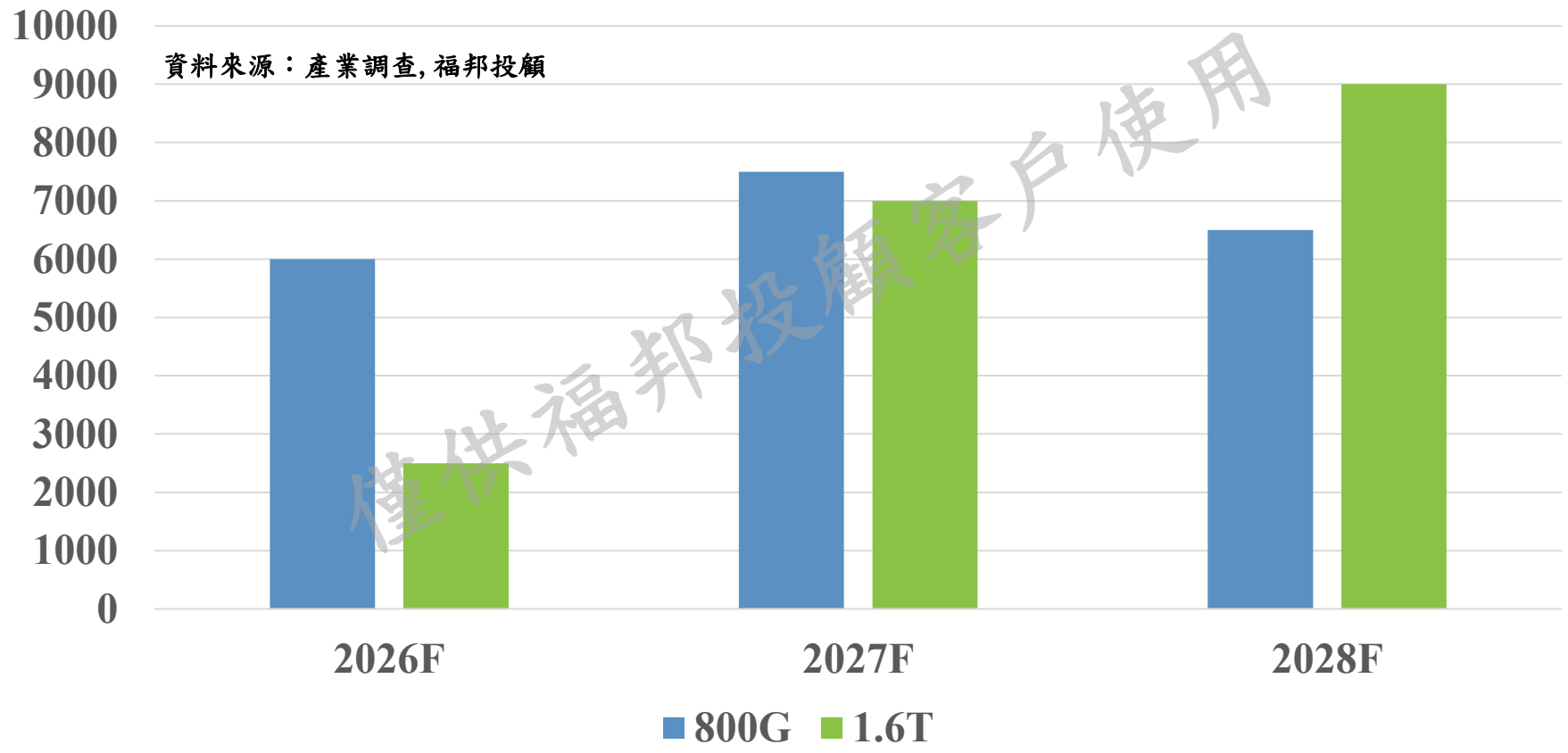
僅供福邦投顧客戶使用

可插拔光通訊模組:800G/1.6T潛在需求上升

■根據福邦投顧產業調查，2026/27/28年800G的採購”需求”可望達到6,000/7,500/6,500萬支，1.6T需求預估2,500/7,000/9,000萬支。

(隱含終端客戶對於良率的考量造成的Overbooking)

圖13: 2026~2028年光通訊模組需求預期(萬支)



僅供福邦投顧客戶使用

傳統光通訊模組板-HDI/mSAP供應體系的新機會

■因應高度整合和小型化需求，光模組開始大量採用MSAP來縮小產品體積和損耗；進入1.6T，L/S 要求微縮至**20μm-25μm**；因此1.6T開始，強制採用MSAP。

■材料規格上，亦往M7~M8等級以上升級，但在MSAP部分，會大量採用載體銅箔(三井金屬90%市佔率)、日東紡的Low CTE布，目前有面臨缺料的情況。

表27 :400G~3.2T PCB方案比較

速率	400G	800G	1.6T	3.2T
製程	HDI	HDI/mSAP	mSAP	mSAP
層數	10L	12~14L	14~18L	20-22L
L/S	40~50μm	30~40μm	20~30μm	10~15μm
CCL等級	M4/M6	M6/M7	M8 (斗山/台光電)	M9~M10+陶瓷基板?
ASP(USD)	~10	12→14~15	30→35	150+
毛利率	20~30%	40~50%	50~60%	70~80%
供應鏈	SCC、欣興、興森等	臻鼎/SCC/欣興/方正/景旺/華通/WUS等	臻鼎、SCC、欣興、華通等	臻鼎、SCC、欣興

資料來源：產業調查、福邦投顧

僅供福邦投顧客戶使用

傳統光通訊模組PCB供應商概況:27年可望緩解供應

表28 :主要供應商客戶與產能概況　　資料來源：產業調查、福邦投顧整

供應商	關鍵客戶	mSAP月產能規劃預期	客戶2027年月產能(支)
欣興	中:中際旭創 美:Coherent、Mellanox	26Y:7,500平米(=250~300萬支) **27Y: 10,000平米(=400萬支)**	■ 中際旭創: 800G/1.6T 125W/210W ■ 新易盛: 800G/1.6T 80W/35~40W ■ Coherent: 800G/1.6T 125W/125W ■ AAOI:90~100W ■ 華工科技: 800G/1.6T 30W/25W ■ 索爾斯:80W
深南電路	中:中際旭創、新易盛 美:Coherent	26H1:3,200平米(=120~130萬支) 26H2:6,000平米(=240萬支) **2027Y:8,000平米(=320萬支)**	
臻鼎	中:中際旭創	26Y: 2,800平米(=110萬支) **27Y:10,000平米(=400萬支)**	
華通	中:中際旭創 美:AAOI、Coherent	27Y:7,500平米(=300萬支) **28Y: 8,000~8,500平米(=330~340萬支)**	
方正科技	中:新易盛、 華工科技、光迅科技	26Y:1,500~2,000平米(=80萬支) **27Y:5,000平米(=200萬支)**	
景旺電子	中:中際旭創、新易盛 美: Coherent	26Y:1000平米(=40萬支) **27Y:10,000平米(=400萬支)**	
興森科技	中:華工科技、光迅科技	28Y:擴產10,000平米/月產能	
勝宏科技	潛在客戶:中際旭創	規劃20,000平米/月	
AT&S	中 ： 中際旭創 美:Coherent、Mellanox	26Y:2800片/月	

僅供福邦投顧客戶使用

雲端資料中心採用之交換器品牌

■雲端資料中心多採用品牌+白牌的混合方案

■Spine和DCI 因為結構複雜，所以多採用品牌廠，但Google和Amazon 由於架構和軟體上較強，所以選擇白牌來配合他們的結構。

■TOR和 Leaf 結構簡單，可以看到較多白牌(裸機、白盒)的方案。

■xAI的交換機品牌主要採用Juniper的方案，主權AI體系(ex:星際之門、中東等)多會採用Cisco等品牌交換機。

表29:各大雲廠商交換機廠商彙整

	TOR(機櫃)	Leaf（接入）	Spine（匯聚）	DCI（核心）
GOOGLE	白牌(CLS/偉創力)	白牌(CLS/偉創力)	白牌(CLS/偉創力)	白牌/Juniper/CSCO
Amazon(AWS)	白牌(CLS/Accton/CSCO)	白牌(CLS/Accton)	白牌(CLS/Accton)	Juniper
微軟(Azure)	Arista/CSCO/Dell	Arista/CSCO/Dell	Arista	Arista
Meta	白牌(CLS/智邦/CSCO)	Arista/Edgecore（智邦）	Arista/Edgecore（智邦）	Arista/Juniper/白牌
百度	白牌/H3C	華為/H3C	華為/H3C	華為/H3C
阿里巴巴	思科/H3C/銳捷/華為/白牌	思科/H3C/銳捷/華為/白牌	思科/H3C/銳捷/華為/白牌	華為/H3C
騰訊	思科/H3C/銳捷/華為/白牌	思科/H3C/銳捷/華為	華為/H3C	華為/H3C
APPLE	Arista/CSCO/白牌	Arista/CSCO	Arista	CSCO/Juniper

資料來源：650 GROUP、福邦投顧整理

【僅供內部教育訓練使用，嚴禁外流】

交換機板:主流AI層數較低，傳統乙太網多層為主

■ 資料中心需要承載流量大+高速傳輸，在此需求下，高多層板成為此需求的解決方案，其優點在於電子元件間連線縮短下，訊號傳輸速度提高，而且也方遍佈線。

■乙太網交換機為各家CSP+主權主要Scale out的主流，800G報價是400G的2倍以上，1.6T是800G的4倍以上。

表30 :交換機供應鏈劃分

板層數劃分		產品	銅箔基板主要供應商	PCB主要供應商	出貨價格
Cisco	400G:24~36L PTH 800G:38~48L PTH (NPO:28L) 1.6T:46~52L PTH / 18~20L HDI	800G	台光電/松下/Doosan	金像電、TTM、滬士電、ISU	400G :1500 800G:4,000 1.6T:16,000+ (美金/平方米)
		1.6T	台光電/松下		
Arista		800G	台光電、Doosan	TTM、滬士電、金像電、ISU	
		1.6T	台燿、台光電		
Celestica		800G	台光電/Doosan/松下/台燿	金像電、ISU	
		1.6T	台光電/松下/台燿	金像電、ISU	
Accton		800G	Doosan、台光電	TTM、ISU、LCS	
		1.6T	台光電	TTM、ISU、LCS 金像電	
Mellanox		800G	台光電	TTM、滬士電	
		1.6T	台光電	TTM、滬士電	

福邦投顧產業調查

交換機的材料

■在交換機升級的過程中，由於高速傳輸需求，對於耗損也是嚴格規範，**800G材料價格則會較400G材料增加60%+，Tomahawk6 將會使用於2026年的1.6T交換機可使用材料約M8U~M9-(碳氫+LK2)等級，2027年後的規格可望往M9發展。**

表31:各速率材料概況

	100G	400G、**800G**	800G、1.6T	1.6T~2.4T	3.2T~
材料耗損等級	Very low loss	Ultra low loss	Extreme Ultra low loss	Super low loss	?
Df	0.005~0.006	~0.003	~0.002	~0.001	~0.0003
頻率	13Ghz(NRZ)	13Ghz(PAM4)	28Ghz(PAM4)	56Ghz(PAM4)	100+Ghz?
設計	PPO	PPO+Low K Glass	PPO+LowK/K2 +VLP2~4	PPO+CH樹脂+ VLP4 +Q/LK3/LK2	CH/PTFE樹脂+LK4/5 +VLP4/5
平均單價/張	2,000~3,000	**3,000~4,000**	**5.000~6,000**	6,000~7,000/ 10,000	12,000+
代表型號	松下M6	松下M7	松下M8	松下M9	松下M10
競爭同業產品儲備	EM-528/626	EM-528K/EM-890K	EM-892K/K2	EM-896K3	EM-898K4
	TU883	ThunderClad2/3	ThunderClad4	ThunderClad 5	?
	IT 968/968G	IT988G	IT998GSE/2	IT999GSE3	ITQ10G(K4)
	NP186/188U,H	NPG188UK/199	NPG199K/HK	NPG-998Q	NPG-999Q
	Synamic 6/6N	Synamic 8GN	Synamic 9GN	Synamic 10/11GQ	SG5300N
	DS7409DV（N)	DS7409DJG(N)	DS7409DQN/N2/T2	DS7409DYQ/L2/N	?
	NY6300S	NY-P1~3	NY-P4	NY-P5	NY-P6

資料來源：各公司，福邦投顧整理

謝謝指教
Q&A

【揭露事項與免責聲明】

本報告僅提供相關部門的內部教育訓練及相關人員之參考資料，並非針對特定客戶所作的投資建議，且在本報告撰寫過程中，並未考量讀者個別的財務狀況與需求，故本報告所提供的資訊無法適用於所有讀者。本報告係根據本公司所取得的資訊加以彙集及研究分析，本公司並不保證各項資訊之完整性及正確性。本報告中所提出之意見係為本報告出版當時的意見，邇後相關資訊或意見若有變更，本公司將不會另行通知。本公司亦無義務持續更新本報告之內容或追蹤研究本報告所涵蓋之主題。本報告中提及的標的價格、價值及收益隨時可能因各種本公司無法控制之政治、經濟、市場等因素而產生變化。本報告中之各項預測，均係基於對目前所得資訊作合理假設下所完成，所以並不必然實現。本報告不得視為買賣有價證券或其他金融商品的要約或要約之引誘。

可能個別基於特定目的且針對特定人士出具研究報告、提供口頭或書面的市場看法或投資建議（下稱“提供資訊”），鑑於提供資訊之單位、時間、對象及目的不同，本報告與本集團其他單位所提供資訊可能有不一致或相牴觸之情事；本集團各單位對於本報告所涵蓋之標的可能有投資或其他業務往來關係，各單位從事交易之方向亦可能與本報告不一致，讀者應審慎評估自身投資風險，自行決定投資方針，不應以前述不一致或相抵觸為由，主張本公司或本集團其他成員有侵害讀者權益之情事。

【富邦法人】台積電(2330)外資報告彙整

- **富邦&外資 preview 彙整**

券商	Rating	TP	2026F EPS	2027F EPS	2028F EPS	摘要
富邦 (10/12)	Buy	$3,500	$112.44 ↑ (from $108.80)	$161.21 ↑ (from $155.41)	$207.9	- 預期毛利率將超越先前指引，估 **3Q26 毛利率達 67.6%，EPS 估 29.97 元**。 - 預期管理層將指引 **4Q26 營收季增 10%至 12%**，毛利率則維持持平。受惠於下半年強勁表現，預估全年營收將年增 45%至 46%，有望再次超越原先指引。 - 資本支出方面，預期台積電維持全年 620 億美元的指引，估計 2027 年及 2028 年資本支出將大幅提升至 850 億及 1,000 億美元。 - 2027 年供應缺口將進一步擴大，因此認為台積電將於 2027 年再次調漲晶圓價格，平均漲幅超過 10%，其中先進製程節點漲價 10-15%，成熟製程節點漲價 5-10%。隨著 ASP 正向趨勢，認為 FY27 毛利率有望進一步攀升至 70%。
MS (10/5)	OW	$3,088 ↑ (from $2,988)	$115.32 ↑ (from $111.08)	$152.74↑ (from $144.93)	$189.91↑ (from $183.55)	- 預估 **3Q26 毛利率 67.5%，EPS 30.74 元**。 - 預估 **4Q26 營收季增 12.5%，毛利率 68.6%，EPS 估 35.25 元**。 - 預期法說會將全年營收成長指引上調至 low-to-mid-40%，較目前超過 40%的指引進一步提高，意味第 4 季營收可望季增 10%至 15%，主要受強勁的 3 奈米需求帶動。 - 預估 2026 年資本支出維持 540 億美元，2027 年則預估達 750~800 億美元。 - 2027 年晶圓價格方面，預期 7 奈米以下先進製程價格將上漲 5%至 10%，成熟製程則上漲 0%至 3%，帶動台積電 2027 年毛利率有機會接近 70%。
GS (10/5)	Buy	$3,300 ↑ (from $3,100)	$110.77 ↑ (from $109.21)	$150 ↑ (from $140.91)	$195.67 ↑ (from $180.83)	- 預估 **3Q26 毛利率 67.5%，EPS 29.07 元**。 - 預估 **4Q26 營收季增 11.0%**，AI 及 HPC 需求持續強勁，但 N2 製程持續擴產將帶來些微毛利率稀釋，**預估 67.3%，EPS 估 32.38 元**。 - 2027 年毛利率預估小幅提升至 67.5%，受惠於價格提升、有利產品組合、高稼動率及持續生產效率改善。 - 2026 年資本支出預估維持 640 億美元，但 2027 年及 2028 年分別上調至 850 億美元及 980 億美元，主要反映設備供應商成本上升，以及可能新增德州擴廠的初期支出。預估 2027 年底及 2028 年底 N3/N2 產能分別達到每月 20 萬片/14 萬片及 22 萬片/20 萬片。

Citi (10/7)	Buy	$4,000 ↑ (from $3,800)	$110.72 ↓ (from $110.73)	$158.30 ↑ (from $151.93)	$217.02 ↑ (from $207.24)	- 預估 **3Q26 毛利率 67.4%，EPS 33.16 元**。 - 預估 **4Q26 營收季增 16%，毛利率 66.7%，EPS 估 28.24 元**。 - 台積電在 AI 運算需求強勁、Agentic AI 潛在上行空間及 CPO 帶動新一輪網路設備週期的支持下，預期 2027 年營收可望年增超過 40%。隨著 N2 全年貢獻增加，以及 AI 運算工作負載自 2026 年下半年起逐步轉向 N3，並於 2028 年進一步轉向 N2 及 A16。 - 儘管近期市場傳出台積電可能與德州 Terafab 合作，但台積電目前已規劃亞利桑那州 6 期先進製程晶圓廠及 2 期先進封裝產能，管理層過去也曾表示，長期目標是讓亞利桑那州具備約 30%的先進製程產能，以滿足美國需求。第二及第三期預計於 2027 年底至 2028 年開始量產 N3 及 N2，第四至第六期則預計逐步建置至 2030 年，因此未來幾年暫不將德州新增產能及投資納入預估。
JPM (10/8)	OW	$3,300	$108.62 ↑ (from $108.15)	$147.80 ↑ (from $141.37)	$190.42 ↑ (from $180.01)	- 預估 **3Q26 毛利率 66.8%**，主要受惠於 N3/N5 產能利用率維持高檔及晶圓漲價，但部分受到 N2 初期量產、海外晶圓廠稀釋效應及下半年新台幣升值影響。**EPS 估 27.92 元**。 - 估 **4Q26 營收季增 11%，毛利率 66.8%，EPS 31.37 元**。 - 估 2026 年美元營收將年增 41%，新台幣營收年增 44%，全年毛利率約 66.9%。 - 預估 N3 產能將於 2026 至 2028 年底分別達到每月 17.8 萬、22.3 萬及 25 萬片。N2 方面，台積電為避免重演 N3 供不應求的情況，正加速擴產，預估 2027 及 2028 年底產能將分別達每月 11.8 萬及 21 萬片。此外，A14 進度優於原先預期，預估最快可於 2028 年下半年量產，早於市場普遍預期的 2029 年。 - 預估 2026 至 2028 年資本支出將分別達 640 億、860 億及 1,000 億美元，其中 2028 年仍有上修空間。 - 預期 2027 年漲價將涵蓋多數製程，先進製程漲幅約 10%，N7 及以上成熟製程則調漲低至中個位數百分比。預估 2027 年整體平均售價將年增 25%，高於 2026 年預估的 17%。

UBS (9/28)	Buy	$3,650	$109.06 ↑ (from $108.96)	$148.98 ↓ (from $149.14)	$202.52 ↑ (from $198.23)	- 預估 **3Q26 毛利率 67.1%，EPS 28.47 元**。 - 預估 **4Q26 營收季增 10%**，受惠於旗艦手機季節性備貨及 HPC 需求加速放量，但考量 N2 初期量產及海外擴廠的成本稀釋效應，**毛利率預估持平於 67.0%，EPS 估 31.26 元**。 - 雲端 AI 長期趨勢持續推動台積電營收成長，預估 2026 年營收將年增逾 40%，2027 年也可望維持約 40%的成長。近期產業調查顯示，雲端 AI 需求尚未出現降溫跡象，且 2027 至 2028 年 CPU、GPU 及 ASIC 需求展望進一步轉強。 - 資本支出方面，維持 2026 年預估 630 億美元，並將 2027 及 2028 年預估由原先的 800 億及 950 億美元，上調至 900 億及 1,050 億美元。主要考量台積電加速擴充 N3 及 N2 產能，並提前 A14 設備交付時程。將 2028 年底 N3 及 N2 月產能預估分別上調至 22 萬及 21 萬片。
廣發 (10/7)	Hold	$2,960 ↑ (from $2,700)	$110.9 ↑ (from $108.2)	$148.7 ↑ (from $138.3)	$184.6	- 預估 **3Q26 毛利率 67.8%，EPS 29.2 元**。 - 預估 **4Q26 營收季增 12%，毛利率小幅降至 67.2%，EPS 32.4 元**。 預計維持 2026 Capex 指引；考慮設備通脹，上調 2027E/2028E Capex 至 US$83bn/95bn。 - 台積電可能與 Terafab 合作的敘事出現變化，此前報道台積電負責運營晶圓廠或成立 JV，及為 Tesla/SpaceX 提供專屬產能；但 Elon Musk 10/7 表示 Tesla/SpaceX 將自行“build and run the fab”，認為市場預期將重新調整，而 Intel 仍參與 Terafab 項目。

- **上季法說重點(2026/7/16)**

2Q26 法說會重點摘要	
Revenue	**- 3Q26 指引：US$44.6-45.8bn(中值 QoQ+12%、YoY+37%)；匯率假設為 USD/TWD 32** **- 將 2026 年全年美元營收成長預估上調至略高於 40%**
GM	**- 3Q26 指引：65-67%，QoQ-1.7ppts** **- 2Q26 毛利率 67.7%，QoQ+1.5ppts 並略優於公司財測，成本改善措施與整體稼動率略高於預期，但部分被海外廠初期營運稀釋抵銷** **- 2nm 將於 2H26 快速 Ramp-Up，稀釋毛利率 3~4%** **- 海外廠未來幾年前期每年約稀釋公司毛利率 2~3%，後期擴大到 3~4%**
OP Margin	**- 3Q26 指引：56-58%**
2Q26 product mix	**- N2 3%；N3 30%；N5 33%；N7 11%；先進製程技術(N7 及以下)占晶圓營收 77%** **- HPC 66%(QoQ+20%)；Smartphone 22%(QoQ-4%)；IoT 5%(QoQ+4%)；Automotive 4%(QoQ+15%)；DCE 1%(QoQ+5%)**

CAPEX&折舊	**- 2026 年 CAPEX 上調至 US$60-64bn(原為 US$52-56bn，接近上緣)；70~80% 於先進製程，10% 特殊製程、10~20% 先進封裝、測試光罩** **- 資本支出通常對應未來數年更高的成長機會，目前看未來三年 CAPEX 會比過去三年再更為強勁** **- 已提前與設備供應商規畫產能，目前不預期設備供應會成為擴產計畫的重大瓶頸** **- 宣布將在亞利桑那追加 US$100bn 投資(總額提高至 US$265bn)，新增數座 N2 及更先進製程邏輯晶圓廠、先進封裝廠**
AI	**- 先前提出的 2024-29 年 AI 相關營收「中高 50% CAGR」展望，管理層表示需求已變得「更強、更強、再更強」，因為展望持續在調高所以不公布數字，實際趨勢高於先前說法**
先進封裝	**- 目前主流方案仍為 CoWoS，同時持續開發其他替代技術以降低封裝成本** **- 對 Intel EMIB-T 等先進封裝競爭，管理層認為後段和前段是完全不同等級，競爭者取得封裝訂單不代表可切入前段邏輯晶圓**

13 Oct 2026 07:30:25 ET | 36 pages

Voltronic Power (6409.TW)

Recharging Growth: UPS Recovery and HVDC Potential; Initiate at Buy

CITI'S TAKE

We initiate coverage on Voltronic with a Buy rating and NT$1,050 target price. Voltronic is a Taiwan-based UPS ODM, supplying to western power vendors including Schneider, Riello, Vertiv and Eaton. We view 2026E as the earnings trough, with improving online UPS mix and operating leverage driving 28% earnings rebound in 2027E. Structural outsourcing by western power vendors should support sustained share gains, while expansion into HVDC power racks offers a new stream for AIDC growth, supporting a potential re-rating. Our TP applies 25x 2027E EPS against earnings CAGR of 20% in 2026-2028E, broadly consistent with its L-T historical average.

Outsourcing and mix upgrade underpin the recovery — Western vendors' increasing focus on integrated AIDC infrastructure strengthens the rationale for outsourcing select UPS platforms. Voltronic's customization capabilities and flexible manufacturing position it to capture additional opportunities. A rising contribution from online and higher power UPS should support margins, driving EPS CAGR of 20% in 2026-2028E.

Power racks could materially broaden its growth opportunity — Voltronic's high voltage EV-charger experience and longstanding relationships with western power vendors support its entry into HVDC power solutions. It has progressed into customer sampling, with management indicating that HVDC power rack mass production could begin in 2H27E at the earliest. Successful commercialization could broaden its addressable market and support a valuation re-rating.

Earnings recovery and strong cash returns — We view 2026E as the earnings trough, followed by a recovery in 2027-2028E. A richer online UPS mix and operating leverage should lift GPM from 27.1% in 2026E to 28.1% and 29.0% in 2027E and 2028E, respectively. Its relatively asset-light business model, net cash position and high dividend payout should provide additional support for shareholder returns.

Initiate Buy with TP NT$1,050 on 25x 2027E EPS — Our target multiple is broadly consistent with Voltronic's average forward P/E since 2013, supported by its competitive position in UPS outsourcing and durable customer relationships. Clearer evidence of power-rack commercialization could offer further re-rating.

Buy	
Price (13 Oct 26 13:30)	NT$854.00
Target price	NT$1,050.00
Expected share price return	23.0%
Expected dividend yield	3.5%
Expected total return	26.5%
Market Cap	NT$75,369M
	US$2,359M

Price Performance
(RIC: 6409.TW, BB: 6409 TT)

Earnings Summary

Year to 31 Dec	Net Profit (NT$M)	Diluted EPS (NT$)	EPS growth (%)	P/E (x)	P/B (x)	ROE (%)	Yield (%)
2024A	4,204	48.13	15.9	17.7	7.5	46.0	4.8
2025A	3,521	40.23	-16.4	21.2	7.8	36.1	4.1
2026E	2,894	33.00	-18.0	25.9	10.4	34.5	3.4
2027E	3,710	42.30	28.2	20.2	9.7	49.7	4.3
2028E	4,144	47.25	11.7	18.1	9.1	51.9	4.8

Source: Powered by dataCentral

Angela Hsu [AC]
+886-2-8726-9083
angela.hc.hsu@citi.com

See Appendix A-1 for Analyst Certification, Important Disclosures and Research Analyst Affiliations.

6409.TW: Fiscal year end 31-Dec

Price: NT$854.00; TP: NT$1,050.00; Market Cap: NT$75,369m; Recomm: Buy

Profit & Loss (NT$m)	2024	2025	2026E	2027E	2028E
Sales revenue	22,813	20,521	20,374	21,913	23,378
Cost of sales	-15,790	-14,561	-14,860	-15,758	-16,609
Gross profit	7,023	5,960	5,514	6,155	6,768
Gross Margin (%)	30.8	29.0	27.1	28.1	29.0
EBITDA (Adj)	**5,274**	**4,491**	**3,730**	**4,704**	**5,208**
EBITDA Margin (Adj) (%)	23.1	21.9	18.3	21.5	22.3
Depreciation	-349	-313	-295	-262	-225
Amortisation	0	0	0	0	0
EBIT (Adj)	**5,115**	**4,329**	**4,015**	**4,569**	**5,093**
EBIT Margin (Adj) (%)	22.4	21.1	19.7	20.9	21.8
Net interest	164	139	130	127	127
Associates	0	0	0	0	0
Non-Op/Except/Other Adj	-189	-151	-579	-127	-110
Pre-tax profit	**5,090**	**4,317**	**3,566**	**4,569**	**5,110**
Tax	-886	-796	-672	-859	-967
Extraord./Min.Int./Pref.div.	0	0	0	0	0
Reported net profit	**4,204**	**3,521**	**2,894**	**3,710**	**4,144**
Net Margin (%)	18.4	17.2	14.2	16.9	17.7
Core NPAT	4,204	3,521	2,894	3,710	4,144

Per share data	2024	2025	2026E	2027E	2028E
Reported EPS ($)	48.13	40.23	33.00	42.30	47.25
Core EPS ($)	48.13	40.23	33.00	42.30	47.25
DPS ($)	41.00	35.00	28.71	36.80	41.10
CFPS ($)	50.41	52.75	23.72	43.92	48.63
FCFPS ($)	49.41	52.07	22.65	43.01	47.72
BVPS ($)	113.66	109.16	82.39	87.89	94.04
Wtd avg ord shares (m)	87.4	87.5	87.7	87.7	87.7
Wtd avg diluted shares (m)	87.4	87.5	87.7	87.7	87.7

Growth rates	2024	2025	2026E	2027E	2028E
Sales revenue (%)	20.4	-10.0	-0.7	7.6	6.7
EBIT (Adj) (%)	23.6	-15.4	-7.3	13.8	11.5
Core NPAT (%)	16.1	-16.3	-17.8	28.2	11.7
Core EPS (%)	15.9	-16.4	-18.0	28.2	11.7

Balance Sheet (NT$m)	2024	2025	2026E	2027E	2028E
Cash & cash equiv.	6,556	6,738	6,685	7,230	7,918
Accounts receivables	3,450	3,387	2,896	3,125	3,348
Inventory	2,093	1,796	1,723	1,830	1,947
Net fixed & other tangibles	4,390	4,172	4,083	3,901	3,757
Goodwill & intangibles	446	535	701	701	593
Financial & other assets	386	456	422	422	422
Total assets	**17,321**	**17,084**	**16,511**	**17,209**	**17,986**
Accounts payable	4,350	4,796	3,485	3,701	3,939
Short-term debt	0	0	0	0	0
Long-term debt	636	538	489	489	489
Provisions & other liab	2,406	2,197	5,310	5,310	5,310
Total liabilities	**7,392**	**7,531**	**9,285**	**9,501**	**9,739**
Shareholders' equity	9,929	9,553	7,226	7,708	8,247
Minority interests	0	0	0	0	0
Total equity	**9,929**	**9,553**	**7,226**	**7,708**	**8,247**
Net debt (Adj)	**-5,920**	**-6,200**	**-6,196**	**-6,741**	**-7,429**
Net debt to equity (Adj) (%)	-59.6	-64.9	-85.7	-87.4	-90.1

Valuation ratios	2024	2025	2026E	2027E	2028E
PE (x)	17.7	21.2	25.9	20.2	18.1
PB (x)	7.5	7.8	10.4	9.7	9.1
EV/EBITDA (x)	13.3	15.4	18.5	14.6	13.1
FCF yield (%)	5.8	6.1	2.7	5.0	5.6
Dividend yield (%)	4.8	4.1	3.4	4.3	4.8
Payout ratio (%)	85	87	87	87	87
ROE (%)	46.0	36.1	34.5	49.7	51.9

Cashflow (NT$m)	2024	2025	2026E	2027E	2028E
EBITDA	5,464	4,641	4,309	4,830	5,318
Working capital	-413	786	-773	-119	-103
Other	-647	-811	-1,456	-859	-950
Operating cashflow	**4,404**	**4,617**	**2,080**	**3,852**	**4,265**
Capex	-87	-60	-93	-80	-80
Net acq/disposals	-17	-333	-113	0	0
Other	0	-2	-7	0	108
Investing cashflow	**-104**	**-395**	**-214**	**na**	**28**
Dividends paid	-3,290	-3,772	-2,518	-3,227	-3,605
Financing cashflow	**-3,488**	**-3,944**	**-2,610**	**-3,227**	**-3,605**
Net change in cash	**1,512**	**-141**	**-185**	**545**	**688**
Free cashflow to s/holders	**4,317**	**4,556**	**1,987**	**3,772**	**4,185**

For definitions of the items in this table, please click here.

Contents

Executive Summary 4
Valuation Comparison 7
Investment Thesis 8
- Positive #1 – AIDC creates a new growth runway, with power-rack expansion potentially lifting Voltronic into broader power solutions 8
- Positive #2 – UPS outsourcing provides a structural share-gain opportunity 10
- Positive #3 – Online and high-power UPS mix upgrade should support structurally better growth 14

Financial Analysis 17
Valuation Methodology and Risks 22
Company Overview 24
Bull/Bear: Voltronic Power (6409.TW) 27
Voltronic Power 28
- Company description 28
- Investment strategy 28
- Valuation 28
- Risks 28

Appendix A-1 28

Executive Summary

We initiate coverage of Voltronic with a Buy rating and target price of NT$1,050 (25x 2027E EPS), supported by structural UPS outsourcing, improving product mix and emerging opportunities in AI datacenter power infrastructure. Voltronic is a Taiwan-based ODM of uninterrupted power supplies ("UPS") and power conversion products for global brands. UPS system provides backup electricity and protection against power disturbances, helping critical equipment maintain operation during supply disruptions. UPS accounted for 63% of Voltronic's 2025 group sales. Its pure-play ODM model, customization capabilities and longstanding customer relationships underpin its competitive positioning.

Our positive stance on the shares is based on three major reasons:

1. We see expansion into HVDC power racks as a potentially significant new growth opportunity. Its experience in high-voltage EV chargers provides relevant engineering capabilities, while its established relationships with western power vendors support access to customer development programs. Voltronic has progressed into customer sampling, with management indicating that HVDC power rack mass production could begin in 2H27E at the earliest. Successful commercialization could broaden its addressable market and support a valuation re-rating, although meaningful earnings contribution remains subject to qualification, design wins and production schedules.

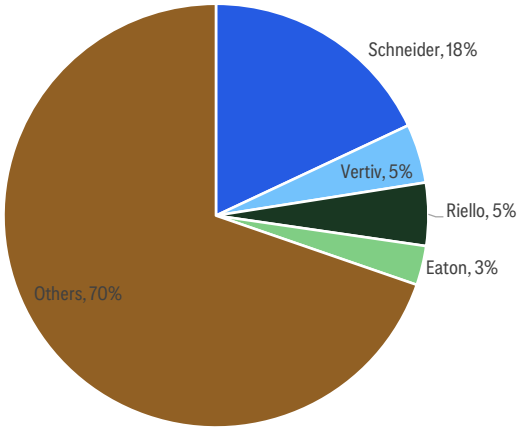
2. Structural UPS outsourcing should support sustained market share gains. Western power vendors are allocating more resources to integrated AIDC infrastructure and energy management, strengthening the rationale for outsourcing selected products. Voltronic's engineering expertise, flexible manufacturing and ability to handle a broad range of customized products in small batches make it well positioned to capture these opportunities.

3. The shift toward online and higher power UPS should improve revenue quality and profitability. We expect online UPS to be the principal driver of the earnings recovery, with online UPS sales growing 12% CAGR in 2026-2028E as outsourcing opportunities and demand for higher power systems expand.

We view 2026E as the earnings trough, followed by a recovery in 2027-2028E. We forecast EPS of NT$33.00/NT$42.30/NT$47.25 in 2026E/2027E/2028E, implying earnings growth of 28% and 12% in 2027E and 2028E. A richer online UPS mix and operating leverage should lift GPM from 27.1% in 2026E to 28.1% and 29.0% in 2027E and 2028E, respectively. Its relatively asset-light business model, net cash position and high dividend payout should provide additional support for shareholder returns.

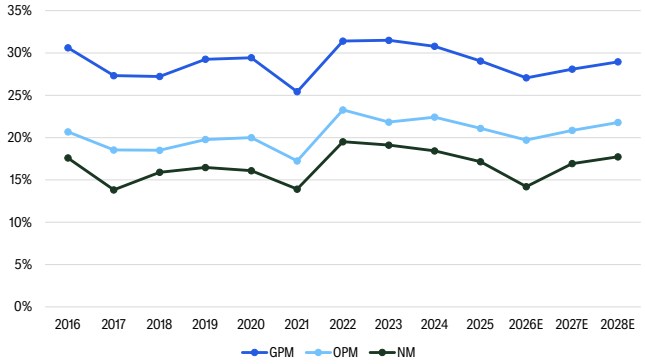
Our target price applies 25x forward P/E to 2027E EPS, broadly consistent with Voltronic's long tern average forward P/E since 2013. We believe its competitive position in UPS outsourcing, durable customer relationships and manufacturing capabilities justify this multiple, with clearer evidence of power rack commercialization offering potential for further re-rating.

Figure 1. Voltronic: Sales Mix



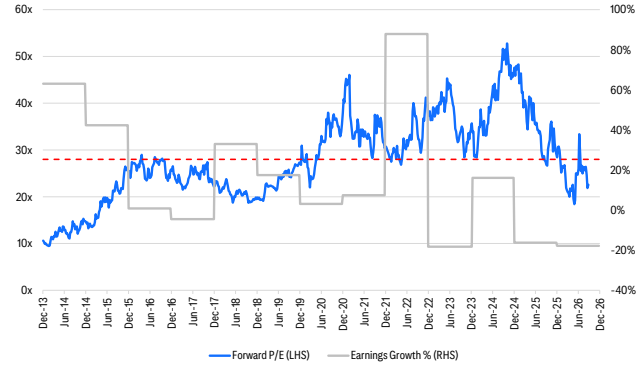
© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Company Reports, Citi Research, Citi Research Estimates

Figure 2. Voltronic: Margins Trend



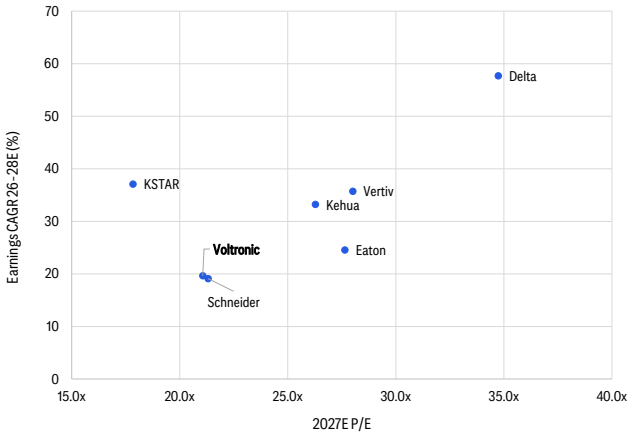
© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Company Reports, Citi Research, Citi Research Estimates

Figure 3. Voltronic: Forward P/E vs. Earnings Growth



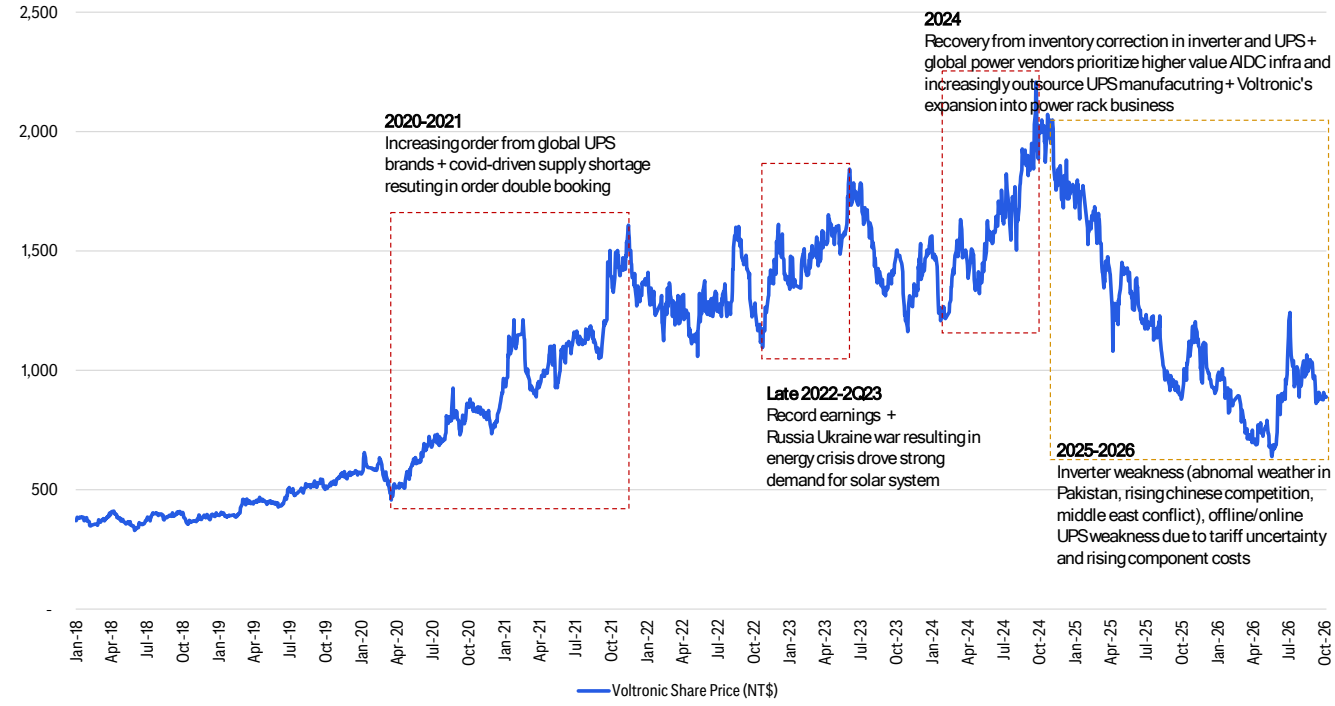
© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, Citi Research Estimates

Figure 4. Peers Comp: Forward P/E vs. Earnings CAGR



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Bloomberg, Citi Research, Citi Research Estimates

Figure 5. Voltronic: Historical Share Price Trend and Key Drivers



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: TEJ , Citi Research

Valuation Comparison

Figure 6. UPS Valuation Comp

Company	RIC	Rating*	Curr	Target Price	Price 12-Oct	Mkt-Cap US$M	3M ADT US$M	P/E (x) FY26E	P/E (x) FY27E	P/E (x) FY28E	EV/EBITDA (x) FY26E	EV/EBITDA (x) FY27E	EV/EBITDA (x) FY28E	P/B (x) FY26E	P/B (x) FY27E	P/B (x) FY28E	Dividend yield (%) FY25E	Dividend yield (%) FY26E	Dividend yield (%) FY27E	Dividend yield (%) FY28E	26-28E 2yr EPS CAGR (%)	ROE FY26E	ROE FY27E	ROE FY28E
Taiwan																								
Voltronic	6409.TW	1	TWD	1,050.0	875.0	2,418.3	14.7	26.5	20.7	18.5	16.5	14.6	13.2	10.6	10.0	9.3	4.0	3.3	4.2	4.7	19.7	34%	50%	52%
Delta	2308.TW	1	TWD	2,280.0	1,900.0	154,557.6	686.7	49.9	32.9	20.1	27.0	17.7	11.0	11.1	8.6	6.2	0.6	1.0	1.6	2.6	57.7	25%	29%	36%
Lite-On	2301.TW	NR	TWD	na	288.0	20,889.4	218.6	26.4	18.9	13.7	17.7	12.1	9.3	6.4	5.7	5.1	1.7	2.6	3.6	5.6	38.7	25%	31%	40%
CyberPower	3617.TW	NR	TWD	na	258.5	767.1	4.2	na	na	na	na	na	na	na	na	na	na	na	na	na	na	na	na	na
Sub-Average								**34.3**	**24.2**	**17.4**	**20.4**	**14.8**	**11.2**	**9.4**	**8.1**	**6.9**	**2.1**	**2.3**	**3.1**	**4.3**	**38.7**	**28%**	**37%**	**42%**
Global																								
Kehua	002335.SZ	NR	CNY	na	28.2	3,184.0	98.9	33.0	24.7	18.7	17.4	13.7	11.5	2.8	2.6	2.3	1.2	1.7	2.4	4.0	32.9	10%	13%	16%
KSTAR	002518.SZ	NR	CNY	na	31.4	2,727.5	78.5	23.8	16.4	12.7	16.3	11.4	8.4	3.4	2.9	2.5	1.4	1.9	2.9	3.7	37.1	14%	18%	20%
Fuji Electric	6504.T	1	JPY	18,000.0	15,015.0	13,844.5	72.3	22.6	17.8	16.7	10.7	9.0	8.1	2.8	2.6	2.4	1.1	1.3	1.6	1.7	16.0	13%	15%	15%
Mitsubishi Electric	6503.T	1	JPY	7,200.0	5,143.0	66,486.3	205.1	25.9	18.8	17.3	14.5	10.1	9.4	2.4	2.1	1.9	1.0	1.1	1.2	1.3	20.9	10%	12%	11%
Vertiv	VRT.N	1	USD	358.0	242.8	93,467.4	307.4	36.3	26.8	20.9	27.3	18.7	14.2	14.9	9.9	6.9	na	na	na	na	35.9	45%	42%	38%
ABB	ABBN.S	2	CHF	85.0	79.8	175,233.6	211.8	28.6	25.6	22.7	22.0	17.7	15.7	8.4	7.2	6.0	1.2	1.3	1.3	1.3	(3.1)	43%	29%	28%
Siemens Energy	ENRtn.DE	2	EUR	185.0	146.8	141,559.8	325.6	30.6	21.6	16.6	18.4	13.1	10.3	11.2	9.7	7.1	0.5	1.5	2.0	2.7	38.0	35%	47%	49%
Schneider Electric	SCHN.PA	1	EUR	370.0	268.5	174,021.5	288.7	24.9	21.0	18.6	17.6	14.8	12.9	5.6	4.9	4.2	1.6	1.6	1.7	2.0	19.1	21%	23%	23%
Eaton	ETN.N	1	USD	485.0	429.7	166,876.1	232.2	31.7	26.8	23.3	23.0	19.7	17.6	7.6	7.1	6.4	0.9	1.0	1.0	1.1	24.5	20%	24%	26%
Emerson Electric	EMR.N	1	USD	182.0	161.8	90,268.8	121.9	24.7	22.5	20.3	18.7	16.8	15.4	25.6	17.2	14.6	1.3	1.4	1.4	1.4	11.2	23%	70%	60%
Sub-Average								**28.9**	**22.5**	**18.6**	**18.7**	**14.6**	**12.2**	**8.5**	**6.6**	**5.4**	**1.0**	**1.3**	**1.5**	**1.9**	**25.3**	**0.2**	**0.3**	**0.3**
Average								**31.6**	**23.3**	**18.0**	**19.6**	**14.7**	**11.7**	**8.9**	**7.4**	**6.1**	**1.6**	**1.8**	**2.3**	**3.1**	**32.0**	**26%**	**33%**	**36%**

*Note: 1=Buy, 2=Neutral, 3=Sell, H=High Risk

Source: Powered by DataCentral, Citi Research estimates and I/B/E/S estimates for Not-Rated (NR) Companies

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Thomson Reuters, Citi Research, Citi Research Estimates

Investment Thesis

Positive #1 – AIDC creates a new growth runway, with power-rack expansion potentially lifting Voltronic into broader power solutions

We see Voltronic's expansion into power racks as a potentially significant extension of its AIDC opportunity in the medium to long term, enabling the company to capture the broader share of datacenter power spending. We believe Voltronic's existing high-voltage engineering capabilities, established relationships with western power vendors and customer collaboration provide a credible foundation for winning business in HVDC power rack.

Rapid growth in AI workloads and rising rack density increase the power capacity required, alongside more demanding requirements for efficiency, reliability and load management. Nvidia's roadmap targets 800VDC infrastructure supporting racks of 1MW and beyond, while its latest architecture includes power racks that supply 800VDC to compute racks within existing AC-powered facilities. We believe this creates scope for greater power-infrastructure, even as the architecture consolidates conversion stages to improve efficiency.

EV-charger experience provides a relevant technical foundation for HVDC development

Voltronic already offers DC fast chargers ranging 30-240kW, with modular designs delivering constant power across a 300-100VDC range and charging controls developed internally. These products demonstrate existing experience with high-voltage conversion, protection, thermal management and modular power-system design. We believe these capabilities should support its development of 800VDC PSUs and reduce the engineering learning curve.

Established westerns customer relationships strengthen its route to market

Voltronic's UPS ODM business provides existing engineering relationships and familiarity with customers' quality, customization and manufacturing requirements. Its pure ODM business model also emphasizes avoiding competition with customers. We believe this positioning makes it a suitable partner for western vendors who are expanding their AI infrastructure offerings. Existing relationships should help Voltronic secure development opportunities and understand product roadmaps early, although each new platform still requires qualification.

Voltronic has already taken concrete steps toward this opportunity. The company has publicly introduced an HVDC PSU for power racks, while its broader development plan extends into solutions combining battery backup unit (BBUs). battery management systems (BMS), and DCDC converters through collaboration with power-system companies.

Management indicated discussions with two Tier 1 vendors with the intended solution ultimately servicing the service providers. This creates a potential path from supplying individual power products to participating in a larger, integrated power-rack offering.

Development has progressed into customer sampling, although specifications continue to evolve with customers' requirements. Management pointed out that shipment of 800VDC products possible from 2H27E at the earliest.

We view its existing relationships with global power-solution vendors as a key advantage, while its commitment to operating as a pure ODM without competing against customers under its own brand supports this partnership model.

That said, customer qualification, confirmed design wins and mass-production timing remain key milestones before we can confidently evaluate a material contribution. Sales potential also depends on Voltronic's scope of supply, which determines content per rack and the share of system value it captures. Progress toward commercial shipments could strengthen earnings visibility and support a re-rating as Voltronic establishes its position in AIDC power infrastructure.

Figure 7. Western Power Vendors' Positioning in 800VDC Power Racks

Company	Schneider Electric	Eaton	ABB	Vertiv
Power Rack Product	NetShelter Power Rack - 800VDC	800VDC/+-400VDC sidecar platform	800VDC power architecture under development.	PowerDirect 5000, a dedicated 800VDC power sidecar/standalone power center.
Power Capacity	>660kW; broader sidecar architecture can support AI racks of up to 1.2MW	900kW max	na	400-900kW per PowerDirect 500 sidecar.
Expected Timeline	Sidecar intended to be available ahead of Rubin Ultra/Kyber deployments beginning in 2027E.	2027 production readiness target	na	800VDC sidecars are under customer validation in 2026 with deployment planned in 2027E.
Work with Nvidia?	Yes. Working directly with Nvidia on 800VDC sidecar	Yes, working with Nvidia on design, reference architectures, and 800VDC power mgmt solutions	Yes, ABB and Nvidia are jointly developing power solutions for 800VDC and gigawatt scale AIDC.	Yes, working closely with Nvidia. Vertiv has aligned the product roadmap with Nvidia's next-gen platforms.
External Partners?	Yes. The mgmt said it will leverage some partners when needed.	Yes. The mgmt said it is actively seeking industry collaboration to refine an d deploy the prototype.	Yes at the ecosystem level.	Yes at the ecosystem level, but less explicit on outsourcing.
Relevant Mgmt Commentary	On 1Q26 call, Schneider said it is investing heavily in 800VDC; the sidecar is the solution that can be deployed quickly. Importantly, mgmt explicitly said Schneider doesn't need to do everything itself and would leverage partners where necessary.	Eaton describes 800VDC as a major part of its "grid-to-chip" strategy. The detailed sidecar roadmap and explicit request for industry collaboration suggest Eaton is actively building an ecosystem around the product rather than developing every element entirely internally.	The mgmt noted ABB was an early investor in UPS, DC and solid-state electronics, which they describe as a future potential demand driver, with its Nvidia collaboration putting these technologies at the center of next-gen datacenter architecture.	Vertiv said 800VDC should become an important portion of the market from 2027E onward. Its programs were on schedule for launches, while mgmt was pleased with feedback from ongoing prototypes and validation activities.

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Company Reports, Citi Research

Figure 8. Voltronic: Prototype of Power Racks

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Voltronic, Citi Research

Figure 9. Voltronic: Prototype of Power Racks

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Voltronic, Citi Research

Positive #2 – UPS outsourcing provides a structural share-gain opportunity

We see structural outsourcing as a durable growth driver for Voltronic, allowing its UPS business to outgrow underlying industry demand. Rising investment in digital infrastructure, commercial applications and industrial automation supports continued demand for power protection, while established brands have incentives to outsource selected hardware platforms.

Voltronic's dedicated ODM model, engineering capabilities and manufacturing efficiency position it to capture this opportunity. In our view, growth can come from both an expanding market and the transfer of existing production from customers' factories to qualified manufacturing partners.

UPS remains a structurally growing power-protection market

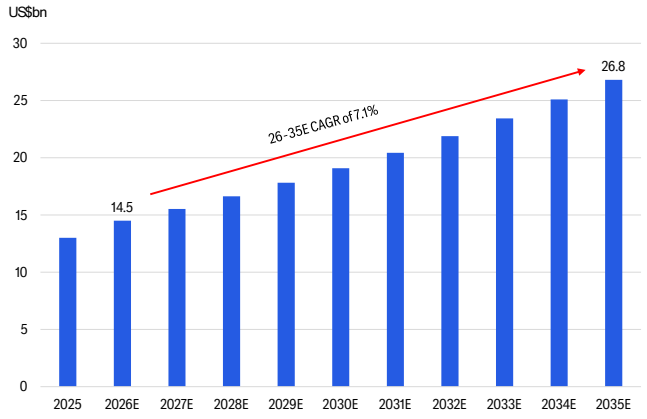
Global Market Insights estimates the worldwide UPS market at US$13bn in 2025, rising to US$14.5bn in 2026E and US$26.8bn by 2035E, implying a 7.1% CAGR over 2026-2035E. Although estimates vary by research provider and market definition, the underlying drivers include increasing dependency on electronic systems, growth in industrial automation and smart manufacturing, increasing power outages and grad instability, rising costs of downtime and greater requirements for reliable electricity.

Datacenters provide an important source of incremental demand, while commercial and industrial applications broaden the market beyond the AI investment cycle. Datacenters expansion supports demand for online, double conversion UPS systems, which continuously condition incoming power and protect critical equipment against supply disturbance. Commercial applications – including offices, retail networks and healthcare facilities – require protection for servers, communications equipment and other essential systems. Industrial demand comes from protecting sensitive control, automation equipment and critical processes where power interruptions can disrupt production.

The competitive landscape is led by established international suppliers. Per Global Market Insights 2025 analysis, Schneider Electric, Eaton, Vertiv are the three leading suppliers, with the top 5 collectively controlling approximately 60-65% of the global UPS market. Other peers include Huawei, KSTAR, Kehua, Delta and Riello.

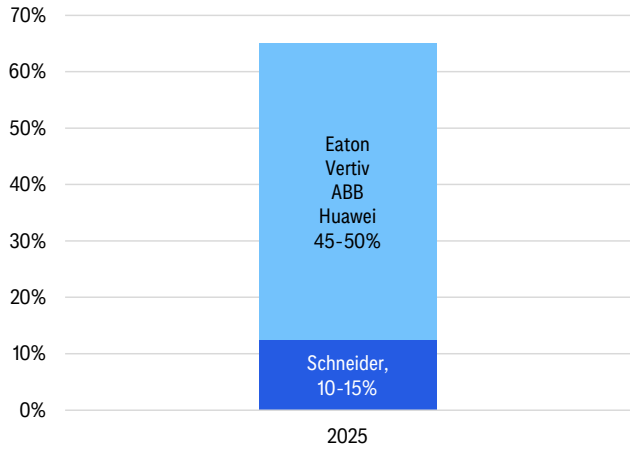
This concentration matters for the outsourcing thesis – securing additional projects from a relatively small number of large brands can open substantial manufacturing opportunities.

Figure 10. Global UPS Market Size 2025-2035E



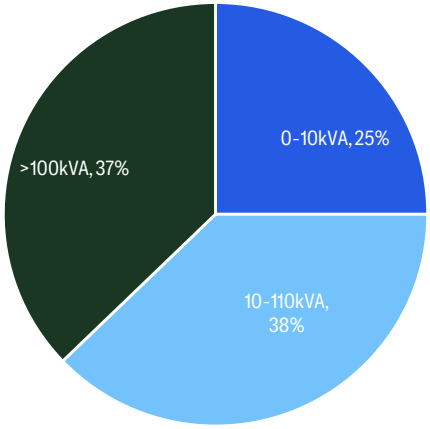
© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Global Market Insights, Citi Research

Figure 11. Global UPS Market Share by Brand (2025)



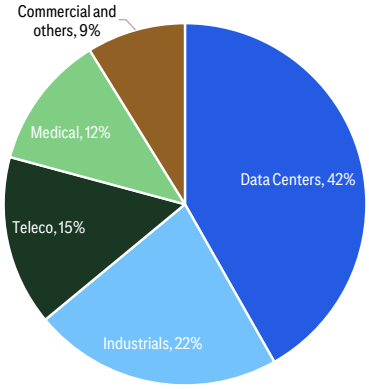
© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Global Market Insights, Citi Research

Figure 12. Global Online UPS by Power Rating



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: PW Consulting Group, Citi Research

Figure 13. Global Online UPS by End Use



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: PW Consulting Group, Citi Research

Western brands' strategic priorities support further selective outsourcing

Per our industry checks, major UPS brands still manufacture a substantial portion of their UPS products internally, while increasingly moving toward energy management and committing resources to broader AIDC infrastructure. This could result in small and medium UPS products to be gradually outsourced as this transition develops. We view this as a multi-year opportunity, particularly where established platforms can be transferred to qualified ODMs while customers retain control over product specifications, branding and distribution.

We identify three economic reasons for this shift –

1. **Outsourcing can improve manufacturing and capital efficiency**: A specialist ODM can aggregate demand across customers, consolidate component purchases and spread manufacturing overhead across a larger production base. For brands, this can reduce incremental factory investment and simplify capacity planning.

2. **Outsourcing can accelerate product development**: access to ODMs' established platforms, component knowledge and manufacturing processes allows brands to adapt existing designs for additional markets and applications. Engineering resources can then concentrate on product differentiation.

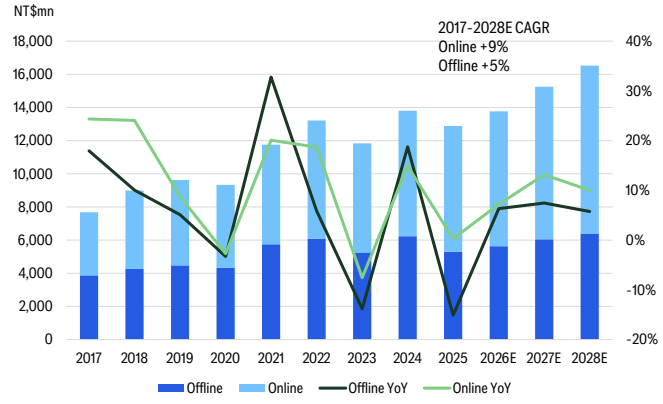
3. **Broader system offerings create competing demands on vendors' resources**. Leading UPS brands emphasize on monitoring, operational management and integrated solutions, strengthening the rationale for outsourcing selected hardware while allocating more resources to system integration, and software.

Voltronic's dedicated ODM model, engineering capabilities and manufacturing efficiency should position it to capture further UPS outsourcing

Voltronic's ODM focus is evident from its commitment to operate without having a brand of its own to compete with its customers. This aligns its interests with customers and differentiates it from suppliers that combine branded sales with contract manufacturing. Long-standing relationships with leading vendors, including Schneider, Riello, Vertiv and Eaton, provide a foundation for securing additional projects. Its engineering capabilities enable it to meet diverse product specifications, while flexible manufacturing supports the small production batches and broad model range characteristic of the UPS industry.

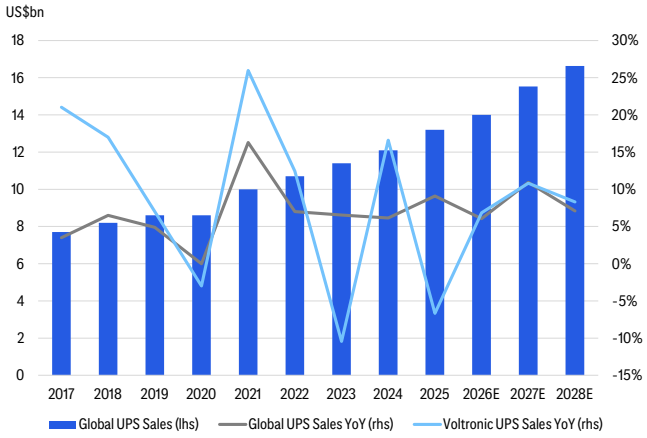
Its track record in product qualification and certification also creates barriers, as moving established programs to other suppliers requires additional validation, audits and testing. We believe this combination of customer trust, execution and cost competitiveness positions Voltronic to secure more outsourcing orders, including higher-power online UPS, as western brands allocate greater resources to AIDC infrastructure and integrated power solutions.

Figure 14. Voltronic: Online UPS and Offline UPS Sales and Y/Y



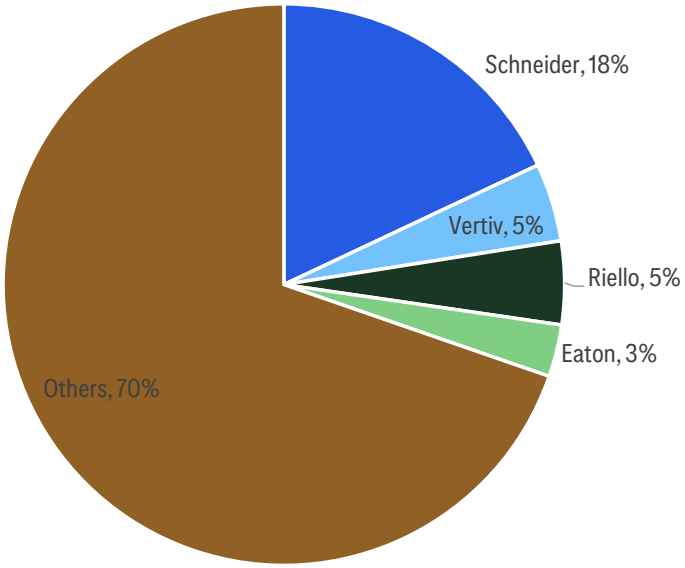
© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Company Reports, Citi Research, Citi Research Estimates

Figure 15. Voltronic UPS Sales vs. Global UPS Sales Growth



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Omdia, IHS Markit, Citi Research, Citi Research Estimates

Figure 16. Voltronic: Customer Mix in 1H26



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Company Reports, Citi Research

Positive #3 – Online and high-power UPS mix upgrade should support structurally better growth

We expect Voltronic's shift toward online and higher-power UPS to become an increasingly important driver of revenue growth and profitability. Moving up the product range increases power capacity and engineering content per system, supporting higher ASPs and greater differentiation. We forecast online UPS sales to rise at a 12% CAGR in 2026-2028E, ahead of offline UPS at 6%, with online UPS gross profit growing faster at 16% as product mix improves.

Offline UPS remains a stable revenue base, yet offers more limited differentiation. Offline UPS typically serves home offices, small businesses and basic power applications. These products feature relatively simple designs and lower costs.

We view this segment as increasingly mature and commoditized, with lower technical barriers exposing suppliers to greater pricing pressure, including competition from Chinese manufacturers. Scale, procurement efficiency and cost control hence are the key competitive factors, while opportunities for sustained ASP and margin expansion are more constrained, in our view. Our model assumes offline UPS gross margin to be at 24-26% in 2025-2028E.

Online UPS offers greater technical differentiation and stronger margin potential. Its double-conversion architecture continuously converts AC power into DC and back into AC, providing stable voltage and frequency, avoiding power disturbances and enabling zero transfer time to battery operation.

It is particularly suited to complex, power-sensitive applications such as datacenters, industrial automation and precision control systems, where even brief interruptions or power quality disturbances can disrupt operations. The business also exhibits high-mix, low-volume characteristics, particularly in customized

commercial and industrial applications, where customers require different power ratings, redundancy configuration and battery interfaces etc. Our model assumes online UPS gross margin to be at 31-34% in 2025-2028E.

We believe this creates opportunities for an experienced ODM partner such as Voltronic to differentiate through customization and reliable delivery, while sharing development and procurement resources across customers and product platforms. Together with demanding certification, customer qualification and reliability testing, these capabilities raise entry barriers and switching costs, supporting higher ASPs, strong customer retention and better margin potential as Voltronic expands into higher power systems.

We project online UPS sales to increase 12% CAGR in 2026-2028E, lifting its contribution to group sales from 37% in 2025 to c.44% in 2028E. The underlying driver comes from a stronger outsourcing opportunity thanks to the AIDC boom. Western vendors, including Schneider, Vertiv are expanding their focus on integrated AIDC infrastructure, power distribution, and infrastructure management. Meanwhile, rising power requirements within conventional AC-based datacenters support demand for higher power UPS systems, prompting vendors to expand UPS production capacity. We believe the combination of stronger demand and competing resource requirements can strain available engineering and manufacturing capabilities, increasing the appeal of qualified ODM partners to supplement internal production. Voltronic's customization capabilities and flexible manufacturing position it to capture this opportunity.

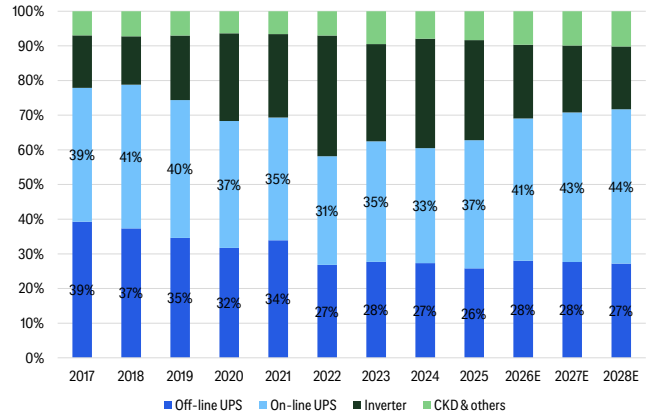
Figure 17. Online UPS vs Offline UPS

	Offline UPS	Online UPS
Voltronic Sales Mix (2025)	26%	37%
Voltronic GPM (2025)	25-26%	31-32%
Operating Principle	Utility power normally feeds the load directly; battery/inverter activates when input power fails or moves our of range	Input AC is continuously converted AC-DC-AC, with inverter continuously supplying the load
Transfer Time	Short transfer time, a few milliseconds	Near-zero transfer time
Typical Power Range	More concentrated in lower-power applications	More prevalent in medium and high power applications
Product Characteristics	More standardized, high volume products	More customized and project based products
Technical Complexity	Lower	Higher
Typical Applications	PCs, home office, POS systems, small networking equipment	Datacenters, server rooms, telecom, industrial equipment, medical and critical infra
Cost	Lower	Higher
Maintenance Needs	Lower	Higher
Main Advantage	Cost effective and efficient for non-critical loads	Best power quality and highest protection for mission-critical loads

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

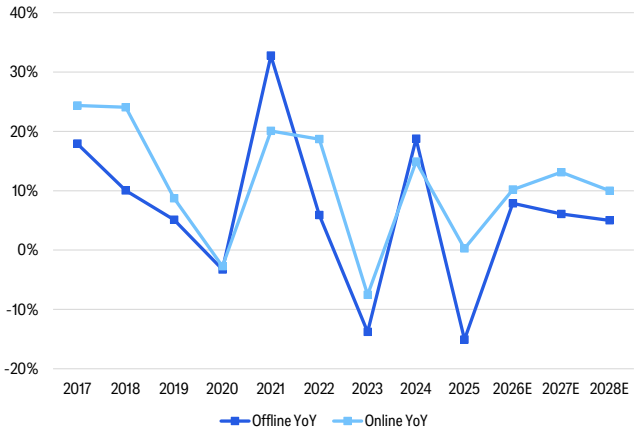
Source: Citi Research

Figure 18. Voltronic: Sales Mix



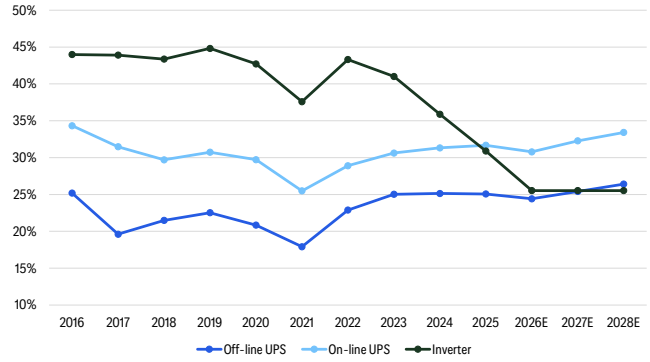
© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Company Reports, Citi Research, Citi Research Estimates

Figure 19. Voltronic: Online UPS vs. Offline UPS Sales Growth



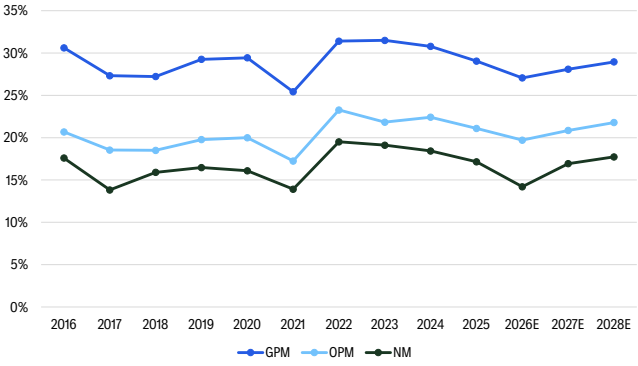
© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Company Reports, Citi Research, Citi Research Estimates

Figure 20. Voltronic: GPM by Product



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Company Reports, Citi Research, Citi Research Estimates

Figure 21. Voltronic: Margins Trend



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Company Reports, Citi Research, Citi Research Estimates

Financial Analysis

Income statement – 2026E marks the earnings trough; online UPS drives the recovery from 2027E

Voltronic delivered an 8.7% sales CAGR and 9.9% net profit CAGR over 2018-2025, although earnings have become more volatile in recent years due largely to swings in the inverter business. We forecast sales to decrease 1% y/y to NT$20.4bn in 2026E, before recovering 8% y/y and 7% y/y to NT$21.9bn and NT$23.4bn in 2027E and 2028E respectively. The key driver is the higher value online UPS.

We forecast online UPS sales to grow 10%/13%/10% y/y in 2026E/2027E/2028E, lifting its sales contribution from 37% in 2025 to 44% by 2028E. Together with offline UPS, we estimate UPS products to make up 72% of group sales in 2028E vs 2025's 63%. Conversely, we forecast inverter sales to decline 27% y/y in 2026E and remain broadly flat thereafter, reducing their sales contribution from 29% in 2025 to 18% in 2028E.

We expect gross margin to bottom at 27.1% in 2026E vs. 29.0% in 2025, reflecting negative impact from material cost inflation and weaker inverter profitability. Thereafter, we forecast gross margin to recover to 28.1% and 29.0% in 2027E and 2028E, supported by a richer online UPS mix and improving segment margins. We estimate online UPS gross margin to increase from 30.8% in 2026E to 33.4% in 2028E driven rising contribution from high power online UPS, while its rising sales mix could provide increasing contribution to group gross profits.

Operating leverage should further support profitability. Opex fell from 8.0% of sales in 2025 to 7.4% in 2026E and could remain at 7.2% thereafter, while R&D investment is maintained at 4.2% of sales. We therefore forecast OPM to recover from 19.7% in 2026E to 20.9% and 21.8% in 2027E and 2028E.

We estimate net profit to decrease 18% y/y in 2026E, partly hampered by a larger FX loss, before rebounding 28% y/y and 12% y/y in 2027E and 2028E. We forecast EPS to register NT$33.00/NT$42.30/NT$47.25 in 2026E/2027E/2028E.

We attribute Voltronic's earnings decline in 2025 and our forecast for a further decline in 2026E to persistent inverter weakness and additional cost and supply pressures. In 2025, adverse weather in Pakistan shortened the inverter peak season and weakened demand, while intensifying Chinese competition pressured pricing and profitability. The weakness extended into 2026E as the Middle East conflict disrupted logistics and delayed shipments, alongside continued customers destocking in Pakistan. In 1H26, RMB appreciation further weighed on margins. Meanwhile, raw material shortages constrained shipments and higher metal and key component prices eroded gross margins. We therefore forecast earnings to decline 18% y/y in 2026E before improving 28% y/y in 2027E.

Figure 22. Voltronic: Income Statement

Income Statement (NT$m)	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Net Sales	**11,408**	**12,936**	**13,653**	**16,957**	**22,725**	**18,951**	**22,813**	**20,521**	**20,374**	**21,913**	**23,378**
COGS	-8,303	-9,151	-9,634	-12,646	-15,588	-12,982	-15,790	-14,561	-14,860	-15,758	-16,609
Gross Profit	**3,105**	**3,785**	**4,018**	**4,311**	**7,137**	**5,968**	**7,023**	**5,960**	**5,514**	**6,155**	**6,768**
Operating Exp.	-994	-1,228	-1,288	-1,386	-1,849	-1,831	-1,908	-1,632	-1,499	-1,586	-1,675
Operating Exp. - Promotion	-269	-322	-351	-387	-434	-372	-385	-305	-262	-282	-301
Operating Exp. - ADM	-308	-339	-362	-398	-509	-525	-561	-482	-386	-382	-384
Operating Exp. - R&D	-418	-567	-574	-601	-906	-933	-962	-845	-851	-922	-990
Operating income	**2,111**	**2,557**	**2,730**	**2,925**	**5,288**	**4,138**	**5,115**	**4,329**	**4,015**	**4,569**	**5,093**
Total non-operating Inc.	**145**	**59**	**-81**	**-34**	**115**	**245**	**-25**	**-12**	**-449**	**0**	**17**
Net Interest Inc.	26	5	19	31	-7	83	164	139	130	127	127
Disposal of Fixed Asset	-1	-5	0	-1	0	-2	-1	-7	0	0	0
Exchange Gain (Loss)	53	18	-154	-92	88	101	-204	-171	-702	0	0
Others Income (Loss)	67	41	54	28	34	63	15	27	123	-127	-110
Pre-tax Income	**2,256**	**2,616**	**2,650**	**2,890**	**5,403**	**4,382**	**5,090**	**4,317**	**3,566**	**4,569**	**5,110**
Income Tax	-443	-485	-452	-531	-970	-760	-886	-796	-672	-859	-967
Net Income	**1,814**	**2,131**	**2,197**	**2,359**	**4,433**	**3,623**	**4,204**	**3,521**	**2,894**	**3,710**	**4,144**
Extraordinary gain (losses), minority interest	0	0	0	0	0	0	0	0	0	0	0
Net profit after extraordinary items	**1,814**	**2,131**	**2,197**	**2,359**	**4,433**	**3,623**	**4,204**	**3,521**	**2,894**	**3,710**	**4,144**
Adj.wtd.avg.shrs (m)	78	82	87	87	87	87	87	88	88	88	88
EPS (NT$)	**23.18**	**25.87**	**25.31**	**27.13**	**50.90**	**41.52**	**48.13**	**40.23**	**33.00**	**42.30**	**47.25**
Effective tax rate	-20%	-19%	-17%	-18%	-18%	-17%	-17%	-18%	-19%	-19%	-19%
Depreciation & Amortization	537	346	238	299	327	362	349	313	295	262	225
EBITDA	**2,767**	**2,957**	**2,869**	**3,159**	**5,736**	**4,661**	**5,274**	**4,491**	**3,730**	**4,704**	**5,208**
Margins											
Gross Margin	27.2%	29.3%	29.4%	25.4%	31.4%	31.5%	30.8%	29.0%	27.1%	28.1%	29.0%
Operating Margin	18.5%	19.8%	20.0%	17.2%	23.3%	21.8%	22.4%	21.1%	19.7%	20.9%	21.8%
Pretax Margin	19.8%	20.2%	19.4%	17.0%	23.8%	23.1%	22.3%	21.0%	17.5%	20.9%	21.9%
Net Margin	15.9%	16.5%	16.1%	13.9%	19.5%	19.1%	18.4%	17.2%	14.2%	16.9%	17.7%
EBITDA margin	24.3%	22.9%	21.0%	18.6%	25.2%	24.6%	23.1%	21.9%	18.3%	21.5%	22.3%
OPEX	-8.7%	-9.5%	-9.4%	-8.2%	8.1%	9.7%	8.4%	8.0%	7.4%	7.2%	7.2%
YoY %											
Turnover	15.7%	13.4%	5.5%	24.2%	34.0%	-16.6%	20.4%	-10.0%	-0.7%	7.6%	6.7%
Operating Profits	15.4%	21.1%	6.8%	7.1%	80.8%	-21.8%	23.6%	-15.4%	-7.3%	13.8%	11.5%
Pretax Profits	36.3%	15.9%	1.3%	9.1%	86.9%	-18.9%	16.1%	-15.2%	-17.4%	28.1%	11.8%
Net Profits	33.0%	17.5%	3.1%	7.4%	87.9%	-18.3%	16.1%	-16.3%	-17.8%	28.2%	11.7%
EPS	32.8%	11.6%	-2.2%	7.2%	87.6%	-18.4%	15.9%	-16.4%	-18.0%	28.2%	11.7%
OPEX	14.8%	23.5%	4.9%	7.7%	33.4%	-1.0%	4.2%	-14.5%	-8.1%	5.8%	5.6%

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Company Reports, Citi Research, Citi Research Estimates

Figure 23. Voltronic: Our Sales and GPM Assumptions

Revenue Assumptions	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
FX (USD/TWD)	30.1	30.9	29.5	27.8	29.8	31.8	32.3	31.0	31.6	31.6	31.6
Sales											
Off-line UPS	4,261	4,479	4,332	5,751	6,090	5,249	6,233	5,293	5,710	6,058	6,361
On-line UPS	4,726	5,138	4,998	6,002	7,124	6,586	7,567	7,587	8,359	9,456	10,402
Inverter	1,592	2,409	3,453	4,080	7,920	5,320	7,202	5,928	4,336	4,233	4,233
CKD & others	828	910	869	1,124	1,591	1,796	1,811	1,713	1,969	2,166	2,383
Total	**11,408**	**12,936**	**13,653**	**16,957**	**22,725**	**18,951**	**22,813**	**20,521**	**20,374**	**21,913**	**23,378**
YoY%											
Off-line UPS	10%	5%	-3%	33%	6%	-14%	19%	-15%	8%	6%	5%
On-line UPS	24%	9%	-3%	20%	19%	-8%	15%	0%	10%	13%	10%
Inverter	7%	51%	43%	18%	94%	-33%	35%	-18%	-27%	-2%	0%
CKD & others	20%	10%	-4%	29%	42%	13%	1%	-5%	15%	10%	10%
Total	**16%**	**13%**	**6%**	**24%**	**34%**	**-17%**	**20%**	**-10%**	**-1%**	**8%**	**7%**
GPM											
Off-line UPS	21.5%	22.5%	20.8%	17.9%	22.9%	25.0%	25.1%	25.1%	24.4%	25.4%	26.4%
On-line UPS	29.7%	30.7%	29.7%	25.5%	28.9%	30.6%	31.3%	31.7%	30.8%	32.3%	33.4%
Inverter	43.4%	44.8%	42.7%	37.6%	43.3%	41.0%	35.9%	30.9%	25.5%	25.5%	25.5%
CKD & others	11.5%	12.8%	17.8%	19.3%	16.0%	25.4%	27.7%	23.3%	22.3%	22.3%	22.3%
Total GM%	**27.2%**	**29.3%**	**29.4%**	**25.4%**	**31.4%**	**31.5%**	**30.8%**	**29.0%**	**27.1%**	**28.1%**	**29.0%**

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Company Reports, Citi Research, Citi Research Estimates

Balance Sheet – Asset light model and net cash suggest high returns

Voltronic maintains a relatively asset-light balance sheet with fixed assets making up 24% of total assets in 2025 despite its sizable sales scale. We estimate the company to remain in a net cash position throughout 2026-2028E. We forecast cash and cash equivalent assets to account for 36-39% of total assets in 2026-2028E while there are no meaningful short-term borrowings and long-term debt.

We forecast ROE to rise from 37% in 2025 to 40% in 2026E and 48% in 2027E. The high ROE is not driven by financial leverage, but primarily by its asset-light business model, strong operating margins and high capital distribution.

Voltronic also maintains high shareholder-return policy. We assume an 87% cash dividend payout ratio through 2026-2028E, implying cash dividends of NT$29/NT$37/NT$41 per share.

Figure 24. Voltronic: Balance Sheet

Balance Sheet (NT$mn)	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Cash	2,629	4,003	4,886	5,155	5,353	5,045	6,556	6,415	6,230	6,775	7,463
Bank Deposit (over 3 mths)	-	-	-	-	-	-	-	323	455	455	455
AR/NR (reported)	2,198	2,278	2,346	2,838	3,117	2,949	3,450	3,387	2,896	3,125	3,348
Inventories	973	1,030	1,294	1,787	2,019	1,412	2,093	1,796	1,723	1,830	1,947
Others	246	251	272	262	328	252	324	354	324	324	324
Current Assets	**6,047**	**7,562**	**8,798**	**10,042**	**10,816**	**9,657**	**12,423**	**12,276**	**11,629**	**12,509**	**13,538**
Long-term Investments	-	-	-	-	-	42	61	101	98	98	98
Fixed Assets	1,964	2,168	2,428	2,730	4,474	4,420	4,390	4,172	4,083	3,901	3,757
Right of Use Assets	-	397	489	390	463	368	301	286	316	316	316
Other Assets	275	213	206	222	332	269	145	249	385	385	277
Total Assets	**8,286**	**10,340**	**11,921**	**13,384**	**16,085**	**14,756**	**17,321**	**17,084**	**16,511**	**17,209**	**17,986**
S-t Borrowings	397	1,091	1,768	2,380	-	-	-	-	-	-	-
AP/NP	2,370	3,003	3,322	3,615	4,173	3,687	4,350	4,796	3,485	3,701	3,939
Other ST liabilities	1,068	1,173	1,234	1,500	2,454	1,798	2,276	2,086	5,078	5,078	5,078
Total Current Liabilities	**3,835**	**5,267**	**6,324**	**7,495**	**6,627**	**5,485**	**6,626**	**6,881**	**8,563**	**8,780**	**9,017**
L-t Debt	-	-	-	-	832	734	636	538	489	489	489
Other Liabilities	1	193	259	181	283	180	130	111	232	232	232
Total Liabilities	**3,836**	**5,460**	**6,583**	**7,676**	**7,742**	**6,398**	**7,392**	**7,531**	**9,285**	**9,501**	**9,739**
Common Shares	787	833	874	874	878	877	877	877	883	883	883
Other Shareholders' Equity	3,663	4,047	4,464	4,833	7,466	7,480	9,052	8,676	6,343	6,826	7,365
Total Equity	**4,450**	**4,880**	**5,339**	**5,708**	**8,343**	**8,358**	**9,929**	**9,553**	**7,226**	**7,708**	**8,247**
Total Liab./Shrhldr's Equity	**8,286**	**10,340**	**11,921**	**13,384**	**16,085**	**14,756**	**17,321**	**17,084**	**16,511**	**17,209**	**17,986**

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Company Reports, Citi Research, Citi Research Estimates

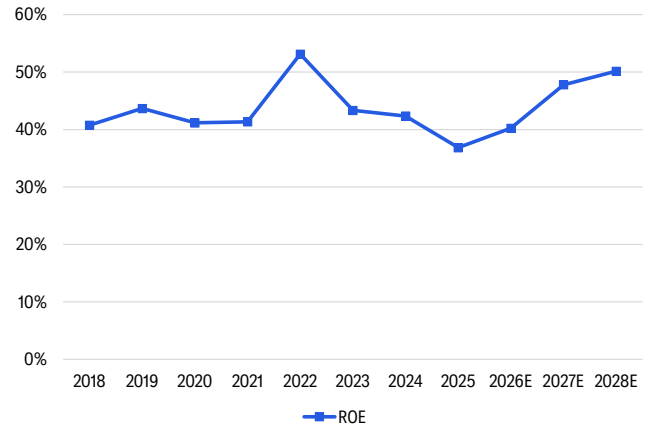
Figure 25. Voltronic: Cash Flow Statement

Cash Flow (NT$ mn)	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Cashflow from Operations	**1,865**	**2,997**	**2,626**	**2,299**	**5,619**	**4,083**	**4,404**	**4,617**	**2,080**	**3,852**	**4,265**
Net profits	1,814	2,131	2,197	2,359	4,433	3,623	4,204	3,521	2,894	3,710	4,144
Depreciation & Amortization	79	178	238	299	327	362	349	313	295	262	225
Inv. Gain/Loss (Equity)	81	110	182	26	164	359	212	85	28	0	0
Disposal of Investment	0	0	0	0	0	0	0	0	0	0	0
Working Capital Change	-468	454	-26	-715	5	173	-413	786	-773	-119	-103
Dec(Inc)-A/R	-402	-111	-81	-512	-313	72	-371	63	480	-229	-223
Dec(Inc)-Inventory	-75	-68	-264	-496	-240	587	-705	278	58	-107	-118
Inc(Dec)-A/P	9	633	319	293	558	-486	663	446	-1,311	216	238
Other adjustments	359	124	35	329	690	-434	52	-88	(364)	-	-
Cashflow from Investing	**-702**	**-438**	**-411**	**-532**	**-1,975**	**-232**	**-104**	**-395**	**-214**	**-80**	**28**
(Purchases) Sale of fixed asset / (capex)	-700	-431	-387	-520	-1,974	-182	-87	-60	(93)	(80)	(80)
(Purchases) Sale of LT investment	0	0	0	0	0	0	0	0	-	-	-
(Purchases) Sale of ST investment	0	0	0	0	0	-45	-17	-333	(113)	-	-
Other adjustments	-3	-7	-24	-12	0	-6	0	-2	(7)	-	108
Cashflow from financing	**-1,175**	**-998**	**-1,301**	**-1,524**	**-3,791**	**-4,009**	**-3,488**	**-3,944**	**-2,610**	**-3,227**	**-3,605**
Increase in L-T debt	0	0	0	0	930	-98	-98	-98	-49	0	0
Increase in S-T debt	398	707	705	622	-2,569	0	0	0	0	0	0
Cash Dividend Paid	-1,574	-1,652	-1,933	-2,055	-2,054	-3,818	-3,290	-3,772	-2,518	-3,227	-3,605
Issuance of stock	0	0	0	0	0	0	0	0	0	0	0
Other adjustments	0	-52	-73	-91	-98	-93	-101	-74	-43	0	0
Exchange rate adjustment	-53	-188	-31	26	345	-151	700	-419	559	0	0
Net change in cash	**-66**	**1,374**	**883**	**269**	**198**	**-308**	**1,512**	**-141**	**-185**	**545**	**688**

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

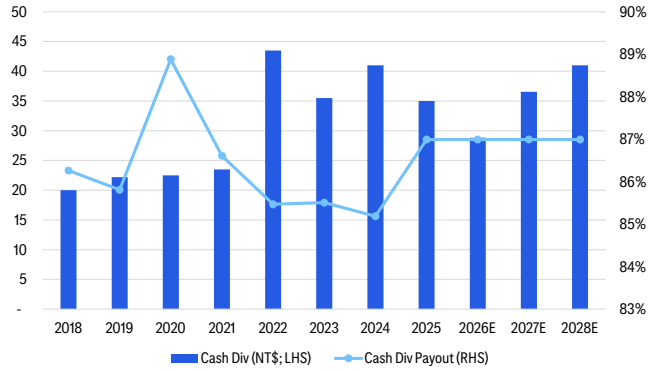
Source: Company Reports, Citi Research, Citi Research Estimates

Figure 26. Voltronic: ROE Trend



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Company Reports, Citi Research, Citi Research Estimates

Figure 27. Voltronic: Cash Dividend Payout



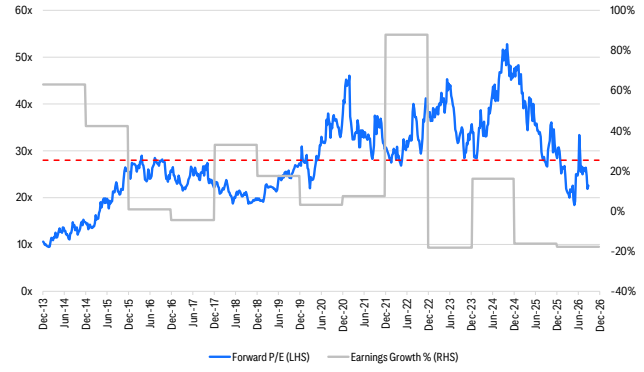
© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Company Reports, Citi Research, Citi Research Estimates

Valuation Methodology and Risks

We value Voltronic using a forward target P/E of 25x our 2027E EPS to derive a target price of NT$1,050, broadly in line with its long-term average forward P/E since 2013. We believe forward P/E is appropriate given the company's established earnings base and the importance of product mix and operating leverage to its profitability. Using 2027E EPS captures a fuller year of anticipated earnings recovery following the softer 2026E base, alongside the ongoing transition toward online and higher power UPS. This also aligns our valuation with the earnings outlook investors are likely to focus on as the recovery progresses.

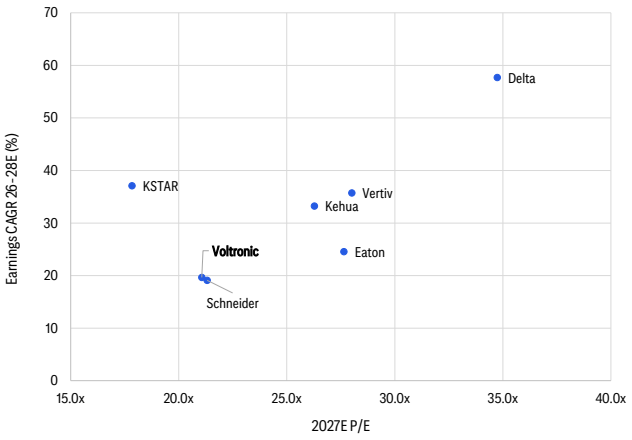
Our target multiple of 25x is supported by Voltronic's competitive position in UPS outsourcing. Its established relationship with western power vendors and ability to accommodate high mix, low volume production creates a solid foundation for the intact share gains, supporting the durability of its earnings and hence justify a valuation broadly consistent with its historical average. Stronger evidence of power rack commercialization and sustained UPS mix improvement could support a further re-rating.

Figure 28. Voltronic: Forward P/E vs. Earnings Growth



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, Citi Research Estimates

Figure 29. Peers Comp: Forward P/E vs. Earnings CAGR



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Bloomberg, Citi Research, Citi Research Estimates

Figure 30. Valuation Comp Table for UPS

Company	RIC	Rating*	Curr	Target Price	Price 12-Oct	Mkt Cap US$M	3M ADT US$M	P/E (x) FY26E	P/E (x) FY27E	P/E (x) FY28E	EV/EBITDA (x) FY26E	EV/EBITDA (x) FY27E	EV/EBITDA (x) FY28E	P/B (x) FY26E	P/B (x) FY27E	P/B (x) FY28E	Dividend yield (%) FY25E	Dividend yield (%) FY26E	Dividend yield (%) FY27E	Dividend yield (%) FY28E	26 - 28E 2yr EPS CAGR (%)	ROE FY26E	ROE FY27E	ROE FY28E
Taiwan																								
Voltronic	6409.TW	1	TWD	1,050.0	875.0	2,498.3	14.7	26.5	20.7	18.5	16.5	14.6	13.2	10.6	10.0	9.3	4.0	3.3	4.2	4.7	19.7	34%	50%	52%
Delta	2308.TW	1	TWD	2,280.0	1,900.0	154,557.6	686.7	49.9	32.9	20.1	27.0	17.7	11.0	11.1	8.6	6.2	0.6	1.0	1.6	2.6	57.7	25%	29%	36%
Lite-On	2301.TW	NR	TWD	na	288.0	20,889.4	218.6	26.4	18.9	13.7	17.7	12.1	9.3	6.4	5.7	5.1	1.7	2.6	3.6	5.6	38.7	25%	31%	40%
CyberPower	3617.TW	NR	TWD	na	258.5	707.1	4.2	na	na	na	na	na	na	na	na	na	na	na	na	na	na	na	na	na
Sub-Average								**34.3**	**24.2**	**17.4**	**20.4**	**14.8**	**11.2**	**9.4**	**8.1**	**6.9**	**2.1**	**2.3**	**3.1**	**4.3**	**38.7**	**28%**	**37%**	**42%**
Global																								
Kehua	002335.SZ	NR	CNY	na	28.2	3,184.0	98.9	33.0	24.7	18.7	17.4	13.7	11.5	2.8	2.6	2.3	1.2	1.7	2.4	4.0	32.9	10%	13%	16%
KSTAR	002518.SZ	NR	CNY	na	34.4	2,727.5	78.5	23.8	16.4	12.7	16.3	11.4	8.4	3.4	2.9	2.5	1.4	1.9	2.9	3.7	37.1	14%	18%	20%
Fuji Electric	6504.T	1	JPY	18,000.0	15,015.0	13,844.5	72.3	22.6	17.8	16.7	10.7	9.0	8.1	2.8	2.6	2.4	1.1	1.3	1.6	1.7	16.0	13%	15%	15%
Mitsubishi Electric	6503.T	1	JPY	7,200.0	5,143.0	66,486.3	205.1	25.9	18.8	17.3	14.5	10.1	9.4	2.4	2.1	1.9	1.0	1.1	1.2	1.3	20.9	10%	12%	11%
Vertiv	VRT.N	1	USD	358.0	242.8	93,467.4	307.4	36.3	26.8	20.9	27.3	18.7	14.2	14.9	9.9	6.9	na	na	na	na	35.9	45%	42%	38%
ABB	ABBN.S	2	CHF	85.0	79.8	175,233.6	211.8	28.6	25.6	22.7	22.0	17.7	15.7	8.4	7.2	6.0	1.2	1.3	1.3	1.3	(3.1)	43%	29%	28%
Siemens Energy	ENPH.DE	2	EUR	185.0	146.8	145,559.8	325.6	30.6	21.6	16.6	18.4	13.1	10.3	11.2	9.7	7.1	0.5	1.5	2.0	2.7	38.0	35%	47%	49%
Schneider Electric	SCHN.PA	1	EUR	370.0	268.5	174,021.5	288.7	24.9	21.0	18.6	17.6	14.8	12.9	5.6	4.9	4.2	1.6	1.6	1.7	2.0	19.1	21%	23%	23%
Eaton	ETN.N	1	USD	485.0	429.7	166,876.1	232.2	31.7	26.8	23.3	23.0	19.7	17.6	7.6	7.1	6.4	0.9	1.0	1.0	1.1	24.5	20%	24%	26%
Emerson Electric	EMR.N	1	USD	182.0	161.8	90,268.8	121.9	24.7	22.5	20.3	18.7	16.8	15.4	25.6	17.2	14.6	1.3	1.4	1.4	1.4	11.2	23%	70%	60%
Sub-Average								**28.9**	**22.5**	**18.6**	**18.7**	**14.6**	**12.2**	**8.5**	**6.6**	**5.4**	**1.0**	**1.3**	**1.5**	**1.9**	**25.3**	**0.2**	**0.3**	**0.3**
Average								**31.6**	**23.3**	**18.0**	**19.6**	**14.7**	**11.7**	**8.9**	**7.4**	**6.1**	**1.6**	**1.8**	**2.3**	**3.1**	**32.0**	**26%**	**33%**	**36%**

*Note: 1=Buy, 2=Neutral, 3=Sell, H=High Risk
Source: Powered by DataCentral, Citi Research estimates and I/B/E/S estimates for Not-Rated (NR) Companies

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Bloomberg, Citi Research, Citi Research Estimates

Key downside risks to our Buy rating and target price include:

1. **Slower UPS demand or outsourcing gain.** Weaker commercial and industrial spending, delayed datacenter projects or customer inventory adjustments could weigh on orders.

2. **Less favorable product mix and margin pressure.** A slower transition toward online and high-power UPS, more aggressive pricing or higher component costs could limit margin improvement.

3. **Persistent PV inverter competition.** Aggressive pricing by Chinese suppliers, weaker residential solar demand or unfavorable subsidy changes could pressure inverter sales and profitability.

Our quantitative model assigns a High Risk rating to Voltronic. However, we believe such a rating is not warranted due to: 1) its established franchise and longstanding relationships with leading power vendors, with demanding qualification requirements and high customization supporting customer retention and repeat business; 2) its track record of profitability and positive operating cash flow enables internally funded investment and reduces reliance on external financing; and 3) the company has a strong balance sheet and has been in net cash position.

Company Overview

Established in 2008 and listed on the Taiwan Stock Exchange in 2014, Voltronic Power is a Taiwan-based design and manufacturing service (DMS) provider specializing in uninterrupted power supplies (UPS) and power-conversion products. The company is headquartered in Taipei and operates manufacturing facilities in Taiwan, China and Vietnam, with approximately 5,000 employees.

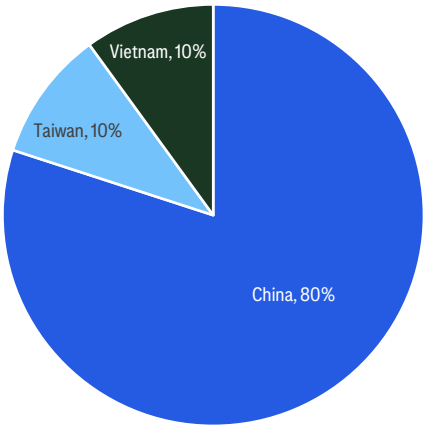
Voltronic adopts a pure-play ODM business model and does not market products under its own brand. This avoids direct competition with customers and has helped the company establish long-term relationships with most global Tier-1 UPS brands, as well as numerous regional brand owners.

The company provides customers with product design, R&D, sourcing, manufacturing and system integration services, with a focus on high-mix, relatively low-volume and customized products.

Voltronic's core products include offline UPS systems ranging from 400VA to 3kVA and on-line UPS systems ranging from 1kVA to 3MVA. The company has progressively expanded from conventional single phase UPS products into three-phase, high-power and modular UPS systems. It has also diversified into off-grid and hybrid solar inverters, energy storage systems, and EV chargers.

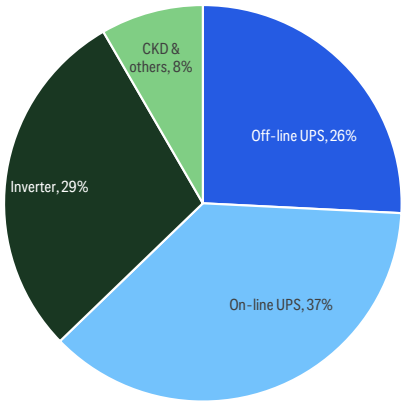
In 2025, online UPS contributed 37% of group sales, offline UPS 26%, inverters 29% and others 8%. By geo, Asia accounted for 60% of group sales, followed by Americas 22%, EU 16% and others 2%.

Figure 31. Voltronic: Capacity Mix



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Company Reports, Citi Research

Figure 32. Voltronic: Sales by Product (2025)



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Company Reports, Citi Research

Figure 33. Voltronic: Photo of Offline UPS

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Voltronic, Citi Research

Figure 34. Voltronic: Photo of Online Power Modular UPS

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Voltronic, Citi Research

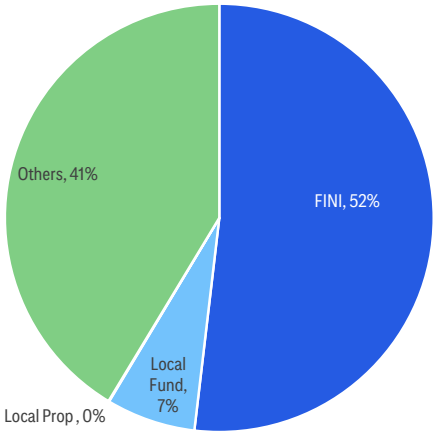
Shareholder Structure

As of March 2026, founder, chairman and general manager Alex Hsieh was the largest individual shareholder, holding 9.6% of outstanding shares. Based on disclosed direct, family-related and investment company holdings, we estimate the founder and related parties collectively controlled c.30% of outstanding shares.

Key Management Team

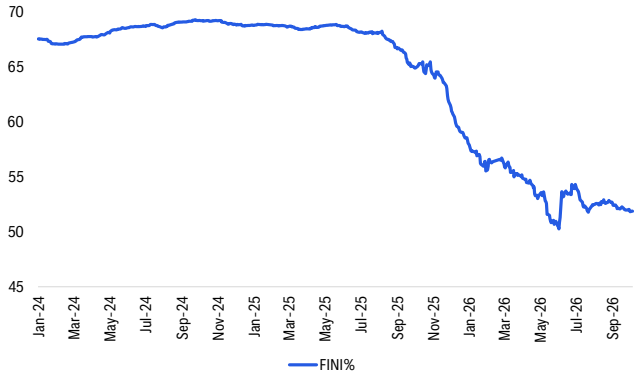
- **Alex Hsieh – Founder, Chairman and General Manager:** Mr. Hsieh has led Voltronic since its establishment in 2008. He holds a bachelor's degree in business management from Tatung University and previously served as Chairman and General Manager of Centralion Industrial and as a director of Phoenixtec Power. His extensive experience in the UPS industry underpins Voltronic's pure-play ODM strategy.
- **Monica Wang – Vice General Manager:** Ms. Wang joined Voltronic at its establishment in 2008. She previously worked as a manager at Centralion Industrial.
- **Chih-Hsin Chin – Vice General Manager:** Mr. Chin joined Voltronic in 2008. Before joining the company, he worked as a manager at Phoenixtec Power.

Figure 35. Voltronic: Shareholder Mix



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: TEJ, Citi Research

Figure 36. Voltronic: FINI Holding %



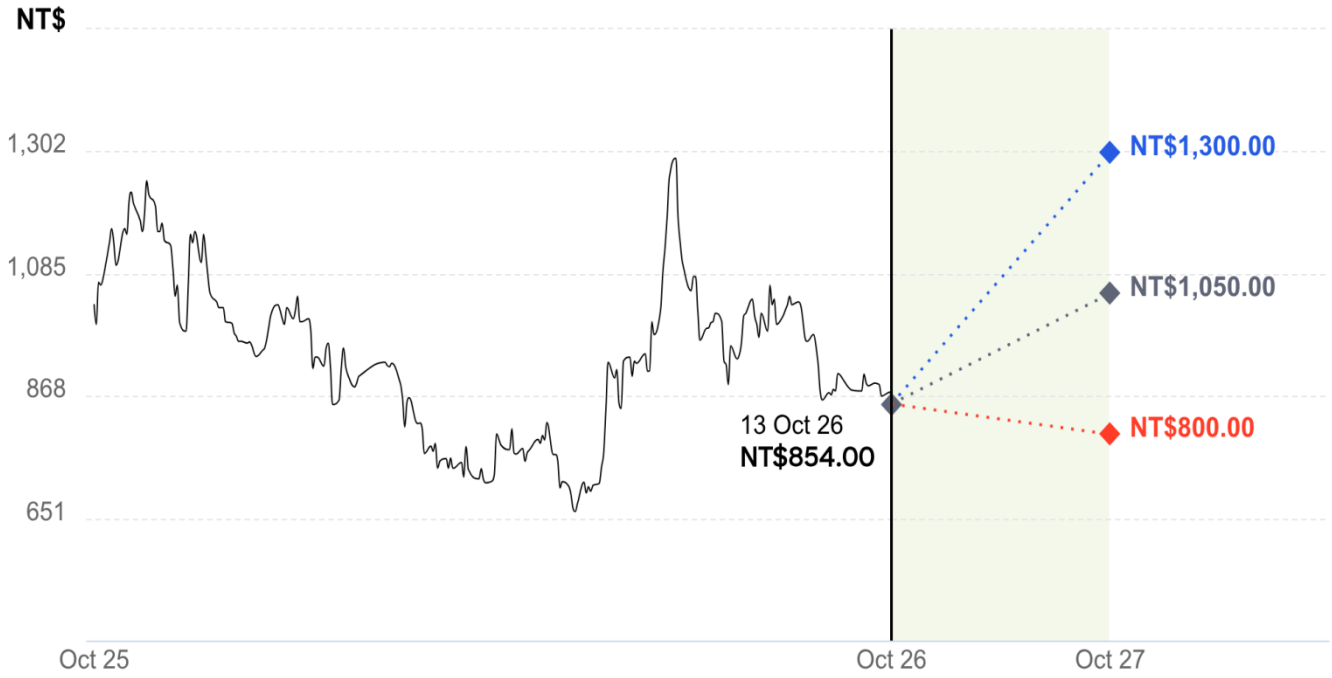
© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: TEJ, Citi Research

Companies Mentioned:

ABB (ABBN.S; SFr80.02; 2; 12 Oct 26; 17:30) | Delta Electronics (2308.TW; NT$1775.0; 1; 13 Oct 26; 13:30) | Eaton Corp (ETN.N; US$426.69; 1; 12 Oct 26; 16:00) | Emerson Electric Co. (EMR.N; US$162.46; 1; 12 Oct 26; 16:00) | Fuji Electric (6504.T; ¥14610.0; 1; 13 Oct 26; 15:30) | Mitsubishi Electric (6503.T; ¥5177.0; 1; 13 Oct 26; 15:30) | Schneider Electric (SCHN.PA; €265.35; 1; 12 Oct 26; 17:30) | Siemens Energy AG (ENR1n.DE; €146.78; 2; 12 Oct 26; 17:30) | Vertiv Holdings Co (VRT.N; US$239.36; 1; 12 Oct 26; 16:00)

Bull/Bear: Voltronic Power (6409.TW)

NT$1,300.00 ▲52% Upside | NT$1,050.00 ▲23% Upside | NT$800.00 ▼6.3% Downside



Spread 59pp
Current Price and expected returns (upside/downside) as of 13 Oct 2026

BULL Assumptions

- Target P/E of 30x

BASE Assumptions

- Sales growth of 5-10% y/y in 2027E
- GPM of 28-28.5% in 2027E
- Target P/E of 25x

BEAR Assumptions

- Target P/E of 19x

Voltronic Power

Company description

Established in 2008 and listed on the Taiwan Stock Exchange in 2014, Voltronic is a Taiwan-based design and manufacturing service (DMS) provider specializing in uninterrupted power supplies (UPS) and power-conversion products. Voltronic adopts a pure-play ODM business model, which avoids direct competition with customers and has helped the company establish long-term relationships with most global Tier-1 UPS brands. The company is headquartered in Taipei and operates manufacturing facilities in Taiwan, China and Vietnam.

Investment strategy

We have a Buy rating on Voltronic, supported by an earnings recovery, structural UPS outsourcing and emerging opportunities in AIDC power infrastructure. Its customization capabilities and longstanding relationships with western power vendors position it to capture additional outsourcing orders and expand into higher power products. HVDC power racks offer a L-T growth opportunity and potential re-rating catalyst. We view 2026E as earnings trough, with a richer online UPS mix and operating leverage driving a 28% earnings rebound in 2027E.

Valuation

We value Voltronic using a forward target P/E of 25x our 2027E EPS to derive a target price of NT$1050, broadly in line with its long term average forward P/E since 2013. We believe forward P/E is appropriate given the company's established earnings base and the importance of product mix and operating leverage to its profitability.

Risks

Key downside risks to our Buy rating and target price include: 1) Slower UPS demand or outsourcing gain. 2) Less favorable product mix and margin pressure. 3) Persistent PV inverter competition. 4) Weaker macro economy

If you are visually impaired and would like to speak to a Citi representative regarding the details of the graphics in this document, please call USA 1-888-500-5008 (TTY: 711), from outside the US +1-210-677-3788

Appendix A-1

ANALYST CERTIFICATION

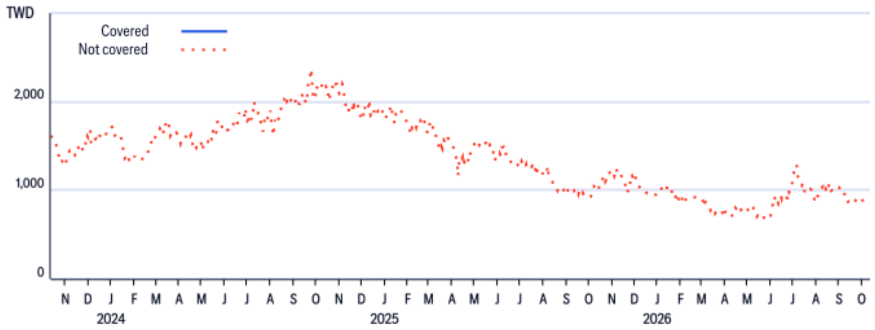
The research analysts primarily responsible for the preparation and content of this research report are either (i) designated by "AC" in the author block or (ii) listed in bold alongside content which is attributable to that analyst. If multiple AC analysts are designated in the author block, each analyst is certifying with respect to the entire research report other than (a) content attributable to another AC certifying analyst listed in bold alongside the content and (b) views expressed solely with respect to a specific issuer which are attributable to another AC certifying analyst identified in the price charts or rating history tables for that issuer shown below. Each of these analysts certify, with respect to the sections of the report

for which they are responsible: (1) that the views expressed therein accurately reflect their personal views about each issuer and security referenced and were prepared in an independent manner, including with respect to Citigroup Global Markets Inc. and its affiliates; and (2) no part of the research analyst's compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed by that research analyst in this report.

IMPORTANT DISCLOSURES

Voltronic Power (6409.TW)

Ratings and Target Price History
Fundamental Research



*Indicates Change

Rating/target price changes above reflect Eastern Time

A member of Citigroup's Board of Directors is also on the Board of Directors of Emerson Electric Co.

Within the past 12 months, Citigroup Global Markets Inc. or its affiliates has acted as manager or co-manager of an offering of securities of Schneider Electric,Vertiv Holdings Co.

Citigroup Global Markets Inc. or its affiliates has received compensation for investment banking services provided within the past 12 months from ABB,Delta Electronics,Eaton Corp,Schneider Electric,Siemens Energy AG,Vertiv Holdings Co.

Citigroup Global Markets Inc. or its affiliates expects to receive or intends to seek, within the next three months, compensation for investment banking services from ABB.

Citigroup Global Markets Inc. or its affiliates received compensation for products and services other than investment banking services from ABB,Delta Electronics,Eaton Corp,Emerson Electric Co.,Fuji Electric,Mitsubishi Electric,Schneider Electric,Siemens Energy AG,Vertiv Holdings Co in the past 12 months.

Citigroup Global Markets Inc. or its affiliates currently has, or had within the past 12 months, the following as investment banking client(s): ABB,Delta Electronics,Eaton Corp,Schneider Electric,Siemens Energy AG,Vertiv Holdings Co.

Citigroup Global Markets Inc. or its affiliates currently has, or had within the past 12 months, the following as clients, and the services provided were non-investment-banking, securities-related: ABB,Delta Electronics,Eaton Corp,Emerson Electric Co.,Mitsubishi Electric,Schneider Electric,Siemens Energy AG,Vertiv Holdings Co.

Citigroup Global Markets Inc. or its affiliates currently has, or had within the past 12 months, the following as clients, and the services provided were non-investment-banking, non-securities-related: ABB,Delta Electronics,Eaton Corp,Emerson Electric Co.,Fuji Electric,Mitsubishi Electric,Schneider Electric,Siemens Energy AG,Vertiv Holdings Co.

Citigroup Global Markets Inc. and/or its affiliates has a significant financial interest in relation to ABB,Delta Electronics,Eaton Corp,Emerson Electric Co.,Schneider Electric,Siemens Energy AG,Vertiv Holdings Co. (For an explanation of the determination of significant financial interest, please refer to the policy for managing conflicts of interest which can be found at www.citiVelocity.com.)

Analysts' compensation is determined by Citi Research management and Citigroup's senior management and is based upon activities and services intended to benefit the investor clients of Citigroup Global Markets Inc. and its affiliates (the "Firm"). Compensation is not linked to specific transactions or recommendations. Like all Firm employees, analysts receive compensation that is impacted by overall Firm profitability which includes investment banking, sales and trading, and principal trading revenues. One factor in equity research analyst compensation is arranging corporate access events between institutional clients and the management teams of covered companies. Typically, company management is more likely to participate when the analyst has a positive view of the company.

For financial instruments recommended in the Product in which the Firm is not a market maker, the Firm is a liquidity provider in such financial instruments (and any underlying instruments) and may act as principal in connection with transactions in such instruments. The Firm is a regular issuer of traded financial instruments linked to securities that may have been recommended in the Product. The Firm regularly trades in the securities of the issuer(s) discussed in the Product. The Firm may engage in securities transactions in a manner inconsistent with the Product and, with respect to securities covered by the Product, will buy or sell from customers on a principal basis.

Unless stated otherwise neither the Research Analyst nor any member of their team has viewed the material operations of the Companies for which an investment view has been provided within the past 12 months.

For important disclosures (including copies of historical disclosures) regarding the companies that are the subject of this Citi Research product ("the Product"), please contact Citi Research, 388 Greenwich Street, 6th Floor, New York, NY, 10013, Attention: Legal/Compliance [E6WYB6412478]. In addition, the same important disclosures, with the exception of the Valuation and Risk assessments and historical disclosures, are contained on the Firm's disclosure website at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures. Valuation and Risk assessments can be found in the text of the most recent research note/report regarding the subject company. Pursuant to the Market Abuse Regulation a history of all Citi Research recommendations published during the preceding 12-month period can be accessed via Citi Velocity (https://www.citivelocity.com/cv2) or your standard distribution portal. Historical disclosures (for up to the past three years) will be provided upon request.

Citi Research Equity Ratings Distribution

	12 Month Rating			Catalyst Watch		
Data current as of 01 Oct 2026	**Buy**	**Hold**	**Sell**	**Buy**	**Hold**	**Sell**
Citi Research Global Fundamental Coverage (Neutral=Hold)	61%	32%	7%	36%	48%	16%
% of companies in each rating category that are investment banking clients	39%	43%	30%	42%	38%	36%

Guide to Citi Research Fundamental Research Investment Ratings:
Citi Research stock recommendations include an investment rating and an optional risk rating to highlight high risk stocks.
Risk rating takes into account both price volatility and fundamental criteria. Stocks will either have no risk rating or a High risk rating assigned.
Investment Ratings: Citi Research investment ratings are Buy, Neutral and Sell. Our ratings are a function of analyst expectations of expected total return ("ETR") and risk. ETR is the sum of the forecast price appreciation (or depreciation) plus the dividend yield for a stock within the next 12 months. The target price is based on a 12 month time horizon. The Investment rating definitions are: Buy (1) ETR of 15% or more or 25% or more for High risk stocks; and Sell (3) for negative ETR. Any covered stock not assigned a Buy or a Sell is a Neutral (2). For stocks rated Neutral (2), if an analyst believes that there are insufficient valuation drivers and/or investment catalysts to derive a positive or negative investment view, they may elect with the approval of Citi Research management not to assign a target price and, thus, not derive an ETR. Citi Research may suspend its rating and target price and assign “Rating Suspended” status for regulatory and/or internal policy reasons. Citi Research may also suspend its rating and target price and assign "Under Review" status for other exceptional circumstances (e.g. lack of information critical to the analyst's thesis, trading suspension) affecting the company and/or trading in the company's securities. In both such situations, the rating and target price will show as “—“ and “-“ respectively in the rating history price chart. Prior to 11 April 2022 Citi Research assigned “Under Review” status to both situations and prior to 11 Nov 2020 only in exceptional circumstances. As soon as practically possible, the analyst will publish a note re-establishing a rating and investment thesis. Investment ratings are determined by the ranges described above at the time of initiation of coverage, a change in investment and/or risk rating, or a change in target price (subject to limited management discretion). At times, the expected total returns may fall outside of these ranges because of market price movements and/or other short-term volatility or trading patterns. Such interim deviations will be permitted but will become subject to review by Research Management. Your decision to buy or sell a security should be based upon your personal investment objectives and should be made only after evaluating the stock's expected performance and risk.

Catalyst Watch/Short Term Views (“STV”) Ratings Disclosure:
Catalyst Watch and STV Upside/Downside calls: Citi Research may also include a Catalyst Watch or STV Upside or Downside call to indicate the analyst expects the share price to rise (fall) in absolute terms over a specified period of 30 or 90 days in reaction to one or more specific near-term catalysts or events impacting the company or the market. A Catalyst Watch will be published when Analyst confidence is high that an impact to share price will occur; it will be a STV when confidence level is moderate. A Catalyst Watch or STV Upside/Downside call will automatically expire at the end of the specified 30/90 day period. The Catalyst Watch will also be automatically removed if share price performance (calculated at market close) exceeds 15% against the direction of the call (unless over-ridden by the analyst). The analyst may also remove a Catalyst Watch or STV call prior to the end of the specified period in a published research note. A Catalyst Watch/STV Upside or Downside call may be different from and does not affect a stock’s fundamental equity rating, which reflects a longer-term total absolute return expectation. For purposes of FINRA ratings-distribution-disclosure rules, a Catalyst Watch/STV Upside call corresponds to a buy recommendation and a Catalyst Watch/STV Downside call corresponds to a sell recommendation. Any stock not assigned to a Catalyst Watch Upside, Catalyst Watch Downside, STV Upside, or STV Downside call is considered Catalyst Watch/STV No View. For purposes of FINRA ratings distribution-disclosure rules, we correspond Catalyst Watch/STV No View to Hold in our ratings distribution table for our Catalyst Watch/STV Upside/Downside rating system. However, we reiterate that we do not consider No View to be a recommendation. For all Catalyst Watch/STV Upside/Downside calls, risk exists that the catalyst(s) and associated share-price movement will not materialize as expected.

Privately held companies:
The legal entities through which Citi conducts its business do not permit investment research recommendations, ratings, target prices, or estimates on any privately held companies (“private companies”) referenced herein. Citi Research does not utilize a rating system for private companies. Any information on private companies in this report is based on generally available information and, although obtained from sources believed by Citi to be reliable, its accuracy and completeness are

not guaranteed. Private companies are generally not subject to reporting obligations, which may result in less publicly available information. The information contained in this report may differ from information a private company may subsequently make public or otherwise share with investors. Continuation of publishing private company reports or private company sections within reports is at Citi's discretion, in accordance with internal policies.

RESEARCH ANALYST AFFILIATIONS / NON-US RESEARCH ANALYST DISCLOSURES

The legal entities employing the authors of this report are listed below (and their regulators are listed further herein). Non-US research analysts who have prepared this report (i.e., all research analysts listed below other than those identified as employed by Citigroup Global Markets Inc.) are not registered/qualified as research analysts with FINRA. Such research analysts may not be associated persons of the member organization (but are employed by an affiliate of the member organization) and therefore may not be subject to the FINRA Rule 2241 restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

Citigroup Global Markets Taiwan Securities Company Limited	Angela Hsu

OTHER DISCLOSURES

Any price(s) of instruments mentioned in recommendations are as of the prior day's market close on the primary market for the instrument, unless otherwise stated.

The completion and first dissemination of any recommendations made within this research report are as of the Eastern date-time displayed at the top of the Product. If the Product references views of other analysts then please refer to the price chart or rating history table for the date/time of completion and first dissemination with respect to that view.

Regulations in various jurisdictions require that where a recommendation differs from any of the author's previous recommendations concerning the same financial instrument or issuer that has been published during the preceding 12-month period that the change(s) and the date of that previous recommendation are indicated. For fundamental coverage please refer to the price chart or rating change history within this disclosure appendix or the issuer disclosure summary at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures.

Citi Research has implemented policies for identifying, considering and managing potential conflicts of interest arising as a result of publication or distribution of investment research. A description of these policies can be found at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures.

The proportion of all Citi Research research recommendations that were the equivalent to "Buy","Hold","Sell" at the end of each quarter over the prior 12 months (with the % of these that had received investment firm services from Citi in the prior 12 months shown in brackets) is as follows; Q2 2026 Buy 33%(63%), Hold 44% (51%), Sell 23% (49%), RV 0.5% (87%): Q1 2026 Buy 33% (63%), Hold 44% (52%), Sell 23% (46%), RV 0.5% (89%); Q4 2025 Buy 33% (63%), Hold 44% (50%), Sell 23% (46%), RV 0.4% (91%); Q3 2025 Buy 33%(61%), Hold 44% (52%), Sell 23% (50%), RV 0.4% (80%). For the purposes of disclosing recommendations other than for equity (whose definitions can be found in the corresponding disclosure sections), "Buy" means a positive directional trade idea; "Sell" means a negative directional trade idea; and "Relative Value" means any trade idea which does not have a clear direction to the investment strategy.

European regulations require a 5 year price history when past performance of a security is referenced. CitiVelocity's Charting Tool (https://www.citivelocity.com/cv2/#go/CHARTING_3_Equities) provides the facility to create customisable price charts including a five year option. This tool can be found in the Data & Analytics section under any of the asset class menus in CitiVelocity (https://www.citivelocity.com/). For further information contact CitiVelocity support (https://www.citivelocity.com/cv2/go/CLIENT_SUPPORT). The source for all referenced prices, unless otherwise stated, is DataCentral, which sources price information from LSEG Data & Analytics. Past performance is not a guarantee or reliable indicator of future results. Forecasts are not a guarantee or reliable indicator of future performance.

Investors should always consider the investment objectives, risks, and charges and expenses of an ETF carefully before investing. The applicable prospectus and key investor information document (as applicable) for an ETF should contain this and other information about such ETF. It is important to read carefully any such prospectus before investing. Clients may obtain prospectuses and key investor information documents for ETFs from the applicable distributor or authorized participant, the exchange upon which an ETF is listed and/or from the applicable website of the applicable ETF issuer. The value of the investments and any accruing income may fall or rise. Any past performance, prediction or forecast is not indicative of future or likely performance. Any information on ETFs contained herein is provided strictly for illustrative purposes and should not be deemed an offer to sell or a solicitation of an offer to purchase units of any ETF either explicitly or implicitly. The opinions expressed are those of the authors and do not necessarily reflect the views of ETF issuers, any of their agents or their affiliates.

Citigroup Global Markets India Private Limited and/or its affiliates may have, from time to time, actual or beneficial ownership of 1% or more in the debt securities of the subject issuer.

Please be advised that pursuant to Executive Order 13959 as amended (the "Order"), U.S. persons are prohibited from investing in securities of any company determined by the United States Government to be the subject of the Order. This research is not intended to be used or relied upon in any way that could result in a violation of the Order. Investors are encouraged to rely upon their own legal counsel for advice on compliance with the Order and other economic sanctions programs administered and enforced by the Office of Foreign Assets Control of the U.S. Treasury Department.

This communication is directed at persons who are "Eligible Clients" as such term is defined in the Israeli Regulation of Investment Advice, Investment Marketing and Investment Portfolio Management law, 1995 (the "Advisory Law"). Within Israel,

this communication is not intended for retail clients and Citi will not make such products or transactions available to retail clients or to non-Eligible Clients. The presenter is not licensed as investment advisor or investment marketer by the Israeli Securities Authority ("ISA") and this communication does not constitute investment or marketing advice. The information contained herein may relate to matters that are not regulated by the ISA. Any securities which are the subject of this communication may not be offered or sold to any Israeli person except pursuant to a security offering exemption according to the Israeli Securities Law, 1968and the public offering rules provided thereunder.

Citi Research broadly and simultaneously disseminates its research content to the Firm's institutional and retail clients via the Firm's proprietary electronic distribution platforms (e.g., Citi Velocity and various Global Wealth platforms). As a convenience, certain, but not all, research content may be distributed through third party aggregators. Clients may receive published research reports by email, on a discretionary basis, and only after such research content has been broadly disseminated. Certain research is made available only to institutional investors to satisfy regulatory requirements. The level and types of services provided by Citi Research analysts to clients may vary depending on various factors such as the client's individual preferences as to the frequency and manner of receiving communications from analysts, the client's risk profile and investment focus and perspective (e.g. market-wide, sector specific, long term, short-term etc.), the size and scope of the overall client relationship with the Firm and legal and regulatory constraints.

Pursuant to Comissão de Valores Mobiliários Resolução 20 and ASIC Regulatory Guide 264, Citi is required to disclose whether a Citi related company or business has a commercial relationship with the subject company. Considering that Citi operates multiple businesses in more than 100 countries around the world, it is likely that Citi has a commercial relationship with the subject company.

Securities recommended, offered, or sold by the Firm: (i) are not insured by the Federal Deposit Insurance Corporation; (ii) are not deposits or other obligations of any insured depository institution (including Citibank); and (iii) are subject to investment risks, including the possible loss of the principal amount invested. The Product is for informational purposes only and is not intended as an offer or solicitation for the purchase or sale of a security. Any decision to purchase securities mentioned in the Product must take into account existing public information on such security or any registered prospectus. Although information has been obtained from and is based upon sources that the Firm believes to be reliable, we do not guarantee its accuracy and it may be incomplete and condensed. Note, however, that the Firm has taken all reasonable steps to determine the accuracy and completeness of the disclosures made in the Important Disclosures section of the Product. The Firm's research department has received assistance from the subject company(ies) referred to in this Product including, but not limited to, discussions with management of the subject company(ies). Firm policy prohibits research analysts from sending draft research to subject companies. However, it should be presumed that the author of the Product has had discussions with the subject company to ensure factual accuracy prior to publication. Statements and views concerning ESG (environmental, social, governance) factors are typically based upon public statements made by the affected company or other public news, which the author may not have independently verified. ESG factors are one consideration that investors may choose to examine when making investment decisions. All opinions, projections and estimates constitute the judgment of the author as of the date of the Product and these, plus any other information contained in the Product, are subject to change without notice. Prices and availability of financial instruments also are subject to change without notice. Notwithstanding other departments within the Firm advising the companies discussed in this Product, information obtained in such role is not used in the preparation of the Product. Although Citi Research does not set a predetermined frequency for publication, if the Product is a fundamental equity or credit research report, it is the intention of Citi Research to provide research coverage of the covered issuers, including in response to news affecting the issuer. For non-fundamental research reports, Citi Research may not provide regular updates to the views, recommendations and facts included in the reports. Notwithstanding that Citi Research maintains coverage on, makes recommendations concerning or discusses issuers, Citi Research may be periodically restricted from referencing certain issuers due to legal or policy reasons. Where a component of a published trade idea is subject to a restriction, the trade idea will be removed from any list of open trade ideas included in the Product. Upon the lifting of the restriction, the trade idea will either be re-instated in the open trade ideas list if the analyst continues to support it or it will be officially closed. Citi Research may provide different research products and services to different classes of customers (for example, based upon long-term or short-term investment horizons) that may lead to differing conclusions or recommendations that could impact the price of a security contrary to the recommendations in the alternative research product, provided that each is consistent with the rating system for each respective product.

Investing in non-U.S. securities, including ADRs, may entail certain risks. The securities of non-U.S. issuers may not be registered with, nor be subject to the reporting requirements of the U.S. Securities and Exchange Commission. There may be limited information available on foreign securities. Foreign companies are generally not subject to uniform audit and reporting standards, practices and requirements comparable to those in the U.S. Securities of some foreign companies may be less liquid and their prices more volatile than securities of comparable U.S. companies. In addition, exchange rate movements may have an adverse effect on the value of an investment in a foreign stock and its corresponding dividend payment for U.S. investors. Net dividends to ADR investors are estimated, using withholding tax rates conventions, deemed accurate, but investors are urged to consult their tax advisor for exact dividend computations. Investors who have received the Product from the Firm may be prohibited in certain states or other jurisdictions from purchasing securities mentioned in the Product from the Firm. Please ask your Financial Consultant for additional details. Citigroup Global Markets Inc. takes responsibility for the

Product in the United States. Any orders by US investors resulting from the information contained in the Product may be placed only through Citigroup Global Markets Inc.

The Citigroup legal entity that takes responsibility for the production of the Product is the legal entity which the first named author is employed by.

The Product is made available in **Australia** through Citigroup Global Markets Australia Pty Limited. (ABN 64 003 114 832 and AFSL No. 240992), participant of the ASX Group and regulated by the Australian Securities & Investments Commission. Citigroup Centre, 2 Park Street, Sydney, NSW 2000. Citigroup Global Markets Australia Pty Limited is not an Authorised Deposit-Taking Institution under the Banking Act 1959, nor is it regulated by the Australian Prudential Regulation Authority.

The Product is made available in **Brazil** by Citigroup Global Markets Brasil - CCTVM SA, which is regulated by CVM - Comissão de Valores Mobiliários ("CVM"), BACEN - Brazilian Central Bank, APIMEC - Associação dos Analistas e Profissionais de Investimento do Mercado de Capitais and ANBIMA – Associação Brasileira das Entidades dos Mercados Financeiro e de Capitais. Av. Paulista, 1111 - 14º andar(parte) - CEP: 01311920 - São Paulo - SP.

This Product is available in **Chile** through Banchile Corredores de Bolsa S.A., an indirect subsidiary of Citigroup Inc., which is regulated by the Comisión Para El Mercado Financiero. Enrique Foster Sur, 20, piso 6, Las Condes, Santiago, Chile.

Disclosure for investors in the Republic of Colombia :This communication or message does not constitute a professional recommendation to make investment in the terms of article 2.40.1.1.2 of Decree 2555 de 2010 or the regulations that modify, substitute or complement it. Para la elaboración y distribución de informes de investigación y de comunicaciones generales de que trata este artículo no se requiere ser una entidad vigilada por la Superintendencia Financiera de Colombia.

The Product is made available in **Germany** by Citigroup Global Markets Europe AG ("CGME"), which is regulated by the European Central Bank and the German Federal Financial Supervisory Authority (Bundesanstalt fur Finanzdienstleistungsaufsicht BaFin). Börsenplatz 9, 60313 Frankfurt am Main, Germany.

Unless otherwise specified, if the analyst who prepared this report is based in Hong Kong and it relates to “securities” (as defined in the Securities and Futures Ordinance (Cap.571 of the Laws of Hong Kong)), the report is issued in Hong Kong by Citigroup Global Markets Asia Limited. Citigroup Global Markets Asia Limited is regulated by Hong Kong Securities and Futures Commission. If the report is prepared by a non-Hong Kong based analyst, please note that such analyst (and the legal entity that the analyst is employed by or accredited to) is not licensed/registered in Hong Kong and they do not hold themselves out as such. Please refer to the section “Research Analyst Affiliations / Non-US Research Analyst Disclosures” for the details of the employment entity of the analysts.

The Product is made available in **India** by Citigroup Global Markets India Private Limited (CGM), which is regulated by the Securities and Exchange Board of India (SEBI), as a Research Analyst (SEBI Registration No. INH000000438). CGM is also actively involved in the business of merchant banking (SEBI Registration No. INM000010718) and stock brokerage ((SEBI Registration No. INZ000263033) in India, and is registered with SEBI in this regard. Registration granted by SEBI, enlistment with Bombay Stock Exchange (BSE) and certification from National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. CGM’s registered office is at 1202, 12th Floor, First International Financial Centre (FIFC), G Block, Bandra Kurla Complex, Bandra East, Mumbai – 400098 & registered Tel: +91 22 61759999. Citi maintains robust policies, procedures,controls, and training to ensure continued compliance with all applicable rulesand regulations. All recommendations contained herein are made by dulyqualified research analysts. CGM’s Corporate Identity Number is U99999MH2000PTC126657, and its Compliance Officer [Vishal Bohra] contact details are: Tel:+91-022-61759994, Fax:+91-022-61759851, Email: cgmcompliance@citi.com. The Investor Charter in respect of Research Analysts, the Compliance Audit Report and Complaints information can be found at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures. The grievance officer [Niraj Mody] contact details are Tel: +91-22-42775002, Email: EMEA.CR.Complaints@citi.com. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. SEBI prescribed Client Terms & Conditions can be found at https://www.citivelocity.com/rendition/authfilelinksvcs/eppublic/V1/file?paramData=ZmlsZU5hbWU9L1MzL0dETVMvcHViRGlzY2xvc3VyZUh0bWxzL2dkbV9kaXNfc2ViaV9UZXJtc29mVXNlLmh0bWw

The Product is made available in **Indonesia** through PT Citigroup Sekuritas Indonesia. Citibank Tower 10/F, Pacific Century Place, SCBD lot 10, Jl. Jend Sudirman Kav 52-53, Jakarta 12190, Indonesia. Neither this Product nor any copy hereof may be distributed in Indonesia or to any Indonesian citizens wherever they are domiciled or to Indonesian residents except in compliance with applicable capital market laws and regulations. This Product is not an offer of securities in Indonesia. The securities referred to in this Product have not been registered with the Capital Market and Financial Services Authority (OJK) pursuant to relevant capital market laws and regulations, and may not be offered or sold within the territory of the Republic of Indonesia or to Indonesian citizens through a public offering or in circumstances which constitute an offer within the meaning of the Indonesian capital market laws and regulations.

The Product is made available in **Japan** by Citigroup Global Markets Japan Inc. ("CGMJ"), which is regulated by Financial Services Agency, Securities and Exchange Surveillance Commission, Japan Securities Dealers Association, Tokyo Stock Exchange and Osaka Securities Exchange. Otemachi Park Building, 1-1-1 Otemachi, Chiyoda-ku, Tokyo 100-8132 Japan. In the event that an error is found in an CGMJ research report, a revised version will be posted on the Firm's Citi Velocity website. If you have questions regarding Citi Velocity, please call (81 3) 6270-3019 for help.

The product is made available in the **Kingdom of Saudi Arabia** in accordance with Saudi laws through Citigroup Saudi Arabia, which is regulated by the Capital Market Authority (CMA) under CMA license (17184-31). 2239 Al Urubah Rd – Al Olaya Dist. Unit No. 18, Riyadh 12214 – 9597, Kingdom Of Saudi Arabia.

The Product is made available in **Korea** by Citigroup Global Markets Korea Securities Ltd. (CGMK), which is regulated by the Financial Services Commission, the Financial Supervisory Service and the Korea Financial Investment Association (KOFIA). The address of CGMK is Citibank Center, 50 Saemunan-ro, Jongno-gu, Seoul 03184, Korea. KOFIA makes available registration information of research analysts on its website. Please visit the following website if you wish to find KOFIA registration information on research analysts of CGMK. http://dis.kofia.or.kr/websquare/index.jsp?w2xPath=/wq/fundMgr/DISFundMgrAnalystList.xml&divisionId=MDIS03002002000000&serviceId=SDIS03002002000. The Product is made available in Korea by Citibank Korea Inc., which is regulated by the Financial Services Commission and the Financial Supervisory Service. Address is Citibank Center, 50 Saemunan-ro, Jongno-gu, Seoul 03184, Korea. This research report is intended to be provided only to Professional Investors as defined in the Financial Investment Services and Capital Market Act and its Enforcement Decree in Korea.

The Product is made available in **Malaysia** by Citigroup Global Markets Malaysia Sdn Bhd (Registration No. 199801004692 (460819-D)) ("CGMM") to its clients and CGMM takes responsibility for its contents as regards CGMM's clients. CGMM is regulated by the Securities Commission Malaysia. Please contact CGMM at Level 43 Menara Citibank, 165 Jalan Ampang, 50450 Kuala Lumpur, Malaysia in respect of any matters arising from, or in connection with, the Product.

The Product is made available in **Mexico** by Citi México Casa de Bolsa, S.A. de C.V., Grupo Financiero Citi México which is a wholly owned subsidiary of Citigroup Inc. and is regulated by Comision Nacional Bancaria y de Valores. Prolongación Reforma 1196, 24 floor, Colonia Santa Fe, Alcaldía Cuajimalpa de Morelos, C.P. 05348, Ciudad de México.

The Product is made available in **Poland** by Biuro Maklerskie Banku Handlowego (DMBH), separate department of Bank Handlowy w Warszawie S.A. a subsidiary of Citigroup Inc., which is regulated by Komisja Nadzoru Finansowego. Biuro Maklerskie Banku Handlowego (DMBH), ul.Senatorska 16, 00-923 Warszawa.

The Product is made available in **Singapore** through Citigroup Global Markets Singapore Pte. Ltd. ("CGMSPL"), a capital markets services and Exempt Financial Advisor license holder, and regulated by Monetary Authority of Singapore. Please contact CGMSPL at 8 Marina View, 21st Floor Asia Square Tower 1, Singapore 018960, in respect of any matters arising from, or in connection with, the analysis of this document. This Product is intended for recipients who are accredited, expert and institutional investors as defined under the Securities and Futures Act 2001. For Citi Private Bank, the Product is made available in Singapore by Citi Private Bank through Citibank, N.A., Singapore Branch. Citibank N.A., Singapore Branch is a licensed bank in Singapore that is regulated by the Monetary Authority of Singapore. Please contact your Private Banker in Citibank N.A., Singapore Branch if you have any queries on or any matters arising from or in connection with this document. The Product is intended for recipients who are accredited, expert and institutional investors as defined under the Securities and Futures Act 2001. For Citibank Singapore Limited ("CSL"), the Product is distributed in Singapore by CSL to selected Citigold/Citigold Private Clients. CSL provides no independent research or analysis of the substance or in preparation of the Product. Please contact your Citigold//Citigold Private Client Relationship Manager in CSL if you have any queries on or any matters arising from or in connection with this document. The Product is intended for recipients who are accredited investors as defined under the Securities and Futures Act (Cap. 289).

Citigroup Global Markets (Pty) Ltd. is incorporated in the **Republic of South Africa** (company registration number 2000/025866/07) and its registered office is at 145 West Street, Sandton, 2196, Saxonwold. Citigroup Global Markets (Pty) Ltd. is regulated by JSE Securities Exchange South Africa, South African Reserve Bank and the Financial Services Board. The investments and services contained herein are not available to private customers in South Africa.

The Product is made available in the **Republic of China (Taiwan)** through Citigroup Global Markets Taiwan Securities Company Ltd. ("CGMTS"), 14F, 15F and 16F, No. 1, Songzhi Road, Taipei 110, Taiwan, subject to the license scope and the applicable laws and regulations in the Republic of China (Taiwan). CGMTS is regulated by the Securities and Futures Bureau of the Financial Supervisory Commission of Taiwan, the Republic of China (Taiwan). No portion of the Product may be reproduced or quoted in the Republic of China (Taiwan) by the press or any third parties [without the written authorization of CGMTS]. Pursuant to the applicable laws and regulations in the Republic of China (Taiwan), the recipient of the Product shall not take advantage of such Product to involve in any matters in which the recipient may have conflicts of interest. If the Product covers securities which are not allowed to be offered or traded in the Republic of China (Taiwan), neither the Product nor any information contained in the Product shall be considered as advertising the securities or making recommendation of the securities in the Republic of China (Taiwan). The Product is for informational purposes only and is not intended as an offer or solicitation for the purchase or sale of a security or financial products. Any decision to purchase securities or financial products mentioned in the Product must take into account existing public information on such security or the financial products or any registered prospectus.

The Product is made available in **Thailand** through Citicorp Securities (Thailand) Ltd., which is regulated by the Securities and Exchange Commission of Thailand. 399 Interchange 21 Building, 18th Floor, Sukhumvit Road, Klongtoey Nua, Wattana ,Bangkok 10110, Thailand.

Disclosure for investors in the Republic of Turkey: The Product is made available in **Turkey** through Citibank AS which is regulated by Capital Markets Board. Tekfen Tower, Eski Buyukdere Caddesi # 209 Kat 2B, 23294 Levent, Istanbul, Turkey.

Under Capital Markets Law of Turkey (Law No: 6362), the investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Comments and recommendations stated here rely on the individual opinions of the ones providing these comments and recommendations. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations. Furthermore, Citi Research is a division of Citigroup Global Markets Inc. (the "Firm"), which does and seeks to do business with companies and/or trades on securities covered in this research reports. As a result, investors should be aware that the Firm may have a conflict of interest that could affect the objectivity of this report, however investors should also note that the Firm has in place organisational and administrative arrangements to manage potential conflicts of interest of this nature.

In the **U.A.E**, these materials (the "Materials") are communicated byCitigroup Global Markets Limited, DIFC branch ("CGML"), an entityregistered in the Dubai International Financial Center ("DIFC") andlicensed and regulated by the Dubai Financial Services Authority ("DFSA",license #CL0221) to Professional Clients and Market Counterparties, as definedin DFSA regulations, only and should not be relied upon or distributed toRetail Clients. Financial products and/or services to which the Materialsrelate will only be made available to Professional Clients and MarketCounterparties. Citigroup Global Markets Limited DIFC Branch registered addressis Level 3, Gate District Building 02, Dubai International Financial Centre andcan be contacted on +971 4 509 97 90.

The Product is made available in **United Kingdom** by Citigroup Global Markets Limited, which is authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA. This material may relate to investments or services of a person outside of the UK or to other matters which are not authorised by the PRA nor regulated by the FCA and the PRA and further details as to where this may be the case are available upon request in respect of this material. Citigroup Centre, Canada Square, Canary Wharf, London, E14 5LB.

The Product is made available in **United States** and **Canada** by Citigroup Global Markets Inc., which is a member of FINRA and registered with the US Securities and Exchange Commission. 388 Greenwich Street, New York, NY 10013.

Unless specified to the contrary, within EU Member States, the Product is made available by Citigroup Global Markets Europe AG ("CGME"), which is regulated by the European Central Bank and the German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht-BaFin).

The Product is not to be construed as providing investment services in any jurisdiction where the provision of such services would not be permitted.
Subject to the nature and contents of the Product, the investments described therein are subject to fluctuations in price and/or value and investors may get back less than originally invested. Certain high-volatility investments can be subject to sudden and large falls in value that could equal or exceed the amount invested.The yield and average life of CMOs (collateralized mortgage obligations) referenced in this Product will fluctuate depending on the actual rate at which mortgage holders prepay the mortgages underlying the CMO and changes in current interest rates. Any government agency backing of the CMO applies only to the face value of the CMO and not to any premium paid. Certain investments contained in the Product may have tax implications for private customers whereby levels and basis of taxation may be subject to change. If in doubt, investors should seek advice from a tax adviser. The Product does not purport to identify the nature of the specific market or other risks associated with a particular transaction. Advice in the Product is general and should not be construed as personal advice given it has been prepared without taking account of the objectives, financial situation or needs of any particular investor. Accordingly, investors should, before acting on the advice, consider the appropriateness of the advice, having regard to their objectives, financial situation and needs. Prior to acquiring any financial product, it is the client's responsibility to obtain the relevant offer document for the product and consider it before making a decision as to whether to purchase the product.

Card Insights. Where this report references Card Insights data, Card Insights consists of selected data from a subset of Citi's proprietary credit card transactions. Such data has undergone rigorous security protocols to keep all customer information confidential and secure; the data is highly aggregated and anonymized so that all unique customer identifiable information is removed from the data prior to receipt by the report's author or distribution to external parties. This data should be considered in the context of other economic indicators and publicly available information. Further, the selected data represents only a subset of Citi's proprietary credit card transactions due to the selection methodology or other limitations and should not be considered as indicative or predictive of the past or future financial performance of Citi or its credit card business.

Citi Research product may source data from dataCentral. dataCentral is a Citi Research proprietary database, which includes the Firm's estimates, data from company reports and feeds from LSEG Data & Analytics. The source for all referenced prices, unless otherwise stated, is DataCentral. Past performance is not a guarantee or reliable indicator of future results. Forecasts are not a guarantee or reliable indicator of future performance. The printed and printable version of the research report may not include all the information <(e.g. certain financial summary information and comparable company data) that is linked to the online version available on the Firm's proprietary electronic distribution platforms.
Where included in this report, MSCI sourced information is the exclusive property of Morgan Stanley Capital International Inc. (MSCI). Without prior written permission of MSCI, this information and any other MSCI intellectual property may not be reproduced, redisseminated or used to create any financial products, including any indices. This information is provided on an "as is" basis. The user assumes the entire risk of any use made of this information. MSCI, its affiliates and any third party

involved in, or related to, computing or compiling the information hereby expressly disclaim all warranties of originality, accuracy, completeness, merchantability or fitness for a particular purpose with respect to any of this information. Without limiting any of the foregoing, in no event shall MSCI, any of its affiliates or any third party involved in, or related to, computing or compiling the information have any liability for any damages of any kind. MSCI, Morgan Stanley Capital International and the MSCI indexes are services marks of MSCI and its affiliates. Where data is attributed to Morningstar that data is © 2026 Morningstar, Inc. All Rights Reserved. That information: (1) is proprietary to Morningstar and/or its content providers; (2) may not be copied or distributed; and (3) is not warranted to be accurate, complete or timely. Neither Morningstar nor its content providers are responsible for any damages or losses arising from any use of this information.

The Firm accepts no liability whatsoever for the actions of third parties. The Product may provide the addresses of, or contain hyperlinks to, websites. Except to the extent to which the Product refers to website material of the Firm, the Firm has not reviewed the linked site. Equally, except to the extent to which the Product refers to website material of the Firm, the Firm takes no responsibility for, and makes no representations or warranties whatsoever as to, the data and information contained therein. Such address or hyperlink (including addresses or hyperlinks to website material of the Firm) is provided solely for your convenience and information and the content of the linked site does not in any way form part of this document. Accessing such website or following such link through the Product or the website of the Firm shall be at your own risk and the Firm shall have no liability arising out of, or in connection with, any such referenced website.

© 2026 Citigroup Global Markets Inc. Citi Research is a division of Citigroup Global Markets Inc. Citi and Citi and Arc Design are trademarks and service marks of Citigroup Inc. and its affiliates and are used and registered throughout the world. All rights reserved. The research data in this report are not intended to be used for the purpose of (a) determining the price of or amounts due in respect of (or to value) one or more financial products or instruments and/or (b) measuring or comparing the performance of, or defining the asset allocation of a financial product, a portfolio of financial instruments, or a collective investment undertaking, and any such use is strictly prohibited without the prior written consent of Citi Research. Any unauthorized use, duplication, redistribution or disclosure of this report (the "Product"), including, but not limited to, redistribution of the Product by electronic mail, posting of the Product on a website or page, and/or providing to a third party a link to the Product, is prohibited by law and will result in prosecution. The information contained in the Product is intended solely for the recipient and may not be further distributed by the recipient to any third party.

ADDITIONAL INFORMATION IS AVAILABLE UPON REQUEST

UMT (3491.TWO): LEO satellite to drive growth ahead; Sep rev +52% YoY while MoM decline on Aug pull-in; Buy

Allen Chang
+852-2978-2930 | allen.k.chang@gs.com
Goldman Sachs (Asia) L.L.C.

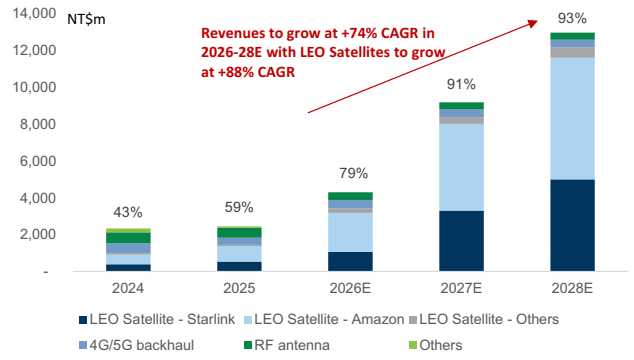
Verena Jeng
+852-2978-1681 | verena.jeng@gs.com
Goldman Sachs (Asia) L.L.C.

Ting Song
+852-2978-6466 | ting.song@gs.com
Goldman Sachs (Asia) L.L.C.

UMT delivered strong 3Q26E revenues growth of 74% YoY (or -7% MoM) to NT$843, while below our estimates, we attribute it to order momentum and the transition period of LEO satellite business towards the new platform. We remain positive and expect to see a rising revenues contribution from the LEO satellites business, and growing dollar content on the new platform, supporting both YoY and QoQ growth in 4Q26E revenue, along with GM improvement. Meanwhile, the company continues to work with multiple LEO satellite clients. We remain positive on UMT given (1) riding on the accelerating LEO satellite launches that enhance the satellite communication services and expand space applications, (2) increasing waveguide for LEO satellite dollar content given more bands are adopted per LEO satellite, and (3) waveguide end market expands from LEO satellite to AI data center for high-speed interconnection in scale up (report link). Maintain Buy.

UMT provides rectangular waveguides, which are conducting pipes with a rectangular cross-section used to guide the propagation of microwave or mmwave signals. Different shapes and combinations of waveguides can form various RF passive components, such as filter / diplexer, coupler, isolator, antenna, power amplifiers (PAs), etc. The company serves leading global LEO satellite operators, and its waveguides can be seen in LEO satellites / payloads (main), and gateways in ground stations.

Exhibit 1: UMT revenues to grow at 74% CAGR in 2026-28E with rising LEO satellite contribution



Source: Company data, Goldman Sachs Global Investment Research

3-month preview: The company's Sep revenues were up 52% YoY, or -9% MoM to NT$292m, while 22% below our estimates, and we attribute it to the product pull-in

Goldman Sachs does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html. Analysts employed by non-US affiliates are not registered/qualified as research analysts with FINRA in the U.S.

during Aug, resulting in 3Q26 revenues up 74% YoY to NT$843m. With the gradual ramp up of clients' new platform, we expect to see sequential MoM growth ahead, with Oct/ Nov revenues up 17%/ 19% MoM to NT$341m/ NT$406m, driven by a rising contribution from the LEO satellites.

Exhibit 2: We expect UMT Oct 2026 revenues to grow 49%/ 17% YoY/ MoM to NT$341m
UMT monthly/ quarterly revenues

	Jul 2026	Aug 2026	Sep 2026	Oct 2026E	Nov 2026E	Dec 2026E	2Q26	3Q26	4Q26E
Revenues (NT$m)	230	321	292	341	406	784	902	843	1,531
YoY	82%	93%	52%	49%	42%	145%	76%	74%	83%
MoM/QoQ	-10%	40%	-9%	17%	19%	93%	-12%	-7%	82%
GS estimates (NT$m)	277	311	374				966	926	
Actual vs. GS	-17%	3%	-22%				-7%	-9%	

Source: Company data, Goldman Sachs Global Investment Research

Earnings revision: We factor in UMT 3Q26 revenues and revise down earnings by 1%/ 1%/ 1% in 2026-28E mainly on lower revenues due to the momentum of product delivery, while we remain positive on the rising LEO satellite business exposure ahead.

Exhibit 3: Earnings revision

	2026E			2027E			2028E		
NT$m	Old	New	Diff %	Old	New	Diff %	Old	New	Diff %
Revenues	4,430	4,296	-3%	9,327	9,179	-2%	13,116	12,967	-1%
GP	2,627	2,569	-2%	6,041	5,974	-1%	8,625	8,557	-1%
OP	1,450	1,429	-1%	3,657	3,628	-1%	5,293	5,263	-1%
Net income	1,273	1,256	-1%	2,893	2,871	-1%	4,185	4,161	-1%
Margins									
GM	59.3%	59.8%		64.8%	65.1%		65.8%	66.0%	
OPM	32.7%	33.3%		39.2%	39.5%		40.4%	40.6%	
NM	28.7%	29.2%		31.0%	31.3%		31.9%	32.1%	

Source: Company data, Goldman Sachs Global Investment Research

Valuation: We continue to derive our target price based on a discounted P/E, and based on 2029E (methodology unchanged). We continue to derive the target P/E multiple from the average PEG&M ratio of peers in the satellite supply chain, and we remove outliers (telecom operators who face competition from LEO satellite operators). Our target 2029E discounted P/E multiple is at 37.1x (vs. previously at 37.0x) which is based on peers' avg. PEG&M, and UMT's 2029-30E avg. NI YoY and OPM, due to higher avg. OPM in 2029-30E at 42% (vs. previously at 41%). We apply the 37.1x multiple to 2029E EPS and discount it back to 2027E via a COE of 7.8% (no change). Our 12m target price is at NT$2,549 (vs. previously at NT$2,557), implying 61x 2027E P/E, which is above the company's average +1 stv. P/E of 40x, reflecting our positive view on UMT's product mix upgrade toward LEO satellites. Maintain Buy.

Exhibit 4: UMT discounted P/E

(NT$ mn)	2022	2023	2024	2025	2026E	2027E	2028E	2029E	2030E
Milestones	Telecom components			LEO satellite launch accelerations				LEO satellite users ramp up	
Satellite revenues %	15%	15%	43%	59%	80%	91%	94%		
Revenue	**1,838**	**1,585**	**2,335**	**2,452**	**4,296**	**9,179**	**12,967**	**16,857**	**19,385**
YoY	*5%*	*-14%*	*47%*	*5%*	*75%*	*114%*	*41%*	*30%*	*15%*
Gross margin	40.9%	40.4%	51.3%	51.1%	59.8%	65.1%	66.0%	66.5%	67.0%
Operating profit	**294**	**202**	**624**	**574**	**1,429**	**3,628**	**5,263**	**6,961**	**8,140**
YoY	33%	-31%	208%	-8%	149%	154%	45%	32%	17%
Operating margin	*16%*	*13%*	*27%*	*23%*	*33%*	*40%*	*41%*	*41%*	*42%*
Net profit	**271**	**200**	**547**	**518**	**1,256**	**2,871**	**4,161**	**5,516**	**6,450**
EPS (NT$, diluted)	**4.34**	**3.12**	**8.27**	**7.56**	**18.16**	**41.53**	**60.19**	**79.79**	**93.30**
YoY	*15%*	*-26%*	*173%*	*-5%*	*142%*	*129%*	*45%*	*33%*	*17%*
TP implied P/E	587	818	308	337	140	61	42	32	27

2029E target P/E	37.1
Target multiple x EPS	2,960
Discounted back to 2027; TP (NTD)	**2,549.0**
Implied 2027 P/E	61

COE assumption	
Beta	1.2
Risk free	1.6%
Market risk premium	5.1%
COE	**7.8%**

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 5: UMT peers comparison

Valuation	2029E P/E	2029-30 avg. NI YoY	2029-30 avg. OPM	PEG&M
UMT	**37.1**	25%	42%	**0.56**
Peers	**2027E P/E**	**2027/28 avg. NI YoY**	**2027/28 avg. OPM**	**PEG&M**
WNC	17	15%	5%	0.85
Compeq	*16*	*60%*	*17%*	*0.21*
Lite-On Tech	*19*	*49%*	*15%*	*0.29*
BT	10	10%	20%	0.32
Verizon	*10*	*9%*	*25%*	*0.27*
T-Mobile	*11*	*37%*	*30%*	*0.16*
Hon Hai	11	15%	3%	0.64
Avg.	**13**	**28%**	**17%**	**0.30**
Avg. w/o outliers	**13**	**13%**	**9%**	**0.56**

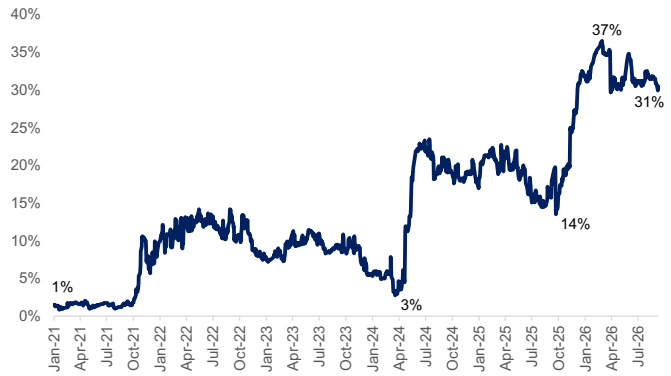
Outliers are in grey; Compeq and Lite-On Tech are not covered and use Bloomberg consensus; price as of Oct 12, 2026

Source: Company data, Goldman Sachs Global Investment Research, Bloomberg

Exhibit 6: UMT 12M forward PE ratio

Source: Company data, Goldman Sachs Global Investment Research, Bloomberg

Exhibit 7: UMT QFII holdings



Source: TEJ

Price target risks and methodology - UMT

Valuation: We use a discounted P/E methodology and apply a 37.1x target P/E multiple to UMT's 2029E EPS, discounting it back to 2027E at a COE of 7.8% (beta 1.2x, risk-free rate 1.6% and market risk premium at 5.1%), which leads to **our 12-month target price of NT$2,549.** The target P/E is based on the average PEG&M ratio of peers in the satellite supply chain. We are Buy-rated on UMT.

Key Risks: Slower-than-expected LEO satellite deployment; Potential competition from new-entrant suppliers; LEO satellite operators manufacturing in-house components.

3491.TWO	12m Price Target: **NT$2,549.00**	Price: **NT$1,525.00**	Upside: **67.1%**

Buy

Market cap: NT$100.9bn / $3.2bn
Enterprise value: NT$100.6bn / $3.1bn
3m ADTV: NT$2.5bn / $77.1mn
Taiwan
Greater China Technology
M&A Rank: 3
Leases incl. in net debt & EV?: Yes

GS Forecast

	12/25	12/26E	12/27E	12/28E
Revenue (NT$ mn) New	**2,452.2**	**4,296.0**	**9,179.3**	**12,966.9**
Revenue (NT$ mn) Old	2,452.2	4,430.4	9,327.0	13,115.7
EBITDA (NT$ mn)	687.4	1,567.3	3,778.1	5,420.9
EPS (NT$) New	**7.56**	**18.16**	**41.53**	**60.19**
EPS (NT$) Old	7.56	18.41	41.84	60.53
P/E (X)	54.8	84.0	36.7	25.3
P/B (X)	7.6	21.7	16.7	12.6
Dividend yield (%)	1.6	0.6	1.4	2.0
CROCI (%)	13.1	31.0	55.5	72.6

	6/26	9/26E	12/26E	3/27E
EPS (NT$)	4.45	3.05	7.00	7.20

Source: Company data, Goldman Sachs Research estimates, FactSet. Price as of 12 Oct 2026 close.

Disclosure Appendix

Reg AC

We, Allen Chang, Verena Jeng and Ting Song, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Allen Chang Goldman Sachs (Asia) L.L.C., Verena Jeng Goldman Sachs (Asia) L.L.C., Ting Song Goldman Sachs (Asia) L.L.C..

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

M&A Rank

Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

The rating(s) for UMT is/are relative to the other companies in its/their coverage universe: AAC, ACM Research, AMEC, ASMPT, AVC, AccoTest, Anji Micro, Asus, Auras, BOE, BYDE, Biren, CR Micro, CXMT, Cambricon, Chenbro, China Mobile (HK), China Telecom, China Tower Corp., China Unicom, Chinasoft Intl, Compal, Desay SV, E Ink, E-Town Semis, EHang, Empyrean, Eoptolink, FOCI, Fositek, Foxconn Industrial Internet, Gigabyte, Gigadevice, Glodon Co., HTC Corp., Hikvision, Hon Hai, Horizon Robotics, Hua Hong, Huace Navigation, Huaqin Co.(A), Huaqin Co.(H), Hwatsing, Iluvatar, InnoScience, Innolight (A), Innolight (H), Inspur, Insta360, Inventec, JCET, Kematek, King Slide, Kingdee, Kingsoft Office, LandMark, Largan, Lenovo, Lingyi, Luxshare (A), Luxshare (H), Maxscend, Meitu, MetaX, Mitac, Montage (A), Montage (H), NAURA, NSIG, Nexchip, OmniVision, Pegatron, Pony AI Inc. (ADR), Pony AI Inc. (H), Quanta, RoboTechnik, Ruijie Networks, SG Micro, SICC, SMIC (A), SMIC (H), SZS, Sangfor, SenseTime, Shengyi Tech, Shennan Circuits, StarPower, Sunny Optical, TFC Optical, Thundersoft, Tongyu Communication, Transsion, UMT, UNIS, VPEC, Vanchip, VeriSilicon, Victory Giant, WNC, WUS, WeRide, Wistron, Wiwynn, YJ Semitech, YOFC, Yonyou, ZTE (A), ZTE (H), iFlytek

Company-specific regulatory disclosures

The following disclosures relate to relationships between The Goldman Sachs Group, Inc. (with its affiliates, "Goldman Sachs") and companies covered by Goldman Sachs Global Investment Research and referred to in this research.

There are no company-specific disclosures for: UMT (NT$1,525.00)

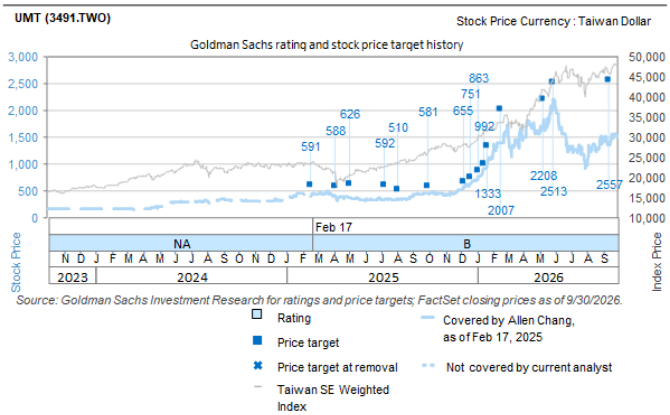
Distribution of ratings/investment banking relationships

Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution			Investment Banking Relationships		
	Buy	Hold	Sell	Buy	Hold	Sell
Global	50%	34%	16%	65%	58%	43%

As of October 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,094 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

Price target and rating history chart(s)



The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.

Target price history table(s)
UMT (3491.TWO)

Date of report	Target price (NT$)	Closing price (NT$)
13-Sep-26	2,557.00	1,415.00
29-May-26	2,513.00	2,200.00
10-May-26	2,208.00	1,655.00
16-Feb-26	2,007.00	1,400.00
22-Jan-26	1,333.00	986.00
15-Jan-26	992.00	890.00
05-Jan-26	863.00	766.00
22-Dec-25	751.00	635.00
07-Dec-25	655.00	529.00
30-Sep-25	581.00	447.50
03-Aug-25	510.00	344.50
09-Jul-25	592.00	335.00
04-May-25	626.00	379.00
06-Apr-25	588.00	415.00
17-Feb-25	591.00	391.50

Price targets shown in table(s) are unadjusted for corporate actions.

Regulatory disclosures

Disclosures required by United States laws and regulations
See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; registered market maker. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States
The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy

Statement are available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at https://www.gs.com/worldwide/brazil/area/gir/index.html. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details, a copy of the annual compliance audit report and other relevant information and disclosures can be found at this link: https://www.goldmansachs.com/worldwide/india/research-analyst. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at https://www.gs.com/disclosures/europeanpolicy.html which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Ratings, coverage universe and related definitions

Buy (B), Neutral (N), Sell (S) Analysts recommend stocks as Buys or Sells for inclusion on various regional Investment Lists. Being assigned a Buy or Sell on an Investment List is determined by a stock's total return potential relative to its coverage universe. Any stock not assigned as a Buy or a Sell on an Investment List with an active rating (i.e., a stock that is not Rating Suspended, Not Rated, Early-Stage Biotech, Coverage Suspended or Not Covered), is deemed Neutral. Each region manages Regional Conviction Lists, which are selected from Buy rated stocks on the respective region's Investment Lists and represent investment recommendations focused on the size of the total return potential and/or the likelihood of the realization of the return across their respective areas of coverage. The addition or removal of stocks from such Conviction Lists are managed by the Investment Review Committee or other designated committee in each respective region and do not represent a change in the analysts' investment rating for such stocks.

Total return potential represents the upside or downside differential between the current share price and the price target, including all paid or anticipated dividends, expected during the time horizon associated with the price target. Price targets are required for all covered stocks. The total return potential, price target and associated time horizon are stated in each report adding or reiterating an Investment List membership.

Coverage Universe: A list of all stocks in each coverage universe is available by primary analyst, stock and coverage universe at https://www.gs.com/research/hedge.html.

Not Rated (NR). The investment rating, target price and earnings estimates (where relevant) are removed pursuant to Goldman Sachs policy when Goldman Sachs is acting in an advisory capacity in a merger or in a strategic transaction involving this company, when there are legal, regulatory or policy constraints due to Goldman Sachs' involvement in a transaction, and in certain other circumstances. **Early-Stage Biotech (ES).** An investment rating and a target price are not assigned pursuant to Goldman Sachs policy when this company has neither a drug, treatment or medical device that has passed a Phase II clinical trial nor a license to distribute a post-Phase II drug, treatment or medical device. **Rating Suspended (RS).** Goldman Sachs Research has suspended the investment rating and price target for this stock, because there is not a sufficient fundamental basis for determining an investment rating or target price. The previous investment rating and target price, if any, are no longer in effect for this stock and should not be relied upon. **Coverage Suspended (CS).** Goldman Sachs has suspended coverage of this company. **Not Covered (NC).** Goldman Sachs does not cover this company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs

do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (https://www.sipc.org).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

The analysts named in this report may have from time to time discussed with our clients, including Goldman Sachs salespersons and traders, or may discuss in this report, trading strategies that reference catalysts or events that may have a near-term impact on the market price of the equity securities discussed in this report, which impact may be directionally counter to the analyst's published price target expectations for such stocks. Any such trading strategies are distinct from and do not affect the analyst's fundamental equity rating for such stocks, which rating reflects a stock's return potential relative to its coverage universe as described herein.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, make a market or act as liquidity provider in, and buy or sell, any securities, derivatives, or other financial instruments referred to in this research (including any financial instruments issued by the companies referred to herein), unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at https://www.theocc.com/about/publications/character-risks.jsp and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to https://research.gs.com.

Disclosure information is also available at https://www.gs.com/research/hedge.html or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs.

You may not use, extract, download or retrieve this information, in whole or in part, to train, finetune, adjust, influence the weights or behavior of, or otherwise develop or improve a machine learning or artificial intelligence model or system, or provide or reproduce this information, in whole or in part, as a prompt or input to any such system without the express written consent of Goldman Sachs. Furthermore, you may not bulk download this information, or employ any robot, spider, crawler, scraper, automated agent, or other electronic means to scrape, harvest, extract, or retrieve this information, without the express written consent of Goldman Sachs. A breach of the above terms may result in termination of your right to access this service.

Accton

New Buy: networking/switch white-box leader riding on AI uptrend; PO NT$3,200

Initiating Coverage: BUY | PO: 3,200 TWD | Price: 2,130 TWD

AI networking/switch white-box leader; PO at NT$3,200

We initiate Accton at Buy with a PO of NT$3,200 (50% potential upside), based on 18x 2028E EPS. We see Accton as the AI networking/switch white-box leader, alongside Celestica (CLS US; covered by Ruplu Bhattacharya), poised to benefit from multi-year AI upcycle, platform migration, and specification upgrade**.** Its valuation/PEG look attractive vs. peers with higher EPS growth/ROE. We believe Street concerns about 3Q26 margin pressure from higher component costs (BofAe: GM/OPM of 18.7%/14.5% vs. consensus' 18.9%/ 14.8%) are well understood and reflected in share-price performance (-28% in 3Q26, vs. TAIEX +4%). We see any potential correction as a particularly good buying opportunity to capture positive long-term outlook. **Near-term catalysts:** monthly sales, Accton/Celestica's 3Q results.

AI platform migration & spec upgrade; more client/project

We project 37%/42% sales/EPS CAGR in 2026-28E, driven by CSPs' switch upgrade to 800G/1.6T, more project/client win, and rising white-box server/switch adoption among CSP clients. We highlight Accton's expanding partnership with its top CSP client in computing (with continued generation migration), optical, liquid cooling, and rack-level integration. We also expect another key CSP customer to contribute a double-digit% of sales in 2027, thanks to share re-gain without 800G project delay. We see further sales/profit upside from more new clients (CSP/enterprise/telco) and JDM/rack business.

Solid BS/CF to support capacity expansion & new business

Accton keeps expanding capacity in Taiwan, Vietnam, Singapore, Malaysia, and US for existing high-end AI business and new optical/CPO/cooling business. We expect annual capex of NT$6-7bn in 2026-28E but believe the historical high capex and higher working capital required can be supported by strong profitability, solid net cash, and positive FCF. **Risks:** (1) AI product/project delay and slower 800G/1.6T adoption; (2) demand push out due to component shortage; (3) GM pressure due to component cost inflation; (4) peer competition; (5) slower capacity expansion; (6) unfavorable FX/macro trend.

Estimates (Dec) (NT$)	2024A	2025A	2026E	2027E	2028E
Net Income (Adjusted - mn)	12,013	26,374	49,449	69,192	99,579
EPS	21.51	47.19	88.27	123.52	177.77
EPS Change (YoY)	34.5%	119.4%	87.1%	39.9%	43.9%
Dividend / Share	11.00	15.00	35.31	49.41	71.11
Free Cash Flow / Share	12.86	53.60	68.86	108.22	160.17
Valuation (Dec)					
P/E	99.03x	45.14x	24.13x	17.24x	11.98x
Dividend Yield	0.516%	0.704%	1.66%	2.32%	3.34%
EV / EBITDA*	77.22x	33.67x	17.78x	12.62x	8.80x
Free Cash Flow Yield*	0.601%	2.51%	3.23%	5.07%	7.51%

* For full definitions of *iQmethod*℠ measures, see page 36.

This research report provides general information only. No part of this report may be used or reproduced or quoted in any manner whatsoever in Taiwan by the press or other persons without the express written consent of BofA Securities.
>> Employed by a non-US affiliate of BofAS and is not registered/qualified as a research analyst under the FINRA rules.
Refer to "Other Important Disclosures" for information on certain BofA Securities entities that take responsibility for the information herein in particular jurisdictions.
BofA Securities does and seeks to do business with issuers covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.
Refer to important disclosures on page 37 to 39. Analyst Certification on page 34. Price

13 October 2026

Equity

Doris Kao >>
Research Analyst
Merrill Lynch (Taiwan)
+886 2 2376 3722
doris.kao@bofa.com

Robert Cheng >>
Research Analyst
Merrill Lynch (Taiwan)
+886 2 2376 3731
robert.cheng@bofa.com

Tal Liani
Research Analyst
BofAS
+1 646 855 5107
tal.liani@bofa.com

Katherine Zhu >>
Research Analyst
Merrill Lynch (Hong Kong)
+852 3508 3374
kexin.zhu@bofa.com

Carol Hu >>
Research Analyst
Merrill Lynch (Taiwan)
+886 2 2376 3721
carol.hu@bofa.com

Stock Data

Price	2,130 TWD
Price Objective	3,200 TWD
Date Established	13-Oct-2026
Investment Opinion	C-1-7
52-Week Range	884.00 TWD-2,825 TWD
Mrkt Val / Shares Out (mn)	37,431 USD / 561.1
Market Value (mn)	1,195,181 TWD
Average Daily Value (mn)	271.23 USD
Free Float	82.2%
BofA Ticker / Exchange	XTWOF / TAI
Bloomberg / Reuters	2345 TT / 2345.TW
ROE (2026E)	63.3%
Net Dbt to Eqty (Dec-2025A)	-47.4%

For a list of acronyms, see Exhibit 58

Related Global Research reports:

Switching Trends: 2Q26 Switching Trends: Market on a roll with growth spreading and accelerating 07 October 2026

US Semiconductors: Scaling AI with Photons: Primer on Optical

iQprofile℠ Accton

Key Income Statement Data (Dec)	2024A	2025A	2026E	2027E	2028E
(NT$ Millions)					
Sales	110,425	248,320	414,359	558,071	772,505
Gross Profit	22,791	44,919	79,933	109,110	155,499
Sell General & Admin Expense	(4,474)	(5,782)	(8,368)	(11,040)	(14,137)
Operating Profit	13,615	32,167	61,344	86,377	124,751
Net Interest & Other Income	1,532	782	1,133	1,110	1,115
Associates	NA	NA	NA	NA	NA
Pretax Income	15,147	32,949	62,478	87,486	125,866
Tax (expense) / Benefit	(3,135)	(6,611)	(13,059)	(18,326)	(26,319)
Net Income (Adjusted)	12,013	26,374	49,449	69,192	99,579
Average Fully Diluted Shares Outstanding	559	559	560	560	560
Key Cash Flow Statement Data					
Net Income	12,013	26,374	49,449	69,192	99,579
Depreciation & Amortization	1,163	1,727	2,829	4,074	4,964
Change in Working Capital	(3,190)	5,187	(6,705)	(5,646)	(8,821)
Deferred Taxation Charge	NA	NA	NA	NA	NA
Other Adjustments, Net	(43)	1,564	0	0	0
Cash Flow from Operations	9,942	34,852	45,573	67,620	95,722
Capital Expenditure	(2,762)	(4,894)	(7,000)	(7,000)	(6,000)
(Acquisition) / Disposal of Investments	(98)	550	0	0	0
Other Cash Inflow / (Outflow)	3,121	(14,805)	0	0	0
Cash Flow from Investing	262	(19,149)	(7,000)	(7,000)	(6,000)
Shares Issue / (Repurchase)	0	0	0	0	0
Cost of Dividends Paid	(5,582)	(6,148)	(8,384)	(19,780)	(27,677)
Cash Flow from Financing	(6,215)	(6,663)	(8,384)	(19,780)	(27,677)
Free Cash Flow	7,181	29,958	38,573	60,620	89,722
Net Debt	(17,359)	(27,314)	(57,504)	(98,344)	(160,389)
Change in Net Debt	(4,839)	(9,235)	(30,189)	(40,840)	(62,045)
Key Balance Sheet Data					
Property, Plant & Equipment	5,445	9,228	13,599	16,726	17,961
Other Non-Current Assets	10,382	12,936	12,736	12,536	12,336
Trade Receivables	20,763	37,275	62,199	83,772	115,960
Cash & Equivalents	18,116	27,891	58,080	98,920	160,965
Other Current Assets	31,760	55,902	75,357	92,364	117,316
Total Assets	86,467	143,231	221,971	304,317	424,538
Long-Term Debt	281	84	84	84	84
Other Non-Current Liabilities	2,831	3,328	3,328	3,328	3,328
Short-Term Debt	476	492	492	492	492
Other Current Liabilities	46,444	81,691	119,366	152,299	200,618
Total Liabilities	50,032	85,596	123,270	156,204	204,523
Total Equity	36,435	57,635	98,700	148,113	220,015
Total Equity & Liabilities	86,467	143,231	221,971	304,317	424,538
iQ*method℠ - Bus Performance					
Return On Capital Employed	33.8%	52.3%	60.2%	54.3%	52.9%
Return On Equity	39.1%	56.2%	63.3%	56.1%	54.1%
Operating Margin	12.3%	13.0%	14.8%	15.5%	16.1%
EBITDA Margin	13.4%	13.6%	15.5%	16.2%	16.8%
iQ*method℠ - Quality of Earnings					
Cash Realization Ratio	0.8x	1.3x	0.9x	1.0x	1.0x
Asset Replacement Ratio	2.6x	3.2x	2.7x	1.8x	1.3x
Tax Rate (Reported)	20.7%	20.1%	20.9%	20.9%	20.9%
Net Debt-to-Equity Ratio	-47.6%	-47.4%	-58.3%	-66.4%	-72.9%
Interest Cover	NM	NM	NM	NM	NM

Key Metrics

* For full definitions of *iQmethod*℠ measures, see page 36.

Company Sector

IT Hardware

Company Description

Accton is the global networking/switch white-box leader with business expanding into optics, cooling, and rack integration. As of 2Q26, switch made up 52% of total sales, followed by networking application (46%), and rest from others. Accton was founded in 1988 and went publicly listed on Taiwan Stock Exchange in 1995. It has facilities in Taiwan, US, Southeast Asia, and Europe. Its headquarter is in Zhubei, Taiwan.

Investment Rationale

We assign a Buy rating to Accton given its leading position in networking/switch white-box market to ride on multi-year AI uptrend, platform migration, and spec upgrade. We expect double-digit% sales/EPS CAGR in 2026-28, driven by switch upgrade to 800G/1.6T and more project/client win. We see further sales/profit upside from more new clients and JDM/rack business.

Stock Data

Price to Book Value	12.1x

Contents

Focus Charts 4

Investment Thesis 5

- AI platform migration & spec upgrade; more client/project 5
- Solid BS/CF to support capacity expansion & new business 5
- Attractive valuation with high sales/EPS growth and ROE 5
- Key investment risks 6

Valuation: PO at NT$3,200 7

Business overview: Switch for datacenter as key driver amid AI uptrend 9

AI networking architecture: scale-up, scale-out, scale-across; CPO/optical as future trend 12

- Scale-up: rising demand for intra-system connectivity 12
- Scale-out: the core architecture for AI cluster expansion 13
- Scale-across: extending AI networking beyond individual clusters/datacenters 13
- CPO: the potential next-gen architecture for AI switches 13
- Potential optical technologies under watch: OCS and OWS 16

Key peer comparison: Celestica (CLS US) 20

- Celestica's overlap with Accton from key CCS segment 20
- Historical sales comparison: larger scale from Celestica 21
- Historical GM/OPM comparison: higher % from Accton 21
- Ongoing R&D efforts from both Accton and Celestica 22
- Balance sheet & CF comparison: Accton vs. Celestica 22

Financial analysis: 37%/42% sales/EPS CAGR in 2026-28 24

- Balance sheet: net cash position and high ROE 26
- Cashflow analysis: negative OCF in 1H26 for higher working capital; higher capex for global expansion 27
- Ongoing capacity expansion in Taiwan and overseas 28

Company Description 30

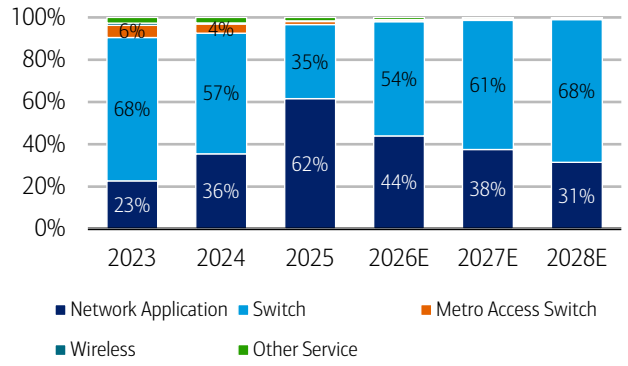
- Management team 30
- Top-10 shareholders and QFII holdings 31

Appendix 31

- What is leaf-spine architecture? 31
- Abbreviation glossary 32

Focus Charts

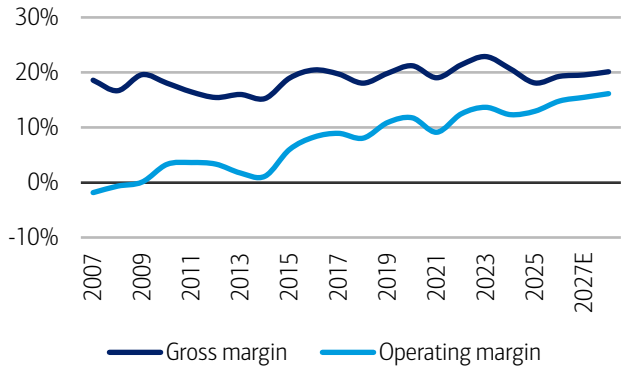
Exhibit 1: We expect rising mix from Switch at 54-68% in 2026-28
Accton sales breakdown by segment, 2023-28



Source: BofA Global Research estimates, company data

BofA GLOBAL RESEARCH

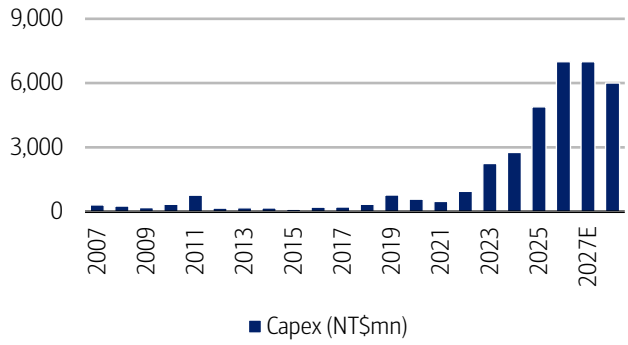
Exhibit 2: We expect GM to sustain in 2026-28E with expanding OPM
Accton's margin trend, 2007-28E



Source: BofA Global Research estimates, company data

BofA GLOBAL RESEARCH

Exhibit 3: We project Accton's annual capex at NT$6-7bn in 2026-28E,
Accton's capex trend (in NT$mn) in 2007-28E



Source: BofA Global Research estimates, company data. Capex refers to purchase of fixed assets

BofA GLOBAL RESEARCH

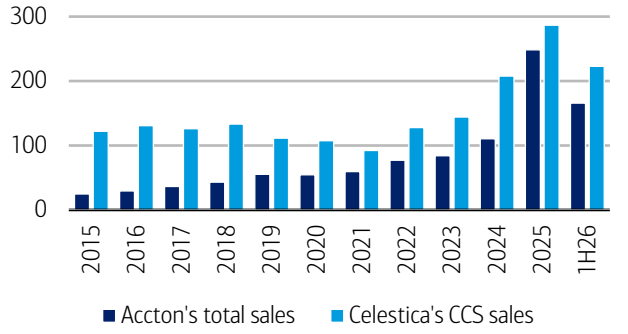
Exhibit 4: Accton's PEG of 0.29 is below peers' average of 0.73
PEG comparison: Accton vs. Greater China and global peers

Ticker	Company	P/E (x) 2028E	EPS CAGR 2026-28E	PEG 2028E	Sales CAGR 2026-28E	ROE 2028E
2345 TT	**Accton**	**12.0**	**42%**	**0.29**	**37%**	**54%**
CLS US	Celestica	14.6	48%	0.31	45%	41%
ANET US	Arista	36.6	21%	1.78	25%	27%
CSCO US	Cisco	20.0	21%	0.97	25%	27%
HPE US	HPE	13.4	12%	1.08	10%	31%
CIEN US	Ciena	26.3	55%	0.48	30%	55%
DELL US	Dell	19.3	72%	0.27	44%	n.a.
NOK US	Nokia	17.1	21%	0.82	9%	13%
2317 TT	Hon Hai	9.5	18%	0.52	17%	16%
6285 TT	Wistron NeWeb *	13.2	26%	0.51	19%	24%
2382 TT	Quanta	9.4	17%	0.57	19%	32%
Average		**17.9**	**31%**	**0.73**	**24%**	**30%**

Source: BofA Global Research estimates. *Non-covered forecasts based on Bloomberg consensus

BofA GLOBAL RESEARCH

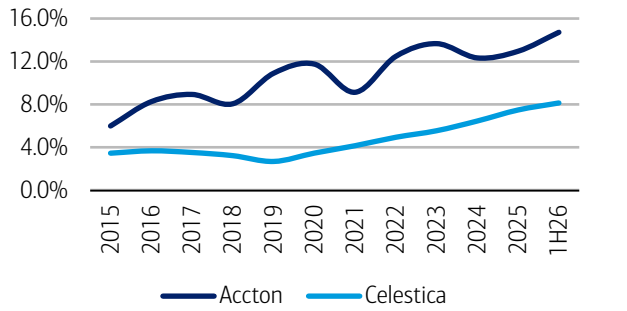
Exhibit 5: Celestica's CCS sales were 1.3x of Accton's total 1H26 sales
Sales (in NT$bn): Accton's total sales vs. Celestica's CCS sales, 2015-1H26



Source: BofA Global Research, TEJ, company data. CCS: Communications, Server, Storage

BofA GLOBAL RESEARCH

Exhibit 6: Accton's OPM of 6.0-14.7% in 2015-1H26 has been much higher than Celestica's 2.7-8.1%
OPM comparison for Accton and Celestica, 2015-1H26



Source: BofA Global Research, Bloomberg, TEJ, company data

BofA GLOBAL RESEARCH

Investment Thesis

We initiate coverage of Accton at Buy with a PO of NT$3,200 (implies 50% upside potential), based on 18x 2028E EPS. We see Accton as the AI networking/switch white-box leader, alongside Celestica (CLS US; covered by Ruplu Bhattacharya), poised to benefit from multi-year AI upcycle. Accton is expanding into optical/CPO, liquid cooling, and rack-level integration business to capture a larger portion of system value in AI.

We forecast Accton to post 37%/42% sales/EPS CAGR in 2026-28, with expanding OPM (Exhibit 41) on top of sustainable/high GM and ROE (Exhibit 48). We expect Accton's biggest switch segment (BofAe: 54-68% of 2026-28E sales; Exhibit 7) to post 52-157% YoY sales growth in 2026-28, thanks to spec upgrade to 800G/1.6T and broader product/client base.

Near-term catalysts: monthly sales, new customer/project win, Accton's 3Q26 results to be announced before mid-Nov (Exhibit 42), Celestica's 3Q26 results scheduled on 26 Oct US Time (with any positive implication/read-through to Accton at Celestica's earnings call on 27 Oct US Time), and clearer project pipeline with spec upgrade for ASIC into 2027.

Exhibit 7: We expect the biggest switch segment to contribute 54-68% of sales in 2026-28E
Key business summary of Accton, including sales mix, sales YoY growth, and GP mix by segments in 2026-28

	Sales mix %			Sales YoY growth %			GP mix %			
	26E	27E	28E	26E	27E	28E	26E	27E	28E	Summary & drivers
Switch	**54%**	**61%**	**68%**	**157%**	**52%**	**53%**	**61%**	**69%**	**75%**	**AI networking spec and content upgrade; new project/client win**
Network Application	44%	38%	31%	19%	15%	16%	37%	30%	24%	AI card as key driver with generation migration
Metro Access Switch	1%	1%	0%	0%	4%	0%	1%	0%	0%	For telecom; less important onwards
Wireless	0%	0%	0%	-11%	2%	0%	0%	0%	0%	Less important onwards
Other Service	1%	1%	0%	0%	-15%	0%	1%	1%	0%	Including NRE sales
Total	**100%**	**100%**	**100%**	**67%**	**35%**	**38%**	**100%**	**100%**	**100%**	

Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

AI platform migration & spec upgrade; more client/project

We project 37%/42% sales/EPS CAGR in 2026-28E, driven by CSPs' switch upgrade to 800G/1.6T, more project/client win, and rising white-box server/switch adoption among CSP clients. We highlight Accton's expanding partnership with its top CSP client in computing (with continued generation migration), optical, liquid cooling, and rack-level integration. We also expect another key CSP client to contribute double-digit% sales in 2027, thanks to share re-gain without 800G project delay. We see further sales/profit upside from more new clients (CSP/enterprise/telco) and JDM/rack business.

Solid BS/CF to support capacity expansion & new business

Accton keeps expanding capacity in Taiwan, Vietnam, Singapore, Malaysia, and US for existing high-end AI business and new optical/CPO/cooling business. We expect annual capex of NT$6-7bn in 2026-28E but believe the historical high capex can be supported by strong profitability, solid net cash position for around two decades, and positive FCF.

Attractive valuation with high sales/EPS growth and ROE

Accton's valuation and PEG both look attractive vs. peers, given higher sales/EPS growth and higher ROE (Exhibit 11). We believe its solid fundamentals are supported by the robust market position as networking/switch white-box leader, long history in networking/switch industry with proven track records, and ongoing R&D efforts. We view the Street's concern about 3Q26 margin pressure (BofA: GM/OPM of 18.7%/14.5%, vs. consensus of 18.9%/14.8%), due to higher component cost, has been well understood and reflected in share-price performance (-28% in 3Q26, vs. TAIEX +4%). We see any correction as a good buying opportunity to capture positive long-term outlook.

Key investment risks

We view the **key upside risks** for Accton include:

- Faster 800G/1.6T deployment by key customers, which may lead to higher sales growth and/or better margins;
- Easing component supply tightness/shortage, such as passive components, memory, processors, PCB/CCL/substrate;
- New client/project win, which may support further sales/margin expansion;
- Faster progress and development in optical/cooling business, which may drive further sales/margin expansion;
- Faster capacity expansion progress in Taiwan and/or overseas, which may support higher sales growth and/or more new project/client win;
- Favorable FX movement (mainly USD/NTD), which may help on GM with potential non-operating gain.

We view the **key downside risks** for Accton include:

- Key customers' AI product or project delay due to supply chain bottleneck, which may lead to slower spec migration to 800G/1.6T;
- Key customers' demand push-out due to component supply tightness/shortage, such as passive components, memory, processors, PCB/CCL/substrate;
- Key customers' demand push-out due to resign in chip/board level;
- GM pressure due to component cost inflation and/or unfavorable FX trend;
- Slower capacity expansion progress in Taiwan and/or overseas, which may cap sales growth or new project/client win;
- More intense peer competition, which may lead to market/order share loss;
- Unfavorable FX movement (mainly USD/NTD) or macro uncertainty, which may weigh on GM with potential non-operating loss.

Valuation: PO at NT$3,200

Our PO of NT$3,200 is based on 18x 2028E EPS (or around 26x 2027E EPS). In our view, the 18x P/E multiple looks justified given Accton's 37%/42% sales/EPS CAGR in 2026-28E, expanding OPM, sustainable/high GM and ROE, besides solid position in networking switch white box market.

The 18x target P/E represents around 1SD above Accton's historical trading average since 2008, the year in which the company first became profitable (Exhibit 2), and the mid-level of historical P/E range of 2-32x. We believe the above-average P/E multiple can be justified as stronger sales/EPS CAGR and higher margins than historical levels. 18x is below 20x our US analyst Ruplu Bhattacharya used for Celestica (CLS US; the key peer to Accton). We made a discount on Accton's P/E multiple given lower revenue share in global network/switching market and lower EPS CAGR in 2026-28E (Exhibit 4).

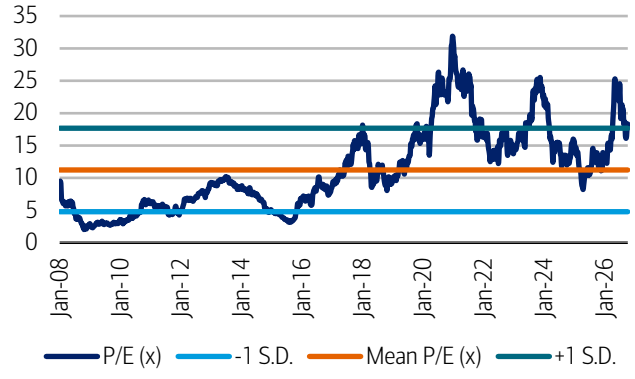
We use 2028 as the valuation base year to capture various future AI-related business opportunities which have been monitored and discussed by investors, including new CPO/cooling/rack business and new customers (CSPs, enterprises, start-ups). Our US analyst Ruplu Bhattacharya also uses 2028 as the valuation base year for Celestica.

In terms of P/E, Accton currently trades at 18x one-year forward P/E, 17x 2027E P/E, or 12x 2028E P/E, vs. the historical P/E trading average of 11x since 2008 when the company started making profits (Exhibit 8). Its 2027E and 2028E P/E are both below overall peers' average of 22x and 18x, respectively (Exhibit 11).

In terms of P/B, Accton currently trades at 8.7x one-year forward P/B, 8.1x 2027E P/B, or 5.4x 2028E P/B, above the historical trading average of 3.3x since 2007 (Exhibit 9). Its 2027E P/B of 8.1x is above overall peers' average of 6.4x, while 2028E P/B of 5.4x is below overall peers' average of 6.4x.

In terms of PEG, Accton's valuation looks attractive, with 0.29 PEG and 54% ROE in 2028E, vs. overall peers' average of 0.73 PEG and 30% ROE (Exhibit 10).

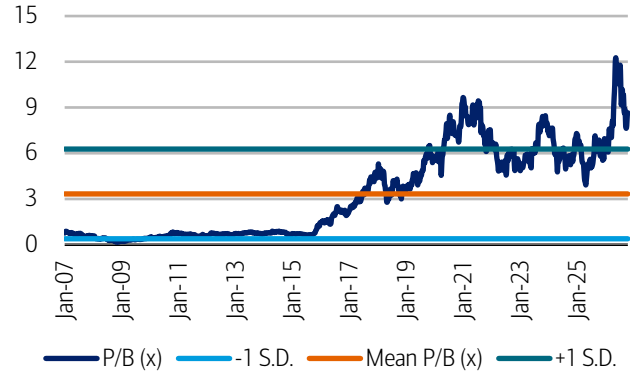
Exhibit 8: Accton's historical P/E trading average since 2008 is around 11x, vs. currently 18x one-year forward P/E
Accton's P/E trend and -1/+1 standard deviation since 2008



Source: BofA Global Research estimates, TEJ, Bloomberg

BofA GLOBAL RESEARCH

Exhibit 9: Accton's historical P/B trading average since 2007 is around 3.3x, vs. currently 8.7x one-year forward P/B
Accton's P/B trend and -1/+1 standard deviation since 2007



Source: BofA Global Research estimates, TEJ, Bloomberg

BofA GLOBAL RESEARCH

Exhibit 10: Accton's PEG of 0.29 is below peers' average of 0.73
PEG comparison: Accton vs. Greater China and global peers

Ticker	Company	P/E (x) 2028E	EPS CAGR 2026-28E	PEG 2028E	Sales CAGR 2026-28E	ROE 2028E
2345 TT	**Accton**	**12.0**	**42%**	**0.29**	**37%**	**54%**
Global peers						
CLS US	Celestica	14.6	48%	0.31	45%	41%
ANET US	Arista	36.6	21%	1.78	25%	27%
CSCO US	Cisco	20.0	21%	0.97	25%	27%
HPE US	HPE	13.4	12%	1.08	10%	31%
CIEN US	Ciena	26.3	55%	0.48	30%	55%
DELL US	Dell	19.3	72%	0.27	44%	n.a.
NOK US	Nokia	17.1	21%	0.82	9%	13%
Greater China peers						
2317 TT	Hon Hai	9.5	18%	0.52	17%	16%
6285 TT	Wistron NeWeb *	13.2	26%	0.51	19%	24%
2382 TT	Quanta	9.4	17%	0.57	19%	32%
Average		**17.9**	**31%**	**0.73**	**24%**	**30%**

Source: BofA Global Research estimates, Bloomberg. ROE refers to average ROE. Market data as of 2026/10/12 (Asia) or 10/09 (US).
*Non-covered company's forecasts based on Bloomberg consensus. PEG = 2028E P/E / 2026-28E EPS CAGR

BofA GLOBAL RESEARCH

Exhibit 11: Accton trades at 12x 2028E P/E, vs. global peers' average of 21x and Greater China peers' average of 11x
Valuation comparison: Accton vs. Greater China and global peers

Ticker	Company	Share price (LC)	Market cap (US$mn)	P/E (x) 26E	P/E (x) 27E	P/E (x) 28E	P/B (x) 26E	P/B (x) 27E	P/B (x) 28E	Dividend yield (%) 26E	Dividend yield (%) 27E	Dividend yield (%) 28E	ROE (%) 26E	ROE (%) 27E	ROE (%) 28E	EV/EBITDA (x) 26E	EV/EBITDA (x) 27E	EV/EBITDA (x) 28E	Net cash (debt) to market cap 26E	Net cash (debt) to market cap 27E	Net cash (debt) to market cap 28E
2345 TT	**Accton**	**2,130.00**	**37,438**	**24.1**	**17.2**	**12.0**	**12.1**	**8.1**	**5.4**	**1.7**	**2.3**	**3.3**	**63.3**	**56.1**	**54.1**	**17.8**	**12.6**	**8.8**	**5%**	**8%**	**13%**
Global peers																					
CLS US	Celestica	362.16	45,695	31.8	18.5	14.6	12.8	7.5	5.0	0.0	0.0	0.0	48.3	51.8	41.3	21.5	12.6	10.0	0%	1%	2%
ANET US	Arista	216.74	273,358	53.3	44.2	36.6	15.5	11.4	8.6	0.0	0.0	0.0	34.6	30.1	27.2	44.4	36.1	29.7	2%	4%	6%
CSCO US	Cisco	118.39	466,763	29.1	24.2	20.0	8.5	6.2	4.7	0.0	0.0	0.0	34.6	30.1	27.2	44.4	36.1	29.7	2%	4%	6%
HPE US	HPE	73.46	97,516	17.0	14.5	13.4	5.8	4.6	3.8	1.4	1.3	1.3	35.5	35.7	31.0	15.6	13.9	13.3	-5%	-3%	0%
CIEN US	Ciena	449.77	63,781	63.3	37.7	26.3	21.3	17.6	11.6	0.0	0.0	0.0	36.2	52.8	55.2	44.3	27.7	19.6	-1%	-1%	1%
DELL US	Dell	586.06	372,624	56.9	22.2	19.3	n.m.	n.m.	21.4	0.4	0.4	0.5	n.m.	n.m.	n.m.	37.9	18.6	16.0	-5%	-4%	-4%
NOK US	Nokia	10.36	59,490	25.0	20.6	17.1	2.5	2.4	2.2	1.5	1.6	1.7	9.7	11.3	13.0	15.7	11.4	9.6	6%	7%	8%
Average				**39.5**	**26.0**	**21.1**	**11.0**	**8.3**	**8.2**	**0.5**	**0.5**	**0.5**	**33.2**	**35.3**	**32.5**	**32.0**	**22.4**	**18.3**	**0%**	**1%**	**3%**
Greater China peers																					
2317 TT	Hon Hai	244.50	107,804	13.3	11.2	9.5	1.7	1.6	1.4	3.8	4.5	5.3	13.8	14.8	15.7	9.8	8.5	7.1	-8%	-12%	-13%
6285 TT	Wistron NeWeb *	263.50	3,990	20.8	15.5	13.2	3.3	2.9	2.6	3.5	4.3	5.0	16.4	20.1	24.2	12.7	11.0	9.3	-8%	-5%	-10%
2382 TT	Quanta	322.00	41,431	12.7	10.9	9.4	3.3	2.9	2.6	5.9	6.9	8.0	33.7	31.0	31.8	10.8	8.7	7.5	-18%	-27%	-33%
Average				**15.6**	**12.5**	**10.7**	**2.8**	**2.5**	**2.2**	**4.4**	**5.2**	**6.1**	**21.3**	**22.0**	**23.9**	**11.1**	**9.4**	**8.0**	**-11%**	**-15%**	**-19%**
Overall average				**32.3**	**21.9**	**17.9**	**8.3**	**6.4**	**6.4**	**1.6**	**1.9**	**2.2**	**29.2**	**30.8**	**29.6**	**25.7**	**18.5**	**15.2**	**-3%**	**-4%**	**-4%**

Source: BofA Global Research estimates, Bloomberg. Market data as of 2026/10/12 (Asia) or 10/09 (US). ROE refers to average ROE. LC: local currency. *Non-covered company's forecasts based on Bloomberg consensus

BofA GLOBAL RESEARCH

Business overview: Switch for datacenter as key driver amid AI uptrend

Accton established a strong position in white-box datacenter switching during the 2010s and has since become a key supplier to hyperscalers. According to third parties' data (including Qyresearch and Intelmarket), white-box datacenter switching TAM is expected to grow to US$7.8-8.1bn in 2032-34 from US$3.2-3.5bn in 2025 with an around 13% CAGR during the period. Accton has been the second largest white-box datacenter switching vendor following Celestica. We noted Accton's rising revenue share in global datacenter switching market, especially for top-4 US cloud datacenter switching market.

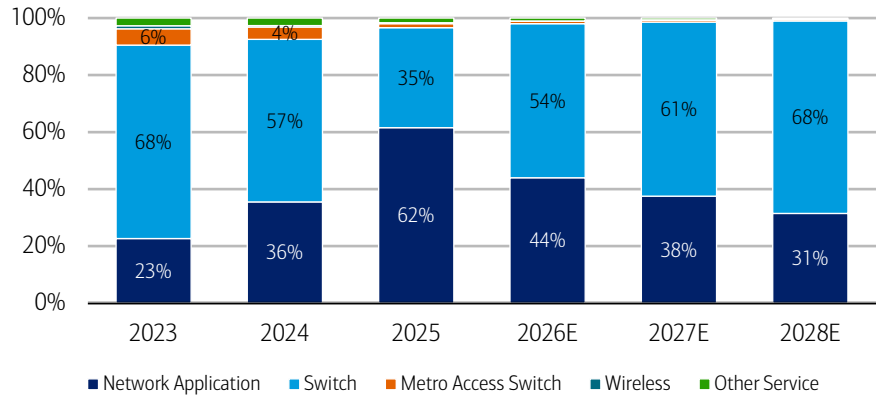
According to Dell'Oro, Accton had 7.0% of revenue share in global datacenter switching market in 2Q26, vs. 7.0%/5.6% in 1Q26/2Q25 and Celestica's 15.1% in 2Q26. In top-4 US cloud datacenter switching market, Accton's revenue share increased to 14.1% in 2Q26, vs. 13.3%/12.6% in 1Q26/2Q25 and Celestica's 36.6% in 2Q26. Its revenue share in tier-2, neo and other cloud also gradually increased to 3.1% in 2Q26, vs. 2.4%/2.8% in in 1Q26/2Q25. Within that hyperscale environment, Accton is also catching up with Celestica and OEM peers in front-end and AI back-end data center switching market. See more details from our US team's latest Switching Trend 2Q26 report published on 7 October 2026.

We see the Switch business for datacenter as the key sales/earnings driver for Accton amid AI uptrend. We expect the Switch business (including datacenter and non-datacenter like enterprise, campus, telecom) to contribute 54-68% of total sales in 2026-28. Switch for datacenter will become the biggest sub-segment in 2027-28 at 53-61% of total sales, vs. 18-42% in 2025-26, surpassing Network Application business.

Switch has historically been Accton's core business, while Network Application business, including AI accelerator cards, smart NICs, and other customized networking platforms, has become increasingly important amid strong hyperscaler AI investments. Metro Access Switch and Wireless are less meaningful in the future, in our view. Sales from Other Services should be relatively volatile as NRE sales are included (Exhibit 7).

We also see Accton's product portfolio increasingly aligned with AI datacenter infrastructure spending. Beyond switch and networking platforms, Accton is expanding into optics, liquid cooling, and rack-level integration, allowing it to capture a larger portion of system value within AI deployments. Management has highlighted optical technologies and liquid-cooled rack-scale solutions as key areas of development, in addition to AI accelerator modules and high-speed switches.

Exhibit 12: We expect rising sales mix from Switch at 54-68% in 2026-28, vs. 35-68% in 2023-25
Accton sales breakdown by segment, 2023-28



Source: BofA Global Research estimates, company data

BofA GLOBAL RESEARCH

Industry-wise, we expect AI uptrend to continue into following years with ongoing spec migration into 800G/1.6T and beyond and continued ASIC platform launches/migration. We believe the ongoing sizable capex investments from top US CSPs indicate increasing datacenter investments and growing demand for higher-speed networking connectivity, which should benefit Accton as the key networking/switch white-box leader.

Our global team expects top-5 US CSPs' total capex to grow 99%/33% YoY in 2026/27, vs. up 55-72% YoY in 2024-25 (Exhibit 13). Our global semi team's supply chain check also reveals AWS Trainium 4 will be launched in 2H27-2028, after Google TPU v7 and AWS Trainium 3 in 2025 and two versions of Google TPU v8 in 2026 (Exhibit 14). They also expect Google TPU and AWS Trainium chip builds to grow to 8.5mn and 3.0mn units in 2027, respectively, vs. 5.1mn and 2.1mn units in 2026 (Exhibit 15).

Exhibit 13: Our global team expects top-5 US CSPs' total capex to grow 99%/33% YoY in 2026/27, vs. -2% to +72% YoY in 2016-25
Top-5 US CSPs' capex trend, 2016-27E

US$mn	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E
Meta	6,732	13,980	15,102	15,115	18,628	31,431	27,266	37,256	69,691	137,728	191,937
Amazon	11,955	13,426	16,861	40,141	61,053	63,645	52,729	82,999	131,819	221,310	302,000
Microsoft	8,696	14,223	13,546	17,592	23,216	24,768	35,202	55,552	83,094	175,427	203,747
Google	13,184	25,139	23,548	22,281	24,640	31,485	32,251	52,535	91,447	199,488	300,573
Oracle	2,037	1,468	1,591	1,833	3,118	6,678	6,935	10,745	35,477	83,576	93,020
Top-5 US CSPs	**42,604**	**68,236**	**70,648**	**96,962**	**130,655**	**158,007**	**154,383**	**239,087**	**411,528**	**817,529**	**1,091,277**
YoY %											
Meta	50%	108%	8%	0%	23%	69%	-13%	37%	87%	98%	39%
Amazon	53%	12%	26%	138%	52%	4%	-17%	57%	59%	68%	36%
Microsoft	-5%	64%	-5%	30%	32%	7%	42%	58%	50%	111%	16%
Google	29%	91%	-6%	-5%	11%	28%	2%	63%	74%	118%	51%
Oracle	27%	-28%	8%	15%	70%	114%	4%	55%	230%	136%	11%
Top-5 US CSPs	**28%**	**60%**	**4%**	**37%**	**35%**	**21%**	**-2%**	**55%**	**72%**	**99%**	**33%**

Source: BofA Global Research estimates, Bloomberg. Historical data from Bloomberg. All data based on calendar

BofA GLOBAL RESEARCH

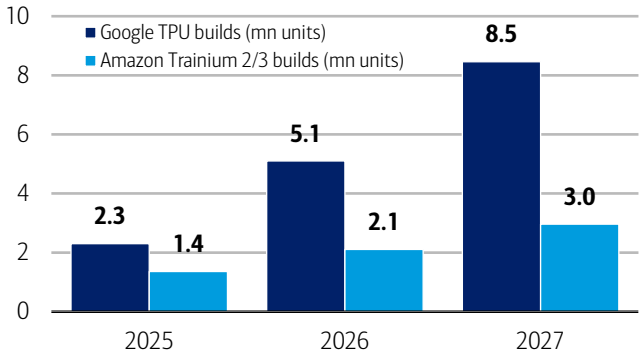
Exhibit 14: Our global semi team expects more ASIC generations from top CSPs, Google and Amazon, into 2027-28
Top CSPs Google and Amazon's ASIC pipeline in 2025-28

CSP	ASIC	Availability
Google	TPU v7 (Ironwood)	2025
	TPU v8t	2026
	TPU v8i	2026
Amazon/AWS	Trainium 2	2024
	Trainium 3	2025
	Trainium 4	2H27-2028

Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 15: Our global semi team expects TPU/Trainium chip builds to grow to 8.5mn/3.0mn units in 2027, vs. 5.1mn/2.1mn units in 2026
Google TPU and AWS Trainium chip builds forecast in 2025-27



Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

Switch business: accelerating shift toward higher speed datacenter networking

We believe Accton's strong sales and demand from datacenter switches will be supported by rising networking bandwidth requirements in AI clusters. The industry is moving from 400G toward 800G/1.6T Ethernet. Dell'Oro expects 2026 to mark the first year of volume deployment for 1.6T switches, driven by growing AI networking requirements, while switch capacity is advancing to 102.4T and beyond from 51.2T.

Accton launched its 51.2T/800G platforms in 2023 (Exhibit 16) and continues to expand its 400G/800G portfolio while developing 102.4T and 1.6T solutions (Exhibit 17).

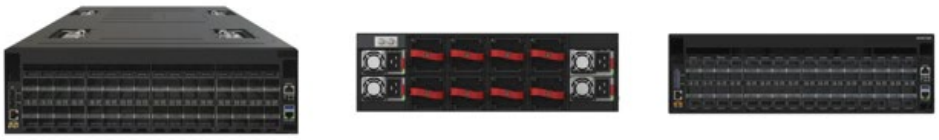
In addition to higher unit demand, next-generation switch platforms should carry higher ASPs, as 800G/1.6T systems require more advanced silicon, faster SerDes, more sophisticated/higher-end PCB and signal-integrity designs, higher-speed optical interfaces, and more demanding power and thermal management.

Exhibit 16: Accton's 51.2T 800G Ethernet switch platform targets AI/ML clusters and provides 51.2Tbps of Ethernet switching capacity, doubling throughput from the previous generation
Accton's 51.2T 800G Ethernet switch product example

Source: Accton

BofA GLOBAL RESEARCH

Exhibit 17: Accton's 102.4T platform represents its next-generation datacenter switching solution, designed to support the rapidly rising bandwidth requirements of AI workloads
Accton's 102.4T Ethernet Switch product example



Source: Accton

BofA GLOBAL RESEARCH

Network Application business: AI platform deployments also drive growth

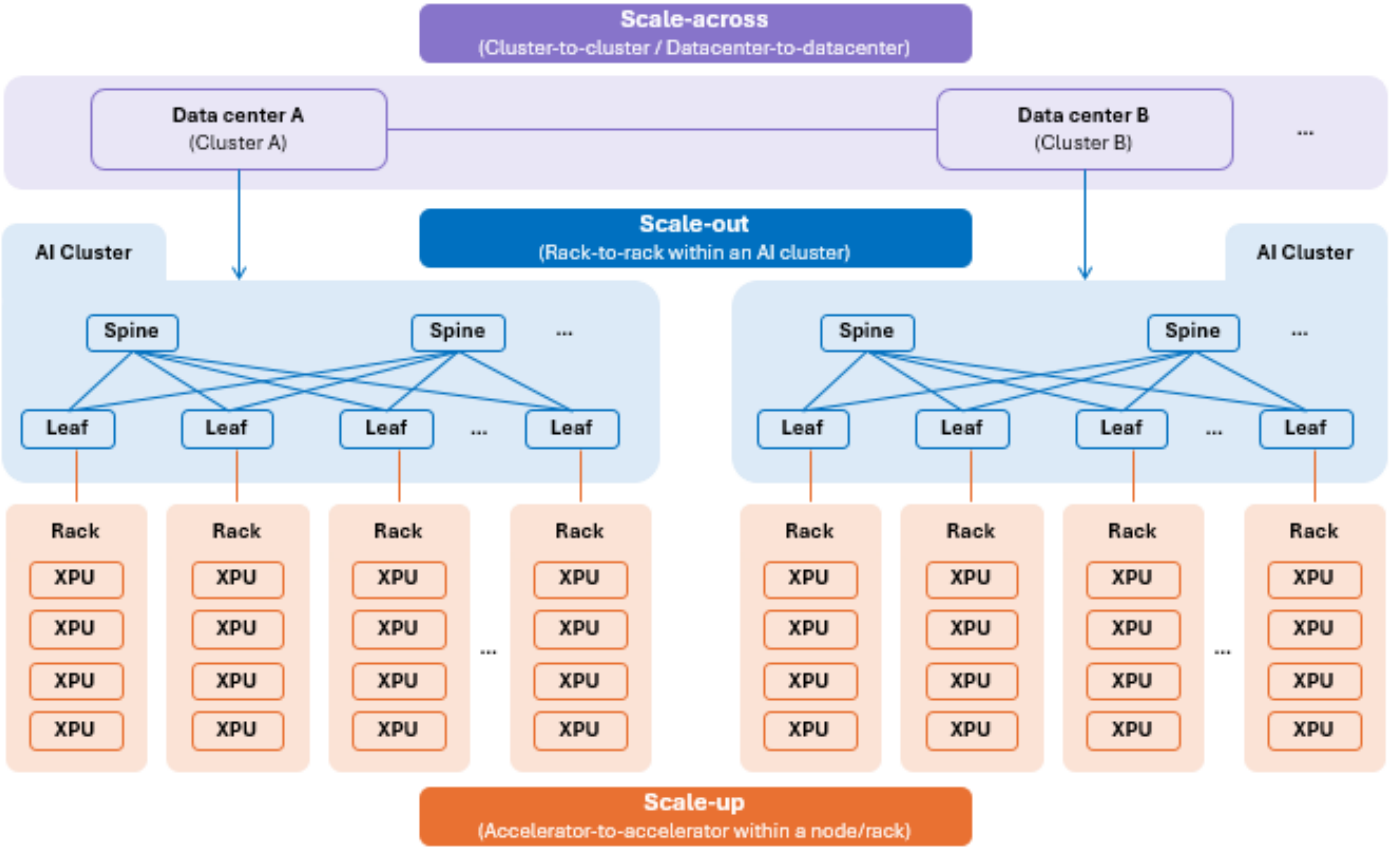
Accton's Network Application business includes AI accelerator cards, smart NICs, network interface products and customized networking and computing platforms. The segment expanded rapidly in 2023-25, with revenue contribution increasing to 62% from 23%, thanks to strong demand for AI-related accelerator cards and customized datacenter platforms for hyperscalers' AI deployments. The growing adoption of custom accelerators by hyperscalers creates attractive opportunities for ODM/JDM suppliers. As cloud customers seek to optimize performance and total cost of ownership, demand for customized hardware platforms, system integration and rack-level solutions has increased alongside investments in AI infrastructure.

We expect Network Application business to continue growing but at a more normalized pace after the initial AI-card ramp. As data center switch demand accelerates, the faster growth of the switch business should increasingly outweigh the contribution from Network Application to Accton's overall revenue growth, making Switch Accton's largest business segment.

AI networking architecture: scale-up, scale-out, scale-across; CPO/optical as future trend

Modern AI infrastructure relies on three complementary networking domains: scale-up connects accelerators within a node or tightly coupled rack domain, scale-out across racks and within a cluster/datacenter, and scale-across between clusters/datacenters. Each domain has different bandwidth, latency, and network requirements, but all three are becoming increasingly important as AI infrastructure grows in scale, complexity, and rising density.

Exhibit 18: AI infrastructure scales from accelerator connectivity within a node or rack, to rack-to-rack networking within a cluster/datacenter, and ultimately to cluster/datacenter interconnection
Scale-up, scale-out, and scale-across in AI networking architecture



Source: BofA Global Research, Broadcom

BofA GLOBAL RESEARCH

Scale-up: rising demand for intra-system connectivity

Scale-up provides tightly coupled connectivity among accelerators within a node or rack-scale computing domain. These networks require extremely high bandwidth and ultra-low latency to support accelerator-to-accelerator communication and collective computing operations. Nvidia's NVLink currently dominates commercial deployments, while Ethernet and UALink are gaining momentum as hyperscalers increasingly deploy custom and merchant accelerators. According to Dell'Oro Group, scale-up networking TAM is expected to be more than double for switches deployed in AI back-end networks and should account for more than half of the AI back-end switch market by 2030.

Scale-up interconnects remain primarily copper-based currently as copper offers low latency, good signal integrity, and high stability in short distance (e.g. below 10 meters) with favorable cost and power characteristics. Our supply chain check suggests that optical connectivity may begin penetrating scale-up more meaningfully after 2H27 as rising bandwidth requirements push copper closer to its physical transmission limits.

For Accton, its scale-up exposure appears to be developing into an incremental business line. We see its business with the biggest CSP customer has expanded into scale-up switch trays, in addition to its existing scale-out switches and AI cards used in compute trays.

Scale-out: the core architecture for AI cluster expansion

Scale-out expands computing capacity by connecting more servers and racks within an AI cluster, typically through high-speed Ethernet or InfiniBand fabrics. In common leaf-spine architecture, leaf or top-of-rack (ToR) switches connect compute racks, while spine switches provide the high-speed backbone between leaf switches. Scale-out is a well-established architecture for AI cluster expansion and remains a major driver of AI back-end networking demand. See more in Appendix: What is leaf-spine architecture?

We believe scale-out represents Accton's largest and most immediate AI networking opportunity. Larger clusters require more leaf and spine switches, higher port speeds, and greater switching capacity, supporting the demand for Accton's 400G/800G/1.6T Ethernet platforms. Scale-out is also the primary near-term application for CPO-enabled Ethernet switches, as CPO can help address power, thermal, and signal-integrity constraints at higher switch bandwidths.

Scale-across: extending AI networking beyond individual clusters/datacenters

Scale-across connects separate AI clusters or datacenters, potentially across different buildings or locations, to support distributed computing and resource pooling. As power, cooling, and space constraints limit expansion within individual facilities, AI infrastructure begins to extend across multiple sites/locations, increasing demand for high-capacity and predictable inter-datacenter connectivity.

While commercial adoption remains at an early stage, scale-across represents the potential market for Accton beyond conventional Ethernet switching, particularly in long-reach optical networking and datacenter interconnection. The company has been developing relevant optical-switching technologies and has used its OWS technology in Taiwan-Japan cross-border IOWN field validation over real computing resources and network infrastructure. We see this development/project as early evidence of Accton's system-integration capabilities and potential role in future scale-across applications.

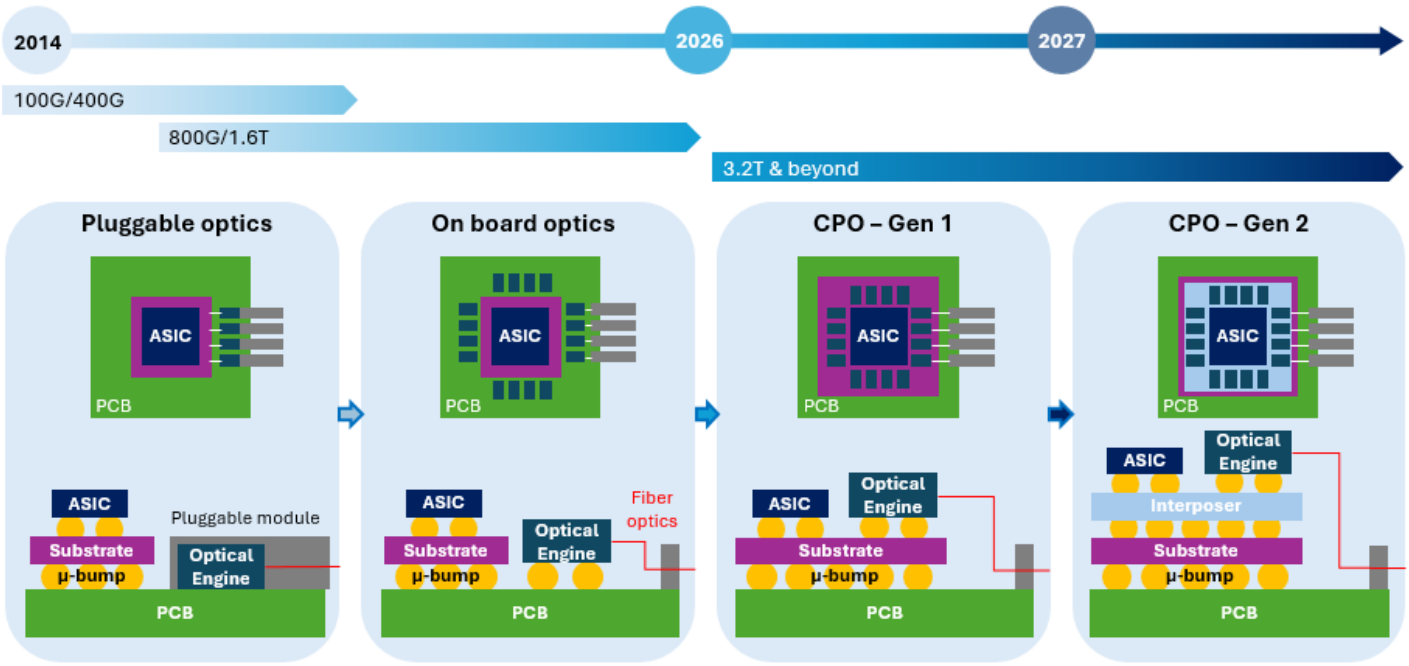
CPO: the potential next-gen architecture for AI switches

AI workloads are driving rapid increases in networking bandwidth and power consumption as infrastructure scales across scale-up, scale-out, and scale-across domains. The tougher requirement has led to bottlenecks in conventional electrical interconnects. We believe the transition toward optical networking could open new growth opportunities for Accton, where its exposure spans different layers of the optical networking stack, including CPO (co-packaged optics) for high-speed switch interconnects, OCS (optical circuit switch) for direct optical connectivity, and OWS (optical wavelength switch) for wavelength-level optical networking and DCI (Datacenter Interconnection).

The rapid growth of AI, cloud, and HPC workloads is driving exponential increases in datacenter computing and networking requirements. As AI clusters scale, traditional electrical interconnects are increasingly constrained by power consumption, signal integrity, and bandwidth density. Even pluggable optical transceivers are gradually approaching their physical and thermal limits at 800G and higher speeds.

To overcome these challenges, the industry is actively developing co-packaged optics (CPO), which integrate switch ASIC and optical engine within a single package, replacing traditional pluggable transceivers. By significantly shortening the electrical path between ASIC and optical engine, CPO can reduce power consumption and latency while improving signal integrity and bandwidth density. As Ethernet networking migrates from 400G to 800G, 1.6T, and eventually 3.2T, CPO theoretically becomes a key technology enabling the next-generation AI data-center networks.

Exhibit 19: CPO integrates optical engine and switch ASIC within a single package
Potential key trend of optical transceiver packaging in high-end data center

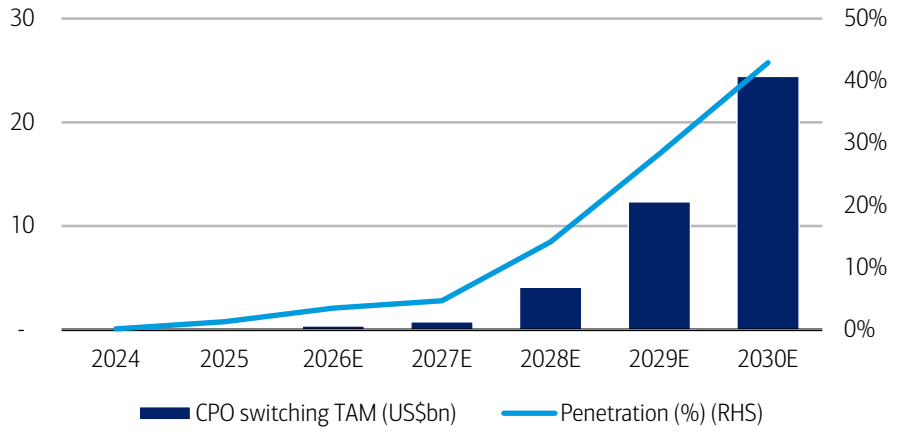


Source: BofA Global Research, IDTechEX

BofA GLOBAL RESEARCH

In terms of addressable market, our US team expects CPO adoption to gain traction from 2027. CPO switch penetration is limited as of aggregate switch market today, with deployments largely limited to engineering samples. However, CPO is expected to begin scaling with more meaningful penetration after 2028. Our US team expects CPO switch TAM to reach US$24.4bn in 2030 from US$0.4bn in 2026, with 14-43% penetration in 2028-30, vs. 1-5% in 2025-27 (Exhibit 20). See our global team's US Semiconductors: Scaling AI with Photons: Primer on Optical Interconnects published on 9 March 2026.

Exhibit 20: Our US team views CPO is at the early stage now but expects a CPO switching TAM to reach US$24.4bn in 2030 (with 43% penetration) from US$0.4bn in 2026 (with 3% penetration)
CPO switching market size (in US$bn) and penetration, 2024-2030E



Source: BofA Global Research, Lightcounting

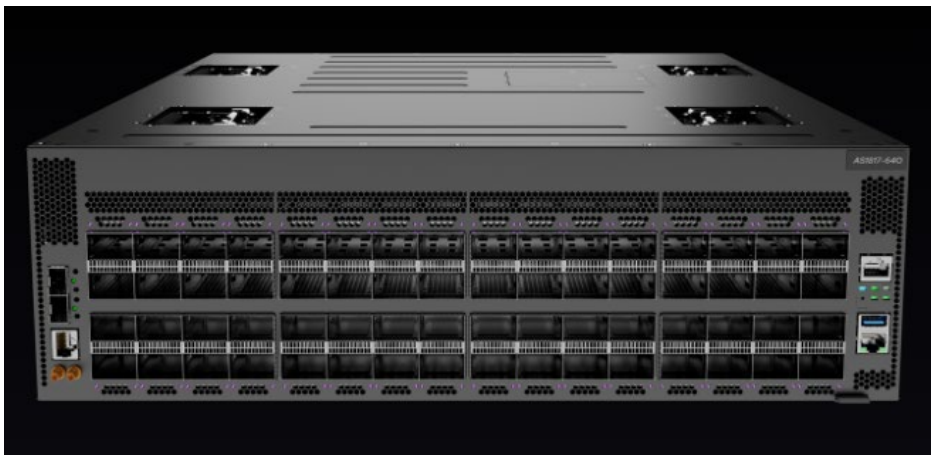
BofA GLOBAL RESEARCH

For Accton, while commercialization remains at an early stage, we believe Accton's early investment in optical integration technologies positions the company favorably to benefit from the industry's migration toward CPO-based networking architectures.

The company has been investing in optical integration technologies for several years, positioning itself for the industry's transition from pluggable optics toward more integrated optical architectures. The company demonstrated an on-board optical engine solution as early as 2020 and has gradually expanded its networking portfolio from 400G to 800G and now 1.6T-class platforms.

We highlight Accton's latest 102.4T Ethernet switch, based on Broadcom's Tomahawk 6 ASIC (Exhibit 21). The platform supports both conventional OSFP1600 pluggable optics and a CPO configuration. Under the CPO architecture, the optical engine is integrated directly adjacent to the switch ASIC, helping mitigate power, thermal, and signal-integrity challenges associated with ultra-high-bandwidth networking.

Exhibit 21: 102.4T Ethernet Switch supports both pluggable optics and a CPO configuration
Accton's 102.4T Ethernet Switch product example



Source: Accton

BofA GLOBAL RESEARCH

Beyond the 102.4T platform, Accton continues to advance toward more tightly integrated optical architectures. In 2025, its subsidiary Edgecore demonstrated a proof-of-concept CPO solution integrating optical engines and switch silicon within a single package, highlighting the group's long-term commitment to CPO development and next-generation optical networking.

We believe CPO could provide Accton with several strategic benefits. First, CPO increases the optical content per switch, creating opportunities for higher-value switch platforms compared with conventional pluggable architectures. Second, Accton's established position as a leading networking switch ODM, combined with its capabilities in switch design, optical integration, thermal management and system-level integration, could allow the company to capture a larger portion of the AI networking value chain as CPO adoption increases. Finally, the company's optical subsidiaries and growing ecosystem partnerships could further strengthen its ability to deliver integrated optical-networking solutions.

Potential optical technologies under watch: OCS and OWS

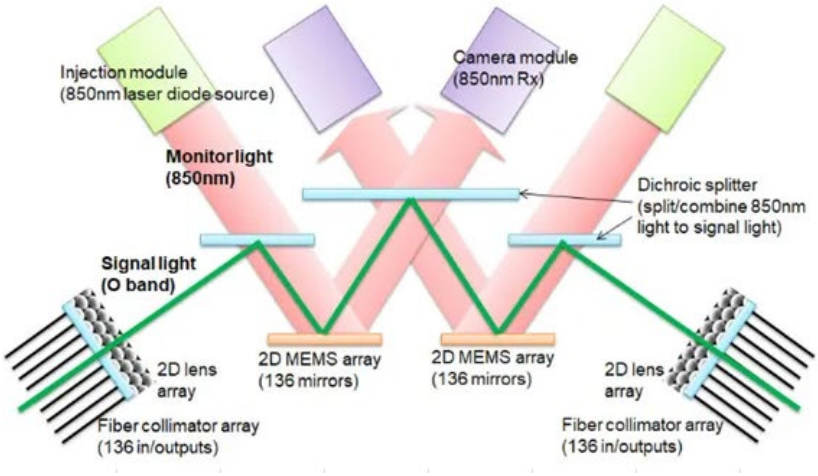
Although CPO significantly shortens the electrical path between switch ASIC and optical engine, data traffic may still require optical-electrical-optical (O-E-O) conversion when it traverses packet switches. As AI clusters continue to expand, the growing volume of data traffic increases network latency, power consumption, and switching overhead, creating bottlenecks beyond the switch level.

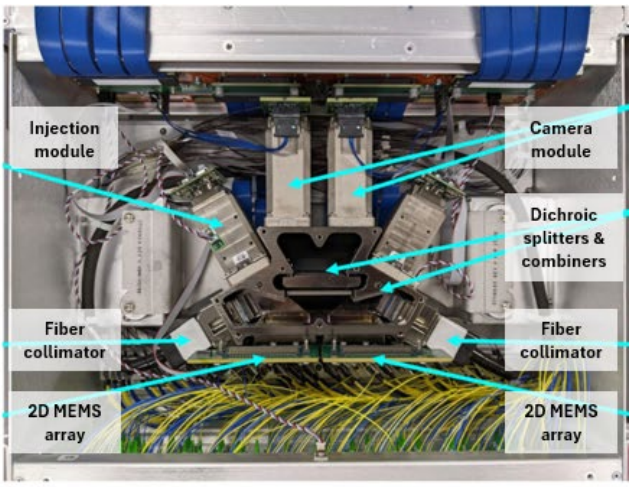
To address these challenges, the industry is increasingly exploring optical switch, which enables optical signals to remain in the optical domain while traversing the network. Compared with traditional Ethernet switching, optical switching reduces intermediate packet-processing stages and minimizes O-E-O conversions, potentially lowering power consumption and latency while improving network efficiency. Two complementary optical switching approaches are Optical Circuit Switching (OCS) and Optical Wavelength Switching (OWS). While OCS establishes dedicated optical paths between endpoints, OWS provides more granular control by dynamically routing individual wavelengths carried within an optical fiber. Our check suggests that OCS is currently for customized adoption only and OWS is at the initial/validation stage. The initial shipment/sales for Accton's OCS and OWS products may start from 2027.

OCS: enabling direct optical connectivity for AI clusters

Unlike traditional Ethernet architectures, OCS establishes dedicated optical paths between endpoints without processing individual packets. When two endpoints need to communicate, a controller dynamically configures the switch matrix to create a dedicated optical circuit between them. By enabling direct optical connectivity, reducing intermediate switching stages, and minimizing O-E-O conversions, OCS can lower latency and power consumption while improving network efficiency, providing a high-bandwidth optical layer for inter-rack and inter-cluster connectivity. Currently the largest OCS use case is from Google (MEMS-based). See more OCS introduction in our global team's US Semiconductors: Scaling AI with Photons: Primer on Optical Interconnects published on 9 March 2026.

Exhibit 22: OCS creates direct optical connections, reducing intermediate packet-switching stages and O-E-O conversions
Optical Circuit Switch diagram and example (MEMS-based example from Google)



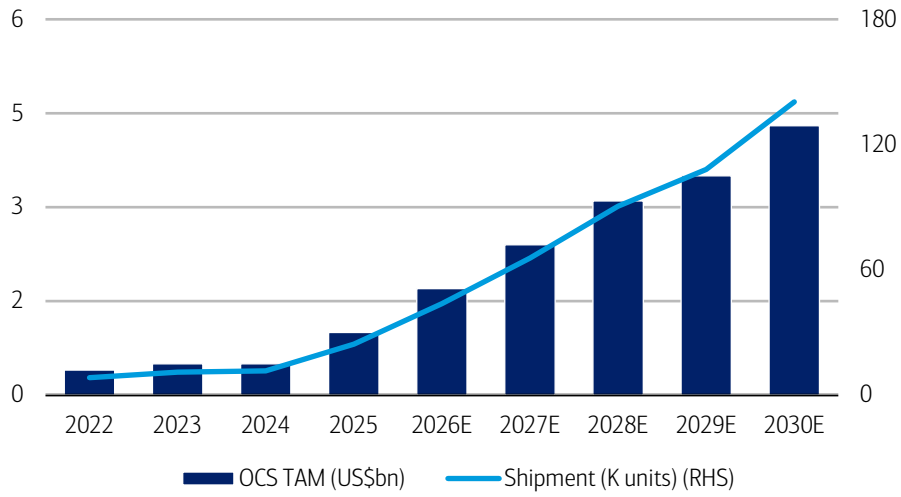


Source: BofA Global Research, Google

BofA GLOBAL RESEARCH

Our US team expects OCS market to expand to US$4.3bn by 2030 from US$1.0bn in 2025, driven by accelerating adoption in AI and HPC networks (Exhibit 23). Despite an expected decline in ASPs as volumes scale, shipment growth is projected to remain robust, increasing to 140K units by 2030 from around 24K units in 2025.

Exhibit 23: Our US team expects OCS TAM to exceed US$4bn by 2030 with robust shipment growth
Global OCS market TAM (US$bn) and shipment (mn units), 2022-30E



Source: BofA Global Research estimates, LightCounting, 650 Group, Dell'Oro

BofA GLOBAL RESEARCH

Accton began building its optical-switching capabilities through its investment in InLC Technology (a Korean private company) in 2024. The two companies collaborate on optical switch development for datacenter and DCI applications. InLC contributes expertise in LCoS-based (liquid crystal on silicon based) optical switching technologies, while Accton's experience in hyperscale networking and switch design provides a pathway toward system-level deployment.

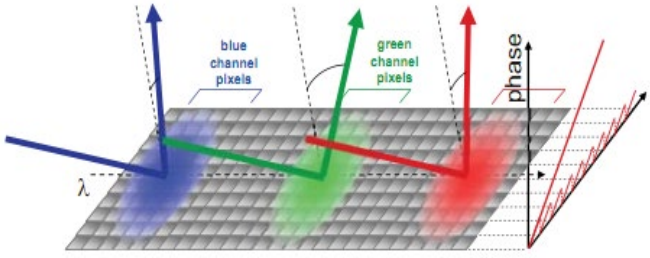
We believe Accton's position in OCS is different as it focuses on complete optical switch systems rather than individual optical components. While Taiwan's CPO ecosystem is mature across foundry, packaging, and optical components, the local supply chain for OCS system remains at an early stage. Accton is among the few Taiwanese networking companies directly developing OCS platforms, creating an opportunity to extend from conventional packet switching into the optical switch layer of future AI networks. Our supply chain check suggests that OCS products are in sampling and pilot-production stages. Combined with its expertise in high-speed Ethernet switches, optical integration, and rack-level infrastructure, we believe Accton could benefit from an early-mover advantage as the optical switch ecosystem develops in the coming years.

OWS: enabling flexible wavelength-level optical switching

While OCS establishes dedicated optical circuits between endpoints, Optical Wavelength Switching (OWS) provides more granular control by routing individual wavelengths within an optical fiber. Through wavelength-division multiplexing (WDM), multiple wavelengths can be transmitted simultaneously over a single fiber. OWS dynamically directs these wavelengths to different destinations, allowing optical bandwidth to be allocated more efficiently without processing traffic in the electrical domain or relying on repeated O-E-O conversions.

Exhibit 24: By controlling the phase of incident light at the pixel level, LCoS enables dynamic wavelength routing and forms the core technology behind many wavelength-switching systems

LCoS modulates and steers individual optical wavelengths

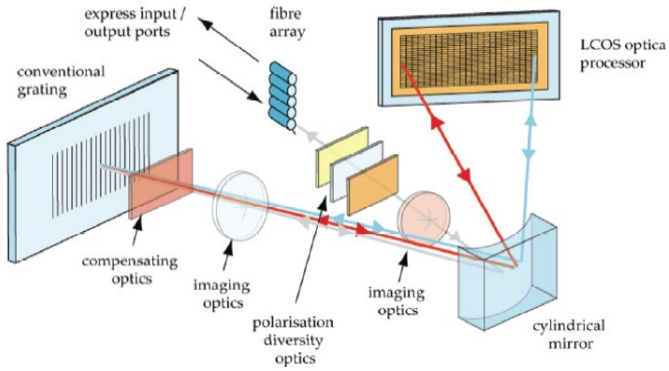


Source: FOSCO

BofA GLOBAL RESEARCH

Exhibit 25: Incoming wavelengths are separated, dynamically redirected by the LCoS processor, and routed to different output ports without O-E-O conversion

LCoS-based OWS architecture



Source: FOSCO

BofA GLOBAL RESEARCH

Accton's OWS development has reportedly progressed beyond the laboratory stage and is moving toward customer validation and commercialization through its subsidiary Edgecore. The company's flagship OWS platform, the IRX3032, supports 32 WDM channels, up to 800G coherent optics, and up to 25.6Tbps of optical bandwidth while operating entirely in the optical domain. According to Edgecore, the platform delivers 160ns end-to-end latency and supports point-to-point, ring, and full-mesh network topologies. The IRX3032 has also reportedly entered validation with a US CSP and NTT from Japan, suggesting continued progress toward commercial deployment.

Beyond standalone switch, Accton is positioning OWS as part of its broader all-photonics networking strategy. The IRX3032 has been incorporated into IOWN (Innovative Optical and Wireless Network) architecture, where optical switching is combined with high-speed Ethernet switches, transport equipment, and AI infrastructure. More importantly, Accton's OWS technology has already been used in Taiwan-Japan

cross-border validation based on real computing resources and network infrastructure, supporting applications such as real-time translation and AI-enabled traffic analytics. In our view, Accton's key advantages in OWS should be its ability to combine optical switch technology and established networking system experiences, via InLC's LCoS technology, Edgecore's OWS platform, and its expertise in high-speed Ethernet switches and system integration. Although the OWS market remains at an early stage, growing interest from hyperscalers and telecom operators in all-optical networking architectures could create new opportunities beyond conventional packet switching.

Exhibit 26: Accton's OWS IRX3032 is a 1U wavelength-programmable LCoS-based optical switch that supports 32 WDM to deliver ultra-low-latency optical routing
Edgecore's OWS IRX3032 product example

Source: Edgecore

BofA GLOBAL RESEARCH

Key peer comparison: Celestica (CLS US)

Celestica (CLS US; covered by Ruplu Bhattacharya) is a leading EMS company based in North America. Celestica provides design, engineering, manufacturing, and hardware-platform solutions to OEMs and cloud service providers. Celestica was incorporated in Ontario, Canada in 1996. Prior to that, the company was an IBM manufacturing unit that provided manufacturing services to IBM for more than 75 years. It was acquired by Onex Corporation and Celestica management in 1996 and went publicly listed in 1998. Headquartered in Toronto, it operates across Americas, Europe, and Asia with around 45 sites globally. It is also expanding capacity in Texas (US) through a new campus. See also: Ruplu Bhattacharya's latest Celestica report published on 29 July 2026.

Celestica's overlap with Accton from key CCS segment

Celestica has two business units: CCS (Communications, Server, Storage) and ATS (Aerospace & Defense, Industrial, Capital Equipment, and HealthTech). The CCS segment is more relevant to Accton's business scope. As of 1H26, CCS contributed 81% of total sales and the remaining 19% came from ATS.

Exhibit 27: CCS segment, which is more relevant to Accton's business scope, contributed 81% of Celestica's total sales in 1H26

Celestica sales breakdown by segment, 2021-1H26

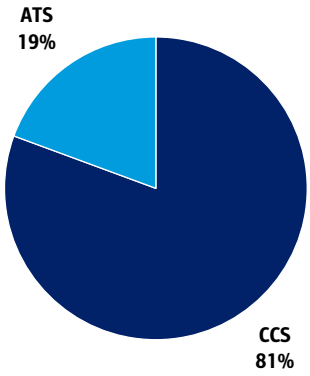
(%)	2021	2022	2023	2024	2025	1H26
CCS (Connectivity & Cloud Solutions)	**59%**	**59%**	**58%**	**67%**	**74%**	**81%**
- Communications	40%	39%	33%	40%	58%	58%
- Enterprise	19%	19%	25%	27%	17%	23%
ATS (Advanced Technology Solutions)	**41%**	**41%**	**42%**	**33%**	**26%**	**19%**

Source: BofA Global Research, company data

BofA GLOBAL RESEARCH

Exhibit 28: CCS contributed 81% of Celestica's total sales in 1H26, vs. 19% from ATS

Celestica's sales breakdown by segment, 1H26

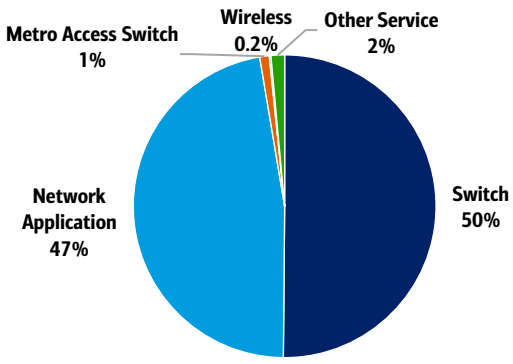


Source: BofA Global Research, company data. CCS: Communications, Server, Storage. ATS: Aerospace & Defense, Industrial, Capital Equipment, and HealthTech

BofA GLOBAL RESEARCH

Exhibit 29: Switch comprised 50% of Accton's total sales in 1H26, vs. 47% from Network Application and the remaining from Metro Access Switch, Wireless, and Other Service

Accton's sales breakdown by segment, 1H26



Source: BofA Global Research, company data

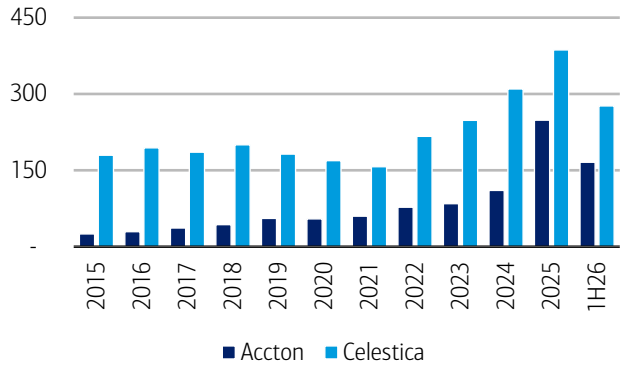
BofA GLOBAL RESEARCH

Historical sales comparison: larger scale from Celestica

In terms of sales, Celestica had a larger scale than Accton, while the gap has narrowed substantially. Celestica's total sales were 1.6x/1.7x of Accton's total sales in 2025/1H26, narrowing from 7.3x in 2015 (Exhibit 30). We believe the sales scale difference reflected Celestica's broader business portfolio. Meanwhile, Celestica's sales from CCS business (which shares the similar end-market/applications with Accton) were 1.2x/1.3x of Accton's total sales in 2025/1H26, vs. 4.9x in 2015 (Exhibit 31).

Exhibit 30: Celestica's total sales were 1.6x/1.7x of Accton's total sales in 2025/1H26, narrowing from 7.3x in 2015

Total sales (in NT$bn) comparison: Accton vs. Celestica, 2015-1H26

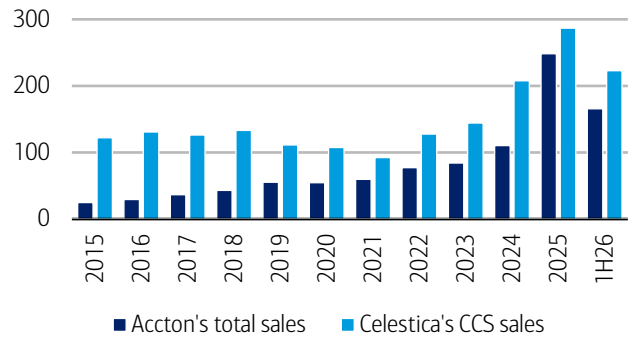


Source: BofA Global Research, TEJ, company data

BofA GLOBAL RESEARCH

Exhibit 31: Celestica's sales from CCS business were 1.2x/1.3x of Accton's total sales in 2025/1H26, vs. 4.9x in 2015

Sales (in NT$bn) comparison: Accton's total sales vs. Celestica's CCS sales, 2015-1H26



Source: BofA Global Research, TEJ, company data. CCS: Communications, Server, Storage

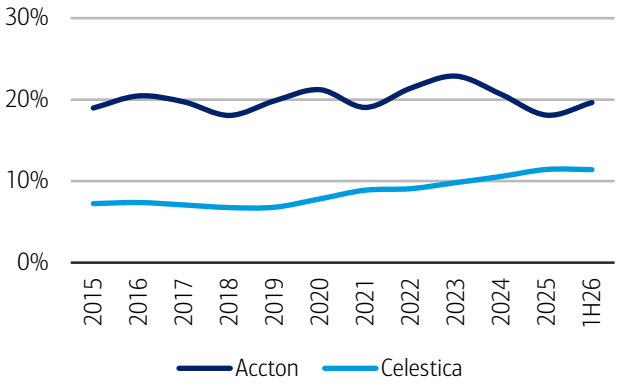
BofA GLOBAL RESEARCH

Historical GM/OPM comparison: higher % from Accton

On the other hand, Accton has consistently had higher margins. Its GM/OPM in 2015-1H26 were 18.1-22.9%/6.0-14.7%, vs. Celestica's 6.7-11.4%/2.7-8.1% (Exhibit 32 and Exhibit 33). Celestica's CCS segment OPM was 7.4-8.2% in 2024-25, vs. 4.5-5.3% for the ATS segment, according to our Celestica analyst Ruplu Bhattacharya's assumptions.

Exhibit 32: Accton had higher GM of 18.1-22.9% in 2015-1H26 vs. Celestica's 6.7-11.4%

GM comparison for Accton and Celestica, 2015-1H26

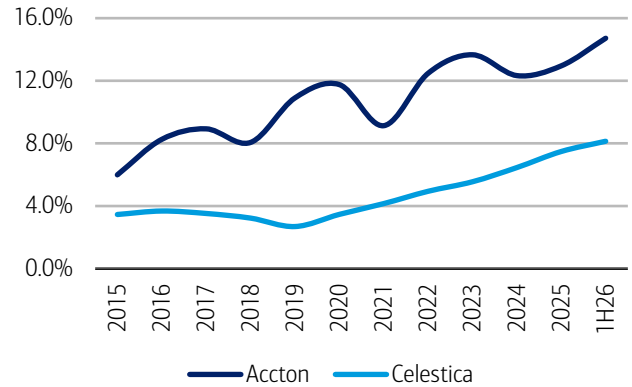


Source: BofA Global Research, Bloomberg, TEJ, company data

BofA GLOBAL RESEARCH

Exhibit 33: Accton's OPM of 6.0-14.7% in 2015-1H26 has been much higher than Celestica's 2.7-8.1%

OPM comparison for Accton and Celestica, 2015-1H26



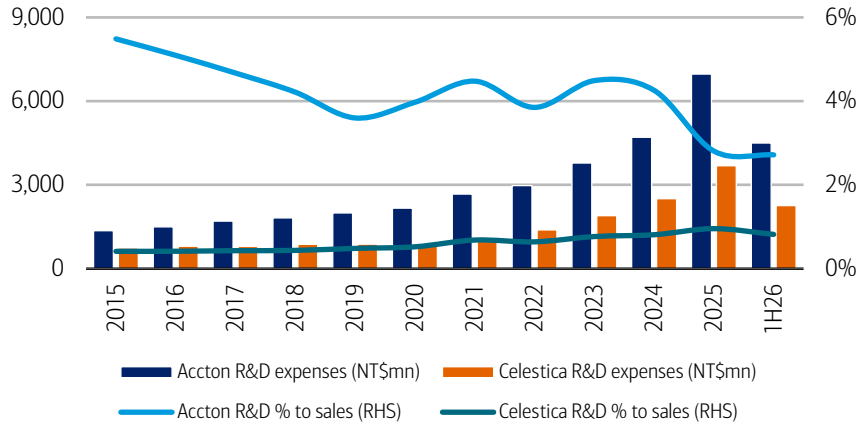
Source: BofA Global Research, Bloomberg, TEJ, company data

BofA GLOBAL RESEARCH

Ongoing R&D efforts from both Accton and Celestica

Accton's R&D expenses had been around 2x of Celestica in absolute terms over 2015-1H26. Accton's R&D expenses to sales were 3-5%, vs. 0-1% for Celestica under Celestica's larger sales base (Exhibit 34). We believe both companies are putting R&D resources into AI rack-scale infrastructure-related business and more design-/engineering-led solution business.

Exhibit 34: Accton's R&D expenses were around 2x of Celestica over 2015-1H26
R&D expenses (in NT$mn) and R&D expenses % to sales: Accton and Celestica, 2015-1H26



Source: BofA Global Research, company data

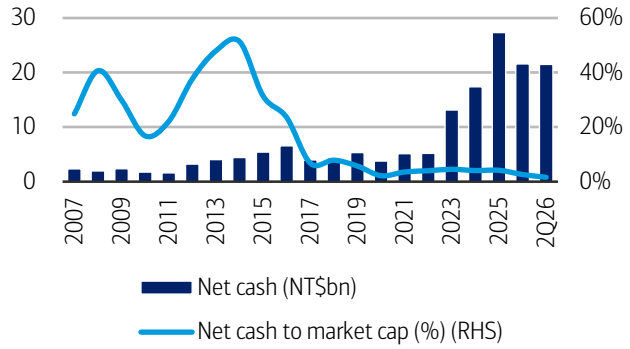
BofA GLOBAL RESEARCH

Balance sheet & CF comparison: Accton vs. Celestica

Net cash position from Accton vs. net debt from Celestica

Accton has maintained a net cash position since 2007 (Exhibit 35), while Celestica has been in a net debt position since 2018 (Exhibit 36). As of 2Q26, Accton's net cash was 2% to market cap, vs. Celestica's 1% net debt to market cap. See more details on Accton's net cash position in Balance sheet: net cash position and high ROE.

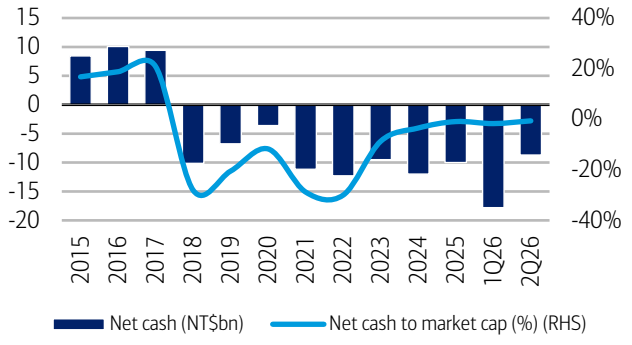
Exhibit 35: Accton has been in a net cash position since 2007. Its net cash was 2% to market cap, as of 2Q26
Accton's net cash (NT$bn) and net cash to market cap, 2007-2Q26



Source: BofA Global Research, TEJ, company data. Net cash = Cash & cash equivalent - Short-term borrowing - Current portion of long-term debt - Long-term liabilities. Market cap based on year-/quarter- end market cap

BofA GLOBAL RESEARCH

Exhibit 36: Celestica has been in a net debt position since 2018. Its net debt was 1% to market cap, as of 2Q26
Celestica's net cash (in NT$bn) and net cash to market cap, 2015-2Q26



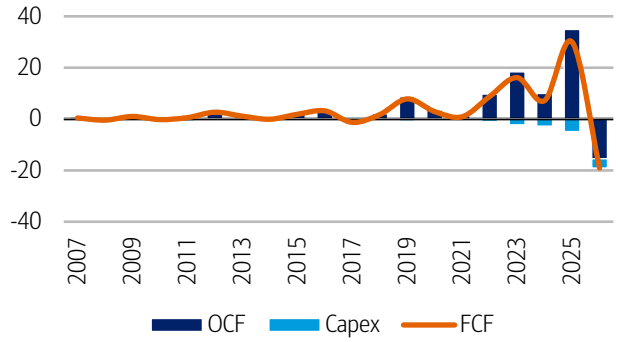
Source: BofA Global Research, Bloomberg, company data. Net cash = Cash & cash equivalent - Short-term borrowing - Current portion of long-term debt - Long-term liabilities. Market cap based on year-/quarter- end market cap

BofA GLOBAL RESEARCH

More volatile FCF from Accton but both had positive FCF most of the time

Accton's historical FCF was more volatile than Celestica but had generated positive FCF in most of the years since 2007 (Exhibit 37). Celestica generated positive FCF every year except 2019 (Exhibit 38). In 1H26, Accton had FCF outflow of NT$19.1bn, vs. Celestica's FCF inflow of US$273.9mn (around NT$8.7bn), with stronger OCF more than offseting the higher capex. See more details on Accton's FCF in Cashflow analysis: negative OCF in 1H26 for higher working capital; higher capex for global expansion

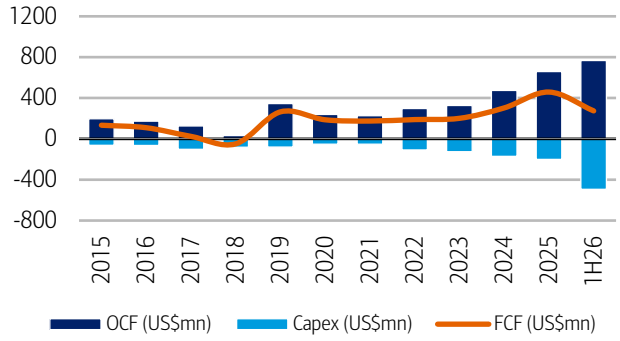
Exhibit 37: Accton has generated positive FCF in most of the years since 2007; it had FCF outflow of NT$19.1bn in 1H26
Accton's OCF, capex, and FCF trend (in NT$bn) in 2007-1H26



Source: BofA Global Research, TEJ, company data. Capex refers to purchase of fixed assets

BofA GLOBAL RESEARCH

Exhibit 38: Celestica generated positive FCF every year except 2019; it had FCF inflow of US$273.9bn in 1H26
Celestica's OCF, capex, and FCF trend (in US$mn) in 2015-1H26



Source: BofA Global Research, Bloomberg, company data. Capex refers to purchase of fixed assets

BofA GLOBAL RESEARCH

Financial analysis: 37%/42% sales/EPS CAGR in 2026-28

We project 37%/42% sales/EPS CAGR for 2026-28. Our 2026-28 EPS estimate is 3-5% above consensus (Exhibit 39). Our key assumptions are the following:

- **Sales:** We estimate 2026-28 sales to grow 35-67% YoY, mainly driven by strong AI market demand and ongoing spec/platform migration. We also reflected 3Q26 preliminary sales of NT$120.7bn (+26% QoQ, +66% YoY) into our forecast. Its monthly sales sustained at historical high levels in 3Q26 with solid YoY trend, indicating still robust market demand amid component shortage (Exhibit 40 and Exhibit 41). Our 3Q26 EPS forecast is 2% above consensus (Exhibit 42).
- **Margins**: We project 2026-28 GM of 19.3-20.1% and OPM of 14.8-16.1%, with improving product mix, rising sales scale, and stringent opex control. We also take high component cost into consideration.
- **Non-operating items, tax rate, and share count for EPS**: We follow the historical patterns for non-operating items and tax rate. We also assume no capital-raising plan in 2026-28.

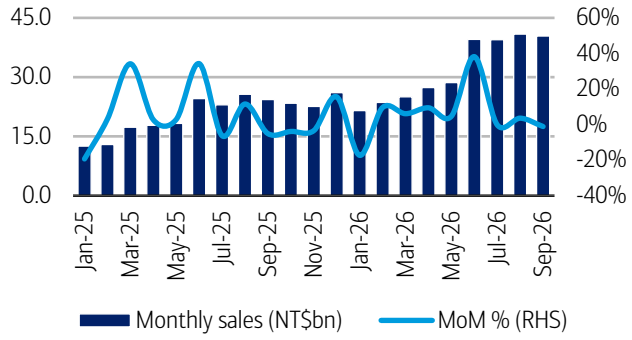
Exhibit 39: Our 2026-28 net income estimates are 3-5% above consensus
BofAe vs Bloomberg consensus, 2026-28E

	2026E			2027E			2028E		
NT$mn	**BofAe**	**Consensus**	**Diff (%)**	**BofAe**	**Consensus**	**Diff (%)**	**BofAe**	**Consensus**	**Diff (%)**
Sales	414,359	409,123	1%	558,071	541,327	3%	772,505	767,520	1%
Gross profit	79,933	78,556	2%	109,110	103,171	6%	155,499	148,047	5%
Operating profit	61,344	60,223	2%	86,377	81,596	6%	124,751	119,731	4%
Pretax profit	62,478	61,079	2%	87,486	82,773	6%	125,866	121,629	3%
Net income	49,449	48,180	3%	69,192	65,727	5%	99,579	96,420	3%
EPS (NT$)	88.27	86.05	3%	123.52	117.52	5%	177.77	172.25	3%
Margin %									
Gross margin	19.3%	19.2%	0.1 ppt	19.6%	19.1%	0.5 ppt	20.1%	19.3%	0.8 ppt
Operating margin	14.8%	14.7%	0.1 ppt	15.5%	15.1%	0.4 ppt	16.1%	15.6%	0.5 ppt
Pre-tax margin	15.1%	14.9%	0.1 ppt	15.7%	15.3%	0.4 ppt	16.3%	15.8%	0.4 ppt
Net margin	11.9%	11.8%	0.2 ppt	12.4%	12.1%	0.3 ppt	12.9%	12.6%	0.3 ppt

Source: BofA Global Research estimates, Bloomberg consensus

BofA GLOBAL RESEARCH

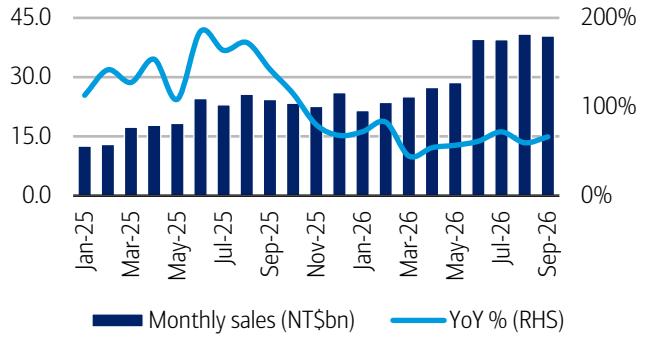
Exhibit 40: Accton's Jul/Aug/Sep'26 sales (flat/+4%/-1% MoM) sustained at historical high levels
Accton's monthly sales and MoM trend, Jan'25-Sep'26



Source: BofA Global Research, TEJ

BofA GLOBAL RESEARCH

Exhibit 41: Accton's Jul/Aug/Sep'26 sales (+72%/+59%/+66% YoY) sustained at historical high levels
Accton's monthly sales and YoY trend, Jan'25-Sep'26



Source: BofA Global Research, TEJ

BofA GLOBAL RESEARCH

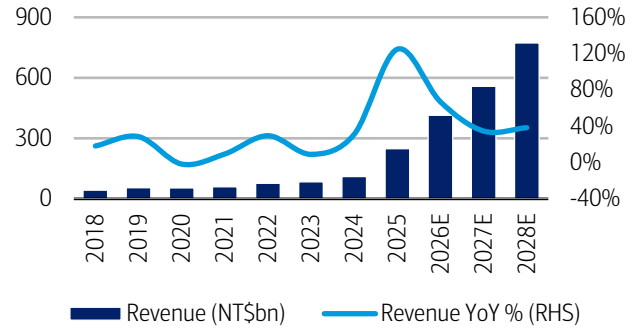
Exhibit 42: Our 3Q26 EPS forecast is 2% above consensus, after reflecting preliminary sales
3Q26 result preview: BofAe vs. consensus

NT$mn	3Q26E	QoQ %	YoY %	Consensus	Diff (%)	2Q26	3Q25
Sales	120,744	26%	66%	118,616	2%	95,538	72,947
Gross profit	22,639	20%	81%	22,384	1%	18,817	12,518
Operating profit	17,567	23%	95%	17,584	0%	14,326	9,001
Pretax profit	17,855	23%	84%	17,606	1%	14,563	9,708
Net income	14,292	29%	83%	14,040	2%	11,048	7,827
EPS (NT$)	25.51	29%	82%	24.92	2%	19.72	14.00
Margin %							
Gross margin	18.7%	-0.9 ppt	1.6 ppt	18.9%	-0.1 ppt	19.7%	17.2%
Operating margin	14.5%	-0.4 ppt	2.2 ppt	14.8%	-0.3 ppt	15.0%	12.3%
Pre-tax margin	14.8%	-0.5 ppt	1.5 ppt	14.8%	-0.1 ppt	15.2%	13.3%
Net margin	11.8%	0.3 ppt	1.1 ppt	11.8%	0.0 ppt	11.6%	10.7%

Source: BofA Global Research estimates, Bloomberg consensus

BofA GLOBAL RESEARCH

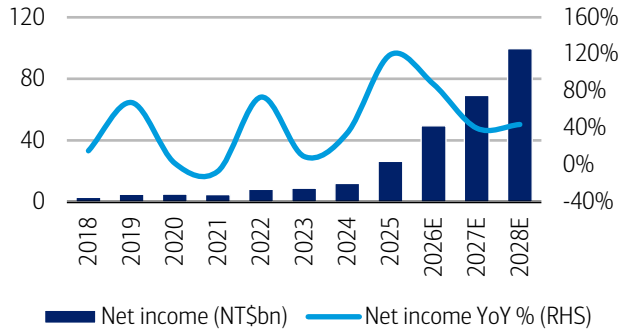
Exhibit 43: We model 2026-28 revenue CAGR at 37%
Accton's revenue and YoY trend, 2018-28E



Source: BofA Global Research estimates, company data

BofA GLOBAL RESEARCH

Exhibit 44: We model 2026-28 net income CAGR at 42%
Accton's net income and YoY trend, 2018-28E



Source: BofA Global Research estimates, company data

BofA GLOBAL RESEARCH

Exhibit 45: We model 2026-28E sales CAGR at 37% and EPS CAGR at 42%
Quarterly earnings summary, 2025-28E

NT$mn	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	1Q28E	2Q28E	3Q28E	4Q28E	2025	2026E	2027E	2028E
Sales	70,121	95,538	120,744	127,956	121,781	132,878	147,894	155,517	156,432	176,898	207,081	232,094	248,320	414,359	558,071	772,505
Gross profit	13,709	18,817	22,639	24,768	23,660	25,904	28,983	30,562	30,799	35,261	41,942	47,497	44,919	79,933	109,110	155,499
Operating profit	10,057	14,326	17,567	19,394	18,667	20,456	23,067	24,186	24,542	28,362	33,866	37,981	32,167	61,344	86,377	124,751
Pretax profit	10,370	14,563	17,855	19,690	18,941	20,729	23,350	24,467	24,819	28,640	34,146	38,260	32,949	62,478	87,486	125,866
Net income	8,349	11,048	14,292	15,760	15,160	15,762	18,688	19,582	19,863	21,775	27,325	30,616	26,374	49,449	69,192	99,579
EPS (NT$)	14.90	19.72	25.51	28.13	27.06	28.14	33.36	34.96	35.46	38.87	48.78	54.66	47.19	88.27	123.52	177.77
Margin %																
Gross margin	19.6%	19.7%	18.7%	19.4%	19.4%	19.5%	19.6%	19.7%	19.7%	19.9%	20.3%	20.5%	18.1%	19.3%	19.6%	20.1%
Operating margin	14.3%	15.0%	14.5%	15.2%	15.3%	15.4%	15.6%	15.6%	15.7%	16.0%	16.4%	16.4%	13.0%	14.8%	15.5%	16.1%
Pretax margin	14.8%	15.2%	14.8%	15.4%	15.6%	15.6%	15.8%	15.7%	15.9%	16.2%	16.5%	16.5%	13.3%	15.1%	15.7%	16.3%
Net margin	11.9%	11.6%	11.8%	12.3%	12.4%	11.9%	12.6%	12.6%	12.7%	12.3%	13.2%	13.2%	10.6%	11.9%	12.4%	12.9%
QoQ %																
Sales	-3%	36%	26%	6%	-5%	9%	11%	5%	1%	13%	17%	12%				
Gross profit	6%	37%	20%	9%	-4%	9%	12%	5%	1%	14%	19%	13%				
Operating profit	11%	42%	23%	10%	-4%	10%	13%	5%	1%	16%	19%	12%				
Pretax profit	1%	40%	23%	10%	-4%	9%	13%	5%	1%	15%	19%	12%				
Net income	0%	32%	29%	10%	-4%	4%	19%	5%	1%	10%	25%	12%				
YoY %																
Sales	64%	58%	66%	78%	74%	39%	22%	22%	28%	33%	40%	49%	125%	67%	35%	38%
Gross profit	59%	73%	81%	92%	73%	38%	28%	23%	30%	36%	45%	55%	97%	78%	37%	43%
Operating profit	69%	76%	95%	113%	86%	43%	31%	25%	31%	39%	47%	57%	136%	91%	41%	44%
Pretax profit	64%	120%	84%	91%	83%	42%	31%	24%	31%	38%	46%	56%	118%	90%	40%	44%
Net income	61%	119%	83%	89%	82%	43%	31%	24%	31%	38%	46%	56%	120%	87%	40%	44%

Source: BofA Global Research estimates, company data

BofA GLOBAL RESEARCH

Balance sheet: net cash position and high ROE

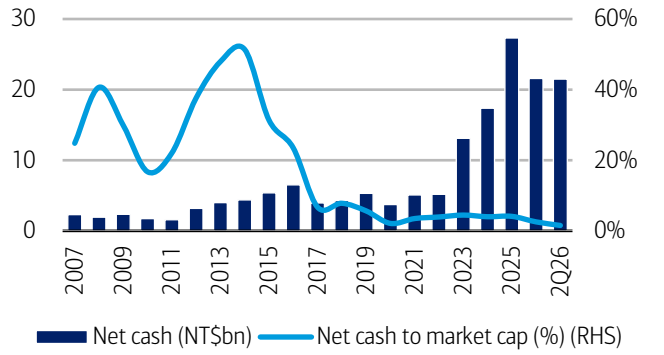
Accton has maintained a net cash position since 2007. We believe it reflects the robust cash-generation capability, supported by sustained profitability and healthy free cash flow. As of 2Q26, its net cash stood at NT$21.5bn or 2% of its market cap, vs. 3% in 1Q26, 2-8% in 2017-25, and 17-51% in 2007-16 (Exhibit 46). Accton had a higher net cash to market cap% before 2016 as its market cap jumped significantly from 2017.

Accton's cash conversion cycle (CCC) days have been low at nearly or below two months. It was 31 days in 2Q26, vs. 15 days in 1Q26 and 11-63 days in 2007-25 (Exhibit 47). We attribute the low CCC days to long A/P days, with A/R days and inventory days sustained at stable levels. We believe its CCC days reflected efficient working-capital management, faster inventory turnover amid robust AI networking demand, strong bargaining power as networking ODM leader, and mature supply chain management.

In terms of DuPont analysis, we expect Accton's ROE to remain high at 45-50% in 2026-28E (Exhibit 48) with continued net margin expansion (Exhibit 49) and lower financial leverage and asset turnover. We view the ROE sustainability as a reflection of a strong balance sheet with retained earnings accumulation and cash build-up.

Exhibit 46: Accton has been in a net cash position and was 2% net cash to market cap in 2Q26, vs. 3% in 1Q26, 2-8% in 2017-25, and 17-51% in 2007-16

Accton's net cash and net cash to market cap, 2007-2Q26

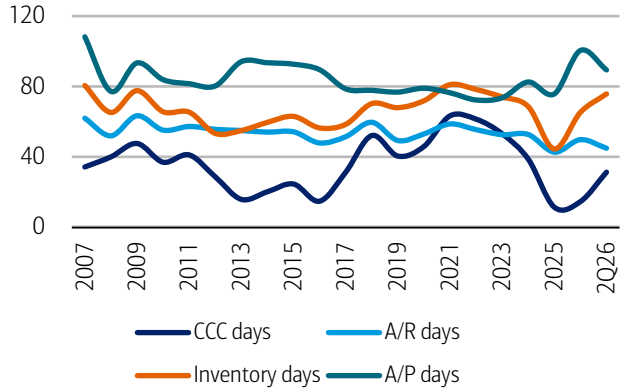


Source: BofA Global Research, TEJ, company data. Net cash = Cash & cash equivalent - Short-term borrowing - Current portion of long-term debt - Long-term liabilities. Market cap based on year-/quarter- end market cap

BofA GLOBAL RESEARCH

Exhibit 47: Accton's CCC days have been low. It was 31 days in 2Q26, vs. 15 days in 1Q26 and 11-63 days in 2007-25

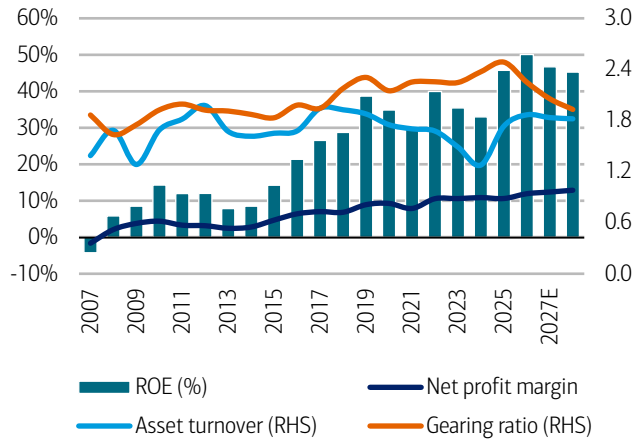
Accton's cash conversion cycle, 2007-2Q26



Source: BofA Global Research, TEJ, company data. Annual A/R days = 365 / (annual sales / average accounts & notes receivable). Quarterly A/R days = 365 / 4 / (quarterly sales / average accounts & notes receivable). Annual inventory days = 365 / (annual COGS / average inventories). Quarterly inventory days = 365 / 4 / (quarterly COGS / average inventories). Annual A/P days = 365 / (annual COGS / average accounts & notes payable). Quarterly A/P days = 365 / 4 / (quarterly COGS / average accounts & notes payable

BofA GLOBAL RESEARCH

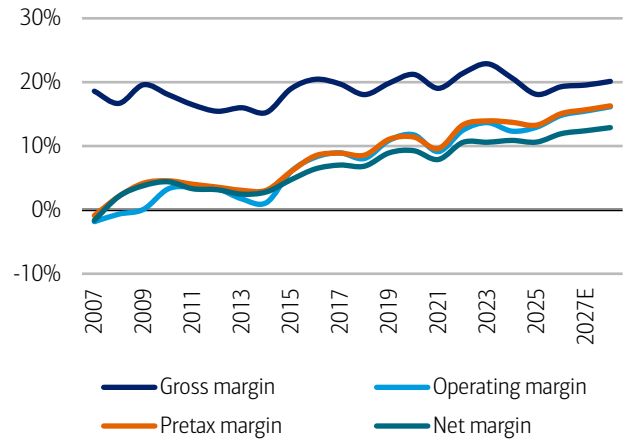
Exhibit 48: We expect Accton's ROE to remain high at 45-50% in 2026-28E with continued net margin expansion
Accton's DuPont analysis, 2007-28E



Source: BofA Global Research estimates, TEJ, company data. ROE refers to average ROE

BofA GLOBAL RESEARCH

Exhibit 49: We expect Accton's GM to sustain at 19.3-20.1% in 2026-28E with expanding operating/pretax/net margins
Accton's margin trend, 2007-28E



Source: BofA Global Research estimates, TEJ, company data

BofA GLOBAL RESEARCH

Cashflow analysis: negative OCF in 1H26 for higher working capital; higher capex for global expansion

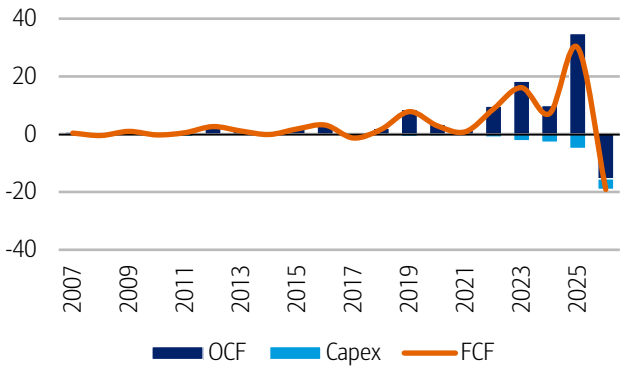
Accton has generated positive FCF in most of the years since 2007 (Exhibit 50), supported by generally positive OCF and relatively disciplined capex. It had FCF outflow of NT$19.1bn in 1H26, with OCF outflow of NT$15.5bn (for higher working capital) and high capex of NT$3.7bn (for ongoing capacity expansion). We believe the higher working capital in 1H26 was for strategic raw material procurement/preparation amid supply shortage and cost inflation. Historically in 2007-25, its FCF ranged from outflow of NT$1.3bn to inflow of NT$30.0bn, with OCF inflow except 2008 and 2017.

In 2026-28, we expect Accton to generate FCF inflow of NT$38.6-89.7bn, supported by strong profitability, which more than offsets the higher working capital and capex required for global capacity expansion.

Looking at capex, Accton's annual capex has accelerated from 2023, reaching NT$2.2-4.9bn during 2023-2025, vs. NT$112-952mn in 2007-2022 (Exhibit 51). The increase primarily reflects the company's multi-year global capacity expansion plans and R&D efforts. We forecast capex to remain at a historically high NT$6.0-7.0bn annually in 2026-28E, mainly supporting new capacity in Vietnam, Singapore, Malaysia, and Dallas (US), as well as continued expansion/renewals for the existing facilities.

As for financing activities, historically Accton has undertaken limited external financing with only CB issuances. No capital raising or capital reduction was conducted. As of end-2025, no corporate bond was outstanding. We thus assume no capital-raising plan in 2026-28. We believe bank borrowing will be the key method for financing needs.

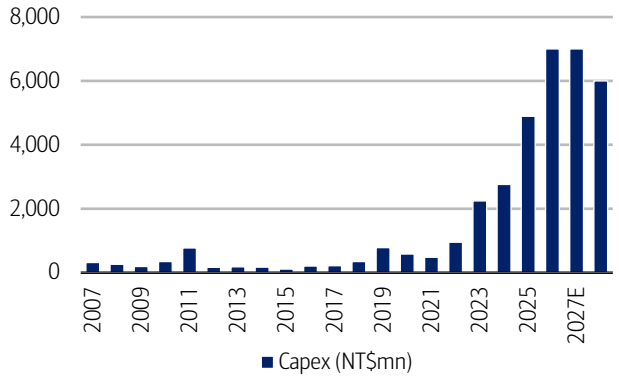
Exhibit 50: Accton has generated positive FCF in most of the years since 2007; it had FCF outflow of NT$19.1bn in 1H26, vs. outflow of NT$1.3bn to inflow of NT$30.0bn in 2007-25
Accton's OCF, capex, and FCF trend (in NT$bn) in 2007-1H26



Source: BofA Global Research, TEJ, company data. Capex refers to purchase of fixed assets

BofA GLOBAL RESEARCH

Exhibit 51: We project Accton's annual capex at NT$6.0-7.0bn in 2026-28E, vs. NT$0.1-4.9bn in 2007-25
Accton's capex trend (in NT$mn) in 2007-28E



Source: BofA Global Research estimates, company data. Capex refers to purchase of fixed assets

BofA GLOBAL RESEARCH

Ongoing capacity expansion in Taiwan and overseas

Accton currently has 4 sites in Taiwan, 3 in US, 1 in Vietnam and 1 in Netherlands (Exhibit 51). The company continues to expand its global manufacturing network to support local demand. Based on our checks, five sites are under expansion now.

In Taiwan, production at Taoyuan facility started ramp-up in 3Q26. Hsinchu site is being converted into a new production line. In August 2026, Accton announced to rent lands in Nanliao (南寮) in Hsinchu with around 30,865.90 pings (or 102,036 square meters). We believe its Taiwan facilities will be more dedicated to high-end AI products.

Overseas, Accton also announced to rent lands and buildings in Vietnam and Singapore. It plans to add the third manufacturing facility in Vietnam, to ramp its Dallas (US) site for rack-level business, and to commence production at new facilities in Singapore and Malaysia. We believe Malaysia will be a key hub for cooling products from 2027, vs. optical/CPO in Singapore.

Exhibit 52: Accton is expanding capacity across Taiwan, Asia, and the US
Accton's global footprint and capacity expansion plan

Location	Site	Main function	Expansion plan
Taiwan	Zhubei	Manufacturing, tech support, RMA, warehouse	
	Zhunan	Manufacturing	
	Hsinchu	R&D, tech support, RMA, warehouse	Warehouse space to be converted for a new production line, starting production in 2027
	Taoyuan	R&D, manufacturing	Small-volume production has commenced; production started ramp-up in 3Q26
China	Shenzhen (JoyTech)	Manufacturing, RMA	
Vietnam	Vinh Phuc	Manufacturing	A third facility is planned to support additional high-end product capacity
Singapore	Singapore	Primarily R&D labs	Expansion planned for 2026-27 to support optical/CPO business
Malaysia	Malaysia	Primarily manufacturing	Expansion planned for 2026-27 to support liquid cooling business
US	San Jose, California	Tech support, R&D, sales	
	Sunnyvale, California	R&D, tech support	
	Irvine, California	Manufacturing, R&D, sales, RMA, warehouse	
	Dallas, Texas	Primarily manufacturing, rack-level integration	Equipment installation expected by end-2026, followed by mass production in 2027
Netherlands	Eindhoven	RMA, warehouse	

Source: BofA Global Research estimates, company data

BofA GLOBAL RESEARCH

Exhibit 53: We expect Accton to stay in a net cash position with stronger cash-generation capability in 2026-28E
Accton's balance sheet, 2018-28E; we assume no capital-raising plan in 2026-28E

Balance sheet (NT$mn)	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Cash & deposits	4,296	6,222	4,895	6,258	8,695	14,070	18,116	27,891	58,080	98,920	160,965
A/R & N/R	7,986	6,971	8,848	10,316	13,139	11,119	20,763	37,275	62,199	83,772	115,960
Inventory	7,814	8,717	8,179	13,246	12,788	13,551	19,371	30,202	49,657	66,664	91,616
Other current assets	778	5,139	6,519	1,963	7,260	11,251	12,389	25,700	25,700	25,700	25,700
Total current assets	20,874	27,049	28,440	31,782	41,883	49,991	70,639	121,067	195,636	275,056	394,241
L-T investments	17	14	13	12	10	8	71	1,012	1,012	1,012	1,012
Tangible fixed assets	967	1,324	1,502	1,487	1,804	3,181	5,445	9,228	13,599	16,726	17,961
Goodwill & intangible assets	36	55	76	109	168	178	1,016	771	571	371	171
Total other assets	485	1,078	1,091	1,631	2,195	3,217	9,295	11,153	11,153	11,153	11,153
Total assets	**22,378**	**29,521**	**31,123**	**35,021**	**46,061**	**56,576**	**86,467**	**143,231**	**221,971**	**304,317**	**424,538**
Short term debt	0	0	0	0	0	133	175	333	492	492	492
Payables	8,751	9,935	8,639	11,567	12,478	13,681	25,955	58,486	96,160	129,094	177,413
Current portion of long-term debt	0	0	0	175	926	301	302	159	0	0	0
Other current liabilities	3,318	5,302	6,192	5,901	8,164	14,964	20,489	23,206	23,206	23,206	23,206
Total current liabilities	12,068	15,238	14,831	17,643	21,568	29,080	46,920	82,184	119,858	152,792	201,111
Long term debt	0	921	1,163	1,003	2,605	538	281	84	84	84	84
Other liabilities	31	560	645	822	1,460	1,771	2,831	3,328	3,328	3,328	3,328
Total other liabilities	31	1,480	1,808	1,825	4,065	2,308	3,112	3,412	3,412	3,412	3,412
Total liabilities	**12,100**	**16,718**	**16,640**	**19,468**	**25,633**	**31,388**	**50,032**	**85,596**	**123,270**	**156,204**	**204,523**
Share capital	5,576	5,581	5,595	5,599	5,601	5,604	5,611	5,611	5,611	5,611	5,611
Reserves	1,848	2,154	2,668	3,191	3,676	4,512	5,428	6,781	9,419	14,364	21,283
Retained earnings	3,206	5,655	6,742	7,308	11,664	15,562	22,443	41,330	79,758	124,225	189,208
Treasury stock	-51	-51	-51	-51	-51	-51	-51	-51	-51	-51	-51
Minority equity	7	3	3	0	0	0	108	96	96	96	96
Capital adjustment	-307	-538	-473	-495	-462	-438	2,896	3,868	3,868	3,868	3,868
Total equity	**10,279**	**12,803**	**14,483**	**15,553**	**20,428**	**25,188**	**36,435**	**57,635**	**98,700**	**148,113**	**220,015**
Total liabilities and equity	**22,378**	**29,521**	**31,123**	**35,021**	**46,061**	**56,576**	**86,467**	**143,231**	**221,971**	**304,317**	**424,538**
Net cash (debt)	**4,296**	**5,301**	**3,731**	**5,079**	**5,164**	**13,099**	**17,359**	**27,314**	**57,504**	**98,344**	**160,389**

Source: BofA Global Research estimates, company data. Net cash = Cash & cash equivalent - short-term borrowing - long-term liabilities

BofA GLOBAL RESEARCH

Exhibit 54: Accton's cash increase in 2026-28E mainly reflects increasing profits despite ongoing capex investment and dividend payout
Accton's cash flow statement, 2018-28E; we also assume no capital-raising plan in 2026-28E

Cash flow statement (NT$mn)	2018	2019	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
Net profit	2,952	4,950	5,048	4,705	8,166	8,920	12,013	26,374	49,449	69,192	99,579
Depreciation	266	508	636	722	805	878	1,071	1,551	2,629	3,874	4,764
Amortization	27	31	39	44	62	75	91	176	200	200	200
Total gross cash flow	3,246	5,490	5,723	5,472	9,032	9,874	13,176	28,101	52,277	73,266	104,543
Change in total working capital	-1,476	1,297	-2,635	-3,607	-1,455	2,461	-3,190	5,187	-6,705	-5,646	-8,821
Other adjustments	285	1,825	384	-496	2,146	6,036	-43	1,564	0	0	0
Total operating cash flow	**2,055**	**8,611**	**3,472**	**1,368**	**9,723**	**18,371**	**9,942**	**34,852**	**45,573**	**67,620**	**95,722**
Capex	-346	-779	-581	-480	-952	-2,245	-2,762	-4,894	-7,000	-7,000	-6,000
Change in other assets	848	-4,284	-917	4,380	-5,123	-3,421	3,121	-14,805	0	0	0
ST & LT investment	-27	0	0	0	-30	0	-98	550	0	0	0
Total investment cash flow	**474**	**-5,063**	**-1,498**	**3,900**	**-6,104**	**-5,666**	**262**	**-19,149**	**-7,000**	**-7,000**	**-6,000**
Change in gross debt	0	921	243	-160	1,602	-1,934	-216	-38	0	0	0
Others	40	-83	-164	-160	462	-937	-418	-477	0	0	0
Dividend	-2,291	-2,222	-3,449	-3,622	-3,346	-4,185	-5,582	-6,148	-8,384	-19,780	-27,677
Total financing cash flow	**-2,250**	**-1,384**	**-3,370**	**-3,943**	**-1,282**	**-7,056**	**-6,215**	**-6,663**	**-8,384**	**-19,780**	**-27,677**
Exchange influence	73	-238	68	38	101	-275	635	156	0	0	0
Change in cash	**352**	**1,926**	**-1,327**	**1,363**	**2,438**	**5,375**	**4,623**	**9,197**	**30,189**	**40,840**	**62,045**
FCF	**1,709**	**7,832**	**2,891**	**888**	**8,772**	**16,126**	**7,181**	**29,958**	**38,573**	**60,620**	**89,722**

Source: BofA Global Research estimates, company data. Free cash flow (FCF) = operating cash flow (OCF) - capital expenditure (capex)

BofA GLOBAL RESEARCH

Company Description

Accton is a leading Taiwan-based networking/switch white-box ODM/JDM/solution provider. Its product offering includes datacenter/server switch, enterprise switch, metro Ethernet switch, smart network interface card, router, 5G gateway, edge-computing server, and wireless networking equipment. Accton is also expanding into high-speed optics, liquid cooling, and rack integration for next-generation AI infrastructure.

Accton was founded on 9 February 1988 and went publicly listed on Taiwan Stock Exchange on 15 November 1995. The company has operations/facilities across Taiwan, US, Southeast Asia, and Europe, with ongoing capacity expansion, and its headquarter is in Zhubei, Taiwan.

Management team

Kuo-Hsiu Huang (黃國修), Chairman

Kuo-Hsiu Huang has served as the Chairman of Accton Technology since 2024 and the Director since 2014. Currently, he also holds key positions in several subsidiaries, including the Chairman of Accton Tech (China) Co., Ltd., E-Direct Corporation, and MuXi Technology (Nanjing) Co., Ltd., as well as the Director of Accton Century Holding (BVI) Co., Ltd., Accton Asia Investments Corp., Accton Investment Corporation, and Edgecore Networks Singapore Pte. Ltd. In addition, he serves as the Chairman of Golden Tengis Co., Ltd. and Kuan Xin Investment Co., Ltd. Previously, he served as an Assistant Manager of Accton Technology. Mr. Huang holds an MBA from Peking University.

Chiu-Hsia Wei (韋秋霞), Vice Chairman

Chiu-Hsia Wei has served as the Vice Chairman of Accton Technology since 2024. Currently, she is also the Director of two subsidiaries, Accton Logistics Corporation and Accton Global Inc. In addition, she serves as the Legal Representative of Kuan Xin Investment Co., Ltd. Previously, she served as the Senior Associate Vice President of Procurement at Accton Technology. Ms. Wei graduated from Shih Chien Junior College.

Jackal (Hsun-Te) Li (李訓德), President

Jackal Li has served as the President of Accton Technology since 2025. Currently, he also serves as the Director of several subsidiaries, including Accton Logistics Corporation, Accton Global, Inc., Vietnam Accton Technology Co., Ltd., and Accton Manufacturing and Service, Inc. Mr. Li holds a Bachelor's Degree in Business Management from Chung Hua University.

Fanny (Fang-I) Chen (陳芳儀), Chief Finance Officer

Fanny Chen has served as the CFO of Accton Technology since 2023. Currently, she also serves as the Supervisor of several subsidiaries, including Edgecore Networks Corporation, E-Direct Corporation, and GoldiLink Technology Corp. Previously, she served as the Accounting Manager and Assistant Manager of Accton Technology. Ms. Chen holds a Bachelor of Science in Finance and Taxation from National Chengchi University.

Edward (Hsing-Hsuan) Lin (林信玄), Corporate Governance Manager

Edward Lin has served as the Corporate Governance Manager of Accton Technology since 2021. Currently, he also serves as the Chief Legal Officer and the Spokesman of the company. Mr. Lin holds a Master of Science in Technology Law from National Yang Ming Chiao Tung University.

Top-10 shareholders and QFII holdings

As of 9 Oct 2026, Accton's top-10 shareholders held a combined 34.21% of its outstanding shares (Exhibit 55). The largest shareholder is Chin Chieh Min Co., Ltd. (金捷敏股份有限公司), chaired by Accton Chairman Kuo-Hsiu Huang, with an 8.04% stake. Rest of the top-10 shareholders are all institutional investors, including Vanguard Group (5.01%), Blackrock (4.52%), New Labor Pension Fund (4.19%), JPMorgan Chase (2.53%), Sheng Yang Investment Consulting (2.31%), Uni-President Asset Management (2.30%), Yuanta Financial Holding (2.11%), Fubon Financial Holding (1.76%), and Norges Bank (1.45%).

Currently, Accton's QFII holding is 56.9% vs. TAIEX's 48.2%. Since IPO, its historically highest QFII holding was 71.8% in Feb-2021 and the lowest at 0.0% in November 1995, vs. TAIEX's lowest/highest level of 50.3%/8.9% in the same period (Exhibit 56).

Exhibit 55: Top-10 shareholders own total 34.2% of Accton's outstanding shares

Accton's top-10 shareholders list

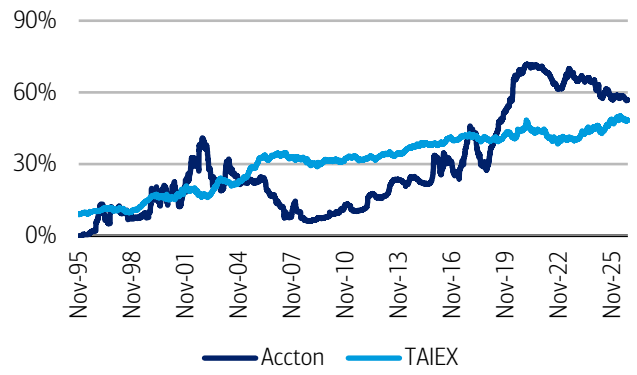
Shareholder name	Holding %
Chin Chieh Min Co Ltd	8.04%
Vanguard Group Inc	5.01%
Blackrock Inc	4.52%
NEW LABOR PENSION FUND	4.19%
JPMorgan Chase & Co	2.53%
Sheng Yang Investment Consulting Co Ltd	2.31%
Uni-President Asset Management Corp/Taiwan	2.30%
Yuanta Financial Holding Co Ltd	2.11%
Fubon Financial Holding Co Ltd	1.76%
Norges Bank	1.45%
Subtotal	**34.22%**

Source: BofA Global Research, Bloomberg. Data as of 2026/10/09

BofA GLOBAL RESEARCH

Exhibit 56: Accton's current/lowest/highest QFII holdings since IPO is 56.9%/0.0%/71.81%, vs. TAIEX's 48.2%/8.9%/50.3%

Accton's QFII holding vs. TAIEX since Accton's IPO in Nov 1995



Source: BofA Global Research, TEJ

BofA GLOBAL RESEARCH

Appendix

What is leaf-spine architecture?

Leaf-spine is a modern data center network architecture designed to provide high bandwidth, low latency, and predictable performance, especially for workloads that generate heavy server-to-server (east-west) traffic such as AI training, cloud services, and distributed computing. In a leaf-spine design, the network is simplified into two layers: the leaf layer and the spine layer. Every leaf switch connects to every spine switch, creating a full-mesh topology between the two layers (Exhibit 57).

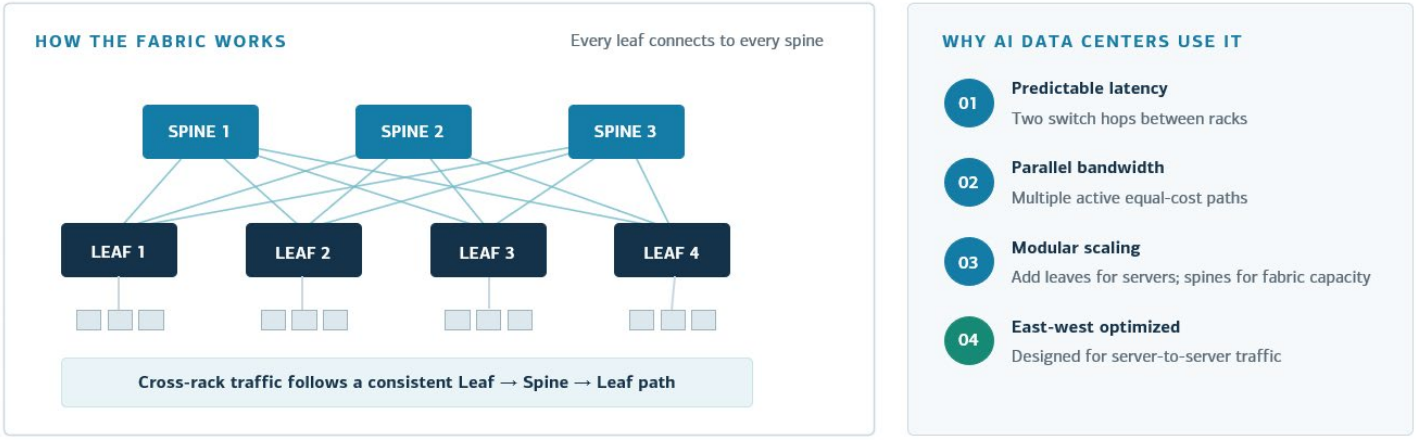
The leaf switches sit at the edge of the network and connect directly to servers, storage systems, and other endpoints. The spine switches act as the backbone, interconnecting all leaf switches but never connecting directly to endpoints. As a result, traffic between any two servers attached to different leaf switches always traverses the same number of hops: Leaf → Spine → Leaf. This predictable path helps maintain consistent latency and simplifies network scaling.

Leaf-spine architecture became popular because traditional three-tier networks (access, distribution, core) were optimized for north-south traffic between users and servers. Modern datacenters, however, generate significant east-west traffic, where servers communicate heavily with one another. AI clusters are a prime example, as GPUs frequently exchange data during model training. A leaf-spine fabric minimizes bottlenecks by enabling multiple equal-cost paths and load-balancing traffic across all available links.

Another major advantage is scalability. As capacity requirements grow, operators can add more leaf switches to connect additional servers, or add more spine switches to increase backbone bandwidth. As every leaf connects to every spine, the architecture scales in a modular and predictable way without requiring a significant redesign. The flexibility serves as one of key reasons behind hyperscale cloud providers and AI infrastructure operators' wide adoption in leaf-spine topologies.

Leaf-spine is particularly important because it drives demand for high-speed Ethernet switching (100G, 400G, 800G and beyond). AI datacenters often dedicate large numbers of spine and leaf switches to provide the massive bandwidth required by GPU clusters, making the leaf-spine fabric a foundational building block of modern AI infrastructure.

Exhibit 57: Leaf-spine is a modern data center network architecture designed to provide high bandwidth, low latency, and predictable performance, especially for workloads that generate heavy server-to-server (east-west) traffic such as AI training, cloud services, and distributed computing
Illustration of leaf-spine architecture and advantages of using leaf-spine architecture in AI



Source: BofA Global Research

BofA GLOBAL RESEARCH

Abbreviation glossary

Exhibit 58: Glossary
Abbreviations used throughout the report

Abbreviation		Abbreviation	
A/P	Accounts Payable	ML	Machine Learning
A/R	Accounts Receivable	N/R	Notes Receivable
AI	Artificial Intelligence	NIC	Network Interface Card
ASIC	Application-Specific Integrated Circuit	NRE	Non-Recurring Engineering
ASP	Average Selling Price	NS	Nanoseconds
ATS	Advanced Technology Solutions	NT$ / NTD	New Taiwan Dollar
AWS	Amazon Web Services	OCF	Operating Cashflow
BS	Balance Sheet	OCS	Optical Circuit Switch
BVI	British Virgin Islands	ODM	Original Design Manufacturer
CAGR	Compound Annual Growth Rate	OEM	Original Equipment Manufacturer
Capex	Capital Expenditure	O-E-O	Optical-Electrical-Optical
CB	Convertible Bond	OPM	Operating Profit Margin
CCC	Cash Conversion Cycle	OWS	Optical Wavelength Switch
CCS	Connectivity & Cloud Solutions	P/B	Price to Book
CF	Cash Flow	P/E	Price to Earnings
COGS	Cost of Goods Sold	PCB	Printed Circuit Board
CPO	Co-packaged Optics	PEG	P/E growth
CSP	Cloud Service Provider	PO	Price Objective
DCI	Datacenter Interconnection	QFII	Qualified Foreign Institutional Investor
Diff.	Difference	R&D	Research and Development
EBITDA	Earnings Before Interest, Taxes, Depreciation, and Amortization	RHS	Right Hand Side

Exhibit 58: Glossary
Abbreviations used throughout the report

Abbreviation		Abbreviation	
EMS	Electronics Manufacturing Services	RMA	Return Merchandise Authorization
EPS	Earnings Per Share	ROE	Return on Equity
EV	Enterprise Value	SD	Standard Deviation
FCF	Free Cashflow	SerDes	Serializer/Deserializer
FX	Foreign Exchange	TAIEX	Taiwan Stock Exchange Capitalization Weighted Stock Index
GM	Gross Margin	TAM	Total Addressable Market
HPC	High Performance Computing	Tbps	Terabits per second
HPS	Hardware Platform Solutions	TEJ	Taiwan Economic Journal
IDC	International Data Corporation	Telco	Telecommunication Company
IOWN	Innovative Optical and Wireless Network	ToR	Top-of-Rack
IPO	Initial Public Offering	TPU	Tensor Processing Unit
JDM	Joint Design and Manufacturing	U	Rack Unit
LC	Local Currency	UALink	Ultra Accelerator Link
LCoS	Liquid Crystal on Silicon	US	United States
MBA	Master of Business Administration	US$ / USD	US Dollar
MEMS	Micro-electromechanical Systems	WDM	Wavelength-division Multiplexing

Source: BofA Global Research

BofA GLOBAL RESEARCH

Exhibit 59: Stock mentioned
Market information for stocks mentioned in the report

BofA Ticker	Bloomberg Ticker	Company Name	Price (LC)	Rating
XTWOF	2345 TT	Accton	2,130.00	C-1-7
CLS	CLS US	Celestica	362.16	C-1-9

Source: BofA Global Research. LC: Local Currency. Market data as of 2026/10/12 (Accton) or 10/09 (Celestica)

BofA GLOBAL RESEARCH

Price objective basis & risk

Accton (XTWOF)
Our PO of NT$3,200 is based on 18x 2028E EPS. We use 18x P/E multiple (around 1SD above historical P/E average since 2008) in view of strong AI uptrend, leading position in AI networking/switch white-box, expanding OPM with sustainable/high GM, and double-digit sales/EPS CAGR in 2026-28E.

Upside risks to our PO are: (1) faster 800G/1.6T deployment by key customers, (2) easing component supply shortage, (3) new customer/project win, (4) accelerating optical/cooling business development, (5) faster capacity expansion progress, (6) favorable FX movement (mainly USD/NTD).

Downside risks to our PO are: (1) AI product/project delay and slower 800G/1.6T adoption, (2) key customers' demand push out due to component supply shortage, (3) higher GM pressure due to component cost inflation, (4) pricing pressure and/or market/order share loss due to peer competition, (5) slower capacity expansion, (6) unfavorable FX movement (mainly USD/NTD) or macro uncertainty.

Analyst Certification

I, Doris Kao, hereby certify that the views expressed in this research report accurately reflect my personal views about the subject securities and issuers. I also certify that no part of my compensation was, is, or will be, directly or indirectly, related to the specific recommendations or view expressed in this research report.

Special Disclosures

Information on securities which are listed on the exchanges where ML Securities (Taiwan) Limited is not permitted to trade or solicit trades for clients is for informational purposes only and is not a recommendation or a solicitation to trade such securities. ML Securities (Taiwan) Limited will not execute transactions for nor accept orders from clients to trade such securities. Foreign investment in Taiwan securities is regulated and restricted. Currently, foreign investment in Taiwan securities is permitted by investment through: (1) global depository receipts, (2) convertible bonds, (3) mutual funds issued offshore of Taiwan, and (4) a special foreign institutional investors (FINIs) and foreign individual investors (FIDIs) program supervised by the Taiwan SFB whereunder FINIs/FIDIs may apply for investment ID to invest in Taiwan securities by registration with Taiwan Stock Exchange. FINIs will additionally need consent from the foreign exchange authority, ie, the Central Bank of China. In addition to the limitations above, various industry-specific percentage-based limitations on foreign ownership of Taiwan companies (and in some cases prohibitions) may apply. Investments are subject to exchange rate and currency conversion restrictions and risks. Dividends and interest earned by foreign investors' Taiwan securities/instruments are generally subject to a 20% withholding tax. Ordinary shares are not available to ML private client accounts in the U.S.

This report is distributed in Taiwan by Merrill Lynch Securities (Taiwan) Limited, which is regulated by the Taiwan SFB.

APR - Technology Hardware Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
BUY				
	AAC Technologies	AACAF	2018 HK	Katherine Zhu
	Accton	XTWOF	2345 TT	Doris Kao
	Advantech	ADTEF	2395 TT	Robert Cheng

APR - Technology Hardware Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
	Asia Vital Components	AVMPF	3017 TT	Doris Kao
	Asustek	AKCPF	2357 TT	Robert Cheng
	BizLink	BIZLF	3665 TT	Doris Kao
	Chenbro	CEOMF	8210 TT	Doris Kao
	Cowell	CWLLF	1415 HK	Katherine Zhu
	Crystal-Optech	XMKAF	002273 CH	Katherine Zhu
	Delta Electronics (Thailand)	XLETF	DELTA TB	Suppapong Iemkongaek, CFA, CMT
	Delta Electronics Inc.	DLTEF	2308 TT	Robert Cheng
	Duksan Neolux	XDSXF	213420 KS	Simon Woo, CFA
	Dynamic Holding	XLHCF	3715 TT	Mike Yang
	Elite Material	ETMCF	2383 TT	Mike Yang
	Foxconn Industrial Internet	XDAZF	601138 CH	Robert Cheng
	Goertek	XGKCF	002241 CH	Katherine Zhu
	Gold Circuit	GOCCF	2368 TT	Doris Kao
	Hana Microelectronics	XHNNF	HANA TB	Suppapong Iemkongaek, CFA, CMT
	Hon Hai Precision Industry	HNHAF	2317 TT	Robert Cheng
	Huaqin	XHUAF	603296 CH	Katherine Zhu
	Huaqin	XPEZF	3296 HK	Katherine Zhu
	ISU Petasys	IPTYF	007660 KS	Matt Shin
	Jentech	XVFSF	3653 TT	Doris Kao
	KCE Electronics	XKKRF	KCE TB	Suppapong Iemkongaek, CFA, CMT
	King Slide	KGSLF	2059 TT	Doris Kao
	Kingboard Laminates	KGBLF	1888 HK	Katherine Zhu
	Kinsus	KNSUF	3189 TT	Mike Yang
	Largan Precision	LGANF	3008 TT	Robert Cheng
	Lenovo Group	LNVGF	992 HK	Robert Cheng
	Lenovo Group	LNVGY	LNVGY US	Robert Cheng
	Lens Tech	XLTVF	6613 HK	Katherine Zhu
	LG CNS	XLGLF	064400 KS	Simon Woo, CFA
	LG Electronics	LGEAF	066570 KS	Simon Woo, CFA
	LG Innotek	XLGQF	011070 KS	Simon Woo, CFA
	Lotes	ZYZTF	3533 TT	Doris Kao
	Luxshare	XNJQF	002475 CH	Katherine Zhu
	Netweb Technologies	XNTZF	NETWEB IN	Jatin Kalra, CFA
	NYPCB	NANYF	8046 TT	Mike Yang
	Quanta Computer	QUCPF	2382 TT	Robert Cheng
	Samsung Electro-Mechanics	SMSGF	009150 KS	Simon Woo, CFA
	Samsung SDI	SSDIF	006400 KS	Simon Woo, CFA
	Samsung SDS	XWPBF	018260 KS	Simon Woo, CFA
	Shennan Circuits	XAMKF	002916 CH	Katherine Zhu
	Taiwan Union Technology Corporation	TWUNF	6274 TT	Mike Yang
	Unimicron	UMCRF	3037 TT	Mike Yang
	Universal Display	OLED	OLED US	Simon Woo, CFA
	Victory Giant	XENEF	300476 CH	Katherine Zhu
	Wistron	WICOF	3231 TT	Doris Kao
	Wiwynn	XWIQF	6669 TT	Robert Cheng
	Xiaomi Corporation	XIACF	1810 HK	Robert Cheng
	Yageo	YGEQF	2327 TT	Robert Cheng
	Zhen Ding Tech	XFAGF	4958 TT	Doris Kao
	Zhongji Innolight	XZICF	300308 CH	Katherine Zhu
NEUTRAL				
	Auras Technology	AUTCF	3324 TT	Doris Kao
	BYD Electronic	BYDIF	285 HK	Katherine Zhu
	DS Precision	XBASF	002384 CH	Katherine Zhu
	E Ink Holdings	PVWIF	8069 TT	Doris Kao
	Lite-On Tech	LOTZF	2301 TT	Doris Kao
	Sunny Optical	SNPTF	2382 HK	Katherine Zhu
UNDERPERFORM				
	Acer	ASIYF	2353 TT	Robert Cheng
	Catcher Tech	CHERF	2474 TT	Robert Cheng
	Compal Electron	XLCPF	2324 TT	Doris Kao
	Everwin Precision	XSHZF	300115 CH	Katherine Zhu
	Genius Electronic Optical	GNSEF	3406 TT	Robert Cheng
	Lens Tech	XLTGF	300433 CH	Katherine Zhu

APR - Technology Hardware Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
	LG Display	LPHLF	034220 KS	Simon Woo, CFA
	LG Display-ADR	LPL	LPL US	Simon Woo, CFA
	Pegatron	PGTRF	4938 TT	Robert Cheng
	Sanan Opto	XEQYF	600703 CH	Katherine Zhu
	Seoul Semiconductor	SLSOF	046890 KS	Simon Woo, CFA
	Shengyi	DGUNF	600183 CH	Mike Yang
	Sunway Communications	XWSVF	300136 CH	Katherine Zhu
	Transsion	XCINF	688036 CH	Robert Cheng
	ZTE Corporation	SHZZF	000063 CH	Katherine Zhu
	ZTE Corporation	ZTCOF	763 HK	Katherine Zhu

iQmethod℠ Measures Definitions

Business Performance	Numerator	Denominator
Return On Capital Employed	NOPAT = (EBIT + Interest Income) × (1 − Tax Rate) + Goodwill Amortization	Total Assets − Current Liabilities + ST Debt + Accumulated Goodwill Amortization
Return On Equity	Net Income	Shareholders' Equity
Operating Margin	Operating Profit	Sales
Earnings Growth	Expected 5 Year CAGR From Latest Actual	N/A
Free Cash Flow	Cash Flow From Operations − Total Capex	N/A

Quality of Earnings	Numerator	Denominator
Cash Realization Ratio	Cash Flow From Operations	Net Income
Asset Replacement Ratio	Capex	Depreciation
Tax Rate	Tax Charge	Pre-Tax Income
Net Debt-To-Equity Ratio	Net Debt = Total Debt − Cash & Equivalents	Total Equity
Interest Cover	EBIT	Interest Expense

Valuation Toolkit	Numerator	Denominator
Price / Earnings Ratio	Current Share Price	Diluted Earnings Per Share (Basis As Specified)
Price / Book Value	Current Share Price	Shareholders' Equity / Current Basic Shares
Dividend Yield	Annualised Declared Cash Dividend	Current Share Price
Free Cash Flow Yield	Cash Flow From Operations − Total Capex	Market Cap = Current Share Price × Current Basic Shares
Enterprise Value / Sales	EV = Current Share Price × Current Shares + Minority Equity + Net Debt + Other LT Liabilities	Sales
EV / EBITDA	Enterprise Value	Basic EBIT + Depreciation + Amortization

iQmethod℠ is the set of BofA Global Research standard measures that serve to maintain global consistency under three broad headings: Business Performance, Quality of Earnings, and validations. The key features of iQmethod are: A consistently structured, detailed, and transparent methodology. Guidelines to maximize the effectiveness of the comparative valuation process, and to identify some common pitfalls.

iQdatabase® is our real-time global research database that is sourced directly from our equity analysts' earnings models and includes forecasted as well as historical data for income statements, balance sheets, and cash flow statements for companies covered by BofA Global Research.

iQprofile℠, *iQmethod℠* are service marks of Bank of America Corporation. *iQdatabase®* is a registered service mark of Bank of America Corporation.

Disclosures

Important Disclosures

Equity Investment Rating Distribution: Technology Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	255	62.04%	Buy	157	61.57%
Hold	82	19.95%	Hold	46	56.10%
Sell	74	18.00%	Sell	31	41.89%

Equity Investment Rating Distribution: Global Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	2057	57.03%	Buy	1386	67.38%
Hold	806	22.35%	Hold	563	69.85%
Sell	744	20.63%	Sell	446	59.95%

[R1] Issuers that were investment banking clients of BofA Securities or one of its affiliates within the past 12 months. For purposes of this Investment Rating Distribution, the coverage universe includes only stocks. A stock rated Neutral is included as a Hold, and a stock rated Underperform is included as a Sell.

FUNDAMENTAL EQUITY OPINION KEY: Opinions include a Volatility Risk Rating, an Investment Rating and an Income Rating. VOLATILITY RISK RATINGS, indicators of potential price fluctuation, are: A - Low, B - Medium and C - High. INVESTMENT RATINGS reflect the analyst's assessment of both a stock's absolute total return potential as well as its attractiveness for investment relative to other stocks within its Coverage Cluster (defined below). Our investment ratings are: 1 - Buy stocks are expected to have a total return of at least 10% and are the most attractive stocks in the coverage cluster; 2 - Neutral stocks are expected to remain flat or increase in value and are less attractive than Buy rated stocks and 3 - Underperform stocks are the least attractive stocks in a coverage cluster. An investment rating of 6 (No Rating) indicates that a stock is no longer trading on the basis of fundamentals. Analysts assign investment ratings considering, among other things, the 0-12 month total return expectation for a stock and the firm's guidelines for ratings dispersions (shown in the table below). The current price objective for a stock should be referenced to better understand the total return expectation at any given time. The price objective reflects the analyst's view of the potential price appreciation (depreciation).

Investment rating	Total return expectation (within 12-month period of date of initial rating)	Ratings dispersion guidelines for coverage cluster[R2]
Buy	≥ 10%	≤ 70%
Neutral	≥ 0%	≤ 30%
Underperform	N/A	≥ 20%

[R2]Ratings dispersions may vary from time to time where BofA Global Research believes it better reflects the investment prospects of stocks in a Coverage Cluster.

INCOME RATINGS, indicators of potential cash dividends, are: 7 - same/higher (dividend considered to be secure), 8 - same/lower (dividend not considered to be secure) and 9 - pays no cash dividend. *Coverage Cluster* is comprised of stocks covered by a single analyst or two or more analysts sharing a common industry, sector, region or other classification(s). A stock's coverage cluster is included in the most recent BofA Global Research report referencing the stock.

In the US, retail sales and/or distribution of this report may be made only in states where these securities are exempt from registration or have been qualified for sale: Accton.
The country in which this issuer is organized has certain laws or regulations that limit or restrict ownership of the issuer's shares by nationals of other countries: Accton.
BofA Global Research personnel (including the analyst(s) responsible for this report) receive compensation based upon, among other factors, the overall profitability of Bank of America Corporation, including profits derived from investment banking. The analyst(s) responsible for this report may also receive compensation based upon, among other factors, the overall profitability of the Bank's sales and trading businesses relating to the class of securities or financial instruments for which such analyst is responsible.

Other Important Disclosures

From time to time research analysts conduct site visits of covered issuers. BofA Global Research policies prohibit research analysts from accepting payment or reimbursement for travel expenses from the issuer for such visits.
Prices are indicative and for information purposes only. Except as otherwise stated in the report, for any recommendation in relation to an equity security, the price referenced is the publicly traded price of the security as of close of business on the day prior to the date of the report or, if the report is published during intraday trading, the price referenced is indicative of the traded price as of the date and time of the report and in relation to a debt security (including equity preferred and CDS), prices are indicative as of the date and time of the report and are from various sources including BofA Securities trading desks.
The date and time of completion of the production of any recommendation in this report shall be the date and time of dissemination of this report as recorded in the report timestamp.

Recipients who are not institutional investors or market professionals should seek the advice of their independent financial advisor before considering information in this report in connection with any investment decision, or for a necessary explanation of its contents.
Officers of BofAS or one or more of its affiliates (other than research analysts) may have a financial interest in securities of the issuer(s) or in related investments.
Refer to BofA Global Research policies relating to conflicts of interest.
"BofA Securities" includes BofA Securities, Inc. ("BofAS") and its affiliates. Investors should contact their BofA Securities representative or Merrill Global Wealth Management financial advisor if they have questions concerning this report or concerning the appropriateness of any investment idea described herein for such investor. "BofA Securities" is a global brand for BofA Global Research.
Information relating to Non-US affiliates of BofA Securities and Distribution of Affiliate Research Reports:
BofAS and/or Merrill Lynch, Pierce, Fenner & Smith Incorporated ("MLPF&S") may in the future distribute, information of the following non-US affiliates in the US (short name: legal name, regulator): Merrill Lynch (South Africa): Merrill Lynch South Africa (Pty) Ltd., regulated by the Financial Sector Conduct Authority; MLI (UK): Merrill Lynch International, regulated by the Financial Conduct Authority (FCA) and the Prudential Regulation Authority (PRA); BofASE (France): BofA Securities Europe SA is authorized by the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and regulated by the ACPR and the Autorité des Marchés Financiers (AMF). BofA Securities Europe SA ("BofASE") with registered address at 51, rue La Boétie, 75008 Paris is registered under no 842 602 690 RCS Paris. In accordance with the provisions of French Code Monétaire et Financier (Monetary and Financial Code), BofASE is an établissement de crédit et d'investissement (credit and investment institution) that is authorised and supervised by the European Central Bank and the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and regulated by the ACPR and the Autorité des Marchés Financiers. BofASE's share capital can be found at www.bofaml.com/BofASEdisclaimer; BofA Europe (Milan): Bank of America Europe Designated Activity Company, Milan Branch, regulated by the Bank of Italy, the European Central Bank (ECB) and the Central Bank of Ireland (CBI); BofA Europe (Frankfurt): Bank of America Europe Designated Activity Company, Frankfurt Branch regulated by BaFin, the ECB and the CBI; BofA Europe (Zurich): Bank of America Europe Designated Activity Company, Zurich Branch, regulated by the Swiss Financial Market Supervisory Authority FINMA, the ECB and CBI; BofA Europe (Madrid): Bank of America Europe Designated Activity Company, Sucursal en España, regulated by the Bank of Spain, the ECB and the

CBI; Merrill Lynch (Australia): Merrill Lynch Equities (Australia) Limited, regulated by the Australian Securities and Investments Commission; Merrill Lynch (Hong Kong): Merrill Lynch (Asia Pacific) Limited, regulated by the Hong Kong Securities and Futures Commission (HKSFC); Merrill Lynch (Singapore): Merrill Lynch (Singapore) Pte Ltd, regulated by the Monetary Authority of Singapore (MAS); Merrill Lynch (Canada): Merrill Lynch Canada Inc, regulated by the Canadian Investment Regulatory Organization; Merrill Lynch (Mexico): Merrill Lynch Mexico, SA de CV, Casa de Bolsa, regulated by the Comisión Nacional Bancaria y de Valores; BofAS Japan: BofA Securities Japan Co., Ltd., regulated by the Financial Services Agency; Merrill Lynch (Seoul): Merrill Lynch International, LLC Seoul Branch, regulated by the Financial Supervisory Service; Merrill Lynch (Taiwan): Merrill Lynch Securities (Taiwan) Ltd., regulated by the Securities and Futures Bureau; BofAS India: BofA Securities India Limited, regulated by the Securities and Exchange Board of India (SEBI); Merrill Lynch (Israel): Merrill Lynch Israel Limited, regulated by Israel Securities Authority; Merrill Lynch (DIFC): Merrill Lynch International (DIFC Branch), regulated by the Dubai Financial Services Authority (DFSA); Merrill Lynch (Brazil): Merrill Lynch S.A. Corretora de Títulos e Valores Mobiliários, regulated by Comissão de Valores Mobiliários; Merrill Lynch KSA Company: Merrill Lynch Kingdom of Saudi Arabia Company, regulated by the Capital Market Authority.
This information: has been approved for publication and is distributed in the United Kingdom (UK) to professional clients and eligible counterparties (as each is defined in the rules of the FCA and the PRA) by MLI (UK), which is authorized by the PRA and regulated by the FCA and the PRA - details about the extent of our regulation by the FCA and PRA are available from us on request; has been approved for publication and is distributed in the European Economic Area (EEA) by BofASE (France), which is authorized by the ACPR and regulated by the ACPR and the AMF; has been considered and distributed in Japan by BofAS Japan, a registered securities dealer under the Financial Instruments and Exchange Act in Japan, or its permitted affiliates; is issued and distributed in Hong Kong by Merrill Lynch (Hong Kong) which is regulated by HKSFC; is issued and distributed in Taiwan by Merrill Lynch (Taiwan); is issued and distributed in India by BofAS India; and is issued and distributed in Singapore to institutional investors and/or accredited investors (each as defined under the Financial Advisers Regulations) by Merrill Lynch (Singapore) (Company Registration No 198602883D). Merrill Lynch (Singapore) is regulated by MAS. Merrill Lynch Equities (Australia) Limited (ABN 65 006 276 795), AFS License 235132 (MLEA) distributes this information in Australia only to 'Wholesale' clients as defined by s.761G of the Corporations Act 2001. With the exception of Bank of America N.A., Australia Branch, neither MLEA nor any of its affiliates involved in preparing this information is an Authorised Deposit-Taking Institution under the Banking Act 1959 nor regulated by the Australian Prudential Regulation Authority. No approval is required for publication or distribution of this information in Brazil and its local distribution is by Merrill Lynch (Brazil) in accordance with applicable regulations. Merrill Lynch (DIFC) is authorized and regulated by the DFSA. Information prepared and issued by Merrill Lynch (DIFC) is done so in accordance with the requirements of the DFSA conduct of business rules. BofA Europe (Frankfurt) distributes this information in Germany and is regulated by BaFin, the ECB and the CBI. BofA Securities entities, including BofA Europe and BofASE (France), may outsource/delegate the marketing and/or provision of certain research services or aspects of research services to other branches or members of the BofA Securities group. You may be contacted by a different BofA Securities entity acting for and on behalf of your service provider where permitted by applicable law. This does not change your service provider. Please refer to the Electronic Communications Disclaimers for further information.
This information has been prepared and issued by BofAS and/or one or more of its non-US affiliates. The author(s) of this information may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so. BofAS and/or MLPF&S is the distributor of this information in the US and accepts full responsibility for information distributed to BofAS and/or MLPF&S clients in the US by its non-US affiliates. Any US person receiving this information and wishing to effect any transaction in any security discussed herein should do so through BofAS and/or MLPF&S and not such foreign affiliates. Hong Kong recipients of this information should contact Merrill Lynch (Asia Pacific) Limited in respect of any matters relating to dealing in securities or provision of specific advice on securities or any other matters arising from, or in connection with, this information. Singapore recipients of this information should contact Merrill Lynch (Singapore) Pte Ltd in respect of any matters arising from, or in connection with, this information. For clients that are not accredited investors, expert investors or institutional investors Merrill Lynch (Singapore) Pte Ltd accepts full responsibility for the contents of this information distributed to such clients in Singapore.

General Investment Related Disclosures:

Taiwan Readers: Neither the information nor any opinion expressed herein constitutes an offer or a solicitation of an offer to transact in any securities or other financial instrument. No part of this report may be used or reproduced or quoted in any manner whatsoever in Taiwan by the press or any other person without the express written consent of BofA Securities.
This document provides general information only, and has been prepared for, and is intended for general distribution to, BofA Securities clients. Neither the information nor any opinion expressed constitutes an offer or an invitation to make an offer, to buy or sell any securities or other financial instrument or any derivative related to such securities or instruments (e.g., options, futures, warrants, and contracts for differences). This document is not intended to provide personal investment advice and it does not take into account the specific investment objectives, financial situation and the particular needs of, and is not directed to, any specific person(s). This document and its content do not constitute, and should not be considered to constitute, investment advice for purposes of ERISA, the US tax code, the Investment Advisers Act or otherwise. Investors should seek financial advice regarding the appropriateness of investing in financial instruments and implementing investment strategies discussed or recommended in this document and should understand that statements regarding future prospects may not be realized. Any decision to purchase or subscribe for securities in any offering must be based solely on existing public information on such security or the information in the prospectus or other offering document issued in connection with such offering, and not on this document.
Securities and other financial instruments referred to herein, or recommended, offered or sold by BofA Securities, are not insured by the Federal Deposit Insurance Corporation and are not deposits or other obligations of any insured depository institution (including, Bank of America, N.A.). Investments in general and, derivatives, in particular, involve numerous risks, including, among others, market risk, counterparty default risk and liquidity risk. No security, financial instrument or derivative is suitable for all investors. Digital assets are extremely speculative, volatile and are not regulated in a manner similar to U.S. securities markets. In some cases, securities and other financial instruments may be difficult to value or sell and reliable information about the value or risks related to the security or financial instrument may be difficult to obtain. Investors should note that income from such securities and other financial instruments, if any, may fluctuate and that price or value of such securities and instruments may rise or fall and, in some cases, investors may lose their entire principal investment. Past performance is not necessarily a guide to future performance. Levels and basis for taxation may change.
This report may contain a short-term trading idea or recommendation, which highlights a specific near-term catalyst or event impacting the issuer or the market that is anticipated to have a short-term price impact on the equity securities of the issuer. Short-term trading ideas and recommendations are different from and do not affect a stock's fundamental equity rating, which reflects both a longer term total return expectation and attractiveness for investment relative to other stocks within its Coverage Cluster. Short-term trading ideas and recommendations may be more or less positive than a stock's fundamental equity rating.
BofA Securities is aware that the implementation of the ideas expressed in this report may depend upon an investor's ability to "short" securities or other financial instruments and that such action may be limited by regulations prohibiting or restricting "shortselling" in many jurisdictions. Investors are urged to seek advice regarding the applicability of such regulations prior to executing any short idea contained in this report.
Foreign currency rates of exchange may adversely affect the value, price or income of any security or financial instrument mentioned herein. Investors in such securities and instruments, including ADRs, effectively assume currency risk.
BofAS or one of its affiliates is a regular issuer of traded financial instruments linked to securities or digital assets that may have been recommended or discussed in this report. BofAS or one of its affiliates may, at any time, hold a trading position (long or short) in the securities and financial instruments recommended or discussed in this report.
BofA Securities, through business units other than BofA Global Research, may have issued and may in the future issue trading ideas or recommendations that are inconsistent with, and reach different conclusions from, the information presented herein. Such ideas or recommendations may reflect different time frames, assumptions, views and analytical methods of the persons who prepared them, and BofA Securities is under no obligation to ensure that such other trading ideas or recommendations are brought to the attention of any recipient of this information.
In the event that the recipient received this information pursuant to a contract between the recipient and BofAS for the provision of research services for a separate fee, and in connection therewith BofAS may be deemed to be acting as an investment adviser, such status relates, if at all, solely to the person with whom BofAS has contracted directly and does not extend beyond the delivery of this report (unless otherwise agreed specifically in writing by BofAS). If such recipient uses the services of BofAS in connection with the sale or purchase of a security referred to herein, BofAS may act as principal for its own account or as agent for another person. BofAS is and continues to act solely as a broker-dealer in connection with the execution of any transactions, including transactions in any securities referred to herein.

Copyright and General Information:

Copyright 2026 Bank of America Corporation. All rights reserved. iQdatabase® is a registered service mark of Bank of America Corporation. This information is prepared for the use of BofA Securities clients and may not be redistributed, retransmitted or disclosed, in whole or in part, or in any form or manner, without the express written consent of BofA Securities. This document and its content is provided solely for informational purposes and cannot be used for training or developing artificial intelligence (AI) models or as an input in any AI application (collectively, an AI tool). Any attempt to utilize this document or any of its content in connection with an AI tool without explicit written permission from BofA Global Research is strictly prohibited. BofA Global Research utilizes AI, including machine learning and other technologies, to enhance the services we provide to our clients. These technologies assist our analysts in various aspects of their work, including but not limited to data analysis, content extraction, content creation, data aggregation and summarization and identifying relevant information from diverse sources. All AI-driven processes are subject to review by BofA Global Research employees. BofA Global Research information is distributed simultaneously to internal and client websites and other portals by BofA Securities and is not publicly-available material. Any unauthorized use or disclosure is prohibited. Receipt and review of this information constitutes your agreement not to redistribute,

retransmit, or disclose to others the contents, opinions, conclusion, or information contained herein (including any investment recommendations, estimates or price targets) without first obtaining express permission from an authorized officer of BofA Securities.

Materials prepared by BofA Global Research personnel are based on public information. Facts and views presented in this material have not been reviewed by, and may not reflect information known to, professionals in other business areas of BofA Securities, including investment banking personnel. BofA Securities has established information barriers between BofA Global Research and certain business groups. As a result, BofA Securities does not disclose certain client relationships with, or compensation received from, such issuers. To the extent this material discusses any legal proceeding or issues, it has not been prepared as nor is it intended to express any legal conclusion, opinion or advice. Investors should consult their own legal advisers as to issues of law relating to the subject matter of this material. BofA Global Research personnel's knowledge of legal proceedings in which any BofA Securities entity and/or its directors, officers and employees may be plaintiffs, defendants, co-defendants or co-plaintiffs with or involving issuers mentioned in this material is based on public information. Facts and views presented in this material that relate to any such proceedings have not been reviewed by, discussed with, and may not reflect information known to, professionals in other business areas of BofA Securities in connection with the legal proceedings or matters relevant to such proceedings.

This information has been prepared independently of any issuer of securities mentioned herein and not in connection with any proposed offering of securities or as agent of any issuer of any securities. None of BofAS any of its affiliates or their research analysts has any authority whatsoever to make any representation or warranty on behalf of the issuer(s). BofA Global Research policy prohibits research personnel from disclosing a recommendation, investment rating, or investment thesis for review by an issuer prior to the publication of a research report containing such rating, recommendation or investment thesis.

Any information relating to sustainability in this material is limited as discussed herein and is not intended to provide a comprehensive view on any sustainability claim with respect to any issuer or security.

Any information relating to the tax status of financial instruments discussed herein is not intended to provide tax advice or to be used by anyone to provide tax advice. Investors are urged to seek tax advice based on their particular circumstances from an independent tax professional.

The information herein (other than disclosure information relating to BofA Securities and its affiliates) was obtained from various sources and we do not guarantee its accuracy. This information may contain links to third-party websites. BofA Securities is not responsible for the content of any third-party website or any linked content contained in a third-party website. Content contained on such third-party websites is not part of this information and is not incorporated by reference. The inclusion of a link does not imply any endorsement by or any affiliation with BofA Securities. Access to any third-party website is at your own risk, and you should always review the terms and privacy policies at third-party websites before submitting any personal information to them. BofA Securities is not responsible for such terms and privacy policies and expressly disclaims any liability for them.

All opinions, projections and estimates constitute the judgment of the author as of the date of publication and are subject to change without notice. Prices also are subject to change without notice. BofA Securities is under no obligation to update this information and BofA Securities ability to publish information on the subject issuer(s) in the future is subject to applicable quiet periods. You should therefore assume that BofA Securities will not update any fact, circumstance or opinion contained herein.

Subject to the quiet period applicable under laws of the various jurisdictions in which we distribute research reports and other legal and BofA Securities policy-related restrictions on the publication of research reports, fundamental equity reports are produced on a regular basis as necessary to keep the investment recommendation current.

Certain outstanding reports or investment opinions relating to securities, financial instruments and/or issuers may no longer be current. Always refer to the most recent research report relating to an issuer prior to making an investment decision.

In some cases, an issuer may be classified as Restricted or may be Under Review or Extended Review. In each case, investors should consider any investment opinion relating to such issuer (or its security and/or financial instruments) to be suspended or withdrawn and should not rely on the analyses and investment opinion(s) pertaining to such issuer (or its securities and/or financial instruments) nor should the analyses or opinion(s) be considered a solicitation of any kind. Sales persons and financial advisors affiliated with BofAS or any of its affiliates may not solicit purchases of securities or financial instruments that are Restricted or Under Review and may only solicit securities under Extended Review in accordance with firm policies.

Neither BofA Securities nor any officer or employee of BofA Securities accepts any liability whatsoever for any direct, indirect or consequential damages or losses arising from any use of this information.

BofA SECURITIES

ASE Technology Holding

3Q26 sales outperformance affirms solid transformation progress

Reiterate Rating: BUY | PO: 900.00 TWD | Price: 762.00 TWD

3Q26 sales beat on LEAP ramp and mature ASP strength

ASE reported 3Q26 sales +28% QoQ better than its guide, driven by stronger IC ATM (+20% QoQ vs. BofAe +13% QoQ) and mild upside in EMS (+42% QoQ vs. BofAe +40% QoQ). We believe IC ATM strength is led by modest mainstream packaging and testing price hike on industry supply tightness, supplemented by LEAP business ramp. EMS also benefited from Apple product seasonal builds and higher component cost pass through. With improving packaging and testing utilization and ASP and rising LEAP mix, we expect its IC ATM GMs to improve 300bps QoQ to 30% (vs. high-20% guide) though consolidated GMs (+180bps QoQ to 23%) could be partially offset by EMS outgrowth.

AI pipeline diversification anchors capex plan in '26-28

We model ASE's 4Q26 IC ATM sales +7-9% QoQ as its LEAP opportunity diversifying from TSMC outsourcing to direct project wins including AMD's Venice CPU, putting sales on track to +42% YoY in '26 (vs. its +35% guide). Its IC ATM GMs could further expand to 31% as LEAP reaching 30% of its sales (vs. 18%/22% in 1H/3Q26). While we expect ASE to spend US$13bn+ annual capex in '27-28 (vs. US$10.5bn in '26), the accelerating IC ATM sales (BofAe +40% CAGR from '26-28) on advanced expansion and mature price uplift, along with the business mix upgrade to cloud AI/HPC projects carrying 35-45% GMs should offset hiking depreciation (BofAe +39% CAGR from '26-28), lifting its IC ATM GM from 29% in '26 to low 30% in '27 and mid 30% in '28 with operating leverage.

Stronger earnings quality unlocks re-rating potential

We lift our '26/27/28E EPS to NT$20/NT$34/NT$50 on strong LEAP outlook. Keep Buy rating and raise POs from NT$800 (US$51) to NT$900 (US$58) still on 20x 2H27-1H28 P/E (vs. its 5-26x range) backed by its structural business transformation with better sales and GMs growth profile as the industry supply further consolidates amid higher tech entry barrier.

Estimates (Dec) (NT$)	2024A	2025A	2026E	2027E	2028E
Net Income (Adjusted - mn)	36,902	43,283	90,393	156,574	226,147
EPS	7.70	9.30	19.64	34.26	49.78
EPS Change (YoY)	4.0%	20.8%	111.1%	74.5%	45.3%
Consensus EPS (Visible Alpha)			18.46	28.72	38.80
Dividend / Share	5.30	5.70	12.20	21.50	31.20
Free Cash Flow / Share	2.61	(5.16)	(36.00)	(43.18)	(10.66)
ADR EPS (US$)	0.480	0.597	1.23	2.15	3.12
ADR Dividend / Share (US$)	0.330	0.366	0.764	1.35	1.95
Valuation (Dec)					
P/E	98.95x	81.93x	38.80x	22.24x	15.31x
Dividend Yield	0.696%	0.748%	1.60%	2.82%	4.09%
EV / EBITDA*	38.92x	32.62x	19.93x	11.45x	8.12x
Free Cash Flow Yield*	0.331%	-0.657%	-4.64%	-5.58%	-1.38%

* For full definitions of iQmethod℠ measures, see page 7.

This research report provides general information only. No part of this report may be used or reproduced or quoted in any manner whatsoever in Taiwan by the press or other persons without the express written consent of BofA Securities.
>> Employed by a non-US affiliate of BofAS and is not registered/qualified as a research analyst under the FINRA rules.
Refer to "Other Important Disclosures" for information on certain BofA Securities entities that take responsibility for the information herein in particular jurisdictions.
BofA Securities does and seeks to do business with issuers covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.
Refer to important disclosures on page 8 to 11. Analyst Certification on page 5. Price

13 October 2026

Equity

Key Changes

(NT$)	Previous	Current
Price Obj.	800.00	900.00
2026E EPS	18.13	19.64
2027E EPS	30.90	34.26
2028E EPS	45.30	49.78
2026E EBITDA (m)	184,814.6	193,427.9
2027E EBITDA (m)	317,549.5	336,641.2
2028E EBITDA (m)	449,629.5	474,983.2

Haas Liu >>
Research Analyst
Merrill Lynch (Taiwan)
+886 2 2376 3727
haas.liu@bofa.com

Mike Yang >>
Research Analyst
Merrill Lynch (Taiwan)
mike.c.yang@bofa.com

Cathy Hsu >>
Research Analyst
Merrill Lynch (Taiwan)
cathy.hsu3@bofa.com

Stock Data

Price (Common / ADR)	762.00 TWD / 46.43 USD
Price Objective	900.00 TWD / 58.00 USD
Date Established	13-Oct-2026 / 13-Oct-2026
Investment Opinion	C-1-7 / C-1-7
52-Week Range	168.50 TWD-765.00 TWD
Market Value (mn)	106,687 USD
Market Value (mn)	3,406,576 TWD
Shares Outstanding (mn)	4,470.6 / 2,235.3
Average Daily Value (mn)	393.20 USD
Free Float	79.1%
BofA Ticker / Exchange	XSRIF / TAI
BofA Ticker / Exchange	ASX / NYS
Bloomberg / Reuters	3711 TT / 3711.TW
ROE (2026E)	21.8%
Net Dbt to Eqty (Dec-2025A)	41.5%

iQprofile℠ ASE Technology Holding

Key Income Statement Data (Dec)	2024A	2025A	2026E	2027E	2028E
(NT$ Millions)					
Sales	595,410	645,388	864,425	1,151,475	1,422,419
Gross Profit	96,997	114,193	190,153	314,081	429,925
Sell General & Admin Expense	(28,935)	(30,586)	(38,450)	(46,743)	(52,992)
Operating Profit	39,232	50,756	107,198	211,750	312,116
Net Interest & Other Income	2,588	545	3,412	(16,792)	(29,302)
Associates	NA	NA	NA	NA	NA
Pretax Income	41,820	51,301	110,610	194,958	282,814
Tax (expense) / Benefit	(7,756)	(9,460)	(20,502)	(36,664)	(52,837)
Net Income (Adjusted)	36,902	43,283	90,393	156,574	226,147
Average Fully Diluted Shares Outstanding	4,333	4,462	4,506	4,506	4,506
Key Cash Flow Statement Data					
Net Income	32,620	40,658	87,857	154,368	224,274
Depreciation & Amortization	59,815	67,440	86,230	124,891	162,867
Change in Working Capital	(3,706)	(9,449)	(25,495)	(42,718)	(36,340)
Deferred Taxation Charge	NA	NA	NA	NA	NA
Other Adjustments, Net	2,058	43,600	27,147	0	0
Cash Flow from Operations	90,788	142,249	175,738	236,541	350,802
Capital Expenditure	(79,522)	(164,643)	(333,900)	(426,600)	(397,800)
(Acquisition) / Disposal of Investments	1	(1,971)	4,511	1,719	(178)
Other Cash Inflow / (Outflow)	(4,387)	969	(12,022)	0	0
Cash Flow from Investing	(83,909)	(165,644)	(341,411)	(424,881)	(397,978)
Shares Issue / (Repurchase)	1,103	3,480	2,203	0	0
Cost of Dividends Paid	(22,459)	(23,034)	(25,208)	(54,471)	(95,708)
Cash Flow from Financing	(7,271)	45,269	89,859	180,698	80,876
Free Cash Flow	11,266	(22,393)	(158,162)	(190,059)	(46,998)
Net Debt	109,225	155,020	333,215	577,745	720,452
Change in Net Debt	20,146	61,690	195,130	242,811	142,885
Key Balance Sheet Data					
Property, Plant & Equipment	312,531	421,115	699,889	1,063,727	1,384,481
Other Non-Current Assets	152,881	154,423	202,546	224,320	250,667
Trade Receivables	113,420	125,042	155,261	210,110	255,747
Cash & Equivalents	84,883	100,223	38,298	28,937	62,815
Other Current Assets	76,982	88,530	101,726	118,340	132,949
Total Assets	740,698	889,333	1,197,720	1,645,435	2,086,659
Long-Term Debt	140,237	214,510	304,588	544,588	724,588
Other Non-Current Liabilities	23,735	57,108	90,750	90,750	90,750
Short-Term Debt	53,872	40,734	66,926	62,094	58,678
Other Current Liabilities	177,068	203,615	222,390	250,739	266,697
Total Liabilities	394,911	515,966	684,654	948,171	1,140,713
Total Equity	345,787	373,368	513,066	697,263	945,946
Total Equity & Liabilities	740,698	889,333	1,197,720	1,645,435	2,086,659
iQmethod℠ **- Bus Performance***					
Return On Capital Employed	6.2%	6.9%	10.7%	14.6%	15.9%
Return On Equity	11.9%	12.9%	21.8%	27.2%	28.6%
Operating Margin	6.6%	7.9%	12.4%	18.4%	21.9%
EBITDA Margin	16.6%	18.3%	22.4%	29.2%	33.4%
iQmethod℠ **- Quality of Earnings***					
Cash Realization Ratio	2.5x	3.3x	1.9x	1.5x	1.6x
Asset Replacement Ratio	1.4x	2.6x	4.0x	3.5x	2.5x
Tax Rate (Reported)	18.5%	18.4%	18.5%	18.8%	18.7%
Net Debt-to-Equity Ratio	31.6%	41.5%	64.9%	82.9%	76.2%
Interest Cover	5.8x	6.8x	8.7x	8.9x	9.4x

Key Metrics

* For full definitions of *iQmethod*℠ measures, see page 7.

Company Sector

Semiconductors

Company Description

Established in 1984 and headquartered in Taiwan, ASE Technology Holding is the no.1 outsourced assembly and testing (OSAT) firm globally, in terms of market share by revenue. ASE acquired Siliconware Precision (SPIL) in 2018, and had acquired USI in 2010, which became its EMS business segment providing synergies in system-level packaging. 2 shares = 1 ADR.

Investment Rationale

We rate ASE Buy, factoring in the company's opportunity in advanced packaging and testing evidenced in TSMC's outsourcing and its own heavy capex investment. The strategy shift should allow the company to capture fast growing addressable market in high performance computing applications and improves its margins. ASE could maintain its leading position in the Outsourced Semiconductor Assembly and Test (OSAT) universe, owing to its well-established footprints and widespread service offerings.

Stock Data

Shares / ADR	2.00
Price to Book Value	7.1x

Key Changes

(US$)	Previous	Current
Price Obj.	51.00	58.00
2026E EPS	1.14	1.23
2027E EPS	1.94	2.15
2028E EPS	2.84	3.12

Focus charts and tables

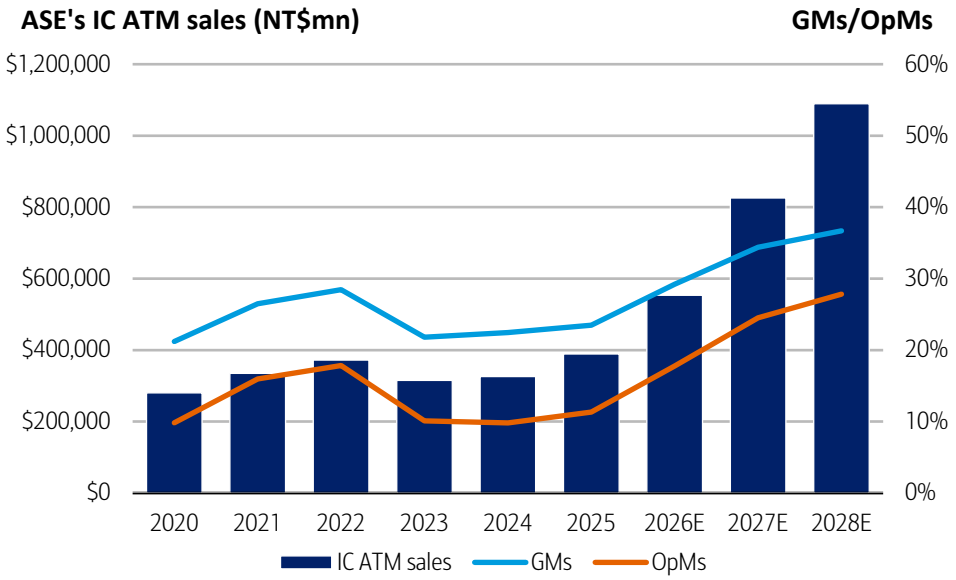
Exhibit 1: ASE 3Q26-4Q26 and 2026-27 estimates BofAe vs. Street
We model 2026/27 EPS of NT$19.64/34.26

	3Q26E			4Q26E			2026E			2027E		
	BofAe	**Old**	**Street**	**BofAe**	**Old**	**Street**	**BofAe**	**Old**	**Street**	**BofAe**	**Old**	**Street**
Sales (NT$mn)	$243,060	$233,274	$233,027	$256,640	$248,543	$241,478	$864,425	$846,543	$837,179	$1,151,475	$1,089,719	$1,043,653
Sales QoQ / YoY (%)	**27.2%**	**22.1%**	**22.0%**	**5.6%**	**6.5%**	**3.6%**	**33.9%**	**31.2%**	**29.7%**	**33.2%**	**28.7%**	**24.7%**
Gross profit (NT$mn)	$55,346	$49,494	$49,292	$59,808	$55,424	$53,676	$190,153	$179,917	$177,390	$314,081	$290,641	$247,335
GMs (%)	**22.8%**	**21.2%**	**21.2%**	**23.3%**	**22.3%**	**22.2%**	**22.0%**	**21.3%**	**21.2%**	**27.3%**	**26.7%**	**23.7%**
Operating profit (NT$mn)	$32,597	$27,638	$28,074	$35,935	$32,281	$31,848	$107,198	$98,585	$98,311	$211,750	$192,658	$155,971
OpMs (%)	**13.4%**	**11.8%**	**12.0%**	**14.0%**	**13.0%**	**13.2%**	**12.4%**	**11.6%**	**11.7%**	**18.4%**	**17.7%**	**14.9%**
Pretax profit (NT$mn)	$31,975	$27,016	$28,489	$34,735	$31,081	$32,092	$110,610	$101,997	$103,760	$194,958	$175,866	$155,822
Pretax margins (%)	13.2%	11.6%	12.2%	13.5%	12.5%	13.3%	12.8%	12.0%	12.4%	16.9%	16.1%	14.9%
Tax expense (NT$mn)	-$6,448	-$5,448	-$5,781	-$7,621	-$6,816	-$6,437	-$22,754	-$20,949	-$20,517	-$40,590	-$36,619	-$31,011
Tax rate (%)	20.2%	20.2%	20.3%	21.9%	21.9%	20.1%	20.6%	20.5%	19.8%	20.8%	20.8%	19.9%
Net profit (NT$mn)	$25,527	$21,568	$22,709	$27,114	$24,265	$25,654	$87,857	$81,048	$83,243	$154,368	$139,247	$124,811
Net margins (%)	10.5%	9.2%	9.7%	10.6%	9.8%	10.6%	10.2%	9.6%	9.9%	13.4%	12.8%	12.0%
EPS (NT$)	**$5.67**	**$4.79**	**$5.11**	**$6.02**	**$5.39**	**$5.76**	**$19.64**	**$18.13**	**$18.78**	**$34.26**	**$30.90**	**$28.24**

Source: BofA Global Research estimates, Bloomberg

BofA GLOBAL RESEARCH

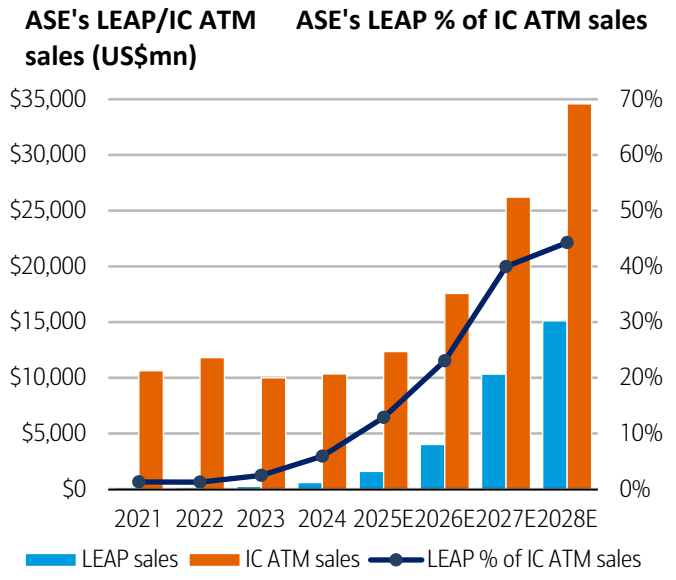
Exhibit 2: ASE IC ATM revenue and GMs/OpMs
ASE's LEAP business to drive IC ATM GM expansion to 37% in 2028



Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

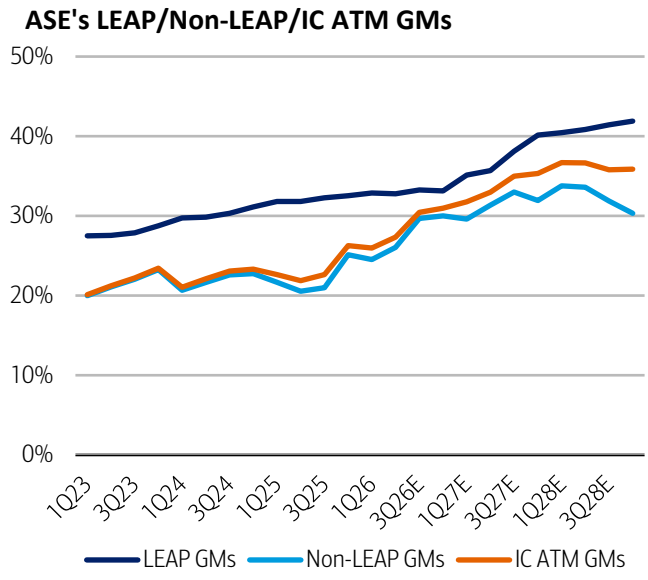
Exhibit 3: ASE LEAP/IC ATM revenue
LEAP will account for 44% of IC ATM sales in 2028



Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

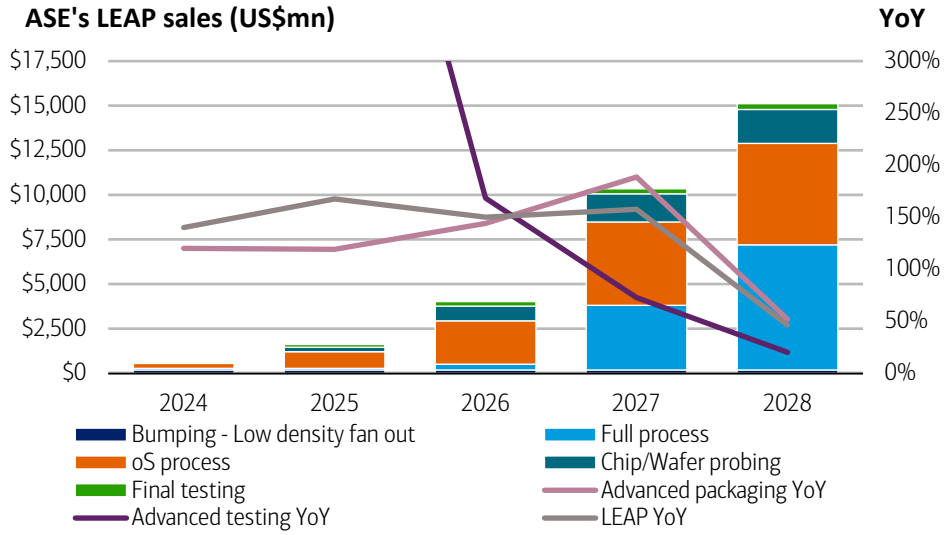
Exhibit 4: ASE LEAP/non-LEAP/IC ATM GMs
LEAP is margin accretive to IC ATM business



Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 5: ASE LEAP sales by subsegments and YoY
ASE's full process packaging will see strong ramp from 2H26



Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 6: Terms and definitions
Glossary of abbreviations/acronyms

Term	Definition
ASIC	Application Specific Integrated Circuit
CPU	Central Processing Unit
EMS	Electronics Manufacturing Services
FOCoS	Fan-Out Chip on Substrate
GPU	Graphic Processing Unit
IC ATM	Integrated Circuit Assembly, Testing, and Material
LEAP	Leading-Edge Advanced Packaging
oS	On Substrate

Source: BofA Global Research

BofA GLOBAL RESEARCH

Price objective basis & risk

ASE Technology Holding (XSRIF / ASX)
We value ASE Technology Holding at NT$900 per share (US$58 for ADR), based on 20x 2H27-1H28E P/E, above its historical range of 5-26x, in view of its solidified industry position in AI era. We use P/E ratio to value ASE as the company has been able to maintain healthy profitability through the cycles in the past 20 years due to consolidating industry landscape supporting a more benign supply/demand and keeping pricing and margins less cyclical.

Downside risks to our PO are 1) share loss and/or ASP erosion owing to Chinese players' cannibalization and/or ASE's worsening execution, 2) weakening end demand due to uncontrollable matters, such as macro halt and/or geopolitical tensions, 3) regulatory issues that could hinder ASE from generating synergies and benefit its competitors.

Analyst Certification

I, Haas Liu, hereby certify that the views expressed in this research report accurately reflect my personal views about the subject securities and issuers. I also certify that no part of my compensation was, is, or will be, directly or indirectly, related to the specific recommendations or view expressed in this research report.

Special Disclosures

Information on securities which are listed on the exchanges where ML Securities (Taiwan) Limited is not permitted to trade or solicit trades for clients is for informational purposes only and is not a recommendation or a solicitation to trade such securities. ML Securities (Taiwan) Limited will not execute transactions for nor accept orders from clients to trade such securities. Foreign investment in Taiwan securities is regulated and restricted. Currently, foreign investment in Taiwan securities is permitted by investment through: (1) global depository receipts, (2) convertible bonds, (3) mutual funds issued offshore of Taiwan, and (4) a special foreign institutional investors (FINIs) and foreign individual investors (FIDIs) program supervised by the Taiwan SFB whereunder FINIs/FIDIs may apply for investment ID to invest in Taiwan securities by registration with Taiwan Stock Exchange. FINIs will additionally need consent from the foreign exchange authority, ie, the Central Bank of China. In addition to the limitations above, various industry-specific percentage-based limitations on foreign ownership of Taiwan companies (and in some cases prohibitions) may apply. Investments are subject to exchange rate and currency conversion restrictions and risks. Dividends and interest earned by foreign investors' Taiwan securities/instruments are generally subject to a 20% withholding tax. Ordinary shares are not available to ML private client accounts in the U.S.

This report is distributed in Taiwan by Merrill Lynch Securities (Taiwan) Limited, which is regulated by the Taiwan SFB.

APR - Semiconductor Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
BUY				
	Alchip	ALCPF	3661 TT	Haas Liu
	All Ring	XNQNF	6187 TT	Haas Liu
	AMEC	XRTOF	688012 CH	Dai Shen
	ASE Technology Holding	XSRIF	3711 TT	Haas Liu
	ASE Technology Holding -ADR	ASX	ASX US	Haas Liu
	ASMPT	ASMVF	522 HK	Simon Woo, CFA
	Aspeed	XLKMF	5274 TT	Mike Yang
	Chroma ATE	CRMJF	2360 TT	Haas Liu

APR - Semiconductor Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
	Doosan Corp	DOSAF	000150 KS	Simon Woo, CFA
	eMemory	XYLWF	3529 TT	Mike Yang
	GigaDevice	XGXIF	603986 CH	Daley Li, CFA
	Global Unichip Corp.	GBUHF	3443 TT	Haas Liu
	Grand Process Technology Corp	XZOWF	3131 TT	Haas Liu
	Hanmi Semiconductor	HNSIF	042700 KS	Simon Woo, CFA
	Hon Precision	XXPIF	7769 TT	Haas Liu
	Horizon Robotics	HRZRF	9660 HK	Daley Li, CFA
	Iluvatar CoreX	XSIBF	9903 HK	Daley Li, CFA
	Ingenic	XISCF	300223 CH	Dai Shen
	InnoScience Technology	XSCHF	2577 HK	Daley Li, CFA
	JCET Group Co Ltd	XJIEF	600584 CH	Dai Shen
	King Yuan Electronics Corp.	KYUFF	2449 TT	Haas Liu
	Kinik	KIKCF	1560 TT	Haas Liu
	Macronix International	MXICF	2337 TT	Dai Shen
	MediaTek	MDTKF	2454 TT	Haas Liu
	Montage Technology	XRDFF	688008 CH	Daley Li, CFA
	MPI Corporation	XMJCF	6223 TT	Mike Yang
	Nanya Technology	NNYAF	2408 TT	Simon Woo, CFA
	NCE Power	XVFFF	605111 CH	Daley Li, CFA
	OmniVision	XXHQF	603501 CH	Dai Shen
	Phison Electronics	PISNF	8299 TT	Simon Woo, CFA
	Powerchip Semiconductor Manufacturing Co	XCHPF	6770 TT	Haas Liu
	Powertech Technology	PWOGF	6239 TT	Simon Woo, CFA
	Rockchip	XRPXF	603893 CH	Daley Li, CFA
	Samsung C&T	XCDTF	028260 KS	Simon Woo, CFA
	Samsung Elec -G	SSNHZ	SMSN LI	Simon Woo, CFA
	Samsung Electronics	SSNLF	005930 KS	Simon Woo, CFA
	Samsung Electronics Preferred	SSNNF	005935 KS	Simon Woo, CFA
	Silergy Corp.	SLEGF	6415 TT	Mike Yang
	Silicon Motion	SIMO	SIMO US	Simon Woo, CFA
	SK Hynix	HXSCF	000660 KS	Simon Woo, CFA
	SK Hynix ADR	SKHY	SKHY US	Simon Woo, CFA
	SK Square	SKSQF	402340 KS	Simon Woo, CFA
	Taiwan Semiconductor Manufacturing Co.	TSM	TSM US	Haas Liu
	Taiwan Semiconductor Manufacturing Co.	TSMWF	2330 TT	Haas Liu
	Vanguard International Semiconductor Co	VGILF	5347 TT	Haas Liu
	Winbond Electronics	WBEKF	2344 TT	Dai Shen
	WinWay Technology	XWCLF	6515 TT	Mike Yang
	WT Microelectronics	XZOPF	3036 TT	Mike Yang
NEUTRAL				
	ASMedia Technology Inc.	XZSFF	5269 TT	Mike Yang
	Black Sesame Intl Holding	BSIHF	2533 HK	Daley Li, CFA
	Hangzhou Silan Microelectronics	XDFRF	600460 CH	Daley Li, CFA
	Lion Electronics	XDHFF	605358 CH	Dai Shen
	MetaX	XMEZF	688802 CH	Daley Li, CFA
	Novatek	NVKMF	3034 TT	Mike Yang
	Realtek	RLTKF	2379 TT	Haas Liu
	Shenzhen Goodix	XQPLF	603160 CH	Daley Li, CFA
	Soulbrain	XSBOF	357780 KS	Simon Woo, CFA
	VeriSilicon	XMLZF	688521 CH	Daley Li, CFA
UNDERPERFORM				
	Crystal Clear	XPPTF	300655 CH	Dai Shen
	Faraday	FDYTF	3035 TT	Haas Liu
	GlobalWafers	XWLFF	6488 TT	Mike Yang
	Hua Hong Grace	HHUSF	1347 HK	Dai Shen
	LX Semicon	XLXSF	108320 KS	Simon Woo, CFA
	M31 Technology	XMTZF	6643 TT	Mike Yang
	Maxscend	XMXSF	300782 CH	Daley Li, CFA
	Parade	PRDWF	4966 TT	Mike Yang
	United Microelectronics Corp.	XUMIF	2303 TT	Haas Liu
	United Microelectronics Corp.	UMC	UMC US	Haas Liu
	Win Semiconductors	WSMIF	3105 TT	Haas Liu
	Wonik IPS	XRHQF	240810 KS	Simon Woo, CFA

APR - Semiconductor Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst

iQmethod℠ Measures Definitions

Business Performance	Numerator	Denominator
Return On Capital Employed	NOPAT = (EBIT + Interest Income) × (1 − Tax Rate) + Goodwill Amortization	Total Assets − Current Liabilities + ST Debt + Accumulated Goodwill Amortization
Return On Equity	Net Income	Shareholders' Equity
Operating Margin	Operating Profit	Sales
Earnings Growth	Expected 5 Year CAGR From Latest Actual	N/A
Free Cash Flow	Cash Flow From Operations − Total Capex	N/A
Quality of Earnings	**Numerator**	**Denominator**
Cash Realization Ratio	Cash Flow From Operations	Net Income
Asset Replacement Ratio	Capex	Depreciation
Tax Rate	Tax Charge	Pre-Tax Income
Net Debt-To-Equity Ratio	Net Debt = Total Debt − Cash & Equivalents	Total Equity
Interest Cover	EBIT	Interest Expense
Valuation Toolkit	**Numerator**	**Denominator**
Price / Earnings Ratio	Current Share Price	Diluted Earnings Per Share (Basis As Specified)
Price / Book Value	Current Share Price	Shareholders' Equity / Current Basic Shares
Dividend Yield	Annualised Declared Cash Dividend	Current Share Price
Free Cash Flow Yield	Cash Flow From Operations − Total Capex	Market Cap = Current Share Price × Current Basic Shares
Enterprise Value / Sales	EV = Current Share Price × Current Shares + Minority Equity + Net Debt + Other LT Liabilities	Sales
EV / EBITDA	Enterprise Value	Basic EBIT + Depreciation + Amortization

iQmethod℠ is the set of BofA Global Research standard measures that serve to maintain global consistency under three broad headings: Business Performance, Quality of Earnings, and validations. The key features of iQmethod are: A consistently structured, detailed, and transparent methodology. Guidelines to maximize the effectiveness of the comparative valuation process, and to identify some common pitfalls.

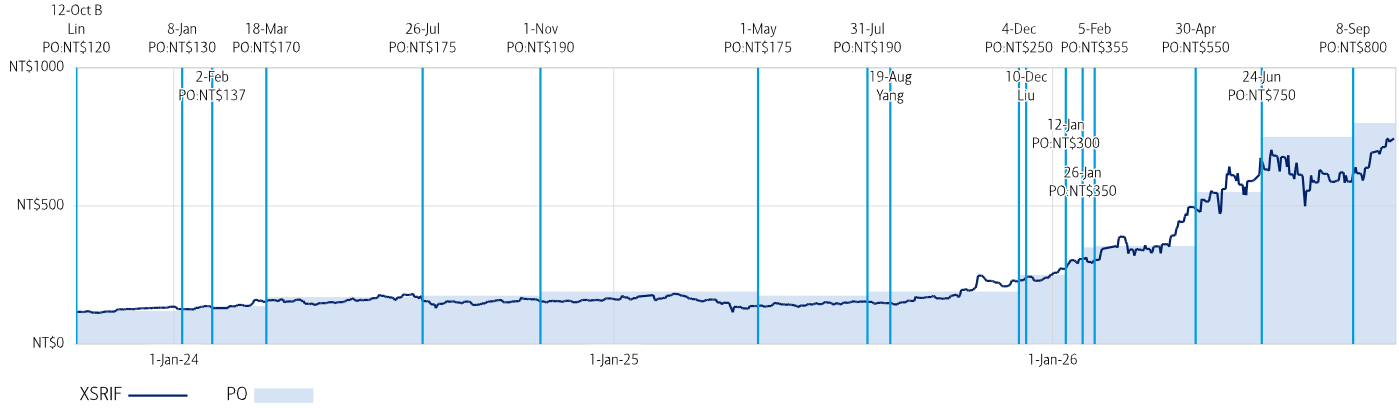
iQdatabase® is our real-time global research database that is sourced directly from our equity analysts' earnings models and includes forecasted as well as historical data for income statements, balance sheets, and cash flow statements for companies covered by BofA Global Research.

iQprofile℠, *iQmethod℠* are service marks of Bank of America Corporation. *iQdatabase®* is a registered service mark of Bank of America Corporation.

Disclosures

Important Disclosures

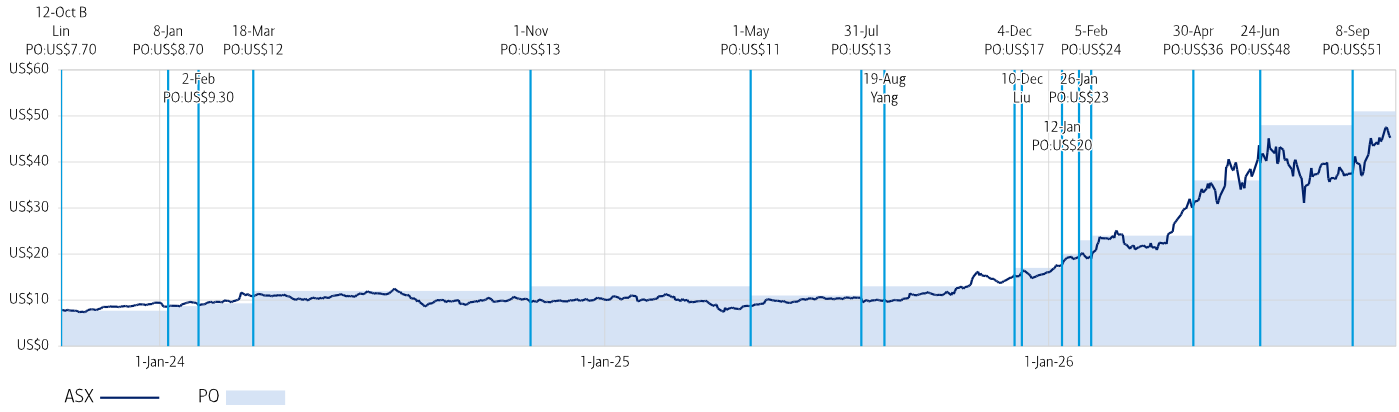
ASE Technology (XSRIF) Price Chart



B: Buy, N: Neutral, U: Underperform, PO: Price Objective, NA: No longer valid, NR: No Rating

The Investment Opinion System is contained at the end of the report under the heading "Fundamental Equity Opinion Key". Dark grey shading indicates the security is restricted with the opinion suspended. Medium grey shading indicates the security is under review with the opinion withdrawn. Light grey shading indicates the security is not covered. Chart is current as of a date no more than one trading day prior to the date of the report.

ASE -ADR (ASX) Price Chart



B: Buy, N: Neutral, U: Underperform, PO: Price Objective, NA: No longer valid, NR: No Rating

The Investment Opinion System is contained at the end of the report under the heading "Fundamental Equity Opinion Key". Dark grey shading indicates the security is restricted with the opinion suspended. Medium grey shading indicates the security is under review with the opinion withdrawn. Light grey shading indicates the security is not covered. Chart is current as of a date no more than one trading day prior to the date of the report.

Equity Investment Rating Distribution: Technology Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	255	62.04%	Buy	157	61.57%
Hold	82	19.95%	Hold	46	56.10%
Sell	74	18.00%	Sell	31	41.89%

Equity Investment Rating Distribution: Global Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	2057	57.03%	Buy	1386	67.38%
Hold	806	22.35%	Hold	563	69.85%
Sell	744	20.63%	Sell	446	59.95%

[R1] Issuers that were investment banking clients of BofA Securities or one of its affiliates within the past 12 months. For purposes of this Investment Rating Distribution, the coverage universe includes only stocks. A stock rated Neutral is included as a Hold, and a stock rated Underperform is included as a Sell.

FUNDAMENTAL EQUITY OPINION KEY: Opinions include a Volatility Risk Rating, an Investment Rating and an Income Rating. VOLATILITY RISK RATINGS, indicators of potential price fluctuation, are: A - Low, B - Medium and C - High. INVESTMENT RATINGS reflect the analyst's assessment of both a stock's absolute total return potential as well as its attractiveness for investment relative to other stocks within its Coverage Cluster (defined below). Our investment ratings are: 1 - Buy stocks are expected to have a total return of at least 10% and are the most attractive stocks in the coverage cluster; 2 - Neutral stocks are expected to remain flat or increase in value and are less attractive than Buy rated stocks and 3 - Underperform stocks are the least attractive stocks in a coverage cluster. An investment rating of 6 (No Rating) indicates that a stock is no longer trading on the basis of fundamentals. Analysts assign investment ratings considering, among other things, the 0-12 month total return expectation for a stock and the firm's guidelines for ratings dispersions (shown in the table below). The current price objective for a stock should be referenced to better understand the total return expectation at any given time. The price objective reflects the analyst's view of the potential price appreciation (depreciation).

Investment rating	Total return expectation (within 12-month period of date of initial rating)	Ratings dispersion guidelines for coverage cluster[R2]
Buy	≥ 10%	≤ 70%
Neutral	≥ 0%	≤ 30%
Underperform	N/A	≥ 20%

[R2]Ratings dispersions may vary from time to time where BofA Global Research believes it better reflects the investment prospects of stocks in a Coverage Cluster.

INCOME RATINGS, indicators of potential cash dividends, are: 7 - same/higher (dividend considered to be secure), 8 - same/lower (dividend not considered to be secure) and 9 - pays no cash dividend. *Coverage Cluster* is comprised of stocks covered by a single analyst or two or more analysts sharing a common industry, sector, region or other classification(s). A stock's coverage cluster is included in the most recent BofA Global Research report referencing the stock.

Price Charts for the securities referenced in this research report are available on the Price Charts website, or call 1-800-MERRILL to have them mailed.
BofAS or one of its affiliates acts as a market maker for the equity securities recommended in the report: ASE -ADR.
The issuer is or was, within the last 12 months, an investment banking client of BofAS and/or one or more of its affiliates: ASE Technology.
BofAS or an affiliate has received compensation from the issuer for non-investment banking services or products within the past 12 months: ASE Technology.
The issuer is or was, within the last 12 months, a non-securities business client of BofAS and/or one or more of its affiliates: ASE Technology.
BofAS or an affiliate expects to receive or intends to seek compensation for investment banking services from this issuer or an affiliate of the issuer within the next three months: ASE Technology.
The country in which this issuer is organized has certain laws or regulations that limit or restrict ownership of the issuer's shares by nationals of other countries: ASE Technology.
BofAS or one of its affiliates is willing to sell to, or buy from, clients the common equity of the issuer on a principal basis: ASE -ADR.
The issuer is or was, within the last 12 months, a securities business client (non-investment banking) of BofAS and/or one or more of its affiliates: ASE Technology.
BofA Global Research personnel (including the analyst(s) responsible for this report) receive compensation based upon, among other factors, the overall profitability of Bank of America Corporation, including profits derived from investment banking. The analyst(s) responsible for this report may also receive compensation based upon, among other factors, the overall profitability of the Bank's sales and trading businesses relating to the class of securities or financial instruments for which such analyst is responsible.

Other Important Disclosures

From time to time research analysts conduct site visits of covered issuers. BofA Global Research policies prohibit research analysts from accepting payment or reimbursement for travel expenses from the issuer for such visits.
Prices are indicative and for information purposes only. Except as otherwise stated in the report, for any recommendation in relation to an equity security, the price referenced is the publicly traded price of the security as of close of business on the day prior to the date of the report or, if the report is published during intraday trading, the price referenced is indicative of the traded price as of the date and time of the report and in relation to a debt security (including equity preferred and CDS), prices are indicative as of the date and time of the report and are from various sources including BofA Securities trading desks.
The date and time of completion of the production of any recommendation in this report shall be the date and time of dissemination of this report as recorded in the report timestamp.

Recipients who are not institutional investors or market professionals should seek the advice of their independent financial advisor before considering information in this report in connection with any investment decision, or for a necessary explanation of its contents.
Officers of BofAS or one or more of its affiliates (other than research analysts) may have a financial interest in securities of the issuer(s) or in related investments.
Refer to BofA Global Research policies relating to conflicts of interest.
"BofA Securities" includes BofA Securities, Inc. ("BofAS") and its affiliates. Investors should contact their BofA Securities representative or Merrill Global Wealth Management financial advisor if they have questions concerning this report or concerning the appropriateness of any investment idea described herein for such investor. "BofA Securities" is a global brand for BofA Global Research.
Information relating to Non-US affiliates of BofA Securities and Distribution of Affiliate Research Reports:
BofAS and/or Merrill Lynch, Pierce, Fenner & Smith Incorporated ("MLPF&S") may in the future distribute, information of the following non-US affiliates in the US (short name: legal name, regulator): Merrill Lynch (South Africa): Merrill Lynch South Africa (Pty) Ltd., regulated by the Financial Sector Conduct Authority; MLI (UK): Merrill Lynch International, regulated by the Financial Conduct Authority (FCA) and the Prudential Regulation Authority (PRA); BofASE (France): BofA Securities Europe SA is authorized by the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and regulated by the ACPR and the Autorité des Marchés Financiers (AMF). BofA Securities Europe SA ("BofASE") with registered address at 51, rue La Boétie, 75008 Paris is registered under no 842 602 690 RCS Paris. In accordance with the provisions of French Code Monétaire et Financier (Monetary and Financial Code), BofASE is an établissement de crédit et d'investissement (credit and investment institution) that is authorised and supervised by the European Central Bank and the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and regulated by the ACPR and the Autorité des Marchés Financiers. BofASE's share capital can be found at www.bofaml.com/BofASEdisclaimer; BofA Europe (Milan): Bank of America Europe Designated Activity Company, Milan Branch, regulated by the Bank of Italy, the European Central Bank (ECB) and the Central Bank of Ireland (CBI); BofA Europe (Frankfurt): Bank of America Europe Designated Activity Company, Frankfurt Branch regulated by BaFin, the ECB and the CBI; BofA Europe (Zurich): Bank of America Europe Designated Activity Company, Zurich Branch, regulated by the Swiss Financial Market Supervisory Authority FINMA, the ECB and CBI; BofA Europe (Madrid): Bank of America Europe Designated Activity Company, Sucursal en España, regulated by the Bank of Spain, the ECB and the CBI; Merrill Lynch (Australia): Merrill Lynch Equities (Australia) Limited, regulated by the Australian Securities and Investments Commission; Merrill Lynch (Hong Kong): Merrill Lynch (Asia Pacific) Limited, regulated by the Hong Kong Securities and Futures Commission (HKSFC); Merrill Lynch (Singapore): Merrill Lynch (Singapore) Pte Ltd, regulated by the Monetary Authority of Singapore (MAS); Merrill Lynch (Canada): Merrill Lynch Canada Inc, regulated by the Canadian Investment Regulatory Organization; Merrill Lynch (Mexico): Merrill Lynch Mexico, SA de CV, Casa de Bolsa, regulated by the Comisión Nacional Bancaria y de Valores; BofAS Japan: BofA Securities Japan Co., Ltd., regulated by the Financial Services Agency; Merrill Lynch (Seoul): Merrill Lynch International, LLC Seoul Branch, regulated by the Financial Supervisory Service; Merrill Lynch (Taiwan): Merrill Lynch Securities (Taiwan) Ltd., regulated by the Securities and Futures Bureau; BofAS India: BofA Securities India Limited, regulated by the Securities and Exchange Board of India (SEBI); Merrill Lynch (Israel): Merrill Lynch Israel Limited, regulated by Israel Securities Authority; Merrill Lynch (DIFC): Merrill Lynch International (DIFC Branch), regulated by the Dubai Financial Services Authority (DFSA); Merrill Lynch (Brazil): Merrill Lynch S.A. Corretora de Títulos e Valores Mobiliários, regulated by Comissão de Valores Mobiliários; Merrill Lynch KSA Company: Merrill Lynch Kingdom of Saudi Arabia Company, regulated by the Capital Market Authority.
This information: has been approved for publication and is distributed in the United Kingdom (UK) to professional clients and eligible counterparties (as each is defined in the rules of the FCA and the PRA) by MLI (UK), which is authorized by the PRA and regulated by the FCA and the PRA - details about the extent of our regulation by the FCA and PRA are available from us on request; has been approved for publication and is distributed in the European Economic Area (EEA) by BofASE (France), which is authorized by the ACPR and regulated by the ACPR and the AMF; has been considered and distributed in Japan by BofAS Japan, a registered securities dealer under the Financial Instruments and Exchange Act in Japan, or its permitted affiliates; is issued and

distributed in Hong Kong by Merrill Lynch (Hong Kong) which is regulated by HKSFC; is issued and distributed in Taiwan by Merrill Lynch (Taiwan); is issued and distributed in India by BofAS India; and is issued and distributed in Singapore to institutional investors and/or accredited investors (each as defined under the Financial Advisers Regulations) by Merrill Lynch (Singapore) (Company Registration No 198602883D). Merrill Lynch (Singapore) is regulated by MAS. Merrill Lynch Equities (Australia) Limited (ABN 65 006 276 795), AFS License 235132 (MLEA) distributes this information in Australia only to 'Wholesale' clients as defined by s.761G of the Corporations Act 2001. With the exception of Bank of America N.A., Australia Branch, neither MLEA nor any of its affiliates involved in preparing this information is an Authorised Deposit-Taking Institution under the Banking Act 1959 nor regulated by the Australian Prudential Regulation Authority. No approval is required for publication or distribution of this information in Brazil and its local distribution is by Merrill Lynch (Brazil) in accordance with applicable regulations. Merrill Lynch (DIFC) is authorized and regulated by the DFSA. Information prepared and issued by Merrill Lynch (DIFC) is done so in accordance with the requirements of the DFSA conduct of business rules. BofA Europe (Frankfurt) distributes this information in Germany and is regulated by BaFin, the ECB and the CBI. BofA Securities entities, including BofA Europe and BofASE (France), may outsource/delegate the marketing and/or provision of certain research services or aspects of research services to other branches or members of the BofA Securities group. You may be contacted by a different BofA Securities entity acting for and on behalf of your service provider where permitted by applicable law. This does not change your service provider. Please refer to the Electronic Communications Disclaimers for further information.

This information has been prepared and issued by BofAS and/or one or more of its non-US affiliates. The author(s) of this information may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so. BofAS and/or MLPF&S is the distributor of this information in the US and accepts full responsibility for information distributed to BofAS and/or MLPF&S clients in the US by its non-US affiliates. Any US person receiving this information and wishing to effect any transaction in any security discussed herein should do so through BofAS and/or MLPF&S and not such foreign affiliates. Hong Kong recipients of this information should contact Merrill Lynch (Asia Pacific) Limited in respect of any matters relating to dealing in securities or provision of specific advice on securities or any other matters arising from, or in connection with, this information. Singapore recipients of this information should contact Merrill Lynch (Singapore) Pte Ltd in respect of any matters arising from, or in connection with, this information. For clients that are not accredited investors, expert investors or institutional investors Merrill Lynch (Singapore) Pte Ltd accepts full responsibility for the contents of this information distributed to such clients in Singapore.

General Investment Related Disclosures:

Taiwan Readers: Neither the information nor any opinion expressed herein constitutes an offer or a solicitation of an offer to transact in any securities or other financial instrument. No part of this report may be used or reproduced or quoted in any manner whatsoever in Taiwan by the press or any other person without the express written consent of BofA Securities.

This document provides general information only, and has been prepared for, and is intended for general distribution to, BofA Securities clients. Neither the information nor any opinion expressed constitutes an offer or an invitation to make an offer, to buy or sell any securities or other financial instrument or any derivative related to such securities or instruments (e.g., options, futures, warrants, and contracts for differences). This document is not intended to provide personal investment advice and it does not take into account the specific investment objectives, financial situation and the particular needs of, and is not directed to, any specific person(s). This document and its content do not constitute, and should not be considered to constitute, investment advice for purposes of ERISA, the US tax code, the Investment Advisers Act or otherwise. Investors should seek financial advice regarding the appropriateness of investing in financial instruments and implementing investment strategies discussed or recommended in this document and should understand that statements regarding future prospects may not be realized. Any decision to purchase or subscribe for securities in any offering must be based solely on existing public information on such security or the information in the prospectus or other offering document issued in connection with such offering, and not on this document.

Securities and other financial instruments referred to herein, or recommended, offered or sold by BofA Securities, are not insured by the Federal Deposit Insurance Corporation and are not deposits or other obligations of any insured depository institution (including, Bank of America, N.A.). Investments in general and, derivatives, in particular, involve numerous risks, including, among others, market risk, counterparty default risk and liquidity risk. No security, financial instrument or derivative is suitable for all investors. Digital assets are extremely speculative, volatile and are not regulated in a manner similar to U.S. securities markets. In some cases, securities and other financial instruments may be difficult to value or sell and reliable information about the value or risks related to the security or financial instrument may be difficult to obtain. Investors should note that income from such securities and other financial instruments, if any, may fluctuate and that price or value of such securities and instruments may rise or fall and, in some cases, investors may lose their entire principal investment. Past performance is not necessarily a guide to future performance. Levels and basis for taxation may change.

This report may contain a short-term trading idea or recommendation, which highlights a specific near-term catalyst or event impacting the issuer or the market that is anticipated to have a short-term price impact on the equity securities of the issuer. Short-term trading ideas and recommendations are different from and do not affect a stock's fundamental equity rating, which reflects both a longer term total return expectation and attractiveness for investment relative to other stocks within its Coverage Cluster. Short-term trading ideas and recommendations may be more or less positive than a stock's fundamental equity rating.

BofA Securities is aware that the implementation of the ideas expressed in this report may depend upon an investor's ability to "short" securities or other financial instruments and that such action may be limited by regulations prohibiting or restricting "shortselling" in many jurisdictions. Investors are urged to seek advice regarding the applicability of such regulations prior to executing any short idea contained in this report.

Foreign currency rates of exchange may adversely affect the value, price or income of any security or financial instrument mentioned herein. Investors in such securities and instruments, including ADRs, effectively assume currency risk.

BofAS or one of its affiliates is a regular issuer of traded financial instruments linked to securities or digital assets that may have been recommended or discussed in this report. BofAS or one of its affiliates may, at any time, hold a trading position (long or short) in the securities and financial instruments recommended or discussed in this report.

BofA Securities, through business units other than BofA Global Research, may have issued and may in the future issue trading ideas or recommendations that are inconsistent with, and reach different conclusions from, the information presented herein. Such ideas or recommendations may reflect different time frames, assumptions, views and analytical methods of the persons who prepared them, and BofA Securities is under no obligation to ensure that such other trading ideas or recommendations are brought to the attention of any recipient of this information.

In the event that the recipient received this information pursuant to a contract between the recipient and BofAS for the provision of research services for a separate fee, and in connection therewith BofAS may be deemed to be acting as an investment adviser, such status relates, if at all, solely to the person with whom BofAS has contracted directly and does not extend beyond the delivery of this report (unless otherwise agreed specifically in writing by BofAS). If such recipient uses the services of BofAS in connection with the sale or purchase of a security referred to herein, BofAS may act as principal for its own account or as agent for another person. BofAS is and continues to act solely as a broker-dealer in connection with the execution of any transactions, including transactions in any securities referred to herein.

Copyright and General Information:

Copyright 2026 Bank of America Corporation. All rights reserved. iQdatabase® is a registered service mark of Bank of America Corporation. This information is prepared for the use of BofA Securities clients and may not be redistributed, retransmitted or disclosed, in whole or in part, or in any form or manner, without the express written consent of BofA Securities. This document and its content is provided solely for informational purposes and cannot be used for training or developing artificial intelligence (AI) models or as an input in any AI application (collectively, an AI tool). Any attempt to utilize this document or any of its content in connection with an AI tool without explicit written permission from BofA Global Research is strictly prohibited. BofA Global Research utilizes AI, including machine learning and other technologies, to enhance the services we provide to our clients. These technologies assist our analysts in various aspects of their work, including but not limited to data analysis, content extraction, content creation, data aggregation and summarization and identifying relevant information from diverse sources. All AI-driven processes are subject to review by BofA Global Research employees. BofA Global Research information is distributed simultaneously to internal and client websites and other portals by BofA Securities and is not publicly-available material. Any unauthorized use or disclosure is prohibited. Receipt and review of this information constitutes your agreement not to redistribute, retransmit, or disclose to others the contents, opinions, conclusion, or information contained herein (including any investment recommendations, estimates or price targets) without first obtaining express permission from an authorized officer of BofA Securities.

Materials prepared by BofA Global Research personnel are based on public information. Facts and views presented in this material have not been reviewed by, and may not reflect information known to, professionals in other business areas of BofA Securities, including investment banking personnel. BofA Securities has established information barriers between BofA Global Research and certain business groups. As a result, BofA Securities does not disclose certain client relationships with, or compensation received from, such issuers. To the extent this material discusses any legal proceeding or issues, it has not been prepared as nor is it intended to express any legal conclusion, opinion or advice. Investors should consult their own legal advisers as to issues of law relating to the subject matter of this material. BofA Global Research personnel's knowledge of legal proceedings in which any BofA Securities entity and/or its directors, officers and employees may be plaintiffs, defendants, co-defendants or co-plaintiffs with or involving issuers mentioned in this material is based on public information. Facts and views presented in this material that relate to any such proceedings have not been reviewed by, discussed with, and may not reflect information known to, professionals in other business areas of BofA Securities in connection with the legal proceedings or matters relevant to such proceedings.

This information has been prepared independently of any issuer of securities mentioned herein and not in connection with any proposed offering of securities or as agent of any issuer of any securities. None of BofAS any of its affiliates or their research analysts has any authority whatsoever to make any representation or warranty on behalf of the issuer(s). BofA Global Research

policy prohibits research personnel from disclosing a recommendation, investment rating, or investment thesis for review by an issuer prior to the publication of a research report containing such rating, recommendation or investment thesis.
Any information relating to sustainability in this material is limited as discussed herein and is not intended to provide a comprehensive view on any sustainability claim with respect to any issuer or security.
Any information relating to the tax status of financial instruments discussed herein is not intended to provide tax advice or to be used by anyone to provide tax advice. Investors are urged to seek tax advice based on their particular circumstances from an independent tax professional.
The information herein (other than disclosure information relating to BofA Securities and its affiliates) was obtained from various sources and we do not guarantee its accuracy. This information may contain links to third-party websites. BofA Securities is not responsible for the content of any third-party website or any linked content contained in a third-party website. Content contained on such third-party websites is not part of this information and is not incorporated by reference. The inclusion of a link does not imply any endorsement by or any affiliation with BofA Securities. Access to any third-party website is at your own risk, and you should always review the terms and privacy policies at third-party websites before submitting any personal information to them. BofA Securities is not responsible for such terms and privacy policies and expressly disclaims any liability for them.
All opinions, projections and estimates constitute the judgment of the author as of the date of publication and are subject to change without notice. Prices also are subject to change without notice. BofA Securities is under no obligation to update this information and BofA Securities ability to publish information on the subject issuer(s) in the future is subject to applicable quiet periods. You should therefore assume that BofA Securities will not update any fact, circumstance or opinion contained herein.
Subject to the quiet period applicable under laws of the various jurisdictions in which we distribute research reports and other legal and BofA Securities policy-related restrictions on the publication of research reports, fundamental equity reports are produced on a regular basis as necessary to keep the investment recommendation current.
Certain outstanding reports or investment opinions relating to securities, financial instruments and/or issuers may no longer be current. Always refer to the most recent research report relating to an issuer prior to making an investment decision.
In some cases, an issuer may be classified as Restricted or may be Under Review or Extended Review. In each case, investors should consider any investment opinion relating to such issuer (or its security and/or financial instruments) to be suspended or withdrawn and should not rely on the analyses and investment opinion(s) pertaining to such issuer (or its securities and/or financial instruments) nor should the analyses or opinion(s) be considered a solicitation of any kind. Sales persons and financial advisors affiliated with BofAS or any of its affiliates may not solicit purchases of securities or financial instruments that are Restricted or Under Review and may only solicit securities under Extended Review in accordance with firm policies.
Neither BofA Securities nor any officer or employee of BofA Securities accepts any liability whatsoever for any direct, indirect or consequential damages or losses arising from any use of this information.

Morgan Stanley | RESEARCH

UPDATE

October 13, 2026 04:27 PM GMT

Greater China Semiconductors | Global

OCP kick off – Optical, Packaging, Memory and Power in focus

Al infrastructure innovation is shifting toward rack-level co-optimization, creating opportunities in optical interconnects, advanced packaging, memory architectures, and power delivery as agentic Al drives inference demand.

Key Takeaways

- Optical interconnect: CPO is moving toward rack-scale scale-up, while pluggables retain a scale-out advantage.
- Custom ASICs and packaging: MediaTek sees 3.5D gaining traction as 2.5D approaches physical limits
- Memory architecture: SK hynix targets inference capacity and KV-cache bottlenecks through MARS arid HBF.
- Power and cooling: Racks above 200kW accelerate 800V DC, vertical power delivery and liquid cooling.

CPO: Optical scale-up is getting closer to reality. A 64-XPU CPO-enabled compute-rack demonstration highlighted the potential for direct optical links between accelerators. As 200G/lane electrical channels approach physical limitations, optical scale-up will become more important, particularly for customized XPUs. However, pluggagle optics should retain scale-out volume leadership given advantages in serviceability and cost. Packaging yield, fiber alignment, and reliability remain key commercialization challenges.

Advanced packaging: Moving beyond 2.5D. MediaTek highlighted custom ASIC designs approaching 10x reticle size, arguing that 3.5D integration could deliver approximately 10x better die-to-die energy efficiency versus conventional 2.5D interconnects. Management expects increasing adoption over the next 3-5 years. MediaTek also emphasized custom HBM, proprietary die-to-die interconnects, and DTCO optimization. We see structural opportunities for advanced packaging and custom ASIC design services as system complexity increases (link).

Memory: Capacity matters alongside bandwidth. SK hynix introduced its MARS disaggregated memory architecture, demonstrating a 3-5x improvement in time-to-first-token through shared KV-cache access versus recomputation. Meanwhile, HBF targets approximately half of HBM bandwidth but 14x higher capacity, with samples expected in early 2028. Inference workloads will increasingly require heterogeneous memory architectures encompassing HBM, DRAM, and NAND, potentially expanding memory content beyond traditional GPU-attached HBM (link).

MORGAN STANLEY TAIWAN LIMITED+
Daniel Yen, CFA
Equity Analyst
Daniel.Yen@morganstanley.com +886 2 2730-2863

Charlie Chan
Equity Analyst
Charlie.Chan@morganstanley.com +886 2 2730-1725

MORGAN STANLEY ASIA LIMITED+
Daisy Dai, CFA
Equity Analyst
Daisy.Dai@morganstanley.com +852 2848-7310

MORGAN STANLEY TAIWAN LIMITED+
Lucas Wang
Equity Analyst
Lucas.Wang@morganstanley.com +886 2 2730-2875

MORGAN STANLEY ASIA LIMITED+
Henry Zhao
Equity Analyst
Henry.Zhao@morganstanley.com +852 2239-7731

Ethan Jia
Research Associate
Ethan.Jia@morganstanley.com +852 3963-2287

GREATER CHINA TECHNOLOGY SEMICONDUCTORS
Asia Pacific
Industry View **Attractive**

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

+= Analysts employed by non-U.S. affiliates are not registered with FINRA, may not be associated persons of the member and may not be subject to FINRA restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

Power and thermal: The next system-level constraints. Demonstrations of 6kw-class double-sided cold plates vertical power delivery, and advanced microchannel cooling highlighted intensifying thermal challenges. The transition toward 800V DC should reduce conversion losses and improve power density. Meanwhile, AMD's Helion agentic Al rack architecture reinforces incremental CPU demand alongside GPU infrastructure. Power delivery, advanced cooling, and system integration capabilities to become increasingly important differentiators.

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley & Co. LLC and/or Morgan Stanley C.T.V.M. S.A. and/or Morgan Stanley México, Casa de Bolsa, S.A. de C.V. and/or Morgan Stanley & Co. International plc and/or Morgan Stanley Europe S.E. and/or RMB Morgan Stanley Proprietary Limited and/or Morgan Stanley MUFG Securities Co., Ltd. and/or Morgan Stanley Capital Group Japan Co., Ltd. and/or Morgan Stanley Asia Limited and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105), Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Charlie Chan; Daisy Dai, CFA; Daniel Yen, CFA; Henry Zhao.

.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictpolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

As of September 30, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: ACM Research Inc, Advanced Micro-Fabrication Equipment Inc, Advanced Wireless Semiconductor Co, Alchip Technologies Ltd, AllRing Tech Co., AP Memory Technology Corp, ASE Technology Holding Co. Ltd., ASMedia Technology Inc, ASMPT Ltd, Aspeed Technology, Cambricon Technology Corporation, eMemory Technology Inc, GigaDevice Semiconductor Beijing Inc, GlobalWafers Co Ltd, Gudeng Precision, Macronix International Co Ltd, Montage Technology Co Ltd, NAURA Technology Group Co Ltd, Novatek, Parade Technologies Ltd, Phison Electronics Corp, Powerchip Semiconductor Manufacturing Co, SG Micro Corp., Shanghai Fudan Microelectronics, SICC Co Ltd, Silergy Corp., Silicon Motion, StarPower Semiconductor Ltd, TSMC, UMC, Vanguard International Semiconductor, WIN Semiconductors Corp, WinWay Technology Co Ltd, WPG Holdings, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Alchip Technologies Ltd, Iluvatar CoreX Semiconductor Co., Ltd., MediaTek, Montage Technology Co Ltd, Powerchip Semiconductor Manufacturing Co, UMC.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from ASMPT Ltd, Iluvatar CoreX Semiconductor Co., Ltd., MediaTek, Montage Technology Co Ltd, Powerchip Semiconductor Manufacturing Co, Silicon Motion, UMC.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from Advanced Micro-Fabrication Equipment Inc, Alchip Technologies Ltd, AP Memory Technology Corp, ASE Technology Holding Co. Ltd., ASMedia Technology Inc, ASMPT Ltd, Aspeed Technology, CXMT Corp, Espressif Systems, GigaDevice Semiconductor Beijing Inc, GlobalWafers Co Ltd, Gudeng Precision, Himax Technologies Inc, Hua Hong Semiconductor Ltd, Iluvatar CoreX Semiconductor Co., Ltd., Innoscience, King Yuan Electronics Co Ltd, Macronix International Co Ltd, MediaTek, MetaX Integrated Circuits, Montage Technology Co Ltd, Novatek, Phison Electronics Corp, Powerchip Semiconductor Manufacturing Co, Realtek Semiconductor, SG Micro Corp., SICC Co Ltd, Silergy Corp., Silicon Motion, TSMC, UMC, Universal Scientific Ind. (Shanghai), Vanguard International Semiconductor, Winbond Electronics Corp, WinWay Technology Co Ltd, WPG Holdings, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from ASE Technology Holding Co. Ltd., King Yuan Electronics Co Ltd, MediaTek, Nanya Technology Corp., Novatek, Nuvoton Technology Corporation, Realtek Semiconductor, Silicon Motion, SMIC, TSMC, UMC, Universal Scientific Ind. (Shanghai), Winbond Electronics Corp, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: Advanced Micro-Fabrication Equipment Inc, Alchip Technologies Ltd, AP Memory Technology Corp, ASE Technology Holding Co. Ltd., ASMedia Technology Inc, ASMPT Ltd, Aspeed Technology, CXMT Corp, Espressif Systems, GigaDevice Semiconductor Beijing Inc, GlobalWafers Co Ltd, Gudeng Precision, Himax Technologies Inc, Hua Hong Semiconductor Ltd, Iluvatar CoreX Semiconductor Co., Ltd., Innoscience, King Yuan Electronics Co Ltd, Macronix International Co Ltd, MediaTek, MetaX Integrated Circuits, Montage Technology Co Ltd, Novatek, Phison Electronics Corp, Powerchip Semiconductor Manufacturing Co, Realtek Semiconductor, SG Micro Corp., SICC Co Ltd, Silergy Corp., Silicon Motion, TSMC, UMC, Universal Scientific Ind. (Shanghai), Vanguard International Semiconductor, Winbond Electronics Corp, WinWay Technology Co Ltd, WPG Holdings, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: ASE Technology Holding Co. Ltd., Iluvatar CoreX Semiconductor Co., Ltd., King Yuan Electronics Co Ltd, MediaTek, Montage Technology Co Ltd, Nanya Technology Corp., Novatek, Nuvoton Technology Corporation, Realtek Semiconductor, Silicon Motion, SMIC, TSMC, UMC, Universal Scientific Ind. (Shanghai), Winbond Electronics Corp, WT Microelectronics Co. Ltd..

Morgan Stanley & Co. LLC makes a market in the securities of ACM Research Inc.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.
Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.
Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of September 30, 2026)
The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.
For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O or Over) - The stock's total return is expected to exceed the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.
Equal-weight (E or Equal) - The stock's total return is expected to be in line with the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis over the next 12-18 months.
Not-Rated (NR) - Currently the analyst does not have adequate conviction about the stock's total return relative to the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Underweight (U or Under) - The stock's total return is expected to be below the total return of the relevant country MSCI Index or the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.
Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.
Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures.
For Morgan Stanley specific disclosures, you may refer to https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch.
Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the

research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (http://www.morganstanley.com/terms.html). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (http://www.morganstanley.com/terms.html). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research.

Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V which is regulated by Comision Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118J); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. Morgan Stanley Research is disseminated in Canada by Morgan Stanley Canada Limited and Morgan Stanley Wealth Management Canada Inc., members of the Canadian Investment Regulatory Organization (CIRO). The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

Morgan Stanley Hong Kong Securities Limited is the liquidity provider/market maker for securities of ASMPT Ltd, GigaDevice Semiconductor Beijing Inc, Hua Hong Semiconductor Ltd, Montage Technology Co Ltd listed on the Stock Exchange of Hong Kong Limited. An updated list can be found on HKEx website: http://www.hkex.com.hk.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

References to any prediction market or event contract do not constitute an offer, solicitation, recommendation, or endorsement to participate in any such market or contract. Morgan Stanley Research may not be distributed or made available in any jurisdiction where such distribution would be unlawful or otherwise prohibited by applicable law.

INDUSTRY COVERAGE: Greater China Technology Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/13/2026)
Charlie Chan		
Advanced Wireless Semiconductor Co (8086.TWO)	U (07/14/2025)	NT$140.00
Alchip Technologies Ltd (3661.TW)	O (05/14/2021)	NT$4,055.00

ASE Technology Holding Co. Ltd. (3711.TW)	O (09/15/2024)	NT$740.00
Cambricon Technology Corporation (688256.SS)	O (04/27/2026)	Rmb976.49
CXMT Corp (688825.SS)	O (08/26/2026)	Rmb46.99
Global Unichip Corp (3443.TW)	O (06/24/2026)	NT$8,095.00
GlobalWafers Co Ltd (6488.TWO)	E (05/19/2026)	NT$1,015.00
Gudeng Precision (3680.TWO)	O (11/25/2025)	NT$553.00
Hua Hong Semiconductor Ltd (1347.HK)	E (03/12/2026)	HK$85.10
Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK)	O (04/27/2026)	HK$84.90
MediaTek (2454.TW)	O (11/28/2025)	NT$4,525.00
MetaX Integrated Circuits (688802.SS)	E (04/27/2026)	Rmb436.90
Nanya Technology Corp. (2408.TW)	O (05/28/2026)	NT$488.00
OmniVision Integrated Circuits Group Inc (603501.SS)	E (11/17/2025)	Rmb75.96
Phison Electronics Corp (8299.TWO)	E (02/25/2026)	NT$1,965.00
SG Micro Corp. (300661.SZ)	E (11/03/2025)	Rmb95.24
Silergy Corp. (6415.TW)	U (05/19/2026)	NT$428.50
SMIC (0981.HK)	O (10/21/2025)	HK$55.55
TSMC (2330.TW)	O (02/07/2022)	NT$2,490.00
UMC (2303.TW)	O (05/19/2026)	NT$141.50
Vanguard International Semiconductor (5347.TWO)	E (01/14/2026)	NT$175.00
WIN Semiconductors Corp (3105.TWO)	E (10/09/2026)	NT$556.00
Daisy Dai, CFA		
ASMPT Ltd (0522.HK)	O (07/24/2025)	HK$158.40
China Resources Microelectronics Limited (688396.SS)	U (03/02/2026)	Rmb49.82
Empyrean Technology Co Ltd (301269.SZ)	E (01/17/2025)	Rmb97.82
Hangzhou Silan Microelectronics Co. Ltd. (600460.SS)	U (08/25/2025)	Rmb27.75
Hygon Information Technology Co., Ltd. (688041.SS)	O (07/03/2026)	Rmb209.74
Innoscience (2577.HK)	E (10/13/2025)	HK$34.54
JCET Group Co Ltd (600584.SS)	O (09/16/2026)	Rmb60.16
Shanghai Fudan Microelectronics (1385.HK)	O (03/07/2025)	HK$23.20
SICC Co Ltd (688234.SS)	O (03/20/2026)	Rmb93.85
SJ Semiconductor Corp (688820.SS)	E (09/16/2026)	Rmb104.39
StarPower Semiconductor Ltd (603290.SS)	E (05/14/2026)	Rmb74.23
Unigroup Guoxin Microelectronics Co Ltd (002049.SZ)	U (01/10/2023)	Rmb55.38
Universal Scientific Ind. (Shanghai) (601231.SS)	O (11/05/2025)	Rmb22.50
Yangjie Technology (300373.SZ)	O (06/10/2022)	Rmb80.60
Daniel Yen, CFA		
AP Memory Technology Corp (6531.TW)	O (07/11/2025)	NT$926.00
ASMedia Technology Inc (5269.TW)	U (10/03/2025)	NT$1,540.00
Aspeed Technology (5274.TWO)	O (06/09/2025)	NT$17,350.00
Egis Technology Inc (6462.TWO)	E (01/28/2026)	NT$97.00
Elan Microelectronics Corp (2458.TW)	O (10/03/2025)	NT$133.00
eMemory Technology Inc (3529.TWO)	O (08/18/2026)	NT$2,810.00
Espressif Systems (688018.SS)	O (05/15/2023)	Rmb107.42
GigaDevice Semiconductor Beijing Inc (603986.SS)	O (05/15/2025)	Rmb306.22
Macronix International Co Ltd (2337.TW)	O (09/18/2025)	NT$106.50
Montage Technology Co Ltd (6809.HK)	O (03/18/2026)	HK$241.80
Montage Technology Co Ltd (688008.SS)	O (03/18/2026)	Rmb184.80
Novatek (3034.TW)	U (02/04/2026)	NT$546.00
Nuvoton Technology Corporation (4919.TW)	U (11/10/2025)	NT$142.50
Parade Technologies Ltd (4966.TWO)	O (05/27/2026)	NT$547.00
Powerchip Semiconductor Manufacturing Co (6770.TW)	O (10/27/2025)	NT$67.50
Realtek Semiconductor (2379.TW)	E (01/30/2026)	NT$758.00
Shenzhen Goodix Technology Co Ltd (603160.SS)	U (07/14/2025)	Rmb49.33
Winbond Electronics Corp (2344.TW)	O (05/28/2026)	NT$161.50
WPG Holdings (3702.TW)	O (03/16/2026)	NT$117.50

WT Microelectronics Co. Ltd. (3036.TW)	O (01/27/2026)	NT$211.50
Henry Zhao		
ACM Research Inc (ACMR.O)	O (03/07/2023)	US$68.10
Advanced Micro-Fabrication Equipment Inc (688012.SS)	O (11/06/2023)	Rmb307.69
Maxscend Microelectronics Co Ltd (300782.SZ)	U (01/11/2021)	Rmb60.11
NAURA Technology Group Co Ltd (002371.SZ)	O (11/06/2023)	Rmb591.00
Lucas Wang		
AllRing Tech Co. (6187.TWO)	O (09/23/2025)	NT$1,600.00
FOCI Fiber Optic Communications Inc (3363.TWO)	E (10/09/2026)	NT$547.00
Hon Precision (7769.TW)	O (04/17/2026)	NT$6,070.00
King Yuan Electronics Co Ltd (2449.TW)	O (03/03/2023)	NT$305.00
MPI Corporation (6223.TWO)	O (04/17/2026)	NT$5,940.00
Tiffany Yeh		
Himax Technologies Inc (HIMX.O)	E (02/04/2026)	US$12.78
Silicon Motion (SIMO.O)	O (05/06/2024)	US$246.95
WinWay Technology Co Ltd (6515.TW)	O (04/17/2026)	NT$5,700.00

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.

© 2026 Morgan Stanley

First Read

Alchip Technologies

Better allocation into 2027E; N2 project on track

Equities

Taiwan
Semiconductors

12-month rating	**Buy**
12m price target	**NT$6,600.00**
Price (12 Oct 2026)	**NT$4,395**

RIC: 3661.TW **BBG:** 3661 TT

Trading data and key metrics

52-wk range	NT$5,375.00-2,485.00
Market cap.	NT$357b/US$11.2b
Shares o/s	81.3m (ORD)
Free float	98%
Avg. daily volume ('000)	2,419
Avg. daily value (m)	NT$9,083.3
Common s/h equity (12/26E)	NT$66.0b
P/BV (12/26E)	5.5x
Net debt to EBITDA (12/26E)	NM

EPS (UBS, diluted) (TWD)

	From	To	% ch	Cons.
12/26E	159.82	141.48	-11	141.48
12/27E	221.11	225.64	2	187.91
12/28E	276.82	291.65	5	262.18

Jerry Su
Analyst
jerry.su@ubs.com
+886-28-722 7306

Annie Chen
Associate Analyst
annie.chen@ubs.com
+886-2-8722 7281

Q3 sales up 243% QoQ on Trainium 3 ramp

Alchip reported September sales of NT$9,979 mn, up 14% MoM and 496% YoY, concluding Q326 sales of NT$26,188 mn for +243% QoQ and 299% YoY, slightly ahead of consensus, although below UBSe as we assumed a higher wafer allocation for the Trainium 3 project, while sales contribution for its Chinese automotive customer and NRE might also have came in lower than our expectation. We now expect Alchip's Q326 GM to arrive at 20.1%, down from 34.8% in Q226 on less favourable mix, leading to OPM of 15.9%. We forecast Q326 EPS to grow 132% QoQ to NT$46.34.

Trainium 3 to sustain longer with potential upside; Trainium 4 tape-out on track

We expect Alchip's Trainium 3 monthly shipment run-rate to retain at current level into H127F, and could extend into H227F with potential better allocation for Alchip with a similar margin profile, subject to the final negotiation with its customer. We think Trainium 3 deployments could stretch a little bit longer with a limited gap (vs prior ~3 months) before the next-generation Trainium 4 launch given the 3nm capacity tightness. We continue to forecast overall Trainium 3 shipments at ~1.8mn units for 2026 and will increase to ~2.8 mn units in 2027E, up 55% YoY, although Alchip's Trainium 3 shipment could outgrow at 80%+ YoY given potential better allocation. For the Trainium 4 project (2nm node), management noted it remains on track for tape-out before year-end and still targets mass production by late-2027 or early 2028. Alchip continues to believe it could enjoy a higher value-add given the increasing design complexity on chiplet and higher wafer pricing for the 2nm node, while GM could retain at a similar level compared with the current N3 project.

Engaging new customers for diversification

Alchip noted it remains in talks with another major CSP customer for potential adoption of COT (customer-owned tooling) business model in the future for further cost saving. We believe Alchip's competitive pricing, strength for faster turnaround, and advanced packaging knowledge put Alchip in a good position to capture this opportunity, if the customer decides to change the business model. Nevertheless, we do not expect meaningful revenue contribution in 2027-28E from the new CSP customer. On the other hand, we believe Alchip is in talks with other HPC customers for mid-size physical design projects and this is likely to become a new revenue stream from 2027E. We estimate it could book US$200-300mn revenue in 2027E from the new HPC customer.

Valuation: Maintain Buy with NT$6,600 price target

We lower our 2026E EPS by 11% primarily on lower Trainium 3 and automotive contribution. We raise 2027E/28E EPS by 2%/5% to reflect revenue contribution from new HPC customer, offset by higher opex. We keep our PT unchanged at NT$6,600, based on the same 30x 2027E EPS (vs. a 15-50x PE range). Stock is trading at ~20x 2027 P/E, below its peers and past five year average of 30x.

Highlights (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenues	30,482	51,969	30,926	70,029	129,909	174,676	233,836	255,099
EBIT (UBS)	3,761	6,496	5,017	12,718	21,597	27,969	35,404	37,613
Net earnings (UBS)	3,325	6,446	5,598	11,567	18,478	23,884	29,325	31,860
EPS (UBS, diluted) (NT$)	45.64	81.61	69.22	141.48	225.64	291.65	358.09	389.05
DPS (net) (NT$)	12.61	22.78	36.38	34.18	70.62	112.82	145.82	179.05
Net (debt) / cash	10,160	27,044	32,627	17,456	29,003	11,648	35,717	16,971
Profitability/valuation	**12/23**	**12/24**	**12/25**	**12/26E**	**12/27E**	**12/28E**	**12/29E**	**12/30E**
EBIT (UBS) margin %	12.3	12.5	16.2	18.2	16.6	16.0	15.1	14.7
ROIC (EBIT) %	46.9	64.5	49.1	44.9	44.0	42.6	45.2	40.5
EV/EBITDA (UBS core) x	23.7	22.3	28.1	21.2	13.6	10.9	8.7	8.1
P/E (UBS, diluted) x	43.3	34.8	46.0	31.1	19.5	15.1	12.3	11.3
Equity FCF (UBS) yield %	4.2	0.3	4.9	(6.7)	5.0	(2.1)	10.2	(1.0)
Dividend yield (net) %	0.6	0.8	1.1	0.8	1.6	2.6	3.3	4.1

Source: Company accounts, LSEG Eikon, UBS estimates. Metrics marked as (UBS) have had analyst adjustments applied. Valuations: based on an average share price that year, (E): based on a share price of NT$ 4,395.00 on 12-Oct-2026

Figure 1: UBS vs. VA Consensus Q326/Q426E and 2025-2028E

	2Q26	3Q26E			4Q26E			2025	2026E			2027E			2028E		
(NT$m)	Actual	UBSe	UBSe prior	VA	UBSe	UBSe prior	VA	Actual	UBSe	UBSe prior	VA	UBSe	UBSe prior	VA	UBSe	UBSe prior	VA
Revenue	**7,640**	**26,188**	**30,866**	**25,735**	**32,014**	**38,076**	**34,481**	**30,926**	**70,029**	**80,768**	**72,143**	**129,909**	**131,483**	**116,160**	**174,676**	**170,145**	**173,971**
- QoQ / YoY	82.5%	242.8%	304.0%	236.9%	22.2%	23.4%	34.0%	-40.5%	126.4%	161.2%	133.3%	85.5%	62.8%	61.0%	34.5%	29.4%	49.8%
Gross profit	**2,662**	**5,259**	**6,371**	**5,303**	**6,678**	**7,695**	**6,993**	**8,154**	**16,700**	**18,829**	**16,994**	**26,259**	**26,761**	**23,829**	**33,039**	**32,897**	**33,523**
- Gross margin	34.8%	20.1%	20.6%	20.6%	20.9%	20.2%	20.3%	26.4%	23.8%	23.3%	23.6%	20.2%	20.4%	20.5%	18.9%	19.3%	19.3%
Operating profit	**1,647**	**4,174**	**5,286**	**4,090**	**5,530**	**6,546**	**5,686**	**5,017**	**12,718**	**14,847**	**12,720**	**21,333**	**21,933**	**18,463**	**27,335**	**27,358**	**26,124**
- OP margin	21.6%	15.9%	17.1%	15.9%	17.3%	17.2%	16.5%	16.2%	18.2%	18.4%	17.6%	16.4%	16.7%	15.9%	15.6%	16.1%	15.0%
Non-Op	**388**	**510**	**579**	**400**	**279**	**278**	**333**	**1,703**	**1,594**	**1,624**	**1,561**	**1,202**	**1,359**	**1,304**	**1,363**	**1,804**	**1,405**
Pre-tax profit	**2,035**	**4,684**	**5,865**	**4,490**	**5,809**	**6,824**	**6,020**	**6,720**	**14,312**	**16,509**	**14,275**	**22,535**	**23,292**	**19,903**	**28,697**	**29,162**	**27,812**
Tax	**397**	**890**	**1,114**	**831**	**1,104**	**1,297**	**1,096**	**1,124**	**2,748**	**3,165**	**2,662**	**4,282**	**4,426**	**3,731**	**5,453**	**5,541**	**5,098**
- Tax rate	-20%	-19%	-19%	-18%	-19%	-19%	-18%	-17%	-19%	-19%	-19%	-19%	-19%	-19%	-19%	-19%	-18%
Net profit	**1,639**	**3,795**	**4,751**	**3,648**	**4,706**	**5,528**	**4,906**	**5,598**	**11,567**	**13,346**	**11,588**	**18,256**	**18,869**	**16,085**	**23,247**	**23,624**	**22,577**
EPS (NT$)	**20.01**	**46.34**	**55.68**	**42.58**	**57.46**	**64.78**	**59.83**	**69.22**	**141.48**	**159.82**	**137.88**	**222.92**	**221.11**	**189.21**	**283.88**	**276.82**	**270.68**

Source: Company data, UBSe, Visible Alpha

Figure 2: Alchip annual P&L summary

Ticker		3661.TW	Current Price (NT$)			$4,395
Shares outstanding (mn)		81.9	Mkt cap (NT$ mn)			$359,915
Year	**2025**	**2026E**	**2027E**	**2028E**	**2029E**	**2030E**
Revenues (NT$ mn)	**$30,926**	**$70,029**	**$129,909**	**$174,676**	**$233,836**	**$255,099**
YoY Growth (%)		**126.4%**	**85.5%**	**34.5%**	**33.9%**	**9.1%**
Gross profit (NT$mn)	$8,154	$16,700	$26,259	$33,039	$41,783	$44,656
GM (%)	**26.4%**	**23.8%**	**20.2%**	**18.9%**	**17.9%**	**17.5%**
Operating profit (NT$mn)	$5,017	$12,718	$21,333	$27,335	$35,303	$37,613
OPM (%)	**16.2%**	**18.2%**	**16.4%**	**15.6%**	**15.1%**	**14.7%**
Net profit (NT$ mn)	$5,598	$11,567	$18,256	$23,247	$29,111	$31,739
EPS (NT$)	**$69.22**	**$141.48**	**$222.92**	**$283.88**	**$355.48**	**$387.58**
EPS growth (%)		104.4%	57.6%	27.3%	25.2%	9.0%
P/E (x)	**63.5**	**31.1**	**19.7**	**15.5**	**12.4**	**11.3**
Dividend yield (%)	0.8%	0.8%	1.6%	2.5%	3.2%	4.0%
P/B (x)	**8.7**	**5.5**	**4.6**	**3.9**	**3.3**	**2.8**
ROE (%)	13.7%	17.5%	23.3%	25.1%	26.4%	24.9%

Source: UBSe, Company Data

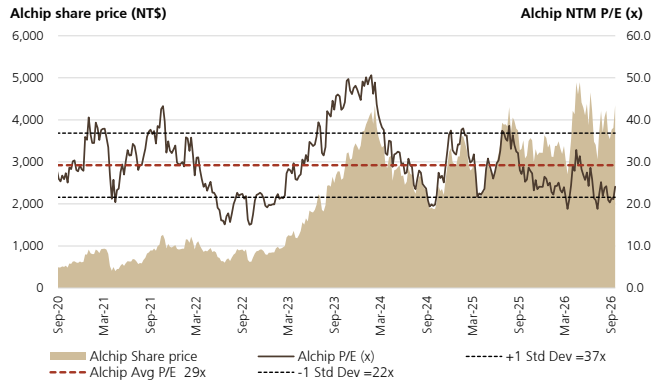
Figure 3: Alchip Annual and Quarterly forecast summary

Quarters	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26E	4Q26E
Revenues (NT$ mn)	**$10,485**	**$9,144**	**$6,570**	**$4,727**	**$4,187**	**$7,640**	**$26,188**	**$32,014**
-- QoQ Growth (%)	*-19.8%*	*-12.8%*	*-28.2%*	*-28.0%*	*-11.4%*	*82.5%*	*242.8%*	*22.2%*
-- YoY Growth (%)	*0.0%*	*-32.7%*	*-55.7%*	*-63.8%*	*-60.1%*	*-16.5%*	*298.6%*	*577.2%*
Gross profit	**$2,429**	**$1,887**	**$1,841**	**$1,997**	**$2,101**	**$2,662**	**$5,259**	**$6,678**
-- Gross Margin (%)	23.2%	20.6%	28.0%	42.3%	50.2%	34.8%	20.1%	20.9%
Operating Profit	**$1,494**	**$1,150**	**$1,157**	**$1,215**	**$1,368**	**$1,647**	**$4,174**	**$5,530**
-- OP Margin (%)	14.3%	12.6%	17.6%	25.7%	32.7%	21.6%	15.9%	17.3%
Non-op	**$299**	**$449**	**$433**	**$522**	**$416**	**$388**	**$510**	**$279**
Pre-tax profit (NT$ mn)	**$1,793**	**$1,599**	**$1,591**	**$1,737**	**$1,784**	**$2,035**	**$4,684**	**$5,809**
Tax	**$332**	**$277**	**$265**	**$250**	**$357**	**$397**	**$890**	**$1,104**
-- tax rate (%)	18.5%	17.3%	16.6%	14.4%	20.0%	19.5%	19.0%	19.0%
Net profit (NT$ mn)	**$1,464**	**$1,324**	**$1,328**	**$1,483**	**$1,428**	**$1,639**	**$3,795**	**$4,706**
-- QoQ Growth (%)	*-20.3%*	*-9.6%*	*0.3%*	*11.7%*	*-3.7%*	*14.8%*	*131.6%*	*24.0%*
-- YoY Growth (%)	*19.0%*	*-16.6%*	*-25.9%*	*-19.2%*	*-2.4%*	*23.8%*	*185.8%*	*217.2%*
EPS (NT$)	**$18.13**	**$16.37**	**$16.40**	**$18.33**	**$17.55**	**$20.01**	**$46.34**	**$57.46**

2024	2025	2026E	2027E	2028E
$51,969	**$30,926**	**$70,029**	**$129,909**	**$174,676**
70.5%	*-40.5%*	*126.4%*	*85.5%*	*34.5%*
$10,201	**$8,154**	**$16,700**	**$26,259**	**$33,039**
19.6%	26.4%	23.8%	20.2%	18.9%
$6,496	**$5,017**	**$12,718**	**$21,333**	**$27,335**
12.5%	16.2%	18.2%	16.4%	15.6%
$1,312	**$1,703**	**$1,594**	**$1,202**	**$1,363**
$7,808	**$6,720**	**$14,312**	**$22,535**	**$28,697**
$1,362	**$1,124**	**$2,748**	**$4,282**	**$5,453**
17.4%	16.7%	19.2%	19.0%	19.0%
$6,446	**$5,598**	**$11,567**	**$18,256**	**$23,247**
93.8%	*-13.1%*	*106.6%*	*57.8%*	*27.3%*
$81.61	**$69.22**	**$141.48**	**$222.92**	**$283.88**

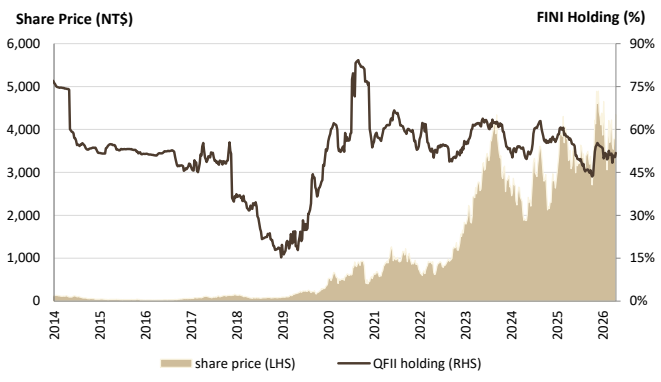
Source: UBSe, Company data

Figure 4: Alchip historical PE range



Source: LSEG

Figure 5: QFII ownership



Source: LSEG, Company data, TEJ

Forecast returns

Forecast price appreciation	50.2%
Forecast dividend yield	0.8%
Forecast stock return	50.9%
Market return assumption	6.3%
Forecast excess return	44.6%

Company Description

Alchip was founded in 2002 by a group of Cadence managers with 20+ years of experience in the semiconductor industry. It is an IC design and production service provider, and management believes it can leverage third-party IP to enable customized chipset solutions for customers cost effectively and fast to market. The company was funded by Cisco Investment and Softbank Investment with VC funding in early 2000, then had its IPO on the emerging board in Taiwan in 2010 and the main board in 2014.

Valuation Method and Risk Statement

We value Alchip using PE methodology.
Downside risks include slower-than-expected advanced node project diversification and ramp, and growing competition in advanced projects from other IC design service companies.

Quantitative Research Review

UBS Global Research publishes a quantitative assessment of its analysts' responses to certain questions about the likelihood of an occurrence of a number of short term factors in a product known as the 'Quantitative Research Review'. The views for this month can be found below. Views contained in this assessment on a particular stock reflect only the views on those short term factors which are a different timeframe to the 12-month timeframe reflected in any equity rating set out in this note. For previous responses please make reference to (i) previous UBS Global Research reports; and (ii) where no applicable research report was published that month, the Quantitative Research Review which can be found at https://neo.ubs.com/quantitative, or contact your UBS sales representative for access to the report or the Quantitative Research Team on ubs-quant-answers@ubs.com. A consolidated report which contains all responses is also available and again you should contact your UBS sales representative for details and pricing or the Quantitative Research Team on the email above.

Alchip Technologies

Question	**Response**
1. Is the industry structure facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting worse, 3 = no change, 5 = getting better, N/A = no view)	3
2. Is the regulatory/government environment facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting tougher 3 = no change, 5 = getting better, N/A = no view)	3
3. Over the last 3-6 months in broad terms have things been improving/no change/getting worse for this stock? Rate on a scale of 1-5 (1 = getting a lot worse, 3 = not much change, 5 = getting a lot better, N/A = no view)	4
4. Relative to the current CONSENSUS EPS forecast, is the next company EPS update likely to lead to: (1 = negative surprise vs consensus, 3 = in-line with consensus, 5 = positive surprise vs consensus expectations, N/A = no view)	3
5. What's driving the difference?	
6. Relative to YOUR current earnings forecast, is there relatively greater risk at the next earnings result of:(1 = downside skew risk to earnings, 3 = equal upside or downside risk to earnings, 5 = upside skew risk to earnings, N/A = no view)	3
7. What's driving the difference?	
8. Is there an upcoming catalyst for the company over the next three months?	
9. Is there an actual or approximate date for the catalyst?	
10. Is the catalyst date an actual or approximate date?	
11. What is the catalyst?	

Required Disclosures

This document has been prepared by UBS Securities Pte. Ltd., Taipei Branch, an affiliate of UBS AG. UBS AG, its subsidiaries, branches and affiliates, including former Credit Suisse AG and its subsidiaries, branches and affiliates are referred to herein as "UBS".

For information on the ways in which UBS manages conflicts and maintains independence of its UBS Global Research product; historical performance information; certain additional disclosures concerning UBS Global Research recommendations; and terms and conditions for certain third party data used in research report, please visit https://www.ubs.com/disclosures. Unless otherwise indicated, information and data in this report are based on company disclosures including but not limited to annual, interim, quarterly reports and other company announcements. The figures contained in performance charts refer to the past; past performance is not a reliable indicator of future results. Additional information will be made available upon request. UBS Securities Co. Limited is licensed to conduct securities investment consultancy businesses by the China Securities Regulatory Commission. UBS acts or may act as principal in the debt securities (or in related derivatives) that may be the subject of this report. This recommendation was finalized on: 12 October 2026 04:09 PM GMT. UBS has designated certain UBS Global Research department members as Derivatives Research Analysts where those department members publish research principally on the analysis of the price or market for a derivative, and provide information reasonably sufficient upon which to base a decision to enter into a derivatives transaction. Where Derivatives Research Analysts co-author research reports with Equity Research Analysts or Economists, the Derivatives Research Analyst is responsible for the derivatives investment views, forecasts, and/or recommendations. **Quantitative Research Review:** UBS Global Research publishes a quantitative assessment of its analysts' responses to certain questions about the likelihood of an occurrence of a number of short term factors in a product known as the 'Quantitative Research Review'. Views contained in this assessment on a particular stock reflect only the views on those short term factors which are a different timeframe to the 12-month timeframe reflected in any equity rating set out in this note. For the latest responses, please see the Quantitative Research Review Addendum at the back of this report, where applicable. For previous responses please make reference to (i) previous UBS Global Research reports; and (ii) where no applicable research report was published that month, the Quantitative Research Review which can be found at https://neo.ubs.com/quantitative, or contact your UBS sales representative for access to the report or the Quantitative Research Team on ubs-quant-answers@ubs.com. A consolidated report which contains all responses is also available and again you should contact your UBS sales representative for details and pricing or the Quantitative Research team on the email above.

Analyst Certification:
Each research analyst primarily responsible for the content of this research report, in whole or in part, certifies that with respect to each security or issuer that the analyst covered in this report: (1) all of the views expressed accurately reflect his or her personal views about those securities or issuers and were prepared in an independent manner, including with respect to UBS, and (2) no part of his or her compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed by that research analyst in the research report.

UBS Global Research: Global Equity Rating Definitions

12-Month Rating	Definition	Coverage[1]	IB Services[2]
Buy	FSR is > 6% above the MRA.	55%	24%
Neutral	FSR is between -6% and 6% of the MRA.	39%	20%
Sell	FSR is > 6% below the MRA.	6%	21%

Source: UBS. Rating allocations are as of 30 September 2026.
1:Percentage of companies under coverage globally within the 12-month rating category.
2:Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

KEY DEFINITIONS:Forecast Stock Return (FSR) is defined as expected percentage price appreciation plus gross dividend yield over the next 12 months. In some cases, this yield may be based on accrued dividends. **Market Return Assumption (MRA)** is defined as the one-year local market interest rate plus 5% (a proxy for, and not a forecast of, the equity risk premium). **Under Review (UR)** Stocks may be flagged as UR by the analyst, indicating that the stock's price target and/or rating are subject to possible change in the near term, usually in response to an event that may affect the investment case or valuation. **Equity Price Targets** have an investment horizon of 12 months.

EXCEPTIONS AND SPECIAL CASES:UK and European Investment Fund ratings and definitions are: Buy: Positive on factors such as structure, management, performance record, discount; **Neutral:** Neutral on factors such as structure, management, performance record, discount; **Sell:** Negative on factors such as structure, management, performance record, discount. **Core Banding Exceptions (CBE):** Exceptions to the standard +/-6% bands may be granted by the Investment Review Consultation (IRC). Factors considered by the IRC include the stock's volatility and the credit spread of the respective company's debt. As a result, stocks deemed to be very high or low risk may be subject to higher or lower bands as they relate to the rating. When such exceptions apply, they will be identified in the Company Disclosures table in the relevant research piece.

Research analysts contributing to this report who are employed by any non-US affiliate of UBS Securities LLC are not registered/ qualified as research analysts with FINRA. Such analysts may not be associated persons of UBS Securities LLC and therefore are not subject to the FINRA restrictions on communications with a subject company, public appearances, and trading securities held by a research analyst account. The name of each affiliate and analyst employed by that affiliate contributing to this report, if any, follows.

UBS Securities Pte. Ltd., Taipei Branch: Annie Chen, Jerry Su.

Company Disclosures

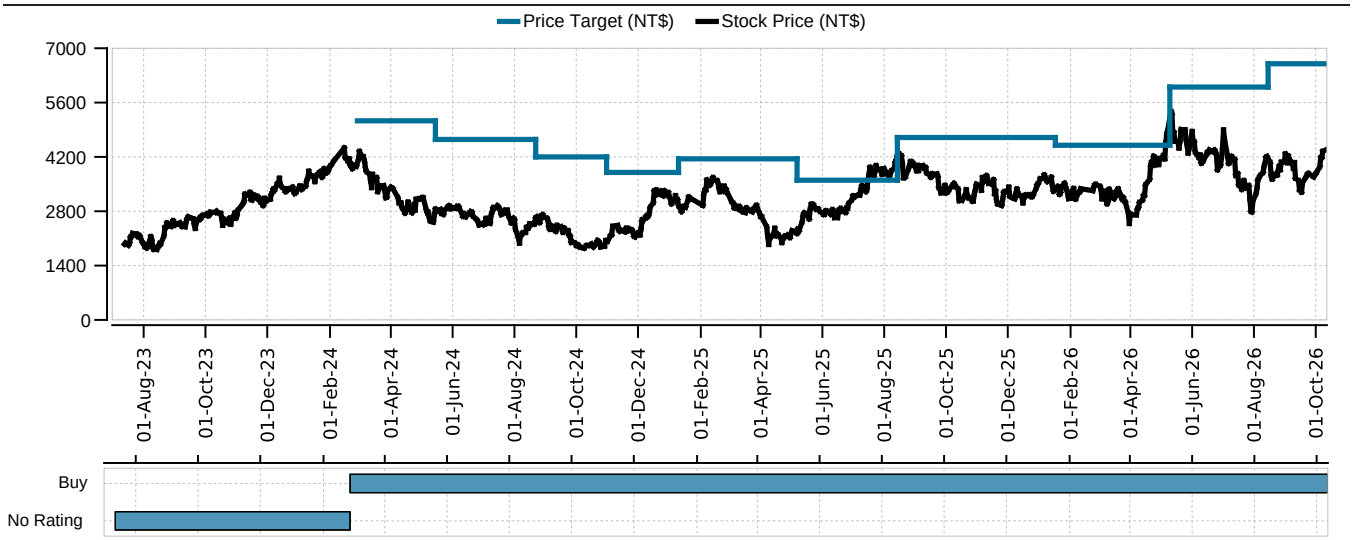
Company Name	Reuters	12-month rating	Price	Price date
Alchip Technologies[13,28]	3661.TW	Buy	NT$4,395	12 Oct 2026

Source: UBS Global Research; LSEG Eikon. All prices as of local market close. Ratings in this table are the most current published ratings prior to this report. They may be more recent than the stock pricing date.

13. UBS beneficially owned 1% or more of a class of this company's common equity securities as of last month's end (or the prior month's end if this report is dated less than 10 days after the most recent month's end).
28. UBS holds a long position of 0.5% or more of the listed shares of this company.

Unless otherwise indicated, please refer to the Valuation and Risk sections within the body of this report. For a complete set of disclosure statements associated with the companies discussed in this report, including information on valuation and risk, please contact UBS Securities LLC, 11 Madison Avenue, New York, NY 10010, USA, Attention: Investment Research.

Alchip Technologies (NT$)



Date	Stock Price (NT$)	Price Target (NT$)	Rating
2023-07-12	1880.00	-	No Rating
2024-02-27	3960.00	5130.00	Buy
2024-05-14	2710.00	4650.00	Buy
2024-08-21	2475.00	4200.00	Buy
2024-10-30	2040.00	3800.00	Buy
2025-01-09	3100.00	4150.00	Buy
2025-05-06	2275.00	3600.00	Buy
2025-08-13	4120.00	4700.00	Buy
2026-01-16	3340.00	4500.00	Buy
2026-05-09	4890.00	6000.00	Buy
2026-08-14	4210.00	6600.00	Buy

Source: UBS Global Research; LSEG Eikon as of 12-Oct-2026. All prices as of local market close. Ratings as of date shown.

The Disclaimer relevant to Global Wealth Management clients follows the Global Research Disclaimer.

UBS Global Research Disclaimer

This document has been prepared by UBS Securities Pte. Ltd., Taipei Branch, an affiliate of UBS AG. UBS AG, its subsidiaries, branches and affiliates, including former Credit Suisse AG and its subsidiaries, branches and affiliates are referred to herein as "UBS".

Any opinions expressed in this document may change without notice and are only current as of the date of publication. Different areas, groups, and personnel within UBS may produce and distribute separate research products independently of each other. For example, research publications from **UBS CIO** are produced by UBS Global Wealth Management. **UBS Global Research** is produced by UBS Investment Bank. Research methodologies and rating systems of each separate research organization may differ, for example, in terms of investment recommendations, investment horizon, model assumptions, and valuation methods. As a consequence, except for certain economic forecasts (for which UBS CIO and UBS Global Research may collaborate), investment recommendations, ratings, price targets, and valuations provided by each of the separate research organizations may be different, or inconsistent. You should refer to each relevant research product for the details as to their methodologies and rating system. Not all clients may have access to all products from every organization. Each research product is subject to the policies and procedures of the organization that produces it.

This document is provided solely to recipients who are expressly authorized by UBS to receive it. If you are not so authorized you must immediately destroy the document.

UBS Global Research is provided to our clients through UBS Neo, and in certain instances, UBS.com and any other system or distribution method specifically identified in one or more communications distributed through UBS Neo or UBS.com (each a system) as an approved means for distributing UBS Global Research. It may also be made available through third party vendors and distributed by UBS and/or third parties via e-mail or alternative electronic means.

All UBS Global Research is available on UBS Neo. Please contact your UBS sales representative if you wish to discuss your access to UBS Neo. Where UBS Global Research refers to "UBS Evidence Lab Inside" or has made use of data provided by UBS Evidence Lab and you would like to access that data please contact your UBS sales representative. UBS Evidence Lab data is available on UBS Neo. The level and types of services provided by UBS Global Research and UBS Evidence Lab to a client may vary depending upon various factors such as a client's individual preferences as to the frequency and manner of receiving communications, a client's risk profile and investment focus and perspective (e.g., market wide, sector specific, long-term, short-term, etc.), the size and scope of the overall client relationship with UBS Global Research and UBS Evidence Lab and legal and regulatory constraints. UBS HOLT and UBS Pharma Values are offerings of UBS Global Research. HOLT Lens is a corporate performance platform offering that provides an objective accounting-led framework for comparing and valuing companies and is available to clients of UBS Global Research; for further details and pricing please contact your UBS Sales representative. In particular, HOLT has a variety of warranted prices based on the scenario chosen; please mail UBS Securities LLC, 11 Madison Avenue, New York, NY 10010, USA, Attention: Investment Research, if you are interested in the warranted price on a particular company, again subject to commercial considerations. UBS Pharma Values is an analytical tool that involves the creation of a number of individual product net present value calculations, based on published forecasts of sales for pharmaceuticals, and is available to clients of UBS Global Research; for further details and pricing please contact your UBS Sales representative. For all other specific disclaimers, please see https://www.ubs.com/disclosures.

When you receive UBS Global Research through a system, your access and/or use of such UBS Global Research is subject to this UBS Global Research Disclaimer and to the UBS Neo Platform Use Agreement (the "Neo Terms") together with any other relevant terms of use governing the applicable System.

When you receive UBS Global Research via a third party vendor, e-mail or other electronic means, you agree that use shall be subject to this UBS Global Research Disclaimer, the Neo Terms and where applicable the UBS Investment Bank terms of business (https://www.ubs.com/global/en/investment-bank/regulatory.html) and to UBS's Terms of Use/Disclaimer (https://www.ubs.com/global/en/legalinfo2/disclaimer.html). In addition, you consent to UBS processing your personal data and using cookies in accordance with our Privacy Statement (https://www.ubs.com/global/en/legalinfo2/privacy.html) and cookie notice (https://www.ubs.com/global/en/legal/privacy/users.html).

If you receive UBS Global Research, whether through a System or by any other means, you agree that you shall not copy, revise, amend, create a derivative work, provide to any third party, or in any way commercially exploit any UBS research provided via UBS Global Research or otherwise, and that you shall not extract data from any research or estimates provided to you via UBS Global Research or otherwise, without the prior written consent of UBS. You agree not to use UBS Global Research in any artificial intelligence system, without the prior written consent of UBS.

In certain circumstances you may receive UBS Global Research otherwise than in the capacity of a client of UBS and you understand and agree that under these circumstances (i) the UBS Global Research is provided to you for information purposes only; (ii) for the purposes of receiving it you are not intended to be and will not be treated as a "client" of UBS for any legal or regulatory purpose; (iii) the UBS Global Research must not be relied on or acted upon for any purpose; and (iv) such content is subject to the relevant disclaimers that follow.

This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject UBS to any registration or licensing requirement within such jurisdiction.

This document is a general communication and is educational in nature; it is not an advertisement nor is it a solicitation or an offer to buy or sell any financial instruments or to participate in any particular trading strategy. Nothing in this document constitutes a representation that any investment strategy or recommendation is suitable or appropriate to an investor's individual circumstances or otherwise constitutes a personal recommendation. By providing this document, none of UBS or its representatives has any responsibility or authority to provide or have provided investment advice in a fiduciary capacity or otherwise. Investments involve risks, and investors should exercise prudence and their own judgment in making their investment decisions. None of UBS or its representatives is suggesting that the recipient or any other person take a specific course of action or any action at all. The recipient should carefully read this document in its entirety and not draw inferences or conclusions from the rating alone. By receiving this document, the recipient acknowledges and agrees with the intended purpose described above and further disclaims any expectation or belief that the information constitutes investment advice to the recipient or otherwise purports to meet the investment objectives of the recipient. The financial instruments described in the document may not be eligible for sale in all jurisdictions or to certain categories of investors.

Options, structured derivative products and futures (including OTC derivatives) are not suitable for all investors. Trading in these instruments is considered risky and may be appropriate only for sophisticated investors. Prior to buying or selling an option, and for the complete risks relating to options, you must receive a copy of "The Characteristics and Risks of Standardized Options." You may read the document at https://www.theocc.com/publications/risks/riskchap1.jsp or ask your salesperson for a copy. Various theoretical explanations of the risks associated with these instruments have been published. Supporting documentation for any claims, comparisons, recommendations, statistics or other technical data will be supplied upon request. Past performance is not necessarily indicative of future results. Transaction costs may be significant in option strategies calling for multiple purchases and sales of options, such as spreads and straddles. Because of the importance of tax considerations to many options transactions, the investor considering options should consult with his/her tax advisor as to how taxes affect the outcome of contemplated options transactions.

Mortgage and asset-backed securities may involve a high degree of risk and may be highly volatile in response to fluctuations in interest rates or other market conditions. Foreign currency rates of exchange may adversely affect the value, price or income of any security or related instrument referred to in the document. For investment advice, trade execution or other enquiries, clients should contact their local sales representative.

UBS notes that no globally accepted framework or definition (legal, regulatory or otherwise) currently exists, nor is there a market consensus as to what constitutes an "ESG" (Environmental, Social or Governance) or an equivalent-label, or as to what precise attributes are required for the Information (as defined below) to be defined as ESG or equivalently-labelled. Any information, data or other content including from a third party source contained, referred to herein or used for whatsoever purpose by UBS or a third party ("Information"), in relation to any actual or potential ESG objective, issue or consideration is not intended to be relied upon for ESG classification, regulatory regime or industry initiative purposes ("ESG Regimes"). Nothing in these materials is intended to convey, suggest or indicate that UBS considers or represents any product, service, person or body mentioned in these materials as meeting or qualifying for any ESG classification, labelling or similar standards that may exist under the ESG Regimes. UBS has not conducted any assessment of compliance with ESG Regimes. Parties are reminded to make their own assessments for these purposes.

The value of any investment or income may go down as well as up, and investors may not get back the full (or any) amount invested. Past performance is not necessarily a guide to future performance. Neither UBS nor any of its directors, employees or agents accepts any liability for any loss (including investment loss) or damage arising out of the use of all or any of the Information.

Prior to making any investment or financial decisions, any recipient of this document or the information should take steps to understand the risk and return of the investment and seek individualized advice from his or her personal financial, legal, tax and other professional advisors that takes into account all the particular facts and circumstances of his or her investment objectives.

Any prices stated in this document are for information purposes only and do not represent valuations for individual securities or other financial instruments. There is no representation that any transaction can or could have been effected at those prices, and any prices do not necessarily reflect UBS's internal books and records or theoretical model-based valuations and may be based on certain assumptions. Different assumptions by UBS or any other source may yield substantially different results.

No representation or warranty, either expressed or implied, is provided in relation to the accuracy, completeness or reliability of the information contained in any materials to which this document relates (the "Information"), except with respect to Information concerning UBS. The Information is not intended to be a

complete statement or summary of the securities, markets or developments referred to in the document. UBS does not undertake to update or keep current the Information. Any statements contained in this report attributed to a third party represent UBS's interpretation of the data, information and/or opinions provided by that third party either publicly or through a subscription service, and such use and interpretation have not been reviewed by the third party. In no circumstances may this document or any of the Information (including any forecast, value, index or other calculated amount ("Values")) be used for any of the following purposes:

(i) valuation or accounting purposes;

(ii) to determine the amounts due or payable, the price or the value of any financial instrument or financial contract; or

(iii) to measure the performance of any financial instrument including, without limitation, for the purpose of tracking the return or performance of any Value or of defining the asset allocation of portfolio or of computing performance fees.

By receiving this document and the Information you will be deemed to represent and warrant to UBS that you will not use this document or any of the Information for any of the above purposes or otherwise rely upon this document or any of the Information.

UBS has policies and procedures, which include, without limitation, independence policies and permanent information barriers, that are intended, and upon which UBS relies, to manage potential conflicts of interest and control the flow of information within divisions of UBS and among its subsidiaries, branches and affiliates. UBS does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For further information on the ways in which UBS Global Research manages conflicts and maintains independence of its research products, historical performance information and certain additional disclosures concerning UBS Global Research recommendations, please visit https://www.ubs.com/disclosures.

UBS Global Research will initiate, update and cease coverage solely at the discretion of UBS Global Research Management, which will also have sole discretion on the timing and frequency of any published research product. The analysis contained in this document is based on numerous assumptions. All material information in relation to published research reports, such as valuation methodology, risk statements, underlying assumptions (including sensitivity analysis of those assumptions), ratings history etc. as required by the Market Abuse Regulation, can be found on UBS Neo. Different assumptions could result in materially different results.

UBS Global Research may utilise artificial intelligence tools ("AI Tools") in the preparation of this document. Notwithstanding any such use of AI Tools, this document has undergone human review.

The analyst(s) responsible for the preparation of this document may interact with trading desk personnel, sales personnel and other parties for the purpose of gathering, applying and interpreting market information. UBS relies on information barriers to control the flow of information contained in one or more areas within UBS into other areas, units, groups or affiliates of UBS. The compensation of the analyst who prepared this document is determined exclusively by UBS Global Research management and senior management (not including investment banking). Analyst compensation is not based on investment banking revenues; however, compensation may relate to the revenues of UBS and/or its divisions as a whole, of which investment banking, sales and trading are a part, and UBS as a whole.

For financial instruments admitted to trading on an EU regulated market: UBS (excluding UBS Securities LLC) acts as a market maker or liquidity provider (in accordance with the interpretation of these terms under English law or, if not carried out by UBS in the UK the law of the relevant jurisdiction in which UBS determines it carries out the activity) in the financial instruments of the issuer save that where the activity of liquidity provider is carried out in accordance with the definition given to it by the laws and regulations of any other EU jurisdictions, such information is separately disclosed in this document. For financial instruments admitted to trading on a non-EU regulated market: UBS may act as a market maker save that where this activity is carried out in the US in accordance with the definition given to it by the relevant laws and regulations, such activity will be specifically disclosed in this document. UBS may have issued a warrant the value of which is based on one or more of the financial instruments referred to in the document. UBS and its affiliates and employees may have long or short positions, trade as principal and buy and sell in instruments or derivatives identified herein; such transactions or positions may be inconsistent with the opinions expressed in this document.

Within the past 12 months UBS may have received or provided investment services and activities or ancillary services as per MiFID II which may have given rise to a payment or promise of a payment in relation to these services from or to this company.

Please note that all transactions conducted by UBS are consistent with sanctions regulations imposed by Switzerland, the European Union, the United Nations, the United Kingdom and the United States, per UBS' global sanctions policy. UBS opinion as to future investment worthiness assumes no new sanctions are imposed.

US persons are prohibited from purchasing or selling securities of certain companies designated as being associated with the Chinese Military in accordance with the amended US Presidential Executive Order 13959.

United Kingdom: This material is distributed by UBS AG, London Branch to persons who are eligible counterparties or professional clients. UBS AG, London Branch is authorised by the Prudential Regulation Authority and subject to regulation by the Financial Conduct Authority and limited regulation by the Prudential Regulation Authority. **Europe:** Except as otherwise specified herein, these materials are distributed by UBS Europe SE, a subsidiary of UBS AG, to persons who are eligible counterparties or professional clients (as detailed in the Bundesanstalt fur Finanzdienstleistungsaufsicht (BaFin) Rules and according to MIFID) and are only available to such persons. The information does not apply to, and should not be relied upon by, retail clients. UBS Europe SE is authorised by the European Central Bank (ECB) and regulated by the BaFin and the ECB. **Germany, Luxembourg, the Netherlands, Belgium and Ireland:** Where an analyst of UBS Europe SE has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE. In all cases it is distributed by UBS Europe SE and UBS AG, London Branch. **Turkey:** Distributed by UBS AG, London Branch. No information in this document is provided for the purpose of offering, marketing and sale by any means of any capital market instruments and services in the Republic of Turkey. Therefore, this document may not be considered as an offer made or to be made to residents of the Republic of Turkey. UBS AG, London Branch is not licensed by the Turkish Capital Market Board under the provisions of the Capital Market Law (Law No. 6362). Accordingly, neither this document nor any other offering material related to the instruments/services may be utilized in connection with providing any capital market services to persons within the Republic of Turkey without the prior approval of the Capital Market Board. However, according to article 15 (d) (ii) of the Decree No. 32, there is no restriction on the purchase or sale of the securities abroad by residents of the Republic of Turkey. **Poland:** Distributed by UBS Europe SE (spolka z ograniczona odpowiedzialnoscia) Oddzial w Polsce. Where an analyst of UBS Europe SE (spolka z ograniczona odpowiedzialnoscia) Oddzial w Polsce has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE (spolka z ograniczona odpowiedzialnoscia) Oddzial w Polsce. **Russia:** Prepared and distributed by UBS Bank (OOO). Should not be construed as an individual Investment Recommendation for the purpose of the Russian Law - Federal Law #39-FZ ON THE SECURITIES MARKET Articles 6.1-6.2.**Switzerland:** Distributed by UBS AG to persons who are institutional investors only. UBS AG is regulated by the Swiss Financial Market Supervisory Authority (FINMA). **Italy:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, Italy Branch. Where an analyst of UBS Europe SE, Italy Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, Italy Branch. **France:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, France Branch. Where an analyst of UBS Europe SE, France Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, France Branch. **Spain:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, Spain Branch. Where an analyst of UBS Europe SE, Spain Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, Spain Branch. **Sweden:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, Sweden Branch. Where an analyst of UBS Europe SE, Sweden Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, Sweden Branch. **South Africa:** Distributed by UBS South Africa (Pty) Limited (Registration No. 1995/011140/07), an authorised user of the JSE and an authorised Financial Services Provider (FSP 7328). **Saudi Arabia:** This document has been issued by UBS AG (and/or any of its subsidiaries, branches or affiliates), a public company limited by shares, incorporated in Switzerland with its registered offices at Aeschenvorstadt 1, CH-4051 Basel and Bahnhofstrasse 45, CH-8001 Zurich. This publication has been approved by UBS Saudi Arabia (a subsidiary of UBS AG), a Saudi closed joint stock company incorporated in the Kingdom of Saudi Arabia under commercial register number 1010257812 having its registered office at Tatweer Towers, P.O. Box 75724, Riyadh 11588, Kingdom of Saudi Arabia. UBS Saudi Arabia is authorized and regulated by the Capital Market Authority to conduct securities business under license number 08113-37. **UAE / Dubai:** The information distributed by UBS AG Dubai Branch is only intended for Professional Clients and/or Market Counterparties, as classified under the DFSA rulebook. No other person should act upon this material/communication. The information is not for further distribution within the United Arab Emirates. UBS AG Dubai Branch is regulated by the DFSA in the DIFC. UBS is not licensed to provide banking services in the UAE by the Central Bank of the UAE, nor is it licensed by the UAE Securities and Commodities Authority. **Israel:** This Material is distributed by UBS AG, London Branch. UBS Securities Israel Ltd is a licensed Investment Marketer that is supervised by the Israel Securities Authority (ISA). UBS AG, London Branch and its affiliates incorporated outside Israel are not licensed under the Israeli Advisory Law. UBS may engage among others in issuance of Financial Assets or in distribution of Financial Assets of other issuers for fees or other benefits. UBS AG, London Branch and its affiliates may prefer various Financial Assets to which they have or may have an Affiliation (as such term is defined under the Israeli Advisory Law). Nothing in this Material should be considered as investment advice under the Israeli Advisory Law. This Material is being issued only to and/or is directed only at persons who are Eligible Clients within the meaning of the Israeli Advisory Law, and this Material must not be furnished to, relied on or acted upon by any other persons. **United States:** Distributed to US persons by either UBS Securities LLC or by UBS Financial Services Inc., subsidiaries of UBS AG; or by a group, subsidiary or affiliate of UBS AG that is not registered as a US broker-dealer (a **'non-US affiliate'**) to major US institutional investors only. UBS Securities LLC or UBS Financial Services Inc. accepts responsibility for the content of a report prepared by another non-US affiliate when distributed to US persons by UBS Securities LLC or UBS Financial Services Inc. All transactions by a US person in the securities mentioned in this report must be effected through UBS Securities LLC or UBS Financial Services Inc., and not through a non-US affiliate. UBS Securities LLC is not acting as a municipal advisor to any municipal entity or obligated person within the meaning of Section 15B of the Securities Exchange Act (the "Municipal Advisor Rule"), and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of the Municipal Advisor Rule. **Canada:** Distributed by UBS Securities Canada Inc., a registered investment dealer in Canada

and a Member-Canadian Investor Protection Fund, or by another affiliate of UBS AG that is registered to conduct business in Canada or is otherwise exempt from registration. **Brazil:** Except as otherwise specified herein, this Material is prepared by UBS BB Corretora de Câmbio, Títulos e Valores Mobiliários S.A. (UBS BB CCTVM) to persons who are eligible investors residing in Brazil, which are considered to be Professional Investors (Investidores Profissionais), as designated by the applicable regulation, mainly the CVM Resolution No. 30 from the 11th of May 2021 (determines the duty to verify the suitability of products, services and transactions with regards to the client´s profile). UBS BB CCTVM is a subsidiary of UBS BB Servicos de Assessoria Financeira e Participacoes S.A. ("UBS BB"). UBS BB is an association between UBS AG and Banco do Brasil (through its subsidiary BB – Banco de Investimentos S.A.), of which UBS AG is the majority owner and which provides investment banking services and coverage in Brazil, Argentina, Chile, Paraguay, Peru and Uruguay. UBS BB CCTVM is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. Ombudsman: 0800-940-0266/ https://www.ubs.com/br/pt/ubsbb-investment-bank/ombudsman.html. UBS may hold relevant financial and commercial interest in relation to the company subject to this Research report. **Hong Kong:** Distributed by UBS Securities Asia Limited. Please contact local licensed persons of UBS Securities Asia Limited in respect of any matters arising from, or in connection with, the analysis or document **Singapore:** Distributed by UBS Securities Pte. Ltd. [Co. Reg. No.: 198500648C] or UBS AG, Singapore Branch. Please contact UBS Securities Pte. Ltd., an exempt financial adviser under the Singapore Financial Advisers Act (Cap. 110); or UBS AG, Singapore Branch, an exempt financial adviser under the Singapore Financial Advisers Act (Cap. 110) and a wholesale bank licensed under the Singapore Banking Act (Cap. 19) regulated by the Monetary Authority of Singapore, in respect of any matters arising from, or in connection with, the analysis or document. The recipients of this document represent and warrant that they are accredited and institutional investors as defined in the Securities and Futures Act (Cap. 289). **Japan:** Distributed by UBS Securities Japan Co., Ltd. to professional investors (except as otherwise permitted). Where this report has been prepared by UBS Securities Japan Co., Ltd., UBS Securities Japan Co., Ltd. is the author, publisher and distributor of the report. Distributed by UBS AG, Tokyo Branch to Professional Investors (except as otherwise permitted) in relation to foreign exchange and other banking businesses when relevant. **Australia:** Recipients through a superannuation fund: Distributed by UBS AG, Australia Branch (ABN 47 088 129 613 and holder of Australian Financial Service Licence No. 231087). For all other recipients: Distributed by UBS Securities Australia Limited (ABN 62 008 586 481 and holder of Australian Financial Services Licence No. 231098). This document contains general information and/or general advice only and does not constitute personal financial product advice. As such, the Information in this document has been prepared without taking into account any investor's objectives, financial situation or needs, and investors should, before acting on the Information, consider the appropriateness of the Information, having regard to their objectives, financial situation and needs. If the Information contained in this document relates to the acquisition, or potential acquisition of a particular financial product by a 'Retail' client as defined by section 761G of the Corporations Act 2001 where a Product Disclosure Statement would be required, the retail client should obtain and consider the Product Disclosure Statement relating to the product before making any decision about whether to acquire the product. **New Zealand:** Distributed by UBS New Zealand Ltd. UBS New Zealand Ltd is not a registered bank in New Zealand. You are being provided with this publication or material because you have indicated to UBS that you are a "wholesale client" within the meaning of clause 4 of schedule 5 of the Financial Markets Conduct Act 2013 of New Zealand (Permitted Client). This publication or material is not intended for clients who are not Permitted Clients (non-permitted Clients). If you are a non-permitted Client you must not rely on this publication or material. If despite this warning you nevertheless rely on this publication or material, you hereby (i) acknowledge that you may not rely on the content of this publication or material and that any recommendations or opinions in such this publication or material are not made or provided to you, and (ii) to the maximum extent permitted by law (a) indemnify UBS and its associates or related entities (and their respective Directors, officers, agents and Advisors) (each a 'Relevant Person') for any loss, damage, liability or claim any of them may incur or suffer as a result of, or in connection with, your unauthorised reliance on this publication or material and (b) waive any rights or remedies you may have against any Relevant Person for (or in respect of) any loss, damage, liability or claim you may incur or suffer as a result of, or in connection with, your unauthorised reliance on this publication or material. **Korea:** Distributed in Korea by UBS Securities Pte. Ltd., Seoul Branch. This report may have been edited or contributed to from time to time by affiliates of UBS Securities Pte. Ltd., Seoul Branch. This material is intended for professional/institutional clients only and not for distribution to any retail clients. **Malaysia:** This material is authorized to be distributed in Malaysia by UBS Securities Malaysia Sdn. Bhd (Capital Markets Services License No.: CMSL/A0063/2007). This material is intended for professional/institutional clients only and not for distribution to any retail clients. **India:** Distributed by UBS Securities India Private Ltd. (Corporate Identity Number U67120MH1996PTC097299) Unit 2901, Level 29 Altimus, Pandurang Budhkar Road, Worli, Mumbai – 400 018. It provides brokerage services bearing SEBI Registration Number: INZ000259830; Merchant Banking services bearing SEBI Registration Number: INM000013101; and Research Analyst services bearing SEBI Registration Number: INH000001204. Name of Compliance Officer Mr. Parameshwaran Shivaramakrishnan, Phone : +912261556151, Email : parameshwaran.s@ubs.com, Name of Grievance Officer Parameshwaran Shivaramakrishnan, Phone : +912261556151, Email: ol-ubs-sec-compliance@ubs.com Registration granted by SEBI, and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. UBS may have debt holdings or positions in the subject Indian company/companies. UBS may have financial interests (e.g. loan/derivative products, rights to or interests in investments, etc.) in the subject Indian company / companies from time to time. Within the past 12 months, UBS may have received compensation for non-investment banking securities-related services and/or non-securities services from the subject Indian company/companies. The subject company/companies may have been a client/clients of UBS during the 12 months preceding the date of distribution of the research report with respect to investment banking and/or non-investment banking securities-related services and/or non-securities services. With regard to information on associates, please refer to the Annual Report at: https://www.ubs.com/global/en/about_ubs/investor_relations/annualreporting.html The Research Annual Compliance Report for UBS Securities India Private Limited is available on www.ubs.com/ubssi under Research tab. **Taiwan:** Except as otherwise specified herein, this material may not be distributed in Taiwan. Information and material on securities/instruments that are traded in a Taiwan organized exchange is deemed to be issued and distributed by UBS Securities Pte. LTD., Taipei Branch to professional institutional investors and/or persons permitted under applicable regulation. Unless permitted by applicable Taiwan laws and regulations, this material is for reference and information only and should not constitute "recommendation" to clients or recipients in Taiwan for the covered companies or any companies mentioned in this document. No portion of the document may be reprinted, reproduced or quoted by the press or any other person without authorisation from UBS. **Indonesia:** This report is being distributed by PT UBS Sekuritas Indonesia and is delivered by its licensed employee(s), including marketing/sales person, to its client. PT UBS Sekuritas Indonesia, having its registered office at Sequis Tower Level 22 unit 22-1,Jl.Jend. Sudirman, kav.71, SCBD lot 11B, Jakarta 12190. Indonesia, is a subsidiary company of UBS AG and licensed under Capital Market Law no. 8 year 1995, a holder of broker-dealer and underwriter licenses issued by the Capital Market and Financial Institution Supervisory Agency (now Otoritas Jasa Keuangan/OJK). PT UBS Sekuritas Indonesia is also a member of Indonesia Stock Exchange and supervised by Otoritas Jasa Keuangan (OJK). Neither this report nor any copy hereof may be distributed in Indonesia or to any Indonesian citizens except in compliance with applicable Indonesian capital market laws and regulations. This report is not an offer of securities in Indonesia and may not be distributed within the territory of the Republic of Indonesia or to Indonesian citizens in circumstance which constitutes an offering within the meaning of Indonesian capital market laws and regulations.

The disclosures contained in research documents produced by UBS AG, London Branch or UBS Europe SE shall be governed by and construed in accordance with English law.

UBS specifically prohibits the redistribution of this document in whole or in part without the written permission of UBS and in any event UBS accepts no liability whatsoever for any redistribution of this document or its contents or the actions of third parties in this respect. Images may depict objects or elements that are protected by third party copyright, trademarks and other intellectual property rights. © UBS 2026. The key symbol and UBS are among the registered and unregistered trademarks of UBS. All rights reserved.

Global Wealth Management Disclaimer

You receive this document in your capacity as a client of UBS Global Wealth Management. This publication has been distributed to you by UBS Switzerland AG (regulated by FINMA in Switzerland) or its affiliates (**"UBS"**) with whom you have a banking relationship with. The full name of the distributing affiliate and its competent authority can be found in the country-specific disclaimer at the end of this document. UBS may utilise artificial intelligence tools ("AI Tools") in the preparation of this document. Notwithstanding any such use of AI Tools, this document has undergone human review.

The date and time of the first dissemination of this publication is the same as the date and time of its publication.

Risk information:

You agree that you shall not copy, revise, amend, create a derivative work, provide to any third party, or in any way commercially exploit any UBS research, and that you shall not extract data from any research or estimates, without the prior written consent of UBS.

This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject UBS to any registration or licensing requirement within such jurisdiction.

This document is **for your information only**; it is not an advertisement nor is it a solicitation or an offer to buy or sell any financial instruments or to participate in any particular trading strategy. Nothing in this document constitutes a representation that any investment strategy or recommendation is suitable or appropriate to an investor's individual circumstances or otherwise constitutes a personal recommendation. By providing this document, none of UBS or its representatives has any responsibility or authority to provide or have provided investment advice in a fiduciary capacity or otherwise. Investments involve risks, and investors should exercise prudence and their own judgment in making their investment decisions. None of UBS or its representatives is suggesting that the recipient or any other person take a specific course of action or any action at all. By receiving this document, the recipient acknowledges and agrees with the intended purpose described above and further disclaims any expectation or belief that the information constitutes investment advice to the recipient or otherwise purports to meet the investment objectives of the recipient. The financial instruments described in the document may not be eligible for sale in all jurisdictions or to certain categories of investors.

Options, derivative products and futures are not suitable for all investors, and trading in these instruments is considered risky. Mortgage and asset-backed securities may involve a high degree of risk and may be highly volatile in response to fluctuations in interest rates or other market conditions. Foreign currency rates of exchange may

adversely affect the value, price or income of any security or related instrument referred to in the document. For investment advice, trade execution or other enquiries, clients should contact their local sales representative.

The value of any investment or income may go down as well as up, and investors may not get back the full (or any) amount invested. Past performance is not necessarily a guide to future performance. Neither UBS nor any of its directors, employees or agents accepts any liability for any loss (including investment loss) or damage arising out of the use of all or any of the information (as defined below).

Prior to making any investment or financial decisions, any recipient of this document or the information should take steps to understand the risk and return of the investment and seek individualized advice from his or her personal financial, legal, tax and other professional advisors that takes into account all the particular facts and circumstances of his or her investment objectives.

Any prices stated in this document are for information purposes only and do not represent valuations for individual securities or other financial instruments. There is no representation that any transaction can or could have been effected at those prices, and any prices do not necessarily reflect UBS's internal books and records or theoretical model-based valuations and may be based on certain assumptions. Different assumptions by UBS or any other source may yield substantially different results.

No representation or warranty, either expressed or implied, is provided in relation to the accuracy, completeness or reliability of the information contained in any materials to which this document relates (the "Information"), except with respect to Information concerning UBS. The Information is not intended to be a complete statement or summary of the securities, markets or developments referred to in the document. UBS does not undertake to update or keep current the Information. Any opinions expressed in this document may change without notice and may differ or be contrary to opinions expressed by other business areas or groups, personnel or other representative of UBS. Any statements contained in this report attributed to a third party represent UBS's interpretation of the data, information and/or opinions provided by that third party either publicly or through a subscription service, and such use and interpretation have not been reviewed by the third party. In no circumstances may this document or any of the Information (including any forecast, value, index or other calculated amount ("Values")) be used for any of the following purposes: (i) valuation or accounting purposes; (ii) to determine the amounts due or payable, the price or the value of any financial instrument or financial contract; or (iii) to measure the performance of any financial instrument including, without limitation, for the purpose of tracking the return or performance of any Value or of defining the asset allocation of portfolio or of computing performance fees.

By receiving this document and the Information you will be deemed to represent and warrant to UBS that you will not use this document or any of the Information for any of the above purposes or otherwise rely upon this document or any of the Information.

UBS has policies and procedures, which include, without limitation, independence policies and permanent information barriers, that are intended, and upon which UBS relies, to manage potential conflicts of interest and control the flow of information within divisions of UBS (including between Global Wealth Management and UBS Global Research) and among its subsidiaries, branches and affiliates. For further information on the ways in which UBS manages conflicts and maintains independence of its research products, historical performance information and certain additional disclosures concerning UBS research recommendations, please visit https://www.ubs.com/research-methodology.

Research will initiate, update and cease coverage solely at the discretion of research management, which will also have sole discretion on the timing and frequency of any published research product. The analysis contained in this document is based on numerous assumptions. Different assumptions could result in materially different results.

The analyst(s) responsible for the preparation of this document may interact with trading desk personnel, sales personnel and other parties for the purpose of gathering, applying and interpreting market information. UBS relies on information barriers to control the flow of information contained in one or more areas within UBS into other areas, units, groups or affiliates of UBS. The compensation of the analyst who prepared this document is determined exclusively by research management and senior management (not including investment banking). Analyst compensation is not based on investment banking revenues; however, compensation may relate to the revenues of UBS and/or its divisions as a whole, of which investment banking, sales and trading are a part, and UBS's subsidiaries, branches and affiliates as a whole.

For financial instruments admitted to trading on an EU regulated market: UBS AG, its affiliates or subsidiaries (excluding UBS Securities LLC) acts as a market maker or liquidity provider (in accordance with the interpretation of these terms in the UK) in the financial instruments of the issuer save that where the activity of liquidity provider is carried out in accordance with the definition given to it by the laws and regulations of any other EU jurisdictions, such information is separately disclosed in this document. For financial instruments admitted to trading on a non-EU regulated market: UBS may act as a market maker save that where this activity is carried out in the US in accordance with the definition given to it by the relevant laws and regulations, such activity will be specifically disclosed in this document. UBS may have issued a warrant the value of which is based on one or more of the financial instruments referred to in the document. UBS and its affiliates and employees may have long or short positions, trade as principal and buy and sell in instruments or derivatives identified herein; such transactions or positions may be inconsistent with the opinions expressed in this document.

Options and futures are not suitable for all investors, and trading in these instruments is considered risky and may be appropriate only for sophisticated investors. Prior to buying or selling an option, and for the complete risks relating to options, you must receive a copy of "Characteristics and Risks of Standardized Options". You may read the document at https://www.theocc.com/about/publications/character-risks.jsp or ask your financial advisor for a copy.

Investing in structured investments involves significant risks. For a detailed discussion of the risks involved in investing in any particular structured investment, you must read the relevant offering materials for that investment. Structured investments are unsecured obligations of a particular issuer with returns linked to the performance of an underlying asset. Depending on the terms of the investment, investors could lose all or a substantial portion of their investment based on the performance of the underlying asset. Investors could also lose their entire investment if the issuer becomes insolvent. UBS does not guarantee in any way the obligations or the financial condition of any issuer or the accuracy of any financial information provided by any issuer. Structured investments are not traditional investments and investing in a structured investment is not equivalent to investing directly in the underlying asset. Structured investments may have limited or no liquidity, and investors should be prepared to hold their investment to maturity. The return of structured investments may be limited by a maximum gain, participation rate or other feature. Structured investments may include call features and, if a structured investment is called early, investors would not earn any further return and may not be able to reinvest in similar investments with similar terms. Structured investments include costs and fees which are generally embedded in the price of the investment. The tax treatment of a structured investment may be complex and may differ from a direct investment in the underlying asset. UBS and its employees do not provide tax advice. Investors should consult their own tax advisor about their own tax situation before investing in any securities.

Important Information About Sustainable Investing Strategies: Sustainable investing strategies aim to consider and incorporate environmental, social and governance (ESG) factors into investment process and portfolio construction. Strategies across geographies approach ESG analysis and incorporate the findings in a variety of ways. Incorporating ESG factors or sustainable investing considerations may inhibit the portfolio manager's ability to participate in certain investment opportunities that otherwise would be consistent with its investment objective and other principal investment strategies. The returns on a portfolio incorporating ESG factors or sustainable investing considerations may be lower or higher than portfolios where ESG factors, exclusions, or other sustainability issues are not considered by the portfolio manager, and the investment opportunities available to such portfolios may differ.

Within the past 12 months UBS Switzerland AG, its affiliates or subsidiaries may have received or provided investment services and activities or ancillary services as per MiFID II which may have given rise to a payment or promise of a payment in relation to these services from or to this company.

Disclosures: If you require detailed information on disclosures of interest or conflict of interest as required by Market Abuse Regulation please contact the mailbox MAR_disclosure_twopager@ubs.com. Please note that e-mail communication is unsecured.

External Asset Managers / External Financial Consultants: In case this research or publication is provided to an External Asset Manager or an External Financial Consultant, UBS expressly prohibits that it is redistributed by the External Asset Manager or the External Financial Consultant and is made available to their clients and/or third parties.

Argentina: All securities that will be offered to you by UBS are not authorized by the Argentine Securities and Exchange Commission (CNV) and are not subject to its reporting, periodic information requirements, or oversight. Furthermore, the CNV has not reviewed or endorsed the information provided in any securities offering document, nor the accuracy of any accounting, financial, economic data, or any other information disclosed therein, which remains the sole responsibility of the respective issuer and the other parties involved. **Australia:** This document is provided by UBS Switzerland AG. UBS Switzerland AG does not hold an Australian Financial Services Licence (AFSL) and relies on an exemption to provide financial services to persons in Australia. This document is intended only for distribution to wholesale clients under the Corporations Act 2001 (Cth). UBS Switzerland AG is a related body corporate of UBS AG, Australia Branch and UBS Securities Australia Limited. This document may be distributed to clients by those entities, but it is provided by UBS Switzerland AG and is not provided under any of the other entities' AFSL. The information in this document is general in nature and is not intended to address the objectives, financial situation or needs of any particular individual or entity. Each recipient should consider their own objectives, financial situation or needs before acting on the advice and obtain the relevant Product Disclosure Statement (if required) before making any decision whether to acquire any product. In Australia, UBS entities, other than UBS AG, Australia Branch, are not authorized deposit-taking institutions for the purposes of the Banking Act 1959 (Cth.) and their obligations do not represent deposits or other liabilities of UBS AG, Australia Branch. UBS AG, Australia Branch does not guarantee or otherwise provide assurance in respect of the obligations of such UBS entities or the funds. An investor is exposed to investment risk including possible delays in repayment and loss of income and principal invested, as relevant. If you do not wish to receive marketing materials from UBS, please contact your UBS representative or the contact details listed in the Australia Privacy Notice: https://www.ubs.com/global/en/legal/privacy.html. Your personal data will be processed in accordance with this notice. **Bahrain**: This report is being distributed by UBS AG, Bahrain Branch, duly licensed and regulated by the Central Bank of Bahrain (CBB) as an Investment Business Firm - Category 2 (Branch). Related financial services or products are only made available to Accredited Investors, as defined by the CBB, and are not intended for any other persons. UBS AG, Bahrain Branch is

a Foreign Branch of UBS AG, Zurich/Switzerland and is located on Level 21, East Tower, Bahrain World Trade Centre, Manama, Kingdom of Bahrain. **Brazil**: This report is distributed in Brazil by UBS BB Corretora de Câmbio, Títulos e Valores Mobiliários S.A. or its affiliates ("UBS"). Pursuant to CVM Resolution No. 20/2021, of February 25, 2021, the author(s) of the report hereby certify(ies) that the views expressed in this report solely and exclusively reflect their personal opinions and have been prepared independently, including with respect to UBS. Part of the author(s)'s compensation is based on various factors, including the total revenues of the relevant UBS Group entity of which they are in employment of, but no part of the compensation has been, is, or will be related to the specific recommendations or views expressed in this report. In addition, UBS declares that: UBS has provided, and/or may in the future provide investment banking, brokerage, asset management, commercial banking and other financial services to the subject company/companies or its affiliates, for which they have received or may receive customary fees and commissions, and which constituted or may constitute relevant financial or commercial interests in relation to the subject company/companies or the subject securities. **Canada**: The information contained herein is not, and under no circumstances is to be construed as, a prospectus, an advertisement, a public offering, an offer to sell securities described herein, solicitation of an offer to buy securities described herein, in Canada or any province or territory thereof. Any offer or sale of the securities described herein in Canada will be made only under an exemption from the requirements to file a prospectus with the relevant Canadian securities regulators and only by a dealer properly registered under applicable securities laws or, alternatively, pursuant to an exemption from the dealer registration requirement in the relevant province or territory of Canada in which such offer or sale is made. Under no circumstances is the information contained herein to be construed as investment advice in any province or territory of Canada and is not tailored to the needs of the recipient. To the extent that the information contained herein references securities of an issuer incorporated, formed or created under the laws of Canada or a province or territory of Canada, any trades in such securities must be conducted through a dealer registered in Canada or, alternatively, pursuant to a dealer registration exemption. No securities commission or similar regulatory authority in Canada has reviewed or in any way passed upon these materials, the information contained herein or the merits of the securities described herein and any representation to the contrary is an offence. In Canada, this publication is distributed by UBS Investment Management Canada Inc. **China**: This report and any offering material such as term sheet, research report, other product or service documentation or any other information (the "Material") sent with this report was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the material erroneously, UBS asks that you kindly delete it and inform UBS immediately. This report is prepared by UBS Switzerland AG or its offshore subsidiary or affiliate (collectively as "UBS Offshore"). UBS Offshore is an entity incorporated out of China and is not licensed, supervised or regulated in China to carry out banking or securities business. The recipient should not contact the analysts or UBS Offshore which produced this report for advice as they are not licensed to provide securities investment advice in China. UBS Investment Bank (including Research) has its own wholly independent research and views which at times may vary from the views of UBS Global Wealth Management. The recipient should not use this document or otherwise rely on any of the information contained in this report in making investment decisions and UBS takes no responsibility in this regard. **Denmark**: This publication is not intended to constitute a public offer under Danish law. It is distributed only for information purposes to clients of UBS Europe SE, Denmark Branch, filial af UBS Europe SE, with place of business at Sankt Annae Plads 13, 1250 Copenhagen, Denmark, registered with the Danish Commerce and Companies Agency, under No. 38 17 24 33. UBS Europe SE, Denmark Branch, filial af UBS Europe SE is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the Danish Financial Supervisory Authority (Finanstilsynet), to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Egypt**: Securities or other investment products are not being offered or sold by UBS to the public in Egypt and they have not been and will not be registered with the Egyptian Financial Regulatory Authority (FRA). **France**: This publication is not intended to constitute a public offer under French law, it does not constitute a personal recommendation as it is distributed only for information purposes to clients of UBS Europe SE Succursale de France (a branch of UBS Europe SE), having its registered office at 39, rue du Colisée, 75008 Paris, France, registered with the "Registre du Commerce et des Sociétés" of Paris under N°844 425 629. UBS Europe SE Succursale de France is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the French "Autorité de contrôle prudentiel et de résolution" and "Autorité des marchés financiers", to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Germany**: This publication is not intended to constitute a public offer under German law. It is distributed only for information purposes to clients of UBS Europe SE, Germany, with place of business at Bockenheimer Landstrasse 2-4, 60306 Frankfurt am Main. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin") and supervised jointly by the European Central Bank, the German Central Bank (Deutsche Bundesbank) and BaFin, to which this publication has not been submitted for approval. **Hong Kong SAR**: This publication is distributed to clients of UBS AG Hong Kong Branch by UBS AG Hong Kong Branch, a licensed bank under the Hong Kong Banking Ordinance and a registered institution under the Securities and Futures Ordinance. UBS AG Hong Kong Branch is incorporated in Switzerland with limited liability. The contents of this material have not been reviewed by any regulatory authority in Hong Kong. **India**: UBS Securities India Private Ltd. (Corporate Identity Number U67120MH1996PTC097299) 2/F, 3 North Avenue, Maker Maxity, Bandra Kurla Complex, Bandra (East), Mumbai (India) 400051. Phone: +912261556000. It provides brokerage services bearing SEBI Registration Number: INZ000259830; Merchant Banking services bearing SEBI Registration Number: INM000013101 and Research Analyst services bearing SEBI Registration Number: INH000001204. Name of Compliance Officer Mr. Parameshwaran Shivaramakrishnan, Phone: +912261556151, Email: parameshwaran.s@ubs.com, Name of Grievance Officer Parameshwaran Shivaramakrishnan, Phone: +912261556151, Email: ol-ubs-sec-compliance@ubs.com. Registration granted by SEBI, and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. UBS AG, its affiliates or subsidiaries may have debt holdings or positions in the subject Indian company/companies. UBS AG, its affiliates or subsidiaries may have financial interests (e.g. loan/derivative products, rights to or interests in investments, etc.) in the subject Indian company/companies from time to time. Within the past 12 months, UBS AG, its affiliates or subsidiaries may have received compensation for non-investment banking securities-related services and/or non-securities services from the subject Indian company/companies. The subject company/companies may have been a client/clients of UBS AG, its affiliates or subsidiaries during the 12 months preceding the date of distribution of the research report with respect to investment banking and/or non-investment banking securities-related services and/or non-securities services. With regard to information on associates, please refer to the Annual Report at: https://www.ubs.com/global/en/about_ubs/investor_relations/annualreporting.html. The Research Annual Compliance report for UBS Securities India Private Limited is available on www.ubs.com/ubssi under Research tab. **Indonesia**: This communication and any offering material term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed without the consent of UBS. None of the Material has been registered or filed under the prevailing laws and with any financial or regulatory authority in your jurisdiction. The Material may not have been approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or products or futures contracts). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including futures contracts) nor is it an offer to buy or to sell any securities or products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Israel**: UBS is a premier global financial firm offering wealth management, asset management and investment banking services from its headquarters in Switzerland and its operations in over 50 countries worldwide to individual, corporate and institutional investors. This publication is intended for information only and is not intended as an offer to buy or solicitation of an offer. Furthermore, this publication is not intended as an investment advice. Nothing contrary to the above, no action has been, or will be, taken in Israel that would permit an offering of the product(s) mentioned in this document or a distribution of this document to the public in Israel. In particular, this document has not been reviewed or approved by the Israeli Securities Authority. The product(s) mentioned in this document is/are being offered to a limited number of sophisticated investors who qualify as one of the investors listed in the first supplement to the Israeli Securities Law, 5728-1968. This document may not be reproduced or used for any other purpose, nor be furnished to any other person other than those to whom copies have been sent. Anyone who purchases the product(s) mentioned herein shall do so for its own benefit and for its own account and not with the aim or intention of distributing or offering the product(s) to other parties. Anyone who purchases the product(s) shall do so in accordance with its own understanding and discretion and after it has received any relevant financial, legal, business, tax or other advice or opinion required by it in connection with such purchase(s). The word "advice" and/or any of its equivalent terms shall be read and construed in conjunction with the definition of the term "investment marketing" as defined under the Israeli Regulation of Investment Advice, Investment Marketing and Portfolio Management Law. The Swiss laws and regulations require a number of mandatory disclosures to be made in independent financial research reports or recommendations. Pursuant to the Swiss Financial Market Infrastructure Act and the Financial Market Infrastructure Ordinance-FINMA, banks must disclose the percentage of voting rights they hold in companies being researched, if these holdings are equal to or exceed the statutory thresholds. In addition, the Directives on the Independence of Financial Research, issued by the Swiss Bankers Association, mandate a number of disclosures, including the disclosure of potential conflicts of interest, the participation within previous 12 months in any securities issues on behalf of the company being researched, as well as the fact that remuneration paid to the financial analysts is based generally upon the performance of (i) the new issues department or investment banking; or (ii) securities trading performance (including proprietary trading) or sales. **Italy**: This publication is not intended to constitute a public offer under Italian law. It is distributed only for information purposes to clients of UBS Europe SE, Succursale Italia, with place of business at Via del Vecchio Politecnico, 3-20121 Milano. UBS Europe SE, Succursale Italia is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the Bank of Italy (Banca d'Italia) and the Italian Financial Markets Supervisory Authority (CONSOB - Commissione Nazionale per le Società e la Borsa), to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Japan**: This report is solely distributed in Japan by UBS SuMi TRUST Wealth Management Co., Ltd, Financial Instruments Dealer, Director-General of Kanto Local Finance Bureau (Kinsho) No. 3233, a member of the Japan Securities Dealers Association, Financial Futures Association of Japan, Investment Management Association of Japan. UBS SuMi TRUST Wealth Management Co., Ltd will not distribute or

forward this report outside Japan. **Jersey**: UBS AG, Jersey Branch, is regulated and authorized by the Jersey Financial Services Commission for the conduct of banking, funds and investment business. Where services are provided from outside Jersey, they will not be covered by the Jersey regulatory regime. UBS AG, Jersey Branch is a branch of UBS AG a public company limited by shares, incorporated in Switzerland whose registered offices are at Aeschenvorstadt 1, CH-4051 Basel and Bahnhofstrasse 45, CH 8001 Zurich. UBS AG, Jersey Branch's principal place of business is 1, IFC Jersey, St Helier, Jersey, JE2 3BX. **Luxembourg**: This publication is not intended to constitute a public offer under Luxembourg law. It is distributed only for information purposes to clients of UBS Europe SE, Luxembourg Branch ("UBS Luxembourg"), R.C.S. Luxembourg n° B209123, with registered office at 33A, Avenue J. F. Kennedy, L-1855 Luxembourg. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea (HRB n° 107046), with registered office at Bockenheimer Landstrasse 2-4, D-60306 Frankfurt am Main, Germany, duly authorized by the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin") and subject to the joint prudential supervision of BaFin, the European Central Bank and the central bank of Germany (Deutsche Bundesbank). UBS Luxembourg is furthermore supervised by the Luxembourg prudential supervisory authority (Commission de Surveillance du Secteur Financier), in its role as host member state authority. This publication has not been submitted for approval to any public supervisory authority. **Malaysia**: This communication and any offering material term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or derivatives products). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or derivatives products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Mexico**: This information is distributed by UBS Asesores México, S.A. de C.V. ("UBS Asesores"), an affiliate of UBS Switzerland AG, incorporated as a non-independent investment advisor under the Mexican regulation due to the relation with a Foreign Bank. UBS Asesores is registered under number 30060-001-(14115)-21/06/2016 and subject to the supervision of the Mexican Banking and Securities Commission ("CNBV") exclusively regarding the rendering of (i) portfolio management services, (ii) securities investment advisory services, analysis and issuance of individual investment recommendations, and (iii) anti-money laundering and terrorism financing matters. This UBS publication or any material related thereto is addressed only to Sophisticated or Institutional Investors located in Mexico. Research reports only reflect the views of the analysts responsible for the report. The compensation of the analyst(s) who prepared this report is determined exclusively by research management and senior management of any entity of UBS Group to which such analyst(s) render(s) services. **Monaco**: This document is not intended to constitute a public offering or a comparable solicitation under the Principality of Monaco laws, but might be made available for information purposes to clients of UBS (Monaco) S.A., a regulated bank having its registered office at 2 avenue de Grande Bretagne 98000 Monaco operating under a banking license granted by the "Autorité de Contrôle Prudentiel et de Résolution" (ACPR) and the Monegasque government which authorizes the provision of banking services in Monaco. UBS (Monaco) S.A. is also licensed by the "Commission de Contrôle des Activités Financières" (CCAF) to provide investment services in Monaco. The latter has not approved this publication. **Philippines:** This communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or derivatives products). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or derivatives products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Portugal**: UBS Switzerland AG is not licensed to conduct banking and financial activities in Portugal nor is UBS Switzerland AG supervised by the portuguese regulators (Bank of Portugal "Banco de Portugal" and Portuguese Securities Exchange Commission "Comissão do Mercado de Valores Mobiliários"). **Qatar**: UBS Qatar LLC is licensed by the Qatar Financial Centre Authority and authorized by the QFC Regulatory Authority, with QFC no. 01169, and has its registered office at Building No. 36, 4th Floor, Abdulla Bin Thani Street, Doha Design District 1, Msheireb Downtown Doha, State of Qatar. This material is strictly intended for Eligible Counterparties and/or Business Customers only as classified under the QFCRA's Customer and Investor Protection Rules 2019. No other person should act upon this material. **Russia**: This document or information contained therein is for information purposes only and constitutes neither a public nor a private offering, is not an invitation to make offers, to sell, exchange or otherwise transfer any financial instruments in the Russian Federation to or for the benefit of any Russian person or entity and does not constitute an advertisement or offering of securities in the Russian Federation within the meaning of Russian securities laws. The information contained herein is not an "individual investment recommendation" as defined in Federal Law of 22 April 1996 No 39-FZ "On Securities Market" (as amended) and the financial instruments and operations specified herein may not be suitable for your investment profile or your investment goals or expectations. The determination of whether or not such financial instruments and operations are in your interests or are suitable for your investment goals, investment horizon or the acceptable risk level is your responsibility. We assume no liability for any losses connected with making any such operations or investing into any such financial instruments and we do not recommend to use such information as the only source of information for making an investment decision. **Saudi Arabia**: This material is for marketing and information purposes by UBS Saudi Arabia only. It has been distributed by UBS Saudi Arabia or reviewed and approved by UBS Saudi Arabia if it was issued by UBS Switzerland AG, by UBS AG, or any of its subsidiaries or branches. UBS Switzerland AG is a public company limited by shares, incorporated in Switzerland with its registered offices at Bahnhofstrasse 45, CH-8001 Zurich. UBS Saudi Arabia (a subsidiary of UBS AG) is a Saudi closed joint stock company incorporated in the Kingdom of Saudi Arabia with a paid capital of 110,000,000 Saudi Riyals under commercial register number 1010257812 having its registered office at Laysen Valley, P.O. Box 75724, Riyadh 11588, Kingdom of Saudi Arabia. UBS Saudi Arabia is authorized and regulated by the Capital Market Authority to conduct securities business under license number 08113-37. This document is distributed only under such circumstances as may be permitted by applicable laws and regulations. The information and – if any - opinions contained in this document have been compiled or arrived at based upon information obtained from sources believed to be reliable and in good faith, but is not guaranteed as being accurate, nor is it a complete statement or summary of the securities, markets or developments referred to in the document. All such information and opinions are subject to changes without notice. Past performance of investments (whether simulated or actual) is not necessarily an indicator of future results. Forward looking data is based on estimates that are subject to changes. Should the currency of a financial product or service not match your reference currency, performance may rise or fall due to currency fluctuations. UBS Switzerland AG and / or other members of the UBS Group may have a position in and may make a purchase and/or sale of any of the securities or other financial instruments mentioned in this document While statements that constitute "forward-looking statements", including, but not limited to, statements relating to our future business development represent UBS judgments and future expectations concerning the development of our business, a number of risks, uncertainties and other important factors could cause actual developments and results to differ materially from our expectations. This document may not be forwarded, reproduced, redistributed or republished for any purpose without the written permission of UBS Saudi Arabia. Source for all data and charts (if not indicated otherwise) is UBS Switzerland AG. **Singapore**: Where applicable, this material is distributed in Singapore by UBS AG, Singapore Branch, which is licensed by the Monetary Authority of Singapore under the Banking Act 1970 to carry on banking business. UBS AG is incorporated in Switzerland with limited liability. UBS AG has a branch registered in Singapore (UEN S98FC5560C). This material has been prepared and issued for distribution in Singapore to institutional investors, accredited investors and expert investors (each as defined under the Financial Advisers Regulations (the "FAR")) only. By virtue of your status as an institutional investor, accredited investor, or expert investor, UBS AG is exempted from complying with certain requirements under the Financial Advisers Act 2001 (the "FAA"), the FAR and the relevant Notices and Guidelines issued thereunder, in respect of any financial advisory service which UBS AG may provide to you. These include exemptions from complying with: Section 34 of the FAA (pursuant to Regulation 33(1) of the FAR); Section 36 of the FAA (pursuant to Regulation 34(1) of the FAR); and Section 45 of the FAA (pursuant to Regulation 35(1) of the FAR). Singapore recipients and clients of UBS AG, Singapore Branch should contact UBS AG, Singapore Branch for any matters arising from, or in connection with, this material. Where applicable, this communication and any offering material term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved or endorsed by any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or products or futures contracts). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Spain:** This report is distributed in Spain by UBS AG, Sucursal en España, authorized under number 1460 in the Register by the Banco de España. **Sweden**: This publication is not intended to constitute a public offer under Swedish law. It is distributed only for information purposes to clients of UBS Europe SE, Sweden Bankfilial, with place of business at Regeringsgatan 38, 11153 Stockholm, Sweden, registered

with the Swedish Companies Registration Office under Reg. No 516406-1011. UBS Europe SE, Sweden Bankfilial is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the Swedish supervisory authority (Finansinspektionen), to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Taiwan**: This material is provided by UBS AG, Taipei Branch in accordance with laws of Taiwan, in agreement with or at the request of clients/prospects. **Thailand**: This communication and any offering material, term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication were done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or derivatives products). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or derivatives products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Türkiye**: The information in this document is not provided for the purpose of offering, marketing or sale of any capital market instrument or service in the Republic of Türkiye. Therefore, this document may not be considered as an offer made, or to be made, to residents of the Republic of Türkiye in the Republic of Türkiye. UBS Switzerland AG is not licensed by the Capital Markets Board of Türkiye (the CMB) under the provisions of the Capital Market Law (Law No. 6362). Accordingly, neither this document nor any other offering material related to the instrument/service may be utilized in connection with providing any capital market services to persons within the Republic of Türkiye without the prior approval of the CMB. However, according to article 15 (d) (ii) of the Decree No. 32 residents of the Republic of Türkiye are allowed to purchase or sell the financial instruments traded in financial markets outside of the Republic of Türkiye. Further to this, pursuant to article 9 of the Communiqué on Principles Regarding Investment Services, Activities and Ancillary Services No. III-37.1, investment services provided abroad to residents of the Republic of Türkiye based on their own initiative are not restricted. **United Arab Emirates (UAE) / DIFC / Abu Dhabi**: UBS is not licensed in the UAE by the Central Bank of the UAE nor by the Emirates' Securities and Commodities Authority and does not undertake banking activities in the UAE. This document is provided for your information only and does not constitute financial advice. **DIFC**: UBS AG Dubai Branch is regulated by the DFSA in the DIFC. This material is strictly intended for Professional Clients and/or Market Counterparties only as classified under the DFSA rulebook. It should not be distributed to Retail Clients. The Investment Research is provided for information purposes only and is not a recommendation or offer to buy/sell/hold a particular investment. The investment research may be out of date. You should seek investment advice before acting on the basis of the Investment Research. **Abu Dhabi**: UBS AG Abu Dhabi Branch is licensed and regulated by the Financial Services Regulatory Authority ("FSRA") of the Abu Dhabi Global Market. This material is intended solely for professional clients or market counterparties, as defined in the rules of the FSRA. It is not directed at, nor intended for, retail clients or any person who does not meet the criteria of a professional client or market counterparty. **United Kingdom**: This document is issued by UBS Wealth Management, a division of UBS AG which is authorised and regulated by the Financial Market Supervisory Authority in Switzerland. In the United Kingdom, UBS AG is authorised by the Prudential Regulation Authority and is subject to regulation by the Financial Conduct Authority and limited regulation by the Prudential Regulation Authority. Details about the extent of regulation by the Prudential Regulation Authority are available from us on request. A member of the London Stock Exchange. This publication is distributed to retail clients of UBS Wealth Management. **Ukraine**: UBS is a premier global financial services firm offering wealth management services to individual, corporate and institutional investors. UBS is established in Switzerland and operates under Swiss law and in over 50 countries and from all major financial centers. UBS is not registered and licensed as a bank/financial institution under Ukrainian legislation and does not provide banking and other financial services in Ukraine. UBS has not made, and will not make, any offer of the mentioned products to the public in Ukraine. No action has been taken to authorize an offer of the mentioned products to the public in Ukraine and the distribution of this document shall not constitute financial services for the purposes of the Law of Ukraine "On Financial Services and Financial Companies" dated 14 December 2021. Any offer of the mentioned products shall not constitute an investment advice, public offer, circulation, transfer, safekeeping, holding or custody of securities in the territory of Ukraine. Accordingly, nothing in this document or any other document, information or communication related to the mentioned products shall be interpreted as containing an offer, a public offer or invitation to offer or to a public offer, or solicitation of securities in the territory of Ukraine or investment advice under Ukrainian law. Electronic communication must not be considered as an offer to enter into an electronic agreement or other electronic instrument within the meaning of the Law of Ukraine "On Electronic Commerce" dated 3 September 2015. This document is strictly for private use by its holder and may not be passed on to third parties or otherwise publicly distributed. **USA**: Distributed to US persons only by UBS Financial Services Inc. or UBS Securities LLC, subsidiaries of UBS AG. UBS Switzerland AG, UBS Europe SE, UBS Bank, S.A., UBS BB Corretora de Câmbio, Títulos e Valores Mobiliários S.A., UBS Asesores México, S.A. de C.V., UBS SuMi TRUST Wealth Management Co., Ltd., UBS Wealth Management Israel Ltd. and UBS Menkul Degerler AS are affiliates of UBS AG. **UBS Financial Services Inc. accepts responsibility for the content of a report prepared by a non-US affiliate when it distributes reports to US persons. All transactions by a US person in the securities mentioned in this report should be effected through a US-registered broker dealer affiliated with UBS, and not through a non-US affiliate. The contents of this report have not been and will not be approved by any securities or investment authority in the United States or elsewhere. UBS Financial Services Inc. is not acting as a municipal advisor to any municipal entity or obligated person within the meaning of Section 15B of the Securities Exchange Act (the "Municipal Advisor Rule") and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of the Municipal Advisor Rule.** For information on the ways in which UBS Securities LLC manages conflicts and maintains independence of its UBS Global Research product; historical performance information; certain additional disclosures concerning UBS Global Research recommendations; and terms and conditions for certain third party data used in research report, please visit https://www.ubs.com/disclosures.

© UBS 2026. The key symbol and UBS are among the registered and unregistered trademarks of UBS. All rights reserved.

Sales Trader Commentary: TAIEX 49k Wobble - July De Ja Vu, or Just Pausing Before the Santa Rally? | GSTW

13 October 2026 | 4:11 PM Taiwan Standard Time

Robert Wang
Goldman Sachs (Asia) L.L.C. Taipei Branch
FICC & Equities

Index lost the 49k points threshold and saw the biggest decline on a point basis since the end of July. Market seemed to lose the steam to break through 50k points after the past choppy sessions. Is this July de ja vu or some healthy pullback? Foreigners and domestic investors seem to share different narrative. Check out below for our investor feedback.

Key Investor Feedback and Observations

- **Positioning**. we saw meaningful outflows today from all kinds of investors today, again focusing in ex-TSMC large caps. Today foreigners **net sold US$5.2bn** today, the second largest in the history (simultaneously drove TWD weakness over 32.23 today). Domestic investors, especially retail, on the other hand, anchored the market with the record high margin loan balance (as of 10/12/2026). Local funds remained buyer as well mainly thanks to ETFs inflow.
- **Macro**. CPI headline brought uncertainty and forced global investors to turn cautious, which was exacerbated by high yields and crude. For Taiwan, AI trade is relatively resilient to the Middle Eastern tension and high energy prices. However, at some point, it is going to have a real economic impact either through Fed or some other macro data.
- **Sentiment**. Surprisingly, people are still bullish in Taiwan, though they also acknowledge some choppy trading sessions ahead. Local investor feedback is not calling for capitulation; in fact, most believe this is just another "two-steps forward one-step backward" cadence towards another record high. Nonetheless, some retail start taking chips off the table. This echoes our observation among the foreign investors. Yes, the AI fundamental future is all sunshine, but as we head into the end of year, it is natural for them to protect their great fun thus far this year instead of adding on immediate risk. ***I'd imagine the next few weeks are going to be full of 1) rapid sector rotation; 2) weak or unsustainable momentum before Santa comes into town after the midterm election for our year-end rally Christmas gift.***
- **TSMC Earnings and Others**. We know the numbers are going to be good, and we also know people are going to sell-on-facts after TSMC earnings call; recall most recent one happened right after the meeting that scarred a lot of investors since. Perhaps this time people started trimming ahead (and maybe too early?), which provides some window for a quick rebound into small and mid cap. OCP summit is happening right now, and TPCA Show will start next week. Watch out for single stock movement mentioned in the conferences.

- **Sector**. **1) ABF/CCL** is still the crowded long sectors for both foreign and domestic investor, but echoing on above points, investors need more catalysts to add on risks or drive the re-ratings. **2) Power** is another big issues that data centers need to resolve, power solutions and fuel cells name are also key focus in Taiwan. 3) Let's not forget about **OSATs**. We need more equipment and testing capacity to suffice enlarging demand from both ASIC and GPU.
  - **Key Stocks I Follow**: Unimicron, Kinsus, NYPCB, ZDT, EMC, TUC, Delta, Lite-On, Kaori, ASE, KYEC, All Ring
- **Risk**. Upside looks thin in the short term. Good earnings are not enough - look at Nanya Tech and Memory sector today. Chasing momentum now may get hurt. I also smelled some short unwind today (i.e. funding short names outperformance such as Realtek, Novatek, Shipper, etc) - signs of de-leveraging. Directionless is how I would describe the past few days, but maybe it's time to find some value Buy?

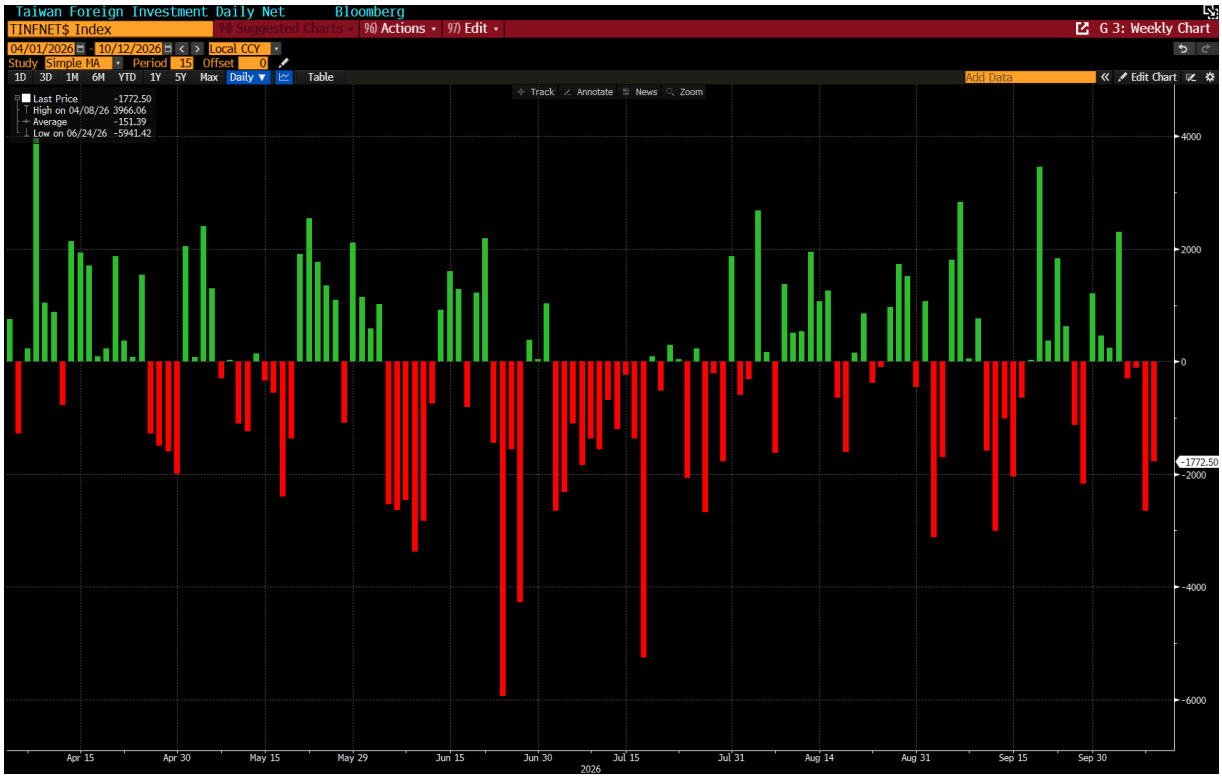


Exhibit 1. Foreign Investors Outflow since 4/1/2026

Please feel free to reach out to your GS representative to discuss any ideas.

All references to “we/us/our” refer to the views and observations of the desk unless otherwise specified

Past performance is not indicative of future results

Max loss amount invested is 100%

Source for all: Goldman Sachs FICC and Equities, Company Data, Bloomberg as of 10/13/2026

The information contained in this document does not constitute, and is not intended to constitute, a public offer of securities in the United Arab Emirates in accordance with the Commercial Companies Law (Federal Law no. 2 of 2015), ESCA Board of Directors' Decision no. (9/r.m.) of 2016, ESCA Chairman Decision no 3/r.m. of 2017 concerning promoting and introducing regulations or otherwise under the laws of the United Arab Emirates. Accordingly, the interests in the securities may not be offered to the public in the UAE (including the Dubai

International Financial Centre and the Abu Dhabi Global Market). This document has not been approved by, or filed with the Central Bank of the United Arab Emirates, the Securities and Commodities Authority, the Dubai Financial Services Authority, the Financial Services Regulatory Authority or any other relevant licensing authorities in the United Arab Emirates. If you do not understand the contents of this document, you should consult with a financial advisor. This document is provided to the recipient only and should not be provided to or relied on by any other person.

© 2026 Goldman Sachs. All rights reserved.

This message has been prepared by personnel in the Goldman Sachs FICC and Equities business units of one or more affiliates of The Goldman Sachs Group, Inc. ("Goldman Sachs") and is not the product of Global Investment Research. It is not a research report and is not intended as such.

Additional Disclaimers

Non-Reliance and Risk Disclosure: This material is for the general information of our clients and is a solicitation of derivatives business generally, only for the purposes of, and to the extent it would otherwise be subject to, CFTC Regulations 1.71 and 23.605. This material should not be construed as an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. We are not soliciting any specific action based on this material. For the purposes of U.S. rules and regulations, it does not constitute a recommendation or take into account the particular investment objectives, financial conditions, or needs of individual clients. Before acting on this material, you should consider whether it is suitable for your particular circumstances and, if necessary, seek professional advice. The price and value of the investments referred to in this material and the income from them may go down as well as up, and investors may realize losses on any investments. Past performance is not a guide to future performance. Future returns are not guaranteed, and a loss of original capital may occur. We do not provide tax, accounting, or legal advice to our clients, and all investors are advised to consult with their tax, accounting, or legal advisers regarding any potential investment. The material is based on information that we consider reliable, but we do not represent that it is accurate, complete and/or up to date, and it should not be relied on as such. Opinions expressed are our current opinions as of the date appearing on this material only and only represent the views of the author and not those of Goldman Sachs, unless otherwise expressly noted.

Conflict of Interest Disclosure: We are a full-service, integrated corporate advisory, investment management, and brokerage firm. The professionals who prepared this material are paid in part based on the profitability of The Goldman Sachs Group, Inc., which includes earnings from the firm's trading, capital markets, corporate advisory and other business. They, along with other salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients that reflect opinions that are contrary to the opinions expressed herein or the opinions expressed in research reports issued by our Research Departments, and our market making, investing and lending businesses may make investment decisions that are inconsistent with the views expressed herein. In addition, the professionals who prepared this material may also produce material for, and from time to time, may advise or otherwise be part of our trading desks that trade as principal in the securities mentioned in this material. This material is therefore not independent from our interests, which may conflict with your interests. We and our affiliates, officers, directors, and employees, including persons involved in the preparation or issuance of this material, may from time to time have "long" or "short" positions in, act as principal in, and buy or sell the securities or derivatives (including options) thereof in, and act as market maker or specialist in, and serve as a director of, companies mentioned in this material. In addition, we may have served as manager or co manager of a public offering of securities by any such company within the past three years

Not a Fiduciary: To the extent this material is provided to an employee benefit plan or account subject to the Employee Retirement Income Security Act of 1974, as amended ("ERISA") or Section 4975 of the Internal Revenue Code of 1986, as amended (the "Code") (each, a "Benefit Plan Investor"), by accepting this material such Benefit Plan Investor will be deemed to have represented, warranted and acknowledged that (1) the Benefit Plan Investor is represented by a fiduciary that is independent of Goldman Sachs and its affiliates (the "Independent Fiduciary"), (2) the Independent Fiduciary is a bank (as defined in section 202 of the Investment Advisers Act of 1940 (the "Advisers Act")) or a similar institution that is regulated and supervised and subject to periodic examination by a State or Federal agency; an insurance carrier which is qualified under the laws of more than one State to perform the services of managing, acquiring or disposing of assets of a plan; an investment adviser registered under the Advisers Act or, if not registered as an investment adviser under the Advisers Act by reason of paragraph (1) of section 203A, is registered as an investment adviser under the laws of the State in which it maintains its principal office and place of business; a broker-dealer registered under the Securities Exchange Act of 1934; or an Independent Fiduciary that holds or has under management or control total assets of at least $50 Million; (3) the Independent Fiduciary is capable of evaluating investment risks independently, both in general and with regard to the information contained in this material; (4) the Independent Fiduciary is a fiduciary under ERISA and/or the Code with respect to, and is responsible for exercising independent judgment in evaluating, the information contained in this material; (5) neither Goldman Sachs nor its affiliates are undertaking to provide impartial investment advice, or to give advice in a fiduciary capacity, in connection with the information contained in this material or with respect to any Benefit Plan Investor's subsequent investment decisions; and (6) any financial interest of Goldman Sachs and its affiliates is or has been disclosed. To the extent this material is provided to any other recipient, this material is provided solely on the basis that the recipient has the capability to independently evaluate investment risk and is exercising independent judgment in evaluating investment decisions in that its investment decisions will be based on its own independent assessment of the opportunities and risks presented by a potential investment, market factors and other investment considerations.

Not a Municipal Advisor: Except in circumstances where Goldman Sachs expressly agrees otherwise in writing, Goldman Sachs is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice, including within the meaning of Section 15B of the Securities Exchange Act of 1934.

Phone recording: Telephone conversations with Goldman Sachs personnel may be recorded and retained.

Legal Entities Disseminating this Material: This material is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Canada by either Goldman Sachs Canada Inc. or Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs(Asia) L.L.C. or by Goldman Sachs Asia Bank Limited, a restricted licence bank; in Japan by Goldman Sachs Japan Co.,Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Singapore by Goldman Sachs(Singapore) Pte. (Company Number: 198602165W), by Goldman Sachs Futures Pte.Ltd (Company Number: 199004153Z) or by J.Aron &Company (Singapore) Pte (Company Number:198902119H); in India by Goldman Sachs (India) Securities Private Limited, Mumbai Branch; in the United Kingdom, Belgium, Denmark, Finland, Ireland, Italy, Liechtenstein and Luxembourg by Goldman Sachs International ("GSI"); in the United Kingdom, Belgium, France, Germany, Ireland, Luxembourg and the Netherlands by Goldman Sachs Bank USA, London Branch (Financial Services Register number 582328); in France by Goldman Sachs Paris Inc. et Cie or GSI Paris branch; in Spain by GSI Madrid branch; in Sweden by GSI Stockholm branch; in all European Economic Area jurisdictions by Goldman Sachs Bank Europe SE, including through its authorised branches ("GSBE"); in Switzerland by GSI; in the Cayman Islands by Goldman Sachs (Cayman) Trust, Limited; in Brazil by Goldman Sachs do Brasil Banco Múltiplo S.A.; in South Africa Goldman Sachs International is operating under an exemption in respect of certain clients in terms of the Financial Advisory and Intermediary Services Act (FAIS), and in the United States of America by Goldman Sachs & Co. LLC (which is a member of FINRA, NYSE and SIPC) and by Goldman Sachs Bank USA. You may obtain information about SIPC, including the SIPC brochure, by contacting SIPC (website: http://www.sipc.org/;phone: 202-371-8300). GSI, which is authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this material in connection with its distribution in the United Kingdom. GSI, whose registered office is at Plumtree Court, 25 Shoe Lane, London EC4A 4AU, appears in the FCA's Register (Registration No.: 142888). GSI is registered as a Private Unlimited Company in England and Wales (Company Number: 2263951) and its VAT registration number is GB 447 2649 28. GSI is subject to the FCA and PRA rules and guidance, details of which can be found on the websites of the FCA and PRA at www.fca.org.uk and www.bankofengland.co.uk/pra.The FCA is located at 12 Endeavour Square, London E20 1JN, and the PRA is located at Bank of England, 20 Moorgate, London EC2R 6DA. GSBE is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism established between those Member States of the European Union whose official currency is the Euro, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank. Unless governing law permits otherwise, you must contact a Goldman Sachs entity in your home jurisdiction if you want to use our services in effecting a transaction in the securities mentioned in this material. This material is not for distribution to retail clients, as that term is defined under The European Union Markets in Financial Instruments Directive(2014/65/EU) (including as implemented in the United Kingdom), and any investments, including derivatives, mentioned in this material will not be made available by us to any such retail client.

Reproduction and Re-Distribution: Without our prior written consent, no part of this material may be (i) copied, photocopied or duplicated in any form by any means or (ii) redistributed. Notwithstanding anything herein to the contrary, and except as required to enable compliance with applicable securities law, you (and each of your employees, representatives and other agents) may disclose to any and all persons the U.S. federal income and state tax treatment and tax structure of the transaction and all materials of any kind (including tax opinions and other tax analyses) that are provided to you relating to such tax treatment and tax structure, without Goldman Sachs imposing any limitation of any kind.

Information Not for Further Dissemination: To the extent this communication contains Goldman Sachs' pricing information, such pricing information is proprietary and/or confidential and is provided solely for the internal use of the intended recipient(s). You are notified that any unauthorized use, dissemination, distribution or copying of this communication or its contents, including pricing information, in whole or in part, is strictly prohibited. Further, unless prohibited by local law, any use, review or acceptance of this information is subject to and manifests your agreement with Goldman Sachs to use such information only in accordance with the terms set forth above. Goldman Sachs has caused its proprietary information to be delivered to you in reliance upon such agreement.

Not a Valuation: Values herein are not customer valuations and should not be used in lieu of a customer valuation statement or account statement. These values may not reflect the value of the positions carried on the books and records of Goldman Sachs or its affiliates and should not be relied upon for the maintenance of your books and records or for any tax, accounting, legal or other purposes. The information provided herein does not supersede any customer statements, confirmations or other similar notifications.

Receipt of Orders: An order sent to Goldman Sachs by email or instant message is not deemed to be received by Goldman Sachs until a Goldman Sachs representative verifies the order details with a phone call to the client or acknowledges receipt of the order via email or instant message to the client. Goldman Sachs does not accept client orders sent via fax or voicemail systems.

Indicative Terms/Pricing Levels: This material may contain indicative terms only, including but not limited to pricing levels. There is no representation that any transaction can or could have been effected at such terms or prices. Proposed terms and conditions are for discussion purposes only. Finalized terms and conditions are subject to further discussion and negotiation.

Special Risk Disclosure related to Exchange-Traded Funds ("ETFs") and Exchange-Traded Notes ("ETNs"): To the extent this communication contains information pertaining to ETFs or ETNs, consider the investment objectives, risks, and charges and expenses of the ETFs and ETNs carefully before investing. Each U.S. Registered ETF and ETN has filed a registration statement (including a prospectus) with the SEC which contains this and other information about the ETF or ETN as applicable. Before you invest in an ETF or ETN, you should obtain and read carefully the prospectus in the registration statement and other documents the issuer has filed with the SEC for more complete information about the product. You may get these documents for free by visiting EDGAR on the SEC website at www.sec.gov. Alternatively, you may obtain a copy of the prospectus for each of the ETFs and ETNs mentioned in these materials by (i)

contacting your Goldman Sachs sales representative, or (ii) sending a request by calling toll-free 1-866-471-2526, or (iii) sending a request by email to Prospectus-ny@ny.email.gs.com, by fax to (212) 902-9316, or by mail to Goldman Sachs, 200 West Street, NY, NY 10282, Attn: Prospectus Department. ETFs are redeemable only in Creation Unit size aggregations and may not be individually redeemed; are redeemable only through Authorized Participants; and are redeemable on an "in-kind" basis. The public trading price of a redeemable lot of the ETFs may be different from its net asset value. These ETFs can trade at a discount or premium to the net asset value. There is always a fundamental risk of declining stock prices, which can cause losses to your investment. Some leveraged and inverse ETFs and ETNs "reset" daily, meaning that they are designed to achieve their stated objectives on a daily basis. They will therefore realize each day a loss or a gain which is a multiple (for leveraged ETFs and ETNs) or the inverse (for inverse ETFs and ETNs) of the performance on that day of their underlying index or benchmark. For this reason, if they are held for a period longer than one day, their performance over such longer periods of time can differ significantly from the stated multiple of the performance (or the inverse of the performance) of their underlying index or benchmark during the same period of time. This effect can be magnified in volatile markets. Prior to entering into a transaction in leveraged or inverse ETFs, you should be aware of the general risks associated with such transactions. You should not enter into leveraged or inverse ETFs transactions unless you understand the nature and extent of your risk exposure. You should also be satisfied that the leveraged or inverse ETFs transaction is appropriate for you in light of your circumstances and financial condition. ETFs and ETNs that are linked to commodity futures do not offer direct exposure to the commodity's spot price, and may perform differently than the spot price for the commodity itself. Performance differential can be magnified if a specific condition persists in the market for a commodity that creates a disparity between near-term future prices and long-term future prices and may lead to unexpected performance results. Other factors, such as roll yield, transaction costs, management fees, and taxes may cause deviation in performance between the spot price of a commodity and commodity futures. You should not assume that an ETF or ETN that is linked to commodity futures will provide an effective hedge because of a negative correlation with equities or other asset classes. You should always be aware of the general risks associated with investing in the commodities market and the futures market before investing in an ETF or ETN that is linked to commodity futures.

Risk Disclosure Regarding futures, options, equity swaps, and other derivatives as well as non-investment-grade securities and ADRs: Please ensure that you have read and understood the current options, futures and security futures disclosure document before entering into any such transactions. Current United States listed options, futures and security futures disclosure documents are available from our sales representatives or at http://www.theocc.com/components/docs/riskstoc.pdf, http://www.goldmansachs.com/disclosures/risk-disclosure-for-futures.pdf and https://www.nfa.futures.org/investors/investor-resources/files/security-futures-disclosure.pdf, respectively. Certain transactions - including those involving futures, options, equity swaps, and other derivatives as well as non-investment-grade securities - give rise to substantial risk and are not available to nor suitable for all investors. If you have any questions about whether you are eligible to enter into these transactions with Goldman Sachs, please contact your sales representative. Foreign-currency-denominated securities are subject to fluctuations in exchange rates that could have an adverse effect on the value or price of, or income derived from, the investment. In addition, investors in securities such as ADRs, the values of which are influenced by foreign currencies, effectively assume currency risk.

Please refer to https://marquee.gs.com/studio/ for price information of corporate equity securities.

Notice to Australian Investors: When this document is disseminated in Australia by Goldman Sachs & Co. LLC ("GSCO"), Goldman Sachs International ("GSI"), Goldman Sachs (Asia) L.L.C. ("GSALLC"), or Goldman Sachs (Singapore) Pte ("GSSP") (collectively the "GS entities"), this document, and any access to it, is intended only for a person that has first satisfied the GS entities that:
▪ the person is a Sophisticated or Professional Investor for the purposes of section 708 of the Corporations Act of Australia; and
▪ the person is a wholesale client for the purpose of section 761G of the Corporations Act of Australia.
To the extent that the GS entities are providing a financial service in Australia, the GS entities are each exempt from the requirement to hold an Australian financial services licence for the financial services they provide in Australia. Each of the GS entities are regulated by a foreign regulator under foreign laws which differ from Australian laws, specifically:
▪ GSCO is regulated by the US Securities and Exchange Commission under US laws;
▪ GSI is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, under UK laws;
▪ GSALLC is regulated by the Hong Kong Securities and Futures Commission under Hong Kong laws; and
▪ GSSP is regulated by the Monetary Authority of Singapore under Singapore laws.

Notice to Brazilian Investors

Marquee is not meant for the general public in Brazil. The services or products provided by or through Marquee, at any time, may not be offered or sold to the general public in Brazil. You have received a password granting access to Marquee exclusively due to your existing relationship with a GS business located in Brazil. The selection and engagement with any of the offered services or products through Marquee, at any time, will be carried out directly by you. Before acting to implement any chosen service or products, provided by or through Marquee you should consider, at your sole discretion, whether it is suitable for your particular circumstances and, if necessary, seek professional advice. Any steps necessary in order to implement the chosen service or product, including but not limited to remittance of funds, shall be carried out at your discretion. Accordingly, such services and products have not been and will not be publicly issued, placed, distributed, offered or negotiated in the Brazilian capital markets and, as a result, they have not been and will not be registered with the Brazilian Securities and Exchange Commission (Comissão de Valores Mobiliários), nor have they been submitted to the foregoing agency for approval. Documents relating to such services or products, as well as the information contained therein, may

not be supplied to the general public in Brazil, as the offering of such services or products is not a public offering in Brazil, nor used in connection with any offer for subscription or sale of securities to the general public in Brazil.

The offer of any securities mentioned in this message may not be made to the general public in Brazil. Accordingly, any such securities have not been nor will they be registered with the Brazilian Securities and Exchange Commission (Comissão de Valores Mobiliários) nor has any offer been submitted to the foregoing agency for approval. Documents relating to the offer, as well as the information contained therein, may not be supplied to the public in Brazil, as the offer is not a public offering of securities in Brazil. These terms will apply on every access to Marquee.

Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 380 5764 e/ou contatogoldmanbrasil@gs.com.
Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h. Canal de Denúncias: 0800 047 4065 e/ou Formulário eletrônico: https://gs.tnwreports.com/
Public Communication Channel Goldman Sachs Brazil: 0800 380 5764 and/or contatogoldmanbrasil@gs.com.
Available Weekdays (except holidays), from 9am to 6pm. Business Integrity Issues Channel: 0800 047 4065 and/or Web Form: https://gs.tnwreports.com/

Please note that none of Goldman Sachs and its affiliates including Goldman Sachs (Asia) LLC maintains any licenses, authorisations or registrations in the People's Republic of China ("PRC") nor are the funds discussed in this material registered in the PRC.

The distribution of this material and the placement of interests in the funds in Hong Kong is restricted. Accordingly, unless permitted by the securities laws of Hong Kong, no person may issue or have in its possession for the purposes of issue, this material, or any advertisement, invitation or document relating to the interests in the funds, whether in Hong Kong or elsewhere, which is directed at, or the contents of which are likely to be accessed or read by, the public in Hong Kong, other than with respect to the interests in the funds which are intended to be disposed of only to persons outside Hong Kong or only to —professional investorsǁ within the meaning of the Securities and Futures Ordinance (Cap. 571 of the Laws of Hong Kong) and any rules made thereunder.

The interests in the limited partnership / funds may not be offered or sold in the Republic of Indonesia or to Indonesian citizens or residents in any manner which may constitute a public offer of securities under the Law Regarding Capital Markets.

Note to Investors in Israel: GS is not licensed to provide investment advice or investment management services under Israeli law.

Notice to Investors in Japan
Marquee is made available in Japan by Goldman Sachs Japan Co., Ltd.

本書は情報の提供を目的としております。また、売却・購入が違法となるような法域での有価証券その他の売却若しくは購入を勧めるものでもありません。ゴールドマン・サックスは本書内の取引又はストラクチャーの勧誘を行うものではございません。これらの取引又はストラクチャーは、社内及び法規制等の承認等次第で実際にはご提供できない場合がございます。

<適格機関投資家限定　転売制限>
ゴールドマン・サックス証券株式会社が適格機関投資家のみを相手方として取得申込みの勧誘（取得勧誘）又は売付けの申込み若しくは買付けの申込みの勧誘(売付け勧誘等)を行う本有価証券には、適格機関投資家に譲渡する場合以外の譲渡が禁止される旨の制限が付されています。本有価証券は金融商品取引法第４条に基づく財務局に対する届出が行われておりません。なお、本告知はお客様によるご同意のもとに、電磁的に交付させていただいております。

＜適格機関投資家用資料＞
本資料は、適格機関投資家のお客さまのみを対象に作成されたものです。本資料における金融商品は適格機関投資家のお客さまのみがお取引可能であり、適格機関投資家以外のお客さまからのご注文等はお受けできませんので、ご注意ください。 商号等/ゴールドマン・サックス証券株式会社 金融商品取引業者　関東財務局長（金商）第６９号
加入協会/　日本証券業協会、一般社団法人金融先物取引業協会、一般社団法人第二種金融商品取引業協会、一般社団法人資産運用業協会
本書又はその添付資料に信用格付が記載されている場合、日本格付研究所（JCR）及び格付投資情報センター（R&I）による格付は、登録信用格付業者による格付（登録格付）です。その他の格付は登録格付である旨の記載がない場合は、無登録格付です。無登録格付を投資判断に利用する前に、「無登録格付に関する説明書」（http://www.goldmansachs.com/disclaimer/ratings.html）を十分にお読みください。

If any credit ratings are contained in this material or any attachments, those that have been issued by Japan Credit Rating Agency, Ltd. (JCR) or Rating and Investment Information, Inc. (R&I) are credit ratings that have been issued by a credit rating agency registered in Japan (registered credit ratings). Other credit ratings are unregistered unless denoted as being registered. Before using unregistered credit ratings to make investment decisions, please carefully read "Explanation Regarding Unregistered Credit Ratings" (http://www.goldmansachs.com/disclaimer/ratings.html).

Notice to Mexican Investors: Information contained herein is not meant for the general public in Mexico. The services or products provided by or through Goldman Sachs Mexico, Casa de Bolsa, S.A. de C.V. (GS Mexico) may not be offered or sold to the general public in Mexico. You have received information herein exclusively due to your existing relationship with a GS Mexico or any other Goldman Sachs business. The selection and engagement with any of the offered services or products through GS Mexico will be carried out directly by you at your own risk. Before acting to implement any chosen service or product provided by or through GS Mexico you should consider, at your sole discretion, whether it is suitable for your particular circumstances and, if necessary, seek professional advice. Information contained herein related to GS Mexico services or products, as well as any other information, shall not be considered as a product coming from research, nor it contains any recommendation to invest, not to invest, hold or sell any security and may not be supplied to the general public in Mexico.

Notice to New Zealand Investors: When this document is disseminated in New Zealand by Goldman Sachs & Co. LLC ("GSCO") , Goldman Sachs International ("GSI"), Goldman Sachs (Asia) L.L.C. ("GSALLC") or Goldman Sachs (Singapore) Pte ("GSSP") (collectively the "GS entities"), this document, and any access to it, is intended only for a person that has first satisfied; the GS entities that the person is someone:
(i) who is an investment business within the meaning of clause 37 of Schedule 1 of the Financial Markets Conduct Act 2013 (New Zealand) (the "FMC Act");
(ii) who meets the investment activity criteria specified in clause 38 of Schedule 1 of the FMC Act;
(iii) who is large within the meaning of clause 39 of Schedule 1 of the FMC Act; or
(iv) is a government agency within the meaning of clause 40 of Schedule 1 of the FMC Act.
No offer to acquire the interests is being made to you in this document. Any offer will only be made in circumstances where disclosure is not required under the Financial Markets Conducts Act 2013 or the Financial Markets Conduct Regulations 2014.

Notice to Swiss Investors: This is marketing material for financial instruments or services. The information contained in this material is for general informational purposes only and does not constitute an offer, solicitation, invitation or recommendation to buy or sell any financial instruments or to provide any investment advice or service of any kind.

Not all services or products can be made available in Taiwan. Any particular offer of securities may not have been and may not be registered with the Securities and Futures Bureau, Financial Supervisory Commission of Taiwan pursuant to relevant securities laws and regulations and may therefore not be capable of being sold or publicly offered in Taiwan. The Goldman Sachs companies involved in any such promotion may not maintain any licenses, authorisations or registrations in Taiwan. Therefore, for such securities, any subscriptions into the securities must be received in a currency foreign to, and to an account offshore, Taiwan. Any issue of such securities will not take effect until a valid subscription is received and accepted in accordance with the offering materials, outside Taiwan.

Notice to Investors in Korea: Marquee is made available in Korea by Goldman Sachs (Asia) L.L.C. Seoul Branch.

Please note that none of Goldman Sachs and its affiliates ("Goldman Sachs"), that provides material and information or enters or will enter into the relevant transactions, maintains any licenses, authorisations or registrations in Thailand nor is any of the material and information contained, or the relevant securities or products specified, herein approved or registered in Thailand. Interests in the relevant securities or products are not being offered or sold and may not be offered or sold in Thailand, except as permitted by the relevant laws and regulations. The attached information is intended to be read by the addressee and has been provided at your request for information purposes only and must not be copied or redistributed to any other person without the prior consent of Goldman Sachs and in no way constitutes an offer, solicitation, advertisement or advice of, or in relation to, the relevant securities or products by Goldman Sachs in Thailand.

The information contained in this document does not constitute, and is not intended to constitute, a public offer of securities in the United Arab Emirates in accordance with the Commercial Companies Law (Federal Law no. 2 of 2015), ESCA Board of Directors' Decision no. (9/r.m.) of 2016, ESCA Chairman Decision no 3/r.m. of 2017 concerning promoting and introducing regulations or otherwise under the laws of the United Arab Emirates. Accordingly, the interests in the securities may not be offered to the public in the UAE (including the Dubai International Financial Centre and the Abu Dhabi Global Market). This document has not been approved by, or filed with the Central Bank of the United Arab Emirates, the Securities and Commodities Authority, the Dubai Financial Services Authority, the Financial Services Regulatory Authority or any other relevant licensing authorities in the United Arab Emirates. If you do not understand the contents of this document, you should consult with a financial advisor. This document is provided to the recipient only and should not be provided to or relied on by any other person.

© 2026 Goldman Sachs. All rights reserved.

國泰期貨 國泰證券子公司

產業更新 – 半導體測試產業

Insertion 4 為 CPO 量產化的「最後一公里」

What's New – CPO 測試流程逐步走向「系統級出貨驗證」

CPO 測試流程逐步從「單體功能驗證」走向「系統級出貨驗證」，測試對象由 PIC / EIC 等單一元件，逐步擴展至 Optical Engine，最終進入完整 CPO 系統的功能、效能與可靠度驗證。Insertion 1–3 的核心目的，是在 CPO 整合程度與元件價值逐步提高的過程中，盡可能提早攔截不良品，為後續 Insertion 4 的系統級出貨驗證建立良率基礎。Insertion 4 之所以重要，在於它是 CPO 從「單體功能驗證」走向「系統級出貨驗證」的最後一道關鍵測試關卡，與前段測試目的大不相同。

Trends/Outlook – In 4 競爭關鍵轉向完整光電測試系統整合

Insertion 4 的測試重點，在於完成 ASIC 與 Optical Engine 整合後，驗證整體 CPO 的光電訊號傳輸是否符合規格。Insertion 4 必須同時確認兩條關鍵路徑：1) ASIC 與 Optical Engine 之間的高速電性互聯。2) Optical Engine 與 FA / FAU 光纖介面之間的光訊號發射(Tx)與接收(Rx)。在測試架構上，In4 可依光學與電性測試的整合程度，分為「光電分開測」與「光電同測」。光電分開測較容易沿用既有的半導體與光學測試設備，但較難在相同條件下直接驗證完整的端到端光電訊號鏈路；光電同測則是在同一測試環境中，同時建立高速電性與光學介面，可更完整掌握 Tx/Rx、BER、訊號品質以及溫度變化對整體系統的影響。

Investment Summary

綜觀 Insertion 4 供應鏈，海內外廠商皆有參與。Advantest 與 Teradyne 皆已將 CPO 最終封裝測試納入光電測試布局。Advantest （6857 JP） 主要以既有 V93000 ATE 平台為核心建立整合式測試架構；Teradyne (TER US) 的 Photon 100 明確定位為矽光與 CPO 的光電自動化量產測試平台。台廠方面，致茂 (2360 TT) 憑藉既有光學對位與量測能力，先切入 In4o 領域，佈局涵蓋自動光學連接、校正清潔與插入損耗量測，將受惠於 CPO 封裝後光學測試需求持續增加；鴻勁 (7769 TT) 則聚焦 Handler 與溫控系統 (ATC)，In4 光電同測設備將從 FY26 進入量產型驗證階段，FY27 將持續隨客戶產品開發進度推進，雖 FY26 CPO 相關營收貢獻仍有限，但將成為中長期成長動能，研究部預期鴻勁於高階 ATC Handler 具領先市場地位，將能受惠於後段測試對精密操作、溫控及自動化整合需求提升。

王筠婷
特化、半導體週邊
+886-2-2326-9899#7961
vanessawang@cathayfut.com.tw

個股	代碼	預估每股盈餘 (以當地幣別計算)		每股盈餘成長率(%)		預估本益比		目標價	最新收盤價 (以當地幣別計算)	評等
		2026E	2027E	2026E	2027E	2026E	2027E			
鴻勁	7769 TT	128.87	200.23	87.58	55.37	50.36	32.41	8,400	6,490	買進
致茂	2360 TT	44.74	65.5	117.82	46.39	52.86	36.11	N/A	2,365	Non-rated
Advantest	6857 JP	937.29	1,209.34	83.45	29.02	44.39	34.41	N/A	41,610	Non-rated
Teradyne	TER US	8.84	11.45	126.90	29.59			N/A	402.68	Non-rated

國泰期貨 國泰證券子公司

Insertion4 為 CPO 量產化的「最後一公里」

CPO 測試流程逐步從「單體功能驗證」走向「系統級出貨驗證」

CPO 測試流程逐步從「單體功能驗證」走向「系統級出貨驗證」，測試對象由 PIC / EIC 等單一元件，逐步擴展至 Optical Engine，最終進入完整 CPO 系統的功能、效能與可靠度驗證。Insertion 4 之所以重要，在於它是 CPO 從「單體功能驗證」走向「系統級出貨驗證」的最後一道關鍵測試關卡，與前段測試目的大不相同。

在進入 Insertion 4 系統級測試之前，CPO 前段主要透過 Insertion 1–3 逐步篩選不良元件並確認光電整合品質。Insertion 1 主要針對尚未整合的 PIC 與 EIC 進行晶圓 / 裸晶層級測試，分別確認光學與電性功能是否符合規格，以篩選 Known Good Die（KGD），避免不良晶粒進入後續高成本封裝。Insertion 2 則發生在 PIC 與 EIC 完成鍵合後，測試重點由單一元件轉向光電整合功能，透過電性探針與光學耦合，同步驗證電轉光（E/O）與光轉電（O/E）訊號是否正常，並檢查鍵合後的高速訊號與光學性能。Insertion 3 則進一步來到 Optical Engine 階段，在晶片切割並完成 FAU / 光纖等光學介面組裝後，針對完整光引擎進行 Tx/Rx、光功率、BER、多通道與高速介面等測試，篩選出 Known Good Optical Engine（KGOE）。因此，Insertion 1–3 的核心目的，是在 CPO 整合程度與元件價值逐步提高的過程中，盡可能提早攔截不良品，為後續 Insertion 4 的系統級出貨驗證建立良率基礎。

表一. Insertion 1/2/3/4 測試重點

測試階段	測試重點	重要性及面對難題	相關廠商
Insertion 1	PIC/EIC 晶圓級單面測試	1. 在 PIC / EIC 尚未整合前先篩選 KGD，避免不良晶粒進入高成本封裝 2. 光學測試為了找出最佳的光耦合點，需要精密設備進行多軸微調，測試時間大幅增加	FormFactor、Advantest、TEL、Teradyne、旺矽
Insertion 2	EPIC 混合鍵合後雙面測試	1. 驗證 PIC 與 EIC 鍵合後的光電整合功能，確認 E/O、O/E 轉換是否正常 2. 光學與電性兩套系統必須同步對位	Teradyne、Advantest、旺矽、ficonTEC+漢測
Insertion 3	Optical Engine 測試	1. 在 Optical Engine 階段篩選 KGOE，避免不良光引擎與高價值 ASIC 整合	Advantest、Teradyne、ficonTEC+漢測、旺矽、致茂
Insertion 4	Module 測試	1. 需同時驗證高速電性互聯、光訊號 Tx/Rx、BER、熱管理與多通道一致性，若失效，報廢成本最高	Advantest、Teradyne、鴻勁、致茂

資料來源：國泰證期研究部整理

Insertion 4 競爭關鍵逐漸由單一設備性能，轉向完整光電測試系統的整合與自動化能力

Insertion 4 的測試重點，在於完成 ASIC 與 Optical Engine 整合後，驗證整體 CPO 的光電訊號傳輸是否符合規格。相較於前段測試著重於個別元件或 Optical Engine，Insertion 4 必須同時確認兩條關鍵路徑：1) ASIC 與 Optical Engine 之間的高速電性互聯。2) Optical Engine 與 FA / FAU 光纖介面之間的光訊號發射（Tx）與接收（Rx）。在 Tx 路徑中，需確認 ASIC 輸出的高速電訊號能否經由 EIC 與 PIC 正確轉換為光訊號並透過光纖送出；Rx 路徑則需驗證外部光訊號能否經由光偵測器與 EIC 轉換回電訊號，再正確傳送至 ASIC。隨著 CPO 傳輸速率與通道數持續提升，測試過程

還需進一步考量 BER、訊號完整性、通道間干擾、功耗及溫度變化等因素，使 I4 從單一元件測試提升至完整光電鏈路的系統級驗證。

在測試架構上，I4 可依光學與電性測試的整合程度，分為「光電分開測」與「光電同測」。光電分開測是將電性與光學測試安排於不同測試步驟或測試站，較容易沿用既有的半導體與光學測試設備，但較難在相同條件下直接驗證完整的端到端光電訊號鏈路；光電同測則是在同一測試環境中，同時建立高速電性與光學介面，直接驗證從 ASIC、Optical Engine 到 Fiber 的訊號傳輸，可更完整掌握 Tx/Rx、BER、訊號品質以及溫度變化對整體系統的影響。

因此，若 In4 朝光電同測發展，其技術門檻將不再只是 Tester 本身的測試能力，而是如何將 Tester、Handler、Test Interface、Optical Interface 與 Thermal Control 整合至同一套量產測試平台。在兼顧測試精度的同時，設備還必須維持穩定的光學耦合、高速電性接觸與溫度控制，並縮短測試時間、提高平行測試能力，以滿足量產所需的 throughput。這使 In4 的競爭關鍵逐漸由單一設備性能，轉向完整光電測試系統的整合與自動化能力。

競爭重點延伸至光學連接自動化、Handler / Interface 整合

綜觀 Insertion 4 供應鏈，海內外廠商皆有參與。Advantest 與 Teradyne 皆已將 CPO 最終封裝測試納入光電測試布局。Advantest 主要以既有 V93000 ATE 平台為核心，透過與光學探針、FAU、Handler 與 Test Interface 生態系夥伴合作，建立可涵蓋 wafer、die 到最終 NPO / CPO package test 的整合式測試架構；Teradyne 的 Photon 100 明確定位為矽光與 CPO 的光電自動化量產測試平台，將光學與電性量測整合進既有 UltraFLEXplus ATE，並可支援 wafer、Optical Engine 以及 co-packaged module 等不同 insertion。整體而言，兩家公司在 I4 的競爭方向，都是從傳統「電性 Tester」走向「光電整合測試單元」，隨 CPO 進入量產，兩家的競爭重點預期將不只在 Tester 規格，而會進一步延伸至光學連接自動化、Handler / Interface 整合、平行測試能力與整體 throughput。

台廠方面，致茂（2360 TT）憑藉既有光學對位與量測能力，先切入 In4o 領域，佈局涵蓋自動光學連接、校正清潔與插入損耗量測，佈局領域完善，將受惠於 CPO 封裝後光學測試需求持續增加；鴻勁（7769 TT）則聚焦 Handler 與溫控系統（ATC），In4 光電同測設備將從 FY26 進入量產型驗證階段，FY27 將持續隨客戶產品開發進度推進，雖 FY26 CPO 相關營收貢獻仍有限，但將成為中長期成長動能，研究部預期鴻勁於高階 ATC Handler 具領先市場地位，將能受惠於後段測試對精密操作、溫控及自動化整合需求提升。

研究團隊

李朋潤
研究部主管
+886-2-2326-9899#9814
pengjunlee@cathayfut.com.tw

藍文彥
金融、消費、工業
+886-2-2326-9899#1145
michael@cathayfut.com.tw

呂旻旂
紡織、製鞋、電子組裝
+886-2-2326-9899#7960
minchilu@cathayfut.com.tw

呂理舜
綠色能源、電動車
+886-2-2326-9899#9844
dufflu@cathayfut.com.tw

林裕禮
IP、IC設計
+886-2-2326-9899#1944
henrylin@cathayfut.com.tw

璩文浩 CA
ESG
+886-2-2326-9899#9856
whchu@cathayfut.com.tw

王筠婷
特用化學、半導體週邊
+886-2-2326-9899#7961
vanessawang@cathayfut.com.tw

姜仁匡
PCB
+886-2-2326-9899#1601
kenchiang@cathayfut.com.tw

紀彥呈
半導體、資訊軟體
+886-2-2326-9899#1940
ianchi@cathayfut.com.tw

劉耀文
IPC、鋼鐵
+886-2-2326-9899#1751
yaowen@cathayfut.com.tw

鍾沛宸
IC設計
+886-2-2326-9899#1750
parisphchung@cathayfut.com.tw

王韋傑
被動元件
+886-2-2326-9899#3954
jerrywang@cathayfut.com.tw

張雅倩
研究員
+886-2-2326-9899#1122
val.chang@cathayfut.com.tw

林美君
研究員
+886-2-2326-9899#1752
meira.lin@cathayfut.com.tw

賴振宇
資深研究員
+886-2-2326-9899#9817
alex.lai@cathayfut.com.tw

郭經緯 CPA
顧問部主管
+886-2-2326-9899#9812
raymondkuo@cathayfut.com.tw

何緯婷
總經
+886-2-2326-9899#9835
lauraho@cathayfut.com.tw

陳玉庭
總經、ETF
+886-2-2326-9899#7964
yutingchen@cathayfut.com.tw

張立民
研究員
+886-2-2326-9899#1731
changlimin@cathayfut.com.tw

陳宇亮
研究員
+886-2-2326-9899#9846
ivor0704@cathayfut.com.tw

林威廷
研究員
+886-2-2326-9899#3956
adamlin@cathayfut.com.tw

國泰期貨 國泰證券子公司

投資評等定義

英文評等	中文評等	潛在報酬率
STRONG BUY	強力買進	> +30%
BUY	買進	介於+15% ~ +30%
NEUTRAL	中立	介於-15% ~ +15%
SELL	賣出	< -15%
註:潛在報酬率=(6 個月內目標價/本次報告評等最近一天收盤價 – 1) * 100		

免責聲明

本報告僅為本公司提供客戶之一般參考資料， 並非針對客戶之特定需求所作的投資建議， 且在本報告撰寫過程中， 並未考量讀者個別的財務狀況與需求， 故本報告所提供的資訊無法適用於所有客戶或投資人， 讀者應審慎考量本身之投資風險， 並就投資結果自行負責。

本報告係根據本公司所取得資訊加以彙集及研究分析， 本公司雖盡力使用可靠且廣泛的資訊，但本公司並不保證各項資訊之完整性及正確性。本報告中所提出之意見係為本報告出版當時的意見， 邇後相關資訊或意見若有變更， 本公司將不會另行通知。本公司亦無義務持續更新本報告之內容或追蹤研究本報告所涵蓋之主題。本報告中提及的標的物價格、價值及收益隨時可能因各種本公司無法控制之政治、經濟、市場等因素而產生變化。本報告中之各項預測， 如營收及目標價格， 均係基於對目前所得資訊作合理假設下所完成， 所以並不必然實現。從事特定交易， 如涉及期貨、選擇權及衍生性金融商品， 因具有高度風險並不適合每一個投資人。本報告不得視為買賣有價證券或其他金融商品的要約或要約之引誘。

國泰金融控股公司所屬子公司暨關係企業 (以下稱 “ 國泰金融集團”) 從事廣泛金融業務，包括但不限於銀行、保險、證券經紀、自有資金投資與交易、資產管理及證券投資信託等。國泰金融集團對於本報告所涵蓋之標的公司可能有投資或其他的業務往來關係。國泰金融集團及其所屬員工， 可能會投資本報告所涵蓋之標的公司， 或以該等公司為標的的衍生性金融商品， 而其交易的方向與本報告中所提及的交易方向可能不一致。除此之外， 國泰金融集團於法令許可的範圍內， 亦有可能以本報告所涵蓋之標的公司作為發行衍生性

金融商品之標的。國泰金融集團轄下的銷售人員、交易員及其他業務人員可能會為他們客戶或自營部門提供口頭或書面的市場看法或交易策略， 然而該等看法與策略可能與本報告的意見不盡一致。國泰金融集團的資產管理、自營及其他投資業務所做的投資決策也可能與本報告所做的建議或看法不一致。

非經本公司事先書面同意， 不得發送或轉送本報告予第三人轉載或使用。

國泰期貨 國泰證券子公司

漢磊 3707 TT （電子–半導體）

AI 挹注，終迎春燕

投資與評價

1. AI 需求強勁，2H26 逐步復甦，SiC 競爭壓力減緩，去庫存亦已完成，預估 2026 年營收年增 29.0%。
2. 新增產能陸續開出，應用多點開花，產品價量齊升，預估 2027 年營收年增 23.8%。
3. 預估漢磊 2026/2027 年 EPS 為 0.99/4.77 元，考量整體產業趨勢顯著下，以同業本益比區間中間偏上位置 21 倍為評價基礎(21X FY27E EPS)，目標價 100 元，由中立轉為買進投資建議。

重點更新

2026 年 Si 逐步回溫，化合物半導體需求跳升，預估營收年增 29.0%

根據調研，漢磊在六吋需求與產能利用率均有改善，四、五吋廠過去產能利用率長期偏低，包含原物料取得不易，8 月已停止生產，10 月完成設備與人員轉移，將資源集中於高價值的六吋與八吋產品，在 Si 產品方面，TVS 受惠 AI 伺服器保護電路需求增加及轉單效應，目前產能處於滿載狀態(使用六吋)，推估 2H26 較 1H26 成長約 32%，化合物半導體部分，GaN 亦受惠 AI 資料中心需求，產能滿載，HoH 研究部推估成長 26%，目前客戶需求已達可供應量的兩倍水準，SiC 部分先前因大陸廠商競爭激烈，亦面臨去庫存，產能利用率上半年僅約三成水準，現因其他廠商減產以及去庫存完成，3Q26 已回升至六成，研究部預估 4Q26 將進一步回升至 65%水準，低基期下 2H26 營收 HoH 將成長約 123.5%，整體預估 2026 年營收至年增 29.0%。

AI 帶動 2027 年維持高速成長，預估營收年增 23.8%

TVS 部分由於新增產能陸續開出，加上 AI 伺服器、高速介面及晶片微縮後增加的靜電保護需求，預估 2027 年將成長 19.8%，GaN 應用由消費性快充擴大至 AI 伺服器 PSU、48V / 800V 直流電源轉換架構，考量生產效率以及供應能力，在客戶需求維持強勁下，研究部預估 2027 年成長約 15%水準，至於 SiC 應用由電動車主驅逆變器，延伸至車載充電器，AI 資料中心的高功率 PSU 與固態變壓器部分客戶已在開發相關產品，在基期低下，明年 SiC 營收相較今年將翻倍成長，目前與世界先進合作的八吋的月產能約 1,500 片，漢磊自有六吋 SiC 月產能約 5,000 片，合計約 8,000 片六吋等效月產能。整體 2027 年研究部預估營收將年增 23.75%，達 92.1 億元。

買進 – 中立轉買進

目標價：100.0 元　+21.6%
收盤價：82.2 元

報告日期	評等	目標價
2026/10/13	買進	100 元
2025/10/09	中立	54 元
2025/01/15	中立	43 元

	國泰	市場	
目標價	100	--	--
2026 EPS	0.99	--	--
2027 EPS	4.77	--	--
現在 P/E	17.25	--	--

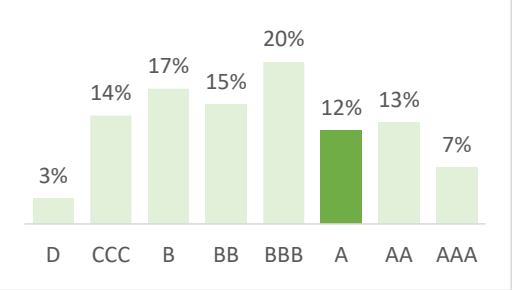
公司資訊

市值(百萬元)	32,970
自由流通股數比例(free float ; %)	78.93
20 日平均成交量(百萬元)	1,613
每股淨值(元,2026E)	24.05
股價淨值比(倍)	3.42
現金股利率(%)	1.58
股東權益報酬率(%,2021E)	18.24
外資持股比率(%)	19.42

漢磊 ESG 評等

ESG 評等	社會(S)	經濟(E)	環境(E)	揭露(D)
A	55%	55%	41%	48%

電子製造業 ESG 評等分配圖



資料來源：台灣指數公司，國泰證期研究部整理

林裕禮 /半導體
+886-2-2326-9899#1944 / henrylin@cathayfut.com.tw

項目(百萬元)	2Q'26	3Q'26(E)	4Q'26(E)	1Q'27(E)	2025	2026(E)	2027(E)
營業收入	1,795	1,969	2,139	2,041	5,766	7,438	9,205
稅前盈餘	34	132	171	165	-738	267	836
稅後盈餘(母公司業主)	-87	261	360	374	-757	397	1,909
營收成長率(%)(QoQ/YoY)	16.97	9.67	8.67	-4.60	-0.88	28.99	23.75
毛利率(%)	11.30	16.10	17.20	18.00	-3.30	12.98	18.82
營益率(%)	1.10	5.84	7.20	7.27	-15.07	2.64	8.34
稅後盈餘成長率(%)(QoQ/YoY)	N/A	虧轉盈	38.24	3.87	N/A	虧轉盈	380.71
每股盈餘 EPS(元)	-0.22	0.65	0.90	0.93	-1.89	0.99	4.77

EPS 以調整後股數 34.87 百萬股計算

訪談內容

SiC 合作優於預期，需求逐步爬坡

SiC 目前 AI 伺服器功率需求逾 3kW 至 8kW 以上，SiC MOSFET 用於 3kW 以上 PSU，可縮小體積及降低散熱成本，另外太陽能逆變器、儲能與其他高壓系統逐步由 650V、750V 往 1200V、1700V 及更高電壓發展，漢磊客戶已有 3300V 產品，SiC 在價格達到甜蜜點下亦往消費性電子及小家電滲透。

而與世界先進合作的八吋 SiC 設備已全數到位，首批試產片良率優於預期，目標年底前進入小量量產，首波五個客戶準備 Tape-out，預估 4Q26 至 1Q27 陸續推進，漢磊負責客戶與技術平台，將技術導入合作產線，世界先進提供生產線及製造服務，客戶關係及技術仍由漢磊負責。八吋目前預估月產能約 1,500 片，漢磊自有六吋 SiC 月產能約 5,000 片，八吋面積約為六吋的 1.7~1.8 倍，良率也較高，估算八吋 1,500 片約相當於六吋 3,000 片，初期合計約 8,000 片六吋等效月產能，透過較小規模投資，合作八吋產線有機會擴至月產 3,000 片，合計六吋等效月產能可超過 1 萬片。

表一、漢磊 2026 年修正預估與前次預估比較

Items	2025/10/09 2026 Forecast	Revised 2026 Forecast	Difference
Revenue (NT$mn)	7,447	7,438	-0.12%
GM (%)	15.62	12.98	-2.60ppt
OPM (%)	4.91	2.64	-2.27ppt
Pretax Profit (NT$mn)	474	267	-43.67%
Net Profit (NT$mn)	378	397	5.03%
EPS (NT$)	0.99	0.99	0.00%

資料來源：國泰證期研究部預估

漢磊 ESG 政策

漢磊主要供應元件及積體電路代工事業：雙載子積體電路(Bipolar IC Service)、高功率場效電晶體(Power MOSFET)、混合型積體電路(邏輯元件)、高壓 CMOS、高壓 BCD、GaN 及 SiC 分離式元件等。產業類別歸類在半導體行業中，依據專門為投資人設計的永續揭露框架 SASB 準則，半導體行業關注重點為下列幾項：**溫室氣體排放、能源管理、水和廢水管理、廢棄物和危險材料管理、員工健康與安全、員工多元化和包容性、產品設計與生命週期管理、材料採購與效率**。因漢磊企業社會責任報告書尚未更新至 2021 年以及 2020 年，相關資訊揭露較少，故從已知資訊作分析。

1. 節省能源消耗政策：

漢磊所使用於辦公室及生產廠區的能源，包含外購電力及各類化石燃料相關統計如下：

a. 天然氣：主要用於有機廢氣吸附及燃燒處理系統與餐廳。

b. 柴油：主要用於緊急發電機，當停電、定期保養及緊急狀況時才啟動，非工廠生產所需直接能源。

c. 外購電力：提供公司內部照明、空調、動力等的系統運作。

因應產品特性，新增 6 吋 SiC 製程產線及車用產品品質要求，2019 年單位用電量較前 2 年增加，但總用電量仍呈現下降趨勢。能源的使用不僅會消耗地球之資源，同時亦會產生二氧化碳，造成溫室效應。漢磊科歷年來持續進行各項節能技術，並進行相關設施之節能改善計畫，節能成效顯著。

圖一、漢磊節能策略

類別	作法
空調	調整無塵室溫度及壓差，達到效能最佳化
	更新空調箱盤管，提升熱交換及冷凍能力系統效能
	冰水主機定期保養維護及管線測漏，維持設備高效率
	循環空調箱加裝變頻器節能控制
照明	逐步將耗能傳統燈具改成T5或LED燈具
CDA	淘換效率不佳機台進而提升效能(如變頻式空壓機、加熱式乾燥機等)
排氣	無塵室排氣抽力調整，降低空調負載
	機台廢熱回收節省能源消耗
泵浦	增設變頻器調整最佳化

資料來源:漢磊、國泰證期研究部整理

2. 水資源管理及水汙染防治：

漢磊用水均來自於自來水廠供應，對水源並無顯著影響。其主要提供製程、空調及民生消耗所需。2019 年因應產品特性，新增 6 吋 SiC 製程產線及車用產品品質要求，單位用水量較前 2 年增，但總用水量仍呈現下降趨勢。受全球氣候變遷影響，漢磊科更積極推動節能減碳活動，除了積極節水外，也持續推動節水專案及引進各方面節水技術，2019 年節水比率逐步成長至 10.19%。漢磊科廠區之用水量除蒸發損耗及綠化用水外，其餘生活廢水及製程循環廢水皆經由適當的污水處理設施處理後，排入污水下水道，流入管理局廢水處理廠。廢水排放水質皆符合主管機關之要求，對生態環境及當地社區無顯著衝擊。

圖二、漢磊節水策略

類別	作法
生產機台節水	製程機台排水回收再利用
	晶舟盒清洗機排水回收再利用
	純水製程回收系統再生廢水回收
	調整機台待機時水流量調整
	回收控制系統優化修改
	UF再生快沖水回收
	研磨機與純水真空泵冷卻水回收
	FAB A L/S 使用微酸性廢水回收
民生用水	廁所馬桶及便斗使用次級用水
開發水源	雨水回收

資料來源:漢磊、國泰證期研究部整理

圖三、漢磊 2017~2019 年用水回收率

	2017年	2018年	2019年	回收率標準
製程用水回收率	69%	69%	73%	>50%
全廠用水回收率	56%	58%	58%	>30%

資料來源:漢磊、國泰證期研究部整理

3. 碳排放管理：

漢磊廠內配合國家溫室氣體減量法推動及相關政策致力於降低溫室氣體的衝擊。依循 CNS 14064-1 及行政院環保署溫室氣體查驗指引、溫室氣體排放量盤查登錄作業指引，並撰寫溫室氣體盤查報告書，對外揭露溫室氣體排放量數據。半導體廠的主要溫室氣體來源為全氟化物（Perfluorocompound, PFC），以≤150mm (六吋)晶圓最佳之技術選擇是製程最佳化（Process Optimization）及取代化學品（Alternative Chemistries）方式進行 PFC 排放減量。另外漢磊之溫室氣體排放主要為使用電力間接排放之溫室氣體約佔總排放量之 58.87%，對溫室氣體減量的努力將著重於用電設備能源效率之提升。2019 年溫室氣體增加主要是因應產能提升需求，採購新機器設備而至用電量增加，故溫室氣體排放量較基準年高。

圖四、溫室氣體統計圖

溫室氣體盤查統計	2016年(基準年)	2019年
溫室氣體排放量(範疇1)(Tons CO2e)	29,136	29,687
溫室氣體排放量(範疇2)(Tons CO2e)	41,593	42,489
範疇1及2所涵蓋之溫室氣體種類及個別排放量	CO_2、CH_4、N_2O、PFCs、HFCs、SF_6、NF_3	

資料來源:漢磊、國泰證期研究部整理

4. 廢棄物管理：

漢磊 2019 年產出廢棄物數量為 566 噸，較 2018 年減少 26 噸，其中回收及再利用廢棄物為 110 噸佔總量 19%；一般事業廢棄物為 199 噸佔總量 35%；有害事業廢棄物為 258 噸佔總量 46%。將廠區廢棄物種類進行分析，2019 年回收及再利用部分較 2018 年減少 13%，一般事業廢棄物則減少 4%，有害事業廢棄物則增加 16%，當中 42%為專管回收有害高濃度氨氮廢液，降低環境廢水的污染，並更有效率的使廢棄物再進入生產鏈的循環中。因應環保理念，漢磊本著永續經營的理念與遵守法規之要求進行廢棄物妥善處理(如.將原本以廢水處理系統處理的含氟廢液，轉為回收及再利用方式處理)，以避免生產活動造成對環境之衝擊。

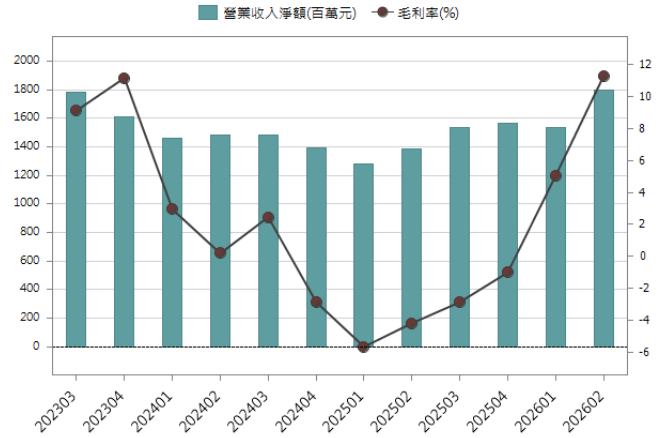
漢磊 3707 TT

評價表

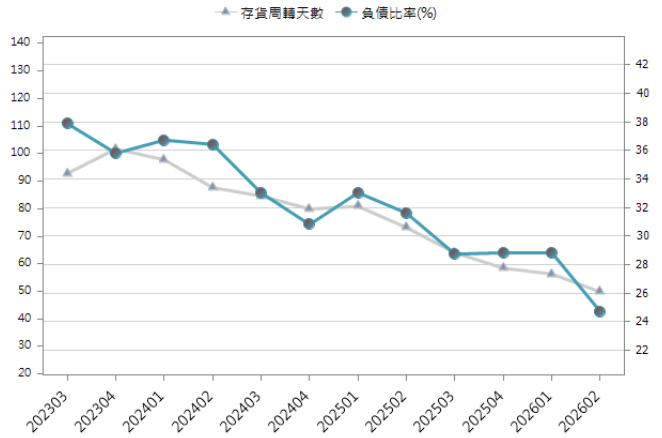
			每股盈餘(元)		每股盈餘成長率(%)		每股淨值(元)		本益比(倍)		股價淨值比(倍)	
公司	代號	2026/10/12	2026E	2027E	2026E	2027E	2026E	2027E	2026E	2027E	2026E	2027E
M31	6643 TT	82.20	0.99	4.77	虧轉盈	380.71	22.4	24.1	82.9	17.2	3.7	3.4

資料來源:CMoney、國泰預估

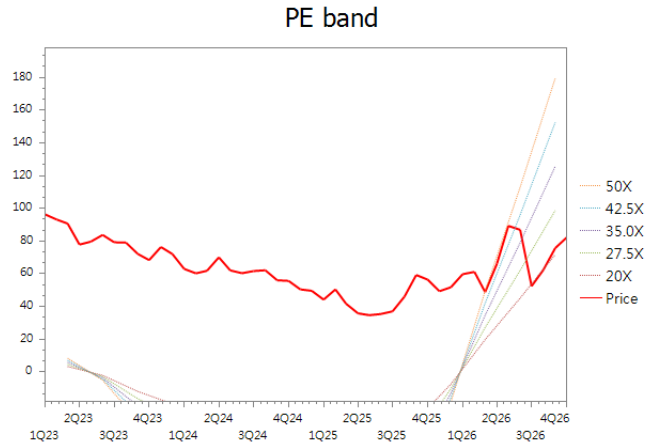
近三年單季營收與毛利率走勢圖



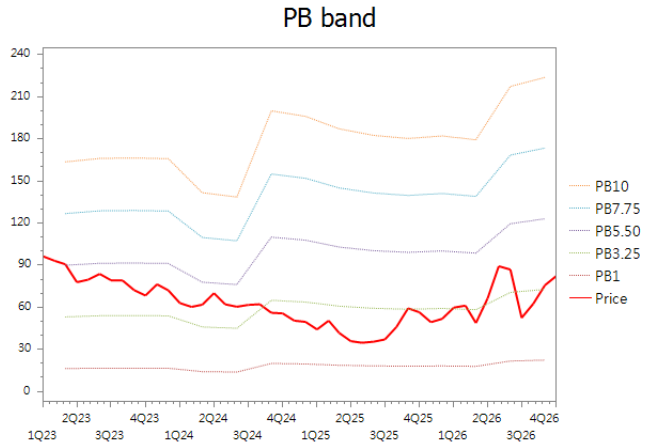
近三年負債比率與存貨週轉天數



未來 12 個月 P/E(本益比)區間圖



未來 12 個月 P/B(股價淨值比)區間圖



漢磊 3707 TT

損益表

至 12 月 31 日(百萬元)	2023	2024	2025	2026E	2027E
營業收入	7,080	5,817	5,766	7,438	9,205
營業成本	6,271	5,775	5,956	6,473	7,473
營業毛利	809	42	-190	965	1,732
營業費用	674	717	679	769	964
營業利益	136	-675	-869	196	768
營業外收入與支出	44	315	131	70	68
其他收入	47	39	50	19	12
其他利益與損失	-13	255	-5	30	36
財務成本	45	36	44	41	40
權益法認列之投資損益	2	-4	54	0	0
其他項目	53	62	75	61	60
稅前純益	180	-360	-738	267	836
所得稅費用	44	54	11	127	167
稅後純益	136	-414	-749	140	669
母公司業主- 稅後純益	67	-525	-757	397	1,909
非控制權益- 稅後純益	69	112	9	-257	-1,240
稅前息前折舊攤銷前淨利	845	36	-32	1,085	1,732
每股盈餘 (元)	0.17	-1.31	-1.89	0.99	4.77

資料來源:公司資料及國泰預估

資產負債表

12 月 31 日(百萬元)	2023	2024	2025	2026E	2027E
資產總額	11,793	13,915	12,880	14,795	14,589
流動資產	6,474	8,189	6,346	7,967	7,240
現金及約當現金	3,532	5,545	4,006	4,592	3,500
短期投資	0	160	14	17	17
存貨	1,594	1,234	959	1,524	1,576
應收帳款及票據	1,151	1,035	1,107	1,604	1,917
其他流動資產	198	214	261	230	230
非流動資產	5,319	5,726	6,534	6,828	7,350
長期投資	402	45	23	24	28
固定資產	4,142	4,796	5,680	6,063	6,581
其他非流動資產	775	885	831	740	740
負債總額	4,223	4,289	3,714	3,760	4,125
流動負債	2,025	3,046	1,928	2,453	2,823
應付帳款及票據	1,112	1,170	1,203	1,404	1,607
短期借款	356	1,541	493	760	927
其他流動負債	557	335	232	289	289
長期負債	2,199	1,242	1,786	1,307	1,302
長期借款	1,481	469	1,119	-4	-9
其他非流動負債	718	773	667	1,311	1,311
股東權益總額	7,570	9,626	9,166	11,035	10,464
股本	3,332	3,832	3,901	4,005	4,005
資本公積	1,538	3,539	3,818	5,054	5,054
保留盈餘	755	248	-503	18	687
非控制權益	2,043	2,110	2,061	2,071	831
其他權益	-99	-104	-112	-113	-113

資料來源:公司資料及國泰預估

漢磊 3707 TT

現金流量表

至 12 月 31 日(百萬元)	2023	2024	2025	2026E	2027E
營業活動之現金流量	760	336	20	1,971	1,471
稅前純益	180	-360	-738	267	836
折舊及攤提	709	711	837	889	964
本期營運資金變動	294	645	216	896	-162
其他營業活動現金流量	-422	-660	-295	-80	-167
投資活動之現金流量	-1,443	-908	-1,401	-946	-1,050
資本支出淨額	-1,460	-1,400	-1,592	-1,273	-1,482
本期長期投資變動	-5	357	22	-3	-3
其他投資活動現金流量	22	136	169	330	435
籌資活動之現金流量	-519	2,585	-158	270	162
現金增資	0	2,480	0	0	0
本期借款變動	103	173	-398	263	162
發放現金股利	-333	0	0	0	-521
其他籌資活動現金流量	-289	-68	240	7	0
匯率調整數	-1	0	-1	-709	-1,675
自由現金流量	-699	-1,064	-1,572	698	-11
本期產生現金流量	-1,203	2,014	-1,540	586	-1,092

資料來源:公司資料及國泰預估

資產負債表主要財務報表分析

至 12 月 31 日(百萬元)	2023	2024	2025	2026E	2027E
年成長率(%)					
營業收入	-20.3	-17.8	-0.9	29.0	23.8
營業利益	-87.7	盈轉虧	N/A	虧轉盈	291.02
稅前息前折舊攤銷前淨利	-51.7	-95.75	盈轉虧	虧轉盈	59.6
稅後純益	-91.83	盈轉虧	N/A	虧轉盈	380.71
每股盈餘	-91.83	盈轉虧	N/A	虧轉盈	380.71
獲利能力分析(%)					
營業毛利率	11.4	0.7	-3.3	13.0	18.8
營業利益率	1.9	-11.6	-15.1	2.6	8.3
稅前息前折舊攤銷前淨利率	11.9	0.6	-0.6	14.6	18.8
稅後純益率	1.0	-9.0	-13.1	5.3	20.7
資產報酬率	0.5	-4.1	-5.7	2.9	13.1
股東權益報酬率	1.2	-8.1	-10.4	3.9	18.2
穩定 / 償債能力分析					
負債/資產（%）	35.8	30.8	28.8	25.4	28.3
淨負債比率（%）	-30.7	-49.2	-33.9	-34.9	-24.8
利息保障倍數（X）	5.0	-8.9	-15.9	7.6	21.9
利息及短期債保障倍數	0.7	-4.0	-1.3	0.4	0.9
現金流量 / 利息保障倍數(X)	17.0	9.2	0.5	48.4	36.8
現金流量 / 利息,短債保障倍數(X)	2.4	4.1	0.0	2.5	1.5
流動比率(%)	319.8	268.8	329.1	324.7	256.4
速動比率(%)	234.4	223.0	270.3	253.3	192.5
淨負債(百萬元)	-1,694	-3,695	-2,408	-3,853	-2,599
每股資料分析(元)					
每股盈餘	0.2	-1.3	-1.9	1.0	4.8
每股營業現金流量	1.9	0.8	0.1	4.9	3.7
每股淨值	13.8	18.8	17.7	22.4	24.1
每股營收	17.7	14.5	14.4	18.6	23.0
每股稅前息前折舊攤銷前淨利	2.1	0.1	-0.1	2.7	4.3
每股現金股利	0.0	0.0	0.0	1.3	0.1

資料來源:公司資料及國泰預估

漢磊 3707 TT

研究團隊

李朋潤
研究部主管
+886-2-2326-9899#9814
pengjunlee@cathayfut.com.tw

藍文彥
金融、消費、工業
+886-2-2326-9899#1145
michael@cathayfut.com.tw

呂旻旂
紡織、製鞋、電子組裝
+886-2-2326-9899#7960
minchilu@cathayfut.com.tw

呂理舜
綠色能源、電動車
+886-2-2326-9899#9844
dufflu@cathayfut.com.tw

林裕禮
IP、IC設計
+886-2-2326-9899#1944
henrylin@cathayfut.com.tw

璩文浩 CA
ESG
+886-2-2326-9899#9856
whchu@cathayfut.com.tw

王筠婷
特用化學、半導體週邊
+886-2-2326-9899#7961
vanessawang@cathayfut.com.tw

姜仁匡
PCB
+886-2-2326-9899#1601
kenchiang@cathayfut.com.tw

紀彥呈
半導體、資訊軟體
+886-2-2326-9899#1940
ianchi@cathayfut.com.tw

劉耀文
IPC、鋼鐵
+886-2-2326-9899#1751
yaowen@cathayfut.com.tw

鍾沛宴
IC設計
+886-2-2326-9899#1750
parisphchung@cathayfut.com.tw

王韋傑
被動元件
+886-2-2326-9899#3954
jerrywang@cathayfut.com.tw

張雅倩
研究員
+886-2-2326-9899#1122
val.chang@cathayfut.com.tw

林美君
研究員
+886-2-2326-9899#1752
meira.lin@cathayfut.com.tw

賴振宇
資深研究員
+886-2-2326-9899#9817
alex.lai@cathayfut.com.tw

郭經緯 CPA
顧問部主管
+886-2-2326-9899#9812
raymondkuo@cathayfut.com.tw

何緯婷
總經
+886-2-2326-9899#9835
lauraho@cathayfut.com.tw

陳玉庭
總經、ETF
+886-2-2326-9899#7964
yutingchen@cathayfut.com.tw

張立民
研究員
+886-2-2326-9899#1731
changlimin@cathayfut.com.tw

陳宇亮
研究員
+886-2-2326-9899#9846
ivor0704@cathayfut.com.tw

林威廷
研究員
+886-2-2326-9899#3956
adamlin@cathayfut.com.tw

投資評等定義

英文評等	中文評等	潛在報酬率
STRONG BUY	強力買進	> +30%
BUY	買進	介於+15% ~ +30%
NEUTRAL	中立	介於-15% ~ +15%
SELL	賣出	< -15%
註:潛在報酬率=(6 個月內目標價/本次報告評等最近一天收盤價 – 1) * 100		

免責聲明

本報告僅為本公司提供客戶之一般參考資料， 並非針對客戶之特定需求所作的投資建議， 且在本報告撰寫過程中， 並未考量讀者個別的財務狀況與需求， 故本報告所提供的資訊無法適用於所有客戶或投資人， 讀者應審慎考量本身之投資風險， 並就投資結果自行負責。

本報告係根據本公司所取得資訊加以彙集及研究分析， 本公司雖盡力使用可靠且廣泛的資訊， 但本公司並不保證各項資訊之完整性及正確性。本報告中所提出之意見係為本報告出版當時的意見，邇後相關資訊或意見若有變更， 本公司將不會另行通知。本公司亦無義務持續更新本報告之內容或追蹤研究本報告所涵蓋之主題。本報告中提及的標的物價格、價值及收益隨時可能因各種本公司無法控制之政治、經濟、市場等因素而產生變化。本報告中之各項預測， 如營收及目標價格， 均係基於對目前所得資訊作合理假設下所完成， 所以並不必然實現。從事特定交易， 如涉及期貨、選擇權及衍生性金融商品， 因具有高度風險並不適合每一個投資人。本報告不得視為買賣有價證券或其他金融商品的要約或要約之引誘。

國泰金融控股公司所屬子公司暨關係企業(以下稱 “ 國泰金融集團”) 從事廣泛金融業務，包括但不限於銀行、保險、證券經紀、自有資金投資與交易、資產管理及證券投資信託等。國泰金融集團對於本報告所涵蓋之標的公司可能有投資或其他的業務往來關係。國泰金融集團及其所屬員工， 可能會投資本報告所涵蓋之標的公司， 或以該等公司為標的的衍生性金融商品， 而其交易的方向與本報告中所提及的交易方向可能不一致。除此之外， 國泰金融集團於法令許可的範圍內，亦有可能以本報告所涵蓋之標的公司作為發行衍生性

金融商品之標的。國泰金融集團轄下的銷售人員、交易員及其他業務人員可能會為他們客戶或自營部門提供口頭或書面的市場看法或交易策略， 然而該等看法與策略可能與本報告的意見不盡一致。國泰金融集團的資產管理、自營及其他投資業務所做的投資決策也可能與本報告所做的建議或看法不一致。

非經本公司事先書面同意， 不得發送或轉送本報告予第三人轉載或使用。

譜瑞-KY

(4966.TWO/4966 TT)

3Q26 財報預覽：PC/NB 需求較預期疲弱

持有 · 維持

收盤價 October 13 (NT$)	547
12 個月目標價 (NT$)	515
前次目標價 (NT$)	610
調降 (%)	15.6
下跌空間 (%)	-5.9

焦點內容

1. PC/NB 需求不如預期，下修 4Q26 獲利預估。
2. PC/NB 需求疲弱，2027 年成長仰賴非出貨量因素之動能。
3. 成本提升使 2027 年毛利率回升幅度有限。

交易資料表

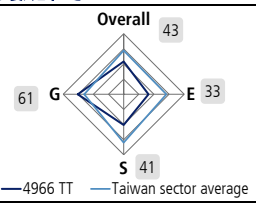
市值：(NT$bn/US$mn)	43.19 / 1,356
流通在外股數 (百萬股)：	78.96
外資持有股數 (百萬股)：	31
3M 平均成交量 (百萬股)：	0.52
52 週股價 (低 \ 高) (NT$)：	463.5 –870

股價表現	3M	6M	12M
絕對表現 (%)	-15.5	5.2	-29.2
相對表現 (%)	-22	-31.1	-108.8

每股盈餘

NT$	1Q	2Q	3Q	4Q
2025	8.38A	8.95A	10.45A	6.89A
2026	6.77A	7.66A	8.45F	7.50F
2027	6.48F	8.20F	9.58F	7.97F

ESG 分數評等



資料來源：Refinitiv

重要訊息

我們根據近期與管理層的交流，更新對公司營運與展望的看法。

評論及分析

PC/NB 需求不如預期，下修 4Q26 獲利預估。譜瑞-KY 3Q26 營收達 41.8 億元，季減 1.6%、年減 4.4%，低於公司財測，主因：(1) DRAM 價格提升抑制筆電終端需求，使旺季不旺；(2) GPU 供給短缺，進一步壓抑高階與電競筆電需求。根據我們的供應鏈訪查，客戶自 7 月下旬起削減下單，8 月降幅擴大，主要以非 Standard-plus 筆電客戶為主，Standard-plus 需求則大致正常，惟亦有所調整。考量記憶體成本壓力較我們預期嚴峻、客戶拉貨減少，且記憶體價格尚無緩解跡象，我們下修 4Q26 營收預估至 39.8 億元，季減 4.9%、年增 2.4%。

PC/NB 需求疲弱，2027 年成長仰賴非出貨量因素之動能。我們對 PC/NB 需求轉趨保守，主因我們認為 2027 年 DRAM 價格將持續上漲，與管理層預期價格壓力將緩解之看法不同。不過我們預期成長動能將來自：(1) TED/tTED 及其他整合式顯示方案取代獨立方案；(2)新一代 AI PC 平台於 4Q26 進入量產，單機搭載多顆 retimer，帶動 USB4 retimer 滲透率提升；(3)車用 DP hub 於 4Q26 於電動車平台進入量產；(4) ASIC-like 業務預期將於 2H27 開始貢獻營收，但在 2028 年前獲利貢獻較不顯著。此外，由於目前尚無實際客戶，且 2H27 潛在營收能見度仍低，我們並未將資料中心 ACC redriver 之營收貢獻納入財務模型。綜上，為反映 PC/NB 展望轉弱，我們下修 2027 年營收預估至 167 億元，年增 2.0%。

成本提升使 2027 年毛利率回升幅度有限。我們下修 3Q-4Q26 毛利率預估至 40.1%與 40.0%，分別季減 0.1 個百分點，反映封測成本與金價上揚，且各產品與客戶的定價能力不一，成本僅能部分轉嫁。儘管如此，我們預期封測產能持續移往中國，加上選擇性調漲價格，2H26 毛利率仍可維持在 40%以上。展望 2027 年，考量封測成本壓力持續、成熟製程晶圓代工可能漲價，以及成本僅能部分轉嫁，我們下調毛利率預估 0.6 個百分點至 40.2%。隨後段產能持續移往中國，我們預期 2027 年毛利率將逐季回升，但保守看待其改善速度。

投資建議

我們下修 2026-27 年 EPS 預估至 30.38 元與 32.22 元，分別年減 12.0%與年增 6.1%。我們對 PC/NB 展望較先前更為悲觀，但認為中長期主要成長機會仍在 ASIC-like 業務。我們將目標價由 610 元下調至 515 元，基於 16 倍 2027 年 EPS 預估不變換算得出，維持「持有」評等。

投資風險

PC/NB 需求疲弱；高速傳輸介面 IC 競爭加劇。

主要財務數據及估值

	Dec-23A	Dec-24A	Dec-25A	Dec-26F	Dec-27F
營業收入 (NT$百萬)	13,769	16,246	16,531	16,402	16,729
營業毛利 (NT$百萬)	6,019	6,903	7,035	6,586	6,729
營業利益 (NT$百萬)	1,967	2,452	2,709	2,370	2,471
稅後淨利 (NT$百萬)	2,033	2,592	2,727	2,357	2,495
每股盈餘 (NT$)	25.73	32.55	34.51	30.38	32.22
每股現金股利 (NT$)	12.77	16.12	17.66	16.53	15.89
每股盈餘成長率 (%)	(58.1)	26.5	6.0	(12.0)	6.1
本益比 (x)	21.3	16.8	15.9	18.0	17.0
股價淨值比 (x)	2.3	2.0	2.0	1.9	1.8
EV/EBITDA (x)	13.2	10.5	9.3	10.0	9.2
淨負債比率 (%)	Net cash	Net cash	Net cash	Net cash	Net cash
殖利率 (%)	2.3	2.9	3.2	3.0	2.9
股東權益報酬率 (%)	10.9	12.4	12.3	10.5	10.7

資料來源：公司資料，凱基預估

凱基投顧

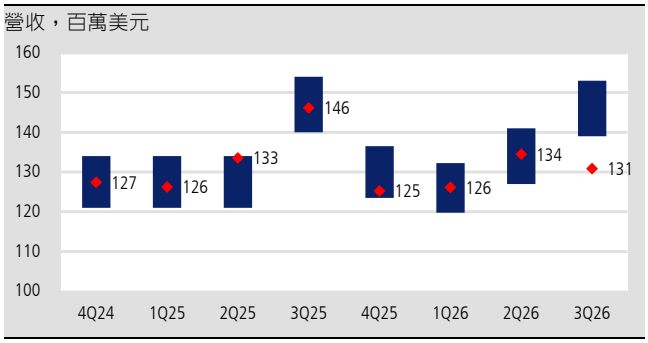
劉宇程
886.2.2181.8727
lucas.liu@kgi.com

潘俊宏
886.2.2181.8706
Felix.Pan@kgi.com

王婉懿
886.2.2181.8715
vera.wy.wang@kgi.com

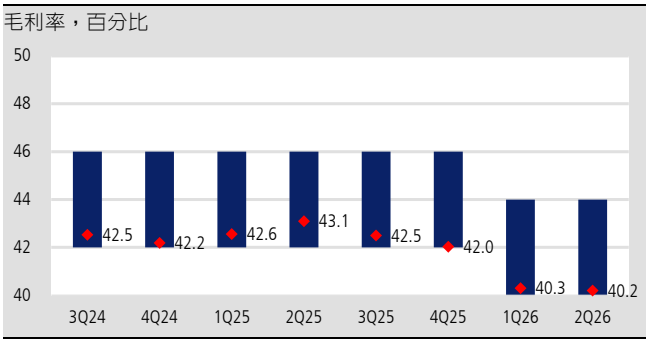
重要免責聲明，詳見最終頁

圖 1：單季營收暨公司財測比較



資料來源：公司資料，凱基

圖 2：單季毛利率暨公司財測比較



資料來源：公司資料，凱基

圖 3：3Q-4Q26 財測修正與市場共識比較

	3Q26F							4Q26F						
百萬元	修正後	修正前	調幅(%)	QoQ (%)	YoY (%)	市場共識	差異(%)	修正後	修正前	調幅(%)	QoQ (%)	YoY (%)	市場共識	差異(%)
營收	4,183	4,618	(9.4)	(1.6)	(4.4)	4,628	(9.6)	3,980	4,475	(11.1)	(4.9)	2.4	4,553	(12.6)
毛利	1,678	1,862	(9.9)	(1.7)	(9.8)	1,887	(11.1)	1,591	1,816	(12.4)	(5.2)	(2.6)	1,866	(14.7)
營業利益	658	740	(11.1)	10.4	(19.0)	768	(14.4)	576	696	(17.2)	(12.4)	8.5	772	(25.4)
稅後淨利	654	734	(10.8)	10.3	(19.8)	753	(13.2)	581	692	(16.1)	(11.3)	6.7	763	(23.9)
每股盈餘 (元)	8.45	9.48	(10.8)	10.3	(19.2)	9.47	(10.8)	7.50	8.94	(16.1)	(11.3)	8.8	9.44	(20.6)
毛利率 (%)	40.1	40.3	(0.2)ppts	(0.1)ppts	(2.4)ppts	40.8	(0.6)ppts	40.0	40.6	(0.6)ppts	(0.1)ppts	(2.0)ppts	41.0	(1.0)ppts
營利率 (%)	15.7	16.0	(0.3)ppts	1.7 ppts	(2.8)ppts	16.6	(0.9)ppts	14.5	15.6	(1.1)ppts	(1.2)ppts	0.8 ppts	17.0	(2.5)ppts
淨利率 (%)	15.6	15.9	(0.2)ppts	1.7 ppts	(3.0)ppts	16.3	(0.6)ppts	14.6	15.5	(0.9)ppts	(1.1)ppts	0.6 ppts	16.8	(2.2)ppts

資料來源：Bloomberg，凱基預估

圖 4：2026-27 年財測修正暨市場共識比較

	2026F						2027F					
百萬元	修正後	修正前	調幅(%)	YoY (%)	市場共識	差異(%)	修正後	修正前	調幅(%)	YoY (%)	市場共識	差異(%)
營收	16,402	17,332	(5.4)	(0.8)	17,421	(5.8)	16,729	18,482	(9.5)	2.0	19,346	(13.5)
毛利	6,586	6,994	(5.8)	(6.4)	7,070	(6.8)	6,729	7,540	(10.8)	2.2	8,010	(16.0)
營業利益	2,370	2,572	(7.9)	(12.5)	2,679	(11.5)	2,471	2,941	(16.0)	4.3	3,376	(26.8)
稅後淨利	2,357	2,548	(7.5)	(13.5)	2,624	(10.2)	2,495	2,938	(15.1)	5.8	3,290	(24.2)
每股盈餘 (元)	30.38	32.84	(7.5)	(12.0)	33.41	(9.1)	32.22	37.95	(15.1)	6.1	42.03	(23.3)
毛利率 (%)	40.2	40.4	(0.2)ppts	(2.4)ppts	40.6	(0.4)ppts	40.2	40.8	(0.6)ppts	0.1 ppts	41.4	(1.2)ppts
營利率 (%)	14.4	14.8	(0.4)ppts	(1.9)ppts	15.4	(0.9)ppts	14.8	15.9	(1.1)ppts	0.3 ppts	17.5	(2.7)ppts
淨利率 (%)	14.4	14.7	(0.3)ppts	(2.1)ppts	15.1	(0.7)ppts	14.9	15.9	(1.0)ppts	0.5 ppts	17.0	(2.1)ppts

資料來源：Bloomberg，凱基預估

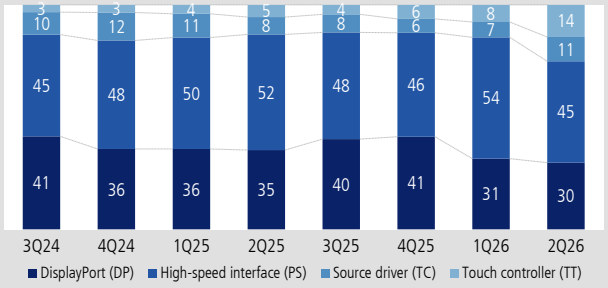
圖 5：公司概況

成立於 2005 年並於 2011 年上櫃交易的譜瑞-KY 是混合訊號 IC 設計廠中的佼佼者，產品用於各種顯示與高速介面標準下的電腦、消費性電子產品與顯示面板中。總部設於加州矽谷而設計中心則位於上海，譜瑞-KY 另外設有子公司於南京與首爾，分公司則開立於香港與台灣。作為 VESA 的共同創辦人與現任的董事，譜瑞-KY 參與了 DP 與 eDP 的標準開發。

資料來源：公司資料，凱基

圖 6：營收佔比

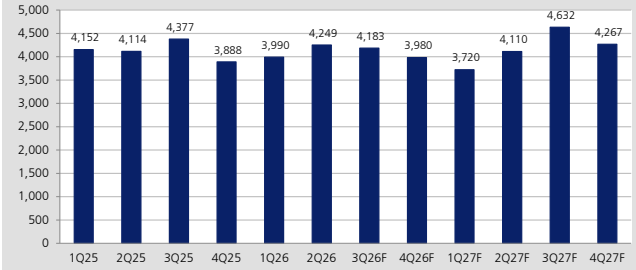
營收按事業群，百分比



資料來源：公司資料，凱基

圖 7：季營業收入

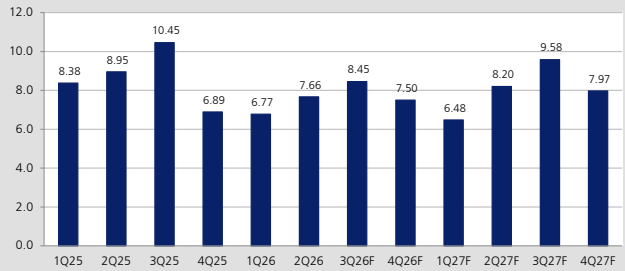
季營業收入，百萬元



資料來源：凱基

圖 8：每股盈利

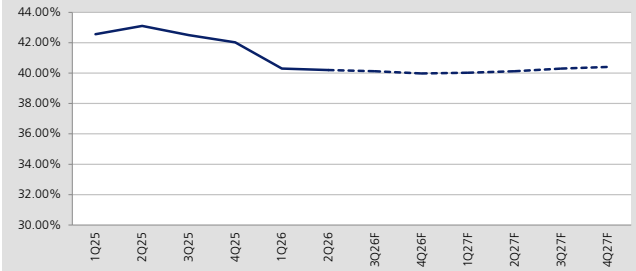
每股盈利，元



資料來源：凱基

圖 9：毛利率

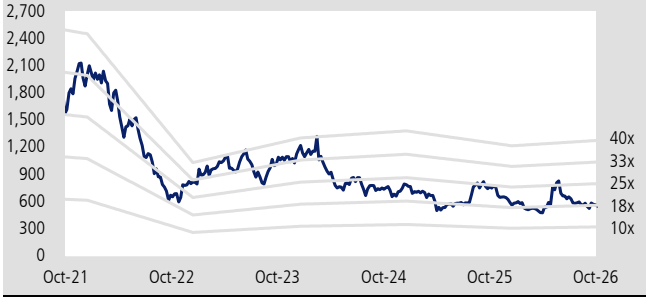
毛利率，百分比



資料來源：凱基

圖 10：未來一年預估本益比區間

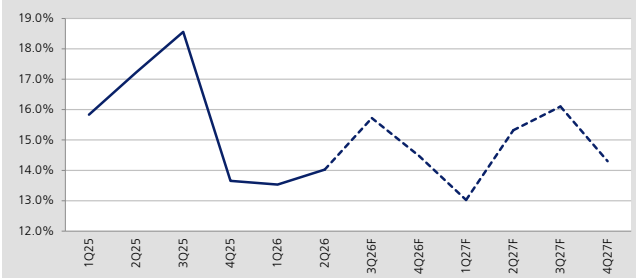
股價，元(左軸)；本益比，倍(右軸)



資料來源：TEJ，凱基預估

圖 11：營業利潤率

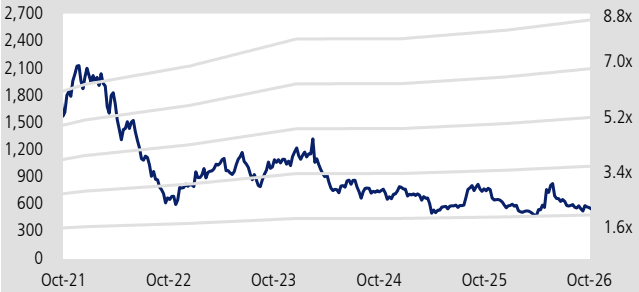
營業利潤率，百分比



資料來源：凱基

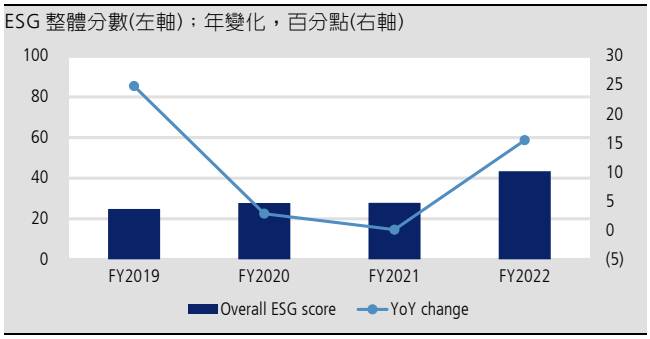
圖 12：未來一年預估股價淨值比區間

股價，元(左軸)；股價淨值比，倍(右軸)



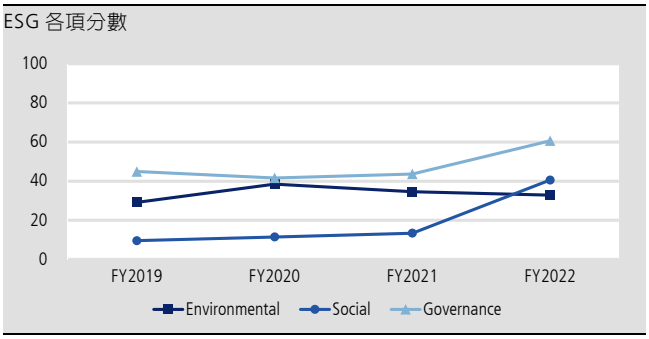
資料來源：TEJ，凱基預估

圖 13：ESG 整體分數



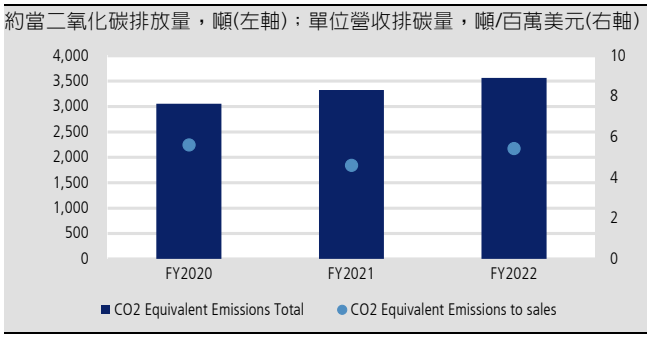
資料來源：Refinitiv，公司資料

圖 14：ESG 各項分數



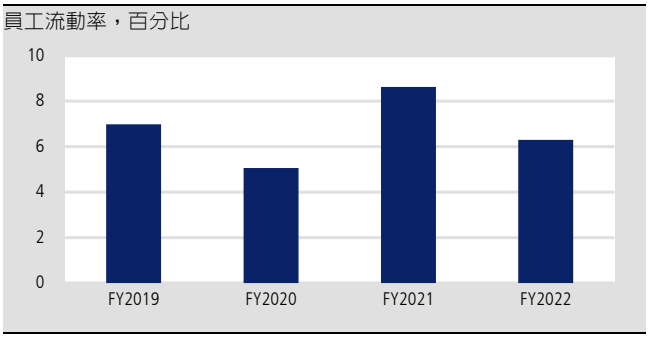
資料來源：Refinitiv，公司資料

圖 15：碳排量



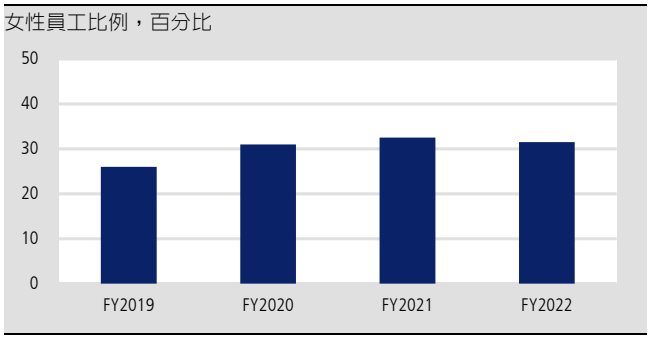
資料來源：Refinitiv，公司資料

圖 16：員工流動率



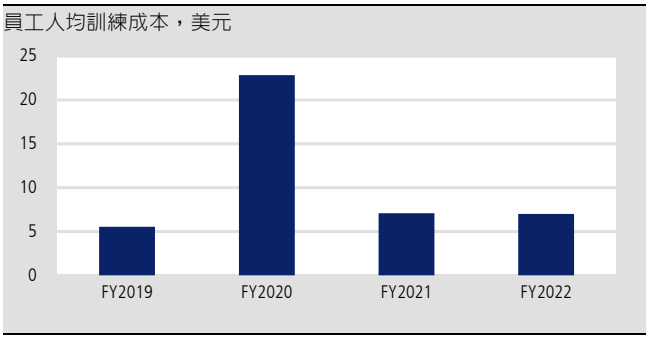
資料來源：Refinitiv，公司資料

圖 17：性別多樣性



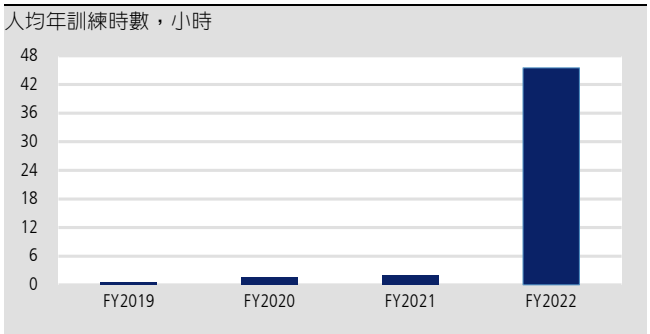
資料來源：Refinitiv，公司資料

圖 18：員工人均訓練成本



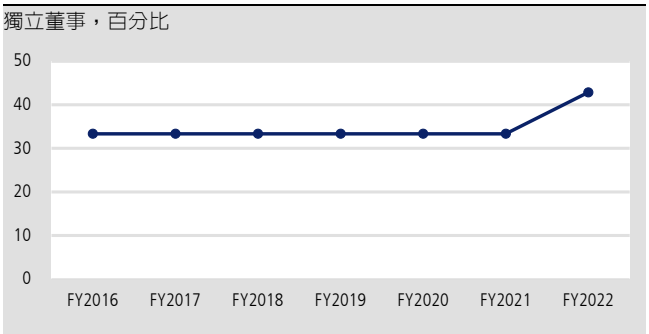
資料來源：Refinitiv，公司資料

圖 19：人均訓練時數



資料來源：Refinitiv，公司資料

圖 20：獨立董事



資料來源：Refinitiv，公司資料

ESG 圖表定義

項目	定義	說明
能源使用	直接與間接能源消耗總量(十億焦耳) - 公司運營範圍內消耗之能源總量 - 能源使用總量 = 直接能源消耗總量 + 間接能源消耗量 - 總能源使用量包括購買的能源、生產的能源 - 就公用事業而言，輸電/電網損耗為其業務活動之一部分，被視為總能源消耗，數據不包括為滿足能源使用而生產的電力(公用事業為出售而生產) - 就公用事業而言，用於能源生產的煤、天然氣或核能等原料不屬於「總能源使用」項下	
購買之再生能源	Primary 再生能源購買總量(十億焦耳) - 公司各種來源之能源消耗量與購買的能源中屬於自然界可再生者(太陽能、風能、水能、生物質能、地熱能)之量 - 如無證據顯示再生能源由公司生產，則所報告的能源數據視為購買的再生能源	
再生能源使用率	再生能源占總能源消耗量之比例	
CO2 約當排放量	直接 CO2 與 CO2 約當排放量(公噸) - 公司擁有或控制的來源的直接排放量(範圍 1 排放量) - 相關氣體：二氧化碳(CO2)、甲烷(CH4)、一氧化二氮(N2O)、氫氟碳化物(HFCS)、全氟化合物(PFCS)、六氟化硫(SF6)、三氟化氮(NF3)	譜瑞-KY 首次於 2020 年 CSR 報告書中揭露該資訊。
CO2 約當排放量營收比	直接 CO2 與 CO2 約當排放量(公噸)銷售(百萬元)比 - 公司擁有或控制的來源的直接排放量(範圍 1 排放量) - 相關氣體：二氧化碳(CO2)、甲烷(CH4)、一氧化二氮(N2O)、氫氟碳化物(HFCS)、全氟化合物(PFCS)、六氟化硫(SF6)、三氟化氮(NF3)	譜瑞-KY 首次於 2020 年 CSR 報告書中揭露該資訊。
廢棄物總量	廢棄物總量(公噸) - 總廢棄物 = 無害廢棄物 + 有害廢棄物 - 僅考慮固體廢棄物，但如液體廢棄物以公噸為單位呈報，則會將其加入求得包含液體廢棄物之總量 - 對於採礦、石油與天然氣等行業，tailings、廢石、煤、飛灰等廢棄物亦考慮在內	
廢棄物回收率	公司呈報的廢棄物回收率 - 廢棄物回收率 = 廢棄物回收量/總廢棄物*100 - 廢棄物轉化為能源或經由廢棄物焚燒產生能源視為廢棄物回收 - 經由堆肥回收的廢棄物視為回收之廢棄物	
總取水量	總取水量(立方公尺) - 由呈報組織直接或經水公司等中介機構從任何水源抽取的總水量 - 水井、城鎮/公用事業/市政用水、河水、地表水等不同之水源均予以考慮	
環保支出	環保支出總金額 - 所有用於環境保護，或防止、減少、控制環境因素、影響、危害的投資與支出，亦包括處置、處理、衛生、清理支出	
員工流動率	員工流動率 - 包括任何原因(自願或非自願)離開公司的員工，如辭職、退休、自然離職/死亡、醫療失能、冗員、裁員、重組、解僱、裁減或定期合約到期 - 員工流動率 = (離開之員工/平均員工人數)*100 - 平均員工人數 = (本年末員工人數 + 去年末員工人數)/2 - 本財年末員工數 = 上個財年末員工數 + 新員工數 - 離開之員工數	
女性經理	女性經理百分比 - 女性經理占公司經理的百分比 - 如有不同階層之百分比，如最高階、高階、中階、初階，則會考慮中階女性經理之百分比 - 女性經理百分比 = 女性經理人數/經理總數*100	
女性員工	女性員工百分比 - 女性員工占公司員工總數的百分比 - 女性員工百分比 = 女性人數/員工總數*100	
教育訓練總時數	所有員工教育訓練總時數 - 僅考慮員工教育訓練時數 - 包括一般員工所有類型的教育訓練(如健康與安全、環境、急難事件救援、技能與職業發展) - 如果數據以天為單位，則乘以 8，係假設 1 天 = 8 小時工作	
每位員工教育訓練時數	平均每年每位員工總教育訓練時數	

資料來源：Refinitiv，公司資料，凱基

損益表

	季度								年度		
	Mar-26A	Jun-26A	Sep-26F	Dec-26F	Mar-27F	Jun-27F	Sep-27F	Dec-27F	Dec-25A	Dec-26F	Dec-27F
損益表 （NT$百萬）											
營業收入	**3,990**	**4,249**	**4,183**	**3,980**	**3,720**	**4,110**	**4,632**	**4,267**	**16,531**	**16,402**	**16,729**
營業成本	(2,382)	(2,541)	(2,505)	(2,389)	(2,231)	(2,461)	(2,765)	(2,543)	(9,496)	(9,817)	(10,000)
營業毛利	**1,608**	**1,708**	**1,678**	**1,591**	**1,489**	**1,649**	**1,867**	**1,724**	**7,035**	**6,586**	**6,729**
營業費用	(1,068)	(1,112)	(1,021)	(1,015)	(1,004)	(1,019)	(1,121)	(1,114)	(4,326)	(4,216)	(4,258)
營業利益	**540**	**596**	**658**	**576**	**485**	**630**	**746**	**610**	**2,709**	**2,370**	**2,471**
折舊	(76)	(92)	(95)	(81)	(112)	(117)	(118)	(37)	(315)	(345)	(383)
攤提	(106)	(107)	(131)	(80)	(130)	(130)	(130)	(31)	(458)	(423)	(421)
EBITDA	**722**	**795**	**883**	**737**	**726**	**876**	**994**	**678**	**3,482**	**3,138**	**3,274**
利息收入	59	60	58	59	64	65	66	65	279	236	259
投資利益淨額	-	-	-	-	-	-	-	-	-	-	-
其他營業外收入	2	0	-	-	-	-	-	-	13	-	-
總營業外收入	**61**	**60**	**58**	**59**	**64**	**65**	**66**	**65**	**292**	**236**	**259**
利息費用	-	-	-	-	-	-	-	-	-	-	-
投資損失	-	-	-	-	-	-	-	-	-	-	-
其他營業外費用	(19)	(11)	3	3	3	3	3	3	(19)	(21)	12
總營業外費用	**(19)**	**(11)**	**3**	**3**	**3**	**3**	**3**	**3**	**(19)**	**(21)**	**12**
稅前純益	**582**	**646**	**719**	**638**	**551**	**697**	**815**	**678**	**2,982**	**2,585**	**2,742**
所得稅費用[利益]	(53)	(52)	(65)	(57)	(50)	(63)	(73)	(61)	(256)	(227)	(247)
少數股東損益	-	-	-	-	-	-	-	-	-	-	-
非常項目前稅後純益	**529**	**593**	**654**	**581**	**502**	**635**	**742**	**617**	**2,727**	**2,357**	**2,495**
非常項目	0	(0)	-	-	-	-	-	-	-	0	-
稅後淨利	**529**	**593**	**654**	**581**	**502**	**635**	**742**	**617**	**2,727**	**2,357**	**2,495**
每股盈餘 (NT$)	6.77	7.66	8.45	7.50	6.48	8.20	9.58	7.97	34.51	30.38	32.22
獲利率 （%）											
營業毛利率	40.3	40.2	40.1	40.0	40.0	40.1	40.3	40.4	42.6	40.2	40.2
營業利益率	13.5	14.0	15.7	14.5	13.0	15.3	16.1	14.3	16.4	14.4	14.8
EBITDA Margin	18.1	18.7	21.1	18.5	19.5	21.3	21.5	15.9	21.1	19.1	19.6
稅前純益率	14.6	15.2	17.2	16.0	14.8	17.0	17.6	15.9	18.0	15.8	16.4
稅後純益率	13.3	14.0	15.6	14.6	13.5	15.4	16.0	14.5	16.5	14.4	14.9
季成長率 （%）											
營業收入	2.6	6.5	(1.6)	(4.9)	(6.5)	10.5	12.7	(7.9)			
營業毛利	(1.6)	6.3	(1.7)	(5.2)	(6.4)	10.8	13.2	(7.7)			
營業收益增長	1.7	10.4	10.4	(12.4)	(15.9)	30.0	18.5	(18.2)			
EBITDA	0.5	10.1	11.1	(16.6)	(1.5)	20.7	13.4	(31.8)			
稅前純益	(2.3)	10.9	11.4	(11.3)	(13.6)	26.5	16.9	(16.8)			
稅後純益	(2.8)	12.1	10.3	(11.3)	(13.6)	26.5	16.9	(16.8)			
年成長率 （%）											
營業收入	(3.9)	3.3	(4.4)	2.4	(6.8)	(3.3)	10.7	7.2	1.8	(0.8)	2.0
營業毛利	(9.0)	(3.7)	(9.8)	(2.6)	(7.4)	(3.5)	11.2	8.4	1.9	(6.4)	2.2
營業收益	(17.9)	(15.9)	(19.0)	8.5	(10.3)	5.7	13.4	5.9	10.5	(12.5)	4.3
EBITDA	(16.0)	(13.4)	(10.4)	2.6	0.5	10.2	12.5	(8.0)	10.2	(9.9)	4.3
稅前純益	(20.0)	(17.0)	(18.4)	7.1	(5.3)	8.0	13.4	6.3	6.2	(13.3)	6.1
稅後純益	(20.3)	(15.5)	(19.8)	6.7	(5.2)	6.9	13.4	6.3	5.2	(13.5)	5.8

資料來源：公司資料，凱基預估

資產負債表

NT$百萬	Dec-23A	Dec-24A	Dec-25A	Dec-26F	Dec-27F
資產總額	**23,369**	**26,205**	**25,956**	**26,410**	**27,676**
流動資產	**14,519**	**16,723**	**16,969**	**17,572**	**19,002**
現金及短期投資	8,488	10,532	10,900	11,210	12,341
存貨	3,774	3,983	3,622	3,666	3,865
應收帳款及票據	1,662	1,564	1,858	1,735	1,835
其他流動資產	596	644	589	961	961
非流動資產	**8,849**	**9,481**	**8,987**	**8,839**	**8,674**
長期投資	-	-	-	-	-
固定資產	392	348	298	259	94
什項資產	8,457	9,133	8,689	8,580	8,580
負債總額	**3,768**	**3,843**	**3,877**	**3,791**	**3,786**
流動負債	**3,609**	**3,646**	**3,622**	**3,563**	**3,558**
應付帳款及票據	1,567	1,321	1,240	1,295	1,371
短期借款	-	-	-	-	-
什項負債	2,042	2,325	2,381	2,269	2,187
長期負債	**159**	**198**	**255**	**228**	**228**
長期借款	-	-	-	-	-
其他負債及準備	-	-	-	228	228
股東權益總額	**19,601**	**22,361**	**22,080**	**22,619**	**23,890**
普通股本	812	812	801	790	790
保留盈餘	14,582	15,979	16,846	17,496	18,767
少數股東權益	-	-	-	-	-
優先股股東資金	-	-	-	-	-

主要財務比率

	Dec-23A	Dec-24A	Dec-25A	Dec-26F	Dec-27F
年成長率					
營業收入	(31.3%)	18.0%	1.8%	(0.8%)	2.0%
營業收益增長	(62.7%)	24.7%	10.5%	(12.5%)	4.3%
EBITDA	(55.3%)	19.5%	10.2%	(9.9%)	4.3%
稅後純益	(58.5%)	27.5%	5.2%	(13.5%)	5.8%
每股盈餘成長率	(58.1%)	26.5%	6.0%	(12.0%)	6.1%
獲利能力分析					
營業毛利率	43.7%	42.5%	42.6%	40.2%	40.2%
營業利益率	14.3%	15.1%	16.4%	14.4%	14.8%
EBITDA Margin	19.2%	19.5%	21.1%	19.1%	19.6%
稅後純益率	14.8%	16.0%	16.5%	14.4%	14.9%
平均資產報酬率	8.9%	10.5%	10.5%	9.0%	9.2%
股東權益報酬率	10.9%	12.4%	12.3%	10.5%	10.7%
穩定 \ 償債能力分析					
毛負債比率 (%)	0.0%	0.0%	0.0%	0.0%	0.0%
淨負債比率	Net cash	Net cash	Net cash	Net cash	Net cash
利息保障倍數 (x)					
利息及短期債保障倍數 (x)	1.0	1.0	1.0	1.0	1.0
Cash Flow Int. Coverage (x)					
Cash Flow/Int. & ST Debt (x)					
流動比率 (x)	4.0	4.6	4.7	4.9	5.3
速動比率 (x)	3.0	3.5	3.7	3.9	4.3
淨負債 (NT$百萬)	(8,488)	(10,532)	(10,900)	(11,210)	(12,341)
每股資料分析					
每股盈餘 (NT$)	25.73	32.55	34.51	30.38	32.22
每股現金盈餘 (NT$)	46.89	41.67	43.27	33.48	39.72
每股淨值 (NT$)	241.49	275.52	275.80	286.47	302.57
調整後每股淨值 (NT$)	248.07	280.83	279.42	291.45	308.54
每股營收 (NT$)	174.27	204.03	209.20	211.34	216.06
EBITDA/Share (NT$)	33.49	39.70	44.07	40.43	42.29
每股現金股利 (NT$)	12.77	16.12	17.66	16.53	15.89
資產運用狀況					
資產周轉率 (x)	0.61	0.66	0.63	0.63	0.62
應收帳款周轉天數	44.0	35.2	41.0	38.6	40.0
存貨周轉天數	177.7	156.0	139.2	136.3	141.1
應付帳款周轉天數	73.8	51.8	47.7	48.1	50.0
現金轉換周轉天數	148.0	139.5	132.6	126.8	131.1

資料來源：公司資料，凱基預估

損益表

NT$百萬	Dec-23A	Dec-24A	Dec-25A	Dec-26F	Dec-27F
營業收入	**13,769**	**16,246**	**16,531**	**16,402**	**16,729**
營業成本	(7,750)	(9,342)	(9,496)	(9,817)	(10,000)
營業毛利	**6,019**	**6,903**	**7,035**	**6,586**	**6,729**
營業費用	(4,053)	(4,451)	(4,326)	(4,216)	(4,258)
營業利益	**1,967**	**2,452**	**2,709**	**2,370**	**2,471**
總營業外收入	**229**	**356**	**292**	**236**	**259**
利息收入	223	349	279	236	259
投資利益淨額	-	-	-	-	-
其他營業外收入	6	8	13	-	-
總營業外費用	**28**	**(1)**	**(19)**	**(21)**	**12**
利息費用	-	-	-	-	-
投資損失	-	-	-	-	-
其他營業外費用	28	(1)	(19)	(21)	12
稅前純益	**2,224**	**2,807**	**2,982**	**2,585**	**2,742**
所得稅費用[利益]	(191)	(215)	(256)	(227)	(247)
少數股東損益	-	-	-	-	-
非常項目	-	0	0	0	-
稅後淨利	**2,033**	**2,592**	**2,727**	**2,357**	**2,495**
EBITDA	**2,646**	**3,161**	**3,482**	**3,138**	**3,274**
每股盈餘 (NT$)	25.73	32.55	34.51	30.38	32.22

現金流量

NT$百萬	Dec-23A	Dec-24A	Dec-25A	Dec-26F	Dec-27F
營運活動之現金流量	**3,704**	**3,318**	**3,419**	**2,598**	**3,075**
本期純益	2,033	2,592	2,727	2,357	2,495
折舊及攤提	680	709	773	768	804
本期運用資金變動	1,266	(96)	(189)	134	(223)
其他營業資產及負債變動	(275)	112	108	(661)	-
投資活動之現金流量	**(363)**	**(606)**	**(614)**	**(466)**	**(639)**
投資用短期投資出售[新購]	-	-	-	-	-
本期長期投資變動	-	-	-	-	-
資本支出淨額	(127)	(140)	(139)	(223)	(218)
其他資產變動	(236)	(466)	(474)	(243)	(421)
自由現金流	**3,274**	**2,358**	**2,464**	**2,417**	**2,190**
融資活動之現金流量	**(1,781)**	**(1,261)**	**(2,163)**	**(1,867)**	**(1,305)**
短期借款變動	-	-	-	-	-
長期借款變動	-	-	-	-	-
現金增資	-	-	-	-	-
已支付普通股股息	(2,439)	(1,016)	(1,295)	(1,386)	(1,305)
其他融資現金流	(103)	(127)	(129)	(78)	-
匯率影響數	**(48)**	**593**	**(275)**	**45**	**-**
本期產生現金流量	**1,512**	**2,044**	**368**	**311**	**1,131**

投資回報率

	Dec-23A	Dec-24A	Dec-25A	Dec-26F	Dec-27F
1 - 營業成本/營業收入					
- 銷管費用/營業收入	29.4%	27.4%	26.2%	25.7%	25.5%
= 營業利益率	**14.3%**	**15.1%**	**16.4%**	**14.4%**	**14.8%**
1 /（營業運用資金/營業收入	0.2	0.2	0.1	0.2	0.2
+ 淨固定資產/營業收入	0.0	0.0	0.0	0.0	0.0
+ 什項資產/營業收入）	0.3	0.3	0.2	0.3	0.3
= 資本周轉率	**2.0**	**2.3**	**2.5**	**2.2**	**2.2**
營業利益率	14.3%	15.1%	16.4%	14.4%	14.8%
x 資本周轉率	2.0	2.3	2.5	2.2	2.2
x（1 - 有效現金稅率）	91.4%	92.3%	91.4%	91.2%	91.0%
= 稅後 ROIC	**26.2%**	**32.0%**	**37.4%**	**28.4%**	**29.0%**

資料來源：公司資料，凱基預估

譜瑞-KY – 以往評級及目標價

日期	評級	目標價	收盤價
2026-08-05	持有	610	618
2026-02-10	持有	580	515
2025-10-29	持有	765	747
2025-08-07	持有	735	669
2025-02-05	持有	810	707
2024-10-30	持有	810	716
2024-08-07	增加持股	970	703
2024-04-24	增加持股	970	803
2024-02-14	增加持股	1,375	1,155
2023-11-21	增加持股	1,250	1,040

資料來源：TEJ，凱基

凱基證券集團據點

中國	上海	上海市靜安區南京西路 1601 號越洋國際廣場 1507 室 郵政編號：200040
台灣	台北	104 台北市明水路 700 號 電話 886.2.2181.8888，傳真 886.2.8501.1691
香港		香港灣仔港灣道十八號中環廣場四十一樓 電話 852.2878.6888，傳真 852.2878.6800
泰國	曼谷	8th - 11th floors, Asia Centre Building 173 South Sathorn Road, Bangkok 10120, Thailand 電話 66.2658.8888，傳真 66.2658.8014
新加坡		珊頓大道 4 號#13-01 新交所第二大廈 郵政編號：068807 電話 65.6202.1188，傳真 65.6534.4826
印尼		Sona Topas Tower Fl.11 Jl. Jend. Sudirman kav.26 JakartaSelatan 12920 Indonesia 電話 62 21 250 6337

股價說明

等級	定義
增加持股 (OP)	對個股持正面看法，預期個股未來十二個月的表現超越凱基證券集團所追蹤的相關市場的總報酬。
持有 (N)	對個股持中性看法，預期個股未來十二個月的表現符合凱基證券集團所追蹤的相關市場的總報酬。
降低持股 (U)	對個股持負面看法，預期個股未來十二個月的表現低於凱基證券集團所追蹤的相關市場的總報酬。
未評等 (NR)	凱基證券未對該個股加以評等。
受法規限制未評等 (R)	受凱基證券集團內部政策和/或相關法令限制使凱基證券集團無法進行某些形式的資訊交流，其中包括提供評等給投資人參考。
	*總報酬 = (十二個月目標價-現價)/現價

免責聲明

凱基證券投資顧問股份有限公司（以下簡稱本公司）為凱基金控集團之成員。本報告之內容皆來自本公司認可之資料來源，但不保證其完整性及精確性。報告內容所提及之各項業務、財務等相關檔案資料及所有的意見及預估皆基於本公司於特定日期所做之判斷，故有其時效性限制，邇後若有變更時，本公司將不做預告或更新。本報告內容僅供參考，並不提供或遊說客戶為買賣股票之投資依據。投資人應審慎考量本身之投資風險，並就投資結果自行負責。本公司及所屬集團成員，暨其主管或員工皆有可能持有報告中所提及的證券。本公司及所屬集團成員並可能經常提供投資銀行或其他服務給報告中提及之公司或向其爭取相關業務。本報告之著作權為本公司所有，非經本公司同意，本報告全文或部份內容，不得以任何形式或方式引用、轉載或轉寄。

2026/10/13

電子上游–IC-DRAM 製造　　維持評等

南亞科 (2408 TT)　　增加持股(Overweight)

長約重訂價帶動獲利創高，積極卡位 AI 基礎設施

- **3Q26 毛利率高達 85.7%，歷史新高**
- **資本支出計畫維持不變，規劃在屏東建置 3D 晶圓封測廠**
- **AI 基礎建設營收占比目標在 2029~2030 年提升至 40%~50%**

3Q26 稅後 EPS 為 25.40 元，優於預期：雖然 3Q26 位元出貨量季減中個位數百分比，但在平均銷售單價季增超過六十位數百分比及匯率影響有利低個位數百分比下，季營收增為 1,336.49 億元(QoQ+61.9%，YoY+611.7%)，持續受惠於報價上漲，毛利率由前季 79.5%增為 85.7%，營業利益 1,068.94 億元(QoQ+75.7%，YoY+9,451.3%)，稅前淨利為 1,093.32 億元(QoQ+75.4%，YoY+5,304.6%)，稅後 EPS 為 25.40 元，優於我們預估的 23.69 元。

AI 帶動 DRAM 應用領域持續擴大：對於市況及供需看法方面，公司表示 AI 運算持續主導 DRAM 需求，並使手機、PC 及一般消費性記憶體供給受限，供應商依照 LTA，進行長期產能規劃。供給端庫存水位偏低，新廠建置、設備交期等都需要較長的前置時間。AI 推升更高容量、更高頻寬及客製化記憶體需求，應用也擴大至網路、存儲及系統管理等。公司目前 AI 基礎建設相關占營收比重繼續超過 20%，將持續協助客戶開發客製化 AI/Wafer-to-Wafer 產品。也因 AI 排擠效應，傳統應用因供給不足及漲價影響，導致產品降規及延長生命週期。

2026 年資本支出維持 697 億元(YoY+420%)：1Q26~3Q26 合計資本支出為 276 億元，2026 年維持 697 億元不變(其中生產設備占約 50%)。2026 年位元銷售量目標成長高十位數百分比，預期在 4Q27 新廠投產前，出貨量維持平穩。泰山 5A 新廠長期投資計劃也維持不變(2026~2029 年資本支出預算以不超過 3,466 億元為上限)，新廠在 1Q27 開始裝機，2028 年底月產能 30K，2029 年拉升至 35.9K，全產能目標 2030 年底達 50K。1C 製程預計在 1Q27 客戶驗證，1D/1E/EUV 開發進度如期。

考量公司在長約保護及報價支撐下，2026~2027 年獲利強勁成長態勢不變，故維持增加持股建議，目標價為 652.0 元(2027 年 PERx6.00 倍)。

風險因子：DDR4/DDR5 需求不佳，導致報價開始反轉向下。

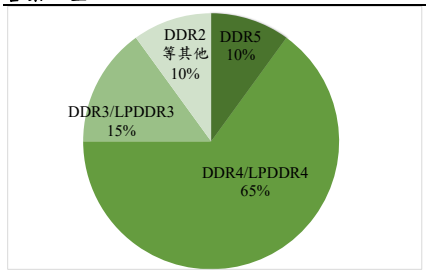
臺灣 ESG 永續指數成分股

潛在漲幅	23.7%
前日收盤價	527.00 元
目標價	652.00 元
前次目標價	608.00 元
前次評等	增加持股

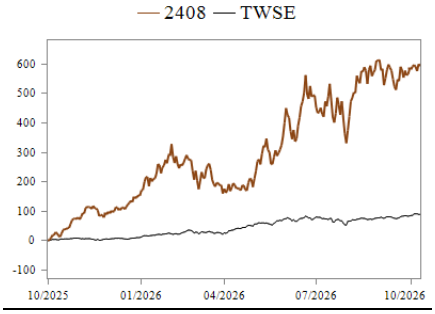
公司基本資訊

目前股本(百萬元)	34,502
市值(億元)	18,183
目前每股淨值(元)	93.49
外資持股比(%)	17.33
董監持股比(%)	26.33

營業比重



個股與大盤走勢(%)



徐千智
Chienchih.hsu@ctbcsis.com

(百萬元)	2022	2023	2024	2025	2026F	2027F
營業收入淨額	56,952	29,892	34,132	66,587	408,386	611,134
營業毛利淨額	21,342	-4,483	-421	14,986	337,264	503,892
營業利益	11,002	-14,460	-10,555	5,243	313,778	467,330
稅後純益	14,619	-7,440	-5,083	6,614	257,092	375,098
公告基本 EPS(元)	4.72	-2.40	-1.64	2.13	--	--
調整後 EPS(元)	4.72	-2.40	-1.64	2.13	74.51	108.72
營業收入 YoY(%)	-33.47%	-47.51%	14.18%	95.09%	513.32%	49.65%
稅後純益 YoY(%)	-36.02%	--	--	--	3,787.22%	45.90%
本益比	10.49	-32.40	-17.78	90.32	7.07	4.85
本淨比	0.85	1.44	0.55	3.50	3.61	2.23
股息(元)	2.13	0.00	0.00	1.35	29.80	43.50
殖利率(%)	3.01%	0.00%	0.00%	0.32%	5.65%	8.25%
ROE(%)	8.26%	-4.28%	-3.06%	3.94%	76.31%	56.95%

集邦預估 4Q26 DRAM 合約價季增 10%~15%：根據市調機構集邦(TrendForce)在 2026 年 9 月底的調查資料(見圖表一)，DRAM 原廠在 4Q26 持續將先進製程投入生產 server 相關應用之高效能記憶體產品，整體維持供不應求格局，惟合約價漲勢放緩，預計 conventional DRAM 合約價將季增 10%~15%。

公司表示 3Q26 公司平均銷售單價季增超過六十位數百分比，主要是反映半年期及一年期長約重新換約生效，貢獻超過 50%，出貨量因為庫存見底，而出現季減中個位數百分比。公司預估 4Q26 報價可望維持逐步往上態勢，2027 年整體市場供需也認為將維持吃緊狀態。公司持續提供客戶多樣產品，如：DDR5、LPDDR5/5X、DDR4、LPDDR4/4X、DDR3 及 LPDDR3。目前產品占營收比重，DDR5 維持占約 10%、DDR4 及 LPDDR4/4X 占超過 60%、其餘為 DDR3/LPDDR3/DDR2 等其他產品。公司目前庫存水位極低，在新廠 4Q27 開出產能前，未來數季度之出貨量將維持平穩，但受惠於正向的 ASP 環境，預期營運成長將具延續性，整體 2026 年位元銷售量年增高十位數百分比。

圖表一、3Q26 及 4Q26 記憶體合約價格預測

	3Q26E	4Q26F
Total DRAM	Conventional DRAM: up 13~18% HBM Blended: up 8~13%	Conventional DRAM: up 10~15% HBM Blended: up 15~20%
Total NAND Flash	up 18~23%	up 15~20%

資料來源：TrendForce

長約涵蓋率 50%~70%，合約期限從季度到長達 3 年：公司表示目前 DDR4 需求仍相當強勁，占全球市場需求仍超過 10%，在公司市占率僅低個位數下，客戶需求穩健，但公司會依照客戶需求，彈性調整切換至生產 DDR5 等產品。關於長約方面，目前長約涵蓋率大約落在 50%~70%，期限從季度合約到 3 年期合約都有，其中 3 年期長約目前已占公司產出 20%以上，正積極依客戶意願擴大長約比例，長約具有雙向法律與商業約束力，公司方面承諾進行資本支出、擴建產能並採購設備，客戶也必須在長期採購提貨方面承擔相對應的擔保與責任。至於策略私募客戶採購方面，主要應用於 eSSD、KV Cache、Switch 及系統管理晶片。公司出貨型態中，Known Good Die 約占 20%、其餘 80%為模組及封裝顆粒。

5A 新廠第一階段月產能目標 35.9K：在 5A 新廠規劃方面，維持與上次法說相同，1Q27 開始裝機，預計 2028 年底前達到月產能 30K，2029 年達 35.9K(總投入以不超過 3,466 億元為上限)。5A 全產能目標為 2030 年底達 50K，包含建築興建、無塵室、設備採購及未來 EUV 導入，預計 2026~2030 年長期整體資本支出預計將高達 160 億美元。

屏東科學園區新廠將以 3D 封測業務為主：規劃將在屏東科學園區建立 3D 晶圓級封裝與測試產能，其中晶圓級加工由公司本身主導，後段晶圓研磨、切割、封裝組裝與成品測試則與福懋科分工合作。目前建廠規劃尚未完全定案，整體

資本支出最高達 100 億美元，預計 2H28 開始進駐設備，希望能滿足客戶在 AI 應用上的強勁需求。至於 Wafer-to-Wafer 進度方面，不只與一家客戶合作，應用從 AI 雲端到邊緣運算及實體 AI 皆有，且目前已有小部分營收貢獻。

圖表二、季度財報與預估差異

損益表	3Q26					4Q26(F)				
(單位：百萬元)	實際數	中信預估	差異	市場共識	差異	調整後	調整前	差異	市場共識	差異
營業收入	133,649	133,126	0.4%	123,227	8.5%	143,101	133,126	7.5%	130,590	9.6%
營業毛利	114,482	107,886	6.1%	103,919	10.2%	123,839	107,886	14.8%	110,900	11.7%
營業利益	106,894	101,793	5.0%	95,444	12.0%	115,946	101,793	13.9%	102,205	13.4%
稅前純益	109,332	102,163	7.0%	99,287	10.1%	116,516	102,163	14.0%	106,132	9.8%
稅後淨利	87,627	81,730	7.2%	83,437	5.0%	93,213	81,730	14.0%	87,994	5.9%
EPS(元)	25.40	23.69	7.2%	24.18	5.0%	27.02	23.69	14.0%	25.50	5.9%
財務比率(%/百分點)										
毛利率	85.7%	81.0%	4.6 ppts	84.3%	1.3 ppts	86.5%	81.0%	5.5 ppts	84.9%	1.6 ppts
營益率	80.0%	76.5%	3.5 ppts	77.5%	2.5 ppts	81.0%	76.5%	4.6 ppts	78.3%	2.8 ppts
稅後淨利率	65.6%	61.4%	4.2 ppts	67.7%	-2.1 ppts	65.1%	61.4%	3.7 ppts	67.4%	-2.2 ppts

資料來源：Cmoney、Bloomberg，中信投顧整理

圖表三、年度財報與預估差異

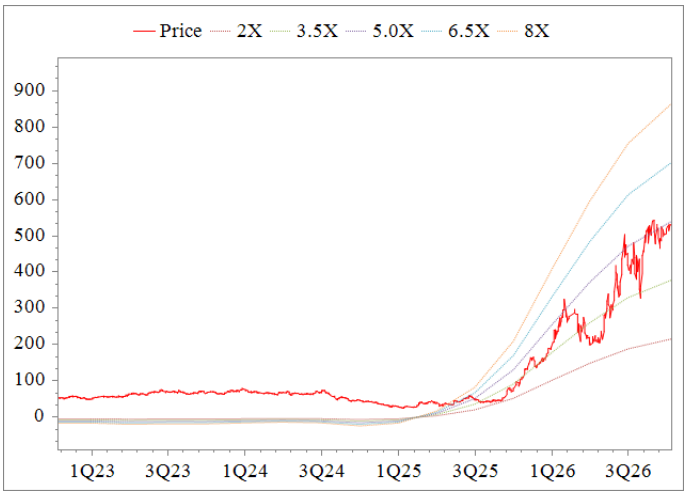
損益表	2026(F)					2027(F)				
(單位：百萬元)	調整後	調整前	差異	市場共識	差異	調整後	調整前	差異	市場共識	差異
營業收入	408,386	398,411	2.5%	383,960	6.4%	611,134	582,531	4.9%	606,508	0.8%
營業毛利	337,257	314,529	7.2%	311,836	8.2%	503,892	462,318	9.0%	508,515	-0.9%
營業利益	313,778	294,754	6.5%	286,326	9.6%	467,330	435,966	7.2%	475,316	-1.7%
稅前純益	319,896	298,738	7.1%	297,543	7.5%	468,872	437,198	7.2%	486,401	-3.6%
稅後淨利	257,092	240,004	7.1%	240,173	7.0%	375,098	349,759	7.2%	390,740	-4.0%
EPS(元)	74.51	69.56	7.1%	69.61	7.0%	108.72	101.37	7.2%	113.25	-4.0%
財務比率(%/百分點)										
毛利率	82.6%	78.9%	3.6 ppts	81.2%	1.4 ppts	82.5%	79.4%	3.1 ppts	83.8%	-1.4 ppts
營益率	76.8%	74.0%	2.9 ppts	74.6%	2.3 ppts	76.5%	74.8%	1.6 ppts	78.4%	-1.9 ppts
稅後淨利率	63.0%	60.2%	2.7 ppts	62.6%	0.4 ppts	61.4%	60.0%	1.3 ppts	64.4%	-3.0 ppts

資料來源：Cmoney、Bloomberg，中信投顧整理

2026/10/13

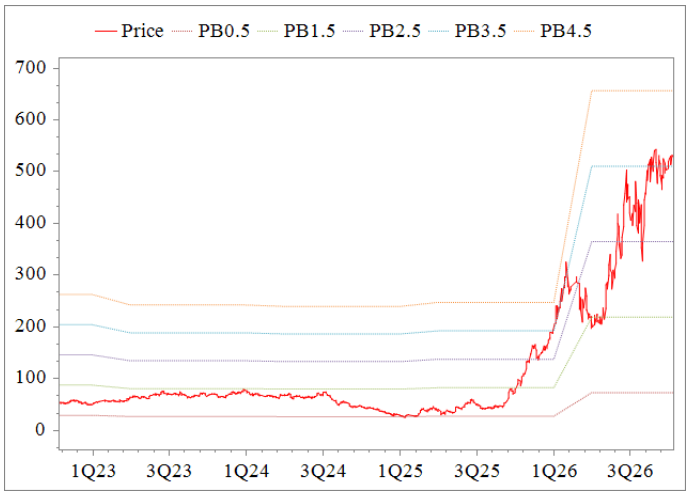
(百萬元)	1Q26	2Q26	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F
營業收入	49,087	82,549	133,649	143,101	147,251	148,635	154,625	160,624
營業成本	15,761	16,932	19,167	19,261	21,719	24,763	29,085	31,675
營業毛利	33,325	65,617	114,482	123,839	125,531	123,872	125,540	128,949
營業費用	3,205	4,793	7,588	7,893	8,102	8,830	9,515	10,115
營業利益	30,111	60,826	106,894	115,946	117,429	115,042	116,025	118,834
營業外收支淨額	1,607	1,503	2,438	570	473	394	455	220
稅前純益	31,718	62,329	109,332	116,516	117,902	115,436	116,480	119,054
所得稅	5,660	12,138	21,705	23,303	23,580	23,087	23,296	23,811
稅後純益	26,059	50,192	87,627	93,213	94,322	92,349	93,184	95,243
調整後 EPS(元)	7.55	14.55	25.40	27.02	27.34	26.77	27.01	27.61
毛利率(%)	67.89%	79.49%	85.66%	86.54%	85.25%	83.34%	81.19%	80.28%
營業利益率(%)	61.34%	73.68%	79.98%	81.02%	79.75%	77.40%	75.04%	73.98%
稅後純益率(%)	53.09%	60.80%	65.57%	65.14%	64.06%	62.13%	60.26%	59.30%
營業收入								
QoQ(%)	63.11%	68.17%	61.90%	7.07%	2.90%	0.94%	4.03%	3.88%
YoY(%)	582.91%	684.23%	611.70%	375.52%	199.98%	80.06%	15.69%	12.25%
營業利益								
QoQ(%)	155.60%	102.00%	75.74%	8.47%	1.28%	-2.03%	0.85%	2.42%
YoY(%)	--	--	9,451.29%	884.21%	289.98%	89.13%	8.54%	2.49%
稅後純益								
QoQ(%)	134.91%	92.61%	74.58%	6.37%	1.19%	-2.09%	0.90%	2.21%
YoY(%)	--	--	5,505.14%	740.27%	261.95%	83.99%	6.34%	2.18%

PER Band



資料來源：中信、CMoney

PBR Band



資料來源：CMoney

2026/10/13

資產負債表

(百萬元)	2022	2023	2024	2025	2Q26
資產總計	202,236	192,351	206,706	208,453	387,758
流動資產	105,512	96,510	107,946	108,541	284,521
現金及約當現金	73,593	58,812	61,903	58,074	217,603
應收帳款與票據	4,360	5,220	4,132	16,558	40,279
存貨	23,384	27,634	35,318	27,288	21,530
採權益法之投資	5,386	5,120	4,645	5,904	7,063
不動產、廠房設備	84,897	81,838	84,327	85,031	89,804
負債總計	21,278	25,427	41,653	37,914	65,060
流動負債	16,179	20,118	32,780	19,020	41,174
應付帳款及票據	5,395	3,298	5,009	6,368	7,980
非流動負債	5,099	5,309	8,873	18,894	23,885
權益總計	180,958	166,924	165,053	170,538	322,699
普通股股本	30,980	30,981	30,986	30,986	34,502
保留盈餘	116,541	102,516	97,478	104,102	175,705
母公司業主權益	180,958	166,924	165,053	170,400	322,561
負債及權益總計	202,236	192,351	206,706	208,453	387,758

損益表

(百萬元)	2023	2024	2025	2026F	2027F
營業收入淨額	29,892	34,132	66,587	408,386	611,134
營業成本	34,375	34,552	51,600	71,121	107,242
營業毛利淨額	-4,483	-421	14,986	337,264	503,892
營業費用	9,977	10,134	9,741	23,479	36,562
營業利益	-14,460	-10,555	5,243	313,778	467,330
EBITDA	1,607	6,612	20,256	331,260	484,977
業外收入及支出	3,756	3,998	2,639	6,118	1,542
稅前純益	-10,705	-6,557	7,882	319,896	468,872
所得稅	-3,265	-1,474	1,269	62,806	93,774
稅後純益	-7,440	-5,083	6,614	257,092	375,098
公告基本 EPS(元)	-2.40	-1.64	2.13	- -	- -
調整後 EPS(元)	-2.40	-1.64	2.13	74.51	108.72

註：稅後純益係指本期淨利歸屬於母公司業主。

現金流量表

(百萬元)	2022	2023	2024	2025	2Q26
營業活動現金	21,082	-6,095	1,952	18,568	55,859
稅前純益	16,877	-10,705	-6,557	7,882	62,329
折舊及攤銷	15,247	15,326	16,144	14,281	2,960
營運資金變動	-1,810	-7,271	-4,859	-3,095	-11,439
其他營運現金	-9,232	-3,445	-2,776	-500	2,009
投資活動現金	-20,868	-12,786	-16,188	-14,110	-4,242
資本支出淨額	-20,588	-13,148	-16,141	-13,441	-4,042
長期投資變動	-729	-6	-6	-614	0
其他投資現金	449	369	-40	-55	-201
籌資活動現金	-11,771	4,196	13,707	-6,317	79,415
長借/公司債變動	0	0	4,000	10,200	600
現金增資	0	0	0	0	0
發放現金股利	-11,470	-6,600	0	0	0
其他籌資現金	-301	10,796	9,707	-16,517	78,815
淨現金流量	-7,107	-14,781	3,091	-3,829	131,322
期初現金	80,700	73,593	58,812	61,903	86,281
期末現金	73,593	58,812	61,903	58,074	217,603

資料來源：中信、CMoney

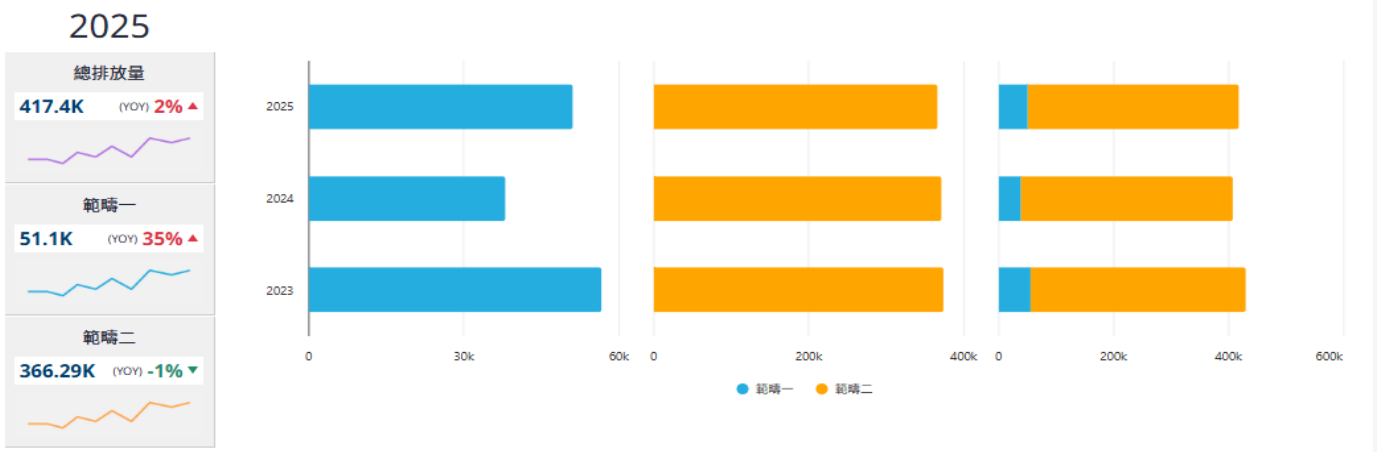
比率分析

(百萬元)	2023	2024	2025	2026F	2027F
成長力分析(%)					
營業收入淨額	-47.51%	14.18%	95.09%	513.32%	49.65%
營業毛利淨額	-121.00%	-90.62%	-3663.18%	2150.50%	49.41%
營業利益	--	--	--	5884.37%	48.94%
稅後純益	--	--	--	3787.22%	45.90%
獲利能力分析(%)					
毛利率	-15.00%	-1.23%	22.51%	82.58%	82.45%
EBITDA(%)	5.38%	19.37%	30.42%	81.11%	79.36%
營益率	-48.37%	-30.92%	7.87%	76.83%	76.47%
稅後純益率	-24.89%	-14.89%	9.93%	62.95%	61.38%
ROE(%)	-4.28%	-3.06%	3.94%	76.31%	56.95%

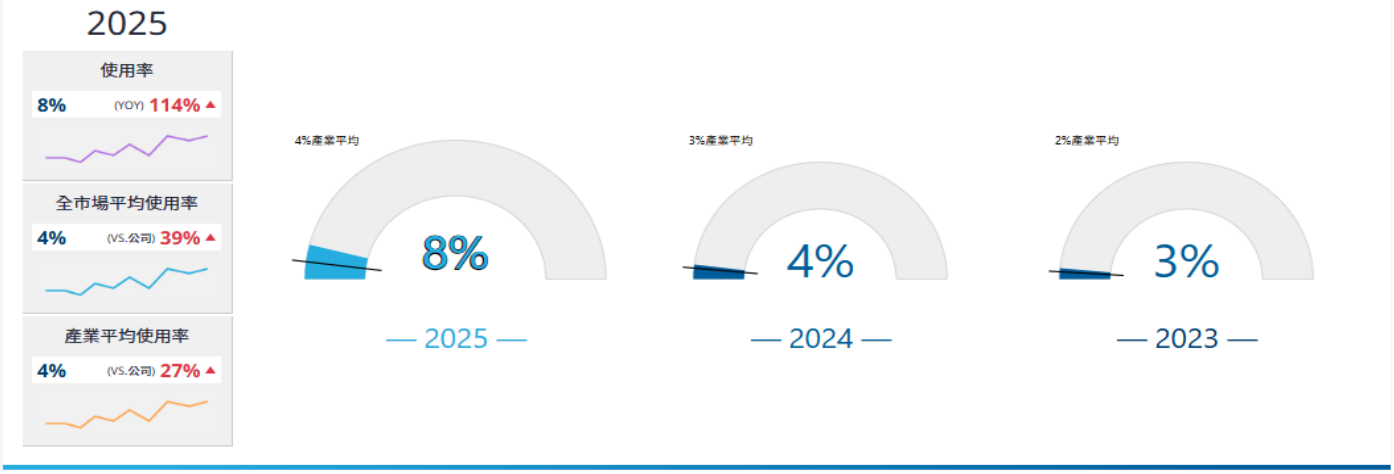
2026/10/13

ESG 資訊-環境保護(Environmental)

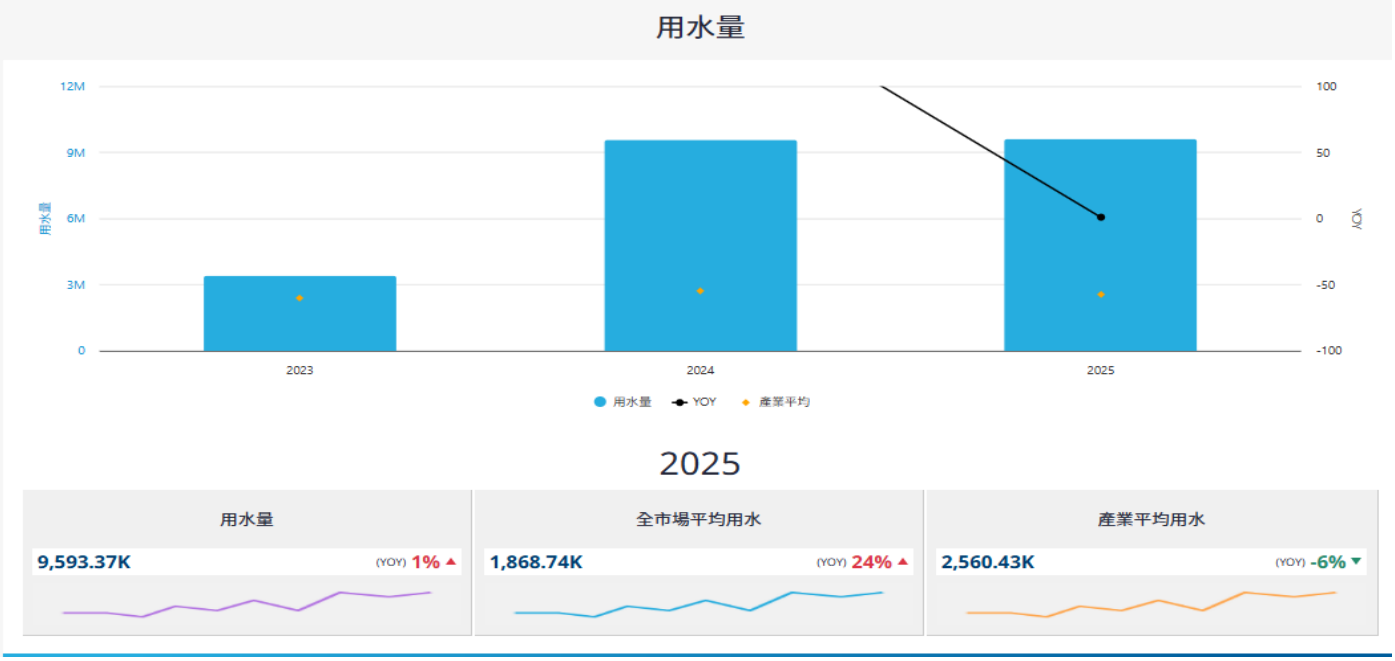
溫室氣體(範疇一/二)排放量



再生能源使用率

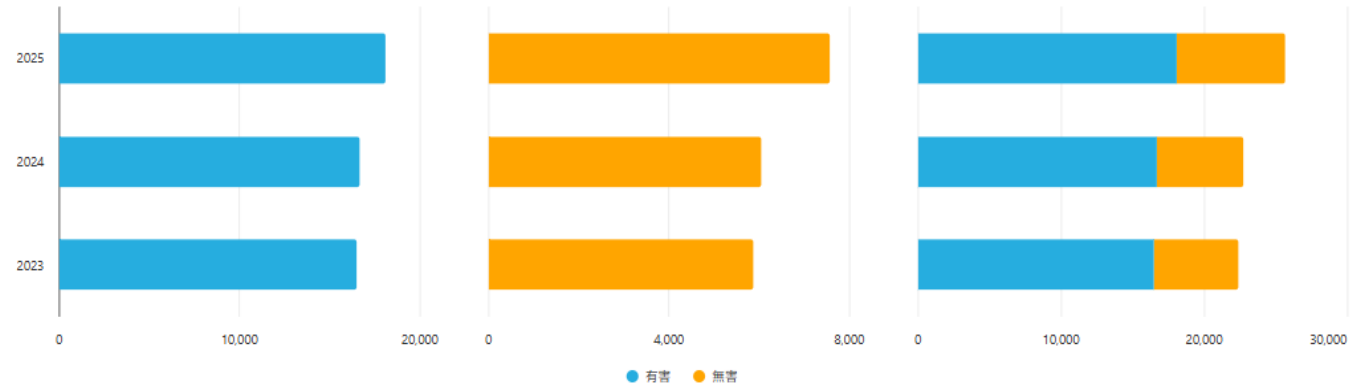


用水量



2026/10/13

廢棄物量



2025

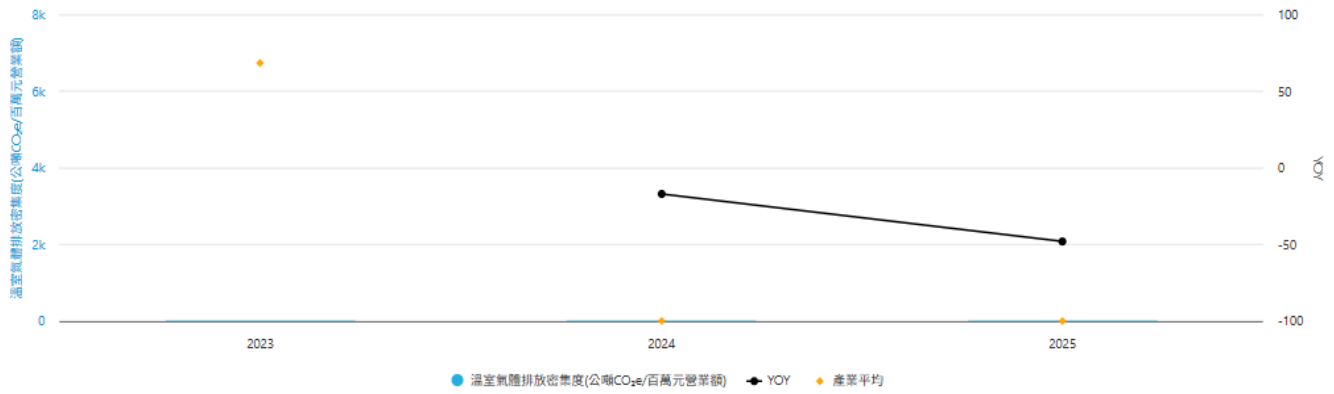
有害	無害	總量
18.07K (YOY) 8% ▲	7.58K (YOY) 25% ▲	25.65K (YOY) 25% ▲

密集度

2025

溫室氣體排放密集度	用水密集度	廢棄物密集度
6.27	3.71	0.023

溫室氣體排放密集度(公噸CO_2e/百萬元營業額)



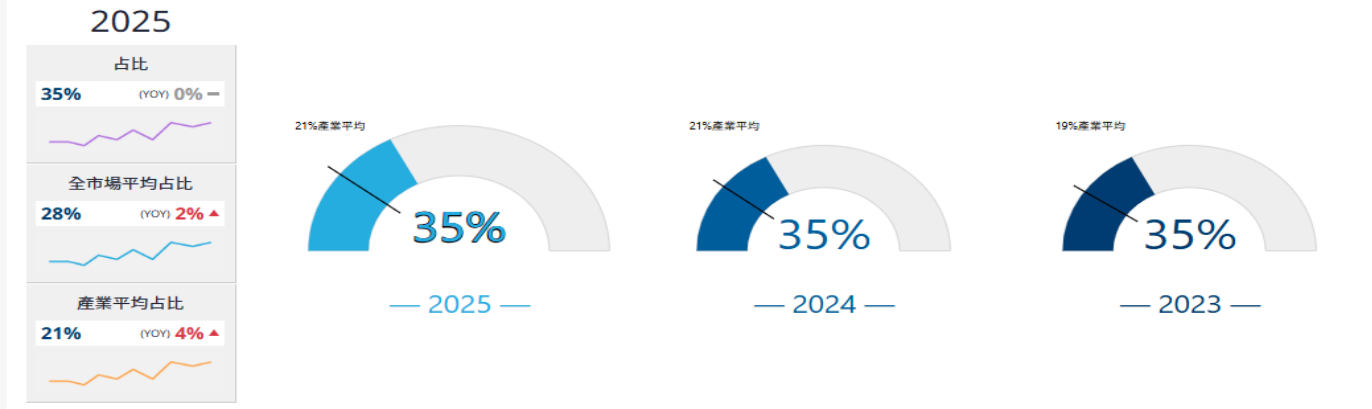
2025

溫室氣體排放密集度(公噸CO_2e/百萬元營業額)	全市場平均溫室氣體排放密集度(公噸CO_2e/百萬元營業額)	產業平均溫室氣體排放密集度(公噸CO_2e/百萬元營業額)
0.01K (YOY) -48% ▼	0.04K (YOY) 10% ▲	0K (YOY) -46% ▼

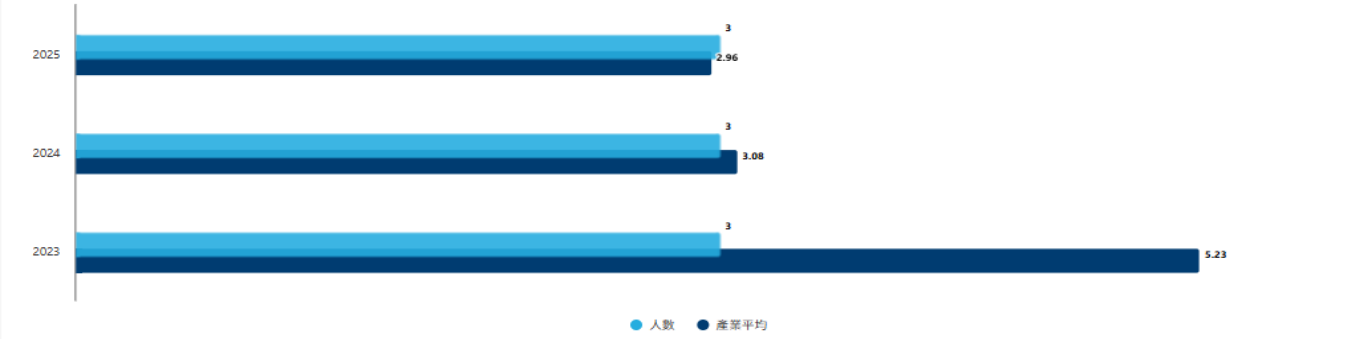
2026/10/13

ESG 資訊-社會責任(Social)

管理職女性主管占比



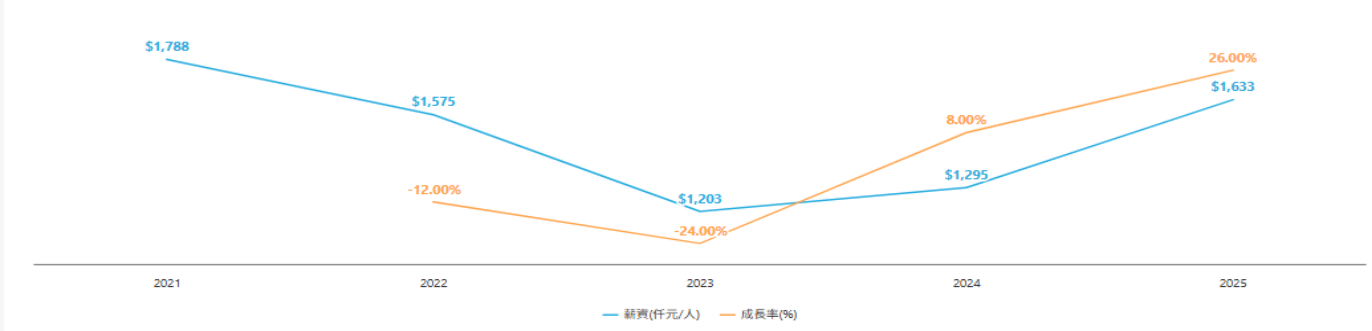
職業災害人數



2025

人數	全市場平均人數	產業平均人數
3 (YOY) 0%	5 (VS.公司) 0%	3 (VS.公司) -4%

員工薪資年度趨勢

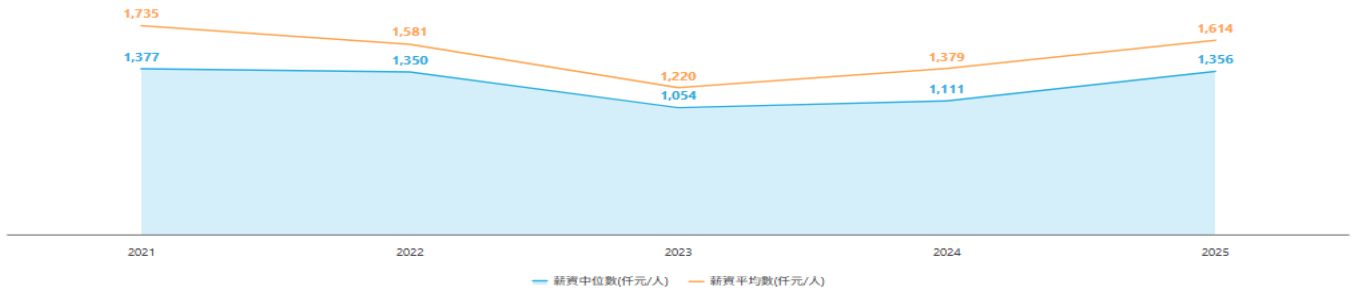


2025

員工薪資平均數	與去年同期比	與同產業薪資比
1,633	26% ▲	-36.72%

2026/10/13

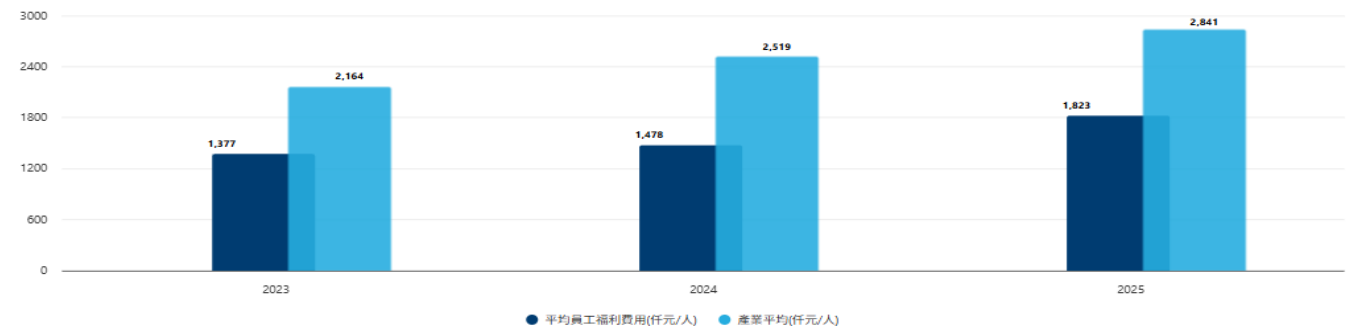
非擔任主管職務之全時員工薪資



2025

平均數	中位數	差異%
1,614 (YOY) 17% ▲	1,356 (YOY) 22% ▲	0% (YOY) -21% ▼

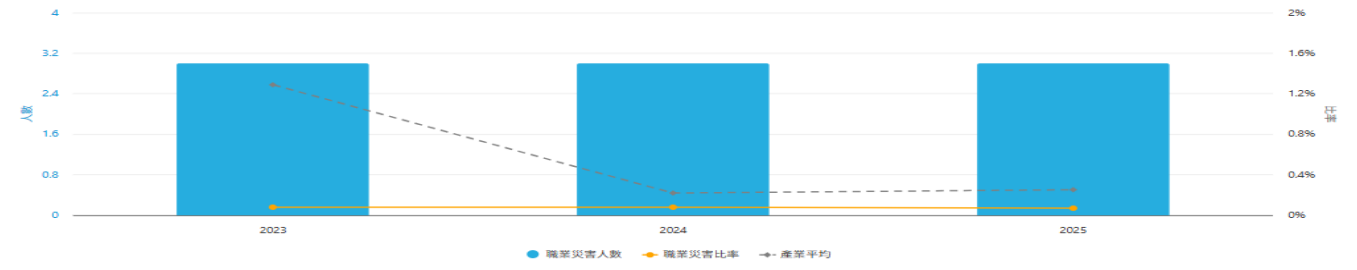
平均員工福利費用



2025

平均福利	全市場平均福利	產業平均福利
1,823 (YOY) 23% ▲	1,350 (YOY) 3% ▲	2,841 (YOY) 13% ▲

職業災害人數與比率

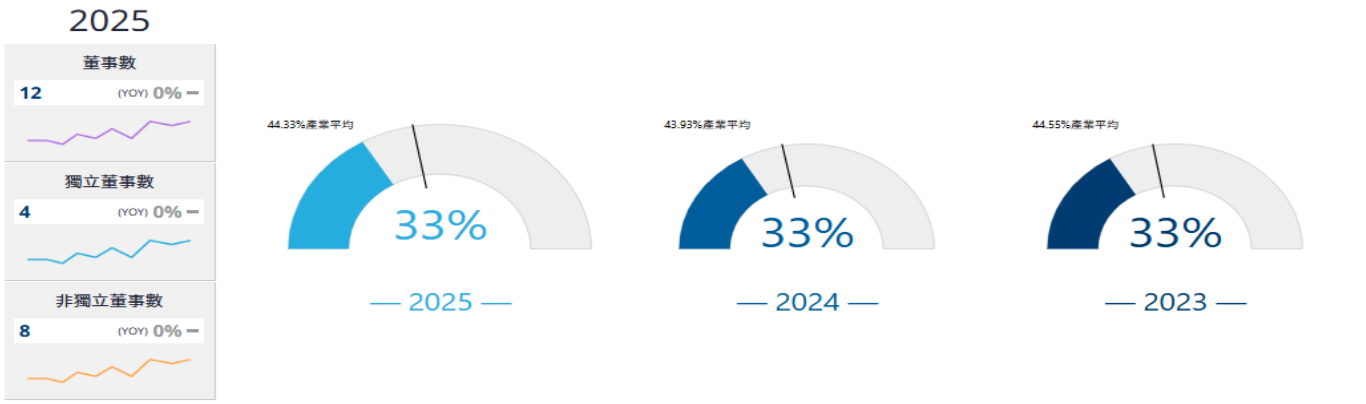


火災件數與死傷人數

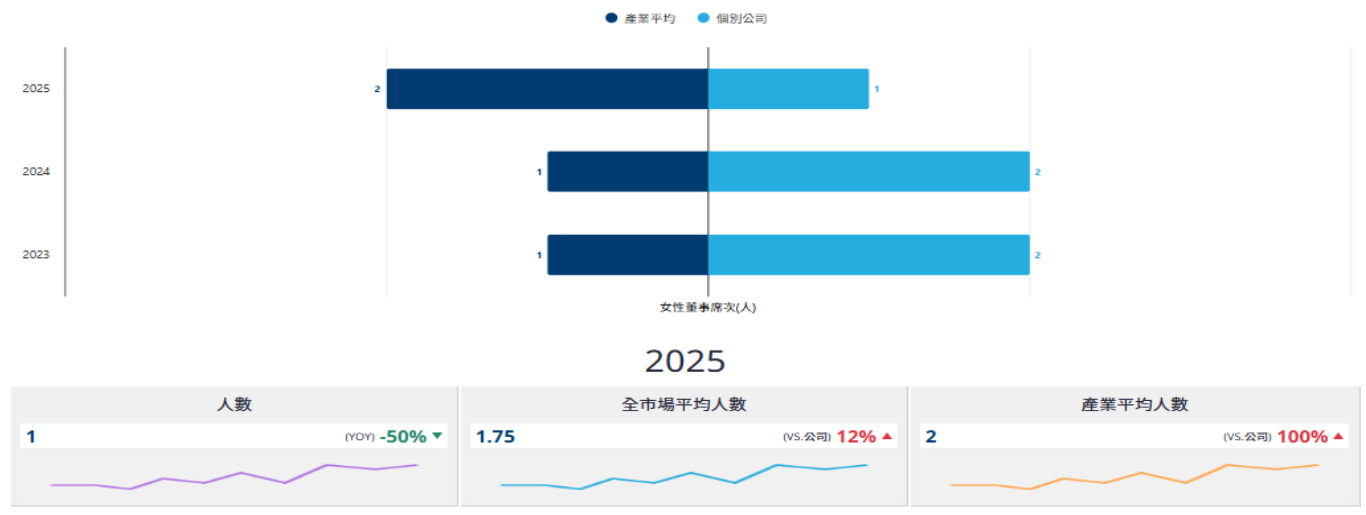
2026/10/13

ESG 資訊-公司治理(Governance)

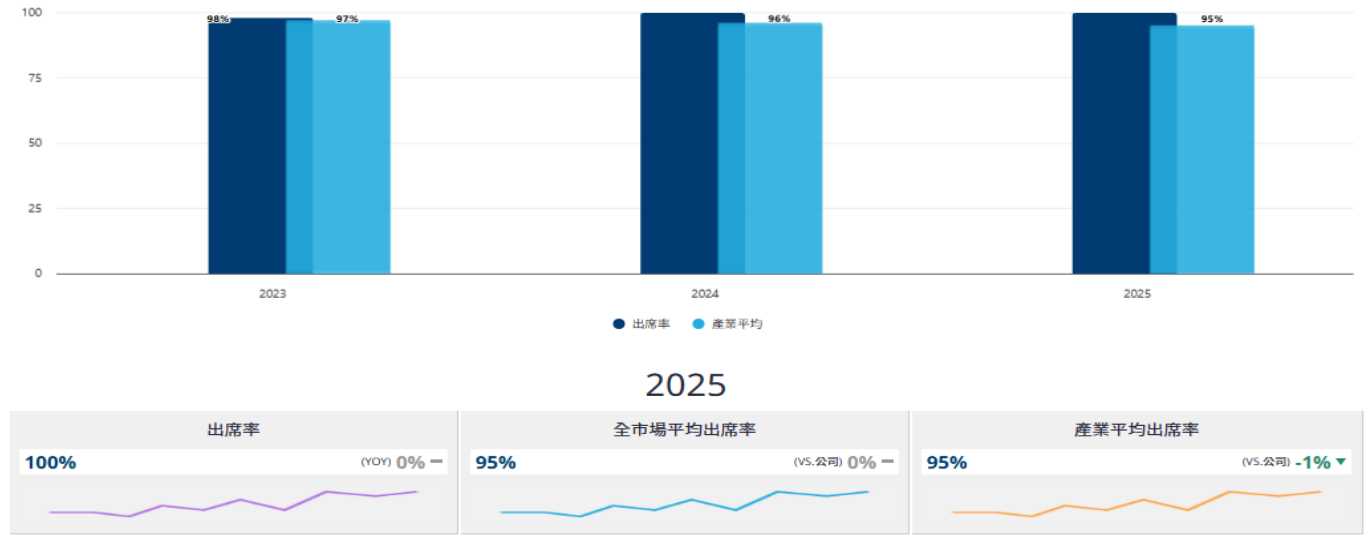
獨立董事vs.董事席次占比



女性董事席次



董事之董事會出席率



2026/10/13

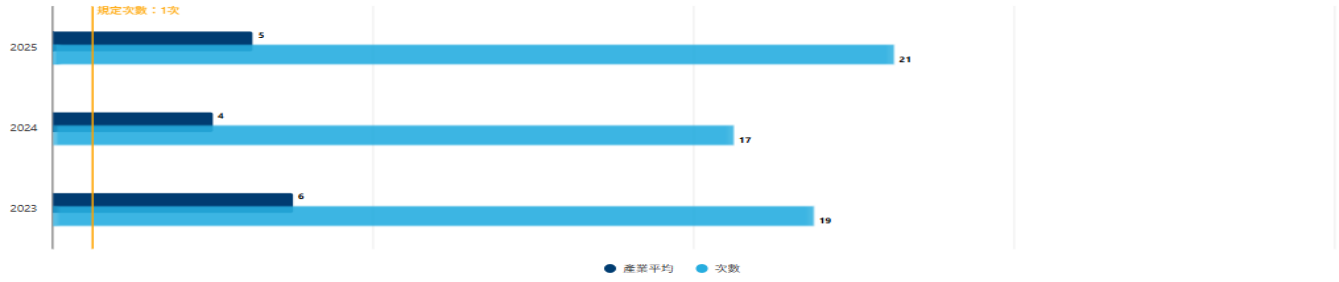
董事進修符合進修要點比率

內圈:個別公司　外圈:產業平均

2025

進修率	全市場平均進修率	產業平均進修率
100% (YOY) 0% —	94% (VS.公司) 1% ▲	95% (VS.公司) 0% —

法說會次數



2025

次數	全市場平均次數	產業平均次數
21 (YOY) 24% ▲	2 (VS.公司) 0% —	5 (VS.公司) 25% ▲

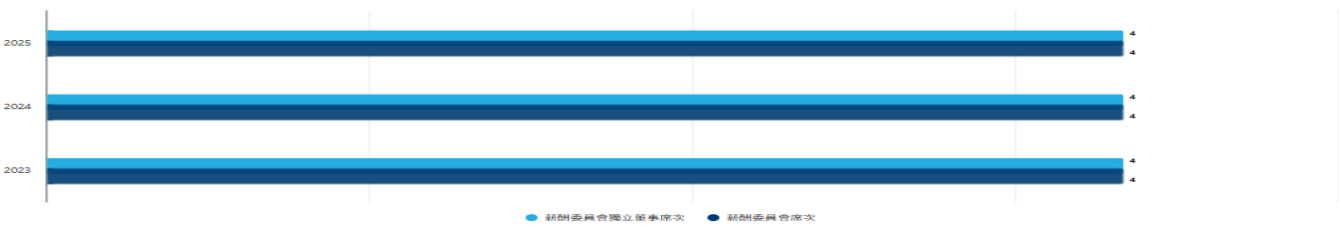
薪酬委員會出席率

內圈:個別公司出席率　外圈:產業平均

2025

進修率	全市場平均進修率	產業平均進修率
100% (YOY) 0% —	97% (VS.公司) 1% —	97% (VS.公司) 0% ▼

薪酬委員會席次



資料來源：證交所

2026/10/13

投資評等說明

個股評等	Rating	定義
買進	Buy (B)	相較於市場指數之回報潛力極具吸引力
增加持股	Overweight (OW)	相較於市場指數之回報潛力略有吸引力
中立	Neutral (N)	相較於市場指數之回報潛力相似
降低持股	Underweight (UW)	相較於市場指數之回報潛力稍低
賣出	Sell (S)	相較於市場指數之回報潛力較低
未評等	Not Rated (NR)	

免責聲明

【中國信託證券投資顧問股份有限公司獨立經營管理】

本報告係本公司於特定日期就現有資訊所提出，雖已力求正確與完整，但因時間及市場客觀因素改變所造成產業、市場或個股之相關條件改變，本公司將不做預告或更新；而且本報告並非依特定投資目的或特定對象之財務狀況所完成，因此，本報告之目的並非針對客戶於買賣證券、選擇權、期貨或其他證券相關商品提供詢價或推介服務，亦非進行交易之要約。本公司出具之短、中、長期策略性報告所做出之投資分析與建議，亦可能會隨市場波動隨時調整，而與個股研究報告之意見未必一致。投資人應自行考量本身之需求、投資風險及承受度，並就投資結果自行負責，本公司不做任何獲利保證，亦不就投資盈虧負擔任何法律責任。非經本公司同意，任何人不得將本報告加以引用或轉載。

中國信託金融控股(股)公司旗下子公司暨關係企業（以下稱"中信金集團"）從事廣泛金融業務，包括但不限於銀行、信託、證券經紀、證券承銷、保險、證券投資信託、全權委託投資、自有資金投資等。中信金集團對於本報告所涵蓋之標的公司可能有投資或其他的業務往來關係。中信金集團及其所屬員工，可能會投資本報告所涵蓋之標的公司，或以該等公司為標的的衍生性金融商品，而其交易的方向與本報告中所提及的交易方向可能不一致。中信金集團於法令許可的範圍內，亦有可能以本報告所涵蓋之標的公司作為發行衍生性金融商品之標的。中信金集團轄下的銷售人員、交易員及其他業務人員可能會為他們客戶或自營部門提供口頭或書面的市場看法或交易策略，然該等看法與策略可能與本報告的意見不盡一致。中信金集團之自營、資產管理及其他投資業務所做之投資決策也可能與本報告所做的建議或看法不一致。

中國信託證券投資顧問股份有限公司

台北市南港區經貿二路 188 號 5 樓及 18 樓 / (02) 5569-1988

營業執照字號：114 金管投顧新字第 014 號

12 Oct 2026 14:45:07 ET | 19 pages

Taiwan Electronics & Semiconductors

Taiwan monthly tracker and what's new in AI – Positive into earnings season

Laura (Chia Yi) Chen [AC]
+886-2-8726-9090
laura.cy.chen@citi.com

Jack Chen
+886-2-8726-9091
jack1.chen@citi.com

Angela Hsu
+886-2-8726-9083
angela.hc.hsu@citi.com

Michael Hung
+886-2-8726-9092
michael.hung@citi.com

Nicholas Lai
+886-2-8726-9093
nicholas.lai@citi.com

CITI'S TAKE

Taiwan technology supply chain concluded a robust 3Q26 revenue. Other than solid momentum in foundry, OSAT and testing interface/equipment, we see accelerating revenue growth for substrate and CCL makers. Server ODMs are also benefiting from new Rubin system and ASIC rack ramping. Heading into 3Q26 earnings season, we expect companies to provide better visibility and outlook into 4Q26 and 2027. We expect TSMC's earnings call to be an important indicator and catalyst for AI supply chain into next year (see our TSMC preview note).

Semiconductor foundry and OSAT- On the back of strong AI compute demand, TSMC and ASEH reported solid 18% and 28% QoQ revenue growth, respectively, in 3Q26. We believe there is a great chance that TSMC beats its full year guidance and delivers mid-40% YoY growth for 2026, and we expect more upside for ASEH's leap business outlook into next year. For trailing edge foundry, we believe the high UTR will support better GM recovery; VIS may see more revenue and GM upside on its interposer business outsourcing from TSMC.

On testing interface and equipment, we expect the rising chip complexity and testing process to persist. Both Chroma and Hon Precision posted stronger than expected Sep and 3Q26 revenue. With smooth capacity ramp for Hon, we see sequential QoQ sales growth momentum into next year. MPI and Winway are also ahead of consensus on 3Q26 revenue while WinWay's MoM decline in Sep was mainly driven by lower probe card revenues as major project shipments have temporarily wrapped up. We expect Rubin Ultra's expandable scale-up CPO rack design to be the next catalyst for CPO testing sector into late 2026/early 2027.

IC fabless - ASIC momentum picking up – Thanks to ARM-based CPU and Trainium 3 momentum pickup respectively at GUC and Alchip, both companies' 3Q26 revenues are ahead of consensus estimate with 35%/243% QoQ growth, respectively. We believe the robust order flow will continue into 4Q26/2027. We see multiple growth on GUC's CPU shipment growth into next year thanks to agentic AI demand and Alchip's Trainium chip ramping up into next year also on track. On Mediatek, 3Q26 revenue is ahead of company's guidance and Street's estimate thanks to its market share gain in smartphone, and we expect TPU to start contributing in 4Q26. We expect Mediatek to further lift its AI ASIC business opportunity thanks to solid product pipeline in the next 2 years (see our recent report on Mediatek).

PCB and substrate- 3Q26 was a strong quarter for substrate and CCL sectors, with demand continuously piling up. Unimicron posted +23% QoQ sales growth in 3Q26, beating Citi/consensus forecasts by 4%/6%. We expect 4Q26E ABF sales momentum to be even stronger vs 3Q considering all the AI chips ramping up together. CCL names also saw strong momentum across the board thanks to enhanced product mix and benefit from price hike in 2Q26. We expect CCL sector to see another round of price hikes from Nov onwards. For PCB, GCE's Sept sales were impacted by flooding in Thailand, which led to logistics disruption, while Tripod posted record monthly sales backed by memory demand and early pull-in before China Golden Week. With TPCA Show coming next week, we think substrate/CCL/PCB sectors might see some event catalysts in the near term. However, from a fundamental perspective, we still prefer substrate (Unimicron>NYPCB>Kinsus) > CCL (EMC>TUC) > PCB.

PC/NB Supply chain - For NB ODMs, overall NB shipments see MoM increase in Sep (+32% MoM) due to the quarter end inventory build; combined 3Q NB shipments were down 17% QoQ due to the early pull-in in 1H. We forecast 4Q26E NB shipments to further decline -7% QoQ/-28% YoY given the muted demand on rising retails prices and the increasing constraint in components procurement, While the price hike effect in the NB products should partially offset shipment decline, we expect increasing margins dilution impact to the NB ODM supply chains.

For lens components - Largan reported relatively muted sales results in Sep at NT$5,779m (+15% MoM/-7% YoY); combined 3Q revenues are below market estimates while Genius' 3Q revenues were slightly better than expectation. Despite the near-term sales performance, we see more upside opportunity coming from their new development in optical business in the longer term. We expect Genius' project with a leading US company on optical connector and ferrule to start shipment in 1H27. For Largan, we expect it to benefit most from FA/FAU for CPO market. We see its current engagement with the leading US client progressing smoothly, as evidenced by its continued new land expansion for future capacity build-out (see our recent report).

Thermal – Sep-26 sales as reported by thermal players (AVC & Auras) were up from Aug at +6% MoM/+42% YoY. While AVC delivered another record month thanks to solid ASIC demand and initial Vera Rubin shipments, Auras experienced MoM decline due to client deferring pull-ins and Thailand flooding. Going forward, we preliminarily expect both companies to see sequential revenue growth in 4Q26E, driven by the AWS Trainium transition from air to liquid cooling, ramp-up of the TPU and Vera Rubin projects.

Power – Delta reported MoM revenue recovery in Sep-26 (+1% MoM/+14% YoY), with power electronics down 4% MoM. 3Q26 revenues were up 8% QoQ but below our and Street's expectation. We expect Delta to regain momentum with double-digit QoQ growth in 4Q26E as ±400V enters the initial transition period, while in the long term, HVDC is still progressing gradually while Delta maintains its technology leadership. **For BBU,** we see Sep sales showing mixed momentum yet resilient BBU demand remains intact. Dynapack's Sep-26 sales declined 34% y/y and 54% m/m to NT$776mn, reflecting 1) weaker PC shipments amid quarter-end inventory adjustments and 2) deferred shipments of one BBU product, broadly consistent with the near-term weakness flagged in our Aug-26 sales note. We expect a meaningful sales rebound from Oct helped by shipments of new higher-power BBU products. Meanwhile, AES maintained solid momentum with Sep-26 sales rising 25% y/y (+1% m/m), suggesting resilient underlying BBU demand. Similarly, STL (4931.TWO; not covered) delivered robust 176% y/y growth, further reinforcing our positive view on the sector. We retain our constructive stance on BBU names, supported by sustained demand and upcoming product ramps

Passive components - Yageo reported trended up monthly sales in Sep-26, as demand in Europe and US recovered after the summer holiday, on top of sustained AI demand. 3Q26 revenues were slightly above cons forecasts. Current quarter top line still beats mgmt.'s "moderate growth" guidance. We believe that growth was again broad-based across commodity and specialty products, and driven by both volume and price. We expect a more visible ASP uplift in 4Q26E as specialty and contract-based business could slow vs the spot and distributor channels.

Figure 1. Overall Taiwan Technology Supply Chains – Revenues Trend

Ticker	Company	Sep revenues (TWDm)	YoY	MoM	Actual 3Q26 QoQ	Citi 3Q26E QoQ	Cons. 3Q26E QoQ	Citi 3Q26E Tracking	Cons. 3Q26E Tracking	9M26 (TWDm)	YoY	Aug-26 YoY	Jul-26 YoY	Jun-26 YoY	May-26 YoY	Apr-26 YoY	Closing Prices (NT$)
Wafer/materials																	
6488.TWO	GWC	5,129	-13%	8%	-2%	6%	7%	92%	92%	44,070	-4%	8%	20%	-2%	-3%	-10%	1,115
3532.TW	Formosa Sumco	1,349	29%	1%	10%	na	10%	na	100%	10,903	20%	29%	29%	24%	24%	14%	486
5434.TW	Topco Scientific	7,701	26%	-1%	14%	na	17%	na	97%	63,029	25%	36%	43%	33%	19%	12%	593
6182.TWO	Wafer Works	1,011	14%	1%	11%	na	na	na	na	8,212	12%	20%	19%	14%	12%	3%	124
Sub-Total		**15,190**	**9%**	**2%**	**8%**					**126,214**	**12%**	**24%**	**32%**	**16%**	**10%**	**3%**	
Semi materials/equipment																	
1560.TW	Kinik	909	30%	1%	8%	na	9%	na	99%	7,480	25%	30%	27%	16%	18%	19%	1,000
8028.TW	Phoenix Silicon	512	33%	1%	7%	na	11%	na	97%	4,262	30%	26%	37%	34%	46%	31%	282
3131.TW	Grand Process	747	62%	10%	29%	na	30%	na	99%	5,544	27%	43%	42%	15%	7%	-4%	2,525
3583.TW	Scientech	1,041	3%	-3%	3%	na	6%	na	97%	9,198	7%	18%	7%	-3%	1%	11%	755
3680.TWO	Gudeng	994	68%	18%	17%	na	5%	na	112%	6,885	36%	50%	72%	38%	16%	53%	569
6187.TWO	All Ring	1,160	109%	5%	41%	na	26%	na	112%	7,090	58%	103%	73%	87%	44%	26%	1,675
6196.TW	Marketech	7,174	88%	-1%	3%	na	4%	na	99%	57,421	56%	84%	101%	121%	65%	46%	657
6438.TW	Syrntek	695	31%	14%	7%	na	na	na	na	5,513	14%	16%	5%	-3%	10%	33%	174
8091.TWO	Feedback Tech	254	27%	-17%	0%	na	-3%	na	104%	2,433	54%	72%	85%	72%	82%	35%	220
Sub-Total		**13,486**	**64%**	**2%**	**8%**					**105,825**	**47%**	**61%**	**66%**	**65%**	**42%**	**35%**	
Probe card/testing equipment																	
6223.TWO	MPI	2,048	82%	0%	13%	6%	10%	106%	102%	15,068	58%	70%	66%	65%	58%	53%	6,130
6515.TW	Winway	1,250	88%	-27%	29%	25%	21%	103%	107%	11,060	97%	233%	156%	288%	120%	51%	6,025
6510.TWO	CHPT	652	56%	2%	16%	19%	20%	97%	97%	4,902	36%	54%	50%	40%	34%	30%	3,420
7769.TW	Hon Precision	5,729	100%	10%	17%	1%	8%	117%	109%	40,713	94%	92%	101%	107%	84%	112%	6,490
2360.TW	Chroma	6,860	206%	22%	32%	6%	8%	125%	123%	43,315	120%	156%	175%	73%	133%	129%	2,365
Sub-Total		**16,539**	**126%**	**8%**	**23%**					**115,058**	**93%**	**116%**	**119%**	**94%**	**94%**	**97%**	
Design service/Semi IP																	
3443.TW	GUC	7,145	98%	22%	35%	17%	15%	115%	117%	44,099	103%	111%	158%	104%	132%	157%	8,100
3661.TW	Alchip	9,979	496%	14%	243%	156%	239%	134%	101%	37,878	45%	274%	182%	15%	-33%	-32%	4,395

3529.TWO	eMemory	177	13%	-50%	10%	na	7%	na	103%	3,401	21%	43%	24%	47%	12%	13%	2,850
3035.TW	Faraday	1,418	4%	20%	11%	na	-2%	na	113%	9,566	-37%	30%	10%	27%	-45%	-43%	198
6533.TW	Andes	193	-10%	69%	-25%	na	-16%	na	90%	1,320	30%	-26%	22%	247%	2%	23%	345
6643.TWO	M31	153	23%	-13%	17%	na	28%	na	92%	1,312	6%	73%	29%	-24%	-18%	31%	477
Sub-Total		**19,066**	**167%**	**16%**	**88%**					**97,575**	**43%**	**152%**	**130%**	**51%**	**13%**	**14%**	
IC design																	
2454.TW	Mediatek	55,399	2%	-14%	10%	5%	3%	105%	108%	469,391	5%	44%	12%	3%	5%	-4%	4,625
2379.TW	Realtek	12,378	23%	-1%	0%	-3%	3%	104%	98%	111,708	16%	28%	33%	21%	22%	11%	746
2401.TW	Sunplus	638	21%	-7%	-3%	na	na	na	na	5,954	24%	41%	38%	35%	29%	14%	26
2388.TW	VIA	940	-7%	2%	-14%	na	na	na	na	8,607	32%	77%	43%	34%	88%	85%	69
3034.TW	Novatek	10,347	25%	0%	8%	8%	7%	100%	101%	82,777	6%	26%	27%	19%	9%	1%	529
6462.TWO	EGIS	496	0%	-7%	11%	na	na	na	na	5,740	53%	11%	64%	-24%	136%	12%	100
5269.TW	ASMedia	995	-30%	13%	-2%	8%	0%	91%	98%	8,881	-10%	-33%	-38%	-22%	-19%	-18%	1,540
5274.TWO	Aspeed	1,636	101%	1%	24%	21%	19%	102%	104%	11,804	78%	118%	98%	67%	69%	82%	17,695
4966.TWO	Parade	1,503	-8%	13%	-2%	na	9%	na	90%	12,423	-2%	0%	-4%	12%	1%	-4%	553
2458.TW	Elan	602	-47%	-19%	-31%	na	-5%	na	73%	8,650	-9%	-37%	-13%	7%	4%	6%	133
6104.TWO	Genesys	311	-1%	-3%	-11%	na	na	na	na	3,084	1%	3%	4%	6%	2%	-7%	87
Sub-Total		**85,246**	**7%**	**-9%**	**8%**					**729,019**	**8%**	**37%**	**17%**	**8%**	**10%**	**0%**	
Foundry																	
2330.TW	TSMC	511,857	55%	-1%	18%	15%	15%	102%	103%	3,898,727	41%	53%	45%	68%	30%	17%	2,540
2303.TW	UMC	25,350	27%	1%	8%	12%	9%	97%	99%	204,010	16%	31%	19%	23%	18%	11%	146
5347.TW	VIS	5,904	16%	16%	10%	11%	8%	99%	102%	42,479	18%	40%	30%	34%	19%	10%	181
6770.TW	PSMC	7,665	94%	9%	23%	na	11%	na	112%	52,205	52%	76%	71%	69%	59%	32%	71
Sub-Total		**550,777**	**53%**	**0%**	**17%**					**4,197,421**	**40%**	**52%**	**43%**	**64%**	**30%**	**17%**	
OSAT																	
3711.TW	ASEH	88,130	46%	7%	28%	22%	22%	105%	105%	608,887	30%	46%	43%	33%	29%	19%	762
2449.TW	KYEC	4,071	24%	0%	9%	9%	9%	100%	100%	33,476	34%	32%	37%	29%	37%	35%	300
6147.TWO	Chipbond	2,439	28%	-7%	14%	3%	6%	111%	108%	19,969	24%	11%	11%	11%	11%	11%	229
8150.TW	Chipmos	2,792	34%	0%	14%	na	9%	na	104%	22,719	30%	-9%	-9%	-9%	-9%	-9%	123
6239.TW	Powertech	8,507	27%	-1%	10%	8%	6%	101%	103%	69,764	30%	-5%	-5%	-5%	-5%	-5%	305
2441.TW	Greatek	1,957	40%	-4%	14%	na	na	na	na	16,004	28%	9%	9%	9%	9%	9%	130

3374.TWO	Xintec	932	21%	3%	24%	na	23%	na	101%	6,800	31%	1%	1%	1%	1%	1%	482
Sub-Total		**108,828**	**42%**	**5%**	**24%**					**777,620**	**30%**	**42%**	**41%**	**32%**	**28%**	**21%**	
PCB																		
3037.TW	Unimicron	18,651	65%	5%	23%	18%	16%	104%	106%	132,986	38%	56%	44%	36%	32%	28%	1,255
3044.TW	Tripod	10,648	54%	4%	23%	13%	15%	109%	108%	76,516	41%	65%	57%	42%	47%	29%	579
8046.TW	Nanya PCB	5,586	58%	4%	21%	19%	20%	102%	100%	41,131	42%	41%	50%	50%	36%	39%	1,440
3189.TW	Kinsus	5,181	44%	8%	16%	20%	15%	97%	101%	38,141	34%	41%	36%	32%	33%	27%	1,060
6213.TW	ITEQ	5,824	111%	15%	35%	na	28%	na	106%	36,290	50%	98%	97%	48%	29%	14%	812
6274.TWO	TUC	6,126	115%	4%	26%	17%	20%	108%	105%	42,304	99%	132%	122%	106%	124%	96%	1,735
2383.TW	Elite Materials	21,532	169%	7%	29%	17%	18%	110%	109%	141,227	104%	130%	129%	121%	115%	94%	6,150
6153.TW	Career	539	1%	-5%	8%	na	na	na	na	4,404	5%	16%	21%	25%	10%	-14%	18
8039.TW	Taiflex	840	-19%	4%	-8%	na	na	na	na	7,605	-3%	-14%	-6%	-7%	1%	-1%	296
2313.TW	Compeq	8,617	15%	23%	13%	na	21%	na	94%	62,125	13%	7%	16%	21%	13%	-3%	252
2367.TW	Unitech Print	1,480	4%	1%	11%	na	na	na	na	12,579	0%	26%	25%	-4%	16%	-22%	48
2368.TW	Gold Circuit	9,320	55%	-10%	22%	16%	26%	105%	97%	73,379	69%	76%	77%	78%	87%	58%	1,235
8358.TWO	Co-Tech	1,104	65%	5%	13%	na	12%	na	100%	8,506	49%	60%	55%	43%	41%	36%	522
8155.TWO	Allied Circuit	563	31%	0%	5%	na	6%	na	99%	4,811	35%	35%	52%	32%	30%	29%	320
6672.TW	Ventec	1,021	202%	27%	54%	na	25%	na	123%	5,352	69%	133%	91%	68%	35%	37%	598
Sub-Total		**97,032**	**70%**	**6%**	**23%**					**687,356**	**50%**	**67%**	**64%**	**55%**	**52%**	**36%**	
Distributor																		
3036.TW	WT Micro	215,933	60%	9%	1%	2%	1%	99%	100%	1,682,249	101%	98%	95%	188%	153%	79%	217
3702.TW	WPG Holdings	219,595	146%	54%	11%	na	-8%	na	120%	1,282,707	72%	80%	91%	132%	83%	39%	116
2347.TW	Synnex	90,309	159%	22%	18%	na	9%	na	108%	564,075	98%	122%	129%	136%	130%	84%	99
3010.TW	Wah Lee	9,012	41%	13%	7%	na	na	na	na	71,221	22%	24%	31%	14%	28%	13%	133
Sub-Total		**534,849**	**102%**	**26%**	**19%**					**3,600,253**	**87%**	**93%**	**97%**	**147%**	**117%**	**77%**	
Compound Semi																		
3105.TWO	Win Semi	2,030	30%	5%	11%	na	12%	na	99%	15,706	33%	31%	31%	30%	40%	49%	580
2455.TW	VPEC	449	29%	-5%	39%	na	24%	na	112%	3,267	35%	71%	45%	24%	47%	38%	564
8086.TW	AWSC	422	8%	0%	-3%	na	-3%	na	100%	3,783	32%	13%	18%	16%	37%	49%	149
3081.TWO	LandMark	540	190%	4%	28%	na	28%	na	100%	3,688	136%	181%	177%	128%	119%	115%	3,025
4968.TW	Richwave	330	0%	2%	-2%	na	10%	na	89%	2,880	3%	-2%	-2%	-1%	7%	0%	95

Sub-Total		**3,771**	**34%**	**3%**	**13%**					**29,324**	**36%**	**39%**	**37%**	**30%**	**42%**	**46%**	
iPhone																	
3008.TW	Largan	5,779	-7%	15%	15%	15%	33%	100%	86%	44,860	2%	-16%	-10%	-10%	42%	23%	7,325
3406.TW	Genius	2,672	1%	-2%	37%	9%	35%	127%	102%	19,636	10%	1%	2%	5%	30%	25%	1,070
6269.TW	Flexium	2,099	-3%	5%	14%	18%	29%	97%	89%	16,989	1%	1%	4%	-1%	-6%	0%	64
4958.TW	ZDT	26,358	36%	28%	33%	33%	31%	100%	102%	153,783	22%	41%	32%	33%	37%	12%	581
2317.TW	Hon Hai	1,158,633	38%	26%	20%	15%	13%	105%	107%	7,669,773	40%	52%	54%	52%	40%	30%	245
Sub-Total		**1,410,294**	**44%**	**36%**	**23%**					**8,812,666**	**36%**	**49%**	**48%**	**47%**	**36%**	**23%**	
Server																	
5274.TWO	Aspeed	1,636	101%	1%	24%	21%	19%	102%	104%	11,804	78%	118%	98%	67%	69%	82%	17,695
6669.TW	Wiwynn	172,620	100%	20%	56%	31%	30%	120%	121%	989,278	50%	50%	39%	30%	18%	30%	2,275
2059.TW	King Slide	4,464	196%	-31%	60%	na	62%	na	99%	33,618	168%	344%	356%	221%	172%	79%	10,580
2368.TW	Gold Circuit	9,320	55%	-10%	22%	16%	26%	105%	97%	73,379	69%	76%	77%	78%	87%	58%	1,235
8210.TW	Chenbro Micom	2,476	14%	1%	-5%	na	11%	na	85%	22,323	46%	41%	42%	98%	37%	5%	861
Sub-Total		**190,517**	**97%**	**15%**	**52%**					**1,130,402**	**54%**	**56%**	**47%**	**36%**	**26%**	**32%**	
ODM/OEM																	
2324.TW	Compal	94,434	35%	40%	2%	10%	6%	92%	97%	682,376	20%	15%	38%	58%	22%	16%	37
3231.TW	Wistron	380,806	87%	-17%	28%	10%	19%	117%	108%	2,890,836	97%	166%	61%	54%	39%	112%	190
6669.TW	Wiwynn	172,620	100%	20%	56%	31%	30%	120%	121%	989,278	50%	50%	39%	30%	18%	30%	2,275
2317.TW	Hon Hai	1,158,633	38%	26%	20%	15%	13%	105%	107%	7,669,773	40%	52%	54%	52%	40%	30%	245
2382.TW	Quanta	325,300	77%	-23%	8%	7%	7%	101%	101%	2,961,358	99%	177%	131%	103%	94%	121%	322
2356.TW	Inventec	98,636	63%	18%	1%	0%	-5%	101%	106%	742,261	43%	36%	66%	62%	35%	37%	58
Sub-Total		**2,445,182**	**57%**	**12%**	**20%**					**16,843,508**	**53%**	**80%**	**61%**	**57%**	**42%**	**46%**	
PC																	
2357.TW	Asus	97,243	18%	2%	7%	13%	10%	95%	98%	742,432	39%	51%	53%	56%	9%	46%	977
2353.TW	Acer	34,737	18%	15%	8%	na	0%	na	108%	249,548	24%	38%	22%	-3%	37%	68%	31
2379.TW	Realtek	12,378	23%	-1%	0%	-3%	3%	104%	98%	111,708	16%	28%	33%	21%	22%	11%	746
6104.TWO	Genesys	311	-1%	-3%	-11%	na	na	na	na	3,084	1%	3%	4%	6%	2%	-7%	87
2458.TW	Elan	602	-47%	-19%	-31%	na	-5%	na	73%	8,650	-9%	-37%	-13%	7%	4%	6%	133
5269.TW	Asmedia	995	-30%	13%	-2%	8%	0%	91%	98%	8,881	-10%	-33%	-38%	-22%	-19%	-18%	1,540
4966.TWO	Parade	1,503	-8%	13%	-2%	na	9%	na	90%	12,423	-2%	0%	-4%	12%	1%	-4%	553

2324.TW	Compal	94,434	35%	40%	2%	10%	6%	92%	97%	682,376	20%	15%	38%	58%	22%	16%	37
3533.TW	Lotes	3,357	16%	5%	4%	8%	7%	96%	97%	28,835	18%	20%	19%	17%	9%	15%	1,880
Sub-Total		**245,560**	**23%**	**16%**	**5%**					**1,847,937**	**27%**	**32%**	**39%**	**43%**	**18%**	**50%**	
Motherboards																	
2376.TW	Giga-Byte	63,099	125%	33%	7%	8%	-2%	99%	110%	403,982	63%	98%	60%	69%	5%	74%	356
2377.TW	Micro-Star	19,910	2%	13%	14%	15%	11%	99%	103%	163,116	-5%	5%	5%	5%	5%	5%	155
3515.TW	ASRock	6,518	27%	71%	68%	na	50%	na	112%	40,414	18%	3%	131%	48%	-16%	-47%	226
2425.TW	Chaintech	1,971	289%	-1%	0%	na	na	na	na	14,845	134%	118%	144%	376%	95%	88%	36
6150.TWO	Tul	648	-12%	13%	66%	na	na	na	na	4,294	-25%	-21%	17%	-30%	-54%	-69%	69
Sub-Total		**92,146**	**71%**	**29%**	**12%**					**626,651**	**34%**	**49%**	**42%**	**38%**	**-4%**	**27%**	
Networking																	
6285.TW	Wistron Neweb	16,079	64%	2%	23%	na	14%	na	108%	117,291	41%	81%	90%	84%	35%	16%	264
5388.TW	Sercomm	6,807	47%	-3%	6%	na	4%	na	102%	59,094	54%	45%	51%	71%	70%	66%	74
3596.TW	Acrcadyan Tech	5,064	9%	7%	5%	na	6%	na	100%	41,309	3%	2%	8%	4%	4%	1%	157
4906.TW	Gemtek Tech	1,643	24%	-1%	34%	na	na	na	na	11,560	-10%	56%	32%	-5%	-12%	-23%	42
2345.TW	Accton	40,399	66%	-1%	26%	14%	20%	111%	106%	286,402	62%	59%	72%	61%	57%	54%	2,130
Sub-Total		**69,993**	**56%**	**0%**	**22%**					**515,655**	**47%**	**56%**	**65%**	**59%**	**44%**	**36%**	
Panel																	
2409.TW	AUO	23,404	-4%	1%	-6%	13%	-2%	84%	96%	206,798	-2%	-6%	-3%	14%	-1%	-4%	36
3481.TW	Innolux	19,995	1%	7%	-9%	na	-3%	na	93%	187,926	11%	0%	-2%	17%	10%	12%	51
Sub-Total		**43,398**	**-2%**	**4%**	**10%**					**394,724**	**4%**	**-3%**	**-2%**	**15%**	**4%**	**-48%**	
Memory																	
2408.TW	Nanya Tech	45,091	577%	1%	62%	na	42%	na	114%	265,285	627%	561%	720%	621%	730%	717%	527
3006.TW	Elite Semi	7,461	470%	-6%	59%	na	na	na	na	43,363	344%	605%	491%	328%	297%	349%	261
2337.TW	Macronix	8,367	188%	3%	27%	na	30%	na	97%	53,831	155%	219%	180%	216%	176%	154%	116
8299.TWO	Phison	30,639	370%	8%	27%	na	21%	na	105%	194,932	291%	377%	378%	301%	301%	237%	2,055
6485.TWO	Asolid	127	8%	2%	24%	na	na	na	na	1,000	-5%	21%	11%	-7%	-6%	-19%	78
2451.TW	Transcend	6,035	290%	25%	-15%	na	14%	na	75%	48,511	370%	275%	308%	382%	470%	594%	287
3260.TWO	A-Data	15,730	200%	-17%	39%	na	23%	na	113%	117,306	216%	280%	331%	212%	210%	170%	368
8271.TW	Apacer	3,001	158%	8%	-11%	na	na	na	na	25,300	217%	177%	165%	201%	224%	361%	197
4967.TW	Team Group	4,081	110%	16%	34%	na	119%	na	61%	26,772	90%	275%	99%	35%	113%	79%	270

Sub-Total		**148,782**	**321%**	**2%**	**37%**					**956,727**	**282%**	**362%**	**375%**	**278%**	**293%**	**277%**	
MCU/PMIC/MOSFET																	
6415.TW	Silergy	2,511	51%	4%	21%	na	10%	na	110%	18,141	35%	53%	59%	38%	32%	28%	440
3707.TWO	Episil Tech	663	24%	1%	10%	na	na	na	na	5,300	26%	30%	31%	30%	28%	31%	82
2481.TW	Pan Jit	1,525	38%	11%	10%	na	6%	na	103%	11,726	20%	30%	31%	27%	14%	9%	171
3016.TW	Episil-Precision	513	53%	1%	15%	na	15%	na	100%	3,869	37%	58%	51%	41%	37%	33%	179
5425.TW	Taiwan Semi	1,800	15%	5%	3%	na	na	na	na	14,716	8%	22%	24%	9%	12%	3%	101
8261.TW	Advanced Power	266	-2%	-7%	-4%	na	4%	na	93%	2,502	6%	-3%	10%	16%	1%	5%	177
2342.TW	Mosel Vitelic	175	1%	-3%	3%	na	na	na	na	1,547	-3%	4%	11%	5%	-3%	-3%	47
5299.TWO	Excelliance Mos	108	-2%	-9%	-13%	na	na	na	na	1,139	-3%	-3%	-6%	-4%	2%	0%	97
6138.TWO	Anpec Elec	609	-9%	-8%	-4%	na	9%	na	87%	6,017	8%	2%	9%	11%	15%	14%	265
8081.TW	Global Mixed-Mod	828	14%	3%	10%	na	9%	na	101%	6,820	3%	11%	10%	7%	-3%	-4%	263
Sub-Total		**8,999**	**26%**	**3%**	**11%**					**71,778**	**18%**	**28%**	**30%**	**21%**	**16%**	**15%**	
Power Supply																	
2308.TW	Delta	65,400	15%	1%	8%	18%	14%	91%	95%	539,668	37%	35%	48%	55%	44%	44%	1,900
2301.TW	Lite-on	18,848	22%	-4%	9%	na	14%	na	96%	153,559	26%	25%	38%	37%	30%	25%	288
6412.TW	Chicony Power	2,413	-21%	-2%	-12%	na	6%	na	83%	23,641	-11%	-20%	-13%	-1%	-10%	-12%	73
Sub-Total		**86,661**	**15%**	**0%**	**7%**					**716,867**	**32%**	**30%**	**43%**	**48%**	**38%**	**36%**	
Heat Dissipation																	
3653.TW	Jentech	3,054	80%	-5%	-28%	na	34%	na	97%	22,073	47%	91%	91%	58%	38%	27%	7,160
3017.TW	AVC	21,018	45%	8%	-3%	18%	17%	102%	103%	157,248	71%	54%	57%	66%	61%	72%	3,385
3324.TWO	Auras	2,884	23%	-8%	-13%	17%	18%	96%	96%	27,040	72%	67%	117%	62%	94%	41%	1,540
Sub-Total		**26,956**	**45%**	**4%**	**20%**					**206,361**	**68%**	**60%**	**68%**	**65%**	**62%**	**61%**	
Passive Components																	
2327.TW	Yageo	17,068	46%	5%	11%	10%	9%	101%	102%	132,154	36%	52%	52%	39%	47%	22%	636
2492.TW	Walsin Tech	4,889	54%	7%	22%	na	na	na	na	34,830	26%	46%	42%	26%	16%	16%	418
Sub-Total		**21,957**	**48%**	**5%**	**14%**					**166,984**	**34%**	**50%**	**49%**	**36%**	**40%**	**21%**	
BBU																	
6781.TW	AES-KY	1,760	25%	1%	9%	0%	8%	109%	100%	14,572	26%	27%	28%	26%	21%	17%	1,015
3211.TWO	Dynapack	776	-34%	-54%	5%	33%	21%	79%	87%	10,517	14%	17%	21%	21%	-11%	19%	382.5
Sub-Total		**2,537**	**-2%**	**-26%**	**7%**					**25,090**	**21%**	**22%**	**25%**	**23%**	**5%**	**17%**	

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, TEJ, Bloomberg

Companies Mentioned:

Accton (2345.TW; NT$2130.0; 1; 12 Oct 26; 13:30) | Acer Incorporated (2353.TW; NT$31.35; Not Rated; 12 Oct 26; 13:30) | ADATA Technology Co Ltd (3260.TWO; NT$367.5; Not Rated; 12 Oct 26; 15:00) | Advanced Energy Solution (6781.TW; NT$1015.0; 1; 12 Oct 26; 13:30) | Advanced Power Electronics Corp (8261.TW; NT$176.5; Not Rated; 12 Oct 26; 13:30) | Advanced Wireless Semiconductor Co (8086.TWO; NT$148.5; Not Rated; 12 Oct 26; 15:00) | Alchip (3661.TW; NT$4395.0; 1; 12 Oct 26; 13:30) | All Ring Tech Co., Ltd. (6187.TWO; NT$1675.0; Not Rated; 12 Oct 26; 15:00) | Allied Circuit Co Ltd (8155.TWO; NT$319.5; Not Rated; 12 Oct 26; 15:00) | Amazon.com, Inc. (AMZN.O; US$262.43; 1; 09 Oct 26; 16:00) | Andes Technology Corporation (6533.TW; NT$345.0; Not Rated; 12 Oct 26; 13:30) | ANPEC ELECTRONICS CORPORATION (6138.TWO; NT$265.0; Not Rated; 12 Oct 26; 15:00) | Apacer Technology Inc (8271.TW; NT$197.0; Not Rated; 12 Oct 26; 13:30) | Arcadyan Technology Corp (3596.TW; NT$157.0; Not Rated; 12 Oct 26; 13:30) | ASE Technology Holding (3711.TW; NT$762.0; 1; 12 Oct 26; 13:30) | Asia Vital Components (3017.TW; NT$3385.0; 1; 12 Oct 26; 13:30) | ASMedia (5269.TW; NT$1540.0; 2; 12 Oct 26; 13:30) | ASolid Technology Co Ltd (6485.TWO; NT$78.4; Not Rated; 12 Oct 26; 15:00) | Aspeed Technology (5274.TWO; NT$17695.0; 1; 12 Oct 26; 15:00) | ASRock Incorporation (3515.TW; NT$225.5; Not Rated; 12 Oct 26; 13:30) | ASUSTeK Computer (2357.TW; NT$977.0; 1; 12 Oct 26; 13:30) | AUO (2409.TW; NT$35.55; 2; 12 Oct 26; 13:30) | Auras Technology (3324.TWO; NT$1540.0; 1; 12 Oct 26; 15:00) | Career Technology MFG Co Ltd (6153.TW; NT$17.6; Not Rated; 12 Oct 26; 13:30) | Chaintech Technology Corp (2425.TW; NT$35.7; Not Rated; 12 Oct 26; 13:30) | CHENBRO MICOM CO., LTD. (8210.TW; NT$861.0; Not Rated; 12 Oct 26; 13:30) | Chicony Power Technology (6412.TW; NT$73.3; Not Rated; 12 Oct 26; 13:30) | Chipbond Technology (6147.TWO; NT$229.0; 1; 12 Oct 26; 15:00) | ChipMOS Technologies Inc (8150.TW; NT$123.0; Not Rated; 12 Oct 26; 13:30) | CHPT (6510.TWO; NT$3420.0; 1H; 12 Oct 26; 15:00) | Chroma ATE (2360.TW; NT$2365.0; 1; 12 Oct 26; 13:30) | Co-Tech Development Corp (8358.TWO; NT$522.0; Not Rated; 12 Oct 26; 15:00) | Compal Electronics (2324.TW; NT$36.6; 3; 12 Oct 26; 13:30) | Compeq Manufacturing Co Ltd (2313.TW; NT$252.0; Not Rated; 12 Oct 26; 13:30) | Delta Electronics (2308.TW; NT$1900.0; 1; 12 Oct 26; 13:30) | Dynapack International (3211.TWO; NT$382.5; 1; 12 Oct 26; 15:00) | EGIS TECHNOLOGY INC. (6462.TWO; NT$100.0; Not Rated; 12 Oct 26; 15:00) | ELAN MICROELECTRONICS CORPORATION (2458.TW; NT$133.0; Not Rated; 12 Oct 26; 13:30) | Elite Material (2383.TW; NT$6150.0; 1; 12 Oct 26; 13:30) | ELITE SEMICONDUCTOR MICROELECTRONICS TECHNOLOGY INC. (3006.TW; NT$260.5; Not Rated; 12 Oct 26; 13:30) | eMemory Technology Inc (3529.TWO; NT$2850.0; Not Rated; 12 Oct 26; 15:00) | EPISIL TECHNOLOGIES INC. (3707.TWO; NT$82.2; Not Rated; 12 Oct 26; 15:00) | Episil-Precision Inc (3016.TW; NT$179.0; Not Rated; 12 Oct 26; 13:30) | Excelliance MOS Corp (5299.TWO; NT$96.6; Not Rated; 12 Oct 26; 15:00) | Faraday (3035.TW; NT$197.5; Not Rated; 12 Oct 26; 13:30) | Feedback Technology Corp (8091.TWO; NT$219.5; Not Rated; 12 Oct 26; 15:00) | Flexium Interconnect (6269.TW; NT$63.5; 2; 12 Oct 26; 13:30) | Formosa Sumco Technology Corp (3532.TW; NT$485.5; Not Rated; 12 Oct 26; 13:30) | Gemtek Technology (4906.TW; NT$42.2; Not Rated; 12 Oct 26; 13:30) | Genesys Logic Inc (6104.TWO; NT$87.2; Not Rated; 12 Oct 26; 15:00) | Genius Electronic Optical (3406.TW; NT$1070.0; 1; 12 Oct 26; 13:30) | Gigabyte Technology (2376.TW; NT$356.0; 1; 12 Oct 26; 13:30) | Global Mixed Mode Technology Inc (8081.TW; NT$262.5; Not Rated; 12 Oct 26; 13:30) | Globalwafers Co (6488.TWO; NT$1115.0; 1; 12 Oct 26; 15:00) | Gold Circuit Electronics (2368.TW; NT$1235.0; 1; 12 Oct 26; 13:30) | Grand Process Technology Corp (3131.TWO; NT$2525.0; Not Rated; 12 Oct 26; 15:00) | Greatek Electronics Inc (2441.TW; NT$130.0; Not Rated; 12 Oct 26; 13:30) | GUC (3443.TW; NT$8100.0; 1H; 12 Oct 26; 13:30) | GUDENG PRECISION INDUSTRIAL CO., LTD. (3680.TWO; NT$569.0; Not Rated; 12 Oct 26; 15:00) | Hon Hai Precision (2317.TW; NT$244.5; 1; 12 Oct 26; 13:30) | Innolux Corp (3481.TW; NT$51.3; Not Rated; 12 Oct 26; 13:30) | Inventec (2356.TW; NT$58.1; 1; 12 Oct 26; 13:30) | ITEQ Corp (6213.TW; NT$812.0; Not Rated; 12 Oct 26; 13:30) | Jentech Precision Industrial Co Ltd (3653.TW; NT$7160.0; Not Rated; 12 Oct 26; 13:30) | KING SLIDE WORKS CO., LTD. (2059.TW; NT$10580.0; Not Rated; 12 Oct 26; 13:30) | King Yuan Electronics Co (2449.TW; NT$300.0; 1; 12 Oct 26; 13:30) | KINIK COMPANY (1560.TW; NT$1000.0; Not Rated; 12 Oct 26; 13:30) | Kinsus Interconnect Technology (3189.TW; NT$1060.0; 1; 12 Oct 26; 13:30) | LandMark Optoelectronics Corp (3081.TWO; NT$3025.0; Not Rated; 12 Oct 26; 15:00) | Largan Precision (3008.TW; NT$7325.0;

1; 12 Oct 26; 13:30) | LITE-ON TECHNOLOGY CORPORATION (2301.TW; NT$288.0; Not Rated; 12 Oct 26; 13:30) | Lotes (3533.TW; NT$1880.0; 1; 12 Oct 26; 13:30) | M31 Technology Corporation (6643.TWO; NT$476.5; Not Rated; 12 Oct 26; 15:00) | Macronix International Co., Ltd. (2337.TW; NT$115.5; Not Rated; 12 Oct 26; 13:30) | Marketech International Corp. (6196.TW; NT$657.0; Not Rated; 12 Oct 26; 13:30) | Mediatek (2454.TW; NT$4625.0; 1; 12 Oct 26; 13:30) | Micro-Star (2377.TW; NT$155.0; 2; 12 Oct 26; 13:30) | Mosel Vitelic Inc (2342.TW; NT$46.95; Not Rated; 12 Oct 26; 13:30) | MPI (6223.TWO; NT$6130.0; 1H; 12 Oct 26; 15:00) | Nanya Technology (2408.TW; NT$527.0; 3; 12 Oct 26; 13:30) | Novatek (3034.TW; NT$529.0; 2; 12 Oct 26; 13:30) | NYPCB (8046.TW; NT$1440.0; 1; 12 Oct 26; 13:30) | Panjit International Inc (2481.TW; NT$171.0; Not Rated; 12 Oct 26; 13:30) | Parade (4966.TWO; NT$553.0; Not Rated; 12 Oct 26; 15:00) | Phison Electronics (8299.TWO; NT$2055.0; Not Rated; 12 Oct 26; 15:00) | Phoenix Silicon International Corporation (8028.TW; NT$281.5; Not Rated; 12 Oct 26; 13:30) | Powertech Technology (6239.TW; NT$304.5; 1; 12 Oct 26; 13:30) | PSMC (6770.TW; NT$71.4; Not Rated; 12 Oct 26; 13:30) | Quanta Computer (2382.TW; NT$322.0; 1; 12 Oct 26; 13:30) | Realtek (2379.TW; NT$746.0; 1; 12 Oct 26; 13:30) | RichWave Technology Corporation (4968.TW; NT$95.2; Not Rated; 12 Oct 26; 13:30) | Scientech Corp (3583.TW; NT$755.0; Not Rated; 12 Oct 26; 13:30) | Sercomm (5388.TW; NT$74.1; Not Rated; 12 Oct 26; 13:30) | Silergy Corp. (6415.TW; NT$440.0; Not Rated; 12 Oct 26; 13:30) | STL Technology Co Ltd (4931.TWO; NT$241.5; Not Rated; 12 Oct 26; 15:00) | Sunplus Technology Co Ltd (2401.TW; NT$25.6; Not Rated; 12 Oct 26; 13:30) | Symtek Automation Asia Co Ltd (6438.TW; NT$174.0; Not Rated; 12 Oct 26; 13:30) | Synnex Technology (2347.TW; NT$98.9; Not Rated; 12 Oct 26; 13:30) | Taiflex Scientific Co Ltd (8039.TW; NT$296.0; Not Rated; 12 Oct 26; 13:30) | Taiwan Semiconductor Co Ltd (5425.TWO; NT$100.5; Not Rated; 12 Oct 26; 15:00) | Team Group Inc (4967.TW; NT$269.5; Not Rated; 12 Oct 26; 13:30) | Topco Scientific Co Ltd (5434.TW; NT$593.0; Not Rated; 12 Oct 26; 13:30) | Transcend Information (2451.TW; NT$286.5; Not Rated; 12 Oct 26; 13:30) | Tripod Technology (3044.TW; NT$579.0; 1; 12 Oct 26; 13:30) | TSMC (2330.TW; NT$2540.0; 1; 12 Oct 26; 13:30) | TUC (6274.TWO; NT$1735.0; 1; 12 Oct 26; 15:00) | Tul Corp (6150.TWO; NT$69.0; Not Rated; 12 Oct 26; 15:00) | Unimicron (3037.TW; NT$1255.0; 1; 12 Oct 26; 13:30) | UNITECH PRINTED CIRCUIT BOARD CORP. (2367.TW; NT$47.5; Not Rated; 12 Oct 26; 13:30) | Vanguard International Semi. (5347.TWO; NT$181.0; 1; 12 Oct 26; 15:00) | Ventec International Group Co Ltd (6672.TW; NT$598.0; Not Rated; 12 Oct 26; 13:30) | VIA Technologies, Inc. (2388.TW; NT$69.1; Not Rated; 12 Oct 26; 13:30) | Visual Photonics Epitaxy Co., Ltd. (2455.TW; NT$564.0; Not Rated; 12 Oct 26; 13:30) | Wafer Works Corp (6182.TWO; NT$124.0; Not Rated; 12 Oct 26; 15:00) | WAH LEE INDUSTRIAL CORP. (3010.TW; NT$132.5; Not Rated; 12 Oct 26; 13:30) | Walsin Technology Corporation (2492.TW; NT$418.0; Not Rated; 12 Oct 26; 13:30) | WIN Semiconductors Corp (3105.TWO; NT$580.0; Not Rated; 12 Oct 26; 15:00) | WinWay Technology (6515.TW; NT$6025.0; 1H; 12 Oct 26; 13:30) | Wistron (3231.TW; NT$189.5; 1; 12 Oct 26; 13:30) | Wiwynn (6669.TW; NT$2275.0; 1; 12 Oct 26; 13:30) | WNC Corporation (6285.TW; NT$263.5; Not Rated; 12 Oct 26; 13:30) | WPG HOLDINGS LIMITED (3702.TW; NT$116.0; Not Rated; 12 Oct 26; 13:30) | WT Micro (3036.TW; NT$216.5; 1; 12 Oct 26; 13:30) | Xintec Inc (3374.TWO; NT$481.5; Not Rated; 12 Oct 26; 15:00) | Yageo (2327.TW; NT$636.0; 1; 12 Oct 26; 13:30) | Zhen Ding Technology (4958.TW; NT$581.0; 1; 12 Oct 26; 13:30)

If you are visually impaired and would like to speak to a Citi representative regarding the details of the graphics in this document, please call USA 1-888-500-5008 (TTY: 711), from outside the US +1-210-677-3788

Appendix A-1

ANALYST CERTIFICATION

The research analysts primarily responsible for the preparation and content of this research report are either (i) designated by "AC" in the author block or (ii) listed in bold alongside content which is attributable to that analyst. If multiple AC analysts are designated in the author block, each analyst is certifying with respect to the entire research report other than (a) content attributable to another AC certifying analyst listed in bold alongside the content and (b) views expressed solely with respect to a specific issuer which are attributable to another AC certifying analyst identified in the price charts or rating history tables for that issuer shown below. Each of these analysts certify, with respect to the sections of the report for which they are responsible: (1) that the views expressed therein accurately reflect their personal views about each issuer and security referenced and were prepared in an independent manner, including with respect to Citigroup Global Markets Inc. and its affiliates; and (2) no part of the research analyst's compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed by that research analyst in this report.

IMPORTANT DISCLOSURES

Within the past 12 months, Citigroup Global Markets Inc. or its affiliates has acted as manager or co-manager of an offering of securities of ASUSTeK Computer,Amazon.com, Inc.,Gigabyte Technology,PSMC,Quanta Computer,Unimicron,WT Micro,Wistron,Zhen Ding Technology.

Citigroup Global Markets Inc. or its affiliates has received compensation for investment banking services provided within the past 12 months from ASUSTeK Computer,AUO,Amazon.com, Inc.,Delta Electronics,Gigabyte Technology,Hon Hai Precision,PSMC,Quanta Computer,TSMC,Unimicron,WT Micro,Wistron,Zhen Ding Technology.

Citigroup Global Markets Inc. or its affiliates expects to receive or intends to seek, within the next three months, compensation for investment banking services from ASUSTeK Computer,Gigabyte Technology,Globalwafers Co,Hon Hai Precision,Quanta Computer,Wistron.

Citigroup Global Markets Inc. or its affiliates received compensation for products and services other than investment banking services from ASE Technology Holding,ASMedia,ASUSTeK Computer,AUO,Accton,Advanced Energy Solution,Alchip,Amazon.com, Inc.,Asia Vital Components,Aspeed Technology,Auras Technology,CHPT,Chipbond Technology,Compal Electronics,Delta Electronics,GUC,GUDENG PRECISION INDUSTRIAL CO., LTD.,Genius Electronic Optical,Gigabyte Technology,Globalwafers Co,Gold Circuit Electronics,Hon Hai Precision,Inventec,KINIK COMPANY,King Yuan Electronics Co,Largan Precision,Lotes ,Mediatek,Micro-Star,NYPCB,Nanya Technology,PSMC,Quanta Computer,Realtek,TSMC,Tripod Technology,Vanguard International Semi.,WT Micro,WinWay Technology,Wistron,Wiwynn,Yageo,Zhen Ding Technology in the past 12 months.

Citigroup Global Markets Inc. or its affiliates currently has, or had within the past 12 months, the following as investment banking client(s): ASUSTeK Computer,AUO,Amazon.com, Inc.,Delta Electronics,Gigabyte Technology,Globalwafers Co,Hon Hai Precision,PSMC,Quanta Computer,TSMC,Unimicron,WT Micro,Wistron,Zhen Ding Technology.

Citigroup Global Markets Inc. or its affiliates currently has, or had within the past 12 months, the following as clients, and the services provided were non-investment-banking, securities-related: ASE Technology Holding,ASUSTeK Computer,AUO,Accton,Advanced Energy Solution,Amazon.com, Inc.,Asia Vital Components,Aspeed Technology,Auras Technology,CHPT,Chipbond Technology,Compal Electronics,Delta Electronics,GUC,Gigabyte Technology,Globalwafers Co,Gold Circuit Electronics,Hon Hai Precision,Inventec,Lotes ,Mediatek,Micro-Star,NYPCB,Nanya Technology,PSMC,Quanta Computer,Realtek,TSMC,Tripod Technology,WT Micro,WinWay Technology,Wistron,Wiwynn,Yageo,Zhen Ding Technology.

Citigroup Global Markets Inc. or its affiliates currently has, or had within the past 12 months, the following as clients, and the services provided were non-investment-banking, non-securities-related: ASE Technology Holding,ASMedia,ASUSTeK Computer,AUO,Accton,Advanced Energy Solution,Alchip,Amazon.com, Inc.,Asia Vital Components,Aspeed Technology,Auras Technology,CHPT,Chipbond Technology,Compal Electronics,Delta Electronics,GUC,GUDENG PRECISION INDUSTRIAL CO., LTD.,Genius Electronic Optical,Gigabyte Technology,Globalwafers Co,Gold Circuit Electronics,Hon Hai Precision,Inventec,KINIK COMPANY,King Yuan Electronics Co,Largan Precision,Lotes ,Mediatek,Micro-Star,NYPCB,Nanya Technology,PSMC,Quanta Computer,Realtek,TSMC,Tripod Technology,Vanguard International Semi.,WT Micro,WinWay Technology,Wistron,Wiwynn,Yageo,Zhen Ding Technology.

Citigroup Global Markets Inc. and/or its affiliates has a significant financial interest in relation to ASE Technology Holding,ASUSTeK Computer,AUO,Accton,Amazon.com, Inc.,Compal Electronics,Delta Electronics,Globalwafers Co,Hon Hai Precision,Inventec,Mediatek,NYPCB,Quanta Computer,TSMC,Tripod Technology,WT Micro,Wistron,Wiwynn,Yageo. (For an explanation of the determination of significant financial interest, please refer to the policy for managing conflicts of interest which can be found at www.citiVelocity.com.)

Analysts' compensation is determined by Citi Research management and Citigroup's senior management and is based upon activities and services intended to benefit the investor clients of Citigroup Global Markets Inc. and its affiliates (the "Firm"). Compensation is not linked to specific transactions or recommendations. Like all Firm employees, analysts receive compensation that is impacted by overall Firm profitability which includes investment banking, sales and trading, and principal trading revenues. One factor in equity research analyst compensation is arranging corporate access events between institutional clients and the management teams of covered companies. Typically, company management is more likely to participate when the analyst has a positive view of the company.

For financial instruments recommended in the Product in which the Firm is not a market maker, the Firm is a liquidity provider in such financial instruments (and any underlying instruments) and may act as principal in connection with transactions in such instruments. The Firm is a regular issuer of traded financial instruments linked to securities that may have been recommended in the Product. The Firm regularly trades in the securities of the issuer(s) discussed in the Product. The Firm may engage in securities transactions in a manner inconsistent with the Product and, with respect to securities covered by the Product, will buy or sell from customers on a principal basis.

Unless stated otherwise neither the Research Analyst nor any member of their team has viewed the material operations of the Companies for which an investment view has been provided within the past 12 months.

For important disclosures (including copies of historical disclosures) regarding the companies that are the subject of this Citi Research product ("the Product"), please contact Citi Research, 388 Greenwich Street, 6th Floor, New York, NY, 10013, Attention: Legal/Compliance [E6WYB6412478]. In addition, the same important disclosures, with the exception of the Valuation and Risk assessments and historical disclosures, are contained on the Firm's disclosure website at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures. Valuation and Risk assessments can be found in the text of the most recent research note/report regarding the subject company. Pursuant to the Market Abuse Regulation a history of all Citi Research recommendations published during the preceding 12-month period can be accessed via Citi Velocity (https://www.citivelocity.com/cv2) or your standard distribution portal. Historical disclosures (for up to the past three years) will be provided upon request.

Citi Research Equity Ratings Distribution

	12 Month Rating			Catalyst Watch		
Data current as of 01 Oct 2026	**Buy**	**Hold**	**Sell**	**Buy**	**Hold**	**Sell**
Citi Research Global Fundamental Coverage (Neutral=Hold)	61%	32%	7%	36%	48%	16%
% of companies in each rating category that are investment banking clients	39%	43%	30%	42%	38%	36%

Guide to Citi Research Fundamental Research Investment Ratings:
Citi Research stock recommendations include an investment rating and an optional risk rating to highlight high risk stocks.
Risk rating takes into account both price volatility and fundamental criteria. Stocks will either have no risk rating or a High risk rating assigned.
Investment Ratings: Citi Research investment ratings are Buy, Neutral and Sell. Our ratings are a function of analyst expectations of expected total return ("ETR") and risk. ETR is the sum of the forecast price appreciation (or depreciation) plus the dividend yield for a stock within the next 12 months. The target price is based on a 12 month time horizon. The Investment rating definitions are: Buy (1) ETR of 15% or more or 25% or more for High risk stocks; and Sell (3) for negative ETR. Any covered stock not assigned a Buy or a Sell is a Neutral (2). For stocks rated Neutral (2), if an analyst believes that there are insufficient valuation drivers and/or investment catalysts to derive a positive or negative investment view, they may elect with the approval of Citi Research management not to assign a target price and, thus, not derive an ETR. Citi Research may suspend its rating and target price and assign "Rating Suspended" status for regulatory and/or internal policy reasons. Citi Research may also suspend its rating and target price and assign "Under Review" status for other exceptional circumstances (e.g. lack of information critical to the analyst's thesis, trading suspension) affecting the company and/or trading in the company's securities. In both such situations, the rating and target price will show as "—" and "-" respectively in the rating history price chart. Prior to 11 April 2022 Citi Research assigned "Under Review" status to both situations and prior to 11 Nov 2020 only in exceptional circumstances. As soon as practically possible, the analyst will publish a note re-establishing a rating and investment thesis. Investment ratings are determined by the ranges described above at the time of initiation of coverage, a change in investment and/or risk rating, or a change in target price (subject to limited management discretion). At times, the expected total returns may fall outside of these ranges because of market price movements and/or other short-term volatility or trading patterns. Such interim deviations will be permitted but will become subject to review by Research Management. Your decision to buy or sell a security should be based upon your personal investment objectives and should be made only after evaluating the stock's expected performance and risk.

Catalyst Watch/Short Term Views ("STV") Ratings Disclosure:
Catalyst Watch and STV Upside/Downside calls: Citi Research may also include a Catalyst Watch or STV Upside or Downside call to indicate the analyst expects the share price to rise (fall) in absolute terms over a specified period of 30 or 90 days in reaction to one or more specific near-term catalysts or events impacting the company or the market. A Catalyst Watch will be published when Analyst confidence is high that an impact to share price will occur; it will be a STV when confidence level is moderate. A Catalyst Watch or STV Upside/Downside call will automatically expire at the end of the specified 30/90 day period. The Catalyst Watch will also be automatically removed if share price performance (calculated at market close) exceeds 15% against the direction of the call (unless over-ridden by the analyst). The analyst may also remove a Catalyst Watch or STV call prior to the end of the specified period in a published research note. A Catalyst Watch/STV Upside or Downside call may be

different from and does not affect a stock's fundamental equity rating, which reflects a longer-term total absolute return expectation. For purposes of FINRA ratings-distribution-disclosure rules, a Catalyst Watch/STV Upside call corresponds to a buy recommendation and a Catalyst Watch/STV Downside call corresponds to a sell recommendation. Any stock not assigned to a Catalyst Watch Upside, Catalyst Watch Downside, STV Upside, or STV Downside call is considered Catalyst Watch/STV No View. For purposes of FINRA ratings distribution-disclosure rules, we correspond Catalyst Watch/STV No View to Hold in our ratings distribution table for our Catalyst Watch/STV Upside/Downside rating system. However, we reiterate that we do not consider No View to be a recommendation. For all Catalyst Watch/STV Upside/Downside calls, risk exists that the catalyst(s) and associated share-price movement will not materialize as expected.

Privately held companies:

The legal entities through which Citi conducts its business do not permit investment research recommendations, ratings, target prices, or estimates on any privately held companies ("private companies") referenced herein. Citi Research does not utilize a rating system for private companies. Any information on private companies in this report is based on generally available information and, although obtained from sources believed by Citi to be reliable, its accuracy and completeness are not guaranteed. Private companies are generally not subject to reporting obligations, which may result in less publicly available information. The information contained in this report may differ from information a private company may subsequently make public or otherwise share with investors. Continuation of publishing private company reports or private company sections within reports is at Citi's discretion, in accordance with internal policies.

RESEARCH ANALYST AFFILIATIONS / NON-US RESEARCH ANALYST DISCLOSURES

The legal entities employing the authors of this report are listed below (and their regulators are listed further herein). Non-US research analysts who have prepared this report (i.e., all research analysts listed below other than those identified as employed by Citigroup Global Markets Inc.) are not registered/qualified as research analysts with FINRA. Such research analysts may not be associated persons of the member organization (but are employed by an affiliate of the member organization) and therefore may not be subject to the FINRA Rule 2241 restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

Citigroup Global Markets Taiwan Securities Company Limited	Angela Hsu; Jack Chen; Michael Hung; Laura (Chia Yi) Chen; Nicholas Lai

OTHER DISCLOSURES

Any price(s) of instruments mentioned in recommendations are as of the prior day's market close on the primary market for the instrument, unless otherwise stated.

The completion and first dissemination of any recommendations made within this research report are as of the Eastern date-time displayed at the top of the Product. If the Product references views of other analysts then please refer to the price chart or rating history table for the date/time of completion and first dissemination with respect to that view.

Regulations in various jurisdictions require that where a recommendation differs from any of the author's previous recommendations concerning the same financial instrument or issuer that has been published during the preceding 12-month period that the change(s) and the date of that previous recommendation are indicated. For fundamental coverage please refer to the price chart or rating change history within this disclosure appendix or the issuer disclosure summary at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures.

Citi Research has implemented policies for identifying, considering and managing potential conflicts of interest arising as a result of publication or distribution of investment research. A description of these policies can be found at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures.

The proportion of all Citi Research research recommendations that were the equivalent to "Buy","Hold","Sell" at the end of each quarter over the prior 12 months (with the % of these that had received investment firm services from Citi in the prior 12 months shown in brackets) is as follows; Q2 2026 Buy 33%(63%), Hold 44% (51%), Sell 23% (49%), RV 0.5% (87%): Q1 2026 Buy 33% (63%), Hold 44% (52%), Sell 23% (46%), RV 0.5% (89%); Q4 2025 Buy 33% (63%), Hold 44% (50%), Sell 23% (46%), RV 0.4% (91%); Q3 2025 Buy 33%(61%), Hold 44% (52%), Sell 23% (50%), RV 0.4% (80%). For the purposes of disclosing recommendations other than for equity (whose definitions can be found in the corresponding disclosure sections), "Buy" means a positive directional trade idea; "Sell" means a negative directional trade idea; and "Relative Value" means any trade idea which does not have a clear direction to the investment strategy.

European regulations require a 5 year price history when past performance of a security is referenced. CitiVelocity's Charting Tool (https://www.citivelocity.com/cv2/#go/CHARTING_3_Equities) provides the facility to create customisable price charts including a five year option. This tool can be found in the Data & Analytics section under any of the asset class menus in CitiVelocity (https://www.citivelocity.com/). For further information contact CitiVelocity support (https://www.citivelocity.com/cv2/go/CLIENT_SUPPORT). The source for all referenced prices, unless otherwise stated, is DataCentral, which sources price information from LSEG Data & Analytics. Past performance is not a guarantee or reliable indicator of future results. Forecasts are not a guarantee or reliable indicator of future performance.

Investors should always consider the investment objectives, risks, and charges and expenses of an ETF carefully before investing. The applicable prospectus and key investor information document (as applicable) for an ETF should contain this and other information about such ETF. It is important to read carefully any such prospectus before investing. Clients may obtain prospectuses and key investor information documents for ETFs from the applicable distributor or authorized participant, the exchange upon which an ETF is listed and/or from the applicable website of the applicable ETF issuer. The value of the investments and any accruing income may fall or rise. Any past performance, prediction or forecast is not indicative of future or

likely performance. Any information on ETFs contained herein is provided strictly for illustrative purposes and should not be deemed an offer to sell or a solicitation of an offer to purchase units of any ETF either explicitly or implicitly. The opinions expressed are those of the authors and do not necessarily reflect the views of ETF issuers, any of their agents or their affiliates.

Citigroup Global Markets India Private Limited and/or its affiliates may have, from time to time, actual or beneficial ownership of 1% or more in the debt securities of the subject issuer.

Please be advised that pursuant to Executive Order 13959 as amended (the “Order”), U.S. persons are prohibited from investing in securities of any company determined by the United States Government to be the subject of the Order. This research is not intended to be used or relied upon in any way that could result in a violation of the Order. Investors are encouraged to rely upon their own legal counsel for advice on compliance with the Order and other economic sanctions programs administered and enforced by the Office of Foreign Assets Control of the U.S. Treasury Department.

This communication is directed at persons who are "Eligible Clients" as such term is defined in the Israeli Regulation of Investment Advice, Investment Marketing and Investment Portfolio Management law, 1995 (the "Advisory Law"). Within Israel, this communication is not intended for retail clients and Citi will not make such products or transactions available to retail clients or to non-Eligible Clients. The presenter is not licensed as investment advisor or investment marketer by the Israeli Securities Authority (“ISA”) and this communication does not constitute investment or marketing advice. The information contained herein may relate to matters that are not regulated by the ISA. Any securities which are the subject of this communication may not be offered or sold to any Israeli person except pursuant to a security offering exemption according to the Israeli Securities Law, 1968and the public offering rules provided thereunder.

Citi Research broadly and simultaneously disseminates its research content to the Firm’s institutional and retail clients via the Firm’s proprietary electronic distribution platforms (e.g., Citi Velocity and various Global Wealth platforms). As a convenience, certain, but not all, research content may be distributed through third party aggregators. Clients may receive published research reports by email, on a discretionary basis, and only after such research content has been broadly disseminated. Certain research is made available only to institutional investors to satisfy regulatory requirements. The level and types of services provided by Citi Research analysts to clients may vary depending on various factors such as the client’s individual preferences as to the frequency and manner of receiving communications from analysts, the client’s risk profile and investment focus and perspective (e.g. market-wide, sector specific, long term, short-term etc.), the size and scope of the overall client relationship with the Firm and legal and regulatory constraints.

Pursuant to Comissão de Valores Mobiliários Resolução 20 and ASIC Regulatory Guide 264, Citi is required to disclose whether a Citi related company or business has a commercial relationship with the subject company. Considering that Citi operates multiple businesses in more than 100 countries around the world, it is likely that Citi has a commercial relationship with the subject company.

Securities recommended, offered, or sold by the Firm: (i) are not insured by the Federal Deposit Insurance Corporation; (ii) are not deposits or other obligations of any insured depository institution (including Citibank); and (iii) are subject to investment risks, including the possible loss of the principal amount invested. The Product is for informational purposes only and is not intended as an offer or solicitation for the purchase or sale of a security. Any decision to purchase securities mentioned in the Product must take into account existing public information on such security or any registered prospectus. Although information has been obtained from and is based upon sources that the Firm believes to be reliable, we do not guarantee its accuracy and it may be incomplete and condensed. Note, however, that the Firm has taken all reasonable steps to determine the accuracy and completeness of the disclosures made in the Important Disclosures section of the Product. The Firm's research department has received assistance from the subject company(ies) referred to in this Product including, but not limited to, discussions with management of the subject company(ies). Firm policy prohibits research analysts from sending draft research to subject companies. However, it should be presumed that the author of the Product has had discussions with the subject company to ensure factual accuracy prior to publication. Statements and views concerning ESG (environmental, social, governance) factors are typically based upon public statements made by the affected company or other public news, which the author may not have independently verified. ESG factors are one consideration that investors may choose to examine when making investment decisions. All opinions, projections and estimates constitute the judgment of the author as of the date of the Product and these, plus any other information contained in the Product, are subject to change without notice. Prices and availability of financial instruments also are subject to change without notice. Notwithstanding other departments within the Firm advising the companies discussed in this Product, information obtained in such role is not used in the preparation of the Product. Although Citi Research does not set a predetermined frequency for publication, if the Product is a fundamental equity or credit research report, it is the intention of Citi Research to provide research coverage of the covered issuers, including in response to news affecting the issuer. For non-fundamental research reports, Citi Research may not provide regular updates to the views, recommendations and facts included in the reports. Notwithstanding that Citi Research maintains coverage on, makes recommendations concerning or discusses issuers, Citi Research may be periodically restricted from referencing certain issuers due to legal or policy reasons. Where a component of a published trade idea is subject to a restriction, the trade idea will be removed from any list of open trade ideas included in the Product. Upon the lifting of the restriction, the trade idea will either be re-instated in the open trade ideas list if the analyst continues to support it or it will be officially closed. Citi Research may provide different research products and services to different classes of customers (for example, based upon long-term or short-term investment horizons) that may lead to differing conclusions or recommendations that could impact the price of a security contrary to the recommendations in the alternative research product, provided that each is consistent with the rating system for each respective product.

Investing in non-U.S. securities, including ADRs, may entail certain risks. The securities of non-U.S. issuers may not be registered with, nor be subject to the reporting requirements of the U.S. Securities and Exchange Commission. There may be limited information available on foreign securities. Foreign companies are generally not subject to uniform audit and reporting standards, practices and requirements comparable to those in the U.S. Securities of some foreign companies may be less liquid and their prices more volatile than securities of comparable U.S. companies. In addition, exchange rate movements may have an adverse effect on the value of an investment in a foreign stock and its corresponding dividend payment for U.S. investors. Net dividends to ADR investors are estimated, using withholding tax rates conventions, deemed accurate, but investors are urged to consult their tax advisor for exact dividend computations. Investors who have received the Product from the Firm may be prohibited in certain states or other jurisdictions from purchasing securities mentioned in the Product from the Firm. Please ask your Financial Consultant for additional details. Citigroup Global Markets Inc. takes responsibility for the Product in the United States. Any orders by US investors resulting from the information contained in the Product may be placed only through Citigroup Global Markets Inc.

The Citigroup legal entity that takes responsibility for the production of the Product is the legal entity which the first named author is employed by.

The Product is made available in **Australia** through Citigroup Global Markets Australia Pty Limited. (ABN 64 003 114 832 and AFSL No. 240992), participant of the ASX Group and regulated by the Australian Securities & Investments Commission. Citigroup Centre, 2 Park Street, Sydney, NSW 2000. Citigroup Global Markets Australia Pty Limited is not an Authorised Deposit-Taking Institution under the Banking Act 1959, nor is it regulated by the Australian Prudential Regulation Authority.

The Product is made available in **Brazil** by Citigroup Global Markets Brasil - CCTVM SA, which is regulated by CVM - Comissão de Valores Mobiliários ("CVM"), BACEN - Brazilian Central Bank, APIMEC - Associação dos Analistas e Profissionais de Investimento do Mercado de Capitais and ANBIMA – Associação Brasileira das Entidades dos Mercados Financeiro e de Capitais. Av. Paulista, 1111 - 14º andar(parte) - CEP: 01311920 - São Paulo - SP.

This Product is available in **Chile** through Banchile Corredores de Bolsa S.A., an indirect subsidiary of Citigroup Inc., which is regulated by the Comisión Para El Mercado Financiero. Enrique Foster Sur, 20, piso 6, Las Condes, Santiago, Chile.

Disclosure for investors in the Republic of Colombia :This communication or message does not constitute a professional recommendation to make investment in the terms of article 2.40.1.1.2 of Decree 2555 de 2010 or the regulations that modify, substitute or complement it. Para la elaboración y distribución de informes de investigación y de comunicaciones generales de que trata este artículo no se requiere ser una entidad vigilada por la Superintendencia Financiera de Colombia.

The Product is made available in **Germany** by Citigroup Global Markets Europe AG ("CGME"), which is regulated by the European Central Bank and the German Federal Financial Supervisory Authority (Bundesanstalt fur Finanzdienstleistungsaufsicht BaFin). Börsenplatz 9, 60313 Frankfurt am Main, Germany.

Unless otherwise specified, if the analyst who prepared this report is based in Hong Kong and it relates to “securities” (as defined in the Securities and Futures Ordinance (Cap.571 of the Laws of Hong Kong)), the report is issued in Hong Kong by Citigroup Global Markets Asia Limited. Citigroup Global Markets Asia Limited is regulated by Hong Kong Securities and Futures Commission. If the report is prepared by a non-Hong Kong based analyst, please note that such analyst (and the legal entity that the analyst is employed by or accredited to) is not licensed/registered in Hong Kong and they do not hold themselves out as such. Please refer to the section “Research Analyst Affiliations / Non-US Research Analyst Disclosures” for the details of the employment entity of the analysts.

The Product is made available in **India** by Citigroup Global Markets India Private Limited (CGM), which is regulated by the Securities and Exchange Board of India (SEBI), as a Research Analyst (SEBI Registration No. INH000000438). CGM is also actively involved in the business of merchant banking (SEBI Registration No. INM000010718) and stock brokerage ((SEBI Registration No. INZ000263033) in India, and is registered with SEBI in this regard. Registration granted by SEBI, enlistment with Bombay Stock Exchange (BSE) and certification from National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. CGM’s registered office is at 1202, 12th Floor, First International Financial Centre (FIFC), G Block, Bandra Kurla Complex, Bandra East, Mumbai – 400098 & registered Tel: +91 22 61759999. Citi maintains robust policies, procedures,controls, and training to ensure continued compliance with all applicable rulesand regulations. All recommendations contained herein are made by dulyqualified research analysts. CGM’s Corporate Identity Number is U99999MH2000PTC126657, and its Compliance Officer [Vishal Bohra] contact details are: Tel:+91-022-61759994, Fax:+91-022-61759851, Email: cgmcompliance@citi.com. The Investor Charter in respect of Research Analysts, the Compliance Audit Report and Complaints information can be found at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures. The grievance officer [Niraj Mody] contact details are Tel: +91-22-42775002, Email: EMEA.CR.Complaints@citi.com. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. SEBI prescribed Client Terms & Conditions can be found at https://www.citivelocity.com/rendition/authfilelinksvcs/eppublic/V1/file?paramData=ZmlsZU5hbWU9L1MzL0dETVMvcHViRGlzY2xvc3VyZUh0bWxzL2dkbV9kaXNfc2ViaV9UZXJtc29mVXNlLmh0bWw

The Product is made available in **Indonesia** through PT Citigroup Sekuritas Indonesia. Citibank Tower 10/F, Pacific Century Place, SCBD lot 10, Jl. Jend Sudirman Kav 52-53, Jakarta 12190, Indonesia. Neither this Product nor any copy hereof may be distributed in Indonesia or to any Indonesian citizens wherever they are domiciled or to Indonesian residents except in compliance with applicable capital market laws and regulations. This Product is not an offer of securities in Indonesia. The securities referred to in this Product have not been registered with the Capital Market and Financial Services Authority (OJK) pursuant to relevant capital market laws and regulations, and may not be offered or sold within the territory of the Republic of

Indonesia or to Indonesian citizens through a public offering or in circumstances which constitute an offer within the meaning of the Indonesian capital market laws and regulations.

The Product is made available in **Japan** by Citigroup Global Markets Japan Inc. ("CGMJ"), which is regulated by Financial Services Agency, Securities and Exchange Surveillance Commission, Japan Securities Dealers Association, Tokyo Stock Exchange and Osaka Securities Exchange. Otemachi Park Building, 1-1-1 Otemachi, Chiyoda-ku, Tokyo 100-8132 Japan. In the event that an error is found in an CGMJ research report, a revised version will be posted on the Firm's Citi Velocity website. If you have questions regarding Citi Velocity, please call (81 3) 6270-3019 for help.

The product is made available in the **Kingdom of Saudi Arabia** in accordance with Saudi laws through Citigroup Saudi Arabia, which is regulated by the Capital Market Authority (CMA) under CMA license (17184-31). 2239 Al Urubah Rd – Al Olaya Dist. Unit No. 18, Riyadh 12214 – 9597, Kingdom Of Saudi Arabia.

The Product is made available in **Korea** by Citigroup Global Markets Korea Securities Ltd. (CGMK), which is regulated by the Financial Services Commission, the Financial Supervisory Service and the Korea Financial Investment Association (KOFIA). The address of CGMK is Citibank Center, 50 Saemunan-ro, Jongno-gu, Seoul 03184, Korea. KOFIA makes available registration information of research analysts on its website. Please visit the following website if you wish to find KOFIA registration information on research analysts of CGMK. http://dis.kofia.or.kr/websquare/index.jsp?w2xPath=/wq/fundMgr/DISFundMgrAnalystList.xml&divisionId=MDIS03002002000000&serviceId=SDIS03002002000. The Product is made available in Korea by Citibank Korea Inc., which is regulated by the Financial Services Commission and the Financial Supervisory Service. Address is Citibank Center, 50 Saemunan-ro, Jongno-gu, Seoul 03184, Korea. This research report is intended to be provided only to Professional Investors as defined in the Financial Investment Services and Capital Market Act and its Enforcement Decree in Korea.

The Product is made available in **Malaysia** by Citigroup Global Markets Malaysia Sdn Bhd (Registration No. 199801004692 (460819-D)) ("CGMM") to its clients and CGMM takes responsibility for its contents as regards CGMM's clients. CGMM is regulated by the Securities Commission Malaysia. Please contact CGMM at Level 43 Menara Citibank, 165 Jalan Ampang, 50450 Kuala Lumpur, Malaysia in respect of any matters arising from, or in connection with, the Product.

The Product is made available in **Mexico** by Citi México Casa de Bolsa, S.A. de C.V., Grupo Financiero Citi México which is a wholly owned subsidiary of Citigroup Inc. and is regulated by Comision Nacional Bancaria y de Valores. Prolongación Reforma 1196, 24 floor, Colonia Santa Fe, Alcaldía Cuajimalpa de Morelos, C.P. 05348, Ciudad de México.

The Product is made available in **Poland** by Biuro Maklerskie Banku Handlowego (DMBH), separate department of Bank Handlowy w Warszawie S.A. a subsidiary of Citigroup Inc., which is regulated by Komisja Nadzoru Finansowego. Biuro Maklerskie Banku Handlowego (DMBH), ul.Senatorska 16, 00-923 Warszawa.

The Product is made available in **Singapore** through Citigroup Global Markets Singapore Pte. Ltd. ("CGMSPL"), a capital markets services and Exempt Financial Advisor license holder, and regulated by Monetary Authority of Singapore. Please contact CGMSPL at 8 Marina View, 21st Floor Asia Square Tower 1, Singapore 018960, in respect of any matters arising from, or in connection with, the analysis of this document. This Product is intended for recipients who are accredited, expert and institutional investors as defined under the Securities and Futures Act 2001. For Citi Private Bank, the Product is made available in Singapore by Citi Private Bank through Citibank, N.A., Singapore Branch. Citibank N.A., Singapore Branch is a licensed bank in Singapore that is regulated by the Monetary Authority of Singapore. Please contact your Private Banker in Citibank N.A., Singapore Branch if you have any queries on or any matters arising from or in connection with this document. The Product is intended for recipients who are accredited, expert and institutional investors as defined under the Securities and Futures Act 2001. For Citibank Singapore Limited ("CSL"), the Product is distributed in Singapore by CSL to selected Citigold/Citigold Private Clients. CSL provides no independent research or analysis of the substance or in preparation of the Product. Please contact your Citigold//Citigold Private Client Relationship Manager in CSL if you have any queries on or any matters arising from or in connection with this document. The Product is intended for recipients who are accredited investors as defined under the Securities and Futures Act (Cap. 289).

Citigroup Global Markets (Pty) Ltd. is incorporated in the **Republic of South Africa** (company registration number 2000/025866/07) and its registered office is at 145 West Street, Sandton, 2196, Saxonwold. Citigroup Global Markets (Pty) Ltd. is regulated by JSE Securities Exchange South Africa, South African Reserve Bank and the Financial Services Board. The investments and services contained herein are not available to private customers in South Africa.

The Product is made available in the **Republic of China (Taiwan)** through Citigroup Global Markets Taiwan Securities Company Ltd. ("CGMTS"), 14F, 15F and 16F, No. 1, Songzhi Road, Taipei 110, Taiwan, subject to the license scope and the applicable laws and regulations in the Republic of China (Taiwan). CGMTS is regulated by the Securities and Futures Bureau of the Financial Supervisory Commission of Taiwan, the Republic of China (Taiwan). No portion of the Product may be reproduced or quoted in the Republic of China (Taiwan) by the press or any third parties [without the written authorization of CGMTS]. Pursuant to the applicable laws and regulations in the Republic of China (Taiwan), the recipient of the Product shall not take advantage of such Product to involve in any matters in which the recipient may have conflicts of interest. If the Product covers securities which are not allowed to be offered or traded in the Republic of China (Taiwan), neither the Product nor any information contained in the Product shall be considered as advertising the securities or making recommendation of the securities in the Republic of China (Taiwan). The Product is for informational purposes only and is not intended as an offer or solicitation for the purchase or sale of a security or financial products. Any decision to purchase securities or financial products mentioned in the Product must take into account existing public information on such security or the financial products or any registered prospectus.

The Product is made available in **Thailand** through Citicorp Securities (Thailand) Ltd., which is regulated by the Securities and Exchange Commission of Thailand. 399 Interchange 21 Building, 18th Floor, Sukhumvit Road, Klongtoey Nua, Wattana ,Bangkok 10110, Thailand.

Disclosure for investors in the Republic of Turkey: The Product is made available in **Turkey** through Citibank AS which is regulated by Capital Markets Board. Tekfen Tower, Eski Buyukdere Caddesi # 209 Kat 2B, 23294 Levent, Istanbul, Turkey. Under Capital Markets Law of Turkey (Law No: 6362), the investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Comments and recommendations stated here rely on the individual opinions of the ones providing these comments and recommendations. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations. Furthermore, Citi Research is a division of Citigroup Global Markets Inc. (the "Firm"), which does and seeks to do business with companies and/or trades on securities covered in this research reports. As a result, investors should be aware that the Firm may have a conflict of interest that could affect the objectivity of this report, however investors should also note that the Firm has in place organisational and administrative arrangements to manage potential conflicts of interest of this nature.

In the **U.A.E**, these materials (the "Materials") are communicated byCitigroup Global Markets Limited, DIFC branch ("CGML"), an entityregistered in the Dubai International Financial Center ("DIFC") andlicensed and regulated by the Dubai Financial Services Authority ("DFSA",license #CL0221) to Professional Clients and Market Counterparties, as definedin DFSA regulations, only and should not be relied upon or distributed toRetail Clients. Financial products and/or services to which the Materialsrelate will only be made available to Professional Clients and MarketCounterparties. Citigroup Global Markets Limited DIFC Branch registered addressis Level 3, Gate District Building 02, Dubai International Financial Centre andcan be contacted on +971 4 509 97 90.

The Product is made available in **United Kingdom** by Citigroup Global Markets Limited, which is authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA. This material may relate to investments or services of a person outside of the UK or to other matters which are not authorised by the PRA nor regulated by the FCA and the PRA and further details as to where this may be the case are available upon request in respect of this material. Citigroup Centre, Canada Square, Canary Wharf, London, E14 5LB.

The Product is made available in **United States** and **Canada** by Citigroup Global Markets Inc., which is a member of FINRA and registered with the US Securities and Exchange Commission. 388 Greenwich Street, New York, NY 10013.

Unless specified to the contrary, within EU Member States, the Product is made available by Citigroup Global Markets Europe AG ("CGME"), which is regulated by the European Central Bank and the German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht-BaFin).

The Product is not to be construed as providing investment services in any jurisdiction where the provision of such services would not be permitted.
Subject to the nature and contents of the Product, the investments described therein are subject to fluctuations in price and/or value and investors may get back less than originally invested. Certain high-volatility investments can be subject to sudden and large falls in value that could equal or exceed the amount invested.The yield and average life of CMOs (collateralized mortgage obligations) referenced in this Product will fluctuate depending on the actual rate at which mortgage holders prepay the mortgages underlying the CMO and changes in current interest rates. Any government agency backing of the CMO applies only to the face value of the CMO and not to any premium paid. Certain investments contained in the Product may have tax implications for private customers whereby levels and basis of taxation may be subject to change. If in doubt, investors should seek advice from a tax adviser. The Product does not purport to identify the nature of the specific market or other risks associated with a particular transaction. Advice in the Product is general and should not be construed as personal advice given it has been prepared without taking account of the objectives, financial situation or needs of any particular investor. Accordingly, investors should, before acting on the advice, consider the appropriateness of the advice, having regard to their objectives, financial situation and needs. Prior to acquiring any financial product, it is the client's responsibility to obtain the relevant offer document for the product and consider it before making a decision as to whether to purchase the product.

Card Insights. Where this report references Card Insights data, Card Insights consists of selected data from a subset of Citi's proprietary credit card transactions. Such data has undergone rigorous security protocols to keep all customer information confidential and secure; the data is highly aggregated and anonymized so that all unique customer identifiable information is removed from the data prior to receipt by the report's author or distribution to external parties. This data should be considered in the context of other economic indicators and publicly available information. Further, the selected data represents only a subset of Citi's proprietary credit card transactions due to the selection methodology or other limitations and should not be considered as indicative or predictive of the past or future financial performance of Citi or its credit card business.

Citi Research product may source data from dataCentral. dataCentral is a Citi Research proprietary database, which includes the Firm's estimates, data from company reports and feeds from LSEG Data & Analytics. The source for all referenced prices, unless otherwise stated, is DataCentral. Past performance is not a guarantee or reliable indicator of future results. Forecasts are not a guarantee or reliable indicator of future performance. The printed and printable version of the research report may not include all the information <(e.g. certain financial summary information and comparable company data) that is linked to the online version available on the Firm's proprietary electronic distribution platforms.
Where included in this report, MSCI sourced information is the exclusive property of Morgan Stanley Capital International Inc. (MSCI). Without prior written permission of MSCI, this information and any other MSCI intellectual property may not be

reproduced, redisseminated or used to create any financial products, including any indices. This information is provided on an "as is" basis. The user assumes the entire risk of any use made of this information. MSCI, its affiliates and any third party involved in, or related to, computing or compiling the information hereby expressly disclaim all warranties of originality, accuracy, completeness, merchantability or fitness for a particular purpose with respect to any of this information. Without limiting any of the foregoing, in no event shall MSCI, any of its affiliates or any third party involved in, or related to, computing or compiling the information have any liability for any damages of any kind. MSCI, Morgan Stanley Capital International and the MSCI indexes are services marks of MSCI and its affiliates. Where data is attributed to Morningstar that data is © 2026 Morningstar, Inc. All Rights Reserved. That information: (1) is proprietary to Morningstar and/or its content providers; (2) may not be copied or distributed; and (3) is not warranted to be accurate, complete or timely. Neither Morningstar nor its content providers are responsible for any damages or losses arising from any use of this information.

The Firm accepts no liability whatsoever for the actions of third parties. The Product may provide the addresses of, or contain hyperlinks to, websites. Except to the extent to which the Product refers to website material of the Firm, the Firm has not reviewed the linked site. Equally, except to the extent to which the Product refers to website material of the Firm, the Firm takes no responsibility for, and makes no representations or warranties whatsoever as to, the data and information contained therein. Such address or hyperlink (including addresses or hyperlinks to website material of the Firm) is provided solely for your convenience and information and the content of the linked site does not in any way form part of this document. Accessing such website or following such link through the Product or the website of the Firm shall be at your own risk and the Firm shall have no liability arising out of, or in connection with, any such referenced website.

© 2026 Citigroup Global Markets Inc. Citi Research is a division of Citigroup Global Markets Inc. Citi and Citi and Arc Design are trademarks and service marks of Citigroup Inc. and its affiliates and are used and registered throughout the world. All rights reserved. The research data in this report are not intended to be used for the purpose of (a) determining the price of or amounts due in respect of (or to value) one or more financial products or instruments and/or (b) measuring or comparing the performance of, or defining the asset allocation of a financial product, a portfolio of financial instruments, or a collective investment undertaking, and any such use is strictly prohibited without the prior written consent of Citi Research. Any unauthorized use, duplication, redistribution or disclosure of this report (the “Product”), including, but not limited to, redistribution of the Product by electronic mail, posting of the Product on a website or page, and/or providing to a third party a link to the Product, is prohibited by law and will result in prosecution. The information contained in the Product is intended solely for the recipient and may not be further distributed by the recipient to any third party.

ADDITIONAL INFORMATION IS AVAILABLE UPON REQUEST

Daiwa Capital Markets Hong Kong Limited

earnings

12 October 2026

MEMO

Nanya Technology (2408 TT)

Share price (12 Oct): **TWD527.00** 12-mth rating: **Buy (1)** **Information Technology: Taiwan**

3Q26 results review: favourable ASP likely to continue in 2027

SK Kim
(82) 2 787 9173
sk.kim@kr.daiwacm.com

Summary: Nanya Technology (Nanya) announced its 3Q26 results after market hours on 12 October. Its revenue of TWD133.6m (+62% QoQ, +612% YoY) and operating profit of TWD106.9m (+76% QoQ, +95.5x YoY) were robust. While shipments decreased by a mid-single-digit % QoQ, ASP increased by > 60% QoQ mainly due to renewed LTA contracts. While DDR4 accounted for >60% of revenue, management expects the favourable pricing environment to persist on robust demand and supply constraints. However, due to massive capex (c.USD16bn) on the build-out of a new 50k wpm capacity, management didn't announce an enhanced shareholder return plan or a higher dividend payout.

Key highlights

- **3Q26 results review.** Nanya's 3Q26 revenue of TWD133.6m (+62% QoQ, +612% YoY) and operating profit of TWD106.9m (+76% QoQ, +95.5x YoY) were slightly below our and recent market estimates due to higher employee compensation expenses. Its DRAM ASP increased by >60% QoQ in 3Q26 on LTA price adjustments, while shipments decreased by a mid-single-digit % QoQ. DDR5 accounted for >10% of 3Q26 revenue, while DDR4 accounted for >60%. Management guided for a persistent market shortage and a continued price uptrend in 4Q26, while shipments are expected to remain flat until wafer output from the new fab comes online in 4Q27.

- **Key takeaways from conference call.** Management expects AI infrastructure to keep driving DRAM demand. While AI infrastructure and server applications currently account for >20% of revenue, it expects this exposure to grow to >30% through 2029-30. In particular, DDR4 demand now accounts for 12-14% of total DRAM market demand, maintaining the higher ASP trend on strong demand. With major DRAM players having almost completed their HBM supply negotiations for 2027, management believes it would be hard to increase supply given capacity allocation and longer lead times. Its LTA output portion was >20%, and management expects this to increase going forward. The company announced a new fab construction plan for 30k wpm by the end of 2028 and up to 50k wpm by 2030 to respond to LTA demand from AI infrastructure. Despite the solid 3Q26 earnings, there was no announcement of an increased dividend payout, due to the capex burden (c.USD16bn for 50k wpm). It did not guide for any share buyback.

Recommendation

We have a Buy (1) call on Nanya with a 12M TP of TWD700, based on a 7.4x PER on the 2026-27E EPS. While the robust 3Q26 earnings led by the significant ASP hike was positive, we view the delay in announcing a shareholder return plan as disappointing. Key risk: weaker-than-expected commodity DRAM demand due to delays in data centre build-outs.

Important disclosures, including any required research certifications, are provided on the last three pages of this report.

To see our recent flash note on Nanya, please ***click here***

Robust revenue likely to drive earnings beat

- 3Q26E earnings to beat consensus estimate, led by DRAM prices
- DDR4 supply shortages to persist in 2027E; upcoming dividend policy
- Reaffirming our Buy (1) call and lifting our 12M TP to TWD700

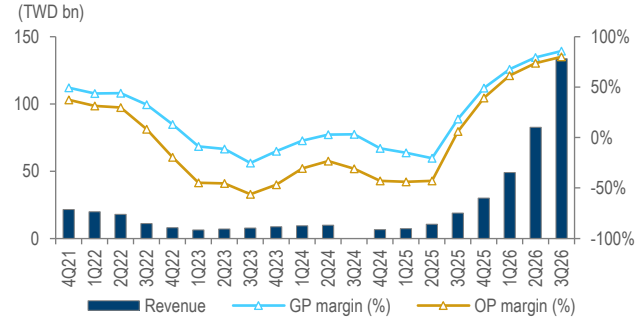
Nanya: 3Q26 results overview

(TWDm)	3Q26	2Q26	*QoQ growth*	3Q25	*YoY growth*	Daiwa estimates	*Diff*	BBG consensus	*Diff*
Revenue	133,649	82,549	*62%*	18,779	*612%*	133,649	*0%*	133,354	*0%*
Operating Profit	106,894	60,826	*76%*	1,119	*95.5x*	108,647	*-2%*	107,724	*-1%*
OP margin	*80.0%*	*73.7%*		*6.0%*		*81.3%*		*80.8%*	

Source: Company, Bloomberg

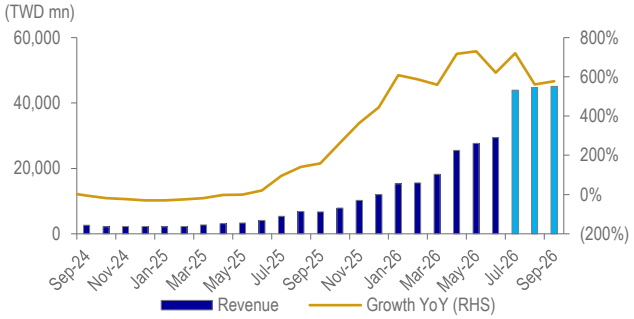
Note: Bloomberg standard consensus as of 12 October 2026

Nanya: revenue and gross/operating margin trend



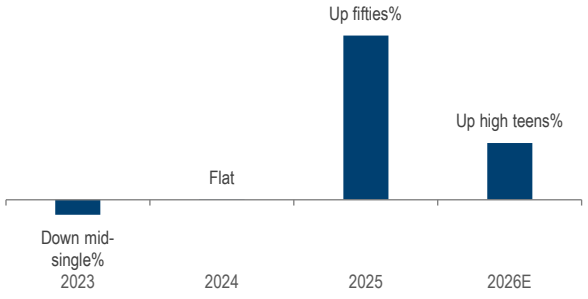
Source: Company

Nanya: monthly revenue and growth YoY trend



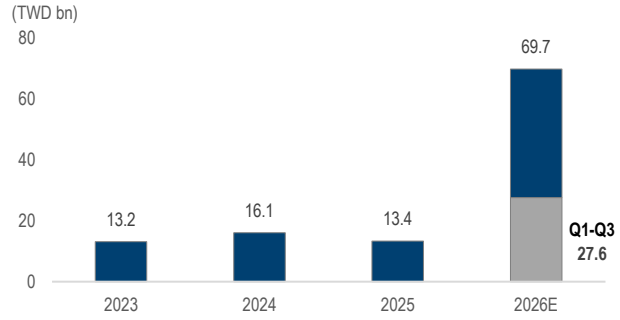
Source: Company

Nanya: bit shipment growth YoY trend



Source: Company

Nanya: annual capex trend



Source: Company

In the interests of timeliness, this document has not been edited.

Important Disclosures and Disclaimer

This publication is produced by Daiwa Securities Group Inc. and/or its non-U.S. affiliates, and distributed by Daiwa Securities Group Inc. and/or its non-U.S. affiliates, except to the extent expressly provided herein. This publication and the contents hereof are intended for information purposes only, and may be subject to change without further notice. Any use, disclosure, distribution, dissemination, copying, printing or reliance on this publication for any other purpose without our prior consent or approval is strictly prohibited. Neither Daiwa Securities Group Inc. nor any of its respective parent, holding, subsidiaries or affiliates, nor any of its respective directors, officers, servants and employees, represent nor warrant the accuracy or completeness of the information contained herein or as to the existence of other facts which might be significant, and will not accept any responsibility or liability whatsoever for any use of or reliance upon this publication or any of the contents hereof. Neither this publication, nor any content hereof, constitute, or are to be construed as, an offer or solicitation of an offer to buy or sell any of the securities or investments mentioned herein in any country or jurisdiction nor, unless expressly provided, any recommendation or investment opinion or advice. Any view, recommendation, opinion or advice expressed in this publication may not necessarily reflect those of Daiwa Securities Group Inc., and/or its affiliates nor any of its respective directors, officers, servants and employees except where the publication states otherwise. This research report is not to be relied upon by any person in making any investment decision or otherwise advising with respect to, or dealing in, the securities mentioned, as it does not take into account the specific investment objectives, financial situation and particular needs of any person.

Daiwa Securities Group Inc., its subsidiaries or affiliates, or its or their respective directors, officers and employees from time to time have trades as principals, or have positions in, or have other interests in the securities of the company under research including market making activities, derivatives in respect of such securities or may have also performed investment banking and other services for the issuer of such securities. Daiwa Securities Group Inc., its subsidiaries or affiliates do and seek to do business with the company(s) covered in this research report. Therefore, investors should be aware that a conflict of interest may exist. The following are additional disclosures.

Licensed and proprietary artificial intelligence (AI) tools may be utilized, where appropriate, as part of the research production process. Such tools may assist with activities including information gathering, data compilation and drafting support. Any AI-assisted output is subject to appropriate review and professional judgment. Research reports are prepared and published in accordance with applicable laws, regulations, and professional standards.

Ownership of Securities

For "Ownership of Securities" information, please visit BlueMatrix disclosure link at *https://daiwa3.bluematrix.com/sellside/Disclosures.action.*

Investment Banking Relationship

For "Investment Banking Relationship", please visit BlueMatrix disclosure link at *https://daiwa3.bluematrix.com/sellside/Disclosures.action.*

Daiwa Securities Co. Ltd. and Daiwa Securities Group Inc.

Daiwa Securities Co. Ltd. is a subsidiary of Daiwa Securities Group Inc.

Investment Banking Relationship

Within the preceding 12 months, Daiwa Capital Markets Hong Kong Limited has lead-managed public offerings and/or secondary offerings of the equities or relevant securities of the following companies: WeRide Inc (800 HK/ WRD US), SEYOND (2665 HK), SENASIC Electronics Technology Co Ltd (6675 HK) and Mech-Mind Robotics Technologies Co Ltd (9615 HK).

*Subsidiaries of Daiwa Securities Group Inc. for the purposes of this section shall mean any one or more of: Daiwa Capital Markets Hong Kong Limited (大和資本市場香港有限公司), Daiwa Capital Markets Singapore Limited, Daiwa Capital Markets Australia Limited, Daiwa-Cathay Capital Markets Co., Ltd., Daiwa Securities Capital Markets Korea Co., Ltd.

Within the last 12 months, the subsidiaries and/or affiliates of Daiwa Securities Group Inc. * has received compensation for investment banking services from KT Corporation (030200 KS) and Hyundai Department Store Co Ltd (069960 KS).

Hong Kong

This research is distributed in Hong Kong by Daiwa Capital Markets Hong Kong Limited (大和資本市場香港有限公司) ("DHK") which is regulated by the Hong Kong Securities and Futures Commission. Recipients of this research in Hong Kong may contact DHK in respect of any matter arising from or in connection with this research.

Investment Banking Relationship

Within the preceding 12 months, Daiwa Capital Markets Hong Kong Limited has lead-managed public offerings and/or secondary offerings of the equities or relevant securities of the following companies: WeRide Inc (800 HK/ WRD US), SEYOND (2665 HK), SENASIC Electronics Technology Co Ltd (6675 HK) and Mech-Mind Robotics Technologies Co Ltd (9615 HK).

Within the preceding 12 months, Daiwa Capital Markets Hong Kong Limited has received compensation for investment banking services from -.

Relevant Relationship (DHK)

DHK may from time to time have an individual employed by or associated with it serves as an officer of any of the companies under its research coverage.

Singapore

This research is distributed in Singapore by Daiwa Capital Markets Singapore Limited and it may only be distributed in Singapore to accredited investors, expert investors and institutional investors as defined in the Financial Advisers Regulations of Singapore and the Securities and Futures Act 2001 of Singapore, both as amended from time to time. By virtue of distribution to these category of investors, Daiwa Capital Markets Singapore Limited and its representatives are not required to comply with Section 45 of the Financial Advisers Act 2001 of Singapore (Section 45 relates to disclosure of Daiwa Capital Markets Singapore Limited's interest and/or any associated or connected person's interest in securities). Recipients of this research in Singapore may contact Daiwa Capital Markets Singapore Limited in respect of any matter arising from or in connection with this research.

Australia

This research is distributed in Australia by Daiwa Capital Markets Australia Limited and it may only be distributed in Australia to wholesale investors within the meaning of the Corporations Act. Recipients of this research in Australia may contact Daiwa Capital Markets Stockbroking Limited in respect of any matter arising from or in connection with the research.

Taiwan

This research is solely for reference and not intended to provide tailored investment recommendations. This research is distributed in Taiwan by Daiwa-Cathay Capital Markets Co., Ltd. and it may only be distributed in Taiwan to specific customers who have signed recommendation contracts with Daiwa-Cathay Capital Markets Co., Ltd. and non-customers including (i) professional institutional investors, (ii) TWSE or TPEx listed companies, upstream and downstream vendors, and specialists that offer or seek advice, and (iii) potential customers with an actual need for business development in accordance with the Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers. Recipients of this research including non-customer recipients of this research shall not provide it to others or engage in any activities in connection with this research which may involve conflicts of interests. Neither Daiwa-Cathay Capital Markets Co., Ltd. nor its personnel who writes or reviews the research report has any conflict of interest in this research. Since Daiwa-Cathay Capital Markets Co., Ltd. does not operate brokerage trading business in foreign markets, **this research is prepared on a "without recommendation" to any foreign securities basis** and Daiwa-Cathay Capital Markets Co., Ltd. does not accept orders from customers to trade in such foreign securities. Recipients of this research shall carefully judge their own investment risk and take full responsibility for the results of any resulting investments in the companies and/or sectors featured in this research. Without the prior written permission of Daiwa-Cathay Capital Markets Co., Ltd., recipients of this research are prohibited from disclosing the research to the media, reprinting the research, or quoting from the research to other parties. Recipients of this research in Taiwan may contact Daiwa-Cathay Capital Markets Co., Ltd. in respect of any matter arising from or in connection with the research.

Thailand

This research is distributed to only institutional investors in Thailand. This report is provided to you for informational purposes only. Information contained herein is not intended to be, and shall not be construed as, an offer or an invitation for subscription or purchase of securities in Thailand. This document has not been and will not be registered with, or approved by, the Office of the Securities and Exchange Commission of Thailand. As neither Daiwa Securities Group Inc. nor any of its subsidiaries or affiliates, nor any of the foregoing entities' respective directors, officers, servants and employees are licensed to carry on any securities business in Thailand, this document is intended to be read by the addressee who is an institutional investor only and may not be circulated or distributed, whether directly or indirectly, to the public or any members of the public in Thailand, unless to the extent permitted by applicable laws and regulations.

This report is prepared by analysts who are employed by Daiwa Securities Group Inc. and/or its non-U.S. affiliates. Neither Daiwa Securities Group Inc. nor any of its subsidiaries or affiliates, nor any of the foregoing entities' respective directors, officers, servants and employees accept any liability whatsoever for any direct or consequential loss arising from any use of this research or its contents.

The information and opinions contained herein have been compiled or arrived at from sources believed to be reliable. However, neither Daiwa Securities Group Inc. nor any of its subsidiaries or affiliates, nor any of the foregoing entities' respective directors, officers, servants and employees make any representation or warranty, express or implied, as to their accuracy or completeness. Expressions of opinion herein are subject to change without notice. The use of any information, forecasts and opinions contained in this report shall be at the sole discretion and risk of the user.

Daiwa Securities Group Inc., its subsidiaries and affiliates and their respective directors, officers, servants and employees may have positions and financial interest in securities mentioned in this research. Daiwa Securities Group Inc. and its subsidiaries and affiliates may from time to time perform investment banking or other services for, or solicit investment banking or other business from, any entity mentioned in this research. Therefore, investors should be aware of conflict of interest that may affect the objectivity of this research.

United Kingdom

These materials constitute independent research for the purposes of the Financial Conduct Authority ("**FCA**") Conduct of Business Rules. These materials have been produced by Daiwa Securities Co. Ltd ("**DSCL**") and/or its affiliates and are distributed by Daiwa Capital Markets Europe Limited ("**DCME**") to persons classified as (i) eligible counterparties; and (ii) professional clients for the purposes of the Conduct of Business Rules of the FCA. These materials should not be disseminated to any retail clients in the United Kingdom. DCME is authorised and regulated by the FCA and is a member of the London Stock Exchange. DCME's FCA registration number is 124490 and the FCA's registered address is 25 The North Colonnade, London, E14 5HS.

DCME makes no representation as to the completeness, accuracy or timeliness of any of the materials produced by DSCL, and does not accept any liability for any losses, costs, liabilities or expenses which may arise directly or indirectly from any use of, or reliance on the materials.

DCME and/or its affiliates may, from time to time, to the extent permitted by law, participate or invest in, or be mandated in respect of transactions with the issuer(s) referred to in the materials prepared by DSCL, perform services for or solicit business from such issuers, and/or have a position or effect transactions in a particular issuer's securities and/or may have acted as an underwriter during the past twelve months in respect of a particular issuer of its securities. DCME does not review DSCL's materials prior to publication on DCME's platforms and has no influence over its content. DCME has in place organisational arrangements for the prevention and avoidance of conflicts of interest, including information barriers to control the flow of information between the private and public sides of the firm. A summary of DCME's conflict management policies are available at *http://www.uk.daiwacm.com/about-us/corporate-governance-regulatory*.

DSCL's materials are distributed by DCME only as permitted by law. The materials are not directed to, or intended for distribution to or use by, any person or entity located in any jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject DCME to any regulation or licensing requirement within such jurisdiction.

Germany

Research reports are produced by Daiwa Securities Co. Ltd. and/or its affiliates and are distributed by Daiwa Capital Markets Deutschland GmbH in the European Union. Daiwa Capital Markets Deutschland GmbH is authorised and regulated by Bundesanstalt für Finanzdienstleistungsaufsicht ("BaFin") under the reference number 149361. Daiwa Capital Markets Deutschland GmbH and its affiliates may, from time to time, to the extent permitted by law, participate or invest in, or be mandated in respect of, other financing transactions with the issuer(s) of the securities referred to herein (the "Securities"), perform services for or solicit business from such issuers, and/or have a position or effect transactions in the Securities or options thereof.

This publication is intended for investors who are Professional Clients or Eligible Counterparties in the European Union within the meaning of Directive 2014/65/EU ("MiFID II") and should therefore not be distributed to Retail Clients in the European Union.

Daiwa Capital Markets Deutschland GmbH has in place organisational arrangements for the identification, prevention and management of conflicts of interest. Information regarding our conflict management are available at: *https://www.de.daiwacm.com/policies/*

Additional disclosures regarding specific company names and related financial instruments are available at: *https://daiwa3.bluematrix.com/sellside/Disclosures.action*.

Bahrain

This research material is distributed in Bahrain by Daiwa Capital Markets Europe Limited, Bahrain Branch, regulated by The Central Bank of Bahrain and holds Investment Business Firm – Category 2 license and having its official place of business at the Bahrain World Trade Centre, South Tower, 7th floor, P.O. Box 30069, Manama, Kingdom of Bahrain. Tel No. +973 17534452 Fax No. +973 535113

United States

This research is distributed into the United States directly by Daiwa Capital Markets Hong Kong Limited (DCMHK) and in certain cases indirectly by Daiwa Capital Markets America Inc. (DCMA), a U.S. Securities and Exchange Commission registered broker-dealer and FINRA member firm, exclusively to "major U.S. institutional investors", as defined under Rule 15a-6 promulgated under the U.S. Securities Exchange Act of 1934, as amended, and as interpreted by the staff of the U.S. Securities and Exchange Commission (SEC). Where this is co-branded research published by Daiwa's partners in collaboration with other parties not affiliated with Daiwa, this research is distributed in the United States by DCMHK only. This report is not an offer to sell or the solicitation of any offer to buy securities. U.S. customers wishing to effect transactions in any designated investment discussed in this report should do so through a qualified salesperson of DCMA. Non-U.S. customers wishing to effect transactions in any designated investment discussed in this report should contact a Daiwa entity in their local jurisdiction. The securities or other investment products discussed in this report may not be eligible for sale in some jurisdictions.

Analysts employed outside the U.S., as specifically indicated elsewhere in this report, are not registered as research analysts with FINRA. These analysts may not be associated persons of DCMA, and therefore may not be subject to FINRA Rule 2241 restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

ADDITIONAL IMPORTANT DISCLOSURES CAN BE FOUND AT:
https://daiwa3.bluematrix.com/sellside/Disclosures.action

Ownership of Securities

For "Ownership of Securities" information please visit BlueMatrix disclosure link at *https://daiwa3.bluematrix.com/sellside/Disclosures.action*.

Investment Banking Relationships

For "Investment Banking Relationships" please visit BlueMatrix disclosure link at *https://daiwa3.bluematrix.com/sellside/Disclosures.action*.

DCMA Market Making

For "DCMA Market Making" please visit BlueMatrix disclosure link at *https://daiwa3.bluematrix.com/sellside/Disclosures.action*.

Research Analyst Conflicts

For updates on "Research Analyst Conflicts" please visit BlueMatrix disclosure link at *https://daiwa3.bluematrix.com/sellside/Disclosures.action*. The principal research analysts who prepared this report have no financial interest in securities of the issuers covered in the report, are not (nor are any members of their household) an officer, director or advisory board member of the issuer(s) covered in the report, and are not aware of any material relevant conflict of interest involving the analyst or DCMA, and did not receive any compensation from the issuer during the past 12 months except as noted: no exceptions.

Research Analyst Certification

For updates on "Research Analyst Certification" and "Rating System" please visit BlueMatrix disclosure link at *https://daiwa3.bluematrix.com/sellside/Disclosures.action*. The views about any and all of the subject securities and issuers expressed in this Research Report accurately reflect the personal views of the research analyst(s) primarily responsible for this report (or the views of the firm producing the report if no individual analyst is named on the report); and no part of the compensation of such analyst (or no part of the compensation of the firm if no individual analyst is named on the report) was, is, or will be directly or indirectly related to the specific recommendations or views contained in this Research Report.

The following explains the rating system in the report as compared to relevant local indices, unless otherwise stated, based on the beliefs of the author of the report.

"1": the security could outperform the local index by more than 15% over the next 12 months.
"2": the security is expected to outperform the local index by 5-15% over the next 12 months.
"3": the security is expected to perform within 5% of the local index (better or worse) over the next 12 months.
"4": the security is expected to underperform the local index by 5-15% over the next 12 months.
"5": the security could underperform the local index by more than 15% over the next 12 months.

Disclosure of investment ratings

Rating	Percentage of total
Buy*	73.96%
Hold**	19.58%
Sell***	6.46%

Source: Daiwa

Notes: data is for single-branded Daiwa research in Asia (ex Japan) and correct as of 30 September 2026.
** comprised of Daiwa's Buy and Outperform ratings.*
*** comprised of Daiwa's Hold ratings.*
**** comprised of Daiwa's Underperform and Sell ratings.*

Additional information may be available upon request.

Japan - Notification items pursuant to Article 37 of the Financial Instruments and Exchange Law

(This Notification is only applicable to where report is distributed by Daiwa Securities Co. Ltd.)

If you decide to enter into a business arrangement with our company based on the information described in this report, we ask you to pay close attention to the following items.

- In addition to the purchase price of a financial instrument, our company will collect a trading commission* for each transaction as agreed beforehand with you. Since commissions may be included in the purchase price or may not be charged for certain transactions, we recommend that you confirm the commission for each transaction. In some cases, our company also may charge a maximum of ¥2 million per year as a standing proxy fee for our deposit of your securities, if you are a non-resident.
- For derivative and margin transactions etc., our company may require collateral or margin requirements in accordance with an agreement made beforehand with you. Ordinarily in such cases, the amount of the transaction will be in excess of the required collateral or margin requirements**.
- There is a risk that you will incur losses on your transactions due to changes in the market price of financial instruments based on fluctuations in interest rates, exchange rates, stock prices, real estate prices, commodity prices, and others. In addition, depending on the content of the transaction, the loss could exceed the amount of the collateral or margin requirements.
- There may be a difference between bid price etc. and ask price etc. of OTC derivatives handled by our company.
- Before engaging in any trading, please thoroughly confirm accounting and tax treatments regarding your trading in financial instruments with such experts as certified public accountants.

* The amount of the trading commission cannot be stated here in advance because it will be determined between our company and you based on current market conditions and the content of each transaction etc.

** The ratio of margin requirements etc. to the amount of the transaction cannot be stated here in advance because it will be determined between our company and you based on current market conditions and the content of each transaction etc.

When making an actual transaction, please be sure to carefully read the materials presented to you prior to the execution of agreement, and to take responsibility for your own decisions regarding the signing of the agreement with our company.

Corporate Name:	Daiwa Securities Co. Ltd.
Registered:	Financial Instruments Business Operator, Chief of Kanto Local Finance Bureau (Kin-sho) No.108
Memberships:	Japan Securities Dealers Association, The Financial Futures Association of Japan, Investment Management Association of Japan, Type II Financial Instruments Firms Association, Japan Security Token Offering Association

Airtac International Group (1590.TW): Sep Monthly revenue -3% vs. GSe; New orders continue to exceed shipments despite low season; Buy

Chao Wang
+886(2)2730-4195 | kuan-chao.wang@gs.com
Goldman Sachs (Asia) L.L.C., Taipei Branch

Yuichiro Isayama
+81(3)4587-9806 | yuichiro.isayama@gs.com
Goldman Sachs Japan Co., Ltd.

Al Wang
+886(2)2730-4081 | al.wang@gs.com
Goldman Sachs (Asia) L.L.C., Taipei Branch

On October 12th, AirTAC reported September revenue of Rmb806mn (+3% MoM, +14% YoY) / NT$3,811mn (+3% MoM, +27% YoY), which was 3% below GSe in Rmb terms.

In September, AirTAC's revenue was 3% below GSe in Rmb terms, up 3% MoM with 2% more working days. This came in slightly below our estimates, primarily due to (1) normal low season effect with daily pneumatic revenue increasing by 1% MoM at Rmb29.4mn (vs. Rmb35.4mn in 2Q26), which is tracking lower than our estimates of Rmb32.1mn and (2) slightly weaker-than-expected linear guide revenue contribution (Rmb64mn slightly missed GSe of Rmb68mn), but the daily average linear guides revenue is higher at Rmb2.7mn (vs. Rmb2.5mn in August).

Moreover, September revenue was up 14% YoY (in Rmb) with 4% less working days vs. September 2025. We believe this solid YoY revenue growth was driven by (1) the strong performance in electronics continues, which accounts for the highest % of revenue with YoY growth of 27%, and (2) continuous improvement in China pneumatic demand environment, as YoY growth across nearly all end-markets reached double digits except for the battery segment.

In 3Q26, AirTAC's revenue was Rmb2,427mn (-7% QoQ, +23% YoY)/NT$11,525mn (-5% QoQ, +40% YoY), which was 3% below GSe in NT$ terms; we believe this miss was mainly driven by the deceleration in YoY growth for the battery segment (Rmb340mn missed GSe of Rmb482mn), offsetting the better than expected contribution from electronics (Rmb671mn beat GSe of Rmb648mn) and energy lighting (Rmb81mn beat GSe of Rmb64mn). We now expect 3Q26 GM/OPM to reach 48.5/14.5%.

In terms of revenue growth by end-market segment for September (see Exhibit 1), the machine tool sector outgrew all other sectors (+35% YoY), followed by energy lighting (+30% YoY), electronics (+27% YoY), general machinery (+12% YoY), packaging (+11% YoY), automotive (+11% YoY), and battery (-10% YoY).

Goldman Sachs does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html. Analysts employed by non-US affiliates are not registered/qualified as research analysts with FINRA in the U.S.

Exhibit 1: Strength/weakness in major segments of AirTAC in September
AirTAC's monthly revenue by segment at a glance

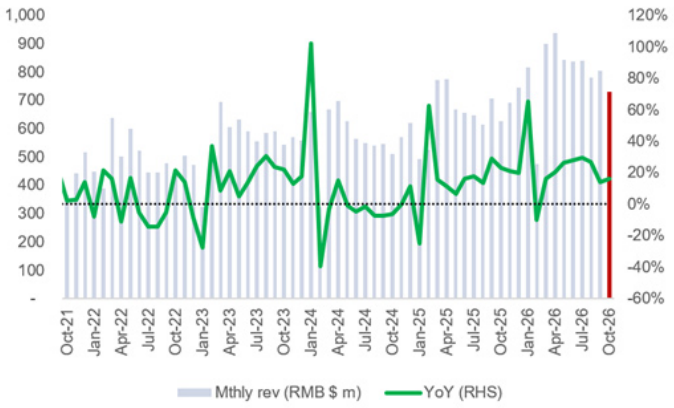
End market	% of revenue	YoY (%)
General machinery	5%	12%
Auto	11%	11%
Machine tools	8%	35%
Textiles	5%	25%
Packaging	8%	11%
Electronics	27%	27%
Battery	12%	-10%
Energy lighting	4%	30%

Revenue breakdown represents the contribution by each end-market customer for AirTAC's "direct" sales, which represent more than 80%+ of its total revenue. Breakdown of distribution sales, which are c.20% of total sales, is not included in this data.

Source: Company data

For the month of October 2026, we estimate AirTAC's monthly revenue will decrease to Rmb729mn (-10% MoM, +16% YoY)/NT$3,447mn (-10% MoM, +28% YoY)
We expect October revenue will go down by 10% MoM to Rmb729mn/NT$3,447mn, mainly due to 19% fewer working days in Mainland China. However, we expect to see improvement in daily pneumatic revenue (Rmb32.3mn in Oct vs. Rmb 29.4mn in Sept) as the improvement in daily pneumatic revenue is driven by the strong growth from key customers, including electronics, followed by machine tools and battery, and we further expect demand to improve following the end of low season.

Exhibit 2: We estimate AirTAC's October revenue will increase by 16% YoY



Source: Company data, Goldman Sachs Global Investment Research

Earnings and TP revisions

We revise down our 2026E earnings estimates by 1% to factor in actual 3Q26 revenue (revise down 2026 revenue by 1%). We revise down our 2027E/2028E earnings estimate by 1%/1% to factor in a more conservative assumption on battery and linear guide business (revise down AirTAC revenue by 1%/1% in 2027E/2028E). We continue to see the company focusing on cost control as well as experiencing a better product mix outlook, but we expect some offset by rising material cost (revise down 2026E/2028E OPM by 0.1/0.1ppt).

Maintain our Buy rating on AirTAC, with a new 12m TP of NT$1,895 (from NT$1,925), based on an unchanged target P/E of 26x on 2027 EPS.

Exhibit 3: Earnings revision table

AirTAC P&L (NT$ mn)	2026E New	2026E Old	Diff.	2027E New	2027E Old	Diff.	2028E New	2028E Old	Diff.
Sales	45,096	45,644	-1%	52,655	53,446	-1%	59,393	60,126	-1%
Gross Profit	21,925	22,216	-1%	26,057	26,443	-1%	29,821	30,144	-1%
EBIT	15,192	15,400	-1%	18,541	18,814	-1%	21,487	21,708	-1%
Net Income	12,225	12,389	-1%	14,587	14,800	-1%	16,884	17,057	-1%
EPS (NT$)	**61.12**	**61.95**	**-1%**	**72.93**	**74.00**	**-1%**	**84.42**	**85.29**	**-1%**
Ratio analysis									
Gross margin	48.6%	48.7%	-0.1pp	49.5%	49.5%	0.0pp	50.2%	50.1%	0.1pp
EBIT margin	33.7%	33.7%	-0.1pp	35.2%	35.2%	0.0pp	36.2%	36.1%	0.1pp
Net margin	27.1%	27.1%	0.0pp	27.7%	27.7%	0.0pp	28.4%	28.4%	0.1pp

Source: Goldman Sachs Global Investment Research

Exhibit 4: P&L table

NT$mn	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025	2026E	2027E	2028E
Revenue	8,096	8,953	8,246	9,035	10,014	12,164	11,512	11,406	11,939	13,918	13,610	13,188	30,660	34,330	45,096	52,655	59,393
Gross profit	3,574	4,119	3,797	4,292	4,804	6,048	5,580	5,493	5,777	7,059	6,798	6,423	14,314	15,782	21,925	26,057	29,821
Operating expense	(1,302)	(1,387)	(1,336)	(1,469)	(1,479)	(1,783)	(1,669)	(1,802)	(1,755)	(1,879)	(1,851)	(2,031)	(5,298)	(5,494)	(6,734)	(7,516)	(8,335)
Operating income	2,272	2,732	2,460	2,823	3,325	4,265	3,911	3,691	4,022	5,180	4,947	4,392	9,016	10,288	15,192	18,541	21,487
Pretax income	2,466	2,570	2,652	2,957	3,424	4,238	4,061	3,841	4,062	5,220	4,987	4,432	9,634	10,645	15,564	18,701	21,647
Taxes expense	(530)	(533)	(553)	(629)	(753)	(927)	(853)	(807)	(894)	(1,148)	(1,097)	(975)	(2,010)	(2,246)	(3,340)	(4,114)	(4,762)
Net income	1,936	2,037	2,099	2,328	2,671	3,311	3,208	3,034	3,168	4,072	3,890	3,457	7,624	8,400	12,225	14,587	16,884
EPS, NT$	**9.68**	**10.19**	**10.50**	**11.64**	**13.35**	**16.55**	**16.04**	**15.17**	**15.84**	**20.36**	**19.45**	**17.29**	**38.12**	**42.00**	**61.12**	**72.93**	**84.42**
Ratio analysis and assumption																	
As % of sales																	
Gross margin	44.1%	46.0%	46.0%	47.5%	48.0%	49.7%	48.5%	48.2%	48.4%	50.7%	50.0%	48.7%	46.7%	46.0%	48.6%	49.5%	50.2%
Operating expense ratio	16.1%	15.5%	16.2%	16.3%	14.8%	14.7%	14.5%	15.8%	14.7%	13.5%	13.6%	15.4%	17.3%	16.0%	14.9%	14.3%	14.0%
Operating margin	28.1%	30.5%	29.8%	31.2%	33.2%	35.1%	34.0%	32.4%	33.7%	37.2%	36.4%	33.3%	29.4%	30.0%	33.7%	35.2%	36.2%
Net margin	23.9%	22.8%	25.5%	25.8%	26.7%	27.2%	27.9%	26.6%	26.5%	29.3%	28.6%	26.2%	24.9%	24.5%	27.1%	27.7%	28.4%
QoQ growth (%)																	
Revenue	6%	11%	-8%	10%	11%	21%	-5%	-1%	5%	17%	-2%	-3%					
Gross profit	2%	15%	-8%	13%	12%	26%	-8%	-2%	5%	22%	-4%	-6%					
Operating income	6%	20%	-10%	15%	18%	28%	-8%	-6%	9%	29%	-4%	-11%					
Net income	7%	5%	3%	11%	15%	24%	-3%	-5%	4%	29%	-4%	-11%					
YoY growth (%)																	
Revenue	12%	6%	12%	18%	24%	36%	40%	26%	19%	14%	18%	16%	3%	12%	31%	17%	13%
Gross profit	5%	3%	11%	22%	34%	47%	47%	28%	20%	17%	22%	17%	4%	10%	39%	19%	14%
OPEX	6%	3%	-1%	7%	14%	29%	25%	23%	19%	5%	11%	13%	8%	4%	23%	12%	11%
Operating income	5%	3%	19%	32%	46%	56%	59%	31%	21%	21%	26%	19%	2%	14%	48%	22%	16%
Net income	6%	-10%	21%	29%	38%	63%	53%	30%	19%	23%	21%	14%	9%	10%	46%	19%	16%

Source: Company data, Goldman Sachs Global Investment Research

Investment view: Accelerating monthly YoY from smaller segments could be a surprise. Maintain positive view on AirTAC

We reiterate our Buy rating on AirTAC as we continue to believe the company is a key beneficiary of the factory automation/industrial automation localization trend in mainland China, and we expect the company will also benefit from the prolonged upcycle supported by the ongoing government stimulus policies. We see multiple key end-segments' performance tracking ahead of the company's 2026 full year guidance of 20%+ YoY growth (see here), with battery/textile/machine tools/electronics delivering 55%/46%/41%/23% YTD YoY growth while total revenue in Rmb is tracking at 23% YTD YoY growth.

We continue to see solid monthly YoY growth from smaller segments, such as energy lighting and textile. However, the battery segment, which is the key growth driver in 2025 (with 89% YoY in 2025), is seeing its first negative monthly YoY growth since Oct 2024, while other major contribution sectors, such as electronics and automotive are seeing decelerating monthly YoY growth. As a result, we remain cautious on the potential of exaggerated orders (including advance or duplicate orders) despite new orders intake continuing to outpace shipments, and we will continue to monitor the underlying end-demand closely.

Maintain Buy on AirTAC with a lower 12m TP of NT$1,895 (from NT$1,925).

Investment Thesis - AirTAC

AirTAC is the 2nd-largest pneumatic supplier in mainland China with 28% market share in 2025, outpacing most local competitors, owing to its low-cost yet high quality product, and service quality reputation in most key cities in mainland China. We forecast its market share will reach 30% in 2026E and expect AirTAC to be the structural winner in a deflationary market, due to its: (1) strong market growth and leadership: AirTAC continues to expand its market share in mainland China driven by competitive pricing that is more than 10% lower vs. international peers (with 45%+ GM / 30-35% OPM) and high demand elasticity among mainland China customers; and (2) superior cost structure and financial health: with best-in-class cash cost and a solid balance sheet, featuring positive net cash over FCF since 2023/2024, we believe AirTAC is well-equipped to weather industry downturns and capitalize on market restructuring.

Overall, we believe AirTAC will see better revenue/profitability in the long term given its share gainer position in the China pneumatic market and potential to gain share in the linear guide market (improved demand in the next 1-2 years). This should lead to strengthened valuation, vs. its recent trading range (trading below its 10-year avg. P/E multiple). We are Buy rated on AirTAC.

Price target risks and methodology - AirTAC

Valuation methodology: Our 12m TP of NT$1,895 is based on a target P/E of 26.0x on our 2027E EPS (0.25x STDV above avg of the last pneumatic upcycle).

Key downside risks: (1) Stronger competition from peers; (2) sharper-than-expected Rmb depreciation, and (3) a longer-than-expected end-market demand downtrend in mainland China.

1590.TW	12m Price Target: **NT$1,895.00**	Price: **NT$1,210.00**	Upside: **56.6%**

Buy

Market cap: NT$231.4bn / $7.2bn
Enterprise value: NT$221.0bn / $6.9bn
3m ADTV: NT$1.2bn / $36.0mn
Taiwan
Taiwan Electronic Components
M&A Rank: 3
Leases incl. in net debt & EV?: No

GS Forecast

	12/25	12/26E	12/27E	12/28E
Revenue (NT$ mn) New	**34,329.9**	**45,096.4**	**52,655.1**	**59,393.1**
Revenue (NT$ mn) Old	34,329.9	45,643.9	53,446.1	60,126.1
EBITDA (NT$ mn)	12,865.9	17,437.2	20,950.4	23,958.4
EPS (NT$) New	**42.00**	**61.12**	**72.93**	**84.42**
EPS (NT$) Old	42.00	61.95	74.00	85.29
P/E (X)	20.9	19.8	16.6	14.3
P/B (X)	3.3	4.3	3.9	3.6
Dividend yield (%)	3.3	3.3	3.9	4.5
CROCI (%)	19.2	23.8	27.5	30.0

	6/26	9/26E	12/26E	3/27E
EPS (NT$)	16.55	16.04	15.17	15.84

Source: Company data, Goldman Sachs Research estimates, FactSet. Price as of 12 Oct 2026 close.

Disclosure Appendix

Reg AC

We, Chao Wang, Yuichiro Isayama and Al Wang, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Chao Wang Goldman Sachs (Asia) L.L.C., Taipei Branch, Yuichiro Isayama Goldman Sachs Japan Co., Ltd., Al Wang Goldman Sachs (Asia) L.L.C., Taipei Branch.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

M&A Rank

Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

The rating(s) for Airtac International Group is/are relative to the other companies in its/their coverage universe: AWSC, Airtac International Group, Co-Tech Development Corp., Delta Electronics, Elite Material, GCE, Hiwin Corp., ITEQ Corp, Kinsus, Lotes, NYPCB, Taiwan Union Technology Corp., Unimicron Technology, Win Semiconductors Corp., Yageo Corp., Zhen Ding Technology Holding

Company-specific regulatory disclosures

The following disclosures relate to relationships between The Goldman Sachs Group, Inc. (with its affiliates, "Goldman Sachs") and companies covered by Goldman Sachs Global Investment Research and referred to in this research.

Goldman Sachs expects to receive or intends to seek compensation for investment banking services in the next 3 months: Airtac International Group (NT$1,210.00)

Goldman Sachs had an investment banking services client relationship during the past 12 months with: Airtac International Group (NT$1,210.00)

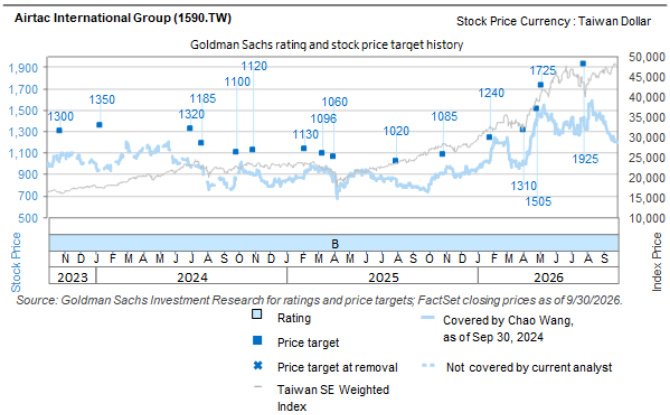
Distribution of ratings/investment banking relationships

Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution			Investment Banking Relationships		
	Buy	Hold	Sell	Buy	Hold	Sell
Global	50%	34%	16%	65%	58%	43%

As of October 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,094 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

Price target and rating history chart(s)



The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.

Target price history table(s)

Airtac International Group (1590.TW)

Date of report	Target price (NT$)	Closing price (NT$)
28-Jul-26	1,925.00	1,365.00
07-May-26	1,725.00	1,490.00
29-Apr-26	1,505.00	1,490.00
02-Apr-26	1,310.00	996.00
29-Jan-26	1,240.00	1,040.00
30-Oct-25	1,085.00	904.00
31-Jul-25	1,020.00	856.00
03-Apr-25	1,060.00	900.00
13-Mar-25	1,096.00	938.00
07-Feb-25	1,130.00	934.00
31-Oct-24	1,120.00	894.00
29-Sep-24	1,100.00	939.00
24-Jul-24	1,185.00	970.00
02-Jul-24	1,320.00	972.00
11-Jan-24	1,350.00	958.00
25-Oct-23	1,300.00	1,060.00

Price targets shown in table(s) are unadjusted for corporate actions.

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; registered market maker. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such

advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at https://www.gs.com/worldwide/brazil/area/gir/index.html. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details, a copy of the annual compliance audit report and other relevant information and disclosures can be found at this link: https://www.goldmansachs.com/worldwide/india/research-analyst. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at https://www.gs.com/disclosures/europeanpolicy.html which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Ratings, coverage universe and related definitions

Buy (B), Neutral (N), Sell (S) Analysts recommend stocks as Buys or Sells for inclusion on various regional Investment Lists. Being assigned a Buy or Sell on an Investment List is determined by a stock's total return potential relative to its coverage universe. Any stock not assigned as a Buy or a Sell on an Investment List with an active rating (i.e., a stock that is not Rating Suspended, Not Rated, Early-Stage Biotech, Coverage Suspended or Not Covered), is deemed Neutral. Each region manages Regional Conviction Lists, which are selected from Buy rated stocks on the respective region's Investment Lists and represent investment recommendations focused on the size of the total return potential and/or the likelihood of the realization of the return across their respective areas of coverage. The addition or removal of stocks from such Conviction Lists are managed by the Investment Review Committee or other designated committee in each respective region and do not represent a change in the analysts' investment rating for such stocks.

Total return potential represents the upside or downside differential between the current share price and the price target, including all paid or anticipated dividends, expected during the time horizon associated with the price target. Price targets are required for all covered stocks. The total return potential, price target and associated time horizon are stated in each report adding or reiterating an Investment List membership.

Coverage Universe: A list of all stocks in each coverage universe is available by primary analyst, stock and coverage universe at https://www.gs.com/research/hedge.html.

Not Rated (NR). The investment rating, target price and earnings estimates (where relevant) are removed pursuant to Goldman Sachs policy when Goldman Sachs is acting in an advisory capacity in a merger or in a strategic transaction involving this company, when there are legal, regulatory or policy constraints due to Goldman Sachs' involvement in a transaction, and in certain other circumstances. **Early-Stage Biotech (ES).** An investment rating and a target price are not assigned pursuant to Goldman Sachs policy when this company has neither a drug, treatment or medical device that has passed a Phase II clinical trial nor a license to distribute a post-Phase II drug, treatment or medical device. **Rating Suspended (RS).** Goldman Sachs Research has suspended the investment rating and price target for this stock, because there is not a sufficient fundamental basis for determining an investment rating or target price. The previous investment rating and target price, if any, are no longer in effect for this stock and should not be relied upon. **Coverage Suspended (CS).** Goldman Sachs has suspended coverage of this company. **Not Covered (NC).** Goldman Sachs does not cover this company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based

in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (https://www.sipc.org).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

The analysts named in this report may have from time to time discussed with our clients, including Goldman Sachs salespersons and traders, or may discuss in this report, trading strategies that reference catalysts or events that may have a near-term impact on the market price of the equity securities discussed in this report, which impact may be directionally counter to the analyst's published price target expectations for such stocks. Any such trading strategies are distinct from and do not affect the analyst's fundamental equity rating for such stocks, which rating reflects a stock's return potential relative to its coverage universe as described herein.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, make a market or act as liquidity provider in, and buy or sell, any securities, derivatives, or other financial instruments referred to in this research (including any financial instruments issued by the companies referred to herein), unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at https://www.theocc.com/about/publications/character-risks.jsp and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to https://research.gs.com.

Disclosure information is also available at https://www.gs.com/research/hedge.html or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative

works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. You may not use, extract, download or retrieve this information, in whole or in part, to train, finetune, adjust, influence the weights or behavior of, or otherwise develop or improve a machine learning or artificial intelligence model or system, or provide or reproduce this information, in whole or in part, as a prompt or input to any such system without the express written consent of Goldman Sachs. Furthermore, you may not bulk download this information, or employ any robot, spider, crawler, scraper, automated agent, or other electronic means to scrape, harvest, extract, or retrieve this information, without the express written consent of Goldman Sachs. A breach of the above terms may result in termination of your right to access this service.

Nanya Technology

Upbeat 3Q margin/ASP, and 2027 guidance also optimistic: lift PO to NT$800

Reiterate Rating: BUY | PO: 800.00 TWD | Price: 527.00 TWD

3Q results confirm super-cycle even with legacy DRAM

Nanya Tech, a pure DRAM supplier, reported very strong 3Q results (sales: NT$134bn, +62% QoQ / +612% YoY; OPM: 80% - similar to big-3 DRAM makers'; net margin: 66%). The key contributor should be upbeat ASP (up 60%+ QoQ vs consensus of 30-40%). In fact, Nanya Tech had already reported more than doubled 1H ASP (60%+ QoQ for two consecutive quarters). Thus, Street expected limited QoQ price increase in 3Q (vs 1H). According to Nanya Tech President Pei Ing Lee, demand for legacy DRAM products (DDR3 & DDR4) remained strong across IT, industrial, and enterprise applications. Mgmt also said legacy DRAM accounted for 80%+ of 3Q sales (vs 10% range for DDR5). Net-net, the 3Q results confirmed super-cycle even selling mainly legacy/conventional DRAM.

Guidance broadly bullish; no sign of a price cut

Lee provided bullish guidance (earnings call). What we learned: (1) rising 4Q ASP, (2) ongoing shortage of legacy DRAM through 2027 (no immediate demand shift from DDR4 to DDR5), (3) new fab mass production (2027-28), (4) significant FCF generation (with high margin profile) despite large capex spend (NT$70bn in 2026), (5) record-high net cash (NT$270bn in 3Q), (6) new node development (in-house) and migration (1c, 1d, 1e) on track, and (7) new construction plan for 3D wafer packaging/testing facilities in Ping Tung Science Park, Taiwan. Overall, new guidance presents further growth (capacity expansion) and high margin profile (favorable ASP trend).

Lift PO to NT$800 on revisions to 2027/28 EPS estimates

We lift our PO from NT$750 to NT$800 using the same target P/E vs earlier (9x 2027-28E EPS). Thus, our PO revision is mostly based on 7-8% revisions to 2027-28 EPS estimates. We assume 5%/1ppt higher sales/OPM vs our earlier forecasts given more favorable conventional DRAM ASP trend. To be conservative, we assume negative EPS growth for 2028 but it can turn positive if the ASP remains stable. We reiterate Buy.

Estimates (Dec) (NT$)	2024A	2025A	2026E	2027E	2028E
Net Income (Adjusted - mn)	-5,083	6,616	251,840	360,077	293,015
EPS	-1.6	2.1	74.5	103.7	84.4
EPS Change (YoY)	31.7%	NM	NM	39.1%	-18.6%
Consensus EPS (Visible Alpha)			70.74	112.55	118.06
Dividend / Share	0	1.35	22.36	41.47	42.18
Free Cash Flow / Share	-4.58	1.65	59.11	86.13	63.96
Valuation (Dec)					
P/E	NM	248.7x	7.1x	5.1x	6.2x
Dividend Yield	0%	0.3%	4.2%	7.9%	8.0%
EV / EBITDA*	270.5x	77.4x	4.6x	3.1x	3.7x
Free Cash Flow Yield*	-0.8%	0.3%	10.9%	16.3%	12.1%

* For full definitions of iQmethod℠ measures, see page 18.

This research report provides general information only. No part of this report may be used or reproduced or quoted in any manner whatsoever in Taiwan by the press or other persons without the express written consent of BofA Securities.
>> Employed by a non-US affiliate of BofAS and is not registered/qualified as a research analyst under the FINRA rules.
Refer to "Other Important Disclosures" for information on certain BofA Securities entities that take responsibility for the information herein in particular jurisdictions.
BofA Securities does and seeks to do business with issuers covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.
Refer to important disclosures on page 19 to 21. Analyst Certification on page 16. Price

12 October 2026

Equity

Key Changes

(NT$)	Previous	Current
Price Obj.	750.00	800.00
2026E EPS	70.5	74.5
2027E EPS	96.5	103.7
2028E EPS	77.8	84.4
2026E EBITDA (m)	314,569	325,705
2027E EBITDA (m)	455,240	488,807
2028E EBITDA (m)	371,834	404,165

Simon Woo, CFA >>
Research Analyst
Merrill Lynch (Seoul)
+82 2 3707 0554
simon.woo@bofa.com

Dai Shen >>
Research Analyst
Merrill Lynch (Hong Kong)
dai.shen@bofa.com

Matt Shin >>
Research Analyst
Merrill Lynch (Seoul)
matt.shin2@bofa.com

Stock Data

Price	527.00 TWD
Price Objective	800.00 TWD
Date Established	12-Oct-2026
Investment Opinion	B-1-7
52-Week Range	85.00 TWD-569.00 TWD
Mrkt Val / Shares Out (mn)	56,944 USD / 3,450.2
Market Value (mn)	1,818,259 TWD
Average Daily Value (mn)	962.33 USD
Free Float	46.5%
BofA Ticker / Exchange	NNYAF / TAI
Bloomberg / Reuters	2408 TT / 2408.TW
ROE (2026E)	75.3%
Net Dbt to Eqty (Dec-2025A)	-22.8%

ASP: Average selling price
DDR3/4/5: 3rd/4th/5th gen DRAM
DRAM: Dynamic random-access memory
OPM: Operating profit margin

iQprofile℠ Nanya Technology

Key Income Statement Data (Dec)	2024A	2025A	2026E	2027E	2028E
(NT$ Millions)					
Sales	34,132	66,587	410,616	633,254	617,223
Gross Profit	(421)	14,986	336,626	508,978	423,265
Sell General & Admin Expense	(10,134)	(9,741)	(23,086)	(34,795)	(38,065)
Operating Profit	(10,555)	5,245	313,540	474,183	385,201
Net Interest & Other Income	3,998	2,639	7,122	5,920	5,487
Associates	NA	NA	NA	NA	NA
Pretax Income	(6,557)	7,884	320,662	480,103	390,687
Tax (expense) / Benefit	1,474	(1,269)	(68,823)	(120,026)	(97,672)
Net Income (Adjusted)	(5,083)	6,616	251,840	360,077	293,015
Average Fully Diluted Shares Outstanding	3,122	3,122	3,379	3,474	3,474
Key Cash Flow Statement Data					
Net Income	(5,083)	6,616	251,840	360,077	293,015
Depreciation & Amortization	16,144	14,281	12,165	14,624	18,964
Change in Working Capital	(5,772)	(1,973)	5,993	2,456	(1,296)
Deferred Taxation Charge	NA	NA	NA	NA	NA
Other Adjustments, Net	(3,337)	(355)	(1,935)	0	0
Cash Flow from Operations	1,952	18,568	268,063	377,157	310,684
Capital Expenditure	(16,143)	(13,446)	(69,700)	(80,000)	(90,000)
(Acquisition) / Disposal of Investments	NA	NA	NA	NA	NA
Other Cash Inflow / (Outflow)	(45)	(664)	(201)	0	0
Cash Flow from Investing	(16,188)	(14,110)	(69,901)	(80,000)	(90,000)
Shares Issue / (Repurchase)	13	246	79,556	0	0
Cost of Dividends Paid	0	0	(4,658)	(77,148)	(143,065)
Cash Flow from Financing	13,707	(6,317)	72,653	(80,228)	(159,531)
Free Cash Flow	(14,191)	5,123	198,363	297,157	220,684
Net Debt	(36,638)	(38,818)	(311,802)	(528,730)	(603,280)
Change in Net Debt	10,993	(2,181)	(272,983)	(216,929)	(74,550)
Key Balance Sheet Data					
Property, Plant & Equipment	84,327	85,031	148,067	216,794	291,049
Other Non-Current Assets	14,433	14,881	13,405	13,358	13,323
Trade Receivables	4,132	16,546	46,486	42,150	38,218
Cash & Equivalents	61,903	58,074	330,199	547,127	608,280
Other Current Assets	41,911	33,921	27,425	26,615	25,869
Total Assets	206,706	208,453	565,582	846,045	976,740
Long-Term Debt	3,995	14,196	18,397	18,397	5,000
Other Non-Current Liabilities	4,878	4,698	5,500	5,723	5,838
Short-Term Debt	21,270	5,060	0	0	0
Other Current Liabilities	11,510	13,960	43,397	40,707	34,734
Total Liabilities	41,653	37,914	67,294	64,827	45,572
Total Equity	165,053	170,538	498,287	781,217	931,168
Total Equity & Liabilities	206,706	208,453	565,582	846,045	976,740
iQmethod℠ **- Bus Performance***					
Return On Capital Employed	-4.3%	2.3%	68.7%	53.6%	33.1%
Return On Equity	-3.1%	3.9%	75.3%	56.3%	34.2%
Operating Margin	-30.9%	7.9%	76.4%	74.9%	62.4%
EBITDA Margin	16.4%	29.3%	79.3%	77.2%	65.5%
iQmethod℠ **- Quality of Earnings***					
Cash Realization Ratio	NM	2.8x	1.1x	1.0x	1.1x
Asset Replacement Ratio	1.0x	0.9x	5.7x	5.5x	4.7x
Tax Rate (Reported)	22.5%	16.1%	21.5%	25.0%	25.0%
Net Debt-to-Equity Ratio	-22.2%	-22.8%	-62.6%	-67.7%	-64.8%
Interest Cover	NA	NA	NA	NA	NA

Key Metrics

* For full definitions of *iQmethod*℠ measures, see page 18.

Company Sector

Semiconductors

Company Description

Nanya Technology is an affiliate of Formosa Plastics Group. The firm manufactures and distributes DRAM chips. Management focuses on non-PC applications such as consumer, mobile, server, autos, etc. Key customers are based in China and Taiwan. Samsung, SK Hynix and Micron are major competitors.

Investment Rationale

Our Buy rating is based on the DRAM industry's supercycle, which should warrant a high-margin profile for Nanya Tech even in the long term. Its 3Q26 results proved record-high revenue (NT$134bn, more than 5x higher than the previous peak cycle) and 80% operating margin. Management focus on long-term contracts, including DDR4 for eSSD or servers, should also act as a catalyst. Significantly rising net cash even after capex hike could also support higher dividend payouts.

Stock Data

Price to Book Value	3.7x

Key charts/tables after 3Q26 results

Exhibit 1: Exceptionally strong 3Q results (sales: NT$134bn; +62% QoQ or +612% YoY; OPM: 80% - similar to big-3 DRAM makers; net margin: 66%). The key contributor should be upbeat ASP (up by 60%+ QoQ vs consensus of 30-40%).

Nanya Tech – 3Q26 earnings vs consensus and our estimate

(NT$mn, NT$)	Actual	QoQ	YoY	Cons	Act vs Cons	Our est	Act vs our est
Sales	**133,649**	**62%**	**612%**	**133,354**	**0%**	**132,157**	**1%**
GP	114,482	74%	3204%	113,837	1%	110,181	4%
OP	**106,894**	**76%**	**9451%**	**107,724**	**-1%**	**105,081**	**2%**
Pre-tax profit	109,332	75%	5305%	109,629	0%	106,417	3%
Effective tax rate	20%	2%	-13%	19%	2%	24%	-17%
NP	**87,627**	**75%**	**5505%**	**88,266**	**-1%**	**80,877**	**8%**
EPS (FD)	**25.23**	**73%**	**4938%**	**25.65**	**-2%**	**23.28**	**8%**
Implied shares (mn)	3,474	1%	11%	3,441	1%	3,474	0%
Gross margin	85.7%	6.2%pt	67.2%pt	85.4%	0.3%pt	83.4%	2.3%pt
OP margin	80.0%	18.6%pt	74.0%pt	80.8%	-0.8%pt	79.5%	0.5%pt
Net margin	65.6%	12.5%pt	57.2%pt	66.2%	-0.6%pt	61.2%	4.4%pt

Source: Company, Bloomberg, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 2: Our more positive 4Q26 earnings estimates compared to that of consensus are primarily based on stronger price assumptions (DRAM ASP up 10% QoQ assumed)

Nanya Tech – 2026 quarterly earnings; consensus vs our estimate

(NT$mn, NT$)	1Q26 Actual	2Q26 Actual	3Q26 Actual	4Q26E Our est	4Q26E Cons	Diff
Sales	**49,087**	**82,549**	**133,649**	**145,331**	**138,726**	**5%**
GP	33,316	65,619	114,482	123,208	118,750	4%
OP	**30,111**	**60,826**	**106,894**	**115,708**	**112,321**	**3%**
Pre-tax profit	31,718	62,329	109,332	117,283	114,486	2%
Effective tax rate	18%	19%	20%	25%	20%	5%
NP	**26,059**	**50,192**	**87,627**	**87,962**	**91,735**	**-4%**
EPS (FD)	**8.35**	**14.56**	**25.23**	**25.32**	**26.76**	**-5%**
Implied shares (mn)	3122	3446	3474	3474	3428	1%
Gross margin	*67.9%*	*79.5%*	*85.7%*	*84.8%*	*85.6%*	*-0.8%pt*
OP margin	*61.3%*	*73.7%*	*80.0%*	*79.6%*	*81.0%*	*-1.3%pt*
Net margin	*53.1%*	*60.8%*	*65.6%*	*60.5%*	*66.1%*	*-5.6%pt*

Source: Company, Bloomberg, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 3: We expect record-high margins to continue in 2026-27E (OPM 75%+) and mild erosion assumed for 2028E (62%)

Nanya Tech – 2026-28 annual earnings; consensus vs our estimate

(NT$mn, NT$)	2025	2026E			2027E			2028E		
	Actual	Our est	Cons	Diff	Our est	Cons	Diff	Our est	Cons	Diff
Sales	**66,587**	**410,616**	**403,146**	**2%**	**633,254**	**632,924**	**0%**	**617,223**	**818,254**	**-25%**
GP	14,986	336,626	331,092	2%	508,978	535,479	-5%	423,265	656,813	-36%
OP	**5,245**	**313,540**	**311,121**	**1%**	**474,183**	**512,340**	**-7%**	**385,201**	**616,299**	**-37%**
Pre-tax profit	7,884	320,662	317,056	1%	480,103	515,978	-7%	390,687	657,969	-41%
Effective tax rate	16%	21%	19%	2%	25%	20%	5%	25%	19%	6%
NP	**6,616**	**251,840**	**256,071**	**-2%**	**360,077**	**414,193**	**-13%**	**293,015**	**534,271**	**-45%**
EPS (FD)	**2.12**	**74.53**	**73.77**	**1%**	**103.66**	**115.77**	**-10%**	**84.36**	**155.37**	**-46%**
Implied shares (mn)	3122	3379	3471	-3%	3474	3578	-3%	3474	3439	1%
Gross margin	*22.5%*	*82.0%*	*82.1%*	*-0.1%pt*	*80.4%*	*84.6%*	*-4.2%pt*	*68.6%*	*80.3%*	*-11.7%pt*
OP margin	*7.9%*	*76.4%*	*77.2%*	*-0.8%pt*	*74.9%*	*80.9%*	*-6.1%pt*	*62.4%*	*75.3%*	*-12.9%pt*
Net margin	*9.9%*	*61.3%*	*63.5%*	*-2.2%pt*	*56.9%*	*65.4%*	*-8.6%pt*	*47.5%*	*65.3%*	*-17.8%pt*

Source: Company, Bloomberg, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 4: Our 7-8% higher 2027-28E EPS estimates are mostly based on our 5%/1%pt higher sales/OPM assumptions vs our earlier forecasts given more favorable conventional DRAM ASP trend

Nanya Tech – Earnings revisions (2026-28E)

(NT$bn, NT$)	1Q26	2Q26	3Q26	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2022	2023	2024	2025	2026E	2027E	2028E
EPS (FD)															
New	8.3	14.6	25.2	25.3	25.9	26.2	26.5	25.1	4.7	-2.4	-1.6	2.1	74.5	103.7	84.4
Old	8.3	14.6	23.3	23.3	24.0	24.5	24.7	23.4	4.7	-2.4	-1.6	2.1	70.5	96.5	77.8
Diff	n/a	n/a	8%	9%	8%	7%	7%	7%	n/a	n/a	n/a	n/a	6%	7%	8%
Dividend/share															
New	0.0	1.4	0.0	0.0	0.0	22.4	0.0	0.0	2.1	0.0	0.0	1.4	22.4	41.5	42.2
Old	0.0	1.4	0.0	0.0	0.0	21.1	0.0	0.0	2.1	0.0	0.0	1.4	21.1	38.6	38.9
Difference	n/a	n/a	nm	nm	nm	nm	nm	nm	n/a	n/a	n/a	n/a	6%	7%	8%
Sales total															
New	49.1	82.5	133.6	145.3	151.6	157.2	164.1	160.3	57.0	29.9	34.1	66.6	410.6	633.3	617.2
Old	49.1	82.5	132.2	137.3	143.8	149.0	155.6	152.0	57.0	29.9	34.1	66.6	401.1	600.4	585.2
Diff	n/a	n/a	1%	6%	5%	5%	5%	5%	n/a	n/a	n/a	n/a	2%	5%	5%
OP margin															
New	61.3%	73.7%	80.0%	79.6%	78.0%	76.3%	73.8%	71.7%	19.3%	-48.4%	-30.9%	7.9%	76.4%	74.9%	62.4%
Old	61.3%	73.7%	79.5%	77.7%	76.4%	75.2%	72.7%	70.4%	19.3%	-48.4%	-30.9%	7.9%	75.5%	73.6%	60.8%
Diff	n/a	n/a	0.5%	1.9%	1.6%	1.0%	1.1%	1.2%	n/a	n/a	n/a	n/a	0.9%	1.2%	1.6%
EBITDA															
New	33.0	63.8	110.0	118.9	121.6	123.4	124.9	118.9	26.2	0.9	5.6	19.5	325.7	488.8	404.2
Old	33.0	63.8	108.1	109.7	112.9	115.3	116.5	110.6	26.2	0.9	5.6	19.5	314.6	455.2	371.8
Diff	n/a	n/a	2%	8%	8%	7%	7%	8%	n/a	n/a	n/a	n/a	4%	7%	9%
Capex															
New	2.8	4.0	20.8	42.1	20.0	18.0	17.0	25.0	20.7	13.2	16.1	13.4	69.7	80.0	90.0
Old	2.8	4.0	15.0	28.1	18.0	16.0	17.0	19.0	20.7	13.2	16.1	13.4	50.0	70.0	70.0
Diff	n/a	n/a	39%	49%	11%	13%	0%	32%	n/a	n/a	n/a	n/a	39%	14%	29%
DRAM sales (US$mn)															
New	1,551	2,612	4,176	4,509	4,742	4,935	5,175	5,069	1,935	958	1,062	2,156	12,847	19,920	19,602
Old	1,551	2,612	4,004	4,175	4,397	4,581	4,809	4,710	1,935	958	1,062	2,156	12,342	18,497	18,210
Diff	n/a	n/a	4%	8%	8%	8%	8%	8%	n/a	n/a	n/a	n/a	4%	8%	8%
Shipments (8Gb, mn)															
New	220	220	209	205	211	219	237	239	447	429	450	705	853	906	1,043
Old	220	220	235	238	245	254	275	278	447	429	450	705	912	1,052	1,211
Diff	n/a	n/a	-11%	-14%	-14%	-14%	-14%	-14%	n/a	n/a	n/a	n/a	-6%	-14%	-14%
ASP US$/8b equiv															
New	7.90	12.70	20.38	22.42	22.87	22.87	22.18	21.52	4.49	2.52	2.86	3.54	15.68	22.33	19.08
Old	7.90	12.70	17.78	18.31	18.68	18.68	18.12	17.58	4.49	2.52	2.86	3.54	14.31	18.24	15.58
Diff	n/a	n/a	15%	22%	22%	22%	22%	22%	n/a	n/a	n/a	n/a	10%	22%	22%
ASP chg (QoQ/YoY)															
New	71%	61%	61%	10%	2%	0%	-3%	-3%	-10%	-44%	13%	24%	343%	42%	-15%
Old	71%	61%	40%	3%	2%	0%	-3%	-3%	-10%	-44%	13%	24%	305%	27%	-15%
Diff	n/a	n/a	21%	7%	0%	0%	0%	0%	n/a	n/a	n/a	n/a	39%	15%	0%
Cost US$/8Gb equiv															
New	3.05	3.34	4.08	4.57	5.03	5.43	5.81	6.10	3.61	3.75	3.74	3.30	3.74	5.61	7.17
Old	3.05	3.34	3.64	4.08	4.41	4.63	4.95	5.20	3.61	3.75	3.74	3.30	3.54	4.81	6.11
Diff	n/a	n/a	44%	49%	62%	80%	86%	90%	n/a	n/a	n/a	n/a	20%	80%	106%
Bit growth (QoQ/YoY)															
New	-5%	0%	-5%	-2%	3%	4%	8%	1%	-27%	-4%	5%	57%	21%	6%	15%
Old	-5%	0%	7%	1%	3%	4%	8%	1%	-27%	-4%	5%	57%	29%	15%	15%
Diff	n/a	n/a	-12%	-3%	0%	0%	0%	0%	n/a	n/a	n/a	n/a	-8%	-9%	0%

Source: Company, BofA Global Research estimates

Exhibit 5: We expect DRAM ASP to rise 10% QoQ in 4Q and eventually low-single% hike in 1H27 and mild decline in 2H27 (down low-single%) after 40%+ QoQ growth each quarter during 2H25-3Q26

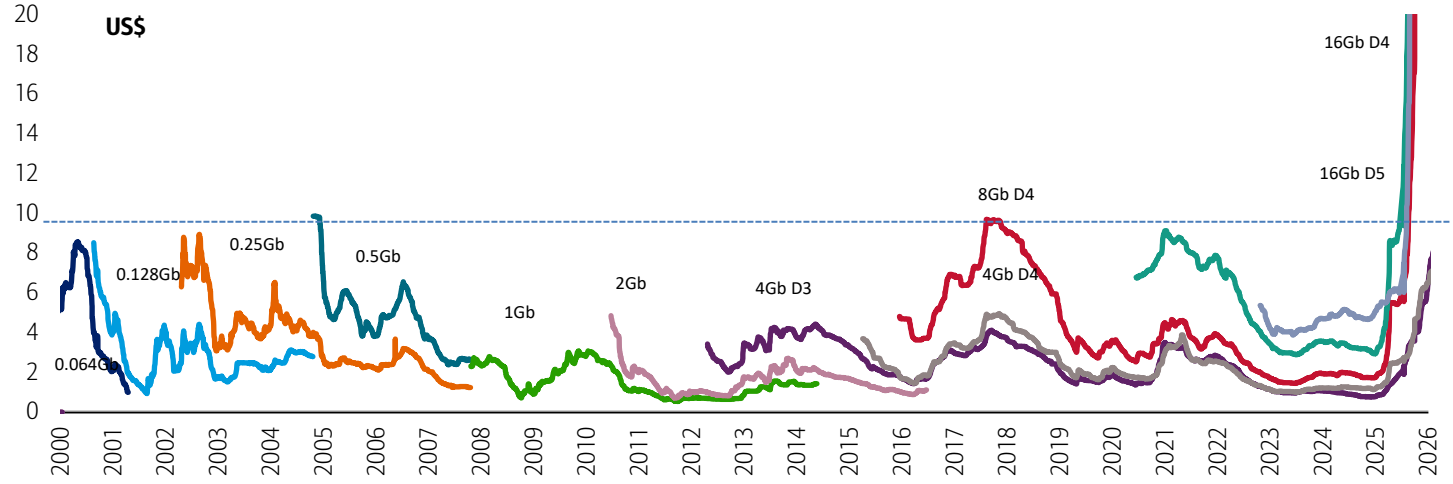
Nanya Tech – DRAM assumptions (2026-28E)

(8Gb equiv)	1Q26	2Q26	3Q26	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2021	2022	2023	2024	2025	2026E	2027E	2028E
Sales US$mn	**1,551**	**2,612**	**4,176**	**4,509**	**4,742**	**4,935**	**5,175**	**5,069**	**3,066**	**1,935**	**958**	**1,062**	**2,156**	**12,847**	**19,920**	**19,602**
Units 8Gb equiv mn	220	220	209	205	211	219	237	239	612	447	429	450	705	853	906	1,043
ASP US$/8Gb	**7.9**	**12.7**	**20.4**	**22.4**	**22.9**	**22.9**	**22.2**	**21.5**	**5.0**	**4.5**	**2.5**	**2.9**	**3.5**	**15.7**	**22.3**	**19.1**
Per unit full cost US$/8Gb	3.1	3.3	4.1	4.6	5.0	5.4	5.8	6.1	3.4	3.6	3.7	3.7	3.3	3.7	5.6	7.2
Full cost US$mn (COGS+SG&A)	671	735	852	935	1,060	1,190	1,376	1,459	2,083	1,617	1,607	1,681	2,325	3,193	5,084	7,482
OP margin	**61%**	**74%**	**80%**	**80%**	**78%**	**76%**	**74%**	**72%**	**32%**	**20%**	**-48%**	**-31%**	**7%**	**76%**	**75%**	**62%**
Depreciation+amort US$mn	92	94	97	101	107	112	118	130	548	511	492	538	476	405	487	632
EBITDA US$mn	1,043	2,019	3,437	3,755	3,868	3,936	3,995	3,818	1,521	879	28	186	651	10,857	16,294	13,472
EBITDA margin	67%	77%	82%	83%	82%	80%	77%	75%	50%	45%	3%	18%	30%	85%	82%	69%
Capex spend US$mn	89	128	649	1328	636	574	544	803	403	694	425	538	448	2,323	2,667	3,000
%QoQ																
Sales	62%	61%	52%	8%	5%	4%	5%	-2%								
Bit growth	-5%	0%	-5%	-2%	3%	4%	8%	1%								
ASP	71%	61%	61%	10%	2%	0%	-3%	-3%								
Per unit full cost	9%	9%	22%	12%	10%	8%	7%	5%								
%YoY																
Sales	564%	561%	476%	329%	178%	80%	23%	12%	47%	-34%	-46%	19%	94%	437%	51%	-2%
Bit growth	124%	32%	0%	-12%	-4%	0%	13%	17%	0%	-27%	-4%	5%	57%	21%	6%	15%
ASP	196%	401%	475%	385%	190%	80%	9%	-4%	47%	-10%	-44%	13%	24%	343%	42%	-15%
Per unit full cost	-20%	-8%	22%	63%	65%	62%	42%	33%	16%	6%	4%	0%	-12%	13%	50%	28%
DRAM wafer capa (K wpm)																
30nm & above	10	9	9	8	8	7	7	6	20	18	16	14	11	9	7	5
20nm (incl. new upgrade)	26	25	24	22	21	20	19	18	55	51	39	38	28	24	20	16
10nm-class (1b or below)	27	31	35	38	46	53	61	66	0	0	0	4	18	33	56	76
Total capacity, k wpm	**62**	**65**	**67**	**68**	**74**	**80**	**86**	**90**	**75**	**69**	**55**	**55**	**57**	**66**	**83**	**96**
% of 20nm or below to total	85%	86%	87%	88%	90%	91%	92%	93%	73%	74%	71%	75%	81%	87%	92%	95%

Source: Company, BofA Global Research estimates

DRAM spot/contract price trend

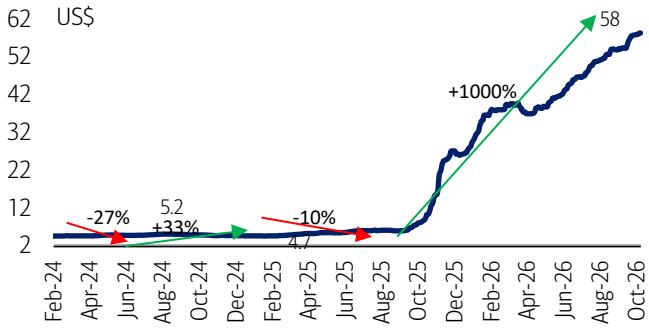
Exhibit 6: DRAM spot prices rebounded gradually through June/July/Aug/Sep, to hit all-time high levels. Prices have reached multi-decade highs, with 16Gb DDR5 (for PC) at ~$58 and DDR4 at ~$85, driven by supply reallocation toward HBM and server DRAM—well above the prior ~$10 peak in Oct-17
DRAM spot price – long-term trend (2000-2026)



Source: DRAMeXchange

BofA GLOBAL RESEARCH

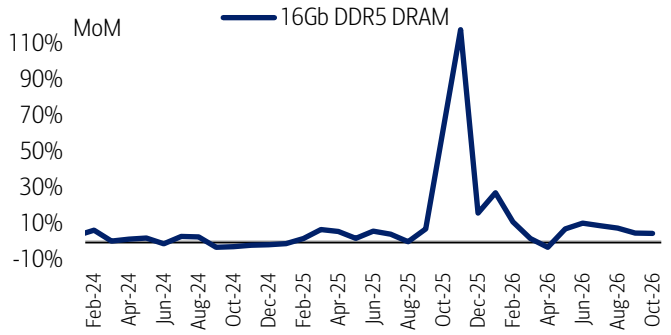
Exhibit 7: Price recovery extended from May to Sep and early-Oct, reaching a record high of $58, driven by sharp gains in Oct (+70%), Nov (+60%), and Jan (+25%), before moderating during Feb–Apr.
16Gb DDR5 spot price – daily, Feb '24 - Oct '26



Source: DRAMeXchange

BofA GLOBAL RESEARCH

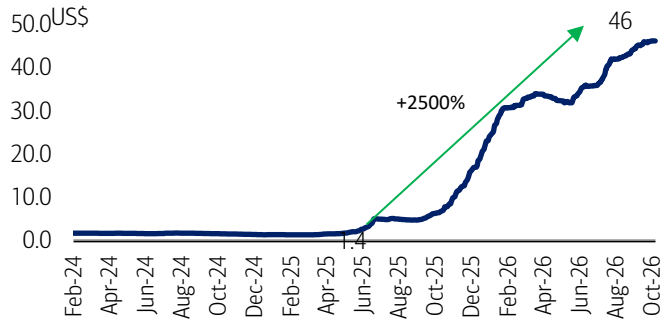
Exhibit 8: Still up 4-5% each in Sep/early-Oct following a modest uptick observed through May/Jun/Jul/Aug (+7-10%)
16Gb DDR5 spot month-average price – MoM change, Feb '24 - Oct '26



Source: DRAMeXchange

BofA GLOBAL RESEARCH

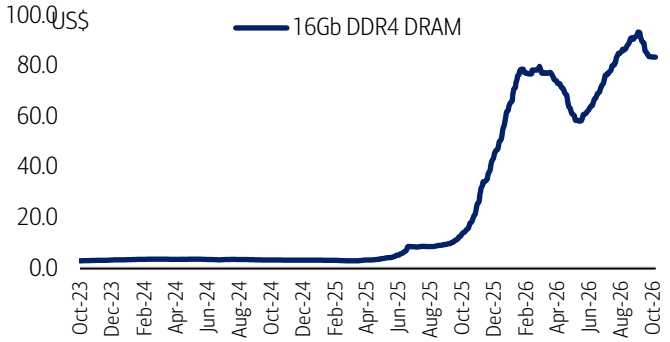
Exhibit 9: 8Gb DDR4 spot prices stable at $45 level, still well above the prior ~$10 peak in October 2017, driven by ongoing supply cuts from major vendors
8Gb DDR4 spot price trend, Feb '24 - Oct '26



Source: DRAMeXchange

BofA GLOBAL RESEARCH

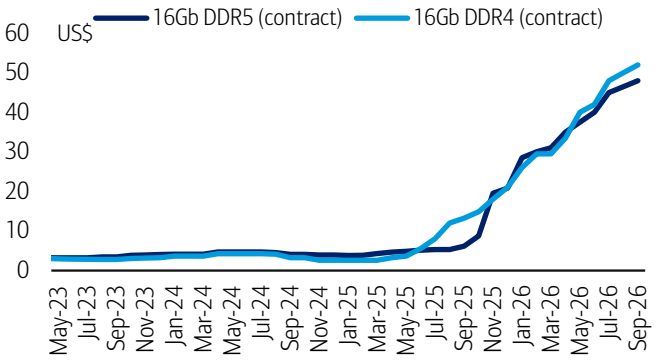
Exhibit 10: Remained flat in early-Oct, but price down 10% after hitting peak in early-Sep, price still ~2000% above the Oct-25 low (~$3)
16Gb DDR4 spot price trend, Oct '23 - Oct '26



Source: DRAMeXchange

BofA GLOBAL RESEARCH

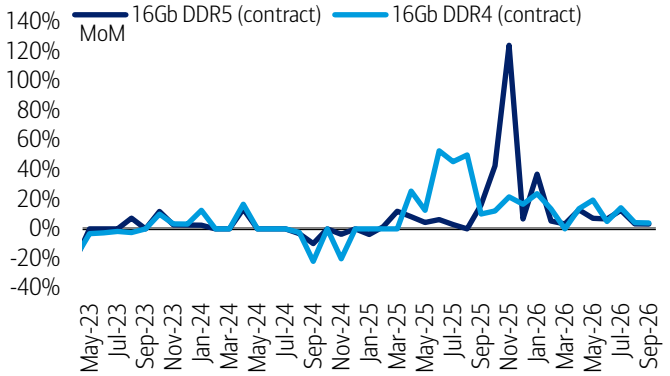
Exhibit 11: DDR4 and DDR5 (16Gb) contract prices are similar at US$50 level – DDR5 price premium no longer exists due to DDR4 shortage
16Gb DDR5 vs 16Gb DDR4 contract price trend, May '23- Sep '26



Source: TrendForce

BofA GLOBAL RESEARCH

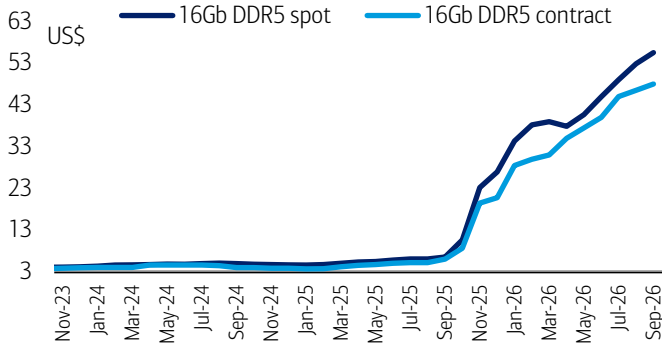
Exhibit 12: DDR5 and DDR4 contract prices up only low-single in Aug/Sep vs +10-15% MoM in July; price strong through 2H25-1H26
16Gb DDR5 vs 16Gb DDR4 contract price change - MoM, May '23- Sep'26



Source: TrendForce

BofA GLOBAL RESEARCH

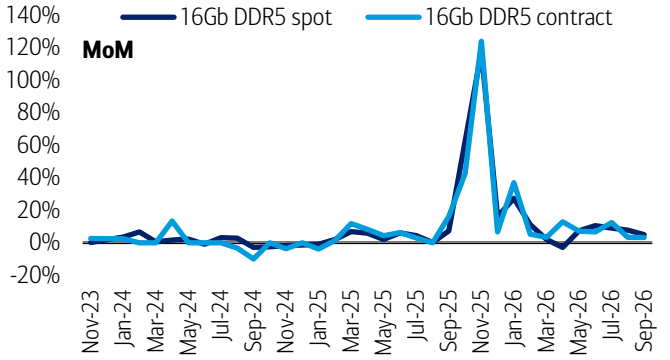
Exhibit 13: 16Gb DDR5 spot and contract prices are in the range of US$50-55 vs historical range of US$3-5
16Gb DDR5 spot vs contract price trend (US$), Nov '23-Sep '26



Source: DRAMeXchange, TrendForce

BofA GLOBAL RESEARCH

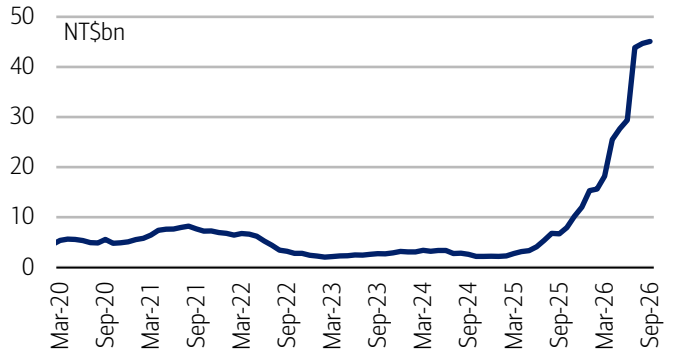
Exhibit 14: Sep price broadly flat following Jun-Aug recovery; prices had witnessed a sharp rally in Oct–Dec'25
16Gb DDR5 spot vs contract price trend – MoM change, Nov '23-Sep '26



Source: DRAMeXchange, TrendForce

BofA GLOBAL RESEARCH

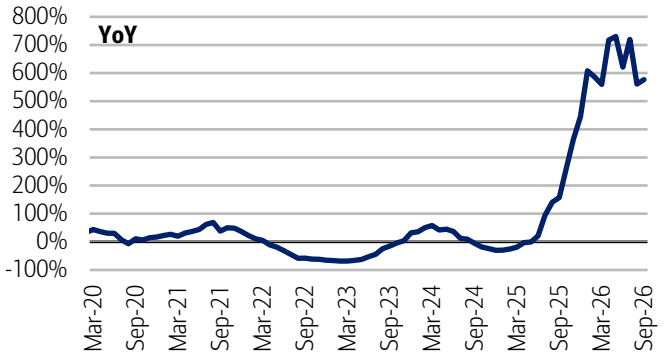
Exhibit 15: Exceptionally strong sales growth continued in 3Q26 (NT$134bn; up 62% QoQ and 612% YoY)
Nanya Tech – Monthly sales trend



Source: Company

BofA GLOBAL RESEARCH

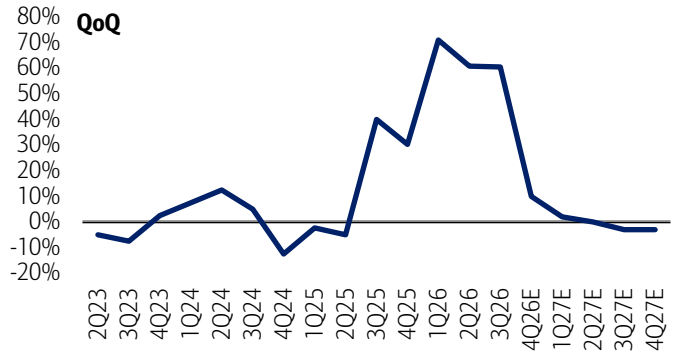
Exhibit 16: YoY monthly sales growth still robust in Sep (up 577%); already 14 consecutive months of triple-digit rebound
Nanya Tech – Monthly sales YoY



Source: Company, BofA Global Research

BofA GLOBAL RESEARCH

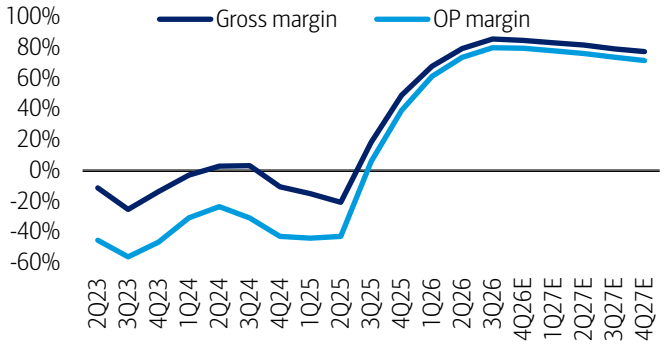
Exhibit 17: Exceptionally strong ASP hike in 3Q (up >60% QoQ); we assume decelerating growth into 4Q (up 10%) and 1H27 (up low-single%) and mild decline in 2H27
Nanya Tech DRAM ASP trend – quarterly; 8Gb equiv.



Source: Company, BofA Global Research estimates

BofA GLOBAL RESEARCH

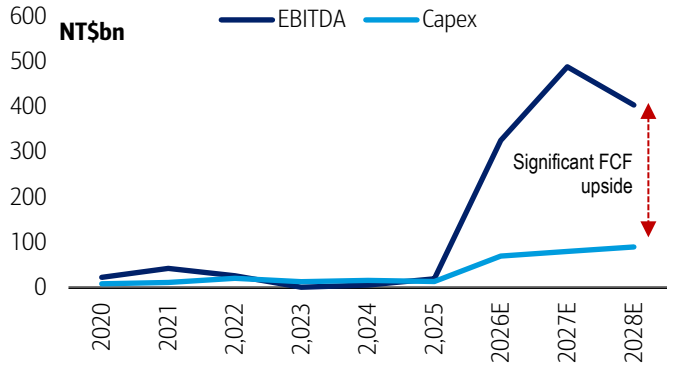
Exhibit 18: We expect robust margins to continue in 4Q26-27 (GM/OPM 70%+); record-high margins well-proved in 3Q (GM 86%, OPM 80%)
Nanya Tech – Gross and OP margin trend



Source: Company, BofA Global Research estimates

BofA GLOBAL RESEARCH

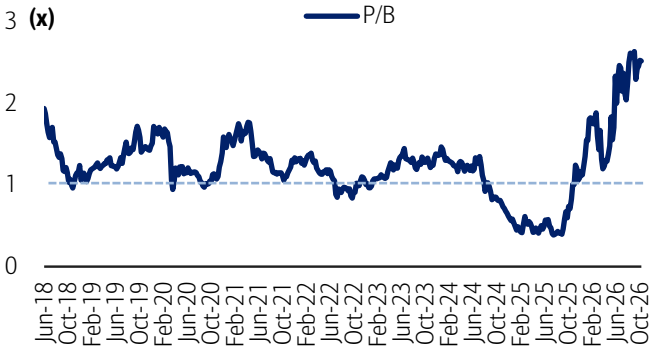
Exhibit 19: Record-high FCF generation well-expected in 2026-28E on the back of much stronger-than-expected commodity DRAM price hike
Nanya Tech – EBITDA vs capex trend



Source: Company, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 20: Currently trading at low-2.0x P/B vs our implied target of low-3.0x
Nanya Tech – 12-month forward P/B



Source: Company, Bloomberg, BofA Global Research estimates

BofA GLOBAL RESEARCH

Valuation

Exhibit 21: Our PO of NT$800 (old PO NT$750) is derived from 9x 2027-28E P/E (unchanged multiple)
Nanya Tech – Valuation multiples at current price and PO

(x)	2024	2025	2026E	2027E	2028E	2027-28E Avg
Current shr price based						
P/E	**n/a**	**n/a**	**7.1**	**5.1**	**6.2**	**5.7**
P/B	**9.9**	**9.6**	**3.7**	**2.3**	**2.0**	**2.1**
EV/EBITDA	285.2	81.6	4.9	3.3	3.9	3.6
EV/Sales	46.7	23.9	3.9	2.5	2.6	2.6
PO based multiples						
PO/EPS	**n/a**	**377.5**	**10.7**	**7.7**	**9.5**	**8.6**
PO/BV	**15.0**	**14.5**	**5.5**	**3.5**	**3.0**	**3.2**
EV/EBITDA	436.6	125.0	7.5	5.0	6.0	5.5
EV/Sales	71.5	36.6	5.9	3.9	4.0	3.9

Source: Company, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 22: Long-term historical average (2014-25) P/E shows: year-avg:13x, year-high: 17x, year-low: 9x
Nanya Tech – Historical valuation multiples

	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	LT avg 2014-25
Yr avg shr price based													
P/E	**5.2**	**7.7**	**4.5**	**4.4**	**6.1**	**21.2**	**26.3**	**31.4**	**8.3**	**14.3**	**nm**	**nm**	**12.9**
P/B	**3.5**	**2.4**	**1.2**	**1.4**	**1.4**	**1.4**	**1.3**	**1.6**	**1.1**	**1.1**	**1.1**	**1.3**	**1.6**
EV/EBITDA	8.2	8.9	9.2	5.2	3.3	6.5	6.5	7.1	2.7	6.1	160.3	31.1	8.6
EV/Sales	4.1	3.9	3.2	2.6	2.0	3.0	2.5	2.7	1.3	2.8	4.6	5.1	3.2
Yr high shr price based													
P/E	**7.3**	**12.0**	**5.7**	**6.5**	**8.3**	**26.8**	**36.9**	**41.1**	**11.4**	**17.0**	**nm**	**nm**	**17.3**
P/B	**5.0**	**3.7**	**1.6**	**2.1**	**1.9**	**1.7**	**1.8**	**2.1**	**1.5**	**1.4**	**1.4**	**3.7**	**2.3**
EV/EBITDA	10.7	12.8	11.0	8.1	4.9	9.1	10.1	10.4	4.4	7.6	231.6	100.9	17.3
EV/Sales	5.3	5.6	3.9	4.0	3.0	4.2	3.8	3.9	2.2	3.5	6.7	16.5	5.2
Yr low shr price based													
P/E	**3.0**	**4.3**	**3.9**	**3.2**	**3.7**	**16.2**	**17.7**	**23.8**	**6.2**	**11.3**	**nm**	**nm**	**9.3**
P/B	**2.1**	**1.3**	**1.1**	**1.0**	**0.8**	**1.0**	**0.9**	**1.2**	**0.8**	**0.9**	**0.5**	**0.5**	**1.0**
EV/EBITDA	5.6	5.9	8.5	3.4	1.6	4.8	3.7	4.5	1.6	4.4	60.8	7.1	4.7
EV/Sales	2.8	2.6	3.0	1.7	1.0	2.2	1.4	1.7	0.8	2.0	1.8	1.2	1.8

Source: Company, Bloomberg, BofA Global Research

BofA GLOBAL RESEARCH

Exhibit 23: We expect Nanya Tech's P/E to recover to near-9x on the back of record-high 2026-27E OP (e.g., legacy DRAM price strengths)
Nanya Tech – Valuation comparison

	BBG Ticker	Rating	Price (Local)	Mcap ($bn)	P/E			P/Book			EV/EBITDA			EV/Sales			ROE		
					FY26E	FY27E	FY28E	FY26E	FY27E	FY28E	FY26E	FY27E	FY28E	FY26E	FY27E	FY28E	FY26E	FY27E	FY28E
Nanya Tech	2408 TT	BUY	527.0	56.9	7.1	5.1	6.2	3.7	2.3	2.0	4.6	3.1	3.7	3.7	2.4	2.4	75.3%	56.3%	34.2%
Powertech	6239 TT	BUY	304.5	7.2	23.2	17.9	14.4	3.6	3.1	2.7	9.1	7.8	6.8	2.7	2.3	2.0	16.1%	18.3%	19.9%
SIMO	SIMO US	BUY	263.9	8.9	22.6	24.8	20.0	7.3	5.9	4.8	19.8	17.3	14.1	4.7	4.2	3.6	39.1%	27.2%	27.4%
Phison	8299 TT	BUY	2,055.0	14.2	5.7	9.7	15.1	3.4	2.8	2.6	4.2	6.1	9.5	1.6	1.4	1.6	92.3%	37.0%	20.7%
SK Hynix	000660 KS	BUY	1,681,000	873.3	4.8	4.7	4.6	3.1	2.0	1.5	3.9	2.9	2.8	3.2	2.3	2.2	98.2%	52.9%	38.1%
Micron	MU US	BUY	1,029.0	1,164.2	13.6	6.0	5.2	8.4	4.2	2.9	10.4	4.6	4.1	8.0	3.9	3.4	89.7%	95.8%	68.8%
Samsung	005930 KS	BUY	262,000	1,090.9	6.1	4.5	4.6	2.6	1.8	1.4	3.7	2.8	2.9	2.3	1.8	1.6	52.4%	46.1%	32.9%
TSMC	2330 TT	BUY	2,540.0	2,062.9	22.6	15.7	12.5	8.4	5.8	4.1	15.1	10.6	8.6	11.3	8.0	6.5	43.8%	43.3%	38.5%

Source: Companies, Bloomberg, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 24: 10% WACC, 45%+ OP margin and 4.0% perpetual growth suggests a fair value of NT$800
Nanya Tech – Discounted cash flow (DCF) valuation as a cross-check

NT$mn	2025	2026E	2027E	2028E	2029E	2030E	2031E
Sales	66,587	410,616	633,254	617,223	536,984	590,682	620,216
Operating profit	5,245	313,540	474,183	385,201	268,492	277,621	285,299
Pre-tax income	7,884	320,662	480,103	390,687	273,862	283,173	291,005
Taxes	1,269	68,823	120,026	97,672	68,465	70,793	72,751
Net profit	6,616	251,840	360,077	293,015	205,396	212,380	218,254
EBITDA (ex-non-cash items)	**19,526**	**325,705**	**488,807**	**404,165**	**289,492**	**302,621**	**315,299**
Depreciation & amortization	14,281	12,165	14,624	18,964	21,000	25,000	30,000
Capex	13,446	69,700	80,000	90,000	70,000	70,000	70,000
NWC increase	1,973	-5,993	-2,456	1,296	1,000	1,000	1,000
Net working capital	36,506	30,514	28,058	29,353	30,353	31,353	32,353
Free cash flow	**2,838**	**193,174**	**291,237**	**215,197**	**150,026**	**160,827**	**171,548**
Adj FCF without NWC change	**4,811**	**187,182**	**288,781**	**216,493**	**151,026**	**161,827**	**172,548**
Sales growth	95%	517%	54%	-3%	-13%	10%	5%
OPM	8%	76%	75%	62%	50%	47%	46%
Pre-tax income growth	-220%	3967%	50%	-19%	-30%	3%	3%
Capex growth	-17%	418%	15%	13%	-22%	0%	0%
Effective tax rate	16%	21%	25%	25%	25%	25%	25%
Shareholder value analysis							
Discount factor	1.00	0.91	0.83	0.75	0.68	0.62	0.56
PV of annual FCF		175,617	240,703	161,692	102,479	99,872	96,848
Sum of PV FCF (a)	**877,211**						
Terminal value							
PV of TV (b)	**1,580,187**						2,799,016
Perpetual growth	4.0%						2,974,740
Fair EV/EBITDA 8x	8.0						2,623,291
Enterprise value total (a+b)	**2,457,398**						
Net cash (end-2025)	38,818						
Fair value for shareholders	**2,496,216**						
Shares - fully diluted (mn)	3,118						
Per share value	**800.58**						
Current share price	527.00						
Upside	52%						
PO	**800.00**						
PO vs DCF based FV	0%						

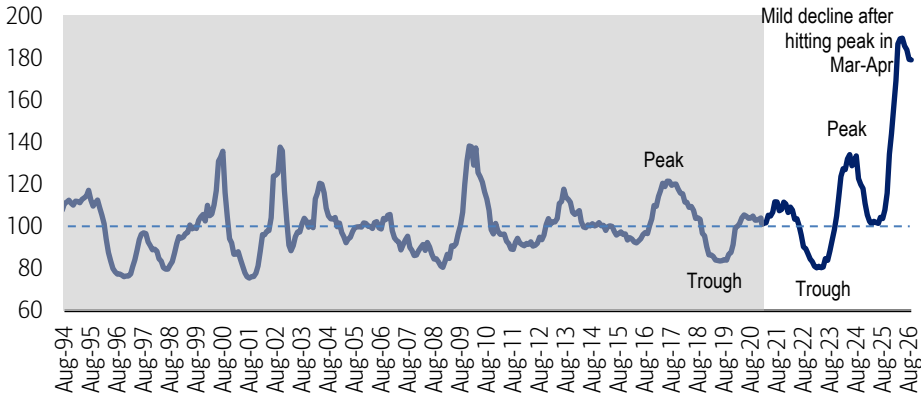
Source: Company, BofA Global Research estimates

BofA GLOBAL RESEARCH

BofA Memory Indicator

Exhibit 25: The BofA Memory Indicator still presents super-cycle – much higher than 2017-18 peak level; of course, it slightly declined to 179 in Aug-26 (due to lower YoY growth) after hitting peak of 190 in Mar/Apr-26, driven by exceptionally strong DRAM/NAND spot pricing, rising ASPs and billings, along with Korea's ~200%+ export growth.

BofA Memory Indicator – back to upturn level since Oct-2025 and hit highest peak in Mar/Apr-26 (back-tested)



Note: Our indicator had previously been capped at ~140, but the recent unprecedented surge—driven by memory spot prices, ASPs, and billings—has led us to raise the ceiling (up to 240 levels) to reflect a more accurate relative comparison. This exceptional strength is further validated by strong earnings from memory chipmakers.

Source: DRAMeXchange, WSTS, MoTIR Korea, BofA Global Research

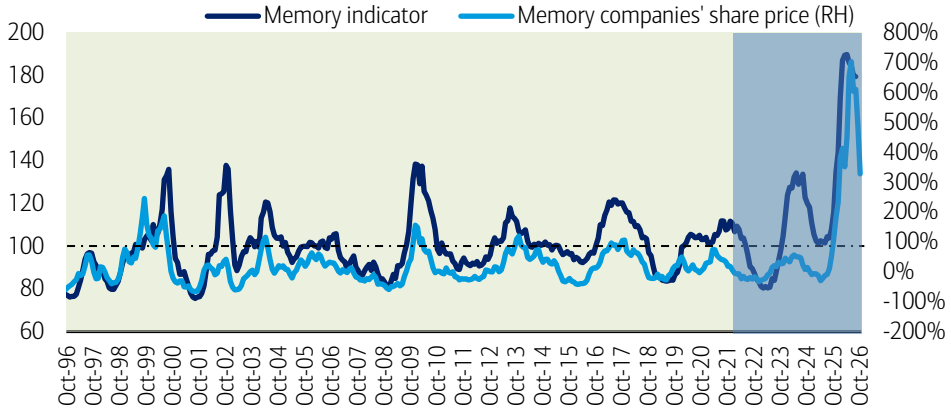
*The shaded area represents back-tested results from January 1991 to March 2021. The unshaded area represents actual performance since April 2021. This performance is back-tested up to March-2021, and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein.

Disclaimer: The BofA Memory Indicator is intended to be an indicative metric only and may not be used for reference purposes or as a measure of performance for any financial instrument or contract, or otherwise relied upon by third parties for any other purpose, without the prior written consent of BofA Global Research. This indicator was not created to act as a benchmark. For methodology, see the 5 Aug 2024 Global Memory Tech report.

BofA GLOBAL RESEARCH

Exhibit 26: Sharp correction in memory companies' stock prices in Jul-Aug due to memory peak-out concerns, global uncertainty and AI concerns, but steady recovery seen in Sep-early Oct

BofA Memory Indicator is highly correlated with stock performance (back-tested)



Note: Memory companies share price is average of Samsung, Hynix, Micron, and Nanya share price YoY changes

Source: Bloomberg, BofA Global Research

The light green shaded area represents back-tested results from January 1991 to March 2021. The blue shaded area represents actual data since April 2021. This performance is back-tested and does not represent the actual performance of any account or fund. Back-tested performance depicts the theoretical (not actual) performance of a particular strategy over the time period indicated. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein. For methodology, see the 5 Aug 2024 Global Memory Tech report.

BofA GLOBAL RESEARCH

The analysis of BofA Memory Indicator in this report is back-tested and does not represent the actual performance of any account or fund. Back-tested performance

depicts the hypothetical back-tested performance of a particular strategy over the time period indicated. In future periods, market and economic conditions will differ and the same strategy will not necessarily produce the same results. No representation is being made that any actual portfolio is likely to have achieved returns similar to those shown herein. In fact, there are frequently sharp differences between back-tested returns and the actual results realized in the actual management of a portfolio. Back-tested performance results are created by applying an investment strategy or methodology to historical data and attempts to give an indication as to how a strategy might have performed during a certain period in the past if the product had been in existence during such time. Back-tested results have inherent limitations including the fact that they are calculated with the full benefit of hindsight, which allows the security selection methodology to be adjusted to maximize the returns. Further, the results shown do not reflect actual trading or the impact that material economic and market factors might have had on a portfolio manager's decision-making under actual circumstances. Back-tested returns do not reflect advisory fees, trading costs, or other fees or expenses.

Earnings outlook in 2026-28E

Exhibit 27: Accelerated topline growth well-expected in 2026 (>500% YoY), even after solid growth in 2025 (+95%) on the back of commodity DRAM price recovery/upturn; solid turnaround well-observed in 1Q26 (61%) and we expect further enhancement throughout 2026-27E (75%+ range)
Nanya Tech – Income statement

(NT$mn)	1Q26	2Q26	3Q26	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2021	2022	2023	2024	2025	2026E	2027E	2028E
Sales total	**49,087**	**82,549**	**133,649**	**145,331**	**151,624**	**157,188**	**164,144**	**160,298**	**85,604**	**56,952**	**29,892**	**34,132**	**66,587**	**410,616**	**633,254**	**617,223**
DRAM	49,087	82,549	133,649	145,331	151,624	157,188	164,144	160,298	85,604	56,952	29,892	34,132	66,587	410,616	633,254	617,223
Others & adj	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
FX (NT$/US$)	32	32	32	32	31	31	31	31	28	30	31	32	31	32	31	31
US$ equiv DRAM sales (mn)	1,551	2,612	4,176	4,589	4,821	5,014	5,253	5,146	3,066	1,935	958	1,062	2,156	12,928	20,234	19,903
COGS	15,771	16,930	19,167	22,123	25,474	28,716	34,035	36,050	48,560	35,610	34,375	34,552	51,600	73,990	124,276	193,958
Gross profit	**33,316**	**65,619**	**114,482**	**123,208**	**126,150**	**128,471**	**130,109**	**124,248**	**37,044**	**21,342**	**-4,483**	**-421**	**14,986**	**336,626**	**508,978**	**423,265**
Opex (SG&A, R&D)	3,205	4,793	7,588	7,500	7,855	8,600	8,950	9,390	9,858	10,340	9,977	10,134	9,741	23,086	34,795	38,065
Operating profit	**30,111**	**60,826**	**106,894**	**115,708**	**118,295**	**119,871**	**121,159**	**114,858**	**27,186**	**11,002**	**-14,460**	**-10,555**	**5,245**	**313,540**	**474,183**	**385,201**
Depreciation & amort	2,897	2,960	3,108	3,200	3,350	3,518	3,693	4,063	15,294	15,247	15,326	16,144	14,281	12,165	14,624	18,964
EBITDA	33,008	63,786	110,002	118,908	121,645	123,389	124,853	118,921	42,480	26,249	865	5,589	19,526	325,705	488,807	404,165
Capital spending	2,809	4,046	20,783	42,062	20,000	18,000	17,000	25,000	11,260	20,711	13,245	16,143	13,446	69,700	80,000	90,000
Net non-operating expenses	1,607	1,503	2,438	1,574	1,502	1,487	1,473	1,459	581	5,875	3,756	3,998	2,639	7,122	5,920	5,487
Pre-tax income	**31,718**	**62,329**	**109,332**	**117,283**	**119,796**	**121,358**	**122,632**	**116,316**	**27,767**	**16,876**	**-10,705**	**-6,557**	**7,884**	**320,662**	**480,103**	**390,687**
Taxes	5,660	12,138	21,705	29,321	29,949	30,340	30,658	29,079	4,918	2,258	-3,265	-1,474	1,269	68,823	120,026	97,672
Effective tax rate	18%	19%	20%	25%	25%	25%	25%	25%	18%	13%	31%	22%	16%	21%	25%	25%
Net profit	**26,058**	**50,192**	**87,627**	**87,962**	**89,847**	**91,019**	**91,974**	**87,237**	**22,849**	**14,618**	**-7,440**	**-5,083**	**6,615**	**251,839**	**360,077**	**293,015**
Minority income	-1	0	0	0	0	0	0	0	0	0	0	0	-1	-2	0	0
Net profit after minority	**26,059**	**50,192**	**87,627**	**87,962**	**89,847**	**91,019**	**91,974**	**87,237**	**22,849**	**14,618**	**-7,440**	**-5,083**	**6,616**	**251,840**	**360,077**	**293,015**
EPS (FD)	**8.35**	**14.56**	**25.23**	**25.32**	**25.87**	**26.20**	**26.48**	**25.12**	**7.35**	**4.68**	**-2.39**	**-1.63**	**2.12**	**74.53**	**103.66**	**84.36**
% to sales total																
DRAM revenue	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Gross profit	68%	79%	86%	85%	83%	82%	79%	78%	43%	37%	-15%	-1%	23%	82%	80%	69%
SG&A	7%	6%	6%	5%	5%	5%	5%	6%	12%	18%	33%	30%	15%	6%	5%	6%
Operating profit	61%	74%	80%	80%	78%	76%	74%	72%	32%	19%	-48%	-31%	8%	76%	75%	62%
Depreciation & amort	6%	4%	2%	2%	2%	2%	2%	3%	18%	27%	51%	47%	21%	3%	2%	3%
EBITDA	67%	77%	82%	82%	80%	78%	76%	74%	50%	46%	3%	16%	29%	79%	77%	65%
Pre-tax income	65%	76%	82%	81%	79%	77%	75%	73%	32%	30%	-36%	-19%	12%	78%	76%	63%
Net profit	53%	61%	66%	61%	59%	58%	56%	54%	27%	26%	-25%	-15%	10%	61%	57%	47%

Source: Company, BofA Global Research estimates

Exhibit 28: Balance sheet broadly healthy with solid net cash (NT$270bn as of 3Q26) and declining inventory burden
Nanya Tech – Balance sheet

NT$mn	1Q26	2Q26	3Q26	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2021	2022	2023	2024	2025	2026E	2027E	2028E
Assets total	**240,968**	**387,758**	**479,685**	**565,582**	**655,210**	**668,868**	**760,633**	**846,045**	**191,421**	**202,236**	**192,351**	**206,706**	**208,453**	**565,582**	**846,045**	**976,740**
Current assets total	141,607	284,521	357,920	404,109	476,250	474,595	552,232	615,892	106,460	105,512	96,510	107,946	108,541	404,109	615,892	672,368
Cash & cash equivalent	86,281	217,603	288,507	330,199	404,273	401,672	478,179	547,127	80,700	73,593	58,812	61,903	58,074	330,199	547,127	608,280
A/R	25,247	40,247	42,260	46,486	44,161	44,603	46,833	42,150	11,569	4,360	5,220	4,132	16,546	46,486	42,150	38,218
Inventories	24,331	21,530	21,961	22,180	22,624	23,077	21,923	21,265	11,611	23,384	27,634	35,318	27,288	22,180	21,265	20,413
Others	5,748	5,142	5,193	5,245	5,192	5,244	5,297	5,350	2,580	4,175	4,844	6,593	6,634	5,245	5,350	5,457
Non-current assets total	99,361	103,237	121,765	161,472	178,960	194,272	208,401	230,153	84,961	96,723	95,841	98,760	99,912	161,472	230,153	304,372
LT investments	6,867	8,006	8,046	8,087	8,127	8,168	8,209	8,250	5,350	6,131	6,009	5,396	6,659	8,087	8,250	8,416
Fixed assets	87,349	89,804	108,347	148,067	165,568	180,892	195,032	216,794	76,207	84,897	81,838	84,327	85,031	148,067	216,794	291,049
Others	5,145	5,426	5,372	5,318	5,265	5,212	5,160	5,109	3,404	5,695	7,994	9,037	8,221	5,318	5,109	4,907
Liabilities total	**47,931**	**65,060**	**69,360**	**67,294**	**67,075**	**66,862**	**66,652**	**64,827**	**18,443**	**21,278**	**25,427**	**41,653**	**37,914**	**67,294**	**64,827**	**45,572**
Current liabilities total	25,207	41,175	45,423	43,397	43,123	42,854	42,589	40,707	15,692	16,179	20,118	32,780	19,020	43,397	40,707	34,734
ST debts	0	0	0	0	0	0	0	0	0	0	11,181	21,270	5,060	0	0	0
A/P	7,298	8,090	8,171	8,008	8,088	8,169	8,251	8,086	2,850	5,604	3,443	5,180	6,481	8,008	8,086	8,164
Others	17,910	33,085	37,252	35,389	35,036	34,685	34,338	32,621	12,842	10,575	5,493	6,330	7,480	35,389	32,621	26,570
Non-current liabilities	22,724	23,885	23,937	23,897	23,952	24,008	24,064	24,120	2,751	5,099	5,309	8,873	18,894	23,897	24,120	10,838
LT debts	17,796	18,397	18,394	18,397	18,397	18,397	18,397	18,397	0	0	0	3,995	14,196	18,397	18,397	5,000
Others	4,927	5,488	5,543	5,500	5,555	5,611	5,667	5,723	2,751	5,099	5,309	4,878	4,698	5,500	5,723	5,838
Equity total	**192,900**	**322,561**	**410,188**	**498,149**	**587,997**	**601,868**	**693,842**	**781,079**	**172,978**	**180,958**	**166,924**	**165,053**	**170,400**	**498,149**	**781,079**	**931,030**
Paid in capital	30,986	34,502	34,502	34,502	34,502	34,502	34,502	34,502	30,973	30,980	30,981	30,986	30,986	34,502	34,502	34,502
Capital surplus	51,794	127,747	127,747	127,747	127,747	127,747	127,747	127,747	50,696	54,098	51,453	51,461	51,707	127,747	127,747	127,747
Retained earnings	125,513	175,705	263,332	351,294	441,142	455,013	546,987	634,224	95,426	95,267	83,890	78,852	104,102	351,294	634,224	784,175
Treasury stocks	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Others	-15,393	-15,394	-15,394	-15,394	-15,394	-15,394	-15,394	-15,394	-4,117	613	601	3,755	-16,395	-15,394	-15,394	-15,394
Minority	137	138	138	138	138	138	138	138	0	0	0	0	139	138	138	138
Net Asset	**193,037**	**322,699**	**410,326**	**498,287**	**588,135**	**602,006**	**693,980**	**781,217**	**172,978**	**180,958**	**166,924**	**165,053**	**170,538**	**498,287**	**781,217**	**931,168**
D/E Analysis																
Cash & cash equivalent	86,281	217,603	288,507	330,199	404,273	401,672	478,179	547,127	80,700	73,593	58,812	61,903	58,074	330,199	547,127	608,280
Gross debts	17,796	18,397	18,394	18,397	18,397	18,397	18,397	18,397	0	0	11,181	25,265	19,256	18,397	18,397	5,000
Net debts	-68,485	-199,206	-270,113	-311,802	-385,876	-383,275	-459,782	-528,730	-80,700	-73,593	-47,631	-36,638	-38,818	-311,802	-528,730	-603,280
Net D/E ratio	-36%	-62%	-66%	-63%	-66%	-64%	-66%	-68%	-47%	-41%	-29%	-22%	-23%	-63%	-68%	-65%
BVPS	62.3	93.5	118.9	144.4	170.4	174.4	201.1	226.4	55.9	58.4	53.9	53.3	55.0	144.4	226.4	269.8

Source: Company, BofA Global Research estimates

Exhibit 29: Strong capex assumed for 2026-28E (NT$70bn+ pa); 2026 and 1H27 spend mainly for new shell fab construction and then more equipment centric in 2H27-28

Nanya Tech – Cash flow statement

NT$mn	1Q26	2Q26	3Q26	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2021	2022	2023	2024	2025	2026E	2027E	2028E
I. CF flows from operation	**31,549**	**55,859**	**96,017**	**84,638**	**94,857**	**16,173**	**171,421**	**94,706**	**44,107**	**21,082**	**-6,095**	**1,952**	**18,568**	**268,063**	**377,157**	**310,684**
Net income	26,058	50,192	87,627	87,962	89,847	91,019	91,974	87,237	22,849	14,618	-7,440	-5,083	6,615	251,839	360,077	293,015
Depreciation and amortization	2,897	2,960	3,108	3,200	3,350	3,518	3,693	4,063	15,294	15,247	15,326	16,144	14,281	12,165	14,624	18,964
Change in net working capital	6,388	-283	6,412	-6,524	1,659	-78,363	75,754	3,406	6,172	-5,672	-13,021	-5,772	-1,973	5,993	2,456	-1,296
Others	-3,794	2,991	-1,130	0	0	0	0	0	-207	-3,111	-960	-3,337	-354	-1,933	0	0
II. CF from investments	**-2,814**	**-4,242**	**-20,783**	**-42,062**	**-20,000**	**-18,000**	**-17,000**	**-25,000**	**-10,930**	**-20,868**	**-12,786**	**-16,188**	**-14,110**	**-69,901**	**-80,000**	**-90,000**
Capital expenditures	-2,809	-4,046	-20,783	-42,062	-20,000	-18,000	-17,000	-25,000	-11,260	-20,711	-13,245	-16,143	-13,446	-69,700	-80,000	-90,000
Others	-5	-196	0	0	0	0	0	.	330	-157	459	-45	-664	-201	0	0
III. CF from financing	**-1,547**	**79,415**	**-4,330**	**-885**	**-783**	**-774**	**-77,914**	**-758**	**-3,060**	**-11,771**	**4,196**	**13,707**	**-6,317**	**72,653**	**-80,228**	**-159,531**
Change in debts	-1,459	600	-3	3	0	0	0	0	0	0	11,181	14,084	-6,009	-859	0	-13,397
Change in capital stocks	87	79,469	0	0	0	0	0	0	18,281	3,409	-2,645	13	246	79,556	0	0
Net share repurchases	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Dividends	0	0	-4,658	0	0	0	-77,148	0	-4,011	-11,468	-6,600	0	0	-4,658	-77,148	-143,065
Others	-174	-655	330	-888	-783	-774	-766	-758	-17,331	-3,712	2,259	-390	-554	-1,386	-3,081	-3,069
IV. FX impact	**1,019**	**290**	**0**	**0**	**0**	**0**	**0**	**0**	**-1,143**	**4,451**	**-96**	**3,619**	**-1,970**	**1,309**	**0**	**0**
IV. Cash flows (I+II+III)	**28,207**	**131,322**	**70,904**	**41,692**	**74,074**	**-2,601**	**76,507**	**68,948**	**28,974**	**-7,107**	**-14,781**	**3,091**	**-3,829**	**272,124**	**216,929**	**61,153**
Cash at beginning of the period	58,074	86,281	217,603	288,507	330,199	404,273	401,672	478,179	51,726	80,700	73,593	58,812	61,903	58,074	330,199	547,127
Adjustment for cash balance	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Cash at end of the period	86,281	217,603	288,507	330,199	404,273	401,672	478,179	547,127	80,700	73,593	58,812	61,903	58,074	330,199	547,127	608,280

Source: Company, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 30: 2026-28E EBITDA large enough to fully cover increasing capex spend; we assume more meaningful FCF generation in 2026 (around NT$200bn+ pa) after two years of OP loss during 2023-24, while steadily improving net cash position to continue (NT$500-600bn range in 2027-28)

Nanya Tech – Free cash flow (FCF) analysis

NT$mn	1Q26	2Q26	3Q26	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2021	2022	2023	2024	2025	2026E	2027E	2028E
Sales	49,087	82,549	133,649	145,331	151,624	157,188	164,144	160,298	85,604	56,952	29,892	34,132	66,587	410,616	633,254	617,223
Operating income	30,111	60,826	106,894	115,708	118,295	119,871	121,159	114,858	27,186	11,002	-14,460	-10,555	5,245	313,540	474,183	385,201
Depreciation & amort	2,897	2,960	3,108	3,200	3,350	3,518	3,693	4,063	15,294	15,247	15,326	16,144	14,281	12,165	14,624	18,964
EBITDA	**33,008**	**63,786**	**110,002**	**118,908**	**121,645**	**123,389**	**124,853**	**118,921**	**42,480**	**26,249**	**865**	**5,589**	**19,526**	**325,705**	**488,807**	**404,165**
Capex	**2,809**	**4,046**	**20,783**	**42,062**	**20,000**	**18,000**	**17,000**	**25,000**	**11,260**	**20,711**	**13,245**	**16,143**	**13,446**	**69,700**	**80,000**	**90,000**
Increase of NWC	**-6,388**	**283**	**-6,412**	**6,524**	**-1,659**	**78,363**	**-75,754**	**-3,406**	**-6,172**	**5,672**	**13,021**	**5,772**	**1,973**	**-5,993**	**-2,456**	**1,296**
Net working capital (NWC)	30,119	30,402	23,990	30,514	28,854	107,217	31,464	28,058	10,068	15,740	28,761	34,533	36,506	30,514	28,058	29,353
Ex-cash current assets	55,326	66,919	69,413	73,911	71,978	72,924	74,052	68,764	25,760	31,919	37,698	46,043	50,467	73,911	68,764	64,088
Ex-debts current liabilities	25,207	36,517	45,423	43,397	43,123	-34,294	42,589	40,707	15,692	16,179	8,937	11,510	13,960	43,397	40,707	34,734
Taxes	5,660	12,138	21,705	29,321	29,949	30,340	30,658	29,079	4,918	2,258	-3,265	-1,474	1,269	68,823	120,026	97,672
Free cash flow (FCF)	**30,927**	**47,319**	**73,926**	**41,003**	**73,355**	**-3,314**	**152,948**	**68,248**	**32,473**	**-2,393**	**-22,135**	**-14,851**	**2,838**	**193,174**	**291,237**	**215,197**
FCF without NWC changes	**24,539**	**47,602**	**67,514**	**47,526**	**71,696**	**75,049**	**77,195**	**64,841**	**26,301**	**3,280**	**-9,114**	**-9,080**	**4,811**	**187,182**	**288,781**	**216,493**
US$ equivalent (mn)																
Sales	1,551	2,612	4,176	4,589	4,821	5,014	5,253	5,146	3,065	1,907	959	1,138	2,220	13,687	21,108	20,574
Operating income	951	1,925	3,340	3,654	3,761	3,824	3,877	3,687	973	368	-464	-352	175	10,451	15,806	12,840
EBITDA	1,043	2,019	3,437	3,755	3,868	3,936	3,995	3,818	1,521	879	28	186	651	10,857	16,294	13,472
Capex	89	128	649	1,328	636	574	544	803	403	694	425	538	448	2,323	2,667	3,000
Free cash flow (FCF)	**977**	**1,497**	**2,310**	**1,295**	**2,332**	**-106**	**4,894**	**2,191**	**1,163**	**-80**	**-710**	**-495**	**95**	**6,439**	**9,708**	**7,173**
FCF without NWC changes	**775**	**1,506**	**2,109**	**1,501**	**2,280**	**2,394**	**2,470**	**2,082**	**942**	**110**	**-292**	**-303**	**160**	**6,239**	**9,626**	**7,216**
Cash & cash equivalent	86,281	217,603	288,507	330,199	404,273	401,672	478,179	547,127	80,700	73,593	58,812	61,903	58,074	330,199	547,127	608,280
Debts	17,796	18,397	18,394	18,397	18,397	18,397	18,397	18,397	0	0	11,181	25,265	19,256	18,397	18,397	5,000
Net debts	-68,485	-199,206	-270,113	-311,802	-385,876	-383,275	-459,782	-528,730	-80,700	-73,593	-47,631	-36,638	-38,818	-311,802	-528,730	-603,280

Source: Company, BofA Global Research estimates

BofA GLOBAL RESEARCH

Price objective basis & risk

Nanya Technology (NNYAF)
Our PO of NT$800 is derived from 9x 2027-28E P/E, which is c.30% below the long-term historical average (13x during 2014-25), but near-50% above the previous upturn period (6x in 2014-18). Memory stocks usually trade at low valuation multiples during peak cycle due to downturn concerns. However, the current DRAM cycle looks structurally much better than ever and we expect 2027-28 earnings to exceed 2026 level due to the AI growth theme. This could lead to a re-rating (e.g., 9x P/E or higher), with solid earnings outlook.

Upside risks: DRAM price increases even after the 1H26 upturn, Big-3 memory chipmakers' significant supply cut for commodity DRAM due to HBM, and Nanya Tech's more successful ramp-up of DDR4 or DDR5 (better cost or margin improvement).

Downside risks: weaker legacy DDR4 demand (quicker shift to DDR5), and China's more successful memory production increase.

Analyst Certification

We, Simon Woo, CFA and Matt Shin, hereby certify that the views each of us has expressed in this research report accurately reflect each of our respective personal views about the subject securities and issuers. We also certify that no part of our respective compensation was, is, or will be, directly or indirectly, related to the specific recommendations or view expressed in this research report.

Special Disclosures

Information on securities which are listed on the exchanges where ML Securities (Taiwan) Limited is not permitted to trade or solicit trades for clients is for informational purposes only and is not a recommendation or a solicitation to trade such securities. ML Securities (Taiwan) Limited will not execute transactions for nor accept orders from clients to trade such securities. Foreign investment in Taiwan securities is regulated and restricted. Currently, foreign investment in Taiwan securities is permitted by investment through: (1) global depository receipts, (2) convertible bonds, (3) mutual funds issued offshore of Taiwan, and (4) a special foreign institutional investors (FINIs) and foreign individual investors (FIDIs) program supervised by the Taiwan SFB whereunder FINIs/FIDIs may apply for investment ID to invest in Taiwan securities by registration with Taiwan Stock Exchange. FINIs will additionally need consent from the foreign exchange authority, ie, the Central Bank of China. In addition to the limitations above, various industry-specific percentage-based limitations on foreign ownership of Taiwan companies (and in some cases prohibitions) may apply. Investments are subject to exchange rate and currency conversion restrictions and risks. Dividends and interest earned by foreign investors' Taiwan securities/instruments are generally subject to a 20% withholding tax. Ordinary shares are not available to ML private client accounts in the U.S.

This report is distributed in Taiwan by Merrill Lynch Securities (Taiwan) Limited, which is regulated by the Taiwan SFB.

APR - Semiconductor Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
BUY				
	Alchip	ALCPF	3661 TT	Haas Liu
	All Ring	XNQNF	6187 TT	Haas Liu
	AMEC	XRTOF	688012 CH	Dai Shen

APR - Semiconductor Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
	ASE Technology Holding	XSRIF	3711 TT	Haas Liu
	ASE Technology Holding -ADR	ASX	ASX US	Haas Liu
	ASMPT	ASMVF	522 HK	Simon Woo, CFA
	Aspeed	XLKMF	5274 TT	Mike Yang
	Chroma ATE	CRMJF	2360 TT	Haas Liu
	Doosan Corp	DOSAF	000150 KS	Simon Woo, CFA
	eMemory	XYLWF	3529 TT	Mike Yang
	GigaDevice	XGXIF	603986 CH	Daley Li, CFA
	Global Unichip Corp.	GBUHF	3443 TT	Haas Liu
	Grand Process Technology Corp	XZOWF	3131 TT	Haas Liu
	Hanmi Semiconductor	HNSIF	042700 KS	Simon Woo, CFA
	Hon Precision	XXPIF	7769 TT	Haas Liu
	Horizon Robotics	HRZRF	9660 HK	Daley Li, CFA
	Iluvatar CoreX	XSIBF	9903 HK	Daley Li, CFA
	Ingenic	XISCF	300223 CH	Dai Shen
	InnoScience Technology	XSCHF	2577 HK	Daley Li, CFA
	JCET Group Co Ltd	XJIEF	600584 CH	Dai Shen
	King Yuan Electronics Corp.	KYUFF	2449 TT	Haas Liu
	Kinik	KIKCF	1560 TT	Haas Liu
	Macronix International	MXICF	2337 TT	Dai Shen
	MediaTek	MDTKF	2454 TT	Haas Liu
	Montage Technology	XRDFF	688008 CH	Daley Li, CFA
	MPI Corporation	XMJCF	6223 TT	Mike Yang
	Nanya Technology	NNYAF	2408 TT	Simon Woo, CFA
	NCE Power	XVFFF	605111 CH	Daley Li, CFA
	OmniVision	XXHQF	603501 CH	Dai Shen
	Phison Electronics	PISNF	8299 TT	Simon Woo, CFA
	Powerchip Semiconductor Manufacturing Co	XCHPF	6770 TT	Haas Liu
	Powertech Technology	PWOGF	6239 TT	Simon Woo, CFA
	Rockchip	XRPXF	603893 CH	Daley Li, CFA
	Samsung C&T	XCDTF	028260 KS	Simon Woo, CFA
	Samsung Elec -G	SSNHZ	SMSN LI	Simon Woo, CFA
	Samsung Electronics	SSNLF	005930 KS	Simon Woo, CFA
	Samsung Electronics Preferred	SSNNF	005935 KS	Simon Woo, CFA
	Silergy Corp.	SLEGF	6415 TT	Mike Yang
	Silicon Motion	SIMO	SIMO US	Simon Woo, CFA
	SK Hynix	HXSCF	000660 KS	Simon Woo, CFA
	SK Hynix ADR	SKHY	SKHY US	Simon Woo, CFA
	SK Square	SKSQF	402340 KS	Simon Woo, CFA
	Taiwan Semiconductor Manufacturing Co.	TSM	TSM US	Haas Liu
	Taiwan Semiconductor Manufacturing Co.	TSMWF	2330 TT	Haas Liu
	Vanguard International Semiconductor Co	VGILF	5347 TT	Haas Liu
	Winbond Electronics	WBEKF	2344 TT	Dai Shen
	WinWay Technology	XWCLF	6515 TT	Mike Yang
	WT Microelectronics	XZOPF	3036 TT	Mike Yang
NEUTRAL				
	ASMedia Technology Inc.	XZSFF	5269 TT	Mike Yang
	Black Sesame Intl Holding	BSIHF	2533 HK	Daley Li, CFA
	Hangzhou Silan Microelectronics	XDFRF	600460 CH	Daley Li, CFA
	Lion Electronics	XDHFF	605358 CH	Dai Shen
	MetaX	XMEZF	688802 CH	Daley Li, CFA
	Novatek	NVKMF	3034 TT	Mike Yang
	Realtek	RLTKF	2379 TT	Haas Liu
	Shenzhen Goodix	XQPLF	603160 CH	Daley Li, CFA
	Soulbrain	XSBOF	357780 KS	Simon Woo, CFA
	VeriSilicon	XMLZF	688521 CH	Daley Li, CFA
UNDERPERFORM				
	Crystal Clear	XPPTF	300655 CH	Dai Shen
	Faraday	FDYTF	3035 TT	Haas Liu
	GlobalWafers	XWLFF	6488 TT	Mike Yang
	Hua Hong Grace	HHUSF	1347 HK	Dai Shen
	LX Semicon	XLXSF	108320 KS	Simon Woo, CFA
	M31 Technology	XMTZF	6643 TT	Mike Yang
	Maxscend	XMXSF	300782 CH	Daley Li, CFA

APR - Semiconductor Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
	Parade	PRDWF	4966 TT	Mike Yang
	United Microelectronics Corp.	XUMIF	2303 TT	Haas Liu
	United Microelectronics Corp.	UMC	UMC US	Haas Liu
	Win Semiconductors	WSMIF	3105 TT	Haas Liu
	Wonik IPS	XRHQF	240810 KS	Simon Woo, CFA

*iQ*method℠ Measures Definitions

Business Performance	**Numerator**	**Denominator**
Return On Capital Employed	NOPAT = (EBIT + Interest Income) × (1 − Tax Rate) + Goodwill Amortization	Total Assets − Current Liabilities + ST Debt + Accumulated Goodwill Amortization
Return On Equity	Net Income	Shareholders' Equity
Operating Margin	Operating Profit	Sales
Earnings Growth	Expected 5 Year CAGR From Latest Actual	N/A
Free Cash Flow	Cash Flow From Operations − Total Capex	N/A
Quality of Earnings	**Numerator**	**Denominator**
Cash Realization Ratio	Cash Flow From Operations	Net Income
Asset Replacement Ratio	Capex	Depreciation
Tax Rate	Tax Charge	Pre-Tax Income
Net Debt-To-Equity Ratio	Net Debt = Total Debt − Cash & Equivalents	Total Equity
Interest Cover	EBIT	Interest Expense
Valuation Toolkit	**Numerator**	**Denominator**
Price / Earnings Ratio	Current Share Price	Diluted Earnings Per Share (Basis As Specified)
Price / Book Value	Current Share Price	Shareholders' Equity / Current Basic Shares
Dividend Yield	Annualised Declared Cash Dividend	Current Share Price
Free Cash Flow Yield	Cash Flow From Operations − Total Capex	Market Cap = Current Share Price × Current Basic Shares
Enterprise Value / Sales	EV = Current Share Price × Current Shares + Minority Equity + Net Debt + Other LT Liabilities	Sales
EV / EBITDA	Enterprise Value	Basic EBIT + Depreciation + Amortization

iQmethod℠ is the set of BofA Global Research standard measures that serve to maintain global consistency under three broad headings: Business Performance, Quality of Earnings, and validations. The key features of iQmethod are: A consistently structured, detailed, and transparent methodology. Guidelines to maximize the effectiveness of the comparative valuation process, and to identify some common pitfalls.

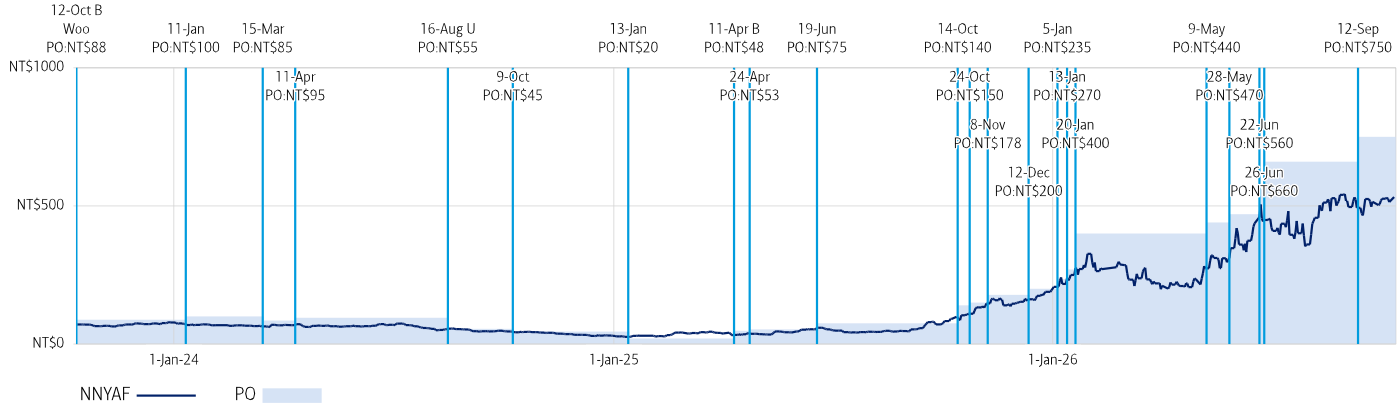
iQdatabase® is our real-time global research database that is sourced directly from our equity analysts' earnings models and includes forecasted as well as historical data for income statements, balance sheets, and cash flow statements for companies covered by BofA Global Research.

iQprofile℠, *iQmethod℠* are service marks of Bank of America Corporation. *iQdatabase®* is a registered service mark of Bank of America Corporation.

Disclosures

Important Disclosures

Nanya Tech (NNYAF) Price Chart



B: Buy, N: Neutral, U: Underperform, PO: Price Objective, NA: No longer valid, NR: No Rating

The Investment Opinion System is contained at the end of the report under the heading "Fundamental Equity Opinion Key". Dark grey shading indicates the security is restricted with the opinion suspended. Medium grey shading indicates the security is under review with the opinion withdrawn. Light grey shading indicates the security is not covered. Chart is current as of a date no more than one trading day prior to the date of the report.

Equity Investment Rating Distribution: Technology Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	255	62.04%	Buy	157	61.57%
Hold	82	19.95%	Hold	46	56.10%
Sell	74	18.00%	Sell	31	41.89%

Equity Investment Rating Distribution: Global Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	2057	57.03%	Buy	1386	67.38%
Hold	806	22.35%	Hold	563	69.85%
Sell	744	20.63%	Sell	446	59.95%

[R1] Issuers that were investment banking clients of BofA Securities or one of its affiliates within the past 12 months. For purposes of this Investment Rating Distribution, the coverage universe includes only stocks. A stock rated Neutral is included as a Hold, and a stock rated Underperform is included as a Sell.

FUNDAMENTAL EQUITY OPINION KEY: Opinions include a Volatility Risk Rating, an Investment Rating and an Income Rating. VOLATILITY RISK RATINGS, indicators of potential price fluctuation, are: A - Low, B - Medium and C - High. INVESTMENT RATINGS reflect the analyst's assessment of both a stock's absolute total return potential as well as its attractiveness for investment relative to other stocks within its Coverage Cluster (defined below). Our investment ratings are: 1 - Buy stocks are expected to have a total return of at least 10% and are the most attractive stocks in the coverage cluster; 2 - Neutral stocks are expected to remain flat or increase in value and are less attractive than Buy rated stocks and 3 - Underperform stocks are the least attractive stocks in a coverage cluster. An investment rating of 6 (No Rating) indicates that a stock is no longer trading on the basis of fundamentals. Analysts assign investment ratings considering, among other things, the 0-12 month total return expectation for a stock and the firm's guidelines for ratings dispersions (shown in the table below). The current price objective for a stock should be referenced to better understand the total return expectation at any given time. The price objective reflects the analyst's view of the potential price appreciation (depreciation).

Investment rating	Total return expectation (within 12-month period of date of initial rating)	Ratings dispersion guidelines for coverage cluster[R2]
Buy	≥ 10%	≤ 70%
Neutral	≥ 0%	≤ 30%
Underperform	N/A	≥ 20%

[R2] Ratings dispersions may vary from time to time where BofA Global Research believes it better reflects the investment prospects of stocks in a Coverage Cluster.

INCOME RATINGS, indicators of potential cash dividends, are: 7 - same/higher (dividend considered to be secure), 8 - same/lower (dividend not considered to be secure) and 9 - pays no cash dividend. *Coverage Cluster* is comprised of stocks covered by a single analyst or two or more analysts sharing a common industry, sector, region or other classification(s). A stock's coverage cluster is included in the most recent BofA Global Research report referencing the stock.

Price Charts for the securities referenced in this research report are available on the Price Charts website, or call 1-800-MERRILL to have them mailed.
The issuer is or was, within the last 12 months, an investment banking client of BofAS and/or one or more of its affiliates: Nanya Tech.
BofAS or an affiliate has received compensation from the issuer for non-investment banking services or products within the past 12 months: Nanya Tech.
The issuer is or was, within the last 12 months, a non-securities business client of BofAS and/or one or more of its affiliates: Nanya Tech.
In the US, retail sales and/or distribution of this report may be made only in states where these securities are exempt from registration or have been qualified for sale: Nanya Tech.
BofAS or an affiliate expects to receive or intends to seek compensation for investment banking services from this issuer or an affiliate of the issuer within the next three months: Nanya Tech.
The country in which this issuer is organized has certain laws or regulations that limit or restrict ownership of the issuer's shares by nationals of other countries: Nanya Tech.
The issuer is or was, within the last 12 months, a securities business client (non-investment banking) of BofAS and/or one or more of its affiliates: Nanya Tech.
BofA Global Research personnel (including the analyst(s) responsible for this report) receive compensation based upon, among other factors, the overall profitability of Bank of America Corporation, including profits derived from investment banking. The analyst(s) responsible for this report may also receive compensation based upon, among other factors, the overall

profitability of the Bank's sales and trading businesses relating to the class of securities or financial instruments for which such analyst is responsible.

Other Important Disclosures

From time to time research analysts conduct site visits of covered issuers. BofA Global Research policies prohibit research analysts from accepting payment or reimbursement for travel expenses from the issuer for such visits.
Prices are indicative and for information purposes only. Except as otherwise stated in the report, for any recommendation in relation to an equity security, the price referenced is the publicly traded price of the security as of close of business on the day prior to the date of the report or, if the report is published during intraday trading, the price referenced is indicative of the traded price as of the date and time of the report and in relation to a debt security (including equity preferred and CDS), prices are indicative as of the date and time of the report and are from various sources including BofA Securities trading desks.
The date and time of completion of the production of any recommendation in this report shall be the date and time of dissemination of this report as recorded in the report timestamp.

Recipients who are not institutional investors or market professionals should seek the advice of their independent financial advisor before considering information in this report in connection with any investment decision, or for a necessary explanation of its contents.
Officers of BofAS or one or more of its affiliates (other than research analysts) may have a financial interest in securities of the issuer(s) or in related investments.
Refer to BofA Global Research policies relating to conflicts of interest.
"BofA Securities" includes BofA Securities, Inc. ("BofAS") and its affiliates. Investors should contact their BofA Securities representative or Merrill Global Wealth Management financial advisor if they have questions concerning this report or concerning the appropriateness of any investment idea described herein for such investor. "BofA Securities" is a global brand for BofA Global Research.
Information relating to Non-US affiliates of BofA Securities and Distribution of Affiliate Research Reports:
BofAS and/or Merrill Lynch, Pierce, Fenner & Smith Incorporated ("MLPF&S") may in the future distribute, information of the following non-US affiliates in the US (short name: legal name, regulator): Merrill Lynch (South Africa): Merrill Lynch South Africa (Pty) Ltd., regulated by the Financial Sector Conduct Authority; MLI (UK): Merrill Lynch International, regulated by the Financial Conduct Authority (FCA) and the Prudential Regulation Authority (PRA); BofASE (France): BofA Securities Europe SA is authorized by the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and regulated by the ACPR and the Autorité des Marchés Financiers (AMF). BofA Securities Europe SA ("BofASE") with registered address at 51, rue La Boétie, 75008 Paris is registered under no 842 602 690 RCS Paris. In accordance with the provisions of French Code Monétaire et Financier (Monetary and Financial Code), BofASE is an établissement de crédit et d'investissement (credit and investment institution) that is authorised and supervised by the European Central Bank and the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and regulated by the ACPR and the Autorité des Marchés Financiers. BofASE's share capital can be found at www.bofaml.com/BofASEdisclaimer; BofA Europe (Milan): Bank of America Europe Designated Activity Company, Milan Branch, regulated by the Bank of Italy, the European Central Bank (ECB) and the Central Bank of Ireland (CBI); BofA Europe (Frankfurt): Bank of America Europe Designated Activity Company, Frankfurt Branch regulated by BaFin, the ECB and the CBI; BofA Europe (Zurich): Bank of America Europe Designated Activity Company, Zurich Branch, regulated by the Swiss Financial Market Supervisory Authority FINMA, the ECB and CBI; BofA Europe (Madrid): Bank of America Europe Designated Activity Company, Sucursal en España, regulated by the Bank of Spain, the ECB and the CBI; Merrill Lynch (Australia): Merrill Lynch Equities (Australia) Limited, regulated by the Australian Securities and Investments Commission; Merrill Lynch (Hong Kong): Merrill Lynch (Asia Pacific) Limited, regulated by the Hong Kong Securities and Futures Commission (HKSFC); Merrill Lynch (Singapore): Merrill Lynch (Singapore) Pte Ltd, regulated by the Monetary Authority of Singapore (MAS); Merrill Lynch (Canada): Merrill Lynch Canada Inc, regulated by the Canadian Investment Regulatory Organization; Merrill Lynch (Mexico): Merrill Lynch Mexico, SA de CV, Casa de Bolsa, regulated by the Comisión Nacional Bancaria y de Valores; BofAS Japan: BofA Securities Japan Co., Ltd., regulated by the Financial Services Agency; Merrill Lynch (Seoul): Merrill Lynch International, LLC Seoul Branch, regulated by the Financial Supervisory Service; Merrill Lynch (Taiwan): Merrill Lynch Securities (Taiwan) Ltd., regulated by the Securities and Futures Bureau; BofAS India: BofA Securities India Limited, regulated by the Securities and Exchange Board of India (SEBI); Merrill Lynch (Israel): Merrill Lynch Israel Limited, regulated by Israel Securities Authority; Merrill Lynch (DIFC): Merrill Lynch International (DIFC Branch), regulated by the Dubai Financial Services Authority (DFSA); Merrill Lynch (Brazil): Merrill Lynch S.A. Corretora de Títulos e Valores Mobiliários, regulated by Comissão de Valores Mobiliários; Merrill Lynch KSA Company: Merrill Lynch Kingdom of Saudi Arabia Company, regulated by the Capital Market Authority.
This information: has been approved for publication and is distributed in the United Kingdom (UK) to professional clients and eligible counterparties (as each is defined in the rules of the FCA and the PRA) by MLI (UK), which is authorized by the PRA and regulated by the FCA and the PRA - details about the extent of our regulation by the FCA and PRA are available from us on request; has been approved for publication and is distributed in the European Economic Area (EEA) by BofASE (France), which is authorized by the ACPR and regulated by the ACPR and the AMF; has been considered and distributed in Japan by BofAS Japan, a registered securities dealer under the Financial Instruments and Exchange Act in Japan, or its permitted affiliates; is issued and distributed in Hong Kong by Merrill Lynch (Hong Kong) which is regulated by HKSFC; is issued and distributed in Taiwan by Merrill Lynch (Taiwan); is issued and distributed in India by BofAS India; and is issued and distributed in Singapore to institutional investors and/or accredited investors (each as defined under the Financial Advisers Regulations) by Merrill Lynch (Singapore) (Company Registration No 198602883D). Merrill Lynch (Singapore) is regulated by MAS. Merrill Lynch Equities (Australia) Limited (ABN 65 006 276 795), AFS License 235132 (MLEA) distributes this information in Australia only to 'Wholesale' clients as defined by s.761G of the Corporations Act 2001. With the exception of Bank of America N.A., Australia Branch, neither MLEA nor any of its affiliates involved in preparing this information is an Authorised Deposit-Taking Institution under the Banking Act 1959 nor regulated by the Australian Prudential Regulation Authority. No approval is required for publication or distribution of this information in Brazil and its local distribution is by Merrill Lynch (Brazil) in accordance with applicable regulations. Merrill Lynch (DIFC) is authorized and regulated by the DFSA. Information prepared and issued by Merrill Lynch (DIFC) is done so in accordance with the requirements of the DFSA conduct of business rules. BofA Europe (Frankfurt) distributes this information in Germany and is regulated by BaFin, the ECB and the CBI. BofA Securities entities, including BofA Europe and BofASE (France), may outsource/delegate the marketing and/or provision of certain research services or aspects of research services to other branches or members of the BofA Securities group. You may be contacted by a different BofA Securities entity acting for and on behalf of your service provider where permitted by applicable law. This does not change your service provider. Please refer to the Electronic Communications Disclaimers for further information.
This information has been prepared and issued by BofAS and/or one or more of its non-US affiliates. The author(s) of this information may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so. BofAS and/or MLPF&S is the distributor of this information in the US and accepts full responsibility for information distributed to BofAS and/or MLPF&S clients in the US by its non-US affiliates. Any US person receiving this information and wishing to effect any transaction in any security discussed herein should do so through BofAS and/or MLPF&S and not such foreign affiliates. Hong Kong recipients of this information should contact Merrill Lynch (Asia Pacific) Limited in respect of any matters relating to dealing in securities or provision of specific advice on securities or any other matters arising from, or in connection with, this information. Singapore recipients of this information should contact Merrill Lynch (Singapore) Pte Ltd in respect of any matters arising from, or in connection with, this information. For clients that are not accredited investors, expert investors or institutional investors Merrill Lynch (Singapore) Pte Ltd accepts full responsibility for the contents of this information distributed to such clients in Singapore.
General Investment Related Disclosures:
Taiwan Readers: Neither the information nor any opinion expressed herein constitutes an offer or a solicitation of an offer to transact in any securities or other financial instrument. No part of this report may be used or reproduced or quoted in any manner whatsoever in Taiwan by the press or any other person without the express written consent of BofA Securities.
This document provides general information only, and has been prepared for, and is intended for general distribution to, BofA Securities clients. Neither the information nor any opinion expressed constitutes an offer or an invitation to make an offer, to buy or sell any securities or other financial instrument or any derivative related to such securities or instruments (e.g., options, futures, warrants, and contracts for differences). This document is not intended to provide personal investment advice and it does not take into account the specific investment objectives, financial situation and the particular needs of, and is not directed to, any specific person(s). This document and its content do not constitute, and should not be considered to constitute, investment advice for purposes of ERISA, the US tax code, the Investment Advisers Act or otherwise. Investors should seek financial advice regarding the appropriateness of investing in financial instruments and implementing investment strategies discussed or recommended in this document and should understand that statements regarding future prospects may not be realized. Any decision to purchase or subscribe for securities in any offering must be based solely on existing public information on such security or the information in the prospectus or other offering document issued in connection with such offering, and not on this document.
Securities and other financial instruments referred to herein, or recommended, offered or sold by BofA Securities, are not insured by the Federal Deposit Insurance Corporation and are not deposits or other obligations of any insured depository institution (including, Bank of America, N.A.). Investments in general and, derivatives, in particular, involve numerous risks, including, among others, market risk, counterparty default risk and liquidity risk. No security, financial instrument or derivative is suitable for all investors. Digital assets are extremely speculative, volatile and are not regulated in a manner similar to U.S. securities markets. In some cases, securities and other financial instruments may be difficult to value or sell and reliable information about the

value or risks related to the security or financial instrument may be difficult to obtain. Investors should note that income from such securities and other financial instruments, if any, may fluctuate and that price or value of such securities and instruments may rise or fall and, in some cases, investors may lose their entire principal investment. Past performance is not necessarily a guide to future performance. Levels and basis for taxation may change.

This report may contain a short-term trading idea or recommendation, which highlights a specific near-term catalyst or event impacting the issuer or the market that is anticipated to have a short-term price impact on the equity securities of the issuer. Short-term trading ideas and recommendations are different from and do not affect a stock's fundamental equity rating, which reflects both a longer term total return expectation and attractiveness for investment relative to other stocks within its Coverage Cluster. Short-term trading ideas and recommendations may be more or less positive than a stock's fundamental equity rating.

BofA Securities is aware that the implementation of the ideas expressed in this report may depend upon an investor's ability to "short" securities or other financial instruments and that such action may be limited by regulations prohibiting or restricting "shortselling" in many jurisdictions. Investors are urged to seek advice regarding the applicability of such regulations prior to executing any short idea contained in this report.

Foreign currency rates of exchange may adversely affect the value, price or income of any security or financial instrument mentioned herein. Investors in such securities and instruments, including ADRs, effectively assume currency risk.

BofAS or one of its affiliates is a regular issuer of traded financial instruments linked to securities or digital assets that may have been recommended or discussed in this report. BofAS or one of its affiliates may, at any time, hold a trading position (long or short) in the securities and financial instruments recommended or discussed in this report.

BofA Securities, through business units other than BofA Global Research, may have issued and may in the future issue trading ideas or recommendations that are inconsistent with, and reach different conclusions from, the information presented herein. Such ideas or recommendations may reflect different time frames, assumptions, views and analytical methods of the persons who prepared them, and BofA Securities is under no obligation to ensure that such other trading ideas or recommendations are brought to the attention of any recipient of this information.

In the event that the recipient received this information pursuant to a contract between the recipient and BofAS for the provision of research services for a separate fee, and in connection therewith BofAS may be deemed to be acting as an investment adviser, such status relates, if at all, solely to the person with whom BofAS has contracted directly and does not extend beyond the delivery of this report (unless otherwise agreed specifically in writing by BofAS). If such recipient uses the services of BofAS in connection with the sale or purchase of a security referred to herein, BofAS may act as principal for its own account or as agent for another person. BofAS is and continues to act solely as a broker-dealer in connection with the execution of any transactions, including transactions in any securities referred to herein.

Copyright and General Information:

Copyright 2026 Bank of America Corporation. All rights reserved. iQdatabase® is a registered service mark of Bank of America Corporation. This information is prepared for the use of BofA Securities clients and may not be redistributed, retransmitted or disclosed, in whole or in part, or in any form or manner, without the express written consent of BofA Securities. This document and its content is provided solely for informational purposes and cannot be used for training or developing artificial intelligence (AI) models or as an input in any AI application (collectively, an AI tool). Any attempt to utilize this document or any of its content in connection with an AI tool without explicit written permission from BofA Global Research is strictly prohibited. BofA Global Research utilizes AI, including machine learning and other technologies, to enhance the services we provide to our clients. These technologies assist our analysts in various aspects of their work, including but not limited to data analysis, content extraction, content creation, data aggregation and summarization and identifying relevant information from diverse sources. All AI-driven processes are subject to review by BofA Global Research employees. BofA Global Research information is distributed simultaneously to internal and client websites and other portals by BofA Securities and is not publicly-available material. Any unauthorized use or disclosure is prohibited. Receipt and review of this information constitutes your agreement not to redistribute, retransmit, or disclose to others the contents, opinions, conclusion, or information contained herein (including any investment recommendations, estimates or price targets) without first obtaining express permission from an authorized officer of BofA Securities.

Materials prepared by BofA Global Research personnel are based on public information. Facts and views presented in this material have not been reviewed by, and may not reflect information known to, professionals in other business areas of BofA Securities, including investment banking personnel. BofA Securities has established information barriers between BofA Global Research and certain business groups. As a result, BofA Securities does not disclose certain client relationships with, or compensation received from, such issuers. To the extent this material discusses any legal proceeding or issues, it has not been prepared as nor is it intended to express any legal conclusion, opinion or advice. Investors should consult their own legal advisers as to issues of law relating to the subject matter of this material. BofA Global Research personnel's knowledge of legal proceedings in which any BofA Securities entity and/or its directors, officers and employees may be plaintiffs, defendants, co-defendants or co-plaintiffs with or involving issuers mentioned in this material is based on public information. Facts and views presented in this material that relate to any such proceedings have not been reviewed by, discussed with, and may not reflect information known to, professionals in other business areas of BofA Securities in connection with the legal proceedings or matters relevant to such proceedings.

This information has been prepared independently of any issuer of securities mentioned herein and not in connection with any proposed offering of securities or as agent of any issuer of any securities. None of BofAS any of its affiliates or their research analysts has any authority whatsoever to make any representation or warranty on behalf of the issuer(s). BofA Global Research policy prohibits research personnel from disclosing a recommendation, investment rating, or investment thesis for review by an issuer prior to the publication of a research report containing such rating, recommendation or investment thesis.

Any information relating to sustainability in this material is limited as discussed herein and is not intended to provide a comprehensive view on any sustainability claim with respect to any issuer or security.

Any information relating to the tax status of financial instruments discussed herein is not intended to provide tax advice or to be used by anyone to provide tax advice. Investors are urged to seek tax advice based on their particular circumstances from an independent tax professional.

The information herein (other than disclosure information relating to BofA Securities and its affiliates) was obtained from various sources and we do not guarantee its accuracy. This information may contain links to third-party websites. BofA Securities is not responsible for the content of any third-party website or any linked content contained in a third-party website. Content contained on such third-party websites is not part of this information and is not incorporated by reference. The inclusion of a link does not imply any endorsement by or any affiliation with BofA Securities. Access to any third-party website is at your own risk, and you should always review the terms and privacy policies at third-party websites before submitting any personal information to them. BofA Securities is not responsible for such terms and privacy policies and expressly disclaims any liability for them.

All opinions, projections and estimates constitute the judgment of the author as of the date of publication and are subject to change without notice. Prices also are subject to change without notice. BofA Securities is under no obligation to update this information and BofA Securities ability to publish information on the subject issuer(s) in the future is subject to applicable quiet periods. You should therefore assume that BofA Securities will not update any fact, circumstance or opinion contained herein.

Subject to the quiet period applicable under laws of the various jurisdictions in which we distribute research reports and other legal and BofA Securities policy-related restrictions on the publication of research reports, fundamental equity reports are produced on a regular basis as necessary to keep the investment recommendation current.

Certain outstanding reports or investment opinions relating to securities, financial instruments and/or issuers may no longer be current. Always refer to the most recent research report relating to an issuer prior to making an investment decision.

In some cases, an issuer may be classified as Restricted or may be Under Review or Extended Review. In each case, investors should consider any investment opinion relating to such issuer (or its security and/or financial instruments) to be suspended or withdrawn and should not rely on the analyses and investment opinion(s) pertaining to such issuer (or its securities and/or financial instruments) nor should the analyses or opinion(s) be considered a solicitation of any kind. Sales persons and financial advisors affiliated with BofAS or any of its affiliates may not solicit purchases of securities or financial instruments that are Restricted or Under Review and may only solicit securities under Extended Review in accordance with firm policies.

Neither BofA Securities nor any officer or employee of BofA Securities accepts any liability whatsoever for any direct, indirect or consequential damages or losses arising from any use of this information.

Taiwan Semiconductor Manufacturing Co.

3Q26 earnings preview: steeper 2nm ramp lifting sales upside at resilient margins

Reiterate Rating: BUY | PO: 3,260 TWD | Price: 2,550 TWD

Solid HPC and flagship mobile pipeline solidifies 2H26

TSMC's 3Q26 sales were above street +18% QoQ in NT$ and reached US$46.7bn (vs. its guide of US$44.6-45.8bn) which we believe should be driven by the builds for Apple's mobile SoC and AMD's server CPU on 2nm along with multiple GPU and ASIC product refreshes migrating to 3nm. The mix upgrades could lift 4Q26 sales by +10-15% QoQ, putting it on track to the guidance for 40%+ YoY in '26. While utilization and pricing are favorable, GMs could hold steady in 3Q/4Q26 (BofAe 67.0%/67.8%) rather than expand (recent street 67-68%/68-69%) factoring dilution from 2nm and overseas fab ramps.

Tight supply enables greater value capture and investment

On solid AI demand outlook, TSMC's capex should be on track to its lifted guidance of US$60-64bn in '26 (BofAe US$63bn). The investment could see another surge in '27 (BofAe US$83bn) for further N3, N2, A16 and CoWoS-L expansion and new technologies ramp (e.g. CoPoS and CPO). The tool makers' modest price hike could require TSMC to spend 10% more capex though should be reflected in its value when the capacity starts production in '28 on top of a 5-10% leading edge ASP lift in 1Q27 on supply tightness.

Checkpoints: AI demand, capex & competition/partnership

1) TSMC's view on cloud AI investment amid rate hike + challenging FCF, 2) comment on the market and its growth and capex commitment, 3) update on the competition and collaboration (see Exhibit 29 for our analysis on Terafab/SpaceX) and US investment.

AI-led earnings outlook firmer; valuation should catch up

We lift our 26/27/28 EPS by 2-9% to NT$112/NT$161/NT$203 and keep our Buy rating on TSMC and raise PO to NT$3,260/US$620 on 20x '27E P/E. Its current valuation is undemanding at 16x '27E P/E (vs. its long term 11-21x) backed by its advanced technology leadership.

Estimates (Dec) (NT$)	2024A	2025A	2026E	2027E	2028E
Net Income (Adjusted - mn)	1,173,268	1,717,886	2,916,545	4,184,666	5,276,738
EPS	45.24	66.24	112.46	161.36	203.48
EPS Change (YoY)	39.9%	46.4%	69.8%	43.5%	26.1%
Consensus EPS (Visible Alpha)			107.72	146.40	183.73
Dividend / Share	18.00	21.50	24.00	24.00	24.00
Free Cash Flow / Share	33.55	38.66	69.93	96.56	138.58
ADR EPS (US$)	7.04	10.62	17.61	25.27	31.86
ADR Dividend / Share (US$)	2.80	3.45	3.76	3.76	3.76
Valuation (Dec)					
P/E	56.36x	38.49x	22.67x	15.80x	12.53x
Dividend Yield	0.706%	0.843%	0.941%	0.941%	0.941%
EV / EBITDA*	31.70x	23.98x	15.14x	10.62x	8.61x
Free Cash Flow Yield*	1.32%	1.52%	2.74%	3.79%	5.43%

* For full definitions of *iQmethod*℠ measures, see page 24.

This research report provides general information only. No part of this report may be used or reproduced or quoted in any manner whatsoever in Taiwan by the press or other persons without the express written consent of BofA Securities.
>> Employed by a non-US affiliate of BofAS and is not registered/qualified as a research analyst under the FINRA rules.
Refer to "Other Important Disclosures" for information on certain BofA Securities entities that take responsibility for the information herein in particular jurisdictions.
BofA Securities does and seeks to do business with issuers covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision.
Refer to important disclosures on page 25 to 28. Analyst Certification on page 22. Price

12 October 2026

Equity

Key Changes

(NT$)	Previous	Current
Price Obj.	3,100.00	3,260.00
2026E EPS	110.20	112.46
2027E EPS	155.85	161.36
2028E EPS	187.89	203.48
2026E EBITDA (m)	4,083,248.5	4,155,258.3
2027E EBITDA (m)	5,745,945.1	5,927,984.9
2028E EBITDA (m)	6,793,119.0	7,307,336.4

Haas Liu >>
Research Analyst
Merrill Lynch (Taiwan)
+886 2 2376 3727
haas.liu@bofa.com

Mike Yang >>
Research Analyst
Merrill Lynch (Taiwan)
+886 2 2376 3729
mike.c.yang@bofa.com

Robert Cheng >>
Research Analyst
Merrill Lynch (Taiwan)
+886 2 2376 3731
robert.cheng@bofa.com

Cathy Hsu >>
Research Analyst
Merrill Lynch (Taiwan)
+886 2 2376 3726
cathy.hsu3@bofa.com

Stock Data

Price (Common / ADR)	2,550 TWD / 453.31 USD
Price Objective	3,260 TWD / 620.00 USD
Date Established	12-Oct-2026 / 12-Oct-2026
Investment Opinion	B-1-7 / B-1-7
52-Week Range	1,375 TWD-2,590 TWD
Market Value (mn)	2,070,984 USD
Market Value (mn)	66,127,544 TWD
Shares Outstanding (mn)	25,932.4 / 5,186.5
Average Daily Value (mn)	1,767 USD
Free Float	92.2%
BofA Ticker / Exchange	TSMWF / TAI
BofA Ticker / Exchange	TSM / NYS
Bloomberg / Reuters	2330 TT / 2330.TW
ROE (2026E)	43.8%
Net Dbt to Eqty (Dec-2025A)	-37.3%

iQprofile℠ Taiwan Semiconductor Manufacturing Co.

Key Income Statement Data (Dec)	2024A	2025A	2026E	2027E	2028E
(NT$ Millions)					
Sales	2,894,308	3,809,054	5,551,264	7,790,371	9,656,518
Gross Profit	1,624,354	2,281,294	3,733,064	5,371,382	6,653,851
Sell General & Admin Expense	(96,888)	(99,222)	(110,416)	(131,671)	(145,125)
Operating Profit	1,322,053	1,936,092	3,323,549	4,866,929	6,097,850
Net Interest & Other Income	83,786	105,574	193,432	175,282	261,328
Associates	NA	NA	NA	NA	NA
Pretax Income	1,405,839	2,041,666	3,516,980	5,042,211	6,359,178
Tax (expense) / Benefit	(233,407)	(326,266)	(601,062)	(860,526)	(1,086,199)
Net Income (Adjusted)	1,173,268	1,717,886	2,916,545	4,184,666	5,276,738
Average Fully Diluted Shares Outstanding	25,932	25,933	25,933	25,933	25,933
Key Cash Flow Statement Data					
Net Income	1,173,268	1,717,886	2,916,545	4,184,666	5,276,738
Depreciation & Amortization	662,797	688,096	831,710	1,061,056	1,209,487
Change in Working Capital	(143,010)	(72,100)	(102,100)	(161,737)	(113,049)
Deferred Taxation Charge	NA	NA	NA	NA	NA
Other Adjustments, Net	133,123	(58,907)	154,974	77,281	95,310
Cash Flow from Operations	1,826,177	2,274,976	3,801,128	5,161,266	6,468,486
Capital Expenditure	(956,007)	(1,272,411)	(1,987,705)	(2,657,265)	(2,874,767)
(Acquisition) / Disposal of Investments	(58,000)	(37,014)	(42,792)	7,594	475
Other Cash Inflow / (Outflow)	149,164	165,031	76,317	411	2,594
Cash Flow from Investing	(864,843)	(1,144,393)	(1,954,180)	(2,649,259)	(2,871,699)
Shares Issue / (Repurchase)	(9)	(2)	(2)	0	0
Cost of Dividends Paid	(337,129)	(440,855)	(531,614)	(622,377)	(622,377)
Cash Flow from Financing	(346,301)	(440,345)	(564,551)	(624,253)	(622,956)
Free Cash Flow	870,171	1,002,565	1,813,423	2,504,002	3,593,719
Net Debt	(1,403,733)	(2,035,607)	(3,419,363)	(5,306,194)	(8,280,429)
Change in Net Debt	(556,872)	(647,079)	(1,277,442)	(1,889,630)	(2,974,410)
Key Balance Sheet Data					
Property, Plant & Equipment	3,234,980	3,691,841	5,015,572	6,578,562	8,037,977
Other Non-Current Assets	368,606	424,052	481,020	478,736	478,718
Trade Receivables	272,088	281,791	341,182	430,143	481,869
Cash & Equivalents	2,422,020	3,068,595	4,463,340	6,348,294	9,321,950
Other Current Assets	394,245	466,745	527,406	634,603	734,512
Total Assets	6,691,938	7,933,024	10,828,520	14,470,338	19,055,027
Long-Term Debt	958,429	896,062	895,282	892,832	892,526
Other Non-Current Liabilities	145,408	118,147	176,911	209,050	236,572
Short-Term Debt	59,858	136,926	148,695	149,268	148,995
Other Current Liabilities	1,204,667	1,321,094	1,682,008	1,732,227	1,786,768
Total Liabilities	2,368,362	2,472,229	2,902,896	2,983,378	3,064,861
Total Equity	4,323,576	5,460,795	7,925,624	11,486,960	15,990,166
Total Equity & Liabilities	6,691,938	7,933,024	10,828,520	14,470,338	19,055,027
iQmethod℠ **- Bus Performance***					
Return On Capital Employed	23.2%	28.4%	36.3%	38.3%	35.2%
Return On Equity	30.3%	35.4%	43.8%	43.3%	38.5%
Operating Margin	45.7%	50.8%	59.9%	62.5%	63.1%
EBITDA Margin	68.6%	68.9%	74.9%	76.1%	75.7%
iQmethod℠ **- Quality of Earnings***					
Cash Realization Ratio	1.6x	1.3x	1.3x	1.2x	1.2x
Asset Replacement Ratio	1.4x	1.8x	2.4x	2.5x	2.4x
Tax Rate (Reported)	16.6%	16.0%	17.1%	17.1%	17.1%
Net Debt-to-Equity Ratio	-32.5%	-37.3%	-43.1%	-46.2%	-51.8%
Interest Cover	NM	NM	NM	NM	NM

Key Metrics

* For full definitions of *iQmethod*℠ measures, see page 24.

Company Sector

Semiconductors

Company Description

TSMC was founded in 1987 and is listed in Taiwan (2330TT) and the US (TSM/NYS).TSMC is the largest and global leader in integrated circuit (IC) manufacturing. As a build-to-order foundry, it provides a wide range of value-add activities: IC manufacturing, mask-making, IC design services, turnkey solutions, and process development. We attribute its success to its proven, winning business model, unparalleled scale advantage, optimized execution, and technology scope and depth. 5 shares = 1 ADR.

Investment Rationale

The ongoing semi content growth in mobile, rise of artificial intelligence (AI), and proliferation of Internet of Things (IoT) should result in sustainable upside in aggregate computing power globally. TSMC, as the leading contract manufacturer of semiconductor chips, is in a good position to capitalize. We are structurally positive on TSMC and expect it to sustain over 15% growth with rising FCF from 2026 onwards.

Stock Data

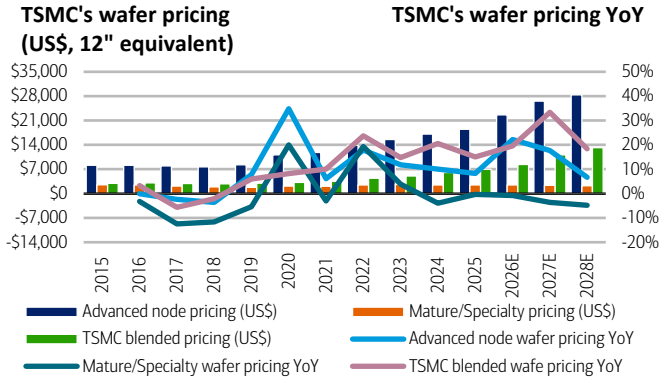
Shares / ADR	5.00
Price to Book Value	8.4x

Key Changes

(US$)	Previous	Current
Price Obj.	590.00	620.00
2026E EPS	17.26	17.61
2027E EPS	24.40	25.27
2028E EPS	29.42	31.86

Focus charts and tables

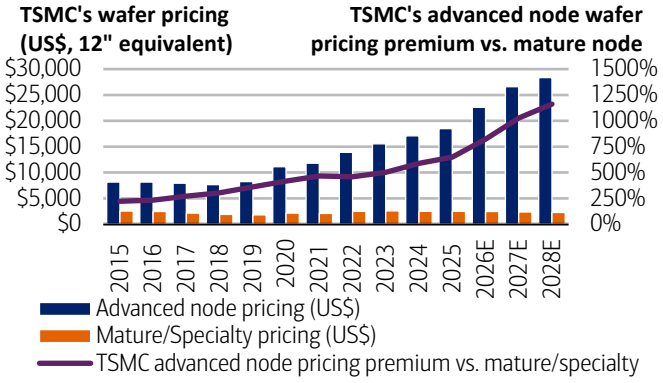
Exhibit 1: TSMC's wafer pricing by advanced vs. mature nodes
TSMC's wafer pricing by advanced vs. mature nodes



Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

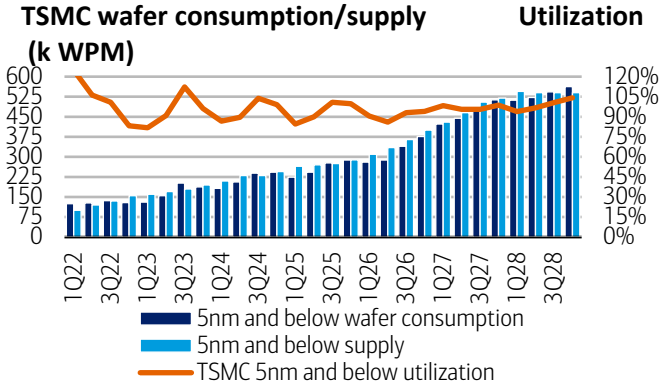
Exhibit 2: TSMC's wafer pricing gap for advanced vs. mature nodes
TSMC's advanced node pricing is now 9x higher than mature ones



Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

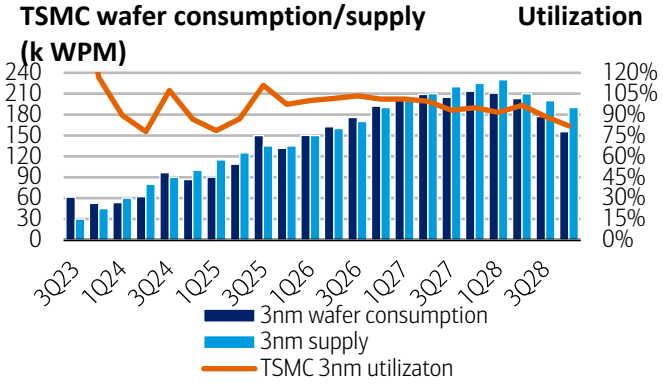
Exhibit 3: TSMC 5nm and below wafer consumption/supply/utilization
Utilization of 5nm and blow to remain high



Source: Company Data, BofA Global Research estimates

BofA GLOBAL RESEARCH

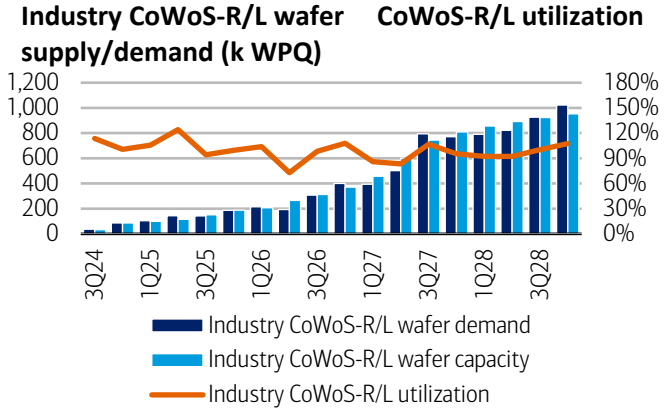
Exhibit 4: TSMC 3nm wafer consumption/supply/utilization
TSMC 3nm to see utilization staying close to full



Source: Company Data, BofA Global Research estimates

BofA GLOBAL RESEARCH

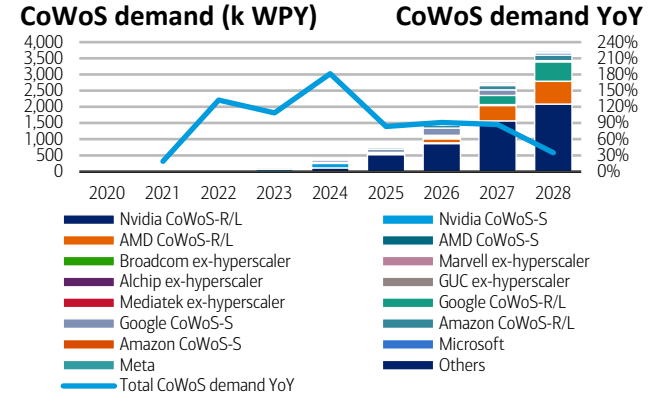
Exhibit 5: Industry CoWoS-R/L supply/demand
CoWoS-R/L utilization should stay at high levels



Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 6: Industry CoWoS demand
CoWoS demand should remain strong through 2028



Source: Global Research estimates

BofA GLOBAL RESEARCH

3Q26 sales topped guidance; GMs tracking well but recent street could be too bullish

We profile our expectation for TSMC's 3Q26 earnings on October 15th. 3Q26 sales were reported +18% QoQ in NT$, above BofAe/street expectations of +14%/+13% QoQ. Based on the average FX of NT$31.99/US$ (vs. its NT$32US$ guide), its sales in US$ term would be US$46.7bn, also above its guidance range of US$44.6-45.8bn. we believe the strength is led by strong ramp on 2nm potentially growing to 11% of its sales in 3Q26 (vs. 3% in 2Q26) for Apple's builds for its product launches and surging demand for AMD's Venice server CPU. These growth drivers on the node are supplemented by the incremental inventory stocking from Mediatek and Qualcomm for their flagship smartphone SoCs albeit at a low base and below seasonal due to weak Android demand.

On top of the 2nm scale up, we expect the utilization for 5nm and 3nm to stay above 100% backed by the continued HPC/AI build strength especially with multiple product upgrades including **1) accelerators** (GPUs and ASICs): strong demand for Nvidia's Rubin GPU, AMD's MI400, AWS's Trainium3 ramp and initial ramp for Google's TPUv8t/i more than offsetting a mild wind down in Google's Ironwood TPU, **2) CPUs:** solid demand for general purpose servers and AI data centers as noted in our deep dive report quantifying CPU demand through '30 across x86 (AMD's Turin CPUs) and ARM based CPUs (e.g. Nvidia's Vera CPU, Google's Axion CPU, AWS Graviton CPU along with incremental contribution from Microsoft Cobalt CPU) and **3) networking switch**: more and faster data transmission in the complex AI data centers requires more advanced networking switch solutions from Nvidia/Mellanox, Broadcom and Marvell.

The company's better than expected sales in 3Q26 backed by healthy shipment and pricing trend on the advanced nodes should be supportive to its GMs. However, in contrast to the recent aggressive street expectations of 67-68% GMs in 3Q26, we only model it at the high-end of its guidance of 65-67%, balancing more favorable utilization and gradual 3nm depreciation roll off partially offset by a 3-4% GMs dilution from 2nm ramp which we believe is driver for the incremental sales during the quarter.

Exhibit 7: TSMC 3Q26-4Q26 and 2026-27 estimates BofAe vs. street
We lift our EPS estimates for '26/27 to NT$112.47/NT$161.37

	3Q26E			4Q26E			2026E			2027E		
	BofAe	BofA Old	Street	BofAe	Old	Street	BofAe	Old	Street	BofAe	Old	Street
Sales (NT$mn)	$1,494,243	$1,447,447	$1,452,942	$1,652,537	$1,625,222	$1,593,320	$5,551,264	$5,477,153	$5,434,682	$7,790,371	$7,711,575	$7,323,473
Sales QoQ / YoY (%)	17.6%	13.9%	14.4%	10.6%	12.3%	9.7%	45.7%	43.8%	42.7%	40.3%	40.8%	34.8%
Gross profit (NT$mn)	$1,000,465	$961,703	$963,882	$1,120,993	$1,081,275	$1,050,189	$3,733,064	$3,654,584	$3,595,640	$5,371,382	$5,191,137	$4,844,770
GMs (%)	67.0%	66.4%	66.3%	67.8%	66.5%	65.9%	67.2%	66.7%	66.2%	68.9%	67.3%	66.2%
Operating profit (NT$mn)	$891,981	$854,803	$842,101	$1,005,999	$971,167	$922,996	$3,323,549	$3,251,539	$3,154,235	$4,866,929	$4,689,032	$4,232,893
OpMs (%)	59.7%	59.1%	58.0%	60.9%	59.8%	57.9%	59.9%	59.4%	58.0%	62.5%	60.8%	57.8%
Pretax profit (NT$mn)	$924,044	$887,585	$873,964	$1,042,707	$1,008,753	$952,145	$3,516,980	$3,446,566	$3,357,001	$5,042,211	$4,870,519	$4,433,422
Pretax margins (%)	61.8%	61.3%	60.2%	63.1%	62.1%	59.8%	63.4%	62.9%	61.8%	64.7%	63.2%	60.5%
Tax expense (NT$mn)	-$154,691	-$148,588	-$148,114	-$174,556	-$168,872	-$159,351	-$600,435	-$588,648	-$597,694	-$857,545	-$828,896	-$786,143
Tax rate (%)	16.7%	16.7%	16.9%	16.7%	16.7%	16.7%	17.1%	17.1%	17.8%	17.0%	17.0%	17.7%
Net profit (NT$mn)	$769,353	$738,997	$725,850	$868,150	$839,881	$792,794	$2,916,545	$2,857,919	$2,759,307	$4,184,666	$4,041,623	$3,647,279
Net margins (%)	51.5%	51.1%	50.0%	52.5%	51.7%	49.8%	52.5%	52.2%	50.8%	53.7%	52.4%	49.8%
EPS (NT$)	$29.67	$28.50	$28.08	$33.48	$32.39	$30.67	$112.47	$110.21	$107.37	$161.37	$155.85	$140.92

Source: BofA Global Research estimates, Bloomberg

HPC/Apple lifting 4Q26 sales outlook; resilient GM expectation but below street

For 4Q26, based on its slightly above +40% YoY full year sales guide, TSMC's sales could grow 10-15% QoQ (BofAe +11%), better than its 10-year seasonality of 7% QoQ even off a high base from 1Q-3Q26. The blended ASPs strength (BofAe +11% QoQ) should be the key driver for 4Q26 sales as shipment mix continues to shift toward advanced nodes, with revenue from 5nm and below processes representing 75% of the company's sales (vs. 66%/71% in 2Q/3Q26). Specifically, 2nm with 15%+ higher wafer pricing than 3nm could further grow its contribution to 18% of TSMC's sales during the quarter. On the other hand, we model a stable wafer shipment (BofAe +9%/-5% QoQ for advanced/mature nodes), balancing HPC demand strength and Apple's seasonal chipset builds for its new product launches and a modest drop on the specialty process output and effective capacity loss due to equipment movement for the node conversion.

While the blended pricing is even more favorable and utilization on the advanced and mature nodes in general stay healthy, we model a relatively stable GMs only up to ~67.8% levels in 4Q26 (vs. recent street and investors' expectations at 68-69%). The key factors we consider as mildly negative offset would be 1) higher depreciation: +25-30% HoH in 2H26 due to more overseas and 2nm capacity ramp and 2) 2nm still going through yield learning curve and could see more meaningful impact in 4Q26 (vs. its 3-4% GMs dilution guide in 2H26) as the sales contribution grows.

Exhibit 8: Our 2026-28E earnings are 5-15% above consensus as we model a higher sales growth
TSMC 2026-28 estimates BofAe vs. Street

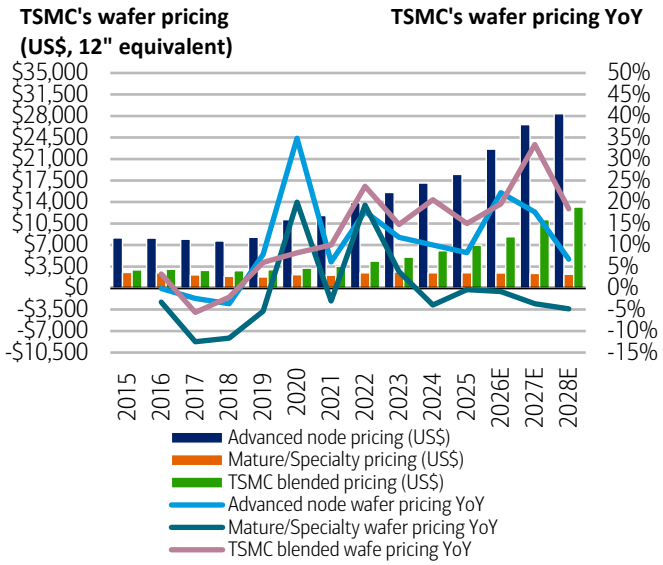
	2026E			2027E			2028E		
	BofAe	**Street**	**vs. street**	**BofAe**	**Street**	**vs. street**	**BofAe**	**Street**	**vs. street**
Sales (NT$mn)	$5,551,264	$5,434,682	2%	$7,790,371	$7,323,473	6%	$9,656,518	$9,152,144	6%
Sales QoQ / YoY (%)	**45.7%**	**42.7%**		**40.3%**	**34.8%**		**24.0%**	**25.0%**	
Gross profit (NT$mn)	$3,733,064	$3,595,640	4%	$5,371,382	$4,844,770	11%	$6,653,851	$6,033,826	10%
GMs (%)	**67.2%**	**66.2%**	**1%**	**68.9%**	**66.2%**	**3%**	**68.9%**	**65.9%**	**3%**
Operating profit (NT$mn)	$3,323,549	$3,154,235	5%	$4,866,929	$4,232,893	15%	$6,097,850	$5,255,681	16%
OpMs (%)	**59.9%**	**58.0%**	**2%**	**62.5%**	**57.8%**	**5%**	**63.1%**	**57.4%**	**6%**
Pretax profit (NT$mn)	$3,516,980	$3,357,001	5%	$5,042,211	$4,433,422	14%	$6,359,178	$5,553,349	15%
Pretax margins (%)	63.4%	61.8%	2%	64.7%	60.5%	4%	65.9%	60.7%	5%
Tax expense (NT$mn)	-$600,435	-$597,694	0%	-$857,545	-$786,143	9%	-$1,082,440	-$998,175	8%
Tax rate (%)	17.1%	17.8%	-1%	17.0%	17.7%	-1%	17.0%	18.0%	-1%
Net profit (NT$mn)	$2,916,545	$2,759,307	6%	$4,184,666	$3,647,279	15%	$5,276,738	$4,555,174	16%
Net margins (%)	52.5%	50.8%	2%	53.7%	49.8%	4%	54.6%	49.8%	5%
EPS (NT$)	**$112.47**	**$107.37**	**5%**	**$161.37**	**$140.92**	**15%**	**$203.48**	**$176.56**	**15%**

Source: BofA Global Research estimates, Bloomberg

2026-27 sales and margins outlook solidified by healthy AI demand

Factoring in stronger CPU and AI GPU/ASIC demand outlook as noted in our Semicon Taiwan takeaways report where we updated our industry models, we are lifting our US$ sales expectation for TSMC in '26/'27 to +42%/+40% YoY respectively (vs. BofA prior +41%/+40% YoY). More importantly, with utilization staying tighter for longer along with inflationary cost pass through (e.g. equipment and consumables), TSMC should raise its advanced node and packaging pricing by 5-10% in the beginning of 2027 to reflect its value. This along with a mild 3-5% price hike on the mature nodes where the utilization is back to 90-95% as it optimizes the product mix following the strategic 6"/8" capacity retirement should lift its GMs to 67%/69% in 2026/27 (vs. BofA prior 67%/67%).

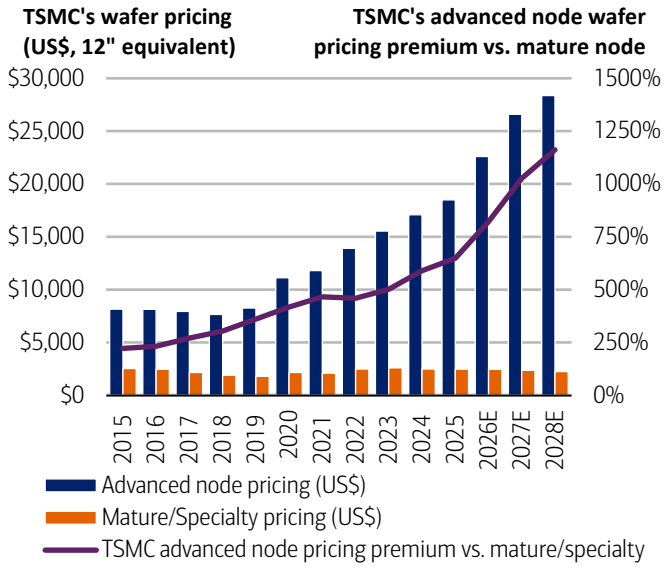
Exhibit 9: TSMC's wafer pricing by advanced vs. mature nodes
TSMC's wafer pricing by advanced vs. mature nodes



Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 10: TSMC's wafer pricing gap for advanced vs. mature nodes
TSMC's advanced node pricing is now 9x higher than mature ones



Source: BofA Global Research estimates

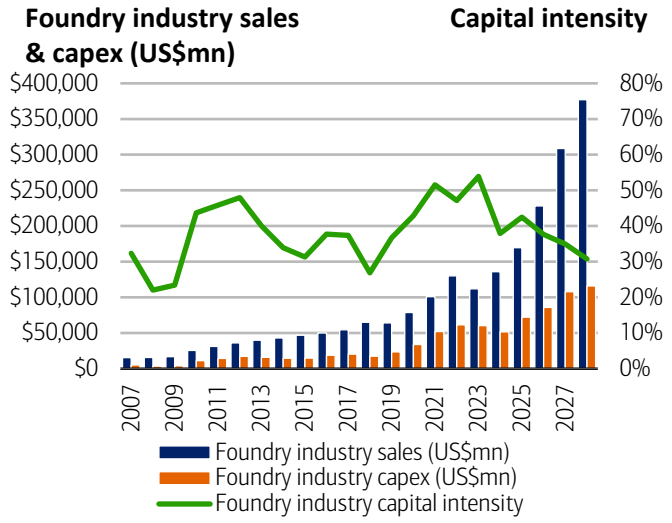
BofA GLOBAL RESEARCH

Capex on track for lifted guide for 2026 and will grow 20%+ CAGR from '26-28

We lifted our front-end foundry industry capex growth expectation to 19%/26% YoY for '26/27 to US$86bn/US$108bn, supporting a CAGR of 16% from '26-28. This is mostly led by our higher expectation on TSMC's capex on stronger wafer demand on N3 and N2 from GPUs, ASICs, CPUs, networking switches and HBM base dies and initial ramp of A16, along with incremental spending reacceleration from tier 2 foundries.

Separately, we also raised the back-end packaging and testing industry spending estimates from 66%/36% YoY to 82%/21% YoY for '26/27 to US$27bn/US$33bn, supporting a CAGR of 15% from '26-28. The upward revision is driven by our higher expectation of TSMC's more aggressive capacity addition on CoWoS-R/L to accommodate more demand from its customers amid growing level of chipset complexity and reticle size and higher level of capacity readiness of SoIC for higher 3D packaging adoption for logic and memory dies integration and CPO. The more aggressive spending from ASE is set to grow by another 30-50% YoY in 2027 off a high base in 2026 on its own FoCoS capacity ramp and TSMC's CoWoS licensing.

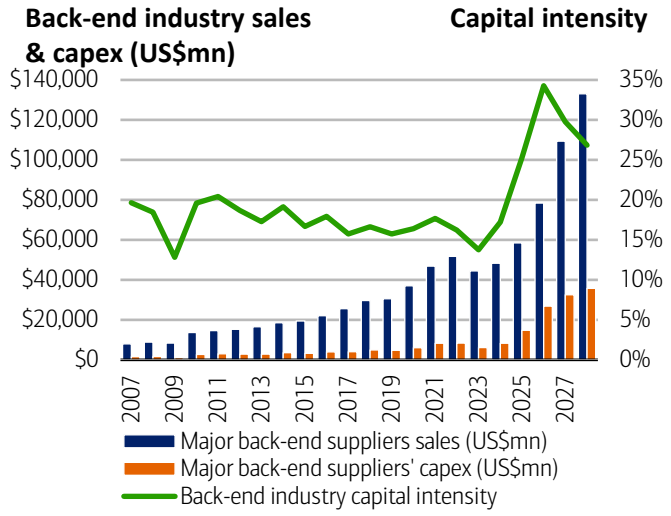
Exhibit 11: Front-end sales, capex, and capital intensity
Front-end capex should reach US$86bn/US$108bn in '26/27



Source: Company Data, BofA Global Research estimates, Bloomberg

BofA GLOBAL RESEARCH

Exhibit 12: Back-end sales, capex, and capital intensity
Back-end capex should reach US$27bn/US$33bn in '26/27



Source: Company Data, BofA Global Research estimates, Bloomberg

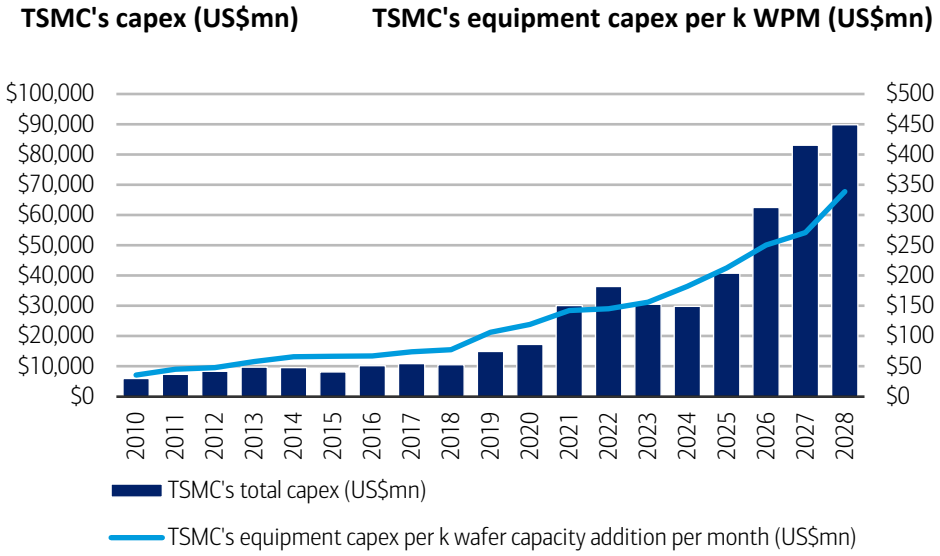
BofA GLOBAL RESEARCH

Specifically on TSMC, based on our bottom up capacity analysis across front-end manufacturing and back-end packaging, TSMC's capex spending is tracking to US$63bn for 2026 (vs. its latest guidance of US$60-64bn) mostly skewed toward advanced front-end manufacturing including 55-60k WPM 3nm capacity addition, 60k WPM 2nm capacity ramp, initial 3-5k WPM A16 capacity installation. The back-end packaging investment should be largely driven by its CoWoS capacity growth while the spending on the equipment pull-in for CPO and CoPoS remains limited as most of the equipment are still under sampling process for qualification and meaningful shipment will be in 2027-28.

In addition to the earlier tool pull-in schedule due to tighter equipment supply with memory makers expanding, the fast-growing capex should reflect the rising cost per 1k WPM on the advanced nodes vs. mature nodes, with the equipment cost per 1k WPM advanced wafer capacity expansion has grown from US$156mn in 2023 to US$241mn in 2025. Even factoring in more common use of equipment on the advanced nodes, the capex/1k WPM advanced capacity could further rise by 20% to US$289mn in 2027 with the technology architecture upgrade.

On top of the investment in the core front-end manufacturing, we expect the company to slightly outspend in the back-end packaging and estimate it is at the upper half of its guidance for 10-20% total capex, mainly for more aggressive capacity expansion to meet its customers' demand amid growing competition from Intel's EMIB-T solution and technology innovation. On the other hand, the maintenance capex could be reduced for specialty technologies as the company gradually retires the equipment.

Exhibit 13: TSMC's total Capex vs. equipment spending per k WPM capacity addition
TSMC spend 1-2x more on 1k WPM expansion due to its shift to advanced nodes in recent years

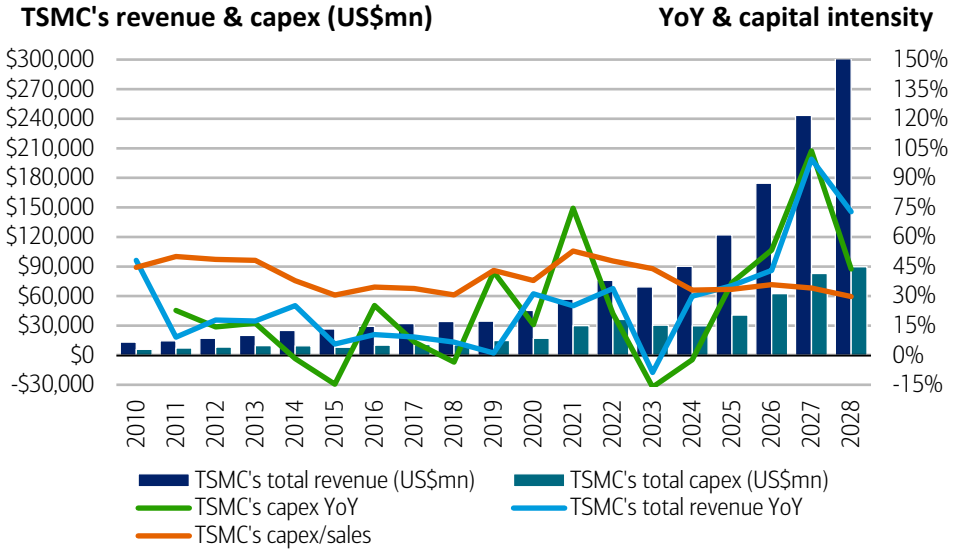


Source: Company Data, BofA Global Research estimates

BofA GLOBAL RESEARCH

Considering the capex required for further 3nm capacity expansion in the additional phases in its Tainan megafab, Arizona and the recently announced 2nd fab in Kumamoto on strong AI/HPC/automotive demand, ramping scale for 2nm family including A16, R&D on A14 supplemented by multiple back-end packaging technologies in progress (e.g. CoWoS, CoPoS, SoIC, WMCM) and lower capex efficiency in its US operation, our bottom up model shows TSMC's capex could be US$83bn/US$90bn in 2027/28 at a CAGR of 20% CAGR from 2026-28.

Exhibit 14: TSMC's revenue and Capex trend
TSMC's sales should still outgrow Capex to keep its capital intensity manageable



Source: Company Data, BofA Global Research estimates

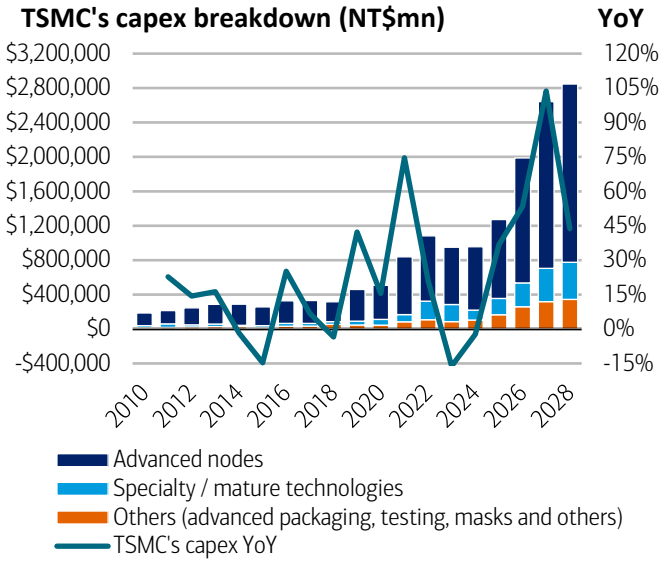
BofA GLOBAL RESEARCH

Beyond the technology upgrades and capacity expansion, we would note the equipment price hike would drive additional spending required by TSMC in the next few years. Our feedback from the tool suppliers noted they plan to hike the price by 10-15% for the front-end manufacturing equipment and 5-10% for the back-end packaging equipment effective from 2027 to reflect higher component and labor cost.

With TSMC's stronger bargaining power against the equipment supply chain vs. its peers, it should be able to fully pass on the ~10% higher nominal capex to its customers by lifting its foundry pricing when the tools are delivered for its production ramp in 2H27, setting up a good start for the pricing outlook in 2028.

While the surge in headline capex hike could lead to concern on accelerating depreciation, we estimate it will only grow at 21% CAGR from 2026-28 due to modestly higher portion of the spending towards facility and clean room with much longer depreciation cycle than equipment. This should keep the depreciation still undergrowing its sales of +32% CAGR, sustaining GMs expansion to 68-69% levels by 2028.

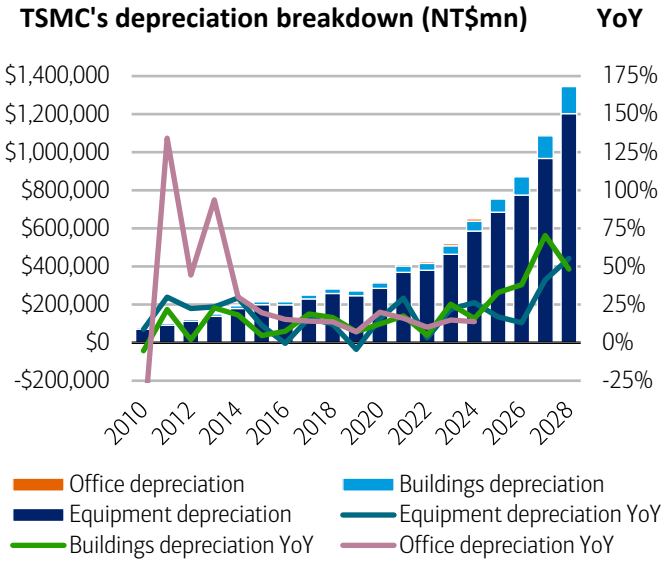
Exhibit 15: TSMC Capex spending breakdown by technology
TSMC spends the most on advanced front-end but packaging is outgrowing



Source: Company Data, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 16: TSMC's depreciation trend
TSMC's depreciation growth will be led by equipment



Source: Company Data, BofA Global Research estimates

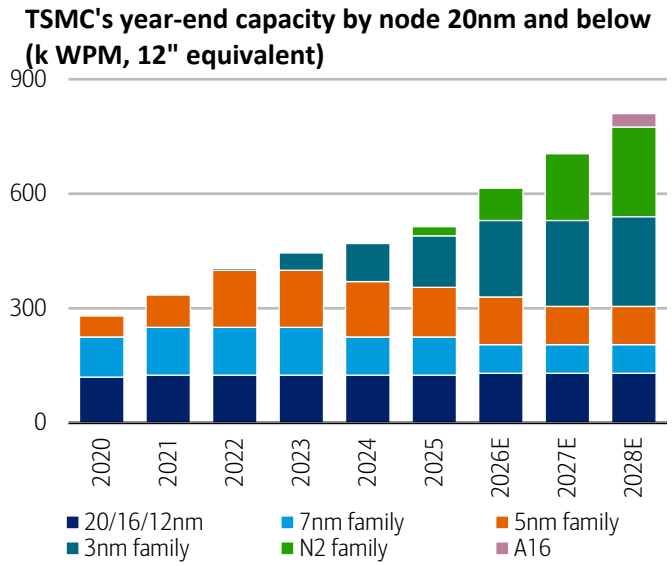
BofA GLOBAL RESEARCH

We keep our Buy rating on TSMC and lift PO from NT$3,100 to NT$3,260 on unchanged 20x '27E P/E and US$ADR PO from $590 to $620. Our EPS for 2026/27/28E are now NT$112/161/203 (vs. NT$110/156/188 prior) as we factor in stronger CPU and AI GPU/ASIC demand outlook. Its current valuation seems undemanding at 16x '27E P/E (vs. its long term P/E average of 11-21x).

Capacity expansion accelerates across advanced front-end and back-end

As TSMC is in the process to retire a portion of its equipment on 6", 8" and mature 12" processes while still adding the capacity on the advanced nodes, we estimate the company's total front-end manufacturing capacity will drop by 2% YoY in 2026.

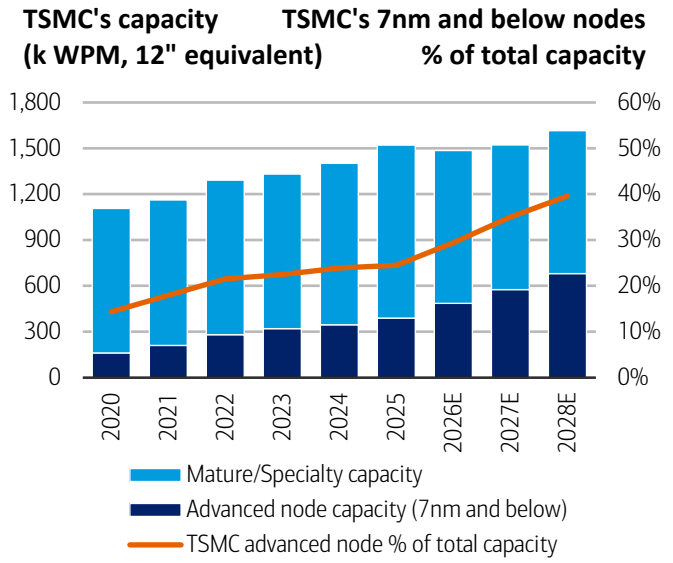
Exhibit 17: TSMC's year-end capacity by node for 20nm and below
TSMC will expand aggressively on 3nm and 2nm through 2027



Source: Company Data, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 18: TSMC's advanced node capacity as % of its installed base
TSMC's advanced nodes will reach 40% of its total capacity



Source: Company Data, BofA Global Research estimates

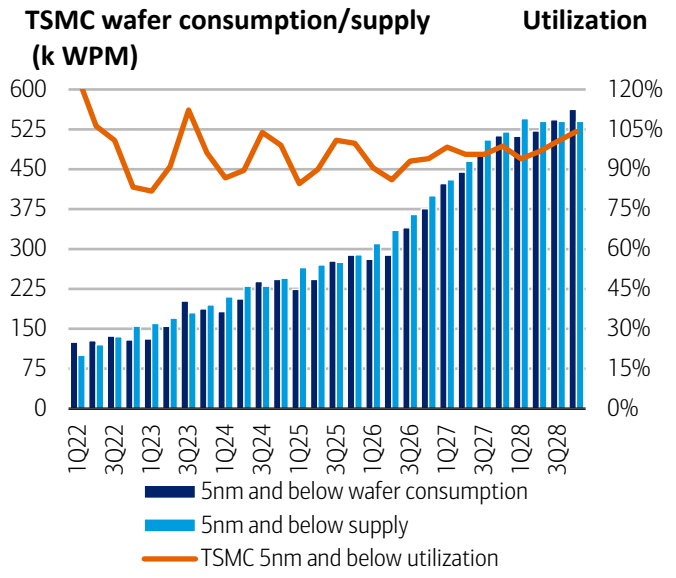
BofA GLOBAL RESEARCH

The trend could be more stable into 2027-28 as TSMC strives to optimize its capacity mix and across node conversion to accommodate its customers' demand especially on the leading-edge technologies, with its advanced capacity set to grow by 10% CAGR from 2025-28 balanced by the mature capacity reduction at -8% CAGR.

Based on our estimate, TSMC's capacity for 7nm and below nodes will grow from 320k WPM per month in 2023 to ~700k WPM by 2028 at a 16% CAGR, through expansion and efficient conversion from higher equipment commonality, cross technology planning and AI productivity gains (e.g. tool idle time and chamber cleaning schedule optimization). This will also support the mix of 7nm and below capacity climb toward 40-45% of its total capacity (vs. 24% in 2023). By region, we expect the capacity expansion through 2027 to enable its overseas fabs to account for 17% of TSMC's total capacity by 2028 (vs. 7% in 2020) and similarly for the advanced node capacity (vs. ~0% in 2023), mainly in Arizona and Kumamoto.

Specifically, we believe the company is on track to its target on 25% CAGR capacity expansion for its 5nm and 3nm from '22-27. We model its 3nm capacity will reach 190k/230k WPM by 4Q26/27. On N2 family, TSMC plans to grow capacity by 70% CAGR from '26-28, with 1st year ramp delivering 45% more capacity vs. 3nm as it rolls out 5 fabs at once.

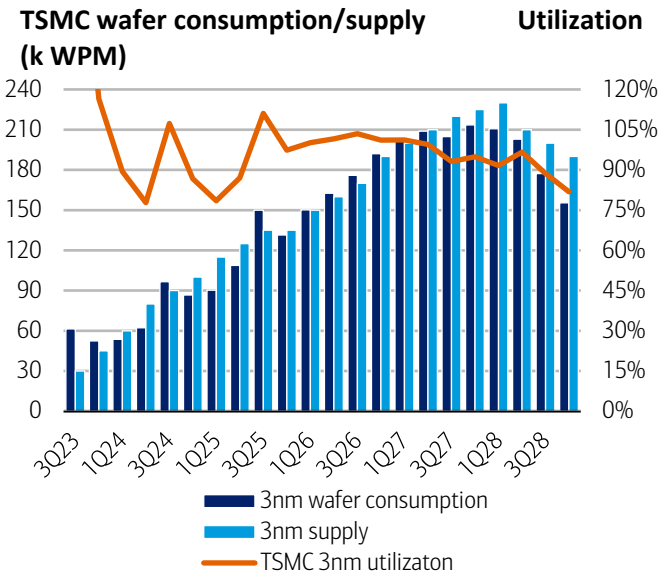
Exhibit 19: TSMC 5nm and below wafer consumption/supply/utilization
Utilization of 5nm and blow to remain high



Source: Company Data, BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 20: TSMC 3nm wafer consumption/supply/utilization
TSMC 3nm to see utilization staying close to full

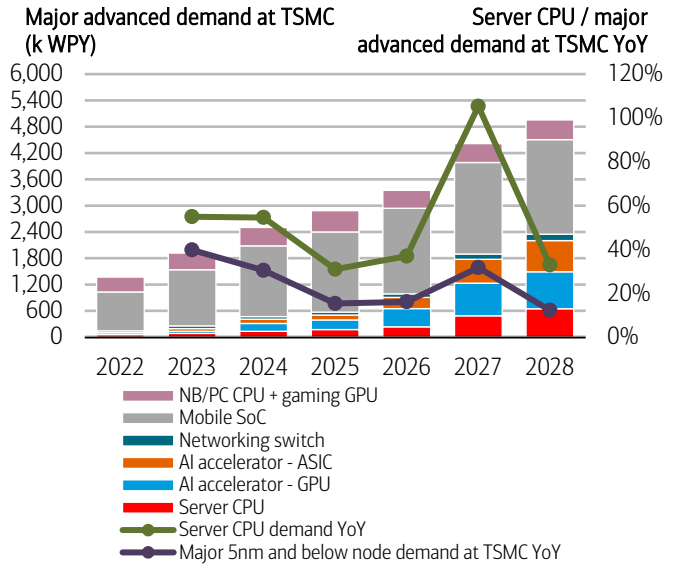


Source: Company Data, BofA Global Research estimates

BofA GLOBAL RESEARCH

On the demand, with 5nm and 3nm generating ~50% of TSMC's revenue in 2025, we analyze the supply and demand landscape through our bottom-up capacity model and wafer consumption by the company's major customers for their major products on these nodes. Based on our estimate, the utilization for 5nm and below nodes could stay high at least 90% if we only consider the effective capacity which is especially important during the period of frequent capacity conversion and new node ramp in the next few years.

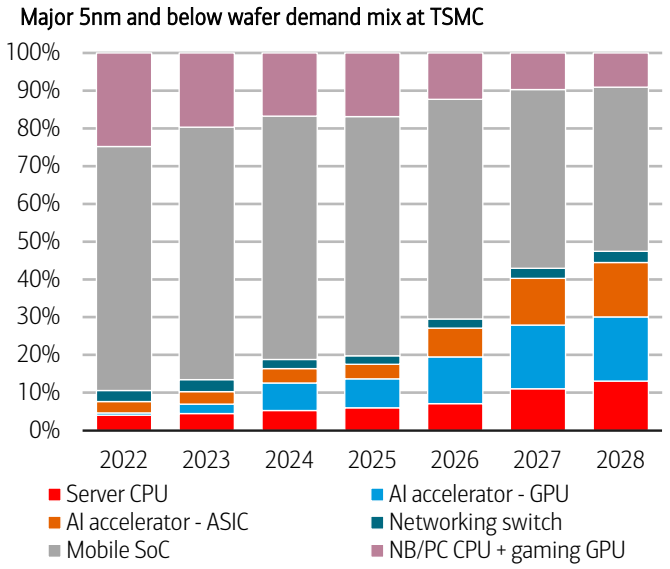
Exhibit 21: Major 5nm and below wafer demand at TSMC
Server CPU demand outgrows overall major advanced chipsets



Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 22: Major 5nm and below wafer demand mix at TSMC
Server CPU accounts for 13% in 2028

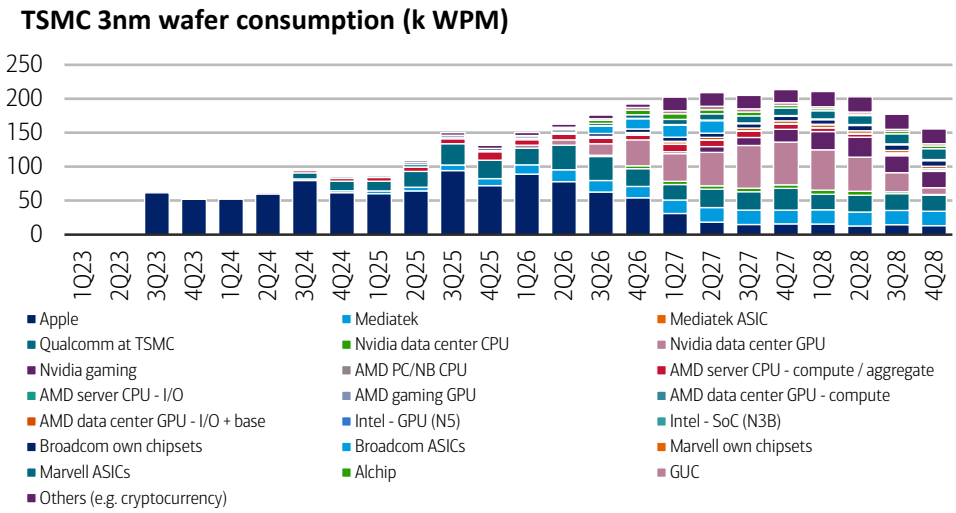


Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

We would note 3nm, as a sweet spot for high performance computing, networking and flagship smartphone chipsets, could see the utilization staying close to full even with continued expansion on strong project pipeline, sustaining continued price hike even in 2027 following the price surge in 2025-26. The node could be even more profitable from 2028-29 when the fixed cost starts rolling off following 5 year depreciation cycle.

Exhibit 23: TSMC 3nm wafer consumption by customers
Mediatek/Qualcomm/Nvidia should remain the largest customers following Apple's migration to 2nm

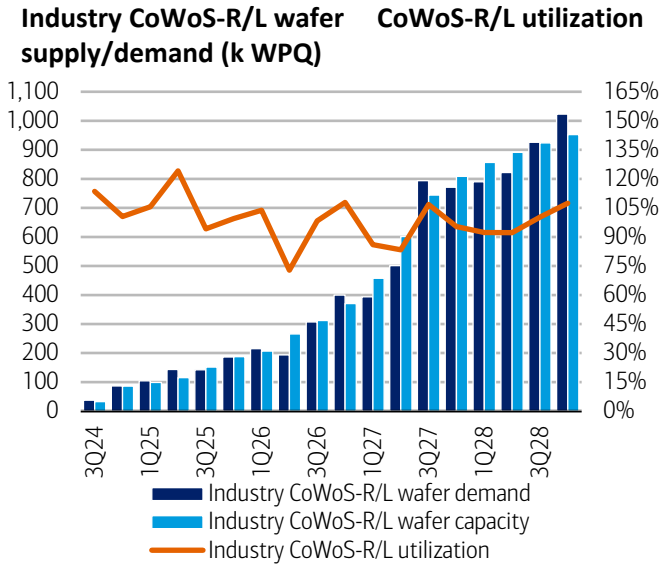


Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

On the back-end packaging, with surging server CPU demand on top of the already strong order backlog for GPUs and ASICs, we estimate the industry's CoWoS supply will need to grow by 52% CAGR from 2026-28. TSMC will still lead the capacity expansion from 120k WPM in 4Q26 to 210k WPM in 4Q27, while the OSATs will also grow their portion of the supply with ASE and Amkor adding capacity to 70-75k+ WPM (vs. 30-35k in 4Q26) and 25-30k WPM (vs. 15-20k by 4Q26) in 4Q27 respectively. Specifically, while TSMC should still lead the supply for complex GPU and ASIC packaging, ASE grows as a major 2nd source with its capacity share growing from 18% of the industry in '26 to 29% in '28 and proven track record as the major CoWoS full process service provider for AMD's Venice server CPU.

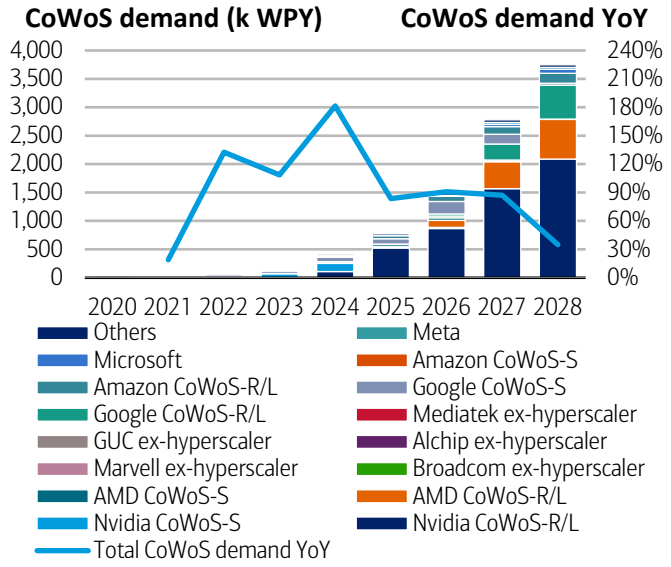
Exhibit 24: Industry CoWoS-R/L supply/demand
CoWoS-R/L utilization should stay at high levels



Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

Exhibit 25: Industry CoWoS demand
CoWoS demand should remain strong through 2028



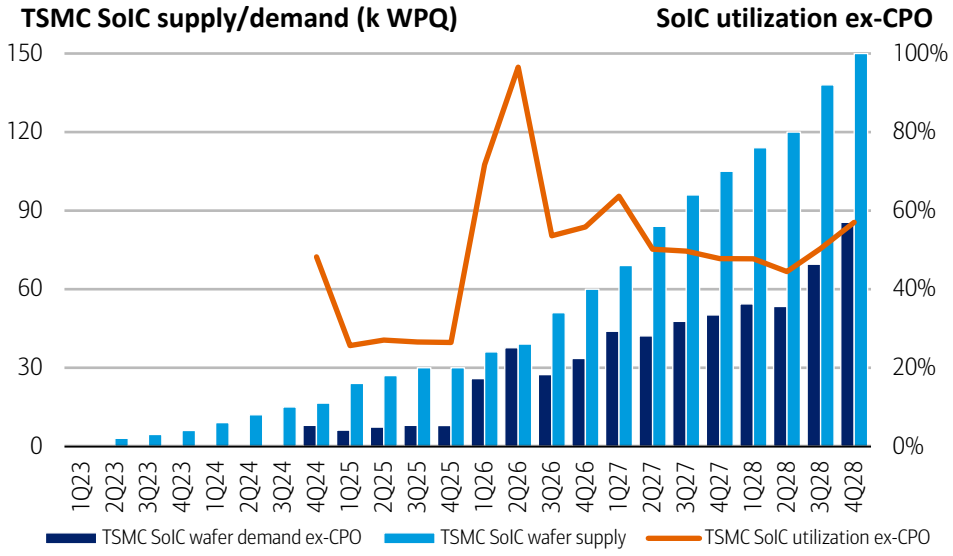
Source: Global Research estimates

BofA GLOBAL RESEARCH

We estimate TSMC's SoIC capacity will reach 20k WPM by 4Q26 and will need to grow to 35k WPM by 4Q27 and further to 50k WPM by 4Q28 with the applications proliferating from AMD's GPUs to Apple's M-series CPUs in 2026, hyperscalers' customized ARM based server CPUs in 2027 and adoption by Nvidia for its Feynman GPU along with the inflection of CPO penetration in 2028.

Similar to CoWoS, the SoIC technology could be even more important for TSMC going forward and the capacity ramp could be steep given much faster data transmission required for the HPC chipsets across CPUs, GPUs, AI accelerators and networking switches. At full utilization, we estimate SoIC could contribute US$2.9bn annually to TSMC in 2028 (1% of the company's total sales)

Exhibit 26: TSMC SoIC supply/demand trend
TSMC's SoIC capacity will reach 20k WPM by 4Q26 and will need to grow to 35k WPM by 4Q27



Source: Company Data, BofA Global Research estimates

BofA GLOBAL RESEARCH

Solid technology roadmap and execution easing competition concern

TSMC is pursuing a dual track strategy for its front-end manufacturing, advancing the solution for both its consumer and HPC customers. The nodes targeted for consumer chipsets requiring cost efficiency and IP reuse will be introduced on an annual basis to match its customers' yearly product launch schedule. For the HPC applications, the company targets to roll out a new node with biennial cadence, supporting major leaps in performance, transistor density, and power density.

Exhibit 27: Front-end manufacturing roadmap comparison
TSMC's advanced node technology roadmap advancing faster vs. Intel and Samsung and has the widest offerings targeted for different applications

Company	End Market / Segment	2023	2024	2025	2026	2027	2028	2029
TSMC	Data Center	N4P & N4X		N3X & N2	N2P & A16	N2X		A12
	High-end consumer/mobile	N3	N3E	N3P & N2	N2P		N2U & A14	A13
	Automotive		N5A		N3A			
	Mainstream		N4P	N4C	N3C		N2U	
Intel Foundry	Data center and high-end client	3	3	3 & 18A		18A-P		14A & 14A-E
	Mature	16	16-E		Intel/UMC 12			
Samsung Foundry	Data Center	SF4P	SF4X				SF2X	SF2Z
	High-end consumer/mobile	SF4P	SF3	SF2	SF2P	SF2P+		SF1.4
	Automotive	SF5A		SF4A		SF2A		
	Mainstream			SF4U				

Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

This is evidenced in its N2 family and below nodes, with N2U ramping in '28 for mobile applications and A13 node in '29 which will be backward compatible to A14. It will also have A12 node in '29 integrated SPR to address power integrity and current delivery constraint suitable for AI/HPC systems. The strategy along with the continued expansion of the derivative offerings for main nodes should position TSMC well to defend its market share with its competitors claiming their progress on the advanced manufacturing.

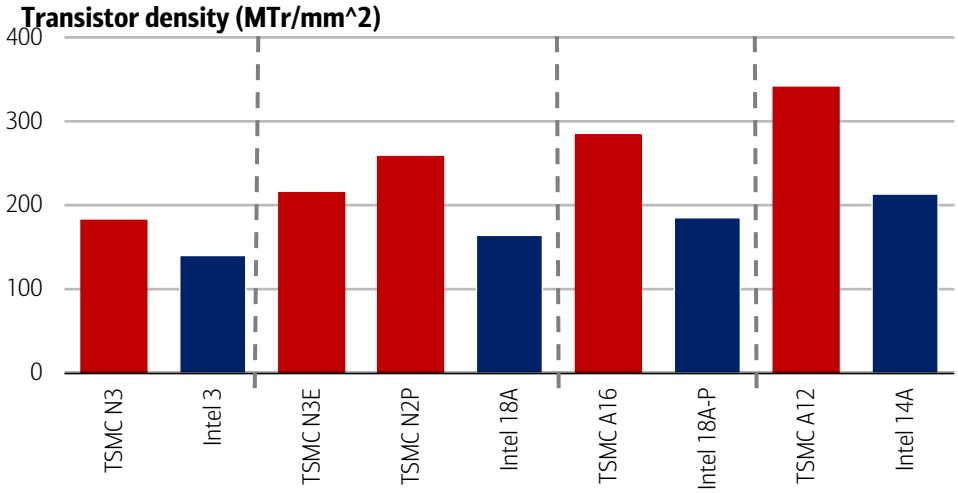
Specifically, Intel's 18A has been in mass production since 4Q25 for its Panther Lake and Clearwater Forest chipsets though yield will not be fully fine tuned until 1Q27 at 25-30k WPM capacity scale in its Oregon and Arizona fabs. The follow on derivative technology, 18A-P, has recently started risk production featuring 9% performance gains though may not enter into mass production until 2027 at the earliest.

Longer term, Intel's next major upgrade will be on 14A which is guided to start risk production in 2028 for commercialization in 2029. It is important to note A14 will be the first major node Intel provides with its external foundry customers and 0.9 PDK is targeted to be shipped in 4Q26.

For Samsung, at its recent SAFE Forum in 2Q26, the company reaffirmed four generations of 2nm roadmap, including SF2 which has been in production since 2025, SF2P in 2026, SF2P+ in 2027-28 and SF2X (no firm date). Notably, the previously announced derivative technologies, including SF2Z featuring backside power for HPC applications and SF2A for automotive applications, were not included in its latest roadmap which should be folded as part of its SF2P+ offerings. With a modest delay for its 2nm progress, the company also pushed out its SF1.4 mass production to 2029 (vs. original 2027).

On the capacity and yield planning for its advanced nodes, Samsung noted its SF2 yield is still at 50-60% levels but confirmed its SF2P+ tooling has been in installation through 2026 for mass production in 2027 in its Taylor fab. Its capacity expansion in the US along with the existing 20k WPM capacity on SF2 in Korea put the company's capacity installed base on the node at 35k WPM by the end of 2027.

Exhibit 28: TSMC's front-end manufacturing transistor density vs. Intel
TSMC is widening its transistor density advantage against Intel as technology advances



Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

Based on the capacity planning and technology roadmap, we believe TSMC retains the leadership through its roadmap and yield execution and scale. Specifically, the company's transistor density has been outperforming by ~30% for its N3/N3E vs. Intel 3/18A. With similar production timing and targeted applications, we estimate the transistor density gap could be even widened to ~50%/~60% for TSMC's upcoming A16/A12 nodes when comparing with Intel 18A-P/14A.

In addition to delivering superior specs on the power, performance and area, TSMC's execution reliability has become self-reinforcing: superior yield and on-time delivery on the leading edge technologies win customer trust. The profitable business opportunity then translates the cash flow funds to the next leg of capex cycle required for further technology and capacity lead. In contrast to TSMC's 150-200k+ WPM capacity scale on every major node at mature yield, Samsung and Intel running their business at 15-20% of the company's scale at 60-70% yield is not sufficient to meet their own demand nor the stringent requirement from the external customers especially as the opportunity cost gets higher on the advanced manufacturing.

On the back-end packaging and testing solutions, TSMC provides its customers with higher complexity of the system level integration across advanced logic dies, CPO components and HBM through its CoWoS and SoIC technologies for the compute density and efficiency optimization. To accommodate larger size package, it is rolling out 5.5-reticle CoWoS in '26 with yield reaching 98%+ and will push the technology to 9.5x/14x reticle limit in '27/'28

Longer term, the company is developing SoW for up to 40-reticle packaging and CoPoS potentially for late '28 / early '29 mass deployment. The company is also upgrading CPO with COUPE solutions from PCB and substrate to interposer for 10x power efficiency uplift and 20x latency reduction though the contribution would still be limited based on our estimate at least through '27 before accelerating to 1% of its sales in '28 with more scale up switch adoption.

While CoPoS mass ramp is toward '29, the CoWoS upgrade at good yield should keep its position vs. Intel's EMIB-T currently at 80-85% pilot yield seeing risk of losing business if it fails to deliver 95%+ mass production yield by mid-'27.

SpaceX could be an anchor customer for TSMC, but not co-owning capacity/tech

On top of the competition risk from the incumbent players, the headline of Elon Musk's aggressiveness to build Terafab targeting long term 1 terawatt equivalent chipset capacity for AI compute deployment in the space term also triggers the investor concerns on TSMC's positioning. We believe the risk is overstated as there could be a delay in the qualification process of Intel's 18A for external customers and 14A due to IP and manufacturing flow issues based on the recent supply chain feedback. In addition, the equipment vendors also noted Terafab is only evaluating on equipment spec and potential procurement rather than placing firm orders. With extending equipment delivery lead time, this could put the company in an unfavorable position to receive tools for the initial production ramp in '29 it scheduled.

In the mid-term, Terafab's applications under Texas's JETI program earlier this year outlined ~US$119bn of planned investment across 4 phases from '26-36, with ~60% of it on tool procurement. The spending in '26-28 would be highly skewed toward facility construction before it starts to pull in US$6-8bn equipment per annum from '29-36.

We estimate the company will need to spend US$600mn+ for every 1k WPM expansion on the advanced nodes it targets in the US. Factoring the investment it targets to spend in the next decade, Terafab could only grow its capacity installed base by ~10k WPM per annum at maximum to 96k WPM front-end manufacturing capacity installed by '36, well below TSMC's scale, even if it can successfully deliver the technology roadmap through its own R&D effort and Intel partnership. This would prevent Terafab's operation from the economies of scale as the node migration continues, keeping TSMC as the major foundry partner for Tesla/SpaceX.

Exhibit 29: Wafer supply/demand analysis on Elon Musk's AI deployment target
With Terafab's near term delay and limited long term scale, TSMC will continue to be critical for Musk's ambitious AI compute deployment plan

Wafer supply/demand analysis on Elon Musk's AI deployment target	
SpaceX's annual orbital AI deployment target in 2028 (GW)	10
SpaceX's annual orbital AI deployment target in 2029 (GW)	100
Musk's long term target to deploy AI compute annual run rate (GW)	1,000
TDP of Nvidia's Feynman GPU (W)	6,000
GPU die output per front-end wafer (units)	55
Wafer demand for Feynman compute die for SpaceX's in '28 (k WPM)	10
Wafer demand for Feynman compute die for SpaceX's in '29 (k WPM)	101
Wafer demand for Feynman compute die to meet Musk's LT target (k WPM)	1,006
TSMC's 3nm and below capacity in 2028 (k WPM)	505
Terafab advanced wafer capacity scale by 2028 (k WPM)	0
Terafab capacity by 2036 based on its US$119bn investment plan (k WPM)	96

Source: BofA Global Research estimates

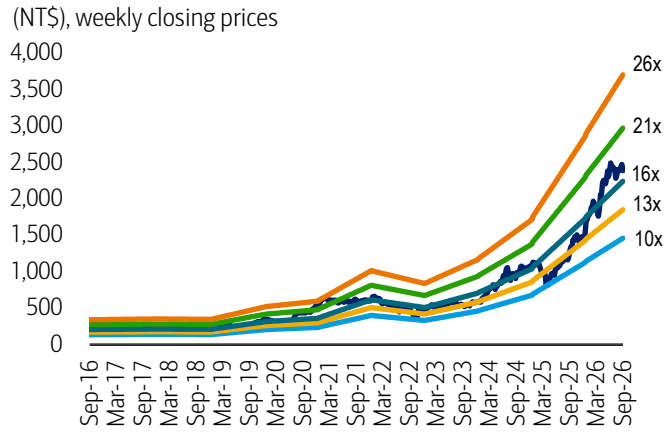
BofA GLOBAL RESEARCH

We also analyze the wafer consumption from SpaceX's annual orbital AI deployment target in '28/'29 of 10GW/100GW and Musk's long term 1TW target. Assuming the compute it plans to deploy in '28-29 is all powered by the most advanced Nvidia's Feynman GPUs delivering 6kW per package, this would drive 1.7mn/16.7mn chipset demand in '28/'29 respectively which requires 10k/101k monthly advanced wafer production. From the supply perspective, as Feynman GPU will be built on TSMC's A16 node, the chipset demand should be fully accommodated. With limited support from Terafab, SpaceX in this scenario could become one of the largest customers for TSMC consuming ~20% of its advanced node capacity in '29 (vs. ~1% in '26 and ~2% in '28).

Longer term, Musk's ambitious target to deploy 1 terawatt AI compute power annually would consume 1mn+ wafers per month on the advanced nodes. To have higher level of the control on capacity sufficiency, Terafab could consider to partner with TSMC though the likelihood of TSMC licensing technology or operating a fab with Terafab holding a meaningful share should be low. Specifically, we would note TSMC's JV fab record is limited only with WaferTech, SSMC, VIS, JASM, ESMC and AVSM mostly on mature nodes and it did not co-own a fab with Apple who contributed ~21% of its revenue from '15-25 mostly on the advanced processes as it keeps full business control and preserves customer neutrality.

TSMC valuation reasonable and has room to re-rate with its HPC customers

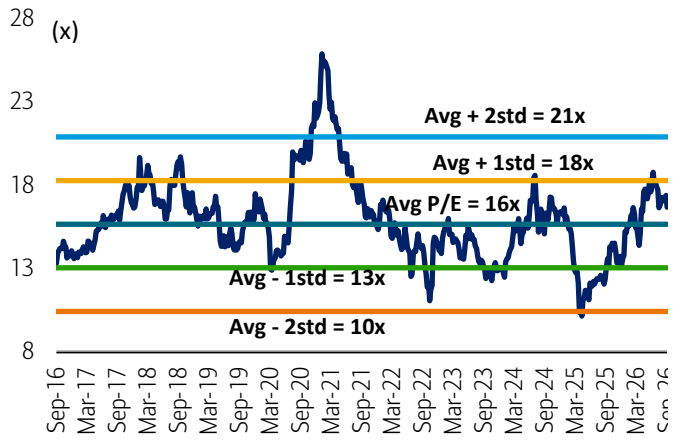
Exhibit 30: TSMC's P/E valuation band
TSMC's share at 16x '27E P/E is undemanding



Source: BofA Global Research estimates, Bloomberg

BofA GLOBAL RESEARCH

Exhibit 31: TSMC's valuation based on BofA estimates
TSMC's valuation of 16x '27E P/E is at the middle of its historical 10-21x range

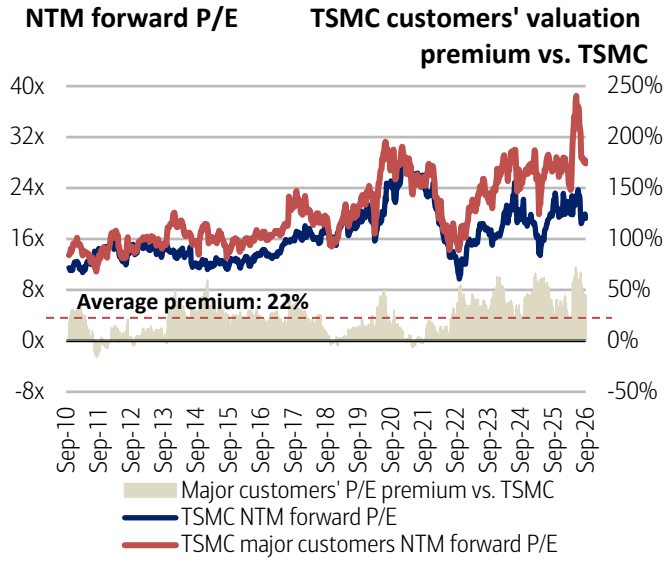


Source: BofA Global Research estimates

BofA GLOBAL RESEARCH

TSMC's current valuation remains well supported when viewed through both its historical context and its relative positioning vs. its customers. In the past decade, TSMC's forward P/E has averaged ~16x with a range of 10-21x through the cycles.

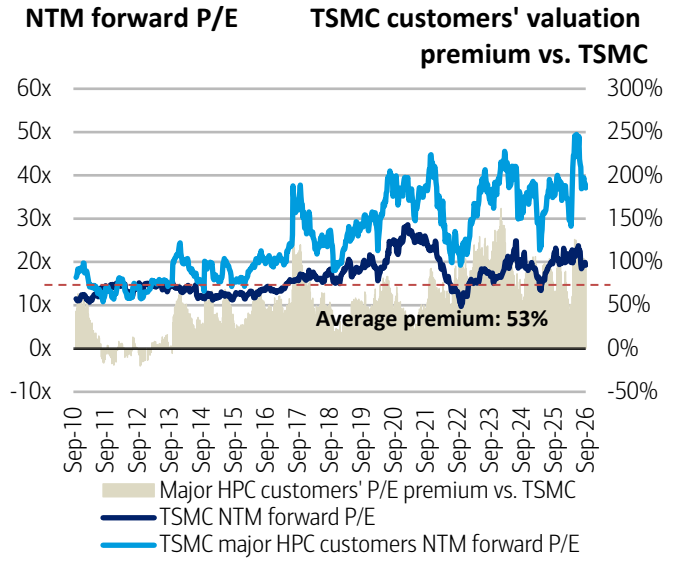
Exhibit 32: TSMC's P/E trend vs. its customers
TSMC customers' P/E is now at 45% premium to TSMC vs. 22% average



Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

Exhibit 33: TSMC's P/E trend vs. its HPC customers
HPC fabless are the main driver for valuation premium at 100%



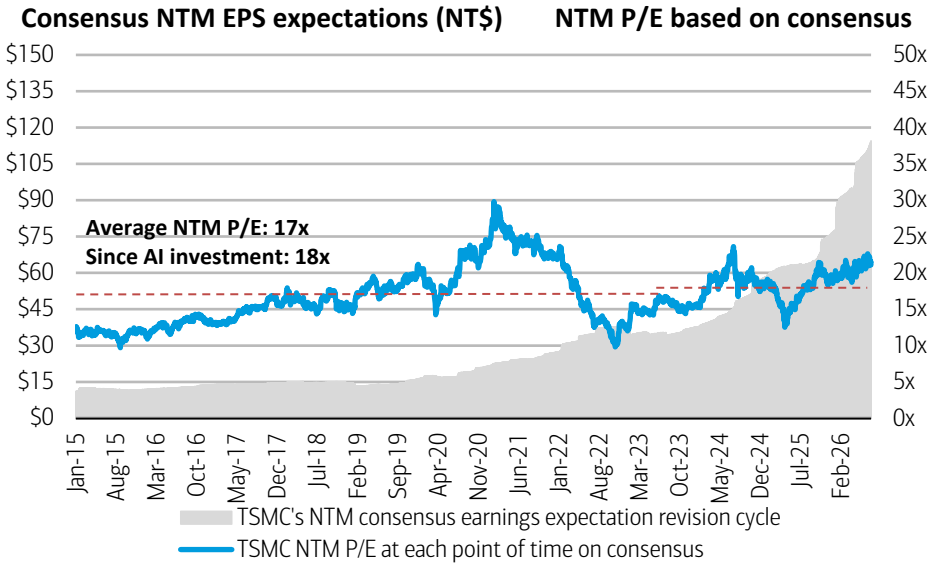
Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

The relatively more modest valuation improvement for TSMC also contrasts with its customers' more pronounced re-ratings. Based on our analysis of forward P/E spreads, TSMC's major customers usually have a 22% average premium vs. TSMC in the long term though the current level is at 45%, suggesting TSMC's valuation has been lagging.

Specifically, the valuation deviation for TSMC's HPC customers vs. the company has widened with 100-150% valuation premium vs. TSMC at this stage (vs. historical average of 53%). We believe this is driven by a surge in valuation for the direct beneficiaries from AI data center deployment, including the companies designing GPUs, ASICs and networking chipsets.

Exhibit 34: TSMC's consensus earnings vs. P/E trend
TSMC's share price strength only driven by earnings upgrade so far



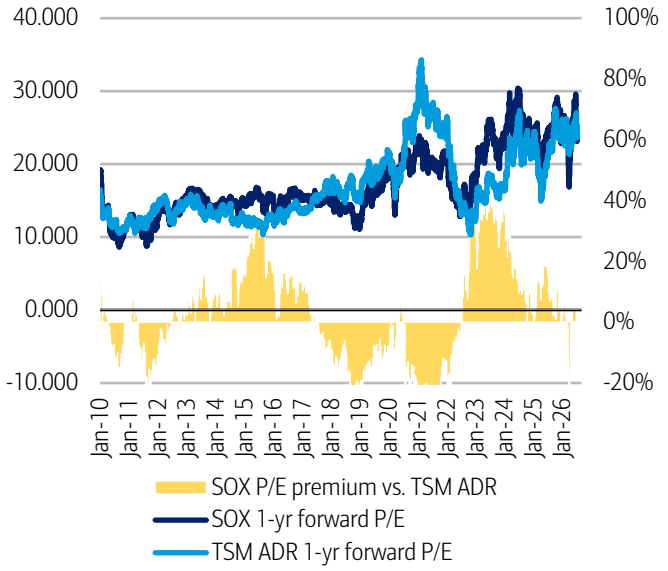
Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

It is also noteworthy that, while TSMC's share performance since 2H23 has been solid, it is predominantly driven by the earnings upward revisions rather than multiple expansion. The company's NTM P/E has remained largely stable, only lifted to 18x in the past 2 years vs. its long-term ~17x levels. The company's ADR though has been performing better as it attracts a broader investor base vs. common share, with its premium still at 10% (vs. the long term average 8%) and at 20x 2027 consensus P/E.

With its firmer technology leadership, this should bode well for the company's earnings upgrade cycle and its share should also be re-rated at least 20x NTM P/E (24x P/E for ADR factoring the premium), back closer to its customers and the broader ecosystem.

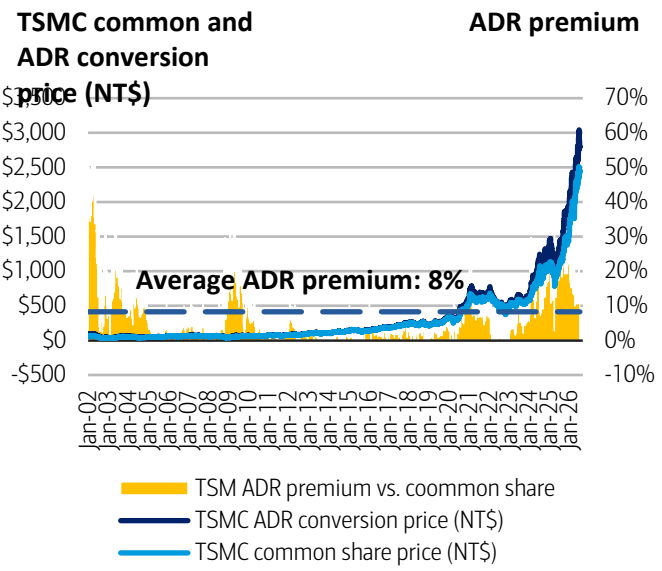
Exhibit 35: TSM ADR 1-year forward P/E vs. SOX premium
SOX index P/E is at -8% premium vs. TSM ADR



Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

Exhibit 36: TSM ADR conversion price vs. common share and premium
TSM ADR premium is higher than common historical level at 10%



Source: BofA Global Research, Bloomberg

BofA GLOBAL RESEARCH

Exhibit 37: TSMC's quarterly and annual income statement
TSMC's quarterly and annual income statement

(NT$bn, %)	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2025	2026E	2027E	2028E
Revenues	$1,134	$1,270	$1,494	$1,653	$1,703	$1,811	$2,065	$2,212	$3,809	$5,551	$7,790	$9,657
Cost of Sales	-$383	-$410	-$494	-$532	-$547	-$584	-$622	-$666	-$1,528	-$1,818	-$2,419	-$3,003
Gross Profit	$751	$860	$1,000	$1,121	$1,156	$1,226	$1,443	$1,545	$2,281	$3,733	$5,371	$6,654
Operating Expenses	-$92	-$94	-$108	-$115	-$118	-$121	-$131	-$135	-$345	-$410	-$504	-$556
Operating Income	$659	$767	$892	$1,006	$1,039	$1,105	$1,312	$1,411	$1,936	$3,324	$4,867	$6,098
Non-operating Income (Exp)	$29	$96	$32	$37	$39	$41	$45	$50	$106	$193	$175	$261
Pretax Income	$688	$862	$924	$1,043	$1,078	$1,146	$1,358	$1,460	$2,042	$3,517	$5,042	$6,359
Net tax	-$115	-$156	-$155	-$175	-$180	-$207	-$228	-$245	-$326	-$601	-$861	-$1,086
Net Income	$573	$707	$769	$868	$898	$939	$1,130	$1,215	$1,718	$2,917	$4,185	$5,277
Fully Diluted EPS - Local (NT$)	$22.08	$27.25	$29.67	$33.48	$34.65	$36.24	$43.59	$46.88	$66.24	$112.46	$161.36	$203.48
% of Revenue												
Gross Profit	66.2	67.7	67.0	67.8	67.9	67.7	69.9	69.9	59.9	67.2	68.9	68.9
Operating Expenses	-8.1	-7.4	-7.3	-7.0	-6.9	-6.7	-6.3	-6.1	-9.1	-7.4	-6.5	-5.8
Operating Income	58.1	60.3	59.7	60.9	61.0	61.0	63.6	63.8	50.8	59.9	62.5	63.1
Pretax Income	60.6	67.9	61.8	63.1	63.3	63.3	65.8	66.0	53.6	63.4	64.7	65.9
Net Income	50.5	55.6	51.5	52.5	52.7	51.9	54.7	54.9	45.1	52.5	53.7	54.6
QoQ Growth %												
Revenues	8.4	12.0	17.6	10.6	3.1	6.3	14.0	7.1				
Gross Profit	15.2	14.5	16.3	12.0	3.1	6.1	17.7	7.1				
Operating Income	16.7	16.3	16.4	12.8	3.2	6.4	18.8	7.5				
Net Income	13.3	23.4	8.8	12.8	3.5	4.6	20.3	7.6				
YoY Growth %												
Revenues	35.1	36.0	50.9	58.0	50.2	42.5	38.2	33.8	31.6	45.7	40.3	24.0
Gross Profit	52.3	57.2	70.0	71.9	53.9	42.5	44.3	37.9	40.4	63.6	43.9	23.9
Operating Income	61.9	65.4	78.2	78.1	57.6	44.2	47.1	40.2	46.4	71.7	46.4	25.3
Net Income	58.8	77.8	70.2	71.6	56.7	32.9	46.9	40.0	46.4	69.8	43.5	26.1
EPS	58.3	77.4	70.1	71.7	56.9	33.0	46.9	40.0	46.4	69.8	43.5	26.1

Source: Company data, BofA Global Research

BofA GLOBAL RESEARCH

Exhibit 38: TSMC's quarterly and annual balance sheet
TSMC's quarterly and annual balance sheet

(NT$bn)	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2025	2026E	2027E	2028E
Cash	$3,036	$3,134	$3,761	$4,089	$4,283	$4,625	$5,181	$5,976	$2,768	$4,089	$5,976	$8,950
Short Term Investment	$348	$384	$366	$375	$370	$373	$371	$372	$301	$375	$372	$372
Accounts receivable	$363	$441	$354	$341	$428	$439	$469	$430	$282	$341	$430	$482
Inventory	$311	$386	$366	$357	$422	$445	$474	$457	$288	$357	$457	$552
Other current assets	$207	$221	$157	$171	$187	$189	$185	$178	$179	$171	$178	$182
Current Assets	$4,266	$4,566	$5,005	$5,332	$5,690	$6,071	$6,680	$7,413	$3,817	$5,332	$7,413	$10,538
Long Term Investments	$167	$228	$197	$213	$205	$209	$207	$208	$172	$213	$208	$207
Fixed Assets	$3,955	$4,303	$4,587	$5,016	$5,417	$5,824	$6,211	$6,579	$3,692	$5,016	$6,579	$8,038
Other Assets	$274	$279	$269	$268	$273	$272	$270	$271	$252	$268	$271	$271
Long-term Assets	$4,395	$4,810	$5,053	$5,497	$5,894	$6,305	$6,689	$7,057	$4,116	$5,497	$7,057	$8,517
Total Assets	**$8,661**	**$9,376**	**$10,057**	**$10,829**	**$11,584**	**$12,376**	**$13,369**	**$14,470**	**$7,933**	**$10,829**	**$14,470**	**$19,055**
Short term loans	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Notes & Accounts Payable	$100	$111	$97	$102	$121	$130	$137	$137	$84	$102	$137	$175
Current portion of long term debt	$156	$167	$134	$149	$152	$150	$146	$149	$137	$149	$149	$149
Other current Liabilities	$1,458	$1,580	$1,580	$1,580	$1,596	$1,596	$1,596	$1,596	$1,237	$1,580	$1,596	$1,611
Current Liabilities	$1,714	$1,858	$1,811	$1,831	$1,868	$1,876	$1,879	$1,881	$1,458	$1,831	$1,881	$1,936
Long term debt	$901	$864	$902	$895	$893	$889	$895	$893	$896	$895	$893	$893
Other non-current liabilities	$113	$179	$134	$177	$151	$160	$155	$209	$118	$177	$209	$237
Long-term Liabilities	$1,014	$1,043	$1,036	$1,072	$1,044	$1,049	$1,050	$1,102	$1,014	$1,072	$1,102	$1,129
Total Liabilities	**$2,729**	**$2,901**	**$2,847**	**$2,903**	**$2,912**	**$2,925**	**$2,929**	**$2,983**	**$2,472**	**$2,903**	**$2,983**	**$3,065**
Total parent shareholders' equity	$5,891	$6,433	$7,170	$7,884	$8,631	$9,410	$10,399	$11,446	$5,420	$7,884	$11,446	$15,949
Minority Interest	$41	$42	$40	$41	$41	$41	$41	$41	$41	$41	$41	$41
Total Liabilities and Equity	**$8,661**	**$9,376**	**$10,057**	**$10,829**	**$11,584**	**$12,376**	**$13,369**	**$14,470**	**$7,933**	**$10,829**	**$14,470**	**$19,055**

Source: Company data, BofA Global Research

BofA GLOBAL RESEARCH

Exhibit 39: TSMC's quarterly and annual cash flow
TSMC's quarterly and annual cash flow

(NT$bn)	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2025	2026E	2027E	2028E
Net Profit	$572	$707	$769	$868	$898	$940	$1,130	$1,216	$1,718	$2,917	$4,185	$5,277
Depreciation & Amortization	$165	$199	$233	$235	$244	$255	$265	$297	$688	$832	$1,061	$1,209
Change in Working Capital	-$118	-$155	$156	$15	-$150	-$27	-$48	$63	-$72	-$102	-$162	-$113
Other adjustment	$79	$33	$17	$25	$12	$6	$20	$40	-$59	$155	$77	$95
Operating Cash Flow	**$699**	**$783**	**$1,176**	**$1,143**	**$1,005**	**$1,173**	**$1,368**	**$1,615**	**$2,275**	**$3,801**	**$5,161**	**$6,468**
Capital Expenditure	-$351	-$496	-$497	-$644	-$664	-$664	-$664	-$664	-$1,272	-$1,988	-$2,657	-$2,875
Proceeds from sale of non-current assets	$2	$0	$1	$1	$1	$1	$1	$1	$1	$3	$3	$3
Acquisitions/Disposals of investments	-$30	-$37	$49	-$24	$12	-$6	$3	-$2	-$37	-$43	$8	$0
Other investment items	$22	$40	$10	$0	-$4	$0	$2	$0	$164	$73	-$3	$0
Investing Cash Flow	**-$357**	**-$493**	**-$437**	**-$667**	**-$656**	**-$669**	**-$659**	**-$666**	**-$1,144**	**-$1,954**	**-$2,649**	**-$2,872**
Net Share issue/repurchase	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Dividends paid	-$130	-$156	-$117	-$156	-$156	-$156	-$156	-$156	-$467	-$558	-$622	-$622
Change in debt	$14	-$22	$5	$8	$0	-$5	$2	$1	$43	$5	-$2	-$1
Other financing cash flow	-$5	-$7	$0	$0	$0	$0	$0	$0	-$17	-$12	$0	$0
Financing Cash Flow	**-$120**	**-$185**	**-$112**	**-$148**	**-$155**	**-$161**	**-$154**	**-$154**	**-$440**	**-$565**	**-$624**	**-$623**
Net Change in Cash	$222	$106	$627	$328	$194	$343	$555	$795	$690	$1,282	$1,888	$2,974
Exchange effect	$46	-$7	$0	$0	$0	$0	$0	$0	-$50	$38	$0	$0
Cash, Beginning of Year	$2,768	$3,036	$3,134	$3,761	$4,089	$4,283	$4,625	$5,181	$2,128	$2,768	$4,089	$5,976
Cash, End of Year	$3,036	$3,134	$3,761	$4,089	$4,283	$4,625	$5,181	$5,976	$2,768	$4,089	$5,976	$8,950

Source: Company data, BofA Global Research

BofA GLOBAL RESEARCH

Exhibit 40: Terms and definitions
Glossary of abbreviations/acronyms

Term	Definition
AI	Artificial Intelligence
AMD	Advanced Micro Devices
ASIC	Application Specific Integrated Circuit
AVSM	Advanced Vision Semiconductor Manufacturing
CoPoS	Chip-on-Panel-on-Substrate
CoWoS-L	Chip-on-Wafer-on-Substrate-Local Silicon Bridge
CoWoS-R	Chip-on-Wafer-on-Substrate-Redistribution Layer
CPO	Co-Packaged Optics
CPU	Central Processing Unit
EMIB	Embedded Multi-die Interconnect Bridge
ESMC	European Semiconductor Manufacturing Company
FoCoS	Fan-Out Chip on Substrate
GPU	Graphic Processing Unit
HBM	High Bandwidth Memory
HPC	High-Performance Computing
JASM	Japan Advanced Semiconductor Manufacturing
JV	Joint Venture
SoC	System on Chip
SoIC	System on Integrated Chips
SoW	System-on-Wafer
SPR	Super Power Rail
SSMC	Systems on Silicon Manufacturing Company
TPU	Tensor Processing Unit
VIS	Vanguard
WMCM	Wafer-level-multi-chip-module
WPM	Wafer per month

Source: BofA Global Research

Price objective basis & risk

Taiwan Semiconductor Manufacturing Co. (TSMWF / TSM)
Our PO for TSMC is NT$3,260 per share (US$620 per ADR), based on 20x 2027E P/E. TSMC's valuation has traded in the range of 10-26x PE from 2017. 20x is in the mid to high end of the historical range, above the average P/E of 16x. We think a PE above the average is justified given TSMC's enhanced industry position and mid-to-high 50% 5-year CAGR in AI should underpin a rerating

Downside risks to our price objective are: (1) greater-than-expected slowdown in global smartphone/consumer electronics demand, (2) Intel's potential insourcing strategy and ambitions in foundry service, (3) execution risks on advanced nodes, and (4) higher than expected tariffs.

Analyst Certification

I, Haas Liu, hereby certify that the views expressed in this research report accurately reflect my personal views about the subject securities and issuers. I also certify that no part of my compensation was, is, or will be, directly or indirectly, related to the specific recommendations or view expressed in this research report.

Special Disclosures

Information on securities which are listed on the exchanges where ML Securities (Taiwan) Limited is not permitted to trade or solicit trades for clients is for informational purposes only and is not a recommendation or a solicitation to trade such securities. ML Securities (Taiwan) Limited will not execute transactions for nor accept orders from clients to trade such securities. Foreign investment in Taiwan securities is regulated and restricted. Currently, foreign investment in Taiwan securities is permitted by investment through: (1) global depository receipts, (2) convertible bonds, (3) mutual funds issued offshore of Taiwan, and (4) a special foreign institutional investors (FINIs) and foreign individual investors (FIDIs) program supervised by the Taiwan SFB whereunder FINIs/FIDIs may apply for investment ID to invest in Taiwan securities by registration with Taiwan Stock Exchange. FINIs will additionally need consent from the foreign exchange authority, ie, the Central Bank of China. In addition to the limitations above, various industry-specific percentage-based limitations on foreign ownership of Taiwan companies (and in some cases prohibitions) may apply. Investments are subject to exchange rate and currency conversion restrictions and risks. Dividends and interest earned by foreign investors' Taiwan securities/instruments are generally subject to a 20% withholding tax. Ordinary shares are not available to ML private client accounts in the U.S.

This report is distributed in Taiwan by Merrill Lynch Securities (Taiwan) Limited, which is regulated by the Taiwan SFB.

APR - Semiconductor Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
BUY				
	Alchip	ALCPF	3661 TT	Haas Liu
	All Ring	XNQNF	6187 TT	Haas Liu
	AMEC	XRTOF	688012 CH	Dai Shen
	ASE Technology Holding	XSRIF	3711 TT	Haas Liu
	ASE Technology Holding -ADR	ASX	ASX US	Haas Liu
	ASMPT	ASMVF	522 HK	Simon Woo, CFA
	Aspeed	XLKMF	5274 TT	Mike Yang
	Chroma ATE	CRMJF	2360 TT	Haas Liu
	Doosan Corp	DOSAF	000150 KS	Simon Woo, CFA

APR - Semiconductor Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
	eMemory	XYLWF	3529 TT	Mike Yang
	GigaDevice	XGXIF	603986 CH	Daley Li, CFA
	Global Unichip Corp.	GBUHF	3443 TT	Haas Liu
	Grand Process Technology Corp	XZOWF	3131 TT	Haas Liu
	Hanmi Semiconductor	HNSIF	042700 KS	Simon Woo, CFA
	Hon Precision	XXPIF	7769 TT	Haas Liu
	Horizon Robotics	HRZRF	9660 HK	Daley Li, CFA
	Iluvatar CoreX	XSIBF	9903 HK	Daley Li, CFA
	Ingenic	XISCF	300223 CH	Dai Shen
	InnoScience Technology	XSCHF	2577 HK	Daley Li, CFA
	JCET Group Co Ltd	XJIEF	600584 CH	Dai Shen
	King Yuan Electronics Corp.	KYUFF	2449 TT	Haas Liu
	Kinik	KIKCF	1560 TT	Haas Liu
	Macronix International	MXICF	2337 TT	Dai Shen
	MediaTek	MDTKF	2454 TT	Haas Liu
	Montage Technology	XRDFF	688008 CH	Daley Li, CFA
	MPI Corporation	XMJCF	6223 TT	Mike Yang
	Nanya Technology	NNYAF	2408 TT	Simon Woo, CFA
	NCE Power	XVFFF	605111 CH	Daley Li, CFA
	OmniVision	XXHQF	603501 CH	Dai Shen
	Phison Electronics	PISNF	8299 TT	Simon Woo, CFA
	Powerchip Semiconductor Manufacturing Co	XCHPF	6770 TT	Haas Liu
	Powertech Technology	PWOGF	6239 TT	Simon Woo, CFA
	Rockchip	XRPXF	603893 CH	Daley Li, CFA
	Samsung C&T	XCDTF	028260 KS	Simon Woo, CFA
	Samsung Elec -G	SSNHZ	SMSN LI	Simon Woo, CFA
	Samsung Electronics	SSNLF	005930 KS	Simon Woo, CFA
	Samsung Electronics Preferred	SSNNF	005935 KS	Simon Woo, CFA
	Silergy Corp.	SLEGF	6415 TT	Mike Yang
	Silicon Motion	SIMO	SIMO US	Simon Woo, CFA
	SK Hynix	HXSCF	000660 KS	Simon Woo, CFA
	SK Hynix ADR	SKHY	SKHY US	Simon Woo, CFA
	SK Square	SKSQF	402340 KS	Simon Woo, CFA
	Taiwan Semiconductor Manufacturing Co.	TSM	TSM US	Haas Liu
	Taiwan Semiconductor Manufacturing Co.	TSMWF	2330 TT	Haas Liu
	Vanguard International Semiconductor Co	VGILF	5347 TT	Haas Liu
	Winbond Electronics	WBEKF	2344 TT	Dai Shen
	WinWay Technology	XWCLF	6515 TT	Mike Yang
	WT Microelectronics	XZOPF	3036 TT	Mike Yang
NEUTRAL				
	ASMedia Technology Inc.	XZSFF	5269 TT	Mike Yang
	Black Sesame Intl Holding	BSIHF	2533 HK	Daley Li, CFA
	Hangzhou Silan Microelectronics	XDFRF	600460 CH	Daley Li, CFA
	Lion Electronics	XDHFF	605358 CH	Dai Shen
	MetaX	XMEZF	688802 CH	Daley Li, CFA
	Novatek	NVKMF	3034 TT	Mike Yang
	Realtek	RLTKF	2379 TT	Haas Liu
	Shenzhen Goodix	XQPLF	603160 CH	Daley Li, CFA
	Soulbrain	XSBOF	357780 KS	Simon Woo, CFA
	VeriSilicon	XMLZF	688521 CH	Daley Li, CFA
UNDERPERFORM				
	Crystal Clear	XPPTF	300655 CH	Dai Shen
	Faraday	FDYTF	3035 TT	Haas Liu
	GlobalWafers	XWLFF	6488 TT	Mike Yang
	Hua Hong Grace	HHUSF	1347 HK	Dai Shen
	LX Semicon	XLXSF	108320 KS	Simon Woo, CFA
	M31 Technology	XMTZF	6643 TT	Mike Yang
	Maxscend	XMXSF	300782 CH	Daley Li, CFA
	Parade	PRDWF	4966 TT	Mike Yang
	United Microelectronics Corp.	XUMIF	2303 TT	Haas Liu
	United Microelectronics Corp.	UMC	UMC US	Haas Liu
	Win Semiconductors	WSMIF	3105 TT	Haas Liu
	Wonik IPS	XRHQF	240810 KS	Simon Woo, CFA

APR - Semiconductor Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst

iQmethod℠ Measures Definitions

Business Performance	Numerator	Denominator
Return On Capital Employed	NOPAT = (EBIT + Interest Income) × (1 − Tax Rate) + Goodwill Amortization	Total Assets − Current Liabilities + ST Debt + Accumulated Goodwill Amortization
Return On Equity	Net Income	Shareholders' Equity
Operating Margin	Operating Profit	Sales
Earnings Growth	Expected 5 Year CAGR From Latest Actual	N/A
Free Cash Flow	Cash Flow From Operations − Total Capex	N/A
Quality of Earnings	**Numerator**	**Denominator**
Cash Realization Ratio	Cash Flow From Operations	Net Income
Asset Replacement Ratio	Capex	Depreciation
Tax Rate	Tax Charge	Pre-Tax Income
Net Debt-To-Equity Ratio	Net Debt = Total Debt − Cash & Equivalents	Total Equity
Interest Cover	EBIT	Interest Expense
Valuation Toolkit	**Numerator**	**Denominator**
Price / Earnings Ratio	Current Share Price	Diluted Earnings Per Share (Basis As Specified)
Price / Book Value	Current Share Price	Shareholders' Equity / Current Basic Shares
Dividend Yield	Annualised Declared Cash Dividend	Current Share Price
Free Cash Flow Yield	Cash Flow From Operations − Total Capex	Market Cap = Current Share Price × Current Basic Shares
Enterprise Value / Sales	EV = Current Share Price × Current Shares + Minority Equity + Net Debt + Other LT Liabilities	Sales
EV / EBITDA	Enterprise Value	Basic EBIT + Depreciation + Amortization

iQmethod℠ is the set of BofA Global Research standard measures that serve to maintain global consistency under three broad headings: Business Performance, Quality of Earnings, and validations. The key features of iQmethod are: A consistently structured, detailed, and transparent methodology. Guidelines to maximize the effectiveness of the comparative valuation process, and to identify some common pitfalls.

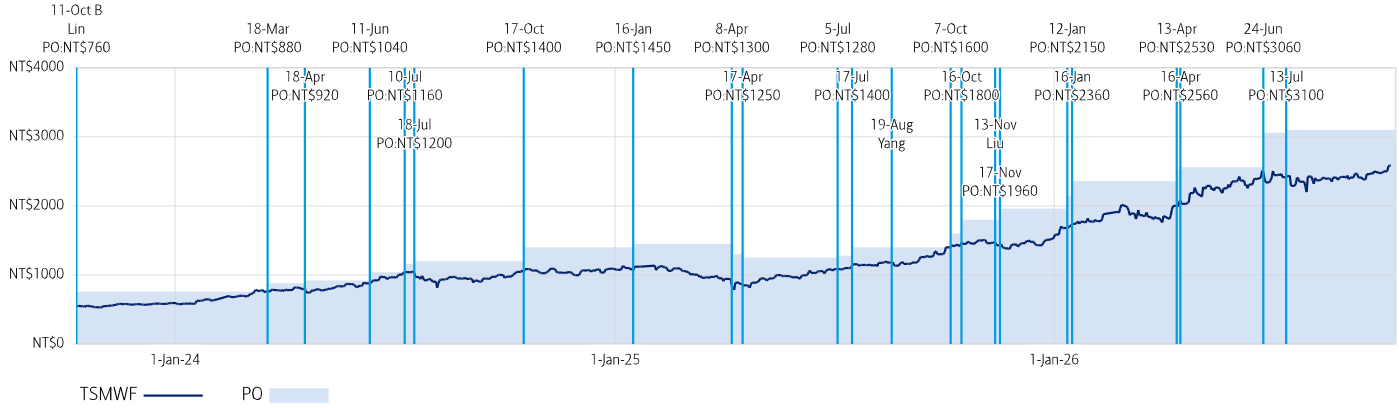
iQdatabase® is our real-time global research database that is sourced directly from our equity analysts' earnings models and includes forecasted as well as historical data for income statements, balance sheets, and cash flow statements for companies covered by BofA Global Research.

iQprofile℠, *iQmethod℠* are service marks of Bank of America Corporation. *iQdatabase®* is a registered service mark of Bank of America Corporation.

Disclosures

Important Disclosures

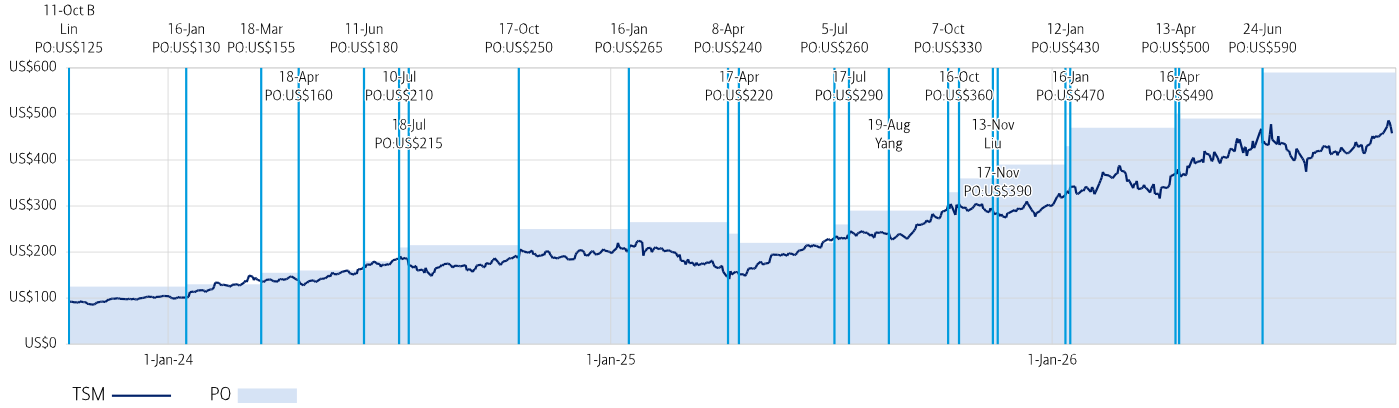
Taiwan Semi Mfg Co (TSMWF) Price Chart



B: Buy, N: Neutral, U: Underperform, PO: Price Objective, NA: No longer valid, NR: No Rating

The Investment Opinion System is contained at the end of the report under the heading "Fundamental Equity Opinion Key". Dark grey shading indicates the security is restricted with the opinion suspended. Medium grey shading indicates the security is under review with the opinion withdrawn. Light grey shading indicates the security is not covered. Chart is current as of a date no more than one trading day prior to the date of the report.

Taiwan Semi Mfg Co (TSM) Price Chart



B: Buy, N: Neutral, U: Underperform, PO: Price Objective, NA: No longer valid, NR: No Rating

The Investment Opinion System is contained at the end of the report under the heading "Fundamental Equity Opinion Key". Dark grey shading indicates the security is restricted with the opinion suspended. Medium grey shading indicates the security is under review with the opinion withdrawn. Light grey shading indicates the security is not covered. Chart is current as of a date no more than one trading day prior to the date of the report.

Equity Investment Rating Distribution: Technology Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	255	62.04%	Buy	157	61.57%
Hold	82	19.95%	Hold	46	56.10%
Sell	74	18.00%	Sell	31	41.89%

Equity Investment Rating Distribution: Global Group (as of 30 Sep 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships [R1]	Count	Percent
Buy	2057	57.03%	Buy	1386	67.38%
Hold	806	22.35%	Hold	563	69.85%
Sell	744	20.63%	Sell	446	59.95%

[R1] Issuers that were investment banking clients of BofA Securities or one of its affiliates within the past 12 months. For purposes of this Investment Rating Distribution, the coverage universe includes only stocks. A stock rated Neutral is included as a Hold, and a stock rated Underperform is included as a Sell.

FUNDAMENTAL EQUITY OPINION KEY: Opinions include a Volatility Risk Rating, an Investment Rating and an Income Rating. VOLATILITY RISK RATINGS, indicators of potential price fluctuation, are: A - Low, B - Medium and C - High. INVESTMENT RATINGS reflect the analyst's assessment of both a stock's absolute total return potential as well as its attractiveness for investment relative to other stocks within its Coverage Cluster (defined below). Our investment ratings are: 1 - Buy stocks are expected to have a total return of at least 10% and are the most attractive stocks in the coverage cluster; 2 - Neutral stocks are expected to remain flat or increase in value and are less attractive than Buy rated stocks and 3 - Underperform stocks are the least attractive stocks in a coverage cluster. An investment rating of 6 (No Rating) indicates that a stock is no longer trading on the basis of fundamentals. Analysts assign investment ratings considering, among other things, the 0-12 month total return expectation for a stock and the firm's guidelines for ratings dispersions (shown in the table below). The current price objective for a stock should be referenced to better understand the total return expectation at any given time. The price objective reflects the analyst's view of the potential price appreciation (depreciation).

Investment rating	Total return expectation (within 12-month period of date of initial rating)	Ratings dispersion guidelines for coverage cluster[R2]
Buy	≥ 10%	≤ 70%
Neutral	≥ 0%	≤ 30%
Underperform	N/A	≥ 20%

[R2]Ratings dispersions may vary from time to time where BofA Global Research believes it better reflects the investment prospects of stocks in a Coverage Cluster.

INCOME RATINGS, indicators of potential cash dividends, are: 7 - same/higher (dividend considered to be secure), 8 - same/lower (dividend not considered to be secure) and 9 - pays no cash dividend. *Coverage Cluster* is comprised of stocks covered by a single analyst or two or more analysts sharing a common industry, sector, region or other classification(s). A stock's coverage cluster is included in the most recent BofA Global Research report referencing the stock.

Price Charts for the securities referenced in this research report are available on the Price Charts website, or call 1-800-MERRILL to have them mailed.
BofAS or one of its affiliates acts as a market maker for the equity securities recommended in the report: Taiwan Semi Mfg Co.
The issuer is or was, within the last 12 months, an investment banking client of BofAS and/or one or more of its affiliates: Taiwan Semi Mfg Co.
BofAS or an affiliate has received compensation from the issuer for non-investment banking services or products within the past 12 months: Taiwan Semi Mfg Co.
The issuer is or was, within the last 12 months, a non-securities business client of BofAS and/or one or more of its affiliates: Taiwan Semi Mfg Co.
BofAS or an affiliate expects to receive or intends to seek compensation for investment banking services from this issuer or an affiliate of the issuer within the next three months: Taiwan Semi Mfg Co.
The country in which this issuer is organized has certain laws or regulations that limit or restrict ownership of the issuer's shares by nationals of other countries: Taiwan Semi Mfg Co.
BofAS or one of its affiliates is willing to sell to, or buy from, clients the common equity of the issuer on a principal basis: Taiwan Semi Mfg Co.
The issuer is or was, within the last 12 months, a securities business client (non-investment banking) of BofAS and/or one or more of its affiliates: Taiwan Semi Mfg Co.
BofA Global Research personnel (including the analyst(s) responsible for this report) receive compensation based upon, among other factors, the overall profitability of Bank of America Corporation, including profits derived from investment banking. The analyst(s) responsible for this report may also receive compensation based upon, among other factors, the overall profitability of the Bank's sales and trading businesses relating to the class of securities or financial instruments for which such analyst is responsible.

Other Important Disclosures

From time to time research analysts conduct site visits of covered issuers. BofA Global Research policies prohibit research analysts from accepting payment or reimbursement for travel expenses from the issuer for such visits.
Prices are indicative and for information purposes only. Except as otherwise stated in the report, for any recommendation in relation to an equity security, the price referenced is the publicly traded price of the security as of close of business on the day prior to the date of the report or, if the report is published during intraday trading, the price referenced is indicative of the traded price as of the date and time of the report and in relation to a debt security (including equity preferred and CDS), prices are indicative as of the date and time of the report and are from various sources including BofA Securities trading desks.
The date and time of completion of the production of any recommendation in this report shall be the date and time of dissemination of this report as recorded in the report timestamp.

Recipients who are not institutional investors or market professionals should seek the advice of their independent financial advisor before considering information in this report in connection with any investment decision, or for a necessary explanation of its contents.
Officers of BofAS or one or more of its affiliates (other than research analysts) may have a financial interest in securities of the issuer(s) or in related investments.
Refer to BofA Global Research policies relating to conflicts of interest.
"BofA Securities" includes BofA Securities, Inc. ("BofAS") and its affiliates. Investors should contact their BofA Securities representative or Merrill Global Wealth Management financial advisor if they have questions concerning this report or concerning the appropriateness of any investment idea described herein for such investor. "BofA Securities" is a global brand for BofA Global Research.
Information relating to Non-US affiliates of BofA Securities and Distribution of Affiliate Research Reports:
BofAS and/or Merrill Lynch, Pierce, Fenner & Smith Incorporated ("MLPF&S") may in the future distribute, information of the following non-US affiliates in the US (short name: legal name, regulator): Merrill Lynch (South Africa): Merrill Lynch South Africa (Pty) Ltd., regulated by the Financial Sector Conduct Authority; MLI (UK): Merrill Lynch International, regulated by the Financial Conduct Authority (FCA) and the Prudential Regulation Authority (PRA); BofASE (France): BofA Securities Europe SA is authorized by the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and regulated by the ACPR and the Autorité des Marchés Financiers (AMF). BofA Securities Europe SA ("BofASE") with registered address at 51, rue La Boétie, 75008 Paris is registered under no 842 602 690 RCS Paris. In accordance with the provisions of French Code Monétaire et Financier (Monetary and Financial Code), BofASE is an établissement de crédit et d'investissement (credit and investment institution) that is authorised and supervised by the European Central Bank and the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and regulated by the ACPR and the Autorité des Marchés Financiers. BofASE's share capital can be found at www.bofaml.com/BofASEdisclaimer; BofA Europe (Milan): Bank of America Europe Designated Activity Company, Milan Branch, regulated by the Bank of Italy, the European Central Bank (ECB) and the Central Bank of Ireland (CBI); BofA Europe (Frankfurt): Bank of America Europe Designated Activity Company, Frankfurt Branch regulated by BaFin, the ECB and the CBI; BofA Europe (Zurich): Bank of America Europe Designated Activity Company, Zurich Branch, regulated by the Swiss Financial Market Supervisory Authority FINMA, the ECB and CBI; BofA Europe (Madrid): Bank of America Europe Designated Activity Company, Sucursal en España, regulated by the Bank of Spain, the ECB and the CBI; Merrill Lynch (Australia): Merrill Lynch Equities (Australia) Limited, regulated by the Australian Securities and Investments Commission; Merrill Lynch (Hong Kong): Merrill Lynch (Asia Pacific) Limited, regulated by the Hong Kong Securities and Futures Commission (HKSFC); Merrill Lynch (Singapore): Merrill Lynch (Singapore) Pte Ltd, regulated by the Monetary Authority of Singapore (MAS); Merrill Lynch (Canada): Merrill Lynch Canada Inc, regulated by the Canadian Investment Regulatory Organization; Merrill Lynch (Mexico): Merrill Lynch Mexico, SA de CV, Casa de Bolsa, regulated by the Comisión Nacional Bancaria y de Valores; BofAS Japan: BofA Securities Japan Co., Ltd., regulated by the Financial Services Agency; Merrill Lynch (Seoul): Merrill Lynch International, LLC Seoul Branch, regulated by the Financial Supervisory Service; Merrill Lynch (Taiwan): Merrill Lynch Securities (Taiwan) Ltd., regulated by the Securities and Futures Bureau; BofAS India: BofA Securities India Limited, regulated by the Securities and Exchange Board of India (SEBI); Merrill Lynch (Israel): Merrill Lynch Israel Limited, regulated by Israel Securities Authority; Merrill Lynch (DIFC): Merrill Lynch International (DIFC Branch), regulated by the Dubai Financial Services Authority (DFSA); Merrill Lynch (Brazil): Merrill Lynch S.A. Corretora de Títulos e Valores Mobiliários, regulated by Comissão de Valores Mobiliários; Merrill Lynch KSA Company: Merrill Lynch Kingdom of Saudi Arabia Company, regulated by the Capital Market Authority.
This information: has been approved for publication and is distributed in the United Kingdom (UK) to professional clients and eligible counterparties (as each is defined in the rules of the FCA and the PRA) by MLI (UK), which is authorized by the PRA and regulated by the FCA and the PRA - details about the extent of our regulation by the FCA and PRA are available from us on request; has been approved for publication and is distributed in the European Economic Area (EEA) by BofASE (France), which is authorized by the ACPR and regulated by the ACPR and the AMF; has been considered and distributed in Japan by BofAS Japan, a registered securities dealer under the Financial Instruments and Exchange Act in Japan, or its permitted affiliates; is issued and

distributed in Hong Kong by Merrill Lynch (Hong Kong) which is regulated by HKSFC; is issued and distributed in Taiwan by Merrill Lynch (Taiwan); is issued and distributed in India by BofAS India; and is issued and distributed in Singapore to institutional investors and/or accredited investors (each as defined under the Financial Advisers Regulations) by Merrill Lynch (Singapore) (Company Registration No 198602883D). Merrill Lynch (Singapore) is regulated by MAS. Merrill Lynch Equities (Australia) Limited (ABN 65 006 276 795), AFS License 235132 (MLEA) distributes this information in Australia only to 'Wholesale' clients as defined by s.761G of the Corporations Act 2001. With the exception of Bank of America N.A., Australia Branch, neither MLEA nor any of its affiliates involved in preparing this information is an Authorised Deposit-Taking Institution under the Banking Act 1959 nor regulated by the Australian Prudential Regulation Authority. No approval is required for publication or distribution of this information in Brazil and its local distribution is by Merrill Lynch (Brazil) in accordance with applicable regulations. Merrill Lynch (DIFC) is authorized and regulated by the DFSA. Information prepared and issued by Merrill Lynch (DIFC) is done so in accordance with the requirements of the DFSA conduct of business rules. BofA Europe (Frankfurt) distributes this information in Germany and is regulated by BaFin, the ECB and the CBI. BofA Securities entities, including BofA Europe and BofASE (France), may outsource/delegate the marketing and/or provision of certain research services or aspects of research services to other branches or members of the BofA Securities group. You may be contacted by a different BofA Securities entity acting for and on behalf of your service provider where permitted by applicable law. This does not change your service provider. Please refer to the Electronic Communications Disclaimers for further information.

This information has been prepared and issued by BofAS and/or one or more of its non-US affiliates. The author(s) of this information may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so. BofAS and/or MLPF&S is the distributor of this information in the US and accepts full responsibility for information distributed to BofAS and/or MLPF&S clients in the US by its non-US affiliates. Any US person receiving this information and wishing to effect any transaction in any security discussed herein should do so through BofAS and/or MLPF&S and not such foreign affiliates. Hong Kong recipients of this information should contact Merrill Lynch (Asia Pacific) Limited in respect of any matters relating to dealing in securities or provision of specific advice on securities or any other matters arising from, or in connection with, this information. Singapore recipients of this information should contact Merrill Lynch (Singapore) Pte Ltd in respect of any matters arising from, or in connection with, this information. For clients that are not accredited investors, expert investors or institutional investors Merrill Lynch (Singapore) Pte Ltd accepts full responsibility for the contents of this information distributed to such clients in Singapore.

General Investment Related Disclosures:

Taiwan Readers: Neither the information nor any opinion expressed herein constitutes an offer or a solicitation of an offer to transact in any securities or other financial instrument. No part of this report may be used or reproduced or quoted in any manner whatsoever in Taiwan by the press or any other person without the express written consent of BofA Securities.

This document provides general information only, and has been prepared for, and is intended for general distribution to, BofA Securities clients. Neither the information nor any opinion expressed constitutes an offer or an invitation to make an offer, to buy or sell any securities or other financial instrument or any derivative related to such securities or instruments (e.g., options, futures, warrants, and contracts for differences). This document is not intended to provide personal investment advice and it does not take into account the specific investment objectives, financial situation and the particular needs of, and is not directed to, any specific person(s). This document and its content do not constitute, and should not be considered to constitute, investment advice for purposes of ERISA, the US tax code, the Investment Advisers Act or otherwise. Investors should seek financial advice regarding the appropriateness of investing in financial instruments and implementing investment strategies discussed or recommended in this document and should understand that statements regarding future prospects may not be realized. Any decision to purchase or subscribe for securities in any offering must be based solely on existing public information on such security or the information in the prospectus or other offering document issued in connection with such offering, and not on this document.

Securities and other financial instruments referred to herein, or recommended, offered or sold by BofA Securities, are not insured by the Federal Deposit Insurance Corporation and are not deposits or other obligations of any insured depository institution (including, Bank of America, N.A.). Investments in general and, derivatives, in particular, involve numerous risks, including, among others, market risk, counterparty default risk and liquidity risk. No security, financial instrument or derivative is suitable for all investors. Digital assets are extremely speculative, volatile and are not regulated in a manner similar to U.S. securities markets. In some cases, securities and other financial instruments may be difficult to value or sell and reliable information about the value or risks related to the security or financial instrument may be difficult to obtain. Investors should note that income from such securities and other financial instruments, if any, may fluctuate and that price or value of such securities and instruments may rise or fall and, in some cases, investors may lose their entire principal investment. Past performance is not necessarily a guide to future performance. Levels and basis for taxation may change.

This report may contain a short-term trading idea or recommendation, which highlights a specific near-term catalyst or event impacting the issuer or the market that is anticipated to have a short-term price impact on the equity securities of the issuer. Short-term trading ideas and recommendations are different from and do not affect a stock's fundamental equity rating, which reflects both a longer term total return expectation and attractiveness for investment relative to other stocks within its Coverage Cluster. Short-term trading ideas and recommendations may be more or less positive than a stock's fundamental equity rating.

BofA Securities is aware that the implementation of the ideas expressed in this report may depend upon an investor's ability to "short" securities or other financial instruments and that such action may be limited by regulations prohibiting or restricting "shortselling" in many jurisdictions. Investors are urged to seek advice regarding the applicability of such regulations prior to executing any short idea contained in this report.

Foreign currency rates of exchange may adversely affect the value, price or income of any security or financial instrument mentioned herein. Investors in such securities and instruments, including ADRs, effectively assume currency risk.

BofAS or one of its affiliates is a regular issuer of traded financial instruments linked to securities or digital assets that may have been recommended or discussed in this report. BofAS or one of its affiliates may, at any time, hold a trading position (long or short) in the securities and financial instruments recommended or discussed in this report.

BofA Securities, through business units other than BofA Global Research, may have issued and may in the future issue trading ideas or recommendations that are inconsistent with, and reach different conclusions from, the information presented herein. Such ideas or recommendations may reflect different time frames, assumptions, views and analytical methods of the persons who prepared them, and BofA Securities is under no obligation to ensure that such other trading ideas or recommendations are brought to the attention of any recipient of this information.

In the event that the recipient received this information pursuant to a contract between the recipient and BofAS for the provision of research services for a separate fee, and in connection therewith BofAS may be deemed to be acting as an investment adviser, such status relates, if at all, solely to the person with whom BofAS has contracted directly and does not extend beyond the delivery of this report (unless otherwise agreed specifically in writing by BofAS). If such recipient uses the services of BofAS in connection with the sale or purchase of a security referred to herein, BofAS may act as principal for its own account or as agent for another person. BofAS is and continues to act solely as a broker-dealer in connection with the execution of any transactions, including transactions in any securities referred to herein.

Copyright and General Information:

Copyright 2026 Bank of America Corporation. All rights reserved. iQdatabase® is a registered service mark of Bank of America Corporation. This information is prepared for the use of BofA Securities clients and may not be redistributed, retransmitted or disclosed, in whole or in part, or in any form or manner, without the express written consent of BofA Securities. This document and its content is provided solely for informational purposes and cannot be used for training or developing artificial intelligence (AI) models or as an input in any AI application (collectively, an AI tool). Any attempt to utilize this document or any of its content in connection with an AI tool without explicit written permission from BofA Global Research is strictly prohibited. BofA Global Research utilizes AI, including machine learning and other technologies, to enhance the services we provide to our clients. These technologies assist our analysts in various aspects of their work, including but not limited to data analysis, content extraction, content creation, data aggregation and summarization and identifying relevant information from diverse sources. All AI-driven processes are subject to review by BofA Global Research employees. BofA Global Research information is distributed simultaneously to internal and client websites and other portals by BofA Securities and is not publicly-available material. Any unauthorized use or disclosure is prohibited. Receipt and review of this information constitutes your agreement not to redistribute, retransmit, or disclose to others the contents, opinions, conclusion, or information contained herein (including any investment recommendations, estimates or price targets) without first obtaining express permission from an authorized officer of BofA Securities.

Materials prepared by BofA Global Research personnel are based on public information. Facts and views presented in this material have not been reviewed by, and may not reflect information known to, professionals in other business areas of BofA Securities, including investment banking personnel. BofA Securities has established information barriers between BofA Global Research and certain business groups. As a result, BofA Securities does not disclose certain client relationships with, or compensation received from, such issuers. To the extent this material discusses any legal proceeding or issues, it has not been prepared as nor is it intended to express any legal conclusion, opinion or advice. Investors should consult their own legal advisers as to issues of law relating to the subject matter of this material. BofA Global Research personnel's knowledge of legal proceedings in which any BofA Securities entity and/or its directors, officers and employees may be plaintiffs, defendants, co-defendants or co-plaintiffs with or involving issuers mentioned in this material is based on public information. Facts and views presented in this material that relate to any such proceedings have not been reviewed by, discussed with, and may not reflect information known to, professionals in other business areas of BofA Securities in connection with the legal proceedings or matters relevant to such proceedings.

This information has been prepared independently of any issuer of securities mentioned herein and not in connection with any proposed offering of securities or as agent of any issuer of any securities. None of BofAS any of its affiliates or their research analysts has any authority whatsoever to make any representation or warranty on behalf of the issuer(s). BofA Global Research

policy prohibits research personnel from disclosing a recommendation, investment rating, or investment thesis for review by an issuer prior to the publication of a research report containing such rating, recommendation or investment thesis.
Any information relating to sustainability in this material is limited as discussed herein and is not intended to provide a comprehensive view on any sustainability claim with respect to any issuer or security.
Any information relating to the tax status of financial instruments discussed herein is not intended to provide tax advice or to be used by anyone to provide tax advice. Investors are urged to seek tax advice based on their particular circumstances from an independent tax professional.
The information herein (other than disclosure information relating to BofA Securities and its affiliates) was obtained from various sources and we do not guarantee its accuracy. This information may contain links to third-party websites. BofA Securities is not responsible for the content of any third-party website or any linked content contained in a third-party website. Content contained on such third-party websites is not part of this information and is not incorporated by reference. The inclusion of a link does not imply any endorsement by or any affiliation with BofA Securities. Access to any third-party website is at your own risk, and you should always review the terms and privacy policies at third-party websites before submitting any personal information to them. BofA Securities is not responsible for such terms and privacy policies and expressly disclaims any liability for them.
All opinions, projections and estimates constitute the judgment of the author as of the date of publication and are subject to change without notice. Prices also are subject to change without notice. BofA Securities is under no obligation to update this information and BofA Securities ability to publish information on the subject issuer(s) in the future is subject to applicable quiet periods. You should therefore assume that BofA Securities will not update any fact, circumstance or opinion contained herein.
Subject to the quiet period applicable under laws of the various jurisdictions in which we distribute research reports and other legal and BofA Securities policy-related restrictions on the publication of research reports, fundamental equity reports are produced on a regular basis as necessary to keep the investment recommendation current.
Certain outstanding reports or investment opinions relating to securities, financial instruments and/or issuers may no longer be current. Always refer to the most recent research report relating to an issuer prior to making an investment decision.
In some cases, an issuer may be classified as Restricted or may be Under Review or Extended Review. In each case, investors should consider any investment opinion relating to such issuer (or its security and/or financial instruments) to be suspended or withdrawn and should not rely on the analyses and investment opinion(s) pertaining to such issuer (or its securities and/or financial instruments) nor should the analyses or opinion(s) be considered a solicitation of any kind. Sales persons and financial advisors affiliated with BofAS or any of its affiliates may not solicit purchases of securities or financial instruments that are Restricted or Under Review and may only solicit securities under Extended Review in accordance with firm policies.
Neither BofA Securities nor any officer or employee of BofA Securities accepts any liability whatsoever for any direct, indirect or consequential damages or losses arising from any use of this information.

Morgan Stanley | RESEARCH

UPDATE

October 12, 2026 04:24 PM GMT

AirTAC International | Asia Pacific

September Sales Rose 3% MoM, 27% YoY

Here we focus on Airtac's monthly sales, which we believe could be a catalyst for its share price.

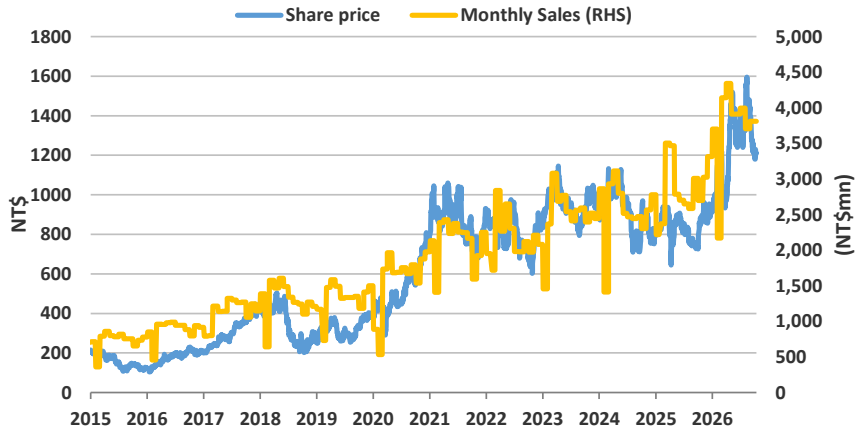
What's new:

- September sales came in at NT$3,811mn (+3% MoM/+27% YoY).
- 3Q26 revenue was NT$11,512mn (-5% QoQ/+40% YoY), in line with our estimate of NT$11,586mn (-5% QoQ/+41% YoY) but slightly better than consensus at NT$11,215mn (-8% QoQ/+36% YoY).
- By segment, shipments of machine tool (+35% YoY), energy & lighting (+30% YoY), electronics (+27% YoY) and textiles (+25% YoY) outperformed during the month.

Outlook:

- Both order booking and shipments continued to exceed company expectations in the month.
- Looking ahead, management remains positive on its growth outlook, driven by share gains and stimulus policies, which could help sustain the upcycle at least into 2027.
- We remain OW as current valuation at 18x 2027e P/E is undemanding vs. its mid-cycle average of 25x, though we think Hiwin (2049.TW) might outperform on share price on the back of sequential growth in the top line and potentially GM in 3Q26.

Exhibit 1: AirTAC monthly sales vs. its share price since 2015



Source: Company data, TEJ, Morgan Stanley Research. Note: Past performance is no guarantee of future results. Results shown do not include transaction costs.

MORGAN STANLEY TAIWAN LIMITED+

Derrick Yang
Equity Analyst
Derrick.Yang@morganstanley.com +886 2 2730-2862

Vivi Huang
Research Associate
Vivi.Huang@morganstanley.com +886 2 2730-2860

MORGAN STANLEY ASIA LIMITED+

Andy Meng, CFA
Equity Analyst
Andy.Meng@morganstanley.com +852 2239-7689

AirTAC International (1590.TW, 1590 TT)

Greater China Technology Hardware | Taiwan

Stock Rating	**Overweight**
Industry View	**In-Line**
Price target	NT$2,000.00
Up/downside to price target (%)	65
Shr price, close (Oct 12, 2026)	NT$1,210.00
52-Week Range	NT$1,615.00-840.00
Sh out, dil, curr (mn)	200
Mkt cap, curr (mn)	US$7,583
EV, curr (mn)	US$7,433
Avg daily trading value (mn)	US$35

Fiscal Year Ending	12/25	12/26e	12/27e	12/28e
EPS (NT$)**	41.97	60.77	68.77	77.77
EPS (NT$)§	41.43	58.80	68.02	74.62
Revenue, net (NT$ mn)	34,330	45,007	50,917	57,428
EBITDA (NT$ mn)	12,866	17,543	19,669	22,130
ModelWare net inc (NT $ mn)	8,400	12,165	13,765	15,567
P/E	22.2	19.9	17.6	15.6
P/BV	3.5	4.1	3.6	3.2
RNOA (%)	17.7	25.1	25.9	27.3
ROE (%)	17.5	23.1	23.4	23.4
EV/EBITDA	14.1	13.4	11.8	10.3
Div yld (%)	2.2	2.4	2.5	2.8
FCF yld ratio (%)	3.4	3.0	4.0	4.4
Leverage (EOP) (%)	(9.1)	(11.0)	(15.2)	(18.7)

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework
** = Based on consensus methodology
§ = Consensus data is provided by Refinitiv Estimates
e = Morgan Stanley Research estimates

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

+= Analysts employed by non-U.S. affiliates are not registered with FINRA, may not be associated persons of the member and may not be subject to FINRA restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

AlphaSignals Catalyst Event Reaction

Company / Catalyst	Date	Impact to thesis	Catalyst versus expectation
AirTAC International 1590.TW			
September 2026 Sales	12 Oct 2026	Unchanged	In-line

Source: Company data, Morgan Stanley Research

Valuation Methodology and Risks

AirTAC International (1590.TW)

Base case, 29x 2027e P/E. We believe this methodology better captures earnings growth momentum for industrial automation component plays (market sentiment moves along with business cycles). As the industry enters an up-cycle, we think applying a near-peak cycle valuation of 29x P/E is reasonable given the stronger business outlook.

Risks to Upside

- Stronger macroeconomic climate in China
- Stronger-than-expected demand from industrial automation

Risks to Downside

- Sharper economic downturn in China
- Miniature linear guideway business development taking more time
- Larger-than-expected investment in new products but weaker-than-expected demand

Hiwin Technologies Corp. (2049.TW)

Base case, P/E. We think this valuation methodology better captures Hiwin's earnings growth momentum, as its valuation tends to move with the business cycle. We apply a target multiple of 38x to our 2027 EPS estimate. We view this multiple as justified by the 47% operating profit CAGR we project for 2025-28.

Risks to Upside

- Stronger industrial automation demand pickup.
- Market share gains.
- Earlier-than-expected revenue contribution from humanoids.

Risks to Downside

- Weaker industrial automation demand.
- More pricing pressure.
- Market share loss.
- More broad-based impact on the global economy from geopolitical risks.

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley Asia Limited (which accepts the responsibility for its contents) and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research), and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Andy Meng, CFA; Derrick Yang.

.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictpolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

As of September 30, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., **AirTAC International**, Asustek Computer Inc., AU Optronics, Auras Technology Co Ltd, Bizlink, Catcher Technology, Chenbro, Co-Tech Development Corp., Compal Electronics, Delta Electronics Inc., E Ink Holdings Inc., Elite Material Co., Ennostar Inc, Eoptolink Technology Inc Ltd, Fositek Corp, Genius Electronic Optical Co. Ltd., Giga-Byte Technology Co. Ltd., Hiwin Technologies Corp., LandMark Optoelectronics Corporation, Largan Precision, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Lotes Co. Ltd., Nan Ya PCB, Quectel Wireless Solutions Co Ltd, Sunny Optical, Suzhou TFC Optical Communication Co Ltd., Taiwan Union Technology Corporation, Unimicron, Visual Photonics Epitaxy Co Ltd, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhejiang Crystal-Optech Co Ltd, Zhen Ding, Zhongji Innolight Co Ltd, ZTE Corporation.

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Asustek Computer Inc., Unimicron, Wistron Corporation, Wiwynn Corp, Zhen Ding, Zhongji Innolight Co Ltd.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Asustek Computer Inc., Lenovo, Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Zhen Ding, Zhongji Innolight Co Ltd.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Advantech, **AirTAC International**, Asustek Computer Inc., AU Optronics, Bizlink, Catcher Technology, Chenbro, Compal Electronics, Delta Electronics Inc., E Ink Holdings Inc., Elite Material Co., Ennostar Inc, Eoptolink Technology Inc Ltd, FIT Hon Teng Ltd, Giga-Byte Technology Co. Ltd., Gold Circuit Electronics Ltd., Hon Hai Precision, Innolux, Lenovo, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Lotes Co. Ltd., Luxshare Precision Industry Co., Ltd., Pegatron Corporation, Q Technology (Group) Company Ltd, Quanta Computer Inc., Shanghai Conant Optical Co Ltd, Shenzhen Transsion Holdings Co Ltd, Suzhou TFC Optical Communication Co Ltd., TCL Corp., Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhen Ding, Zhongji Innolight Co Ltd.

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Asustek Computer Inc., AU Optronics, BYD Electronics, Compal Electronics, E Ink Holdings Inc., Eoptolink Technology Inc Ltd, Giga-Byte Technology Co. Ltd., GoerTek Inc, Hon Hai Precision, Innolux, Lenovo, Lingyi Itech Guangdong Co, Quanta Computer Inc., Xiaomi Corp, Yageo Corp..

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Advantech, **AirTAC International**, Asustek Computer Inc., AU Optronics, Bizlink, Catcher Technology, Chenbro, Compal Electronics, Delta Electronics Inc., E Ink Holdings Inc., Elite Material Co., Ennostar Inc, Eoptolink Technology Inc Ltd, FIT Hon Teng Ltd, Giga-Byte Technology Co. Ltd., Gold Circuit Electronics Ltd., Hon Hai Precision, Innolux, Lenovo, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Lotes Co. Ltd., Luxshare Precision Industry Co., Ltd., Pegatron Corporation, Q Technology (Group) Company Ltd, Quanta Computer Inc., Shanghai Conant Optical Co Ltd, Shenzhen Transsion Holdings Co Ltd, Suzhou TFC Optical Communication Co Ltd., TCL Corp., Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhen Ding, Zhongji Innolight Co Ltd.

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Asustek Computer Inc., AU Optronics, BYD Electronics, Compal Electronics, E Ink Holdings Inc., Eoptolink Technology Inc Ltd, Giga-Byte Technology Co. Ltd., GoerTek Inc, Hon Hai Precision, Innolux, Lenovo, Lingyi Itech Guangdong Co, Quanta Computer Inc., Xiaomi Corp, Yageo Corp., Zhen Ding.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality

of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.
Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.
Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of September 30, 2026)
The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.
For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

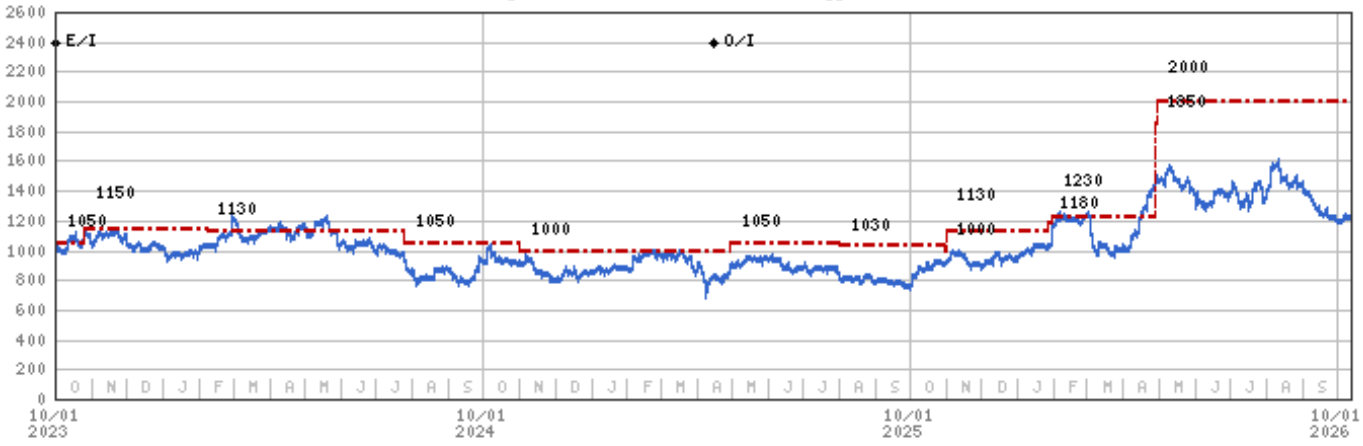
Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.
Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.
Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

AirTAC International (1590.TW) - As of 10/12/26 GMT in TWD
Industry : Greater China Technology Hardware



Stock Rating History: 10/1/21 : O/I; 11/8/21 : E/I; 1/11/22 : O/I; 8/4/22 : E/I; 4/16/25 : O/I

Price Target History: 8/5/21 : 1379.98; 10/4/21 : 1300; 11/8/21 : 870; 1/11/22 : 1250; 3/24/22 : 1150; 8/4/22 : 850; 10/27/22 : 600; 1/16/23 : 1050; 2/8/23 : 1100; 4/26/23 : 1150; 7/28/23 : 1050; 10/26/23 : 1150; 2/7/24 : 1130; 7/25/24 : 1050; 11/1/24 : 1000; 4/30/25 : 1050; 8/1/25 : 1030; 10/30/25 : 1000; 10/31/25 : 1130; 1/27/26 : 1180; 1/30/26 : 1230; 4/28/26 : 1850; 4/30/26 : 2000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Hiwin Technologies Corp. (2049.TW) - As of 10/12/26 GMT in TWD
Industry : Greater China Technology Hardware

Stock Rating History: 10/1/21 : O/I; 8/11/22 : E/I; 2/21/23 : O/I; 8/11/23 : E/I; 5/12/25 : U/I; 2/27/26 : E/I; 3/30/26 : O/I

Price Target History: 8/5/21 : 480.04; 10/5/21 : 479.6; 2/28/22 : 341.16; 3/24/22 : 301.61; 7/20/22 : 295; 8/11/22 : 230; 11/21/22 : 200; 2/21/23 : 295; 2/27/23 : 283; 5/10/23 : 280; 8/11/23 : 205; 11/13/23 : 220; 2/27/24 : 210; 8/13/24 : 205; 11/12/24 : 225; 2/27/25 : 310; 5/12/25 : 160; 2/27/26 : 210; 3/30/26 : 300; 4/28/26 : 380; 5/11/26 : 400; 8/12/26 : 450

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures.
For Morgan Stanley specific disclosures, you may refer to https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch.
Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.
Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.
Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.
Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.
Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (http://www.morganstanley.com/terms.html). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (http://www.morganstanley.com/terms.html). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).
If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research.
Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.
The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.
The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.
With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.
Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.
Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.
To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V which is regulated by Comision Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118J); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. Morgan Stanley Research is disseminated in Canada by Morgan Stanley Canada Limited and Morgan Stanley Wealth Management Canada Inc., members of the Canadian Investment Regulatory Organization (CIRO). The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

Morgan Stanley Hong Kong Securities Limited is the liquidity provider/market maker for securities of AAC Technologies Holdings, BYD Electronics, FIT Hon Teng Ltd, Hon Hai Precision, Kingboard Laminates Holdings Ltd., Lenovo, Lens Technology, Luxshare Precision Industry Co., Ltd., Sunny Optical, Xiaomi Corp, Yangtze Optical Fibre and Cable JSC Ltd, Zhongji Innolight Co Ltd, ZTE Corporation listed on the Stock Exchange of Hong Kong Limited. An updated list can be found on HKEx website: http://www.hkex.com.hk.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

References to any prediction market or event contract do not constitute an offer, solicitation, recommendation, or endorsement to participate in any such market or contract. Morgan Stanley Research may not be distributed or made available in any jurisdiction where such distribution would be unlawful or otherwise prohibited by applicable law.

INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/12/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$36.06
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb140.17
BYD Electronics (0285.HK)	O (04/28/2023)	HK$22.80
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.44
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.51
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb347.60
Genius Electronic Optical Co. Ltd. (3406.TW)	U (09/10/2026)	NT$1,070.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb20.85
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.17
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb33.24
Largan Precision (3008.TW)	E (09/10/2026)	NT$7,325.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.73
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb10.88
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$4.97
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$49.50
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb46.32
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.36
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$4.79
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb69.76
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$34.68
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb49.36
Sunny Optical (2382.HK)	E (03/23/2026)	HK$58.80
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb229.54
Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb14.89
Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$24.52
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb362.18
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$156.10
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb99.88
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb21.36
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb724.47
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$903.00
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$20.36
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb28.44
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,130.00
Advantech (2395.TW)	O (01/20/2021)	NT$721.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,210.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,385.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$35.55
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,540.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,355.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.33
Chenbro (8210.TW)	O (07/23/2025)	NT$861.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,365.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$1,900.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$149.00
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$63.80
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,270.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$338.00
Innolux (3481.TW)	E (04/07/2025)	NT$51.30
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$10,580.00

Lens Technology (300433.SZ)	E (07/22/2020)	Rmb26.73
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$288.00
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.52
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.08
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb167.00
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.35
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$977.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$194.50
Compal Electronics (2324.TW)	U (04/23/2025)	NT$36.60
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$6,150.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$4.43
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb52.05
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$356.00
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,235.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$244.50
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb62.05
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$46.80
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$3,025.00
Lenovo (0992.HK)	O (07/10/2026)	HK$32.34
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,880.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,440.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$98.60
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$322.00
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb120.00
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb327.78
Unimicron (3037.TW)	O (02/23/2026)	NT$1,255.00
Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$564.00
Wistron Corporation (3231.TW)	O (07/12/2023)	NT$189.50
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,275.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$636.00
Zhen Ding (4958.TW)	++	NT$581.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$522.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,735.00

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.

© 2026 Morgan Stanley

October 12, 2026 06:26 PM GMT

Greater China Technology Hardware | Asia Pacific

Automation – Read-across from Preliminary JMTBA September 2026 Orders

In this report, we focus on preliminary JMTBA orders, which we believe could be a catalyst for AirTAC's and Hiwin's share prices.

According to the Japan Machine Tool Builders' Association (JMTBA), headline orders grew 13% MoM and 60% YoY in September 2026 (vs. +2% MoM/+65% YoY in August 2026), to ¥223bn.

- Domestic orders were +24% MoM/+46% YoY (vs. -3% MoM/+62% YoY in August), at ¥64bn.
- Overseas orders were +9% MoM/+67% YoY (vs. +5% MoM/+66% YoY in August), at ¥159bn.

Our view:

- We think the sequential growth in overseas orders was consistent with seasonality, while the strength on a YoY basis continued to point to an upcycle scenario.
- Within overseas orders, we think China and Europe may have outperformed, as manufacturing PMIs held up better (50.1/52.9 for China/Europe in September vs. 49.8/52.7 in August). N.A is likely to be softer compared to other regions, as its manufacturing PMI edged down slightly to 54.5 in September (vs. 54.6 in August).
- We remain OW on both AirTAC (1590.TW) and Hiwin (2049.TW) as we see company-specific drivers on top of the industry recovery trend:
  - Airtac should continue to benefit from its own market share gains, and valuation appears undemanding at 18x 2027e P/E (vs. 25x average since 2020).
  - Hiwin should see margin expansion in the coming quarters, driven by higher utilization and price hikes, which we believe could be share price catalysts. Current valuation of 29x 2027 P/E implies upside vs. peak cycle valuation at 35-40x P/E.

MORGAN STANLEY TAIWAN LIMITED+

Derrick Yang
Equity Analyst
Derrick.Yang@morganstanley.com +886 2 2730-2862

Vivi Huang
Research Associate
Vivi.Huang@morganstanley.com +886 2 2730-2860

MORGAN STANLEY ASIA LIMITED+

Andy Meng, CFA
Equity Analyst
Andy.Meng@morganstanley.com +852 2239-7689

GREATER CHINA TECHNOLOGY HARDWARE

Asia Pacific
Industry View **In-Line**

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

+= Analysts employed by non-U.S. affiliates are not registered with FINRA, may not be associated persons of the member and may not be subject to FINRA restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

AlphaSignals Catalyst Event Reaction

Company / Catalyst	Date	Impact to thesis	Catalyst versus expectation
AirTAC International 1590.TW			
Preliminary JMTBA September 2026 Orders	12 Oct 2026	Unchanged	In-line
Hiwin Technologies Corp. 2049.TW			
Preliminary JMTBA September 2026 Orders	12 Oct 2026	Unchanged	In-line

Source: Company data, Morgan Stanley Research

Valuation Methodology and Risks

AirTAC International (1590.TW)

Base case, 29x 2027e P/E. We believe this methodology better captures earnings growth momentum for industrial automation component plays (market sentiment moves along with business cycles). As the industry enters an up-cycle, we think applying a near-peak cycle valuation of 29x P/E is reasonable given the stronger business outlook.

Risks to Upside

- Stronger macroeconomic climate in China
- Stronger-than-expected demand from industrial automation

Risks to Downside

- Sharper economic downturn in China
- Miniature linear guideway business development taking more time
- Larger-than-expected investment in new products but weaker-than-expected demand

Hiwin Technologies Corp. (2049.TW)

Base case, P/E. We think this valuation methodology better captures Hiwin's earnings growth momentum, as its valuation tends to move with the business cycle. We apply a target multiple of 38x to our 2027 EPS estimate. We view this multiple as justified by the 47% operating profit CAGR we project for 2025-28.

Risks to Upside

- Stronger industrial automation demand pickup.
- Market share gains.
- Earlier-than-expected revenue contribution from humanoids.

Risks to Downside

- Weaker industrial automation demand.
- More pricing pressure.
- Market share loss.
- More broad-based impact on the global economy from geopolitical risks.

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley Asia Limited (which accepts the responsibility for its contents) and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research), and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Andy Meng, CFA; Derrick Yang.

.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictpolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

As of September 30, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., AirTAC International, Asustek Computer Inc., AU Optronics, Auras Technology Co Ltd, Bizlink, Catcher Technology, Chenbro, Co-Tech Development Corp., Compal Electronics, Delta Electronics Inc., E Ink Holdings Inc., Elite Material Co., Ennostar Inc, Eoptolink Technology Inc Ltd, Fositek Corp, Genius Electronic Optical Co. Ltd., Giga-Byte Technology Co. Ltd., Hiwin Technologies Corp., LandMark Optoelectronics Corporation, Largan Precision, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Lotes Co. Ltd., Nan Ya PCB, Quectel Wireless Solutions Co Ltd, Sunny Optical, Suzhou TFC Optical Communication Co Ltd., Taiwan Union Technology Corporation, Unimicron, Visual Photonics Epitaxy Co Ltd, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhejiang Crystal-Optech Co Ltd, Zhen Ding, Zhongji Innolight Co Ltd, ZTE Corporation.

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Asustek Computer Inc., Unimicron, Wistron Corporation, Wiwynn Corp, Zhen Ding, Zhongji Innolight Co Ltd.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Asustek Computer Inc., Lenovo, Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Zhen Ding, Zhongji Innolight Co Ltd.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Advantech, AirTAC International, Asustek Computer Inc., AU Optronics, Bizlink, Catcher Technology, Chenbro, Compal Electronics, Delta Electronics Inc., E Ink Holdings Inc., Elite Material Co., Ennostar Inc, Eoptolink Technology Inc Ltd, FIT Hon Teng Ltd, Giga-Byte Technology Co. Ltd., Gold Circuit Electronics Ltd., Hon Hai Precision, Innolux, Lenovo, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Lotes Co. Ltd., Luxshare Precision Industry Co., Ltd., Pegatron Corporation, Q Technology (Group) Company Ltd, Quanta Computer Inc., Shanghai Conant Optical Co Ltd, Shenzhen Transsion Holdings Co Ltd, Suzhou TFC Optical Communication Co Ltd., TCL Corp., Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhen Ding, Zhongji Innolight Co Ltd.

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Asustek Computer Inc., AU Optronics, BYD Electronics, Compal Electronics, E Ink Holdings Inc., Eoptolink Technology Inc Ltd, Giga-Byte Technology Co. Ltd., GoerTek Inc, Hon Hai Precision, Innolux, Lenovo, Lingyi Itech Guangdong Co, Quanta Computer Inc., Xiaomi Corp, Yageo Corp..

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Advantech, AirTAC International, Asustek Computer Inc., AU Optronics, Bizlink, Catcher Technology, Chenbro, Compal Electronics, Delta Electronics Inc., E Ink Holdings Inc., Elite Material Co., Ennostar Inc, Eoptolink Technology Inc Ltd, FIT Hon Teng Ltd, Giga-Byte Technology Co. Ltd., Gold Circuit Electronics Ltd., Hon Hai Precision, Innolux, Lenovo, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Lotes Co. Ltd., Luxshare Precision Industry Co., Ltd., Pegatron Corporation, Q Technology (Group) Company Ltd, Quanta Computer Inc., Shanghai Conant Optical Co Ltd, Shenzhen Transsion Holdings Co Ltd, Suzhou TFC Optical Communication Co Ltd., TCL Corp., Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhen Ding, Zhongji Innolight Co Ltd.

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Asustek Computer Inc., AU Optronics, BYD Electronics, Compal Electronics, E Ink Holdings Inc., Eoptolink Technology Inc Ltd, Giga-Byte Technology Co. Ltd., GoerTek Inc, Hon Hai Precision, Innolux, Lenovo, Lingyi Itech Guangdong Co, Quanta Computer Inc., Xiaomi Corp, Yageo Corp., Zhen Ding.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality

of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.
Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.
Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of September 30, 2026)
The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.
For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

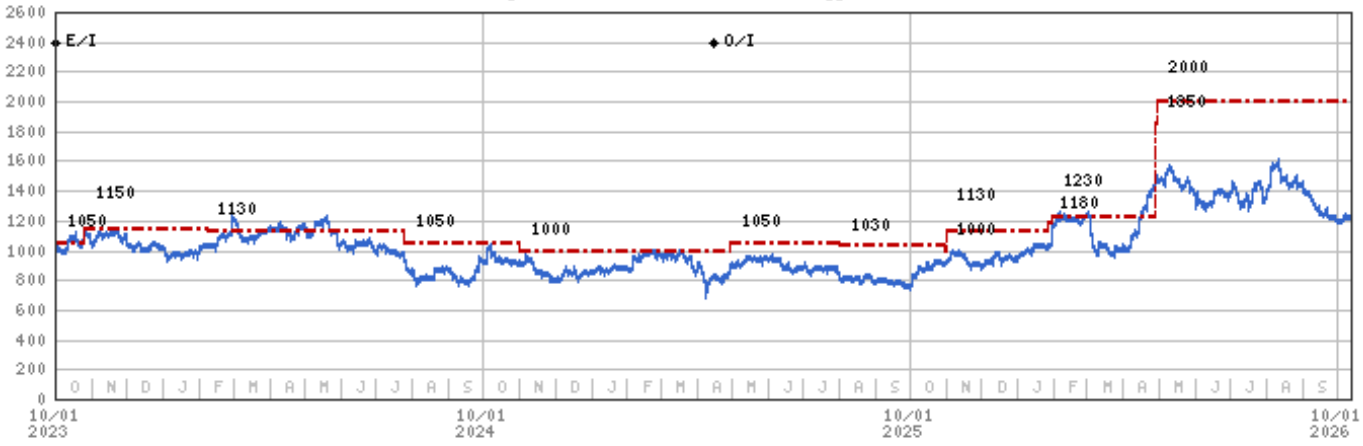
Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.
Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.
Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

AirTAC International (1590.TW) - As of 10/12/26 GMT in TWD
Industry : Greater China Technology Hardware



Stock Rating History: 10/1/21 : O/I; 11/8/21 : E/I; 1/11/22 : O/I; 8/4/22 : E/I; 4/16/25 : O/I

Price Target History: 8/5/21 : 1379.98; 10/4/21 : 1300; 11/8/21 : 870; 1/11/22 : 1250; 3/24/22 : 1150; 8/4/22 : 850; 10/27/22 : 600; 1/16/23 : 1050; 2/8/23 : 1100; 4/26/23 : 1150; 7/28/23 : 1050; 10/26/23 : 1150; 2/7/24 : 1130; 7/25/24 : 1050; 11/1/24 : 1000; 4/30/25 : 1050; 8/1/25 : 1030; 10/30/25 : 1000; 10/31/25 : 1130; 1/27/26 : 1180; 1/30/26 : 1230; 4/28/26 : 1850; 4/30/26 : 2000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Hiwin Technologies Corp. (2049.TW) - As of 10/12/26 GMT in TWD
Industry : Greater China Technology Hardware

Stock Rating History: 10/1/21 : O/I; 8/11/22 : E/I; 2/21/23 : O/I; 8/11/23 : E/I; 5/12/25 : U/I; 2/27/26 : E/I; 3/30/26 : O/I

Price Target History: 8/5/21 : 480.04; 10/5/21 : 479.6; 2/28/22 : 341.16; 3/24/22 : 301.61; 7/20/22 : 295; 8/11/22 : 230; 11/21/22 : 200; 2/21/23 : 295; 2/27/23 : 283; 5/10/23 : 280; 8/11/23 : 205; 11/13/23 : 220; 2/27/24 : 210; 8/13/24 : 205; 11/12/24 : 225; 2/27/25 : 310; 5/12/25 : 160; 2/27/26 : 210; 3/30/26 : 300; 4/28/26 : 380; 5/11/26 : 400; 8/12/26 : 450

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures.
For Morgan Stanley specific disclosures, you may refer to https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch.
Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.
Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.
Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.
Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.
Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (http://www.morganstanley.com/terms.html). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (http://www.morganstanley.com/terms.html). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).
If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research.
Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.
The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.
The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.
With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.
Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.
Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.
To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V which is regulated by Comision Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118J); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. Morgan Stanley Research is disseminated in Canada by Morgan Stanley Canada Limited and Morgan Stanley Wealth Management Canada Inc., members of the Canadian Investment Regulatory Organization (CIRO). The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

Morgan Stanley Hong Kong Securities Limited is the liquidity provider/market maker for securities of AAC Technologies Holdings, BYD Electronics, FIT Hon Teng Ltd, Hon Hai Precision, Kingboard Laminates Holdings Ltd., Lenovo, Lens Technology, Luxshare Precision Industry Co., Ltd., Sunny Optical, Xiaomi Corp, Yangtze Optical Fibre and Cable JSC Ltd, Zhongji Innolight Co Ltd, ZTE Corporation listed on the Stock Exchange of Hong Kong Limited. An updated list can be found on HKEx website: http://www.hkex.com.hk.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

References to any prediction market or event contract do not constitute an offer, solicitation, recommendation, or endorsement to participate in any such market or contract. Morgan Stanley Research may not be distributed or made available in any jurisdiction where such distribution would be unlawful or otherwise prohibited by applicable law.

INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/12/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$36.06
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb140.17
BYD Electronics (0285.HK)	O (04/28/2023)	HK$22.80
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.44
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.51
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb347.60
Genius Electronic Optical Co. Ltd. (3406.TW)	U (09/10/2026)	NT$1,070.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb20.85
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.17
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb33.24
Largan Precision (3008.TW)	E (09/10/2026)	NT$7,325.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.73
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb10.88
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$4.97
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$49.50
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb46.32
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.36
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$4.79
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb69.76
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$34.68
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb49.36
Sunny Optical (2382.HK)	E (03/23/2026)	HK$58.80
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb229.54
Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb14.89
Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$24.52
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb362.18
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$156.10
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb99.88
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb21.36
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb724.47
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$903.00
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$20.36
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb28.44
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,130.00
Advantech (2395.TW)	O (01/20/2021)	NT$721.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,210.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,385.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$35.55
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,540.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,355.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.33
Chenbro (8210.TW)	O (07/23/2025)	NT$861.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,365.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$1,900.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$149.00
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$63.80
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,270.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$338.00
Innolux (3481.TW)	E (04/07/2025)	NT$51.30
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$10,580.00

Lens Technology (300433.SZ)	E (07/22/2020)	Rmb26.73
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$288.00
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.52
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.08
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb167.00
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.35
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$977.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$194.50
Compal Electronics (2324.TW)	U (04/23/2025)	NT$36.60
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$6,150.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$4.43
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb52.05
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$356.00
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,235.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$244.50
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb62.05
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$46.80
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$3,025.00
Lenovo (0992.HK)	O (07/10/2026)	HK$32.34
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,880.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,440.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$98.60
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$322.00
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb120.00
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb327.78
Unimicron (3037.TW)	O (02/23/2026)	NT$1,255.00
Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$564.00
Wistron Corporation (3231.TW)	O (07/12/2023)	NT$189.50
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,275.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$636.00
Zhen Ding (4958.TW)	++	NT$581.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$522.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,735.00

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.

© 2026 Morgan Stanley

October 12, 2026 01:01 AM GMT

Delta Electronics Inc. | Asia Pacific

Our Thoughts on USITC Investigation No. 337

The near-term impact should be manageable, but we'll closely monitor developments on this matter.

What's new:

- The US International Trade Commission (USITC) formally launched a patent infringement investigation on October 9, 2026, following a complaint by Vicor Corporation, a US-based power electronics company, against 20 entities, including Delta and others across the power supply, server assembly and semiconductor space.
- Vicor alleges that certain imported vertical power delivery systems (used mostly in HPC chips) have infringed US Patent 10,903,734.
- Vicor requested two types of remedy: 1) a Limited Exclusion Order (LEO); and 2) Cease and Desist Orders (CDOs), which could restrict US imports and sales of infringing products if granted.

Our view:

- We note that this is not the first time Delta and Vicor have had legal disputes over patent infringement.
- We will closely monitor any developments in this matter, though we think the impact on Delta's operations should be manageable (the products in dispute account for high single digit % of its total revenue for 2026, based on our estimates).
- We remain OW.

MORGAN STANLEY TAIWAN LIMITED+

Derrick Yang
Equity Analyst
Derrick.Yang@morganstanley.com +886 2 2730-2862

Vivi Huang
Research Associate
Vivi.Huang@morganstanley.com +886 2 2730-2860

MORGAN STANLEY ASIA LIMITED+

Andy Meng, CFA
Equity Analyst
Andy.Meng@morganstanley.com +852 2239-7689

Delta Electronics Inc. (2308.TW, 2308 TT)

Greater China Technology Hardware | Taiwan

Stock Rating	**Overweight**
Industry View	**In-Line**
Price target	NT$2,500.00
Up/downside to price target (%)	27
Shr price, close (Oct 8, 2026)	NT$1,965.00
52-Week Range	NT$2,585.00-886.00
Sh out, dil, curr (mn)	2,598
Mkt cap, curr (mn)	US$159,810
EV, curr (mn)	US$158,149
Avg daily trading value (mn)	US$615

Fiscal Year Ending	12/25	12/26e	12/27e	12/28e
EPS (NT$)**	23.14	39.77	64.77	100.77
Prior EPS (NT$)**	-	-	-	-
EPS (NT$)§	23.91	40.39	64.11	99.50
Revenue, net (NT$ bn)	554.9	775.2	1,131.9	1,659.5
EBITDA (NT$ bn)	111.8	169.2	267.5	404.0
ModelWare net inc (NT $ bn)	60.1	103.3	168.2	261.7
P/E	41.6	49.4	30.3	19.5
P/BV	7.1	11.8	9.1	6.8
RNOA (%)	27.3	42.2	70.3	101.3
ROE (%)	21.5	31.7	42.2	50.8
EV/EBITDA	20.1	29.4	18.3	11.7
Div yld (%)	0.8	0.6	1.0	1.7
FCF yld ratio (%)**	2.1	0.9	2.1	4.0
Leverage (EOP) (%)	(41.9)	(51.7)	(57.8)	(64.0)

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework
** = Based on consensus methodology
§ = Consensus data is provided by Refinitiv Estimates
e = Morgan Stanley Research estimates

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

+= Analysts employed by non-U.S. affiliates are not registered with FINRA, may not be associated persons of the member and may not be subject to FINRA restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

Key Table

Exhibit 1: Historical litigation between Vicor and Delta

Filing date	Plaintiff(s) / Defendant(s) / Case Number	Reason for Filing	Latest Progress
2023-07-12	Vicor v. Delta et al. | E.D. Tex., 2:23-cv-00323	Patent infringement: US 6,930,893; 9,166,481; 9,516,761; 10,199,950.	Settled; dismissed with prejudice (May 13, 2025).
2023-07-12	Vicor v. Delta, Cyntec, Foxconn, Quanta et al. | ITC 337-TA-1370	ITC import complaint: US 9,166,481; 9,516,761; 10,199,950.	Closed (Feb 13, 2025); partial infringement finding; LEO/CDOs issued.
2023-10-23	Delta v. Vicor | W.D. Tex., 6:23-cv-00726	Patent infringement: US 9,819,263 (power converters).	Settled; dismissed with prejudice (May 7, 2025).
2023-11-01	Delta v. Vicor | D. Del., 1:23-cv-01246	Patent infringement: US 8,711,580 and 10,877,534.	Settled; dismissed with prejudice (May 8, 2025).
2024-01-02	Delta v. Vicor | China patent case (number unverified)	Patent infringement: CN 106571354 B (power-converter manufacturing).	Final case disposition not confirmed.
2024-01-26	Delta (applicant) / Vicor | D. Del., 1:24-cv-00107	Section 1782 evidence request for China patent litigation.	Jan 10, 2025: main discovery request denied; limited relief granted.
2026-01-09	Vicor v. Delta | E.D. Tex., 2:26-cv-00017	Patent infringement: US 12,395,087 (bus-converter design).	Stayed (Feb 24, 2026) pending ITC 337-TA-1484.
2026-01-12	Vicor v. Delta et al. | ITC 337-TA-1484	ITC import complaint: US 12,395,087 (power modules and servers).	Ongoing; instituted Feb 11, 2026 (published Feb 17); target completion Sep 20, 2027.
2026-07-23	Delta v. Vicor | W.D. Tex., 1:26-cv-02057	Alleged trade-secret misuse and interference with business relations.	Ongoing; no merits ruling reported.
2026-07-24	Delta Shanghai v. Vicor | D. Del., 1:26-cv-00913	Patent infringement: US 10,845,853 (processor-board power delivery).	Ongoing; no final infringement ruling reported.
2026-07-31	Delta v. Vicor | W.D. Tex., 7:26-cv-00288	Patent infringement: US 11,923,772 (DC/DC converters).	Pending; Vicor moved to dismiss or transfer (Sep 28, 2026).
2026-09-08	Vicor v. Delta, Infineon et al. | E.D. Tex., 2:26-cv-00796	Patent infringement: US 10,903,734 (vertical power delivery).	Ongoing; early stage, no infringement finding.
2026-09-08	Delta (applicant) / Vicor | D. Del., 1:26-cv-01135	Section 1782 evidence request for overseas proceedings.	Pending; no final order confirmed.
2026-09-09	Vicor v. Delta et al. | ITC 337-TA-1526	ITC import complaint: US 10,903,734 (vertical power delivery).	Instituted (Oct 9, 2026); no final ruling.

Source: US Federal Court, US International Trade Commission (ITC), USPTO/PTAB filings, Company data, Morgan Stanley research

Valuation Methodology and Risks

Delta Electronics Inc. (2308.TW)

Base case: P/E multiple of 30x based on average 2027-28 EPS. We believe our target multiple is justified by the 63% earnings CAGR we estimate for 2025-28, implying a PEG of 0.5, versus the stock's ~23x P/E and 23% earnings CAGR during 2022-25, which implied a PEG of 1.0.

Risks to Upside

- Faster-than-expected global economic growth
- Falling raw material prices
- Margin improvement

Risks to Downside

- Global economic slowdown
- Ongoing component supply tightness
- Rising raw material prices
- Margin contraction as a result of higher operating expenses or faster ASP erosion

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley Asia Limited (which accepts the responsibility for its contents) and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research), and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Andy Meng, CFA; Derrick Yang.

.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictpolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

As of September 30, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., AirTAC International, Asustek Computer Inc., AU Optronics, Auras Technology Co Ltd, Bizlink, Catcher Technology, Chenbro, Co-Tech Development Corp., Compal Electronics, **Delta Electronics Inc.**, E Ink Holdings Inc., Elite Material Co., Ennostar Inc, Eoptolink Technology Inc Ltd, Fositek Corp, Genius Electronic Optical Co. Ltd., Giga-Byte Technology Co. Ltd., Hiwin Technologies Corp., LandMark Optoelectronics Corporation, Largan Precision, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Lotes Co. Ltd., Nan Ya PCB, Quectel Wireless Solutions Co Ltd, Sunny Optical, Suzhou TFC Optical Communication Co Ltd., Taiwan Union Technology Corporation, Unimicron, Visual Photonics Epitaxy Co Ltd, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhejiang Crystal-Optech Co Ltd, Zhen Ding, Zhongji Innolight Co Ltd, ZTE Corporation.

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Asustek Computer Inc., Unimicron, Wistron Corporation, Wiwynn Corp, Zhen Ding, Zhongji Innolight Co Ltd.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Asustek Computer Inc., Lenovo, Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Zhen Ding, Zhongji Innolight Co Ltd.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Advantech, AirTAC International, Asustek Computer Inc., AU Optronics, Bizlink, Catcher Technology, Chenbro, Compal Electronics, **Delta Electronics Inc.**, E Ink Holdings Inc., Elite Material Co., Ennostar Inc, Eoptolink Technology Inc Ltd, FIT Hon Teng Ltd, Giga-Byte Technology Co. Ltd., Gold Circuit Electronics Ltd., Hon Hai Precision, Innolux, Lenovo, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Lotes Co. Ltd., Luxshare Precision Industry Co., Ltd., Pegatron Corporation, Q Technology (Group) Company Ltd, Quanta Computer Inc., Shanghai Conant Optical Co Ltd, Shenzhen Transsion Holdings Co Ltd, Suzhou TFC Optical Communication Co Ltd., TCL Corp., Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhen Ding, Zhongji Innolight Co Ltd.

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Asustek Computer Inc., AU Optronics, BYD Electronics, Compal Electronics, E Ink Holdings Inc., Eoptolink Technology Inc Ltd, Giga-Byte Technology Co. Ltd., GoerTek Inc, Hon Hai Precision, Innolux, Lenovo, Lingyi Itech Guangdong Co, Quanta Computer Inc., Xiaomi Corp, Yageo Corp..

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Advantech, AirTAC International, Asustek Computer Inc., AU Optronics, Bizlink, Catcher Technology, Chenbro, Compal Electronics, **Delta Electronics Inc.**, E Ink Holdings Inc., Elite Material Co., Ennostar Inc, Eoptolink Technology Inc Ltd, FIT Hon Teng Ltd, Giga-Byte Technology Co. Ltd., Gold Circuit Electronics Ltd., Hon Hai Precision, Innolux, Lenovo, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Lotes Co. Ltd., Luxshare Precision Industry Co., Ltd., Pegatron Corporation, Q Technology (Group) Company Ltd, Quanta Computer Inc., Shanghai Conant Optical Co Ltd, Shenzhen Transsion Holdings Co Ltd, Suzhou TFC Optical Communication Co Ltd., TCL Corp., Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhen Ding, Zhongji Innolight Co Ltd.

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Asustek Computer Inc., AU Optronics, BYD Electronics, Compal Electronics, E Ink Holdings Inc., Eoptolink Technology Inc Ltd, Giga-Byte Technology Co. Ltd., GoerTek Inc, Hon Hai Precision, Innolux, Lenovo, Lingyi Itech Guangdong Co, Quanta Computer Inc., Xiaomi Corp, Yageo Corp., Zhen Ding.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality

of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.
Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.
Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of September 30, 2026)
The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.
For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

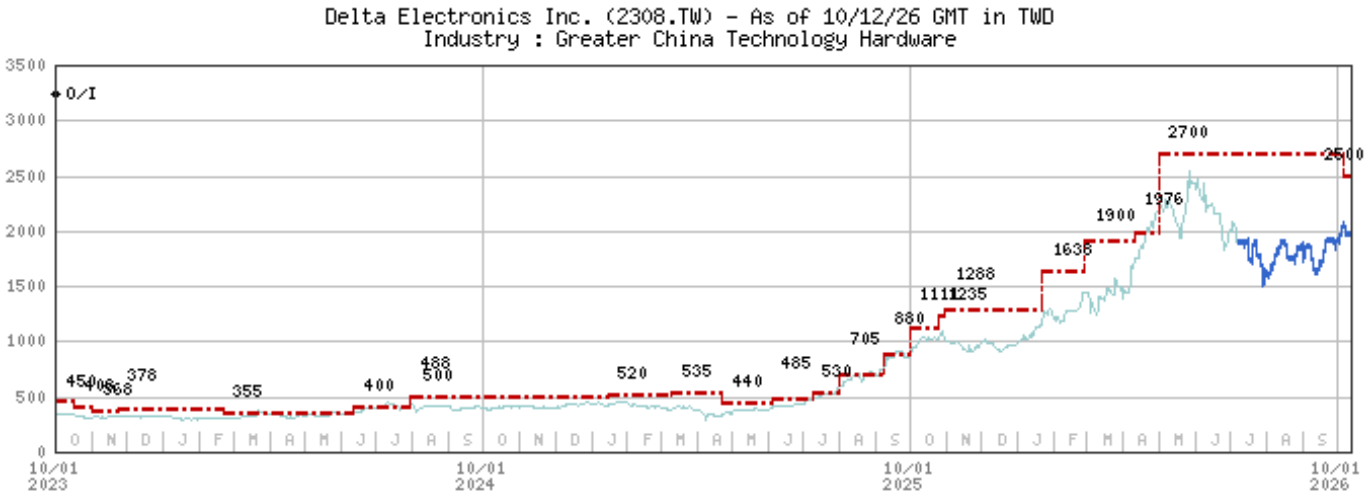
Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.
Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.
Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)



Stock Rating History: 10/1/21 : O/I

Price Target History: 9/9/21 : 345; 10/29/21 : 325; 2/25/22 : 303; 7/29/22 : 312; 9/21/22 : 326; 10/6/22 : 350; 11/23/22 : 368; 8/2/23 : 450; 10/17/23 : 408; 11/1/23 : 368; 11/23/23 : 378; 2/21/24 : 355; 6/12/24 : 400; 7/30/24 : 488; 8/1/24 : 500; 1/15/25 : 520; 3/11/25 : 535; 4/23/25 : 440; 6/4/25 : 485; 7/9/25 : 530; 8/1/25 : 705; 9/8/25 : 880; 9/30/25 : 1111; 10/24/25 : 1235; 10/30/25 : 1288; 1/21/26 : 1638; 2/27/26 : 1900; 4/10/26 : 1976; 5/1/26 : 2700; 10/6/26 : 2500

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (http://www.morganstanley.com/terms.html). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (http://www.morganstanley.com/terms.html). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research.

Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V which is regulated by Comision Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118J); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. Morgan Stanley Research is disseminated in Canada by Morgan Stanley Canada Limited and Morgan Stanley Wealth Management Canada Inc., members of the Canadian

Investment Regulatory Organization (CIRO). The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

Morgan Stanley Hong Kong Securities Limited is the liquidity provider/market maker for securities of AAC Technologies Holdings, BYD Electronics, FIT Hon Teng Ltd, Hon Hai Precision, Kingboard Laminates Holdings Ltd., Lenovo, Lens Technology, Luxshare Precision Industry Co., Ltd., Sunny Optical, Xiaomi Corp, Yangtze Optical Fibre and Cable JSC Ltd, Zhongji Innolight Co Ltd, ZTE Corporation listed on the Stock Exchange of Hong Kong Limited. An updated list can be found on HKEx website: http://www.hkex.com.hk.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

References to any prediction market or event contract do not constitute an offer, solicitation, recommendation, or endorsement to participate in any such market or contract. Morgan Stanley Research may not be distributed or made available in any jurisdiction where such distribution would be unlawful or otherwise prohibited by applicable law.

INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/09/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$37.36
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb148.71
BYD Electronics (0285.HK)	O (04/28/2023)	HK$23.16
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.40
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.46
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb370.20
Genius Electronic Optical Co. Ltd. (3406.TW)	U (09/10/2026)	NT$1,060.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb21.51
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.20
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb32.97
Largan Precision (3008.TW)	E (09/10/2026)	NT$6,660.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.90
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb11.15
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$5.05
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$50.55
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb47.59
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.46
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$4.90
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb72.20
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$35.60
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb48.95
Sunny Optical (2382.HK)	E (03/23/2026)	HK$60.50
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb235.04

Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb15.64
Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$25.96
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb387.87
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$168.30
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb98.38
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb22.37
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb775.29
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$974.50
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$20.76
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb29.03
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,070.00
Advantech (2395.TW)	O (01/20/2021)	NT$727.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,205.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,440.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$36.60
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,520.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,360.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.50
Chenbro (8210.TW)	O (07/23/2025)	NT$874.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,330.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$1,965.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$144.00
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$67.00
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,370.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$332.50
Innolux (3481.TW)	E (04/07/2025)	NT$51.00
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$10,615.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb28.41
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$297.50
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.64
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.23
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb168.22
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.30
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$996.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$195.50
Compal Electronics (2324.TW)	U (04/23/2025)	NT$35.50
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$6,000.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$4.63
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb56.07
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$363.00
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,285.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$249.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb64.69
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$49.00
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$2,960.00
Lenovo (0992.HK)	O (07/10/2026)	HK$32.84
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,915.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,475.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$89.70
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$325.50
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb124.14
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb347.25
Unimicron (3037.TW)	O (02/23/2026)	NT$1,270.00

Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$558.00
Wistron Corporation (3231.TW)	O (07/12/2023)	NT$188.50
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,240.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$635.00
Zhen Ding (4958.TW)	++	NT$592.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$535.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,740.00

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.

© 2026 Morgan Stanley

October 12, 2026 09:53 PM GMT

Delta Electronics Inc. | Asia Pacific

September Sales +1% MoM/ +15% YoY

In this report, we focus on Delta's monthly sales, which we believe could be a catalyst for its share price.

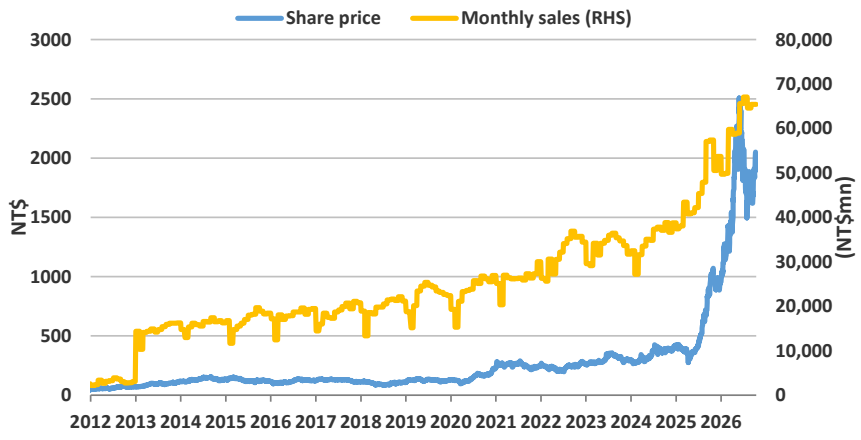
Details:

- September sales were NT$65,400mn (+1% MoM/+15% YoY).
- 3Q26 sales came in at NT$197,059mn (+8% QoQ/+31% YoY), largely in line with our estimate at NT$201,379mn (+10% QoQ/+34% YoY), but slightly below the consensus at NT$208,250mn (+14% QoQ/+39% YoY).

Our view:

- We believe the modest shortfall in 3Q26 revenue vs. the market expectation of ~10% QoQ, is due to some component supply shortages and the customer's pull-in schedule.
- Looking ahead, we expect 4Q26 revenue to grow by double-digit % QoQ on alleviated component supply and solid demand.
- Regarding the USITC Investigation No. 337, we believe near-term impact should be manageable, but we will continue to monitor the developments on this matter (see our report here).
- Stay OW.

Exhibit 1: Delta's monthly sales vs. its share price



Source: Past performance is no guarantee of future results. Results shown do not include transaction costs. Source: Company data, TEJ, Morgan Stanley Research.

MORGAN STANLEY TAIWAN LIMITED+

Derrick Yang
Equity Analyst
Derrick.Yang@morganstanley.com +886 2 2730-2862

Vivi Huang
Research Associate
Vivi.Huang@morganstanley.com +886 2 2730-2860

MORGAN STANLEY ASIA LIMITED+

Andy Meng, CFA
Equity Analyst
Andy.Meng@morganstanley.com +852 2239-7689

Delta Electronics Inc. (2308.TW, 2308 TT)

Greater China Technology Hardware | Taiwan

Stock Rating	**Overweight**
Industry View	**In-Line**
Price target	NT$2,500.00
Up/downside to price target (%)	32
Shr price, close (Oct 12, 2026)	NT$1,900.00
52-Week Range	NT$2,585.00-886.00
Sh out, dil, curr (mn)	2,598
Mkt cap, curr (mn)	US$154,524
EV, curr (mn)	US$152,862
Avg daily trading value (mn)	US$616

Fiscal Year Ending	12/25	12/26e	12/27e	12/28e
EPS (NT$)**	23.14	39.77	64.77	100.77
Prior EPS (NT$)**	-	-	-	-
EPS (NT$)§	23.91	40.39	64.11	99.50
Revenue, net (NT$ bn)	554.9	775.2	1,131.9	1,659.5
EBITDA (NT$ bn)	111.8	169.2	267.5	404.0
ModelWare net inc (NT $ bn)	60.1	103.3	168.2	261.7
P/E	41.6	47.8	29.3	18.9
P/BV	7.1	11.4	8.8	6.5
RNOA (%)	27.3	42.2	70.3	101.3
ROE (%)	21.5	31.7	42.2	50.8
EV/EBITDA	20.1	28.4	17.7	11.3
Div yld (%)	0.8	0.6	1.0	1.7
FCF yld ratio (%)**	2.1	0.9	2.2	4.1
Leverage (EOP) (%)	(41.9)	(51.7)	(57.8)	(64.0)

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework
** = Based on consensus methodology
§ = Consensus data is provided by Refinitiv Estimates
e = Morgan Stanley Research estimates

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

+= Analysts employed by non-U.S. affiliates are not registered with FINRA, may not be associated persons of the member and may not be subject to FINRA restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

AlphaSignals Catalyst Event Reaction

Company / Catalyst	Date	Impact to thesis	Catalyst versus expectation
Delta Electronics Inc. 2308.TW			
September 2026 Sales	13 Oct 2026	Unchanged	↓ Modest downside surprise

Source: Company data, Morgan Stanley Research

Valuation Methodology and Risks

Delta Electronics Inc. (2308.TW)

Base case: P/E multiple of 30x based on average 2027-28 EPS. We believe our target multiple is justified by the 63% earnings CAGR we estimate for 2025-28, implying a PEG of 0.5, versus the stock's ~23x P/E and 23% earnings CAGR during 2022-25, which implied a PEG of 1.0.

Risks to Upside

- Faster-than-expected global economic growth
- Falling raw material prices
- Margin improvement

Risks to Downside

- Global economic slowdown
- Ongoing component supply tightness
- Rising raw material prices
- Margin contraction as a result of higher operating expenses or faster ASP erosion

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley Asia Limited (which accepts the responsibility for its contents) and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research), and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Andy Meng, CFA; Derrick Yang.

.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictpolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

As of September 30, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., AirTAC International, Asustek Computer Inc., AU Optronics, Auras Technology Co Ltd, Bizlink, Catcher Technology, Chenbro, Co-Tech Development Corp., Compal Electronics, **Delta Electronics Inc.**, E Ink Holdings Inc., Elite Material Co., Ennostar Inc, Eoptolink Technology Inc Ltd, Fositek Corp, Genius Electronic Optical Co. Ltd., Giga-Byte Technology Co. Ltd., Hiwin Technologies Corp., LandMark Optoelectronics Corporation, Largan Precision, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Lotes Co. Ltd., Nan Ya PCB, Quectel Wireless Solutions Co Ltd, Sunny Optical, Suzhou TFC Optical Communication Co Ltd., Taiwan Union Technology Corporation, Unimicron, Visual Photonics Epitaxy Co Ltd, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhejiang Crystal-Optech Co Ltd, Zhen Ding, Zhongji Innolight Co Ltd, ZTE Corporation.

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Asustek Computer Inc., Unimicron, Wistron Corporation, Wiwynn Corp, Zhen Ding, Zhongji Innolight Co Ltd.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Asustek Computer Inc., Lenovo, Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Zhen Ding, Zhongji Innolight Co Ltd.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Advantech, AirTAC International, Asustek Computer Inc., AU Optronics, Bizlink, Catcher Technology, Chenbro, Compal Electronics, **Delta Electronics Inc.**, E Ink Holdings Inc., Elite Material Co., Ennostar Inc, Eoptolink Technology Inc Ltd, FIT Hon Teng Ltd, Giga-Byte Technology Co. Ltd., Gold Circuit Electronics Ltd., Hon Hai Precision, Innolux, Lenovo, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Lotes Co. Ltd., Luxshare Precision Industry Co., Ltd., Pegatron Corporation, Q Technology (Group) Company Ltd, Quanta Computer Inc., Shanghai Conant Optical Co Ltd, Shenzhen Transsion Holdings Co Ltd, Suzhou TFC Optical Communication Co Ltd., TCL Corp., Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhen Ding, Zhongji Innolight Co Ltd.

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Asustek Computer Inc., AU Optronics, BYD Electronics, Compal Electronics, E Ink Holdings Inc., Eoptolink Technology Inc Ltd, Giga-Byte Technology Co. Ltd., GoerTek Inc, Hon Hai Precision, Innolux, Lenovo, Lingyi Itech Guangdong Co, Quanta Computer Inc., Xiaomi Corp, Yageo Corp..

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Advantech, AirTAC International, Asustek Computer Inc., AU Optronics, Bizlink, Catcher Technology, Chenbro, Compal Electronics, **Delta Electronics Inc.**, E Ink Holdings Inc., Elite Material Co., Ennostar Inc, Eoptolink Technology Inc Ltd, FIT Hon Teng Ltd, Giga-Byte Technology Co. Ltd., Gold Circuit Electronics Ltd., Hon Hai Precision, Innolux, Lenovo, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Lotes Co. Ltd., Luxshare Precision Industry Co., Ltd., Pegatron Corporation, Q Technology (Group) Company Ltd, Quanta Computer Inc., Shanghai Conant Optical Co Ltd, Shenzhen Transsion Holdings Co Ltd, Suzhou TFC Optical Communication Co Ltd., TCL Corp., Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhen Ding, Zhongji Innolight Co Ltd.

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Asustek Computer Inc., AU Optronics, BYD Electronics, Compal Electronics, E Ink Holdings Inc., Eoptolink Technology Inc Ltd, Giga-Byte Technology Co. Ltd., GoerTek Inc, Hon Hai Precision, Innolux, Lenovo, Lingyi Itech Guangdong Co, Quanta Computer Inc., Xiaomi Corp, Yageo Corp., Zhen Ding.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality

of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.
Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.
Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of September 30, 2026)
The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.
For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

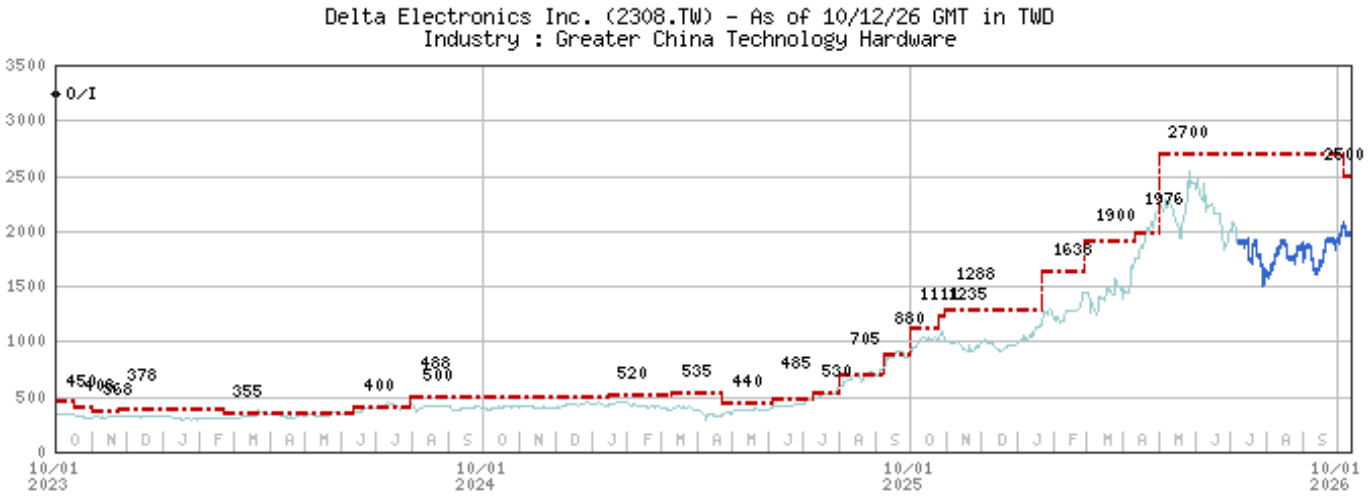
Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.
Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.
Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)



Stock Rating History: 10/1/21 : O/I

Price Target History: 9/9/21 : 345; 10/29/21 : 325; 2/25/22 : 303; 7/29/22 : 312; 9/21/22 : 326; 10/6/22 : 350; 11/23/22 : 368; 8/2/23 : 450; 10/17/23 : 408; 11/1/23 : 368; 11/23/23 : 378; 2/21/24 : 355; 6/12/24 : 400; 7/30/24 : 488; 8/1/24 : 500; 1/15/25 : 520; 3/11/25 : 535; 4/23/25 : 440; 6/4/25 : 485; 7/9/25 : 530; 8/1/25 : 705; 9/8/25 : 880; 9/30/25 : 1111; 10/24/25 : 1235; 10/30/25 : 1288; 1/21/26 : 1638; 2/27/26 : 1900; 4/10/26 : 1976; 5/1/26 : 2700; 10/6/26 : 2500

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (http://www.morganstanley.com/terms.html). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (http://www.morganstanley.com/terms.html). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research.

Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V which is regulated by Comision Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118J); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. Morgan Stanley Research is disseminated in Canada by Morgan Stanley Canada Limited and Morgan Stanley Wealth Management Canada Inc., members of the Canadian

Investment Regulatory Organization (CIRO). The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

Morgan Stanley Hong Kong Securities Limited is the liquidity provider/market maker for securities of AAC Technologies Holdings, BYD Electronics, FIT Hon Teng Ltd, Hon Hai Precision, Kingboard Laminates Holdings Ltd., Lenovo, Lens Technology, Luxshare Precision Industry Co., Ltd., Sunny Optical, Xiaomi Corp, Yangtze Optical Fibre and Cable JSC Ltd, Zhongji Innolight Co Ltd, ZTE Corporation listed on the Stock Exchange of Hong Kong Limited. An updated list can be found on HKEx website: http://www.hkex.com.hk.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

References to any prediction market or event contract do not constitute an offer, solicitation, recommendation, or endorsement to participate in any such market or contract. Morgan Stanley Research may not be distributed or made available in any jurisdiction where such distribution would be unlawful or otherwise prohibited by applicable law.

INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/12/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$36.06
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb140.17
BYD Electronics (0285.HK)	O (04/28/2023)	HK$22.80
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.44
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.51
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb347.60
Genius Electronic Optical Co. Ltd. (3406.TW)	E (10/13/2026)	NT$1,070.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb20.85
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.17
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb33.24
Largan Precision (3008.TW)	O (10/13/2026)	NT$7,325.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.73
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb10.88
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$4.97
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$49.50
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb46.32
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.36
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$4.79
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb69.76
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$34.68
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb49.36
Sunny Optical (2382.HK)	E (03/23/2026)	HK$58.80
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb229.54

Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb14.89
Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$24.52
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb362.18
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$156.10
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb99.88
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb21.36
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb724.47
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$903.00
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$20.36
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb28.44
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,130.00
Advantech (2395.TW)	O (01/20/2021)	NT$721.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,210.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,385.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$35.55
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,540.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,355.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.33
Chenbro (8210.TW)	O (07/23/2025)	NT$861.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,365.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$1,900.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$149.00
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$63.80
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,270.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$338.00
Innolux (3481.TW)	E (04/07/2025)	NT$51.30
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$10,580.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb26.73
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$288.00
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.52
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.08
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb167.00
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.35
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$977.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$194.50
Compal Electronics (2324.TW)	U (04/23/2025)	NT$36.60
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$6,150.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$4.43
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb52.05
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$356.00
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,235.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$244.50
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb62.05
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$46.80
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$3,025.00
Lenovo (0992.HK)	O (07/10/2026)	HK$32.34
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,880.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,440.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$98.60
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$322.00
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb120.00
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb327.78
Unimicron (3037.TW)	O (02/23/2026)	NT$1,255.00

Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$564.00
Wistron Corporation (3231.TW)	O (07/12/2023)	NT$189.50
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,275.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$636.00
Zhen Ding (4958.TW)	++	NT$581.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$522.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,735.00

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.

© 2026 Morgan Stanley

Morgan Stanley | RESEARCH

IDEA

October 12, 2026 09:00 PM GMT

Greater China Technology Hardware | Asia Pacific

Higher OE Estimates, Upgrade Largan and Genius

Following our semi team's increase in optical engine (OE) estimates, we raise our fiber array unit (FAU) estimates accordingly and thus upgrade Largan to OW and Genius to EW.

Key Takeaways

- We trim Largan's traditional business value to NT$2,800/share but raise FAU business to NT$6,240/share, from which we derive a new PT of NT$9,040.
- We value Genius's traditional business at NT$600/share and FAU business at NT $470/share, from which we derive a new PT of NT$1,070.
- We are more positive on FAU's TAM but, we think FAU's coupling/matching could be an overhang to our higher OE/FAU shipment estimates in 2027-28.

Raising FAU estimates after upstream semi team's OE estimates changes: Our OE shipment forecasts increase by 128%/52% to 17.73mn/73.87mn in 2027/28 (see full report). As OE and FAU are 1:1, we raise our FAU volume forecasts accordingly, while key other assumptions are unchanged, which resulted in an FAU TAM increase from US$778mn to US$1.773bn in 2027e and from US$4.86bn to US$7.387bn in 2028e. This significant FAU TAM increase leads to the changes below.

We now value Largan at NT$9,040/share, or 36% upside potential: Our analysis suggests that Largan's share price already embeds >30% global market share in FAU before formal production starts in 2H27 or even 2028. But our revised assumptions suggest further upside if Largan successfully ramps up production and gains meaningful market share. To reflect 3Q earnings miss, we trim our intrinsic value estimates from NT$3,000 to NT$2,800 per share for the company's traditional business, while our TAM analysis derives a value of NT$6,240/share for CPO/FAU new business, bringing the price target to NT$9,040. Considering the stronger growth potential in 2028, we upgrade the stock to OW from EW.

We now value Genius at NT$1,070/share, upgrade to EW: Similarly, we believe Genius's share price is also affected by hopes for FAU components. Using a similar approach, we conclude that the aggregate value is NT$1,070/share (NT$600 for existing business and NT$470/share for the FAU component business). We therefore see a more balanced risk/reward and hence move to EW.

Key risks to our call: While upstream semi capacity and yield might have already improved, we believe the yield of FAU coupling/matching will be a key overhang. Any lower-than-expected yield could result in shipment misses for FAU and OE. If the yield is not ideal, key customers are less likely to place large orders despite upstream capacity being ready. On the upside, earlier-than-expected mass production and more rapid volume growth could drive upside to our estimates.

MORGAN STANLEY ASIA LIMITED+

Andy Meng, CFA
Equity Analyst
Andy.Meng@morganstanley.com +852 2239-7689

Betty Chen
Research Associate
Betty.H.Chen@morganstanley.com +852 2239-7213

GREATER CHINA TECHNOLOGY HARDWARE

Asia Pacific
Industry View **In-Line**

WHAT'S CHANGED

Genius Electronic Optical Co. Ltd. (3406.TW)	From	To
Rating	Underweight	**Equal-weight**
Price Target	NT$910.00	**NT$1,070.00**

Largan Precision (3008.TW)	From	To
Rating	Equal-weight	**Overweight**
Price Target	NT$7,100.00	**NT$9,040.00**

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

+= Analysts employed by non-U.S. affiliates are not registered with FINRA, may not be associated persons of the member and may not be subject to FINRA restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

Executive Summary

Higher optical engine shipment assumptions as yields improve

Our semi team has lifted their optical engine shipment assumptions following their latest supply-chain checks. While PIC production remains a bottleneck, improving SoIC yields and downstream assembly yields should support a faster OE ramp. They now expect OE production to reach 2mn/59mn/148mn units in 2026/27/28, with actual OE shipments reaching 0.39mn/17.73mn/73.87mn units, respectively. In particular, they assume the SoIC yield improves from 50% in 2026 to 95% in 2027 onwards, while downstream assembly yield rises from 20% in 2026 to 30%/50% in 2027/28. We believe this provides greater confidence in the CPO volume ramp, although downstream assembly and testing remain key gating factors for how quickly OE shipments can scale.

Exhibit 1: Optical engine assumptions: New vs. Old

Metric	Assumption	2026E	2027E	2028E
TSMC PIC capacity (kwpm)	**Latest**	0.5	8.0	20.0
	Old	0.5	10.0	25.0
	% change	0%	-20%	-20%
Die per wafer	**Latest**	648	648	648
	Old	648	648	648
Annual PIC production (mn)	**Latest**	4.0	62.0	156.0
	Old	4.0	78.0	194.0
	% change	0%	-21%	-20%
SoIC yield rate	**Latest**	50%	95%	95%
	Old	50%	50%	50%
	% change	0%	45%	45%
Optical engine production (mn)	**Latest**	2.0	59.0	148.0
	Old	2.0	39.0	97.0
	% change	0%	51%	53%
Downstream assembly yield rate	**Latest**	20%	30%	50%
	Old	20%	20%	50%
	% change	0%	10%	0%
Actual optical engine shipment (mn)	**Latest**	0.39	17.73	73.87
	Old	0.39	7.78	48.60
	% change	0%	128%	52%

Source: Company data, Morgan Stanley Research (E) estimates

Largan: Larger CPO/FAU upside; upgrade to OW

Based on the semi team's assumption revision, we lift our CPO optical engine volume assumptions for 2027-28e while keeping other assumptions unchanged:

1. We forecast CPO optical engine volume to increase from 0.39mn in 2026 to 17.73mn in 2027, with a further surge to 73.87mn in 2028. Consequently, FAU (Fiber Array Units – vital optical interconnect components that align multiple optical fibers with silicon chips, making them critical for Co-Packaged Optics) volumes for CPO also grow from 0.39mn to 73.87mn in the same period;
2. We assume FAU ASP of US$100 in 2026-28 (average market level);
3. We forecast Largan's FAU market share to increase from 0% in 2026 to 5% in 2027 and 40% in 2028.
4. Assuming net margins of 10% in 2027 and 30% in 2028, we forecast FAU earnings of US$9mn in 2027 and US$882mn in 2028.

If we value the FAU business at 35x 2028e P/E (industry peers are at 40-50x P/E, but we use a more conservative multiple to reflect the first year of scalable production), this implies an incremental market cap of US$30.862bn. Discounting this back to 2026 at a 10% rate, we derive a per share positive impact of NT$6,240.

Exhibit 2: Base Case – Potential Positive Impact from FAU New Business for Largan

	2026E	2027E	2028E
Optical Engine Volume (mn)	0.39	17.73	73.87
FAU Volume (mn units)	0.39	17.73	73.87
ASP (USD)	100	100	100
Total TAM (US$ mn)	39	1,773	7,387
Largan's Market Share	0%	5%	40%
FAU Revenue for Largan (US$ mn)	0	89	2,939
Net Margin	n.a.	10%	30%
FAU Earnings for Largan (US$ mn)	n..a	9	882
Assumed Forward P/E			35
Market Cap in USD (mn)			30,862
Market Cap in TWD (mn)			987,578
Per share impact	6,240	6,864	7,550

Source: Morgan Stanley Research (E) estimates.

3Q26 results miss; revising down traditional business estimates. Largan reported 3Q26 revenue of NT$15.65bn, down 11% YoY, with gross margin declining to 42.2% and net income of NT$5.18bn, down 27% YoY. The weaker-than-expected gross margin, mainly due to lower new-product yields and higher structural green-power costs, resulted in a meaningful earnings miss. Together with softer smartphone order momentum, we lower our 2026-28 revenue estimates by 5-6% and gross margin assumptions to 45.8%/44.5%/47.5%, resulting in 12%/16%/9% downward revisions to our 2026-28 earnings estimates, respectively. For more details, please refer to Earnings Estimate Revision section.

We remain positive on Largan's CPO/FAU opportunity given the company's strong track record in high-precision manufacturing relative to most Taiwan peers. Following our semi team's higher optical engine shipment assumptions, we see a meaningfully larger FAU addressable market in 2027-28. More recently, Largan indicated that its first FA project has effectively passed customer verification and that it is expanding from FA into FAU, including coupling and potential FAU-to-carrier assembly, with dedicated capacity expected to support mass production from around mid-2027/2H27. While the current share price already reflects some CPO/FAU expectations, our revised assumptions suggest further upside if Largan successfully ramps production and gains meaningful market share. We set our price target at NT$9,040, comprising NT$2,800 from the traditional business and NT$6,240 from the CPO/FAU business.

Genius: Higher CPO/FAU optionality; upgrade to EW

Similar to Largan, following semi-team's higher assumptions, we also lift our CPO optical engine volume assumptions for 2027-2028e while keeping other assumptions unchanged.

1. We apply the same FAU volume and ASP assumptions as we do for Largan.
2. We expect Genius to be involved in FAU components, rather than overall solutions; therefore, we assume components TAM to be 15% of the FAU TAM.
3. We forecast Genius's market share to increase from 0% in 2026 to 5% in 2027 and 30% in 2028.
4. Assuming net margins of 10% in 2027 and 25% in 2028, we forecast FAU component earnings of US$1mn in 2027 and US$81mn in 2028.

If we value the FAU component business at 25x 2028e P/E (lower multiple than Largan because Genius has smaller scale), it could introduce incremental market cap of US $2,004mn. Discounting it back to 2026 at a 10% discounted rate, we derive a per-share positive impact of NT$470.

Exhibit 3: Base Case – Potential Positive Impact from FAU New Business for Genius

	2026E	2027E	2028E
Optical Engine Volume (mn)	0.39	17.73	73.87
FAU Volume (mn units)	0.39	17.73	73.87
ASP (USD)	100	100	100
Total TAM (US$ mn)	39	1,773	7,387
15% Share of Components TAM	6	266	1,108
Genius's Market Share	0%	5%	30%
FAU Component Revenue for Genius (US$ mn)	0	13	331
Net Margin	n.a.	10%	25%
FAU Component Earnings for Genius (US$ mn)	n..a	1	81
Assumed Forward P/E			25
Market Cap in USD (mn)			2,004
Market Cap in TWD (mn)			64,114
Per share impact	470	517	569

Source: Morgan Stanley Research estimates

Similar to Largan, our residual income model derives an intrinsic value of NT$600/share for the traditional business, while the TAM analysis suggests CPO/FAU-related opportunities could be valued at NT$470/share. Adding these together, we raise our price target materially to NT$1,070.

Given Largan's traditional business has faced margin headwinds in 3Q26 due to the application of green power, it's reasonable to expect that Genius could also face similar customer requirements on green power. However, as Genius has more production capacity in mainland China, we believe it could face less significant cost pressure. We therefore maintain our estimates unchanged at this stage and will monitor margin development in the upcoming quarterly results.

Earnings Estimate Revision

Largan reported 3Q26 revenue of NT$15.65bn (-11% YoY) and net income of NT$5.18bn (-27% YoY), with gross margin declining to 42.2%. Following weaker-than-expected 3Q26 results, softer smartphone order momentum and a lower gross margin outlook due to new-product yield pressure and higher green-power costs, we lower our revenue and GPM estimates for 2026-28, resulting in 9-16% downward earnings revisions over the period.

Exhibit 4: Earnings revisions: New vs. Old

	New			Old			New vs Old		
NT$ mn	2026E	2027E	2028E	2026E	2027E	2028E	2026E	2027E	2028E
Net sales	62,665	70,887	72,868	66,920	74,757	77,025	-6%	-5%	-5%
% Growth YoY	2%	13%	3%	9%	12%	3%			
Gross profit	28,696	31,569	34,629	33,103	36,834	37,952	-13%	-14%	-9%
% margin	45.8%	44.5%	47.5%	49.5%	49.3%	49.3%	-4%	-5%	-2%
Net income	21,308	22,536	25,639	24,179	26,707	28,204	-12%	-16%	-9%
% Growth YoY	0%	6%	14%	14%	10%	6%			
% margin	34%	32%	35%	36%	36%	37%	-2%	-4%	-1%
EPS (NT$)	162.90	172.30	196.02	184.85	204.18	215.63	-12%	-16%	-9%

Source: Morgan Stanley Research (E) estimates.

Valuation and Price Target Revisions

Largan: For the traditional business, we lower our earnings and medium-term growth rate; as a result, our intrinsic value in our base case scenario is adjusted down to NT$2,800 from NT$3,000. We use RIM valuation methodology for Greater China Hardware Tech companies in our coverage. We apply an unchanged CoE of 8.5%, a medium-term growth rate of 8% (from 7% previously on higher 2026e estimates), and a long-term growth rate of 3% (unchanged). However, we believe the upward revision to optical engine shipment assumptions materially increases the addressable market and earnings potential for Largan's FAU business. Therefore, we add the base case per share positive value from FAU based on the analysis in the previous section. The new SOTP valuation derives a new price target at NT$9,040 (from NT$7,100). We also raise our bull case and bear case values to NT$11,426 (from NT$11,142.80) and NT$4,620 (from NT$3,649.20).

Genius: For the traditional business, we make no earnings and intrinsic value revisions; as a result, our intrinsic value under our base case scenario is unchanged at NT$600. We use RIM valuation methodology for Greater China Hardware Tech companies in our coverage. We apply a CoE of 8.5%, a medium-term growth rate of 4%, and a long-term growth rate of 3%. However, we believe the upward revision to optical engine shipment assumptions increases the potential value of Genius's FAU opportunity, although execution visibility remains lower than for more established suppliers. Therefore, we add the base case per share positive value from FAU based on the analysis in the previous section. The new SOTP valuation derives a new price target at NT$1,070 (from NT$910). We also raise our bull case and bear case values to NT$1,834 (from NT$1,705.20) and NT$658 (from NT $592.40).

Exhibit 5: SOTP/RI valuation: Largan

RI Valuation - Base case

Rmb mn	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E
Total equity	212,176	234,712	260,351	273,729	288,178	303,783	320,636	338,834	358,484	379,510	401,772
Revenue	62,665	70,887	72,868	78,697	84,993	91,793	99,136	107,047	115,589	123,681	130,953
Net profit	21,308	22,536	25,639	26,757	28,898	31,209	33,706	36,396	39,300	42,051	44,524
YoY %	0%	6%	14%	4%	8%	8%	8%	8%	8%	7%	6%
ROE	11%	11%	11%	10%	11%	11%	11%	11%	12%	12%	12%
Residual income	**5,080**	**4,497**	**5,683**	**4,622**	**5,625**	**6,708**	**7,878**	**9,135**	**10,492**	**11,572**	**12,257**
Terminal value											**229,453**
Discount factor	1.00	0.92	0.85	0.78	0.72	0.66	0.61	0.56	0.52	0.48	0.44
Equity value (NT$)	**366,249**										
Implied P/E (X)	**17.2**	**16.3**									
# of shares	131										
Per share value (NT$)	**2,800**										
FAU Positive Impact/Share	**6,240**										
Aggregate Value/Share	**9,040**										

Source: Morgan Stanley Research (E) estimates.

Exhibit 6: SOTP/RI valuation: Genius

Genius (3406.TW)
Residual income model

Rmb mn	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E
Total equity	28,984	31,852	34,894	37,855	40,963	44,227	47,487	50,845	54,304	57,866	61,502
Revenue	29,825	32,165	34,176	35,884	37,679	39,563	40,750	41,972	43,231	44,528	45,864
Net profit	**5,000**	**5,736**	**6,086**	**5,921**	**6,217**	**6,528**	**6,520**	**6,716**	**6,917**	**7,124**	**7,273**
YoY%	35%	15%	6%	-3%	5%	5%	0%	3%	3%	3%	2%
ROE%	19%	20%	19%	17%	16%	16%	15%	14%	14%	13%	13%
Residual income	**2,748**	**3,272**	**3,378**	**2,954**	**2,998**	**3,045**	**2,760**	**2,678**	**2,594**	**2,507**	**2,353**
Equity value (NT$)	**67,642**										
# of shares	113										
Implied P/E (x)	**13.5**										
Per share value (NT$)	**600.0**										
FAU Positive Impact/Share	**470.0**										
Aggregate Value/Share	**1,070**										

Source: Morgan Stanley Research (E) estimates.

Genius: Financial Summary

Exhibit 7: Genius: Financial Summary

Genius (3406.TW)

Financial Summary

Profit and Loss Statement	2025	2026E	2027E	2028E
Net Sales	**24,989**	**29,825**	**32,165**	**34,176**
COGS	(16,864)	(19,219)	(20,710)	(22,004)
Gross Profit	**8,125**	**10,606**	**11,455**	**12,171**
Selling	(277)	(328)	(396)	(421)
G&A	(986)	(1,154)	(1,253)	(1,318)
R&D	(2,089)	(2,455)	(2,601)	(2,762)
Operating Income	**4,773**	**6,668**	**7,206**	**7,671**
Finance costs, net	66	256	256	256
Others	152	(199)	200	200
Pre-tax Income	**4,990**	**6,725**	**7,661**	**8,127**
Income Tax	(1,282)	(1,710)	(1,910)	(2,026)
Profit for the year	**3,708**	**5,015**	**5,751**	**6,101**
Minority interests	6	15	15	15
Net Income	**3,702**	**5,000**	**5,736**	**6,086**
EPS for consensus (NT$)	**32.8**	**44.3**	**50.9**	**54.0**
EBITDA	**4,773**	**6,668**	**7,206**	**7,671**

Balance Sheet	2025	2026E	2027E	2028E
CCE	12,894	12,964	15,620	16,301
Inventory	2,115	3,173	4,759	7,139
Trade receivables	5,640	5,843	6,541	6,617
Other current assets	728	728	728	728
Total current assets	**21,378**	**22,708**	**27,649**	**30,786**
Net PPE	18,991	17,639	15,984	14,025
Other non-current assets	3,271	5,814	8,728	11,820
Total non-current assets	**22,263**	**23,453**	**24,712**	**25,845**
Trade payables	2,631	1,926	2,984	2,233
Borrowings - short term	383	383	383	383
Other current liabilities	5,632	3,858	3,263	2,200
Total current liabilities	**8,646**	**6,167**	**6,630**	**4,815**
Borrowings - long term	2,759	2,759	2,759	2,759
Other non-current liabilities	5,752	5,752	5,752	5,752
Total non-current liabilities	**8,511**	**8,511**	**8,511**	**8,511**
Net Assets	**26,484**	**31,484**	**37,220**	**43,305**
Paid-in capital	1,127	1,127	1,127	1,127
Capital reserves	25,356	30,356	36,092	42,178
Total shareholder's equity	**26,484**	**31,484**	**37,220**	**43,305**

Cash Flow Statement	2025	2026E	2027E	2028E
Cashflow from operations	**7,312**	**5,569**	**8,525**	**6,724**
Net income	3,702	5,000	5,736	6,086
Depreciation+Amortization	4,009	4,309	4,609	4,909
Others	(400)	(3,740)	(1,821)	(4,271)
Cashflow from investing	**(2,992)**	**(3,000)**	**(3,000)**	**(3,000)**
Capex	(1,602)	(3,000)	(3,000)	(3,000)
Others	(1,389)	-	-	-
Cashflow from financing	**(4,816)**	**(2,500)**	**(2,868)**	**(3,043)**
Dividend paid	(2,029)	(2,500)	(2,868)	(3,043)
Others	(2,787)	-	-	-
Changes in cash	**(379)**	**69**	**2,657**	**681**

Ratio Analysis	2025	2026E	2027E	2028E
Growth %				
Net Sales	8%	19%	8%	6%
Gross Profit	1%	31%	8%	6%
EBIT	0%	40%	8%	6%
EBITDA	0%	40%	8%	6%
Net income	-14%	35%	15%	6%
Margins %				
Gross Margin	32.5%	35.6%	35.6%	35.6%
EBIT Margin	19.1%	22.4%	22.4%	22.4%
EBITDA Margin	19.1%	22.4%	22.4%	22.4%
Net Profit Margin	14.8%	16.8%	17.8%	17.8%
Return%				
ROE	14%	16%	15%	14%
ROA	8%	11%	11%	11%
Gearing %				
Net Debt/Equity	-29%	-25%	-28%	-26%
Liabilities/Equity	65%	47%	41%	31%

Valuation	2025	2026E	2027E	2028E
Multiples				
P/E	32.1	23.8	20.7	19.5
P/B	4.5	3.8	3.2	2.7
P/S	4.8	4.0	3.7	3.5
EV/EBITDA	22.8	16.3	15.1	14.2

Source: Company data, Morgan Stanley Research (E) estimates

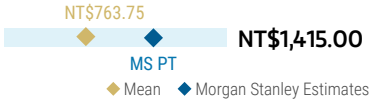
Risk Reward – Genius Electronic Optical Co. Ltd. (3406.TW)

Higher FAU optionality; EW

PRICE TARGET NT$1,070.00

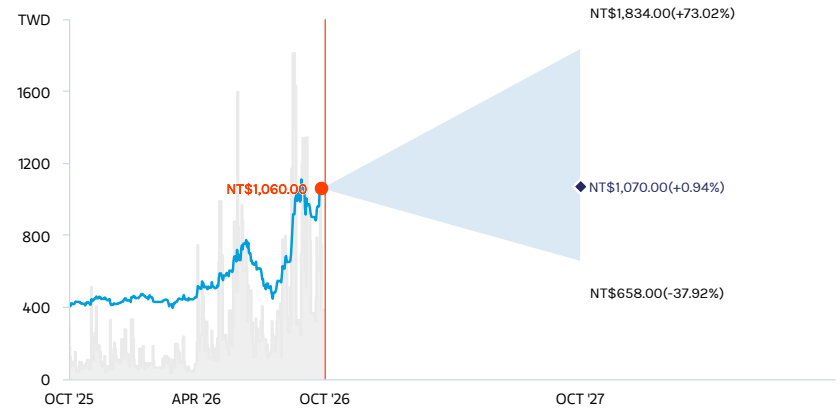
Base case: Our price target is based on an SOTP valuation, comprising the traditional business and potential FAU opportunity. We value the traditional business using a residual income model, consistent with our Greater China Tech coverage, assuming an 8.5% cost of equity, a medium-term net profit CAGR of 4% and a 3% terminal growth rate. We separately value the FAU component opportunity based on our TAM and market-share assumptions.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



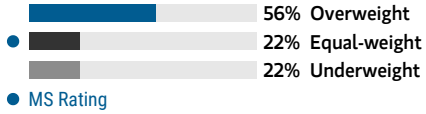
Key: Historical Stock Performance Current Stock Price Price Target

Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

- Genius should benefit from iPhone 18 variable-aperture, foldable and periscope camera upgrades as a major Apple optical-lens supplier.
- Higher OE shipment assumptions increase the addressable FAU market and potential value of Genius's FAU component opportunity.
- However, Genius is exposed to only around 15% of the overall FAU TAM, while execution visibility remains lower than for more established suppliers. We therefore see a more balanced risk/reward and hence we are EW.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Secular Growth: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE NT$1,834.00

36x 2027E base case EPS

Stronger iPhone demand and faster FAU development: We assume: 1) stronger-than-expected iPhone orders; 2) faster penetration of variable-aperture, foldable and periscope lenses; and 3) faster FAU component adoption and greater market-share gains.

BASE CASE NT$1,070.00

21x 2027E base case EPS

iPhone camera upgrades support the traditional business, with incremental FAU contribution: We assume: 1) benefits from variable-aperture, foldable and periscope upgrades; 2) Genius's FAU component market share reaches 5% in 2027 and 30% in 2028; and 3) FAU components represent 15% of the overall FAU TAM.

BEAR CASE NT$658.00

13x 2027E base case EPS

Weaker iPhone demand and slower FAU development: We assume: 1) weaker-than-expected iPhone demand; 2) slower camera-upgrade penetration and greater pricing pressure; and 3) delays in FAU component commercialization and market-share gains.

Risk Reward – Genius Electronic Optical Co. Ltd. (3406.TW)

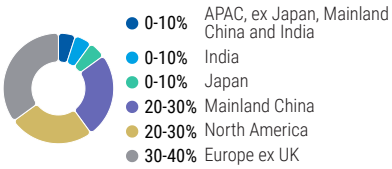
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Total smartphone shipment YoY (%)	2.1	(12.7)	3.3	1.0
iPhone revenue YoY (%)	0.0	0.0	0.0	0.0
VR revenue YoY (%)	0.0	0.0	0.0	0.0

INVESTMENT DRIVERS

- VR shipments
- iPhone shipments

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

RISKS TO PT/RATING

RISKS TO UPSIDE

- More share gains in iPhones
- Better-than-expected mixed reality demand from Apple
- Better-than-expected VR/MR shipments
- Faster-than-expected business development in CPO/NPO

RISKS TO DOWNSIDE

- Delays in Apple's mixed reality shipments
- Share loss in iPhones
- Execution/yield
- Slower VR/MR shipments

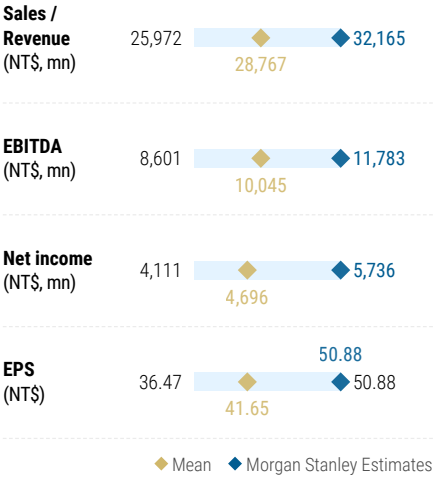
OWNERSHIP POSITIONING

Inst. Owners, % Active 60.9%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2027e



Source: Refinitiv, Morgan Stanley Research

Largan: Financial Summary

Exhibit 8: Largan: Financial Summary

Financial Summary

Profit and Loss Statement	2025	2026E	2027E	2028E
Net Sales	**61,148**	**62,665**	**70,887**	**72,868**
COGS	(30,311)	(33,969)	(39,318)	(38,239)
Gross Profit	**30,837**	**28,696**	**31,569**	**34,629**
Selling	(376)	(387)	(441)	(453)
G&A	(1,609)	(1,530)	(1,731)	(1,779)
R&D	(3,123)	(3,183)	(3,607)	(3,709)
Employee Bonus	(2,171)	(2,215)	(2,539)	(2,609)
Operating Income	**23,558**	**21,380**	**23,252**	**26,078**
Finance costs, net	4,260	4,106	5,000	6,000
Investment income	2	-	-	-
Others	(1,920)	678	-	-
Pre-tax Income	**25,900**	**26,163**	**28,252**	**32,078**
Income Tax	(4,340)	(4,586)	(5,446)	(6,169)
Profit for the year	**21,560**	**21,578**	**22,806**	**25,909**
Minority interests	(285)	(270)	(270)	(270)
Net Income	**21,275**	**21,308**	**22,536**	**25,639**
EPS for consensus (NT$)	**159.4**	**162.9**	**172.3**	**196.0**
EBITDA	**23,558**	**21,380**	**23,252**	**26,078**

Balance Sheet	2025	2026E	2027E	2028E
CCE	131,295	132,127	135,131	141,206
Inventory	6,713	10,070	15,105	22,657
Trade receivables	10,462	9,649	9,824	9,088
Other current assets	5,816	5,816	5,816	5,816
Total current assets	**154,285**	**157,662**	**165,876**	**178,767**
Net PPE	51,472	50,855	49,430	47,196
Other non-current assets	15,029	25,792	36,579	47,990
Total non-current assets	**66,502**	**76,647**	**86,009**	**95,187**
Trade payables	1,728	2,324	3,228	3,767
Borrowings - short term	-	-	-	-
Other current liabilities	28,021	19,639	13,774	9,666
Total current liabilities	**29,749**	**21,962**	**17,002**	**13,433**
Borrowings - long term	-	-	-	-
Other non-current liabilities	171	171	171	171
Total non-current liabilities	**171**	**171**	**171**	**171**
Net Assets	**190,868**	**212,176**	**234,712**	**260,351**
Paid-in capital	1,335	1,335	1,335	1,335
Capital reserves	189,533	210,841	233,377	259,016
Total shareholder's equity	**190,868**	**212,176**	**234,712**	**260,351**

Cash Flow Statement	2025	2026E	2027E	2028E
Cashflow from operations	**26,080**	**19,510**	**21,698**	**25,385**
Net income	21,275	21,308	22,536	25,639
Depreciation+Amortization	7,732	8,532	9,332	10,132
Others	(2,927)	(10,330)	(10,170)	(10,386)
Cashflow from investing	**(9,236)**	**(8,000)**	**(8,000)**	**(8,000)**
Capex	(12,338)	(8,000)	(8,000)	(8,000)
Others	3,102	-	-	-
Cashflow from financing	**(13,458)**	**(10,677)**	**(10,694)**	**(11,310)**
Net borrowings	(212)	-	-	-
Dividend paid	(11,412)	(10,677)	(10,694)	(11,310)
Proceeds from new issues	-	-	-	-
Others	(1,834)	-	-	-
Changes in cash	**2,283**	**832**	**3,004**	**6,075**

Ratio Analysis	2025	2026E	2027E	2028E
Growth %				
Net Sales	3%	2%	13%	3%
Gross Profit	-1%	-7%	10%	10%
EBIT	-2%	-9%	9%	12%
EBITDA	-2%	-9%	9%	12%
Net income	-18%	0%	6%	14%
Margins %				
Gross Margin	50.4%	45.8%	44.5%	47.5%
EBIT Margin	38.5%	34.1%	32.8%	35.8%
EBITDA Margin	38.5%	34.1%	32.8%	35.8%
Net Profit Margin	34.8%	34.0%	31.8%	35.2%
Return%				
ROE	11%	11%	11%	11%
ROA	10%	9%	9%	9%
Gearing %				
Net Debt/Equity	-61%	-55%	-51%	-48%
Liabilities/Equity	16%	10%	7%	5%

Segment Analysis	2025	2026E	2027E	2028E
Revenue				
Apple	33,634	34,874	41,866	42,689
Others (Android)	27,514	27,791	29,021	30,179
Revenue YoY%				
Apple	3%	4%	20%	2%
Others (Android)	3%	1%	4%	4%
Revenue mix%				
Apple	55%	56%	59%	59%
Others (Android)	45%	44%	41%	41%

Valuation	2025	2026E	2027E	2028E
Multiples				
P/E	41.8	41.7	39.4	34.7
P/B	4.7	4.2	3.8	3.4
P/S	14.5	14.2	12.5	12.2
EV/EBITDA	32.9	36.2	33.3	29.7
Yields				
FCF yield	2.0%	2.2%	2.3%	2.5%
Dividend yield	1.2%	1.2%	1.3%	1.5%

Efficiency	2025	2026E	2027E	2028E
AR days	63	62	59	50
AP days	23	22	22	26
Inventory days	67	75	90	117
Cash conversion cycle	106	116	127	141

Source: Company data, Morgan Stanley Research (E) estimates

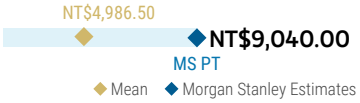
Risk Reward – Largan Precision (3008.TW)

Higher OE estimates expand FAU opportunity; OW

PRICE TARGET NT$9,040.00

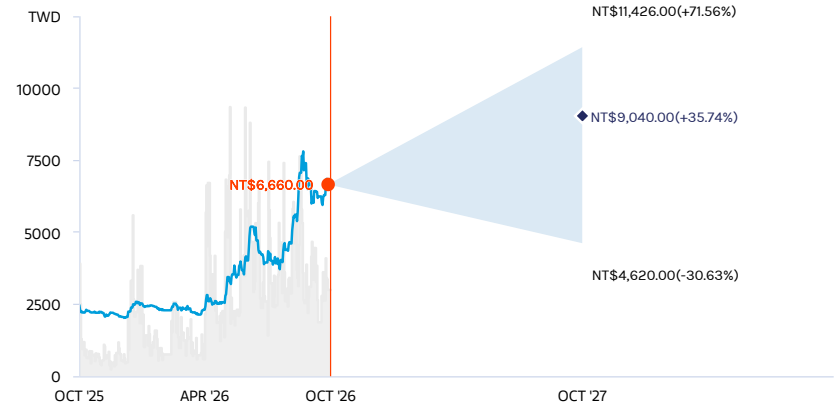
Base case: Our price target is based on an SOTP valuation, comprising the traditional business and potential FAU opportunity. We value the traditional business using a residual income model, consistent with our Greater China Hardware Tech coverage, assuming an 8.5% cost of equity, an 8% net profit CAGR over 2026-36E and a 3% terminal growth rate. We separately value the FAU business, assuming a 40% market share in 2028.

Consensus Price Target Distribution NT$2,330.00



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



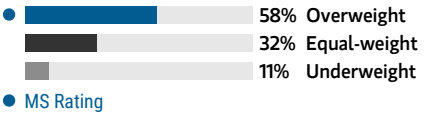
Key: Historical Stock Performance Current Stock Price Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- Largan should continue to benefit from iPhone camera upgrades, including variable aperture, foldable and periscope opportunities.
- Our semi team's higher OE shipment assumptions materially expand the addressable FAU market, supporting a larger potential contribution from Largan's CPO/FAU business.
- We see further upside if Largan successfully ramps mass production and gains meaningful FAU share, although coupling/matching yield and execution remain key risks.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE NT$11,426.00

65x 2027e bull case EPS

Stronger-than-expected client demand and FAU development: We assume: 1) stronger-than-expected iPhone orders; 2) faster penetration of variable-aperture and periscope lenses; 3) faster-than-expected margin recovery; and 4) Largan's FAU market share reaches 45% in 2028.

BASE CASE NT$9,040.00

52x 2027e base case EPS

iPhone upgrades support the traditional business, while FAU provides incremental upside: We assume: 1) Largan benefits from variable-aperture, foldable and periscope camera upgrades; and 2) Largan's FAU market share increases from 0% in 2026 to 5% in 2027 and 40% in 2028, supporting meaningful incremental earnings contribution from the FAU business.

BEAR CASE NT$4,620.00

26x 2027e bear case EPS

Weaker client demand and slower FAU development: We assume: 1) weaker-than-expected iPhone demand; 2) slower penetration of variable-aperture and periscope lenses; 3) weaker-than-expected margin recovery; and 4) delays in FAU mass production and slower market-share gains.

Risk Reward – Largan Precision (3008.TW)

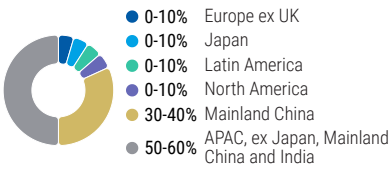
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Total smartphone shipment YoY (%)	2.1	(12.7)	3.3	1.0

INVESTMENT DRIVERS

- New iPhone sell-through
- Monthly sales
- Upgrade trend for smartphone cameras
- FAU business development

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

3/5 MOST — 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Better-than-expected high-end smartphone demand
- Faster-than-expected penetration of periscopes
- Stronger-than-expected CPO volume growth

RISKS TO DOWNSIDE

- Greater-than-expected competition, which would erode Largan's ASP and gross margin
- Weaker-than-expected smartphone demand, especially for high-end models
- Market share gains and more pricing pressure from Sunny
- Weaker-than-expected CPO volume growth

OWNERSHIP POSITIONING

Inst. Owners, % Active 71%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2027e

Sales / Revenue (NT$, mn)	63,436	72,397	70,887 / 81,971
EBITDA (NT$, mn)	31,769	36,427	32,552 / 41,768
Net income (NT$, mn)	21,410	26,572	22,536 / 29,666
EPS (NT$)	163.69	200.72	172.30 / 226.80

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - Options_Probabilities_Exhibit_Link.pdf

2. View descriptions of Risk Rewards Themes - RR_Themes_Exhibit_Link.pdf

3. View explanation of regional hierarchies - GEG_Exhibit_Link.pdf

4. View explanation of Theme/Exposure methodology - ESG_Sustainable_Solutions_External_Link.pdf

5. View explanation of HERS methodology - ESG_HERS_External_Link.pdf

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley Asia Limited (which accepts the responsibility for its contents) and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research), and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Andy Meng, CFA.

.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictpolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

As of September 30, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., AirTAC International, Asustek Computer Inc., AU Optronics, Auras Technology Co Ltd, Bizlink, Catcher Technology, Chenbro, Co-Tech Development Corp., Compal Electronics, Delta Electronics Inc., E Ink Holdings Inc., Elite Material Co., Ennostar Inc, Eoptolink Technology Inc Ltd, Fositek Corp, Genius Electronic Optical Co. Ltd., Giga-Byte Technology Co. Ltd., Hiwin Technologies Corp., LandMark Optoelectronics Corporation, Largan Precision, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Lotes Co. Ltd., Nan Ya PCB, Quectel Wireless Solutions Co Ltd, Sunny Optical, Suzhou TFC Optical Communication Co Ltd., Taiwan Union Technology Corporation, Unimicron, Visual Photonics Epitaxy Co Ltd, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhejiang Crystal-Optech Co Ltd, Zhen Ding, Zhongji Innolight Co Ltd, ZTE Corporation.

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Asustek Computer Inc., Unimicron, Wistron Corporation, Wiwynn Corp, Zhen Ding, Zhongji Innolight Co Ltd.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Asustek Computer Inc., Lenovo, Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Zhen Ding, Zhongji Innolight Co Ltd.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Advantech, AirTAC International, Asustek Computer Inc., AU Optronics, Bizlink, Catcher Technology, Chenbro, Compal Electronics, Delta Electronics Inc., E Ink Holdings Inc., Elite Material Co., Ennostar Inc, Eoptolink Technology Inc Ltd, FIT Hon Teng Ltd, Giga-Byte Technology Co. Ltd., Gold Circuit Electronics Ltd., Hon Hai Precision, Innolux, Lenovo, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Lotes Co. Ltd., Luxshare Precision Industry Co., Ltd., Pegatron Corporation, Q Technology (Group) Company Ltd, Quanta Computer Inc., Shanghai Conant Optical Co Ltd, Shenzhen Transsion Holdings Co Ltd, Suzhou TFC Optical Communication Co Ltd., TCL Corp., Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhen Ding, Zhongji Innolight Co Ltd.

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Asustek Computer Inc., AU Optronics, BYD Electronics, Compal Electronics, E Ink Holdings Inc., Eoptolink Technology Inc Ltd, Giga-Byte Technology Co. Ltd., GoerTek Inc, Hon Hai Precision, Innolux, Lenovo, Lingyi Itech Guangdong Co, Quanta Computer Inc., Xiaomi Corp, Yageo Corp..

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Advantech, AirTAC International, Asustek Computer Inc., AU Optronics, Bizlink, Catcher Technology, Chenbro, Compal Electronics, Delta Electronics Inc., E Ink Holdings Inc., Elite Material Co., Ennostar Inc, Eoptolink Technology Inc Ltd, FIT Hon Teng Ltd, Giga-Byte Technology Co. Ltd., Gold Circuit Electronics Ltd., Hon Hai Precision, Innolux, Lenovo, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Lotes Co. Ltd., Luxshare Precision Industry Co., Ltd., Pegatron Corporation, Q Technology (Group) Company Ltd, Quanta Computer Inc., Shanghai Conant Optical Co Ltd, Shenzhen Transsion Holdings Co Ltd, Suzhou TFC Optical Communication Co Ltd., TCL Corp., Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhen Ding, Zhongji Innolight Co Ltd.

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Asustek Computer Inc., AU Optronics, BYD Electronics, Compal Electronics, E Ink Holdings Inc., Eoptolink Technology Inc Ltd, Giga-Byte Technology Co. Ltd., GoerTek Inc, Hon Hai Precision, Innolux, Lenovo, Lingyi Itech Guangdong Co, Quanta Computer Inc., Xiaomi Corp, Yageo Corp., Zhen Ding.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality

of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.
Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.
Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of September 30, 2026)
The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.
For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

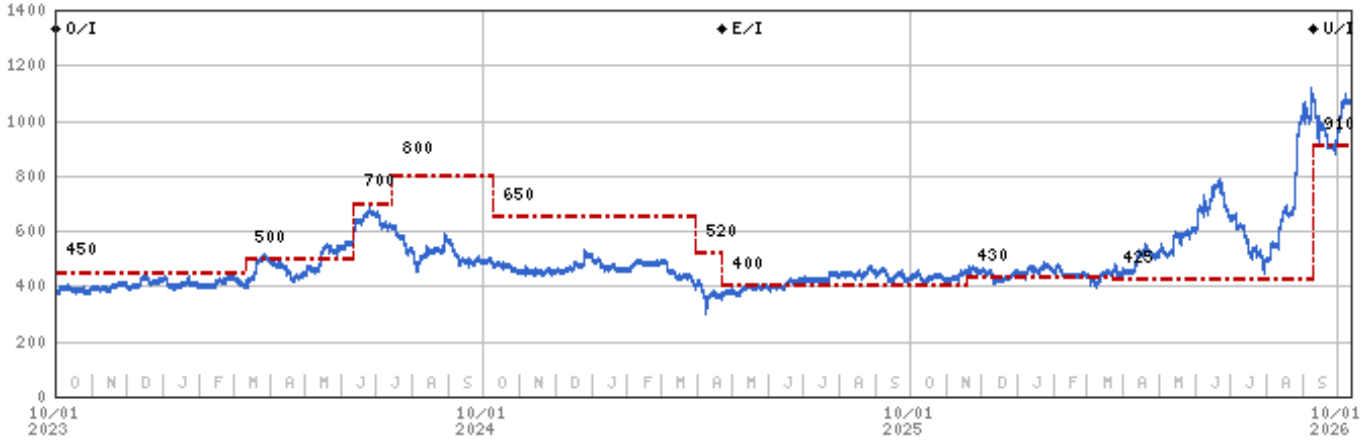
Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.
Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.
Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Genius Electronic Optical Co. Ltd. (3406.TW) - As of 10/12/26 GMT in TWD
Industry : Greater China Technology Hardware



Stock Rating History: 4/5/22 : O/I; 1/9/23 : E/I; 5/16/23 : O/I; 4/23/25 : E/I; 9/10/26 : U/I

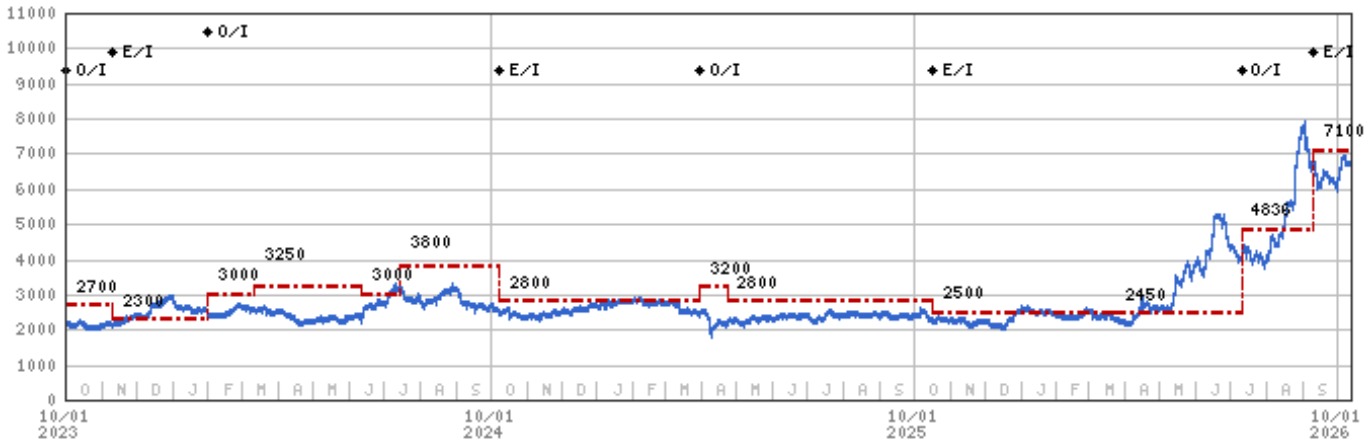
Price Target History: 4/5/22 : 600; 7/25/22 : 630; 9/13/22 : 600; 11/15/22 : 430; 1/9/23 : 400; 5/16/23 : 450; 3/11/24 : 500; 6/12/24 : 700; 7/15/24 : 800; 10/9/24 : 650; 3/31/25 : 520; 4/23/25 : 400; 11/18/25 : 430; 3/23/26 : 425; 9/10/26 : 910

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Largan Precision (3008.TW) - As of 10/12/26 GMT in TWD
Industry : Greater China Technology Hardware



Stock Rating History: 10/1/21 : E/I; 12/17/21 : O/I; 9/13/22 : E/I; 1/9/23 : U/I; 5/16/23 : O/I; 11/10/23 : E/I; 1/31/24 : O/I; 10/9/24 : E/I; 3/31/25 : O/I; 10/17/25 : E/I; 7/10/26 : O/I; 9/10/26 : E/I

Price Target History: 9/7/21 : 3050; 10/7/21 : 2300; 12/17/21 : 2600; 4/5/22 : 2300; 7/25/22 : 2400; 9/13/22 : 2200; 10/14/22 : 1900; 1/9/23 : 1700; 5/16/23 : 2700; 11/10/23 : 2300; 1/31/24 : 3000; 3/11/24 : 3250; 6/12/24 : 3000; 7/15/24 : 3800; 10/9/24 : 2800; 3/31/25 : 3200; 4/23/25 : 2800; 10/17/25 : 2500; 3/23/26 : 2450; 7/10/26 : 4836; 9/10/26 : 7100

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures.
For Morgan Stanley specific disclosures, you may refer to https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch.
Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.
Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.
Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.
Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.
Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (http://www.morganstanley.com/terms.html). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (http://www.morganstanley.com/terms.html). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).
If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research.
Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.
The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.
The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.
With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.
Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.
Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.
To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V which is regulated by Comision Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118J); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. Morgan Stanley Research is disseminated in Canada by Morgan Stanley Canada Limited and Morgan Stanley Wealth Management Canada Inc., members of the Canadian Investment Regulatory Organization (CIRO). The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

Morgan Stanley Hong Kong Securities Limited is the liquidity provider/market maker for securities of AAC Technologies Holdings, BYD Electronics, FIT Hon Teng Ltd, Hon Hai Precision, Kingboard Laminates Holdings Ltd., Lenovo, Lens Technology, Luxshare Precision Industry Co., Ltd., Sunny Optical, Xiaomi Corp, Yangtze Optical Fibre and Cable JSC Ltd, Zhongji Innolight Co Ltd, ZTE Corporation listed on the Stock Exchange of Hong Kong Limited. An updated list can be found on HKEx website: http://www.hkex.com.hk.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

References to any prediction market or event contract do not constitute an offer, solicitation, recommendation, or endorsement to participate in any such market or contract. Morgan Stanley Research may not be distributed or made available in any jurisdiction where such distribution would be unlawful or otherwise prohibited by applicable law.

INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/09/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$37.36
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb148.71
BYD Electronics (0285.HK)	O (04/28/2023)	HK$23.16
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.40
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.46
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb370.20
Genius Electronic Optical Co. Ltd. (3406.TW)	E (10/12/2026)	NT$1,060.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb21.51
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.20
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb32.97
Largan Precision (3008.TW)	O (10/12/2026)	NT$6,660.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.90
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb11.15
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$5.05
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$50.55
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb47.59
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.46
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$4.90
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb72.20
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$35.60
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb48.95
Sunny Optical (2382.HK)	E (03/23/2026)	HK$60.50
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb235.04
Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb15.64
Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$25.96
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb387.87
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$168.30
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb98.38
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb22.37
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb775.29
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$974.50
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$20.76
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb29.03
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,070.00
Advantech (2395.TW)	O (01/20/2021)	NT$727.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,205.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,440.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$36.60
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,520.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,360.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.50
Chenbro (8210.TW)	O (07/23/2025)	NT$874.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,330.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$1,965.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$144.00
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$67.00
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,370.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$332.50
Innolux (3481.TW)	E (04/07/2025)	NT$51.00
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$10,615.00

Lens Technology (300433.SZ)	E (07/22/2020)	Rmb28.41
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$297.50
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.64
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.23
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb168.22
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.30
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$996.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$195.50
Compal Electronics (2324.TW)	U (04/23/2025)	NT$35.50
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$6,000.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$4.63
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb56.07
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$363.00
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,285.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$249.00
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb64.69
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$49.00
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$2,960.00
Lenovo (0992.HK)	O (07/10/2026)	HK$32.84
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,915.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,475.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$89.70
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$325.50
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb124.14
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb347.25
Unimicron (3037.TW)	O (02/23/2026)	NT$1,270.00
Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$558.00
Wistron Corporation (3231.TW)	O (07/12/2023)	NT$188.50
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,240.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$635.00
Zhen Ding (4958.TW)	++	NT$592.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$535.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,740.00

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.

© 2026 Morgan Stanley

October 12, 2026 01:46 PM GMT

Greater China Technology Hardware | Asia Pacific

GB200 and GB300 NVL72 Racks in September 2026; Introduce 2027 Rack Forecast

We provide monthly/quarterly rack shipment forecasts for ODMs. Within the major Server ODMs, our order of preference is Wistron > Hon Hai > Quanta, based on upside to PT.

Key Takeaways

- We estimate GB200/300 rack output in September at ~6.5K (-23% m/m).
- Quanta shipped 1.55-1.60K GB200/GB300 racks.
- Wistron shipped 1.25-1.30K GB200/GB300 racks.
- Hon Hai shipped ~2.7K GB200/300 racks.
- We also introduce our 2027 forecast at 90.4K racks (+10% y/y).

We introduce 2027 forecast at 90.4K racks (+10% y/y) vs our 2026 forecast now at 82K racks: This year is a strong one for downstream rack assemblers, with rack shipments up ~209% y/y, but we expect rack shipments to grow another ~10% y/y to reach 90.4K in CY27. For CY27, our forecast implies Hon Hai will be a share gainer with rack share up to 43% (from 37% in CY26), with Quanta staying unchanged at 26% and Wistron's share up slightly to 24% (from 23% in CY26).

Quanta (2382.TW, OW): September revenue was NT$325.3B (-23% m/m, +77% y/y). We attribute the m/m decline mainly to lower GB200/300 rack shipments at 1.55-1.60K racks (-38% m/m), despite being offset slightly by higher NB shipments at 4.2M units (+62% m/m, -9% y/y). In 3Q, Quanta shipped ~6.2K racks (+3% q/q), which came in slightly lower than our forecast of ~6.4K racks. Total 3Q revenue reached ~NT$1,116M (+8% q/q, +125% y/y), which is 15% above MSe but in-line with the Street. For 2026, we forecast Quanta to ship 20.9K racks, and we introduce our preliminary 2027 forecast at 23.5K (+12% y/y).

Wistron (3231.TW, OW): September revenue was NT$380.8B (-17% m/m, +87% y/y). We attribute the m/m decline to: 1) lower GB200/300 server computing tray shipments (1.25-1.30K rack equivalents, by our estimates, down approximately 46% m/m), despite higher Wiwynn revenue at ~NT$172.6B (+20% m/m, +100% y/y) and higher NB shipments at 1.5M units (+7% m/m, -32% y/y), flat DT shipments at 500K (flat m/m, -44% y/y), and higher monitor shipments at 950K (+27% m/m, +6% y/y). In 3Q, Wistron shipped ~4.8K racks (+22% q/q), which came in slightly lower than our forecast of ~5.1K racks. Total 3Q revenue reached ~NT$1,149M (+28% q/q, +102% y/y), which is 10% above MSe and 9% above the Street. For 2026, we forecast Wistron to ship 19.13K racks, and we introduce our preliminary 2027 forecast at 22K (+15% y/y).

MORGAN STANLEY TAIWAN LIMITED+

Howard Kao
Equity Analyst
Howard.Kao@morganstanley.com +886 2 2730-2989

Irene Yen
Equity Analyst
Irene.Yen@morganstanley.com +886 2 2730-2869

MORGAN STANLEY ASIA LIMITED+

Andy Meng, CFA
Equity Analyst
Andy.Meng@morganstanley.com +852 2239-7689

GREATER CHINA TECHNOLOGY HARDWARE

Asia Pacific
Industry View **In-Line**

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

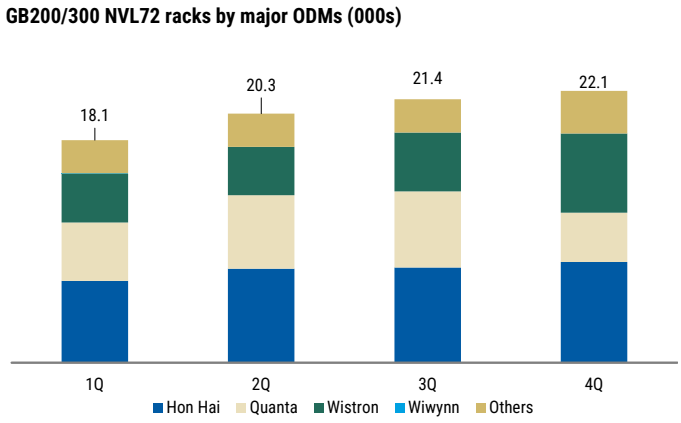
+= Analysts employed by non-U.S. affiliates are not registered with FINRA, may not be associated persons of the member and may not be subject to FINRA restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

Hon Hai (2317.TW, OW): September revenue was NT$1,158.6B (+26% m/m, +38% y/y), with consumer segment revenue seeing the strongest m/m growth, followed by components and other products, cloud and networking products, and lastly computing. We model HH's GB200/300 rack shipments at 2.7K (+4% m/m). This brings total 3Q rack shipments at 7.8K racks (+1% q/q). For 2026, we forecast Foxconn to ship 30.3K racks, and we introduce our preliminary 2027 forecast at 38.8K (+28% y/y).

NVL72 Exhibits

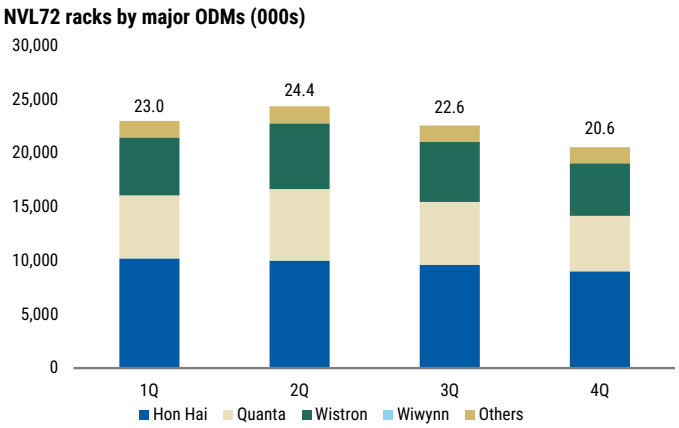
We believe actual rack deliveries to end-customers are likely lower than these numbers because we include Wistron's computing tray (L10) rack equivalent (without accounting for rack assembly and test times for L11).

Exhibit 1: Industry-wide NVL72-equivalent quarterly rack output, by major ODMs (2026)



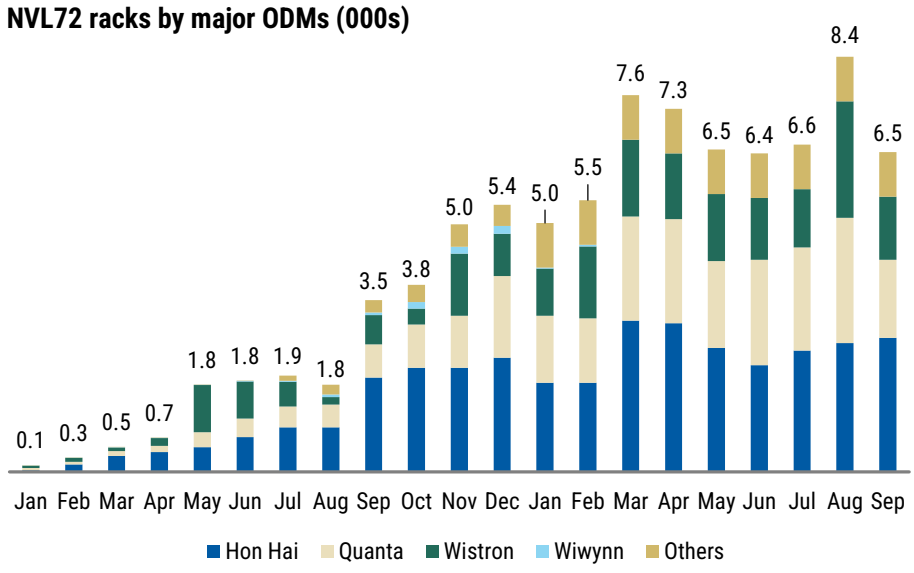
Source: Morgan Stanley Research estimates.

Exhibit 2: Industry-wide NVL72-equivalent quarterly rack output, by major ODMs (2027)



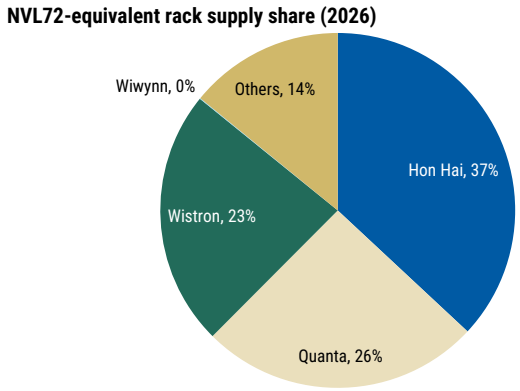
Source: Morgan Stanley Research estimates.

Exhibit 3:
Industry-wide NVL72-equivalent monthly rack output, by major ODMs (2025-2026)



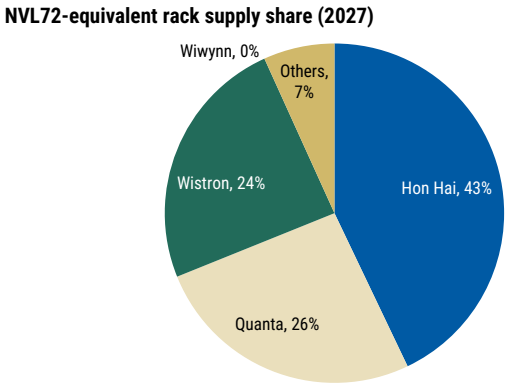
Source: Morgan Stanley Research estimates.

Exhibit 4: NVL72 rack supply share, by major ODMs (2026)



Source: Morgan Stanley Research estimates.

Exhibit 5: NVL72 rack supply share, by major ODMs (2027)



Source: Morgan Stanley Research estimates.

Valuation Methodology and Risks

Hon Hai Precision (2317.TW)

Base case scenario value, derived from a residual income valuation model. Our key assumptions include a cost of equity of 8.5%, a medium-term growth rate of 13%, and a terminal growth rate of 3%.

Risks to Upside

- Better-than-expected iPhone sell-through
- Faster progress in AI server business
- Faster EV business development progress
- Any new M&A activity that could improve sentiment

Risks to Downside

- Lower iPhone sell-through
- Smaller profit contribution from AI business
- Slower EV business development progress
- Geopolitical developments that could negatively affect foreign investment

Quanta Computer Inc. (2382.TW)

Base case, residual income model. Key assumptions include a cost of equity of 9.0% (beta of 1.2, equity premium of 6.0% and risk-free rate of 1.5%), an 8.5% medium-term growth rate, and a 3% terminal growth rate.

Risks to Upside

- Stronger-than-expected NB demand
- Stronger-than-expected Apple Watch demand
- Stronger-than-expected server demand
- Faster-than-expected AI server penetration

Risks to Downside

- Weaker-than-expected NB demand
- Softer-than-expected Apple Watch demand
- Weak margin performance owing to rising labor costs and sales shortfalls
- Fierce price competition in the mega data center segment
- Slower-than-expected AI server penetration

Wistron Corporation (3231.TW)

Base case, residual income valuation. Key assumptions: 8.7% cost of equity, 7.0% medium-term growth rate and 3% terminal growth rate.

Risks to Upside

- Faster-than-expected divestiture of consumer electronics business
- Stronger-than-expected NB demand
- Margin expansion from better product mix

- Faster-than-expected AI server penetration

Risks to Downside

- Slower-than-expected divestiture of consumer electronics business
- Weaker-than-expected NB demand
- Margin contraction from sales shortfall and fierce competition
- Slower-than-expected AI server penetration

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley Asia Limited (which accepts the responsibility for its contents) and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research), and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Howard Kao; Andy Meng, CFA.

.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictpolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

As of September 30, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., AirTAC International, Asustek Computer Inc., AU Optronics, Auras Technology Co Ltd, Bizlink, Catcher Technology, Chenbro, Co-Tech Development Corp., Compal Electronics, Delta Electronics Inc., E Ink Holdings Inc., Elite Material Co., Ennostar Inc, Eoptolink Technology Inc Ltd, Fositek Corp, Genius Electronic Optical Co. Ltd., Giga-Byte Technology Co. Ltd., Hiwin Technologies Corp., LandMark Optoelectronics Corporation, Largan Precision, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Lotes Co. Ltd., Nan Ya PCB, Quectel Wireless Solutions Co Ltd, Sunny Optical, Suzhou TFC Optical Communication Co Ltd., Taiwan Union Technology Corporation, Unimicron, Visual Photonics Epitaxy Co Ltd, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhejiang Crystal-Optech Co Ltd, Zhen Ding, Zhongji Innolight Co Ltd, ZTE Corporation.

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Asustek Computer Inc., Unimicron, Wistron Corporation, Wiwynn Corp, Zhen Ding, Zhongji Innolight Co Ltd.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from Asustek Computer Inc., Lenovo, Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Zhen Ding, Zhongji Innolight Co Ltd.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Advantech, AirTAC International, Asustek Computer Inc., AU Optronics, Bizlink, Catcher Technology, Chenbro, Compal Electronics, Delta Electronics Inc., E Ink Holdings Inc., Elite Material Co., Ennostar Inc, Eoptolink Technology Inc Ltd, FIT Hon Teng Ltd, Giga-Byte Technology Co. Ltd., Gold Circuit Electronics Ltd., Hon Hai Precision, Innolux, Lenovo, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Lotes Co. Ltd., Luxshare Precision Industry Co., Ltd., Pegatron Corporation, Q Technology (Group) Company Ltd, Quanta Computer Inc., Shanghai Conant Optical Co Ltd, Shenzhen Transsion Holdings Co Ltd, Suzhou TFC Optical Communication Co Ltd., TCL Corp., Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhen Ding, Zhongji Innolight Co Ltd.

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Asustek Computer Inc., AU Optronics, BYD Electronics, Compal Electronics, E Ink Holdings Inc., Eoptolink Technology Inc Ltd, Giga-Byte Technology Co. Ltd., GoerTek Inc, Hon Hai Precision, Innolux, Lenovo, Lingyi Itech Guangdong Co, Quanta Computer Inc., Xiaomi Corp, Yageo Corp..

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Advantech, AirTAC International, Asustek Computer Inc., AU Optronics, Bizlink, Catcher Technology, Chenbro, Compal Electronics, Delta Electronics Inc., E Ink Holdings Inc., Elite Material Co., Ennostar Inc, Eoptolink Technology Inc Ltd, FIT Hon Teng Ltd, Giga-Byte Technology Co. Ltd., Gold Circuit Electronics Ltd., Hon Hai Precision, Innolux, Lenovo, Lens Technology, Lingyi Itech Guangdong Co, Lite-On Technology, Lotes Co. Ltd., Luxshare Precision Industry Co., Ltd., Pegatron Corporation, Q Technology (Group) Company Ltd, Quanta Computer Inc., Shanghai Conant Optical Co Ltd, Shenzhen Transsion Holdings Co Ltd, Suzhou TFC Optical Communication Co Ltd., TCL Corp., Unimicron, Wistron Corporation, Wiwynn Corp, Xiaomi Corp, Yageo Corp., Yangtze Optical Fibre and Cable JSC Ltd, Zhen Ding, Zhongji Innolight Co Ltd.

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: AAC Technologies Holdings, Accton Technology Corporation, Acer Inc., Asustek Computer Inc., AU Optronics, BYD Electronics, Compal Electronics, E Ink Holdings Inc., Eoptolink Technology Inc Ltd, Giga-Byte Technology Co. Ltd., GoerTek Inc, Hon Hai Precision, Innolux, Lenovo, Lingyi Itech Guangdong Co, Quanta Computer Inc., Xiaomi Corp, Yageo Corp., Zhen Ding.

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality

of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.
Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.
Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of September 30, 2026)
The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.
For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

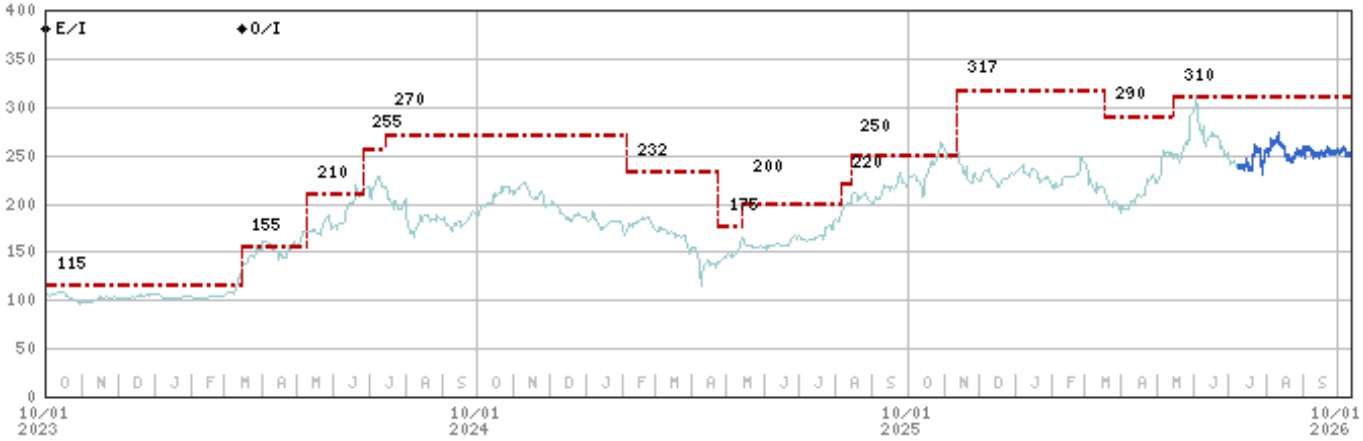
Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.
Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.
Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Hon Hai Precision (2317.TW) - As of 10/12/26 GMT in TWD
Industry : Greater China Technology Hardware



Stock Rating History: 10/1/21 : O/I; 1/9/23 : E/I; 3/15/24 : O/I

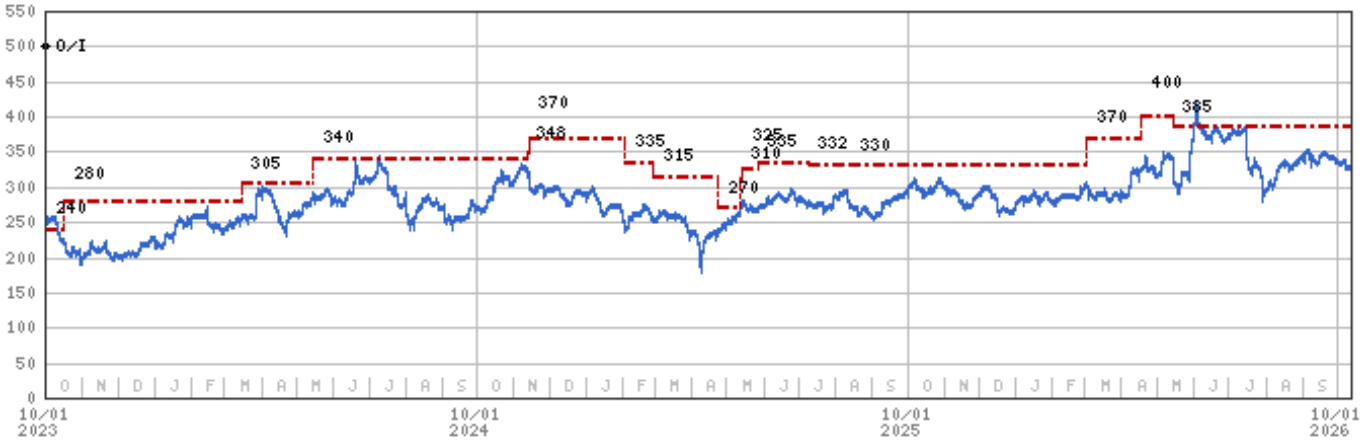
Price Target History: 3/22/21 : 168; 3/8/22 : 145; 1/9/23 : 116; 5/11/23 : 115; 3/15/24 : 155; 5/10/24 : 210; 6/26/24 : 255; 7/15/24 : 270; 2/5/25 : 232; 4/23/25 : 175; 5/14/25 : 200; 8/6/25 : 220; 8/14/25 : 250; 11/12/25 : 317; 3/17/26 : 290; 5/15/26 : 310

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Quanta Computer Inc. (2382.TW) - As of 10/12/26 GMT in TWD
Industry : Greater China Technology Hardware



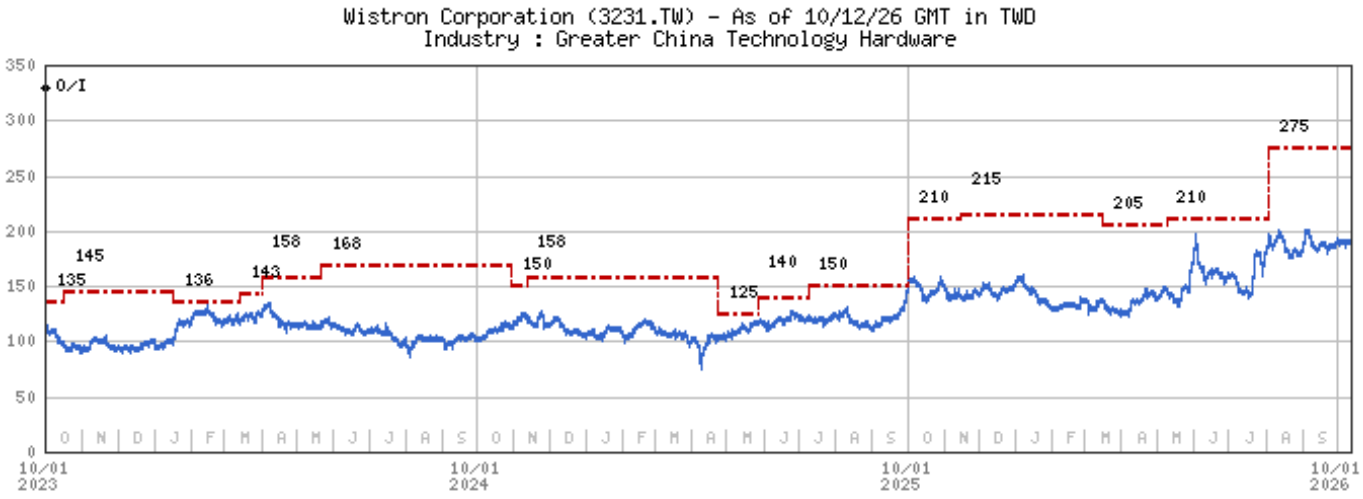
Stock Rating History: 10/1/21 : E/I; 5/1/23 : O/I

Price Target History: 7/28/21 : 85; 11/12/21 : 94; 3/7/22 : 99; 3/16/22 : 98; 5/13/22 : 85; 7/26/22 : 80; 8/13/22 : 82.5; 5/1/23 : 105; 5/12/23 : 110; 7/12/23 : 215; 8/12/23 : 240; 10/17/23 : 280; 3/15/24 : 305; 5/15/24 : 340; 11/12/24 : 348; 11/14/24 : 370; 2/4/25 : 335; 2/28/25 : 315; 4/23/25 : 270; 5/13/25 : 310; 5/14/25 : 325; 5/27/25 : 335; 7/9/25 : 332; 8/13/25 : 330; 3/2/26 : 370; 4/17/26 : 400; 5/14/26 : 385

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.



Stock Rating History: 10/1/21 : E/I; 7/12/23 : O/I

Price Target History: 7/28/21 : 30; 11/6/21 : 32; 5/9/22 : 31; 7/26/22 : 29; 8/8/22 : 30; 11/7/22 : 29; 1/17/23 : 31; 3/15/23 : 34; 5/1/23 : 48.5; 5/12/23 : 47; 7/12/23 : 135; 10/17/23 : 145; 1/18/24 : 136; 3/14/24 : 143; 4/1/24 : 158; 5/22/24 : 168; 10/30/24 : 150; 11/12/24 : 158; 4/23/25 : 125; 5/27/25 : 140; 7/9/25 : 150; 10/2/25 : 210; 11/16/25 : 215; 3/16/26 : 205; 5/9/26 : 210; 8/4/26 : 275

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (http://www.morganstanley.com/terms.html). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (http://www.morganstanley.com/terms.html). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research.

Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V which is regulated by Comision Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118J); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. Morgan Stanley Research is disseminated in Canada by Morgan Stanley Canada Limited and Morgan Stanley Wealth Management Canada Inc., members of the Canadian

Investment Regulatory Organization (CIRO). The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

Morgan Stanley Hong Kong Securities Limited is the liquidity provider/market maker for securities of AAC Technologies Holdings, BYD Electronics, FIT Hon Teng Ltd, Hon Hai Precision, Kingboard Laminates Holdings Ltd., Lenovo, Lens Technology, Luxshare Precision Industry Co., Ltd., Sunny Optical, Xiaomi Corp, Yangtze Optical Fibre and Cable JSC Ltd, Zhongji Innolight Co Ltd, ZTE Corporation listed on the Stock Exchange of Hong Kong Limited. An updated list can be found on HKEx website: http://www.hkex.com.hk.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

References to any prediction market or event contract do not constitute an offer, solicitation, recommendation, or endorsement to participate in any such market or contract. Morgan Stanley Research may not be distributed or made available in any jurisdiction where such distribution would be unlawful or otherwise prohibited by applicable law.

INDUSTRY COVERAGE: Greater China Technology Hardware

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/12/2026)
Andy Meng, CFA		
AAC Technologies Holdings (2018.HK)	O (01/29/2024)	HK$36.06
Accelink Technologies Co. Ltd. (002281.SZ)	U (05/12/2022)	Rmb140.17
BYD Electronics (0285.HK)	O (04/28/2023)	HK$22.80
China TransInfo Technology Co Ltd (002373.SZ)	E (07/18/2023)	Rmb7.44
Dahua Technology Co. Ltd. (002236.SZ)	E (12/12/2024)	Rmb15.51
Eoptolink Technology Inc Ltd (300502.SZ)	O (02/27/2026)	Rmb347.60
Genius Electronic Optical Co. Ltd. (3406.TW)	U (09/10/2026)	NT$1,070.00
GoerTek Inc (002241.SZ)	E (08/14/2026)	Rmb20.85
Gosuncn Technology Group Co Ltd (300098.SZ)	U (11/07/2022)	Rmb4.17
HIKVision Digital Technology (002415.SZ)	E (12/12/2024)	Rmb33.24
Largan Precision (3008.TW)	E (09/10/2026)	NT$7,325.00
LianChuang Electronic Technology Co Ltd (002036.SZ)	U (06/12/2024)	Rmb5.73
Lingyi Itech Guangdong Co (002600.SZ)	U (04/23/2025)	Rmb10.88
Lingyi Itech Guangdong Co (1688.HK)	O (08/14/2026)	HK$4.97
Luxshare Precision Industry Co., Ltd. (2475.HK)	O (08/14/2026)	HK$49.50
Luxshare Precision Industry Co., Ltd. (002475.SZ)	O (10/24/2016)	Rmb46.32
OFILM Group Co Ltd (002456.SZ)	U (06/12/2024)	Rmb6.36
Q Technology (Group) Company Ltd (1478.HK)	U (03/23/2026)	HK$4.79
Quectel Wireless Solutions Co Ltd (603236.SS)	E (10/09/2024)	Rmb69.76
Shanghai Conant Optical Co Ltd (2276.HK)	O (04/10/2026)	HK$34.68
Shenzhen Transsion Holdings Co Ltd (688036.SS)	E (03/23/2026)	Rmb49.36
Sunny Optical (2382.HK)	E (03/23/2026)	HK$58.80
Suzhou TFC Optical Communication Co Ltd. (300394.SZ)	E (11/06/2025)	Rmb229.54

Wingtech Technology Co Ltd (600745.SS)	U (05/18/2026)	Rmb14.89
Xiaomi Corp (1810.HK)	O (04/14/2021)	HK$24.52
Yangtze Optical Fibre and Cable JSC Ltd (601869.SS)	U (10/13/2021)	Rmb362.18
Yangtze Optical Fibre and Cable JSC Ltd (6869.HK)	O (07/15/2026)	HK$156.10
Yongxin Optics Co Ltd (603297.SS)	E (11/15/2022)	Rmb99.88
Zhejiang Crystal-Optech Co Ltd (002273.SZ)	O (11/15/2022)	Rmb21.36
Zhongji Innolight Co Ltd (300308.SZ)	O (09/07/2026)	Rmb724.47
Zhongji Innolight Co Ltd (3308.HK)	O (09/07/2026)	HK$903.00
ZTE Corporation (0763.HK)	O (06/04/2026)	HK$20.36
ZTE Corporation (000063.SZ)	E (06/04/2026)	Rmb28.44
Derrick Yang		
Accton Technology Corporation (2345.TW)	O (06/06/2024)	NT$2,130.00
Advantech (2395.TW)	O (01/20/2021)	NT$721.00
AirTAC International (1590.TW)	O (04/16/2025)	NT$1,210.00
Asia Vital Components Co. Ltd. (3017.TW)	O (07/30/2024)	NT$3,385.00
AU Optronics (2409.TW)	E (02/10/2026)	NT$35.55
Auras Technology Co Ltd (3324.TWO)	E (05/04/2023)	NT$1,540.00
Bizlink (3665.TW)	O (03/10/2025)	NT$2,355.00
BOE Technology (000725.SZ)	O (09/06/2019)	Rmb5.33
Chenbro (8210.TW)	O (07/23/2025)	NT$861.00
Chroma Ate Inc. (2360.TW)	O (10/05/2021)	NT$2,365.00
Delta Electronics Inc. (2308.TW)	O (07/13/2017)	NT$1,900.00
E Ink Holdings Inc. (8069.TWO)	E (08/10/2026)	NT$149.00
Ennostar Inc (3714.TW)	U (09/23/2022)	NT$63.80
Fositek Corp (6805.TW)	O (06/25/2025)	NT$2,270.00
Hiwin Technologies Corp. (2049.TW)	O (03/30/2026)	NT$338.00
Innolux (3481.TW)	E (04/07/2025)	NT$51.30
King Slide Works Co. Ltd. (2059.TW)	O (11/08/2023)	NT$10,580.00
Lens Technology (300433.SZ)	E (07/22/2020)	Rmb26.73
Lite-On Technology (2301.TW)	E (01/15/2025)	NT$288.00
TCL Corp. (000100.SZ)	E (04/07/2025)	Rmb4.52
Tianma Microelectronics (000050.SZ)	U (01/24/2018)	Rmb6.08
Wuhan Jingce Electronic Group Co Ltd (300567.SZ)	E (11/26/2021)	Rmb167.00
Howard Kao		
Acer Inc. (2353.TW)	U (04/23/2025)	NT$31.35
Asustek Computer Inc. (2357.TW)	U (11/16/2025)	NT$977.00
Catcher Technology (2474.TW)	U (11/17/2025)	NT$194.50
Compal Electronics (2324.TW)	U (04/23/2025)	NT$36.60
Elite Material Co. (2383.TW)	O (09/14/2026)	NT$6,150.00
FIT Hon Teng Ltd (6088.HK)	O (11/03/2025)	HK$4.43
Foxconn Industrial Internet Co. Ltd. (601138.SS)	O (07/10/2019)	Rmb52.05
Giga-Byte Technology Co. Ltd. (2376.TW)	E (11/16/2025)	NT$356.00
Gold Circuit Electronics Ltd. (2368.TW)	O (10/06/2022)	NT$1,235.00
Hon Hai Precision (2317.TW)	O (03/15/2024)	NT$244.50
Inspur Electronic Information (000977.SZ)	E (08/28/2023)	Rmb62.05
Kingboard Laminates Holdings Ltd. (1888.HK)	O (09/14/2026)	HK$46.80
LandMark Optoelectronics Corporation (3081.TWO)	E (03/26/2026)	NT$3,025.00
Lenovo (0992.HK)	O (07/10/2026)	HK$32.34
Lotes Co. Ltd. (3533.TW)	E (05/12/2025)	NT$1,880.00
Nan Ya PCB (8046.TW)	O (02/23/2026)	NT$1,440.00
Pegatron Corporation (4938.TW)	E (03/25/2026)	NT$98.60
Quanta Computer Inc. (2382.TW)	O (05/01/2023)	NT$322.00
Shengyi Technology Co Ltd. (600183.SS)	E (05/26/2022)	Rmb120.00
Shennan Circuits Co Ltd (002916.SZ)	E (08/24/2023)	Rmb327.78
Unimicron (3037.TW)	O (02/23/2026)	NT$1,255.00

Visual Photonics Epitaxy Co Ltd (2455.TW)	E (09/11/2023)	NT$564.00
Wistron Corporation (3231.TW)	O (07/12/2023)	NT$189.50
Wiwynn Corp (6669.TW)	O (11/10/2025)	NT$2,275.00
Yageo Corp. (2327.TW)	O (10/28/2025)	NT$636.00
Zhen Ding (4958.TW)	++	NT$581.00
Irene Yen		
Co-Tech Development Corp. (8358.TWO)	O (09/14/2026)	NT$522.00
Taiwan Union Technology Corporation (6274.TWO)	O (09/14/2026)	NT$1,735.00

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.

© 2026 Morgan Stanley

October 12, 2026 03:39 PM GMT

Greater China Semiconductors – DRAM | Asia Pacific

Nanya 3Q beat; CXMT updates

DRAM supply constraints continue. Contract renewals drove Nanya's 3Q beat. CXMT's G5 milestone strengthens its tech roadmap, but yield ramp and customer qualification take time.

Nanya 3Q beat; LTAs in spotlight: Nanya reported 3Q revenue of NT$134bn, +62% Q/Q, +612% Y/Y, 8% above consensus. 3Q gross profit was NT$114bn, +114% Q/Q, 12% above consensus, implying a GM of 85.7%. 3Q net profit was NT$88bn, +75% Q/Q, 9% above consensus, implying a net margin of 65.1%. The 3Q Q/Q revenue growth was driven mainly by ASP increase, which primarily reflects renewal of the one-year and six-month LTAs, which were previously negotiated when ASPs were lower; most expired in 3Q26. Currently LTAs cover 50-70% of output; only 20% of output is covered by three-year contracts; Nanya intends to extend the LTA durations for longer-term supply according to customer demand, and there are discussions on price floors and ceilings for the new contracts.

Nanya management sees continued DRAM shortage in 4Q and expects that ASP may continue to improve: The supply constraint is the long lead time for new capacity, and the undersupply is unlikely to be resolved in the near term. Nanya targets capex of NT$69.7bn in 2026 and NT$346bn for the 35.9kwpm capacity of its new Fab 5A from 2026-29. Management estimates that AI infra currently accounts for ~20% of its revenue mix, but that could increase to 40-50% by 2029/30.

CXMT updates: On September 7 during Q&A, CXMT reiterated expectations for tight DRAM supply in 2H26; its response on Apple expressed openness to global customers without confirming a supply agreement. On September 21, after its G5 mass production announcement, CXMT clarified that G5 sales are not yet at scale, while yield ramp-up and customer qualification would take time. Based on our checks, CXMT's HBM3E yield is already 40%, and DRAM wafer input should ramp to 30kwpm by end-2027.

Constrained DRAM supply is the industry consensus: The 4QF26 earnings call for Micron (covered by Joseph Moore) highlighted that persistent supply constraints are expected through CY2027-28. It cited industry-wide clean room constraints owing to lengthy construction; most of the increase in Micron's F27 capex (vs. its previous plan) is allocated to construction, targeting clean room availability in late CY28 and beyond. Micron also expects rising industry HBM wafer requirements to limit bit growth, forecasting industry DRAM bit shipment growth of low-20s (%) in CY27-28. Samsung (covered by Shawn Kim) also expects HBM to absorb nearly 30% of industry DRAM wafer capacity in 2027, vs ~20% currently.

Stock implications:

- **Nanya Tech (OW):** Price target raised to NT$600; 2026e, 2027e, and 2028e EPS raised 22%, 18%, and 19%, respectively.
- **CXMT (OW):** Base case value/price target unchanged at Rmb88.

MORGAN STANLEY TAIWAN LIMITED+
Charlie Chan
Equity Analyst
Charlie.Chan@morganstanley.com +886 2 2730-1725

MORGAN STANLEY ASIA LIMITED+
Ethan Jia
Research Associate
Ethan.Jia@morganstanley.com +852 3963-2287

MORGAN STANLEY TAIWAN LIMITED+
Daniel Yen, CFA
Equity Analyst
Daniel.Yen@morganstanley.com +886 2 2730-2863

MORGAN STANLEY ASIA LIMITED+
Daisy Dai, CFA
Equity Analyst
Daisy.Dai@morganstanley.com +852 2848-7310

Henry Zhao
Equity Analyst
Henry.Zhao@morganstanley.com +852 2239-7731

MORGAN STANLEY TAIWAN LIMITED+
Lucas Wang
Equity Analyst
Lucas.Wang@morganstanley.com +886 2 2730-2875

GREATER CHINA TECHNOLOGY SEMICONDUCTORS

Asia Pacific Industry View	**Attractive**

WHAT'S CHANGED

Nanya Technology Corp. (2408.TW)	From	To
Price Target	NT$580.00	**NT$600.00**

Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. Investors should consider Morgan Stanley Research as only a single factor in making their investment decision.

For analyst certification and other important disclosures, refer to the Disclosure Section, located at the end of this report.

+= Analysts employed by non-U.S. affiliates are not registered with FINRA, may not be associated persons of the member and may not be subject to FINRA restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

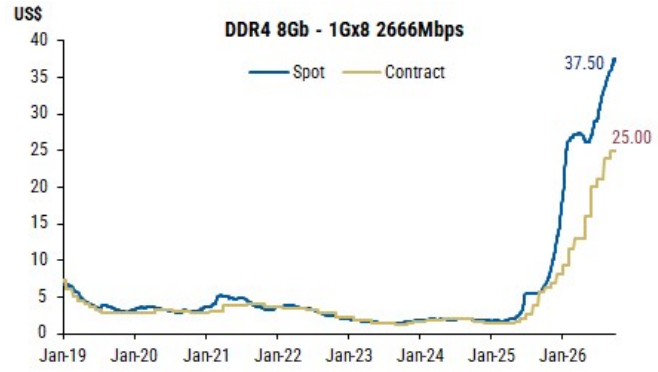
AlphaSignals Earnings Reaction

Impact to our thesis	Financial results versus consensus	Direction of next 12-month consensus EPS
Nanya Technology Corp. (2408.TW)		
Strengthens our thesis	↑ Modest upside	↑ Modest revision higher

Source: Company data, Morgan Stanley Research

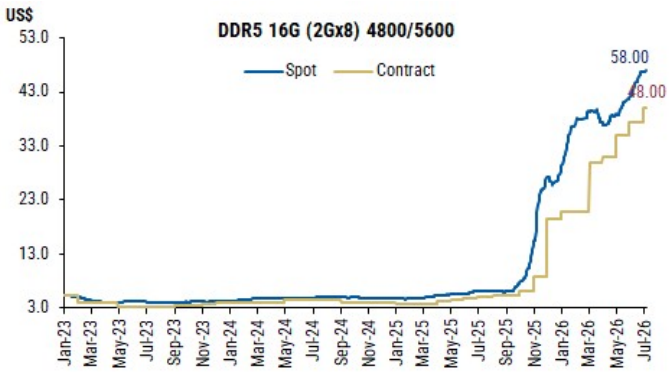
Key Charts and Exhibits

Exhibit 1: DDR4 8Gb (1Gx8) pricing chart



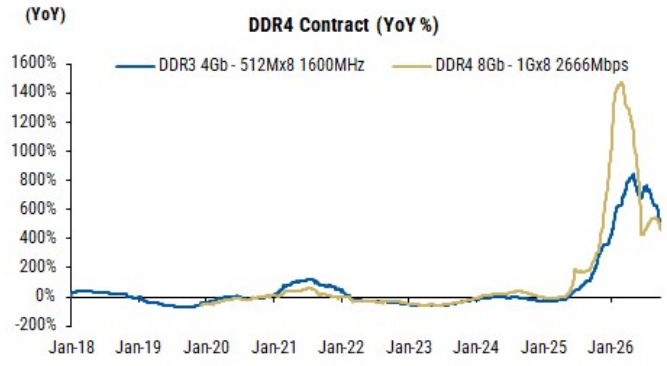
Source: DRAMeXchange, Morgan Stanley Research. Data as of Sep 2026.

Exhibit 2: DDR5 16Gb (2Gx8) pricing chart



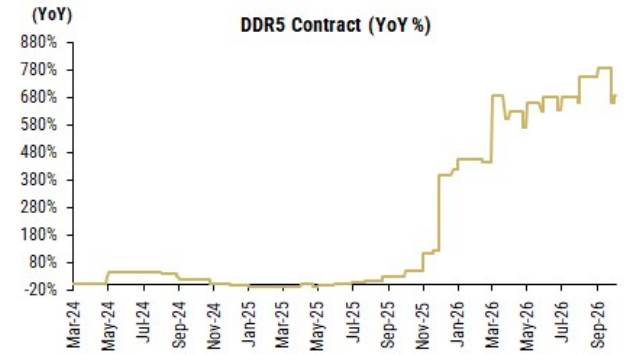
Source: DRAMeXchange, Morgan Stanley Research. Data as of Sep 2026.

Exhibit 3: DDR4 Contract (YoY%)



Source: DRAMeXchange, Morgan Stanley Research. Data as of Sep 2026

Exhibit 4: DDR5 Contract (YoY%)



Source: DRAMeXchange, Morgan Stanley Research. Data as of Sep 2026

Exhibit 5:

DRMA/NAND ASP Projection

	4Q26E - New	1Q27F	2Q27F	3Q27F	4Q27
PC DRAM	DDR4: up 13-18% DDR5: up 13-18% Blended ASP: up 13-18%	up 3-8%	up 3-8%	up 0-5%	up 0-5%
Server DRAM	DDR4: up 10-15% DDR5: up 10-15% Blended ASP: up 10-15%	up 3-8%	up 3-8%	up 0-5%	up 0-5%
Mobile DRAM	LPDDR4X: up 0-5% LPDDR5(x): up 0-5%	LPDDR4X: up 5-10% LPDDR5(x): up 13-18%	LPDDR4X: up 0-5% LPDDR5(x): up 5-10%	LPDDR4X: up 0-5% LPDDR5(x): up 3-8%	LPDDR4X: up 0-5% LPDDR5(x): up 0-5%
Graphics DRAM	GDDR6: up 5-10% GDDR7: up 8-13%	up 3-8%	up 3-8%	up 0-5%	up 0-5%
Consumer DRAM	DDR3: up 3-8% DDR4: up 3-8%	up 3-8%	up 3-8%	up 0-5%	up 0-5%
Total DRAM	Conventional DRAM: up 10-15% HBM Blended: up 15-20%	Conventional DRAM: up 5-10%	Conventional DRAM: up 3-8%	Conventional DRAM: up 0-5%	Conventional DRAM: up 0-5%

	4Q26E - New	1Q27F	2Q27F	3Q27F	4Q27
eMMC	up 5-10%	up 0-5%	mostly flat	mostly flat	down 3-8%
UFS	up 23-28%	up 13-18%	up 8-13%	up 3-8%	mostly flat
Enterprise SSD	up 5-10%	up 0-5%	mostly flat	mostly flat	down 3-8%
Client SSD	up 3-8%	mostly flat	mostly flat	down 5-10%	down 10-15%
3D NAND Wafers (TLC & QLC)					
Total NAND Flash	up 15-20%	up 13-18%	up 5-10%	up 0-5%	down 0-5%

Source: DRAMeXchange, Morgan Stanley Research. Data as of Sep 2026

Nanya Tech: Estimate Revisions and Quarterly Financials

We raise our EPS estimates 22% for 2026, 18% for 2027, and 19% for 2028: We incorporate 3Q26 actual results, which were driven by ASP increases. We also expect DDR4 pricing to remain elevated into 1H27, helped by the continued industry-wide DRAM supply constraint.

Exhibit 6: Nanya Tech: Estimate revisions

NT$ mn	New '26E	Old '26E	Diff.	New '27E	Old '27E	Diff.	New '28E	Old '28E	Diff.
Net sales	405,725	348,048	17%	728,810	638,966	14%	585,384	518,763	13%
Gross profit	333,069	275,802	21%	609,702	520,060	17%	426,727	360,346	18%
Operating profit	310,209	255,102	22%	559,124	473,137	18%	374,295	310,712	20%
Pretax Income	316,087	260,265	21%	560,644	477,413	17%	375,827	315,001	19%
Net income	257,292	209,579	23%	455,960	384,461	19%	305,892	254,439	20%
EPS (NT$)	**75.96**	**62.13**	**22%**	**132.17**	**112.32**	**18%**	**88.67**	**74.33**	**19%**
Margins									
Gross margin	82.1%	79.2%		83.7%	81.4%		72.9%	69.5%	
Operating margin	76.5%	73.3%		76.7%	74.0%		63.9%	59.9%	
Pretax margin	77.9%	74.8%		76.9%	74.7%		64.2%	60.7%	
Net margin	63.4%	60.2%		62.6%	60.2%		52.3%	49.0%	

NT$	New '26E	Old '26E	Diff.	New '27E	Old '27E	Diff.	New '28E	Old '28E	Diff.
BVPS (NT$)	146.00	121.52	20%	278.16	245.60	13%	366.81	327.71	12%

Source: Morgan Stanley Research (E) estimates

Exhibit 7: Nanya Tech: Quarterly financials

NT$ in million	1Q26	2Q26	3Q26P	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025	2026e	2027e	2028e
Total Revenues	**49,087**	**82,549**	**133,649**	**140,440**	**171,301**	**191,350**	**190,679**	**175,480**	**34,132**	**66,587**	**405,725**	**728,810**	**585,384**
Sequential Change	*63.1%*	*68.2%*	*61.9%*	*5.1%*	*22.0%*	*11.7%*	*-0.4%*	*-8.0%*					
Change vs Year Ago	*582.9%*	*684.2%*	*611.7%*	*366.7%*	*249.0%*	*131.8%*	*42.7%*	*24.9%*	*14.2%*	*95.1%*	*509.3%*	*79.6%*	*-19.7%*
Cost of Sales	(15,771)	(16,930)	(19,167)	(20,788)	(22,805)	(25,384)	(28,086)	(42,833)	(32,292)	(50,841)	(72,656)	(119,107)	(158,657)
Percent of Revenues	*32.1%*	*20.5%*	*14.3%*	*14.8%*	*13.3%*	*13.3%*	*14.7%*	*24.4%*	*94.6%*	*76.4%*	*17.9%*	*16.3%*	*27.1%*
Variable costs	(12,874)	(13,970)	(16,102)	(17,449)	(19,924)	(22,502)	(25,204)	(28,007)	(18,408)	(37,321)	(60,394)	(95,637)	(133,967)
as a % of revenue	*26.2%*	*16.9%*	*12.0%*	*12.4%*	*11.6%*	*11.8%*	*13.2%*	*16.0%*	*53.9%*	*56.0%*	*14.9%*	*13.1%*	*22.9%*
Dep. & Amort. expense	(2,897)	(2,960)	(3,065)	(3,340)	(2,882)	(2,882)	(2,882)	(14,825)	(13,883)	(13,520)	(12,262)	(23,470)	(24,690)
as a % of COGS	*18.4%*	*17.5%*	*16.0%*	*16.1%*	*12.6%*	*11.4%*	*10.3%*	*34.6%*	*43.0%*	*26.6%*	*16.9%*	*19.7%*	*15.6%*
as a % of revenue	*5.9%*	*3.6%*	*2.3%*	*2.4%*	*1.7%*	*1.5%*	*1.5%*	*8.4%*	*40.7%*	*20.3%*	*3.0%*	*3.2%*	*4.2%*
Gross Margin	**33,316**	**65,619**	**114,482**	**119,652**	**148,496**	**165,966**	**162,593**	**132,647**	**1,840**	**15,745**	**333,069**	**609,702**	**426,727**
Percent of Revenues	*67.9%*	*79.5%*	*85.7%*	*85.2%*	*86.7%*	*86.7%*	*85.3%*	*75.6%*	*5.4%*	*23.6%*	*82.1%*	*83.7%*	*72.9%*
Total Opex	(3,205)	(4,793)	(7,588)	(7,275)	(11,352)	(12,604)	(13,263)	(13,360)	(10,134)	(9,741)	(22,860)	(50,579)	(52,432)
Percent of Revenues	*6.5%*	*5.8%*	*5.7%*	*5.2%*	*6.6%*	*6.6%*	*7.0%*	*7.6%*	*29.7%*	*14.6%*	*5.6%*	*6.9%*	*9.0%*
R&D	(2,109)	(3,230)	(5,145)	(4,334)	(4,500)	(4,950)	(5,445)	(5,990)	(7,685)	(7,041)	(14,818)	(20,885)	(27,846)
Percent of Revenues	*4.3%*	*3.9%*	*3.8%*	*3.1%*	*2.6%*	*2.6%*	*2.9%*	*3.4%*	*22.5%*	*10.6%*	*3.7%*	*2.9%*	*4.8%*
Sales and Marketing	(347)	(503)	(594)	(856)	(1,713)	(1,913)	(1,907)	(1,755)	(665)	(832)	(2,301)	(7,288)	(7,025)
Percent of Revenues	*0.7%*	*0.6%*	*0.4%*	*0.6%*	*1.0%*	*1.0%*	*1.0%*	*1.0%*	*1.9%*	*1.2%*	*0.6%*	*1.0%*	*1.2%*
General and Admin	(748)	(1,060)	(1,849)	(2,084)	(5,139)	(5,740)	(5,911)	(5,615)	(1,784)	(1,868)	(5,742)	(22,406)	(17,562)
Percent of Revenues	*1.5%*	*1.3%*	*1.4%*	*1.5%*	*3.0%*	*3.0%*	*3.1%*	*3.2%*	*5.2%*	*2.8%*	*1.4%*	*3.1%*	*3.0%*
Operating Income	**30,111**	**60,826**	**106,894**	**112,377**	**137,144**	**153,362**	**149,330**	**119,287**	**(8,294)**	**6,004**	**310,209**	**559,124**	**374,295**
Percent of Revenues	*61.3%*	*73.7%*	*80.0%*	*80.0%*	*80.1%*	*80.1%*	*78.3%*	*68.0%*	*-24.3%*	*9.0%*	*76.5%*	*76.7%*	*63.9%*
EBITDA	33,008	63,786	109,959	115,717	140,026	156,243	152,212	134,113	5,589	19,524	322,470	582,594	398,985
EBITDA margin	*67%*	*77%*	*82%*	*82%*	*82%*	*82%*	*80%*	*76%*	*16%*	*29%*	*79%*	*80%*	*68%*
Non-operating Income(Loss)	**1,607**	**1,503**	**2,438**	**330**	**381**	**381**	**381**	**378**	**3,998**	**2,639**	**5,878**	**1,520**	**1,532**
Profit Before Taxes	**31,718**	**62,329**	**109,332**	**112,707**	**137,525**	**153,743**	**149,711**	**119,665**	**(4,297)**	**8,643**	**316,087**	**560,644**	**375,827**
Percent of Revenues	*64.6%*	*75.5%*	*81.8%*	*80.3%*	*80.3%*	*80.3%*	*78.5%*	*68.2%*	*-12.6%*	*13.0%*	*77.9%*	*76.9%*	*64.2%*
Taxes	(5,660)	(12,138)	(21,705)	(19,292)	(24,540)	(29,939)	(29,721)	(20,483)	1,474	(1,269)	(58,795)	(104,684)	(69,934)
Tax Rate	*17.8%*	*19.5%*	*19.9%*	*17.1%*	*17.8%*	*19.5%*	*19.9%*	*17.1%*	*34.3%*	*14.7%*	*18.6%*	*18.7%*	*18.6%*
Reported Income (TW GAAP)	**26,058**	**50,192**	**87,627**	**93,415**	**112,985**	**123,804**	**119,990**	**99,182**	**(2,823)**	**7,373**	**257,292**	**455,960**	**305,892**
Percent of Revenues	*53.1%*	*60.8%*	*65.6%*	*66.5%*	*66.0%*	*64.7%*	*62.9%*	*56.5%*	*-8.3%*	*11.1%*	*63.4%*	*62.6%*	*52.3%*
Change vs Year Ago	*-1443%*	*-1324%*	*5505%*	*742%*	*334%*	*147%*	*37%*	*6%*	*-59%*	*-361%*	*3389%*	*77%*	*-33%*
Reported EPS (NT$, TW GAAP)	**8.08**	**14.66**	**25.40**	**27.08**	**32.75**	**35.89**	**34.78**	**28.75**	**(0.91)**	**2.36**	**75.96**	**132.17**	**88.67**
Sequential Change	*129%*	*81%*	*73%*	*7%*	*21%*	*10%*	*-3%*	*-17%*	*-59%*	*-359%*	*3117%*	*74%*	*-33%*
Book value per share									**53.3**	**54.6**	**148.7**	**278.2**	**366.8**

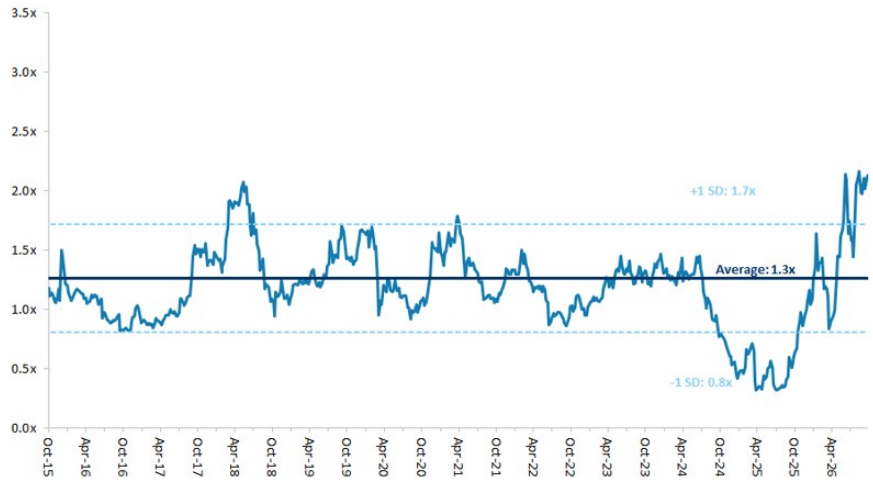
Source: Company data, Morgan Stanley Research (E) estimates

Nanya Tech: Valuation Methodology

We raise our price target (base case scenario value) from NT$580 to NT$600: Our price target reflects a BVPS multiple of 4.11x for 2026 (was 4.77x) versus ROE of 76% for 2026. We lower our target multiple to reflect recent memory segment volatility. Our 2026e BVPS estimates is raised to NT$146.00 from NT$121.52. We expect DDR4 pricing strength and LTAs to drive a re-rating.

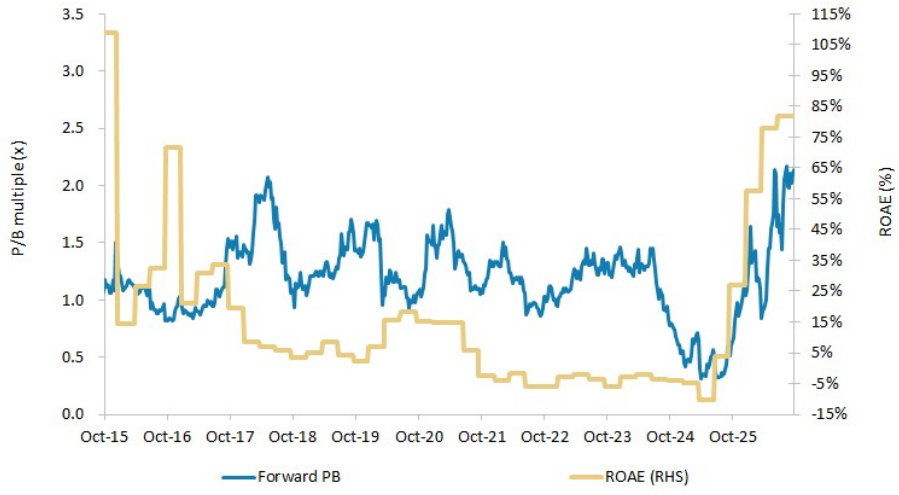
We raise our bull and bear case values from NT$1,230 and NT$320 to NT$1,270 and NT $330, respectively, implying 8.70x and 2.26x 2026e BVPS.

Exhibit 8: Nanya Tech – Historical P/B band



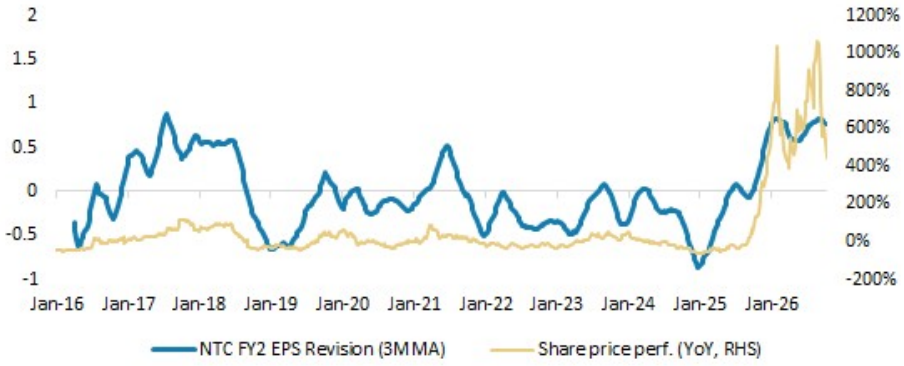
Source: Company data, Morgan Stanley Research estimates

Exhibit 9: Nanya Tech: Historical forward P/B vs. ROE



Source: Company data, Morgan Stanley Research estimates

Exhibit 10: Nanya Tech: Earnings estimate revision breadth



Source: Company data, Morgan Stanley Research estimates

Risk Reward – Nanya Technology Corp. (2408.TW)

DDR4 supply-demand becoming more favorable; OW

PRICE TARGET NT$600.00

Base case, P/B. We expect the stock to trade at 4.11x 2026e BVPS, above its average since 2015, justified by 2026e ROE of 76%. We think NTC will benefit from a supply-driven up-cycle as major memory players are exiting the DDR4 market. That should help NTC more than offset the competitive impact from CXMT in the near term.

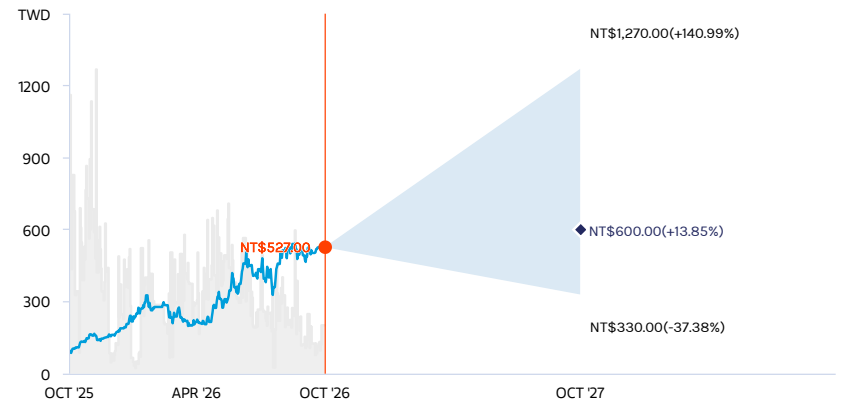
Consensus Price Target Distribution NT$150.00 — NT$612.67 — NT$800.00

MS PT

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- Major memory suppliers are gradually exiting the DDR4 market, benefiting existing Taiwanese players, including Nanya Tech.
- We expect DDR4 shortage into 4Q26 to help NTC more than offset the competitive impact from CXMT in the near term. The supply shortage appears set to widen into 1H27.
- Our price target reflects BVPS of 4.11x for 2026e and 2.16x for 2027e, and 1.64x 2028e BVPS. We expect re-rating.

Consensus Rating Distribution

93%	Overweight
0%	Equal-weight
7%	Underweight

● MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power: *Positive*

View descriptions of Risk Rewards Themes here

BULL CASE NT$1,270.00

8.70x 2026e BVPS

DRAM prices strong; faster-than-expected 1a/1b nm ramp; competition from Korea and China is a non-factor: DRAM pricing remains elevated in into 1H27 thanks to production cuts and relief of inventory pressure. NTC's utilization rate returns to the 100% level. Korean and Chinese mainstream vendors exit the consumer DRAM market.

BASE CASE NT$600.00

4.11x 2026e BVPS

Strong DRAM cycle: Major DRAM players now plan to exit the DDR4 market, which positions Taiwanese players to benefit. We expect price -hike momentum to continue through 4Q26, with widened supply shortage into 1H27.

BEAR CASE NT$330.00

2.26x 2026e BVPS

Specialty DRAM demand becomes weak in 1H27 with intensifying Chinese competition: Nanya Tech could face more pressure from pricing and technological competition amid a weak cycle. Inventory remains at historical peak levels. SK hynix and Chinese DRAM competitors take all the market share in China. Nanya Tech's utilization rate falls to the 40% level.

Risk Reward – Nanya Technology Corp. (2408.TW)

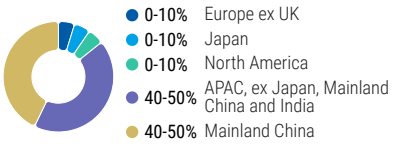
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
12" Wafers Out per quarter (000s)	825.0	820.0	932.5	1,112.5
Average Wafer Yield (%)	81.8	102.8	106.2	95.0
Total Die shipments - 2 gb equivalent (mn)	2,368.0	2,975.7	3,604.0	3,985.1

INVESTMENT DRIVERS

- Specialty DRAM pricing
- Monthly sales momentum
- Industry outlook reported by industry peers

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

MS ALPHA MODELS

1/5 MOST — 3 Month Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Sustained DRAM prices from more disciplined supply and stronger demand
- Stronger pricing environment, enabling revenue guidance to be beaten
- Faster-than-expected 1a/1b nm ramp-up progress

RISKS TO DOWNSIDE

- Slower-than-expected 1a/1b nm ramp-up
- Less demand for specialty DRAM from 4K2K TVs and smart set-top boxes

OWNERSHIP POSITIONING

Inst. Owners, % Active 73.3%

Source: Refinitiv, Morgan Stanley Research

MS ESTIMATES VS. CONSENSUS

FY Dec 2026e

Sales / Revenue (NT$, mn)	195,412	405,725 / 366,494	422,497
EBITDA (NT$, mn)	131,299	322,470 / 292,542	338,976
Net income (NT$, mn)	97,986	257,292 / 228,511	267,005
EPS (NT$)	31.62	75.96 / 67.67	79.89

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Nanya Tech: Financial Summary

Income Statement

NT$mn (Years End Dec)	2025	2026E	2027E	2028E
Net sales	66,587	405,725	728,810	585,384
COGS	(50,841)	(72,656)	(119,107)	(158,657)
Depreciation Expense	(13,520)	(12,262)	(23,470)	(24,690)
Variable Cost	(37,321)	(60,394)	(95,637)	(133,967)
Gross profit	15,745	333,069	609,702	426,727
Operating expenses	(9,741)	(22,860)	(50,579)	(52,432)
Operating income	**6,004**	**310,209**	**559,124**	**374,295**
Non-operating income	2,639	5,878	1,520	1,532
Pre-tax income	8,643	316,087	560,644	375,827
Income tax	(1,269)	(58,795)	(104,684)	(69,934)
Reported net Income	**7,373**	**257,292**	**455,960**	**305,892**
Adj.wtd.avg.shrs(m)	3,123	3,387	3,450	3,450
Reported EPS (NT$)	**2.36**	**75.96**	**132.17**	**88.67**
EPS for consensus (NT$)	**2.36**	**75.96**	**132.17**	**88.67**

Balance Sheet

NT$mn (Years End Dec)	2025	2026E	2027E	2028E
Cash	58,074	468,374	880,681	1,189,638
Mkt Securities	0	0	0	0
AR/NR	16,558	71,688	89,574	64,618
Inventory	27,288	26,437	54,472	56,422
Other	6,621	20,033	24,134	18,411
Current Assets	108,541	586,532	1,048,861	1,329,089
Long-term investments	5,904	8,476	9,415	10,354
Fixed assets	85,031	128,544	135,074	140,384
Other assets	8,977	5,426	5,426	5,426
Total Assets	**208,453**	**728,978**	**1,198,776**	**1,485,253**
S/T borrowings	5,060	160,000	165,000	145,000
AP/NP	6,481	8,308	17,117	17,730
Other ST liabilities	7,480	33,045	33,074	33,076
LT debt	14,196	18,397	18,397	18,397
Other LT liabilities	4,698	5,488	5,488	5,488
Total Liabilities	**37,914**	**225,237**	**239,076**	**219,690**
Common stocks	30,986	34,502	34,502	34,502
Preferred stocks	0	0	0	0
Capital Reserve	33,081	108,459	108,459	108,459
Retained earning	106,333	360,642	816,601	1,122,464
Other shareholders' equity	0	0	0	0
Total Equity	170,538	503,741	959,700	1,265,562
Total Liab. & Shrhldr's Equity	**208,453**	**728,978**	**1,198,776**	**1,485,253**
BVPS	**54.61**	**148.72**	**278.18**	**366.84**

Cash Flow Statement

NT$mn (Years	2025	2026E	2027E	2028E
Cashflow fror	**18,568**	**223,324**	**437,308**	**358,986**
Net profits	7,373	257,292	455,960	305,892
Depreciation	14,034	12,262	23,470	24,690
Working Capital	5,697	(52,453)	(37,111)	23,618
Other adjustmer	(8,536)	6,223	(5,011)	4,785
Cashflow fror	**(14,110)**	**(52,201)**	**(30,000)**	**(30,000)**
Capex	(13,441)	(51,996)	(30,000)	(30,000)
Change of LT In	(612)	0	0	0
Change of ST Ir	0	0	0	0
Other adjustmer	(57)	(206)	0	0
Cashflow fror	**(6,317)**	**237,868**	**4,999**	**(20,030)**
Increase in L/T (	(6,750)	4,201	0	0
Increase in S/T (	(9,477)	154,941	5,000	(20,000)
Cash Dividend F	0	0	(1)	(30)
Dir& Emp Bonus	(7,000)	0	0	0
Proceed fmNew	0	78,718	0	0
Other adjustmer	16,909	9	0	0
Exchange rate (	(1,970)	1,309	0	(0)
Net change ir	**-3,829**	**410,300**	**412,307**	**308,956**

Financial Ratios

	2025	2026E	2027E	2028E
Growth(%)				
Turnover	95.1	509.3	79.6	-19.7
Margins (%)				
Gross Margin	23.6	82.1	83.7	72.9
Operating Marg	9.0	76.5	76.7	63.9
EBITA Margin	-11.3	73.4	73.5	59.7
Pretax Margin	13.0	77.9	76.9	64.2
Net Profit	11.1	63.4	62.6	52.3
Return (%)				
ROAE	4.4	76.3	62.3	27.5
ROAA	3.6	54.9	47.3	22.8
Gearing (%)				
Net Debt/Equity	(22.7)	(57.6)	(72.7)	(81.1)
Liabilities/Equity	22.2	44.7	24.9	17.4
Ratios (X)				
Current ratio	5.7	2.9	4.9	6.8
Quick ratio	3.9	2.7	4.5	6.4
Others				
AR/NR Turnove	50	57	40	40
Inventory Turno	337	172	24	20
AP Turnover (da	59	47	42	52
Cash Conversio	328	182	22	8

Source: Company data, TEJ, Morgan Stanley ResearchE= Morgan Stanley Research Estimates

CXMT: Estimate Revisions and Quarterly Financials

We raise our EPS estimates 28% for 2026 and 29% for 2027; we keep our 2028 estimate virtually unchanged: This incorporates 2Q26 actual results and the stronger outlook for mainstream DRAM driven by continued supply constraint. We also factor in CXMT's G5 breakthrough, which reduces risks around its technology roadmap.

Exhibit 11: CXMT: Estimate revisions

	Current	Previous		Current	Previous		Current	Previous	
Rmb mn	**2026E**	**2026E**	**Diff.**	**2027E**	**2027E**	**Diff.**	**2028E**	**2028E**	**Diff.**
Net sales	438,929	388,869	13%	718,172	643,077	12%	870,554	850,096	2%
COGS	56,776	92,599		84,300	144,853		204,734	185,478	
Gross profit	382,153	296,271	29%	633,871	498,224	27%	665,820	664,618	0%
Operating expenses	24,152	22,789		42,372	37,942		55,803	54,491	
Operating profit	358,001	273,482	31%	591,499	460,282	29%	610,017	610,127	0%
Non-op. income (exp.)	(3,577)	(1,078)		(941)	(1,600)		(941)	(1,600)	
Pretax Income	354,424	272,404	30%	590,558	458,682	29%	609,077	608,527	0%
Taxes	(26,680)	(19,485)		(42,520)	(33,025)		(43,854)	(43,814)	
Net income	242,751	189,693	28%	411,029	319,243	29%	423,917	423,535	0%
Reported EPS (Rmb)	**3.84**	**3.00**	**28%**	**6.15**	**4.77**	**29%**	**6.34**	**6.33**	**0%**
Margins									
Gross margin	87.1%	76.2%	10.9 ppt	88.3%	77.5%	10.8 ppt	76.5%	78.2%	-1.7 ppt
Operating margin	81.6%	70.3%	11.2 ppt	82.4%	71.6%	10.8 ppt	70.1%	71.8%	-1.7 ppt
Pretax margin	80.7%	70.1%	10.7 ppt	82.2%	71.3%	10.9 ppt	70.0%	71.6%	-1.6 ppt
Net margin	55.3%	48.8%	6.5 ppt	57.2%	49.6%	7.6 ppt	48.7%	49.8%	-1.1 ppt
BVPS (Rmb)	**8.53**	**7.35**	**16%**	**16.26**	**13.31**	**22%**	**24.71**	**21.76**	**14%**

Source: Morgan Stanley Research (E) estimates

Exhibit 12: CXMT: Quarterly financials

Rmb mn	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	1Q28E	2Q28E	3Q28E	4Q28E	2026E	2027E	2028E
Total Revenues	**50,800**	**99,510**	**126,347**	**162,271**	**153,054**	**172,641**	**188,204**	**204,273**	**210,668**	**220,600**	**219,190**	**220,096**	**438,929**	**718,172**	**870,554**
Sequential Change	*71.0%*	*95.9%*	*27.0%*	*28.4%*	*-5.7%*	*12.8%*	*9.0%*	*8.5%*	*3.1%*	*4.7%*	*-0.6%*	*0.4%*			
Change vs Year Ago	*719.1%*	*977.4%*	*659.0%*	*446.1%*	*201.3%*	*73.5%*	*49.0%*	*25.9%*	*37.6%*	*27.8%*	*16.5%*	*7.7%*	*610.2%*	*63.6%*	*21.2%*
Cost of Sales	10,588	12,346	16,022	17,819	19,751	20,730	21,508	22,312	37,664	47,581	56,137	63,352	56,776	84,300	204,734
Percent of Revenues	*21%*	*12%*	*13%*	*11%*	*13%*	*12%*	*11%*	*11%*	*18%*	*22%*	*26%*	*29%*	*13%*	*12%*	*24%*
Gross Profit	**40,212**	**87,163**	**110,325**	**144,453**	**133,303**	**151,911**	**166,696**	**181,961**	**173,003**	**173,019**	**163,053**	**156,744**	**382,153**	**633,871**	**665,820**
Percent of Revenues	*79.2%*	*87.6%*	*87.3%*	*89.0%*	*87.1%*	*88.0%*	*88.6%*	*89.1%*	*82.1%*	*78.4%*	*74.4%*	*71.2%*	*87.1%*	*88.3%*	*76.5%*
Incremental Margin	*0%*	*96%*	*86%*	*95%*	*NM*	*95%*	*95%*	*95%*	*-140%*	*0%*	*NM*	*-696%*	*95%*	*90%*	*21%*
Total Opex	2,715	4,408	7,454	9,574	9,030	10,186	11,104	12,052	13,504	14,140	14,050	14,108	24,152	42,372	55,803
Percent of Revenues	*5.3%*	*4.4%*	*5.9%*	*5.9%*	*5.9%*	*5.9%*	*5.9%*	*5.9%*	*6.4%*	*6.4%*	*6.4%*	*6.4%*	*5.5%*	*5.9%*	*6.4%*
R&D	2,154	3,383	4,422	5,679	5,357	6,042	6,587	7,150	8,005	8,383	8,329	8,364	15,639	25,136	33,081
Percent of Revenues	*4.2%*	*3.4%*	*3.5%*	*3.5%*	*3.5%*	*3.5%*	*3.5%*	*3.5%*	*3.8%*	*3.8%*	*3.8%*	*3.8%*	*3.6%*	*3.5%*	*3.8%*
General & administrative	483	880	2,906	3,732	3,520	3,971	4,329	4,698	5,267	5,515	5,480	5,502	8,001	16,518	21,764
Percent of Revenues	*1.0%*	*0.9%*	*2.3%*	*2.3%*	*2.3%*	*2.3%*	*2.3%*	*2.3%*	*2.5%*	*2.5%*	*2.5%*	*2.5%*	*1.8%*	*2.3%*	*2.5%*
Selling & marketing	78	145	126	162	153	173	188	204	232	243	241	242	512	718	958
Percent of Revenues	*0.2%*	*0.1%*	*0.1%*	*0.1%*	*0.1%*	*0.1%*	*0.1%*	*0.1%*	*0.1%*	*0.1%*	*0.1%*	*0.1%*	*0.1%*	*0.1%*	*0.1%*
Operating Income	**37,497**	**82,755**	**102,870**	**134,879**	**124,273**	**141,725**	**155,592**	**169,909**	**159,500**	**158,879**	**149,003**	**142,636**	**358,001**	**591,499**	**610,017**
Percent of Revenues	*73.8%*	*83.2%*	*81.4%*	*83.1%*	*81.2%*	*82.1%*	*82.7%*	*83.2%*	*75.7%*	*72.0%*	*68.0%*	*64.8%*	*81.6%*	*82.4%*	*70.1%*
Change vs Year Ago	*0.0%*	*0.0%*	*0.0%*	*885.3%*	*231.4%*	*71.3%*	*51.3%*	*26.0%*	*28.3%*	*12.1%*	*-4.2%*	*-16.1%*	*3981%*	*65%*	*3%*
Total Non-operating Income(Loss)	(2,063)	(1,044)	(235)	(235)	(235)	(235)	(235)	(235)	(235)	(235)	(235)	(235)	(3,577)	(941)	(941)
Profit Before Taxes	**35,434**	**81,711**	**102,635**	**134,643**	**124,038**	**141,490**	**155,356**	**169,674**	**159,264**	**158,644**	**148,768**	**142,401**	**354,424**	**590,558**	**609,077**
Percent of Revenues	*70%*	*82%*	*81%*	*83%*	*81%*	*82%*	*83%*	*83%*	*76%*	*72%*	*68%*	*65%*	*81%*	*82%*	*70%*
Taxes	(2,423)	(7,173)	(7,390)	(9,694)	(8,931)	(10,187)	(11,186)	(12,217)	(11,467)	(11,422)	(10,711)	(10,253)	(26,680)	(42,520)	(43,854)
Tax Rate	*6.8%*	*8.8%*	*7.2%*	*7.2%*	*7.2%*	*7.2%*	*7.2%*	*7.2%*	*7.2%*	*7.2%*	*7.2%*	*7.2%*	*7.5%*	*7.2%*	*7.2%*
Net Income, Cont Ops	33,012	74,538	95,245	124,949	115,107	131,303	144,171	157,458	147,797	147,221	138,057	132,148	327,744	548,038	565,223
Percent of Revenues	*65%*	*75%*	*75%*	*77%*	*75%*	*76%*	*77%*	*77%*	*70%*	*67%*	*63%*	*60%*	*75%*	*76%*	*65%*
Reported Income (Rmb)	**24,762**	**52,843**	**71,434**	**93,712**	**86,330**	**98,477**	**108,128**	**118,093**	**110,848**	**110,416**	**103,542**	**99,111**	**242,751**	**411,029**	**423,917**
Percent of Revenues	*49%*	*53%*	*57%*	*58%*	*56%*	*57%*	*57%*	*58%*	*53%*	*50%*	*47%*	*45%*	*55%*	*57%*	*49%*
Change vs Year Ago	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*	*12848%*	*69%*	*3%*
Reported EPS (Rmb, cent)	**0.41**	**0.88**	**1.09**	**1.40**	**1.29**	**1.47**	**1.62**	**1.77**	**1.66**	**1.65**	**1.55**	**1.48**	**3.84**	**6.15**	**6.34**
Change vs Year Ago	*0%*	*0%*	*0%*	*1079%*	*214%*	*68%*	*48%*	*26%*	*28%*	*12%*	*-4%*	*-16%*	*11715%*	*60%*	*3%*

Source: Company data, Morgan Stanley Research (E) estimates

CXMT: Valuation Methodology

We keep our base case value/price target unchanged: We continue to apply RI methodology to derive our price target. The price target did not change despite our 2026/27 earnings estimate increases because the contribution to calculated value from our RI model comes mostly from the subsequent forecast period and the continuing value. Key RI assumptions remain unchanged, including cost of equity of 9.9%, , intermediate growth rate of 11.0%, terminal growth rate of 3.0%, and cash dividend payout ratio of 51%, .

Our base case value/price target now implies 14.3x 2027e EPS vs. our prior target multiple of 18.4x 2027e BVPS, as we see a stronger DRAM market outlook given the continued supply constraint; we also believe the G5 breakthrough reduces risks around CXMT's technology roadmap.

Our bull case value remains unchanged at Rmb132; it now implies 21.5x our 2027e EPS.

We adjust our bear case value to Rmb40 from Rmb70 to account for recent volatility; it now implies 6.5x 2027e EPS.

Exhibit 13: CXMT: RI model

Rmb mn	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E
Total Equity	**357,143**	**768,172**	**1,192,089**	**1,422,658**	**1,678,589**	**1,962,673**	**2,278,006**	**2,628,025**	**3,016,547**	**3,447,806**	**3,926,503**	**4,457,857**
Net Profit	**242,751**	**411,029**	**423,917**	**470,548**	**522,309**	**579,763**	**643,536**	**714,325**	**792,901**	**880,120**	**976,934**	**1,084,396**
ROAE	**117.3%**	**73.1%**	**43.3%**	**36.0%**	**33.7%**	**31.8%**	**30.4%**	**29.1%**	**28.1%**	**27.2%**	**26.5%**	**25.9%**
Residual Income	60,942	225,469	256,040	310,800	338,077	368,015	400,987	437,384	477,625	522,163	571,494	626,164
	237,121											
Spread	**107.4%**	**63.1%**	**33.3%**	**26.1%**	**23.8%**	**21.9%**	**20.4%**	**19.2%**	**18.2%**	**17.3%**	**16.6%**	**15.9%**
Ending Equity Capital	357,143											
PV of Forecast Period	2,233,870											
PV of Continuing Value	3,292,876											
Equity Value	**5,883,890**											
No. of Shares	66,881											
Projected Price	**88**											

Source: Morgan Stanley Research (E) estimates

Risk Reward – CXMT Corp (688825.SS)

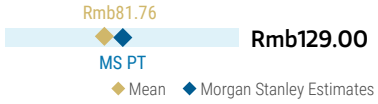
Chinese DRAM Going Mainstream; OW

PRICE TARGET Rmb88.00

Base case, residual income model. Key assumptions:
- 9.9% CoE (1.2 beta, 2.0% risk-free rate, and 6.6% risk premium)
- 51% payout ratio
- 11.0% medium-term growth rate
- 3% terminal growth rate

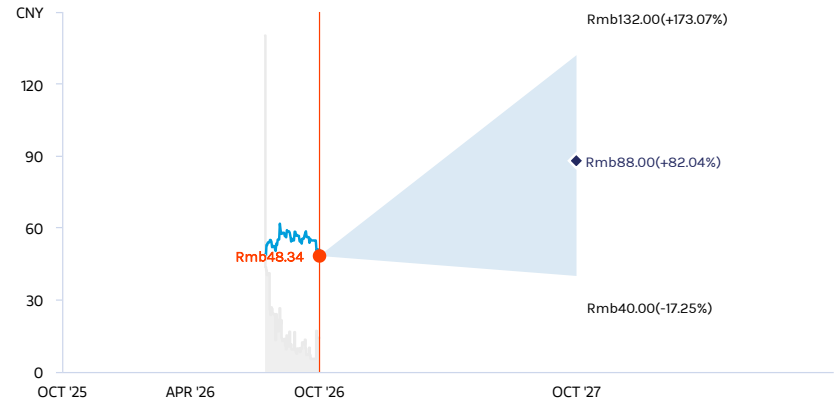
We believe these assumptions are justified, given CXMT's fast growth, capacity expansion, localization tailwinds and AI-memory exposure. Target implies 14.3x 2027e P/E vs global peers' 4.4x 2027e P/E; the premium reflects CXMT's stronger growth profile.

Consensus Price Target Distribution



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



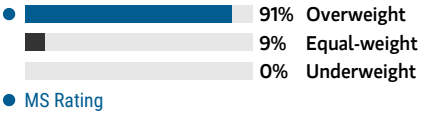
Key: Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- CXMT is now the world's #4 DRAM maker and a key beneficiary of China's memory localization push.
- We forecast that DRAM capacity will rise from 180k wpm in 2025 to 500k by 2028, with G4b, DDR5, LPDDR5/5X, server DRAM and HBM3e/HBM4 improving mix.
- We expect 2025–28E revenue CAGR of 140%+, 2026-28E GM >76%, and OPM of ~70–80%, supported by pricing, scale and higher-value products.
- Our Rmb88 price target is based on a residual income model and implies 14.3x 2027E P/E. We see the premium to global memory peers as justified by CXMT's faster growth, capacity expansion, localization tailwinds, and AI-memory exposure.
- **Risks:** DRAM pricing, capex/depreciation, export controls and migration execution.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Pricing Power:	*Positive*
Secular Growth:	*Positive*

View descriptions of Risk Rewards Themes here

BULL CASE Rmb132.00

21.5x 2027e P/E

We assume stronger sales/earnings growth, with faster global market share expansion:

(1) 200% revenue CAGR, 2025-28e, as we assume a stronger-than-expected DRAM commodity cycle, supported by AI inference demand moving into 2027 and beyond;

(2) Faster-than-expected HBM and G5 ramp-up;

(3) Gross margin improves to over 90%, driven by economies of scale and CXMT's technology competitiveness;

(4) >800% earnings CAGR, 2025-28e.

BASE CASE Rmb88.00

14.3x 2027e P/E

We forecast strong revenue/earnings growth, with global market share expansion.

(1) 140%+ revenue CAGR, 2025-28e, as we assume DRAM pricing remains strong, supported by AI demand;

(2) Moderate pace of HBM and G5 ramp-up;

(3) 2026-28e gross margin of over 76%, driven by economies of scale and CXMT's technology competitiveness;

(4) ~480% earnings CAGR, 2025-28e

BEAR CASE Rmb40.00

6.5x 2027e P/E

We assume revenue/earnings growth is slower, with slower global market share expansion.

(1) 80% revenue CAGR, 2025-28e, as we assume a period of AI computing digestion in 2028, leading to more aggressive margin erosion for HBM owing to competitors' aggressive capacity expansion and pricing competition, as well as faster DRAM pricing drops with muted demand recovery;

(2) Slower than expected G5 ramp;

(2) Customer bargaining power pushes gross margin below 40%;

(3) 200% earnings CAGR, 2025-28e

Risk Reward – CXMT Corp (688825.SS)

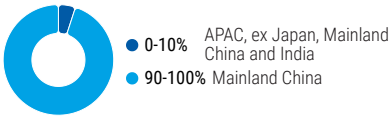
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Revenue (Rmb, mn)	180	300	400	500
Gross profit (Rmb, mn)	25,334	382,153	633,871	665,820
EPS (Rmb, mn)	0	4	6	6

INVESTMENT DRIVERS

- Commodity DRAM pricing
- HBM capacity expansion and tech migration
- Demand from major customers

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

RISKS TO PT/RATING

RISKS TO UPSIDE

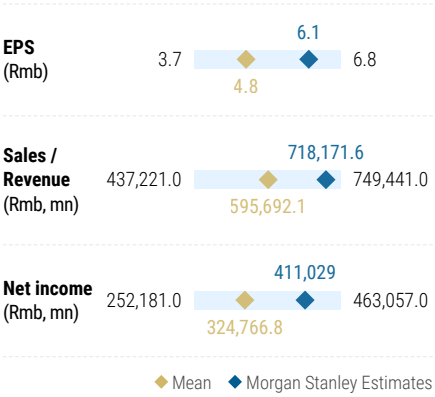
- resilient DRAM pricing, supported by AI demand
- faster-than-expected tech migration
- export control relief
- stronger demand from major customers

RISKS TO DOWNSIDE

- earlier-than-expected global DRAM price declines
- CXMT delays in new products, HBM or 3D DRAM development
- tighter restrictions on China's fabs for semi equipment, materials, or servicing
- weaker demand from major China customers

MS ESTIMATES VS. CONSENSUS

FY Dec 2027e



Source: Refinitiv, Morgan Stanley Research

CXMT: Financial Summary

Exhibit 14: Financial summary

Income Statement Rmb mn (Years End Dec)	2025	2026E	2027E	2028E
Net sales	61,799	438,929	718,172	870,554
COGS	(36,465)	(56,776)	(84,300)	(204,734)
Gross profit	25,334	382,153	633,871	665,820
Operating expenses	(16,561)	(24,152)	(42,372)	(55,803)
Operating income	**8,773**	**358,001**	**591,499**	**610,017**
Non-operating income	(1,071)	(3,577)	(941)	(941)
Pre-tax income	7,701	354,424	590,558	609,077
Income tax	(557)	(26,680)	(42,520)	(43,854)
Minority interests	5,269	84,993	137,010	141,306
Reported net Income	**1,875**	**242,751**	**411,029**	**423,917**
Adj.wtd.avg.shrs(m)	59,018	63,216	66,881	66,881
MW EPS (Rmb)	0.03	3.84	6.15	6.34
Reported EPS (Rmb)	0.03	3.84	6.15	6.34
EPS for consensus (Rmb)	**0.03**	**3.84**	**6.15**	**6.34**

Balance Sheet Rmb mn (Years End Dec)	2025	2026E	2027E	2028E
Cash	37,798	364,853	874,560	1,373,537
Mkt Securities	0	0	0	0
AR/NR	1,524	11,966	19,579	23,734
Inventory	28,568	36,680	54,462	132,269
Other	43,096	43,096	43,096	43,096
Current Assets	**110,986**	**456,596**	**991,698**	**1,572,636**
Long-term investments	4,384	4,384	4,384	4,384
Fixed assets	183,024	225,774	244,382	253,419
Deferred assets	1,006	1,006	1,006	1,006
Other assets	37,385	37,385	37,385	37,385
Total Assets	**336,785**	**725,145**	**1,278,855**	**1,868,830**
S/T borrowings	9,676	9,686	9,696	9,706
AP/NP	8,702	11,659	17,311	42,043
Other ST liabilities	34,465	34,465	34,465	34,465
LT debt	118,825	118,835	118,845	118,855
Other LT liabilities	11,017	11,017	11,017	11,017
Total Liabilities	**182,685**	**185,662**	**191,335**	**216,086**
Common shares	60,193	66,881	66,881	66,881
Additional capital	33,104	84,054	84,054	84,054
Retained earning	(36,650)	206,101	617,129	1,041,047
Other shareholders' equity	97,454	182,447	319,456	460,762
Total Equity	**154,100**	**539,482**	**1,087,521**	**1,652,744**
Total Liab. & Shrhldr's Equity	**336,785**	**725,145**	**1,278,855**	**1,868,830**

E = Morgan Stanley Research Estimates
Source: Morgan Stanley Research, Company Data

Cash Flow Statement Rmb mn (Years End Dec)	2025	2026E	2027E	2028E
Cashflow from Operations	**36,520**	**350,967**	**576,687**	**565,957**
Net profits	7,144	327,744	548,038	565,223
Depreciation	24,680	38,820	48,392	57,963
Working Capital Change	(48,395)	(15,598)	(19,743)	(57,229)
Other adjustments	53,090	0	0	0
Cashflow from Investing	**(86,823)**	**(81,570)**	**(67,000)**	**(67,000)**
Capex	(49,739)	(81,570)	(67,000)	(67,000)
Change of LT Investment	2,434	0	0	0
Change of ST Investment	194	0	0	0
Other adjustments	(39,712)	0	0	0
Cashflow from financing	**46,121**	**57,658**	**20**	**20**
Increase in L/T debt	22,832	10	10	10
Increase in S/T debt	7,879	10	10	10
Cash Dividend Paid	0	0	0	0
Dir& Emp Bonus Paid	0	0	0	0
Issuance of stock	2,422	57,638	0	0
Other adjustments	12,988	0	0	0
Exchange rate adjustment	(528)	0	0	0
Net change in cash	**-4,710**	**327,055**	**509,707**	**498,977**

Financial Ratios	2025	2026E	2027E	2028E
Growth(%)				
Turnover	155.6	610.2	63.6	21.2
Operating profits	-250.2	3980.9	65.2	3.1
EPS	-125.7	11715.5	60.0	3.1
Margins (%)				
Gross Margin	41.0	87.1	88.3	76.5
Operating Margin	14.2	81.6	82.4	70.1
Pretax Margin	12.5	80.7	82.2	70.0
Net Margin	3.0	55.3	57.2	48.7
Return (%)				
ROAE	1.2	70.0	50.5	30.9
ROAA	0.6	45.7	41.0	26.9
Gearing (%)				
Net Debt/Equity	58.9	(43.8)	(68.6)	(75.3)
Liabilities/Equity	118.5	34.4	17.6	13.1
Ratios (X)				
Current ratio	2.1	8.2	16.1	18.2
Quick ratio	0.7	6.8	14.5	16.2
Others				
AR/NR Turnover (days)	10	10	10	10
Inventory Turnover (days)	236	236	236	236
AP Turnover (days)	75	75	75	75
Cash Conversion (days)	171	171	171	171

Source: Company data, Morgan Stanley Research (E) estimates

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - Options_Probabilities_Exhibit_Link.pdf

2. View descriptions of Risk Rewards Themes - RR_Themes_Exhibit_Link.pdf

3. View explanation of regional hierarchies - GEG_Exhibit_Link.pdf

4. View explanation of Theme/Exposure methodology - ESG_Sustainable_Solutions_External_Link.pdf

5. View explanation of HERS methodology - ESG_HERS_External_Link.pdf

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley Asia Limited (which accepts the responsibility for its contents) and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research), and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Charlie Chan; Daisy Dai, CFA; Daniel Yen, CFA; Henry Zhao.

.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictpolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

As of September 30, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: ACM Research Inc, Advanced Micro-Fabrication Equipment Inc, Advanced Wireless Semiconductor Co, Alchip Technologies Ltd, AllRing Tech Co., AP Memory Technology Corp, ASE Technology Holding Co. Ltd., ASMedia Technology Inc, ASMPT Ltd, Aspeed Technology, Cambricon Technology Corporation, eMemory Technology Inc, GigaDevice Semiconductor Beijing Inc, GlobalWafers Co Ltd, Gudeng Precision, Macronix International Co Ltd, Montage Technology Co Ltd, NAURA Technology Group Co Ltd, Novatek, Parade Technologies Ltd, Phison Electronics Corp, Powerchip Semiconductor Manufacturing Co, SG Micro Corp., Shanghai Fudan Microelectronics, SICC Co Ltd, Silergy Corp., Silicon Motion, StarPower Semiconductor Ltd, TSMC, UMC, Vanguard International Semiconductor, WIN Semiconductors Corp, WinWay Technology Co Ltd, WPG Holdings, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Alchip Technologies Ltd, Iluvatar CoreX Semiconductor Co., Ltd., MediaTek, Montage Technology Co Ltd, Powerchip Semiconductor Manufacturing Co, UMC.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from ASMPT Ltd, Iluvatar CoreX Semiconductor Co., Ltd., MediaTek, Montage Technology Co Ltd, Powerchip Semiconductor Manufacturing Co, Silicon Motion, UMC.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from Advanced Micro-Fabrication Equipment Inc, Alchip Technologies Ltd, AP Memory Technology Corp, ASE Technology Holding Co. Ltd., ASMedia Technology Inc, ASMPT Ltd, Aspeed Technology, CXMT Corp, Espressif Systems, GigaDevice Semiconductor Beijing Inc, GlobalWafers Co Ltd, Gudeng Precision, Himax Technologies Inc, Hua Hong Semiconductor Ltd, Iluvatar CoreX Semiconductor Co., Ltd., Innoscience, King Yuan Electronics Co Ltd, Macronix International Co Ltd, MediaTek, MetaX Integrated Circuits, Montage Technology Co Ltd, Novatek, Phison Electronics Corp, Powerchip Semiconductor Manufacturing Co, Realtek Semiconductor, SG Micro Corp., Shenzhen Longsys Electronics Co Ltd, SICC Co Ltd, Silergy Corp., Silicon Motion, TSMC, UMC, Universal Scientific Ind. (Shanghai), Vanguard International Semiconductor, Winbond Electronics Corp, WinWay Technology Co Ltd, WPG Holdings, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from ASE Technology Holding Co. Ltd., King Yuan Electronics Co Ltd, MediaTek, Nanya Technology Corp., Novatek, Nuvoton Technology Corporation, Realtek Semiconductor, Silicon Motion, SMIC, TSMC, UMC, Universal Scientific Ind. (Shanghai), Winbond Electronics Corp, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: Advanced Micro-Fabrication Equipment Inc, Alchip Technologies Ltd, AP Memory Technology Corp, ASE Technology Holding Co. Ltd., ASMedia Technology Inc, ASMPT Ltd, Aspeed Technology, CXMT Corp, Espressif Systems, GigaDevice Semiconductor Beijing Inc, GlobalWafers Co Ltd, Gudeng Precision, Himax Technologies Inc, Hua Hong Semiconductor Ltd, Iluvatar CoreX Semiconductor Co., Ltd., Innoscience, King Yuan Electronics Co Ltd, Macronix International Co Ltd, MediaTek, MetaX Integrated Circuits, Montage Technology Co Ltd, Novatek, Phison Electronics Corp, Powerchip Semiconductor Manufacturing Co, Realtek Semiconductor, SG Micro Corp., Shenzhen Longsys Electronics Co Ltd, SICC Co Ltd, Silergy Corp., Silicon Motion, TSMC, UMC, Universal Scientific Ind. (Shanghai), Vanguard International Semiconductor, Winbond Electronics Corp, WinWay Technology Co Ltd, WPG Holdings, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: ASE Technology Holding Co. Ltd., Iluvatar CoreX Semiconductor Co., Ltd., King Yuan Electronics Co Ltd, MediaTek, Montage Technology Co Ltd, Nanya Technology Corp., Novatek, Nuvoton Technology Corporation, Realtek Semiconductor, Silicon Motion, SMIC, TSMC, UMC, Universal Scientific Ind. (Shanghai), Winbond Electronics Corp, WT Microelectronics Co. Ltd..

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not

linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.
Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.
Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of September 30, 2026)
The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.
For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1599	43%	455	48%	28%	783	44%
Equal-weight/Hold	1563	42%	394	42%	25%	763	43%
Not-Rated/Hold	2	0%	1	0%	50%	1	0%
Underweight/Sell	570	15%	96	10%	17%	227	13%
Total	3,734		946			1774	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

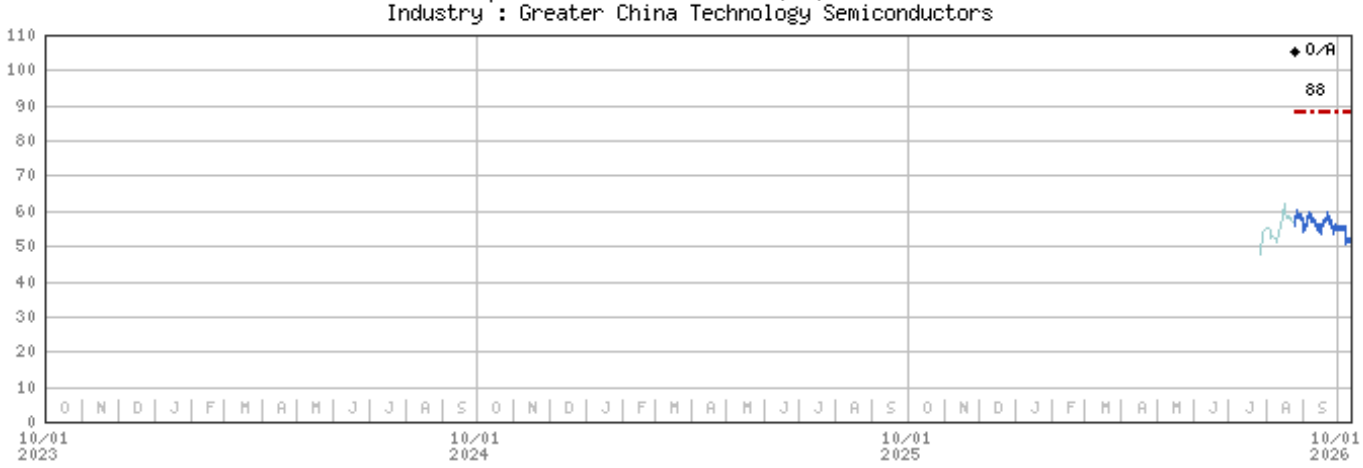
Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.
Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.
Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.
Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

CXMT Corp (688825.SS) - As of 10/12/26 GMT in CNY
Industry : Greater China Technology Semiconductors



Stock Rating History: 8/26/26 : O/A

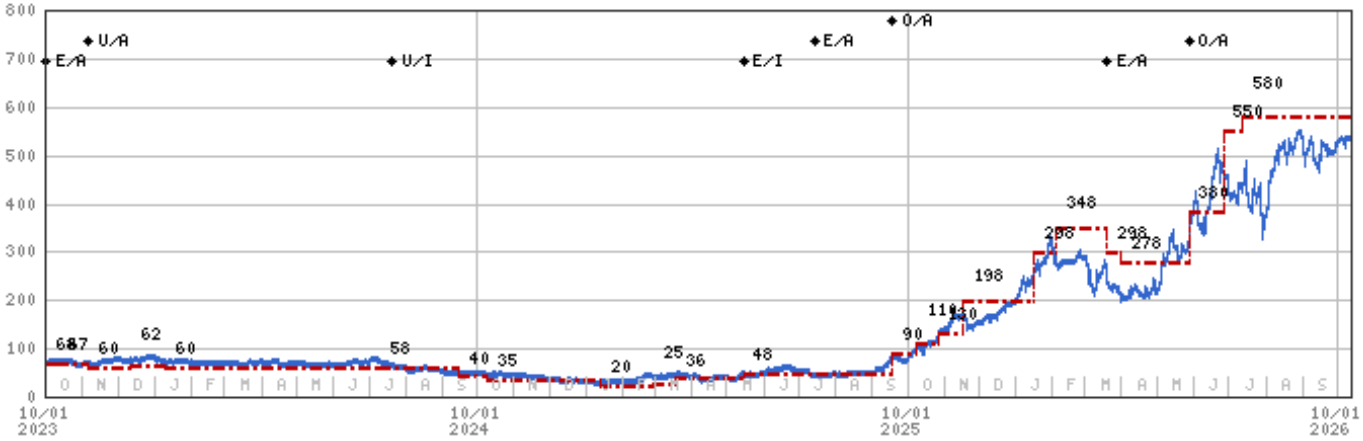
Price Target History: 8/26/26 : 88

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -·- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) ▬
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Nanya Technology Corp. (2408.TW) - As of 10/12/26 GMT in TWD
Industry : Greater China Technology Semiconductors



Stock Rating History: 10/1/21 : E/I; 10/12/21 : E/C; 12/3/21 : U/C; 10/4/22 : E/A; 2/22/23 : E/I; 7/7/23 : E/A; 11/6/23 : U/A; 7/21/24 : U/I; 5/15/25 : E/I; 7/14/25 : E/A; 9/18/25 : O/A; 3/20/26 : E/A; 5/28/26 : O/A

Price Target History: 8/12/21 : 75; 10/5/21 : 68; 3/24/22 : 65; 7/1/22 : 48; 1/11/23 : 46; 3/24/23 : 52; 4/11/23 : 60; 5/2/23 : 58; 7/11/23 : 62; 9/8/23 : 68; 10/12/23 : 67; 11/6/23 : 60; 12/13/23 : 62; 1/11/24 : 60; 7/10/24 : 58; 9/15/24 : 40; 10/9/24 : 35; 1/14/25 : 20; 2/27/25 : 25; 3/19/25 : 36; 5/15/25 : 48; 9/18/25 : 90; 10/9/25 : 110; 10/27/25 : 130; 11/18/25 : 198; 1/16/26 : 298; 2/5/26 : 348; 3/20/26 : 298; 3/31/26 : 278; 5/28/26 : 380; 6/26/26 : 550; 7/13/26 : 580

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -·- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) — Stock Price (Covered by Current Analyst) ▬
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures.
For Morgan Stanley specific disclosures, you may refer to https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch.
Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.
Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.
Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.
Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.
Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (http://www.morganstanley.com/terms.html). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (http://www.morganstanley.com/terms.html). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).
If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research.
Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.
The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.
The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.
With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.
Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.
Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.
To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V which is regulated by Comision Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118J); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. Morgan Stanley Research is disseminated in Canada by Morgan Stanley Canada Limited and Morgan Stanley Wealth Management Canada Inc., members of the Canadian Investment Regulatory Organization (CIRO). The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

Morgan Stanley Hong Kong Securities Limited is the liquidity provider/market maker for securities of ASMPT Ltd, GigaDevice Semiconductor Beijing Inc, Hua Hong Semiconductor Ltd, Montage Technology Co Ltd listed on the Stock Exchange of Hong Kong Limited. An updated list can be found on HKEx website: http://www.hkex.com.hk.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

References to any prediction market or event contract do not constitute an offer, solicitation, recommendation, or endorsement to participate in any such market or contract. Morgan Stanley Research may not be distributed or made available in any jurisdiction where such distribution would be unlawful or otherwise prohibited by applicable law.

INDUSTRY COVERAGE: Greater China Technology Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (10/12/2026)
Charlie Chan		
Advanced Wireless Semiconductor Co (8086.TWO)	U (07/14/2025)	NT$148.50
Alchip Technologies Ltd (3661.TW)	O (05/14/2021)	NT$4,395.00
ASE Technology Holding Co. Ltd. (3711.TW)	O (09/15/2024)	NT$762.00
Cambricon Technology Corporation (688256.SS)	O (04/27/2026)	Rmb971.88
CXMT Corp (688825.SS)	O (08/26/2026)	Rmb48.34
Global Unichip Corp (3443.TW)	O (06/24/2026)	NT$8,100.00
GlobalWafers Co Ltd (6488.TWO)	E (05/19/2026)	NT$1,115.00
Gudeng Precision (3680.TWO)	O (11/25/2025)	NT$569.00
Hua Hong Semiconductor Ltd (1347.HK)	E (03/12/2026)	HK$89.85
Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK)	O (04/27/2026)	HK$88.00
MediaTek (2454.TW)	O (11/28/2025)	NT$4,625.00
MetaX Integrated Circuits (688802.SS)	E (04/27/2026)	Rmb451.55
Nanya Technology Corp. (2408.TW)	O (05/28/2026)	NT$527.00
OmniVision Integrated Circuits Group Inc (603501.SS)	E (11/17/2025)	Rmb77.29
Phison Electronics Corp (8299.TWO)	E (02/25/2026)	NT$2,055.00
SG Micro Corp. (300661.SZ)	E (11/03/2025)	Rmb93.41
Silergy Corp. (6415.TW)	U (05/19/2026)	NT$440.00
SMIC (0981.HK)	O (10/21/2025)	HK$56.20
TSMC (2330.TW)	O (02/07/2022)	NT$2,540.00
UMC (2303.TW)	O (05/19/2026)	NT$145.50
Vanguard International Semiconductor (5347.TWO)	E (01/14/2026)	NT$181.00
WIN Semiconductors Corp (3105.TWO)	E (10/09/2026)	NT$580.00
Daisy Dai, CFA		
ASMPT Ltd (0522.HK)	O (07/24/2025)	HK$160.70
China Resources Microelectronics Limited (688396.SS)	U (03/02/2026)	Rmb50.15
Empyrean Technology Co Ltd (301269.SZ)	E (01/17/2025)	Rmb97.90
Hangzhou Silan Microelectronics Co. Ltd. (600460.SS)	U (08/25/2025)	Rmb27.85
Hygon Information Technology Co., Ltd. (688041.SS)	O (07/03/2026)	Rmb213.50
Innoscience (2577.HK)	E (10/13/2025)	HK$35.62
JCET Group Co Ltd (600584.SS)	O (09/16/2026)	Rmb59.99
Shanghai Fudan Microelectronics (1385.HK)	O (03/07/2025)	HK$23.54
SICC Co Ltd (688234.SS)	O (03/20/2026)	Rmb93.57
SJ Semiconductor Corp (688820.SS)	E (09/16/2026)	Rmb104.09
StarPower Semiconductor Ltd (603290.SS)	E (05/14/2026)	Rmb74.06
Unigroup Guoxin Microelectronics Co Ltd (002049.SZ)	U (01/10/2023)	Rmb55.70
Universal Scientific Ind. (Shanghai) (601231.SS)	O (11/05/2025)	Rmb22.39
Yangjie Technology (300373.SZ)	O (06/10/2022)	Rmb78.79
Daniel Yen, CFA		
AP Memory Technology Corp (6531.TW)	O (07/11/2025)	NT$947.00
ASMedia Technology Inc (5269.TW)	U (10/03/2025)	NT$1,540.00
Aspeed Technology (5274.TWO)	O (06/09/2025)	NT$17,695.00
Egis Technology Inc (6462.TWO)	E (01/28/2026)	NT$100.00
Elan Microelectronics Corp (2458.TW)	O (10/03/2025)	NT$133.00
eMemory Technology Inc (3529.TWO)	O (08/18/2026)	NT$2,850.00
Espressif Systems (688018.SS)	O (05/15/2023)	Rmb111.70
GigaDevice Semiconductor Beijing Inc (603986.SS)	O (05/15/2025)	Rmb310.84
Macronix International Co Ltd (2337.TW)	O (09/18/2025)	NT$115.50
Montage Technology Co Ltd (6809.HK)	O (03/18/2026)	HK$254.80
Montage Technology Co Ltd (688008.SS)	O (03/18/2026)	Rmb191.84
Novatek (3034.TW)	U (02/04/2026)	NT$529.00
Nuvoton Technology Corporation (4919.TW)	U (11/10/2025)	NT$154.00
Parade Technologies Ltd (4966.TWO)	O (05/27/2026)	NT$553.00

Powerchip Semiconductor Manufacturing Co (6770.TW)	O (10/27/2025)	NT$71.40
Realtek Semiconductor (2379.TW)	E (01/30/2026)	NT$746.00
Shenzhen Goodix Technology Co Ltd (603160.SS)	U (07/14/2025)	Rmb50.08
Winbond Electronics Corp (2344.TW)	O (05/28/2026)	NT$174.00
WPG Holdings (3702.TW)	O (03/16/2026)	NT$116.00
WT Microelectronics Co. Ltd. (3036.TW)	O (01/27/2026)	NT$216.50
Duan Liu		
Dosilicon Co Ltd (688110.SS)	U (09/06/2024)	Rmb65.71
Shenzhen Longsys Electronics Co Ltd (301308.SZ)	E (02/25/2026)	Rmb271.81
Henry Zhao		
ACM Research Inc (ACMR.O)	O (03/07/2023)	US$71.37
Advanced Micro-Fabrication Equipment Inc (688012.SS)	O (11/06/2023)	Rmb311.21
Maxscend Microelectronics Co Ltd (300782.SZ)	U (01/11/2021)	Rmb60.48
NAURA Technology Group Co Ltd (002371.SZ)	O (11/06/2023)	Rmb587.98
Lucas Wang		
AllRing Tech Co. (6187.TWO)	O (09/23/2025)	NT$1,675.00
FOCI Fiber Optic Communications Inc (3363.TWO)	E (10/09/2026)	NT$602.00
Hon Precision (7769.TW)	O (04/17/2026)	NT$6,490.00
King Yuan Electronics Co Ltd (2449.TW)	O (03/03/2023)	NT$300.00
MPI Corporation (6223.TWO)	O (04/17/2026)	NT$6,130.00
Tiffany Yeh		
Himax Technologies Inc (HIMX.O)	E (02/04/2026)	US$13.30
Silicon Motion (SIMO.O)	O (05/06/2024)	US$263.90
WinWay Technology Co Ltd (6515.TW)	O (04/17/2026)	NT$6,025.00

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.

© 2026 Morgan Stanley

Largan Precision

Core weakness outweighs progress in AI optics

Equities

Taiwan

Electric Components & Equipment

12-month rating	**Sell**
12m price target	**NT$5,000.00**
Price (08 Oct 2026)	**NT$6,660**

RIC: 3008.TW **BBG:** 3008 TT

Trading data and key metrics

52-wk range	NT$7,810.00-2,025.00
Market cap.	NT$875b/US$27.4b
Shares o/s	131m (ORD)
Free float	70%
Avg. daily volume ('000)	2,379
Avg. daily value (m)	NT$13,134.2
Common s/h equity (12/26E)	NT$199b
P/BV (12/26E)	4.4x
Net debt to EBITDA (12/26E)	NM

EPS (UBS, diluted) (TWD)

	From	To	% ch	Cons.
12/26E	188.21	160.04	-15	184.91
12/27E	214.00	180.00	-16	204.02
12/28E	250.66	220.00	-12	238.78

Randy Abrams
Analyst
randy.abrams@ubs.com
+886-2-8722 7338

Irene Chang
Associate Analyst
irene.chang@ubs.com
+886-2-8722 7280

CPO progress is more tangible but substantial success is already priced in

We maintain Sell and our NT$5,000 target price as erosion in the core smartphone business outweighs CPO progress. The Q326 call confirmed that Largan's fiber array has passed customer qualification although its customer qualified another vendor's prism micro-lens assembly (PMLA) that Largan would now buy in. If Largan qualifies the full FAU plus carrier assembly for 2H27, content rises but with more pass-through cost and less margin upside. We roll our valuation to 2028E, the first full year of CPO production, though 25% below the NT$6,660 close on October 8.

Full FAU assembly would lift content, but with more pass-through

We believe Largan's FA customer is one of two module suppliers on NVIDIA's 2H27 program, which we estimate could reach 40% share. The program is still subject to successful yield ramp and could move to metalens in 2028, which would reopen supplier qualification. Assembling the full FAU could lift Largan's ASP to US$125-150 leveraging its precision alignment and automation but the PMLA, V-groove and carrier parts would be pass through that cuts into margin leverage. We expand our FAU TAM analysis (Figure 9) with FAU market in 2028/30 expanding to 12.3mn/35.6mn units, with Largan base case set at 25% of the TAM.

Core smartphone business stays soft into 2027

Q326 sales of NT$15.7bn, +15% QoQ/-11% YoY missedUBSe/Street +36%/+33% on weaker flagship smartphone demand. GMs fell 7.2ppts QoQ to 42.2%, lowest in 13 years on renewable energy costs and low variable-aperture yields, dropping OpM to 17 year lows at 30.3% and EPS to NT$39.6, below UBS/street NT$51.50/NT$49.69. We cut Q4 to NT$16.7bn +7% QoQ, 11% below consensus as its customer trimmed orders after a pricier iPhone 18 launch and possible lag to Genius which is picking up periscope share and outgrew in Q3 (+1% YoY vs. -11% YoY). We cut 2027 from NT$77bn to NT $67bn, before a recovery to NT$79bn in 2028 with the next periscope upgrade. We see GM recovering to 47.5%/48.9% in 2027-28 as yields improve and energy costs pass through in 2028 pricing.

Maintain Sell and our NT$5,000 target

We cut 2026E/27E EPS by 14%/16% to NT$160/NT$180 and introduce 2028E EPS of NT$220 for the core smartphone business. We value the lens business at NT$3,300 (15x 2028E ex-CPO EPS of NT$220, the middle of Largan's 13–18x historical range) and CPO option at NT$1,700 or US$7bn (49x our base-case 2028E CPO EPS of NT$35, near the top of optical peers' 14–54x). Our base case gives Largan 25% of a 12.3mn-unit, US $1.85bn FAU market at US$150 per unit and a 50% gross margin. That credits it with yields and margins it still needs to prove in a quarter when yields and energy costs cut 7.2ppts from its lens margin. At ~NT$6,660, the shares imply ~NT$69 of 2028E CPO EPS, about twice our base case.

Highlights (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenues	48,842	59,458	61,148	61,550	67,438	79,464	78,546	81,569
EBIT (UBS)	17,807	24,033	23,558	21,139	24,150	30,110	29,487	30,648
Net earnings (UBS)	18,047	22,480	23,613	20,953	23,567	28,805	28,397	29,473
EPS (UBS, diluted) (NT$)	135.22	168.43	176.92	160.04	180.00	220.00	216.89	225.11
DPS (net) (NT$)	85.50	67.50	97.50	80.00	81.97	90.34	110.41	108.85
Net (debt) / cash	111,655	117,781	122,309	109,131	120,267	127,842	144,348	160,697

Profitability/valuation	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
EBIT (UBS) margin %	36.5	40.4	38.5	34.3	35.8	37.9	37.5	37.6
ROIC (EBIT) %	39.3	46.2	40.9	30.3	30.1	35.1	33.1	35.3
EV/EBITDA (UBS core) x	7.7	7.1	6.0	24.7	21.7	18.1	17.7	16.6
P/E (UBS, diluted) x	16.2	15.1	13.3	41.6	37.0	30.3	30.7	29.6
Equity FCF (UBS) yield %	2.0	4.3	3.0	2.1	2.5	2.2	3.6	3.5
Dividend yield (net) %	3.9	2.7	4.1	1.2	1.2	1.4	1.7	1.6

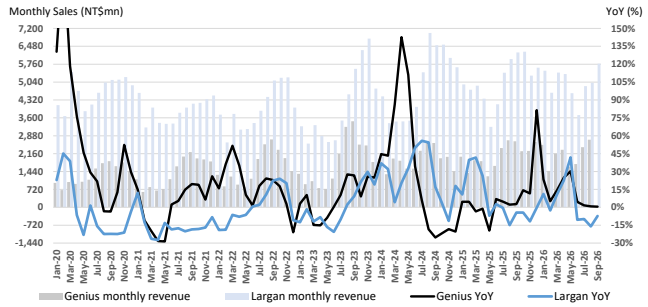
Source: Company accounts, LSEG Eikon, UBS estimates. Metrics marked as (UBS) have had analyst adjustments applied. Valuations: based on an average share price that year, (E): based on a share price of NT$ 6,660.00 on 12-Oct-2026 06:29:19 CST

Core smartphone lens seeing weakness with declining margins

Q326 sales was reported light at NT$15.7bn, +15% QoQ missing UBSe/Street +36%/ +33% amid weaker flagship smartphone demand. GMs fell 720bp to 42.2%, lowest in 13 years on renewable energy costs and low variable-aperture yields, dropping OpM to 17 year lows at 30.3% and EPS to NT$39.6, below UBS/street NT$51.50/NT$49.69. Renewable-energy costs are structural under Largan's long-term power commitments and rise with Apple's share of the mix; with 2027 pricing set, pass-through is more likely in 2028.

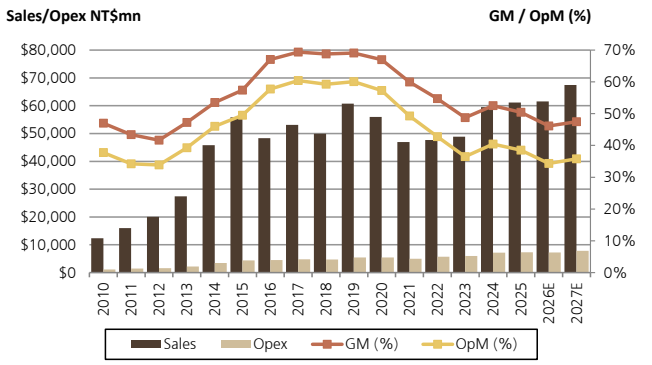
We cut Q4 to NT$16.7bn +7% QoQ, 11% below consensus as its customer trimmed orders after a pricier iPhone 18 launch and possible lag to Genius which is picking up periscope share and outgrew in Q3 (flat vs. -11% YoY). October revenue should be flat MoM and November lower as the key customer has shortened its forecast after cutting flagship orders in September. The spring-2027 model starts shipping in October but will not offset the flagship decline.

Figure 1: Genius outpacing Largan in recent months



Source: Company data

Figure 2: Largan margins dropping in recent years



Source: Company data, UBS estimates

We cut 2026E/27E revenue 9%/13% to NT$61.6bn/NT$67.4bn and EPS 14%/16% to NT$163/NT$180, and introduce 2028E revenue of NT$79.5bn, +18% YoY for the new smartphone lens upgrade, taking EPS to NT$220. Our estimates exclude CPO, which we value separately under base, bull and bear scenarios. Including our base case, 2008E EPS would be NT$255, 6% above the NT$240 consensus which likely includes some CPO. We see GM recovering to 47.5%/48.9% in 2027-28 as yields improve and energy costs pass-through in 2028 pricing. With a still a range of outcomes on 2028 CPO, we keep it in our valuation as a scenario for our upside/downside/base cases.

Figure 3: UBS vs. VA Consensus Q126A-Q426E and 2025A-28E

	Q126	Q226	Q326E			Q426E		
(NT$m)	Actual	Actual	UBSe	UBSe prior	VA	UBSe	UBSe prior	VA
Revenue	**15,544**	**13,665**	**15,651**	**18,566**	**18,147**	**16,690**	**20,094**	**16,173**
- QoQ / YoY	-9.7%	-12.1%	14.5%	35.9%	32.8%	6.6%	28.4%	3.3%
Gross profit	**7,680**	**6,753**	**6,607**	**9,242**	**8,968**	**7,346**	**10,044**	**6,980**
- Gross margin	49.4%	49.4%	42.2%	49.8%	49.4%	44.0%	50.0%	43.2%
Operating profit	**5,812**	**5,148**	**4,735**	**7,144**	**6,819**	**5,443**	**7,833**	**5,105**
- OP margin	37.4%	37.7%	30.3%	38.5%	37.6%	32.6%	39.0%	31.6%
Non-op profit	**1,476**	**951**	**1,357**	**1,027**	**1,071**	**1,064**	**1,064**	**1,477**
Pre-tax profit	**7,288**	**6,099**	**6,092**	**8,171**	**7,889**	**6,507**	**8,897**	**6,582**
Tax	**-1,112**	**-1,373**	**-853**	**-1,385**	**-1,250**	**-1,041**	**-1,499**	**-1,306**
- Tax rate	15.3%	22.5%	14.0%	17.0%	15.8%	16.0%	16.9%	19.8%
Net profit	**6,123**	**4,670**	**5,182**	**6,736**	**6,614**	**5,409**	**7,348**	**5,275**
EPS (NT$)	**46.63**	**35.70**	**39.62**	**51.50**	**49.69**	**41.36**	**56.18**	**40.04**

2025	2026E			2027E			2028E		
Actual	UBSe	UBSe prior	VA	UBSe	UBSe prior	VA	UBSe	UBSe prior	VA
61,148	**61,550**	**67,869**	**61,033**	**67,438**	**77,189**	**68,359**	**79,464**	**83,044**	**115,261**
2.8%	0.7%	11.0%	-0.2%	9.6%	13.7%	12.0%	17.8%	7.6%	68.6%
30,837	**28,386**	**33,718**	**28,019**	**32,008**	**38,405**	**30,319**	**38,823**	**42,004**	**53,520**
50.4%	46.1%	49.7%	45.9%	47.5%	49.8%	44.4%	48.9%	50.6%	46.4%
23,558	**21,139**	**25,937**	**20,800**	**24,150**	**29,796**	**21,482**	**30,110**	**34,945**	**39,917**
38.5%	34.3%	38.2%	34.1%	35.8%	38.6%	31.4%	37.9%	42.1%	34.6%
2,342	**4,847**	**4,518**	**5,261**	**4,467**	**4,467**	**4,734**	**4,595**	**4,595**	**4,937**
25,900	**25,986**	**30,455**	**26,061**	**28,618**	**34,263**	**26,543**	**34,705**	**39,540**	**44,854**
-4,340	**-4,379**	**-5,369**	**-4,763**	**-5,050**	**-6,244**	**-5,150**	**-5,900**	**-6,722**	**-8,572**
16.8%	16.9%	17.6%	18.3%	17.6%	18.2%	19.4%	17%	17%	19%
21,275	**21,385**	**24,877**	**21,298**	**23,567**	**28,019**	**21,414**	**28,805**	**32,818**	**36,282**
159.41	**163.33**	**190.00**	**161.25**	**180.00**	**214.00**	**161.64**	**220.00**	**250.66**	**275.28**

Source: Company data, Visible Alpha, UBS estimates. Note: EPS based on reported and may differ with cover page diluted EPS.

CPO fiber array qualified, but rising pass-through as it upgrades to FAU production

Largan's fiber array (FA) has passed qualification with a volume customer, and management expects mass-production capacity by mid-2027 in a newly completed fab dedicated to FA/FAU, using in-house equipment. The company's next step is to produce the full detachable FAU mounted on a carrier, a board with three alignment beams that sits on the optical engine's receptor. Largan can build FAUs now and says the customer

would prefer it to supply the full assembly, but qualification is pending with the customer preferring FAU and carrier sampled separately.

Even as its potential to supply the full FAU improve, Largan will buy in most of the content. The customer has qualified another PMLA vendor and prefers semiconductor-made optics; V-grooves and carrier beams and boards are outsourced and the customer does not want Largan to make MT ferrules. Management described its role as closer to a contractor and said margins hinge on yields, as the FA carries a high ASP but also a large bill of materials. Supplying the complete FAU could raise billed content, but a larger purchased bill of materials also dilutes the relationship between revenue and value added. Our full-FAU ASP assumption requires Largan to capture that invoice scope.

Next year's product is grating-coupled and works for both CPO and NPO while edge-coupled designs are being explored with the customer. The current product does not go directly to end customers, but management expects to work directly with end customers on next-generation designs (likely 2028) as it progresses toward multi-row FAs.

To ramp production, Largan has bought several land parcels and will build new fabs from scratch, with construction finishing around 2030. It may outsource some capacity in the interim and said it cannot yet be sure capacity will meet demand.

Annual FAU TAM reaches US$1.85bn in 2028 and US$4.5bn in 2030 in our base case

We update our FAU (fiber array unit) TAM factoring CPO/NPO penetration into AI networking switches, NVIDIA multi-rack systems, scale-up on compute and switch boards, and non-NVIDIA accelerators. We assume one passive FAU assembly connecting a fiber array to a photonic engine with one unit per optical engine.

We assume one FAU per optical engine and 32 per switch ASIC in our base case, matching NVIDIA's 32x3.2T engines on its 102.4T Spectrum-X package. Broadcom's 102.4Tb/s Davisson uses 16 engines at 6.4Tb/s, so our bear case assumes and even mix of these two designs (24 per ASIC). We note that higher switch bandwidth does not automatically increase FAU count if engine bandwidth rises at the same rate. Our 2030 assumption maintains the 32 FAUs per switch based on bandwidth rising to 204.8Tb/s ASICs using 6.4Tb/s optical engines.

Figure 4: UBS scenario based on 32 FAUs per switch although switches to date carry a range of specifications

	TH5 Bailly	TH6 Davisson	Quantum-X Q3450	Spectrum-X SN6810	Spectrum-X SN6800
Year of Introduction	**2023**	**H225**	**H225**	**H226**	**H226**
Switch ASIC	Tomahawk 5	Tomahawk 6	Quantum-3	Spectrum 6	Spectrum 6
Bandwidth per Switch box	51.2T	102.4T	115.2T	102.4T	409.6T
Switch ASICs per box	1	1	4	1	4
OE bandwidth (Tb/s each way)	6.4T	6.4T	1.6T	3.2T	3.2T
Optical engines per ASIC	8	16	18	32	32
FAU per Engine	1	1	1	1	1
FAU per Switch	**8**	**16**	**72**	**32**	**128**

Source: Company Data, UBS

Our base case for the FAU market sums up 4 opportunities for CPO/NPO attach:

1. Scale-out and scale-across switches

We estimate annual switch ASIC shipments x CPO + NPO penetration x optical engines per ASIC x FAUs per engine. In 2028, 25% CPO/NPO penetration or 1.0m 102.4T switch ASICs at 32 FAUs each produces 8.0m FAUs. In 2030, 55% of 1.1mn 204.8T switch ASICs at 32 FAUs drives a market size of 19.4mn FAUs.

Figure 5: Scale-out/scale-across could drive 8.0/19.4mn FAUs in 2028/2030

	2028			2030			Assumption
Scale out/Scale across	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**	
Switch ASIC shipments (k units)	800	1,000	1,050	880	1,100	1,155	
NPO penetration	3%	5%	7%	8%	10%	12%	Optics on the board beside the ASIC
CPO penetration	10%	20%	25%	35%	45%	55%	Optics inside the switch package.
NPO/CPO switch ASICs (k units)	**104**	**250**	**336**	**378**	**605**	**774**	
Bandwidth per Switch (Tbps)	102.4	102.4	102.4	204.8	204.8	204.8	
Optical engine (Tb)	3.2	3.2	3.2	6.4	6.4	6.4	
FAUs per switch ASIC	24	32	32	24	32	32	NVIDIA: 32 OE per ASIC, Broadcom 16
Scale out/Scale across # of FAUs (mn)	**2.5**	**8.0**	**10.8**	**9.1**	**19.4**	**24.8**	

Source: UBS

2. NVIDIA's multi-rack connections for Rubin Ultra and Feynman

NVIDIA at its 2026 GTC San Jose proposed a multi-rack option to network together 8 Oberon NVL72 racks for the Rubin Ultra generation and 8 Kyber NVL144 racks. Feedback at the time showed possible transceiver or NPO connections for Rubin Ultra and CPO connections for Feynman. Our 2028 base case uses 110,000 racks, 15% multi-rack adoption and 75% CPO/NPO penetration, giving approximately 12,400 connected racks.

For 2030, our rack count falls to 35,000 racks as we assume Kyber would have 4x the number of GPUs, equivalent to 140k Oberon equivalents. We assume the 35,000 racks at 55% multi-rack adoption x 80% CPO/NPO attach gives 15,400 connected racks. We assume about 216 and 576 inter-rack FAUs per rack, respectively, driving a market opportunity of 2.7mn FAUs in 2028 and 8.9mn in 2030.

Figure 6: NVIDIA rack connected pods could drive 2.7mn/8.9mn FAUs in 2028/2030

	2028			2030			Assumption
NVIDIA Pod Connected GPUs	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**	
NV platform	Oberon / Rubin Ultra			Kyber / Feynman			
NV racks shipments (k)	95	110	125	30	35.0	40	UBS rack assumptions
8-rack pods penetration (%)	10%	15%	20%	40%	55%	70%	
NPO/CPO penetration within pods (%)	50%	75%	80%	65%	80%	95%	Pluggable FAUs excluded from this model
NPO/CPO connected racks (k)	**4.8**	**12.4**	**20.0**	**7.8**	**15.4**	**26.6**	
GPUs per rack	72	72	72	144	144	144	
Switches per rack	72	72	72	144	144	144	72 ASICs in Oberon, 144 per Kyber
Optical engine (Tb) bandwidth	3.2	3.2	3.2	6.4	6.4	6.4	
FAUs attached per switch	2.5	3.0	4.0	2.5	4.0	4.5	3.2T OE in 2028; 6.4T in 2030 (each way)
NVLink bandwidth per NVIDIA GPU	14.4	14.4	14.4	28.8	28.8	28.8	
Optical build-out (% of full)	56%	67%	89%	56%	89%	100%	% of rack to rack bandwidth wired to optics
FAUs per connected racks	**180**	**216**	**288**	**360**	**576**	**648**	
NV scale up # of FAUs (mn)	**0.9**	**2.7**	**5.8**	**2.8**	**8.9**	**17.2**	

Source: UBS. Switch count increases to connect to other racks

3. Initial CPO/NPO optical connections on compute and switch boards by 2030

We include by 2030 CPO native optical connections on compute and switch boards in 5% of racks each adding 1,152 FAUs across the GPUs and switches or 2.0mn FAUs by 2030.

Figure 7: NVIDIA compute board optics could drive 2.0mn FAUs in 2030

	2028			2030			Assumption
NVIDIA Compute Board Optics	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**	
NV racks shipments (k)	95	110	125	30	35.0	40	UBS rack assumptions
Optical GPU penetration	0%	0%	0%	3%	5%	15%	CPU-native GPUs
Compute tray FAUs #	0	0	0	144	144	144	36 trays x 16
GPUs per rack	72	72	72	144	144	144	
Optical engine (Tb)	3.2	3.2	3.2	6.4	6.4	6.4	
Optical build-out (% of full)	56%	67%	89%	56%	89%	100%	% of rack to rack bandwidth wired to optics
NVLink bandwidth per NVIDIA GPU	14.4	14.4	14.4	28.8	28.8	28.8	
FAUs per connected racks	0	0	0	720	1,152	1,296	
NV optical GPU # of FAUs (mn)	**0.0**	**0.0**	**0.0**	**0.6**	**2.0**	**7.8**	

Source: UBS.

4. Non NVIDIA scale-up opportunity

Non-NVIDIA scale-up uses accelerator rack shipments x optical attach rate x FAUs per rack, producing about 0.5m units in 2028 and 3.7m in 2030.

Figure 8: ASIC market could drive additional compute board optics could drive 3.7mn FAUs in 2030

	2028			2030			Assumption
Non-NV scale up	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**	
ASIC rack shipments (k)	43.6	48.5	50.9	76.6	85.1	89.3	*Trainium 4/5/6, MI500, MI600, Jalapeno
Addressable NPO/CPO attach rate (%)	0%	5%	10%	5%	15%	25%	
NPO/CPO connected racks (k)	**0.0**	**2.4**	**5.1**	**3.8**	**12.8**	**22.3**	
Accelerators per rack	72	72	72	72	72	72	Average accelerators per rack
FAUs per accelerator	2.5	3.0	4.0	2.5	4.0	4.5	
Optical build-out (% of full)	56%	67%	89%	56%	89%	100%	% of rack to rack bandwidth wired to optics
Scale-up per non-NVIDIA accelerator (Tb)	14.4	14.4	14.4	28.8	28.8	28.8	
FAUs per connected racks	180	216	288	180	288	324	
Non-NV # of FAUs (mn)	**0.0**	**0.5**	**1.5**	**0.7**	**3.7**	**7.2**	

Source: UBS

Overall market opportunity: 12.3mn/35.6mn generates a US$1.85bn/US $4.5bn TAM in 2028/2030

We sum four pools: scale-out/scale-across switches, NVIDIA multi-rack pods, CPO native NVIDIA GPUs and non-NVIDIA scale-up. We add 10%/5% for FAUs lost at downstream attach and test and for field spares, driving a market size of in our base case of 12.3mn FAUs in 2028 and 35.6mn in 2030. At US$150/US$125 per FAU, that is a US$1.85bn market in 2028 and US$4.5bn in 2030 (bear/bull US$0.5bn/US$3.5bn in 2028; US $1.4bn/US$9.0bn in 2030). The expansion to 2030 comes from wider optical adoption and greater content in scale-up systems, despite an assumed FAU ASP decline from US $150 to US$125.

Figure 9: Summary of FAU drivers: 12.3mn/35.6mn base case in 2028-2030

	2028			2030		
Summary of FAU drivers	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**
Scale out/Scale across # of FAUs (mn)	2.5	8.0	10.8	9.1	19.4	24.8
NV scale up # of FAUs (mn)	0.9	2.7	5.8	2.8	8.9	17.2
NV optical GPU # of FAUs (mn)	0.0	0.0	0.0	0.6	2.0	7.8
Non-NV # of FAUs (mn)	0.0	0.5	1.5	0.7	3.7	7.2
Downstream yield loss and spares	10%	10%	10%	5%	5%	5%
FAUs (mn units)	**3.7**	**12.3**	**19.8**	**13.9**	**35.6**	**59.9**
FAU ASP	$125	$150	$175	$100	$125	$150
Total FAU TAM (US$bn)	**$0.5**	**$1.8**	**$3.5**	**$1.4**	**$4.5**	**$9.0**

Source: UBS

Scenario analysis on Largan's CPO opportunity

Our base case gives Largan the benefit of the doubt for successfully ramping up a full FAU production and capturing 25% of the overall FAU TAM, yielding NT$35 of 2028E EPS, about 16% of our 2028E lens EPS. A 25% share means 3.1mn FAUs in its first full year of production: about 29% of NVIDIA-linked units and roughly three-quarters of the allocation our checks suggest its module partner holds. We book these at the full US $150 FAU price and 50% gross margin, allowing for an early-supply premium even though the optics, V-grooves and carrier parts are bought in. That gives 2028E CPO sales of US$462mn, rising to US$1.1bn and NT$84 of EPS in 2030 on a larger market.

Our bear case (15% share, 45% GM) gives NT$5 of CPO EPS in 2028E and NT$14 in 2030E; our bull case (35% share, 55% GM) gives NT$103 and NT$267. Each 5ppt of share moves 2028E CPO EPS by ~NT$6.9 (~NT$340/share at 49x), and each 5ppt of gross margin by ~NT$4.6 (~NT$225/share). If gross margin were 35% rather than 50%, holding operating expenses at 12% of sales, EPS would fall to approximately NT$21. A next-generation shift in lens or coupling design could also change content and alignment requirements including a potential metalens design which could invite in different suppliers.

Our 2030 base case implies ~8.9mn Largan FAUs a year (21mn in the upside case), though with risks around competition in a crowded field, yields and technology. A shift to semiconductor-made optics or metalens would loosen alignment tolerances and erode the precision premium behind Largan's pitch. The upside lies in direct work with cloud providers on multi-row FAs from 2028, where Largan's multi-layer alignment is most differentiated.

Figure 10: Largan downside/base/upside cases for 2028 and 2030

Largan CPO	2028			2030		
Scenario analysis	**Downside**	**Base**	**Upside**	**Downside**	**Base**	**Upside**
FAUs (mn units)	3.7	12.3	19.8	13.9	35.6	59.9
FAU ASP	125	150	175	100	125	150
Total FAU TAM (US$mn)	**$461**	**$1,847**	**$3,461**	**$1,389**	**$4,452**	**$8,980**
Largan Sales (US$mn)	**$69.1**	**$461.9**	**$1,211.3**	**$208.3**	**$1,113.1**	**$3,142.8**
Largan share of the TAM (%)	15%	25%	35%	15%	25%	35%
GM %	45%	50%	55%	45%	50%	55%
Pretax %	33%	38%	43%	33%	38%	43%
Pretax income (US$mn)	**$22.8**	**$175.5**	**$520.8**	**$68.7**	**$423.0**	**$1,351.4**
Tax	18%	18%	18%	18%	18%	18%
Net income (US$mn)	**$18.7**	**$143.9**	**$427.1**	**$56.4**	**$346.8**	**$1,108.2**
Net income (NT$mn)	**$589**	**$4,533**	**$13,453**	**$1,776**	**$10,925**	**$34,907**
EPS (NT$)	**$5**	**$35**	**$103**	**$14**	**$84**	**$267**
Shares	130.8	130.8	130.8	130.8	130.8	130.8

Source: UBS

Valuation and investment implications

We roll our valuation forward to 2028E and value Largan on a sum of the parts. We value the lens business at NT$3,300 (15x 2028E EPS, the middle of Largan's 13–18x historical range) and CPO at NT$1,700 (49x our base-case 2028E CPO EPS of NT$35, near the top of optical peers' 14–54x). Our base case gives Largan 25% of a 12.3mn-unit, US$1.85bn FAU market at US$150 per unit and a 50% gross margin. The 49x multiple also assigns a substantial value to a business whose complete assembly still requires qualification and whose margins still need to demonstrate can reach similar levels to its mature smartphone business. At ~NT$6,660 recent share price level, the shares imply ~NT$69 of 2028E CPO EPS, about twice our base case.

Upside risks include faster full-FAU qualification, firm volume commitments from additional customers, stronger lens demand and better yields. Near-term catalysts are the customer's capacity forecast and its visit to the new fab in the coming month. A large forecast or faster FAU plus carrier qualification would be upside risks to our sell.

Figure 11: UBS SOTP valuation for Largan

(NT$)		UBSe			Prior UBS		Upside Case			Downside Case		
	2025	**2026**	**2027**	**2028**	**2026**	**2027**	**2026**	**2027**	**2028**	**2026**	**2027**	**2028**
Lens Business ex-CPO												
Sales	$61,148	$61,550	$67,438	$79,464	$67,869	$77,189	$62,166	$70,135	$82,642	$59,704	$64,066	$75,491
YoY Growth	2.8%	0.7%	9.6%	17.8%	11.0%	13.7%	1.7%	12.8%	17.8%	-2.4%	7.3%	17.8%
Gross Profit	**$30,837**	**$28,386**	**$32,008**	**$38,823**	**$33,718**	**$38,405**	**$28,981**	**$34,410**	**$41,699**	**$26,997**	**$29,446**	**$35,750**
GM%	**50.4%**	**46.1%**	**47.5%**	**48.9%**	**49.7%**	**49.8%**	**46.6%**	**49.1%**	**50.5%**	**45.2%**	**46.0%**	**47.4%**
Opex	$7,279	$7,247	$7,857	$8,714	$7,781	$8,609	$7,283	$8,015	$8,888	$7,138	$7,661	$8,496
Op Profit	**$23,558**	**$21,139**	**$24,150**	**$30,110**	**25,937**	**29,796**	**21,697**	**$26,396**	**$32,811**	**19,858**	**$21,785**	**$27,254**
OpM%	**38.5%**	**34.3%**	**35.8%**	**37.9%**	**38.2%**	**38.6%**	**34.9%**	**37.6%**	**39.7%**	**33.3%**	**34.0%**	**36.1%**
Non-Op	$2,342	$4,847	$4,467	$4,595	$4,518	$4,467	$4,847	$4,467	$4,595	$4,847	$4,467	$4,595
Pretax	$25,900	$25,986	$28,618	$34,705	$30,455	$34,263	$26,545	$30,863	$37,405	$24,706	$26,253	$31,849
Tax	($4,340)	($4,379)	($5,050)	($5,900)	($5,369)	($6,244)	($4,473)	($5,447)	($6,601)	($4,163)	($4,633)	($5,621)
Net income	**$21,275**	**$21,385**	**$23,567**	**$28,805**	**$24,877**	**$28,019**	**$21,849**	**$25,416**	**$30,804**	**$20,321**	**$21,620**	**$26,228**
EPS	**$159.41**	**$163.33**	**$180.00**	**$220.00**	**$190.00**	**$214.00**	**$166.88**	**$194.12**	**$235.28**	**$155.20**	**$165.13**	**$200.32**
Shares	133	131	131	131	131	131	131	131	131	131	131	131
EPS vs. base case							**-94.1%**	**7.8%**	**6.9%**	**-5.0%**	**-8.3%**	**-8.9%**
Current Price		**$6,660**										
P/E				15		15			16			14
Implied Valuation				**$3,300**		**$3,270**			**$3,670**			**$2,810**
CPO Scenario												
EPS				**$35**		**$35**			**$103**			**$5**
P/E				49		50			52			47
Implied Valuation				**$1,700**		**$1,730**			**$5,300**			**$210**
Total Largan												
Implied Share Price				**$5,000**		**$5,000**			**$8,970**			**$3,020**
Upside/Downside				**-25%**					**35%**			**-55%**

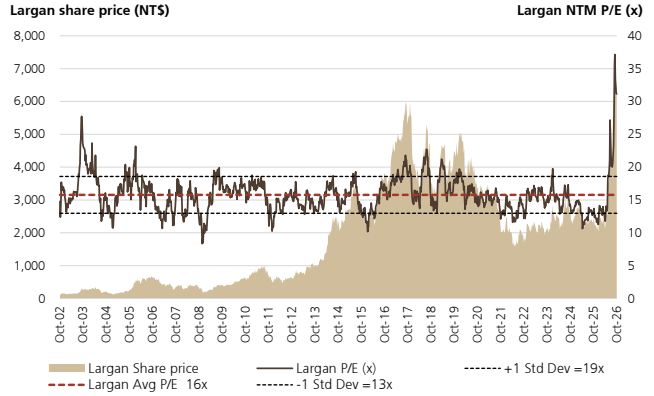
Source: Company data, UBS estimates

Figure 12: Optical component peers average 27x 2028 with a broad range of 12-75x 2028 P/E

Optical		Market	Price	Price	P/E		EPS (LC$)				Earnings growth (%)			PE (x)			ROE (%)			Historical P/E (x)		
Company	Ticker	cap (US$m)	11-Oct (LC$)	target (LC$)	target (x)	Rating	2025	2026E	2027E	2028E	2026E	2027E	2028E	2026E	2027E	2028E	2026E	2027E	2028E	-1 Std Dev	Avg.	+1 Std Dev
Lens																						
Largan	3008.TW	27,378	6,660.0	5,000	23.4	Sell	176.92	188.21	214.00	250.66	4.4	13.7	17.1	35.4	31.1	26.6	12.7	13.7	14.8	13.1	15.9	18.8
Genius	3406.TW	3,741	1,060.0	1,250	16.9	Buy	36.28	38.72	41.47	47.75	6.7	7.1	15.1	27.4	25.6	22.2	15.8	15.6	16.4	1.1	2.3	3.5
Average											**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**	**5.5**
CPO Supply Chain																						
Lumentum	LITE.O	93,344	1,103.4	960	40.4	Neutral	2.12	10.52	23.78	35.81	434.4	160.4	51.7	52.0	56.0	37.9	27.1	36.1	37.9	8.5	19.4	30.3
Corning	GLW.N	137,095	156.7	201	42.1	Buy	2.57	3.34	4.78	7.12	30.6	43.5	47.3	47.7	33.4	22.4	22.9	31.8	41.1	-168.2	32.9	234.0
Coherent	COHR.N	123,129	312.6	-	-	-	4.49	7.42	11.74	16.82	86.0	64.0	42.0	42.1	26.6	18.6	13.5	19.7	23.8	14.1	19.6	25.1
Senko	9069.T	416,040	2,368.0	-	-	-	119.98	140.89	159.70	171.34	19.5	12.8	7.4	16.8	14.8	13.8	10.4	11.3	11.6	9.2	11.6	14.0
TFC	300394.SZ	256,388	235.0	-	-	-	2.02	3.32	5.36	6.88	65.5	60.7	28.5	70.8	43.9	34.2	47.0	47.4	42.1	21.2	33.9	46.5
FOCI	3363.TWO	75,798	667.0	-	-	-	0.85	0.22	13.89	43.01	(66.3)	6684.0	210.2	3,102.3	48.0	15.5	-1.6	39.2	74.6	51.6	115.2	178.8
EZconn	6442.TW	146,044	1,785.0	-	-	-	20.95	62.01	130.00	NULL	206.9	109.6	n.a.	28.8	13.7	n.a.	56.1	63.9	NULL	12.9	23.7	34.5
Himax	HIMX.N	2,320	13.3	-	-	-	0.32	0.52	1.01	1.96	62.3	88.2	93.5	25.5	13.2	6.8	10.5	14.5	26.1	0.6	21.5	42.3
Browave	3163.TWO	65,204	689.0	-	-	-	4.36	9.33	21.17	NULL	n.a.	n.a.	n.a.	73.8	32.5	n.a.	22.2	38.6	NULL	10.1	32.2	54.2
VisEra	6789.TW	153,102	481.0	-	-	-	3.60	6.13	7.45	8.95	70.9	21.5	20.2	78.4	64.6	53.7	10.7	13.3	12.2	31.3	50.4	69.6
Average											**101.1**	**805.0**	**62.6**	**353.8**	**34.7**	**25.4**	**21.9**	**31.6**	**33.7**	**-0.9**	**36.0**	**72.9**
Optical Average											**76.4**	**606.4**	**50.7**	**325.1**	**35.5**	**27.2**	**19.9**	**27.5**	**28.6**	**1.2**	**30.5**	**59.8**

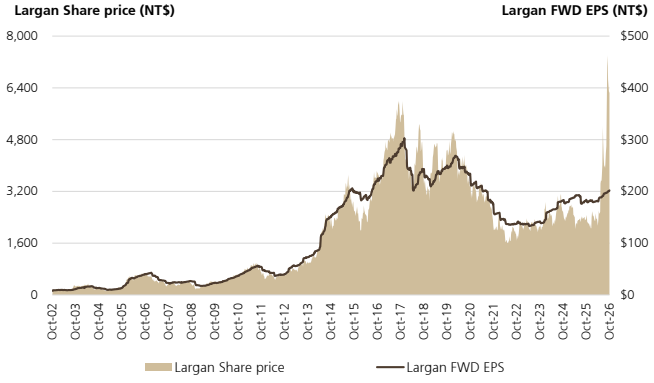
Source: UBSe, Eikon

Figure 13: Share price rally has re-rated Largan well above its historical range



Source: LSEG Workspace

Figure 14: Largan's share price has run well ahead of forward EPS as investors price in CPO



Source: LSEG Workspace

Figure 15: UBS P&L for Largan - mild growth at lower margins in recent years

Ticker		3008.TW	Current Price (NT$)			$6,660
Shares outstanding (mn)		130.8	Mkt cap (NT$ mn)			$871,115
Year	**2022**	**2023**	**2024**	**2025**	**2026E**	**2027E**
Revenues (NT$ mn)	**$47,675**	**$48,842**	**$59,458**	**$61,148**	**$61,550**	**$67,438**
YoY Growth (%)		**2.4%**	**21.7%**	**2.8%**	**0.7%**	**9.6%**
Gross profit (NT$mn)	$26,083	$23,794	$31,209	$30,837	$28,386	$32,008
GM (%)	**54.7%**	**48.7%**	**52.5%**	**50.4%**	**46.1%**	**47.5%**
Operating profit (NT$mn)	$20,384	$17,807	$24,033	$23,558	$21,139	$24,150
OPM (%)	**42.8%**	**36.5%**	**40.4%**	**38.5%**	**34.3%**	**35.8%**
Net profit (NT$ mn)	$22,625	$17,902	$25,915	$21,275	$21,385	$23,567
EPS (NT$)	**$169.52**	**$134.13**	**$194.17**	**$159.41**	**$163.33**	**$180.00**
EPS growth (%)		-21%	45%	-18%	2%	10%
P/E (x)	**39.3**	**49.7**	**34.3**	**41.8**	**40.8**	**37.0**
Dividend yield (%)	2%	1.1%	1.2%	1.3%	2.0%	1.2%
P/B (x)	**5.7**	**5.4**	**4.8**	**4.7**	**4.4**	**4.1**
ROE (%)	15%	11%	14%	11%	11%	12%

Quarters	Q325	Q425	Q126	Q226	Q326E	Q426E
Revenues (NT$ mn)	**$17,677**	**$17,219**	**$15,544**	**$13,665**	**$15,651**	**$16,690**
QoQ Growth (%)		**-2.6%**	**-9.7%**	**-12.1%**	**14.5%**	**6.6%**
Gross profit (NT$mn)	$8,352	$8,260	$7,680	$6,753	$6,607	$7,346
GM (%)	**47.2%**	**48.0%**	**49.4%**	**49.4%**	**42.2%**	**44.0%**
Operating profit (NT$mn)	$6,265	$6,329	$5,812	$5,148	$4,735	$5,443
OPM (%)	**35.4%**	**36.8%**	**37.4%**	**37.7%**	**30.3%**	**32.6%**
Net profit (NT$ mn)	$7,080	$6,720	$6,123	$4,670	$5,182	$5,409
EPS (NT$)	**$53.05**	**$50.35**	**$45.88**	**$34.99**	**$39.58**	**$41.32**

Source: Company data, UBS estimates. Note: 2026E EPS based on reported.

Figure 16: UBS Cash Flow Summary - staying positive to invest in CPO

Annual (NT$mn)	2021	2022	2023	2024	2025	2026E	2027E	21-25 Avg
Revenue	$46,962	$47,675	$48,842	$59,458	$61,148	$61,550	$67,438	$52,817
Capital spending	$5,850	$8,256	$8,080	$11,126	$12,338	$11,721	$11,135	$9,130
Capex/revenue (%)	**12%**	**17%**	**17%**	**19%**	**20%**	**19%**	**17%**	**17%**
Dep and amort	$4,650	$5,031	$5,299	$6,038	$7,511	$9,221	$10,460	$5,706
Depr/revenue (%)	**10%**	**11%**	**11%**	**10%**	**12%**	**15%**	**16%**	**11%**
Operating cash flow	$20,878	$44,207	$18,199	$31,579	$26,080	$20,614	$33,003	$28,188
Free cash flow	**$15,028**	**$35,951**	**$10,118**	**$20,453**	**$13,743**	**$8,893**	**$21,868**	**$19,058**

FCF and Dividends	2021	2022	2023	2024	2025	2026E	2027E	21-25 Avg
FCF / share (NT$)	$112	$269	$76	$153	$103	$68	$167	$143
FCF / EV Yield (%)	**2%**	**5%**	**1%**	**3%**	**2%**	**1%**	**3%**	**3%**
FCF Yield (%)	**2%**	**4%**	**1%**	**2%**	**2%**	**1%**	**3%**	**2%**
Dividend/share (NT$)	$91.51	$109.66	$72.50	$81.00	$85.50	$131.95	$81.89	$88.03
Payout rate (%)	**50%**	**79%**	**43%**	**60%**	**44%**	**83%**	**50%**	**55%**
Dividend amount (NT$)	$12,274	$14,636	$9,676	$10,811	$11,412	$17,276	$10,722	$11,762
Pre-Dividend close (NT$)	$3,040.00	$1,990.00	$2,310.00	$3,120.00	$2,260.00	$6,660.00	$6,660.00	
Dividend Yield (%)	**3.0%**	**5.5%**	**3.1%**	**2.6%**	**3.8%**	**2.0%**	**1.2%**	**3.6%**

Source: Company data, UBS estimates

Largan Precision (3008.TW)

Income Statement (NT$m)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Revenues	**48,842**	**59,458**	**61,148**	**61,550**	*0.7*	**67,438**	*9.6*	**79,464**	**78,546**	**81,569**
Gross profit	23,794	31,209	30,837	28,386	*-7.9*	32,008	*12.8*	38,823	38,398	39,903
EBITDA (UBS)	**23,106**	**30,070**	**31,069**	**30,360**	*-2.3*	**34,610**	*14.0*	**41,028**	**41,108**	**42,938**
Depreciation & amortisation	(5,299)	(6,038)	(7,511)	(9,221)	*-22.8*	(10,460)	*-13.4*	(10,918)	(11,621)	(12,290)
EBIT (UBS)	**17,807**	**24,033**	**23,558**	**21,139**	*-10.3*	**24,150**	*14.2*	**30,110**	**29,487**	**30,648**
Associates & investment income	212	0	3	12	*352.1*	11	*-7.1*	11	11	11
Other non-operating income	294	303	419	458	*9.5*	200	*-56.4*	200	200	200
Net interest	3,934	4,404	4,259	3,946	*-7.3*	4,256	*7.9*	4,384	4,516	4,651
Exceptionals (incl goodwill)	(145)	3,435	(2,338)	432	*-*	0	*-*	0	0	0
Pre-tax profit	**22,102**	**32,174**	**25,900**	**25,986**	*0.3*	**28,618**	*10.1*	**34,705**	**34,213**	**35,510**
Tax	(4,200)	(5,963)	(4,340)	(4,379)	*-0.9*	(5,050)	*-15.3*	(5,900)	(5,816)	(6,037)
Profit after tax	**17,902**	**26,211**	**21,560**	**21,607**	*0.2*	**23,567**	*9.1*	**28,805**	**28,397**	**29,473**
Preference dividends	0	0	0	0	*-*	0	*-*	0	0	0
Minorities	0	(296)	(285)	(222)	*22.1*	0	*-*	0	0	0
Extraordinary items	0	0	0	0	*-*	0	*-*	0	0	0
Net earnings (local GAAP)	**17,902**	**25,915**	**21,275**	**21,385**	*0.5*	**23,567**	*10.2*	**28,805**	**28,397**	**29,473**
Net earnings (UBS)	**18,047**	**22,480**	**23,613**	**20,953**	*-11.3*	**23,567**	*12.5*	**28,805**	**28,397**	**29,473**
Tax rate (%)	19.0	18.5	16.8	16.9	*0.6*	17.6	*4.7*	17.0	17.0	17.0

Per Share (NT$)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
EPS (UBS, diluted)	135.22	168.43	176.92	160.04	*-9.5*	180.00	*12.5*	220.00	216.89	225.11
EPS (local GAAP, diluted)	134.13	194.17	159.41	163.33	*2.5*	180.00	*10.2*	220.00	216.89	225.11
EPS (UBS, basic)	135.22	168.43	176.92	160.04	*-9.5*	180.00	*12.5*	220.00	216.89	225.11
DPS (net) (NT$)	85.50	67.50	97.50	80.00	*-17.9*	81.97	*2.5*	90.34	110.41	108.85
Cash EPS (UBS, diluted) [1]	174.92	213.67	233.20	230.47	*-1.2*	259.89	*12.8*	303.39	305.65	318.97
Book value per share	1,234.64	1,382.91	1,430.07	1,518.08	*6.2*	1,616.11	*6.5*	1,745.87	1,849.64	1,963.23
Average shares (diluted)	133	133	133	131	*-1.9*	131	*0.0*	131	131	131

Balance Sheet (NT$m)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Cash and equivalents	111,655	117,985	122,309	109,131	*-10.8*	120,267	*10.2*	127,842	144,348	160,697
Other current assets	22,666	27,779	31,977	42,218	*32.0*	45,034	*6.7*	56,368	55,712	57,851
Total current assets	**134,321**	**145,764**	**154,285**	**151,350**	*-1.9*	**165,301**	*9.2*	**184,210**	**200,060**	**218,547**
Net tangible fixed assets	41,135	46,936	51,472	54,428	*5.7*	55,103	*1.2*	54,763	53,191	50,448
Net intangible fixed assets	0	0	0	0	*-*	0	*-*	0	0	0
Investments / other assets	19,682	23,827	15,029	23,254	*54.7*	24,367	*4.8*	24,215	24,225	24,236
Total assets	**195,138**	**216,527**	**220,787**	**229,032**	*3.7*	**244,770**	*6.9*	**263,188**	**277,476**	**293,232**
Trade payables & other ST liabilities	29,517	30,375	29,749	30,078	*1.1*	32,982	*9.7*	34,573	35,276	36,158
Short term debt	0	203	0	0	*-*	0	*-*	0	0	0
Total current liabilities	**29,517**	**30,578**	**29,749**	**30,078**	*1.1*	**32,982**	*9.7*	**34,573**	**35,276**	**36,158**
Long term debt	0	0	0	0	*-*	0	*-*	0	0	0
Other long term liabilities	110	561	171	194	*13.7*	194	*0.0*	31	31	31
Preferred shares	0	0	0	0	*-*	0	*-*	0	0	0
Total liabilities (incl pref shares)	**29,627**	**31,139**	**29,919**	**30,272**	*1.2*	**33,176**	*9.6*	**34,604**	**35,307**	**36,189**
Common s/h equity	**165,510**	**185,388**	**190,868**	**198,760**	*4.1*	**211,594**	*6.5*	**228,584**	**242,170**	**257,042**
Minority interests	0	0	0	0	*-*	0	*-*	0	0	0
Total liabilities & equity	**195,138**	**216,527**	**220,787**	**229,032**	*3.7*	**244,770**	*6.9*	**263,188**	**277,476**	**293,232**

Cash Flow (NT$m)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Net income (before pref divs)	17,902	25,915	21,275	21,385	*0.5*	23,567	*10.2*	28,805	28,397	29,473
Depreciation & amortisation	5,299	6,038	7,511	9,221	*22.8*	10,460	*13.4*	10,918	11,621	12,290
Net change in working capital	(8,990)	(6,633)	(7,328)	(1,082)	*85.2*	(6,064)	*NM*	(9,743)	1,359	(1,256)
Other operating	(212)	295	282	210	*-25.5*	5,040	*NM*	(11)	(11)	(11)
Operating cash flow	**13,999**	**25,615**	**21,740**	**29,735**	*36.8*	**33,003**	*11.0*	**29,968**	**41,366**	**40,496**
Tangible capital expenditure	(8,080)	(11,126)	(12,338)	(11,721)	*5.0*	(11,135)	*5.0*	(10,578)	(10,049)	(9,547)
Intangible capital expenditure	0	0	0	0	*-*	0	*-*	0	0	0
Net (acquisitions) & disposals	(2,111)	(13)	(1,200)	962	*-*	0	*-*	0	0	0
Other investing	(959)	(4,927)	4,302	(965)	*-*	0	*-*	0	0	0
Investing cash flow	**(11,151)**	**(16,066)**	**(9,236)**	**(11,724)**	*-26.9*	**(11,135)**	*5.0*	**(10,578)**	**(10,049)**	**(9,547)**
Equity dividends paid	(9,676)	(10,811)	(11,412)	(17,276)	*-51.4*	(10,732)	*37.9*	(11,816)	(14,811)	(14,601)
Share issues / (buybacks)	0	0	0	0	*-*	0	*-*	0	0	0
Other financing	(34)	1,255	(2,937)	(4,133)	*-40.7*	0	*-*	0	0	0
Change in debt & pref shares	(19)	212	(212)	0	*-*	0	*-*	0	0	0
Financing cash flow	**(9,729)**	**(9,344)**	**(14,561)**	**(21,408)**	*-47.0*	**(10,732)**	*49.9*	**(11,816)**	**(14,811)**	**(14,601)**
Cash flow inc/(dec) in cash	**(6,881)**	**205**	**(2,057)**	**(3,398)**	*-65.2*	**11,136**	*-*	**7,575**	**16,506**	**16,349**
FX / non cash items	6,823	6,125	6,380	(9,780)	*-*	0	*-*	0	0	0
Balance sheet inc/(dec) in cash	**(59)**	**6,330**	**4,324**	**(13,177)**	*-*	**11,136**	*-*	**7,575**	**16,506**	**16,349**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.[1] Cash EPS (UBS, diluted) is calculated using UBS net income adding back depreciation and amortization.

Largan Precision (3008.TW)

Valuation (x)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
P/E (local GAAP, diluted)	16.4	13.1	14.8	40.8	37.0	30.3	30.7	29.6
P/E (UBS, diluted)	16.2	15.1	13.3	41.6	37.0	30.3	30.7	29.6
P/CEPS	12.6	11.9	10.1	28.9	25.6	22.0	21.8	20.9
Equity FCF (UBS) yield %	2.0	4.3	3.0	2.1	2.5	2.2	3.6	3.5
Dividend yield (net) %	3.9	2.7	4.1	1.2	1.2	1.4	1.7	1.6
P/BV	1.8	1.8	1.6	4.4	4.1	3.8	3.6	3.4
EV/revenues (core)	3.6	3.6	3.1	NM	NM	9.3	9.3	8.7
EV/EBITDA (UBS core)	7.7	7.1	6.0	24.7	21.7	18.1	17.7	16.6
EV/EBIT (core)	10.0	8.9	8.0	35.4	31.1	24.6	24.7	23.2
EV/OpFCF (core)	8.1	7.4	6.3	25.1	22.0	18.0	17.7	16.5
EV/op. invested capital	3.9	4.1	3.3	10.7	9.3	8.6	8.2	8.2

Enterprise value (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Market cap.	294,465	340,820	315,906	874,578	874,578	874,578	874,578	874,578
Net debt (cash)	(111,675)	(114,718)	(120,045)	(115,720)	(114,699)	(124,054)	(136,095)	(152,522)
Buy out of minorities	0	0	0	0	0	0	0	0
Pension provisions/other	0	0	0	0	0	0	0	0
Total enterprise value	**182,790**	**226,102**	**195,861**	**758,858**	**759,879**	**750,524**	**738,483**	**722,055**
Non core assets	(4,916)	(12,460)	(8,517)	(9,934)	(9,945)	(9,955)	(9,966)	(9,976)
Core enterprise value	**177,874**	**213,642**	**187,344**	**748,924**	**749,934**	**740,568**	**728,517**	**712,079**

Growth (%)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenue	2.4	21.7	2.8	0.7	9.6	17.8	(1.2)	3.8
EBITDA (UBS)	(9.1)	30.1	3.3	(2.3)	14.0	18.5	0.2	4.5
EBIT (UBS)	(12.6)	35.0	(2.0)	(10.3)	14.2	24.7	(2.1)	3.9
EPS (UBS, diluted)	3.8	24.6	5.0	(9.5)	12.5	22.2	(1.4)	3.8
Net DPS	21.9	(21.1)	44.4	(17.9)	2.5	10.2	22.2	(1.4)

Margins & Profitability (%)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Gross profit margin	48.7	52.5	50.4	46.1	47.5	48.9	48.9	48.9
EBITDA margin	47.3	50.6	50.8	49.3	51.3	51.6	52.3	52.6
EBIT (UBS) margin	36.5	40.4	38.5	34.3	35.8	37.9	37.5	37.6
Net earnings (UBS) margin	37.0	37.8	38.6	34.0	34.9	36.2	36.2	36.1
ROIC (EBIT)	39.3	46.2	40.9	30.3	30.1	35.1	33.1	35.3
ROIC post tax	31.8	36.6	34.6	25.1	24.8	29.1	27.5	29.3
ROE (UBS)	11.3	12.8	12.6	10.8	11.5	13.1	12.1	11.8

Capital structure & Coverage (x)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Net debt / EBITDA	(4.8)	(3.9)	(3.9)	(3.6)	(3.5)	(3.1)	(3.5)	(3.7)
Net debt / total equity %	(67.5)	(63.5)	(64.1)	(54.9)	(56.8)	(55.9)	(59.6)	(62.5)
Net debt / (net debt + total equity) %	NM	NM	NM	NM	NM	NM	NM	NM
Net debt/EV %	(61.1)	(50.7)	(61.3)	(15.2)	(15.1)	(16.5)	(18.4)	(21.1)
Capex / depreciation %	152.5	184.3	164.3	127.1	106.5	96.9	86.5	77.7
Capex / revenue %	16.5	18.7	20.2	19.0	16.5	13.3	12.8	11.7
EBIT / net interest	-	-	-	-	-	-	-	-
Dividend cover (UBS)	1.6	2.5	1.8	2.0	2.2	2.4	2.0	2.1
Div. payout ratio (UBS) %	63.2	40.1	55.1	50.0	45.5	41.1	50.9	48.4

Revenues by division (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	48,842	59,458	61,148	61,550	67,438	79,464	78,546	81,569
Total	**48,842**	**59,458**	**61,148**	**61,550**	**67,438**	**79,464**	**78,546**	**81,569**

EBIT (UBS) by division (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	17,807	24,033	23,558	21,139	24,150	30,110	29,487	30,648
Total	**17,807**	**24,033**	**23,558**	**21,139**	**24,150**	**30,110**	**29,487**	**30,648**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.

Forecast returns

Forecast price appreciation	-24.9%
Forecast dividend yield	1.2%
Forecast stock return	-23.7%
Market return assumption	6.3%
Forecast excess return	-30.0%

Company Description

Founded in 1987, Largan Precision (Largan) is one of the world's leading manufacturers of lenses used in cameras and IT products. It specialises in plastic/hybrid lenses and, to a lesser extent, glass lenses and modules. In 2022, 20MP+ camera lenses were 10-20% of Largan's shipment mix, 10MP lenses were 50-60%, 8MP lenses were 0-10% and other lenses were 30-40%.

Valuation Method and Risk Statement

Our price target is based on SOTP valuation on the CPO opportunity and traditional lens business.
We think Largan's downside risks include: 1) ongoing pricing pressure from smartphone OEM clients and increasingly intense competition among lens suppliers; and 2) a slower-than-expected smartphone demand recovery in China, and 3) CPO commercial success below the market's elevated expectations.
We believe the key upside risk is a faster-than-expected CPO qualification process at Largan. In addition, if the company demonstrates exceptionally efficient and highly accurate automation capabilities, it could support greater market share gains within the industry than we currently anticipate.

Quantitative Research Review

UBS Global Research publishes a quantitative assessment of its analysts' responses to certain questions about the likelihood of an occurrence of a number of short term factors in a product known as the 'Quantitative Research Review'. The views for this month can be found below. Views contained in this assessment on a particular stock reflect only the views on those short term factors which are a different timeframe to the 12-month timeframe reflected in any equity rating set out in this note. For previous responses please make reference to (i) previous UBS Global Research reports; and (ii) where no applicable research report was published that month, the Quantitative Research Review which can be found at https://neo.ubs.com/quantitative, or contact your UBS sales representative for access to the report or the Quantitative Research Team on ubs-quant-answers@ubs.com. A consolidated report which contains all responses is also available and again you should contact your UBS sales representative for details and pricing or the Quantitative Research Team on the email above.

Largan Precision

Question	**Response**
1. Is the industry structure facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting worse, 3 = no change, 5 = getting better, N/A = no view)	3
2. Is the regulatory/government environment facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting tougher 3 = no change, 5 = getting better, N/A = no view)	3
3. Over the last 3-6 months in broad terms have things been improving/no change/getting worse for this stock? Rate on a scale of 1-5 (1 = getting a lot worse, 3 = not much change, 5 = getting a lot better, N/A = no view)	4
4. Relative to the current CONSENSUS EPS forecast, is the next company EPS update likely to lead to: (1 = negative surprise vs consensus, 3 = in-line with consensus, 5 = positive surprise vs consensus expectations, N/A = no view)	2
5. What's driving the difference?	We expect downside to consensus in 2026 due to slower smartphone sales through August. We indicate things getting better for the stock in the past 3-6 months on the smartphone traction.
6. Relative to YOUR current earnings forecast, is there relatively greater risk at the next earnings result of:(1 = downside skew risk to earnings, 3 = equal upside or downside risk to earnings, 5 = upside skew risk to earnings, N/A = no view)	2
7. What's driving the difference?	
8. Is there an upcoming catalyst for the company over the next three months?	Negative Catalyst
9. Is there an actual or approximate date for the catalyst?	
10. Is the catalyst date an actual or approximate date?	
11. What is the catalyst?	

Required Disclosures

This document has been prepared by UBS Securities Pte. Ltd., Taipei Branch, an affiliate of UBS AG. UBS AG, its subsidiaries, branches and affiliates, including former Credit Suisse AG and its subsidiaries, branches and affiliates are referred to herein as "UBS".

For information on the ways in which UBS manages conflicts and maintains independence of its UBS Global Research product; historical performance information; certain additional disclosures concerning UBS Global Research recommendations; and terms and conditions for certain third party data used in research report, please visit https://www.ubs.com/disclosures. Unless otherwise indicated, information and data in this report are based on company disclosures including but not limited to annual, interim, quarterly reports and other company announcements. The figures contained in performance charts refer to the past; past performance is not a reliable indicator of future results. Additional information will be made available upon request. UBS Securities Co. Limited is licensed to conduct securities investment consultancy businesses by the China Securities Regulatory Commission. UBS acts or may act as principal in the debt securities (or in related derivatives) that may be the subject of this report. This recommendation was finalized on: 11 October 2026 10:46 PM GMT. UBS has designated certain UBS Global Research department members as Derivatives Research Analysts where those department members publish research principally on the analysis of the price or market for a derivative, and provide information reasonably sufficient upon which to base a decision to enter into a derivatives transaction. Where Derivatives Research Analysts co-author research reports with Equity Research Analysts or Economists, the Derivatives Research Analyst is responsible for the derivatives investment views, forecasts, and/or recommendations. **Quantitative Research Review:** UBS Global Research publishes a quantitative assessment of its analysts' responses to certain questions about the likelihood of an occurrence of a number of short term factors in a product known as the 'Quantitative Research Review'. Views contained in this assessment on a particular stock reflect only the views on those short term factors which are a different timeframe to the 12-month timeframe reflected in any equity rating set out in this note. For the latest responses, please see the Quantitative Research Review Addendum at the back of this report, where applicable. For previous responses please make reference to (i) previous UBS Global Research reports; and (ii) where no applicable research report was published that month, the Quantitative Research Review which can be found at https://neo.ubs.com/quantitative, or contact your UBS sales representative for access to the report or the Quantitative Research Team on ubs-quant-answers@ubs.com. A consolidated report which contains all responses is also available and again you should contact your UBS sales representative for details and pricing or the Quantitative Research team on the email above.

Analyst Certification:
Each research analyst primarily responsible for the content of this research report, in whole or in part, certifies that with respect to each security or issuer that the analyst covered in this report: (1) all of the views expressed accurately reflect his or her personal views about those securities or issuers and were prepared in an independent manner, including with respect to UBS, and (2) no part of his or her compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed by that research analyst in the research report.

UBS Global Research: Global Equity Rating Definitions

12-Month Rating	Definition	Coverage[1]	IB Services[2]
Buy	FSR is > 6% above the MRA.	55%	24%
Neutral	FSR is between -6% and 6% of the MRA.	39%	20%
Sell	FSR is > 6% below the MRA.	6%	21%

Source: UBS. Rating allocations are as of 30 September 2026.
1:Percentage of companies under coverage globally within the 12-month rating category.
2:Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

KEY DEFINITIONS:Forecast Stock Return (FSR) is defined as expected percentage price appreciation plus gross dividend yield over the next 12 months. In some cases, this yield may be based on accrued dividends. **Market Return Assumption (MRA)** is defined as the one-year local market interest rate plus 5% (a proxy for, and not a forecast of, the equity risk premium). **Under Review (UR)** Stocks may be flagged as UR by the analyst, indicating that the stock's price target and/or rating are subject to possible change in the near term, usually in response to an event that may affect the investment case or valuation. **Equity Price Targets** have an investment horizon of 12 months.

EXCEPTIONS AND SPECIAL CASES:UK and European Investment Fund ratings and definitions are: Buy: Positive on factors such as structure, management, performance record, discount; **Neutral:** Neutral on factors such as structure, management, performance record, discount; **Sell:** Negative on factors such as structure, management, performance record, discount. **Core Banding Exceptions (CBE):** Exceptions to the standard +/-6% bands may be granted by the Investment Review Consultation (IRC). Factors considered by the IRC include the stock's volatility and the credit spread of the respective company's debt. As a result, stocks deemed to be very high or low risk may be subject to higher or lower bands as they relate to the rating. When such exceptions apply, they will be identified in the Company Disclosures table in the relevant research piece.

Research analysts contributing to this report who are employed by any non-US affiliate of UBS Securities LLC are not registered/ qualified as research analysts with FINRA. Such analysts may not be associated persons of UBS Securities LLC and therefore are not subject to the FINRA restrictions on communications with a subject company, public appearances, and trading securities held by a research analyst account. The name of each affiliate and analyst employed by that affiliate contributing to this report, if any, follows.

UBS Securities Pte. Ltd., Taipei Branch: Irene Chang, Randy Abrams.

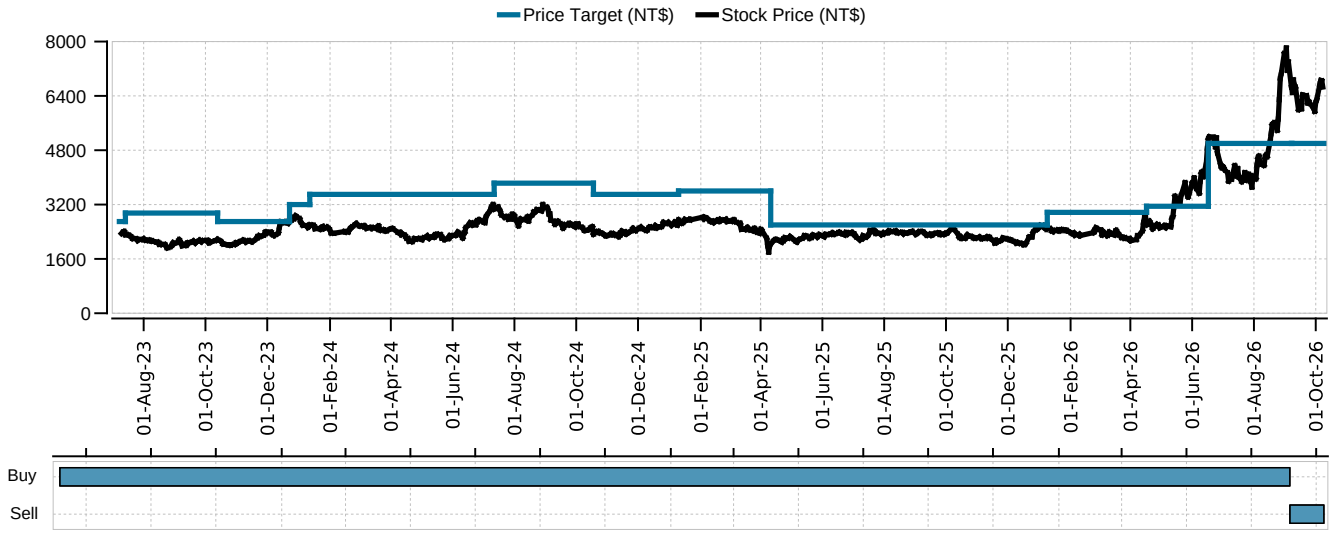
Company Disclosures

Company Name	Reuters	12-month rating	Price	Price date
Largan Precision[28]	3008.TW	Sell	NT$6,660	08 Oct 2026

Source: UBS Global Research; LSEG Eikon. All prices as of local market close. Ratings in this table are the most current published ratings prior to this report. They may be more recent than the stock pricing date.
28. UBS holds a long position of 0.5% or more of the listed shares of this company.

Unless otherwise indicated, please refer to the Valuation and Risk sections within the body of this report. For a complete set of disclosure statements associated with the companies discussed in this report, including information on valuation and risk, please contact UBS Securities LLC, 11 Madison Avenue, New York, NY 10010, USA, Attention: Investment Research.

Largan Precision (NT$)



Date	Stock Price (NT$)	Price Target (NT$)	Rating
2023-07-07	2340.00	2700.00	Buy
2023-07-13	2405.00	2950.00	Buy
2023-10-12	2150.00	2700.00	Buy
2023-12-22	2650.00	3200.00	Buy
2024-01-11	2550.00	3500.00	Buy
2024-07-11	3200.00	3830.00	Buy
2024-10-17	2545.00	3500.00	Buy
2025-01-09	2595.00	3600.00	Buy
2025-04-10	1980.00	2600.00	Buy
2026-01-08	2475.00	2970.00	Buy
2026-04-16	2820.00	3150.00	Buy
2026-06-16	4860.00	5000.00	Buy
2026-09-06	7400.00	5000.00	Sell

Source: UBS Global Research; LSEG Eikon as of 08-Oct-2026. All prices as of local market close. Ratings as of date shown.

The Disclaimer relevant to Global Wealth Management clients follows the Global Research Disclaimer.

UBS Global Research Disclaimer

This document has been prepared by UBS Securities Pte. Ltd., Taipei Branch, an affiliate of UBS AG. UBS AG, its subsidiaries, branches and affiliates, including former Credit Suisse AG and its subsidiaries, branches and affiliates are referred to herein as "UBS".

Any opinions expressed in this document may change without notice and are only current as of the date of publication. Different areas, groups, and personnel within UBS may produce and distribute separate research products independently of each other. For example, research publications from **UBS CIO** are produced by UBS Global Wealth Management. **UBS Global Research** is produced by UBS Investment Bank. Research methodologies and rating systems of each separate research organization may differ, for example, in terms of investment recommendations, investment horizon, model assumptions, and valuation methods. As a consequence, except for certain economic forecasts (for which UBS CIO and UBS Global Research may collaborate), investment recommendations, ratings, price targets, and valuations provided by each of the separate research organizations may be different, or inconsistent. You should refer to each relevant research product for the details as to their methodologies and rating system. Not all clients may have access to all products from every organization. Each research product is subject to the policies and procedures of the organization that produces it.

This document is provided solely to recipients who are expressly authorized by UBS to receive it. If you are not so authorized you must immediately destroy the document.

UBS Global Research is provided to our clients through UBS Neo, and in certain instances, UBS.com and any other system or distribution method specifically identified in one or more communications distributed through UBS Neo or UBS.com (each a system) as an approved means for distributing UBS Global Research. It may also be made available through third party vendors and distributed by UBS and/or third parties via e-mail or alternative electronic means.

All UBS Global Research is available on UBS Neo. Please contact your UBS sales representative if you wish to discuss your access to UBS Neo. Where UBS Global Research refers to "UBS Evidence Lab Inside" or has made use of data provided by UBS Evidence Lab and you would like to access that data please contact your UBS sales representative. UBS Evidence Lab data is available on UBS Neo. The level and types of services provided by UBS Global Research and UBS Evidence Lab to a client may vary depending upon various factors such as a client's individual preferences as to the frequency and manner of receiving communications, a client's risk profile and investment focus and perspective (e.g., market wide, sector specific, long-term, short-term, etc.), the size and scope of the overall client relationship with UBS Global Research and UBS Evidence Lab and legal and regulatory constraints. UBS HOLT and UBS Pharma Values are offerings of UBS Global Research. HOLT Lens is a corporate performance platform offering that provides an objective accounting-led framework for comparing and valuing companies and is available to clients of UBS Global Research; for further details and pricing please contact your UBS Sales representative. In particular, HOLT has a variety of warranted prices based on the scenario chosen; please mail UBS Securities LLC, 11 Madison Avenue, New York, NY 10010, USA, Attention: Investment Research, if you are interested in the warranted price on a particular company, again subject to commercial considerations. UBS Pharma Values is an analytical tool that involves the creation of a number of individual product net present value calculations, based on published forecasts of sales for pharmaceuticals, and is available to clients of UBS Global Research; for further details and pricing please contact your UBS Sales representative. For all other specific disclaimers, please see https://www.ubs.com/disclosures.

When you receive UBS Global Research through a system, your access and/or use of such UBS Global Research is subject to this UBS Global Research Disclaimer and to the UBS Neo Platform Use Agreement (the "Neo Terms") together with any other relevant terms of use governing the applicable System.

When you receive UBS Global Research via a third party vendor, e-mail or other electronic means, you agree that use shall be subject to this UBS Global Research Disclaimer, the Neo Terms and where applicable the UBS Investment Bank terms of business (https://www.ubs.com/global/en/investment-bank/regulatory.html) and to UBS's Terms of Use/Disclaimer (https://www.ubs.com/global/en/legalinfo2/disclaimer.html). In addition, you consent to UBS processing your personal data and using cookies in accordance with our Privacy Statement (https://www.ubs.com/global/en/legalinfo2/privacy.html) and cookie notice (https://www.ubs.com/global/en/legal/privacy/users.html).

If you receive UBS Global Research, whether through a System or by any other means, you agree that you shall not copy, revise, amend, create a derivative work, provide to any third party, or in any way commercially exploit any UBS research provided via UBS Global Research or otherwise, and that you shall not extract data from any research or estimates provided to you via UBS Global Research or otherwise, without the prior written consent of UBS. You agree not to use UBS Global Research in any artificial intelligence system, without the prior written consent of UBS.

In certain circumstances you may receive UBS Global Research otherwise than in the capacity of a client of UBS and you understand and agree that under these circumstances (i) the UBS Global Research is provided to you for information purposes only; (ii) for the purposes of receiving it you are not intended to be and will not be treated as a "client" of UBS for any legal or regulatory purpose; (iii) the UBS Global Research must not be relied on or acted upon for any purpose; and (iv) such content is subject to the relevant disclaimers that follow.

This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject UBS to any registration or licensing requirement within such jurisdiction.

This document is a general communication and is educational in nature; it is not an advertisement nor is it a solicitation or an offer to buy or sell any financial instruments or to participate in any particular trading strategy. Nothing in this document constitutes a representation that any investment strategy or recommendation is suitable or appropriate to an investor's individual circumstances or otherwise constitutes a personal recommendation. By providing this document, none of UBS or its representatives has any responsibility or authority to provide or have provided investment advice in a fiduciary capacity or otherwise. Investments involve risks, and investors should exercise prudence and their own judgment in making their investment decisions. None of UBS or its representatives is suggesting that the recipient or any other person take a specific course of action or any action at all. The recipient should carefully read this document in its entirety and not draw inferences or conclusions from the rating alone. By receiving this document, the recipient acknowledges and agrees with the intended purpose described above and further disclaims any expectation or belief that the information constitutes investment advice to the recipient or otherwise purports to meet the investment objectives of the recipient. The financial instruments described in the document may not be eligible for sale in all jurisdictions or to certain categories of investors.

Options, structured derivative products and futures (including OTC derivatives) are not suitable for all investors. Trading in these instruments is considered risky and may be appropriate only for sophisticated investors. Prior to buying or selling an option, and for the complete risks relating to options, you must receive a copy of "The Characteristics and Risks of Standardized Options." You may read the document at https://www.theocc.com/publications/risks/riskchap1.jsp or ask your salesperson for a copy. Various theoretical explanations of the risks associated with these instruments have been published. Supporting documentation for any claims, comparisons, recommendations, statistics or other technical data will be supplied upon request. Past performance is not necessarily indicative of future results. Transaction costs may be significant in option strategies calling for multiple purchases and sales of options, such as spreads and straddles. Because of the importance of tax considerations to many options transactions, the investor considering options should consult with his/her tax advisor as to how taxes affect the outcome of contemplated options transactions.

Mortgage and asset-backed securities may involve a high degree of risk and may be highly volatile in response to fluctuations in interest rates or other market conditions. Foreign currency rates of exchange may adversely affect the value, price or income of any security or related instrument referred to in the document. For investment advice, trade execution or other enquiries, clients should contact their local sales representative.

UBS notes that no globally accepted framework or definition (legal, regulatory or otherwise) currently exists, nor is there a market consensus as to what constitutes an "ESG" (Environmental, Social or Governance) or an equivalent-label, or as to what precise attributes are required for the Information (as defined below) to be defined as ESG or equivalently-labelled. Any information, data or other content including from a third party source contained, referred to herein or used for whatsoever purpose by UBS or a third party ("Information"), in relation to any actual or potential ESG objective, issue or consideration is not intended to be relied upon for ESG classification, regulatory regime or industry initiative purposes ("ESG Regimes"). Nothing in these materials is intended to convey, suggest or indicate that UBS considers or represents any product, service, person or body mentioned in these materials as meeting or qualifying for any ESG classification, labelling or similar standards that may exist under the ESG Regimes. UBS has not conducted any assessment of compliance with ESG Regimes. Parties are reminded to make their own assessments for these purposes.

The value of any investment or income may go down as well as up, and investors may not get back the full (or any) amount invested. Past performance is not necessarily a guide to future performance. Neither UBS nor any of its directors, employees or agents accepts any liability for any loss (including investment loss) or damage arising out of the use of all or any of the Information.

Prior to making any investment or financial decisions, any recipient of this document or the information should take steps to understand the risk and return of the investment and seek individualized advice from his or her personal financial, legal, tax and other professional advisors that takes into account all the particular facts and circumstances of his or her investment objectives.

Any prices stated in this document are for information purposes only and do not represent valuations for individual securities or other financial instruments. There is no representation that any transaction can or could have been effected at those prices, and any prices do not necessarily reflect UBS's internal books and records or theoretical model-based valuations and may be based on certain assumptions. Different assumptions by UBS or any other source may yield substantially different results.

No representation or warranty, either expressed or implied, is provided in relation to the accuracy, completeness or reliability of the information contained in any materials to which this document relates (the "Information"), except with respect to Information concerning UBS. The Information is not intended to be a

complete statement or summary of the securities, markets or developments referred to in the document. UBS does not undertake to update or keep current the Information. Any statements contained in this report attributed to a third party represent UBS's interpretation of the data, information and/or opinions provided by that third party either publicly or through a subscription service, and such use and interpretation have not been reviewed by the third party. In no circumstances may this document or any of the Information (including any forecast, value, index or other calculated amount ("Values")) be used for any of the following purposes:

(i) valuation or accounting purposes;

(ii) to determine the amounts due or payable, the price or the value of any financial instrument or financial contract; or

(iii) to measure the performance of any financial instrument including, without limitation, for the purpose of tracking the return or performance of any Value or of defining the asset allocation of portfolio or of computing performance fees.

By receiving this document and the Information you will be deemed to represent and warrant to UBS that you will not use this document or any of the Information for any of the above purposes or otherwise rely upon this document or any of the Information.

UBS has policies and procedures, which include, without limitation, independence policies and permanent information barriers, that are intended, and upon which UBS relies, to manage potential conflicts of interest and control the flow of information within divisions of UBS and among its subsidiaries, branches and affiliates. UBS does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For further information on the ways in which UBS Global Research manages conflicts and maintains independence of its research products, historical performance information and certain additional disclosures concerning UBS Global Research recommendations, please visit https://www.ubs.com/disclosures.

UBS Global Research will initiate, update and cease coverage solely at the discretion of UBS Global Research Management, which will also have sole discretion on the timing and frequency of any published research product. The analysis contained in this document is based on numerous assumptions. All material information in relation to published research reports, such as valuation methodology, risk statements, underlying assumptions (including sensitivity analysis of those assumptions), ratings history etc. as required by the Market Abuse Regulation, can be found on UBS Neo. Different assumptions could result in materially different results.

UBS Global Research may utilise artificial intelligence tools ("AI Tools") in the preparation of this document. Notwithstanding any such use of AI Tools, this document has undergone human review.

The analyst(s) responsible for the preparation of this document may interact with trading desk personnel, sales personnel and other parties for the purpose of gathering, applying and interpreting market information. UBS relies on information barriers to control the flow of information contained in one or more areas within UBS into other areas, units, groups or affiliates of UBS. The compensation of the analyst who prepared this document is determined exclusively by UBS Global Research management and senior management (not including investment banking). Analyst compensation is not based on investment banking revenues; however, compensation may relate to the revenues of UBS and/or its divisions as a whole, of which investment banking, sales and trading are a part, and UBS as a whole.

For financial instruments admitted to trading on an EU regulated market: UBS (excluding UBS Securities LLC) acts as a market maker or liquidity provider (in accordance with the interpretation of these terms under English law or, if not carried out by UBS in the UK the law of the relevant jurisdiction in which UBS determines it carries out the activity) in the financial instruments of the issuer save that where the activity of liquidity provider is carried out in accordance with the definition given to it by the laws and regulations of any other EU jurisdictions, such information is separately disclosed in this document. For financial instruments admitted to trading on a non-EU regulated market: UBS may act as a market maker save that where this activity is carried out in the US in accordance with the definition given to it by the relevant laws and regulations, such activity will be specifically disclosed in this document. UBS may have issued a warrant the value of which is based on one or more of the financial instruments referred to in the document. UBS and its affiliates and employees may have long or short positions, trade as principal and buy and sell in instruments or derivatives identified herein; such transactions or positions may be inconsistent with the opinions expressed in this document.

Within the past 12 months UBS may have received or provided investment services and activities or ancillary services as per MiFID II which may have given rise to a payment or promise of a payment in relation to these services from or to this company.

Please note that all transactions conducted by UBS are consistent with sanctions regulations imposed by Switzerland, the European Union, the United Nations, the United Kingdom and the United States, per UBS' global sanctions policy. UBS opinion as to future investment worthiness assumes no new sanctions are imposed.

US persons are prohibited from purchasing or selling securities of certain companies designated as being associated with the Chinese Military in accordance with the amended US Presidential Executive Order 13959.

United Kingdom: This material is distributed by UBS AG, London Branch to persons who are eligible counterparties or professional clients. UBS AG, London Branch is authorised by the Prudential Regulation Authority and subject to regulation by the Financial Conduct Authority and limited regulation by the Prudential Regulation Authority. **Europe:** Except as otherwise specified herein, these materials are distributed by UBS Europe SE, a subsidiary of UBS AG, to persons who are eligible counterparties or professional clients (as detailed in the Bundesanstalt fur Finanzdienstleistungsaufsicht (BaFin) Rules and according to MIFID) and are only available to such persons. The information does not apply to, and should not be relied upon by, retail clients. UBS Europe SE is authorised by the European Central Bank (ECB) and regulated by the BaFin and the ECB. **Germany, Luxembourg, the Netherlands, Belgium and Ireland:** Where an analyst of UBS Europe SE has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE. In all cases it is distributed by UBS Europe SE and UBS AG, London Branch. **Turkey:** Distributed by UBS AG, London Branch. No information in this document is provided for the purpose of offering, marketing and sale by any means of any capital market instruments and services in the Republic of Turkey. Therefore, this document may not be considered as an offer made or to be made to residents of the Republic of Turkey. UBS AG, London Branch is not licensed by the Turkish Capital Market Board under the provisions of the Capital Market Law (Law No. 6362). Accordingly, neither this document nor any other offering material related to the instruments/services may be utilized in connection with providing any capital market services to persons within the Republic of Turkey without the prior approval of the Capital Market Board. However, according to article 15 (d) (ii) of the Decree No. 32, there is no restriction on the purchase or sale of the securities abroad by residents of the Republic of Turkey. **Poland:** Distributed by UBS Europe SE (spolka z ograniczona odpowiedzialnoscia) Oddzial w Polsce. Where an analyst of UBS Europe SE (spolka z ograniczona odpowiedzialnoscia) Oddzial w Polsce has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE (spolka z ograniczona odpowiedzialnoscia) Oddzial w Polsce. **Russia:** Prepared and distributed by UBS Bank (OOO). Should not be construed as an individual Investment Recommendation for the purpose of the Russian Law - Federal Law #39-FZ ON THE SECURITIES MARKET Articles 6.1-6.2.**Switzerland:** Distributed by UBS AG to persons who are institutional investors only. UBS AG is regulated by the Swiss Financial Market Supervisory Authority (FINMA). **Italy:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, Italy Branch. Where an analyst of UBS Europe SE, Italy Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, Italy Branch. **France:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, France Branch. Where an analyst of UBS Europe SE, France Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, France Branch. **Spain:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, Spain Branch. Where an analyst of UBS Europe SE, Spain Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, Spain Branch. **Sweden:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, Sweden Branch. Where an analyst of UBS Europe SE, Sweden Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, Sweden Branch. **South Africa:** Distributed by UBS South Africa (Pty) Limited (Registration No. 1995/011140/07), an authorised user of the JSE and an authorised Financial Services Provider (FSP 7328). **Saudi Arabia:** This document has been issued by UBS AG (and/or any of its subsidiaries, branches or affiliates), a public company limited by shares, incorporated in Switzerland with its registered offices at Aeschenvorstadt 1, CH-4051 Basel and Bahnhofstrasse 45, CH-8001 Zurich. This publication has been approved by UBS Saudi Arabia (a subsidiary of UBS AG), a Saudi closed joint stock company incorporated in the Kingdom of Saudi Arabia under commercial register number 1010257812 having its registered office at Tatweer Towers, P.O. Box 75724, Riyadh 11588, Kingdom of Saudi Arabia. UBS Saudi Arabia is authorized and regulated by the Capital Market Authority to conduct securities business under license number 08113-37. **UAE / Dubai:** The information distributed by UBS AG Dubai Branch is only intended for Professional Clients and/or Market Counterparties, as classified under the DFSA rulebook. No other person should act upon this material/communication. The information is not for further distribution within the United Arab Emirates. UBS AG Dubai Branch is regulated by the DFSA in the DIFC. UBS is not licensed to provide banking services in the UAE by the Central Bank of the UAE, nor is it licensed by the UAE Securities and Commodities Authority. **Israel:** This Material is distributed by UBS AG, London Branch. UBS Securities Israel Ltd is a licensed Investment Marketer that is supervised by the Israel Securities Authority (ISA). UBS AG, London Branch and its affiliates incorporated outside Israel are not licensed under the Israeli Advisory Law. UBS may engage among others in issuance of Financial Assets or in distribution of Financial Assets of other issuers for fees or other benefits. UBS AG, London Branch and its affiliates may prefer various Financial Assets to which they have or may have an Affiliation (as such term is defined under the Israeli Advisory Law). Nothing in this Material should be considered as investment advice under the Israeli Advisory Law. This Material is being issued only to and/or is directed only at persons who are Eligible Clients within the meaning of the Israeli Advisory Law, and this Material must not be furnished to, relied on or acted upon by any other persons. **United States:** Distributed to US persons by either UBS Securities LLC or by UBS Financial Services Inc., subsidiaries of UBS AG; or by a group, subsidiary or affiliate of UBS AG that is not registered as a US broker-dealer (a **'non-US affiliate'**) to major US institutional investors only. UBS Securities LLC or UBS Financial Services Inc. accepts responsibility for the content of a report prepared by another non-US affiliate when distributed to US persons by UBS Securities LLC or UBS Financial Services Inc. All transactions by a US person in the securities mentioned in this report must be effected through UBS Securities LLC or UBS Financial Services Inc., and not through a non-US affiliate. UBS Securities LLC is not acting as a municipal advisor to any municipal entity or obligated person within the meaning of Section 15B of the Securities Exchange Act (the "Municipal Advisor Rule"), and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of the Municipal Advisor Rule. **Canada:** Distributed by UBS Securities Canada Inc., a registered investment dealer in Canada

and a Member-Canadian Investor Protection Fund, or by another affiliate of UBS AG that is registered to conduct business in Canada or is otherwise exempt from registration. **Brazil:** Except as otherwise specified herein, this Material is prepared by UBS BB Corretora de Câmbio, Títulos e Valores Mobiliários S.A. (UBS BB CCTVM) to persons who are eligible investors residing in Brazil, which are considered to be Professional Investors (Investidores Profissionais), as designated by the applicable regulation, mainly the CVM Resolution No. 30 from the 11th of May 2021 (determines the duty to verify the suitability of products, services and transactions with regards to the client´s profile). UBS BB CCTVM is a subsidiary of UBS BB Servicos de Assessoria Financeira e Participacoes S.A. ("UBS BB"). UBS BB is an association between UBS AG and Banco do Brasil (through its subsidiary BB – Banco de Investimentos S.A.), of which UBS AG is the majority owner and which provides investment banking services and coverage in Brazil, Argentina, Chile, Paraguay, Peru and Uruguay. UBS BB CCTVM is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. Ombudsman: 0800-940-0266/ https://www.ubs.com/br/pt/ubsbb-investment-bank/ombudsman.html. UBS may hold relevant financial and commercial interest in relation to the company subject to this Research report. **Hong Kong:** Distributed by UBS Securities Asia Limited. Please contact local licensed persons of UBS Securities Asia Limited in respect of any matters arising from, or in connection with, the analysis or document **Singapore:** Distributed by UBS Securities Pte. Ltd. [Co. Reg. No.: 198500648C] or UBS AG, Singapore Branch. Please contact UBS Securities Pte. Ltd., an exempt financial adviser under the Singapore Financial Advisers Act (Cap. 110); or UBS AG, Singapore Branch, an exempt financial adviser under the Singapore Financial Advisers Act (Cap. 110) and a wholesale bank licensed under the Singapore Banking Act (Cap. 19) regulated by the Monetary Authority of Singapore, in respect of any matters arising from, or in connection with, the analysis or document. The recipients of this document represent and warrant that they are accredited and institutional investors as defined in the Securities and Futures Act (Cap. 289). **Japan:** Distributed by UBS Securities Japan Co., Ltd. to professional investors (except as otherwise permitted). Where this report has been prepared by UBS Securities Japan Co., Ltd., UBS Securities Japan Co., Ltd. is the author, publisher and distributor of the report. Distributed by UBS AG, Tokyo Branch to Professional Investors (except as otherwise permitted) in relation to foreign exchange and other banking businesses when relevant. **Australia:** Recipients through a superannuation fund: Distributed by UBS AG, Australia Branch (ABN 47 088 129 613 and holder of Australian Financial Service Licence No. 231087). For all other recipients: Distributed by UBS Securities Australia Limited (ABN 62 008 586 481 and holder of Australian Financial Services Licence No. 231098). This document contains general information and/or general advice only and does not constitute personal financial product advice. As such, the Information in this document has been prepared without taking into account any investor's objectives, financial situation or needs, and investors should, before acting on the Information, consider the appropriateness of the Information, having regard to their objectives, financial situation and needs. If the Information contained in this document relates to the acquisition, or potential acquisition of a particular financial product by a 'Retail' client as defined by section 761G of the Corporations Act 2001 where a Product Disclosure Statement would be required, the retail client should obtain and consider the Product Disclosure Statement relating to the product before making any decision about whether to acquire the product. **New Zealand:** Distributed by UBS New Zealand Ltd. UBS New Zealand Ltd is not a registered bank in New Zealand. You are being provided with this publication or material because you have indicated to UBS that you are a "wholesale client" within the meaning of clause 4 of schedule 5 of the Financial Markets Conduct Act 2013 of New Zealand (Permitted Client). This publication or material is not intended for clients who are not Permitted Clients (non-permitted Clients). If you are a non-permitted Client you must not rely on this publication or material. If despite this warning you nevertheless rely on this publication or material, you hereby (i) acknowledge that you may not rely on the content of this publication or material and that any recommendations or opinions in such this publication or material are not made or provided to you, and (ii) to the maximum extent permitted by law (a) indemnify UBS and its associates or related entities (and their respective Directors, officers, agents and Advisors) (each a 'Relevant Person') for any loss, damage, liability or claim any of them may incur or suffer as a result of, or in connection with, your unauthorised reliance on this publication or material and (b) waive any rights or remedies you may have against any Relevant Person for (or in respect of) any loss, damage, liability or claim you may incur or suffer as a result of, or in connection with, your unauthorised reliance on this publication or material. **Korea:** Distributed in Korea by UBS Securities Pte. Ltd., Seoul Branch. This report may have been edited or contributed to from time to time by affiliates of UBS Securities Pte. Ltd., Seoul Branch. This material is intended for professional/institutional clients only and not for distribution to any retail clients. **Malaysia:** This material is authorized to be distributed in Malaysia by UBS Securities Malaysia Sdn. Bhd (Capital Markets Services License No.: CMSL/A0063/2007). This material is intended for professional/institutional clients only and not for distribution to any retail clients. **India:** Distributed by UBS Securities India Private Ltd. (Corporate Identity Number U67120MH1996PTC097299) Unit 2901, Level 29 Altimus, Pandurang Budhkar Road, Worli, Mumbai – 400 018. It provides brokerage services bearing SEBI Registration Number: INZ000259830; Merchant Banking services bearing SEBI Registration Number: INM000013101; and Research Analyst services bearing SEBI Registration Number: INH000001204. Name of Compliance Officer Mr. Parameshwaran Shivaramakrishnan, Phone : +912261556151, Email : parameshwaran.s@ubs.com, Name of Grievance Officer Parameshwaran Shivaramakrishnan, Phone : +912261556151, Email: ol-ubs-sec-compliance@ubs.com Registration granted by SEBI, and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. UBS may have debt holdings or positions in the subject Indian company/companies. UBS may have financial interests (e.g. loan/derivative products, rights to or interests in investments, etc.) in the subject Indian company / companies from time to time. Within the past 12 months, UBS may have received compensation for non-investment banking securities-related services and/or non-securities services from the subject Indian company/companies. The subject company/companies may have been a client/clients of UBS during the 12 months preceding the date of distribution of the research report with respect to investment banking and/or non-investment banking securities-related services and/or non-securities services. With regard to information on associates, please refer to the Annual Report at: https://www.ubs.com/global/en/about_ubs/investor_relations/annualreporting.html The Research Annual Compliance Report for UBS Securities India Private Limited is available on www.ubs.com/ubssi under Research tab. **Taiwan:** Except as otherwise specified herein, this material may not be distributed in Taiwan. Information and material on securities/instruments that are traded in a Taiwan organized exchange is deemed to be issued and distributed by UBS Securities Pte. LTD., Taipei Branch to professional institutional investors and/or persons permitted under applicable regulation. Unless permitted by applicable Taiwan laws and regulations, this material is for reference and information only and should not constitute "recommendation" to clients or recipients in Taiwan for the covered companies or any companies mentioned in this document. No portion of the document may be reprinted, reproduced or quoted by the press or any other person without authorisation from UBS. **Indonesia:** This report is being distributed by PT UBS Sekuritas Indonesia and is delivered by its licensed employee(s), including marketing/sales person, to its client. PT UBS Sekuritas Indonesia, having its registered office at Sequis Tower Level 22 unit 22-1,Jl.Jend. Sudirman, kav.71, SCBD lot 11B, Jakarta 12190. Indonesia, is a subsidiary company of UBS AG and licensed under Capital Market Law no. 8 year 1995, a holder of broker-dealer and underwriter licenses issued by the Capital Market and Financial Institution Supervisory Agency (now Otoritas Jasa Keuangan/OJK). PT UBS Sekuritas Indonesia is also a member of Indonesia Stock Exchange and supervised by Otoritas Jasa Keuangan (OJK). Neither this report nor any copy hereof may be distributed in Indonesia or to any Indonesian citizens except in compliance with applicable Indonesian capital market laws and regulations. This report is not an offer of securities in Indonesia and may not be distributed within the territory of the Republic of Indonesia or to Indonesian citizens in circumstance which constitutes an offering within the meaning of Indonesian capital market laws and regulations.

The disclosures contained in research documents produced by UBS AG, London Branch or UBS Europe SE shall be governed by and construed in accordance with English law.

UBS specifically prohibits the redistribution of this document in whole or in part without the written permission of UBS and in any event UBS accepts no liability whatsoever for any redistribution of this document or its contents or the actions of third parties in this respect. Images may depict objects or elements that are protected by third party copyright, trademarks and other intellectual property rights. © UBS 2026. The key symbol and UBS are among the registered and unregistered trademarks of UBS. All rights reserved.

Global Wealth Management Disclaimer

You receive this document in your capacity as a client of UBS Global Wealth Management. This publication has been distributed to you by UBS Switzerland AG (regulated by FINMA in Switzerland) or its affiliates (**"UBS"**) with whom you have a banking relationship with. The full name of the distributing affiliate and its competent authority can be found in the country-specific disclaimer at the end of this document. UBS may utilise artificial intelligence tools ("AI Tools") in the preparation of this document. Notwithstanding any such use of AI Tools, this document has undergone human review.

The date and time of the first dissemination of this publication is the same as the date and time of its publication.

Risk information:

You agree that you shall not copy, revise, amend, create a derivative work, provide to any third party, or in any way commercially exploit any UBS research, and that you shall not extract data from any research or estimates, without the prior written consent of UBS.

This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject UBS to any registration or licensing requirement within such jurisdiction.

This document is **for your information only**; it is not an advertisement nor is it a solicitation or an offer to buy or sell any financial instruments or to participate in any particular trading strategy. Nothing in this document constitutes a representation that any investment strategy or recommendation is suitable or appropriate to an investor's individual circumstances or otherwise constitutes a personal recommendation. By providing this document, none of UBS or its representatives has any responsibility or authority to provide or have provided investment advice in a fiduciary capacity or otherwise. Investments involve risks, and investors should exercise prudence and their own judgment in making their investment decisions. None of UBS or its representatives is suggesting that the recipient or any other person take a specific course of action or any action at all. By receiving this document, the recipient acknowledges and agrees with the intended purpose described above and further disclaims any expectation or belief that the information constitutes investment advice to the recipient or otherwise purports to meet the investment objectives of the recipient. The financial instruments described in the document may not be eligible for sale in all jurisdictions or to certain categories of investors.

Options, derivative products and futures are not suitable for all investors, and trading in these instruments is considered risky. Mortgage and asset-backed securities may involve a high degree of risk and may be highly volatile in response to fluctuations in interest rates or other market conditions. Foreign currency rates of exchange may

adversely affect the value, price or income of any security or related instrument referred to in the document. For investment advice, trade execution or other enquiries, clients should contact their local sales representative.

The value of any investment or income may go down as well as up, and investors may not get back the full (or any) amount invested. Past performance is not necessarily a guide to future performance. Neither UBS nor any of its directors, employees or agents accepts any liability for any loss (including investment loss) or damage arising out of the use of all or any of the information (as defined below).

Prior to making any investment or financial decisions, any recipient of this document or the information should take steps to understand the risk and return of the investment and seek individualized advice from his or her personal financial, legal, tax and other professional advisors that takes into account all the particular facts and circumstances of his or her investment objectives.

Any prices stated in this document are for information purposes only and do not represent valuations for individual securities or other financial instruments. There is no representation that any transaction can or could have been effected at those prices, and any prices do not necessarily reflect UBS's internal books and records or theoretical model-based valuations and may be based on certain assumptions. Different assumptions by UBS or any other source may yield substantially different results.

No representation or warranty, either expressed or implied, is provided in relation to the accuracy, completeness or reliability of the information contained in any materials to which this document relates (the "Information"), except with respect to Information concerning UBS. The Information is not intended to be a complete statement or summary of the securities, markets or developments referred to in the document. UBS does not undertake to update or keep current the Information. Any opinions expressed in this document may change without notice and may differ or be contrary to opinions expressed by other business areas or groups, personnel or other representative of UBS. Any statements contained in this report attributed to a third party represent UBS's interpretation of the data, information and/or opinions provided by that third party either publicly or through a subscription service, and such use and interpretation have not been reviewed by the third party. In no circumstances may this document or any of the Information (including any forecast, value, index or other calculated amount ("Values")) be used for any of the following purposes: (i) valuation or accounting purposes; (ii) to determine the amounts due or payable, the price or the value of any financial instrument or financial contract; or (iii) to measure the performance of any financial instrument including, without limitation, for the purpose of tracking the return or performance of any Value or of defining the asset allocation of portfolio or of computing performance fees.

By receiving this document and the Information you will be deemed to represent and warrant to UBS that you will not use this document or any of the Information for any of the above purposes or otherwise rely upon this document or any of the Information.

UBS has policies and procedures, which include, without limitation, independence policies and permanent information barriers, that are intended, and upon which UBS relies, to manage potential conflicts of interest and control the flow of information within divisions of UBS (including between Global Wealth Management and UBS Global Research) and among its subsidiaries, branches and affiliates. For further information on the ways in which UBS manages conflicts and maintains independence of its research products, historical performance information and certain additional disclosures concerning UBS research recommendations, please visit https://www.ubs.com/research-methodology.

Research will initiate, update and cease coverage solely at the discretion of research management, which will also have sole discretion on the timing and frequency of any published research product. The analysis contained in this document is based on numerous assumptions. Different assumptions could result in materially different results.

The analyst(s) responsible for the preparation of this document may interact with trading desk personnel, sales personnel and other parties for the purpose of gathering, applying and interpreting market information. UBS relies on information barriers to control the flow of information contained in one or more areas within UBS into other areas, units, groups or affiliates of UBS. The compensation of the analyst who prepared this document is determined exclusively by research management and senior management (not including investment banking). Analyst compensation is not based on investment banking revenues; however, compensation may relate to the revenues of UBS and/or its divisions as a whole, of which investment banking, sales and trading are a part, and UBS's subsidiaries, branches and affiliates as a whole.

For financial instruments admitted to trading on an EU regulated market: UBS AG, its affiliates or subsidiaries (excluding UBS Securities LLC) acts as a market maker or liquidity provider (in accordance with the interpretation of these terms in the UK) in the financial instruments of the issuer save that where the activity of liquidity provider is carried out in accordance with the definition given to it by the laws and regulations of any other EU jurisdictions, such information is separately disclosed in this document. For financial instruments admitted to trading on a non-EU regulated market: UBS may act as a market maker save that where this activity is carried out in the US in accordance with the definition given to it by the relevant laws and regulations, such activity will be specifically disclosed in this document. UBS may have issued a warrant the value of which is based on one or more of the financial instruments referred to in the document. UBS and its affiliates and employees may have long or short positions, trade as principal and buy and sell in instruments or derivatives identified herein; such transactions or positions may be inconsistent with the opinions expressed in this document.

Options and futures are not suitable for all investors, and trading in these instruments is considered risky and may be appropriate only for sophisticated investors. Prior to buying or selling an option, and for the complete risks relating to options, you must receive a copy of "Characteristics and Risks of Standardized Options". You may read the document at https://www.theocc.com/about/publications/character-risks.jsp or ask your financial advisor for a copy.

Investing in structured investments involves significant risks. For a detailed discussion of the risks involved in investing in any particular structured investment, you must read the relevant offering materials for that investment. Structured investments are unsecured obligations of a particular issuer with returns linked to the performance of an underlying asset. Depending on the terms of the investment, investors could lose all or a substantial portion of their investment based on the performance of the underlying asset. Investors could also lose their entire investment if the issuer becomes insolvent. UBS does not guarantee in any way the obligations or the financial condition of any issuer or the accuracy of any financial information provided by any issuer. Structured investments are not traditional investments and investing in a structured investment is not equivalent to investing directly in the underlying asset. Structured investments may have limited or no liquidity, and investors should be prepared to hold their investment to maturity. The return of structured investments may be limited by a maximum gain, participation rate or other feature. Structured investments may include call features and, if a structured investment is called early, investors would not earn any further return and may not be able to reinvest in similar investments with similar terms. Structured investments include costs and fees which are generally embedded in the price of the investment. The tax treatment of a structured investment may be complex and may differ from a direct investment in the underlying asset. UBS and its employees do not provide tax advice. Investors should consult their own tax advisor about their own tax situation before investing in any securities.

Important Information About Sustainable Investing Strategies: Sustainable investing strategies aim to consider and incorporate environmental, social and governance (ESG) factors into investment process and portfolio construction. Strategies across geographies approach ESG analysis and incorporate the findings in a variety of ways. Incorporating ESG factors or sustainable investing considerations may inhibit the portfolio manager's ability to participate in certain investment opportunities that otherwise would be consistent with its investment objective and other principal investment strategies. The returns on a portfolio incorporating ESG factors or sustainable investing considerations may be lower or higher than portfolios where ESG factors, exclusions, or other sustainability issues are not considered by the portfolio manager, and the investment opportunities available to such portfolios may differ.

Within the past 12 months UBS Switzerland AG, its affiliates or subsidiaries may have received or provided investment services and activities or ancillary services as per MiFID II which may have given rise to a payment or promise of a payment in relation to these services from or to this company.

Disclosures: If you require detailed information on disclosures of interest or conflict of interest as required by Market Abuse Regulation please contact the mailbox MAR_disclosure_twopager@ubs.com. Please note that e-mail communication is unsecured.

External Asset Managers / External Financial Consultants: In case this research or publication is provided to an External Asset Manager or an External Financial Consultant, UBS expressly prohibits that it is redistributed by the External Asset Manager or the External Financial Consultant and is made available to their clients and/or third parties.

Argentina: All securities that will be offered to you by UBS are not authorized by the Argentine Securities and Exchange Commission (CNV) and are not subject to its reporting, periodic information requirements, or oversight. Furthermore, the CNV has not reviewed or endorsed the information provided in any securities offering document, nor the accuracy of any accounting, financial, economic data, or any other information disclosed therein, which remains the sole responsibility of the respective issuer and the other parties involved. **Australia:** This document is provided by UBS Switzerland AG. UBS Switzerland AG does not hold an Australian Financial Services Licence (AFSL) and relies on an exemption to provide financial services to persons in Australia. This document is intended only for distribution to wholesale clients under the Corporations Act 2001 (Cth). UBS Switzerland AG is a related body corporate of UBS AG, Australia Branch and UBS Securities Australia Limited. This document may be distributed to clients by those entities, but it is provided by UBS Switzerland AG and is not provided under any of the other entities' AFSL. The information in this document is general in nature and is not intended to address the objectives, financial situation or needs of any particular individual or entity. Each recipient should consider their own objectives, financial situation or needs before acting on the advice and obtain the relevant Product Disclosure Statement (if required) before making any decision whether to acquire any product. In Australia, UBS entities, other than UBS AG, Australia Branch, are not authorized deposit-taking institutions for the purposes of the Banking Act 1959 (Cth.) and their obligations do not represent deposits or other liabilities of UBS AG, Australia Branch. UBS AG, Australia Branch does not guarantee or otherwise provide assurance in respect of the obligations of such UBS entities or the funds. An investor is exposed to investment risk including possible delays in repayment and loss of income and principal invested, as relevant. If you do not wish to receive marketing materials from UBS, please contact your UBS representative or the contact details listed in the Australia Privacy Notice: https://www.ubs.com/global/en/legal/privacy.html. Your personal data will be processed in accordance with this notice. **Bahrain**: This report is being distributed by UBS AG, Bahrain Branch, duly licensed and regulated by the Central Bank of Bahrain (CBB) as an Investment Business Firm - Category 2 (Branch). Related financial services or products are only made available to Accredited Investors, as defined by the CBB, and are not intended for any other persons. UBS AG, Bahrain Branch is

a Foreign Branch of UBS AG, Zurich/Switzerland and is located on Level 21, East Tower, Bahrain World Trade Centre, Manama, Kingdom of Bahrain. **Brazil**: This report is distributed in Brazil by UBS BB Corretora de Câmbio, Títulos e Valores Mobiliários S.A. or its affiliates ("UBS"). Pursuant to CVM Resolution No. 20/2021, of February 25, 2021, the author(s) of the report hereby certify(ies) that the views expressed in this report solely and exclusively reflect their personal opinions and have been prepared independently, including with respect to UBS. Part of the author(s)'s compensation is based on various factors, including the total revenues of the relevant UBS Group entity of which they are in employment of, but no part of the compensation has been, is, or will be related to the specific recommendations or views expressed in this report. In addition, UBS declares that: UBS has provided, and/or may in the future provide investment banking, brokerage, asset management, commercial banking and other financial services to the subject company/companies or its affiliates, for which they have received or may receive customary fees and commissions, and which constituted or may constitute relevant financial or commercial interests in relation to the subject company/companies or the subject securities. **Canada**: The information contained herein is not, and under no circumstances is to be construed as, a prospectus, an advertisement, a public offering, an offer to sell securities described herein, solicitation of an offer to buy securities described herein, in Canada or any province or territory thereof. Any offer or sale of the securities described herein in Canada will be made only under an exemption from the requirements to file a prospectus with the relevant Canadian securities regulators and only by a dealer properly registered under applicable securities laws or, alternatively, pursuant to an exemption from the dealer registration requirement in the relevant province or territory of Canada in which such offer or sale is made. Under no circumstances is the information contained herein to be construed as investment advice in any province or territory of Canada and is not tailored to the needs of the recipient. To the extent that the information contained herein references securities of an issuer incorporated, formed or created under the laws of Canada or a province or territory of Canada, any trades in such securities must be conducted through a dealer registered in Canada or, alternatively, pursuant to a dealer registration exemption. No securities commission or similar regulatory authority in Canada has reviewed or in any way passed upon these materials, the information contained herein or the merits of the securities described herein and any representation to the contrary is an offence. In Canada, this publication is distributed by UBS Investment Management Canada Inc. **China**: This report and any offering material such as term sheet, research report, other product or service documentation or any other information (the "Material") sent with this report was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the material erroneously, UBS asks that you kindly delete it and inform UBS immediately. This report is prepared by UBS Switzerland AG or its offshore subsidiary or affiliate (collectively as "UBS Offshore"). UBS Offshore is an entity incorporated out of China and is not licensed, supervised or regulated in China to carry out banking or securities business. The recipient should not contact the analysts or UBS Offshore which produced this report for advice as they are not licensed to provide securities investment advice in China. UBS Investment Bank (including Research) has its own wholly independent research and views which at times may vary from the views of UBS Global Wealth Management. The recipient should not use this document or otherwise rely on any of the information contained in this report in making investment decisions and UBS takes no responsibility in this regard. **Denmark**: This publication is not intended to constitute a public offer under Danish law. It is distributed only for information purposes to clients of UBS Europe SE, Denmark Branch, filial af UBS Europe SE, with place of business at Sankt Annae Plads 13, 1250 Copenhagen, Denmark, registered with the Danish Commerce and Companies Agency, under No. 38 17 24 33. UBS Europe SE, Denmark Branch, filial af UBS Europe SE is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the Danish Financial Supervisory Authority (Finanstilsynet), to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Egypt**: Securities or other investment products are not being offered or sold by UBS to the public in Egypt and they have not been and will not be registered with the Egyptian Financial Regulatory Authority (FRA). **France**: This publication is not intended to constitute a public offer under French law, it does not constitute a personal recommendation as it is distributed only for information purposes to clients of UBS Europe SE Succursale de France (a branch of UBS Europe SE), having its registered office at 39, rue du Colisée, 75008 Paris, France, registered with the "Registre du Commerce et des Sociétés" of Paris under N°844 425 629. UBS Europe SE Succursale de France is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the French "Autorité de contrôle prudentiel et de résolution" and "Autorité des marchés financiers", to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Germany**: This publication is not intended to constitute a public offer under German law. It is distributed only for information purposes to clients of UBS Europe SE, Germany, with place of business at Bockenheimer Landstrasse 2-4, 60306 Frankfurt am Main. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin") and supervised jointly by the European Central Bank, the German Central Bank (Deutsche Bundesbank) and BaFin, to which this publication has not been submitted for approval. **Hong Kong SAR**: This publication is distributed to clients of UBS AG Hong Kong Branch by UBS AG Hong Kong Branch, a licensed bank under the Hong Kong Banking Ordinance and a registered institution under the Securities and Futures Ordinance. UBS AG Hong Kong Branch is incorporated in Switzerland with limited liability. The contents of this material have not been reviewed by any regulatory authority in Hong Kong. **India**: UBS Securities India Private Ltd. (Corporate Identity Number U67120MH1996PTC097299) 2/F, 3 North Avenue, Maker Maxity, Bandra Kurla Complex, Bandra (East), Mumbai (India) 400051. Phone: +912261556000. It provides brokerage services bearing SEBI Registration Number: INZ000259830; Merchant Banking services bearing SEBI Registration Number: INM000013101 and Research Analyst services bearing SEBI Registration Number: INH000001204. Name of Compliance Officer Mr. Parameshwaran Shivaramakrishnan, Phone: +912261556151, Email: parameshwaran.s@ubs.com, Name of Grievance Officer Parameshwaran Shivaramakrishnan, Phone: +912261556151, Email: ol-ubs-sec-compliance@ubs.com. Registration granted by SEBI, and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. UBS AG, its affiliates or subsidiaries may have debt holdings or positions in the subject Indian company/companies. UBS AG, its affiliates or subsidiaries may have financial interests (e.g. loan/derivative products, rights to or interests in investments, etc.) in the subject Indian company/companies from time to time. Within the past 12 months, UBS AG, its affiliates or subsidiaries may have received compensation for non-investment banking securities-related services and/or non-securities services from the subject Indian company/companies. The subject company/companies may have been a client/clients of UBS AG, its affiliates or subsidiaries during the 12 months preceding the date of distribution of the research report with respect to investment banking and/or non-investment banking securities-related services and/or non-securities services. With regard to information on associates, please refer to the Annual Report at: https://www.ubs.com/global/en/about_ubs/investor_relations/annualreporting.html. The Research Annual Compliance report for UBS Securities India Private Limited is available on www.ubs.com/ubssi under Research tab. **Indonesia**: This communication and any offering material term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed without the consent of UBS. None of the Material has been registered or filed under the prevailing laws and with any financial or regulatory authority in your jurisdiction. The Material may not have been approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or products or futures contracts). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including futures contracts) nor is it an offer to buy or to sell any securities or products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Israel**: UBS is a premier global financial firm offering wealth management, asset management and investment banking services from its headquarters in Switzerland and its operations in over 50 countries worldwide to individual, corporate and institutional investors. This publication is intended for information only and is not intended as an offer to buy or solicitation of an offer. Furthermore, this publication is not intended as an investment advice. Nothing contrary to the above, no action has been, or will be, taken in Israel that would permit an offering of the product(s) mentioned in this document or a distribution of this document to the public in Israel. In particular, this document has not been reviewed or approved by the Israeli Securities Authority. The product(s) mentioned in this document is/are being offered to a limited number of sophisticated investors who qualify as one of the investors listed in the first supplement to the Israeli Securities Law, 5728-1968. This document may not be reproduced or used for any other purpose, nor be furnished to any other person other than those to whom copies have been sent. Anyone who purchases the product(s) mentioned herein shall do so for its own benefit and for its own account and not with the aim or intention of distributing or offering the product(s) to other parties. Anyone who purchases the product(s) shall do so in accordance with its own understanding and discretion and after it has received any relevant financial, legal, business, tax or other advice or opinion required by it in connection with such purchase(s). The word "advice" and/or any of its equivalent terms shall be read and construed in conjunction with the definition of the term "investment marketing" as defined under the Israeli Regulation of Investment Advice, Investment Marketing and Portfolio Management Law. The Swiss laws and regulations require a number of mandatory disclosures to be made in independent financial research reports or recommendations. Pursuant to the Swiss Financial Market Infrastructure Act and the Financial Market Infrastructure Ordinance-FINMA, banks must disclose the percentage of voting rights they hold in companies being researched, if these holdings are equal to or exceed the statutory thresholds. In addition, the Directives on the Independence of Financial Research, issued by the Swiss Bankers Association, mandate a number of disclosures, including the disclosure of potential conflicts of interest, the participation within previous 12 months in any securities issues on behalf of the company being researched, as well as the fact that remuneration paid to the financial analysts is based generally upon the performance of (i) the new issues department or investment banking; or (ii) securities trading performance (including proprietary trading) or sales. **Italy**: This publication is not intended to constitute a public offer under Italian law. It is distributed only for information purposes to clients of UBS Europe SE, Succursale Italia, with place of business at Via del Vecchio Politecnico, 3-20121 Milano. UBS Europe SE, Succursale Italia is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the Bank of Italy (Banca d'Italia) and the Italian Financial Markets Supervisory Authority (CONSOB - Commissione Nazionale per le Società e la Borsa), to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Japan**: This report is solely distributed in Japan by UBS SuMi TRUST Wealth Management Co., Ltd, Financial Instruments Dealer, Director-General of Kanto Local Finance Bureau (Kinsho) No. 3233, a member of the Japan Securities Dealers Association, Financial Futures Association of Japan, Investment Management Association of Japan. UBS SuMi TRUST Wealth Management Co., Ltd will not distribute or

forward this report outside Japan. **Jersey**: UBS AG, Jersey Branch, is regulated and authorized by the Jersey Financial Services Commission for the conduct of banking, funds and investment business. Where services are provided from outside Jersey, they will not be covered by the Jersey regulatory regime. UBS AG, Jersey Branch is a branch of UBS AG a public company limited by shares, incorporated in Switzerland whose registered offices are at Aeschenvorstadt 1, CH-4051 Basel and Bahnhofstrasse 45, CH 8001 Zurich. UBS AG, Jersey Branch's principal place of business is 1, IFC Jersey, St Helier, Jersey, JE2 3BX. **Luxembourg**: This publication is not intended to constitute a public offer under Luxembourg law. It is distributed only for information purposes to clients of UBS Europe SE, Luxembourg Branch ("UBS Luxembourg"), R.C.S. Luxembourg n° B209123, with registered office at 33A, Avenue J. F. Kennedy, L-1855 Luxembourg. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea (HRB n° 107046), with registered office at Bockenheimer Landstrasse 2-4, D-60306 Frankfurt am Main, Germany, duly authorized by the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin") and subject to the joint prudential supervision of BaFin, the European Central Bank and the central bank of Germany (Deutsche Bundesbank). UBS Luxembourg is furthermore supervised by the Luxembourg prudential supervisory authority (Commission de Surveillance du Secteur Financier), in its role as host member state authority. This publication has not been submitted for approval to any public supervisory authority. **Malaysia**: This communication and any offering material term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or derivatives products). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or derivatives products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Mexico**: This information is distributed by UBS Asesores México, S.A. de C.V. ("UBS Asesores"), an affiliate of UBS Switzerland AG, incorporated as a non-independent investment advisor under the Mexican regulation due to the relation with a Foreign Bank. UBS Asesores is registered under number 30060-001-(14115)-21/06/2016 and subject to the supervision of the Mexican Banking and Securities Commission ("CNBV") exclusively regarding the rendering of (i) portfolio management services, (ii) securities investment advisory services, analysis and issuance of individual investment recommendations, and (iii) anti-money laundering and terrorism financing matters. This UBS publication or any material related thereto is addressed only to Sophisticated or Institutional Investors located in Mexico. Research reports only reflect the views of the analysts responsible for the report. The compensation of the analyst(s) who prepared this report is determined exclusively by research management and senior management of any entity of UBS Group to which such analyst(s) render(s) services. **Monaco**: This document is not intended to constitute a public offering or a comparable solicitation under the Principality of Monaco laws, but might be made available for information purposes to clients of UBS (Monaco) S.A., a regulated bank having its registered office at 2 avenue de Grande Bretagne 98000 Monaco operating under a banking license granted by the "Autorité de Contrôle Prudentiel et de Résolution" (ACPR) and the Monegasque government which authorizes the provision of banking services in Monaco. UBS (Monaco) S.A. is also licensed by the "Commission de Contrôle des Activités Financières" (CCAF) to provide investment services in Monaco. The latter has not approved this publication. **Philippines:** This communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or derivatives products). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or derivatives products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Portugal**: UBS Switzerland AG is not licensed to conduct banking and financial activities in Portugal nor is UBS Switzerland AG supervised by the portuguese regulators (Bank of Portugal "Banco de Portugal" and Portuguese Securities Exchange Commission "Comissão do Mercado de Valores Mobiliários"). **Qatar**: UBS Qatar LLC is licensed by the Qatar Financial Centre Authority and authorized by the QFC Regulatory Authority, with QFC no. 01169, and has its registered office at Building No. 36, 4th Floor, Abdulla Bin Thani Street, Doha Design District 1, Msheireb Downtown Doha, State of Qatar. This material is strictly intended for Eligible Counterparties and/or Business Customers only as classified under the QFCRA's Customer and Investor Protection Rules 2019. No other person should act upon this material. **Russia**: This document or information contained therein is for information purposes only and constitutes neither a public nor a private offering, is not an invitation to make offers, to sell, exchange or otherwise transfer any financial instruments in the Russian Federation to or for the benefit of any Russian person or entity and does not constitute an advertisement or offering of securities in the Russian Federation within the meaning of Russian securities laws. The information contained herein is not an "individual investment recommendation" as defined in Federal Law of 22 April 1996 No 39-FZ "On Securities Market" (as amended) and the financial instruments and operations specified herein may not be suitable for your investment profile or your investment goals or expectations. The determination of whether or not such financial instruments and operations are in your interests or are suitable for your investment goals, investment horizon or the acceptable risk level is your responsibility. We assume no liability for any losses connected with making any such operations or investing into any such financial instruments and we do not recommend to use such information as the only source of information for making an investment decision. **Saudi Arabia**: This material is for marketing and information purposes by UBS Saudi Arabia only. It has been distributed by UBS Saudi Arabia or reviewed and approved by UBS Saudi Arabia if it was issued by UBS Switzerland AG, by UBS AG, or any of its subsidiaries or branches. UBS Switzerland AG is a public company limited by shares, incorporated in Switzerland with its registered offices at Bahnhofstrasse 45, CH-8001 Zurich. UBS Saudi Arabia (a subsidiary of UBS AG) is a Saudi closed joint stock company incorporated in the Kingdom of Saudi Arabia with a paid capital of 110,000,000 Saudi Riyals under commercial register number 1010257812 having its registered office at Laysen Valley, P.O. Box 75724, Riyadh 11588, Kingdom of Saudi Arabia. UBS Saudi Arabia is authorized and regulated by the Capital Market Authority to conduct securities business under license number 08113-37. This document is distributed only under such circumstances as may be permitted by applicable laws and regulations. The information and – if any - opinions contained in this document have been compiled or arrived at based upon information obtained from sources believed to be reliable and in good faith, but is not guaranteed as being accurate, nor is it a complete statement or summary of the securities, markets or developments referred to in the document. All such information and opinions are subject to changes without notice. Past performance of investments (whether simulated or actual) is not necessarily an indicator of future results. Forward looking data is based on estimates that are subject to changes. Should the currency of a financial product or service not match your reference currency, performance may rise or fall due to currency fluctuations. UBS Switzerland AG and / or other members of the UBS Group may have a position in and may make a purchase and/or sale of any of the securities or other financial instruments mentioned in this document While statements that constitute "forward-looking statements", including, but not limited to, statements relating to our future business development represent UBS judgments and future expectations concerning the development of our business, a number of risks, uncertainties and other important factors could cause actual developments and results to differ materially from our expectations. This document may not be forwarded, reproduced, redistributed or republished for any purpose without the written permission of UBS Saudi Arabia. Source for all data and charts (if not indicated otherwise) is UBS Switzerland AG. **Singapore**: Where applicable, this material is distributed in Singapore by UBS AG, Singapore Branch, which is licensed by the Monetary Authority of Singapore under the Banking Act 1970 to carry on banking business. UBS AG is incorporated in Switzerland with limited liability. UBS AG has a branch registered in Singapore (UEN S98FC5560C). This material has been prepared and issued for distribution in Singapore to institutional investors, accredited investors and expert investors (each as defined under the Financial Advisers Regulations (the "FAR")) only. By virtue of your status as an institutional investor, accredited investor, or expert investor, UBS AG is exempted from complying with certain requirements under the Financial Advisers Act 2001 (the "FAA"), the FAR and the relevant Notices and Guidelines issued thereunder, in respect of any financial advisory service which UBS AG may provide to you. These include exemptions from complying with: Section 34 of the FAA (pursuant to Regulation 33(1) of the FAR); Section 36 of the FAA (pursuant to Regulation 34(1) of the FAR); and Section 45 of the FAA (pursuant to Regulation 35(1) of the FAR). Singapore recipients and clients of UBS AG, Singapore Branch should contact UBS AG, Singapore Branch for any matters arising from, or in connection with, this material. Where applicable, this communication and any offering material term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved or endorsed by any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or products or futures contracts). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Spain:** This report is distributed in Spain by UBS AG, Sucursal en España, authorized under number 1460 in the Register by the Banco de España. **Sweden**: This publication is not intended to constitute a public offer under Swedish law. It is distributed only for information purposes to clients of UBS Europe SE, Sweden Bankfilial, with place of business at Regeringsgatan 38, 11153 Stockholm, Sweden, registered

with the Swedish Companies Registration Office under Reg. No 516406-1011. UBS Europe SE, Sweden Bankfilial is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the Swedish supervisory authority (Finansinspektionen), to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Taiwan**: This material is provided by UBS AG, Taipei Branch in accordance with laws of Taiwan, in agreement with or at the request of clients/prospects. **Thailand**: This communication and any offering material, term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication were done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or derivatives products). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or derivatives products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Türkiye**: The information in this document is not provided for the purpose of offering, marketing or sale of any capital market instrument or service in the Republic of Türkiye. Therefore, this document may not be considered as an offer made, or to be made, to residents of the Republic of Türkiye in the Republic of Türkiye. UBS Switzerland AG is not licensed by the Capital Markets Board of Türkiye (the CMB) under the provisions of the Capital Market Law (Law No. 6362). Accordingly, neither this document nor any other offering material related to the instrument/service may be utilized in connection with providing any capital market services to persons within the Republic of Türkiye without the prior approval of the CMB. However, according to article 15 (d) (ii) of the Decree No. 32 residents of the Republic of Türkiye are allowed to purchase or sell the financial instruments traded in financial markets outside of the Republic of Türkiye. Further to this, pursuant to article 9 of the Communiqué on Principles Regarding Investment Services, Activities and Ancillary Services No. III-37.1, investment services provided abroad to residents of the Republic of Türkiye based on their own initiative are not restricted. **United Arab Emirates (UAE) / DIFC / Abu Dhabi**: UBS is not licensed in the UAE by the Central Bank of the UAE nor by the Emirates' Securities and Commodities Authority and does not undertake banking activities in the UAE. This document is provided for your information only and does not constitute financial advice. **DIFC**: UBS AG Dubai Branch is regulated by the DFSA in the DIFC. This material is strictly intended for Professional Clients and/or Market Counterparties only as classified under the DFSA rulebook. It should not be distributed to Retail Clients. The Investment Research is provided for information purposes only and is not a recommendation or offer to buy/sell/hold a particular investment. The investment research may be out of date. You should seek investment advice before acting on the basis of the Investment Research. **Abu Dhabi**: UBS AG Abu Dhabi Branch is licensed and regulated by the Financial Services Regulatory Authority ("FSRA") of the Abu Dhabi Global Market. This material is intended solely for professional clients or market counterparties, as defined in the rules of the FSRA. It is not directed at, nor intended for, retail clients or any person who does not meet the criteria of a professional client or market counterparty. **United Kingdom**: This document is issued by UBS Wealth Management, a division of UBS AG which is authorised and regulated by the Financial Market Supervisory Authority in Switzerland. In the United Kingdom, UBS AG is authorised by the Prudential Regulation Authority and is subject to regulation by the Financial Conduct Authority and limited regulation by the Prudential Regulation Authority. Details about the extent of regulation by the Prudential Regulation Authority are available from us on request. A member of the London Stock Exchange. This publication is distributed to retail clients of UBS Wealth Management. **Ukraine**: UBS is a premier global financial services firm offering wealth management services to individual, corporate and institutional investors. UBS is established in Switzerland and operates under Swiss law and in over 50 countries and from all major financial centers. UBS is not registered and licensed as a bank/financial institution under Ukrainian legislation and does not provide banking and other financial services in Ukraine. UBS has not made, and will not make, any offer of the mentioned products to the public in Ukraine. No action has been taken to authorize an offer of the mentioned products to the public in Ukraine and the distribution of this document shall not constitute financial services for the purposes of the Law of Ukraine "On Financial Services and Financial Companies" dated 14 December 2021. Any offer of the mentioned products shall not constitute an investment advice, public offer, circulation, transfer, safekeeping, holding or custody of securities in the territory of Ukraine. Accordingly, nothing in this document or any other document, information or communication related to the mentioned products shall be interpreted as containing an offer, a public offer or invitation to offer or to a public offer, or solicitation of securities in the territory of Ukraine or investment advice under Ukrainian law. Electronic communication must not be considered as an offer to enter into an electronic agreement or other electronic instrument within the meaning of the Law of Ukraine "On Electronic Commerce" dated 3 September 2015. This document is strictly for private use by its holder and may not be passed on to third parties or otherwise publicly distributed. **USA**: Distributed to US persons only by UBS Financial Services Inc. or UBS Securities LLC, subsidiaries of UBS AG. UBS Switzerland AG, UBS Europe SE, UBS Bank, S.A., UBS BB Corretora de Câmbio, Títulos e Valores Mobiliários S.A., UBS Asesores México, S.A. de C.V., UBS SuMi TRUST Wealth Management Co., Ltd., UBS Wealth Management Israel Ltd. and UBS Menkul Degerler AS are affiliates of UBS AG. **UBS Financial Services Inc. accepts responsibility for the content of a report prepared by a non-US affiliate when it distributes reports to US persons. All transactions by a US person in the securities mentioned in this report should be effected through a US-registered broker dealer affiliated with UBS, and not through a non-US affiliate. The contents of this report have not been and will not be approved by any securities or investment authority in the United States or elsewhere. UBS Financial Services Inc. is not acting as a municipal advisor to any municipal entity or obligated person within the meaning of Section 15B of the Securities Exchange Act (the "Municipal Advisor Rule") and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of the Municipal Advisor Rule.** For information on the ways in which UBS Securities LLC manages conflicts and maintains independence of its UBS Global Research product; historical performance information; certain additional disclosures concerning UBS Global Research recommendations; and terms and conditions for certain third party data used in research report, please visit https://www.ubs.com/disclosures.

© UBS 2026. The key symbol and UBS are among the registered and unregistered trademarks of UBS. All rights reserved.

Nanya Technology

Further upside to expectations likely

Another strong quarter

Nanya Tech (NTC) reported strong 3Q26 revenue of NT$133.6bn (+62% QoQ, +612% YoY) in line with UBSe of NT$132.7bn and 6% above Visible Alpha consensus of NT$126.5bn. EPS came in at $25.40, slightly below UBSe of NT$26.44 and modestly above consensus of NT$24.40. Growth was driven primarily by strong ASPs, up 65% QoQ (vs UBSe of +50%) mainly on the back of the renewal of 6-month and 1-year contracts that had expired, partially offset by lower-than-expected bit growth of -5% QoQ (vs UBSe of +5% QoQ). Profitability, again, improved sequentially, with gross/operating margins rising to 85.7%/80.0% from 79.5%/73.7% in 2Q26.

Fulfilment rates to remain low until Fab 5A ramp

NTC continues to see supportive supply-demand dynamics, with fulfilment rates for demand outside of long-term contracts – which account for close to 70% of bits already – remaining low. That said, NTC sees limited upside to unit shipments in the near term, given low inventory levels and limited floor space: (1) reiterating its full-year bit growth guidance of "up high-teens % YoY," implying 4Q unit shipments will decline MSD % QoQ (UBSe: -6%); and (2) guiding shipments to remain flat thereafter until its new fab begins to ramp in 2H27. NTC's efforts to bring Fab 5A online are progressing well, with equipment move-in scheduled for 1Q27 and initial production expected by 3Q27. NTC continues to expect wafer capacity to reach 30k wpm by the end of 2028 and 50k by the end of 2030. With demand remaining strong and existing capacity not enough to fully address this, NTC continues to see upside to sequential ASP growth in the near term, although at a more moderate pace.

Changes to forecasts

Reflecting 3Q26 results, as well as our new bit growth and ASP forecasts, we keep our 2026E forecasts largely unchanged but slightly raise our 2027E/2028E revenue forecasts by 7%/5% to NT$655bn/NT$735bn (from NT$613bn/NT$702bn previously) and EPS forecasts by 10%/7% to NT$128.16/NT$134.53 (from NT$116.41/NT$125.68 previously). Our 2027E/2028E EPS forecasts now sit 10%/8% above pre-results Visible Alpha consensus.

Valuation: PT NT$640/share; maintain Buy

We value Nanya Tech shares at 2.63x NTM P/BV based on our 2028E-31E ROE estimate of 21.4% and CoE of 8.1%.

Equities

Taiwan
Semiconductors

12-month rating	**Buy**
12m price target	**NT$640.00**
Price (12 Oct 2026)	**NT$527.00**

RIC: 2408.TW **BBG:** 2408 TT

Trading data and key metrics

52-wk range	NT$543.00-86.30
Market cap.	NT$1,818b/US$56.9b
Shares o/s	3,450m (ORD)
Free float	37%
Avg. daily volume ('000)	72,536
Avg. daily value (m)	NT$34,651.5
Common s/h equity (12/26E)	NT$501b
P/BV (12/26E)	3.6x
Net debt to EBITDA (12/26E)	NM

EPS (UBS, diluted) (TWD)

	From	To	% ch	Cons.
12/26E	76.97	76.14	-1	67.67
12/27E	116.41	128.16	10	104.38
12/28E	125.68	134.53	7	108.81

Nicolas Gaudois
Analyst
nicolas.gaudois@ubs.com
+65-6495 5148

Jimmy Yoon
Analyst
jimmy.yoon@ubs.com
+65-6495 4617

Luke Yoo
Analyst
luke.yoo@ubs.com
+82-2-3702 8132

Highlights (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenues	29,892	34,132	66,587	403,604	655,400	734,860	415,701	419,491
EBIT (UBS)	(14,460)	(10,555)	5,242	310,445	519,567	536,428	166,190	130,632
Net earnings (UBS)	(7,440)	(5,083)	6,612	259,333	442,187	464,159	168,716	140,745
EPS (UBS, diluted) (NT$)	(2.40)	(1.64)	2.13	76.14	128.16	134.53	48.90	40.79
DPS (net) (NT$)	0.00	0.00	1.50	31.81	32.04	33.63	12.23	10.20
Net (debt) / cash	47,631	36,638	49,018	339,408	493,617	741,394	806,981	819,328

Profitability/valuation	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
EBIT (UBS) margin %	(48.4)	(30.9)	7.9	76.9	79.3	73.0	40.0	31.1
ROIC (EBIT) %	(13.4)	(8.9)	4.4	229.5	210.8	137.3	38.0	27.6
EV/EBITDA (UBS core) x	>100	23.9	8.2	4.6	2.4	1.8	4.3	4.4
P/E (UBS, diluted) x	(27.9)	(34.4)	32.5	6.9	4.1	3.9	10.8	12.9
Equity FCF (UBS) yield %	(9.3)	(8.1)	2.5	11.7	14.3	19.7	10.0	3.0
Dividend yield (net) %	0.0	0.0	2.2	6.0	6.1	6.4	2.3	1.9

Source: Company accounts, LSEG Eikon, UBS estimates. Metrics marked as (UBS) have had analyst adjustments applied. Valuations: based on an average share price that year, (E): based on a share price of NT$ 527.00 on 12-Oct-2026 21:58:09 CST

This report has been prepared by UBS Securities Pte. Ltd. (Reg. No. 198500648C) **ANALYST CERTIFICATION AND REQUIRED**

UBS Research THESIS MAP a guide to our thinking and what's where in this report

PIVOTAL QUESTIONS

Q: Will DRAM undersupply continue into 2028?
Yes, in our view. Agentic AI is boosting further memory demand on multiple fronts beyond HBM, including DDR5/LPDDR5 for conventional servers and CPU head nodes in AI servers. We now forecast an acceleration of memory demand growth into 2027 with DRAM bit end-consumption growing 40% YoY vs 21% in 2026. As almost all of the incremental wafer capacity addition for DRAM is going to HBM, supply is not catching up with demand. The main risk remains affordability by customers as memory industry revenues are expected to reach US$1.69tn in 2027E.

Q: Can NTC grow as a niche DRAM player?
Yes, we think 2027 is likely to be a year of more constrained growth. Subsequently, though, we think its process tech migration to 1c nm and 1d nm will likely lead to a 30% increase in bits per wafer for each process tech migration (at comparable yields). NTC is also slightly pulling in its new fab ramp, with equipment expected to move in 1Q27. We also see a tail of DDR3 and DDR4 products NTC can address for a while.

UBSVIEW

Nanya Tech shares trade at 2.17x NTM P/BV, above +1SD of its 10-year average. That said, we believe the strength of the unfolding DRAM cycle, as well as the tail of legacy products that Nanya can address as a niche player, should continue to provide support for the stock.

EVIDENCE

Our industry checks suggest HBM demand continues to result in capacity allocation away from DDR, which may moderate possible cyclical corrections, supporting upside to DDR contract pricing for longer.

WHAT'S PRICED IN?

At 2.17x NTM P/BV, we believe Nanya Tech shares are discounting a LT average ROE of 17.6% vs our LT estimate of 21.4%.

Value drivers	DRAM ASP '27E/'28E	OPM % '27E/'28E
NT$800 upside	+80%/-10%	81%/78%
NT$640 base	+62%/-24%	79%/73%
NT$380 downside	+40%/-40%	74%/68%

Source: UBS estimates

COMPANY DESCRIPTION

Nanya Tech is a specialty DRAM supplier in Taiwan focusing on legacy DDR products.

UBS FORECASTS

Figure 1: Nanya Tech – UBSe vs consensus estimates

	4Q26E			1Q27E			2026E			2027E			2028E		
(NT$m)	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**	**UBSe**	**Cons**	**Delta**
Sales	138,319	133,672	3%	146,973	142,479	3%	403,604	391,819	3%	655,400	623,265	5%	734,860	716,632	3%
Operating Profit	112,621	107,365	5%	119,606	113,723	5%	310,445	301,816	3%	519,567	491,832	6%	536,428	520,481	3%
% OP margin	*81.4%*	*80.3%*	*110bps*	*81.4%*	*79.8%*	*156bps*	*76.9%*	*77.0%*	*-11bps*	*79.3%*	*78.9%*	*36bps*	*73.0%*	*72.6%*	*37bps*
EPS (NT$)	**27.67**	**25.15**	**10%**	**29.49**	**26.59**	**11%**	**76.14**	**72.83**	**5%**	**128.16**	**116.75**	**10%**	**134.53**	**124.94**	**8%**

Source: Visible Alpha, UBS estimates

Figure 2: Nanya Tech – UBS estimates - new vs old

	4Q26E			1Q27E			2026E			2027E			2028E		
(NT$m)	**New**	**Old**	**Delta**	**New**	**Old**	**Delta**	**New**	**Old**	**Delta**	**New**	**Old**	**Delta**	**New**	**Old**	**Delta**
Sales	138,319	138,414	0%	146,973	139,476	5%	403,604	402,823	0%	655,400	612,781	7%	734,860	702,216	5%
Operating Profit	112,621	112,174	0%	119,606	110,882	8%	310,445	311,653	0%	519,567	471,498	10%	536,428	502,443	7%
% OP margin	*81.4%*	*81.0%*	*38bps*	*81.4%*	*79.5%*	*188bps*	*76.9%*	*77.4%*	*-45bps*	*79.3%*	*76.9%*	*233bps*	*73.0%*	*71.6%*	*145bps*
EPS	**27.67**	**27.45**	**1%**	**29.49**	**27.36**	**8%**	**76.14**	**76.97**	**-1%**	**128.16**	**116.41**	**10%**	**134.53**	**125.68**	**7%**

Source: UBS estimates

Figure 3: Nanya Tech – DRAM forecasts

	2025	1Q26	2Q26	3Q26	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
DRAM revenues (NT$m)	**66,587**	**49,087**	**82,549**	**133,649**	**138,319**	**403,604**	**146,973**	**153,369**	**168,013**	**187,046**	**655,400**	**734,860**	**415,701**	**419,491**
% QoQ		*63%*	*68%*	*62%*	*3%*		*6%*	*4%*	*10%*	*11%*				
% YoY	*95%*	*583%*	*684%*	*612%*	*360%*	*506%*	*199%*	*86%*	*26%*	*35%*	*62%*	*12%*	*-43%*	*1%*
DRAM revenues (US$m)	**2,158**	**1,579**	**2,647**	**4,148**	**4,344**	**12,718**	**4,608**	**4,808**	**5,267**	**5,864**	**20,547**	**23,039**	**13,033**	**13,151**
% QoQ		*62%*	*68%*	*57%*	*5%*		*6%*	*4%*	*10%*	*11%*				
% YoY	*97%*	*623%*	*677%*	*564%*	*346%*	*489%*	*192%*	*82%*	*27%*	*35%*	*62%*	*12%*	*-43%*	*1%*
DRAM unit shipment (1Gb eq)	4,643	1,441	1,438	1,364	1,286	5,529	1,286	1,292	1,382	1,556	5,516	8,169	10,752	13,603
% QoQ		*-6%*	*0%*	*-5%*	*-6%*		*0%*	*0%*	*7%*	*13%*				
% YoY	*54%*	*123%*	*32%*	*-1%*	*-16%*	*19%*	*-11%*	*-10%*	*1%*	*21%*	*0%*	*48%*	*32%*	*27%*
DRAM ASP (US$ 1Gb eq)	0.46	1.10	1.84	3.04	3.38	2.30	3.58	3.72	3.81	3.77	3.72	2.82	1.21	0.97
% QoQ		*73%*	*68%*	*65%*	*11%*		*6%*	*4%*	*2%*	*-1%*				
% YoY	*28%*	*224%*	*487%*	*568%*	*433%*	*395%*	*227%*	*102%*	*25%*	*12%*	*62%*	*-24%*	*-57%*	*-20%*
Operating Profit (NT$m)	**5,242**	**30,102**	**60,828**	**106,894**	**112,621**	**310,445**	**119,606**	**123,641**	**132,974**	**143,347**	**519,567**	**536,428**	**166,190**	**130,632**
% OP margin	*7.9%*	*61.3%*	*73.7%*	*80.0%*	*81.4%*	*76.9%*	*81.4%*	*80.6%*	*79.1%*	*76.6%*	*79.3%*	*73.0%*	*40.0%*	*31.1%*

Source: UBS estimates

Figure 4: Nanya Tech – Income statement forecasts

(NT$m)	**2025**	**1Q26**	**2Q26**	**3Q26**	**4Q26E**	**2026E**	**1Q27E**	**2Q27E**	**3Q27E**	**4Q27E**	**2027E**	**2028E**	**2029E**	**2030E**
Revenues	**66,587**	**49,087**	**82,549**	**133,649**	**138,319**	**403,604**	**146,973**	**153,369**	**168,013**	**187,046**	**655,400**	**734,860**	**415,701**	**419,491**
% QoQ		*63%*	*68%*	*62%*	*3%*		*6%*	*4%*	*10%*	*11%*				
% YoY	*95%*	*583%*	*684%*	*612%*	*360%*	*506%*	*199%*	*86%*	*26%*	*35%*	*62%*	*12%*	*-43%*	*1%*
COGS	51,600	15,761	16,932	19,167	19,489	71,349	20,669	22,392	26,746	34,449	104,255	157,603	201,028	232,721
Gross Profit	**14,986**	**33,325**	**65,617**	**114,482**	**118,831**	**332,256**	**126,304**	**130,977**	**141,267**	**152,597**	**551,145**	**577,256**	**214,673**	**186,771**
% GP margin	*22.5%*	*67.9%*	*79.5%*	*85.7%*	*85.9%*	*82.3%*	*85.9%*	*85.4%*	*84.1%*	*81.6%*	*84.1%*	*78.6%*	*51.6%*	*44.5%*
Operating Expenses	9,745	3,224	4,789	7,588	6,210	21,811	6,698	7,336	8,293	9,250	31,578	40,828	48,483	56,139
R&D	7,043	2,118	3,228	5,145	4,645	15,136	4,645	5,145	5,645	6,145	21,580	27,580	35,180	38,980
SG&A	2,702	1,105	1,561	2,443	1,565	6,674	2,053	2,191	2,648	3,105	9,998	13,248	13,303	17,159
Opex as % of revenues	*14.6%*	*6.6%*	*5.8%*	*5.7%*	*4.5%*	*5.4%*	*4.6%*	*4.8%*	*4.9%*	*4.9%*	*4.8%*	*5.6%*	*11.7%*	*13.4%*
Operating Profit	**5,242**	**30,102**	**60,828**	**106,894**	**112,621**	**310,445**	**119,606**	**123,641**	**132,974**	**143,347**	**519,567**	**536,428**	**166,190**	**130,632**
% OP margin	*7.9%*	*61.3%*	*73.7%*	*80.0%*	*81.4%*	*76.9%*	*81.4%*	*80.6%*	*79.1%*	*76.6%*	*79.3%*	*73.0%*	*40.0%*	*31.1%*
Non-Operating Income	2,641	1,616	1,501	2,438	3,791	9,347	4,477	5,147	4,609	5,453	19,686	29,621	39,563	41,010
Investment Income	417	239	235	235	235	943	235	235	235	235	939	939	939	939
Interest Income	1,907	527	1,109	2,670	3,556	7,862	4,243	4,913	4,374	5,218	18,748	28,682	38,624	40,071
Others	316	851	158	0	0	542	0	0	0	0	(0)	0	0	0
Earnings Before Tax	**7,882**	**31,718**	**62,329**	**109,332**	**116,412**	**319,792**	**124,083**	**128,788**	**137,583**	**148,799**	**539,253**	**566,049**	**205,752**	**171,642**
Tax Expenses	1,269	5,660	12,138	21,705	20,954	60,457	22,335	23,182	24,765	26,784	97,066	101,889	37,035	30,896
% Tax rate	*16.1%*	*17.8%*	*19.5%*	*19.9%*	*18.0%*	*18.9%*	*18.0%*	*18.0%*	*18.0%*	*18.0%*	*18.0%*	*18.0%*	*18.0%*	*18.0%*
Earnings After Tax	6,613	26,058	50,192	87,627	95,458	259,335	101,748	105,606	112,818	122,015	442,188	464,160	168,717	140,746
Minority Interest	(1)	(1)	(0)	(0)	(0)	(2)	(0)	(0)	(0)	(0)	(1)	(1)	(1)	(1)
Net Profit After MI	6,614	26,059	50,192	87,627	95,458	259,337	101,748	105,606	112,818	122,016	442,189	464,161	168,718	140,747
EPS	**2.13**	**8.41**	**14.66**	**25.40**	**27.67**	**76.14**	**29.49**	**30.61**	**32.70**	**35.36**	**128.16**	**134.53**	**48.90**	**40.79**

Source: Company data, UBS estimates

Figure 5: Nanya Tech – Balance sheet forecasts

(Won bn)	2025	1Q26	2Q26	3Q26	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Cash and Cash Equivalents	58,074	86,281	217,603	288,507	343,405	343,405	397,009	353,948	421,432	497,613	497,613	745,391	810,978	823,325
Accounts Receivable	16,558	25,247	40,279	58,586	68,212	68,212	72,480	75,634	82,856	92,242	92,242	87,207	49,923	70,505
Inventories	27,288	24,331	21,530	6,301	12,814	12,814	13,591	14,723	17,586	22,651	22,651	27,181	38,255	40,269
Other Current Assets	6,621	5,748	5,110	5,110	5,110	5,110	5,110	5,110	5,110	5,110	5,110	5,110	5,110	5,110
Total Current Assets	**108,541**	**141,607**	**284,521**	**358,503**	**429,541**	**429,541**	**488,189**	**449,415**	**526,983**	**617,616**	**617,616**	**864,889**	**904,265**	**939,208**
Tangible Assets	85,031	87,349	89,804	107,302	128,887	128,887	171,518	210,990	247,305	280,881	280,881	388,401	407,051	471,037
Intangible Assets	531	500	1,038	1,214	1,432	1,432	1,863	2,262	2,629	2,968	2,968	4,054	4,242	4,888
Investments	5,904	6,111	7,063	7,063	7,063	7,063	7,063	7,063	7,063	7,063	7,063	7,063	7,063	7,063
Other Non-Current Assets	8,446	5,401	5,332	5,332	5,332	5,332	5,332	5,332	5,332	5,332	5,332	5,332	5,332	5,332
Total Non-Current Assets	**99,912**	**99,361**	**103,237**	**120,912**	**142,715**	**142,715**	**185,776**	**225,647**	**262,328**	**296,244**	**296,244**	**404,850**	**423,688**	**488,320**
Total Assets	**208,453**	**240,968**	**387,758**	**479,415**	**572,256**	**572,256**	**673,965**	**675,062**	**789,311**	**913,860**	**913,860**	**1,269,739**	**1,327,954**	**1,427,528**
Account Payables	6,481	7,298	8,090	9,452	6,834	6,834	6,795	7,362	8,793	11,326	11,326	13,591	19,128	20,134
ST Debts	5,060	0	0	0	0	0	0	0	0	0	0	0	0	0
Other Current Liabilities	7,480	17,910	33,085	40,082	40,082	40,082	40,082	40,082	40,082	40,082	40,082	40,082	40,082	40,082
Current Liabilities	**19,020**	**25,207**	**41,175**	**49,535**	**46,917**	**46,917**	**46,878**	**47,444**	**48,875**	**51,408**	**51,408**	**53,673**	**59,210**	**60,217**
LT Liabilities	3,996	3,996	3,997	3,997	3,997	3,997	3,997	3,997	3,997	3,997	3,997	3,997	3,997	3,997
Other Non-Current Liabilities	35	83	192	192	192	192	192	192	192	192	192	192	192	192
Non-Current Liabilities	**18,894**	**22,724**	**23,885**	**23,885**	**23,885**	**23,885**	**23,885**	**23,885**	**23,885**	**23,885**	**23,885**	**23,885**	**23,885**	**23,885**
Total Liabilities	**37,914**	**47,931**	**65,060**	**73,419**	**70,801**	**70,801**	**70,762**	**71,329**	**72,760**	**75,293**	**75,293**	**77,558**	**83,095**	**84,102**
Total Shareholders' Equity	**170,538**	**193,037**	**322,699**	**405,996**	**501,454**	**501,454**	**603,202**	**603,733**	**716,551**	**838,567**	**838,567**	**1,192,181**	**1,244,859**	**1,343,426**

Source: Company data, UBS estimates

Figure 6: Nanya Tech – Cash flow statement forecasts

(Won bn)	2025	1Q26	2Q26	3Q26	4Q26E	2026E	1Q27E	2Q27E	3Q27E	4Q27E	2027E	2028E	2029E	2030E
Net Profit	**6,614**	**26,059**	**50,192**	**87,627**	**95,458**	**259,337**	**101,748**	**105,606**	**112,818**	**122,016**	**442,189**	**464,161**	**168,718**	**140,747**
Depreciation and Amortization	14,281	2,897	2,960	3,108	3,897	12,862	4,785	7,974	11,164	13,930	37,853	58,853	76,853	94,853
Changes in Assets and Liabilities	(3,610)	(6,911)	(12,436)	(1,716)	(18,757)	(39,820)	(5,083)	(3,720)	(8,653)	(11,919)	(29,375)	2,769	31,748	(21,589)
Changes in A/R	(12,259)	(8,414)	(14,871)	(18,307)	(9,626)	(51,218)	(4,268)	(3,154)	(7,222)	(9,386)	(24,029)	5,034	37,285	(20,582)
Changes in Inventories	8,031	2,957	2,801	15,229	(6,513)	14,473	(776)	(1,133)	(2,863)	(5,065)	(9,837)	(4,530)	(11,074)	(2,013)
Changes in A/P	618	(1,454)	(366)	1,362	(2,618)	(3,076)	(39)	566	1,431	2,533	4,491	2,265	5,537	1,007
Others	1,284	9,503	15,143	6,998	0	31,644	0	0	0	0	0	(0)	0	0
Cash Flow from Operations	**18,568**	**31,549**	**55,859**	**96,017**	**80,598**	**264,023**	**101,450**	**109,861**	**115,329**	**124,027**	**450,667**	**525,784**	**277,319**	**214,011**
Capital Expenditures	(13,245)	(2,851)	(2,587)	(20,783)	(25,700)	(51,921)	(47,846)	(47,846)	(47,846)	(47,846)	(191,382)	(167,459)	(95,691)	(159,485)
Others	(865)	37	(1,656)	0	0	(1,619)	0	0	0	0	0	0	0	0
NOCF	5,324	28,698	53,272	75,234	54,898	212,102	53,604	62,015	67,484	76,182	259,285	358,324	181,628	54,526
Cash Flow from Investments	**(14,110)**	**(2,814)**	**(4,242)**	**(20,783)**	**(25,700)**	**(53,540)**	**(47,846)**	**(47,846)**	**(47,846)**	**(47,846)**	**(191,382)**	**(167,459)**	**(95,691)**	**(159,485)**
Dividend	0	0	0	(4,330)	0	(4,330)	0	(105,076)	0	0	(105,076)	(110,547)	(116,040)	(42,179)
Other	(6,317)	(1,547)	79,415	0	0	77,868	0	0	0	0	0	0	0	0
Cash Flow from Financing	**(6,317)**	**(1,547)**	**79,415**	**(4,330)**	**0**	**73,538**	**0**	**(105,076)**	**0**	**0**	**(105,076)**	**(110,547)**	**(116,040)**	**(42,179)**
Cash Flow from Others	(1,970)	1,019	290	0	0	1,309	0	0	0	0	0	0	0	0
Total Cash flow	**(3,829)**	**28,207**	**131,322**	**70,904**	**54,898**	**285,331**	**53,604**	**(43,061)**	**67,484**	**76,182**	**154,209**	**247,777**	**65,587**	**12,347**

Source: Company data, UBS estimates

Figure 7: Nanya Tech – UBS valuation

UBS price target (NT$)	**640.00**
Current stock price (NT$)	527.00
Number of shares (mil)	3,450
Market cap (NT$m)	1,818,254
Net debt (NT$m)	(284,510)
EV (NT$m)	1,533,745

	2025	2026E	2027E	2028E	2029E	2030E
Current valuation (x)						
EV/Revenues	23.03	3.80	2.34	2.09	3.69	3.66
P/Book	9.58	3.58	2.17	1.53	1.46	1.35
P/E	246.91	6.92	4.11	3.92	10.78	12.92
EV/EBITDA	78.56	4.74	2.75	2.58	6.31	6.80
Target price valuation (x)						
EV/Revenues	28.89	4.77	2.94	2.62	4.63	4.59
P/Book	11.63	4.35	2.63	1.85	1.77	1.64
P/E	299.85	8.41	4.99	4.76	13.09	4.76
EV/EBITDA	98.53	5.95	3.45	3.23	7.91	8.53
Other						
EPS	2.13	76.14	128.16	134.53	48.90	40.79
% ROE	*3.9%*	*77.2%*	*66.0%*	*45.7%*	*13.8%*	*10.9%*
Book value per share (NT$)	55.04	147.22	243.05	345.54	360.81	389.38
Cash per share (NT$)	18.74	100.82	144.23	216.04	235.05	238.63
Free cash flow per share (NT$)	1.44	61.79	75.15	103.86	52.64	15.80
% Free cash flow yield	*0.3%*	*11.7%*	*14.3%*	*19.7%*	*10.0%*	*3.0%*

Source: Company data, UBS estimates. Pricing data as of 12 October 2026.

Nanya Technology (2408.TW)

Income Statement (NT$m)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Revenues	**29,892**	**34,132**	**66,587**	**403,604**	*NM*	**655,400**	*62.4*	**734,860**	**415,701**	**419,491**
Gross profit	(4,483)	(421)	14,986	332,256	*NM*	551,145	*65.9*	577,256	214,673	186,771
EBITDA (UBS)	**865**	**5,589**	**19,523**	**323,307**	*NM*	**557,420**	*72.4*	**595,281**	**243,042**	**225,485**
Depreciation & amortisation	(15,326)	(16,144)	(14,281)	(12,862)	*9.9*	(37,853)	*-194.3*	(58,853)	(76,853)	(94,853)
EBIT (UBS)	**(14,460)**	**(10,555)**	**5,242**	**310,445**	*NM*	**519,567**	*67.4*	**536,428**	**166,190**	**130,632**
Associates & investment income	193	101	417	943	*125.8*	939	*-0.4*	939	939	939
Other non-operating income	549	923	316	542	*71.2*	0	-	0	0	0
Net interest	3,014	2,974	1,907	7,862	*312.4*	18,748	*138.5*	28,682	38,624	40,071
Exceptionals (incl goodwill)	0	0	0	0	-	0	-	0	0	0
Pre-tax profit	**(10,705)**	**(6,557)**	**7,882**	**319,792**	*NM*	**539,253**	*68.6*	**566,049**	**205,752**	**171,642**
Tax	3,265	1,474	(1,269)	(60,457)	*NM*	(97,066)	*-60.6*	(101,889)	(37,035)	(30,896)
Profit after tax	**(7,440)**	**(5,083)**	**6,613**	**259,335**	*NM*	**442,188**	*70.5*	**464,160**	**168,717**	**140,746**
Preference dividends	0	0	0	0	-	0	-	0	0	0
Minorities	0	0	(1)	(2)	*-100.8*	(1)	*47.3*	(1)	(1)	(1)
Extraordinary items	0	0	0	0	-	0	-	0	0	0
Net earnings (local GAAP)	**(7,440)**	**(5,083)**	**6,612**	**259,333**	*NM*	**442,187**	*70.5*	**464,159**	**168,716**	**140,745**
Net earnings (UBS)	**(7,440)**	**(5,083)**	**6,612**	**259,333**	*NM*	**442,187**	*70.5*	**464,159**	**168,716**	**140,745**
Tax rate (%)	0.0	0.0	16.1	18.9	*17.4*	18.0	*-4.8*	18.0	18.0	18.0

Per Share (NT$)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
EPS (UBS, diluted)	(2.40)	(1.64)	2.13	76.14	*NM*	128.16	*68.3*	134.53	48.90	40.79
EPS (local GAAP, diluted)	(2.40)	(1.64)	2.13	76.14	*NM*	128.16	*68.3*	134.53	48.90	40.79
EPS (UBS, basic)	(2.40)	(1.64)	2.13	76.14	*NM*	128.16	*68.3*	134.53	48.90	40.79
DPS (net) (NT$)	0.00	0.00	1.50	31.81	*NM*	32.04	*0.7*	33.63	12.23	10.20
Cash EPS (UBS, diluted) [1]	2.55	3.57	6.74	79.91	*NM*	139.13	*74.1*	151.59	71.18	68.29
Book value per share	53.98	53.37	55.04	147.22	*167.5*	243.05	*65.1*	345.54	360.81	389.38
Average shares (diluted)	3,098	3,099	3,099	3,406	*9.9*	3,450	*1.3*	3,450	3,450	3,450

Balance Sheet (NT$m)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Cash and equivalents	58,812	61,903	58,074	343,405	*NM*	497,613	*44.9*	745,391	810,978	823,325
Other current assets	37,698	46,043	50,467	86,136	*70.7*	120,002	*39.3*	119,498	93,288	115,883
Total current assets	**96,510**	**107,946**	**108,541**	**429,541**	*295.7*	**617,616**	*43.8*	**864,889**	**904,265**	**939,208**
Net tangible fixed assets	81,838	84,327	85,031	128,887	*51.6*	280,881	*117.9*	388,401	407,051	471,037
Net intangible fixed assets	927	688	531	1,432	*169.6*	2,968	*107.2*	4,054	4,242	4,888
Investments / other assets	13,076	13,745	14,349	12,395	*-13.6*	12,395	*0.0*	12,395	12,395	12,395
Total assets	**192,351**	**206,706**	**208,453**	**572,256**	*174.5*	**913,860**	*59.7*	**1,269,739**	**1,327,954**	**1,427,528**
Trade payables & other ST liabilities	8,937	11,510	13,960	46,917	*236.1*	51,408	*9.6*	53,673	59,210	60,217
Short term debt	11,181	21,270	5,060	0	-	0	-	0	0	0
Total current liabilities	**20,118**	**32,780**	**19,020**	**46,917**	*146.7*	**51,408**	*9.6*	**53,673**	**59,210**	**60,217**
Long term debt	0	3,995	3,996	3,997	*0.0*	3,997	*0.0*	3,997	3,997	3,997
Other long term liabilities	5,309	4,878	14,898	19,888	*33.5*	19,888	*0.0*	19,888	19,888	19,888
Preferred shares	0	0	0	0	-	0	-	0	0	0
Total liabilities (incl pref shares)	**25,427**	**41,653**	**37,914**	**70,801**	*86.7*	**75,293**	*6.3*	**77,558**	**83,095**	**84,102**
Common s/h equity	**166,924**	**165,053**	**170,538**	**501,454**	*194.0*	**838,567**	*67.2*	**1,192,181**	**1,244,859**	**1,343,426**
Minority interests	0	0	0	0	-	0	-	0	0	0
Total liabilities & equity	**192,351**	**206,706**	**208,453**	**572,256**	*174.5*	**913,860**	*59.7*	**1,269,739**	**1,327,954**	**1,427,528**

Cash Flow (NT$m)	12/23	12/24	12/25	12/26E	%ch	12/27E	%ch	12/28E	12/29E	12/30E
Net income (before pref divs)	(7,440)	(5,083)	6,612	259,333	*NM*	442,187	*70.5*	464,159	168,716	140,745
Depreciation & amortisation	15,326	16,144	14,281	12,862	*-9.9*	37,853	*194.3*	58,853	76,853	94,853
Net change in working capital	(5,679)	(6,705)	(3,610)	(39,820)	*NM*	(29,375)	*26.2*	2,769	31,748	(21,589)
Other operating	(8,303)	(2,404)	1,284	31,644	*NM*	0	-	0	0	0
Operating cash flow	**(6,095)**	**1,952**	**18,566**	**264,019**	*NM*	**450,665**	*70.7*	**525,782**	**277,316**	**214,009**
Tangible capital expenditure	(13,112)	(15,981)	(13,112)	(51,402)	*-292.0*	(189,468)	*-268.6*	(165,785)	(94,734)	(157,890)
Intangible capital expenditure	(132)	(161)	(132)	(519)	*-292.0*	(1,914)	*-268.6*	(1,675)	(957)	(1,595)
Net (acquisitions) & disposals	0	0	0	0	-	0	-	0	0	0
Other investing	459	(45)	(865)	(1,619)	*-87.2*	0	-	0	0	0
Investing cash flow	**(12,786)**	**(16,188)**	**(14,110)**	**(53,540)**	*-279.5*	**(191,382)**	*-257.5*	**(167,459)**	**(95,691)**	**(159,485)**
Equity dividends paid	(6,600)	0	0	(4,330)	-	(105,076)	*NM*	(110,547)	(116,040)	(42,179)
Share issues / (buybacks)	0	0	0	0	-	0	-	0	0	0
Other financing	(596)	4,051	(97)	78,095	-	0	-	0	0	0
Change in debt & pref shares	11,392	9,656	(6,221)	(226)	*96.4*	0	-	0	0	0
Financing cash flow	**4,196**	**13,707**	**(6,317)**	**73,538**	-	**(105,076)**	-	**(110,547)**	**(116,040)**	**(42,179)**
Cash flow inc/(dec) in cash	**(14,685)**	**(529)**	**(1,860)**	**284,017**	-	**154,207**	*-45.7*	**247,775**	**65,585**	**12,345**
FX / non cash items	(96)	3,619	(1,968)	1,313	-	2	*-99.8*	2	2	2
Balance sheet inc/(dec) in cash	**(14,781)**	**3,091**	**(3,829)**	**285,331**	-	**154,209**	*-46.0*	**247,777**	**65,587**	**12,347**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.[1] Cash EPS (UBS, diluted) is calculated using UBS net income adding back depreciation and amortization.

Nanya Technology (2408.TW)

Valuation (x)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
P/E (local GAAP, diluted)	NM	NM	32.5	6.9	4.1	3.9	10.8	12.9
P/E (UBS, diluted)	(27.9)	(34.4)	32.5	6.9	4.1	3.9	10.8	12.9
P/CEPS	26.3	15.8	10.3	6.6	3.8	3.5	7.4	7.7
Equity FCF (UBS) yield %	(9.3)	(8.1)	2.5	11.7	14.3	19.7	10.0	3.0
Dividend yield (net) %	0.0	0.0	2.2	6.0	6.1	6.4	2.3	1.9
P/BV	1.2	1.1	1.3	3.6	2.2	1.5	1.5	1.4
EV/revenues (core)	5.2	3.9	2.4	3.6	2.0	1.5	2.5	2.4
EV/EBITDA (UBS core)	>100	23.9	8.2	4.6	2.4	1.8	4.3	4.4
EV/EBIT (core)	NM	NM	30.4	4.7	2.5	2.0	6.2	7.6
EV/OpFCF (core)	NM	NM	24.9	5.4	3.6	2.5	7.0	14.8
EV/op. invested capital	1.4	1.1	1.3	10.9	5.3	2.7	2.4	2.1

Enterprise value (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Market cap.	207,317	174,677	214,287	1,818,254	1,818,254	1,818,254	1,818,254	1,818,254
Net debt (cash)	(47,631)	(36,638)	(49,018)	(339,408)	(493,617)	(741,394)	(774,187)	(813,155)
Buy out of minorities	0	0	0	0	0	0	0	0
Pension provisions/other	0	0	0	0	0	0	0	0
Total enterprise value	**159,686**	**138,039**	**165,268**	**1,478,846**	**1,324,638**	**1,076,860**	**1,044,067**	**1,005,100**
Non core assets	(5,120)	(4,645)	(5,904)	(7,063)	(7,063)	(7,063)	(7,063)	(7,063)
Core enterprise value	**154,566**	**133,394**	**159,365**	**1,471,783**	**1,317,574**	**1,069,797**	**1,037,004**	**998,036**

Growth (%)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Revenue	(47.5)	14.2	95.1	NM	62.4	12.1	(43.4)	0.9
EBITDA (UBS)	(96.7)	NM	NM	NM	72.4	6.8	(59.2)	(7.2)
EBIT (UBS)	-	27.0	-	NM	67.4	3.2	(69.0)	(21.4)
EPS (UBS, diluted)	-	31.7	-	NM	68.3	5.0	(63.7)	(16.6)
Net DPS	-	-	-	NM	0.7	5.0	(63.7)	(16.6)

Margins & Profitability (%)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Gross profit margin	NM	NM	22.5	NM	NM	NM	51.6	44.5
EBITDA margin	2.9	16.4	29.3	NM	NM	NM	58.5	53.8
EBIT (UBS) margin	(48.4)	(30.9)	7.9	76.9	79.3	73.0	40.0	31.1
Net earnings (UBS) margin	NM	NM	9.9	64.3	67.5	63.2	40.6	33.6
ROIC (EBIT)	NM	NM	4.4	NM	NM	NM	38.0	27.6
ROIC post tax	NM	NM	3.7	NM	NM	NM	31.2	22.6
ROE (UBS)	(4.3)	(3.1)	3.9	77.2	66.0	45.7	13.8	10.9

Capital structure & Coverage (x)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Net debt / EBITDA	(55.0)	(6.6)	(2.5)	(1.0)	(0.9)	(1.2)	(3.3)	(3.6)
Net debt / total equity %	(28.5)	(22.2)	(28.7)	(67.7)	(58.9)	(62.2)	(64.8)	(61.0)
Net debt / (net debt + total equity) %	(39.9)	(28.5)	(40.3)	NM	NM	NM	NM	NM
Net debt/EV %	(38.0)	(30.5)	(25.9)	(13.1)	(31.4)	(57.3)	(74.2)	(80.9)
Capex / depreciation %	86.4	100.0	92.7	NM	NM	NM	124.5	168.1
Capex / revenue %	NM	NM	19.7	12.7	28.9	22.6	22.8	NM
EBIT / net interest	-	-	-	-	-	-	-	-
Dividend cover (UBS)	-	-	1.4	2.4	4.0	4.0	4.0	4.0
Div. payout ratio (UBS) %	-	-	70.3	41.8	25.0	25.0	25.0	25.0

Revenues by division (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	29,892	34,132	66,587	403,604	655,400	734,860	415,701	419,491
Total	**29,892**	**34,132**	**66,587**	**403,604**	**655,400**	**734,860**	**415,701**	**419,491**

EBIT (UBS) by division (NT$m)	12/23	12/24	12/25	12/26E	12/27E	12/28E	12/29E	12/30E
Others	(14,460)	(10,555)	5,242	310,445	519,567	536,428	166,190	130,632
Total	**(14,460)**	**(10,555)**	**5,242**	**310,445**	**519,567**	**536,428**	**166,190**	**130,632**

Source: Company accounts, UBS estimates. (UBS) metrics use reported figures which have been adjusted by UBS analysts.

Forecast returns

Forecast price appreciation	21.4%
Forecast dividend yield	5.8%
Forecast stock return	27.2%
Market return assumption	6.3%
Forecast excess return	20.9%

Company Description

Nanya Tech is a pure DRAM company based in Taiwan, focusing on legacy products for consumers, broad-based applications and is now growing in servers.

Valuation Method and Risk Statement

Nanya Tech is a pure-play DRAM manufacturer based in Taiwan mainly focused on legacy DDR products for customers across various end applications. We think downside risks to our thesis could include a potential slowdown in end demand and hence the conventional memory cycle as well as Nanya's legacy-centric product portfolio. We value Nanya Tech shares using a target NTM P/BV multiple based on long-term average ROE and cost of equity.

Quantitative Research Review

UBS Global Research publishes a quantitative assessment of its analysts' responses to certain questions about the likelihood of an occurrence of a number of short term factors in a product known as the 'Quantitative Research Review'. The views for this month can be found below. Views contained in this assessment on a particular stock reflect only the views on those short term factors which are a different timeframe to the 12-month timeframe reflected in any equity rating set out in this note. For previous responses please make reference to (i) previous UBS Global Research reports; and (ii) where no applicable research report was published that month, the Quantitative Research Review which can be found at https://neo.ubs.com/quantitative, or contact your UBS sales representative for access to the report or the Quantitative Research Team on ubs-quant-answers@ubs.com. A consolidated report which contains all responses is also available and again you should contact your UBS sales representative for details and pricing or the Quantitative Research Team on the email above.

Nanya Technology

Question	**Response**
1. Is the industry structure facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting worse, 3 = no change, 5 = getting better, N/A = no view)	4
2. Is the regulatory/government environment facing the firm likely to improve or deteriorate over the next six months? Rate on a scale of 1-5 (1 = getting tougher 3 = no change, 5 = getting better, N/A = no view)	3
3. Over the last 3-6 months in broad terms have things been improving/no change/getting worse for this stock? Rate on a scale of 1-5 (1 = getting a lot worse, 3 = not much change, 5 = getting a lot better, N/A = no view)	4
4. Relative to the current CONSENSUS EPS forecast, is the next company EPS update likely to lead to: (1 = negative surprise vs consensus, 3 = in-line with consensus, 5 = positive surprise vs consensus expectations, N/A = no view)	4
5. What's driving the difference?	
6. Relative to YOUR current earnings forecast, is there relatively greater risk at the next earnings result of:(1 = downside skew risk to earnings, 3 = equal upside or downside risk to earnings, 5 = upside skew risk to earnings, N/A = no view)	3
7. What's driving the difference?	
8. Is there an upcoming catalyst for the company over the next three months?	Positive Catalyst
9. Is there an actual or approximate date for the catalyst?	November 13, 2026
10. Is the catalyst date an actual or approximate date?	Approximate
11. What is the catalyst?	Further upside to DDR pricing

Required Disclosures

This document has been prepared by UBS Securities Pte. Ltd., an affiliate of UBS AG. UBS AG, its subsidiaries, branches and affiliates, including former Credit Suisse AG and its subsidiaries, branches and affiliates are referred to herein as "UBS".

For information on the ways in which UBS manages conflicts and maintains independence of its UBS Global Research product; historical performance information; certain additional disclosures concerning UBS Global Research recommendations; and terms and conditions for certain third party data used in research report, please visit https://www.ubs.com/disclosures. Unless otherwise indicated, information and data in this report are based on company disclosures including but not limited to annual, interim, quarterly reports and other company announcements. The figures contained in performance charts refer to the past; past performance is not a reliable indicator of future results. Additional information will be made available upon request. UBS Securities Co. Limited is licensed to conduct securities investment consultancy businesses by the China Securities Regulatory Commission. UBS acts or may act as principal in the debt securities (or in related derivatives) that may be the subject of this report. This recommendation was finalized on: 12 October 2026 02:28 PM GMT. UBS has designated certain UBS Global Research department members as Derivatives Research Analysts where those department members publish research principally on the analysis of the price or market for a derivative, and provide information reasonably sufficient upon which to base a decision to enter into a derivatives transaction. Where Derivatives Research Analysts co-author research reports with Equity Research Analysts or Economists, the Derivatives Research Analyst is responsible for the derivatives investment views, forecasts, and/or recommendations. **Quantitative Research Review:** UBS Global Research publishes a quantitative assessment of its analysts' responses to certain questions about the likelihood of an occurrence of a number of short term factors in a product known as the 'Quantitative Research Review'. Views contained in this assessment on a particular stock reflect only the views on those short term factors which are a different timeframe to the 12-month timeframe reflected in any equity rating set out in this note. For the latest responses, please see the Quantitative Research Review Addendum at the back of this report, where applicable. For previous responses please make reference to (i) previous UBS Global Research reports; and (ii) where no applicable research report was published that month, the Quantitative Research Review which can be found at https://neo.ubs.com/quantitative, or contact your UBS sales representative for access to the report or the Quantitative Research Team on ubs-quant-answers@ubs.com. A consolidated report which contains all responses is also available and again you should contact your UBS sales representative for details and pricing or the Quantitative Research team on the email above.

Analyst Certification:
Each research analyst primarily responsible for the content of this research report, in whole or in part, certifies that with respect to each security or issuer that the analyst covered in this report: (1) all of the views expressed accurately reflect his or her personal views about those securities or issuers and were prepared in an independent manner, including with respect to UBS, and (2) no part of his or her compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed by that research analyst in the research report.

UBS Global Research: Global Equity Rating Definitions

12-Month Rating	Definition	Coverage[1]	IB Services[2]
Buy	FSR is > 6% above the MRA.	55%	24%
Neutral	FSR is between -6% and 6% of the MRA.	39%	20%
Sell	FSR is > 6% below the MRA.	6%	21%

Source: UBS. Rating allocations are as of 30 September 2026.
1:Percentage of companies under coverage globally within the 12-month rating category.
2:Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

KEY DEFINITIONS:Forecast Stock Return (FSR) is defined as expected percentage price appreciation plus gross dividend yield over the next 12 months. In some cases, this yield may be based on accrued dividends. **Market Return Assumption (MRA)** is defined as the one-year local market interest rate plus 5% (a proxy for, and not a forecast of, the equity risk premium). **Under Review (UR)** Stocks may be flagged as UR by the analyst, indicating that the stock's price target and/or rating are subject to possible change in the near term, usually in response to an event that may affect the investment case or valuation. **Equity Price Targets** have an investment horizon of 12 months.

EXCEPTIONS AND SPECIAL CASES:UK and European Investment Fund ratings and definitions are: Buy: Positive on factors such as structure, management, performance record, discount; **Neutral:** Neutral on factors such as structure, management, performance record, discount; **Sell:** Negative on factors such as structure, management, performance record, discount. **Core Banding Exceptions (CBE):** Exceptions to the standard +/-6% bands may be granted by the Investment Review Consultation (IRC). Factors considered by the IRC include the stock's volatility and the credit spread of the respective company's debt. As a result, stocks deemed to be very high or low risk may be subject to higher or lower bands as they relate to the rating. When such exceptions apply, they will be identified in the Company Disclosures table in the relevant research piece.

Research analysts contributing to this report who are employed by any non-US affiliate of UBS Securities LLC are not registered/ qualified as research analysts with FINRA. Such analysts may not be associated persons of UBS Securities LLC and therefore are not subject to the FINRA restrictions on communications with a subject company, public appearances, and trading securities held by a research analyst account. The name of each affiliate and analyst employed by that affiliate contributing to this report, if any, follows.

UBS AG, Singapore Branch: Jimmy Yoon, Nicolas Gaudois.**UBS Securities Pte. Ltd., Seoul Branch:** Luke Yoo.

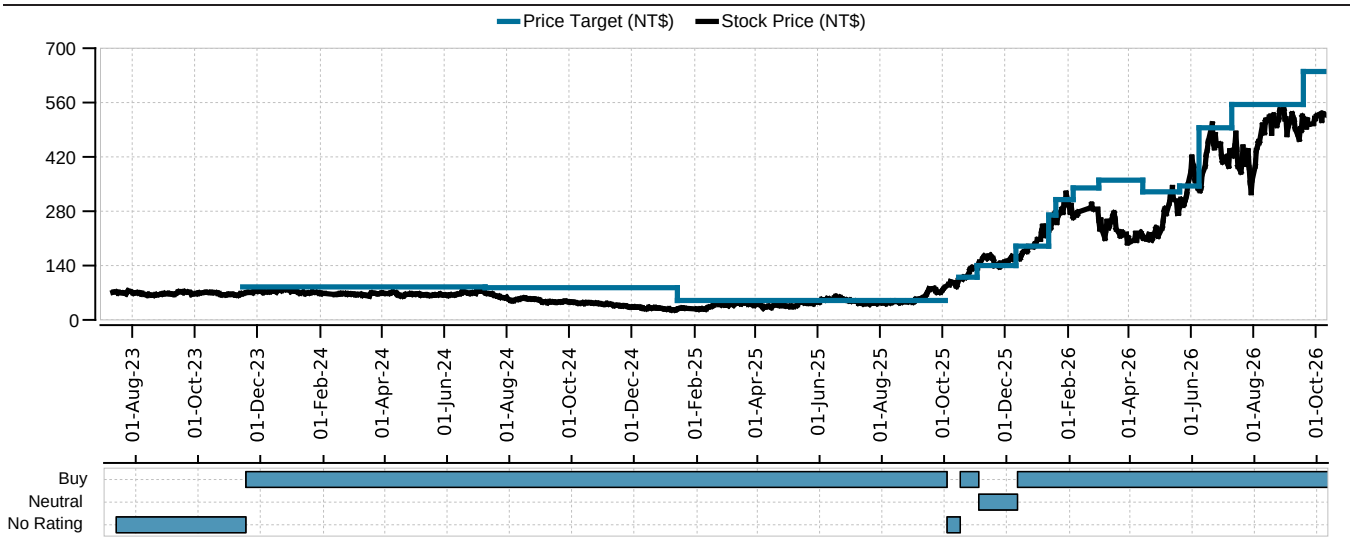
Company Disclosures

Company Name	Reuters	12-month rating	Price	Price date
Nanya Technology	2408.TW	Buy	NT$527.00	12 Oct 2026

Source: UBS Global Research; LSEG Eikon. All prices as of local market close. Ratings in this table are the most current published ratings prior to this report. They may be more recent than the stock pricing date.

Unless otherwise indicated, please refer to the Valuation and Risk sections within the body of this report. For a complete set of disclosure statements associated with the companies discussed in this report, including information on valuation and risk, please contact UBS Securities LLC, 11 Madison Avenue, New York, NY 10010, USA, Attention: Investment Research.

Nanya Technology (NT$)



Date	Stock Price (NT$)	Price Target (NT$)	Rating
2023-07-12	70.10	-	No Rating
2023-11-16	67.20	85.00	Buy
2024-07-10	72.30	83.00	Buy
2025-01-14	26.65	50.00	Buy
2025-10-03	83.00	-	No Rating
2025-10-16	94.90	110.00	Buy
2025-11-03	137.00	140.00	Neutral
2025-12-11	156.00	190.00	Buy
2026-01-12	239.00	270.00	Buy
2026-01-19	275.00	310.00	Buy
2026-02-05	275.00	340.00	Buy
2026-03-02	285.00	360.00	Buy
2026-04-14	220.00	330.00	Buy
2026-05-20	275.00	345.00	Buy
2026-06-08	339.50	495.00	Buy
2026-07-10	435.50	555.00	Buy
2026-09-18	525.00	640.00	Buy

Source: UBS Global Research; LSEG Eikon as of 12-Oct-2026. All prices as of local market close. Ratings as of date shown.

The Disclaimer relevant to Global Wealth Management clients follows the Global Research Disclaimer.

UBS Global Research Disclaimer

This document has been prepared by UBS Securities Pte. Ltd., an affiliate of UBS AG. UBS AG, its subsidiaries, branches and affiliates, including former Credit Suisse AG and its subsidiaries, branches and affiliates are referred to herein as "UBS".

Any opinions expressed in this document may change without notice and are only current as of the date of publication. Different areas, groups, and personnel within UBS may produce and distribute separate research products independently of each other. For example, research publications from **UBS CIO** are produced by UBS Global Wealth Management. **UBS Global Research** is produced by UBS Investment Bank. Research methodologies and rating systems of each separate research organization may differ, for example, in terms of investment recommendations, investment horizon, model assumptions, and valuation methods. As a consequence, except for certain economic forecasts (for which UBS CIO and UBS Global Research may collaborate), investment recommendations, ratings, price targets, and valuations provided by each of the separate research organizations may be different, or inconsistent. You should refer to each relevant research product for the details as to their methodologies and rating system. Not all clients may have access to all products from every organization. Each research product is subject to the policies and procedures of the organization that produces it.

This document is provided solely to recipients who are expressly authorized by UBS to receive it. If you are not so authorized you must immediately destroy the document.

UBS Global Research is provided to our clients through UBS Neo, and in certain instances, UBS.com and any other system or distribution method specifically identified in one or more communications distributed through UBS Neo or UBS.com (each a system) as an approved means for distributing UBS Global Research. It may also be made available through third party vendors and distributed by UBS and/or third parties via e-mail or alternative electronic means.

All UBS Global Research is available on UBS Neo. Please contact your UBS sales representative if you wish to discuss your access to UBS Neo. Where UBS Global Research refers to "UBS Evidence Lab Inside" or has made use of data provided by UBS Evidence Lab and you would like to access that data please contact your UBS sales representative. UBS Evidence Lab data is available on UBS Neo. The level and types of services provided by UBS Global Research and UBS Evidence Lab to a client may vary depending upon various factors such as a client's individual preferences as to the frequency and manner of receiving communications, a client's risk profile and investment focus and perspective (e.g., market wide, sector specific, long-term, short-term, etc.), the size and scope of the overall client relationship with UBS Global Research and UBS Evidence Lab and legal and regulatory constraints. UBS HOLT and UBS Pharma Values are offerings of UBS Global Research. HOLT Lens is a corporate performance platform offering that provides an objective accounting-led framework for comparing and valuing companies and is available to clients of UBS Global Research; for further details and pricing please contact your UBS Sales representative. In particular, HOLT has a variety of warranted prices based on the scenario chosen; please mail UBS Securities LLC, 11 Madison Avenue, New York, NY 10010, USA, Attention: Investment Research, if you are interested in the warranted price on a particular company, again subject to commercial considerations. UBS Pharma Values is an analytical tool that involves the creation of a number of individual product net present value calculations, based on published forecasts of sales for pharmaceuticals, and is available to clients of UBS Global Research; for further details and pricing please contact your UBS Sales representative. For all other specific disclaimers, please see https://www.ubs.com/disclosures.

When you receive UBS Global Research through a system, your access and/or use of such UBS Global Research is subject to this UBS Global Research Disclaimer and to the UBS Neo Platform Use Agreement (the "Neo Terms") together with any other relevant terms of use governing the applicable System.

When you receive UBS Global Research via a third party vendor, e-mail or other electronic means, you agree that use shall be subject to this UBS Global Research Disclaimer, the Neo Terms and where applicable the UBS Investment Bank terms of business (https://www.ubs.com/global/en/investment-bank/regulatory.html) and to UBS's Terms of Use/Disclaimer (https://www.ubs.com/global/en/legalinfo2/disclaimer.html). In addition, you consent to UBS processing your personal data and using cookies in accordance with our Privacy Statement (https://www.ubs.com/global/en/legalinfo2/privacy.html) and cookie notice (https://www.ubs.com/global/en/legal/privacy/users.html).

If you receive UBS Global Research, whether through a System or by any other means, you agree that you shall not copy, revise, amend, create a derivative work, provide to any third party, or in any way commercially exploit any UBS research provided via UBS Global Research or otherwise, and that you shall not extract data from any research or estimates provided to you via UBS Global Research or otherwise, without the prior written consent of UBS. You agree not to use UBS Global Research in any artificial intelligence system, without the prior written consent of UBS.

In certain circumstances you may receive UBS Global Research otherwise than in the capacity of a client of UBS and you understand and agree that under these circumstances (i) the UBS Global Research is provided to you for information purposes only; (ii) for the purposes of receiving it you are not intended to be and will not be treated as a "client" of UBS for any legal or regulatory purpose; (iii) the UBS Global Research must not be relied on or acted upon for any purpose; and (iv) such content is subject to the relevant disclaimers that follow.

This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject UBS to any registration or licensing requirement within such jurisdiction.

This document is a general communication and is educational in nature; it is not an advertisement nor is it a solicitation or an offer to buy or sell any financial instruments or to participate in any particular trading strategy. Nothing in this document constitutes a representation that any investment strategy or recommendation is suitable or appropriate to an investor's individual circumstances or otherwise constitutes a personal recommendation. By providing this document, none of UBS or its representatives has any responsibility or authority to provide or have provided investment advice in a fiduciary capacity or otherwise. Investments involve risks, and investors should exercise prudence and their own judgment in making their investment decisions. None of UBS or its representatives is suggesting that the recipient or any other person take a specific course of action or any action at all. The recipient should carefully read this document in its entirety and not draw inferences or conclusions from the rating alone. By receiving this document, the recipient acknowledges and agrees with the intended purpose described above and further disclaims any expectation or belief that the information constitutes investment advice to the recipient or otherwise purports to meet the investment objectives of the recipient. The financial instruments described in the document may not be eligible for sale in all jurisdictions or to certain categories of investors.

Options, structured derivative products and futures (including OTC derivatives) are not suitable for all investors. Trading in these instruments is considered risky and may be appropriate only for sophisticated investors. Prior to buying or selling an option, and for the complete risks relating to options, you must receive a copy of "The Characteristics and Risks of Standardized Options." You may read the document at https://www.theocc.com/publications/risks/riskchap1.jsp or ask your salesperson for a copy. Various theoretical explanations of the risks associated with these instruments have been published. Supporting documentation for any claims, comparisons, recommendations, statistics or other technical data will be supplied upon request. Past performance is not necessarily indicative of future results. Transaction costs may be significant in option strategies calling for multiple purchases and sales of options, such as spreads and straddles. Because of the importance of tax considerations to many options transactions, the investor considering options should consult with his/her tax advisor as to how taxes affect the outcome of contemplated options transactions.

Mortgage and asset-backed securities may involve a high degree of risk and may be highly volatile in response to fluctuations in interest rates or other market conditions. Foreign currency rates of exchange may adversely affect the value, price or income of any security or related instrument referred to in the document. For investment advice, trade execution or other enquiries, clients should contact their local sales representative.

UBS notes that no globally accepted framework or definition (legal, regulatory or otherwise) currently exists, nor is there a market consensus as to what constitutes an "ESG" (Environmental, Social or Governance) or an equivalent-label, or as to what precise attributes are required for the Information (as defined below) to be defined as ESG or equivalently-labelled. Any information, data or other content including from a third party source contained, referred to herein or used for whatsoever purpose by UBS or a third party ("Information"), in relation to any actual or potential ESG objective, issue or consideration is not intended to be relied upon for ESG classification, regulatory regime or industry initiative purposes ("ESG Regimes"). Nothing in these materials is intended to convey, suggest or indicate that UBS considers or represents any product, service, person or body mentioned in these materials as meeting or qualifying for any ESG classification, labelling or similar standards that may exist under the ESG Regimes. UBS has not conducted any assessment of compliance with ESG Regimes. Parties are reminded to make their own assessments for these purposes.

The value of any investment or income may go down as well as up, and investors may not get back the full (or any) amount invested. Past performance is not necessarily a guide to future performance. Neither UBS nor any of its directors, employees or agents accepts any liability for any loss (including investment loss) or damage arising out of the use of all or any of the Information.

Prior to making any investment or financial decisions, any recipient of this document or the information should take steps to understand the risk and return of the investment and seek individualized advice from his or her personal financial, legal, tax and other professional advisors that takes into account all the particular facts and circumstances of his or her investment objectives.

Any prices stated in this document are for information purposes only and do not represent valuations for individual securities or other financial instruments. There is no representation that any transaction can or could have been effected at those prices, and any prices do not necessarily reflect UBS's internal books and records or theoretical model-based valuations and may be based on certain assumptions. Different assumptions by UBS or any other source may yield substantially different results.

No representation or warranty, either expressed or implied, is provided in relation to the accuracy, completeness or reliability of the information contained in any materials to which this document relates (the "Information"), except with respect to Information concerning UBS. The Information is not intended to be a

complete statement or summary of the securities, markets or developments referred to in the document. UBS does not undertake to update or keep current the Information. Any statements contained in this report attributed to a third party represent UBS's interpretation of the data, information and/or opinions provided by that third party either publicly or through a subscription service, and such use and interpretation have not been reviewed by the third party. In no circumstances may this document or any of the Information (including any forecast, value, index or other calculated amount ("Values")) be used for any of the following purposes:

(i) valuation or accounting purposes;

(ii) to determine the amounts due or payable, the price or the value of any financial instrument or financial contract; or

(iii) to measure the performance of any financial instrument including, without limitation, for the purpose of tracking the return or performance of any Value or of defining the asset allocation of portfolio or of computing performance fees.

By receiving this document and the Information you will be deemed to represent and warrant to UBS that you will not use this document or any of the Information for any of the above purposes or otherwise rely upon this document or any of the Information.

UBS has policies and procedures, which include, without limitation, independence policies and permanent information barriers, that are intended, and upon which UBS relies, to manage potential conflicts of interest and control the flow of information within divisions of UBS and among its subsidiaries, branches and affiliates. UBS does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For further information on the ways in which UBS Global Research manages conflicts and maintains independence of its research products, historical performance information and certain additional disclosures concerning UBS Global Research recommendations, please visit https://www.ubs.com/disclosures.

UBS Global Research will initiate, update and cease coverage solely at the discretion of UBS Global Research Management, which will also have sole discretion on the timing and frequency of any published research product. The analysis contained in this document is based on numerous assumptions. All material information in relation to published research reports, such as valuation methodology, risk statements, underlying assumptions (including sensitivity analysis of those assumptions), ratings history etc. as required by the Market Abuse Regulation, can be found on UBS Neo. Different assumptions could result in materially different results.

UBS Global Research may utilise artificial intelligence tools ("AI Tools") in the preparation of this document. Notwithstanding any such use of AI Tools, this document has undergone human review.

The analyst(s) responsible for the preparation of this document may interact with trading desk personnel, sales personnel and other parties for the purpose of gathering, applying and interpreting market information. UBS relies on information barriers to control the flow of information contained in one or more areas within UBS into other areas, units, groups or affiliates of UBS. The compensation of the analyst who prepared this document is determined exclusively by UBS Global Research management and senior management (not including investment banking). Analyst compensation is not based on investment banking revenues; however, compensation may relate to the revenues of UBS and/or its divisions as a whole, of which investment banking, sales and trading are a part, and UBS as a whole.

For financial instruments admitted to trading on an EU regulated market: UBS (excluding UBS Securities LLC) acts as a market maker or liquidity provider (in accordance with the interpretation of these terms under English law or, if not carried out by UBS in the UK the law of the relevant jurisdiction in which UBS determines it carries out the activity) in the financial instruments of the issuer save that where the activity of liquidity provider is carried out in accordance with the definition given to it by the laws and regulations of any other EU jurisdictions, such information is separately disclosed in this document. For financial instruments admitted to trading on a non-EU regulated market: UBS may act as a market maker save that where this activity is carried out in the US in accordance with the definition given to it by the relevant laws and regulations, such activity will be specifically disclosed in this document. UBS may have issued a warrant the value of which is based on one or more of the financial instruments referred to in the document. UBS and its affiliates and employees may have long or short positions, trade as principal and buy and sell in instruments or derivatives identified herein; such transactions or positions may be inconsistent with the opinions expressed in this document.

Within the past 12 months UBS may have received or provided investment services and activities or ancillary services as per MiFID II which may have given rise to a payment or promise of a payment in relation to these services from or to this company.

Please note that all transactions conducted by UBS are consistent with sanctions regulations imposed by Switzerland, the European Union, the United Nations, the United Kingdom and the United States, per UBS' global sanctions policy. UBS opinion as to future investment worthiness assumes no new sanctions are imposed.

US persons are prohibited from purchasing or selling securities of certain companies designated as being associated with the Chinese Military in accordance with the amended US Presidential Executive Order 13959.

United Kingdom: This material is distributed by UBS AG, London Branch to persons who are eligible counterparties or professional clients. UBS AG, London Branch is authorised by the Prudential Regulation Authority and subject to regulation by the Financial Conduct Authority and limited regulation by the Prudential Regulation Authority. **Europe:** Except as otherwise specified herein, these materials are distributed by UBS Europe SE, a subsidiary of UBS AG, to persons who are eligible counterparties or professional clients (as detailed in the Bundesanstalt fur Finanzdienstleistungsaufsicht (BaFin) Rules and according to MIFID) and are only available to such persons. The information does not apply to, and should not be relied upon by, retail clients. UBS Europe SE is authorised by the European Central Bank (ECB) and regulated by the BaFin and the ECB. **Germany, Luxembourg, the Netherlands, Belgium and Ireland:** Where an analyst of UBS Europe SE has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE. In all cases it is distributed by UBS Europe SE and UBS AG, London Branch. **Turkey:** Distributed by UBS AG, London Branch. No information in this document is provided for the purpose of offering, marketing and sale by any means of any capital market instruments and services in the Republic of Turkey. Therefore, this document may not be considered as an offer made or to be made to residents of the Republic of Turkey. UBS AG, London Branch is not licensed by the Turkish Capital Market Board under the provisions of the Capital Market Law (Law No. 6362). Accordingly, neither this document nor any other offering material related to the instruments/services may be utilized in connection with providing any capital market services to persons within the Republic of Turkey without the prior approval of the Capital Market Board. However, according to article 15 (d) (ii) of the Decree No. 32, there is no restriction on the purchase or sale of the securities abroad by residents of the Republic of Turkey. **Poland:** Distributed by UBS Europe SE (spolka z ograniczona odpowiedzialnoscia) Oddzial w Polsce. Where an analyst of UBS Europe SE (spolka z ograniczona odpowiedzialnoscia) Oddzial w Polsce has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE (spolka z ograniczona odpowiedzialnoscia) Oddzial w Polsce. **Russia:** Prepared and distributed by UBS Bank (OOO). Should not be construed as an individual Investment Recommendation for the purpose of the Russian Law - Federal Law #39-FZ ON THE SECURITIES MARKET Articles 6.1-6.2.**Switzerland:** Distributed by UBS AG to persons who are institutional investors only. UBS AG is regulated by the Swiss Financial Market Supervisory Authority (FINMA). **Italy:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, Italy Branch. Where an analyst of UBS Europe SE, Italy Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, Italy Branch. **France:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, France Branch. Where an analyst of UBS Europe SE, France Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, France Branch. **Spain:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, Spain Branch. Where an analyst of UBS Europe SE, Spain Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, Spain Branch. **Sweden:** Prepared by UBS Europe SE and distributed by UBS Europe SE and UBS Europe SE, Sweden Branch. Where an analyst of UBS Europe SE, Sweden Branch has contributed to this document, the document is also deemed to have been prepared by UBS Europe SE, Sweden Branch. **South Africa:** Distributed by UBS South Africa (Pty) Limited (Registration No. 1995/011140/07), an authorised user of the JSE and an authorised Financial Services Provider (FSP 7328). **Saudi Arabia:** This document has been issued by UBS AG (and/or any of its subsidiaries, branches or affiliates), a public company limited by shares, incorporated in Switzerland with its registered offices at Aeschenvorstadt 1, CH-4051 Basel and Bahnhofstrasse 45, CH-8001 Zurich. This publication has been approved by UBS Saudi Arabia (a subsidiary of UBS AG), a Saudi closed joint stock company incorporated in the Kingdom of Saudi Arabia under commercial register number 1010257812 having its registered office at Tatweer Towers, P.O. Box 75724, Riyadh 11588, Kingdom of Saudi Arabia. UBS Saudi Arabia is authorized and regulated by the Capital Market Authority to conduct securities business under license number 08113-37. **UAE / Dubai:** The information distributed by UBS AG Dubai Branch is only intended for Professional Clients and/or Market Counterparties, as classified under the DFSA rulebook. No other person should act upon this material/communication. The information is not for further distribution within the United Arab Emirates. UBS AG Dubai Branch is regulated by the DFSA in the DIFC. UBS is not licensed to provide banking services in the UAE by the Central Bank of the UAE, nor is it licensed by the UAE Securities and Commodities Authority. **Israel:** This Material is distributed by UBS AG, London Branch. UBS Securities Israel Ltd is a licensed Investment Marketer that is supervised by the Israel Securities Authority (ISA). UBS AG, London Branch and its affiliates incorporated outside Israel are not licensed under the Israeli Advisory Law. UBS may engage among others in issuance of Financial Assets or in distribution of Financial Assets of other issuers for fees or other benefits. UBS AG, London Branch and its affiliates may prefer various Financial Assets to which they have or may have an Affiliation (as such term is defined under the Israeli Advisory Law). Nothing in this Material should be considered as investment advice under the Israeli Advisory Law. This Material is being issued only to and/or is directed only at persons who are Eligible Clients within the meaning of the Israeli Advisory Law, and this Material must not be furnished to, relied on or acted upon by any other persons. **United States:** Distributed to US persons by either UBS Securities LLC or by UBS Financial Services Inc., subsidiaries of UBS AG; or by a group, subsidiary or affiliate of UBS AG that is not registered as a US broker-dealer (a **'non-US affiliate'**) to major US institutional investors only. UBS Securities LLC or UBS Financial Services Inc. accepts responsibility for the content of a report prepared by another non-US affiliate when distributed to US persons by UBS Securities LLC or UBS Financial Services Inc. All transactions by a US person in the securities mentioned in this report must be effected through UBS Securities LLC or UBS Financial Services Inc., and not through a non-US affiliate. UBS Securities LLC is not acting as a municipal advisor to any municipal entity or obligated person within the meaning of Section 15B of the Securities Exchange Act (the "Municipal Advisor Rule"), and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of the Municipal Advisor Rule. **Canada:** Distributed by UBS Securities Canada Inc., a registered investment dealer in Canada

and a Member-Canadian Investor Protection Fund, or by another affiliate of UBS AG that is registered to conduct business in Canada or is otherwise exempt from registration. **Brazil:** Except as otherwise specified herein, this Material is prepared by UBS BB Corretora de Câmbio, Títulos e Valores Mobiliários S.A. (UBS BB CCTVM) to persons who are eligible investors residing in Brazil, which are considered to be Professional Investors (Investidores Profissionais), as designated by the applicable regulation, mainly the CVM Resolution No. 30 from the 11th of May 2021 (determines the duty to verify the suitability of products, services and transactions with regards to the client´s profile). UBS BB CCTVM is a subsidiary of UBS BB Servicos de Assessoria Financeira e Participacoes S.A. ("UBS BB"). UBS BB is an association between UBS AG and Banco do Brasil (through its subsidiary BB – Banco de Investimentos S.A.), of which UBS AG is the majority owner and which provides investment banking services and coverage in Brazil, Argentina, Chile, Paraguay, Peru and Uruguay. UBS BB CCTVM is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. Ombudsman: 0800-940-0266/ https://www.ubs.com/br/pt/ubsbb-investment-bank/ombudsman.html. UBS may hold relevant financial and commercial interest in relation to the company subject to this Research report. **Hong Kong:** Distributed by UBS Securities Asia Limited. Please contact local licensed persons of UBS Securities Asia Limited in respect of any matters arising from, or in connection with, the analysis or document **Singapore:** Distributed by UBS Securities Pte. Ltd. [Co. Reg. No.: 198500648C] or UBS AG, Singapore Branch. Please contact UBS Securities Pte. Ltd., an exempt financial adviser under the Singapore Financial Advisers Act (Cap. 110); or UBS AG, Singapore Branch, an exempt financial adviser under the Singapore Financial Advisers Act (Cap. 110) and a wholesale bank licensed under the Singapore Banking Act (Cap. 19) regulated by the Monetary Authority of Singapore, in respect of any matters arising from, or in connection with, the analysis or document. The recipients of this document represent and warrant that they are accredited and institutional investors as defined in the Securities and Futures Act (Cap. 289). **Japan:** Distributed by UBS Securities Japan Co., Ltd. to professional investors (except as otherwise permitted). Where this report has been prepared by UBS Securities Japan Co., Ltd., UBS Securities Japan Co., Ltd. is the author, publisher and distributor of the report. Distributed by UBS AG, Tokyo Branch to Professional Investors (except as otherwise permitted) in relation to foreign exchange and other banking businesses when relevant. **Australia:** Recipients through a superannuation fund: Distributed by UBS AG, Australia Branch (ABN 47 088 129 613 and holder of Australian Financial Service Licence No. 231087). For all other recipients: Distributed by UBS Securities Australia Limited (ABN 62 008 586 481 and holder of Australian Financial Services Licence No. 231098). This document contains general information and/or general advice only and does not constitute personal financial product advice. As such, the Information in this document has been prepared without taking into account any investor's objectives, financial situation or needs, and investors should, before acting on the Information, consider the appropriateness of the Information, having regard to their objectives, financial situation and needs. If the Information contained in this document relates to the acquisition, or potential acquisition of a particular financial product by a 'Retail' client as defined by section 761G of the Corporations Act 2001 where a Product Disclosure Statement would be required, the retail client should obtain and consider the Product Disclosure Statement relating to the product before making any decision about whether to acquire the product. **New Zealand:** Distributed by UBS New Zealand Ltd. UBS New Zealand Ltd is not a registered bank in New Zealand. You are being provided with this publication or material because you have indicated to UBS that you are a "wholesale client" within the meaning of clause 4 of schedule 5 of the Financial Markets Conduct Act 2013 of New Zealand (Permitted Client). This publication or material is not intended for clients who are not Permitted Clients (non-permitted Clients). If you are a non-permitted Client you must not rely on this publication or material. If despite this warning you nevertheless rely on this publication or material, you hereby (i) acknowledge that you may not rely on the content of this publication or material and that any recommendations or opinions in such this publication or material are not made or provided to you, and (ii) to the maximum extent permitted by law (a) indemnify UBS and its associates or related entities (and their respective Directors, officers, agents and Advisors) (each a 'Relevant Person') for any loss, damage, liability or claim any of them may incur or suffer as a result of, or in connection with, your unauthorised reliance on this publication or material and (b) waive any rights or remedies you may have against any Relevant Person for (or in respect of) any loss, damage, liability or claim you may incur or suffer as a result of, or in connection with, your unauthorised reliance on this publication or material. **Korea:** Distributed in Korea by UBS Securities Pte. Ltd., Seoul Branch. This report may have been edited or contributed to from time to time by affiliates of UBS Securities Pte. Ltd., Seoul Branch. This material is intended for professional/institutional clients only and not for distribution to any retail clients. **Malaysia:** This material is authorized to be distributed in Malaysia by UBS Securities Malaysia Sdn. Bhd (Capital Markets Services License No.: CMSL/A0063/2007). This material is intended for professional/institutional clients only and not for distribution to any retail clients. **India:** Distributed by UBS Securities India Private Ltd. (Corporate Identity Number U67120MH1996PTC097299) Unit 2901, Level 29 Altimus, Pandurang Budhkar Road, Worli, Mumbai – 400 018. It provides brokerage services bearing SEBI Registration Number: INZ000259830; Merchant Banking services bearing SEBI Registration Number: INM000013101; and Research Analyst services bearing SEBI Registration Number: INH000001204. Name of Compliance Officer Mr. Parameshwaran Shivaramakrishnan, Phone : +912261556151, Email : parameshwaran.s@ubs.com, Name of Grievance Officer Parameshwaran Shivaramakrishnan, Phone : +912261556151, Email: ol-ubs-sec-compliance@ubs.com Registration granted by SEBI, and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. UBS may have debt holdings or positions in the subject Indian company/companies. UBS may have financial interests (e.g. loan/derivative products, rights to or interests in investments, etc.) in the subject Indian company / companies from time to time. Within the past 12 months, UBS may have received compensation for non-investment banking securities-related services and/or non-securities services from the subject Indian company/companies. The subject company/companies may have been a client/clients of UBS during the 12 months preceding the date of distribution of the research report with respect to investment banking and/or non-investment banking securities-related services and/or non-securities services. With regard to information on associates, please refer to the Annual Report at: https://www.ubs.com/global/en/about_ubs/investor_relations/annualreporting.html The Research Annual Compliance Report for UBS Securities India Private Limited is available on www.ubs.com/ubssi under Research tab. **Taiwan:** Except as otherwise specified herein, this material may not be distributed in Taiwan. Information and material on securities/instruments that are traded in a Taiwan organized exchange is deemed to be issued and distributed by UBS Securities Pte. LTD., Taipei Branch to professional institutional investors and/or persons permitted under applicable regulation. Unless permitted by applicable Taiwan laws and regulations, this material is for reference and information only and should not constitute "recommendation" to clients or recipients in Taiwan for the covered companies or any companies mentioned in this document. No portion of the document may be reprinted, reproduced or quoted by the press or any other person without authorisation from UBS. **Indonesia:** This report is being distributed by PT UBS Sekuritas Indonesia and is delivered by its licensed employee(s), including marketing/sales person, to its client. PT UBS Sekuritas Indonesia, having its registered office at Sequis Tower Level 22 unit 22-1,Jl.Jend. Sudirman, kav.71, SCBD lot 11B, Jakarta 12190. Indonesia, is a subsidiary company of UBS AG and licensed under Capital Market Law no. 8 year 1995, a holder of broker-dealer and underwriter licenses issued by the Capital Market and Financial Institution Supervisory Agency (now Otoritas Jasa Keuangan/OJK). PT UBS Sekuritas Indonesia is also a member of Indonesia Stock Exchange and supervised by Otoritas Jasa Keuangan (OJK). Neither this report nor any copy hereof may be distributed in Indonesia or to any Indonesian citizens except in compliance with applicable Indonesian capital market laws and regulations. This report is not an offer of securities in Indonesia and may not be distributed within the territory of the Republic of Indonesia or to Indonesian citizens in circumstance which constitutes an offering within the meaning of Indonesian capital market laws and regulations.

The disclosures contained in research documents produced by UBS AG, London Branch or UBS Europe SE shall be governed by and construed in accordance with English law.

UBS specifically prohibits the redistribution of this document in whole or in part without the written permission of UBS and in any event UBS accepts no liability whatsoever for any redistribution of this document or its contents or the actions of third parties in this respect. Images may depict objects or elements that are protected by third party copyright, trademarks and other intellectual property rights. © UBS 2026. The key symbol and UBS are among the registered and unregistered trademarks of UBS. All rights reserved.

Global Wealth Management Disclaimer

You receive this document in your capacity as a client of UBS Global Wealth Management. This publication has been distributed to you by UBS Switzerland AG (regulated by FINMA in Switzerland) or its affiliates (**"UBS"**) with whom you have a banking relationship with. The full name of the distributing affiliate and its competent authority can be found in the country-specific disclaimer at the end of this document. UBS may utilise artificial intelligence tools ("AI Tools") in the preparation of this document. Notwithstanding any such use of AI Tools, this document has undergone human review.

The date and time of the first dissemination of this publication is the same as the date and time of its publication.

Risk information:

You agree that you shall not copy, revise, amend, create a derivative work, provide to any third party, or in any way commercially exploit any UBS research, and that you shall not extract data from any research or estimates, without the prior written consent of UBS.

This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject UBS to any registration or licensing requirement within such jurisdiction.

This document is **for your information only**; it is not an advertisement nor is it a solicitation or an offer to buy or sell any financial instruments or to participate in any particular trading strategy. Nothing in this document constitutes a representation that any investment strategy or recommendation is suitable or appropriate to an investor's individual circumstances or otherwise constitutes a personal recommendation. By providing this document, none of UBS or its representatives has any responsibility or authority to provide or have provided investment advice in a fiduciary capacity or otherwise. Investments involve risks, and investors should exercise prudence and their own judgment in making their investment decisions. None of UBS or its representatives is suggesting that the recipient or any other person take a specific course of action or any action at all. By receiving this document, the recipient acknowledges and agrees with the intended purpose described above and further disclaims any expectation or belief that the information constitutes investment advice to the recipient or otherwise purports to meet the investment objectives of the recipient. The financial instruments described in the document may not be eligible for sale in all jurisdictions or to certain categories of investors.

Options, derivative products and futures are not suitable for all investors, and trading in these instruments is considered risky. Mortgage and asset-backed securities may involve a high degree of risk and may be highly volatile in response to fluctuations in interest rates or other market conditions. Foreign currency rates of exchange may

adversely affect the value, price or income of any security or related instrument referred to in the document. For investment advice, trade execution or other enquiries, clients should contact their local sales representative.

The value of any investment or income may go down as well as up, and investors may not get back the full (or any) amount invested. Past performance is not necessarily a guide to future performance. Neither UBS nor any of its directors, employees or agents accepts any liability for any loss (including investment loss) or damage arising out of the use of all or any of the information (as defined below).

Prior to making any investment or financial decisions, any recipient of this document or the information should take steps to understand the risk and return of the investment and seek individualized advice from his or her personal financial, legal, tax and other professional advisors that takes into account all the particular facts and circumstances of his or her investment objectives.

Any prices stated in this document are for information purposes only and do not represent valuations for individual securities or other financial instruments. There is no representation that any transaction can or could have been effected at those prices, and any prices do not necessarily reflect UBS's internal books and records or theoretical model-based valuations and may be based on certain assumptions. Different assumptions by UBS or any other source may yield substantially different results.

No representation or warranty, either expressed or implied, is provided in relation to the accuracy, completeness or reliability of the information contained in any materials to which this document relates (the "Information"), except with respect to Information concerning UBS. The Information is not intended to be a complete statement or summary of the securities, markets or developments referred to in the document. UBS does not undertake to update or keep current the Information. Any opinions expressed in this document may change without notice and may differ or be contrary to opinions expressed by other business areas or groups, personnel or other representative of UBS. Any statements contained in this report attributed to a third party represent UBS's interpretation of the data, information and/or opinions provided by that third party either publicly or through a subscription service, and such use and interpretation have not been reviewed by the third party. In no circumstances may this document or any of the Information (including any forecast, value, index or other calculated amount ("Values")) be used for any of the following purposes: (i) valuation or accounting purposes; (ii) to determine the amounts due or payable, the price or the value of any financial instrument or financial contract; or (iii) to measure the performance of any financial instrument including, without limitation, for the purpose of tracking the return or performance of any Value or of defining the asset allocation of portfolio or of computing performance fees.

By receiving this document and the Information you will be deemed to represent and warrant to UBS that you will not use this document or any of the Information for any of the above purposes or otherwise rely upon this document or any of the Information.

UBS has policies and procedures, which include, without limitation, independence policies and permanent information barriers, that are intended, and upon which UBS relies, to manage potential conflicts of interest and control the flow of information within divisions of UBS (including between Global Wealth Management and UBS Global Research) and among its subsidiaries, branches and affiliates. For further information on the ways in which UBS manages conflicts and maintains independence of its research products, historical performance information and certain additional disclosures concerning UBS research recommendations, please visit https://www.ubs.com/research-methodology.

Research will initiate, update and cease coverage solely at the discretion of research management, which will also have sole discretion on the timing and frequency of any published research product. The analysis contained in this document is based on numerous assumptions. Different assumptions could result in materially different results.

The analyst(s) responsible for the preparation of this document may interact with trading desk personnel, sales personnel and other parties for the purpose of gathering, applying and interpreting market information. UBS relies on information barriers to control the flow of information contained in one or more areas within UBS into other areas, units, groups or affiliates of UBS. The compensation of the analyst who prepared this document is determined exclusively by research management and senior management (not including investment banking). Analyst compensation is not based on investment banking revenues; however, compensation may relate to the revenues of UBS and/or its divisions as a whole, of which investment banking, sales and trading are a part, and UBS's subsidiaries, branches and affiliates as a whole.

For financial instruments admitted to trading on an EU regulated market: UBS AG, its affiliates or subsidiaries (excluding UBS Securities LLC) acts as a market maker or liquidity provider (in accordance with the interpretation of these terms in the UK) in the financial instruments of the issuer save that where the activity of liquidity provider is carried out in accordance with the definition given to it by the laws and regulations of any other EU jurisdictions, such information is separately disclosed in this document. For financial instruments admitted to trading on a non-EU regulated market: UBS may act as a market maker save that where this activity is carried out in the US in accordance with the definition given to it by the relevant laws and regulations, such activity will be specifically disclosed in this document. UBS may have issued a warrant the value of which is based on one or more of the financial instruments referred to in the document. UBS and its affiliates and employees may have long or short positions, trade as principal and buy and sell in instruments or derivatives identified herein; such transactions or positions may be inconsistent with the opinions expressed in this document.

Options and futures are not suitable for all investors, and trading in these instruments is considered risky and may be appropriate only for sophisticated investors. Prior to buying or selling an option, and for the complete risks relating to options, you must receive a copy of "Characteristics and Risks of Standardized Options". You may read the document at https://www.theocc.com/about/publications/character-risks.jsp or ask your financial advisor for a copy.

Investing in structured investments involves significant risks. For a detailed discussion of the risks involved in investing in any particular structured investment, you must read the relevant offering materials for that investment. Structured investments are unsecured obligations of a particular issuer with returns linked to the performance of an underlying asset. Depending on the terms of the investment, investors could lose all or a substantial portion of their investment based on the performance of the underlying asset. Investors could also lose their entire investment if the issuer becomes insolvent. UBS does not guarantee in any way the obligations or the financial condition of any issuer or the accuracy of any financial information provided by any issuer. Structured investments are not traditional investments and investing in a structured investment is not equivalent to investing directly in the underlying asset. Structured investments may have limited or no liquidity, and investors should be prepared to hold their investment to maturity. The return of structured investments may be limited by a maximum gain, participation rate or other feature. Structured investments may include call features and, if a structured investment is called early, investors would not earn any further return and may not be able to reinvest in similar investments with similar terms. Structured investments include costs and fees which are generally embedded in the price of the investment. The tax treatment of a structured investment may be complex and may differ from a direct investment in the underlying asset. UBS and its employees do not provide tax advice. Investors should consult their own tax advisor about their own tax situation before investing in any securities.

Important Information About Sustainable Investing Strategies: Sustainable investing strategies aim to consider and incorporate environmental, social and governance (ESG) factors into investment process and portfolio construction. Strategies across geographies approach ESG analysis and incorporate the findings in a variety of ways. Incorporating ESG factors or sustainable investing considerations may inhibit the portfolio manager's ability to participate in certain investment opportunities that otherwise would be consistent with its investment objective and other principal investment strategies. The returns on a portfolio incorporating ESG factors or sustainable investing considerations may be lower or higher than portfolios where ESG factors, exclusions, or other sustainability issues are not considered by the portfolio manager, and the investment opportunities available to such portfolios may differ.

Within the past 12 months UBS Switzerland AG, its affiliates or subsidiaries may have received or provided investment services and activities or ancillary services as per MiFID II which may have given rise to a payment or promise of a payment in relation to these services from or to this company.

Disclosures: If you require detailed information on disclosures of interest or conflict of interest as required by Market Abuse Regulation please contact the mailbox MAR_disclosure_twopager@ubs.com. Please note that e-mail communication is unsecured.

External Asset Managers / External Financial Consultants: In case this research or publication is provided to an External Asset Manager or an External Financial Consultant, UBS expressly prohibits that it is redistributed by the External Asset Manager or the External Financial Consultant and is made available to their clients and/or third parties.

Argentina: All securities that will be offered to you by UBS are not authorized by the Argentine Securities and Exchange Commission (CNV) and are not subject to its reporting, periodic information requirements, or oversight. Furthermore, the CNV has not reviewed or endorsed the information provided in any securities offering document, nor the accuracy of any accounting, financial, economic data, or any other information disclosed therein, which remains the sole responsibility of the respective issuer and the other parties involved. **Australia:** This document is provided by UBS Switzerland AG. UBS Switzerland AG does not hold an Australian Financial Services Licence (AFSL) and relies on an exemption to provide financial services to persons in Australia. This document is intended only for distribution to wholesale clients under the Corporations Act 2001 (Cth). UBS Switzerland AG is a related body corporate of UBS AG, Australia Branch and UBS Securities Australia Limited. This document may be distributed to clients by those entities, but it is provided by UBS Switzerland AG and is not provided under any of the other entities' AFSL. The information in this document is general in nature and is not intended to address the objectives, financial situation or needs of any particular individual or entity. Each recipient should consider their own objectives, financial situation or needs before acting on the advice and obtain the relevant Product Disclosure Statement (if required) before making any decision whether to acquire any product. In Australia, UBS entities, other than UBS AG, Australia Branch, are not authorized deposit-taking institutions for the purposes of the Banking Act 1959 (Cth.) and their obligations do not represent deposits or other liabilities of UBS AG, Australia Branch. UBS AG, Australia Branch does not guarantee or otherwise provide assurance in respect of the obligations of such UBS entities or the funds. An investor is exposed to investment risk including possible delays in repayment and loss of income and principal invested, as relevant. If you do not wish to receive marketing materials from UBS, please contact your UBS representative or the contact details listed in the Australia Privacy Notice: https://www.ubs.com/global/en/legal/privacy.html. Your personal data will be processed in accordance with this notice. **Bahrain**: This report is being distributed by UBS AG, Bahrain Branch, duly licensed and regulated by the Central Bank of Bahrain (CBB) as an Investment Business Firm - Category 2 (Branch). Related financial services or products are only made available to Accredited Investors, as defined by the CBB, and are not intended for any other persons. UBS AG, Bahrain Branch is

a Foreign Branch of UBS AG, Zurich/Switzerland and is located on Level 21, East Tower, Bahrain World Trade Centre, Manama, Kingdom of Bahrain. **Brazil**: This report is distributed in Brazil by UBS BB Corretora de Câmbio, Títulos e Valores Mobiliários S.A. or its affiliates ("UBS"). Pursuant to CVM Resolution No. 20/2021, of February 25, 2021, the author(s) of the report hereby certify(ies) that the views expressed in this report solely and exclusively reflect their personal opinions and have been prepared independently, including with respect to UBS. Part of the author(s)'s compensation is based on various factors, including the total revenues of the relevant UBS Group entity of which they are in employment of, but no part of the compensation has been, is, or will be related to the specific recommendations or views expressed in this report. In addition, UBS declares that: UBS has provided, and/or may in the future provide investment banking, brokerage, asset management, commercial banking and other financial services to the subject company/companies or its affiliates, for which they have received or may receive customary fees and commissions, and which constituted or may constitute relevant financial or commercial interests in relation to the subject company/companies or the subject securities. **Canada**: The information contained herein is not, and under no circumstances is to be construed as, a prospectus, an advertisement, a public offering, an offer to sell securities described herein, solicitation of an offer to buy securities described herein, in Canada or any province or territory thereof. Any offer or sale of the securities described herein in Canada will be made only under an exemption from the requirements to file a prospectus with the relevant Canadian securities regulators and only by a dealer properly registered under applicable securities laws or, alternatively, pursuant to an exemption from the dealer registration requirement in the relevant province or territory of Canada in which such offer or sale is made. Under no circumstances is the information contained herein to be construed as investment advice in any province or territory of Canada and is not tailored to the needs of the recipient. To the extent that the information contained herein references securities of an issuer incorporated, formed or created under the laws of Canada or a province or territory of Canada, any trades in such securities must be conducted through a dealer registered in Canada or, alternatively, pursuant to a dealer registration exemption. No securities commission or similar regulatory authority in Canada has reviewed or in any way passed upon these materials, the information contained herein or the merits of the securities described herein and any representation to the contrary is an offence. In Canada, this publication is distributed by UBS Investment Management Canada Inc. **China**: This report and any offering material such as term sheet, research report, other product or service documentation or any other information (the "Material") sent with this report was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the material erroneously, UBS asks that you kindly delete it and inform UBS immediately. This report is prepared by UBS Switzerland AG or its offshore subsidiary or affiliate (collectively as "UBS Offshore"). UBS Offshore is an entity incorporated out of China and is not licensed, supervised or regulated in China to carry out banking or securities business. The recipient should not contact the analysts or UBS Offshore which produced this report for advice as they are not licensed to provide securities investment advice in China. UBS Investment Bank (including Research) has its own wholly independent research and views which at times may vary from the views of UBS Global Wealth Management. The recipient should not use this document or otherwise rely on any of the information contained in this report in making investment decisions and UBS takes no responsibility in this regard. **Denmark**: This publication is not intended to constitute a public offer under Danish law. It is distributed only for information purposes to clients of UBS Europe SE, Denmark Branch, filial af UBS Europe SE, with place of business at Sankt Annae Plads 13, 1250 Copenhagen, Denmark, registered with the Danish Commerce and Companies Agency, under No. 38 17 24 33. UBS Europe SE, Denmark Branch, filial af UBS Europe SE is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the Danish Financial Supervisory Authority (Finanstilsynet), to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Egypt**: Securities or other investment products are not being offered or sold by UBS to the public in Egypt and they have not been and will not be registered with the Egyptian Financial Regulatory Authority (FRA). **France**: This publication is not intended to constitute a public offer under French law, it does not constitute a personal recommendation as it is distributed only for information purposes to clients of UBS Europe SE Succursale de France (a branch of UBS Europe SE), having its registered office at 39, rue du Colisée, 75008 Paris, France, registered with the "Registre du Commerce et des Sociétés" of Paris under N°844 425 629. UBS Europe SE Succursale de France is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the French "Autorité de contrôle prudentiel et de résolution" and "Autorité des marchés financiers", to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Germany**: This publication is not intended to constitute a public offer under German law. It is distributed only for information purposes to clients of UBS Europe SE, Germany, with place of business at Bockenheimer Landstrasse 2-4, 60306 Frankfurt am Main. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin") and supervised jointly by the European Central Bank, the German Central Bank (Deutsche Bundesbank) and BaFin, to which this publication has not been submitted for approval. **Hong Kong SAR**: This publication is distributed to clients of UBS AG Hong Kong Branch by UBS AG Hong Kong Branch, a licensed bank under the Hong Kong Banking Ordinance and a registered institution under the Securities and Futures Ordinance. UBS AG Hong Kong Branch is incorporated in Switzerland with limited liability. The contents of this material have not been reviewed by any regulatory authority in Hong Kong. **India**: UBS Securities India Private Ltd. (Corporate Identity Number U67120MH1996PTC097299) 2/F, 3 North Avenue, Maker Maxity, Bandra Kurla Complex, Bandra (East), Mumbai (India) 400051. Phone: +912261556000. It provides brokerage services bearing SEBI Registration Number: INZ000259830; Merchant Banking services bearing SEBI Registration Number: INM000013101 and Research Analyst services bearing SEBI Registration Number: INH000001204. Name of Compliance Officer Mr. Parameshwaran Shivaramakrishnan, Phone: +912261556151, Email: parameshwaran.s@ubs.com, Name of Grievance Officer Parameshwaran Shivaramakrishnan, Phone: +912261556151, Email: ol-ubs-sec-compliance@ubs.com. Registration granted by SEBI, and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. UBS AG, its affiliates or subsidiaries may have debt holdings or positions in the subject Indian company/companies. UBS AG, its affiliates or subsidiaries may have financial interests (e.g. loan/derivative products, rights to or interests in investments, etc.) in the subject Indian company/companies from time to time. Within the past 12 months, UBS AG, its affiliates or subsidiaries may have received compensation for non-investment banking securities-related services and/or non-securities services from the subject Indian company/companies. The subject company/companies may have been a client/clients of UBS AG, its affiliates or subsidiaries during the 12 months preceding the date of distribution of the research report with respect to investment banking and/or non-investment banking securities-related services and/or non-securities services. With regard to information on associates, please refer to the Annual Report at: https://www.ubs.com/global/en/about_ubs/investor_relations/annualreporting.html. The Research Annual Compliance report for UBS Securities India Private Limited is available on www.ubs.com/ubssi under Research tab. **Indonesia**: This communication and any offering material term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed without the consent of UBS. None of the Material has been registered or filed under the prevailing laws and with any financial or regulatory authority in your jurisdiction. The Material may not have been approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or products or futures contracts). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including futures contracts) nor is it an offer to buy or to sell any securities or products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Israel**: UBS is a premier global financial firm offering wealth management, asset management and investment banking services from its headquarters in Switzerland and its operations in over 50 countries worldwide to individual, corporate and institutional investors. This publication is intended for information only and is not intended as an offer to buy or solicitation of an offer. Furthermore, this publication is not intended as an investment advice. Nothing contrary to the above, no action has been, or will be, taken in Israel that would permit an offering of the product(s) mentioned in this document or a distribution of this document to the public in Israel. In particular, this document has not been reviewed or approved by the Israeli Securities Authority. The product(s) mentioned in this document is/are being offered to a limited number of sophisticated investors who qualify as one of the investors listed in the first supplement to the Israeli Securities Law, 5728-1968. This document may not be reproduced or used for any other purpose, nor be furnished to any other person other than those to whom copies have been sent. Anyone who purchases the product(s) mentioned herein shall do so for its own benefit and for its own account and not with the aim or intention of distributing or offering the product(s) to other parties. Anyone who purchases the product(s) shall do so in accordance with its own understanding and discretion and after it has received any relevant financial, legal, business, tax or other advice or opinion required by it in connection with such purchase(s). The word "advice" and/or any of its equivalent terms shall be read and construed in conjunction with the definition of the term "investment marketing" as defined under the Israeli Regulation of Investment Advice, Investment Marketing and Portfolio Management Law. The Swiss laws and regulations require a number of mandatory disclosures to be made in independent financial research reports or recommendations. Pursuant to the Swiss Financial Market Infrastructure Act and the Financial Market Infrastructure Ordinance-FINMA, banks must disclose the percentage of voting rights they hold in companies being researched, if these holdings are equal to or exceed the statutory thresholds. In addition, the Directives on the Independence of Financial Research, issued by the Swiss Bankers Association, mandate a number of disclosures, including the disclosure of potential conflicts of interest, the participation within previous 12 months in any securities issues on behalf of the company being researched, as well as the fact that remuneration paid to the financial analysts is based generally upon the performance of (i) the new issues department or investment banking; or (ii) securities trading performance (including proprietary trading) or sales. **Italy**: This publication is not intended to constitute a public offer under Italian law. It is distributed only for information purposes to clients of UBS Europe SE, Succursale Italia, with place of business at Via del Vecchio Politecnico, 3-20121 Milano. UBS Europe SE, Succursale Italia is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the Bank of Italy (Banca d'Italia) and the Italian Financial Markets Supervisory Authority (CONSOB - Commissione Nazionale per le Società e la Borsa), to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Japan**: This report is solely distributed in Japan by UBS SuMi TRUST Wealth Management Co., Ltd, Financial Instruments Dealer, Director-General of Kanto Local Finance Bureau (Kinsho) No. 3233, a member of the Japan Securities Dealers Association, Financial Futures Association of Japan, Investment Management Association of Japan. UBS SuMi TRUST Wealth Management Co., Ltd will not distribute or

forward this report outside Japan. **Jersey**: UBS AG, Jersey Branch, is regulated and authorized by the Jersey Financial Services Commission for the conduct of banking, funds and investment business. Where services are provided from outside Jersey, they will not be covered by the Jersey regulatory regime. UBS AG, Jersey Branch is a branch of UBS AG a public company limited by shares, incorporated in Switzerland whose registered offices are at Aeschenvorstadt 1, CH-4051 Basel and Bahnhofstrasse 45, CH 8001 Zurich. UBS AG, Jersey Branch's principal place of business is 1, IFC Jersey, St Helier, Jersey, JE2 3BX. **Luxembourg**: This publication is not intended to constitute a public offer under Luxembourg law. It is distributed only for information purposes to clients of UBS Europe SE, Luxembourg Branch ("UBS Luxembourg"), R.C.S. Luxembourg n° B209123, with registered office at 33A, Avenue J. F. Kennedy, L-1855 Luxembourg. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea (HRB n° 107046), with registered office at Bockenheimer Landstrasse 2-4, D-60306 Frankfurt am Main, Germany, duly authorized by the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin") and subject to the joint prudential supervision of BaFin, the European Central Bank and the central bank of Germany (Deutsche Bundesbank). UBS Luxembourg is furthermore supervised by the Luxembourg prudential supervisory authority (Commission de Surveillance du Secteur Financier), in its role as host member state authority. This publication has not been submitted for approval to any public supervisory authority. **Malaysia**: This communication and any offering material term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or derivatives products). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or derivatives products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Mexico**: This information is distributed by UBS Asesores México, S.A. de C.V. ("UBS Asesores"), an affiliate of UBS Switzerland AG, incorporated as a non-independent investment advisor under the Mexican regulation due to the relation with a Foreign Bank. UBS Asesores is registered under number 30060-001-(14115)-21/06/2016 and subject to the supervision of the Mexican Banking and Securities Commission ("CNBV") exclusively regarding the rendering of (i) portfolio management services, (ii) securities investment advisory services, analysis and issuance of individual investment recommendations, and (iii) anti-money laundering and terrorism financing matters. This UBS publication or any material related thereto is addressed only to Sophisticated or Institutional Investors located in Mexico. Research reports only reflect the views of the analysts responsible for the report. The compensation of the analyst(s) who prepared this report is determined exclusively by research management and senior management of any entity of UBS Group to which such analyst(s) render(s) services. **Monaco**: This document is not intended to constitute a public offering or a comparable solicitation under the Principality of Monaco laws, but might be made available for information purposes to clients of UBS (Monaco) S.A., a regulated bank having its registered office at 2 avenue de Grande Bretagne 98000 Monaco operating under a banking license granted by the "Autorité de Contrôle Prudentiel et de Résolution" (ACPR) and the Monegasque government which authorizes the provision of banking services in Monaco. UBS (Monaco) S.A. is also licensed by the "Commission de Contrôle des Activités Financières" (CCAF) to provide investment services in Monaco. The latter has not approved this publication. **Philippines:** This communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or derivatives products). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or derivatives products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Portugal**: UBS Switzerland AG is not licensed to conduct banking and financial activities in Portugal nor is UBS Switzerland AG supervised by the portuguese regulators (Bank of Portugal "Banco de Portugal" and Portuguese Securities Exchange Commission "Comissão do Mercado de Valores Mobiliários"). **Qatar**: UBS Qatar LLC is licensed by the Qatar Financial Centre Authority and authorized by the QFC Regulatory Authority, with QFC no. 01169, and has its registered office at Building No. 36, 4th Floor, Abdulla Bin Thani Street, Doha Design District 1, Msheireb Downtown Doha, State of Qatar. This material is strictly intended for Eligible Counterparties and/or Business Customers only as classified under the QFCRA's Customer and Investor Protection Rules 2019. No other person should act upon this material. **Russia**: This document or information contained therein is for information purposes only and constitutes neither a public nor a private offering, is not an invitation to make offers, to sell, exchange or otherwise transfer any financial instruments in the Russian Federation to or for the benefit of any Russian person or entity and does not constitute an advertisement or offering of securities in the Russian Federation within the meaning of Russian securities laws. The information contained herein is not an "individual investment recommendation" as defined in Federal Law of 22 April 1996 No 39-FZ "On Securities Market" (as amended) and the financial instruments and operations specified herein may not be suitable for your investment profile or your investment goals or expectations. The determination of whether or not such financial instruments and operations are in your interests or are suitable for your investment goals, investment horizon or the acceptable risk level is your responsibility. We assume no liability for any losses connected with making any such operations or investing into any such financial instruments and we do not recommend to use such information as the only source of information for making an investment decision. **Saudi Arabia**: This material is for marketing and information purposes by UBS Saudi Arabia only. It has been distributed by UBS Saudi Arabia or reviewed and approved by UBS Saudi Arabia if it was issued by UBS Switzerland AG, by UBS AG, or any of its subsidiaries or branches. UBS Switzerland AG is a public company limited by shares, incorporated in Switzerland with its registered offices at Bahnhofstrasse 45, CH-8001 Zurich. UBS Saudi Arabia (a subsidiary of UBS AG) is a Saudi closed joint stock company incorporated in the Kingdom of Saudi Arabia with a paid capital of 110,000,000 Saudi Riyals under commercial register number 1010257812 having its registered office at Laysen Valley, P.O. Box 75724, Riyadh 11588, Kingdom of Saudi Arabia. UBS Saudi Arabia is authorized and regulated by the Capital Market Authority to conduct securities business under license number 08113-37. This document is distributed only under such circumstances as may be permitted by applicable laws and regulations. The information and – if any - opinions contained in this document have been compiled or arrived at based upon information obtained from sources believed to be reliable and in good faith, but is not guaranteed as being accurate, nor is it a complete statement or summary of the securities, markets or developments referred to in the document. All such information and opinions are subject to changes without notice. Past performance of investments (whether simulated or actual) is not necessarily an indicator of future results. Forward looking data is based on estimates that are subject to changes. Should the currency of a financial product or service not match your reference currency, performance may rise or fall due to currency fluctuations. UBS Switzerland AG and / or other members of the UBS Group may have a position in and may make a purchase and/or sale of any of the securities or other financial instruments mentioned in this document While statements that constitute "forward-looking statements", including, but not limited to, statements relating to our future business development represent UBS judgments and future expectations concerning the development of our business, a number of risks, uncertainties and other important factors could cause actual developments and results to differ materially from our expectations. This document may not be forwarded, reproduced, redistributed or republished for any purpose without the written permission of UBS Saudi Arabia. Source for all data and charts (if not indicated otherwise) is UBS Switzerland AG. **Singapore**: Where applicable, this material is distributed in Singapore by UBS AG, Singapore Branch, which is licensed by the Monetary Authority of Singapore under the Banking Act 1970 to carry on banking business. UBS AG is incorporated in Switzerland with limited liability. UBS AG has a branch registered in Singapore (UEN S98FC5560C). This material has been prepared and issued for distribution in Singapore to institutional investors, accredited investors and expert investors (each as defined under the Financial Advisers Regulations (the "FAR")) only. By virtue of your status as an institutional investor, accredited investor, or expert investor, UBS AG is exempted from complying with certain requirements under the Financial Advisers Act 2001 (the "FAA"), the FAR and the relevant Notices and Guidelines issued thereunder, in respect of any financial advisory service which UBS AG may provide to you. These include exemptions from complying with: Section 34 of the FAA (pursuant to Regulation 33(1) of the FAR); Section 36 of the FAA (pursuant to Regulation 34(1) of the FAR); and Section 45 of the FAA (pursuant to Regulation 35(1) of the FAR). Singapore recipients and clients of UBS AG, Singapore Branch should contact UBS AG, Singapore Branch for any matters arising from, or in connection with, this material. Where applicable, this communication and any offering material term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication was done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved or endorsed by any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or products or futures contracts). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided on and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Spain:** This report is distributed in Spain by UBS AG, Sucursal en España, authorized under number 1460 in the Register by the Banco de España. **Sweden**: This publication is not intended to constitute a public offer under Swedish law. It is distributed only for information purposes to clients of UBS Europe SE, Sweden Bankfilial, with place of business at Regeringsgatan 38, 11153 Stockholm, Sweden, registered

with the Swedish Companies Registration Office under Reg. No 516406-1011. UBS Europe SE, Sweden Bankfilial is subject to the joint supervision of the European Central Bank, the German Central Bank (Deutsche Bundesbank), the German Federal Financial Services Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, "BaFin"), as well as of the Swedish supervisory authority (Finansinspektionen), to which this publication has not been submitted for approval. UBS Europe SE is a credit institution constituted under German law in the form of a Societas Europaea, duly authorized by BaFin. **Taiwan**: This material is provided by UBS AG, Taipei Branch in accordance with laws of Taiwan, in agreement with or at the request of clients/prospects. **Thailand**: This communication and any offering material, term sheet, research report, other product or service documentation or any other information (the "Material") sent with this communication were done so as a result of a request received by UBS from you and/or persons entitled to make the request on your behalf. Should you have received the Material erroneously, UBS asks that you kindly delete/destroy the communication and Material and inform UBS immediately. The Material, where provided, was provided for your information only and is not to be further distributed in whole or in part in or into your jurisdiction without the consent of UBS. The Material may not have been reviewed, approved, disapproved, endorsed, registered or filed with any financial or regulatory authority in your jurisdiction. UBS has not, by virtue of the Material, made available, issued any invitation to subscribe for or to purchase any investment (including securities or derivatives products). The Material is neither an offer nor a solicitation to enter into any transaction or contract (including future contracts) nor is it an offer to buy or to sell any securities or derivatives products. The relevant investments will be subject to restrictions and obligations on transfer as set forth in the Material, and by receiving the Material you undertake to comply fully with such restrictions and obligations. You should carefully study and ensure that you understand and exercise due care and discretion in considering your investment objective, risk appetite and personal circumstances against the risk of the investment. You are advised to seek independent professional advice in case of doubt. Any and all advice provided and/or trades executed by UBS pursuant to the Material will only have been provided upon your specific request or executed upon your specific instructions, as the case may be, and may be deemed as such by UBS and you. **Türkiye**: The information in this document is not provided for the purpose of offering, marketing or sale of any capital market instrument or service in the Republic of Türkiye. Therefore, this document may not be considered as an offer made, or to be made, to residents of the Republic of Türkiye in the Republic of Türkiye. UBS Switzerland AG is not licensed by the Capital Markets Board of Türkiye (the CMB) under the provisions of the Capital Market Law (Law No. 6362). Accordingly, neither this document nor any other offering material related to the instrument/service may be utilized in connection with providing any capital market services to persons within the Republic of Türkiye without the prior approval of the CMB. However, according to article 15 (d) (ii) of the Decree No. 32 residents of the Republic of Türkiye are allowed to purchase or sell the financial instruments traded in financial markets outside of the Republic of Türkiye. Further to this, pursuant to article 9 of the Communiqué on Principles Regarding Investment Services, Activities and Ancillary Services No. III-37.1, investment services provided abroad to residents of the Republic of Türkiye based on their own initiative are not restricted. **United Arab Emirates (UAE) / DIFC / Abu Dhabi**: UBS is not licensed in the UAE by the Central Bank of the UAE nor by the Emirates' Securities and Commodities Authority and does not undertake banking activities in the UAE. This document is provided for your information only and does not constitute financial advice. **DIFC**: UBS AG Dubai Branch is regulated by the DFSA in the DIFC. This material is strictly intended for Professional Clients and/or Market Counterparties only as classified under the DFSA rulebook. It should not be distributed to Retail Clients. The Investment Research is provided for information purposes only and is not a recommendation or offer to buy/sell/hold a particular investment. The investment research may be out of date. You should seek investment advice before acting on the basis of the Investment Research. **Abu Dhabi**: UBS AG Abu Dhabi Branch is licensed and regulated by the Financial Services Regulatory Authority ("FSRA") of the Abu Dhabi Global Market. This material is intended solely for professional clients or market counterparties, as defined in the rules of the FSRA. It is not directed at, nor intended for, retail clients or any person who does not meet the criteria of a professional client or market counterparty. **United Kingdom**: This document is issued by UBS Wealth Management, a division of UBS AG which is authorised and regulated by the Financial Market Supervisory Authority in Switzerland. In the United Kingdom, UBS AG is authorised by the Prudential Regulation Authority and is subject to regulation by the Financial Conduct Authority and limited regulation by the Prudential Regulation Authority. Details about the extent of regulation by the Prudential Regulation Authority are available from us on request. A member of the London Stock Exchange. This publication is distributed to retail clients of UBS Wealth Management. **Ukraine**: UBS is a premier global financial services firm offering wealth management services to individual, corporate and institutional investors. UBS is established in Switzerland and operates under Swiss law and in over 50 countries and from all major financial centers. UBS is not registered and licensed as a bank/financial institution under Ukrainian legislation and does not provide banking and other financial services in Ukraine. UBS has not made, and will not make, any offer of the mentioned products to the public in Ukraine. No action has been taken to authorize an offer of the mentioned products to the public in Ukraine and the distribution of this document shall not constitute financial services for the purposes of the Law of Ukraine "On Financial Services and Financial Companies" dated 14 December 2021. Any offer of the mentioned products shall not constitute an investment advice, public offer, circulation, transfer, safekeeping, holding or custody of securities in the territory of Ukraine. Accordingly, nothing in this document or any other document, information or communication related to the mentioned products shall be interpreted as containing an offer, a public offer or invitation to offer or to a public offer, or solicitation of securities in the territory of Ukraine or investment advice under Ukrainian law. Electronic communication must not be considered as an offer to enter into an electronic agreement or other electronic instrument within the meaning of the Law of Ukraine "On Electronic Commerce" dated 3 September 2015. This document is strictly for private use by its holder and may not be passed on to third parties or otherwise publicly distributed. **USA**: Distributed to US persons only by UBS Financial Services Inc. or UBS Securities LLC, subsidiaries of UBS AG. UBS Switzerland AG, UBS Europe SE, UBS Bank, S.A., UBS BB Corretora de Câmbio, Títulos e Valores Mobiliários S.A., UBS Asesores México, S.A. de C.V., UBS SuMi TRUST Wealth Management Co., Ltd., UBS Wealth Management Israel Ltd. and UBS Menkul Degerler AS are affiliates of UBS AG. **UBS Financial Services Inc. accepts responsibility for the content of a report prepared by a non-US affiliate when it distributes reports to US persons. All transactions by a US person in the securities mentioned in this report should be effected through a US-registered broker dealer affiliated with UBS, and not through a non-US affiliate. The contents of this report have not been and will not be approved by any securities or investment authority in the United States or elsewhere. UBS Financial Services Inc. is not acting as a municipal advisor to any municipal entity or obligated person within the meaning of Section 15B of the Securities Exchange Act (the "Municipal Advisor Rule") and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of the Municipal Advisor Rule.** For information on the ways in which UBS Securities LLC manages conflicts and maintains independence of its UBS Global Research product; historical performance information; certain additional disclosures concerning UBS Global Research recommendations; and terms and conditions for certain third party data used in research report, please visit https://www.ubs.com/disclosures.

© UBS 2026. The key symbol and UBS are among the registered and unregistered trademarks of UBS. All rights reserved.

臻鼎科技控股
Zhen Ding Tech. Holding

臻鼎科技控股 (4958 TT)

投資人簡報

2026年10月

臻鼎科技控股
Zhen Ding Tech. Holding

免責聲明

- 本簡報及同時發佈之相關訊息內，含有從公司內部與外部來源所取得的預測性資訊。
- 本公司未來實際所發生的營運結果、財務狀況以及業務展望，可能與這些預測性資訊所明示或暗示的預估有所差異，其原因可能來自於各種本公司所不能掌控的風險。
- 本簡報中對未來的展望，反應本公司截至目前為止對於未來的看法。對於這些看法，未來若有任何變更或調整時，本公司並不負責隨時提醒或更新。

臻鼎科技控股
Zhen Ding Tech. Holding

一分鐘認識臻鼎

全球第1大
印刷電路板製造公司

2006年 成立
(前身華虹電子 1978)

台灣桃園 總部

52,108
2025年底員工數

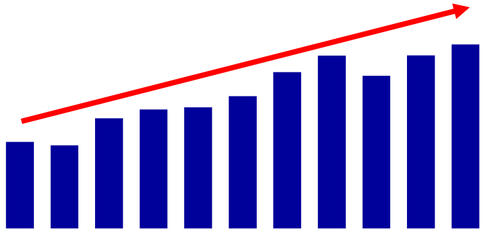
28 棟工廠
位於大陸、台灣、泰國，
其中 **19** 棟為智能工廠

NT$1,825.2億
2025年營收(6.3% YoY)

NT$5,666億
2026年9月30日市值

11.9%
10年平均股東權益報酬率
(2016-2025)

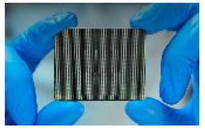
+8%
2015-2025年
營收年複合成長率



未來關鍵成長動能

AI伺服器

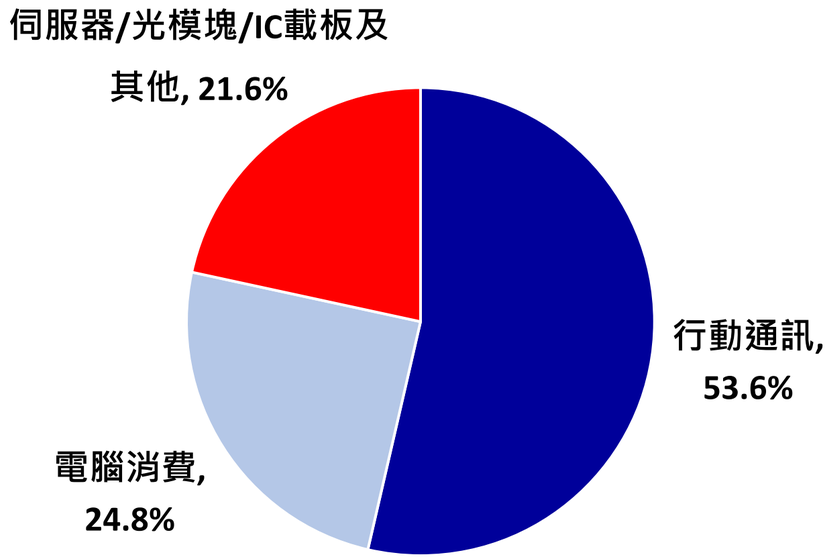
IC載板



光模塊

邊緣AI

銷售分析–產品應用別 (2026 H1)



臻鼎科技控股
Zhen Ding Tech. Holding

營運概況重點更新

營運成果回顧暨未來展望

1. **1H26營收再創同期新高，AI產品組合升級帶動獲利提升：**受惠於高階AI應用需求持續成長，1H26營收達新台幣892億元，年增13.9%，再創歷年同期新高。其中，伺服器 / 光模塊 / IC載板合計營收年增113%，營收占比達21.6%。隨高附加價值產品比重增加，1H26毛利率及營業利益率分別提升至22.4%及7.0%，顯示公司成長動能正由營收規模擴張，進一步轉化為獲利提升。

2. **AI算力建設開啟高階PCB長期成長週期，臻鼎憑藉完整技術布局邁入新一輪成長：**隨下一世代AI平台陸續量產，公司伺服器 / 光模塊業務自第二季起逐季增強，2026年相關營收目標維持成倍成長。中長期而言，AI需求將由伺服器與光模塊進一步擴展至Edge AI及人形機器人等終端應用，持續提升高階PCB的使用量、技術門檻與產品價值，公司對2026-2030 年的成長深具信心。

3. **以業界最大的擴產規模打造下一階段營運動能：**高階PCB產能建置與客戶認證均需較長前置期，唯有糧草先行，才能掌握AI帶來的長期成長機會。公司2026年資本支出為新台幣800億+，2027-2028年亦將維持較高投資水準，聚焦iHDI / mSAP、HLC、IC載板及高階軟板等高成長、高附加價值領域。公司目前共有22棟新廠建設中、9棟新廠規劃中，新增產能將依客戶導入及量產進度，於2026至2030年間分階段開出，逐步將資本投入轉化為營收與獲利貢獻。

4. **禮鼎於2026年7月2日正式向港交所遞交主板上市申請：**上市申請目前仍處於審查程序，相關資訊皆以港交所正式刊載的文件為準。

各產品應用營運概況

光模塊

- 光模塊方面，1.6T及後續世代產品須採用mSAP技術，全球高階光模塊供給缺口高，臻鼎正積極擴大相關產能。
- 公司訂單以 1.6T高速率板為主，預期2026年營收將成倍成長，同時客戶已開始預訂2027年產能。**目標2027年躍升為全球最大高階光模塊PCB供應商。**

AI伺服器

- AI伺服器方面，GPU與ASIC客戶之Intelligent HDI (iHDI) 與HLC產品已陸續轉量產，目標後續份額持續提升。
- 持續與客戶針對未來2-3個世代平台的產品進行開發。

邊緣AI相關應用

AI手機

AI眼鏡/VR/AR

- 2026年手機新機規格升級，帶動軟硬板單機價值同步提升，產品結構有利於毛利率表現。
- AI眼鏡朝更輕薄、高整合發展，不僅軟板使用數量提高，硬板亦導入更高線路等級與mSAP製程，PCB技術門檻顯著提升，預期AI眼鏡營收在2026/2027年皆保持快速成長。

參與SEMICON Taiwan 2026
1.6T光模塊、34層AI Server板齊發，對接次世代AI基建需求

參與SEMICON Taiwan 2026

光模塊

- 展出 800G 及最新 1.6T 高階光模塊展品，涵蓋 XPO 與 NPO 等次世代應用，鎖定新世代資料中心的大頻寬與低延遲需求。
- 產品結合 SLP（mSAP）精密製程與埋孔技術，為市場提供更具節能效益的傳輸解決方案。

AI Server 板

- 展示高達 34層之超高層伺服器板。
- 該系列產品採用 Ultra/Extreme Low Loss 材料與 M8 級技術標準，全面導入 VIPPO、Back drilling 與高厚徑比電鍍等極致工藝。

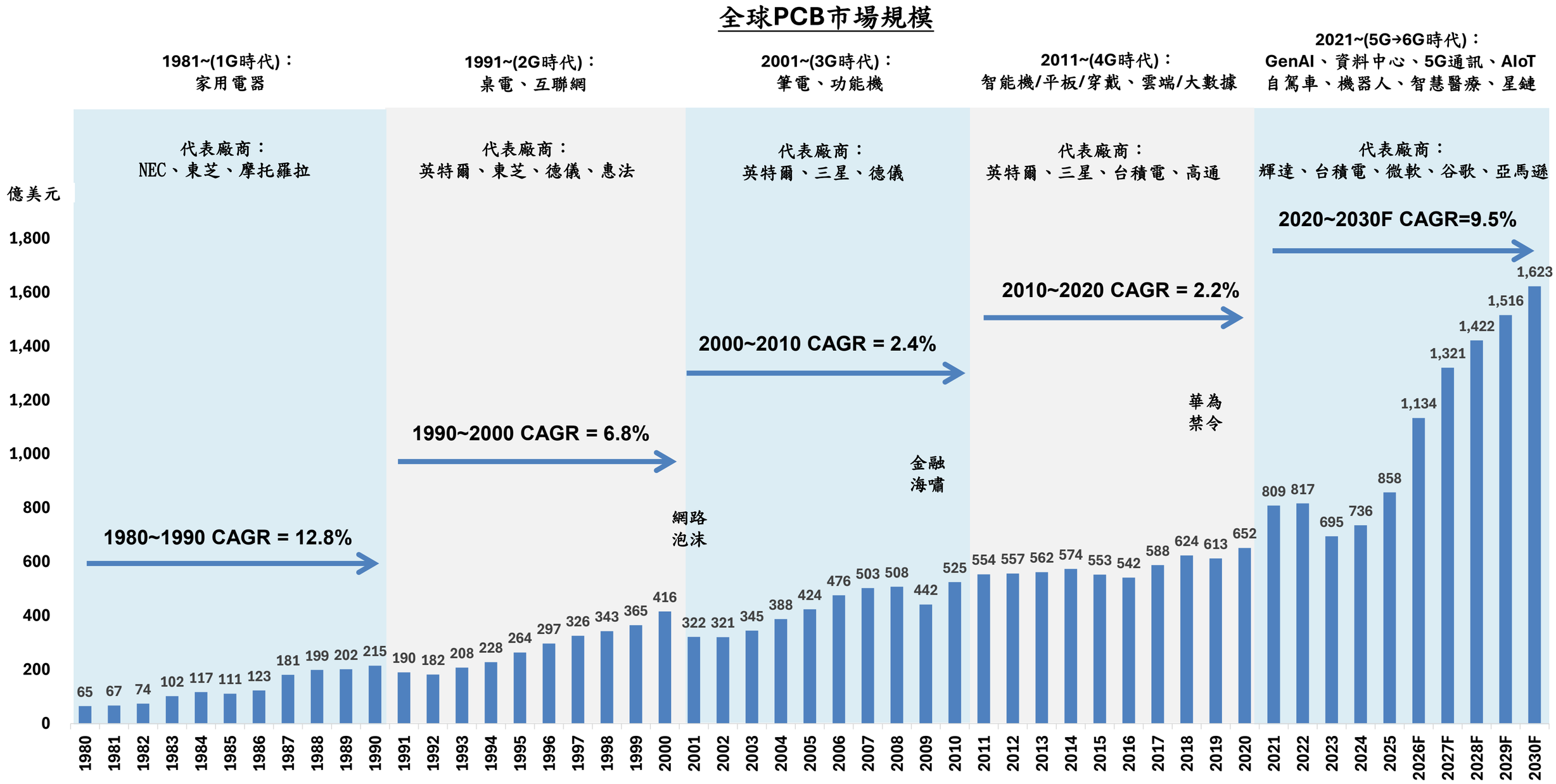
臻鼎科技控股
Zhen Ding Tech. Holding

強化領先優勢，

全面掌握HDI、HLC、與IC載板的黃金時代

臻鼎科技控股 Zhen Ding Tech. Holding

PCB產業隨電子產品迭代更新，朝高階高值化發展



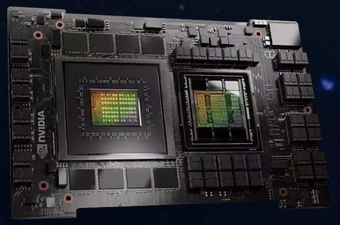
資料來源：Prismark (2026/9)

臻鼎科技集團
Zhen Ding Tech. Group

掌握AI潮流與趨勢

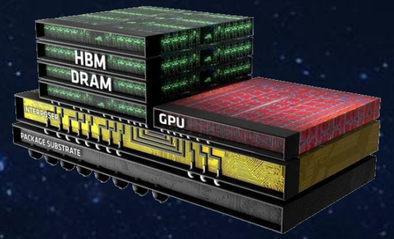
AI + PCB

雲

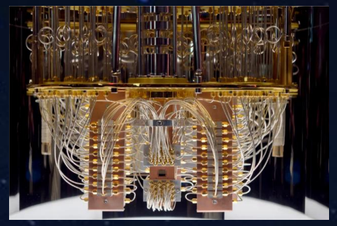


AI 伺服器

邊緣運算



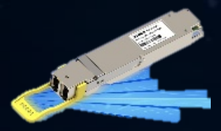
高速運算記憶體



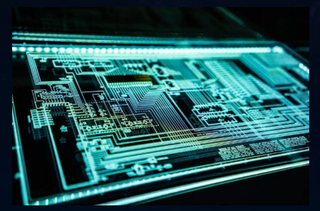
量子計算

管

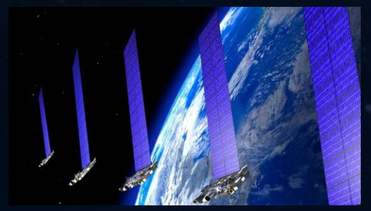
5G通訊設備



光模塊



衛星通信



光通訊

6G通訊感知一體

端

邊緣AI

AI 手機

AI PC/平板

智能穿戴

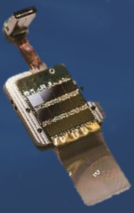
AR/VR

實體AI

智慧座艙

自駕車

人型機器人



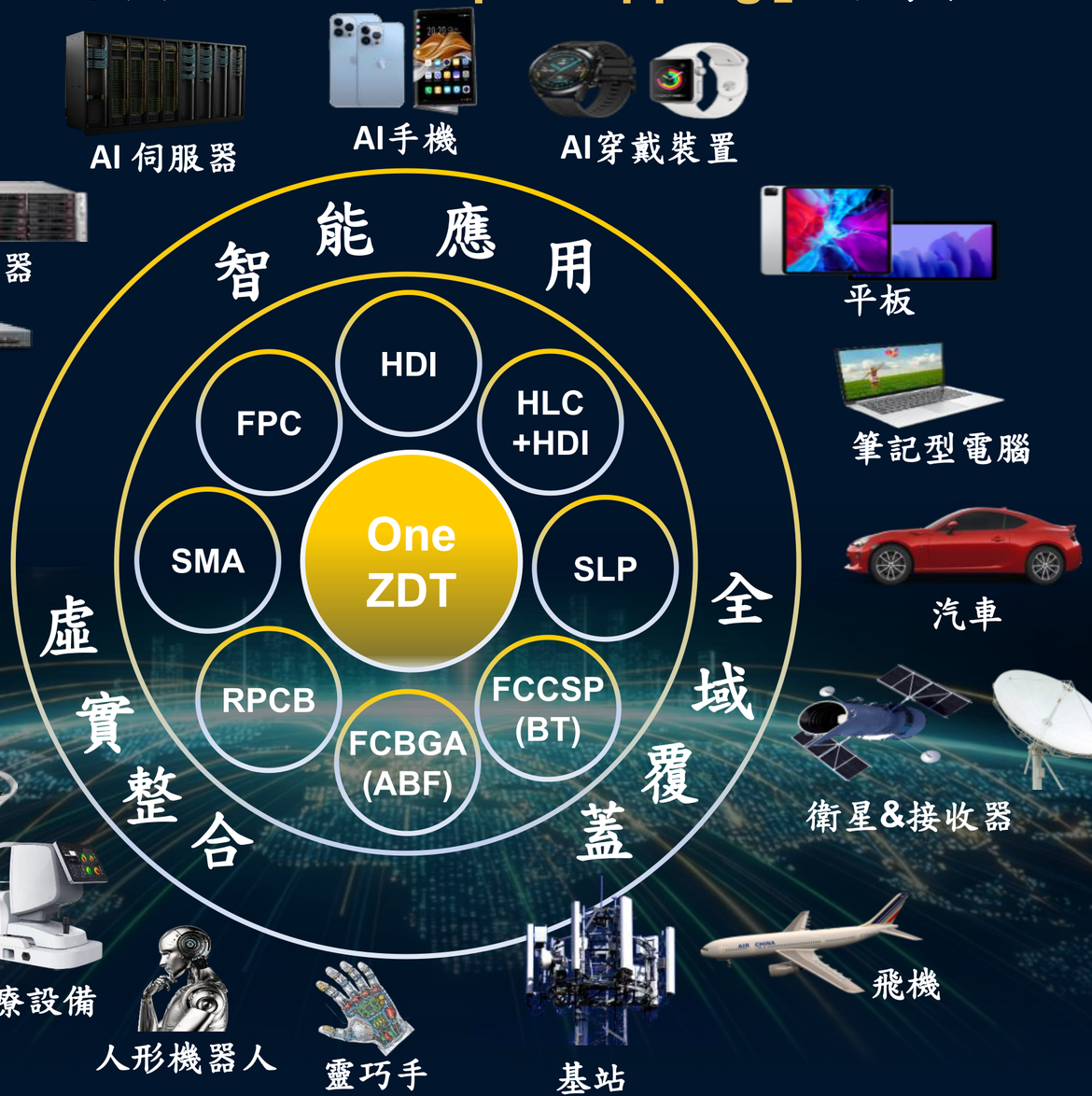
腦機介面

One ZDT, 多元化, 發展高端, 數位轉型, 擁抱AI

臻鼎科技集團
Zhen Ding Tech. Group

打造One ZDT平臺

臻鼎科技集團擁有一流產品的技術、品質、製造服務能力，並提供全方位電路板系列產品，滿足客戶「One Stop Shopping」的需求。

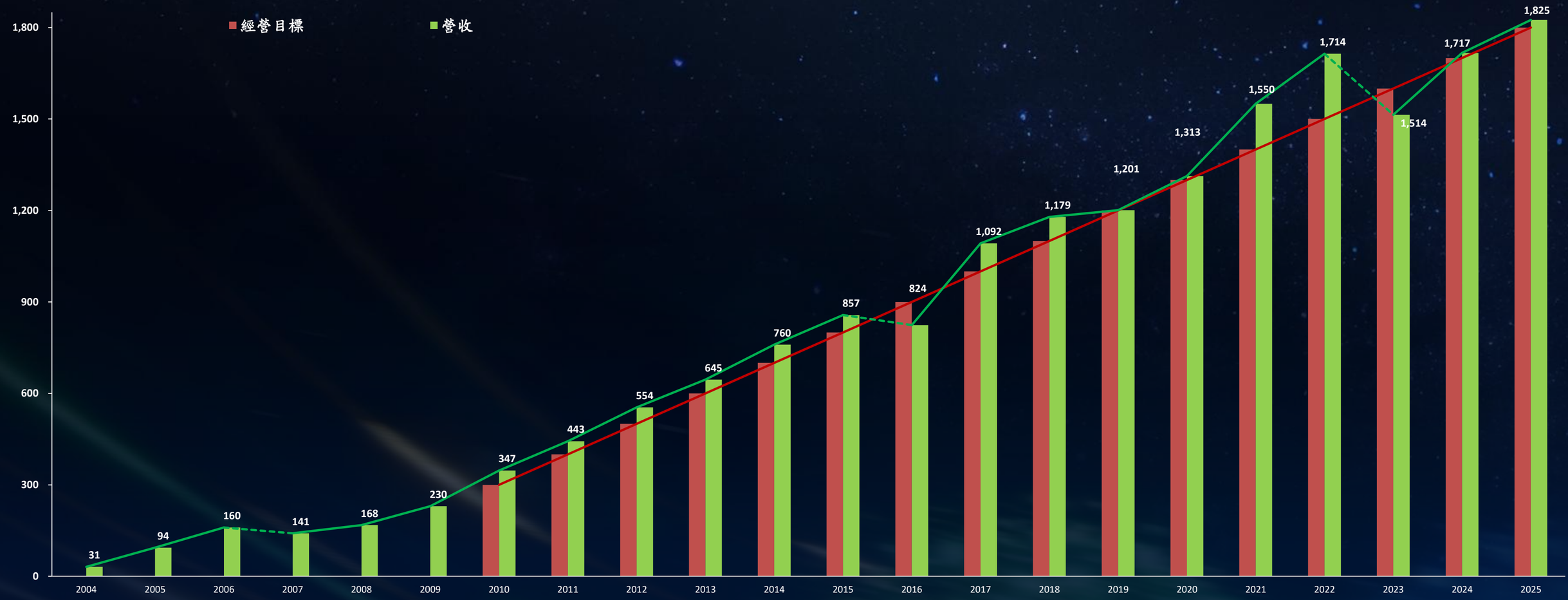


臻鼎科技集團
Zhen Ding Tech. Group

經營目標與成果

2010年4月製表,每年成長百億

單位：新台幣億元



年度	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
經營目標	-	-	-	-	-	-	300	400	500	600	700	800	900	1,000	1,100	1,200	1,300	1,400	1,500	1,600	1,700	1,800
營收	31	94	160	141	168	230	347	443	554	645	760	857	824	1,092	1,179	1,201	1,313	1,550	1,714	1,514	1,717	1,825
員工人數	3,396	4,605	8,805	13,006	17,009	18,142	22,983	23,034	25,392	27,820	33,212	37,801	39,943	40,622	38,588	35,953	40,521	44,330	42,425	41,478	49,149	50,694

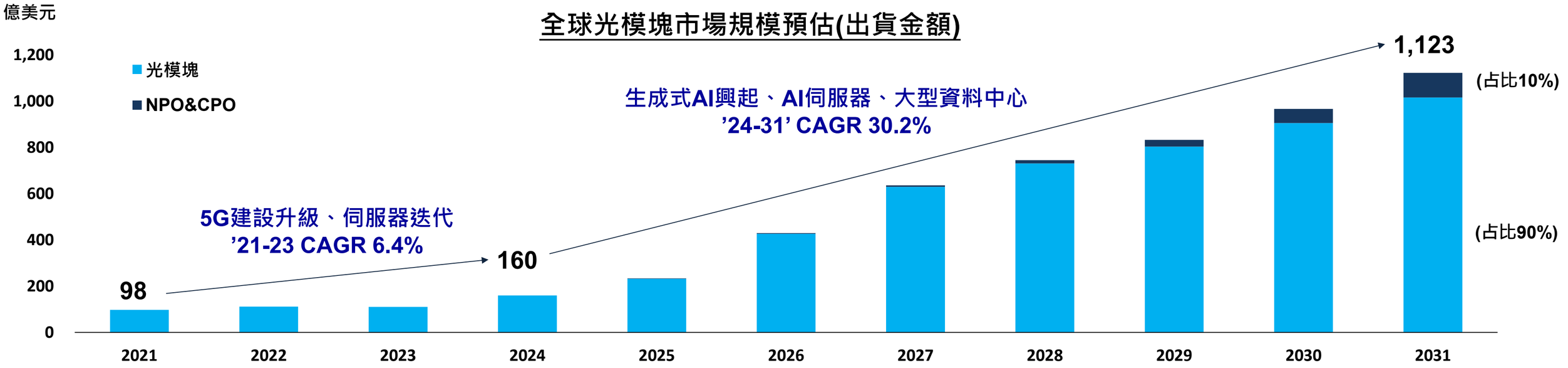
臻鼎科技集團
Zhen Ding Tech. Group

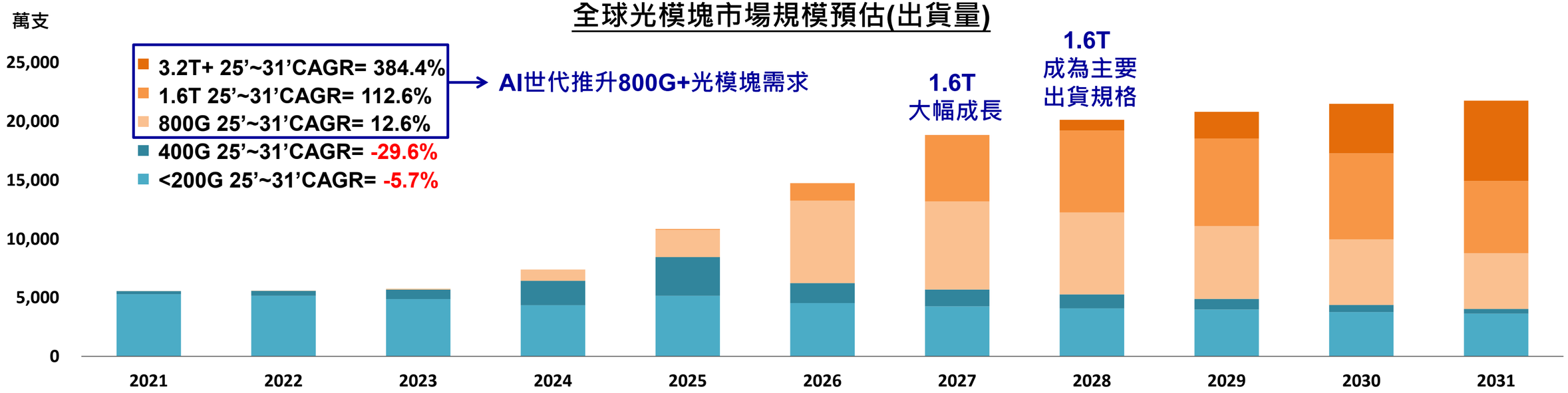
攜手共創，產業共生共榮

臻鼎科技集團於2026年7月24日在深圳舉辦全球供應商大會，匯聚247家供應商、400餘位代表共襄盛舉。

臻鼎科技控股 Zhen Ding Tech. Holding

光模塊市場需求快速攀升 1.6T時代提前來臨





資料來源：Yole Group(2026/5)

AI伺服器 PCB技術要求：臻鼎具備Tier 1技術能力

• GPU Servers

	2022	2023	2024		2025		2026	2027
PCB	NVIDIA HGX A100	NVIDIA HGX H200	NVIDIA HGX B200	NVIDIA NVL72 (GB200)	NVIDIA HGX B300	NVIDIA NVL72 (GB300)	NVIDIA NVL72 (VR200)	NVIDIA NVL576 (Rubin Ultra)
OAM	4+N+4 iHDI	5+N+5 iHDI	5+N+5 iHDI	5+N+5 iHDI (Compute Board)	5+N+5 iHDI	5+N+5 iHDI (Compute Board)	7+N+7 iHDI (Compute Board)	8+N+8 iHDI (Compute Board)
UBB	24-26L	26-28L	28-30L		28-30 L			
Mainboard	16L +	16-18L	16-20L		16-20 L			
Midplane							44L RPCB	44L RPCB
Backplane								
Switch board	20L +	20-22L	22-24L	22L RPCB	22-24L	22L RPCB	5+N+5 iHDI	6+N+6 iHDI
Others				4+N+4 iHDI		4+N+4 iHDI RPCB-網路卡	4+N+4 iHDI RPCB-網路卡 MSAP-光模塊	4+N+4 iHDI

• ASIC Servers

	2026	2027	2025	2026	2027	2025	2026	2027	2026-2027
PCB	Microsoft Maia200	Microsoft Maia300	Google TPU v7	Google TPU v8	Google TPU v9	Amazon Trainium 2+2.5	Amazon Trainium 3	Amazon Trainium 4	China ASIC Servers
OAM					iHDI (TPU board)	3+N+3 iHDI	4+N+4 iHDI	iHDI	5+N+5 iHDI
UBB	26L+ RPCB (ASIC Board)	RPCB (ASIC Board)	28 L RPCB (TPU board)	30/34 L RPCB (TPU board)		24L RPCB	28/30L RPCB		26-28L RPCB
Others	CPU-RPCB 配版-RPCB	CPU-RPCB 配版-RPCB		配版-RPCB			CPU-RPCB Switch-RPCB 網路卡-RPCB	CPU-RPCB Switch-RPCB 網路卡-RPCB	Huawei 光模塊-MSAP NPU-iHDI

資料來源： Goldman Sachs Research(2026/8)

臻鼎科技控股 Zhen Ding Tech. Holding

臻鼎將全面掌握HDI與高多層板(HLC)的黃金時代

Global AI Server PCB Market Opportunity

單位：百萬美元

應 用	2025	2026E	2027E	2028E
GPU Server	2,247	4,728	18,803	42,568
OAM(加速板模組)	966	2,132	9,343	24,168
UBB(通用基板)	555	628	1,183	784
Mainboard(主板)	208	221	223	116
Midplane(中介板)		185	2,451	5,787
Backplane(背板)			225	900
Switch board(交換器板)	439	1,094	3,964	9,010
Others	79	468	1,415	1,803
ASIC Server	2,734	8,832	18,726	41,077
合計	4,981	13,560	37,529	83,645

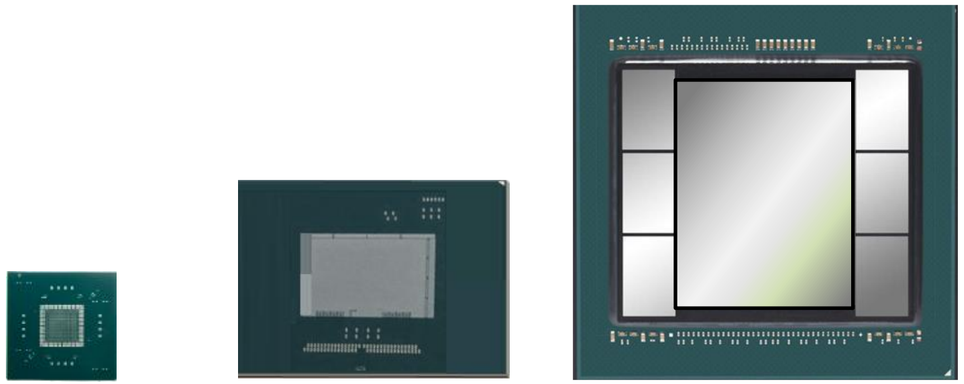
產 品 技 術	2025	2026E	2027E	2028E
RPCB	2,960	7,864	19,609	40,673
<20 layers	509	1,277	2,190	4,980
20-30 layers	2,087	5,171	8,474	16,451
30 layers+	364	1,417	8,946	19,241
HDI	2,021	5,695	17,920	42,973
4+N+4	79	468	1,415	1,803
5+N+5	1,730	3,508	10,195	12,783
6+N+6 or above	212	1,719	6,310	28,386
合計	4,981	13,560	37,529	83,645

資料來源： Goldman Sachs Research(2026/8)

臻鼎ABF載板技術能力與Tier 1競爭者相當，致力爭取全球領先半導體客戶

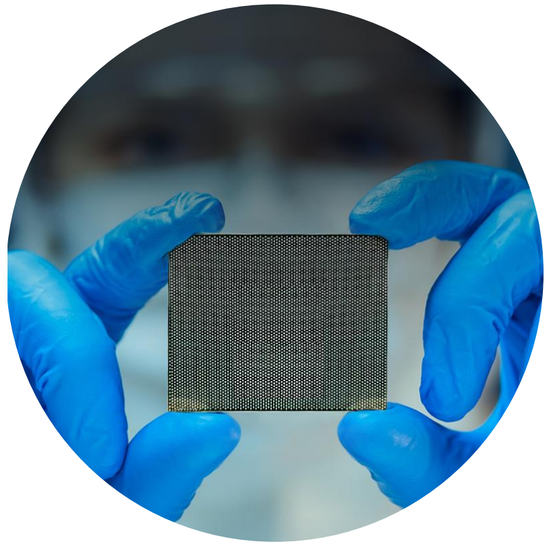
ABF載板發展趨勢

層數高、尺寸大、表面平、產品精度準

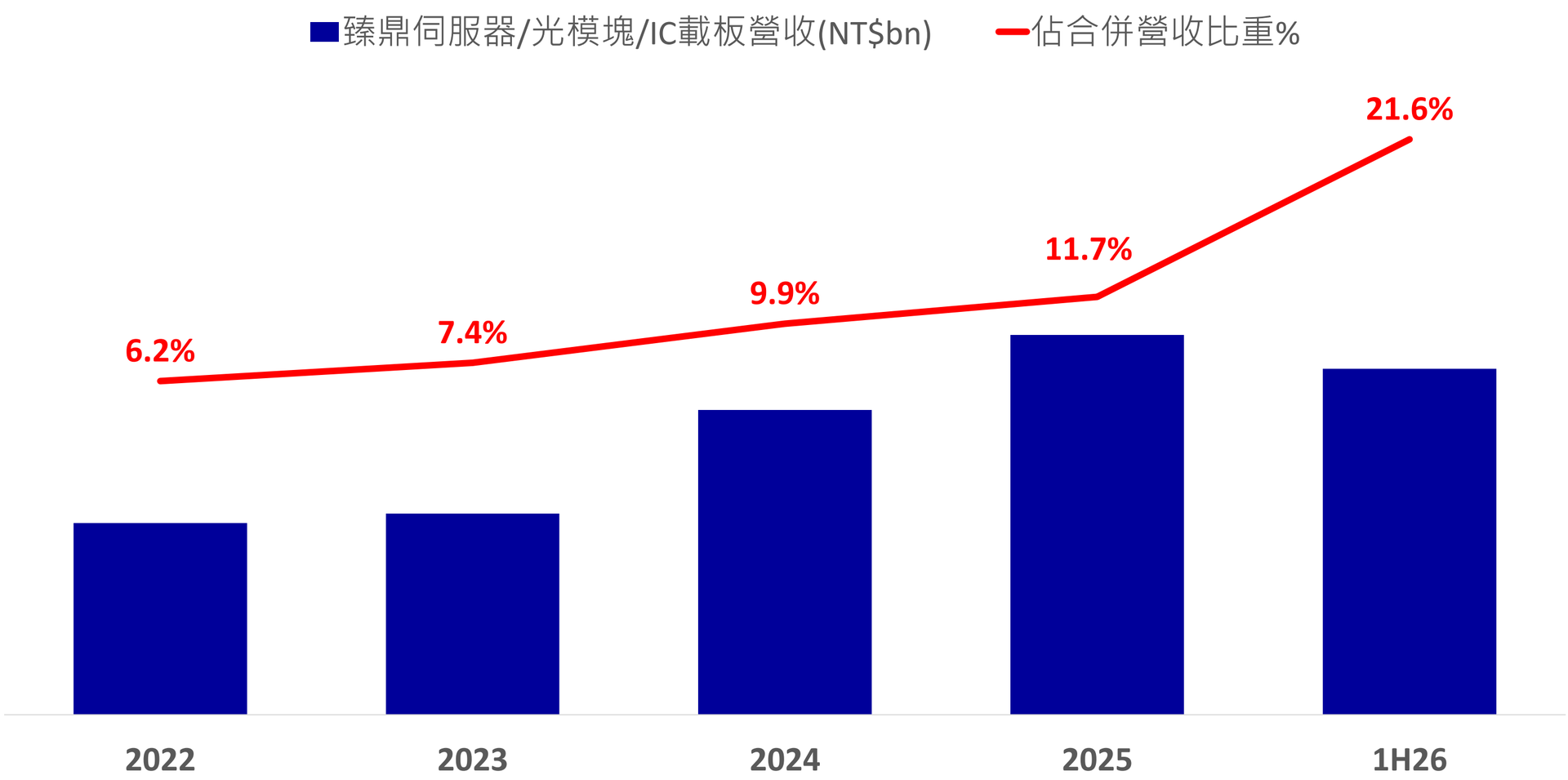


	2005	2020	2026
Body Size (mm)	31 x 31	75 x 60	>150 x 150
Layer count (L)	6	20	28
Bump Count	1K	100K	300~500k

臻鼎專注高階需求，
致力爭取全球領先半導體客戶

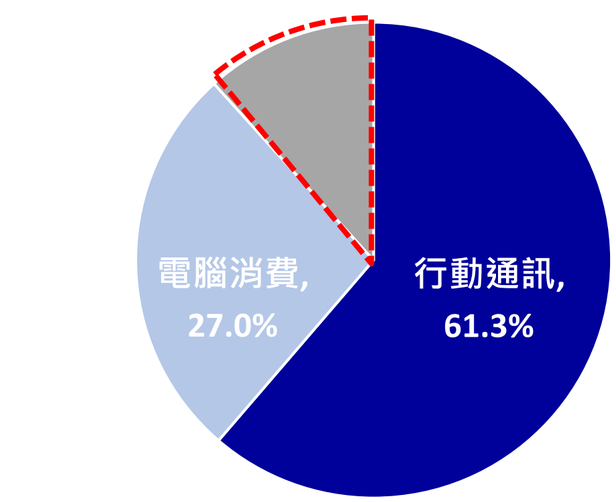


One ZDT 驅動結構升級，
1H26 伺服器 / 光模塊 / IC載板營收佔比突破20%

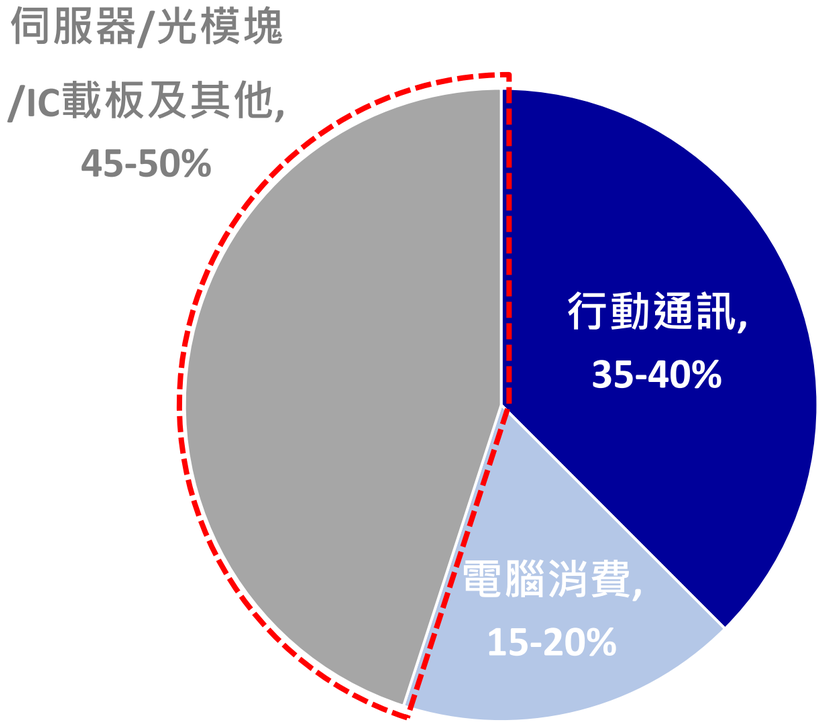


穩步推進伺服器/光模塊與IC載板佈局，持續擴大營收貢獻

臻鼎：2025年產品應用營收佔比



臻鼎：2030年目標產品應用營收佔比



臻鼎科技控股
Zhen Ding Tech. Holding

擴產有依，穩健前行，
搶占市場先機

臻鼎科技集團
Zhen Ding Tech. Group

臻鼎集團戰略佈局：產能藍圖

2026年
274公頃(4,110畝)

深圳第一園區

深圳第二園區

深圳第三園區

深圳禮鼎園區

淮安HA園區

淮安HB園區

淮安HC園區

淮安HD園區

秦皇島園區

秦皇島禮鼎園區

高雄園區

泰國巴真園區

2026年

深圳第一園區

深圳第二園區

深圳第三園區

深圳禮鼎園區

淮安HA園區

淮安園區

淮安園區

淮安HD園區

28座

秦皇島園區

秦皇島禮鼎園區

高雄園區

泰國巴真園區

2030年

59座

(建設中22座、規劃中9座)

全球生產基地加速擴產，效益將於2026年起逐步顯現

擴產規劃

大陸廠區

- 淮安園區：
  - 規劃至2026年底，淮安HA、HB、HC園區將有6棟高階PCB新廠完成建設。
  - 另HD園區已於今年四月動土，規劃新建HDI/MSAP與HLC廠，8月10日封頂，預計明年三月加入生產。
- 深圳園區：
  - 第三園區智慧製造基地於今年七月動土，預計興建三棟智慧製造廠房，重點布局AI伺服器用及光模塊用的高階類載板，及AI終端用高階軟板等高端產品。
  - 禮鼎深圳第二園區於今年九月正式動土，總投資金額不低於人民幣120億元，規劃建設數座工廠，聚焦FCBGA等高階IC載板。新產能主要面向AI、高效能運算及先進封裝需求，進一步強化公司高階FCBGA產品的研發、製造與規模化交付能力。

泰國巴真府廠區

- 泰國一廠：
  - 主要生產高階iHDI、HLC及光模塊產品，於3Q25進入量產。
  - 量產爬坡順利，已通過多家客戶認證及量產，尚有多家伺服器與光模塊全球領先客戶正積極進行認證中。
  - 7月已達損益兩平。
- 目前另有三棟新廠房及一棟機械鑽孔中心同步建置中，規劃將於4Q26起逐步進入試產階段。

臻鼎科技集團
Zhen Ding Tech. Group

260922

臻鼎科技集團淮安科技城

(總用地面積：2,151畝)

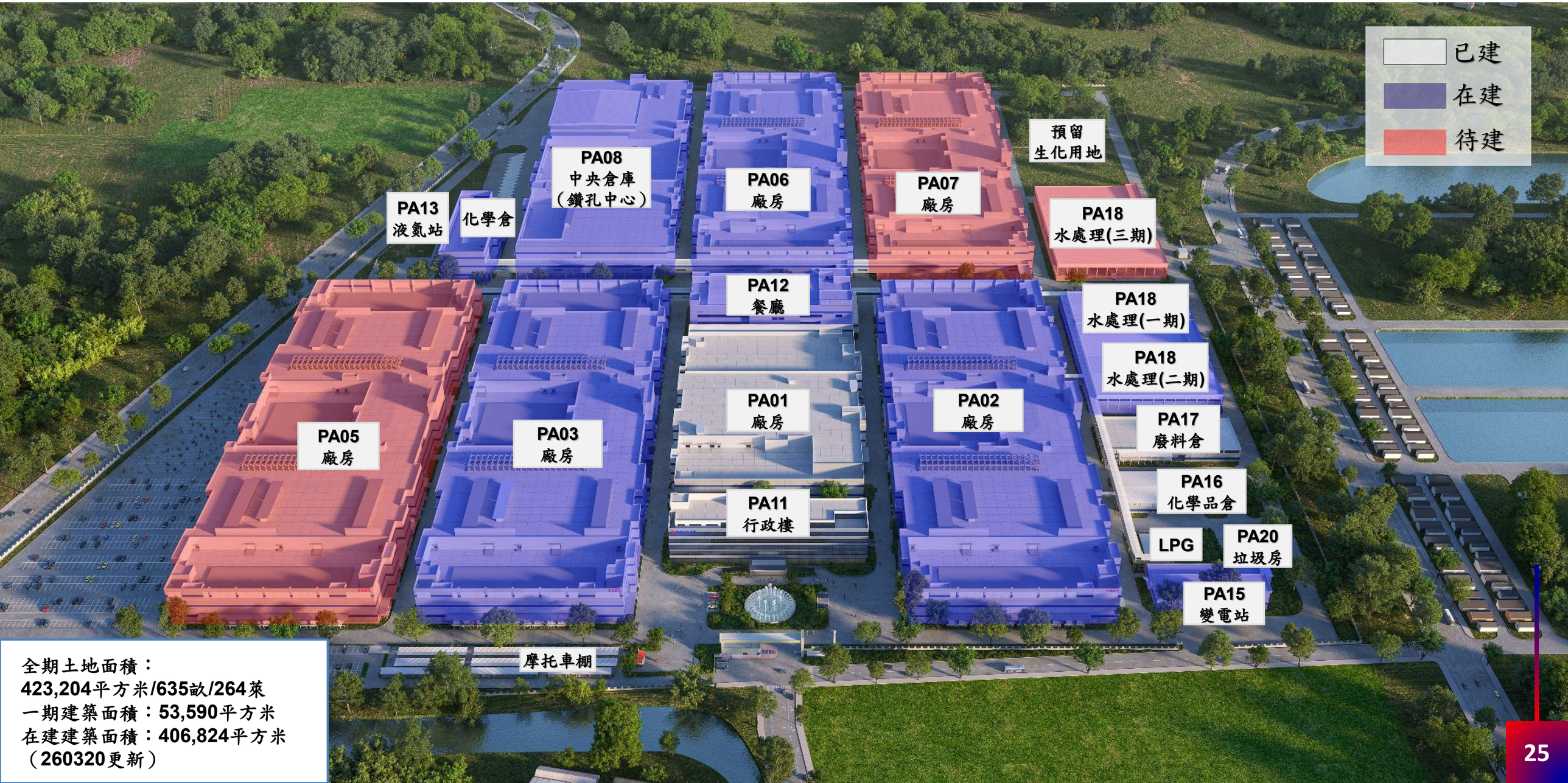
項目總用地面積：1,434,322平方米/2,151畝
項目總建築面積：2,038,683平方米
（260924更新）

臻鼎科技控股
Zhen Ding Tech. Holding

臻鼎深圳廠區規劃

臻鼎科技控股
Zhen Ding Tech. Holding

臻鼎泰國巴真府廠區規劃



全期土地面積：
423,204平方米/635畝/264萊
一期建築面積：53,590平方米
在建建築面積：406,824平方米
（260320更新）

臻鼎集團產品線分佈概覽

V: Will be ready

臻鼎科技控股 Zhen Ding Tech. Holding

禮鼎科技 LEADING TECH　佰鼎科技 Brai Technology　鵬鼎控股 AVARY HOLDING　先豐通訊 BoardTek Elec.

	大陸 深圳廠	大陸 秦皇島廠	台灣 高雄廠	大陸 秦皇島廠	大陸 深圳廠	大陸 淮安廠	泰國 巴真廠	印度 清奈廠	台灣 高雄廠	台灣 高雄廠
FPC				V	V	V	V		V	
SMA				V	V	V	V	V	V	
HDI (含SLP/MSAP)				V	V	V	V			V
HLC					V	V	V			V
Rigid-Flex						V				
IC Substrate	V	V	V							

臻鼎科技控股 Zhen Ding Tech. Holding

透過智慧製造，提高人均產值，創造更佳獲利

智慧製造
Smart Manufacturing for Industry 3.5/4.0

PLM | SCM | ERP | HRM | i-Foundry

非結構化數據系統 | 結構化數據系統 | 資料倉儲系統 | AI 算力平台

MES 製造執行系統與聰明生產
Manufacturing Execution System
APS/Data-driven DSS

QMS 品質管理系統與良率提升
Quality Management System
SPC/APC/AQC/AEC

EAP (Equipment Automation System) / IIOT

半導體設備通訊標準介面SECS / 通用設備模型GEM/ IPC-CFX /WEBAPI/SebService/

壓合

雷射

曝光 LDI

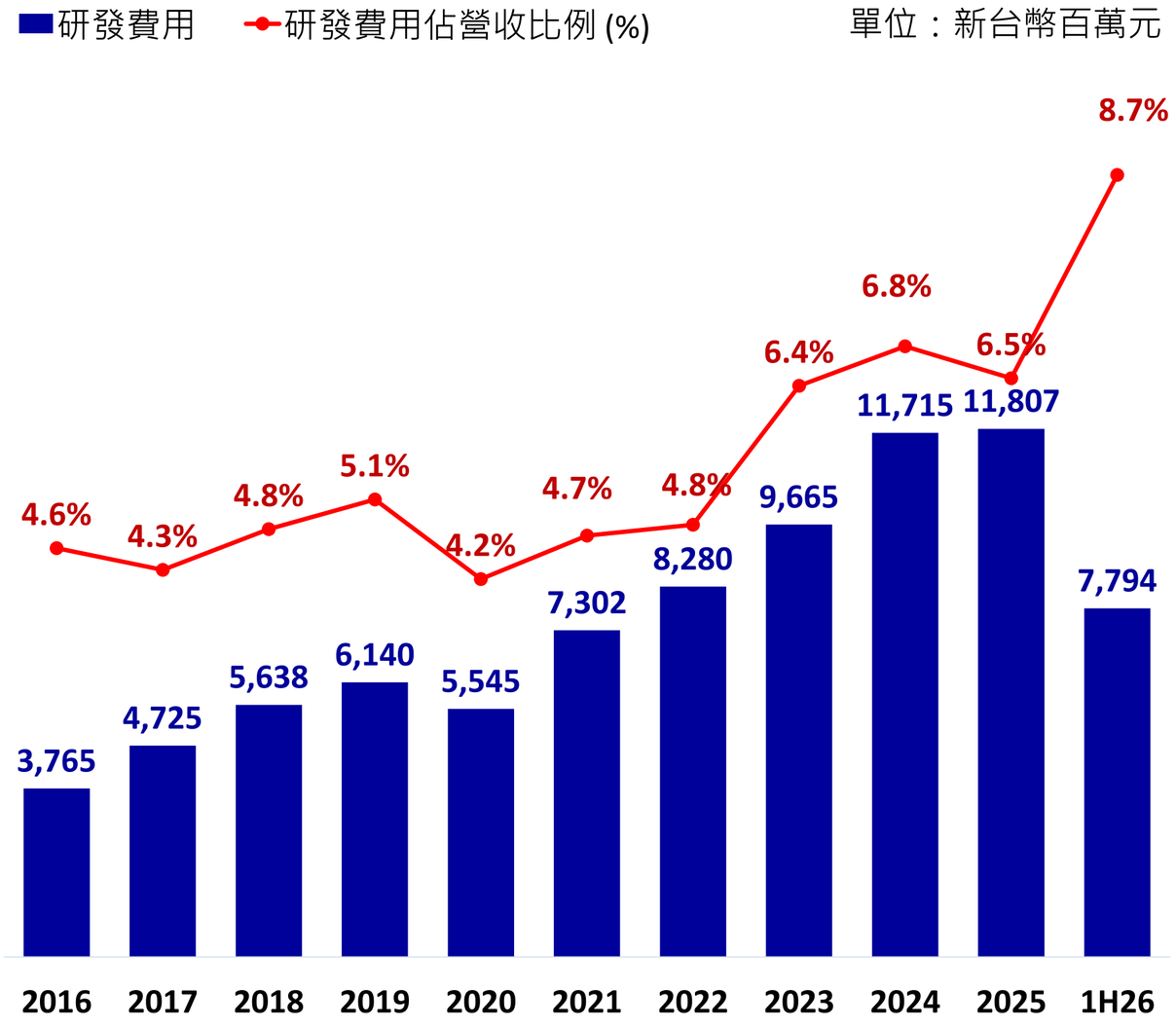
垂直顯影

防焊印刷

電測

作業數位化、數位平台化、平台智能化、智能實戰化

持續投入高階產品研發，鞏固產業領導地位



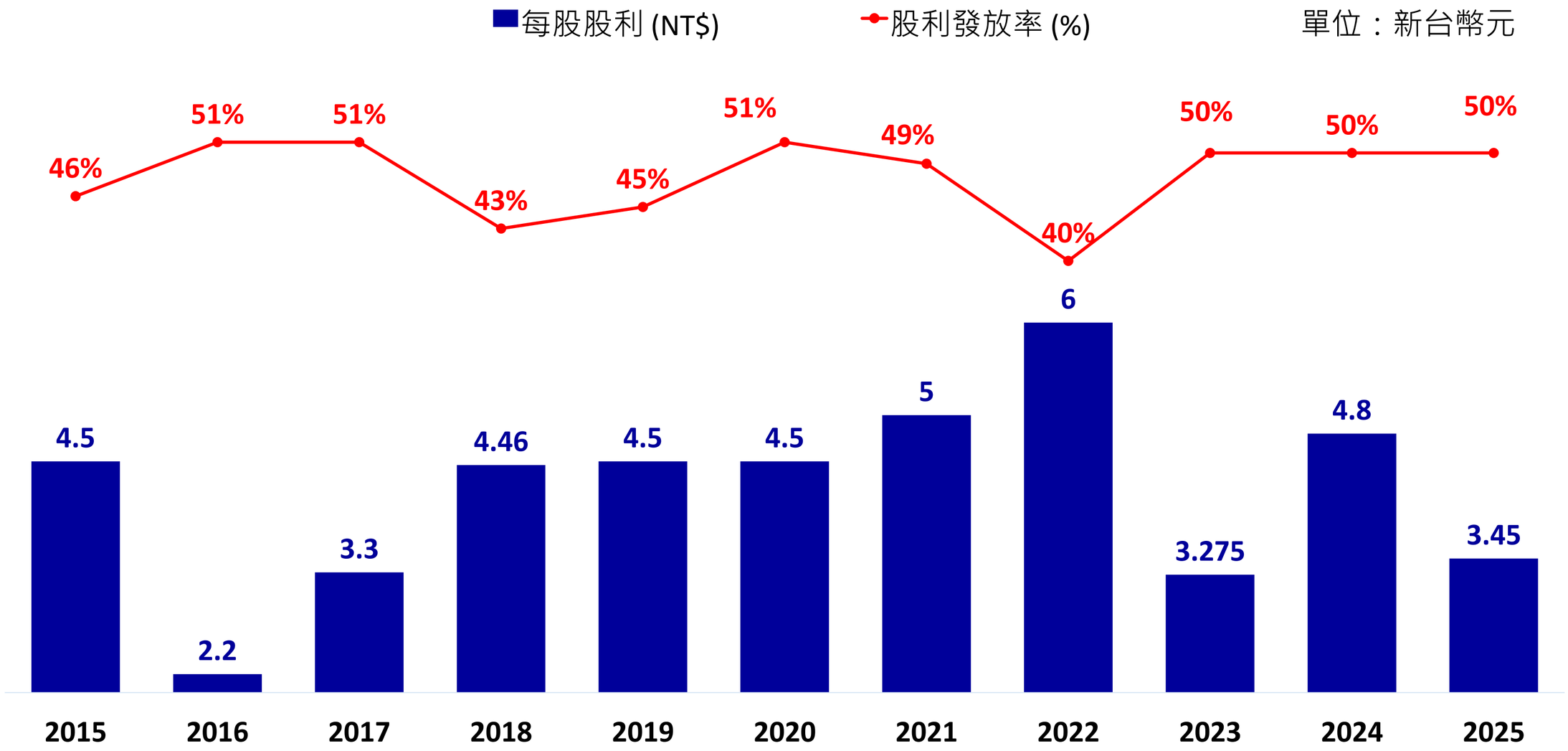
在印刷電路板和半導體相關開發計畫上，臻鼎遵循12 字方針，研發高階技術並引入到各項具有前瞻性之領域應用中。

12 字方針：

輕 輕型化	薄 薄型化	短 流程短	小 小型化
高 高頻、高速、高散熱	低 低污染、低成本、低功耗	多 多功能、多層	快 快速研發、快速製造
精 公差精進	美 美觀	細 細線路	智 智能化

實施台灣及大陸雙研發中心，持續提升新技術開發能力，以保持高端產品及技術之領先優勢。

維持高股息發放率，與股東共享經營成果



致力推動EPS + ESG，國際ESG評比持續提升

▶ 標普全球企業永續評鑑 S&P Global ESG

2025年S&P Global ESG評級進步至88分，並入選**道瓊領先世界指數 (DJBIC World) 及道瓊領先新興市場指數 (DJBIC Emerging Markets)**，亦於2022年至2025年連續4年入選標普全球永續年鑑 (Sustainability Yearbook)。

▶ ISS ESG 評級

最新年度的ISS ESG評級為「C+」，**獲得「優秀 (Prime)」等級**。

▶ 臺灣公司治理評鑑

2025年公司治理評鑑排名上市公司6%-20%，再度**入選臺灣證券交易所公司治理100指數**。

▶ CDP 碳揭露專案

2025年在氣候變遷與水安全管理皆獲得**「A」最高等級殊榮**，其中氣候變遷為首度獲得A級，水安全則已連續兩年獲得A級。

▶ Sustainalytics ESG 風險評級

最新Sustainalytics ESG風險評級為16.1，為**低等風險等級**。

▶ 富時羅素 FTSE Russell ESG

最新年度的FTSE Russell ESG評級獲得4.5分（滿分為5分），成為**台灣PCB企業第一名**。

臻鼎科技控股
Zhen Ding Tech. Holding

營運成果摘要

臻鼎科技控股 Zhen Ding Tech. Holding

季度營運成果

單位: 新台幣佰萬元，除另予註明者外

	2Q26	2Q25	年變化 (%)
營業收入	48,431	38,203	+26.8%
營業毛利	11,186	7,011	+59.6%
營業毛利率	**23.1%**	**18.4%**	**+4.7ppts**
營業費用	7,425	4,586	+61.9%
營業利益	3,761	2,425	+55.1%
營業利益率	**7.8%**	**6.3%**	**+1.5ppts**
營業外收入及支出	1,170	(153)	
匯兌損益	(668)	(742)	
本期淨利	3,727	1,387	+168.6%
淨利率	**7.7%**	**3.6%**	**+4.1ppts**
歸屬於母公司之淨利	2,392	605	+295.3%
每股盈餘 (新台幣元) (1)	**2.19**	**0.63**	
研發費用	4,218	2,611	+61.6%
折舊及攤銷	5,171	4,516	+14.5%
資本支出	17,009	8,751	+94.4%
現金及約當現金(2)	97,098	74,194	+30.9%
股東權益報酬率 (%) (3)	**7.8%**	**3.9%**	

附註：(1) 2026第二季加權平均流通在外股數為1,090,870仟(實際發行股本1,108,887仟股-庫藏股424仟股) 。(2) 現金及約當現金包含按攤銷後成本衡量之金融資產的定期存款等。(3) 股東權益報酬率為以母公司股東平均股權計算的年化數據。

上半年營運成果

單位: 新台幣佰萬元，除另予註明者外

	1H26	1H25	年變化 (%)
營業收入	89,159	78,285	+13.9%
營業毛利	19,998	12,896	+55.1%
營業毛利率	**22.4%**	**16.5%**	**+5.9ppts**
營業費用	13,733	9,415	+45.9%
營業利益	6,265	3,481	+80.0%
營業利益率	**7.0%**	**4.4%**	**+2.6ppts**
營業外收入及支出	1,067	248	+330.4%
匯兌損益	(1,598)	(639)	
本期淨利	5,774	2,413	+139.3%
淨利率	**6.5%**	**3.1%**	**+3.4ppts**
歸屬於母公司之淨利	3,817	1,237	+208.5%
每股盈餘 (新台幣元) (1)	**3.55**	**1.30**	
研發費用	7,794	5,173	+50.7%
折舊及攤銷	10,122	9,255	+9.4%
資本支出	29,688	14,388	+106.3%
現金及約當現金(2)	97,098	74,194	+30.9%
股東權益報酬率 (%) (3)	**6.1%**	**3.4%**	

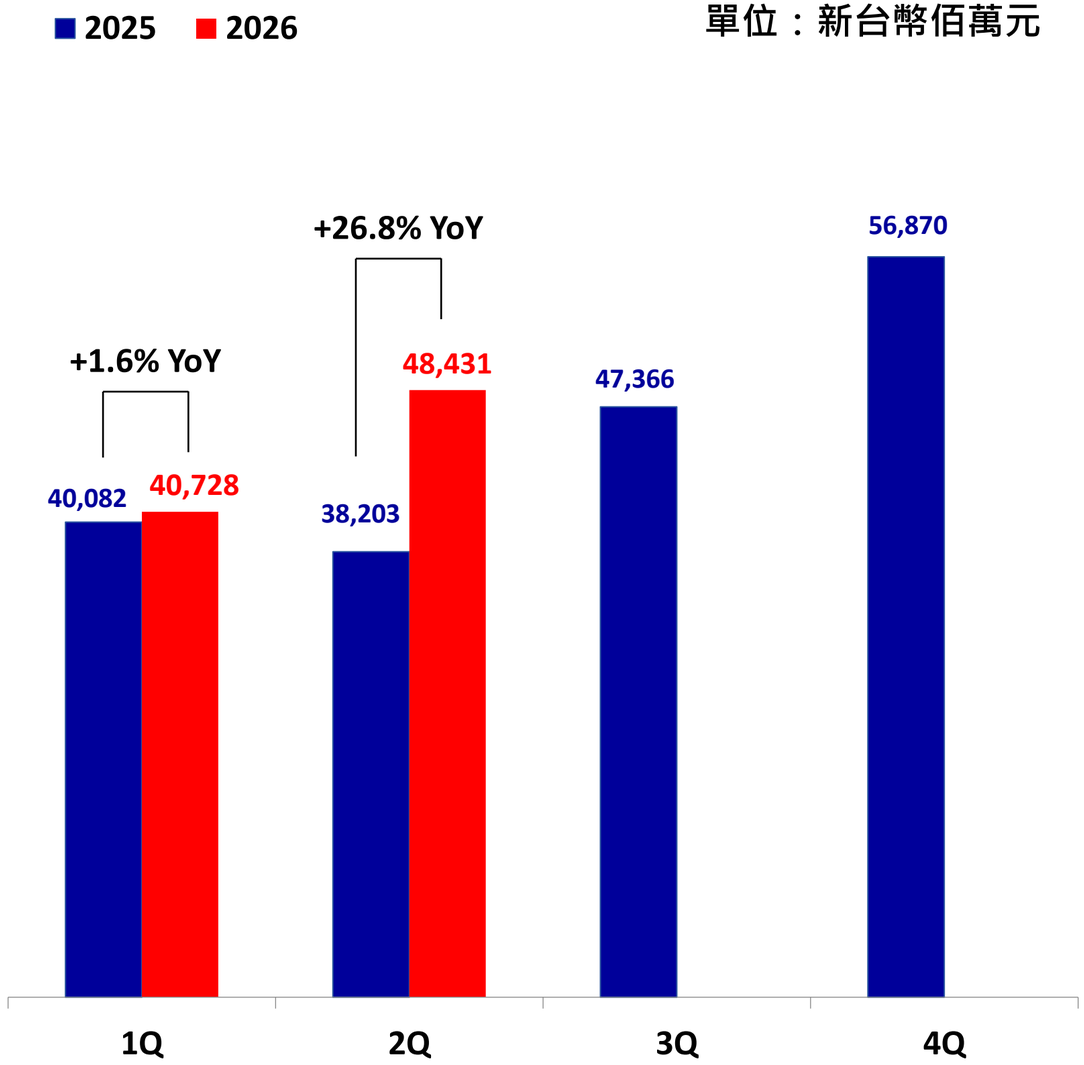
附註：(1) 2026上半年加權平均流通在外股數為1,074,541仟股 (實際發行股本1,108,887仟股-庫藏股424仟股)。(2) 現金及約當現金包含按攤銷後成本衡量之金融資產的定期存款等。(3) 股東權益報酬率為以母公司股東平均股權計算的年化數據。

臻鼎科技控股 Zhen Ding Tech. Holding

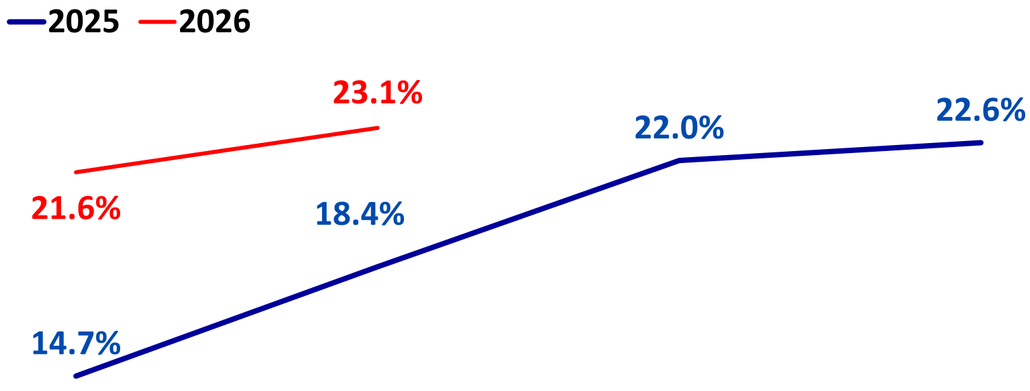
季度營運成果

營業收入

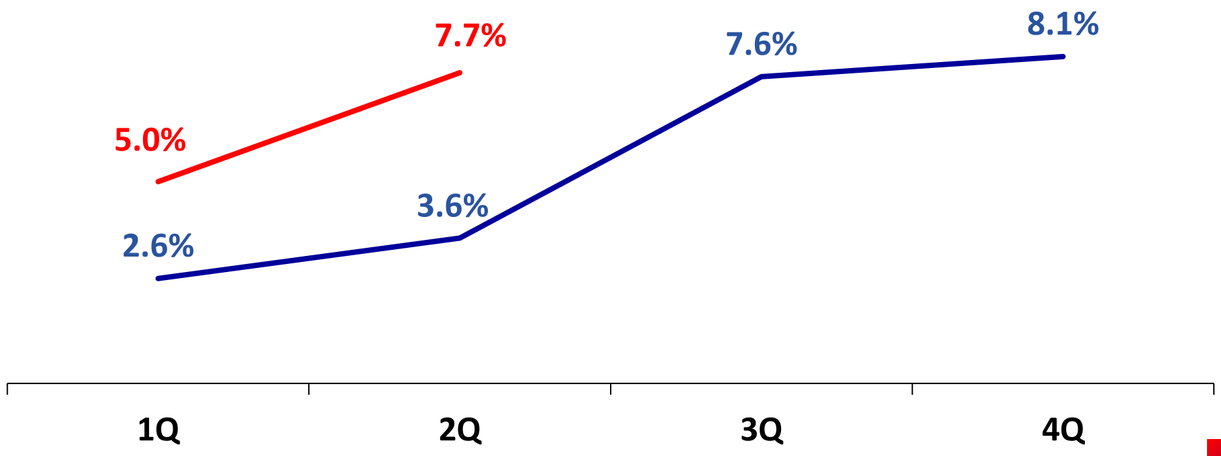
單位：新台幣佰萬元



毛利率 (%)

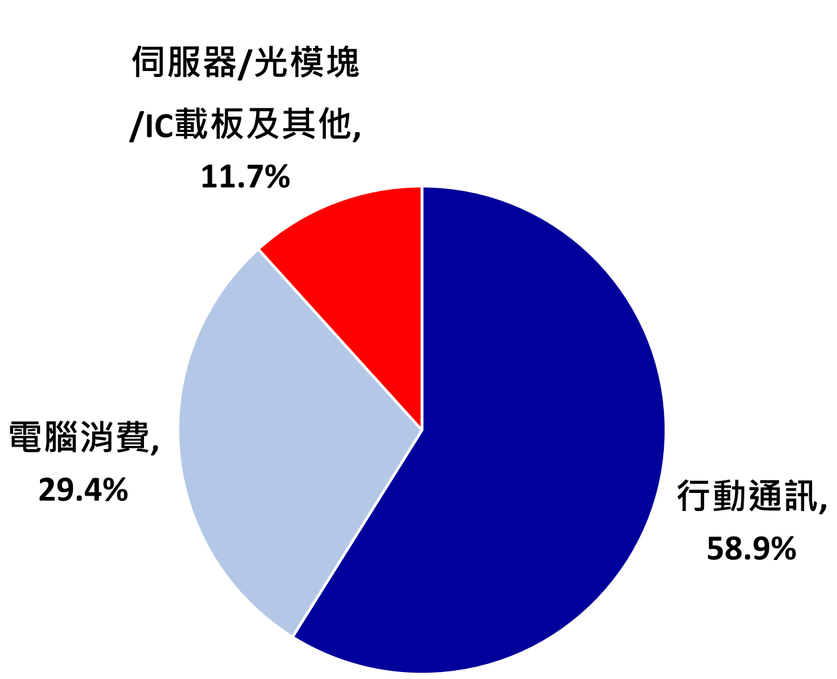


淨利率 (%)

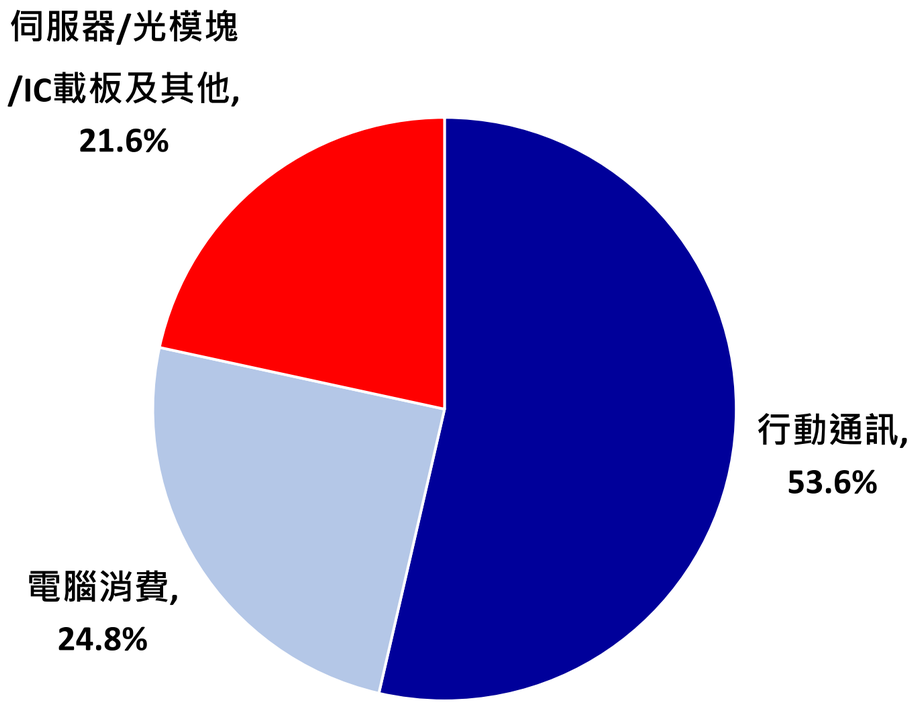


銷售分析–產品應用別

1H25營收：新台幣783億



1H26營收：新台幣892億



資產負債表及重要財務比率

單位：新台幣佰萬元

	2026-6-30		2025-6-30		差異	
	金額	%	金額	%	金額	%
現金及約當現金[2]	97,098	27.8%	74,194	30.6%	22,904	+30.9%
應收款項	30,359	8.7%	18,409	7.6%	11,950	+64.9%
存貨	28,590	8.2%	16,178	6.7%	12,412	+76.7%
不動產、廠房及設備[3]	152,794	43.7%	109,642	45.3%	43,152	+39.4%
資產總額	**349,706**	**100.0%**	**242,172**	**100.0%**	**107,534**	
借款	70,950	51.1%	55,438	22.9%	15,512	+28.0%
應付款項	53,780	15.4%	41,265	17.0%	12,515	+30.3%
負債總額	**138,872**	**39.7%**	**109,026**	**45.0%**	**29,846**	
股東權益總額	**210,834**	**60.3%**	**133,145**	**55.0%**	**77,689**	
重要財務比率						
平均收現日數(天)	62		57		5	
平均銷貨日數(天)	67		51		16	
流動比率(倍)	1.61		1.54		0.07	
資產生產力(倍)[4]	1.28		1.40		(0.13)	

附註：(1) 2026上半年加權平均流通在外股數為1,090,870仟(實際發行股本1,108,887仟股-庫藏股424仟股) 。 (2) 現金及約當現金包含按攤銷後成本衡量之金融資產的定期存款等。(3) 不動產、廠房 及設備 包含 投資性不動產。(4) 資產生產力 =年化營業收入淨額 /平均不動產、廠房及設備淨額。

臻鼎科技控股 Zhen Ding Tech. Holding

歷年營運成果

單位：新台幣佰萬元

年度	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
營業收入	82,393	109,238	117,913	120,068	131,279	155,022	171,356	151,398	171,664	182,522
營業毛利	12,542	17,833	26,061	27,222	26,584	30,537	39,888	27,459	32,461	36,136
本期淨利	3,456	6,772	11,536	12,402	11,508	13,694	20,535	9,432	13,096	10,605
母公司淨利	3,456	5,172	8,448	8,685	8,095	9,651	14,197	6,189	9,180	6,791
折舊及攤銷	5,295	5,679	6,820	7,955	8,405	11,875	14,638	16,323	17,749	18,552
每股盈餘 (元)	4.29	6.43	10.50	9.93	8.90	10.21	15.02	6.55	9.67	6.91
每股股利 (元)	2.20	3.30	4.46	4.50	4.50	5.00	6.00	3.275	4.80	3.45
分派比率 (%)	51%	51%	43%	45%	51%	49%	40%	50%	50%	50%
現金及約當現金 *	30,241	33,296	49,154	43,071	46,775	35,179	57,599	65,970	79,830	71,152
不動產、廠房及設備	32,262	36,681	41,913	46,243	68,177	86,073	104,814	109,965	113,462	126,808
資本額	8,047	8,047	8,047	9,022	9,470	9,470	9,470	9,470	9,567	10,706
資本支出	9,682	12,747	16,224	16,211	21,645	31,149	29,032	26,066	16,431	33,257
股東權益報酬率 (%)	8.59%	14.49%	17.30%	14.72%	11.84%	12.59%	16.67%	7.10%	9.15%	6.57%
負債比率 (%)	59.72%	55.33%	44.25%	35.41%	42.56%	42.01%	42.87%	44.67%	42.85%	39.04%

* 現金及約當現金包含按攤銷後成本衡量之金融資產的定期存款等

臻鼎科技控股
Zhen Ding Tech. Holding

Thank You

長榮 (2603 TT) Evergreen Marine

3Q26 獲利大進，巴拿馬限航至 1H27 支撐運價，2027 年持續成長

買進 (調升評等) | **目標價 (12 個月)：NT$290.0** | 收盤價 (2026/10/08)：NT$232.5 隱含漲幅：24.7%

營收組成 (2025)

美洲線 37%、歐洲線 34%、亞洲線 14%、其他 15%。

本次報告更新重點

項目	本次	前次
評等	買進	持有-超越同業
目標價 (NT$)	290	215
2026年營收 (NT$/十億)	465.4	372.0
2026年EPS	43.5	21.6

交易資料表

市值	NT$503,360百萬元
外資持股比率	25.7%
董監持股比率	2.7%
調整後每股淨值 (2026F)	NT$287.95
負債比	39.8%
ESG評級 (Sustainalytics)	低 (曝險程度共5級)

簡明損益表 (NT$百萬元)

年初至12月	2024A	2025A	2026F	2027F
營業收入	463,568	379,069	465,410	499,341
營業利益	159,945	74,123	118,903	141,901
稅後純益	139,453	68,581	94,241	110,497
EPS (元)	64.41	31.68	43.53	51.04
EPS YoY (%)	287.8	-50.8	37.4	17.2
本益比 (倍)	3.6	7.3	5.3	4.6
股價淨值比 (倍)	0.9	0.9	0.8	0.7
ROE (%)	26.8	11.8	15.5	16.6
現金殖利率 (%)	14.0%	6.9%	9.4%	11.0%
現金股利 (元)	32.50	16.00	21.76	25.52

蔡爵丞 合格證券投資分析人員
Jason.JC.Tsai@Yuanta.com

陳冠成
charleschen24@yuanta.com

元大觀點

- **美伊戰爭及搶出貨潮，巴拿馬運河限航及亞歐塞港等推動運價大漲，3Q26/4Q26 稅後 EPS 上修 177.1%/188.3%至 19.43 元/12.86 元。**
- **美國進口需求增溫，巴拿馬限航延續至 1H27，支撐運價高檔，2027 年歐洲線及美國線長約價將遠高於 2026 年，提振 2027 年營運。**
- **估 2027 年獲利續揚至 51.04 元，年增 17.2%，殖利率 9%具吸引力，調升至買進，目標價 290 元(0.9 倍 PBR X 2027 年淨值 317.22 元)。**

運價大漲，估 3Q26/4Q26 稅後 EPS 上修至 19.43 元/12.86 元

2026 年 9 月營收 497.11 億元，月增 1.5%，年增 65.0%，動能續揚，受惠巴拿馬運河限航，亞歐塞港推動運價盤堅。3Q26 營收 1467.76 億元，季增 39.6%，年增 51.4%，優於預期 47.5%，預估毛利率上修 17.5ppt 至 40.5%，稅後 EPS 上修 177.1%至 19.43 元，季增 162.4%，年增 93.4%。展望 4Q26，營收上修 38.6%至 1269.61 億元，季減 13.5%，年增 48.2%，稅後 EPS 上修 188.3%至 12.86 元，季減 33.8%，年增 226.8%。

美國進口需求增溫，巴拿馬限航延續至 1H27，支撐運價高檔

看好貨櫃產業展望，考量:1)美國零售商及通路商拿到退稅款，於關稅政策空窗期持續回補庫存，美國進口增溫。2)聖嬰導致中美洲降雨減少，影響期間約 12-14 個月，預期 2027 年 5 月後加通湖水位才回升，巴拿馬運河限航將延續至 1H27。3)貨物安全考量，歐洲線去程繞道好望角，回程空櫃走紅海，航商重返紅海衝擊可控。4)2026 年 9 月歐洲線均價年增 121%，美西線及美東線年增超過 250%，預期 2027 年新歐洲線及美國線長約價將遠高於 2026 年長約價，有利 2027 年營運及獲利表現。

受惠運價高檔，2027 年獲利持續成長，殖利率達 9%具吸引力

對長榮投資建議調升至買進，目標價 290 元，運價大漲提振營運，預估長榮 2026 年營收上修 25.1%至 4654.1 億元，年增 22.8%，稅後 EPS 上修 101.9%至 43.53 元，年增 37.4%，預計配息 21.76 元，殖利率達 9.0%以上，極具吸引力。預期巴拿馬運河限航延續至 1H27，維繫 2027 年運價表現，預估 2027 年營收 4993.41 億元，年增 7.3%，稅後 EPS 續增至 51.04 元，年增 17.2%，對長榮投資建議調升至買進，目標價 290 元(0.9 倍 PBR X 2027 年淨值 317.22 元)。

營運分析

3Q26 稅後 EPS 上修 177.1%至 19.43 元，年增 93.4%

長榮公布 2026 年 9 月營收 497.11 億元，月增 1.5%，年增 65.0%，動能續揚，反映 7 月下旬運價終止連 3 週滑落後，再連 8 週上漲盤堅之效應，優於預期，乃巴拿馬運河限航效應，美國線船舶調度吃緊，而颱風頻繁侵襲東亞，及萊茵河水位偏低影響歐洲河運作業，歐洲碼頭亦有罷工事件，造成亞洲及歐洲貨櫃卸貨停擺，導致塞港，約造成全球 10%運力供給流失，推動運價盤堅。累積 3Q26 營收 1467.76 億元(原估 995.39 億元)，季增 39.6%，年增 51.4%，優於預期 47.5%，毛利率上修 17.5ppt 至 40.5%(原估 23%)，季增 17.7ppt，年增 14.4ppt，營益率上修 16.9ppt 至 37.6%(原估 20.7%)，營業利益上修 187.2%至 552.07 億元(原估 192.24 億元)，季增 185.2%，年增 150.6%，稅後 EPS 上修 177.1%至 19.43 元(原估 7.01 元)，季增 162.4%，年增 93.4%，獲利動能優於預期。

圖 1：2026 年第 3 季財測與預估比較

(百萬元)	3Q25A	2Q26A	3Q26F	季增率	年增率	3Q26F		預估差異	
						元大預估	市場預估	元大	市場
營業收入	96,921	105,161	146,776	39.6%	51.4%	99,539	134,803	47.5%	8.9%
營業毛利	25,341	23,978	59,444	147.9%	134.6%	22,894	51,522	159.6%	15.4%
營業利益	22,031	19,358	55,207	185.2%	150.6%	19,224	44,050	187.2%	25.3%
稅前利益	25,053	20,864	59,038	183.0%	135.7%	21,534	45,564	174.2%	29.6%
稅後淨利	21,746	16,034	42,065	162.4%	93.4%	15,181	36,871	177.1%	14.1%
調整後 EPS (元)	10.04	7.41	19.43	162.4%	93.4%	7.01	17.03	177.1%	14.1%
重要比率 (%)				百分點	百分點			百分點	百分點
營業毛利率	26.1%	22.8%	40.5%	17.7	14.4	23.0%	38.2%	17.5	2.3
營業利益率	22.7%	18.4%	37.6%	19.2	14.9	20.7%	32.7%	16.9	4.9
稅後純益率	22.4%	15.3%	28.7%	13.4	6.3	15.3%	27.4%	13.4	1.3

資料來源：公司資料、元大投顧預估、Bloomberg

4Q26 稅後 EPS 上修 188.3%至 12.86 元，年增 226.8%

展望 4Q26 整體貨載需求將轉淡，預期運價短期將呈現季節性淡季修正，不過公司訂單資料顯示 2026 年 11 月之前的船舶貨載狀況樂觀，預期修正幅度可控，而根據歷史經驗，12 月至 2027 年 1 月則會進入中國農曆年前提前備貨的出貨潮，將有季節性旺季需求，有利歐洲線運價止跌回升，美國線運價則可維持高檔整理，預估 4Q26 營收上修 38.6%至 1269.61 億元(原估 915.76 億元)，季減 13.5%，年增 48.2%，毛利率上修 16.3ppt 至 34.5%(原估 18.2%)，營益率上修 16.8ppt 至 27.9%(原估 11.1%)，營業利益上修 248.9%至 353.9 億元(原估 101.42 億元)，季減 35.9%，年增 324.9%，稅後 EPS 上修 188.3%至 12.86 元(原估 4.46 元)，季減 33.8%，年增 226.8%。

圖 2：2026 年第 4 季財測與預估比較

(百萬元)	4Q25A	3Q26F	4Q26F	季增率	年增率	4Q26F		預估差異	
						元大預估	市場預估	元大	市場
營業收入	85,694	146,776	126,961	-13.5%	48.2%	91,576	111,080	38.6%	14.3%
營業毛利	15,353	59,444	43,802	-26.3%	185.3%	16,667	29,044	162.8%	50.8%
營業利益	8,328	55,207	35,390	-35.9%	324.9%	10,142	21,260	248.9%	66.5%
稅前利益	10,539	59,038	41,243	-30.1%	291.3%	14,304	22,844	188.3%	80.5%
稅後淨利	8,518	42,065	27,839	-33.8%	226.8%	9,655	18,153	188.3%	53.4%
調整後 EPS (元)	3.93	19.43	12.86	-33.8%	226.8%	4.46	8.39	188.3%	53.4%
重要比率 (%)				百分點	百分點			百分點	百分點
營業毛利率	17.9%	40.5%	34.5%	-6.0	16.6	18.2%	26.1%	16.3	8.4
營業利益率	9.7%	37.6%	27.9%	-9.7	18.2	11.1%	19.1%	16.8	8.8
稅後純益率	9.9%	28.7%	21.9%	-6.8	12.0	10.5%	16.3%	11.4	5.6

資料來源：公司資料、元大投顧預估、Bloomberg

預估 2026 年稅後 EPS 上修 101.9%至 43.53 元，年增 37.4%

受惠美伊戰爭爆發造成荷姆茲海峽遭封鎖，引發中東港口運輸中斷，全球貨櫃運輸調度混亂，以及美國將各國關稅調整期截止前的搶出貨潮湧現，推動運價大漲效應，長榮 2Q26 本業獲利回升，營收 1051.61 億元，季增 21.6%，年增 21.6%，毛利率季增 7.3ppt 至 22.8%，營益率季增 8.1ppt 至 18.4%，營業利益 193.58 億元，季增 116.3%，年增 34.2%，稅後 EPS 為 7.41 元，季增 93.1%，年增 46.3%。

而 3Q26 巴拿馬運河限航，亞洲颱風密集及歐洲碼頭工人罷工等事件，扭轉 3Q26 貨載需求由旺轉淡的季節性特徵，運價走勢強勁，挹注長榮 2H26 營運向上，預估 2026 年營收上修 25.1%至 4654.1 億元(原估 3720.25 億元)，年增 22.8%，毛利率上修 10.3ppt 至 30.2%(原估 19.9%)，營益率上修 10.6ppt 至 25.6%(原估 15.0%)，營業利益上修 112.6%至 1189.03 億元(原估 559.31 億元)，年增 60.4%，稅後 EPS 上修 101.9%至 43.53 元(原估 21.56 元)，年增 37.4%，年底淨值 287.95 元(原估 265.98 元)。

預估 1Q27/2027 年稅後 EPS 各為 14.26/51.04 元，各自年增 271.9%/17.2%

展望 1Q27，中國農曆年前提前備貨的出貨潮的季節性旺季需求，加上 2027 年初加通湖水位再次回落機率高，巴拿馬運河限航持續，可望維繫運價高檔整理，預估 1Q27 營收 1288.66 億元，季增 1.5%，年增 49%。毛利率季增 0.5ppt 至 35%，而獎金費用減少，營益率季增 3.3ppt 至 31.2%，營業利益 401.45 億元，季增 13.4%，年增 348.6%，稅後 EPS 為 14.26 元，季增 10.9%，年增 271.9%。

圖 3：2027 年第 1 季財測與預估比較

(百萬元)	1Q26A	4Q26F	1Q27F	季增率	年增率	1Q27F		預估差異	
						元大預估	市場預估	元大	市場
營業收入	86,511	126,961	128,866	1.5%	49.0%	92,950	86,970	38.6%	48.2%
營業毛利	13,387	43,802	45,103	3.0%	236.9%	17,103	15,462	163.7%	191.7%
營業利益	8,948	35,390	40,145	13.4%	348.6%	12,969	8,700	209.5%	361.4%
稅前利益	10,639	41,243	43,342	5.1%	307.4%	15,160	10,226	185.9%	323.8%
稅後淨利	8,304	27,839	30,881	10.9%	271.9%	10,688	8,203	188.9%	276.5%
調整後 EPS (元)	3.84	12.86	14.26	10.9%	271.9%	4.94	3.79	188.9%	276.5%
重要比率 (%)				百分點	百分點			百分點	百分點
營業毛利率	15.5%	34.5%	35.0%	0.5	19.5	18.4%	17.8%	16.6	17.2
營業利益率	10.3%	27.9%	31.2%	3.3	20.9	14.0%	10.0%	17.2	21.1
稅後純益率	9.6%	21.9%	24.0%	2.1	14.4	11.5%	9.4%	12.5	14.6

資料來源：公司資料、元大投顧預估、Bloomberg

展望 2027 年，受惠 2Q26 至 3Q26 運價大漲，歐洲線 2026 年 9 月平均運價為 2481 美元，年增 121%，美西線 9 月平均運價 7400 美元，年增 287%，美東線 9 月平均運價為 10470 美元，年增 270%，整體 SCFI 運價指數 9 月均價為 3656 點，年增 184%。由於運價水準遠高於 2025 年同期，預期 2026 年底洽談 2027 年新的歐洲線長約價將遠高於 2026 年，而極端氣候的聖嬰現象恐延續至 1H27，巴拿馬運河限航持續，維繫美國線運價表現，且美伊戰爭變數，全球貨櫃運輸調度干擾仍在，諸多因素拉扯有利支撐運價，對比 2Q26 美西線運價區間 2000-3500 美元，美東線 2800-4500 美元的相對低檔水準，推估 1H27 運價水準有望較 2Q26 大幅年增，預期 2027 年 5 月前議定的美國線新的長約價將可高於 2026 年水準，有利 2027 年營運表現。預估 2027 年營收 4993.41 億元，年增 7.3%，毛利率 33.1%，年增 2.9ppt，營益率 28.4%，年增 2.8ppt，營業利益 1419.01 億元，年增 19.3%，稅後 EPS 為 51.04 元，年增 17.2%，年底淨值 317.22 元。

若以 50%股利配發率及 2026 年稅後 EPS 約 43.53 元計算，推估 2027 年長榮將配息 21.76 元，以目前股價計算，殖利率達 9.0%以上，極具吸引力。

對貨櫃產業看法轉為正向，運價可望維持高檔，2027 年營運向上

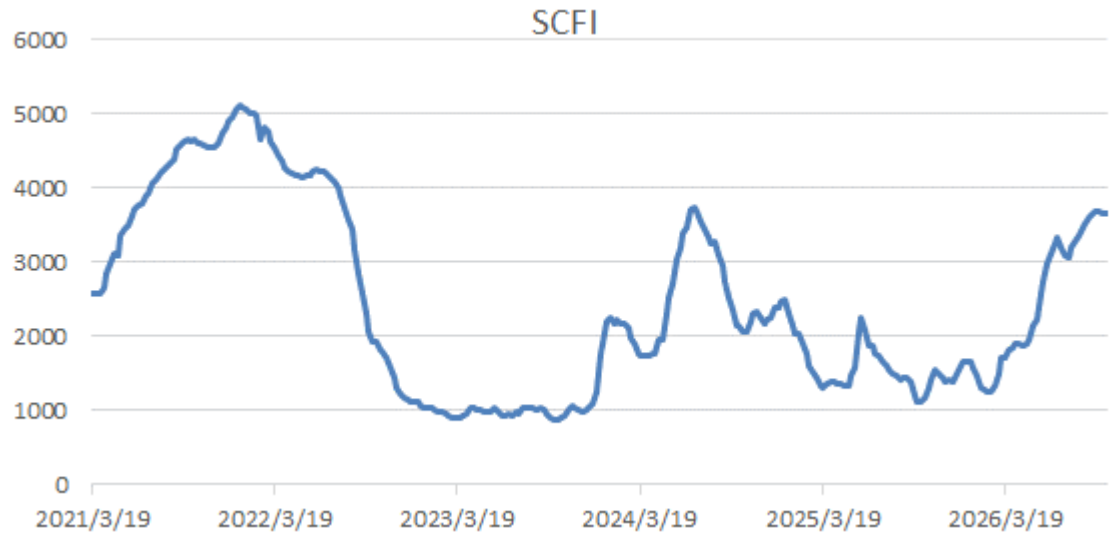
對貨櫃產業及運價看法轉向正面，原因包括:1)美國零售商及通路商拿到川普政府退稅款，採取擴大採購策略，於關稅政策空窗期期間，持續回補庫存，以規避未來的關稅不確定因素，美國貨櫃進口需求持續增溫。2)強聖嬰年的極端氣候導致中美洲降雨減少，根據過往幾次聖嬰年的經驗，加通湖水位降低週期影響期間約 12-14 個月，預期 2027 年 5 月之後加通湖水位才有望回升，預期巴拿馬運河限航將延續至 1H27，可望支撐 1H27 美國線運價維持高檔。3)航商基於貨物安全考量，對於歐洲線航行策略，採取去程繞道好望角，回程空櫃則走紅海航道，因此航商重返紅海對運力需求減少的衝擊相對可控。4)受惠 2Q26 至 3Q26 運價大漲，9 月歐洲線 9 月平均運價年增 121%，美西線年增 287%，美東線年增 270%，SCFI 運價指數 9 月均價年增 184%，遠高於 2025 年同期，預期 2027 年新的歐洲線及美國線長約價也將遠高於 2026 年同期，有利貨櫃航商 2027 年營運及獲利表現。後續觀察重點包括美伊戰爭發展及和談進度，美國是否再推出其他關稅政策等。

產業概況

美伊戰爭加上關稅降低搶運潮推動運價大漲，扭轉 2026 年初之跌勢

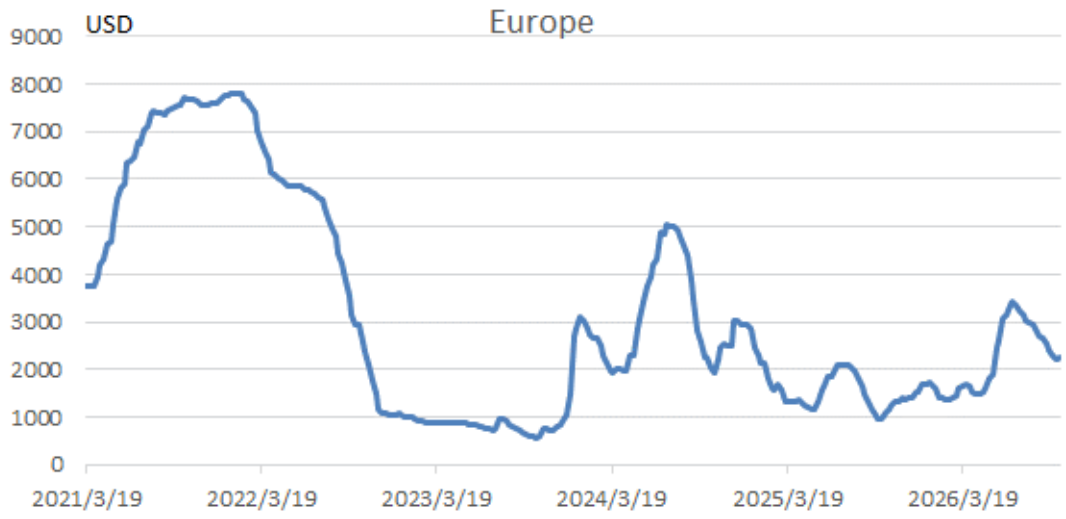
川普關稅政策扭曲 2025 年貨櫃運輸季節性，運價大幅波動，而航商透過縮班減艙維繫運價，加上美國品牌商及零售商回補庫存需求浮現，運價於 2025 年 11 月底觸底回彈，走高至 2025 年 12 月下旬 1656.3 點，波段漲幅 18.9%。進入 2026 年，美國市場回補庫存告一段落，加上航商運力持續增加，以及馬士基宣布將返回紅海航線等利空因素影響，運價於年初再次修正至 2 月中旬的低點 1251.5 點，波段跌幅 24.4%，中國農曆年前回補庫存的小旺季需求提早結束。而 2026/2/28 美伊戰爭爆發，伊朗封鎖荷姆茲海峽，引發中東港口運輸中斷，航商改道，船期拉長，碼頭塞港等狀況發生，亞洲近洋航線船班調度混亂，運價提前於 3 月淡季期間起漲，另外美國對等關稅遭判違法，美國將各國關稅調整至 10%為期 150 天，中國及東南亞的越南、印尼等國稅率降低，回補庫存出貨需求浮現，搭配 5 月下旬至 6 月上旬美國返校需求備貨潮的季節性循環將啟動，提振運價一路走揚至 7 月初的波段高點 3326.87 點，自 2 月下旬起算的波段漲幅達 165.8%，其中以美西線飆漲 271%最強，美東線也勁揚 228.7%，歐洲線上漲 151.1%，地中海線上漲 116.7%，漲勢相對落後。美國線運價領漲反應搶運潮的效應顯著。

圖 4：SCFI 運價指數走勢圖



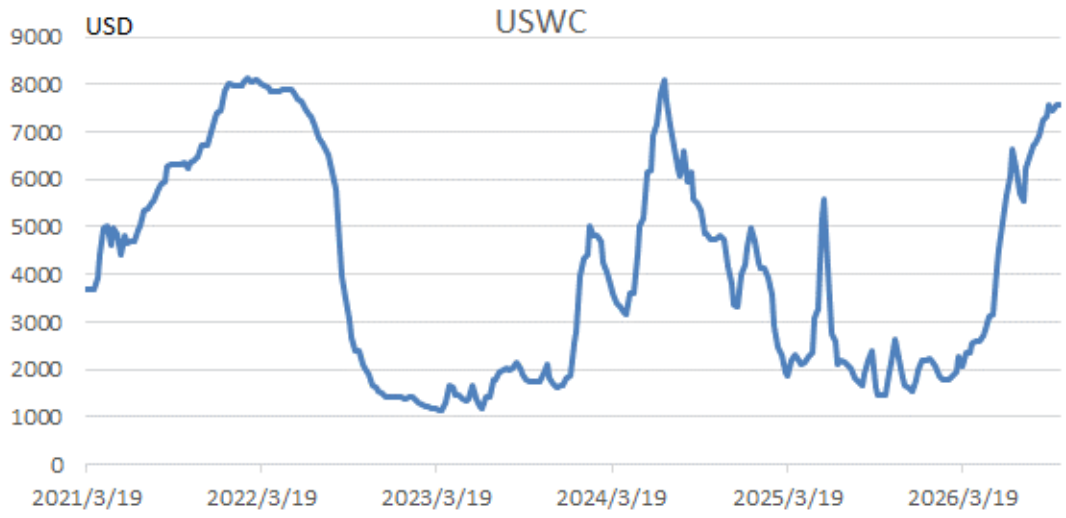
資料來源：SCFI、元大投顧整理

圖 5：歐洲線運價走勢圖



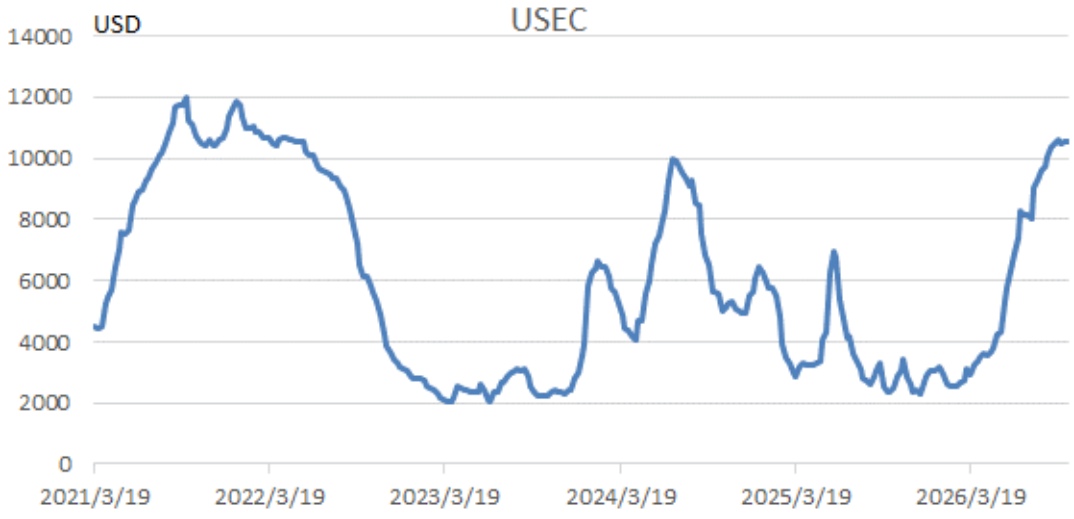
資料來源：SCFI、元大投顧整理

圖 6：美西線運價走勢圖



資料來源：SCFI、元大投顧整理

圖 7：美東線運價走勢圖



資料來源：SCFI、元大投顧整理

巴拿馬運河限航，扭轉運價淡旺季特徵，惟貨載需求轉淡，運價短期有修正壓力

運價於 2026 年 7 月初見高後，呈現連 3 週下跌至 7 月下旬的 3062.95 點，波段跌幅 7.9%，以美西線下挫 16.5%較弱，其餘跌幅則為 3-8%不等，相對平穩。而 8 月初開始運價啟動另一波漲勢，連 8 週上漲至 9 月中下旬的 3687.18 點，波段漲幅 20.4%，惟這一波運價走勢呈現兩極，美西線波段上漲 36.6%至 7560 美元，創 2024 年 8 月中以來新高，美東線波段上漲 31.6%至 10579 美元，突破萬元關卡，突破紅海危機時期 2024 年 7 月中高點，創 2022 年 7 月下旬以來新高，不過歐洲線同期間下跌 23.1%，地中海線下跌 28.2%，強弱分明，美國搶運潮過後，加上返校需求備貨高峰已過，貨載需求開始由旺轉淡，惟 8 月再次發生聖嬰年，巴拿馬運河缺水限航，影響美國線船舶調度，扭曲原本的貨載量及運價的淡旺季特徵，提振美國線運價走勢強勁，加上颱風侵襲東亞，以及萊茵河水位偏低影響歐洲河運作業，歐洲碼頭亦有罷工事件，相繼造成亞洲及歐洲貨櫃卸貨停擺，導致塞港，約造成全球 10%運力供給流失，支撐運價呈現高檔整理。

先前的兩週 SCFI 運價連 2 週下跌，跌幅 0.7%，其中歐洲線下跌 9.3%，地中海線下跌 4.1%較弱，不過美西線小漲 0.2%，美東線僅跌 0.5%，維持相對高檔水準。2026/10/9 最新資料，SCFI 運價上漲 0.1%至 3664.11 點，終止連 2 週滑落，其中以歐洲線上漲 3.4%至 2274 美元領漲最強，地中海線上漲 0.2%至 3002 美元，雙雙終止連 13 週滑落。至於美西線小漲 0.1%至 7584 美元，美東線滑落 0.04%至 10524 美元，均維持高檔整理。歐洲線運價反彈，乃航商嚴格執行空白航班進行控艙，加上歐洲補庫需求轉佳，提振運價反彈。至於美國線運價高檔整理，延續巴拿馬運河限航，影響到美國線船舶調度之效應。

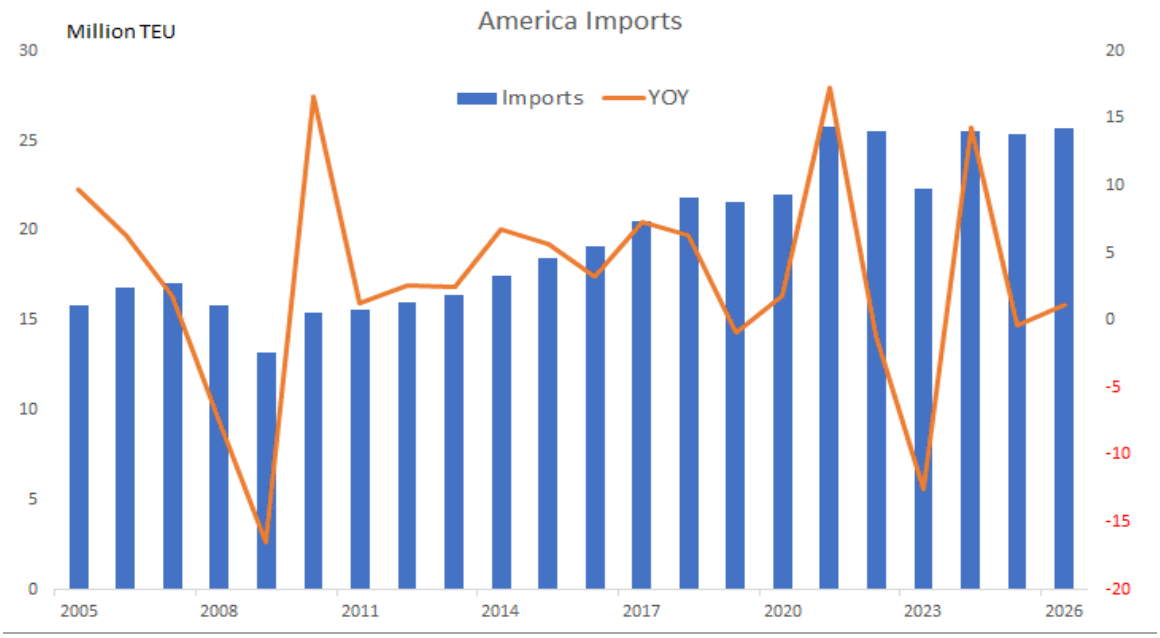
展望後市，目前美東線運價處相對高檔，加上中國十一長假期間工廠停工，出貨量減少，整體貨載需求將轉淡，研判運價有修正壓力，預期未來 4-6 週運價短期將呈現季節性淡季的修正格局，推估修正幅度落在 5-10%之間。而根據歷史經驗，12 月至 2027 年 1 月則會進入中國農曆年前提前備貨的出貨潮，將有季節性旺季需求，有利歐洲線運價止跌回升，至於美國政府的關稅退稅款上看 1600 億美元，美國進口商取得退稅後，採取調降產品終端售價以提振買氣，或是擴大庫存回補，維繫美國商品的進口動能，美國線運價則可望高檔整理，對中期運價看法由保守轉向正面，可期待年底旺季的運價回升動能。

美國回補庫存需求研續，而極端氣候導致巴拿馬運河限航，航商重返紅海的衝擊相對可控，有利運價延續高檔

當前影響運價走勢的因素共有三個面向，包括 1)美國消費動能，2)極端氣候以及 3)航商重返紅海航道等。以下分別探討。

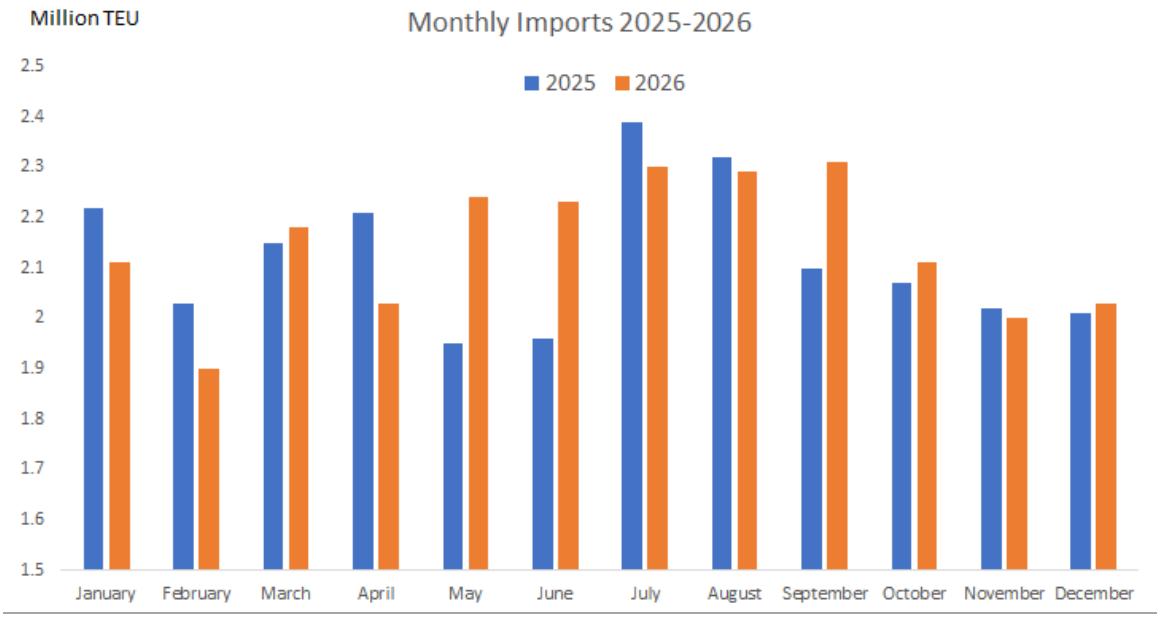
1)關於美國消費動能觀察，根據美國 NRF 資料顯示，2025 年美國貨物進口量將 25.4 百萬 TEU，年減 0.4%，乃對等關稅政策出台衝擊進口表現。進入 2026 年，受制於 2025 年美國加徵關稅預期，提前拉貨效應墊高基期，2026 年前 4 個月貨物進口量 8.22 百萬 TEU，年減 4.5%，不過 5 月開始美國調降關稅期間將至的搶運潮浮現，美國貨物進口量轉趨正成長，5 月年增 14.9%，6 月年增 13.8%，提振 1H26 進口量達 12.69 百萬 TEU，年增 1.4%。目前川普總統執政焦點主要放在中東情勢以及期中選舉，對於關稅政策尚未有新的方向定案，處於關稅政策空窗期，部分零售商及通路商趁關稅稅率相對較友善之際，特別是體積龐大的耐久財(如家電及家具等)的進口商，積極進行庫存回補，另一方面美國進口商取得關稅退稅款後，也擴大庫存回補等，因此 NRF 預估回補庫存需求於 2H26 可望延續，根據 NRF 資料顯示，預估 2026 年美國貨物進口量將可回升至 25.7 百萬 TEU，年增 1.2%，美國需求展望偏向正面。

圖 8：美國進口貨物之貨櫃數及預估量趨勢圖



資料來源：NRF、元大投顧整理

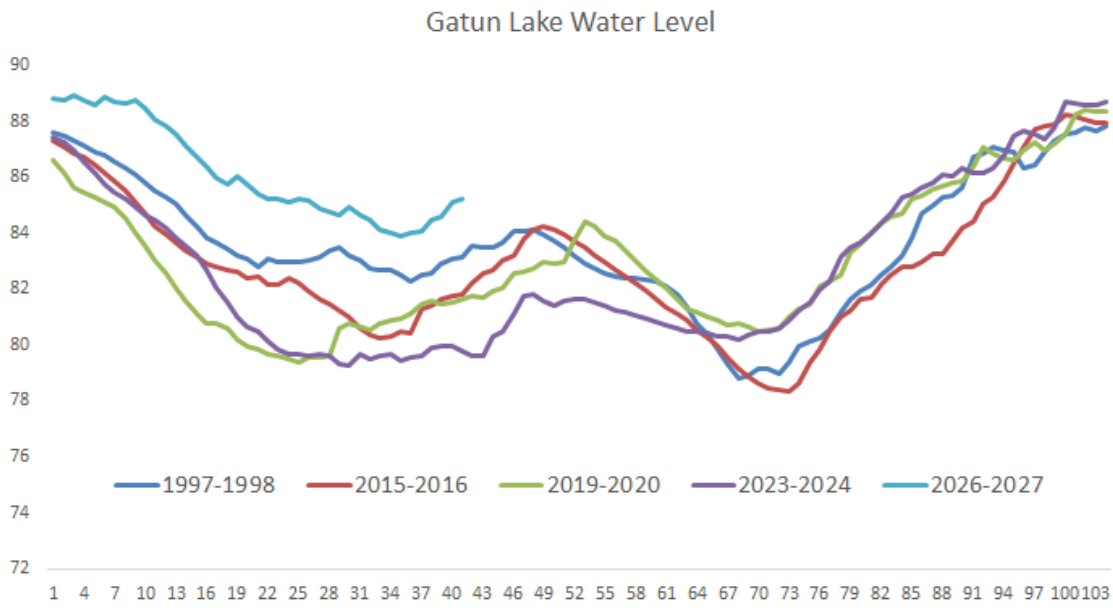
圖 9：美國 2025-2026 年各月進口貨物之貨櫃數及預估量比較圖



資料來源：NRF、元大投顧整理

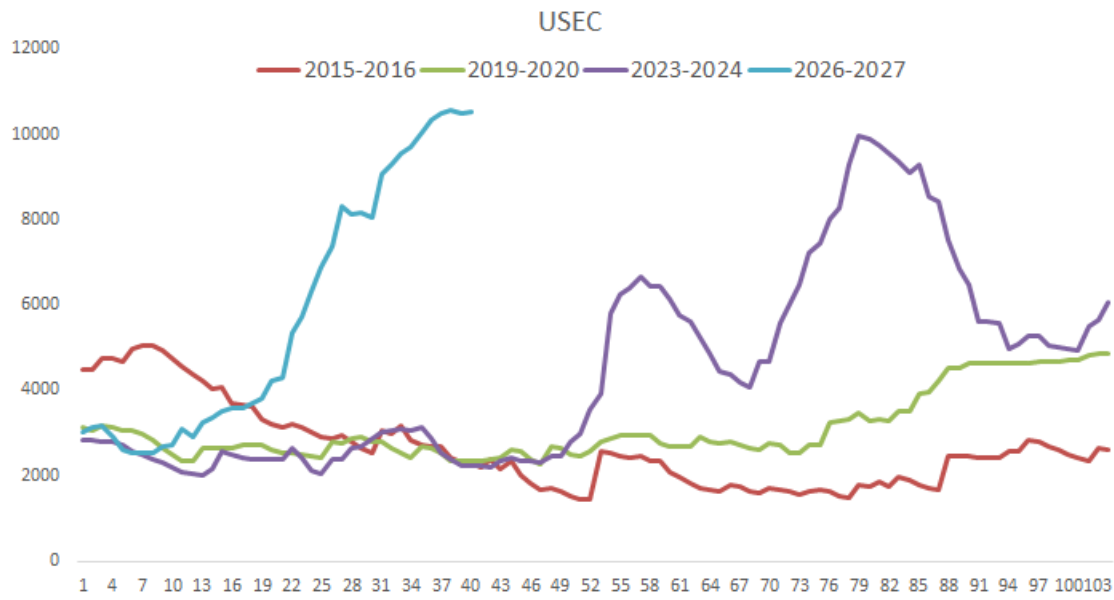
2)極端氣候議題，研調機構判斷 2026 年將是強聖嬰年，2026 年中全球氣候持續處於異常狀態，亞太地區颱風頻傳，但歐洲及中美洲則呈現乾旱少雨，造成歐洲萊茵河以及巴拿馬運河的加通湖水位偏低，造成歐洲依靠萊茵河河運運作停擺，干擾歐洲內陸運輸及歐洲港口的碼頭作業，而巴拿馬運河限航也衝擊美東航線運作。目前針對極端氣候議題，我們整理過去幾次聖嬰效應較明顯年度的加通湖水位的歷史資料，包括 1997-1998 年、2015-2016 年、2019-2020 年、2023-2024 年以及 2026 年至今，當發生聖嬰現象的當年，加通湖水位自當年年中 4 月至 5 月間水位便明顯降低，當年年底水位均曾回升，但隔年年初水位又再次滑落至隔年年中 5 月至 6 月水位達到最低後才逐漸回升，整個水位降低週期影響期間約 12-14 個月，因此根據過去幾次經驗，研判 2026 年這一波聖嬰年造成的加通湖水位降低趨勢將延續到 2027 年 5 月過後才能逐步紓解，亦即巴拿馬限航持續期間可能也將延續至 1H27，這將持續干擾美東線的運力供需情勢。

圖 10：聖嬰現象發生年度之加通湖水位趨勢圖



資料來源：NRF、元大投顧整理

圖 11：聖嬰現象發生年度之美東線運價走勢圖



資料來源：NRF、元大投顧整理

觀察過去幾次聖嬰年度發生時，美東線運價走勢，2015 年及 2019 年兩次經驗，聖嬰現象推動美東線運價大漲的效應不顯著，乃因當時全球貨櫃供需運力長期呈現供過於求的運力過剩格局，因此運價反應平淡，直至 2020 年下半年疫後復甦，需求爆發，運價才開始向上走揚。至於 2023 年的經驗，2023 年效應不顯著，直至年底紅海危機爆發，繞道好望角，全球運力需求增加 9-10%，推動 2024 年運價大幅上漲。至於 2026 年，美東線運價大幅走揚，反映美伊戰爭及美國關稅搶運潮等因素疊加在一起，原本 8 月開始將進入美國線貨櫃載運需求由旺轉淡階段的季節性特徵，因為巴拿滿運河限航，扭轉原本的季節性特徵，近 2 個月美東線運價研續盤堅格局，持續處於 1 萬美元以上的高位。由於聖嬰現象導致中美洲降雨減少，預期加通湖水位 4Q26 至 1H27 仍將處於偏低水準，巴拿馬運河限航恐延續一段時間，有助於支撐美東線運價表現。

3) 關於航商重返紅海航道議題，2024 年紅海危機，航商繞道好望角，航程拉長，運力需求一度因此增加 9-10%，隨著歐系航商於 2026 年中宣布將重返紅海航線，運力需求恐減少，不過基於地緣政治風險考量，目前航商針對歐洲線運作將採取差異化策略，去程載運高價值貨物負擔及高保險費與安全風險，將維持繞道好望角，回程空櫃則走蘇伊士運河及紅海航道，以縮短航程，增加船舶周轉率，因此重返紅海的運力需求影響數推估僅 4-5%左右，衝擊相對可控，低於原先預期。

整體而言，影響貨櫃運價的三大因素，美國零售商及通路商於關稅政策空窗期期間，持續回補庫存，而極端氣候導致中南美洲降雨減少，預期 2027 年 5 月之後加通湖水位才有望回升，預期巴拿馬運河限航將延續至 1H27，而航商基於貨物安全考量，採取去程繞道好望角，回程空櫃則走重返紅海，依此航商重返紅海對運力需求減少的衝擊相對可控，有利運價延續高檔。

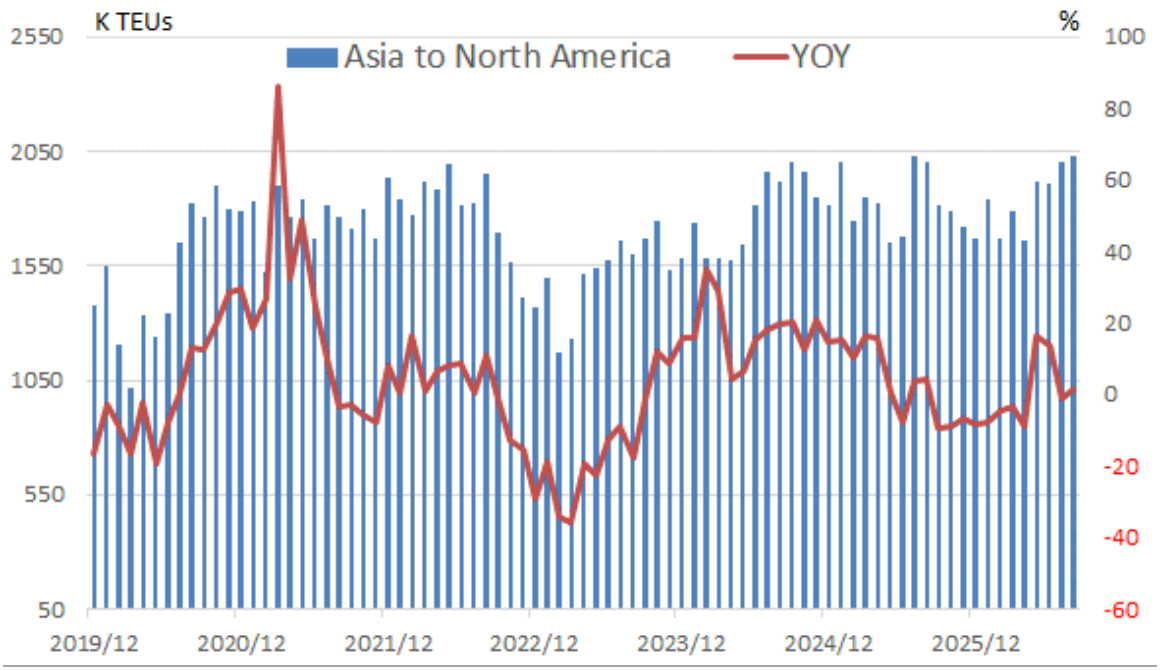
川普關稅政策扭曲 2025 年至 2026 年全球貨櫃運輸季節性及動能表現，惟美國政府關稅退稅，政策不確定性降低

觀察日本海事中心企劃研究部的進出口港貨櫃統計資料，由日本的貨櫃箱裝載量一窺全球貨櫃運輸的表現及淡旺季軌跡。2025 年全年亞洲到美國的貨櫃運輸量 2176.82 萬 TEU，僅年增 1.5%，乃美國關稅政策扭曲全球貿易，也引發通膨預期打壓消費，美國進口量大幅降溫，而關稅政策後遺症延續至 2026 年，2026 年 8 月貨櫃運輸量 203.05 萬 TEU，年增 1.4%，不過 2026 年前 8 個月累積運輸量 1481.53 萬 TEU，年增 0.2%，扭轉原本負成長趨勢，乃 5 月與 6 月期間，美國調降關稅期間截止前的搶運潮湧現，帶動 5 月及 6 月運輸量各自年增 16.2%及 14.0%，維繫整體貨載運輸表現。

至於 2025 年全年亞洲至歐洲的貨櫃運輸量 1983.73 萬 TEU，年增 9.1%，維持 2024 年水準，表現穩健。2026 年 7 月貨櫃運輸量 192.69 萬 TEU，年增 9.4%，2026 年前 7 個月累積運輸量 1271.15 萬 TEU，年增 12.1%，動能優於 2025 年水準，表現強勢，推測乃美中貿易摩擦不斷，中國擴大對歐洲貿易量所致，以去化產能。而亞洲區域內的貨櫃運輸量，2025 年全年貨櫃運輸量 5003.67 萬 TEU，年增 5.5%，動能增溫，乃川普對等關稅出台，各產業陸續將產能從中國往外到亞洲其他國家，供應鏈重組，提振亞洲區域內貿易表現。2026 年 7 月運輸量 471.3 萬 TEU，年增 11.2%，維持穩健，2026 年前 7 個月累積運輸量 3062.54 萬 TEU，年增 8.7%，延續 2025 年動能表現，供應鏈重組效應持續。

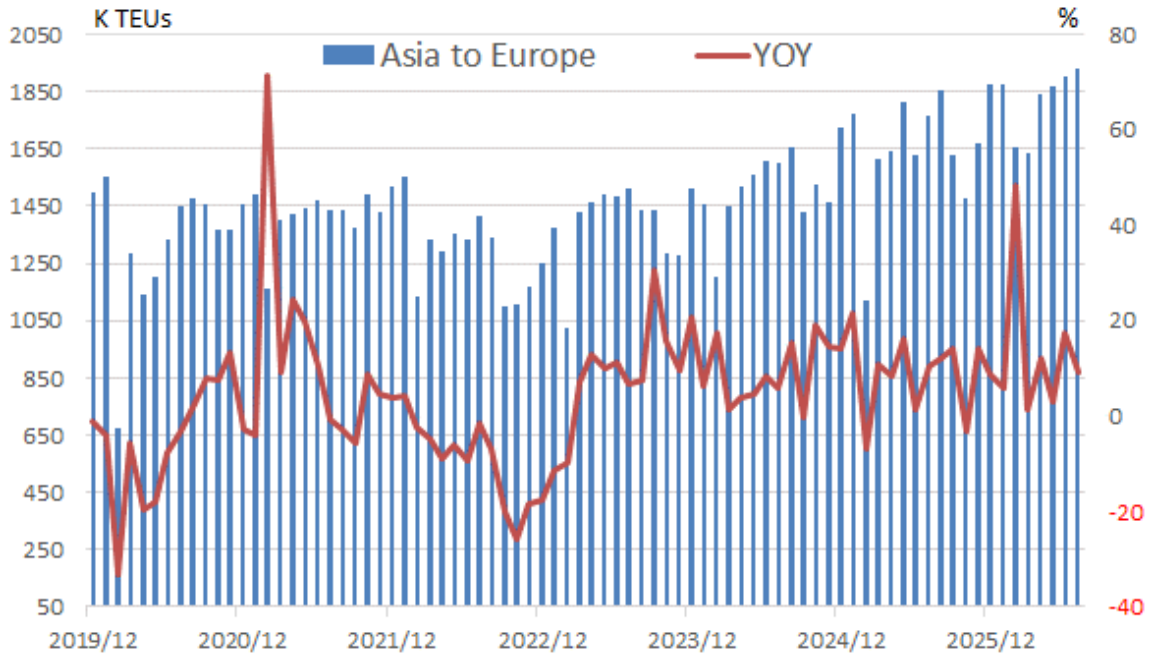
整體而言，川普關稅政策干擾延續，亞洲對美國貨櫃運輸量動能疲軟，至於亞洲到歐洲，以及亞洲區域內貨櫃運輸均表現平穩，支撐貨櫃產業的需求。展望後市，較大隱憂來自川普發動美伊戰爭，油價大漲，推升通膨預期，對於美國消費動能形成打壓效應，不過美國政府的關稅退稅款上看 1600 億美元，美國進口商取得退稅後，採取調降產品終端售價以提振買氣，或是擴大庫存回補，維繫美國商品的進口動能，以及全球貨櫃運輸的貿易量，川普關稅政策不確定性降低，後續則需續觀察美國政府是否再推出其他關稅政策。

圖 12：亞洲到北美的貨櫃運輸量及 YOY



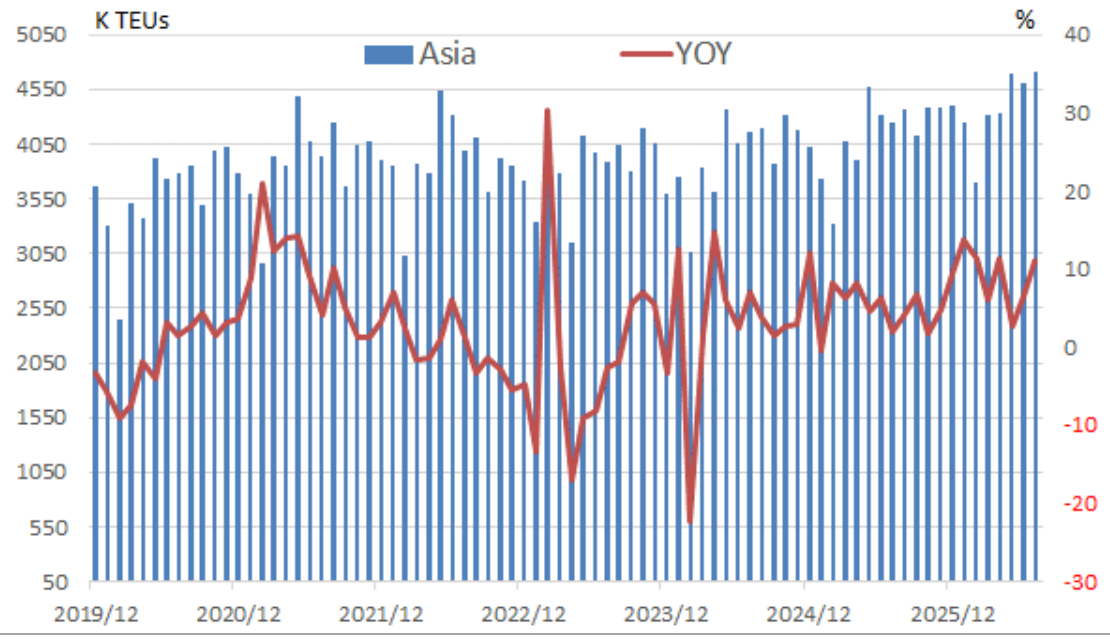
資料來源：元大投顧整理

圖 13：亞洲到歐洲的貨櫃運輸量及 YOY



資料來源：元大投顧整理

圖 14：亞洲區域內貨櫃運輸量及 YOY

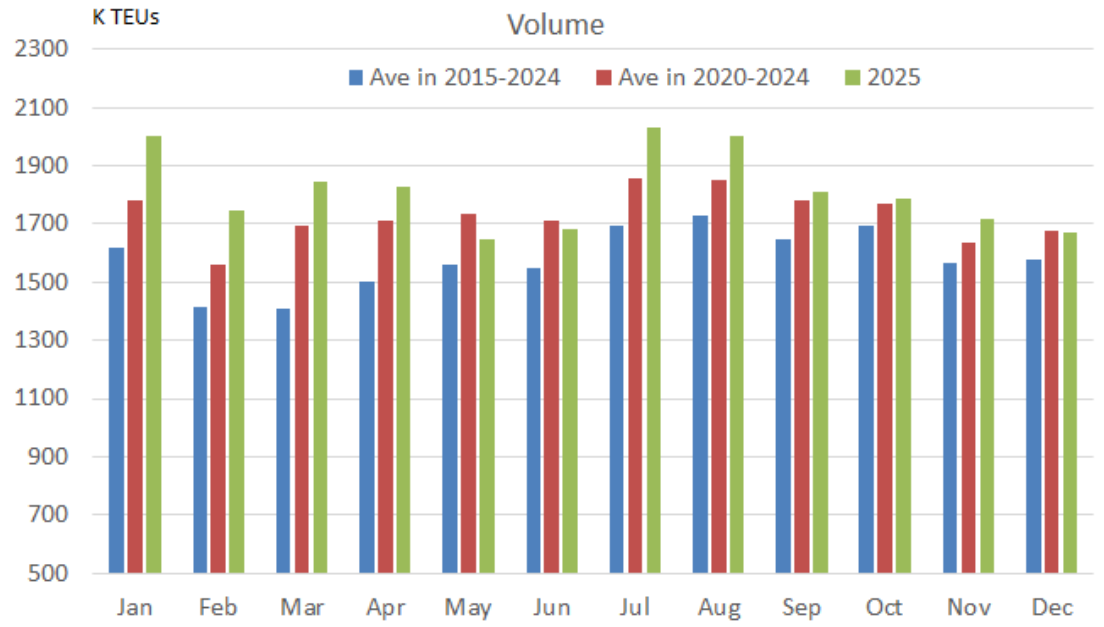


資料來源：元大投顧整理

每年年內各月貨櫃運輸淡旺季顯著，惟關稅政策及搶運潮改變淡旺季節奏

接下來再分析各年度內，每月貨櫃運輸量之表現，取樣資料為 2015 年至 2025 年的每月平均資料，以及 2025 年部分月份資料。亞洲到北美的貨櫃運輸量，在每年 2 月至 3 月期間處於全年低點，是傳統的貨載淡季，乃因中國農曆年期間，工廠處於休長假階段，生產量減少，相對應貨櫃運輸量也減少。而中國農曆年後，隨著工廠陸續恢復生產，產量提高，貨載需求也逐步提升，而因應美國 8 月至 9 月的返校需求，及年底 11 月至 12 月感恩節及耶誕節的消費旺季，美國通路商均需提前備貨因應年底旺季需求，因此貨載需求通常一路走高至 8 月見高點後，才逐步降溫到 11 月。隨著中國農曆年普遍落在 2 月，美國通路商需建置安全庫存，趕在中國農曆年前下單生產備貨，因此每年 12 月到隔年 1 月還有一波貨載出貨旺季。整體而言，每年貨載淡季落在 2 月至 3 月間及 11 月，貨載高峰則落在每年 7 月至 8 月及 12 月至隔年 1 月。

圖 15：亞洲到北美各年度每月貨櫃運輸量淡旺趨勢圖

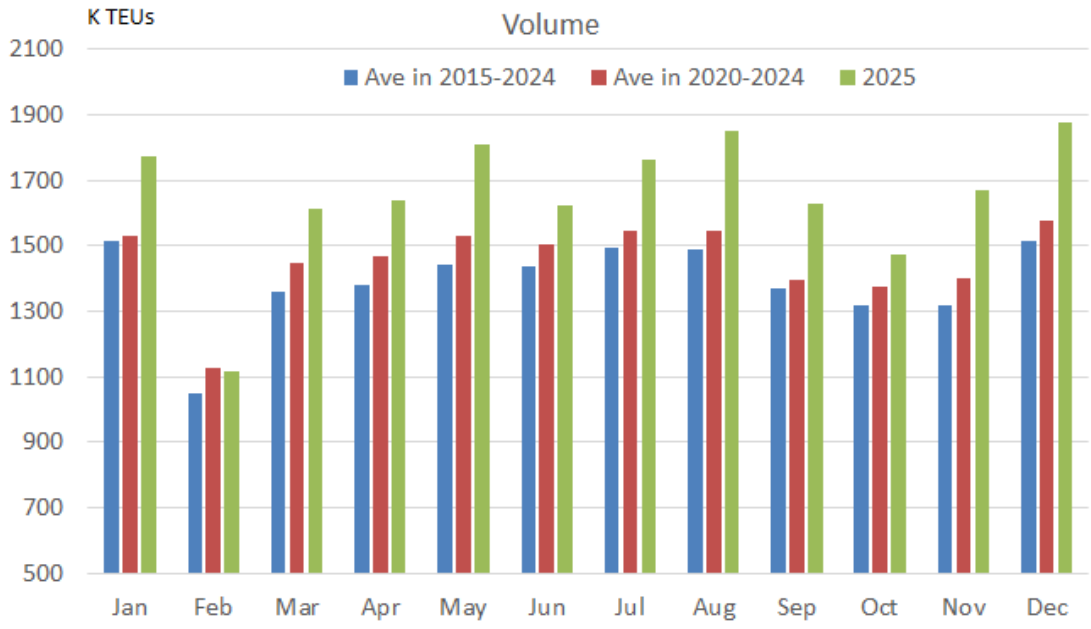


資料來源：元大投顧整理

觀察亞洲到歐洲各年度內，每月貨櫃運輸量之表現，亦有類似亞洲到每週每月貨櫃運輸量的類似特徵，每年 2 月為貨載量最淡的月份，3 月開始貨載量逐步提升至 7 月見高點，8 月開始漸趨滑落至 11 月，年底 12 月至隔年 1 月又有另一波的貨載需求提升。至於亞洲區域內的各年度每月貨載運輸量的淡旺季循環略有差異，每年 2 月一樣是最淡季，而貨載需求於 3 月提高後，約在 5 月就會見到高峰，較歐美的旺季高峰提前約 2-3 個月，之後緩步降溫到每年 9 月，隨後的 10 月 11 月再回升，每年的貨載量淡旺季略與亞洲到北美，亞洲到歐洲有所差異。

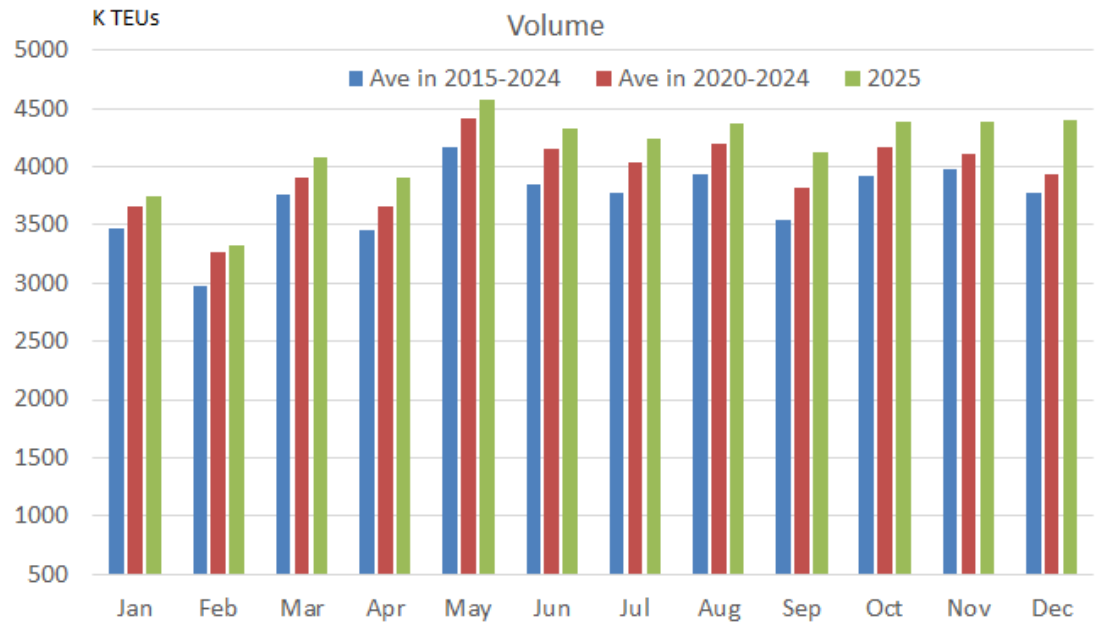
整體而言，觀察過去歷史經驗，北美、歐洲及亞洲區域內等不同地區年內各月貨櫃運輸量淡旺季顯著，不過 2025 年美國關稅政策干擾，引發搶運潮，扭曲全球貿易活動的節奏，打亂原本貨櫃運輸的淡旺季特性。至於 2026 年貨櫃運輸的淡旺季軌跡仍是面臨扭曲狀態，先是美伊戰爭於 2 月底爆發，荷姆茲海峽封鎖，造成船舶滯留波斯灣，衝擊全球運力供給，運價提前於 3 月淡季期間起漲，加上 5 月 6 月美國關稅降低期限將至搶運潮，搭配 5 月下旬至 6 月上旬美國返校需求備貨潮的季節性循環將啟動，提振運價走揚，8 月再次發生聖嬰年，巴拿馬運河缺水限航，影響美國線船舶調度，扭曲原本的貨載量及運價的淡旺季特徵，進一步推升運價。目前觀察貨載量處於由旺轉淡階段，運輸需求將減弱，不過根據歷史經驗，12 月至 2027 年 1 月則會進入中國農曆年前提前備貨的出貨潮，將有季節性旺季需求，至於美國進口商取得政府關稅退稅款後，採取調降產品終端售價以提振買氣，或是擴大庫存回補，維繫美國商品的進口動能，2026 年底至 2027 年初的貨載需求旺季應可期待。

圖 16：亞洲到歐洲各年度每月貨櫃運輸量淡旺趨勢圖



資料來源：元大投顧整理

圖 17：亞洲區域內各年度每月貨櫃運輸量淡旺趨勢圖



資料來源：元大投顧整理

預估 2026 年運力供過於求較 2025 年收斂，2027 年供過於求幅度可控

依據三大研調機構評估 2026 年全球海運市場供需狀況，Clarksons 預估全球運力成長由 2025 年年增 7.1%降至年增 4.6%，需求由年增 4.6%降至年增 3.8%，供過於求由 2.5%收斂至 0.8%，Drewry 預估全球運力成長由 2025 年年增 7.0%降至年增 4.4%，需求由年增 6.5%降至年增 2.1%，供過於求由 0.5%擴大至 2.3%。至於 Alphaliner 預估全球運力成長由 2025 年年增 7.2%降至年增 4.5%，需求由年增 3.5%降至年增 2.5%，供過於求由 3.7%收斂至 2.0%，新增運力供給趨緩，需求仍維持溫和成長，整體供過於求幅度收斂。

展望 2027 年，Clarksons 預估 2027 年全球運力成長將回升至年增 8.8%，需求年增 3.5%，供過於求約 5.3%，而 Drewry 預估 2027 年全球運力成長回升至年增 8.1%，需求年增 5.3%，供過於求約 2.6%，至於 Alphaliner 則尚未發布預測值。整體而言，研調機構預估 2027 年供過於求幅度轉趨惡化，新船下水造成的運力大增是一大隱憂。至於需求端，2024 年紅海危機，航商繞道好望角，航程拉長，運力需求一度因此增加 9-10%，隨著歐系航商宣布將重返紅海航線，運力需求恐減少，都是 2027 年運力供需惡化的原因之一，不過基於地緣政治風險考量，目前航商針對歐洲線運作將採取差異化策略，去程載運高價值貨物負擔及高保險費與安全風險，將維持繞道好望角，回程空櫃則走蘇伊士運河及紅海航道，以縮短航程，增加船舶周轉率，因此重返紅海的運力需求影響數推估僅 4-5%左右，衝擊低於原先預期。另一方面，全球碼頭基礎建設投資金額過去 10 年複合成長率僅 2.7%，低於全球運力複合成長略 5.4%，導致全球碼頭作業負擔量提升，加上極端氣候造成颱風與乾旱頻傳，影響碼頭作業效率，導致主要貨櫃吞吐港塞港嚴重，導致有效運力損失超過 300 萬 TEU，佔全球運力供給約 10%，可舒緩新船下水新增運力大增之衝擊，預期 2027 年運力供過於求幅度有機會收斂。

圖 18：全球 2026 年至 2027 年運力供給與需求成長預估

Institute	YOY	2023	2024	2025	2026F	2027F
Alphaliner	Supply	8.10%	10.30%	7.20%	4.50%	
	Demand	-0.30%	4.50%	3.50%	2.50%	
Clarksons	Supply	8.00%	10.10%	7.10%	4.60%	8.80%
	Demand	1.60%	6.20%	4.60%	3.80%	3.50%
Drewry	Supply	8.10%	10.80%	7.00%	4.40%	8.10%
	Demand	0.70%	7.70%	6.50%	2.10%	5.30%

資料來源：Alphaliner、Clarksons、Drewry、長榮、元大投顧整理

對貨櫃產業看法轉為正向，運價可望維持高檔，2027 年營運向上

對貨櫃產業及運價看法轉向正面，原因包括:1)美國零售商及通路商拿到川普政府退稅款，採取擴大採購策略，於關稅政策空窗期期間，持續回補庫存，以規避未來的關稅不確定因素，美國貨櫃進口需求持續增溫。2)強聖嬰年的極端氣候導致中美洲降雨減少，根據過往幾次聖嬰年的經驗，加通湖水位降低週期影響期間約 12-14 個月，預期 2027 年 5 月之後加通湖水位才有望回升，預期巴拿馬運河限航將延續至 1H27，可望支撐 1H27 美國線運價維持高檔。3)航商基於貨物安全考量，對於歐洲線航行策略，採取去程繞道好望角，回程空櫃則走紅海航道，因此航商重返紅海對運力需求減少的衝擊相對可控。4)受惠 2Q26 至 3Q26 運價大漲，9 月歐洲線 9 月平均運價年增 121%，美西線年增 287%，美東線年增 270%，SCFI 運價指數 9 月均價年增 184%，遠高於 2025 年同期，預期 2027 年新的歐洲線及美國線長約價也將遠高於 2026 年同期，有利貨櫃航商 2027 年營運及獲利表現。後續觀察重點包括美伊戰爭發展及和談進度，美國是否再推出其他關稅政策等。

獲利調整與股票評價

巴拿馬運河限航維繫運價上漲，殖利率達 9%，極具吸引力

受惠美伊戰爭爆發造成荷姆茲海峽遭封鎖，引發中東港口運輸中斷，全球貨櫃運輸調度混亂，以及美國將各國關稅調整期截止前的搶出貨潮湧現，推動運價大漲。而 3Q26 巴拿馬運河限航，亞洲颱風密集及歐洲碼頭工人罷工等事件，扭轉 3Q26 貨載需求由旺轉淡的季節性特徵，運價走勢強勁，挹注長榮營運向上，預估 2026 年營收上修 25.1%至 4654.1 億元(原估 3720.25 億元)，年增 22.8%，毛利率上修 10.3ppt 至 30.2%(原估 19.9%)，營益率上修 10.6ppt 至 25.6%(原估 15.0%)，營業利益上修 112.6%至 1189.03 億元(原估 559.31 億元)，年增 60.4%，稅後 EPS 上修 101.9%至 43.53 元(原估 21.56 元)，年增 37.4%，年底淨值 287.95 元(原估 265.98 元)。

展望 2027 年，受惠 2Q26 至 3Q26 運價大漲，歐洲線 2026 年 9 月平均運價為 2481 美元，年增 121%，美西線 9 月平均運價 7400 美元，年增 287%，美東線 9 月平均運價為 10470 美元，年增 270%，整體 SCFI 運價指數 9 月均價為 3656 點，年增 184%。由於運價水準遠高於 2025 年同期，預期 2026 年底洽談 2027 年新的歐洲線長約價將遠高於 2026 年，而極端氣候的聖嬰現象恐延續至 1H27，巴拿馬運河限航持續，維繫美國線運價表現，且美伊戰爭變數，全球貨櫃運輸調度干擾仍在，諸多因素拉扯有利支撐運價，對比 2Q26 美西線運價區間 2000-3500 美元，美東線 2800-4500 美元的相對低檔水準，推估 1H27 運價水準有望較 2Q26 大幅年增，預期 2027 年 5 月前議定的美國線新的長約價將可高於 2026 年水準，有利 2027 年營運表現。預估 2027 年營收 4993.41 億元，年增 7.3%，毛利率 33.1%，年增 2.9ppt，營益率 28.4%，年增 2.8ppt，營業利益 1419.01 億元，年增 19.3%，稅後 EPS 為 51.04 元，年增 17.2%，年底淨值 317.22 元。

若以 50%股利配發率及 2026 年稅後 EPS 約 43.53 元計算，推估 2027 年長榮將配息 21.76 元，以目前股價計算，殖利率達 9.0%以上，極具吸引力。

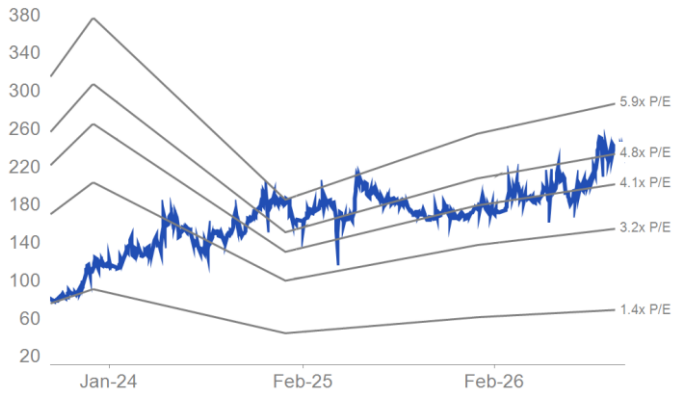
對長榮看法轉趨正向，投資建議由持有調升至買進，目標價 290 元，乃考量:1)受惠美伊戰爭、關稅調整期截止前搶運潮，以及極端氣候干擾，亞洲颱風頻繁與巴拿馬運河限航等因素，2Q26 至 3Q26 運價大漲，長榮 9 月營收 497.11 億元，月增 1.5%，年增 65.0%，動能續揚，累積 3Q26 營收 1467.76 億元，季增 39.6%，年增 51.4%，優於預期，預估 3Q26 稅後 EPS 上修 177.1%至 19.43 元，季增 162.4%，年增 93.4%，獲利動能強勁。2)4Q26 淡季不淡，預估營收上修 38.6%至 1269.61 億元，年增 48.2%，稅後 EPS 上修 188.3%至 12.86 元，年增 226.8%，獲利延續強勁格局。3) 聖嬰現象導致中美洲降雨減少，預期 2027 年 5 月之後加通湖水位才有望回升，預期巴拿馬運河限航將延續至 1H27，美國線運價維持強勢，美國線長約價可往走高，有利 2027 年營運表現，預估 2027 年營收 4993.41 億元，年增 7.3%，稅後 EPS 由 2026 年 43.53 元續增至 51.04 元，年增 17.2%，年底淨值 317.22 元。4) 推估 2027 年長榮將配息 21.76 元，以目前股價計算，殖利率達 9.0%以上，極具吸引力等理由，對長榮投資建議由持有調升至買進，目標價 290 元(約 0.9 倍 PBR X 2027 年淨值 317.22 元)。

圖 19：2026 和 2027 年財務預估調整

(百萬元)	2026 估		2027 估		預估差異	
	調整後	調整前	調整後	調整前	2026	2027
營業收入	465,410	372,025	499,341	360,170	25.1%	38.6%
營業毛利	140,617	73,830	165,131	58,122	90.5%	184.1%
營業利益	118,903	55,931	141,901	41,102	112.6%	245.2%
稅前利益	131,785	66,818	156,936	51,590	97.2%	204.2%
稅後淨利	94,241	46,678	110,497	36,068	101.9%	206.4%
調整後 EPS (元)	43.53	21.56	51.04	16.66	101.9%	206.4%
重要比率 (%)					百分點	百分點
營業毛利率	30.2%	19.9%	33.1%	16.1%	10.3	16.9
營業利益率	25.6%	15.0%	28.4%	11.4%	10.6	17.0
稅後純益率	20.2%	12.5%	22.1%	10.0%	7.7	12.1

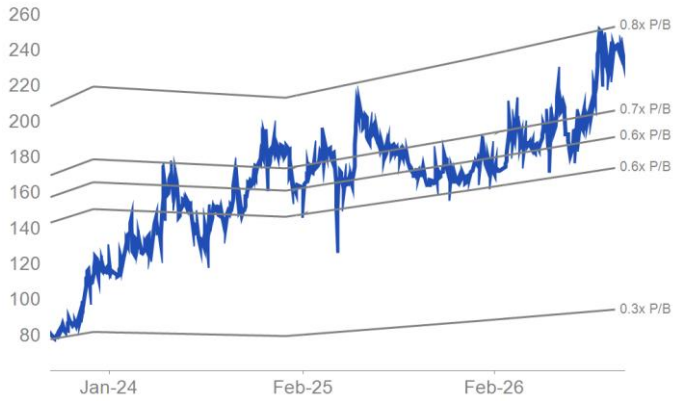
資料來源：公司資料、元大投顧預估

圖 20：12 個月預期本益比區間圖



資料來源：公司資料、元大投顧

圖 21：12 個月預期股價淨值比區間圖



資料來源：公司資料、元大投顧、CMoney

圖 22：同業評價比較表

公司	代碼	評等	股價	市值 (百萬美元)	調整後每股盈餘			本益比(倍)			調整後每股盈餘成長率(%)		
					2025	2026	2027	2025	2026	2027	2025	2026	2027
長榮	**2603 TT**	**買進**	**232.5**	**15,730**	**31.68**	**43.53**	**51.04**	**7.3**	**5.3**	**4.6**	**(50.8)**	**37.4**	**17.2**
國際同業													
Maersk	MAERSKB DC	未評等	24540.0	52,854	1138.1	2892.0	818.0	21.6	8.5	30.0	(59.2)	154.1	(71.7)
COSCO	1919 HK	未評等	16.7	36,308	2.2	2.3	1.4	7.6	7.2	11.9	(32.7)	5.7	(39.6)
Hapag Lloyd	HLAG GR	未評等	134.4	26,445	5.2	5.2	0	26.0	25.7	--	(61.9)	1.2	--
Kawasaki Kisen Kaisha	9107 JP	未評等	3302.0	12,181	460.1	178.5	275.5	7.2	18.5	12.0	216.8	(61.2)	54.3
Mitsui O.S.K. Lines	9104 JP	未評等	7138.0	16,207	1186.6	591.9	800.1	6.0	12.1	8.9	64.2	(50.1)	35.2
Nippon Yusen KK	9101 JP	未評等	7235.0	18,182	1070.3	510.9	683.5	6.8	14.2	10.6	128.6	(52.3)	33.8
國際同業平均					**643.8**	**696.8**	**429.7**	**12.5**	**14.3**	**14.7**	**42.6**	**(0.4)**	**2.4**
國內同業													
陽明海運	2609 TT	未評等	59.7	6,495	4.9	8.3	3.7	12.2	7.2	16.0	(73.4)	68.8	(54.9)
萬海航運	2615 TT	未評等	111.5	9,846	11.2	15.1	11.4	9.9	7.4	9.8	(33.6)	34.8	(24.6)
國內同業平均					**8.1**	**11.7**	**7.6**	**11.1**	**7.3**	**12.9**	**(53.5)**	**51.8**	**(39.8)**

資料來源：公司資料、元大投顧、Reuters；每股盈餘數字以當地貨幣為單位；股價依首頁收盤價日期為準。

圖 23：同業評價比較表 (續)

公司	代碼	評等	股價	市值 (百萬美元)	股東權益報酬率(%)			每股淨值			股價淨值比(倍)		
					2025	2026	2027	2025	2026	2027	2025	2026	2027
長榮	**2603 TT**	**買進**	**232.5**	**15,730**	**11.8**	**15.5**	**16.6**	**260.42**	**287.95**	**317.22**	**0.9**	**0.8**	**0.7**
國際同業													
Maersk	MAERSKB DC	未評等	24540.0	52,854	4.8	9.6	2.2	23214.4	27334.1	27066.9	1.1	0.9	0.9
COSCO	1919 HK	未評等	16.7	36,308	13.2	13.8	8.8	16.6	18.3	19.3	1.0	0.9	0.9
Hapag Lloyd	HLAG GR	未評等	134.4	26,445	4.7	4.9	(1.1)	102.0	102.0	100.0	1.3	1.3	1.3
Kawasaki Kisen Kaisha	9107 JP	未評等	3302.0	12,181	18.8	6.8	9.0	2483.7	2699.4	2966.6	1.3	1.2	1.1
Mitsui O.S.K. Lines	9104 JP	未評等	7138.0	16,207	21.4	7.6	9.0	7492.5	7864.3	8762.5	1.0	0.9	0.8
Nippon Yusen KK	9101 JP	未評等	7235.0	18,182	17.2	7.2	8.9	6539.8	7151.5	7820.4	1.1	1.0	0.9
國際同業平均					**13.4**	**8.3**	**6.1**	**6641.5**	**7528.3**	**7789.3**	**1.1**	**1.0**	**1.0**
國內同業													
陽明海運	2609 TT	未評等	59.7	6,495	5.1	6.2	3.2	93.3	99.9	106.0	0.6	0.6	0.6
萬海航運	2615 TT	未評等	111.5	9,846	12.2	8.9	9.5	98.1	116.5	--	1.1	1.0	--
國內同業平均					**8.7**	**7.5**	**6.4**	**95.7**	**108.2**	**106.0**	**0.9**	**0.8**	**0.6**

資料來源：公司資料、元大投顧、Reuters；每股淨值數字以當地貨幣為單位；股價依首頁收盤價日期為準。

圖 24：季度及年度簡明損益表 (合併)

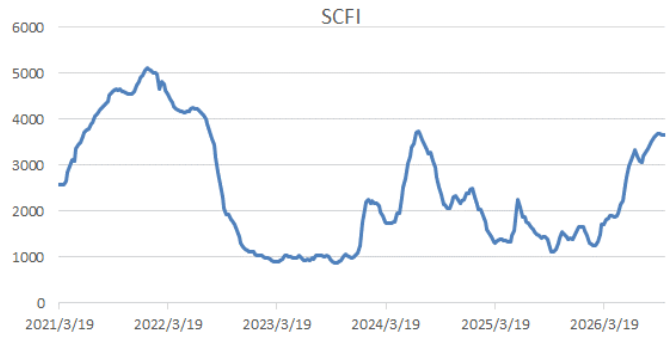
(NT$百萬元)	1Q2026A	2Q2026A	3Q2026F	4Q2026F	1Q2027F	2Q2027F	3Q2027F	4Q2027F	FY2026F	FY2027F
營業收入	86,511	105,161	146,776	126,961	128,866	120,489	129,526	120,459	465,410	499,341
銷貨成本	(73,125)	(81,183)	(87,332)	(83,160)	(83,763)	(83,740)	(84,192)	(82,515)	(324,793)	(334,210)
營業毛利	13,387	23,978	59,444	43,802	45,103	36,749	45,334	37,945	140,617	165,131
營業費用	(4,598)	(4,756)	(5,239)	(9,355)	(5,799)	(5,422)	(6,152)	(9,155)	(23,948)	(26,528)
營業利益	8,948	19,358	55,207	35,390	40,145	32,090	40,051	29,616	118,903	141,901
業外利益	1,691	1,507	3,831	5,853	3,197	3,151	3,096	5,590	12,882	15,035
稅前純益	10,639	20,864	59,038	41,243	43,342	35,241	43,147	35,206	131,785	156,936
所得稅費用	(1,949)	(4,522)	(14,760)	(10,311)	(10,836)	(8,810)	(10,787)	(8,802)	(31,541)	(39,234)
少數股東權益	387	309	0	0	0	0	0	0	6,003	7,205
歸屬母公司稅後純益	8,304	16,034	42,065	27,839	30,881	25,109	30,742	23,764	94,241	110,497
調整後每股盈餘(NT$)	3.84	7.41	19.43	12.86	14.26	11.60	14.20	10.98	43.53	51.04
調整後加權平均股數(百萬股)	2,165	2,165	2,165	2,165	2,165	2,165	2,165	2,165	2,165	2,165
重要比率										
營業毛利率	15.5%	22.8%	40.5%	34.5%	35.0%	30.5%	35.0%	31.5%	30.2%	33.1%
營業利益率	10.3%	18.4%	37.6%	27.9%	31.2%	26.6%	30.9%	24.6%	25.6%	28.4%
稅前純益率	12.3%	19.8%	40.2%	32.5%	33.6%	29.3%	33.3%	29.2%	28.3%	31.4%
稅後純益率	9.6%	15.3%	28.7%	21.9%	24.0%	21.9%	25.0%	21.9%	20.2%	22.1%
有效所得稅率	18.3%	21.7%	25.0%	25.0%	25.0%	25.0%	25.0%	25.0%	23.9%	25.0%
季增率(%)										
營業收入	0.9%	21.6%	39.6%	-13.5%	1.5%	-6.5%	7.5%	-7.0%		
營業利益	7.4%	116.3%	185.2%	-35.9%	13.4%	-20.1%	24.8%	-26.1%		
稅後純益	-2.5%	93.1%	162.4%	-33.8%	10.9%	-18.7%	22.4%	-22.7%		
調整後每股盈餘	-2.5%	93.1%	162.4%	-33.8%	10.9%	-18.7%	22.4%	-22.7%		
年增率(%)										
營業收入	-21.3%	21.6%	51.4%	48.2%	49.0%	14.6%	-11.8%	-5.1%	22.8%	7.3%
營業利益	-69.5%	34.2%	150.6%	324.9%	348.6%	65.8%	-27.5%	-16.3%	60.4%	19.3%
稅後純益	-69.7%	46.3%	93.4%	226.8%	271.9%	56.6%	-26.9%	-14.6%	37.4%	17.2%
調整後每股盈餘	-69.6%	46.3%	93.4%	226.8%	271.9%	56.6%	-26.9%	-14.6%	37.4%	17.2%

資料來源：公司資料、元大投顧、CMoney；標"A"為歷史數據；調整後每股盈餘為根據調整後加權平均股數計算。

公司簡介

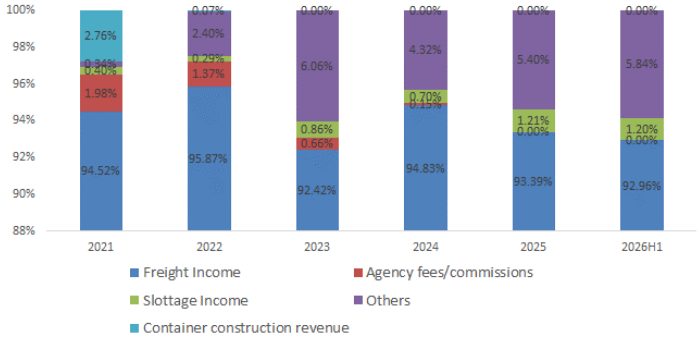
長榮(2603)成立於 1968 年，為台灣最大的貨櫃輪船公司，目前旗下船隊共有 238 艘，合計長榮 2025 年底總運力達到 195.6 萬 TEU，運力年增約 11%，全球運力市佔率約 5.7%，為全球第七大貨櫃航商。依據交船規劃，2026 年底總運力將續增至 207.1 萬 TEU，運力年增約 5.9%，全球運力市佔率可望提升至 5.9%。

圖 25：SCFI 運價指數



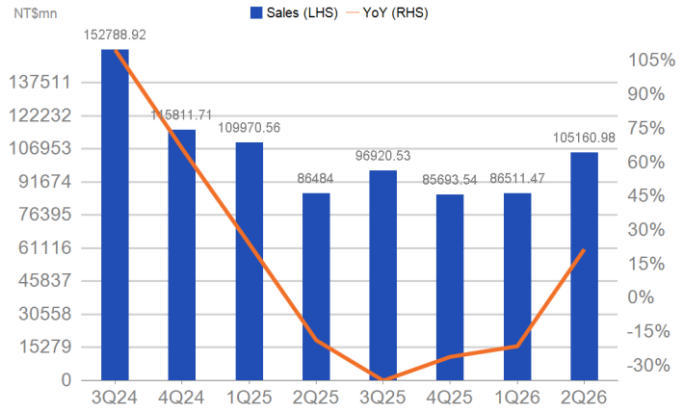
資料來源：公司資料、元大投顧預估

圖 26：營收組成



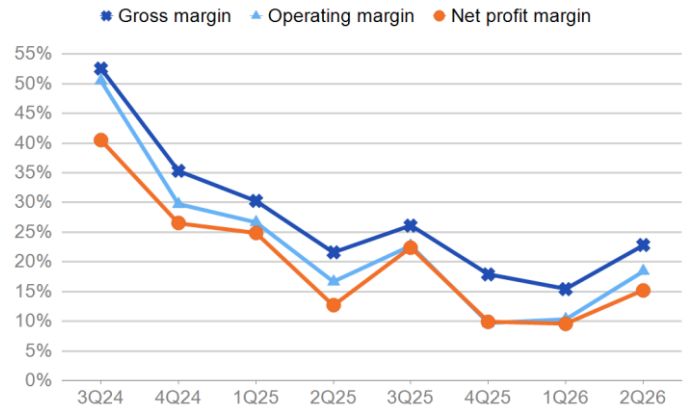
資料來源：公司資料

圖 27：營收趨勢



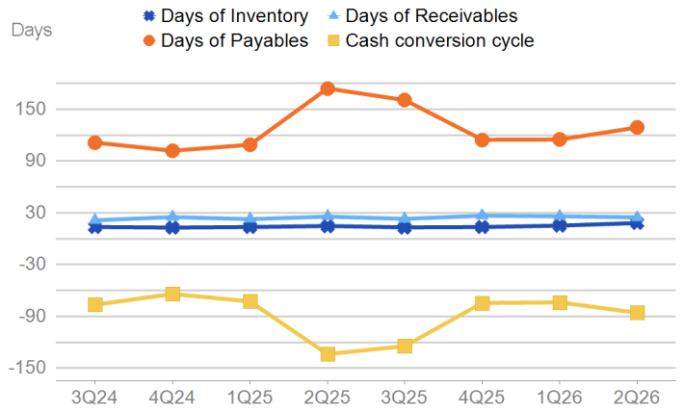
資料來源：CMoney、公司資料

圖 28：毛利率、營益率、淨利率



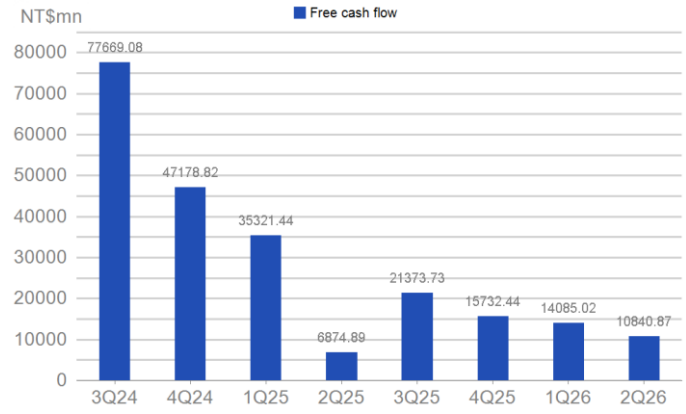
資料來源：CMoney、公司資料

圖 29：經營能力



資料來源：CMoney、公司資料

圖 30：自由現金流量



資料來源：CMoney、公司資料

ESG 分析

分別藉由「在 ESG 議題上的曝險」和「個別公司在 ESG 議題上的執行力」兩個面向來衡量公司於 ESG 上的表現。

- **ESG 總分：** 長榮整體的 ESG 風險評級屬於低風險，於 Sustainalytics 資料庫所覆蓋的公司中排名亦排行於領先的位置，在航運行業中的公司中排名領先同業。
- **在 ESG 議題上的曝險：** 長榮的整體曝險屬於低等水準，且略優於航運行業平均水準。碳（自身運作）、排放、廢水與廢棄物以及職業健康與安全，均屬於顯著的實質 ESG 議題。
- **個別公司在 ESG 議題上的執行力：**長榮在針對重大 ESG 議題的管理水準和執行力屬強勁。該公司憑藉健全的環境政策展現出卓越的 ESG 表現，並遵循高標準的 ESG 報告規範。其 ESG 治理工作由董事會負責監督，同時也建立了完善的舉報機制。

圖 31：ESG 分析

分項	評分/評級
ESG 總分	17.1
在 ESG 議題上的曝險 (A)	31.8
個別公司在 ESG 議題上的執行力 (B)	52.1
風險評級	低 ★
同業排行(1~100，1 為最佳)	13

資料來源：Sustainalytics (2026/10/11)

*註 1：ESG 總分=A-(A*可控風險因子*B/100)*

註 1-1：可控風險因子介於 0-1 之間，越大為越佳。

註 2：ESG 總分風險評級： 極低 (0-10) | 低 (10-20) | 中 (20-30) | 高 (30-40) | 極高 (40+)

註 3：曝險分數評級： 低 (0-35) | 中 (35-55) | 高 (55+)

註 4：執行力分數評級： 弱 (0-25) | 中 (25-50) | 強 (50-100)

資產負債表

年初至 12 月 (NT$ 百萬元)	2023A	2024A	2025A	2026F	2027F
現金與短期投資	170,230	253,370	174,366	201,138	216,516
存貨	10,117	11,018	10,428	11,225	11,798
應收帳款及票據	20,700	28,336	23,567	25,560	26,593
其他流動資產	37,094	20,149	25,368	25,598	27,464
流動資產	**238,140**	**312,872**	**233,728**	**263,521**	**282,371**
採用權益法之投資	38,322	39,789	39,822	44,002	45,002
固定資產	260,244	346,026	398,815	432,831	474,374
無形資產	1,130	894	646	646	646
其他非流動資產	**193,995**	**214,633**	**221,321**	**220,424**	**234,044**
非流動資產	**493,690**	**601,341**	**660,604**	**697,903**	**754,066**
資產總額	**731,831**	**914,214**	**894,332**	**961,423**	**1,036,437**
應付帳款及票據	35,642	44,359	45,804	49,829	51,919
短期借款	0	0	0	0	0
什項負債	76,108	105,986	100,549	100,549	100,549
流動負債	**111,750**	**150,345**	**146,353**	**150,379**	**152,468**
長期借款	31,666	38,137	54,440	53,522	59,921
其他負債及準備	117,620	123,276	107,315	111,698	114,848
長期負債	**149,286**	**161,412**	**161,755**	**165,221**	**174,769**
負債總額	**261,036**	**311,758**	**308,108**	**315,599**	**327,237**
股本	21,273	21,650	21,650	21,650	21,650
資本公積	17,093	20,447	20,345	20,345	20,345
保留盈餘	385,923	504,165	502,181	561,782	625,157
什項權益	15,610	34,066	19,642	19,642	19,642
歸屬母公司之權益	**439,899**	**580,328**	**563,818**	**645,824**	**709,199**
非控制權益	**30,896**	**22,127**	**22,406**	**22,406**	**22,406**
股東權益總額	**470,795**	**602,456**	**586,224**	**645,824**	**709,199**

資料來源：公司資料、元大投顧

現金流量表

年初至 12 月 (NT$ 百萬元)	2023A	2024A	2025A	2026F	2027F
本期純益	39,976	143,984	70,248	100,244	117,702
折舊及攤提	33,864	39,320	42,967	56,200	60,419
本期營運資金變動	(12,561)	(30)	5,807	1,234	484
其他營業資產及負債變動	(75,112)	8,985	(17,820)	5,875	(1,625)
營運活動之現金流量	**(13,833)**	**192,259**	**101,202**	**157,551**	**169,775**
資本支出	(12,405)	(22,232)	(21,900)	(90,217)	(101,962)
本期長期投資變動	(5,327)	1,467	33	(1,000)	(1,000)
其他資產變動	(22,024)	(59,956)	(87,130)	5,875	(1,625)
投資活動之現金流量	**(39,756)**	**(80,721)**	**(108,997)**	**(88,197)**	**(103,624)**
股本變動	109	378	0	0	0
本期負債變動	(2,590)	5,040	20,049	7,491	11,638
現金增減資	0	0	0	0	0
支付現金股利	(148,149)	(21,439)	(70,364)	(34,641)	(47,122)
其他調整數	(17,449)	(16,605)	(15,700)	5,875	(1,625)
融資活動之現金流量	**(168,079)**	**(32,626)**	**(66,015)**	**(42,582)**	**(50,773)**
匯率影響數	**(449)**	**4,229**	**(5,194)**	**0**	**0**
本期產生現金流量	**(222,117)**	**83,140**	**(79,004)**	**26,771**	**15,378**
自由現金流量	**(26,238)**	**170,026**	**79,303**	**67,334**	**67,813**

資料來源：公司資料、元大投顧

損益表

年初至 12 月 (NT$ 百萬元)	2023A	2024A	2025A	2026F	2027F
營業收入	**276,715**	**463,568**	**379,069**	**465,410**	**499,341**
銷貨成本	(226,806)	(287,463)	(286,390)	(324,793)	(334,210)
營業毛利	**49,909**	**176,105**	**92,679**	**140,617**	**165,131**
營業費用	(16,814)	(21,446)	(20,408)	(23,948)	(26,528)
推銷費用	(3,044)	(3,768)	(3,864)	(5,739)	(6,427)
研究費用	0	0	0	0	0
管理費用	(13,760)	(17,693)	(16,453)	(18,179)	(20,102)
其他費用	(10)	15	(91)	(30)	0
營業利益	**34,750**	**159,945**	**74,123**	**118,903**	**141,901**
利息收入	14,379	10,479	8,453	8,453	8,453
利息費用	(4,961)	(5,238)	(5,010)	(7,774)	(8,794)
利息收入淨額	9,418	5,241	3,443	679	(341)
投資利益(損失)淨額	5,981	4,040	4,567	5,354	6,362
匯兌損益	2,152	7,984	(406)	337	0
其他業外收入(支出)淨額	11,870	3,140	1,133	6,512	9,014
稅前純益	64,172	180,351	82,860	131,785	156,936
所得稅費用	(24,196)	(36,366)	(12,613)	(31,541)	(39,234)
少數股權淨利	4,639	4,531	1,667	6,003	7,205
歸屬母公司之稅後純益	**35,337**	**139,453**	**68,581**	**94,241**	**110,497**
稅前息前折舊攤銷前淨利	102,985	224,896	130,805	180,211	208,561
調整後每股盈餘 (NT$)	**16.61**	**64.41**	**31.68**	**43.53**	**51.04**

資料來源：公司資料、元大投顧

主要財務報表分析

年初至 12 月	2023A	2024A	2025A	2026F	2027F
年成長率 (%)					
營業收入	(55.9)	67.5	(18.2)	22.8	7.3
營業利益	(90.7)	360.3	(53.7)	60.4	19.3
稅前息前折舊攤銷前淨利	(76.1)	118.4	(41.8)	37.8	15.7
稅後純益	(88.4)	260.2	(51.2)	37.4	17.2
調整後每股盈餘	(89.5)	287.8	(50.8)	37.4	17.2
獲利能力分析 (%)					
營業毛利率	18.0	38.0	24.5	30.2	33.1
營業利益率	12.6	34.5	19.6	25.6	28.4
稅前息前淨利率	21.4	37.8	20.5	26.7	29.7
稅前息前折舊攤銷前淨利率	37.2	48.5	34.5	38.7	41.8
稅前純益率	23.2	38.9	21.9	28.3	31.4
稅後純益率	12.8	30.1	18.1	20.2	22.1
資產報酬率	4.9	17.5	7.8	10.4	11.4
股東權益報酬率	7.6	26.8	11.8	15.5	16.6
穩定/償債能力分析					
負債權益比 (%)	55.5	51.8	52.6	48.9	46.1
淨負債權益比(%)	(29.4)	(35.7)	(20.5)	(22.9)	(22.1)
利息保障倍數 (倍)	13.9	35.4	17.5	(16.0)	(16.9)
流動比率 (%)	213.1	208.1	159.7	175.2	185.2
速動比率 (%)	202.0	199.2	150.8	167.8	177.5
淨負債 (NT$百萬元)	(138,564)	(215,233)	(119,926)	(147,616)	(156,595)
調整後每股淨值 (NT$)	206.79	268.04	260.42	287.95	317.22
評價指標 (倍)					
本益比	14.0	3.6	7.3	5.3	4.6
股價自由現金流量比	--	3.0	6.3	7.5	7.4
股價淨值比	1.1	0.9	0.9	0.8	0.7
股價稅前息前折舊攤銷前淨	4.8	2.2	3.8	2.8	2.4
股價營收比	1.8	1.1	1.3	1.1	1.0

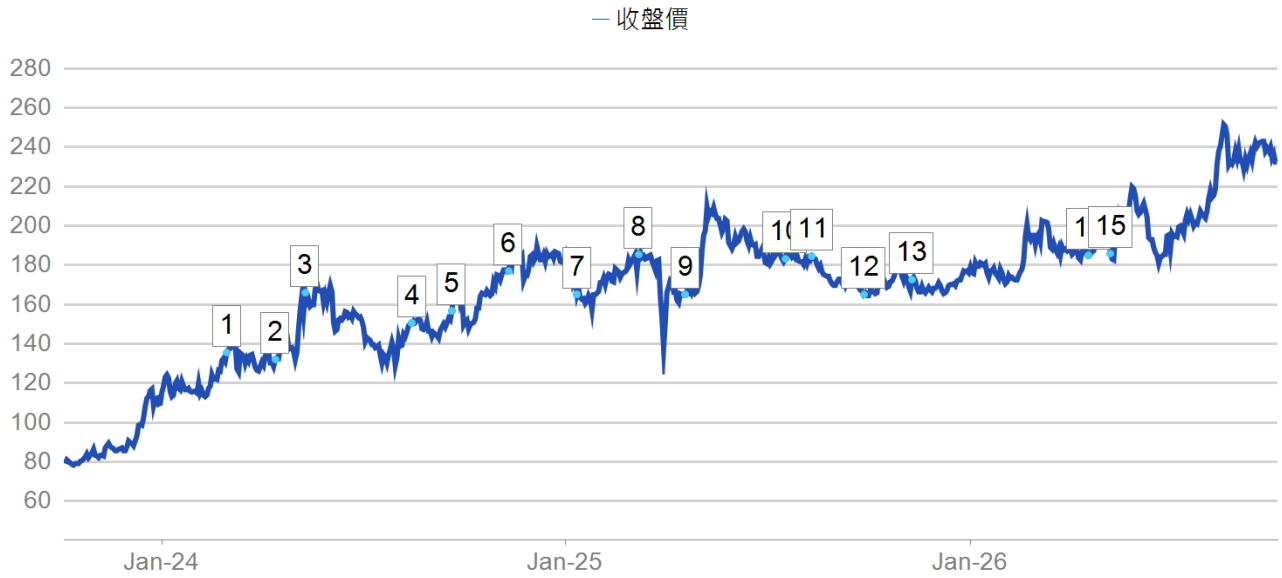
資料來源：公司資料、元大投顧；註：負債為短期債加上長期債。

附錄：重要揭露事項

分析師聲明

主要負責撰寫本研究報告全文或部分內容之分析師，茲針對本報告所載證券或證券發行機構，於此聲明：(1) 文中所述觀點皆準確反映其個人對各證券或證券發行機構之看法；(2) 研究部分析師於本研究報告中所提出之特定投資建議或觀點，與其過去、現在、未來薪酬的任何部份皆無直接或間接關聯。

長榮 (2603 TT)- 投資建議與目標價三年歷史趨勢



#	日期	收盤價 (A)	目標價 (B)	調整後目標價 (C)	評等	分析師
1	20240307	171.50	200	200	買進	蔡爵丞
2	20240419	168.00	200	200	買進	蔡爵丞
3	20240516	203.50	240	240	買進	蔡爵丞
4	20240819	186.00	250	250	買進	蔡爵丞
5	20240923	189.00	250	249	買進	蔡爵丞
6	20241114	219.50	265	265	買進	蔡爵丞
7	20250113	214.00	225	225	持有-超越同業	蔡爵丞
8	20250317	230.50	225	225	持有-超越同業	蔡爵丞
9	20250428	204.50	205	205	持有-超越同業	蔡爵丞
10	20250723	197.50	205	205	持有-超越同業	蔡爵丞
11	20250814	197.00	205	205	持有-超越同業	蔡爵丞
12	20250926	180.00	190	190	持有-超越同業	蔡爵丞
13	20251112	186.00	190	190	持有-超越同業	蔡爵丞
14	20260427	201.50	215	215	持有-超越同業	蔡爵丞
15	20260515	211.50	215	215	持有-超越同業	蔡爵丞

資料來源：CMoney、元大投顧
註：A = 未調整之收盤價；B = 未調整之目標價； C = 依據股本變動調整後之目標價。員工分紅稀釋影響未反映於 A、B 或 C。

投資評等說明

買進：根據本中心對該檔個股投資期間絕對或相對報酬率之預測，我們對該股持正面觀點。此一觀點係基於本中心對該股之發展前景、財務表現、利多題材、評價資訊以及風險概況之分析。建議投資人於投資部位中增持該股。

持有-超越同業：本中心認為根據目前股價，該檔個股基本面吸引力高於同業。此一觀點係基於本中心對該股發展前景、財務表現、利多題材、評價資訊以及風險概況之分析。

持有-落後同業：本中心認為根據目前股價，該檔個股基本面吸引力低於同業。此一觀點係基於本中心對該股發展前景、財務表現、利多題材、評價資訊以及風險概況之分析。

賣出：根據本中心對該檔個股投資期間絕對或相對報酬率之預測，我們對該股持負面觀點。此一觀點係基於本中心對該股之發展前景、財務表現、利多題材、評價資訊以及風險概況之分析。建議投資人於投資部位中減持該股。

評估中：本中心之預估、評等、目標價尚在評估中，但仍積極追蹤該個股。

限制評等：為遵循相關法令規章及/或元大之政策，暫不給予評等及目標價。

註：元大給予個股之目標價係依 12 個月投資期間計算。大中華探索系列報告並無正式之 12 個月目標價，其投資建議乃根據分析師報告中之指定期間分析而得。

總聲明

© 2026 元大版權所有。本報告之內容取材自本公司認可之資料來源，但並不保證其完整性或正確性。報告內容並非任何證券之銷售要約或邀購。報告中所有的意見及預估，皆基於本公司於特定日期所做之判斷，如有變更恕不另行通知。

本報告僅提供一般資訊，文中所載資訊或任何意見，並不構成任何買賣證券或其他投資標的之要約或要約之引誘。報告資料之刊發僅供客戶一般傳閱用途，並非意欲提供專屬之投資建議，亦無考慮任何可能收取本報告之人士的個別財務狀況與目標。對於投資本報告所討論或建議之任何證券、投資標的，或文中所討論或建議之投資策略，投資人應就其是否適合本身而諮詢財務顧問的意見。本報告之內容取材自據信為可靠之資料來源，但概不以明示或默示的方式，對資料之準確性、完整性或正確性作出任何陳述或保證。本報告並非（且不應解釋為）在任何司法管轄區內，任何非依法從事證券經紀或交易之人士或公司，為於該管轄區內從事證券經紀或交易之遊說。

元大研究報告於美國僅發送予美國主要投資法人（依據 1934 年《證券交易法》15a-6 號規則及其修正條文與美國證券交易委員會詮釋定義）。美國投資人若欲進行與本報告所載證券相關之交易，皆必須透過依照 1934 年《證券交易法》第 15 條及其修正條文登記註冊之券商為之。元大研究報告在台灣由元大證券投資顧問股份有限公司發佈，在香港則由元大證券(香港)有限公司發佈。元大證券(香港)係獲香港證券及期貨事務監察委員會核准註冊之券商，並獲許從事受規管活動，包括第 4 類規管活動（就證券提供意見）。非經元大證券(香港)有限公司書面明示同意，本研究報告全文或部份，不得以任何形式或方式轉載、轉寄或揭露。

欲取得任何本報告所載證券詳細資料之台灣人士，應透過下列方式聯絡元大證券投資顧問股份有限公司：

致：聯絡人姓名
元大證券投資顧問股份有限公司
台灣臺北市 106 仁愛路三段 157 號 4 樓

Copyright © 2026 Sustainalytics, a Morningstar company. All rights reserved. The information, data, analyses and opinions contained herein: (1) includes the proprietary information of Sustainalytics and/or its content providers; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product, project, investment strategy or consideration of any particular environmental, social or governance related issues as part of any investment strategy; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. The ESG-related information, methodologies, tools, ratings, data and opinions contained or reflected herein are not directed to or intended for use or distribution to India-based clients or users and their distribution to Indian resident individuals or entities is not permitted. Neither Morningstar Inc., Sustainalytics, Morningstar Asia Ltd. Taiwan Branch nor their content providers accept any liability for the use of the information, for actions of third parties in respect to the information, nor are responsible for any trading decisions, damages or other losses related to the information or its use. The use of the data is subject to conditions available at https://www.sustainalytics.com/legal-disclaimers.

和碩 (4938 TT) Pegatron

AI 伺服器急起直追

買進 (調升評等)

目標價 (12 個月)：NT$116.0

收盤價 (2026/10/08)：NT$89.7
隱含漲幅：29.3%

營收組成 (2Q26)

資訊產品 23%、消費性電子產品 7%、通訊產品 52%、其他 18%。

本次報告更新重點

項目	本次	前次
評等	買進	持有-超越同業
目標價 (NT$)	116	100
2026年營收 (NT$/十億)	1328.8	1141.1
2026年EPS	5.9	4.8

*因應股本變動，調整前次目標價及 EPS 數值

交易資料表

市值	NT$240,030百萬元
外資持股比率	30.8%
董監持股比率	5.9%
調整後每股淨值 (2026F)	NT$76.96
負債比	62.2%
ESG評級 (Sustainalytics)	中 (曝險程度共5級)

簡明損益表 (NT$百萬元)

年初至12月	2024A	2025A	2026F	2027F
營業收入	1,125,349	1,117,160	1,328,848	1,727,754
營業利益	12,501	11,609	25,955	44,820
稅後純益	16,877	14,403	15,871	23,966
EPS (元)	6.34	5.39	5.92	8.95
EPS YoY (%)	7.5	-14.9	9.9	51.0
本益比 (倍)	14.2	16.6	15.1	10.0
股價淨值比 (倍)	1.2	1.2	1.2	1.1
ROE (%)	8.3	7.4	7.7	11.0
現金殖利率 (%)	5.0%	4.5%	4.8%	7.2%
現金股利 (元)	4.50	4.01	4.27	6.44

蘇子錚
Alex.Su@yuanta.com

劉東萌
miketmliu@yuanta.com

元大觀點

- **3Q26 營收優於預期 33%，主因伺服器業務於 9 月開始放量，主要為 GB300 高單價產品。**
- **GB300 訂單將維持強勁態勢至 4Q26，並靜待明年 VR200 產品拉貨，伺服器業務將迎來大成長。**
- **伺服器業務迎爆發期，子公司獲利持續提升，上修 2026/27 年獲利 23%/24%，上調持有至買進，目標價上修至 116 元。**

伺服器大爆發，營收大幅優於預期

和碩 3Q26 營收 3,890 億元，季增 42%、年增 51%，主要受惠於美系客戶的 GB300 產品於 9 月開始拉貨，且不只原先的 NeoCloud 業者，大型業者也開始拉貨，帶動伺服器營收大幅度成長；預估毛利率 4.1%，季減 0.3 個百分點、年減 0.1 個百分點，主因低毛利率的消費性產品營收比重提升，且受到立訊股價大跌影響，預計業外虧損達 6.9 億元，主要來自於金融資產評價減損，EPS 1.22 元，上修 15%。

GB300 伺服器訂單將持續至 4Q26

展望 4Q26，我們認為手機部分，iPhone18 產品開始出貨，低階版本也開始小量產，筆電部分，記憶體價格仍上揚，且筆電價格提升影響終端購買需求，預期出貨量將再次下降，且因成本提升，加工費未成長，毛利率承壓，伺服器部分，美系 NeoCloud 及大型業者持續對 GB300 產品拉貨，並預期 9 月訂單強度將持續。我們認為，儘管消費性產品 4Q26 營收恐衰退，手機營收恐旺季不旺，但在伺服器客戶對於 GB300 需求仍強勁下，營收仍維持成長，預估 4Q26 營收 4,212 億元，季增 8%，EPS 2.45 元，季增 101%。

靜待明年 VR200 持續帶動營運成長

儘管消費性市場仍未起色，但伺服器市場需求仍持續強勁成長，和碩自身的伺服器產品開始出貨，且明年可望取得來自美系大廠 VR200 訂單，帶動伺服器業務大幅成長；另外，永擎及景碩兩子公司營收及獲利持續強勁成長，整體產品組合明顯改善。我們正向看待和碩未來伺服器及其子公司相關伺服器產品的成長動能，上修 2026/27 年獲利 23%/24%，上調評等至買進，目標價上修至 116 元以 2027 年預估每股盈餘及 13 倍本益比推得。

營運分析

伺服器大爆發，營收大幅優於預期

管理層原先預期 3Q26，保守看待筆電市場，手機部分僅 18 Pro 可量產，通訊業務壓力較大，伺服器需求強勁，伺服器營收貢獻將持續成長。整體而言，3Q26 符合管理層的預期，但手機及伺服器業務成長幅度仍大幅度優於管理層預期，尤其是美系大型業者拉貨時程略提前。

和碩 3Q26 營收 3,890 億元，季增 42%、年增 51%，主要受惠於美系客戶的 GB300 產品於 9 月開始拉貨，且不只原先的 NeoCloud 業者，大型業者也開始拉貨，帶動伺服器營收大幅度成長；預估毛利率 4.1%，季減 0.3 個百分點、年減 0.1 個百分點，主因低毛利率的消費性產品營收比重提升，且受到立訊股價大跌影響，預計業外虧損達 6.9 億元，主要來自於金融資產評價減損，EPS 1.22 元，上修 15%。

展望 4Q26，我們認為手機部分，iPhone18 產品開始出貨，低階版本也開始小量產，筆電部分，記憶體價格仍上揚，且筆電價格提升影響終端購買需求，預期出貨量將再次下降，且因成本提升，加工費未成長，毛利率承壓，伺服器部分，美系 NeoCloud 及大型業者持續對 GB300 產品拉貨，並預期 9 月訂單強度將持續。我們認為，儘管消費性產品 4Q26 營收恐衰退，手機營收恐旺季不旺，但在伺服器客戶對於 GB300 需求仍強勁下，營收仍維持成長，預估 4Q26 營收 4,212 億元，季增 8%，EPS 2.45 元，季增 101%。

圖 1：2026 年第 3 季財測與預估比較

(百萬元)	3Q25A	2Q26A	3Q26F	季增率	年增率	3Q26F		預估差異	
						元大預估	市場預估	元大	市場
營業收入	257,863	274,448	389,071	41.8%	50.9%	293,227	303,514	32.7%	28.2%
營業毛利	10,902	12,156	16,133	32.7%	48.0%	13,280	13,798	21.5%	16.9%
營業利益	3,218	4,198	7,983	90.2%	148.1%	5,130	5,025	55.6%	58.9%
稅前利益	7,675	7,744	7,293	-5.8%	-5.0%	6,240	7,604	16.9%	-4.1%
稅後淨利	4,545	4,488	3,264	-27.3%	-28.2%	2,838	1,903	15.0%	71.5%
調整後 EPS (元)	1.69	1.68	1.22	-27.3%	-28.1%	1.06	0.71	15.0%	71.5%
重要比率 (%)				百分點	百分點			百分點	百分點
營業毛利率	4.2%	4.4%	4.1%	-0.3	-0.1	4.5%	4.5%	-0.4	-0.4
營業利益率	1.2%	1.5%	2.1%	0.5	0.8	1.7%	1.7%	0.3	0.4
稅後純益率	1.8%	1.6%	0.8%	-0.8	-0.9	1.0%	0.6%	-0.1	0.2

資料來源：公司資料、元大投顧預估、Bloomberg

圖 2：2026 年第 4 季財測與預估比較

(百萬元)	4Q25A	3Q26F	4Q26F	季增率	年增率	4Q26F		預估差異	
						元大預估	市場預估	元大	市場
營業收入	319,528	389,071	421,224	8.3%	31.8%	329,303	367,107	27.9%	14.7%
營業毛利	13,057	16,133	19,364	20.0%	48.3%	15,359	16,395	26.1%	18.1%
營業利益	4,437	7,983	10,944	37.1%	146.7%	6,939	6,212	57.7%	76.2%
稅前利益	8,859	7,293	11,754	61.2%	32.7%	8,049	9,136	46.0%	28.7%
稅後淨利	5,267	3,264	6,558	100.9%	24.5%	3,976	4,955	64.9%	32.3%
調整後 EPS (元)	1.97	1.22	2.45	100.9%	24.2%	1.48	1.85	64.9%	32.3%
重要比率 (%)				百分點	百分點			百分點	百分點
營業毛利率	4.1%	4.1%	4.6%	0.5	0.5	4.7%	4.5%	-0.1	0.1
營業利益率	1.4%	2.1%	2.6%	0.5	1.2	2.1%	1.7%	0.5	0.9
稅後純益率	1.6%	0.8%	1.6%	0.7	-0.1	1.2%	1.3%	0.3	0.2

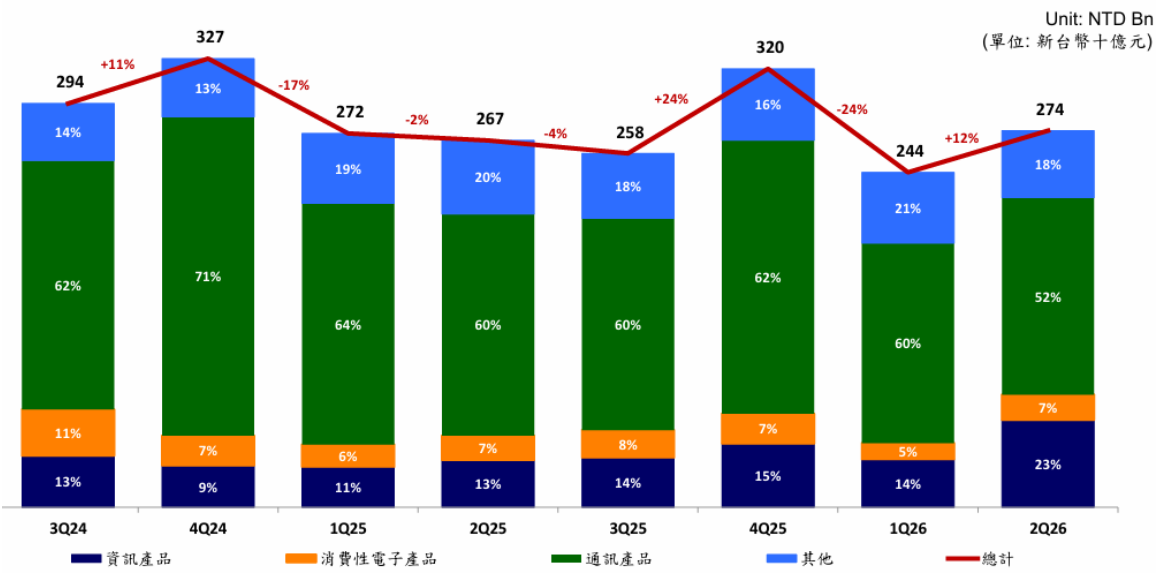
資料來源：公司資料、元大投顧預估、Bloomberg

伺服器業務今明年將迎大成長

和碩的伺服器客戶主要是 CSP 業者及 Neocloud (新興雲端業者)，受惠於今明年大小型業者都將開始大量採購伺服器，不論是 HGX、GB、VR 伺服器都有機會出貨。公司目前為應對 GB300 強勁需求，預計美國德州廠 3 月底開始就會加入生產行列，且 VR200 產品目前已送樣，可望通過客戶認證，並在 2027 年開始量產。

和碩的美系 NeoCloud 及大型業者持續對 GB300 產品拉貨，並預期 9 月訂單強度將持續，10-11 月的單月 GB300 量將小幅優於 9 月。且明年可望取得來自美系大廠 VR200 訂單，帶動伺服器業務大幅成長；另外，永擎及景碩兩子公司營收及獲利持續強勁成長，整體產品組合明顯改善。

圖 3：和碩產品組合



資料來源：公司資料

產業概況

AI、ASIC、通用型強勢出擊，2026 年美系 CSP 資本支出估年增 96%

在 ChatGPT 等 AI 應用陸續推向市場，各大業者紛紛開始研發自有超級大模型，包括 Gemini (Google)、Claude (Anthropic)、Llama (Meta) 等，各大品牌開始大舉投入大型語言模型(LLM)市場，且每一個產品的智慧皆持續提高，部分消費者也陸續從 ChatGPT 陣營轉移至 Gemini 陣營，反映出消費者開始逐漸提升對 AI 的黏著度。根據 Grand View Research 調查，2024 年全球 AI 市場營收約 2,792 億美元，預估 2030 年市場規模超過 3 兆美元，擁有超過 30% 的年複合增長率。

面對這爆發性的長期成長趨勢，CSP 業者(Amazon、Microsoft、Google、Meta)在法說會上一致強調，AI 投資已轉變為一項橫跨 2026 年的結構性戰略，CSP 大廠不僅持續擴大對 GB300 等次世代 AI 伺服器的採購，更將重點放在資料中心基礎設施的長期戰略規劃，以應對 AI 運算對電力及散熱系統強勁成長的需求，確保未來數年內擁有足夠的能源與容量來支持持續迭代的大模型部署。本中心認為，2026 年美國主要 CSP 資本支出將年增 96%，且將有一半的成長來自於資料中心建置，代表了資料中心的不足及未來伺服器需求。明年不僅僅是 NVIDIA GB300 的天下，AWS、Google、Meta ASIC 的崛起也將是一大關鍵，其中又以 Google TPU 需求最為強勁，已經有兩個客戶預計將採購 TPU。

另外，受惠於 AI 訓練與推論需求爆發的連帶效應，本中心預估 2026 年通用型伺服器出貨量將年增 23%，其中以存儲型伺服器需求最為強勁。隨著生成式 AI 應用普及和使用者數量暴增，CSP 等業者必須不斷擴充容量，且不僅限於高階 GPU 伺服器，龐大的 AI 模型訓練和推論在使用狀態中將消耗大量的數據(DATA)，使得存儲型伺服器的需求大增。

圖 4：CSP 資本支出預估

CAPEX (US$mn	2022	2023	2024	2025	2026F	2027F
Amazon	63,645	52,729	82,999	131,819	221,567	281,335
Meta	31,431	27,266	37,256	69,691	139,408	198,568
Microsoft	24,768	35,202	55,552	83,094	158,417	207,930
Google	31,485	32,251	52,535	91,447	201,802	309,988
Oracle	6,678	6,935	10,745	35,477	83,868	94,756
Total	**158,007**	**154,383**	**239,087**	**411,528**	**805,063**	**1,092,578**
YoY (%)	**CY 2022**	**CY 2023**	**CY 2024**	**CY 2025**	**CY 2026**	**CY 2027**
Amazon	58.3%	-17.2%	57.4%	58.8%	68.1%	27.0%
Meta	35.4%	-13.3%	36.6%	87.1%	100.0%	42.4%
Microsoft	-59.4%	42.1%	57.8%	49.6%	90.6%	31.3%
Google	69.0%	2.4%	62.9%	74.1%	120.7%	53.6%
Oracle	97.4%	3.8%	54.9%	230.2%	136.4%	13.0%
Total	**20.3%**	**-2.3%**	**54.9%**	**72.1%**	**95.6%**	**35.7%**

資料來源：元大投顧整理

ASIC 進入高速成長期，GPU 需求仍維持擴張

從 AI 晶片預期出貨來看，GPU 與 ASIC 需求均持續成長，但 2026 年後 ASIC 成長動能明顯加速。Nvidia GPU 出貨量預估由 2025 年 565.7 萬顆增至 2026 年 650 萬顆、2027 年 850 萬顆，年增 17%及 34%；AMD 亦由 2025 年 90 萬顆增至 2027 年 180 萬顆，顯示通用 GPU 算力需求仍維持成長。

相較之下，ASIC 受惠 CSP 加速自研晶片及導入規模擴大，總出貨量預估由 2025 年 380 萬顆大幅增至 2026 年 725 萬顆、2027 年 1,360 萬顆，年增率達 91%及 88%，成長速度明顯高於 GPU。其中 Google 仍為最大 ASIC 使用者，2027 年用量預估達 800 萬顆，AWS 亦增至 320 萬顆，Microsoft 及 Meta 同步擴大部署。我們認為，在資料中心電力逐漸成為算力擴張瓶頸下，CSP 將更重視每瓦運算效能及整體運算成本，ASIC 針對特定運算工作的效率優勢有望推升滲透率，惟 GPU 憑藉通用性及軟體生態系優勢，仍將維持重要地位。

圖 5：GPU/ASIC 出貨量預估

	2023	2024	2025	2026	2027
Nvidia (K units)	1,462	4,056	5,657	6,500	8,500
AMD	200	600	900	1,200	1,800
GPU YoY		180%	41%	17%	34%
MSFT	12	22	0	100	350
Google	1,450	2,050	2,100	4,300	8,000
AWS	650	775	1,600	2,200	3,200
Meta	200	120	100	350	700
Others	0	0	0	300	1,350
Total ASIC (K units)	2,312	2,967	3,800	7,250	13,600
Total ASIC YoY		28%	28%	91%	88%

資料來源：元大投顧

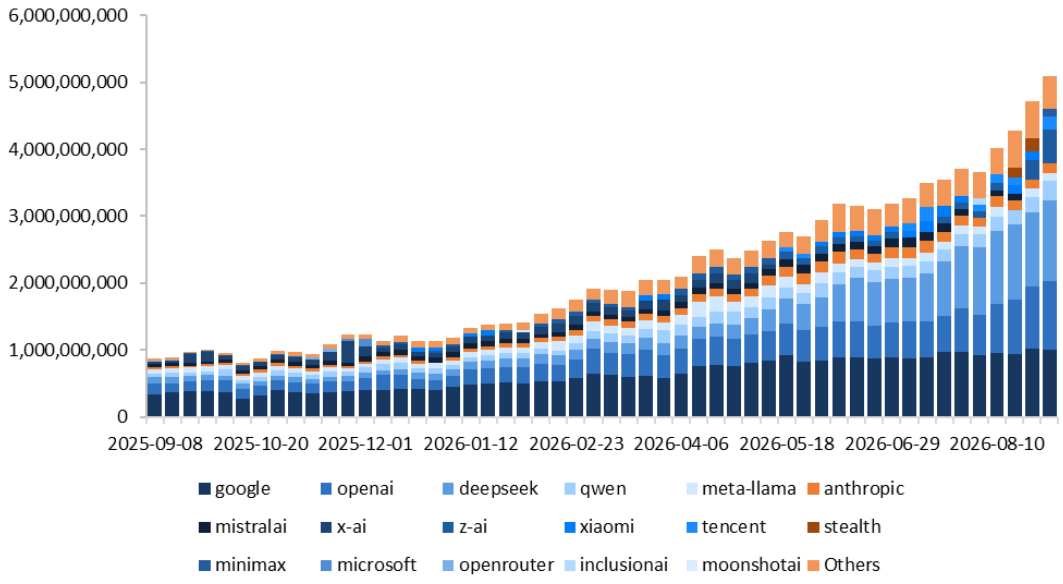
各大模型每週文字請求持續增加，AI 伺服器需求將持續提升

根據 OpenRouter 統計，2025 年 9 月至 2026 年 8 月底，平台內 AI 模型每週文字請求次數由約 8 億次增加至逾 50 億次，年增超過 5 倍，顯示生成式 AI 的實際使用需求仍處於快速成長階段。OpenRouter 主要整合 OpenAI、Google、Anthropic、DeepSeek 及 Qwen 等不同業者的模型，讓使用者可透過單一平台選擇及呼叫多種大型語言模型，雖然 OpenRouter 僅為全球眾多 AI 服務平台之一，但其涵蓋的模型開發商廣泛，可作為觀察 AI 應用活躍度及市場需求變化的重要參考。AI 需求並未因前期高速成長而趨緩，反而隨著模型能力提升、使用成本下降及應用範圍擴大，持續帶動平台請求次數創高。

過去 AI 應用多以簡單問答為主，使用者輸入一個問題，再由模型產生一次回答；隨著 AI Agent 逐步普及，模型開始具備規劃任務、讀取文件、搜尋資料、呼叫工具及執行程式等能力，部分任務還會同時啟動多個子任務，再由主模型彙整結果，即使使用者表面上只提出一次要求，背後實際執行的模型呼叫次數及 Token 消耗量都可能較傳統聊天應用增加數倍。因此，未來 AI 運算需求將同時受到請求次數增加及單次請求 Token 用量提升所推動，實際 Token 需求增幅可能高於圖表所呈現的請求次數成長。

在模型使用頻率與單次運算量同步提高的情況下，全球雲端服務商及大型科技公司仍須持續擴充 AI 伺服器基礎建設。NVIDIA GPU 具備較高的通用性及成熟的軟體生態，能夠較快支援持續更新的模型架構，適合模型種類多元及快速迭代的使用環境；Google TPU 等 ASIC 則是針對特定 AI 運算設計的客製化晶片，當模型架構、運算模式及流量相對穩定，且 Token 規模足以分攤前期軟體最佳化成本時，通常有機會提供較低的每 Token 成本與功耗。由於未來 AI 市場將同時存在快速更新的前沿模型、固定的大規模推論服務及不同企業的客製化應用，GPU 與 ASIC 預期將各自對應不同工作負載，而非由單一架構完全取代另一方。整體 Token 需求持續上升，將推動兩類 AI 伺服器建置同步擴張，並進一步帶動運算晶片、記憶體、高速傳輸、散熱及電源等相關零組件需求。

圖 6：OpenRouter 平台內 AI 模型每週文字請求次數



資料來源：OpenRouter

各 AI 運算需求持續攀升，CSP 與 NeoCloud 加速擴建

Amazon、Google、Meta 及 Microsoft 主要 CSP 目前營運中的 AI 資料中心電力容量合計約 15.18GW，另有 15.71GW 正在建置、及已宣布的 16.08GW，若相關計畫如期落地，整體電力容量可望由目前的 15.18GW 擴大至約 46.97GW，約為現有規模的 3.1 倍。除大型 CSP 持續投入外，CoreWeave、Crusoe、Nebius 及 IREN 等 NeoCloud 業者也積極擴充 AI 基礎設施。NeoCloud 是以 GPU 運算及 AI 雲端服務為主要業務的新型雲端業者，目前營運中的電力容量合計約 7.01GW，加計在建及已宣布項目後，總規模可達約 16.01GW，約為現有規模的 2.3 倍。

從業者的電力容量規劃來看，AI 基礎設施投資仍處於快速擴張階段。4 家主要 CSP 在建及已宣布容量合計達 31.79GW，高於目前已營運容量的 2 倍；NeoCloud 業者後續新增容量也達 9.01GW，顯示 AI 運算需求不僅集中於既有大型雲端服務商，新型 AI 雲端業者也正加快建置速度。隨著生成式 AI 使用量持續增加，以及 AI Agent 帶動單次請求的 Token 消耗、推理步驟與模型呼叫次數提升，資料中心需要配置更多 AI 伺服器以支應新增運算需求。

各業者最終採用的伺服器架構可能因工作負載而異。模型種類多、更新速度快的環境，通常較適合採用 NVIDIA GPU 伺服器；模型架構與運算模式相對固定，且使用規模足夠大的情況下，Google TPU 等 ASIC 伺服器則有機會提供較低的每 Token 成本。無論最終採用 GPU 或 ASIC，電力容量持續擴張均代表 AI 伺服器部署量仍有明顯成長空間，並將同步帶動運算晶片、記憶體、高速傳輸、電源及散熱等相關供應鏈需求。

圖 7：CSP 及 NeoCloud 資料中心電力容量

Company	Category	Operational (GW)	Under Construction (GW)	Announced (GW)	Total (GW)
Amazon / AWS	CSP	12.649	6.596	2.690	21.935
Microsoft	CSP	11.101	0.922	2.350	14.373
Google	CSP	6.937	1.460	4.477	12.874
Oracle	CSP	6.760	2.294	0.500	9.554
Alibaba	CSP	0.880	0.203	0.000	1.083
xAI	NeoCloud	1.925	0.000	0.425	2.350
CoreWeave	NeoCloud	1.767	0.480	0.000	2.247
Crusoe	NeoCloud	0.000	2.200	0.000	2.200
Nebius	NeoCloud	0.175	0.725	1.100	2.000
IREN	NeoCloud	0.900	0.000	0.000	0.900
TeraWulf	NeoCloud	0.102	0.737	0.000	0.839
Nscale	NeoCloud	0.404	0.230	0.000	0.634
Together AI	NeoCloud	0.000	0.000	0.600	0.600
Fluidstack	NeoCloud	0.509	0.000	0.000	0.509
Groq	NeoCloud	0.000	0.000	0.200	0.200
Lambda	NeoCloud	0.000	0.054	0.000	0.054
QumulusAI	NeoCloud	0.008	0.010	0.000	0.018
Vultr	NeoCloud	0.000	0.000	0.000	0.000
CSP Total		**38.327**	**11.475**	**10.017**	**59.819**
NeoCloud Total		**5.790**	**4.436**	**2.325**	**12.551**

資料來源：datacenterindex

獲利調整與股票評價

上調評等至買進，目標價上修至 116 元

和碩持續提升車用及伺服器營收比重，已成為 Nvidia 合作夥伴，可望布局未來 MGX 伺服器產品，並持續推進自家水冷設計技術，GB300 於 3Q25 出貨給客戶認證，預估和碩的伺服器營收爆發成長，並有機會成長 10 倍以上。

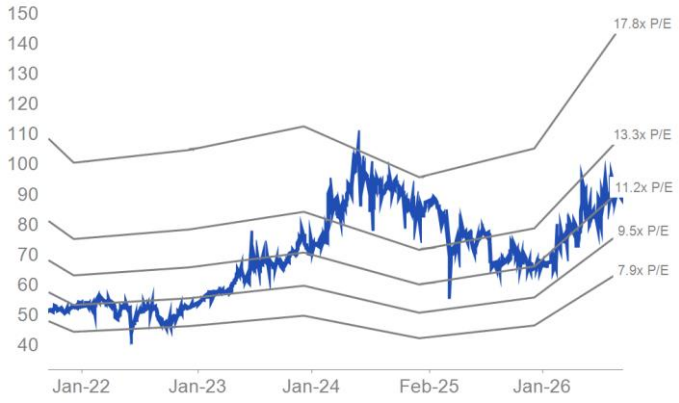
儘管消費性市場仍未起色，但伺服器市場需求仍持續強勁成長，和碩自身的伺服器產品開始出貨，且明年可望取得來自美系大廠 VR200 訂單，帶動伺服器業務大幅成長；另外，永擎及景碩兩子公司營收及獲利持續強勁成長，整體產品組合明顯改善。我們正向看待和碩未來伺服器及其子公司相關伺服器產品的成長動能，上修 2026/27 年獲利 23%/24%，上調評等至買進，目標價上修至 116 元以 2027 年預估每股盈餘及 13 倍本益比推得。

圖 8：2026 和 2027 年財務預估調整

(百萬元)	2026 估		2027 估		預估差異	
	調整後	調整前	調整後	調整前	2026	2027
營業收入	1,328,848	1,141,084	1,727,754	1,336,667	16.5%	29.3%
營業毛利	58,455	51,597	80,580	70,065	13.3%	15.0%
營業利益	25,955	19,096	44,820	35,475	35.9%	26.3%
稅前利益	30,257	25,498	49,160	39,815	18.7%	23.5%
稅後淨利	15,871	12,863	23,966	19,298	23.4%	24.2%
調整後 EPS (元)	5.92	4.80	8.95	7.20	23.4%	24.2%
重要比率 (%)					百分點	百分點
營業毛利率	4.4%	4.5%	4.7%	5.2%	-0.1	-0.6
營業利益率	2.0%	1.7%	2.6%	2.7%	0.3	-0.1
稅後純益率	1.2%	1.1%	1.4%	1.4%	0.1	-0.1

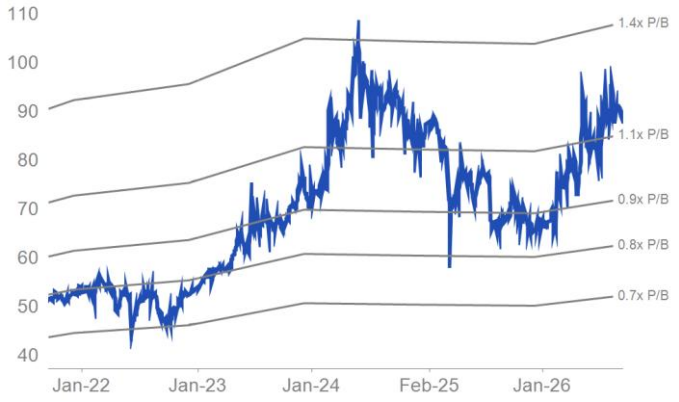
資料來源：公司資料、元大投顧預估

圖 9：12 個月預期本益比區間圖



資料來源：公司資料、元大投顧

圖 10：12 個月預期股價淨值比區間圖



資料來源：公司資料、元大投顧、CMoney

圖 11：同業評價比較表

公司	代碼	評等	股價	市值 (百萬美元)	調整後每股盈餘			本益比(倍)			調整後每股盈餘成長率(%)		
					2025	2026	2027	2025	2026	2027	2025	2026	2027
和碩	**4938 TT**	**買進**	**89.7**	**7,500**	**5.39**	**5.92**	**8.95**	**16.6**	**15.1**	**10.0**	**(14.9)**	**9.9**	**51.0**
國外同業													
立訊精密	002475 CH	未評等	47.6	54,795	2.4	2.8	3.7	20.1	16.8	12.8	27.2	20.0	31.6
歌爾聲學	002241 CH	未評等	21.5	11,418	1.0	1.0	1.2	22.3	20.8	17.3	22.3	7.2	20.0
藍思科技	300433 CH	未評等	28.4	21,904	0.9	0.9	1.3	30.5	33.0	22.0	27.7	(7.7)	50.1
比亞迪電子	285 HK	未評等	85.0	104,461	3.9	4.4	5.7	21.7	19.4	15.0	(15.1)	11.9	29.6
國外同業平均					**2.0**	**2.3**	**3.0**	**23.6**	**22.5**	**16.8**	**15.5**	**7.9**	**32.8**
國內同業													
鴻海	2317 TT	持有-超越同業	249.0	110,032	13.6	18.8	21.2	18.3	13.3	11.8	23.6	38.0	12.7
廣達	2382 TT	買進	325.5	41,975	19.5	26.3	30.3	16.7	12.4	10.7	25.6	35.2	15.3
緯創	3231 TT	未評等	188.5	18,818	9.1	16.9	24.6	20.8	11.2	7.7	48.4	86.0	45.5
仁寶	2324 TT	持有-超越同業	35.5	4,912	1.4	1.9	3.7	25.7	18.4	9.7	(40.0)	39.5	89.6
英業達	2356 TT	持有-超越同業	57.5	6,476	2.3	3.0	--	24.6	19.0	--	15.5	29.1	--
國內同業平均					**9.2**	**13.4**	**19.9**	**21.2**	**14.9**	**10.0**	**14.6**	**45.5**	**40.8**

資料來源：公司資料、元大投顧、Reuters；每股盈餘數字以當地貨幣為單位；股價依首頁收盤價日期為準。

圖 12：同業評價比較表 (續)

公司	代碼	評等	股價	市值 (百萬美元)	股東權益報酬率(%)			每股淨值			股價淨值比(倍)		
					2025	2026	2027	2025	2026	2027	2025	2026	2027
和碩	**4938 TT**	**買進**	**89.7**	**7,500**	**7.4**	**7.7**	**11.0**	**77.30**	**76.96**	**81.15**	**1.2**	**1.2**	**1.1**
國外同業													
立訊精密	002475 CH	未評等	47.6	54,795	19.9	20.1	21.9	12.8	15.0	18.5	3.7	3.2	2.6
歌爾聲學	002241 CH	未評等	21.5	11,418	9.5	9.4	10.1	11.6	14.2	15.2	1.9	1.5	1.4
藍思科技	300433 CH	未評等	28.4	21,904	9.5	8.3	11.5	10.8	11.4	12.2	2.6	2.5	2.3
比亞迪電子	285 HK	未評等	85.0	104,461	16.9	14.7	16.7	25.7	30.3	35.1	3.3	2.8	2.4
國外同業平均					**13.9**	**13.1**	**15.0**	**15.2**	**17.7**	**20.3**	**2.9**	**2.5**	**2.2**
國內同業													
鴻海	2317 TT	持有-超越同業	249.0	110,032	11.3	13.5	14.0	126.6	139.2	151.1	2.0	1.8	1.7
廣達	2382 TT	買進	325.5	41,975	31.2	42.7	42.4	63.1	61.6	71.4	5.2	5.3	4.6
緯創	3231 TT	未評等	188.5	18,818	17.1	25.0	29.7	54.2	75.0	94.2	3.5	2.5	2.0
仁寶	2324 TT	持有-超越同業	35.5	4,912	5.2	6.4	11.2	29.9	30.4	32.7	1.2	1.2	1.1
英業達	2356 TT	持有-超越同業	57.5	6,476	12.7	14.2	--	21.2	21.2	--	2.7	2.7	--
國內同業平均					**15.5**	**20.4**	**24.3**	**59.0**	**65.5**	**87.4**	**2.9**	**2.7**	**2.3**

資料來源：公司資料、元大投顧、Reuters；每股淨值數字以當地貨幣為單位；股價依首頁收盤價日期為準。

圖 13：季度及年度簡明損益表 (合併)

(NT$百萬元)	1Q2026A	2Q2026A	3Q2026F	4Q2026F	1Q2027F	2Q2027F	3Q2027F	4Q2027F	FY2026F	FY2027F
營業收入	244,105	274,448	389,071	421,224	307,303	367,331	496,622	556,498	1,328,848	1,727,754
銷貨成本	(233,304)	(262,292)	(372,938)	(401,859)	(292,947)	(349,436)	(473,317)	(531,474)	(1,270,392)	(1,647,174)
營業毛利	10,801	12,156	16,133	19,364	14,355	17,895	23,305	25,024	58,455	80,580
營業費用	(7,972)	(7,958)	(8,150)	(8,420)	(8,400)	(8,800)	(9,100)	(9,460)	(32,501)	(35,760)
營業利益	2,829	4,198	7,983	10,944	5,955	9,095	14,205	15,564	25,955	44,820
業外利益	636	3,547	(690)	810	1,060	1,060	1,110	1,110	4,302	4,340
稅前純益	3,465	7,744	7,293	11,754	7,015	10,155	15,315	16,674	30,257	49,160
所得稅費用	(605)	(1,431)	(1,605)	(2,586)	(1,543)	(2,234)	(3,369)	(3,668)	(6,227)	(10,815)
少數股東權益	1,298	1,826	2,425	2,611	2,514	3,333	4,113	4,419	8,160	14,378
歸屬母公司稅後純益	1,561	4,488	3,264	6,558	2,958	4,588	7,833	8,587	15,871	23,966
調整後每股盈餘(NT$)	0.58	1.68	1.22	2.45	1.10	1.71	2.92	3.21	5.92	8.95
調整後加權平均股數(百萬股)	2,682	2,679	2,679	2,679	2,679	2,679	2,679	2,679	2,679	2,679
重要比率										
營業毛利率	4.4%	4.4%	4.1%	4.6%	4.7%	4.9%	4.7%	4.5%	4.4%	4.7%
營業利益率	1.2%	1.5%	2.1%	2.6%	1.9%	2.5%	2.9%	2.8%	2.0%	2.6%
稅前純益率	1.4%	2.8%	1.9%	2.8%	2.3%	2.8%	3.1%	3.0%	2.3%	2.9%
稅後純益率	0.6%	1.6%	0.8%	1.6%	1.0%	1.3%	1.6%	1.5%	1.2%	1.4%
有效所得稅率	17.5%	18.5%	22.0%	22.0%	22.0%	22.0%	22.0%	22.0%	20.6%	22.0%
季增率(%)										
營業收入	-23.6%	12.4%	41.8%	8.3%	-27.0%	19.5%	35.2%	12.1%		
營業利益	-36.2%	48.4%	90.2%	37.1%	-45.6%	52.7%	56.2%	9.6%		
稅後純益	-70.3%	187.4%	-27.3%	100.9%	-54.9%	55.1%	70.7%	9.6%		
調整後每股盈餘	-70.5%	187.8%	-27.3%	100.9%	-54.9%	55.1%	70.7%	9.6%		
年增率(%)										
營業收入	-10.4%	2.7%	50.9%	31.8%	25.9%	33.8%	27.6%	32.1%	18.9%	30.0%
營業利益	14.0%	185.4%	148.1%	146.7%	110.5%	116.7%	77.9%	42.2%	123.6%	72.7%
稅後純益	-63.7%	1461.5%	-28.2%	24.5%	89.4%	2.2%	140.0%	30.9%	32.7%	59.6%
調整後每股盈餘	-64.0%	1451.9%	-28.1%	24.2%	89.7%	2.2%	140.0%	30.9%	9.9%	51.0%

資料來源：公司資料、元大投顧、CMoney；標"A"為歷史數據；調整後每股盈餘為根據調整後加權平均股數計算。

公司簡介

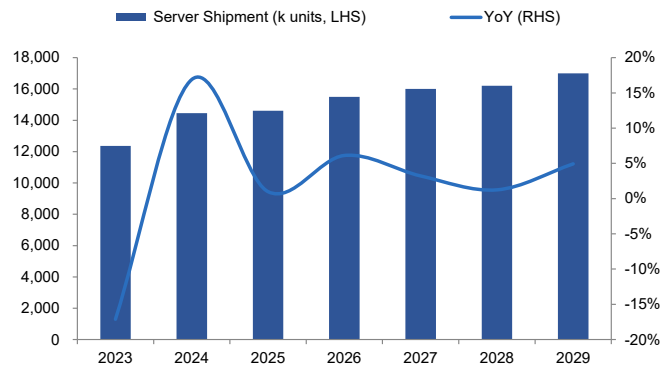
和碩成立於 2007 年，業務包含開發電腦周邊產品、通信技術及消費性電子產品予品牌供應商，從事產品組件的開發、設計和製造，結合 EMS 與 ODM 產業，成為新興 DMS 企業(設計整合服務製造)。主要產品包括手機、筆記型電腦、桌上型電腦、顯示器及車用電子等，營業據點遍布全球，大陸生產廠區包括蘇州、重慶、昆山、上海共四個廠區，台灣則有桃園、新店廠，另有捷克、墨西哥、印尼等廠，此外，北美(墨西哥廠)、印度及越南三座新廠區於 2022 年開始量產。

隨美系手機組裝業務利潤率逐漸降低，和碩近年來積極投入 IoT、AI、5G 等相關技術研發與產品設計進行轉型，相關產品應用裝置技術皆有所提升。和碩長期耕耘的車用電子，也切入美系電動車大廠供應鏈，包括高階 ECU (電子控制元件)、車用平板、機構件、線束、通訊模組等陸續發酵，目前對營收的貢獻仍小，隨著歐洲、美洲、亞洲客戶需求穩定成長，預估未來 5-10 年車用電子產品將保持每年雙位數的成長，成長可期。

圖 14：和碩全球據點

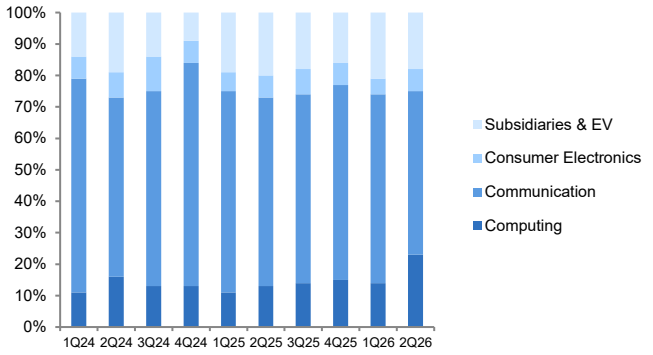
資料來源：公司資料

圖 15：伺服器出貨預估



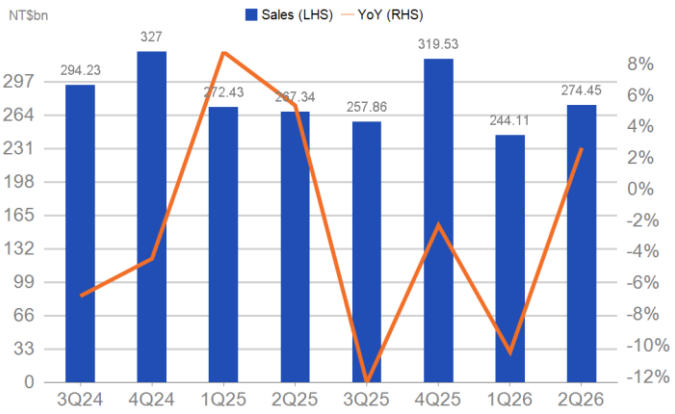
資料來源：公司資料、元大投顧預估

圖 16：營收組成



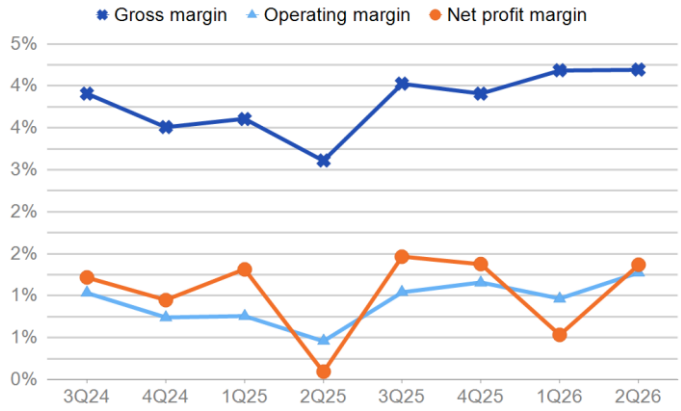
資料來源：公司資料

圖 17：營收趨勢



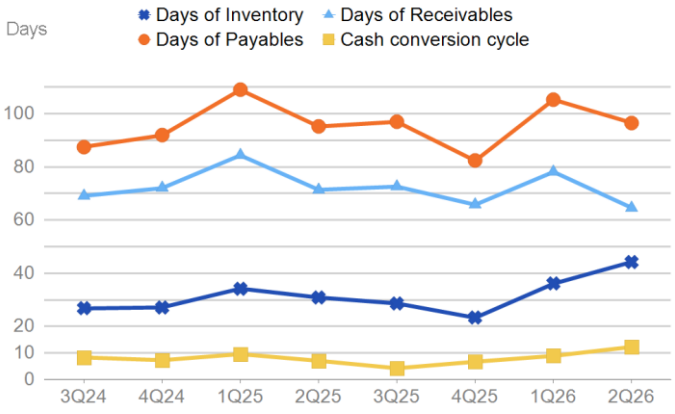
資料來源：CMoney、公司資料

圖 18：毛利率、營益率、淨利率



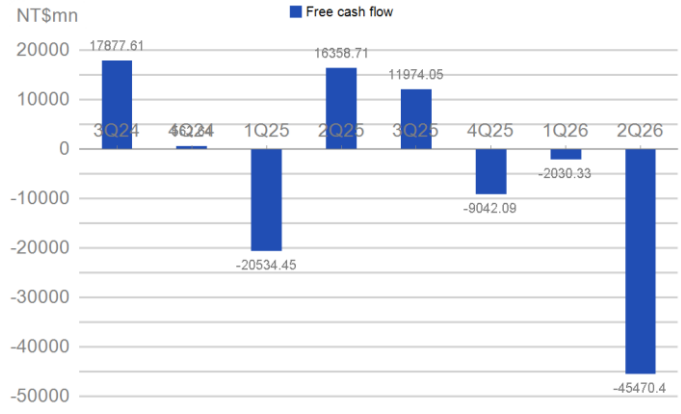
資料來源：CMoney、公司資料

圖 19：經營能力



資料來源：CMoney、公司資料

圖 20：自由現金流量



資料來源：CMoney、公司資料

ESG 分析

分別藉由「在 ESG 議題上的曝險」和「個別公司在 ESG 議題上的執行力」兩個面向來衡量公司於 ESG 上的表現。

- **ESG 總分：** 和碩聯合整體的 ESG 風險評級屬於中風險，於 Sustainalytics 資料庫所覆蓋的公司中排名亦排行於略微領先的位置，在科技硬體行業的公司中排名同樣略微領先同業。
- **在 ESG 議題上的曝險：** 和碩聯合的整體曝險屬於中等水準，但略高於科技硬體行業的平均水準。公司較需要改善的 ESG 議題包含公司數據安全、人力資源、商業道德等。
- **個別公司在 ESG 議題上的執行力：**和碩聯合在針對重大 ESG 議題的管理水準和執行力屬中等。公司清楚揭露 ESG 相關事宜，且積極面對環境保護等議題和確實實施永續供應鏈，例如禁用童工等，表明對投資者和社會的強烈責任感。

圖 21：ESG 分析

分項	評分/評級
ESG 總分	22.2
在 ESG 議題上的曝險 (A)	39.4
個別公司在 ESG 議題上的執行力 (B)	46.7
風險評級	中 ★
同業排行(1~100，1 為最佳)	60

資料來源：Sustainalytics (2026/10/11)

*註 1：ESG 總分=A-(A*可控風險因子*B/100)*

註 1-1：可控風險因子介於 0-1 之間，越大為越佳。

註 2：ESG 總分風險評級： 極低 (0-10) | 低 (10-20) | 中 (20-30) | 高 (30-40) | 極高 (40+)

註 3：曝險分數評級： 低 (0-35) | 中 (35-55) | 高 (55+)

註 4：執行力分數評級： 弱 (0-25) | 中 (25-50) | 強 (50-100)

資產負債表

年初至 12 月 (NT$ 百萬元)	2023A	2024A	2025A	2026F	2027F
現金與短期投資	97,722	121,625	119,380	109,017	123,461
存貨	105,616	101,702	81,235	152,241	153,763
應收帳款及票據	154,906	265,171	221,025	198,381	200,365
其他流動資產	62,157	26,318	37,649	36,849	36,849
流動資產	**420,401**	**514,817**	**459,289**	**496,488**	**514,438**
採用權益法之投資	31,945	49,509	44,727	50,692	50,692
固定資產	75,303	79,017	84,677	89,336	86,695
無形資產	238	284	458	499	524
其他非流動資產	**29,182**	**31,125**	**33,963**	**33,690**	**34,364**
非流動資產	**136,668**	**159,935**	**163,826**	**174,217**	**172,275**
資產總額	**557,069**	**674,752**	**623,115**	**670,705**	**686,713**
應付帳款及票據	176,616	309,314	249,241	250,176	252,677
短期借款	45,164	23,819	42,212	76,141	78,425
什項負債	60,475	52,925	53,779	65,220	65,220
流動負債	**282,256**	**386,057**	**345,232**	**391,537**	**396,323**
長期借款	16,786	14,526	11,447	8,916	8,916
其他負債及準備	36,410	32,469	19,978	17,010	17,010
長期負債	**53,196**	**46,996**	**31,425**	**25,926**	**25,926**
負債總額	**335,451**	**433,053**	**376,657**	**417,463**	**422,249**
股本	26,642	26,632	26,822	26,759	26,759
資本公積	84,196	84,899	86,921	87,070	87,070
保留盈餘	85,082	92,031	95,740	90,178	102,731
什項權益	(7,110)	3,532	(2,140)	2,161	829
歸屬母公司之權益	**188,810**	**207,093**	**207,343**	**206,167**	**217,388**
非控制權益	**32,808**	**34,606**	**39,115**	**47,076**	**47,076**
股東權益總額	**221,618**	**241,699**	**246,458**	**253,243**	**264,464**

資料來源：公司資料、元大投顧

現金流量表

年初至 12 月 (NT$ 百萬元)	2023A	2024A	2025A	2026F	2027F
本期純益	17,413	19,154	18,110	24,030	38,345
折舊及攤提	15,331	15,701	14,730	16,005	16,895
本期營運資金變動	77,932	28,503	5,127	(47,305)	(1,004)
其他營業資產及負債變動	(23,297)	(7,409)	(18,327)	(12,678)	(14,378)
營運活動之現金流量	**87,379**	**55,948**	**19,640**	**(19,948)**	**39,856**
資本支出	(16,908)	(21,200)	(20,883)	(16,975)	(14,000)
本期長期投資變動	4,084	17,564	(4,782)	6,734	0
其他資產變動	(7,976)	(17,630)	8,833	(620)	0
投資活動之現金流量	**(20,800)**	**(21,266)**	**(16,833)**	**(10,861)**	**(14,000)**
股本變動	(34)	(10)	190	0	0
本期負債變動	(35,397)	(7,488)	7,808	22,226	0
現金增減資	0	0	0	0	0
支付現金股利	(13,488)	(11,969)	(13,455)	(10,703)	(11,413)
其他調整數	270	(950)	2,292	7,074	0
融資活動之現金流量	**(48,649)**	**(20,418)**	**(3,165)**	**18,596**	**(11,413)**
匯率影響數	**66**	**6,250**	**(3,348)**	**1,849**	**0**
本期產生現金流量	**17,996**	**20,513**	**(3,706)**	**(10,363)**	**14,443**
自由現金流量	**70,471**	**34,748**	**(1,244)**	**(36,922)**	**25,856**

資料來源：公司資料、元大投顧

損益表

年初至 12 月 (NT$ 百萬元)	2023A	2024A	2025A	2026F	2027F
營業收入	**1,256,783**	**1,125,349**	**1,117,160**	**1,328,848**	**1,727,754**
銷貨成本	(1,210,67	(1,079,316	(1,074,659	(1,270,392	(1,647,174
營業毛利	**46,106**	**46,033**	**42,501**	**58,455**	**80,580**
營業費用	(31,354)	(33,532)	(30,893)	(32,501)	(35,760)
推銷費用	(4,740)	(4,711)	(5,202)	(4,919)	(5,160)
研究費用	(15,889)	(16,125)	(16,101)	(16,730)	(18,800)
管理費用	(10,725)	(12,696)	(9,589)	(10,852)	(11,800)
其他費用	0	0	0	0	0
營業利益	**14,753**	**12,501**	**11,609**	**25,955**	**44,820**
利息收入	4,918	5,562	4,452	3,625	2,000
利息費用	(2,920)	(1,360)	(1,327)	(1,491)	(1,547)
利息收入淨額	1,998	4,202	3,125	2,134	453
投資利益(損失)淨額	4,347	3,534	3,651	45	0
匯兌損益	(348)	579	(968)	67	0
其他業外收入(支出)淨額	3,557	6,408	7,517	2,057	3,887
稅前純益	24,307	27,224	24,934	30,257	49,160
所得稅費用	(6,894)	(8,071)	(6,824)	(6,227)	(10,815)
少數股權淨利	1,700	2,277	3,707	8,160	14,378
歸屬母公司之稅後純益	**15,713**	**16,877**	**14,403**	**15,871**	**23,966**
稅前息前折舊攤銷前淨利	42,558	44,286	40,991	41,960	61,714
調整後每股盈餘 (NT$)	**5.90**	**6.34**	**5.39**	**5.92**	**8.95**

資料來源：公司資料、元大投顧

主要財務報表分析

年初至 12 月	2023A	2024A	2025A	2026F	2027F
年成長率 (%)					
營業收入	(4.7)	(10.5)	(0.7)	18.9	30.0
營業利益	(41.0)	(15.3)	(7.1)	123.6	72.7
稅前息前折舊攤銷前淨利	(4.8)	4.1	(7.4)	2.4	47.1
稅後純益	(17.7)	10.0	(5.4)	32.7	59.6
調整後每股盈餘	4.2	7.5	(14.9)	9.9	51.0
獲利能力分析 (%)					
營業毛利率	3.7	4.1	3.8	4.4	4.7
營業利益率	1.2	1.1	1.0	2.0	2.6
稅前息前淨利率	1.7	2.3	2.1	2.0	2.6
稅前息前折舊攤銷前淨利率	3.4	3.9	3.7	3.2	3.6
稅前純益率	1.9	2.4	2.2	2.3	2.9
稅後純益率	1.3	1.5	1.3	1.2	1.4
資產報酬率	2.9	3.1	2.8	3.6	5.6
股東權益報酬率	8.0	8.3	7.4	7.7	11.0
穩定/償債能力分析					
負債權益比 (%)	151.4	179.2	152.8	164.9	159.7
淨負債權益比(%)	(16.1)	(34.5)	(26.7)	(9.5)	(13.7)
利息保障倍數 (倍)	9.3	21.0	19.8	21.3	32.8
流動比率 (%)	148.9	133.4	133.0	126.8	129.8
速動比率 (%)	111.5	107.0	109.5	87.9	91.0
淨負債 (NT$百萬元)	(35,771)	(83,280)	(65,721)	(23,960)	(36,119)
調整後每股淨值 (NT$)	70.87	77.76	77.30	76.96	81.15
評價指標 (倍)					
本益比	15.2	14.2	16.6	15.1	10.0
股價自由現金流量比	3.4	6.9	--	--	9.3
股價淨值比	1.3	1.2	1.2	1.2	1.1
股價稅前息前折舊攤銷前淨	5.7	5.4	5.9	5.7	3.9
股價營收比	0.2	0.2	0.2	0.2	0.1

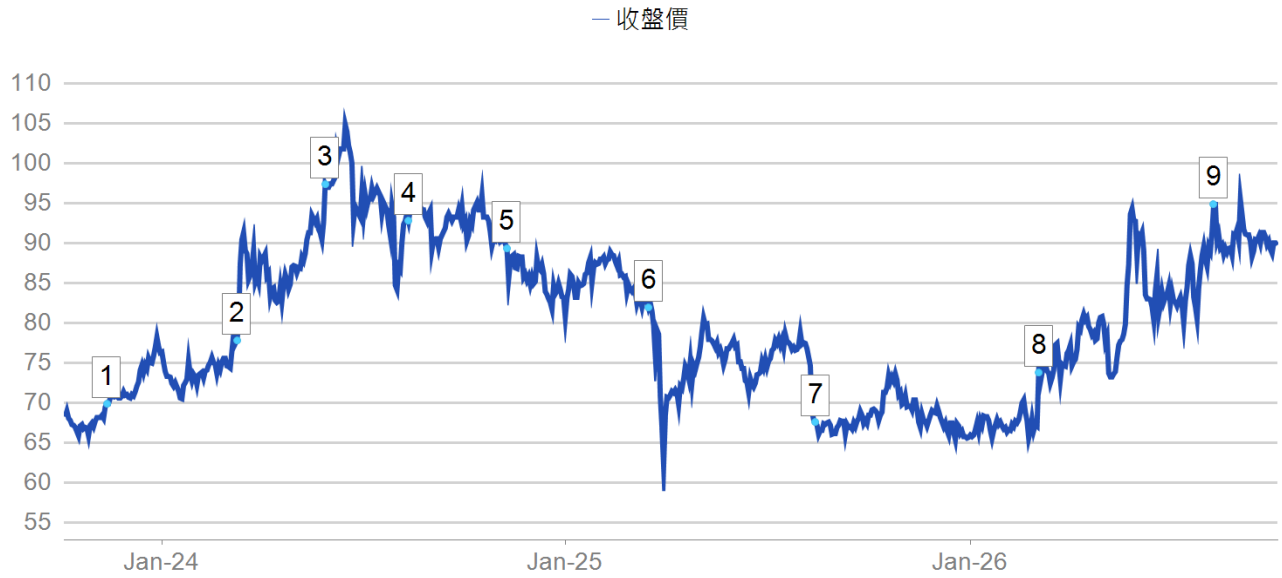
資料來源：公司資料、元大投顧；註：負債為短期債加上長期債。

附錄：重要揭露事項

分析師聲明

主要負責撰寫本研究報告全文或部分內容之分析師，茲針對本報告所載證券或證券發行機構，於此聲明：(1) 文中所述觀點皆準確反映其個人對各證券或證券發行機構之看法；(2) 研究部分析師於本研究報告中所提出之特定投資建議或觀點，與其過去、現在、未來薪酬的任何部份皆無直接或間接關聯。

和碩 (4938 TT)- 投資建議與目標價三年歷史趨勢



#	日期	收盤價 (A)	目標價 (B)	調整後目標價 (C)	評等	分析師
1	20231115	79.50	75	75	持有-落後同業	蘇子錚
2	20240315	88.60	84	84	持有-落後同業	蘇子錚
3	20240603	111.50	115	115	持有-超越同業	蘇子錚
4	20240815	103.50	110	110	持有-超越同業	蘇子錚
5	20241113	98.70	105	105	持有-超越同業	蘇子錚
6	20250325	91.50	92	92	持有-超越同業	蘇子錚
7	20250818	70.90	78	78	持有-超越同業	蘇子錚
8	20260312	77.10	88	88	持有-超越同業	蘇子錚
9	20260813	91.40	100	100	持有-超越同業	蘇子錚

資料來源：CMoney、元大投顧
註：A = 未調整之收盤價；B = 未調整之目標價； C = 依據股本變動調整後之目標價。員工分紅稀釋影響未反映於 A、B 或 C。

投資評等說明

買進：根據本中心對該檔個股投資期間絕對或相對報酬率之預測，我們對該股持正面觀點。此一觀點係基於本中心對該股之發展前景、財務表現、利多題材、評價資訊以及風險概況之分析。建議投資人於投資部位中增持該股。

持有-超越同業：本中心認為根據目前股價，該檔個股基本面吸引力高於同業。此一觀點係基於本中心對該股發展前景、財務表現、利多題材、評價資訊以及風險概況之分析。

持有-落後同業：本中心認為根據目前股價，該檔個股基本面吸引力低於同業。此一觀點係基於本中心對該股發展前景、財務表現、利多題材、評價資訊以及風險概況之分析。

賣出：根據本中心對該檔個股投資期間絕對或相對報酬率之預測，我們對該股持負面觀點。此一觀點係基於本中心對該股之發展前景、財務表現、利多題材、評價資訊以及風險概況之分析。建議投資人於投資部位中減持該股。

評估中：本中心之預估、評等、目標價尚在評估中，但仍積極追蹤該個股。

限制評等：為遵循相關法令規章及/或元大之政策，暫不給予評等及目標價。

註：元大給予個股之目標價係依 12 個月投資期間計算。大中華探索系列報告並無正式之 12 個月目標價，其投資建議乃根據分析師報告中之指定期間分析而得。

總聲明

© 2026 元大版權所有。本報告之內容取材自本公司認可之資料來源，但並不保證其完整性或正確性。報告內容並非任何證券之銷售要約或邀購。報告中所有的意見及預估，皆基於本公司於特定日期所做之判斷，如有變更恕不另行通知。

本報告僅提供一般資訊，文中所載資訊或任何意見，並不構成任何買賣證券或其他投資標的之要約或要約之引誘。報告資料之刊發僅供客戶一般傳閱用途，並非意欲提供專屬之投資建議，亦無考慮任何可能收取本報告之人士的個別財務狀況與目標。對於投資本報告所討論或建議之任何證券、投資標的，或文中所討論或建議之投資策略，投資人應就其是否適合本身而諮詢財務顧問的意見。本報告之內容取材自據信為可靠之資料來源，但概不以明示或默示的方式，對資料之準確性、完整性或正確性作出任何陳述或保證。本報告並非（且不應解釋為）在任何司法管轄區內，任何非依法從事證券經紀或交易之人士或公司，為於該管轄區內從事證券經紀或交易之遊說。

元大研究報告於美國僅發送予美國主要投資法人（依據 1934 年《證券交易法》15a-6 號規則及其修正條文與美國證券交易委員會詮釋定義）。美國投資人若欲進行與本報告所載證券相關之交易，皆必須透過依照 1934 年《證券交易法》第 15 條及其修正條文登記註冊之券商為之。元大研究報告在台灣由元大證券投資顧問股份有限公司發佈，在香港則由元大證券(香港)有限公司發佈。元大證券(香港)係獲香港證券及期貨事務監察委員會核准註冊之券商，並獲許從事受規管活動，包括第 4 類規管活動（就證券提供意見）。非經元大證券(香港)有限公司書面明示同意，本研究報告全文或部份，不得以任何形式或方式轉載、轉寄或揭露。

欲取得任何本報告所載證券詳細資料之台灣人士，應透過下列方式聯絡元大證券投資顧問股份有限公司：

致：聯絡人姓名
元大證券投資顧問股份有限公司
台灣臺北市 106 仁愛路三段 157 號 4 樓

Copyright © 2026 Sustainalytics, a Morningstar company. All rights reserved. The information, data, analyses and opinions contained herein: (1) includes the proprietary information of Sustainalytics and/or its content providers; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product, project, investment strategy or consideration of any particular environmental, social or governance related issues as part of any investment strategy; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. The ESG-related information, methodologies, tools, ratings, data and opinions contained or reflected herein are not directed to or intended for use or distribution to India-based clients or users and their distribution to Indian resident individuals or entities is not permitted. Neither Morningstar Inc., Sustainalytics, Morningstar Asia Ltd. Taiwan Branch nor their content providers accept any liability for the use of the information, for actions of third parties in respect to the information, nor are responsible for any trading decisions, damages or other losses related to the information or its use. The use of the data is subject to conditions available at https://www.sustainalytics.com/legal-disclaimers.

視陽 (6782 TT) Visco Vision

4Q26 營運有望再創高峰，因應亞洲強勁需求積極擴充產能

買進 (維持評等)

目標價 (12 個月)：NT$305.0

收盤價 (2026/10/08)：NT$230.0
隱含漲幅：32.6%

營收組成 (1H26)

日本 60%、中國+台灣 20%、歐洲+美國 20%。

本次報告更新重點

項目	本次	前次
評等	買進	買進
目標價 (NT$)	305	275
2026年營收 (NT$/十億)	5.5	5.2
2026年EPS	18.9	17.4

交易資料表

市值	NT$14,490百萬元
外資持股比率	10.9%
董監持股比率	16.6%
調整後每股淨值 (2026F)	NT$72.47
負債比	35.0%
ESG評級 (Sustainalytics)	-- (曝險程度共5級)

簡明損益表 (NT$百萬元)

年初至12月	2024A	2025A	2026F	2027F
營業收入	3,672	4,223	5,520	6,342
營業利益	790	1,155	1,598	1,859
稅後純益	636	887	1,188	1,409
EPS (元)	10.10	14.08	18.86	22.36
EPS YoY (%)	111.0	39.4	34.0	18.5
本益比 (倍)	22.8	16.3	12.2	10.3
股價淨值比 (倍)	3.9	3.3	3.2	2.8
ROE (%)	19.1	21.7	26.0	26.8
現金殖利率 (%)	2.2%	3.7%	4.9%	5.8%
現金股利 (元)	5.10	8.40	11.32	13.42

蔣欣穎
Jane.Jiang@yuanta.com

陳冠成
charleschen24@yuanta.com

元大觀點

- **日本、中國矽水膠彩片需求暢旺，產能擴充亦使歐洲出貨回升，4Q26 營收預估上調 21%至 15.73 億元，QoQ+8.4%/YoY+47.3%。**
- **目前視陽產品仍處供不應求，預估 2026 年總月產能擴充 29%至 6,300 萬片；1Q27 再擴充 19%至 7,500 萬片，彩片產能佔比提升至 46%。**
- **受惠於亞洲矽水膠鏡片滲透率提升趨勢，營業槓桿效益顯現，以本益比 15 倍及 2H26-1H27F EPS 20.43 元，推得目標價 305 元。**

4Q26 日本、中國市場需求維持強勁，營運有望再創高峰

視陽 3Q26 營收 14.51 億元 (QoQ+14.5%/YoY+28.7%)，優於元大/市場預期 6.4%/5.7%，主因日本、中國矽水膠彩片需求暢旺，產能擴充亦使歐洲出貨回升，4Q26 營收預估上調 21%至 15.73 億元，QoQ+8.4%/YoY+47.3%。整體稼動率維持 90%以上，營收規模擴大有利降低單位成本，3Q26/4Q26 毛利率預估上調 0.2ppt/0.7ppt 至 46.9%/47.4% (QoQ-1.8ppt/+0.5ppt)，稅後淨利同步上調 9%/23%至 3.12 億元/3.44 億元 (YoY+22%/+37%)，預估 3Q26/4Q26 EPS 4.95 元/5.46 元 (QoQ+7%/+10%)。

產品供不應求，2026 年/1Q27 總月產能預估擴充 29%/19%

由於日本、中國客戶對於彩片需求強勁，排擠歐美市場出貨，目前視陽產品仍處供不應求，2Q26 月產能已達 5,500 萬片(彩片產能佔比 36%)，預估 4Q26/1Q27 月產能將進一步提升至 6,300 萬片/7,500 萬片(彩片產能佔比 44%/46%)，近期彩片產能的擴充仍高於一般透片，反映亞洲矽水膠彩片需求仍強，預估 2026 年彩片營收佔比將提升至 40%。產品組合轉佳、營業槓桿效益顯現，預估 2026-2027 年營業利益率達 29%以上。

矽水膠彩片成長強勁，預估 2026/2027 年 EPS 年增 34%/19%

展望 2026-2027 年，視陽營運受惠於 1)日本、中國矽水膠彩片需求強勁成長，終端品牌客戶主推恆定彩拋；2) 日本大客戶散光鏡片滲透率持續成長；及 3) 替代料效益顯現、產品組合轉佳，獲利率逐年改善，2026/2027 年營收預估上調 7%/11%至 55.2 億元/63.42 億元(YoY+31%/+15%)，EPS 預估同步上調 8%/12%至 18.86 元/22.36 元(YoY+34%/+19%)。視陽作為矽水膠鏡片代工龍頭廠商，將受惠於亞洲矽水膠鏡片滲透率提升趨勢，因應訂單成長擴產無虞，並受惠於營業槓桿效益，給予 15 倍目標本益比，基於 2H26-1H27 預估 EPS 20.43 元，推得新目標價 305 元，給予買進評等。

營運分析

4Q26 日本、中國市場需求維持強勁，營運有望再創高峰

視陽 2026 年 9 月營收 4.95 億元 (MoM+1.1%/YoY+14.9%)，累積 3Q26 營收 14.51 億元 (QoQ+14.5%/YoY+28.7%)，優於元大/市場預期 6.4%/5.7%，主因日本、中國矽水膠彩片需求暢旺，產能擴充亦使歐洲出貨回升。展望 4Q26，日本、中國市場對於矽水膠彩片及透片的強勁需求仍為主要成長動能，整體稼動率維持在 90%以上的滿載水位，隨產能釋出，亦逐步滿足歐美訂單需求，帶動 2H26 營收逐季成長，4Q26 營收預估上調 21%至 15.73 億元，QoQ+8.4%/YoY+47.3%。毛利率方面，由於視陽於 2Q26 開始以較高價格追加 PP 原料訂單，管理層評估原料中的 PP 漲價對 3Q26 起毛利率負面影響約 1ppt，惟營收規模擴大有利降低單位成本，3Q26/4Q26 毛利率預估上調 0.2ppt/0.7ppt 至 46.9%/47.4%，QoQ-1.8ppt/+0.5ppt、YoY+0.7ppt/+0.4ppt；同步上調稅後淨利 9.2%/23.3%至 3.12 億元/3.44 億元 (QoQ+6.8%/+10.2%、YoY+22.0%/+37.0%)，預估 3Q26/4Q26 EPS 為 4.95 元/5.46 元，QoQ+7%/+10%、YoY+22%/+37%。

圖 1：2026 年第 3 季獲利預估調整

(百萬元)	3Q25A	2Q26A	3Q26F	季增率	年增率	3Q26F		預估差異	
						元大預估	市場預估	元大	市場
營業收入	1,132	1,268	1,451	14.5%	28.7%	1,363	1,373	6.4%	5.7%
營業毛利	523	619	681	10.1%	30.1%	637	649	6.9%	5.0%
營業利益	337	379	416	9.6%	23.3%	381	398	9.1%	4.4%
稅前利益	336	382	413	8.1%	23.1%	379	392	9.1%	5.4%
稅後淨利	256	292	312	6.8%	22.0%	286	305	9.2%	2.3%
調整後 EPS (元)	4.06	4.64	4.95	6.8%	22.0%	4.54	4.84	9.2%	2.3%
重要比率 (%)				百分點	百分點			百分點	百分點
營業毛利率	46.2%	48.8%	46.9%	-1.8	0.7	46.8%	47.2%	0.2	-0.3
營業利益率	29.8%	29.9%	28.6%	-1.3	-1.1	28.0%	29.0%	0.7	-0.3
稅後純益率	22.6%	23.0%	21.5%	-1.5	-1.1	21.0%	22.2%	0.5	-0.7

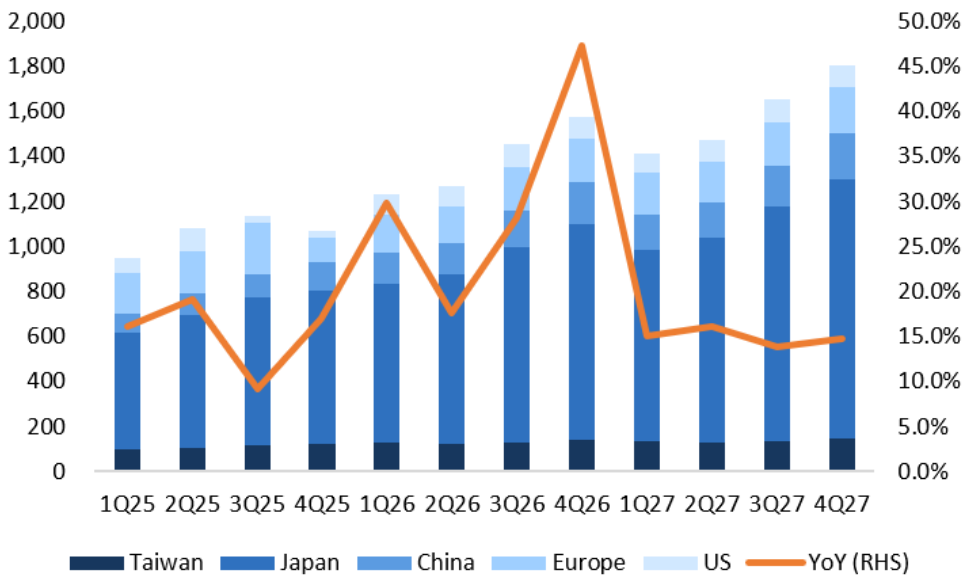
資料來源：公司資料、元大投顧預估、Bloomberg

圖 2：2026 年第 4 季獲利預估調整

(百萬元)	4Q25A	3Q26F	4Q26F	季增率	年增率	4Q26F		預估差異	
						元大預估	市場預估	元大	市場
營業收入	1,068	1,451	1,573	8.4%	47.3%	1,299	1,400	21.0%	12.3%
營業毛利	501	681	746	9.5%	48.7%	607	661	22.8%	12.7%
營業利益	313	416	458	10.2%	46.2%	372	407	23.0%	12.6%
稅前利益	261	413	455	10.1%	74.2%	369	399	23.2%	14.0%
稅後淨利	251	312	344	10.2%	37.0%	279	311	23.3%	10.7%
調整後 EPS (元)	3.98	4.95	5.46	10.2%	37.0%	4.43	4.93	23.3%	10.7%
重要比率 (%)				百分點	百分點			百分點	百分點
營業毛利率	47.0%	46.9%	47.4%	0.5	0.4	46.7%	47.2%	0.7	0.2
營業利益率	29.3%	28.6%	29.1%	0.5	-0.2	28.6%	29.0%	0.5	0.1
稅後純益率	23.5%	21.5%	21.9%	0.4	-1.6	21.5%	22.2%	0.4	-0.3

資料來源：公司資料、元大投顧預估、Bloomberg

圖 3：日本地區為視陽近期營收主要成長動能



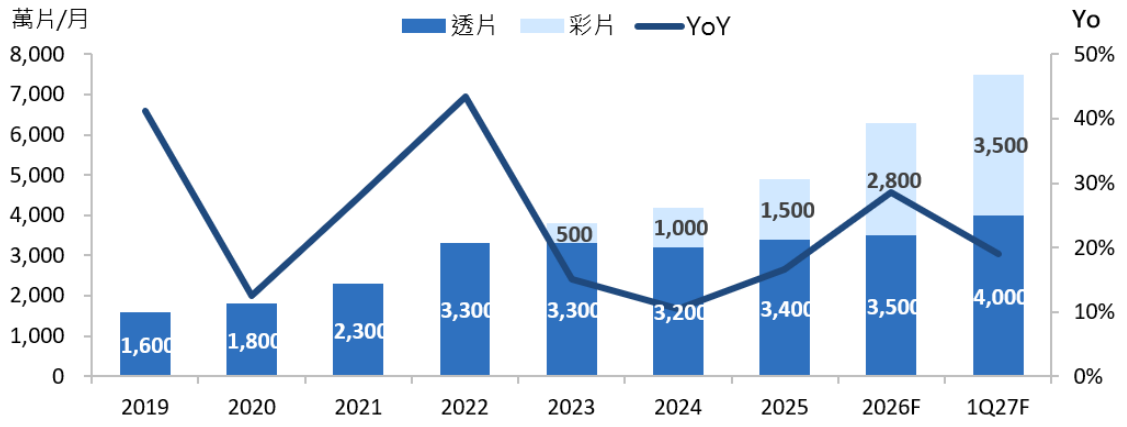
資料來源：公司資料、元大投顧預估

因應亞洲強勁需求積極擴充產能，預計 4Q26/1Q27 總月產能達 6,300/7,500 萬片

視陽透過既有設備優化與良率提升，4Q25 完成擴充 700 萬片至月產能 4,900 萬片(矽水膠彩片月產能提升至 1,500 萬片、彩片產能佔比 30%)，將涵蓋一般透片、老花片與散光片之產線調整，每 100 萬片新產線資本支出落在 6,000~8,000 萬台幣，新產能貢獻的營收已於 1Q26 展現。由於日本、中國客戶對於彩片需求強勁，排擠歐美市場出貨，目前視陽產品仍處供不應求，2Q26 月產能已達 5,500 萬片(矽水膠彩片月產能 2,000 萬片、彩片產能佔比 36%)，預估 4Q26/1Q27 月產能將進一步提升至 6,300 萬片/7,500 萬片(彩片產能佔比 44%/46%)，近期彩片產能的擴充仍高於一般透片，反映亞洲對於矽水膠彩片需求仍強。預估視陽矽水膠彩片營收佔比將自 2025 年底的約 30%，提升至 2026 年底的 40%。

其餘資本支出方面，為配合新產線進駐，並滿足後續擴產需求，公司於馬來西亞建置無塵室之建置費用預估約 3.3 億元；由於公司過去在台灣之辦公室皆為承租，公司於 2026 年 3 月已在中壢購置約 4.68 億元土地，預計未來用於興建自有辦公室。整體而言，預估 2026-2027 年合計資本支出將達 27 億元，相較過往較為積極。

圖 4：2026 年視陽預估月產能擴充 29%至 6,300 萬片，1Q27 月產能再擴充 19%至 7,500 萬片

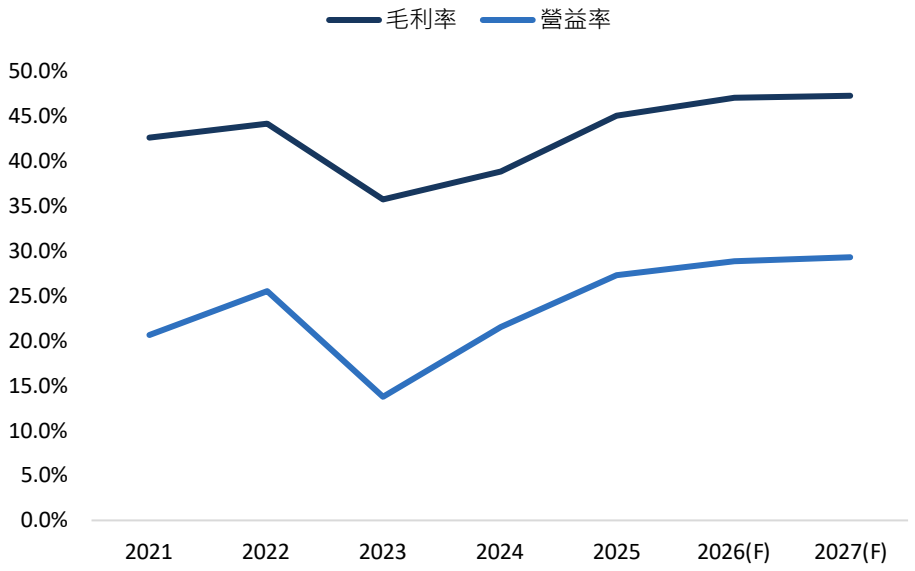


資料來源：公司資料、元大投顧預估

彩片佔比提升使產品組合轉佳，帶動獲利率逐年改善

視陽近期毛利率主要受產能利用率、生產效率、匯率影響，因應強勁需求擴充產能，2026 年產能利用率皆維持在 90%以上，不過快速擴產將負面影響短期毛利率，主因新增人力的生產效率需要時間提升，近期馬幣兌台幣大致持穩，加上替代料效益顯現，整體而言，預估 2026-2027 年毛利率達 47%以上，遠高於 2021-2022 年 43~44%水平，主要得益於彩片營收佔比從過往的 20%提升至目前的 30-40%，帶動產品組合改善，管理層預期 2026 年年底彩片營收佔比將進一步提升至 40%，有利後續毛利率表現。彩片相比於透明片 ASP 增加 40~50%，但成本僅增加 10~20%，故彩拋產品毛利金額相比透明片多 20~30%。除此之外，隨著規模經濟效益顯現、單位成本下降，亦有利後續獲利率表現，2025 年視陽營業利益率達 27.3%，創歷史新高，預估 2026-2027 年視陽營業利益率將擴張至 29%以上。

圖 5：產品組合轉佳、營運槓桿效益顯現，帶動視陽 2025-2027 年毛利率/營業利益率逐年改善



資料來源：公司資料、元大投顧預估

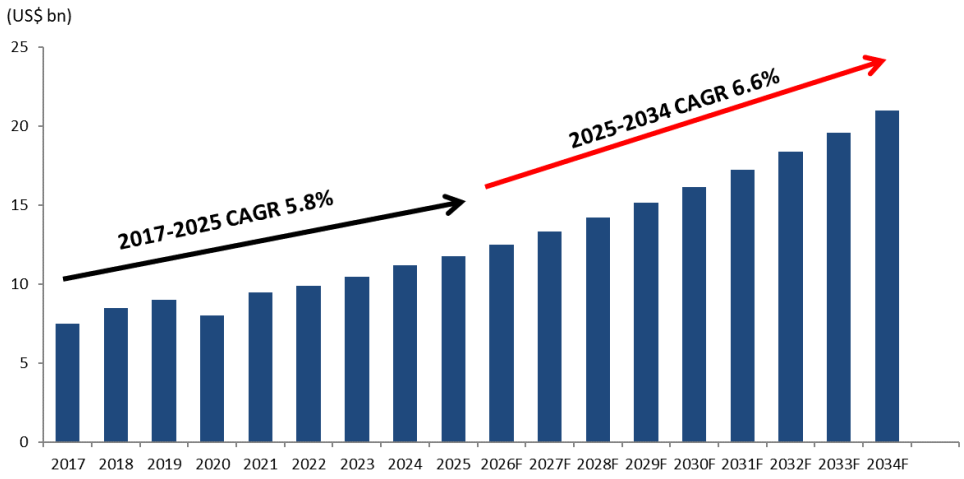
產業概況

全球隱形眼鏡市場以 CAGR 6.6%成長，主要由亞洲需求帶動

2020 年全球隱形眼鏡市場衰退 1 成，主因全球新冠病毒大流行，短期無疫苗、治療藥物可使用下，各國施行不同程度的封控，以降低疾病傳播，外出人群減少，配戴隱形眼鏡的需求降低，不過 2021 年開始隨各地疫苗普及率提升，鬆綁防疫政策，2021 年全球隱形眼鏡市場規模達 94.8 億美元，相比於 2019 年疫情前的 90 億美元成長，顯示後疫情全球隱形眼鏡需求復甦力道強勁。

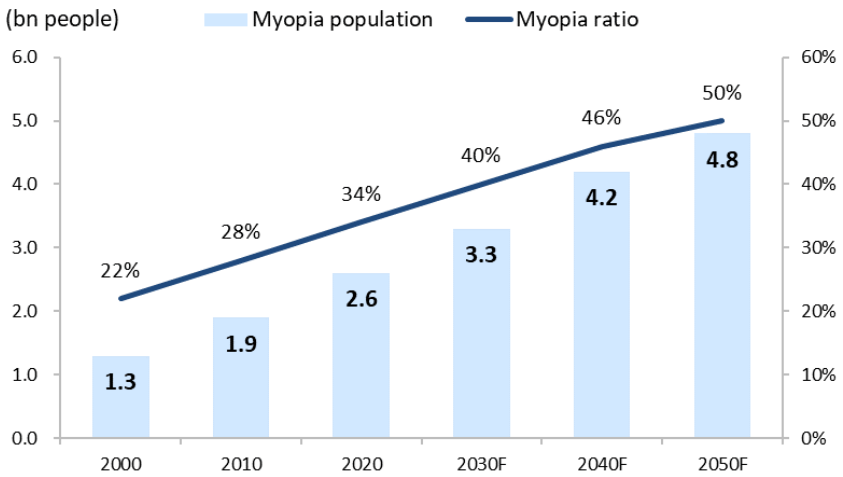
2025 年全球隱形眼鏡市場規模約 118 億美元，YoY+4.9%，預估全球隱形眼鏡市場規模將由 2025 年的 118 億美元成長至 2034 年的 210 億美元，CAGR 為 6.6%。研究指出，全球近視人口將從 2020 年的 26.2 億人(佔總人口 34%)提升至 2050 年的 47.6 億人(佔總人口 50%)，美洲、歐洲、亞洲近視率分別約為 30-40%、30-40%、70-90%，而兒童及青少年的近視率方面，歐美、日本、韓國、中國分別約為 15%、85%、73%、41%，全球屈光不正人口增加及保健意識提升為隱形眼鏡市場成長動能。2025 年美國、歐洲、亞洲在全球隱形眼鏡市佔率分別約為 40%、20-25%、20-25%，其中日本為主要市場。亞洲人口數及近視率皆高於歐美，由此可見亞洲市場隱形眼鏡的使用率仍低，未來主要成長動能將由亞洲需求所帶動。

圖 6：2025-2034 年全球隱形眼鏡市場穩定以 CAGR 6.6%成長



資料來源：Contact Lens Spectrum、Fortune Business Insights、元大投顧

圖 7：2020-2050 年全球近視人口快速成長

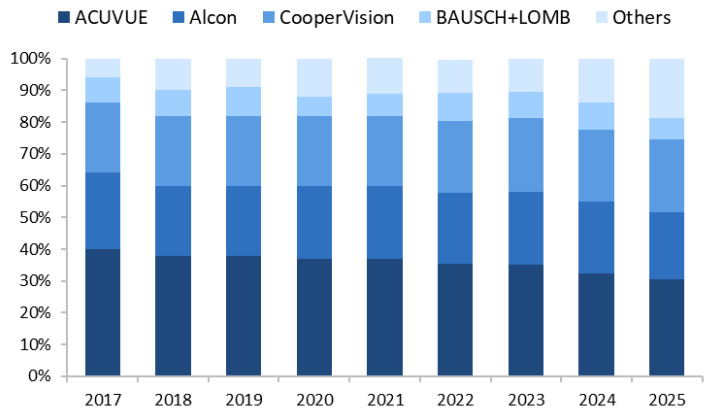


資料來源：World Health Organization、元大投顧

隱形眼鏡四大品牌廠市佔率逐年遭侵蝕

全球隱形眼鏡市場主要由四大廠商 Johnson& Johnson (嬌生)、Alcon (愛爾康)、Cooper Vision (酷柏)、Bausch+Lomb (博士倫)主導，前四大隱形眼鏡公司因經營規模大，且品牌知名度高，導致經營彈性降低，在某些區域通路上有著歷史的包袱，不過四大廠市佔率已從 2017 年的 94%下降到 2025 年的 81.3%，其他廠商逐漸搶佔市佔率，主因 1) 越來越多生產廠有能力提供高性價比產品，隱形眼鏡品牌競爭激烈；2) 國際大廠為維持自身利潤，壓縮經銷通路利潤，使經銷通路推出自有品牌，如 Vision Direct、1-800 等；3) 亞洲彩片盛行，當地品牌廠較有能力根據消費者喜好設計花樣；及 4) 線上通路興起，新銳品牌進入障礙降低。

圖 8：隱形眼鏡四大品牌廠市佔率逐年遭侵蝕

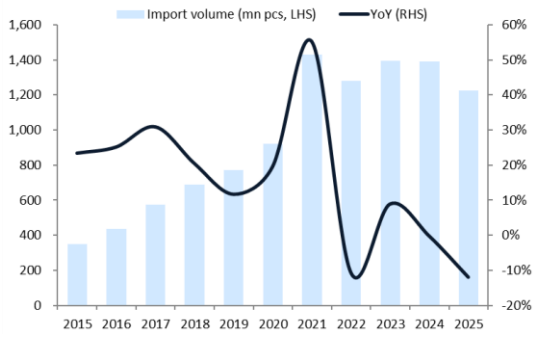


資料來源：CooperVision、元大投顧

中國隱形眼鏡市場高度依賴進口，惟近年中國殺價競爭降低台廠出貨意願

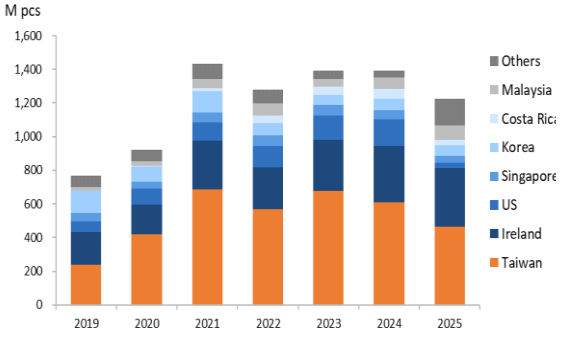
2016-2025 年中國隱形眼鏡進口量 CAGR 達 12%，反映中國隱形眼鏡需求快速成長，且大量仰賴進口。2025 年中國隱形眼鏡進口量為 12.3 億片，YoY-11.9%，其中約 36%來自台灣，顯示中國隱形眼鏡市場對於台灣廠商的依賴性高。然而中國市場近年因中國當地製造商及台廠新進者可提供更低價產品，殺價競爭壓力下，台廠分別應對狀況：1) 晶碩因未加入價格競爭，自 2022 年起壓抑其中國地區營收成長，但有助毛利率的維持；2) 精華 2023 年開始與中國大客戶協議降價，協議結果為降價 10%以上，訂單增加使 2023 年營收年增幅達 60%，然因中國隱形眼鏡代工市場競爭激烈，後續精華在中國市場的營收成長也大幅下降；3) 視陽中國地區佔比本就較低(10-20%)，目前在中國推出的矽水膠彩片中，以視陽為主要供應商，四大品牌在這一塊並未積極涉入。儘管如此，多家研調機構皆指出，中國隱形眼鏡市場持續擴大，矽水膠等高端產品的需求主要依賴進口，隨消費升級，預估 2025/2026 年中國隱形眼鏡進口量將有所增長。

圖 9：中國隱形眼鏡進口量趨勢



資料來源：中國海關總署、元大投顧

圖 10：中國隱形眼鏡進口地區分布

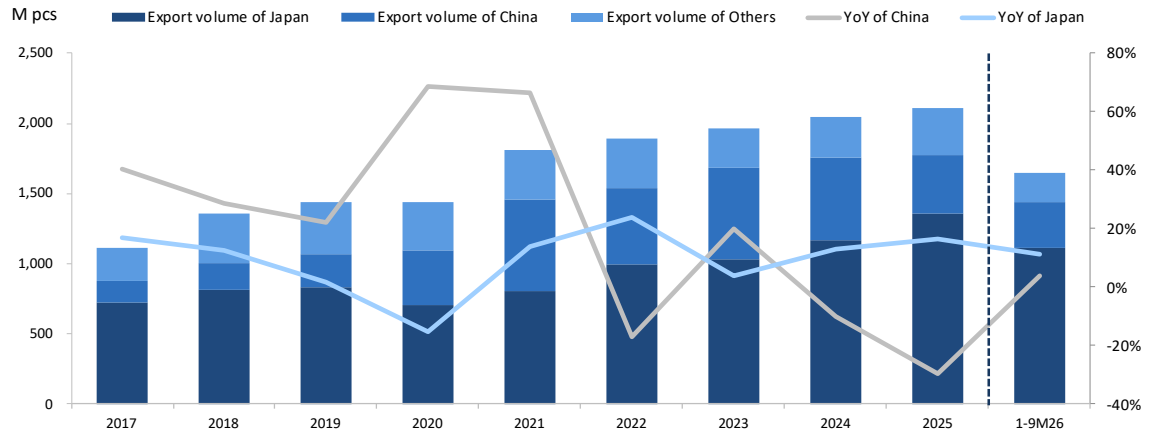


資料來源：中國海關總署、元大投顧

台廠逐漸將重心由中國市場轉往日本市場

目前視陽、晶碩、精華皆已取得日本市場矽水膠產品證，將使台廠出口至日本地區比重逐漸上升，加上日本客戶忠誠度高、願付價格高，且對廠商主要採取獨家採購方案，故較無如中國市場所面臨的削價競爭問題。再者，日本市場屬於隱形眼鏡高滲透率國家，而日本消費者對健康意識的提升，以及日本消費者有能力承擔更高的隱形眼鏡費用，特別是在健康及舒適性有明顯差異的情況下，使矽水膠在日本地區滲透率快速增加。

圖 11：台灣出口至日本比重逐漸上升



資料來源：海關進出口統計、元大投顧

矽水膠軟式隱形眼鏡為未來主流

隱形眼鏡分硬式與軟式兩大類，目前軟式隱形眼鏡為市場主流，因材質軟，可與角膜緊密貼合，不易產生滑動現象，且較無異物感，配戴者能很快地適應，整體配戴舒適度遠勝過硬式隱形眼鏡。軟式隱形眼鏡依材質主要可分為水膠及矽水膠。水膠含水量高，配戴舒適度高，許多人不自覺地超時配戴，但水膠透氧度低，長期配戴會使眼睛缺氧，產生不適的症狀，例如紅眼、酸痛、乾眼等。矽水膠產品則彌補了水膠鏡片透氧不足的問題，藉由在水膠成分中加入矽，大幅提高鏡片得透氧率，具有良好的強韌度。由於矽是高透氧而不含水的原料，也可降低鏡片含水量，減輕鏡片脫水後吸收眼睛表面淚液現象，眼睛較不易感覺乾澀。其高透氧特性，可改善角膜缺氧問題。

在民眾健康意識抬頭之下，矽水膠鏡片的普及率正快速崛起。在開發中或是隱形眼鏡滲透率不高的國家，例如中國、東南亞各國，因價格因素仍會以水膠的材質成長主力；隱形眼鏡滲透率高的地區，成長力道將會轉以矽水膠為主。

圖 12：矽水膠隱形眼鏡透氧度高，可解決水膠乾澀、紅眼、無法久戴等問題

	水膠	矽水膠
Dk/t 值	低，Dk/t 平均約在 20~30	高，可達 Dk/t 145~175 以上
含水量	38-70%	30-50%
材質	傳統水膠	水膠中加入矽分子
價格	較低	較高
健康影響	乾澀、紅眼、疲勞等	較少
特性	長時間配戴易造成角膜缺氧	舒適感較佳，適合長時間配戴，乾眼症患者也可配戴

資料來源：元大投顧整理

獲利調整與股票評價

矽水膠彩片成長強勁，產品組合轉佳，預估 2026/2027 年 EPS 年增 34%/19%

展望 2026-2027 年，視陽營運受惠於 1) 日本市場 20-30 家終端品牌客戶既有產品熱銷，尤其恆定彩片需求大幅優於預期，帶動客戶積極追單與拉貨，同時矽水膠散光片營收佔比已達低個位數，預估 2026/2027 年日本地區營收 32.91 億元/39.49 億元(較前次上調 7%/12%)，YoY+34%/+20%；2) 中國市場矽水膠彩片需求暢旺，目前幾位大客戶皆對視陽推出之恆定彩片具高度興趣，預估 2026/2027 年中國及台灣地區營收合計 11.43 億元/12.45 億元(較前次上調 4%/6%)，YoY+36%/+9%；3) 美國兩大客戶之一在關稅問題趨緩後，1H26 回補庫存力道強勁，預估 2026/2027 年美國地區營收 3.72 億元/3.83 億元(較前次上調 17%/24%)，YoY+71%/+3%；及 4) 隨 2026-2027 年產能陸續開出，歐洲市場出貨回溫，預估 2026/2027 年歐洲地區營收 7.14 億元/7.64 億元，YoY-0.2%/+7%。綜上所述，日本、中國地區為今明年營收主要成長動能，預估 2026/2027 年視陽合併營收 55.2 億元/63.42 億元(YoY+30.7%/+14.9%)，較前次上調 7%/11%。毛利率方面，近期馬幣兌台幣大致持穩，產能快速擴充下稼動率仍維持 90%以上，加上產品組合轉佳，2026/2027 年毛利率上調 0.2ppt/0.1ppt 至 47.3%/47.5% (YoY+2.1ppt/+0.2ppt)；視陽日本子公司自有品牌 Refrear 於 2026 年 5 月引進台灣後，推升行銷費用支出，且公司持續開發新產品及眼藥水等研發投入，預估 2026/2027 年營業費用率為 18.3%/18.2% (較 2025 年的 17.9%略增)；2026 年將執行日本子公司盈餘匯回，預估所得稅率自過去的 18-20%，增加至 20-25%，預估 2026/2027 年 EPS 為 18.86 元/22.36 元(YoY+34.0%/ +18.5%)，較前次上調 8%/12%。

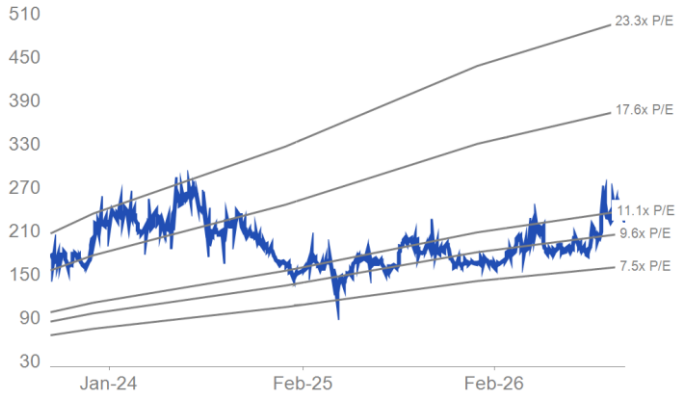
考量 1) 視陽在矽水膠彩片及矽水膠功能型鏡片代工站穩龍頭地位，將受惠於亞洲矽水膠鏡片滲透率提升趨勢；2) 恆定彩拋在日本及中國地區熱銷，彩片營收佔比已達 30-40%，且仍持續成長，有利產品組合轉佳及毛利率擴張；及 3) 營業槓桿效益顯現，2026-2027 年營業利益率將逐年改善，給予 15 倍目標本益比(落於視陽歷史本益比區間 11~18 倍的中值)，基於 2H26-1H27 預估 EPS 20.43 元，推得新目標價 305 元，給予買進評等，後續隨產能釋出、功能型鏡片銷售成長，可關注其矽水膠產品在歐洲市場的成長機會。

圖 13：2026 和 2027 年財務預估調整

(百萬元)	2026 估		2027 估		預估差異	
	調整後	調整前	調整後	調整前	2026	2027
營業收入	5,520	5,158	6,342	5,719	7.0%	10.9%
營業毛利	2,608	2,426	3,010	2,708	7.5%	11.2%
營業利益	1,598	1,478	1,859	1,664	8.1%	11.7%
稅前利益	1,571	1,450	1,853	1,658	8.3%	11.8%
稅後淨利	1,188	1,097	1,409	1,260	8.3%	11.8%
調整後 EPS (元)	18.86	17.42	22.36	20.00	8.3%	11.8%
重要比率 (%)					百分點	百分點
營業毛利率	47.3%	47.0%	47.5%	47.4%	0.2	0.1
營業利益率	29.0%	28.6%	29.3%	29.1%	0.3	0.2
稅後純益率	21.5%	21.3%	22.2%	22.0%	0.3	0.2

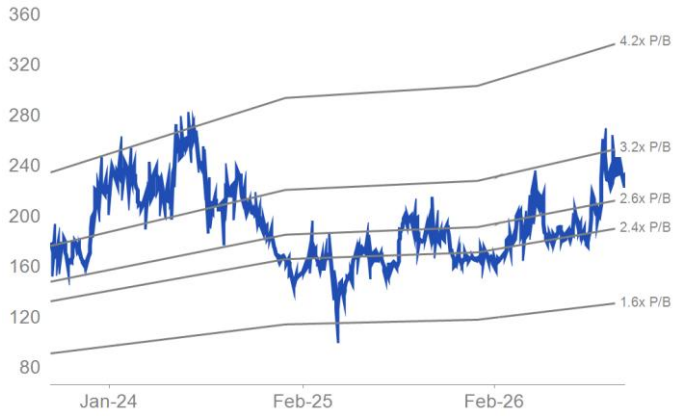
資料來源：公司資料、元大投顧預估

圖 14：12 個月預期本益比區間圖



資料來源：公司資料、元大投顧

圖 15：12 個月預期股價淨值比區間圖



資料來源：公司資料、元大投顧、CMoney

圖 16：同業評價比較表

公司	代碼	評等	股價	市值 (百萬美元)	調整後每股盈餘			本益比(倍)			調整後每股盈餘成長率(%)		
					2025	2026	2027	2025	2026	2027	2025	2026	2027
視陽	**6782 TT**	**買進**	**230.0**	**452**	**14.08**	**18.86**	**22.36**	**16.3**	**12.2**	**10.3**	**39.4**	**34.0**	**18.5**
國際同業													
Seed	7743 JT	未評等	793.0	152	36.1	34.0	41.3	22.0	23.3	19.2	(53.4)	(5.8)	21.5
Menicon	7780 JT	未評等	1852.0	900	74.0	87.3	94.1	25.0	21.2	19.7	24.1	17.9	7.9
Cooper	COO US	未評等	54.8	10,411	1.9	4.5	4.7	29.1	12.1	11.7	(4.5)	140.5	3.7
Alcon	ALCC.SW	未評等	51.9	31,610	1.6	2.9	3.3	33.0	18.0	15.8	(15.7)	83.6	13.4
J&J	JNJ US	未評等	258.5	622,838	11.6	11.1	12.5	22.4	23.3	20.7	97.7	(4.1)	12.6
國際同業平均								**26.3**	**19.6**	**17.4**	**9.6**	**46.4**	**11.8**
國內同業													
晶碩	6491 TT	未評等	403.0	987	20.9	26.5	29.8	19.3	15.2	13.5	(11.1)	27.2	12.1
精華	1565 TT	未評等	97.5	154	9.0	4.9	5.3	10.8	19.8	18.6	(39.1)	(45.4)	6.5
望隼	4771 TT	未評等	203.0	374	12.7	17.5	20.4	16.0	11.6	10.0	7.6	37.4	16.6
國內同業平均								**15.4**	**15.5**	**14.0**	**(14.2)**	**6.4**	**11.7**

資料來源：公司資料、元大投顧、Reuters；每股盈餘數字以當地貨幣為單位；股價依首頁收盤價日期為準。

圖 17：同業評價比較表 (續)

公司	代碼	評等	股價	市值 (百萬美元)	股東權益報酬率(%)			每股淨值			股價淨值比(倍)		
					2025	2026	2027	2025	2026	2027	2025	2026	2027
視陽	**6782 TT**	**買進**	**230.0**	**452**	**21.7**	**26.0**	**26.8**	**70.20**	**72.47**	**83.51**	**3.3**	**3.2**	**2.8**
國際同業													
Seed	7743 JT	未評等	793.0	152	6.2	6.7	6.3	596.8	--	663.0	1.3	--	1.2
Menicon	7780 JT	未評等	1852.0	900	6.8	7.5	7.2	1133.8	1206.8	1306.7	1.6	1.5	1.4
Cooper	COO US	未評等	54.8	10,411	10.1	10.7	9.8	41.4	43.2	43.8	1.3	1.3	1.3
Alcon	ALCC.SW	未評等	51.9	31,610	7.0	6.6	7.9	35.4	38.5	40.2	1.5	1.4	1.3
J&J	JNJ US	未評等	258.5	622,838	34.3	32.5	32.0	33.9	36.9	41.5	7.6	7.0	6.2
國際同業平均					**12.9**	**12.8**	**12.6**				**2.7**	**2.8**	**2.3**
國內同業													
晶碩	6491 TT	未評等	403.0	987	14.4	15.6	16.0	149.1	157.1	173.8	2.7	2.6	2.3
精華	1565 TT	未評等	97.5	154	6.9	--	--	129.9	123.1	125.0	0.8	0.8	0.8
望隼	4771 TT	未評等	203.0	374	19.7	22.3	22.6	68.4	78.2	88.9	3.0	2.6	2.3
國內同業平均					**13.7**	**18.9**	**19.3**				**2.1**	**2.0**	**1.8**

資料來源：公司資料、元大投顧、Reuters；每股淨值數字以當地貨幣為單位；股價依首頁收盤價日期為準。

圖 18：季度及年度簡明損益表 (合併)

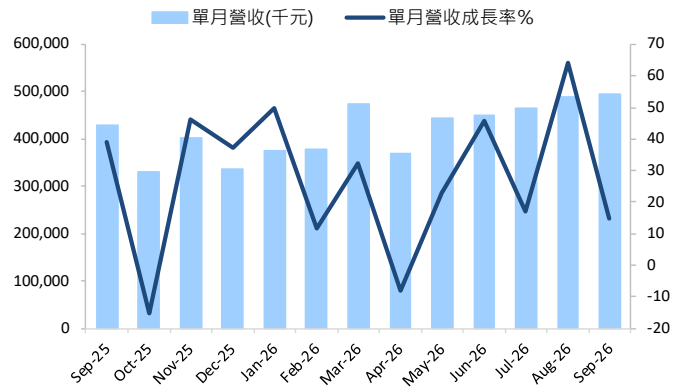
(NT$百萬元)	1Q2026A	2Q2026A	3Q2026F	4Q2026F	1Q2027F	2Q2027F	3Q2027F	4Q2027F	FY2026F	FY2027F
營業收入	1,229	1,268	1,451	1,573	1,414	1,472	1,652	1,804	5,520	6,342
銷貨成本	(666)	(649)	(770)	(827)	(752)	(780)	(864)	(936)	(2,911)	(3,332)
營業毛利	563	619	681	746	662	692	788	868	2,608	3,010
營業費用	(218)	(239)	(265)	(288)	(255)	(268)	(301)	(328)	(1,010)	(1,151)
營業利益	345	379	416	458	407	425	487	540	1,598	1,859
業外利益	(25)	3	(3)	(3)	2	(2)	(2)	(3)	(27)	(6)
稅前純益	321	382	413	455	409	423	485	537	1,571	1,853
所得稅費用	(80)	(89)	(100)	(110)	(98)	(101)	(116)	(126)	(380)	(441)
少數股東權益	0	1	1	1	1	1	1	1	3	3
歸屬母公司稅後純益	240	292	312	344	310	321	368	410	1,188	1,409
調整後每股盈餘(NT$)	3.82	4.64	4.95	5.46	4.92	5.10	5.83	6.51	18.86	22.36
調整後加權平均股數(百萬股)	63	63	63	63	63	63	63	63	63	63
重要比率										
營業毛利率	45.8%	48.8%	46.9%	47.4%	46.8%	47.0%	47.7%	48.1%	47.3%	47.5%
營業利益率	28.1%	29.9%	28.6%	29.1%	28.8%	28.8%	29.5%	29.9%	29.0%	29.3%
稅前純益率	26.1%	30.2%	28.5%	28.9%	28.9%	28.7%	29.3%	29.8%	28.5%	29.2%
稅後純益率	19.6%	23.0%	21.5%	21.9%	21.9%	21.8%	22.3%	22.7%	21.5%	22.2%
有效所得稅率	24.9%	23.4%	24.2%	24.2%	24.0%	23.9%	23.9%	23.5%	24.2%	23.8%
季增率(%)										
營業收入	15.1%	3.2%	14.5%	8.4%	-10.1%	4.1%	12.2%	9.2%		
營業利益	10.3%	9.8%	9.6%	10.2%	-11.1%	4.4%	14.6%	10.9%		
稅後純益	-4.2%	21.5%	6.8%	10.2%	-9.9%	3.5%	14.6%	11.4%		
調整後每股盈餘	-4.2%	21.5%	6.8%	10.2%	-9.9%	3.8%	14.3%	11.5%		
年增率(%)										
營業收入	29.7%	17.8%	28.7%	47.3%	15.1%	16.1%	13.9%	14.7%	30.7%	14.9%
營業利益	73.2%	24.5%	23.3%	46.2%	17.8%	12.0%	17.1%	17.9%	38.4%	16.3%
稅後純益	24.0%	56.6%	22.0%	37.0%	28.9%	9.9%	17.9%	19.2%	34.0%	18.5%
調整後每股盈餘	24.0%	56.6%	22.0%	37.0%	28.8%	10.0%	17.8%	19.2%	34.0%	18.5%

資料來源：公司資料、元大投顧、CMoney；標"A"為歷史數據；調整後每股盈餘為根據調整後加權平均股數計算。

公司簡介

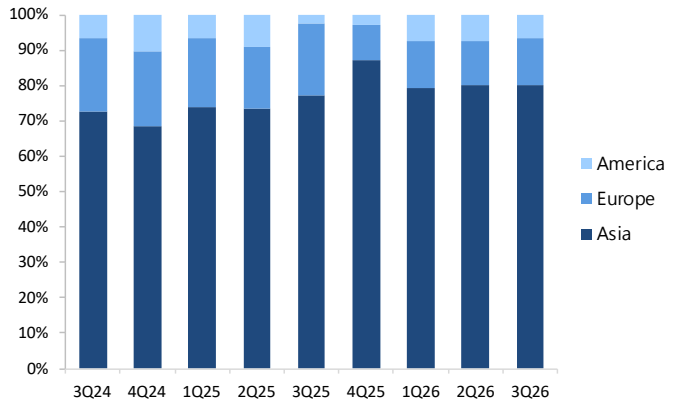
視陽光學 1998 年在台灣成立，主要從事隱形眼鏡的 OEM/ODM 業務，是目前全球第五大、亞洲第一大的矽水膠隱形眼鏡製造商，矽水膠營收佔比達 80%以上。2009 年明基材料轉投資視陽，2013 年發表了世界首款矽水膠彩片。公司除代工生產，亦以 Visco 為品牌銷售產品，且在日本、台灣擁有隱形眼鏡自有品牌 Refrear。

圖 19：視陽月營收走勢圖



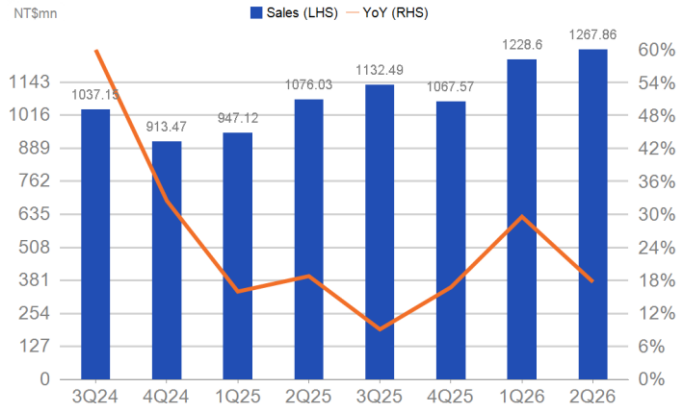
資料來源：公司資料、元大投顧預估

圖 20：營收組成



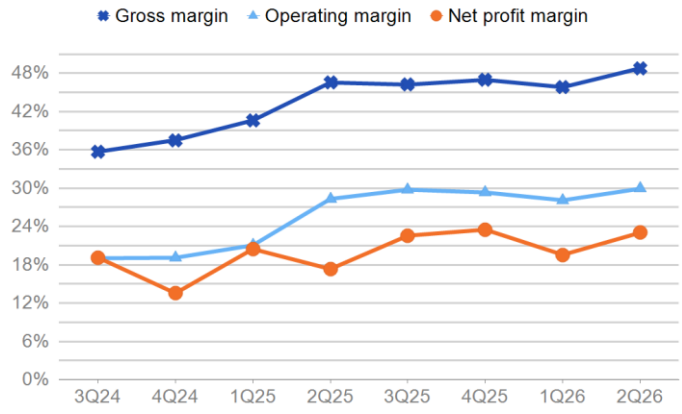
資料來源：公司資料

圖 21：營收趨勢



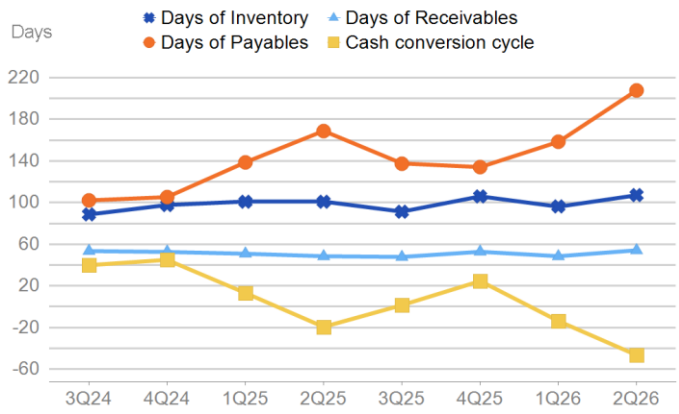
資料來源：CMoney、公司資料

圖 22：毛利率、營益率、淨利率



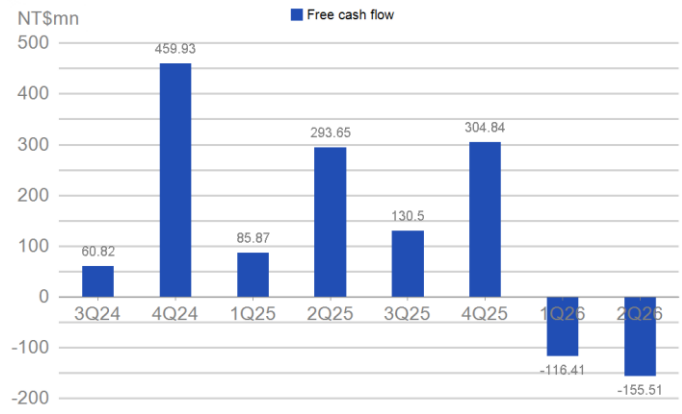
資料來源：CMoney、公司資料

圖 23：經營能力



資料來源：CMoney、公司資料

圖 24：自由現金流量



資料來源：CMoney、公司資料

資產負債表

年初至 12 月 (NT$ 百萬元)	2023A	2024A	2025A	2026F	2027F
現金與短期投資	521	757	708	121	(25)
存貨	606	594	719	1,069	1,476
應收帳款及票據	374	441	577	994	1,264
其他流動資產	282	107	174	254	254
流動資產	**1,784**	**1,899**	**2,179**	**2,439**	**2,969**
採用權益法之投資	0	452	578	616	616
固定資產	1,890	2,197	2,406	3,372	3,122
無形資產	98	87	73	70	70
其他非流動資產	**952**	**669**	**763**	**1,063**	**1,063**
非流動資產	**2,940**	**3,404**	**3,820**	**5,121**	**4,870**
資產總額	**4,724**	**5,303**	**5,999**	**7,560**	**7,839**
應付帳款及票據	172	205	208	325	409
短期借款	44	0	0	240	240
什項負債	669	872	1,020	1,654	1,654
流動負債	**885**	**1,076**	**1,228**	**2,219**	**2,303**
長期借款	833	497	308	723	223
其他負債及準備	24	14	22	32	32
長期負債	**857**	**511**	**330**	**755**	**255**
負債總額	**1,742**	**1,587**	**1,558**	**2,974**	**2,558**
股本	630	630	630	630	630
資本公積	1,431	1,431	1,431	1,431	1,431
保留盈餘	1,098	1,622	2,188	2,318	3,013
什項權益	(194)	14	174	187	187
歸屬母公司之權益	**2,965**	**3,697**	**4,422**	**4,565**	**5,261**
非控制權益	**17**	**19**	**19**	**20**	**20**
股東權益總額	**2,982**	**3,716**	**4,441**	**4,585**	**5,281**

資料來源：公司資料、元大投顧

現金流量表

年初至 12 月 (NT$ 百萬元)	2023A	2024A	2025A	2026F	2027F
本期純益	295	638	887	1,191	1,412
折舊及攤提	405	443	511	639	780
本期營運資金變動	(213)	(18)	(238)	(647)	(592)
其他營業資產及負債變動	(133)	53	193	(4)	(3)
營運活動之現金流量	**353**	**1,116**	**1,352**	**1,179**	**1,597**
資本支出	(449)	(509)	(537)	(1,843)	(530)
本期長期投資變動	0	452	126	(45)	0
其他資產變動	(490)	(455)	(321)	(1)	0
投資活動之現金流量	**(939)**	**(512)**	**(732)**	**(1,911)**	**(530)**
股本變動	0	0	0	0	0
本期負債變動	(299)	(252)	(368)	675	(500)
現金增減資	0	0	0	0	0
支付現金股利	(347)	(151)	(321)	(529)	(713)
其他調整數	(21)	(15)	(10)	(5)	0
融資活動之現金流量	**(667)**	**(418)**	**(699)**	**141**	**(1,213)**
匯率影響數	**(28)**	**51**	**30**	**4**	**0**
本期產生現金流量	**(1,281)**	**237**	**(49)**	**(587)**	**(146)**
自由現金流量	**(96)**	**607**	**815**	**(664)**	**1,067**

資料來源：公司資料、元大投顧

損益表

年初至 12 月 (NT$ 百萬元)	2023A	2024A	2025A	2026F	2027F
營業收入	**2,398**	**3,672**	**4,223**	**5,520**	**6,342**
銷貨成本	(1,541)	(2,245)	(2,313)	(2,911)	(3,332)
營業毛利	**857**	**1,426**	**1,911**	**2,608**	**3,010**
營業費用	(525)	(636)	(756)	(1,010)	(1,151)
推銷費用	(174)	(203)	(281)	(397)	(469)
研究費用	(153)	(201)	(195)	(273)	(285)
管理費用	(195)	(244)	(282)	(343)	(397)
其他費用	(3)	13	2	3	0
營業利益	**332**	**790**	**1,155**	**1,598**	**1,859**
利息收入	13	8	8	4	4
利息費用	(33)	(29)	(21)	(15)	(15)
利息收入淨額	(20)	(21)	(13)	(11)	(11)
投資利益(損失)淨額	0	2	21	13	0
匯兌損益	(10)	(39)	(91)	(29)	0
其他業外收入(支出)淨額	(6)	9	0	0	6
稅前純益	296	740	1,072	1,571	1,853
所得稅費用	(2)	(101)	(186)	(380)	(441)
少數股權淨利	(7)	2	0	3	3
歸屬母公司之稅後純益	**302**	**636**	**887**	**1,188**	**1,409**
稅前息前折舊攤銷前淨利	734	1,211	1,604	2,237	2,639
調整後每股盈餘 (NT$)	**4.79**	**10.10**	**14.08**	**18.86**	**22.36**

資料來源：公司資料、元大投顧

主要財務報表分析

年初至 12 月	2023A	2024A	2025A	2026F	2027F
年成長率 (%)					
營業收入	(13.7)	53.1	15.0	30.7	14.9
營業利益	(53.2)	138.0	46.1	38.4	16.3
稅前息前折舊攤銷前淨利	(32.1)	65.2	32.4	39.5	18.0
稅後純益	(52.0)	116.7	38.9	34.0	18.5
調整後每股盈餘	(56.9)	111.0	39.4	34.0	18.5
獲利能力分析 (%)					
營業毛利率	35.8	38.9	45.2	47.3	47.5
營業利益率	13.9	21.5	27.3	29.0	29.3
稅前息前淨利率	11.0	19.4	24.9	29.0	29.3
稅前息前折舊攤銷前淨利率	30.6	33.0	38.0	40.5	41.6
稅前純益率	12.4	20.2	25.4	28.5	29.2
稅後純益率	12.6	17.3	21.0	21.5	22.2
資產報酬率	5.9	12.7	15.7	17.6	18.3
股東權益報酬率	9.7	19.1	21.7	26.0	26.8
穩定/償債能力分析					
負債權益比 (%)	58.4	42.7	35.1	64.9	48.4
淨負債權益比(%)	11.9	(7.0)	(9.0)	18.4	9.2
利息保障倍數 (倍)	10.0	26.3	53.0	104.7	123.4
流動比率 (%)	201.6	176.4	177.5	109.9	128.9
速動比率 (%)	133.1	121.3	119.0	61.7	64.8
淨負債 (NT$百萬元)	356	(261)	(400)	842	488
調整後每股淨值 (NT$)	47.06	58.68	70.20	72.47	83.51
評價指標 (倍)					
本益比	48.0	22.8	16.3	12.2	10.3
股價自由現金流量比	--	23.9	17.8	--	13.6
股價淨值比	4.9	3.9	3.3	3.2	2.8
股價稅前息前折舊攤銷前淨	19.8	12.0	9.0	6.5	5.5
股價營收比	6.0	4.0	3.4	2.6	2.3

資料來源：公司資料、元大投顧；註：負債為短期債加上長期債。

附錄：重要揭露事項

分析師聲明

主要負責撰寫本研究報告全文或部分內容之分析師，茲針對本報告所載證券或證券發行機構，於此聲明：(1) 文中所述觀點皆準確反映其個人對各證券或證券發行機構之看法；(2) 研究部分析師於本研究報告中所提出之特定投資建議或觀點，與其過去、現在、未來薪酬的任何部份皆無直接或間接關聯。

視陽 (6782 TT)- 投資建議與目標價三年歷史趨勢

#	日期	收盤價 (A)	目標價 (B)	調整後目標價 (C)	評等	分析師
1	20231114	175.50	215	215	買進	蔣欣穎
2	20240108	240.50	300	300	買進	蔣欣穎
3	20240829	221.50	300	300	買進	蔣欣穎
4	20241121	207.00	250	250	買進	蔣欣穎
5	20250317	180.00	220	220	買進	蔣欣穎
6	20250519	181.00	220	220	買進	蔣欣穎
7	20250806	189.00	240	240	買進	蔣欣穎
8	20260114	174.50	230	230	買進	陳穆萱
9	20260225	181.00	230	230	買進	陳穆萱
10	20260416	212.50	260	260	買進	陳穆萱
11	20260518	189.00	235	235	買進	陳穆萱
12	20260806	199.00	275	275	買進	蔣欣穎

資料來源：CMoney、元大投顧
註：A = 未調整之收盤價；B = 未調整之目標價； C = 依據股本變動調整後之目標價。員工分紅稀釋影響未反映於 A、B 或 C。

投資評等說明

買進：根據本中心對該檔個股投資期間絕對或相對報酬率之預測，我們對該股持正面觀點。此一觀點係基於本中心對該股之發展前景、財務表現、利多題材、評價資訊以及風險概況之分析。建議投資人於投資部位中增持該股。

持有-超越同業：本中心認為根據目前股價，該檔個股基本面吸引力高於同業。此一觀點係基於本中心對該股發展前景、財務表現、利多題材、評價資訊以及風險概況之分析。

持有-落後同業：本中心認為根據目前股價，該檔個股基本面吸引力低於同業。此一觀點係基於本中心對該股發展前景、財務表現、利多題材、評價資訊以及風險概況之分析。

賣出：根據本中心對該檔個股投資期間絕對或相對報酬率之預測，我們對該股持負面觀點。此一觀點係基於本中心對該股之發展前景、財務表現、利多題材、評價資訊以及風險概況之分析。建議投資人於投資部位中減持該股。

評估中：本中心之預估、評等、目標價尚在評估中，但仍積極追蹤該個股。

限制評等：為遵循相關法令規章及/或元大之政策，暫不給予評等及目標價。

註：元大給予個股之目標價係依 12 個月投資期間計算。大中華探索系列報告並無正式之 12 個月目標價，其投資建議乃根據分析師報告中之指定期間分析而得。

總聲明

© 2026 元大版權所有。本報告之內容取材自本公司認可之資料來源，但並不保證其完整性或正確性。報告內容並非任何證券之銷售要約或邀購。報告中所有的意見及預估，皆基於本公司於特定日期所做之判斷，如有變更恕不另行通知。

本報告僅提供一般資訊，文中所載資訊或任何意見，並不構成任何買賣證券或其他投資標的之要約或要約之引誘。報告資料之刊發僅供客戶一般傳閱用途，並非意欲提供專屬之投資建議，亦無考慮任何可能收取本報告之人士的個別財務狀況與目標。對於投資本報告所討論或建議之任何證券、投資標的，或文中所討論或建議之投資策略，投資人應就其是否適合本身而諮詢財務顧問的意見。本報告之內容取材自據信為可靠之資料來源，但概不以明示或默示的方式，對資料之準確性、完整性或正確性作出任何陳述或保證。本報告並非（且不應解釋為）在任何司法管轄區內，任何非依法從事證券經紀或交易之人士或公司，為於該管轄區內從事證券經紀或交易之遊說。

元大研究報告於美國僅發送予美國主要投資法人（依據 1934 年《證券交易法》15a-6 號規則及其修正條文與美國證券交易委員會詮釋定義）。美國投資人若欲進行與本報告所載證券相關之交易，皆必須透過依照 1934 年《證券交易法》第 15 條及其修正條文登記註冊之券商為之。元大研究報告在台灣由元大證券投資顧問股份有限公司發佈，在香港則由元大證券(香港)有限公司發佈。元大證券(香港)係獲香港證券及期貨事務監察委員會核准註冊之券商，並獲許從事受規管活動，包括第 4 類規管活動（就證券提供意見）。非經元大證券(香港)有限公司書面明示同意，本研究報告全文或部份，不得以任何形式或方式轉載、轉寄或揭露。

欲取得任何本報告所載證券詳細資料之台灣人士，應透過下列方式聯絡元大證券投資顧問股份有限公司：

致：聯絡人姓名
元大證券投資顧問股份有限公司
台灣臺北市 106 仁愛路三段 157 號 4 樓

Copyright © 2026 Sustainalytics, a Morningstar company. All rights reserved. The information, data, analyses and opinions contained herein: (1) includes the proprietary information of Sustainalytics and/or its content providers; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product, project, investment strategy or consideration of any particular environmental, social or governance related issues as part of any investment strategy; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. The ESG-related information, methodologies, tools, ratings, data and opinions contained or reflected herein are not directed to or intended for use or distribution to India-based clients or users and their distribution to Indian resident individuals or entities is not permitted. Neither Morningstar Inc., Sustainalytics, Morningstar Asia Ltd. Taiwan Branch nor their content providers accept any liability for the use of the information, for actions of third parties in respect to the information, nor are responsible for any trading decisions, damages or other losses related to the information or its use. The use of the data is subject to conditions available at https://www.sustainalytics.com/legal-disclaimers.

個股報告

投資評等	目標價
買進	$9100

前次投資建議：無

大立光(3008)

光通訊成第二成長引擎，業務結構質變推升評價

報告日期：2026.10.12　　研 究 員：陳洞暉

摘 要 及 調 整 原 因

- 手機客戶下修訂單致 11 月拉貨趨緩、27Q1 能見度有限；規格方面 2027 年主鏡頭送樣，2028 年大升級進入試產。
- 大立光 FA 產線 27H2 量產、2028 年多排放量；憑 100%自研對位設備消除公差，下世代專案由機會從單一客戶擴展至多家雲端商。
- 投資評等「買進」，6 個月目標價 9100 (40X2027(F)PER)，潛在上漲空間 36.63%。

基本資料

2026/10/8　收盤價：	6,660.00
普通股股本(百萬股)	131
市值(NT$ 百萬)	871,116
董監持股比例	19.02%

近期股價表現

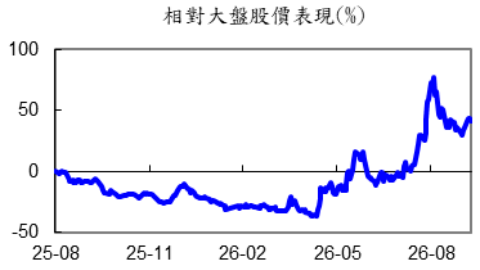
	個股表現	相對大盤
一週	7.33%	5.24%
一月	2.46%	-2.13%
一季	68.61%	56.37%

法人持股

	持股比例
外資	24.16%
投信	7.93%
自營商	4.76%

融資券餘額

(張數)	餘　額
融資餘額	4,830
融券餘額	55



結論與建議

- 公司 26Q3 營收為 156.68 億元，QoQ+14.7%、YoY-11.4%，主要受智慧型手機新機拉貨動能帶動；毛利率 42.2%，QoQ-7.2ppts、YoY-5.1ppts，受到新機良率較差以及綠電成本上升所致；稅後淨利 51.82 億元，QoQ+11%、YoY-26.8%，營業外利益約 13.57 億元（其中匯兌收益約 1.97 億元）；稅後 EPS 39.62 元。
- 受終端銷量不佳影響，手機客戶下修訂單使 11 月起拉貨步入季節性趨緩，27Q1 能見度有限。中長期規格升級方面，2027 年主鏡頭大感光元件目前處於送樣階段，而大幅升級週期預計落在 2028 年，前鏡頭與高階潛望長焦已進入試產。
- 大立光 FA 首條產線預計 27H2 量產，多排產品規劃 2028 年放量。公司憑藉 100%自研主動對位設備消除公差，並調配廠房支應量產，下世代專案由機會從單一客戶擴展至多家雲端商。
- 預估 2027 年營收 825.12 億元，YoY+32.3%；毛利率 48.3%，YoY+2.6ppts；稅後淨利 297.13 億元，YoY+36.8%；稅後 EPS 227.16 元。

公司目前股價位於 29.31X27(F)PER，考量光通業務成長明確，占比提高，所以取 40X，給予投資評等「買進」，6 個月目標價 9100 (40X2027(F)PER)，潛在上漲空間 36.63%。

評等分級：買進（預期未來六個月股價漲幅超過+30%）； 逢低買進（預期未來六個月股價漲幅介於+15%與+30%之間）； 區間操作（預期未來六個月股價漲跌幅介於-15%與+15%之間）； 逢高賣出（預期未來六個月股價跌幅介於-15%與-30%之間）；賣出（預期未來六個月股價跌幅超過-30%）。

本刊所刊載之內容僅做為參考，惟已力求正確與完整，但因時間及市場客觀因素改變所造成產業、市場或個股之相關條件改變，投資人需自行考量投資之需求與風險，本公司恕不負擔任何法律責任。任何轉載或引用本報告內容必須經本公司同意。

簡 要 財 務 預 估 資 訊												
會計年度	營收		營業利益		稅後純益		每股盈餘		每股現金股利(元)	ROE (%)	PER (X)	PBR (X)
	(億元)	YoY(%)	(億元)	YoY(%)	(億元)	YoY(%)	(元)	YoY(%)				
2025	611.5	2.8	235.6	-2.0	212.8	-17.9	159.41	-17.9	97.5	11.4	15.7	1.7
2026(F)	623.9	2.0	216.4	-8.1	217.2	2.1	166.08	4.2	80.0	10.9	40.1	4.3
2027(F)	825.1	32.3	320.9	48.3	297.1	36.8	227.16	36.8	80.0	13.8	29.3	3.9

資料來源：CMoney；兆豐投顧預估

營 運 分 析

大立光為全球光學鏡頭領導龍頭

大立光電成立於 1987 年，為全球光學鏡頭與光電零組件領導廠商，核心技術涵蓋高階模具設計、非球面塑膠鏡片射出成型、多鏡片精密堆疊及全自動化對位組裝等製程研發。因應智慧型手機市場成長放緩與規格升級趨緩，公司近年積極推動產品結構優化，將發展重心由傳統手機鏡頭逐步拓展至高附加價值的共封裝光學、車載鏡頭與醫療光電等領域。透過其微米級精密製造、獨家主動對準製程與設備全自製整合能力，大立光成功切入光纖陣列單元(FAU)、玻塑混合車載鏡頭等高技術門檻供應鏈，致力於開闢第二成長曲線並建立長期穩定的高毛利業務基礎，逐步降低對單一智慧型手機產業的依賴。

公司 26Q3 手機鏡頭出貨量比重：2,000 萬畫素以上佔 10%–20%、1,000 萬畫素佔 60%–70%、800 萬畫素佔約 10%、500 萬畫素以下佔 20%–30%。

圖一、手機鏡頭出貨量比重

手機鏡頭出貨量比重

	手機鏡頭出貨量比重
2000萬畫素以上	10~20%
1000萬畫素	60~70%
800萬畫素	10%
500萬畫素以下	20~30%

資料來源：大立光法說會；兆豐投顧整理

新機拉貨帶動營收季增，惟良率與綠電逆風壓抑毛利

公司 26Q3 營收為 156.68 億元，QoQ+14.7%、YoY-11.4%，主要受智慧型手機新機拉貨動能帶動；毛利率 42.2%，QoQ-7.2ppts、YoY-5.1ppts，受到新機良率較差以及綠電成本上升所致；稅後淨利 51.82 億元，QoQ+11%、YoY-26.8%，營業外利益約 13.57 億元（其中匯兌收益約 1.97 億元）；稅後 EPS 39.62 元。

新品新機拉貨動能提早步入季節性趨緩，靜待後續鏡頭規格升級

新機拉貨動能提早步入季節性趨緩，靜待後續鏡頭規格升級

短期拉貨動能方面，公司 10 月出貨量預估與 9 月相當，11 月起拉貨將步入季節性下滑。雖然 26Q4 產能稼動率預期維持滿載，且可變光圈等高階新機

評等分級：買進（預期未來六個月股價漲幅超過+30%）； 逢低買進（預期未來六個月股價漲幅介於+15%與+30%之間）； 區間操作（預期未來六個月股價漲跌幅介於-15%與+15%之間）； 逢高賣出（預期未來六個月股價跌幅介於-15%與-30%之間）；賣出（預期未來六個月股價跌幅超過-30%）。

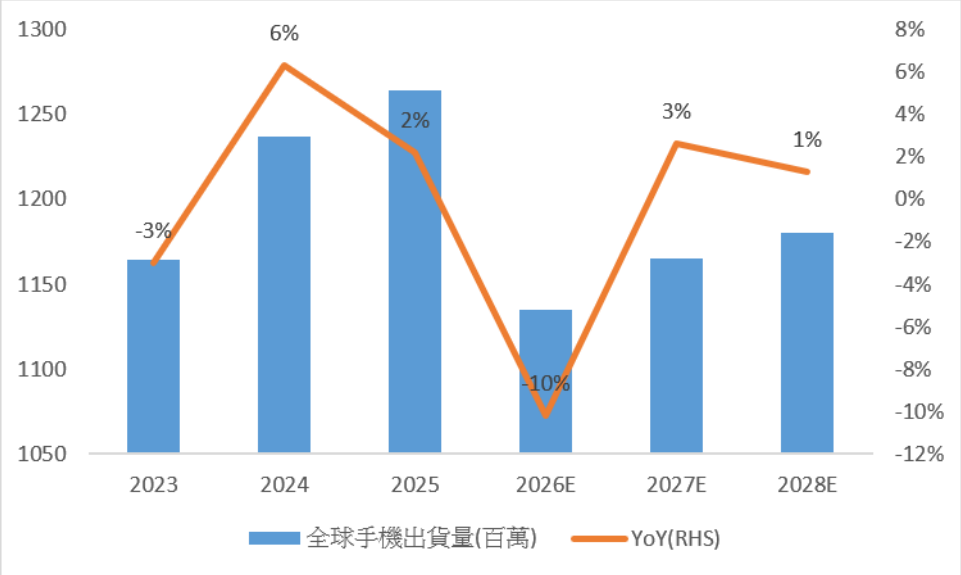
本刊所刊載之內容僅做為參考，惟已力求正確與完整，但因時間及市場客觀因素改變所造成產業、市場或個股之相關條件改變，投資人需自行考量投資之需求與風險，本公司恕不負擔任何法律責任。任何轉載或引用本報告內容必須經本公司同意。

鏡頭良率逐步改善，但拉貨高峰已過。主因新機開賣後終端銷量未達預期，客戶自 9 月即開始下修訂單預測；即便客戶預計 27H1 發表之新機種於 10 月提前啟動拉貨，亦無法彌補主力機種之修正缺口。由於客戶目前未提供長天期訂單預測，27Q1 出貨能見度仍相對有限。

中長期規格升級與競爭策略方面，智慧型手機鏡頭的大幅度升級週期預計落在 2028 年，屆時包含前鏡頭與高階潛望長焦等規格將迎來升級，目前已進入試產階段；2027 年則以大感光元件的主鏡頭升級為主，具備特定技術門檻，目前處於送樣階段，產品單價仍待協商。

圖二、全球手機出貨量以及成長率

全球手機出貨量以及成長率



資料來源：兆豐投顧預測

綠電採購形成結構性成本壓力

綠電採購形成結構性成本壓力

綠電採購方面，大立光依客戶合約規範與其出貨比重採購綠電，已形成長期結構性成本，為 26Q3 毛利率承壓主因。由於 2027 年全年度產品價格已全數底定，綠電成本短期內無法即時轉嫁，最快需待 2028 年新機種報價方能反映，短中期仍對本業毛利形成壓抑。

光通訊佈局全面提速，從 FA 升級直供 FAU 打造第二引擎

光通訊佈局全面提速，從 FA 升級直供 FAU 打造第二引擎

在產品佈局與量產時程方面，大立光光纖陣列（FA）已獲客戶認證通過。為降低零組件組裝之責任爭議，公司順應客戶需求，由單純供應 FA 升級為整合自製 FA、外購微透鏡陣列（PMLA）與稜鏡之完整 FAU 模組，並積極爭取向下整合封裝至具備高精度定位珠載板。首波產品鎖定單排可插拔式與光

評等分級：買進（預期未來六個月股價漲幅超過+30%）； 逢低買進（預期未來六個月股價漲幅介於+15%與+30%之間）； 區間操作（預期未來六個月股價漲跌幅介於-15%與+15%之間）； 逢高賣出（預期未來六個月股價跌幅介於-15%與-30%之間）；賣出（預期未來六個月股價跌幅超過-30%）。

本刊所刊載之內容僅做為參考，惟已力求正確與完整，但因時間及市場客觀因素改變所造成產業、市場或個股之相關條件改變，投資人需自行考量投資之需求與風險，本公司恕不負擔任何法律責任。任何轉載或引用本報告內容必須經本公司同意。

柵耦合架構，試產線已完成驗證；首條自研量產線預計於 27Q2 季底完成建置，並於 27H2 正式量產放量。至於技術難度更高的多排產品目前處於送樣階段，規劃為 2028 年量產專案。

在產能佈局與核心優勢方面，面對光通訊市場供不應求，公司已調配既有第二大廠支應 2027 年量產，新購土地則規劃全數拆除重建為高規格無塵室，預計 2030 年完工以支應長期擴產。核心壁壘上，大立光 FAU 精密對位設備達 100%全段自研自製，憑藉獨家主動對位技術消除外購 V-groove 基板公差，兼具自動化高速除錯與機台調校彈性。客戶結構方面，當前世代以代工模式供應單一客戶，2028 年下一世代專案則已直接對接超過一家雲端服務供應商洽談，確立光通訊為長期第二成長動能。

光通訊成第二成長引擎，業務結構質變推升評價

光通訊成第二成長引擎，業務結構質變推升評價

預估 26Q4 營收為 175.13 億元，QoQ+11.8%、YoY+1.7%；毛利率 42.5%，QoQ+0.3ppts、YoY-7.5ppts；稅後淨利 51.57 億元，QoQ-0.5%、YoY-23.3%；稅後 EPS 39.43 元。

預估 2026 年營收 623.9 億元，YoY+2%；毛利率 45.7%，YoY-4.7ppts；稅後淨利 217.23 億元，YoY+3.2%；稅後 EPS 166.08 元。

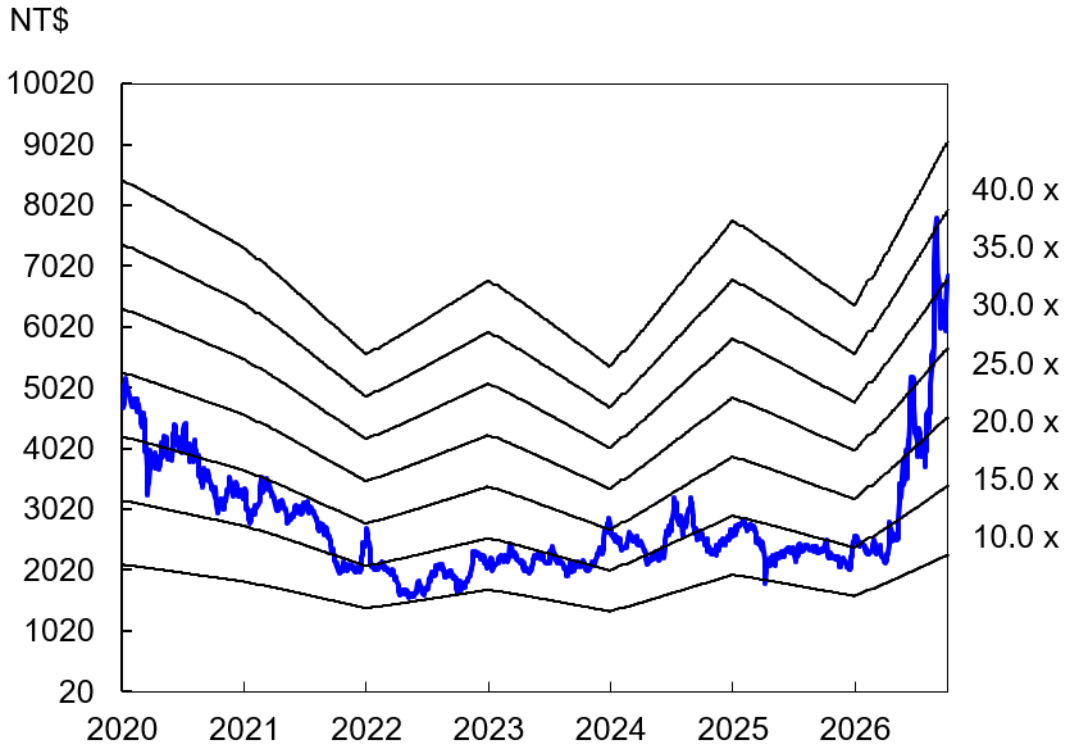
預估 2027 年營收 825.12 億元，YoY+32.3%；毛利率 48.3%，YoY+2.6ppts；稅後淨利 297.13 億元，YoY+36.8%；稅後 EPS 227.16 元。

公司目前股價位於 29.31X27(F)PER，考量光通業務成長明確，占比提高，所以取 40X，給予投資評等「買進」，6 個月目標價 9100 (40X2027(F)PER)，潛在上漲空間 36.63%。

評等分級：買進（預期未來六個月股價漲幅超過+30%）； 逢低買進（預期未來六個月股價漲幅介於+15%與+30%之間）； 區間操作（預期未來六個月股價漲跌幅介於-15%與+15%之間）； 逢高賣出（預期未來六個月股價跌幅介於-15%與-30%之間）；賣出（預期未來六個月股價跌幅超過-30%）。

本刊所刊載之內容僅做為參考，惟已力求正確與完整，但因時間及市場客觀因素改變所造成產業、市場或個股之相關條件改變，投資人需自行考量投資之需求與風險，本公司恕不負擔任何法律責任。任何轉載或引用本報告內容必須經本公司同意。

圖三、大立光(3008)P/E　Band



資料來源：CMoney；兆豐投顧預估

評等分級：買進（預期未來六個月股價漲幅超過+30%）； 逢低買進（預期未來六個月股價漲幅介於+15%與+30%之間）； 區間操作（預期未來六個月股價漲跌幅介於-15%與+15%之間）； 逢高賣出（預期未來六個月股價跌幅介於-15%與-30%之間）；賣出（預期未來六個月股價跌幅超過-30%）。

本刊所刊載之內容僅做為參考，惟已力求正確與完整，但因時間及市場客觀因素改變所造成產業、市場或個股之相關條件改變，投資人需自行考量投資之需求與風險，本公司恕不負擔任何法律責任。任何轉載或引用本報告內容必須經本公司同意。

財務報表

季度損益表

單位：新台幣百萬元

3008 大立光	2025	26Q1	26Q2	26Q3(F)	26Q4(F)	2026(F)	27Q1(F)	27Q2(F)	27Q3(F)	27Q4(F)	2027(F)
營業收入淨額	**61,148**	**15,544**	**13,665**	**15,668**	**17,513**	**62,390**	**16,832**	**15,862**	**23,856**	**25,962**	**82,512**
營業成本	**30,311**	7,865	6,912	9,056	10,070	**33,902**	9,460	8,914	11,642	12,643	**42,659**
營業毛利淨額	**30,837**	**7,680**	**6,753**	**6,612**	**7,443**	**28,487**	**7,372**	**6,948**	**12,214**	**13,319**	**39,853**
營業費用	**7,279**	**1,867**	**1,605**	**1,682**	**1,689**	**6,843**	**1,921**	**1,933**	**1,946**	**1,962**	**7,762**
營業淨利(損)	**23,558**	**5,812**	**5,148**	**4,930**	**5,754**	**21,644**	**5,451**	**5,015**	**10,268**	**11,357**	**32,091**
營業外收支											
利息收入	**4,260**	971	975	1,036	1,029	**4,011**	1,021	982	996	965	**3,964**
財務成本	**2**	1	1	-	-	**1**	-	1	1	0	**2**
其他利益及損失	**(2,045)**	503	(51)	284	215	**952**	246	296	253	244	**1,039**
其他收支淨額	**128**	2	27	37	15	**82**	12	15	16	18	**61**
營業外收支淨額合計	**2,342**	**1,476**	**951**	**1,357**	**1,259**	**5,043**	**1,279**	**1,292**	**1,264**	**1,227**	**5,062**
稅前淨利	**25,900**	**7,288**	**6,099**	**6,287**	**7,013**	**26,687**	**6,730**	**6,307**	**11,532**	**12,584**	**37,153**
所得稅費用(利益)	**4,340**	1,112	1,373	1,049	1,182	**4,716**	1,168	1,153	2,306	2,517	**7,144**
稅後淨利-非控制權益	**285**	53	56	56	83	**248**	71	72	77	76	**296**
稅後淨利-歸屬母公司	**21,275**	**6,123**	**4,670**	**5,182**	**5,748**	**21,723**	**5,491**	**5,082**	**9,149**	**9,991**	**29,713**
普通股股本	**1,335**	1,335	1,308	1,308	1,308	**1,308**	1,308	1,308	1,308	1,308	**1,308**
每股盈餘(元)	**159.41**	**46.63**	**35.70**	**39.62**	**43.95**	**166.08**	**41.98**	**38.85**	**69.95**	**76.38**	**227.16**
季成長分析(QoQ,%)											
營收成長率		-9.7	-12.1	14.7	11.8		-3.9	-5.8	50.4	8.8	
營業毛利成長率		-7.0	-12.1	-2.1	12.6		-1.0	-5.8	75.8	9.0	
營業利益成長率		-8.2	-11.4	-4.2	16.7		-5.3	-8.0	104.8	10.6	
稅前淨利成長率		-11.5	-16.3	3.1	11.6		-4.0	-6.3	82.9	9.1	
稅後淨利成長率		-8.9	-23.7	11.0	10.9		-4.5	-7.5	80.0	9.2	
年成長分析(YoY,%)											
營收成長率	**2.8**	6.6	17.1	-11.4	1.7	**2.0**	8.3	16.1	52.3	48.2	**32.3**
營業毛利成長率	**-1.2**	-3.6	7.9	-20.8	-9.9	**-7.6**	-4.0	2.9	84.7	78.9	**39.9**
營業利益成長率	**-2.0**	-4.5	5.5	-21.3	-9.1	**-8.1**	-6.2	-2.6	108.3	97.4	**48.3**
稅前淨利成長率	**-19.5**	-5.6	240.1	-22.9	-14.8	**3.0**	-7.7	3.4	83.4	79.4	**39.2**
稅後淨利成長率	**-17.9**	-5.0	352.5	-26.8	-14.5	**2.1**	-10.3	8.8	76.6	73.8	**36.8**
獲利分析(%)											
營業毛利率	**50.4**	49.4	49.4	42.2	42.5	**45.7**	43.8	43.8	51.2	51.3	**48.3**
營業利益率	**38.5**	37.4	37.7	31.5	32.9	**34.7**	32.4	31.6	43.0	43.7	**38.9**
稅前淨利率	**42.4**	46.9	44.6	40.1	40.1	**42.8**	40.0	39.8	48.3	48.5	**45.0**
稅後淨利率	**34.8**	39.4	34.2	33.1	32.8	**34.8**	32.6	32.0	38.4	38.5	**36.0**
實質所得稅率	**16.8**	15.3	22.5	16.7	16.9	**17.7**	17.4	18.3	20.0	20.0	**19.2**

資料來源：CMoney；兆豐投顧預估；註：每股盈餘以期末股本計算

評等分級：買進（預期未來六個月股價漲幅超過+30%）； 逢低買進（預期未來六個月股價漲幅介於+15%與+30%之間）； 區間操作（預期未來六個月股價漲跌幅介於-15%與+15%之間）； 逢高賣出（預期未來六個月股價跌幅介於-15%與-30%之間）；賣出（預期未來六個月股價跌幅超過-30%）。

本刊所刊載之內容僅做為參考，惟已力求正確與完整，但因時間及市場客觀因素改變所造成產業、市場或個股之相關條件改變，投資人需自行考量投資之需求與風險，本公司恕不負擔任何法律責任。任何轉載或引用本報告內容必須經本公司同意。

資產負債表

單位：NT$百萬元	2025	2026(F)	2027(F)
流動資產			
現金及約當現金	115,942	112,076	205,063
存貨	6,713	7,334	10,391
應收帳款及票據	10,462	10,799	17,314
其他流動資產	21,169	16,179	37,571
流動資產合計	**154,285**	**146,388**	**270,339**
非流動資產			
長期投資合計	8,517	14,191	-1,126
固定資產	51,472	58,546	63,376
其它非流動資產	6,512	20,581	-70,286
非流動資產合計	**66,502**	**93,318**	**-8,036**
資產總計	**220,787**	**239,706**	**262,303**
流動負債			
應付帳款及票據	1,729	2,397	2,788
短期借款	0	0	0
其他流動負債	28,019	27,712	53,958
流動負債合計	**29,749**	**30,109**	**56,746**
長期負債			
長期借款	0	0	0
其他非流動負債	171	3,345	-19,944
長期負債合計	**171**	**3,345**	**-19,944**
負債總計	**29,919**	**33,454**	**36,802**
權益			
普通股股本	1,335	1,335	1,335
資本公積	1,565	1,534	1,534
保留盈餘	185,818	196,130	215,378
其他項目	2,151	7,254	7,254
權益總計	**190,868**	**206,252**	**225,501**
負債與權益總計	**220,787**	**239,706**	**262,303**

資料來源：CMoney；兆豐投顧預估

現金流量表

單位：NT$百萬元	2025	2026(F)	2027(F)
本期淨利	21,275	21,723	29,713
折舊及攤提	7,732	8,243	9,012
本期營運資金變動	-1,209	-1,626	-9,963
其他營業活動現金流量	-1,718	-671	-1,195
營業活動之淨現金流入(出)	**26,080**	**27,669**	**27,567**
資本支出淨額	-12,340	-16,842	-11,232
本期長期投資變動	3,304	5,675	-15,317
其他投資活動現金流量	-200	-3,073	-1,800
投資活動之淨現金流入(出)	-9,236	-14,240	-28,349
現金增資	0		0
本期借款變動	-212	3,175	-23,290
其他籌資活動現金流量	-13,245		117,059
融資活動之淨現金流入(出)	**-13,458**	**-17,294**	**93,769**
本期現金與約當現金增加數	**2,283**	**-3,865**	**92,987**
期初現金與約當現金餘額	113,658	115,942	112,076
期末現金與約當現金餘額	**115,942**	**112,076**	**205,063**

資料來源：CMoney；兆豐投顧預估

重要財務比率分析

	2025	2026(F)	2027(F)
年成長率(%)			
營業收入	2.8	2.0	32.3
營業利益	-2.0	-8.1	48.3
稅後純益	-17.9	2.1	36.8
獲利能力分析(%)			
營業毛利率	50.4	45.7	48.3
營業利益率	38.5	34.7	38.9
稅後純益率	34.8	34.8	36.0
資產報酬率	9.7	9.4	11.8
股東權益報酬率	11.4	10.9	13.8
償債能力分析			
負債/股東權益(%)	15.7	16.2	16.3
流動比率(X)	5.2	4.9	4.8
每股資料分析			
每股盈餘(NT$元)	159.4	166.1	227.2
本益比(X)	15.7	40.1	29.3
每股淨值(NT$元)	1,422.2	1,545.3	1,689.5
股價淨值比(X)	1.7	4.3	3.9
每股現金股利(NT$元)	97.5	80.0	80.0
現金股利殖利率(%)	3.9	3.3	1.2

資料來源：CMoney；兆豐投顧預估

評等分級：買進（預期未來六個月股價漲幅超過+30%）； 逢低買進（預期未來六個月股價漲幅介於+15%與+30%之間）； 區間操作（預期未來六個月股價漲跌幅介於-15%與+15%之間）； 逢高賣出（預期未來六個月股價跌幅介於-15%與-30%之間）；賣出（預期未來六個月股價跌幅超過-30%）。

本刊所刊載之內容僅做為參考，惟已力求正確與完整，但因時間及市場客觀因素改變所造成產業、市場或個股之相關條件改變，投資人需自行考量投資之需求與風險，本公司恕不負擔任何法律責任。任何轉載或引用本報告內容必須經本公司同意。

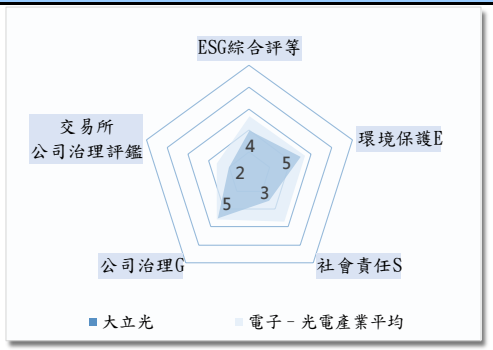
3008 大立光 永續發展概況

CMoney ESG Rating　　評鑑年季：2026Q3

落後			平均				領先		
1	2	3	4	5	6	7	8	9	10

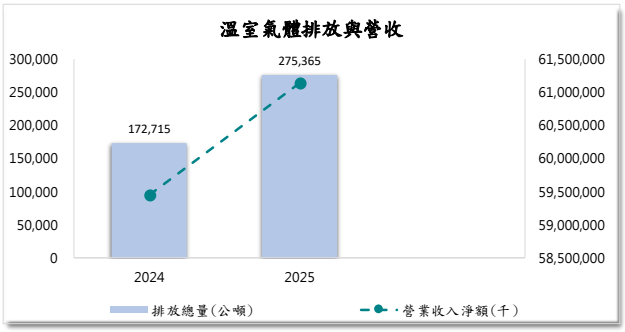
E(環境保護)	S(社會責任)	G(公司治理)	永續報告書連結
5	3	5	https://esggenplus.twse.com.tw/inquiry/info/individual

產業類別	產業樣本數	ESG評等產業排名	TCFD報告書簽署	TCFD報告書連結
電子－光電	116	80	否	否

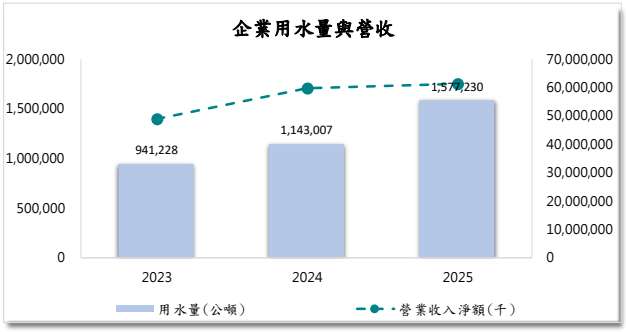


資料來源：CMoney
說明：評等1-10分，10分最高

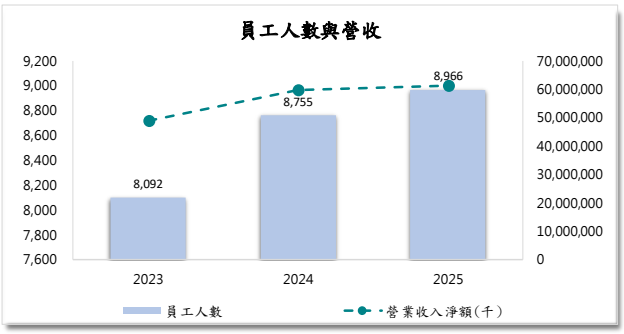
溫室氣體



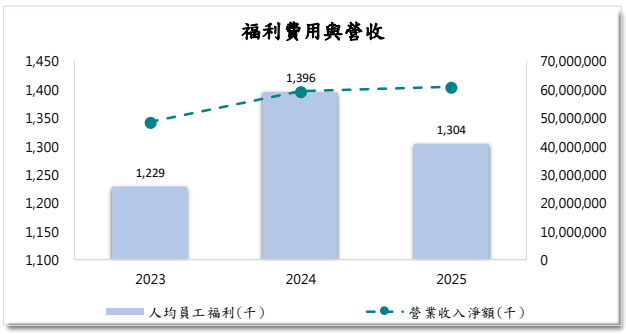
水資源



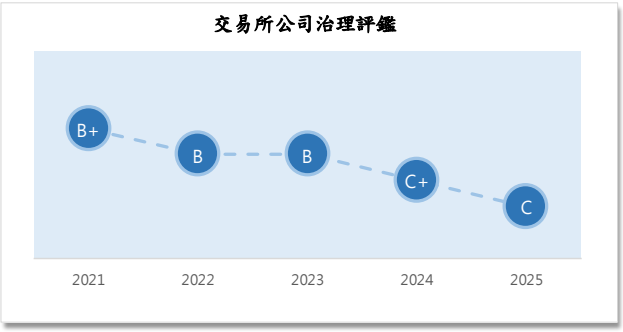
員工人數



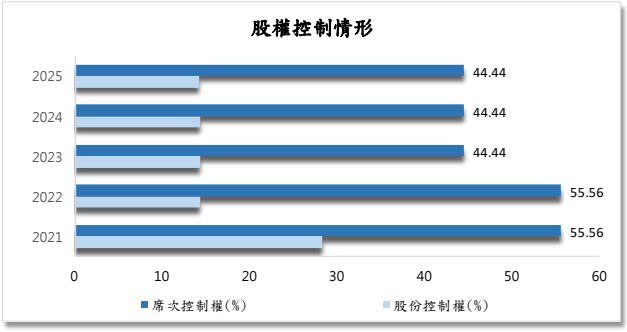
員工福利



公司治理評鑑



股權控制



資料來源：交易所

評等分級：買進（預期未來六個月股價漲幅超過+30%）； 逢低買進（預期未來六個月股價漲幅介於+15%與+30%之間）； 區間操作（預期未來六個月股價漲跌幅介於-15%與+15%之間）； 逢高賣出（預期未來六個月股價跌幅介於-15%與-30%之間）；賣出（預期未來六個月股價跌幅超過-30%）。

本刊所刊載之內容僅做為參考，惟已力求正確與完整，但因時間及市場客觀因素改變所造成產業、市場或個股之相關條件改變，投資人需自行考量投資之需求與風險，本公司恕不負擔任何法律責任。任何轉載或引用本報告內容必須經本公司同意。

個股報告

投資評等	目標價
逢低買進	$327

前次投資建議：2026.06.26
「買進」，目標價: $301

矽格(6257)

湖口二廠進入量產、中興三廠預計 27Q1 接續貢獻

報告日期：2026.10.12　　研 究 員：李彝安

摘 要 及 調 整 原 因

➢ 26Q2 測試需求續強，湖口二廠產能爬坡速度與毛利率優於本研究部原預估，中興三廠預計 27Q1 加入。據此上修 26/27F EPS 至 9.9/13.6 元；目標價由 301 元上調至 327 元(24x2027(F)PER)，潛在上漲空間 15.5%，評等由「買進」調整為「逢低買進」。

基本資料

2026/10/07　收盤價：	283.0
普通股股本(百萬股)	484
市值(NT$ 百萬)	136,947
董監持股比例	4.92%

近期股價表現

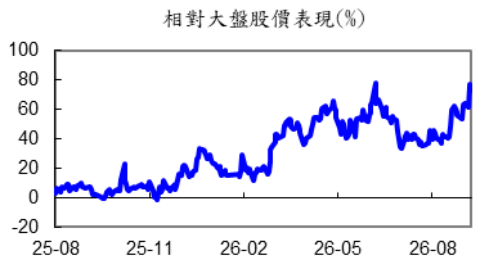
	個股表現	相對大盤
一週	10.33%	8.18%
一月	35.41%	29.34%
一季	15.75%	7.35%

法人持股

	持股比例
外資	32.46%
投信	3.92%
自營商	0.96%

融資券餘額

(張數)	餘　額
融資餘額	13,157
融券餘額	224



結論與建議

➢ 26Q2 合併營收 58.8 億元，季增 11.1%、年增 23.1%，毛利率 34.2%、營益率 25.6%，歸屬母公司淨利 12.2 億元、EPS 2.55 元。營收僅略高於研究部原估，惟毛利率較原估 32.5%高 1.7ppt，為單季獲利優於預期主要原因。26Q3 營收 61.7 億元，季增 4.9%、年增 27.2%，續創單季新高。

➢ 26Q2 毛利率優於預期，主要反映 AI 與高階測試需求成長、部分測試價格調升及折舊增幅有限；湖口二廠 7 月量產後產能開出速度亦快於原預期，帶動下半年營收與稼動率提升。AI 晶片規格升級使測試項目及時間增加，加上高階產品占比提高，也因此上修 2026 年獲利預估。

➢ 湖口二廠，目前產能利用率約四成；中興三廠預計 26Q4 完工、27Q1 開始量產。新增產能主要因應 AI、高階運算及光通訊測試需求，湖口二廠全年貢獻及中興三廠新增產能支撐研究部上修 2027 年營收。

➢ 研究部預估矽格 26F 營收 238.7 億元（年增 21.9%）、毛利率 33.5%、EPS 9.9 元；27F 營收 298 億元（年增 24.9%）、毛利率 35.1%、EPS 13.6 元。湖口二廠產能開出快於預期，加上 AI 及光通訊測試需求成長、測試價格調升，27 年中興三廠支撐營收與獲利續增；惟新增設備陸續到位後折舊增加，毛利率擴張幅度可能收斂，上調目標價 327 元(24x2027(F)PER)，潛在上漲空間 15.5%，投資評等為「逢低買進」。

評等分級：買進（預期未來六個月股價漲幅超過+30%）；逢低買進（預期未來六個月股價漲幅介於+15%與+30%之間）；區間操作（預期未來六個月股價漲跌幅介於-15%與+15%之間）；逢高賣出（預期未來六個月股價跌幅介於-15%與-30%之間）；賣出（預期未來六個月股價跌幅超過-30%）。

本刊所刊載之內容僅做為參考，惟已力求正確與完整，但因時間及市場客觀因素改變所造成產業、市場或個股之相關條件改變，投資人需自行考量投資之需求與風險，本公司恕不負擔任何法律責任。任何轉載或引用本報告內容必須經本公司同意。

簡 要 財 務 預 估 資 訊												
會計年度	營收		營業利益		稅後純益		每股盈餘		每股現金股利(元)	ROE (%)	PER (X)	PBR (X)
	(億元)	YoY(%)	(億元)	YoY(%)	(億元)	YoY(%)	(元)	YoY(%)				
2025	195.9	7.5	39.7	29.1	29.5	6.6	6.14	4.2	4.0	14.2	19.1	2.3
2026(F)	238.6	21.8	59.1	49.1	47.9	62.2	9.90	61.3	4.2	17.4	28.1	4.5
2027(F)	297.9	24.8	81.9	38.5	65.9	37.4	13.61	37.4	0.0	19.7	20.4	3.7

資料來源：CMoney；兆豐投顧預估

營 運 分 析

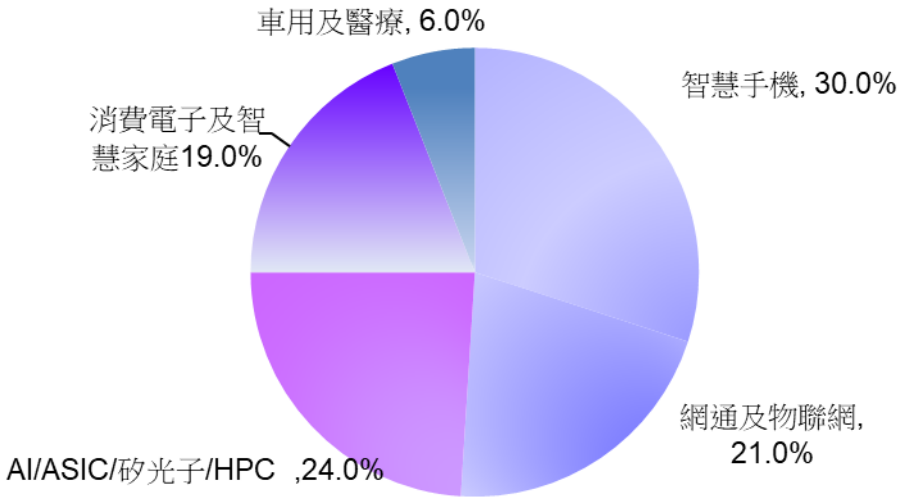
矽格為台灣領先 IC 封測廠，攜手台星科布局矽光子封測

矽格(6257)成立於 1995 年，總部位於新竹，為台灣主要 IC 封裝與測試廠商之一，測試營收占比達 90-95%，封裝占比約 5-10%。公司提供晶圓測試(CP，占 30%）及成品測試（FT，占 70%）服務，並具備 WLCSP、RF QFN 等封裝能力，可提供客戶一站式 turnkey 服務。

產品應用方面，2026 年 1 至 8 月依應用別，智慧手機占營收約 30%、網通與物聯網 21%、AI／ASIC／矽光子／高速運算 24%、消費電子與智慧家庭 19%、車用與醫療 6%；公司積極調整產品結構，提升 AI、高速運算與矽光子等高毛利產品比重，目標中長期 AI 相關營收占比達到與手機業務相當水準。

矽格透過子公司台星科(6335)布局先進封裝領域，台星科專注晶圓級封裝（Bumping），目前 N3/N5 製程已量產、N2 完成認證。在矽光子領域，台星科負責 Bumping 封裝、矽格負責測試，並與客戶共同開發 CPO 相關產品。公司自主研發高速高頻 SoC 測試設備（MAP），切入矽光子與 RF IC 高階測試需求，提供客戶客製化測試解決方案。

圖一、矽格(6257) 2026 1~8 月產品別營收比重。



資料來源：矽格，兆豐投顧彙整

評等分級：買進（預期未來六個月股價漲幅超過+30%）；逢低買進（預期未來六個月股價漲幅介於+15%與+30%之間）；區間操作（預期未來六個月股價漲跌幅介於-15%與+15%之間）；逢高賣出（預期未來六個月股價跌幅介於-15%與-30%之間）；賣出（預期未來六個月股價跌幅超過-30%）。

本刊所刊載之內容僅做為參考，惟已力求正確與完整，但因時間及市場客觀因素改變所造成產業、市場或個股之相關條件改變，投資人需自行考量投資之需求與風險，本公司恕不負擔任何法律責任。任何轉載或引用本報告內容必須經本公司同意。

26Q2 毛利率優於預期，上修 2026 年獲利

26Q2 合併營收 58.8 億元，季增 11.1%、年增 23.1%，毛利率 34.2%，較 26Q1 增加 2.5ppt，營益率 25.6%，歸屬母公司淨利 12.2 億元、EPS 2.55 元。毛利率較研究部原估 32.5%高 1.7ppt，主要受惠 AI 與高階測試比重提高、部分測試價格調升，以及折舊增幅仍低於營收成長。

2026 年 1 至 8 月 AI、CPU/GPU、ASIC 及矽光子營收 36.1 億元，年增 43%，占營收 24%，高於去年同期的 20%；記憶體營收 15.9 億元，年增 32%。AI 晶片腳數、功耗及傳輸頻寬提高，使測試項目及單顆測試時間增加，高階測試營收增幅因而高於晶片出貨量。

湖口二廠於 7 月開始量產後，產能開出速度快於研究部原先預估，26H2 營收貢獻亦高於前次模型。研究部上修 2026 年營收由 233.0 億元至 238.7 億元，毛利率由 32.5%至 33.5%，EPS 由 9.46 元至 9.9 元。

AI 晶片測試時間拉長，高階測試產值持續增加

2026 年 1 至 8 月 AI、CPU/GPU、ASIC 及矽光子營收 36.1 億元，年增 43%，營收占比由去年同期 20%提高至 24%；記憶體營收 15.9 億元，年增 32%，均高於公司整體營收成長。AI 晶片腳數、功耗及頻寬持續提高，使測試項目與單顆測試時間增加，帶動測試產值增幅高於晶片出貨量。

目前 AI、ASIC 及記憶體測試需求維持高檔，公司亦陸續調整部分測試價格。隨高階測試需求增加，公司 2027 年 AI 相關營收占比目標超過三成，仍為主要成長來源。

光通訊測試亦持續擴大，目前測試範圍已由 EIC 延伸至 PIC 及 Optical Engine (Insertion 1-3)，相關產品除電性測試外，亦需進行光訊號測試及光纖對位，測試時間與設備要求高於傳統 IC。預期 2027 年公司光通訊測試營收成長優於 2026。

湖口二廠開始放量，中興三廠接續 2027 年擴產

湖口二廠總建物面積約 8,500 坪，7 月開始量產，10 月產能已開出約四成，目前月營收約 1 億元；以約八成稼動率視為滿載，滿載月營收約 2 至 3 億元。依目前爬坡速度，預計 2027 年初達滿載，較研究部原先預估進度快。

湖口二廠量產後已開始反映於營收。26Q3 合併營收 61.7 億元，較 26Q2 增加 2.9 億元，期間台星科營收下滑，因此營收成長主要來自矽格本體。湖口二廠產能爬坡速度優於預期，不僅提高 26H2 營收，亦墊高 2027 年的營收基期。

中興三廠總樓地板面積約 1.1 萬坪，預計 26Q4 完工、27Q1 量產，設備將依客戶需求陸續進駐，新增產能主要投入 AI、高階運算及光通訊測試。研究部預估 2027 年上半年成長以湖口二廠全年貢獻為主，中興三廠則隨設備到位逐季增加營收。

母公司 2026 年資本支出提高至 88 億元，主要投入新廠及高階測試設備；矽格聯測 7 月另追加 20 億元，用於擴充美系 CPU/GPU 客戶測試產能。設備到位速度將影響 2027 年新增產能開出/營收貢獻節奏。

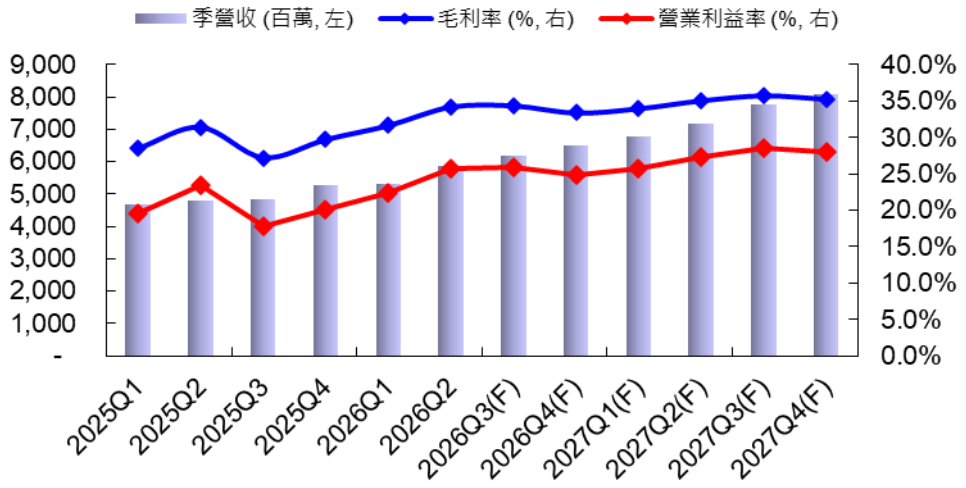
評等分級：買進（預期未來六個月股價漲幅超過+30%）；逢低買進（預期未來六個月股價漲幅介於+15%與+30%之間）；區間操作（預期未來六個月股價漲跌幅介於-15%與+15%之間）；逢高賣出（預期未來六個月股價跌幅介於-15%與-30%之間）；賣出（預期未來六個月股價跌幅超過-30%）。

本刊所刊載之內容僅做為參考，惟已力求正確與完整，但因時間及市場客觀因素改變所造成產業、市場或個股之相關條件改變，投資人需自行考量投資之需求與風險，本公司恕不負擔任何法律責任。任何轉載或引用本報告內容必須經本公司同意。

台星科擴充晶圓測試，新增產能主要自 27H2 貢獻

台星科 2026 年上半年營收 27.2 億元，年增 20%，毛利率 27.5%。因應 AI 及 HPC 客戶需求，公司 2026 年資本支出由原規劃 56.48 億元提高至 134.27 億元，主要投入測試及晶圓級封裝相關設備；2027 年 Bumping、Testing 及 Assembly 產品線均有擴充需求，其中晶圓測試為主要擴產項目之一。先進封裝方面，55×55mm FCBGA 已進入量產，並持續開發 100×100mm 以上大尺寸 FCBGA 及 2.5D、CoWoS 相關製程。由於設備交期較長，新增設備預計主要自 27H2 陸續到位，相關產能將增加 2027 年下半年營收；另一方面，大幅增加的資本支出亦將帶動折舊提高，因此台星科對 2027 年集團獲利的貢獻仍取決於新設備到位及稼動率提升速度，集團毛利率改善仍以 AI、光通訊等高階測試占比提高及測試價格調整為主。

圖二、矽格(6257)各季營收及獲利走勢預估



資料來源：Cmoney；兆豐投顧預估

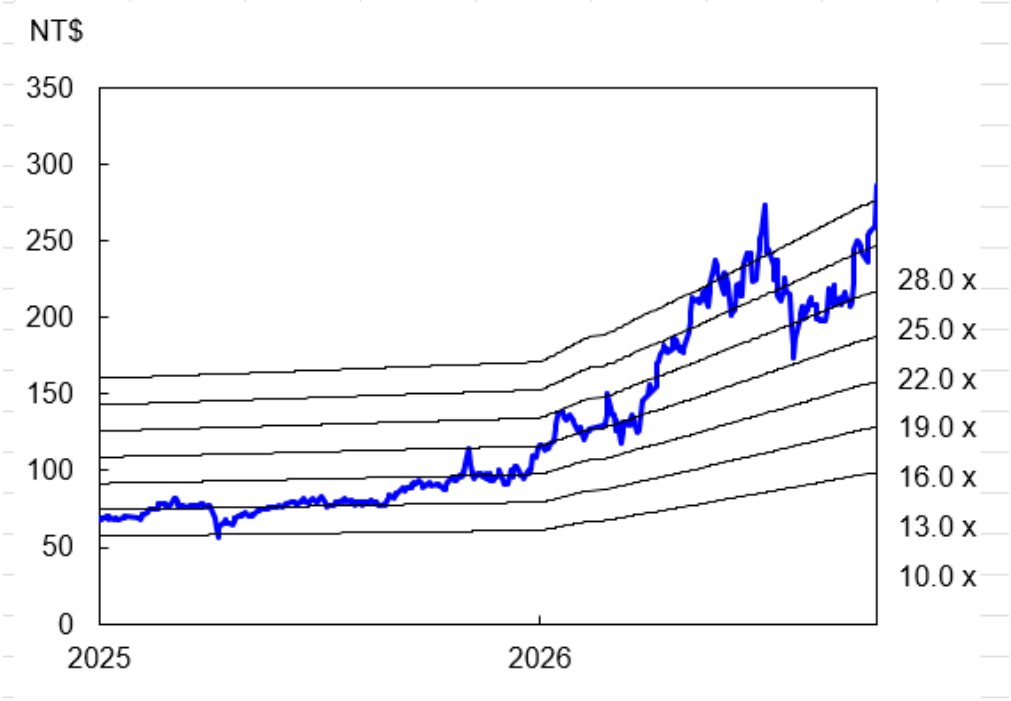
高階測試需求續增，產能擴充效益逐步反映

研究部預估 2026 年營收 238.7 億元，年增 21.9%，毛利率 33.5%、EPS 9.9 元；2027 年營收 298 億元，年增 24.9%，毛利率 35.1%、EPS 13.6 元。評價方面，矽格 26Q2 獲利優於預期，湖口二廠爬坡速度亦快於原估，加上 AI、高階運算及光通訊測試需求維持成長，研究部上修 2026/27F EPS 至 9.9／13.6 元。前次報告考量 AI Connectivity 與矽光子同步推升營收及毛利率，將評價基準上調至 26x 27F PER，已位於公司歷史評價高檔；本次雖進一步上修 27F 獲利，但考量 27H2 中興三廠及台星科新增設備陸續到位後折舊增加，獲利增速與毛利率擴張幅度可能較上半年收斂，因此將評價調整至 24x 27F PER。以 27F EPS 13.6 元計算，目標價由 301 元上調至 327 元，潛在上漲空間 15.5%，投資評等由「買進」調整為「逢低買進」。

評等分級：買進（預期未來六個月股價漲幅超過+30%）；逢低買進（預期未來六個月股價漲幅介於+15%與+30%之間）；區間操作（預期未來六個月股價漲跌幅介於-15%與+15%之間）；逢高賣出（預期未來六個月股價跌幅介於-15%與-30%之間）；賣出（預期未來六個月股價跌幅超過-30%）。

本刊所刊載之內容僅做為參考，惟已力求正確與完整，但因時間及市場客觀因素改變所造成產業、市場或個股之相關條件改變，投資人需自行考量投資之需求與風險，本公司恕不負擔任何法律責任。任何轉載或引用本報告內容必須經本公司同意。

圖三、矽格(4979) P/E Band



資料來源：CMoney；兆豐投顧預估

評等分級：買進（預期未來六個月股價漲幅超過+30%）；逢低買進（預期未來六個月股價漲幅介於+15%與+30%之間）；區間操作（預期未來六個月股價漲跌幅介於-15%與+15%之間）；逢高賣出（預期未來六個月股價跌幅介於-15%與-30%之間）；賣出（預期未來六個月股價跌幅超過-30%）。

本刊所刊載之內容僅做為參考，惟已力求正確與完整，但因時間及市場客觀因素改變所造成產業、市場或個股之相關條件改變，投資人需自行考量投資之需求與風險，本公司恕不負擔任何法律責任。任何轉載或引用本報告內容必須經本公司同意。

財務報表

季度損益表

單位：新台幣百萬元

6257矽格	2025	26Q1	26Q2(F)	26Q3(F)	26Q4(F)	2026(F)	27Q1(F)	27Q2(F)	27Q3(F)	27Q4(F)	2027(F)
營業收入淨額	**19,587**	**5,294**	**5,881**	**6,172**	**6,518**	**23,864**	**6,759**	**7,164**	**7,766**	**8,100**	**29,789**
營業成本	**13,864**	3,618	3,867	4,049	4,341	**15,875**	4,461	4,650	4,986	5,241	**19,337**
營業毛利淨額	**5,723**	**1,676**	**2,013**	**2,123**	**2,177**	**7,989**	**2,298**	**2,515**	**2,780**	**2,859**	**10,452**
營業費用	**1,756**	**492**	**505**	**525**	**554**	**2,076**	**554**	**559**	**567**	**583**	**2,263**
營業淨利(損)	**3,968**	**1,184**	**1,508**	**1,599**	**1,623**	**5,914**	**1,744**	**1,956**	**2,213**	**2,276**	**8,189**
營業外收支											
利息收入	**271**	59	65	70	70	**263**	60	65	70	76	**271**
財務成本	**219**	54	49	58	57	**218**	55	54	58	56	**223**
其他利益及損失	**-291**	136	15	80	100	**331**	70	80	100	75	**325**
其他收支淨額	**165**	55	37	35	35	**163**	35	35	35	35	**140**
營業外收支淨額合計	**-74**	**196**	**68**	**127**	**148**	**538**	**110**	**126**	**147**	**130**	**513**
稅前淨利	**3,893**	**1,380**	**1,576**	**1,726**	**1,771**	**6,452**	**1,854**	**2,082**	**2,360**	**2,406**	**8,702**
所得稅費用(利益)	**563**	258	217	328	336	**1,140**	334	375	425	433	**1,566**
稅後淨利-非控制權益	**376**	116	138	130	136	**520**	131	143	136	140	**550**
稅後淨利-歸屬母公司	**2,955**	**1,006**	**1,220**	**1,268**	**1,298**	**4,792**	**1,389**	**1,564**	**1,799**	**1,833**	**6,586**
普通股股本	**4,839**	4,839	4,839	4,839	4,839	**4,839**	4,839	4,839	4,839	4,839	**4,839**
每股盈餘(元)	**6.14**	**2.10**	**2.55**	**2.62**	**2.68**	**9.90**	**2.87**	**3.23**	**3.72**	**3.79**	**13.61**
季成長分析(QoQ,%)											
營收成長率		0.5	11.1	5.0	5.6		3.7	6.0	8.4	4.3	
營業毛利成長率		7.1	20.1	5.5	2.5		5.6	9.4	10.6	2.8	
營業利益成長率		11.3	27.4	6.0	1.5		7.5	12.2	13.2	2.8	
稅前淨利成長率		9.1	14.2	9.5	2.6		4.7	12.3	13.4	1.9	
稅後淨利成長率		-4.9	21.3	3.9	2.4		7.0	12.6	15.1	1.9	
年成長分析(YoY,%)											
營收成長率	**7.5**	13.0	23.1	27.2	23.7	**21.8**	27.7	21.8	25.8	24.3	**24.8**
營業毛利成長率	**20.3**	25.6	34.0	60.7	39.1	**39.6**	37.1	24.9	31.0	31.4	**30.8**
營業利益成長率	**29.1**	29.2	34.4	84.6	52.6	**49.1**	47.3	29.7	38.5	40.3	**38.5**
稅前淨利成長率	**-0.4**	27.7	266.6	54.4	40.0	**65.7**	34.4	32.1	36.8	35.9	**34.9**
稅後淨利成長率	**6.6**	32.8	263.6	57.8	22.7	**62.2**	38.1	28.2	42.0	41.2	**37.4**
獲利分析(%)											
營業毛利率	**29.2**	31.7	34.2	34.4	33.4	**33.5**	34.0	35.1	35.8	35.3	**35.1**
營業利益率	**20.3**	22.4	25.6	25.9	24.9	**24.8**	25.8	27.3	28.5	28.1	**27.5**
稅前淨利率	**19.9**	26.1	26.8	28.0	27.2	**27.0**	27.4	29.1	30.4	29.7	**29.2**
稅後淨利率	**15.1**	19.0	20.8	20.5	19.9	**20.1**	20.6	21.8	23.2	22.6	**22.1**
實質所得稅率	**14.5**	18.7	13.8	19.0	19.0	**17.7**	18.0	18.0	18.0	18.0	**18.0**

資料來源：CMoney；兆豐投顧預估；註：每股盈餘以期末股本計算

評等分級：買進（預期未來六個月股價漲幅超過+30%）；逢低買進（預期未來六個月股價漲幅介於+15%與+30%之間）；區間操作（預期未來六個月股價漲跌幅介於-15%與+15%之間）；逢高賣出（預期未來六個月股價跌幅介於-15%與-30%之間）；賣出（預期未來六個月股價跌幅超過-30%）。

本刊所刊載之內容僅做為參考，惟已力求正確與完整，但因時間及市場客觀因素改變所造成產業、市場或個股之相關條件改變，投資人需自行考量投資之需求與風險，本公司恕不負擔任何法律責任。任何轉載或引用本報告內容必須經本公司同意。

資產負債表

單位：NT$百萬元	2025	2026(F)	2027(F)
流動資產			
現金及約當現金	12,719	17,043	19,591
存貨	588	456	816
應收帳款及票據	4,733	6,382	7,493
其他流動資產	2,698	4,586	4,428
流動資產合計	**20,738**	**28,466**	**32,328**
非流動資產			
長期投資合計	3,860	4,262	5,356
固定資產	16,816	18,696	21,216
其它非流動資產	1,607	483	3,599
非流動資產合計	**22,284**	**23,441**	**30,172**
資產總計	**43,022**	**51,908**	**62,500**
流動負債			
應付帳款及票據	396	561	604
短期借款	419	770	770
其他流動負債	8,888	9,617	14,020
流動負債合計	**9,702**	**10,949**	**15,394**
長期負債			
長期借款	7,410	10,165	10,165
其他非流動負債	1,033	615	176
長期負債合計	**8,443**	**10,780**	**10,341**
負債總計	**18,146**	**21,729**	**25,735**
權益			
普通股股本	4,839	4,839	4,839
資本公積	1,038	221	221
保留盈餘	14,397	17,254	23,839
其他項目	4,602	7,865	7,865
權益總計	**24,876**	**30,179**	**36,764**
負債與權益總計	**43,022**	**51,908**	**62,500**

資料來源：CMoney；兆豐投顧預估

現金流量表

單位：NT$百萬元	2025	2026(F)	2027(F)
本期淨利	2,955	4,792	6,586
折舊及攤提	4,038	5,048	5,516
本期營運資金變動	-566	-1,682	-1,514
其他營業活動現金流量	-10	438	214
營業活動之淨現金流入(出)	**6,416**	**8,596**	**10,802**
資本支出淨額	-5,311	-4,700	-5,600
本期長期投資變動	1,920	402	1,094
其他投資活動現金流量	-241	760	282
投資活動之淨現金流入(出)	-3,633	-3,539	-4,224
現金增資	0		0
本期借款變動	1,471	2,337	-439
其他籌資活動現金流量	-2,819		-3,591
融資活動之淨現金流入(出)	**-1,348**	**-734**	**-4,030**
本期現金與約當現金增加數	**1,470**	**4,324**	**2,548**
期初現金與約當現金餘額	11,249	12,719	17,043
期末現金與約當現金餘額	**12,719**	**17,043**	**19,591**

資料來源：CMoney；兆豐投顧預估

重要財務比率分析

	2025	2026(F)	2027(F)
年成長率(%)			
營業收入	7.5	21.8	24.8
營業利益	29.1	49.1	38.5
稅後純益	6.6	62.2	37.4
獲利能力分析(%)			
營業毛利率	29.2	33.5	35.1
營業利益率	20.3	24.8	27.5
稅後純益率	15.1	20.1	22.1
資產報酬率	7.2	10.1	11.5
股東權益報酬率	14.2	17.4	19.7
償債能力分析			
負債/股東權益(%)	72.9	72.0	70.0
流動比率(X)	2.1	2.6	2.1
每股資料分析			
每股盈餘(NT$元)	6.1	9.9	13.6
本益比(X)	19.1	28.1	20.4
每股淨值(NT$元)	45.5	62.4	76.0
股價淨值比(X)	2.3	4.5	3.7
每股現金股利(NT$元)	4.0	4.2	0.0
現金股利殖利率(%)	5.1	1.6	0.0

資料來源：CMoney；兆豐投顧預估

評等分級：買進（預期未來六個月股價漲幅超過+30%）；逢低買進（預期未來六個月股價漲幅介於+15%與+30%之間）；區間操作（預期未來六個月股價漲跌幅介於-15%與+15%之間）；逢高賣出（預期未來六個月股價跌幅介於-15%與-30%之間）；賣出（預期未來六個月股價跌幅超過-30%）。

本刊所刊載之內容僅做為參考，惟已力求正確與完整，但因時間及市場客觀因素改變所造成產業、市場或個股之相關條件改變，投資人需自行考量投資之需求與風險，本公司恕不負擔任何法律責任。任何轉載或引用本報告內容必須經本公司同意。

6257 矽格 永續發展概況

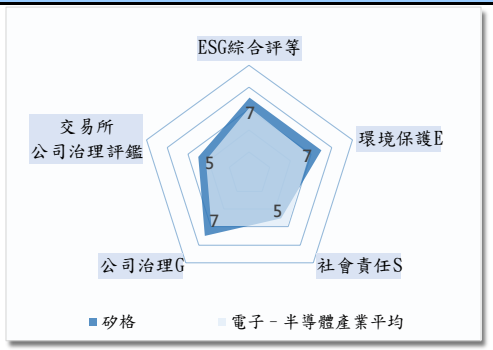
CMoney ESG Rating　　評鑑年季：2026Q3

落後			平均				領先		
1	2	3	4	5	6	7	8	9	10

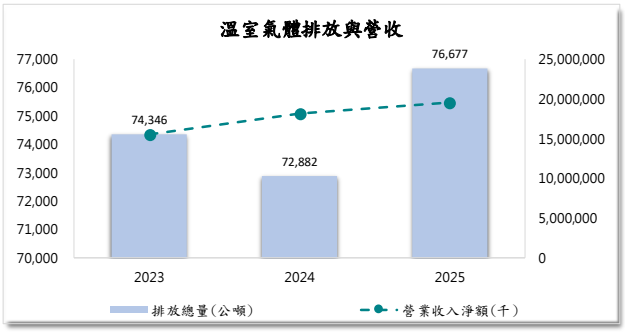
E(環境保護)	S(社會責任)	G(公司治理)	永續報告書連結
7	5	7	https://esggenplus.twse.com.tw/inquiry/info/individual

產業類別	產業樣本數	ESG評等產業排名	TCFD報告書簽署	TCFD報告書連結
電子－半導體	207	37	否	否

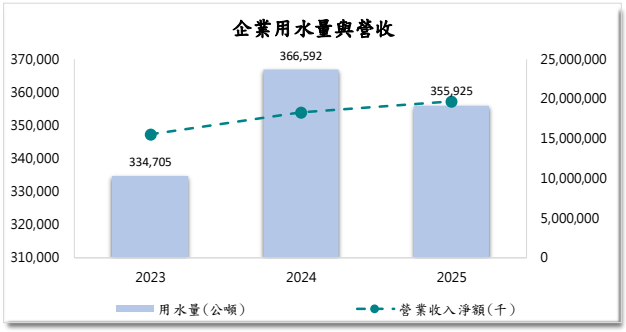
資料來源：CMoney
說明：評等1-10分，10分最高



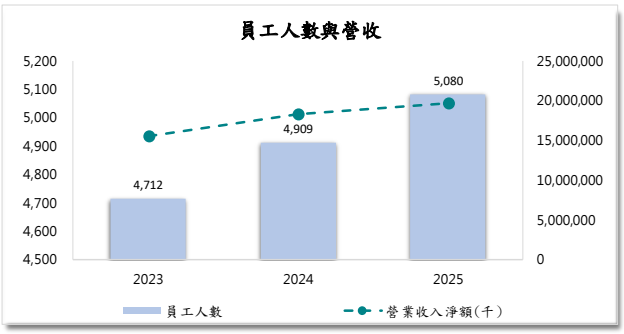
溫室氣體



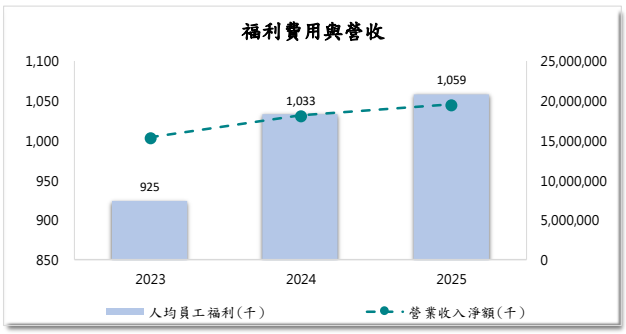
水資源



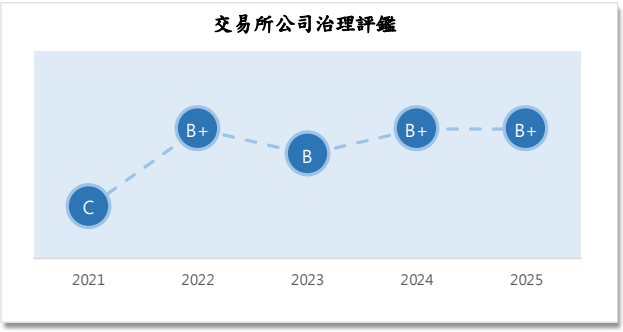
員工人數



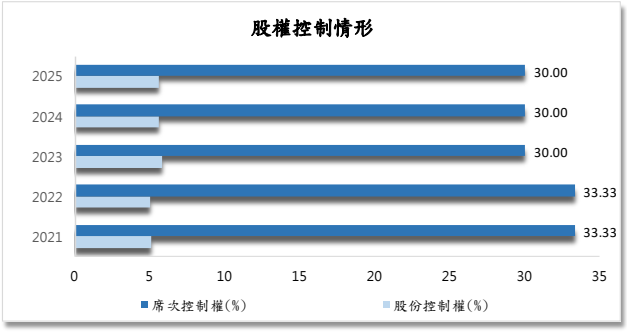
員工福利



公司治理評鑑



股權控制



資料來源：交易所

評等分級：買進（預期未來六個月股價漲幅超過+30%）；逢低買進（預期未來六個月股價漲幅介於+15%與+30%之間）；區間操作（預期未來六個月股價漲跌幅介於-15%與+15%之間）；逢高賣出（預期未來六個月股價跌幅介於-15%與-30%之間）；賣出（預期未來六個月股價跌幅超過-30%）。

本刊所刊載之內容僅做為參考，惟已力求正確與完整，但因時間及市場客觀因素改變所造成產業、市場或個股之相關條件改變，投資人需自行考量投資之需求與風險，本公司恕不負擔任何法律責任。任何轉載或引用本報告內容必須經本公司同意。

91APP*-KY

(6741.TW/6741 TT)

AgentOne 讓企業 AI 從對話走向執行

未評等

收盤價 October 12 (NT$)	54.00
12 個月目標價 (NT$)	N/A
前次目標價 (NT$)	N/A
維持 / 調升 / 調降 (%)	N/A
上漲空間 (%)	N/A

焦點內容

1. 3Q26 營收符合預期。
2. iCHEF 整併快於預期。
3. AgentOne 2027 上線鎖定 Agentic AI 商機。

交易資料表

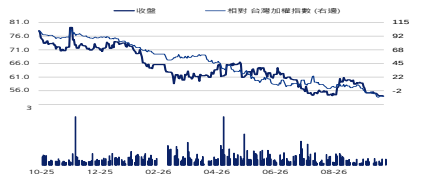
市值：(NT$bn/US$mn)	6.51 / 204.4
流通在外股數 (百萬股)：	120.6
外資持有股數 (百萬股)：	61
3M 平均成交量 (百萬股)：	0.19
52 週股價 (低 \ 高) (NT$):	54.00 –79.20

股價表現	3M	6M	12M
絕對表現 (%)	-10	-11.3	-30.8
相對表現 (%)	-18.7	-50.5	-111.4

每股盈餘

NT$	1Q	2Q	3Q	4Q
2024	1.02	1.00	0.88	1.32
2025	1.15	0.82	0.99	1.45
2026	1.15	1.15	N.A.	N.A.

股價圖



資料來源：TEJ

重要訊息

我們近期拜訪管理層更新產業動態與公司近況。

評論及分析

3Q26 營收符合預期。3Q26 營收 5.59 億元，年增 32.5%，大致符合市場共識 5.64 億元。台灣電商環境成長維持溫和，1H26 電商銷售額年增 11.3%，排除其他零售業的實體電商 D2C 銷售額年增 5.3%，7M-8M26 電商銷售額分別年增 5.5/8.5%。管理層看好中大型品牌因原有小型電商系統無法支撐 OMO 擴張而轉換平台，91APP 以全通路 IMS（線上、門市、倉儲庫存打通）、Super APP 系統整合與在地團隊勝出，同時進行營收結構轉型，GMV 抽成的系統使用費占比由 1H25 59.6% 降至 1H26 的 48.6%。

iCHEF 整併快於預期。iCHEF 團隊已於 7 月遷入總部，快於原訂年底之目標，嚴格的營業費用與人力管理已讓 iCHEF 2Q26 營業利益率顯著提升，透過 91APP 優勢與擅長系統整合之能力，iCHEF 已取得中大型連鎖餐飲集團簽約採用，線上刷卡轉入 91APP Payments 以較低手續費擴大金流服務滲透，管理層看好 1Q27 併表效應結束後仍可提供不俗的營收成長貢獻。

AgentOne 2027 上線鎖定 Agentic AI 商機。 AgentOne 於 7 月客戶大會發表，2H26 與客戶進行 POC，2027 正式上線。首波主打為 MediaOne（廣告企劃到投放的自動化）與 EngageOne（個人化行銷），也已接入 iCHEF 餐飲場景。目標市場是 91APP 既有的 KA/SKA 品牌與通路客戶，優勢在於客戶第一方數據都存放在 91APP 平台，可直接執行複雜工作流並完善治理。AgentOne 變現路徑：(1)GMV 提升帶動系統使用費；(2)訂閱與 token 費；(3)客戶擴大廣告投放預算。為該司 2027 主要成長動能之一。

投資建議

凱基正向看待 AgentOne 直擊企業轉型需求。目前股價交易在 2027 市場共識每股盈餘 6.23 元之 8.7 倍 PE(約當 5 年平均之-1.5 倍標準差)。

投資風險

D2C 成長不如預期，Agent 產品滲透率有限。

主要財務數據及估值

	Dec-21A	Dec-22A	Dec-23A	Dec-24A	Dec-25A
營業收入 (NT$百萬)	1,101	1,263	1,389	1,619	1,807
營業毛利 (NT$百萬)	830	948	1,041	1,205	1,341
營業利益 (NT$百萬)	368	411	423	529	589
稅後淨利 (NT$百萬)	298	338	398	498	508
每股盈餘 (NT$)	2.58	2.83	3.37	4.22	4.40
每股現金股利 (NT$)	0.49	0.70	0.95	1.69	2.05
每股盈餘成長率 (%)	34.4	9.6	19.0	25.0	4.5
本益比 (x)	20.9	19.1	16.0	12.8	12.3
股價淨值比 (x)	1.4	1.3	1.2	1.0	1.0
EV/EBITDA (x)	11.6	2.1	4.7	3.0	2.3
淨負債比率 (%)	Net cash	Net cash	Net cash	Net cash	Net cash
殖利率 (%)	0.9	1.3	1.8	3.1	3.8
股東權益報酬率 (%)	18.3	13.8	15.0	16.4	15.8

資料來源：公司資料，凱基

凱基投顧

鄭傑懋
886.2.2181.8748
andrew.cheng@kgi.com

黃恩定
886.2.2181.8736
andy.et.huang@kgi.com

重要免責聲明，詳見最終頁

圖 1: 公司概況

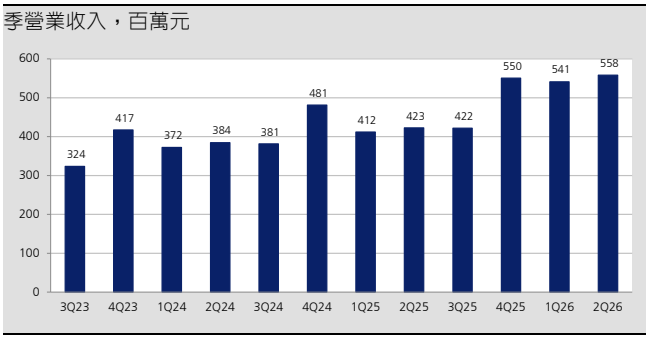
創立於 2013 年的 91APP，專注於提供雲端軟體加值服務，並專精於電商與虛實融合 (OMO) 的領域。它是最早提供 D2C 解決方案的 SaaS 供應商，主力在幫助零售業者完成數位轉型。91APP 主要提供商務解決方案及行銷解決方案，1H26 營收組成分成四大項：系統使用費(占比 45-50%)、軟體訂閱費(25-30%)、廣告 x 電商服務(15-20%)及一次性及專案收入(5-10%)。

資料來源：凱基

圖 2: 91APP 客戶群為中大型品牌商及通路業者

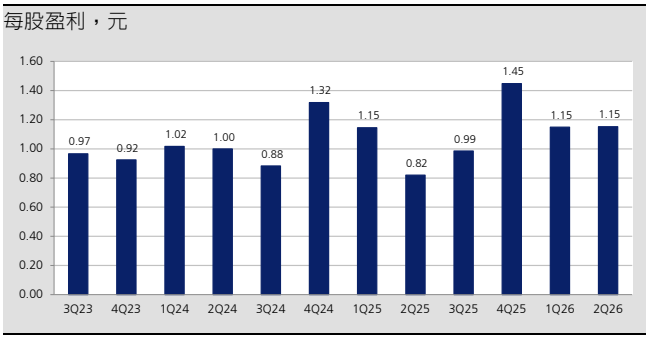
資料來源：凱基

圖 3: 季營業收入



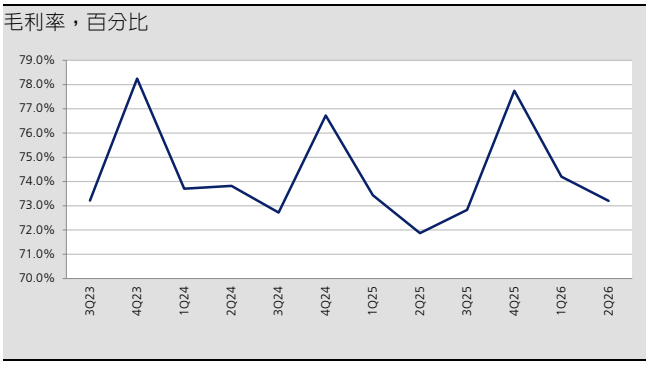
資料來源：凱基

圖 4: 每股盈利



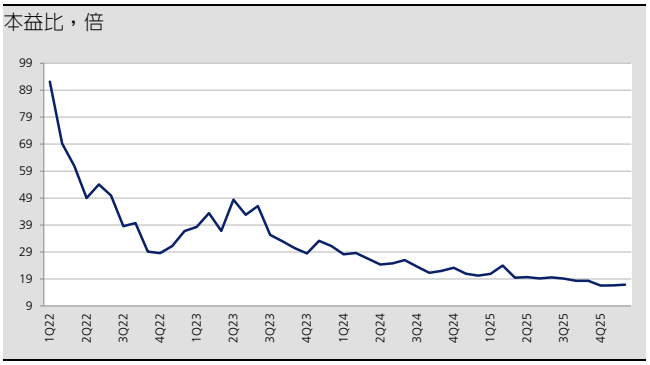
資料來源：凱基

圖 5: 毛利率



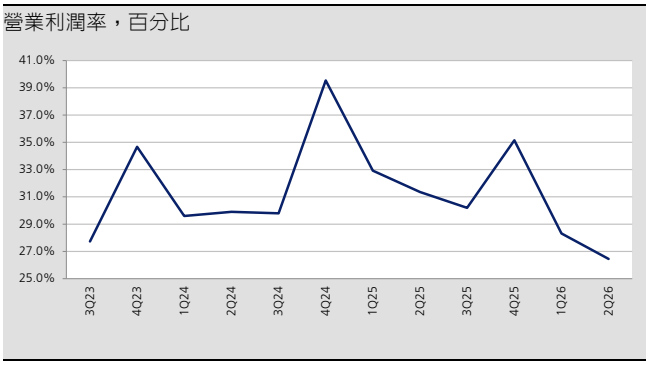
資料來源：凱基

圖 6: 本益比



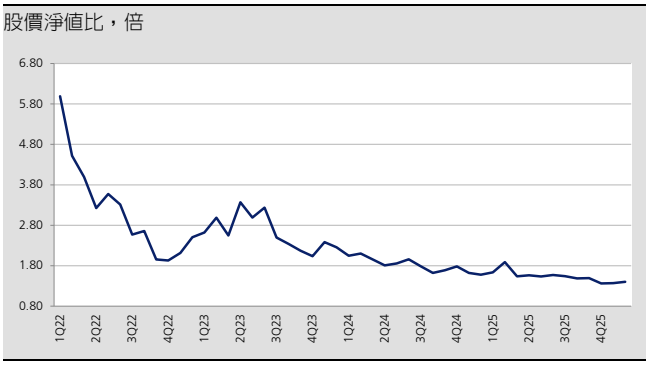
資料來源：凱基

圖 7: 營業利潤率



資料來源：凱基

圖 8: 股價淨值比



資料來源：凱基

損益表

	季度								年度		
	Sep-24A	Dec-24A	Mar-25A	Jun-25A	Sep-25A	Dec-25A	Mar-26A	Jun-26A	Dec-23A	Dec-24A	Dec-25A
損益表 (NT$百萬)											
營業收入	**381**	**481**	**412**	**423**	**422**	**550**	**541**	**558**	**1,389**	**1,619**	**1,807**
營業成本	(104)	(112)	(109)	(119)	(115)	(122)	(140)	(150)	(348)	(415)	(465)
營業毛利	**277**	**369**	**302**	**304**	**307**	**428**	**402**	**409**	**1,041**	**1,205**	**1,341**
營業費用	(164)	(179)	(167)	(171)	(180)	(234)	(248)	(261)	(618)	(676)	(752)
營業利益	**114**	**190**	**136**	**132**	**127**	**193**	**153**	**148**	**423**	**529**	**589**
折舊	(9)	(9)	(8)	(7)	(7)	(9)	(11)	(12)	(38)	(37)	(31)
攤提	(0)	(1)	(1)	(1)	(1)	(2)	(4)	(4)	(1)	(2)	(4)
EBITDA	**124**	**200**	**144**	**140**	**135**	**204**	**169**	**164**	**462**	**568**	**624**
利息收入	20	24	18	20	15	19	12	19	102	89	71
投資利益淨額	0	-	1	0	-	-	0	0	1	0	-
其他營業外收入	1	1	0	1	0	4	1	1	5	4	5
總營業外收入	**21**	**25**	**19**	**20**	**15**	**23**	**14**	**20**	**108**	**93**	**76**
利息費用	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(1)	(1)
投資損失	(5)	(15)	-	-	-	1	-	-	(18)	(24)	(1)
其他營業外費用	(2)	(3)	11	(23)	1	(0)	1	(1)	5	17	(8)
總營業外費用	**(7)**	**(18)**	**10**	**(24)**	**0**	**1**	**0**	**(2)**	**(13)**	**(7)**	**(11)**
稅前純益	**128**	**198**	**165**	**129**	**142**	**217**	**167**	**166**	**518**	**616**	**654**
所得稅費用[利益]	(24)	(41)	(31)	(35)	(29)	(50)	(33)	(34)	(119)	(118)	(145)
少數股東損益	0	(1)	(0)	(0)	(1)	(1)	(2)	(1)	(0)	0	(2)
非常項目前稅後純益	**104**	**156**	**134**	**94**	**113**	**167**	**132**	**131**	**398**	**498**	**508**
非常項目	0	0	(0)	0	0	0	0	(0)	-	0	(0)
稅後淨利	**104**	**156**	**134**	**94**	**113**	**167**	**132**	**131**	**398**	**498**	**508**
每股盈餘 (NT$)	0.88	1.32	1.15	0.82	0.99	1.45	1.15	1.15	3.37	4.22	4.40
獲利率 (%)											
營業毛利率	72.7	76.7	73.4	71.9	72.8	77.7	74.2	73.2	75.0	74.4	74.2
營業利益率	29.8	39.5	32.9	31.4	30.2	35.2	28.3	26.5	30.5	32.7	32.6
EBITDA Margin	32.4	41.5	34.9	33.2	32.1	37.1	31.2	29.3	33.3	35.1	34.5
稅前純益率	33.7	41.1	40.1	30.6	33.7	39.5	30.9	29.7	37.3	38.0	36.2
稅後純益率	27.3	32.3	32.5	22.2	26.7	30.3	24.4	23.4	28.7	30.7	28.1
季成長率 (%)											
營業收入	(0.8)	26.2	(14.4)	2.6	(0.2)	30.5	(1.7)	3.1			
營業毛利	(2.3)	33.1	(18.1)	0.4	1.1	39.3	(6.2)	1.8			
營業收益增長	(1.2)	67.5	(28.7)	(2.3)	(3.9)	51.9	(20.8)	(3.6)			
EBITDA	(0.9)	61.8	(28.1)	(2.3)	(3.7)	51.0	(17.4)	(3.0)			
稅前純益	(9.6)	53.9	(16.5)	(21.7)	10.1	52.8	(23.2)	(0.8)			
稅後純益	(11.6)	49.2	(13.8)	(30.0)	20.1	47.8	(20.9)	(1.0)			
年成長率 (%)											
營業收入	17.8	15.4	10.6	9.9	10.6	14.4	31.4	32.1	10.0	16.6	11.6
營業毛利	17.0	13.1	10.2	7.0	10.8	15.9	32.7	34.5	9.8	15.7	11.3
營業收益	26.6	31.6	23.0	15.3	12.1	1.7	13.0	11.4	3.1	24.9	11.3
EBITDA	24.1	29.5	20.0	12.7	9.5	2.2	17.2	16.5	3.3	22.8	9.8
稅前純益	(6.5)	38.9	11.8	(8.9)	10.9	10.1	1.2	28.1	21.4	18.9	6.3
稅後純益	(8.6)	42.6	11.7	(20.4)	8.2	7.2	(1.7)	39.0	17.9	25.0	2.0

資料來源：公司資料，凱基

資產負債表

NT$百萬	Dec-21A	Dec-22A	Dec-23A	Dec-24A	Dec-25A
資產總額	**4,192**	**3,943**	**4,240**	**5,025**	**5,101**
流動資產	**3,891**	**3,612**	**3,614**	**4,193**	**3,150**
現金及短期投資	2,201	2,477	2,507	2,676	1,672
存貨	-	-	-	-	2
應收帳款及票據	55	77	81	114	133
其他流動資產	1,635	1,058	1,027	1,403	1,344
非流動資產	**301**	**331**	**625**	**832**	**1,951**
長期投資	187	243	564	755	900
固定資產	25	32	21	15	26
什項資產	89	56	40	63	1,025
負債總額	**1,780**	**1,434**	**1,426**	**1,771**	**1,912**
流動負債	**1,761**	**1,406**	**1,409**	**1,724**	**1,867**
應付帳款及票據	0	0	0	-	6
短期借款	-	-	-	-	-
什項負債	1,761	1,406	1,409	1,724	1,861
長期負債	**19**	**28**	**17**	**47**	**45**
長期借款	-	-	-	-	-
其他負債及準備	8	9	11	18	20
股東權益總額	**2,412**	**2,510**	**2,814**	**3,254**	**3,190**
普通股本	603	603	603	603	603
保留盈餘	608	858	1,140	1,487	1,746
少數股東權益	5	6	7	7	8
優先股股東資金	-	-	-	-	-

主要財務比率

	Dec-21A	Dec-22A	Dec-23A	Dec-24A	Dec-25A
年成長率					
營業收入	25.8%	14.7%	10.0%	16.6%	11.6%
營業收益增長	51.0%	11.5%	3.1%	24.9%	11.3%
EBITDA	48.3%	14.5%	3.3%	22.8%	9.8%
稅後純益	44.7%	13.2%	17.9%	25.0%	2.0%
每股盈餘成長率	34.4%	9.6%	19.0%	25.0%	4.5%
獲利能力分析					
營業毛利率	75.4%	75.1%	75.0%	74.4%	74.2%
營業利益率	33.4%	32.5%	30.5%	32.7%	32.6%
EBITDA Margin	35.5%	35.4%	33.3%	35.1%	34.5%
稅後純益率	27.1%	26.8%	28.7%	30.7%	28.1%
平均資產報酬率	9.9%	8.3%	9.7%	10.7%	10.0%
股東權益報酬率	18.3%	13.8%	15.0%	16.4%	15.8%
穩定 \ 償債能力分析					
毛負債比率 (%)	0.0%	0.0%	0.0%	0.0%	0.0%
淨負債比率	Net cash	Net cash	Net cash	Net cash	Net cash
利息保障倍數 (x)	1,290.5	950.6	1,188.6	820.6	469.6
利息及短期債保障倍數 (x)	1.0	1.0	1.0	1.0	1.0
Cash Flow Int. Coverage (x)	(926.5)	1,294.1	1,160.0	637.4	408.3
Cash Flow/Int. & ST Debt (x)	(926.5)	1,294.1	1,160.0	637.4	408.3
流動比率 (x)	2.2	2.6	2.6	2.4	1.7
速動比率 (x)	2.2	2.6	2.6	2.4	1.7
淨負債 (NT$百萬)	(644)	(2,467)	(1,500)	(1,631)	(1,588)
每股資料分析					
每股盈餘 (NT$)	2.58	2.83	3.37	4.22	4.40
每股現金盈餘 (NT$)	(2.33)	4.87	4.28	4.05	4.95
每股淨值 (NT$)	39.93	41.52	46.56	53.87	52.78
調整後每股淨值 (NT$)	20.85	21.00	23.77	27.51	27.61
每股營收 (NT$)	9.54	10.59	11.76	13.71	15.68
EBITDA/Share (NT$)	3.39	3.75	3.92	4.81	5.41
每股現金股利 (NT$)	0.49	0.70	0.95	1.69	2.05
資產運用狀況					
資產周轉率 (x)	0.37	0.31	0.34	0.35	0.36
應收帳款周轉天數	18.1	22.1	21.2	25.7	26.9
存貨周轉天數	-	-	-	-	1.3
應付帳款周轉天數	0.1	0.1	0.0	-	4.5
現金轉換周轉天數	18.0	22.0	21.2	25.7	23.7

資料來源：公司資料，凱基

損益表

NT$百萬	Dec-21A	Dec-22A	Dec-23A	Dec-24A	Dec-25A
營業收入	**1,101**	**1,263**	**1,389**	**1,619**	**1,807**
營業成本	(270)	(315)	(348)	(415)	(465)
營業毛利	**830**	**948**	**1,041**	**1,205**	**1,341**
營業費用	(462)	(538)	(618)	(676)	(752)
營業利益	**368**	**411**	**423**	**529**	**589**
總營業外收入	**11**	**46**	**108**	**93**	**76**
利息收入	5	31	102	89	71
投資利益淨額	-	-	1	0	-
其他營業外收入	6	15	5	4	5
總營業外費用	**(5)**	**(30)**	**(13)**	**(7)**	**(11)**
利息費用	(0)	(0)	(0)	(1)	(1)
投資損失	(5)	(12)	(18)	(24)	(1)
其他營業外費用	1	(17)	5	17	(8)
稅前純益	**374**	**426**	**518**	**616**	**654**
所得稅費用[利益]	(75)	(88)	(119)	(118)	(145)
少數股東損益	(0)	(1)	(0)	0	(2)
非常項目	-	0	0	0	(0)
稅後淨利	**298**	**338**	**398**	**498**	**508**
EBITDA	**391**	**448**	**462**	**568**	**624**
每股盈餘 (NT$)	2.58	2.83	3.37	4.22	4.40

現金流量

NT$百萬	Dec-21A	Dec-22A	Dec-23A	Dec-24A	Dec-25A
營運活動之現金流量	**(269)**	**581**	**506**	**479**	**570**
本期純益	298	338	398	498	508
折舊及攤提	23	37	39	39	35
本期運用資金變動	(13)	(22)	(4)	(33)	(11)
其他營業資產及負債變動	(577)	228	73	(25)	38
投資活動之現金流量	**(1,094)**	**1,525**	**(1,363)**	**(260)**	**(50)**
投資用短期投資出售[新購]	(915)	1,547	(1,362)	(248)	778
本期長期投資變動	(161)	-	-	-	-
資本支出淨額	(20)	(19)	(5)	(8)	(10)
其他資產變動	2	(3)	4	(4)	(818)
自由現金流	**283**	**322**	**355**	**424**	**469**
融資活動之現金流量	**1,257**	**(415)**	**(106)**	**(137)**	**(550)**
短期借款變動	-	-	-	-	-
長期借款變動	-	-	-	-	-
現金增資	1,270	-	-	-	-
已支付普通股股息	-	(58)	(83)	(112)	(200)
其他融資現金流	(13)	(22)	(23)	(24)	(21)
匯率影響數	**(18)**	**132**	**(4)**	**48**	**(13)**
本期產生現金流量	**(124)**	**1,823**	**(967)**	**130**	**(43)**

投資回報率

	Dec-21A	Dec-22A	Dec-23A	Dec-24A	Dec-25A
1 - 營業成本/營業收入					
- 銷管費用/營業收入	42.0%	42.6%	44.5%	41.7%	41.6%
= 營業利益率	**33.4%**	**32.5%**	**30.5%**	**32.7%**	**32.6%**
1 / (營業運用資金/營業收入	(0.1)	(0.2)	(0.2)	(0.1)	(0.2)
+ 淨固定資產/營業收入	0.0	0.0	0.0	0.0	0.0
+ 什項資產/營業收入)	0.1	0.0	0.0	0.0	0.0
= 資本周轉率	**75.9**	**(5.4)**	**(5.1)**	**(8.8)**	**(5.1)**
營業利益率	33.4%	32.5%	30.5%	32.7%	32.6%
x 資本周轉率	75.9	(5.4)	(5.1)	(8.8)	(5.1)
x (1 - 有效現金稅率)	79.9%	79.4%	77.0%	80.8%	77.8%
= 稅後 ROIC	**2026.4%**	**(140.2%)**	**(118.8%)**	**(233.0%)**	**(129.6%)**

資料來源：公司資料，凱基

凱基證券集團據點

中國	上海	上海市靜安區南京西路 1601 號越洋國際廣場 1507 室 郵政編號：200040
台灣	台北	104 台北市明水路 700 號 電話 886.2.2181.8888，傳真 886.2.8501.1691
香港		香港灣仔港灣道十八號中環廣場四十一樓 電話 852.2878.6888，傳真 852.2878.6800
泰國	曼谷	8th - 11th floors, Asia Centre Building 173 South Sathorn Road, Bangkok 10120, Thailand 電話 66.2658.8888，傳真 66.2658.8014
新加坡		珊頓大道 4 號#13-01 新交所第二大廈 郵政編號：068807 電話 65.6202.1188，傳真 65.6534.4826
印尼		Sona Topas Tower Fl.11 Jl. Jend. Sudirman kav.26 JakartaSelatan 12920 Indonesia 電話 62 21 250 6337

股價說明

等級	定義
增加持股 (OP)	對個股持正面看法，預期個股未來十二個月的表現超越凱基證券集團所追蹤的相關市場的總報酬。
持有 (N)	對個股持中性看法，預期個股未來十二個月的表現符合凱基證券集團所追蹤的相關市場的總報酬。
降低持股 (U)	對個股持負面看法，預期個股未來十二個月的表現低於凱基證券集團所追蹤的相關市場的總報酬。
未評等 (NR)	凱基證券未對該個股加以評等。
受法規限制 未評等 (R)	受凱基證券集團內部政策和/或相關法令限制使凱基證券集團無法進行某些形式的資訊交流，其中包括提供評等給投資人參考。
	*總報酬 = (十二個月目標價-現價)/現價

免責聲明

凱基證券投資顧問股份有限公司（以下簡稱本公司）為凱基金控集團之成員。本報告之內容皆來自本公司認可之資料來源，但不保證其完整性及精確性。報告內容所提及之各項業務、財務等相關檔案資料及所有的意見及預估皆基於本公司於特定日期所做之判斷，故有其時效性限制，邇後若有變更時，本公司將不做預告或更新。本報告內容僅供參考，並不提供或遊說客戶為買賣股票之投資依據。投資人應審慎考量本身之投資風險，並就投資結果自行負責。本公司及所屬集團成員，暨其主管或員工皆有可能持有報告中所提及的證券。本公司及所屬集團成員並可能經常提供投資銀行或其他服務給報告中提及之公司或向其爭取相關業務。本報告之著作權為本公司所有，非經本公司同意，本報告全文或部份內容，不得以任何形式或方式引用、轉載或轉寄。

訪談重點摘要

補丁科

(7947 TT, NT$881, 未評等)

焦點內容

1. 記憶體牆為 AI 效能的主要瓶頸，其中推論對延遲與能耗的要求高於訓練。硬體算力約每 2 年成長 3 倍，DRAM 頻寬卻只有約 1.6 倍，而 GPU+HBM 採用的 2.5D 封裝，延遲與功耗也相對較高。公司以 3D 堆疊 near memory 架構切入，有望兼顧接近 SRAM 的延遲，以及遠高於 SRAM 的容量，適合需同時服務百萬級使用者的推論應用。
2. 公司在 2016 年首次投片即成功，4Gb 產品頻寬達 144GB/s、延遲僅 17ns (HBM 約 40-50ns)，並於 2017 年在 ISSCC 發表。3D WoW hybrid bonding 並非公司獨有，真正的門檻在於客戶、晶圓大廠、南亞科與公司需四方深度共同開發，公司負責記憶體架構與介面的協同設計，並以 KGD 技術確保堆疊良率。南亞科持股約 35%，HiBaLL-M 多層系列目前為 4 層，客戶已要求 8 層。
3. AI 產品營收占比由 2024 年的 4%提升至 1H26 的約 40%，3Q26 營收也較 2Q26 成長約 43%。公司自 2025 年起簽訂 SOW，預期明年有客戶進入量產並開始貢獻營收，其中部分客戶將採 turnkey 模式，量產規模放大後獲利率有望提高。公司預期 2028-29 年量持續增加，2 至 3 年內將產生重大影響。

結論

補丁科以自有高頻寬低延遲架構結合 3D 堆疊 DRAM，受惠於 AI 推論對低延遲、低功耗與大容量的需求，以及雲端客戶走向客製化記憶體的趨勢。短期營收來自 AI 客製化專案與傳統 DRAM 業務；中期成長動能為明年起客戶陸續進入量產，帶動客製化專案由設計階段轉向量產階段；長期則有機會由雲端延伸至邊緣推論，並隨客戶增加與規格收斂推出通用化產品。

營運現況

- 營收：3Q26 季增約 43%，過去六季趨勢向上
- EPS：1H26 9.35 元，1H26 營收與獲利同步成長
- 產品組合
  - AI 產品營收占比由 2024 年的 4%提升至 1H26 約 40%
  - 傳統 DRAM 業務占 1H26 營收約 60%，提供 SoC 廠與系統廠，並協助客戶客製化

營運展望

- 明年預期有客戶進入量產並開始貢獻營收，2028-29 年量持續增加
- AI 占比持續提升，預期 2 至 3 年內產生重大影響
- 傳統產品除維持現況外成長有限，AI 為未來重心
- 客戶表示已取得 CSP 大廠訂單，公司目前開發未來產品，產品驗證成功即可進入後續導入，有別於過去先做產品再推市場

凱基投顧

沈漢軒
886.2.2181.8005
hanhsuan.shen@kgi.com

潘俊宏
886.2.2181.8706
Felix.Pan@kgi.com

王婉懿
886.2.2181.8715
vera.wy.wang@kgi.com

重要免責聲明，詳見最終頁

- 公司為無廠房的 IP 與設計公司，生產由晶圓大廠(邏輯)與南亞科(DRAM)負責，由客戶向其下單

成長動能

- AI 記憶體：記憶體牆與 3D 堆疊解決方案
  - 硬體算力約每 2 年成長 3 倍，DRAM 頻寬約每 2 年成長 1.6 倍，二十年累積形成落差
  - 現行 GPU+HBM 採 2.5D 封裝，匯流排較長，存取延遲較高、負載與功耗較高，且台積電 CoWoS 產能大部分由 NVIDIA/AMD 使用
  - 公司採 3D 堆疊 near memory 架構：DRAM 先堆疊，再以 3D 方式與 CPU/GPU 堆疊
  - HBM 採 micro-bump，公司採 3D WoW hybrid bonding，pitch 小一個數量級以上，頻寬等於 IO 數乘以頻率，IO 數增加可大幅提升頻寬，同時達成高頻寬/低延遲/低功耗
- AI 推論需求
  - 訓練端 HBM 容量遠大於 SRAM，對延遲容忍度尚可；推論端延遲非常關鍵，已有業者不採 HBM、改採 SRAM
  - 客戶以同一模型比較 GPU+HBM 與自家 SRAM 平台，SRAM 平台約十幾二十秒出答案，GPU+HBM 明顯較慢，使用者無法接受
  - SRAM 容量在百萬級同時使用者的情境下遠遠不足；3D 堆疊 DRAM 容量遠高於 SRAM，延遲與能耗可做到非常接近 SRAM
  - 單一項目未必勝過 SRAM，但整體 AI 效能優於 HBM 與 SRAM
  - HBM 因容量大仍有其價值，公司目標為超越 HBM，產品目前更適合推論應用
- 技術累積與驗證
  - 2014 年與 Intel 合作高頻寬低延遲記憶體計畫，2016 年投片一次成功，容量 4Gb，頻寬 144GB/s，高於當時 HBM2
  - 延遲 17ns，HBM2 乃至今日 HBM 約 40-50ns，成果 2017 年 2 月於 ISSCC 發表
  - 2017 年起累積高頻寬低延遲架構，持續為不同客戶客製化，並具備多種 interface 設計經驗
  - 專利全部自行開發，部分已核准，部分申請中
- 競爭優勢
  - 邏輯設計能力是 3D 堆疊關鍵：單片良率 80%，不處理的情況下堆疊 4 片剩約 41%，8 片剩約 16.7%，10 片剩約 11%，無法量產，須在底層邏輯設計端克服
  - 3D 堆疊接合前可測，接合後(含溫度循環等熱應力後)仍可測並可修復，客戶系統端亦可再測再修復，測試設計須整合進邏輯，與客戶如同一家公司般共同設計

- 與南亞科策略合作，針對客戶高頻寬需求在技術上突破原有框架
- 3D 堆疊技術本身並非稀缺，包括華邦/力積電/聯電/台積電/南亞科均具備，南亞科屏東廠將投入 3D 堆疊與測試
- 差異化在於客製化共同設計能力與團隊 20 多年合作累積的首次量產成功經驗

■ 商業模式與合作條件

- 過去 DRAM 依 JEDEC 標準，各家依標準自行設計；現今 AI 客戶為超車走客製化，公司從 interface/規格定義起即與客戶共同開發，設計範圍並含處理單元
- 客戶皆為具備足夠規模與資本的 AI 雲端客戶，需使用最先進邏輯製程(全球目前僅台積電能提供)，並需 DRAM 端夥伴
- 三大記憶體廠標準品量大獲利高，目前不願大量提供客製化；中國有其限制，南亞科 DRAM 製程為台灣最先進，現有大客戶都想找南亞科，也是南亞科積極擴產的原因之一
- 公司角色為與客戶/台積電/南亞科三方共同開發，並協同客戶設計介面

■ 與南亞科的關係

- 2010 年南亞科持股超過 50%，2017 年協議結束後仍維持晶圓代工合作；2024 年起重新討論，2025 年因應 AI 記憶體客製化需求再度投資，目前為最大股東，持股約 35%
- 維持晶圓代工關係，合作模式彈性，可深化合作也可各自發展

■ 產品線

- 以 HBLL-RAM 架構為基礎，目前幾乎所有研發量均投入 AI 相關產品，2017 年架構早期以 2D 方案提供低成本選擇
- HiBaLL-M 多層系列：多顆 DRAM 以 TSV 堆疊，目前為 4 層，客戶要求 8 層，未來視技術進步而定
- HiBaLL-S 單層系列：僅一層，相同條件下可提供 10 倍以上頻寬優勢
  - 頻寬>30TBps，延遲<15ns，能耗約 0.2pJ/bit，HBM 為 2–3pJ/bit
  - 有客戶需求達多個 PB 等級的容量，且要求更低能耗
- 兩者取捨取決於客戶規格需求，並非哪一種必然勝出

■ 應用方向

- 目前主要已啓動的開發案件皆為雲端客戶，為其下一世代擴大服務做準備
- 邊緣推論客戶已開始洽談，進度較慢屬 AI 應用由雲端先行再落地邊緣的需求節奏，並非技術問題
- 標準方案無法滿足、對功耗/成本有特殊要求的邊緣客戶，也會來找公司客製化

- 傳統 DRAM 業務
  - 過去 20 年累積的產品仍在量產，因無自有工廠，以授權及客戶關係為主

QA

- 客製化需求增加，是否代表 DRAM 景氣波動減小、價格較為穩定？
  - 是，這也是南亞科約 2 至 3 年前決定投入 AI 客製化 DRAM 的原因，因量體不及三大廠，需走差異化
  - 現階段先挑選優質且能長期合作、降低波動性的客戶，客戶成功則公司與南亞科即成功
  - 客戶目標是超越 HBM，通用化並非優點；未來客戶增加、規格收斂後，有機會推出較通用化的產品
- 與華邦的 CUBE 有何差異？
  - 對方先推出自有標準再尋找客戶導入，並尋求外部協助做標準化與介面，客製化為輔
  - 公司以客戶需求出發，客戶千挑萬選後找來，是希望在製程與設計上取得特別的東西
- 與愛普的技術路線有何差異？
  - 兩家各有 DRAM 架構，公司認為自家架構更具優勢
  - 測試方面，server 等級良率必須測試與修復，公司做法與對方不同
  - 公司深入客戶環境，與客戶緊密共同設計
  - 愛普營收規模大於公司，部分來自公司未涉足的業務，目前 AI 需求強勁，各家隨客戶進展成長
- NVIDIA 若自行在 GPU 上堆疊 SRAM，公司產品是否仍有必要？
  - HBM4 以上也在考慮 3D，NV 可能把部分功能整合進 HBM 的 base die，並與 DRAM 廠討論 3D 整合
  - 採客製化或標準化尚不確定，時程比公司稍慢
  - NVIDIA 偏通用型，需兼顧所有應用，CSP 大廠也希望有更多選擇
  - 即使 NVIDIA 做客製化，全球客製化需求也不會只有一家，公司與南亞科的組合具競爭力
- 3D 堆疊在處理器上，散熱是否會成為更大的問題，是否存在功耗上限？
  - 功耗是 AI 共同指標，設計初期即規劃，也是客戶採購重要標準
  - 公司產品功耗至少為 HBM 的四分之一，目前合作的客戶，整體系統設計尚未看到散熱成為此方案的限制，這類問題可以解決
  - 目前為 4 層，客戶要求 8 層，設計須隨層數調整

- 良率與修復如何看待？
  - 設計本身有面積/測試等 overhead，4 層/8 層的目標需分別設定
  - 客戶要求目標為 95%，目前至少可做到 80%，有機會更高
- DRAM 層與層之間也是用 hybrid bonding？是否提供 2.5D 方案？
  - DRAM 與 SoC 之間採 hybrid bonding，多層 DRAM 間以 TSV 互連，公司提供 DRAM 方案，DRAM 本身仍以 3D 為主
  - 與 SoC 的連接方式(含 2.5D)依客戶系統決定，公司 DRAM 端不需太多改變，部分美國客戶已在測試 2.5D
- 客戶如何選擇商業模式，利潤率如何？
  - 預估部分客戶明年進入量產，量產規模放大後毛利率可望更高，發酵時點稍晚
  - 客戶選擇與自身資源有關：缺乏後段管理外包商資源者傾向委由公司處理，有能力者可能自行處理
- 明年量產的晶圓需求如何與晶圓廠洽談？2027 年量產專案使用現有產線還是新廠？
  - 直接關係為公司的客戶，實際為客戶/公司/南亞科的三方關係
  - 專案啓動前須先與南亞科確認可接受客戶未來需求，各晶圓廠均需如此洽談；南亞科每年固定詢問需求，近期因新廠開出而增加確認次數
  - 確切數量於下單時確認，大客戶通常提前半年至一年安排並持續更新；南亞科擴產時會確認可供應量，無法供應的部分會告知一至兩年後的供應安排
  - 2027 年專案現有產線與新廠皆有機會，新廠 2H27 開出
- 合約負債的性質為何？
  - 客戶 NRE 的收款方式與財報認列方式不同，財報依投入的時間與人力成本認列，部分客戶預付大筆現金，記為合約負債
- 人力擴充計畫？
  - 目前缺人而非缺錢，正積極招募
  - 與客戶合作增加邏輯設計/實體設計/custom cell 等需求，所需人才與過去不同。各階段人力需求不同，前期最多，專案後期釋出的人力可轉投其他專案，部分部門前後期均需支援
  - 客戶洽談先談規格與時程，待人力到位再正式啓動

凱基證券集團據點

中國	上海	上海市靜安區南京西路 1601 號越洋國際廣場 1507 室 郵政編號：200040
台灣	台北	104 台北市明水路 700 號 電話 886.2.2181.8888，傳真 886.2.8501.1691
香港		香港灣仔港灣道十八號中環廣場四十一樓 電話 852.2878.6888，傳真 852.2878.6800
泰國	曼谷	8th - 11th floors, Asia Centre Building 173 South Sathorn Road, Bangkok 10120, Thailand 電話 66.2658.8888，傳真 66.2658.8014
新加坡		珊頓大道 4 號#13-01 新交所第二大廈 郵政編號：068807 電話 65.6202.1188，傳真 65.6534.4826
印尼		Sona Topas Tower Fl.11 Jl. Jend. Sudirman kav.26 JakartaSelatan 12920 Indonesia 電話 62 21 250 6337

股價說明

等級	定義
增加持股 (OP)	對個股持正面看法，預期個股未來十二個月的表現超越凱基證券集團所追蹤的相關市場的總報酬。
持有 (N)	對個股持中性看法，預期個股未來十二個月的表現符合凱基證券集團所追蹤的相關市場的總報酬。
降低持股 (U)	對個股持負面看法，預期個股未來十二個月的表現低於凱基證券集團所追蹤的相關市場的總報酬。
未評等 (NR)	凱基證券未對該個股加以評等。
受法規限制未評等 (R)	受凱基證券集團內部政策和/或相關法令限制使凱基證券集團無法進行某些形式的資訊交流，其中包括提供評等給投資人參考。
	*總報酬 = (十二個月目標價-現價)/現價

免責聲明

凱基證券投資顧問股份有限公司(以下簡稱本公司)為凱基金控集團之成員。本報告之內容皆來自本公司認可之資料來源，但不保證其完整性及精確性。報告內容所提及之各項業務、財務等相關檔案資料及所有的意見及預估皆基於本公司於特定日期所做之判斷，故有其時效性限制，邇後若有變更時，本公司將不做預告或更新。本報告內容僅供參考，並不提供或遊說客戶為買賣股票之投資依據。投資人應審慎考量本身之投資風險，並就投資結果自行負責。本公司及所屬集團成員，暨其主管或員工皆有可能持有報告中所提及的證券。本公司及所屬集團成員並可能經常提供投資銀行或其他服務給報告中提及之公司或向其爭取相關業務。本報告之著作權為本公司所有，非經本公司同意，本報告全文或部份內容，不得以任何形式或方式引用、轉載或轉寄。

電子硬體產業

PC 需求惡化，2027 年持續疲弱

焦點內容

1. 3Q26 全球 PC 出貨量季減 14%至 5,540 萬台，較我們預估低 8%。3Q26 僅華碩 PC 出貨量季增 7%，優於同業因電競 PC 比重較高。然而，蘋果 (美)、HPQ (美)、Dell (美)則季減幅度較大，連帶影響台灣 ODM 出貨量 3Q26 季減幅較大。1Q-3Q26 全球 PC 出貨量年減 8%，僅 Apple 出貨量年增 12%，優於同業，主因消費者 1H26 提前購買。
2. 我們預期零組件價格上漲，將使品牌廠 3Q-4Q26 持續調漲 PC 售價，此將不利 PC 需求。我們同時觀察到 PC 品牌因消費需求疲弱，2H26 削減 PC 零組件 (如 IC) 訂單。我們下修 4Q26 全球 PC 出貨量預估至季減 12%、年減 33%，下修 2026 年全球 PC 出貨量至 2.32 億台，年減 14%。
3. 由於 1Q-4Q27 記憶體價格可能維持每季個位數漲幅，PC 品牌為維持獲利能力，恐將持續調漲 PC 售價，或降低新機規格。此外，2027 年 ODM 筆電新開案數至少年減 10-20%，PC 品牌可能延長既有機種銷售週期。我們保守看待 2027 年 PC 需求，下修 2027 年全球 PC 需求預估至 2.05 億台，年減 12%。

重要訊息

3Q26 全球 PC 出貨量季減 14%至 5,540 萬台，較我們預估低 8%。由於零組件成本上升，促使品牌廠持續調漲 PC 售價，我們預估 4Q26 全球 PC 出貨量將再季減 12%。我們保守看待全球 PC 需求，我們預估 2026 年 PC 出貨量衰退幅度擴大至年減 14%，2027 年則續減 12%。

評論及分析

3Q26 PC 出貨不如預期。根據 Gartner 統計，3Q26 全球 PC 出貨量年減 24%、季減 14%至 5,540 萬台，較我們原估低 8%。華碩 (2357 TT, NT$977, 增加持股) PC 出貨量季增 7%，符合公司展望，優於同業因電競 PC 比重較高，而此類客戶價格敏感度較低。然而，其餘品牌出貨量均明顯季減。其中，Apple (美) 出貨量季減 23%，降幅最大，因消費者趕在 6 月 MacBook 價格調漲 15-20%前，於 1H26 提前購買。HP (美) 因消費比重較高，PC 出貨量季減 20%；Dell (美)也因消費者提前購買而使出貨量季減 19%。主要筆電 ODM 出貨亦受到拖累，其中仁寶 (2324 TT, NT$36.6, 未評等) 3Q26 筆電出貨量季減 21%、緯創 (3231 TT, NT$189.5, 增加持股) 季減 18%、廣達 (2382 TT, NT$322, 增加持股) 季減 17%。整體而言，台灣前五大筆電 ODM 出貨量合計季減 17%。區域別來看，3Q26 亞太地區 (不含日本) PC 出貨量僅季減 4%，優於其他地區，但歐非中東季減 17%、北美季減 22%。1Q-3Q26 全球 PC 出貨量年減 8%，其中亞太地區 (不含日本) 僅年減 1%，美國年減 5%，優於其他區域。1-3Q26 各品牌中，Apple 出貨量年增 12%，優於同業，主因消費者於 1H26 提前購買。

4Q26 PC 出貨量季減，下修 2026 年 PC 需求預估。隨 DRAM 與 NAND 價格 3Q25 以來持續上漲，目前其成本占 PC BOM 比重已超過 40%，較 1Q25 的 15-20%大幅提升。1H26 Windows PC 零售價年增超過 30%，考量記憶體及 CPU 等零組件成本持續攀升，我們預期 3Q-4Q26 PC 價格再季增 5-10%，進一步抑制 PC 需求。同時，我們觀察到 PC 品牌因消費需求疲弱，2H26 削減 PC 零組件 (如 IC) 訂單。多數台灣筆電 ODM 預期 4Q26 筆電出貨量持續季減，我們預估 4Q26 前五大筆電 ODM 筆電出貨量季減 12%。反映消費需求低於預期，我們下修 4Q26 全球 PC 出貨量預估至季減 12%、年減 33%，下修 2026 年全球 PC 出貨量至 2.32 億台，年減 14%。

2027 年 PC 需求仍疲軟。由於 1Q-4Q27 記憶體價格可能維持每季個位數漲幅，PC 品牌為維持獲利能力，恐將持續調漲 PC 售價，或降低新機規格。此外，2027 年 ODM 筆電新開案數至少年減 10-20%，PC 品牌可能延長既有機種銷售週期。綜上，我們保守看待 2027 年 PC 需求，預期 1Q-2Q27 全球 PC 季度出貨量大致與 4Q26 持平，隱含年減幅 28%。我們認為 PC 需求於 3Q-4Q27 才可望逐季復甦，因低基期及新機上市使旺季出貨量逐步回升。我們下修 2027 年全球 PC 需求預估至 2.05 億台，年減 12%。

投資建議

在 PC 需求放緩的環境下，PC 品牌營收因 ASP 上揚，營收與獲利率受影響較少。但我們對 PC 零組件供應鏈較保守看待，除營收面臨 PC 需求不如預期之衝擊外，且因難以將上升成本完全轉嫁予品牌廠，獲利率恐進一步受到侵蝕。

投資風險

消費者需求疲弱、零組件價格攀升。

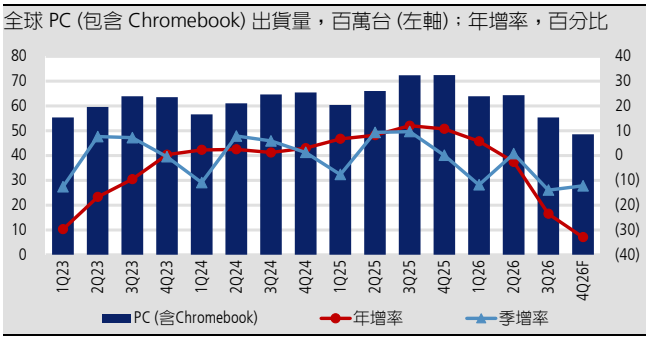
凱基投顧

余昀澄
886 2 2181 8013
alex.a.yu@kgi.com

向子慧
886 2 2181 8726
angelah@kgi.com

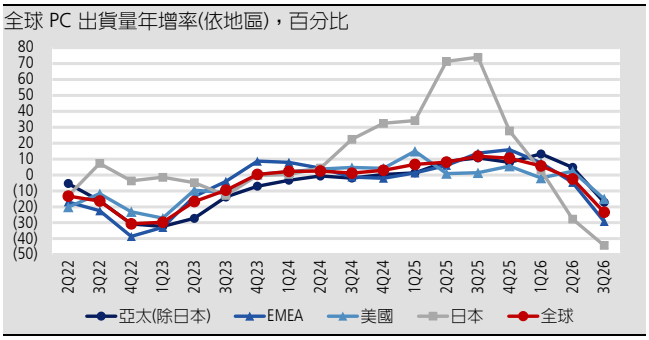
重要免責聲明，詳見最終頁

圖 1：3Q26 全球 PC (含 Chromebook) 出貨量年減 24%、季減 14%，低於我們預估



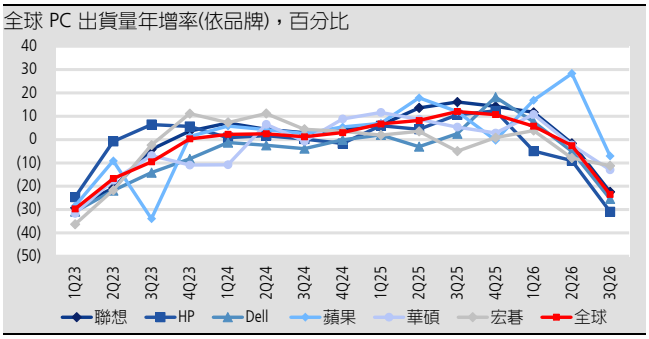
資料來源：Gartner；凱基預估

圖 2：3Q26 美國、亞太地區 (不含日本) 出貨量年減 15-17%，表現優於其他地區



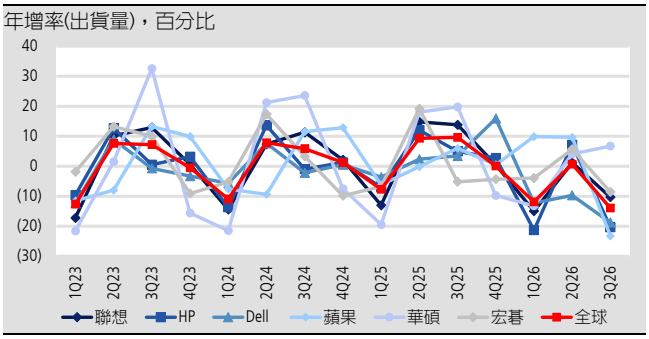
資料來源：Gartner；凱基

圖 3：3Q26 蘋果、華碩、宏碁年減幅度小於其他品牌



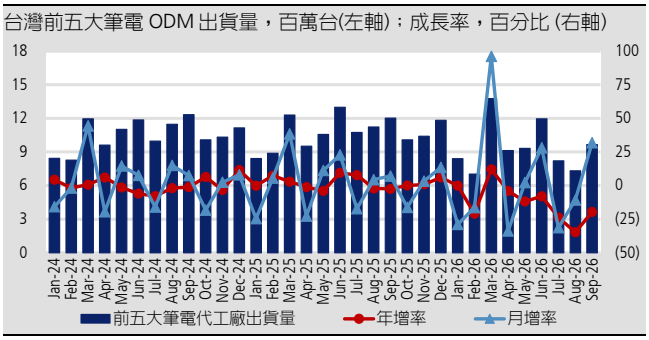
資料來源：Gartner；凱基

圖 4：3Q26 僅華碩出貨量季增，然而其他品牌多數季減雙位數



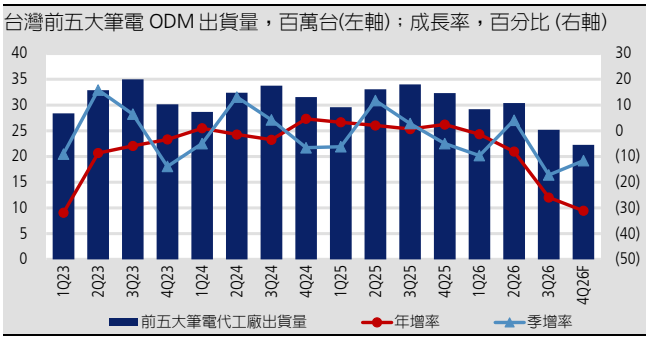
資料來源：Gartner；凱基

圖 5：九月台灣前五大筆電 ODM 出貨量年減 20%，但月增 32%



資料來源：Gartner；凱基

圖 6：因 1H26 已被提前拉貨，3Q26 筆電 ODM 出貨量年減 26%、季減 17%



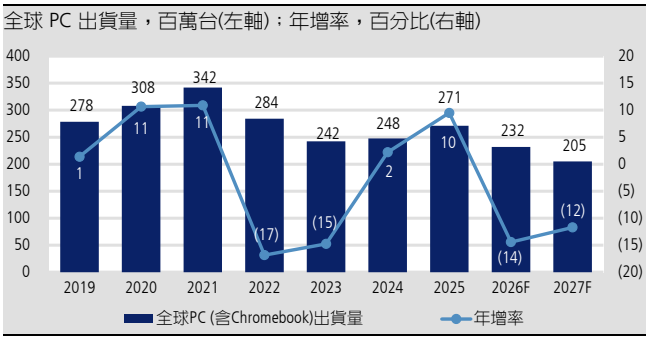
資料來源：公司資料；凱基

圖 7：3Q26 筆電 ODM 出貨量年減 26%、季減 17%，低於我們預估；我們預估 4Q26 ODM 出貨量季減 12%

筆電出貨量 (百萬台)																
股票代號	公司	Jul-26	Aug-26	Sep-26	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26	4Q26F	2024	2025	1-3Q26	2026F	2027F
2324 TT	仁寶	1.8	1.4	1.7	7.0	7.1	7.1	6.8	5.9	6.2	4.9	4.6	32.3	24.1	17.0	21.6	17.9
2356 TT	英業達	1.6	1.5	1.8	5.0	5.6	5.4	5.3	5.4	5.4	4.9	4.5	20.0	21.3	15.7	20.2	18.1
2382 TT	廣達	2.8	2.6	4.2	10.8	12.1	12.7	10.9	10.0	11.5	9.6	8.2	45.9	46.5	31.1	39.3	34.5
3231 TT	緯創	1.6	1.4	1.5	4.9	5.9	6.4	6.9	6.1	5.5	4.5	3.9	20.2	24.1	16.1	20.0	16.2
4938 TT	和碩	0.4	0.4	0.5	1.9	2.4	2.4	2.4	1.8	1.8	1.3	1.1	8.0	9.1	4.9	6.0	5.1
前五大筆電代工廠		**8.2**	**7.3**	**9.7**	**29.6**	**33.1**	**34.0**	**32.3**	**29.2**	**30.4**	**25.2**	**22.3**	**126.4**	**129.0**	**84.8**	**107.1**	**91.9**
年增率 (%)																	
2324 TT	仁寶	(21.7)	(39.1)	(32.0)	(6.7)	(18.4)	(14.5)	(12.8)	(15.7)	(12.7)	(31.0)	(32.4)	(4.7)	(13.3)	(19.8)	(10.4)	(17.0)
2356 TT	英業達	(5.9)	(16.7)	(5.3)	11.1	14.3	3.8	(1.9)	8.0	(3.6)	(9.3)	(16.0)	7.0	6.5	(1.9)	(5.4)	(10.0)
2382 TT	廣達	(30.0)	(36.6)	(8.7)	2.9	3.4	0.8	(1.8)	(7.4)	(5.0)	(24.4)	(24.8)	(2.1)	1.3	(12.6)	(15.5)	(12.2)
3231 TT	緯創	(20.0)	(36.4)	(31.8)	6.5	15.7	23.1	30.2	24.5	(6.8)	(29.7)	(43.5)	5.8	19.3	(6.4)	(17.0)	(19.0)
4938 TT	和碩	(46.7)	(48.5)	(44.2)	22.3	18.5	(3.0)	23.6	(5.0)	(24.2)	(46.5)	(53.7)	(0.3)	13.8	(26.8)	(33.9)	(15.0)
前五大筆電代工廠		**(23.7)**	**(34.7)**	**(19.7)**	**3.3**	**2.1**	**0.7**	**2.4**	**(1.3)**	**(8.1)**	**(25.9)**	**(31.1)**	**(0.2)**	**2.1**	**(12.3)**	**(17.0)**	**(14.2)**
季增率 (%)																	
2324 TT	仁寶				(10.3)	1.4	0.0	(4.2)	(13.2)	5.1	(21.0)	(6.1)					
2356 TT	英業達				(7.4)	12.0	(3.6)	(1.9)	1.9	0.0	(9.3)	(9.2)					
2382 TT	廣達				(2.7)	12.0	5.0	(14.2)	(8.3)	15.0	(16.5)	(14.6)					
3231 TT	緯創				(7.5)	20.4	8.5	7.8	(11.6)	(9.8)	(18.2)	(13.3)					
4938 TT	和碩				(2.8)	25.3	1.1	0.4	(25.3)	0.0	(28.6)	(13.2)					
前五大筆電代工廠					**(6.2)**	**11.8**	**2.8**	**(5.0)**	**(9.6)**	**4.1**	**(17.2)**	**(11.6)**					
月增率 (%)																	
2324 TT	仁寶	(25.0)	(22.2)	21.4													
2356 TT	英業達	(33.3)	(6.3)	20.0													
2382 TT	廣達	(37.8)	(7.1)	61.5													
3231 TT	緯創	(20.0)	(12.5)	7.1													
4938 TT	和碩	(38.5)	6.3	8.2													
前五大筆電代工廠		**(31.4)**	**(10.7)**	**31.9**													

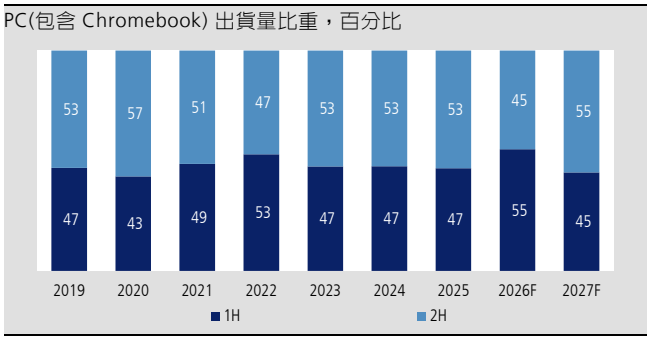
資料來源：公司資料；凱基預估

圖 8： 我們下修 2026 年全球 PC 出貨量至年減 14%，預估 2027 年年減 12%



資料來源：Gartner；凱基預估

圖 9：因提前拉貨，2H26 PC 出貨量將小於上半年，預計 2027 年才有望恢復過往季節性



資料來源：Gartner；凱基預估

圖 10：電子硬體供應鏈投資評價

公司	代碼	相關業務	市值 (美金百萬元)	股價 (當地貨幣)	評等	目標價 (元)	每股盈餘 (當地貨幣)		每股盈餘年增率 (%)		PE (x)		PB (x)		ROE (%)		現金殖利率(%)	
							2026F	2027F	2026F	2027F	2026F	2027F	2026F	2027F	2026F	2027F	2025	2026F
華碩	2357 TT	品牌	23,029	977.0	增加持股	1,100.0	73.28	73.31	22.1	0.1	13.3	13.3	2.6	2.5	19.8	19.0	4.3	6.0
宏碁*	2353 TT	品牌	3,032	31.4	未評等	N.A.	1.72	2.14	36.3	24.6	18.2	14.6	1.2	1.1	6.5	6.6	4.1	4.3
微星	2377 TT	品牌	4,156	155.0	持有	93.0	10.00	10.50	46.9	5.0	15.5	14.8	2.3	2.1	15.0	14.9	2.7	4.0
技嘉	2376 TT	品牌	7,568	356.0	增加持股	515.0	33.00	35.70	81.3	8.2	10.8	10.0	3.6	3.2	35.2	33.8	3.4	6.1
華擎	3515 TT	品牌	884	225.5	增加持股	290.0	18.24	23.18	19.4	27.1	12.4	9.7	2.2	2.0	18.1	21.2	4.9	4.8
英業達	2356 TT	ODM	6,615	58.1	持有	65.0	3.51	4.45	44.6	27.0	16.6	13.1	2.6	2.5	16.2	19.8	3.4	4.8
廣達	2382 TT	ODM	41,974	322.0	增加持股	500.0	27.06	31.75	39.1	17.3	11.9	10.1	4.7	4.3	41.1	44.2	4.8	6.7
緯創	3231 TT	ODM	20,628	189.5	增加持股	308.0	19.96	31.06	120.7	55.6	9.5	6.1	2.8	2.2	31.6	40.0	2.9	6.2
鴻海精密	2317 TT	ODM	109,216	244.5	增加持股	330.0	18.67	22.64	37.2	21.3	13.1	10.8	1.8	1.7	14.2	16.1	2.9	4.1
和碩	4938 TT	ODM	8,372	98.6	增加持股	100.0	5.39	7.43	0.0	37.7	18.3	13.3	1.2	1.2	6.9	9.2	4.1	3.5
仁寶電腦*	2324 TT	ODM	5,119	36.6	未評等	N.A.	2.20	2.89	59.5	31.1	16.6	12.7	1.2	1.1	6.9	8.9	3.0	3.8
群光	2385 TT	周邊	2,515	103.5	持有	106.0	8.16	9.58	(9.9)	17.4	12.7	10.8	1.7	1.6	13.3	15.2	7.0	6.3
致伸	4915 TT	周邊	949	63.4	持有	66.0	5.00	6.00	(12.4)	19.9	12.7	10.6	1.4	1.4	11.1	13.0	7.2	6.3
新普	6121 TT	電池	2,140	364.5	增加持股	460.0	33.76	36.99	10.3	9.6	10.8	9.9	1.7	1.6	15.9	16.6	6.0	6.6
新日興	3376 TT	樞軸	1,075	173.0	增加持股	272.0	6.42	13.58	295.1	111.5	26.9	12.7	1.9	1.9	7.3	15.0	0.8	3.0
建準	2421 TT	散熱模組	1,534	168.5	增加持股	190.0	10.68	13.75	34.5	28.8	15.8	12.3	4.7	4.2	32.4	38.0	0.0	4.6
雙鴻	3324 TT	散熱模組	4,549	1,540.0	增加持股	1,410.0	55.69	81.55	97.1	46.4	27.7	18.9	10.2	7.8	41.2	46.9	0.8	1.6
嘉澤	3533 TT	連接器 / 插槽	6,714	1,880.0	增加持股	2,710.0	82.37	123.18	17.4	49.5	22.8	15.3	4.9	4.2	22.7	29.7	1.9	2.2
凡甲科技	3526 TT	連接器	582	274.5	增加持股	380.0	20.73	28.17	22.2	35.8	13.2	9.7	4.3	4.2	32.3	43.7	6.1	7.5
宏致	3605 TT	連接器	996	179.0	受法規限制	N.A.	5.47	7.51	25.6	37.3	32.7	23.8	3.4	3.1	10.9	13.6	0.9	0.9
優群*	3217 TT	連接器	363	127.0	未評等	N.A.	12.95	14.07	2.0	8.7	9.8	9.0	2.3	2.0	28.9	31.4	7.9	7.9
金像電	2368 TT	PCB	20,495	1,235.0	增加持股	2,000.0	41.49	71.44	112.9	72.2	29.8	17.3	11.5	7.9	48.5	54.8	0.8	1.8
華通	2313 TT	PCB	9,867	252.0	增加持股	360.0	7.72	15.02	40.1	94.5	32.6	16.8	5.9	4.7	18.6	30.9	1.1	1.5
台光電子	2383 TT	CCL	69,934	6,150.0	增加持股	7,625.0	108.84	190.64	161.2	75.2	56.5	32.3	30.1	18.6	63.1	71.4	0.4	1.1
景碩	3189 TT	Substrate	17,725	1,060.0	持有	935.0	10.92	25.93	211.5	137.5	97.1	40.9	11.7	9.7	15.6	30.0	0.2	0.5
欣興電子	3037 TT	Substrate	65,368	1,255.0	增加持股	1,175.0	16.10	32.71	267.5	103.1	77.9	38.4	15.3	11.6	22.6	35.8	0.2	0.6
臻鼎-KY	4958 TT	FPCB	20,605	581.0	持有	685.0	15.61	26.33	126.0	68.6	37.2	22.1	3.8	3.4	12.1	17.0	0.6	1.3
可成	2474 TT	機殼	3,326	194.5	持有	205.0	10.34	12.04	(8.7)	16.5	18.8	16.1	0.8	0.7	3.8	4.1	5.9	4.8
戴爾	Dell US	品牌	372,624	586.1	增加持股	600.0	9.47	25.87	7.8	173.1	61.9	22.7	(262.5)	143.1	N.M.	N.M.	0.4	0.4
惠普	HPQ US	品牌	27,234	30.2	持有	32.0	3.25	3.16	21.5	(2.6)	9.3	9.6	380.7	27.5	N.M.	N.A.	3.8	4.0
聯想	992 HK	品牌	52,954	32.3	持有	N.A.	2.05	3.21	57.1	56.3	15.8	10.1	5.1	3.7	24.3	25.0	1.8	2.5

資料來源：凱基預估；Bloomberg

** Bloomberg 市場共識*

圖 11：全球各品牌 PC (含 Chromebook) 出貨量

廠商	PC (含Chromebook出貨量)																			
	1Q23	2Q23	3Q23	4Q23	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26	2022	2023	2024	2025	1-3Q26
Lenovo	13.0	14.3	16.2	16.3	13.9	15.0	16.6	17.0	14.8	17.0	19.3	19.4	16.5	16.7	15.0	69.0	59.8	62.5	70.5	48.2
HP Inc.	11.9	13.5	13.5	14.0	12.1	13.7	13.6	13.7	12.8	14.3	15.0	15.4	12.1	13.0	10.4	55.4	52.9	53.0	57.5	35.5
Dell	9.5	10.4	10.3	10.0	9.4	10.1	9.9	10.0	9.6	9.8	10.2	11.8	10.3	9.3	7.6	50.0	40.2	39.4	41.4	27.3
Apple	5.2	4.8	5.5	6.0	5.5	5.0	5.6	6.3	5.9	5.9	6.3	6.3	6.9	7.6	5.8	26.8	21.5	22.5	24.4	20.4
Asus	3.8	3.9	5.1	4.3	3.4	4.1	5.1	4.7	3.8	4.5	5.4	4.9	4.2	4.4	4.7	20.7	17.2	17.4	18.5	13.3
Acer Group	3.5	4.0	4.4	4.0	3.8	4.4	4.6	4.1	3.9	4.6	4.4	4.2	4.0	4.2	3.9	18.7	15.9	16.9	17.0	12.1
Top-6	47.1	50.9	55.0	54.6	48.1	52.4	55.4	55.9	50.8	56.1	60.5	62.0	54.1	55.3	47.3	240.6	207.5	211.8	229.3	156.7
其他	8.3	8.7	8.8	9.0	8.5	8.7	9.2	9.5	9.6	9.9	11.9	10.5	9.7	9.0	8.0	43.6	34.8	35.8	41.9	26.8
總計	55.3	59.6	63.8	63.5	56.6	61.0	64.6	65.4	60.4	66.0	72.4	72.4	63.8	64.3	55.4	284.2	242.3	247.6	271.2	183.5
	年增率 (%)																			
Lenovo	(29.2)	(20.7)	(4.2)	3.6	7.1	4.4	2.9	4.5	6.2	13.6	16.1	14.2	11.7	(1.6)	(22.4)	(17.3)	(13.4)	4.6	12.8	(5.6)
HP Inc.	(24.7)	(0.8)	6.4	5.6	0.9	1.7	0.2	(1.7)	5.9	4.4	10.6	12.3	(4.9)	(9.0)	(30.9)	(25.4)	(4.4)	0.2	8.4	(15.6)
Dell	(30.9)	(21.8)	(14.2)	(8.3)	(1.3)	(2.4)	(3.9)	(0.1)	2.0	(3.0)	2.5	18.2	7.6	(5.1)	(25.4)	(16.0)	(19.5)	(2.0)	4.9	(8.0)
Apple	(28.4)	(9.1)	(33.9)	1.3	5.8	4.2	2.6	5.4	7.0	17.9	12.0	(0.3)	16.9	28.3	(7.1)	(0.4)	(19.7)	4.5	8.6	12.3
Asus	(31.7)	(17.4)	(6.8)	(10.9)	(10.8)	6.5	(0.7)	8.9	11.6	8.7	5.4	2.8	10.8	(2.3)	(13.0)	(4.5)	(16.9)	1.1	6.7	(2.9)
Acer Group	(36.3)	(21.7)	(2.4)	11.1	7.3	11.2	4.4	3.6	1.9	3.5	(5.0)	0.8	3.9	(7.9)	(11.2)	(22.9)	(15.1)	6.5	0.2	(5.5)
Top-6	(29.2)	(15.3)	(8.1)	0.7	2.2	3.0	0.7	2.4	5.4	7.1	9.2	10.9	6.7	(1.5)	(21.8)	(17.0)	(13.8)	2.1	8.3	(6.4)
其他	(32.6)	(24.2)	(17.5)	(1.7)	2.3	(0.4)	4.1	6.2	13.8	14.4	29.0	9.8	0.8	(8.7)	(32.3)	(15.7)	(20.2)	3.1	16.8	(14.7)
總計	(29.7)	(16.7)	(9.6)	0.3	2.3	2.5	1.2	3.0	6.7	8.2	12.0	10.7	5.7	(2.6)	(23.5)	(16.8)	(14.7)	2.2	9.5	(7.7)
	季增率 (%)																			
Lenovo	(17.2)	10.2	12.9	0.6	(14.4)	7.3	11.3	2.1	(13.0)	14.8	13.8	0.5	(15.0)	1.2	(10.3)					
HP Inc.	(9.6)	12.7	0.5	3.2	(13.7)	13.6	(1.0)	1.2	(7.0)	12.0	5.0	2.8	(21.3)	7.2	(20.3)					
Dell	(12.3)	8.9	(0.7)	(3.3)	(5.6)	7.7	(2.2)	0.5	(3.6)	2.3	3.4	15.9	(12.3)	(9.7)	(18.8)					
Apple	(11.5)	(8.0)	13.3	9.9	(7.6)	(9.4)	11.5	12.9	(6.2)	(0.1)	6.0	0.5	9.9	9.7	(23.3)					
Asus	(21.5)	1.5	32.6	(15.6)	(21.4)	21.2	23.6	(7.5)	(19.4)	18.0	19.8	(9.7)	(13.2)	4.1	6.7					
Acer Group	(1.8)	13.2	10.0	(9.1)	(5.2)	17.3	3.3	(9.8)	(6.7)	19.2	(5.2)	(4.3)	(3.9)	5.6	(8.5)					
Top-6	(13.1)	8.1	8.1	(0.8)	(11.8)	8.8	5.8	0.8	(9.2)	10.6	7.8	2.4	(12.6)	2.1	(14.4)					
其他	(9.4)	5.1	1.7	1.6	(5.7)	2.3	6.3	3.6	1.1	2.8	19.8	(11.7)	(7.3)	(6.9)	(11.1)					
總計	(12.6)	7.6	7.2	(0.5)	(10.9)	7.8	5.9	1.2	(7.7)	9.3	9.6	0.1	(11.8)	0.7	(13.9)					
	全球市佔率 (%)																			
Lenovo	23.5	24.1	25.3	25.6	24.6	24.5	25.8	26.0	24.5	25.7	26.7	26.8	25.9	26.0	27.1	24.3	24.7	25.3	26.0	26.3
HP Inc.	21.6	22.6	21.2	22.0	21.3	22.4	21.0	21.0	21.1	21.7	20.7	21.3	19.0	20.2	18.7	19.5	21.8	21.4	21.2	19.3
Dell	17.2	17.5	16.2	15.7	16.6	16.6	15.4	15.2	15.9	14.9	14.1	16.3	16.2	14.5	13.7	17.6	16.6	15.9	15.3	14.9
Apple	9.5	8.1	8.6	9.5	9.8	8.2	8.7	9.7	9.8	9.0	8.7	8.7	10.9	11.8	10.5	9.4	8.9	9.1	9.0	11.1
Asus	6.9	6.5	8.0	6.8	6.0	6.8	7.9	7.2	6.3	6.8	7.4	6.7	6.6	6.8	8.4	7.3	7.1	7.0	6.8	7.2
Acer Group	6.4	6.7	6.9	6.3	6.7	7.3	7.1	6.3	6.4	7.0	6.0	5.8	6.3	6.6	7.0	6.6	6.6	6.8	6.3	6.6
Top-6	85.1	85.4	86.2	85.9	85.0	85.8	85.8	85.4	84.0	85.0	83.6	85.5	84.8	85.9	85.5	84.7	85.6	85.5	84.6	85.4
其他	14.9	14.6	13.8	14.1	15.0	14.2	14.2	14.6	16.0	15.0	16.4	14.5	15.2	14.1	14.5	15.3	14.4	14.5	15.4	14.6
總計	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

資料來源：Gartner；凱基

凱基證券集團據點

中國	上海	上海市靜安區南京西路 1601 號越洋國際廣場 1507 室 郵政編號：200040
台灣	台北	104 台北市明水路 700 號 電話 886.2.2181.8888·傳真 886.2.8501.1691
香港		香港灣仔港灣道十八號中環廣場四十一樓 電話 852.2878.6888·傳真 852.2878.6800
泰國	曼谷	8th - 11th floors， Asia Centre Building 173 South Sathorn Road， Bangkok 10120， Thailand 電話 66.2658.8888·傳真 66.2658.8014
新加坡		珊頓大道 4 號#13-01 新交所第二大廈 郵政編號：068807 電話 65.6202.1188·傳真 65.6534.4826
印尼		Sona Topas Tower Fl.11 Jl. Jend. Sudirman kav.26 JakartaSelatan 12920 Indonesia 電話 62 21 250 6337

股價說明

等級	**定義**
增加持股 (OP)	對個股持正面看法，預期個股未來十二個月的表現超越凱基證券集團所追蹤的相關市場的總報酬。
持有 (N)	對個股持中性看法，預期個股未來十二個月的表現符合凱基證券集團所追蹤的相關市場的總報酬。
降低持股 (U)	對個股持負面看法，預期個股未來十二個月的表現低於凱基證券集團所追蹤的相關市場的總報酬。
未評等 (NR)	凱基證券未對該個股加以評等。
受法規限制未評等 (R)	受凱基證券集團內部政策和/或相關法令限制使凱基證券集團無法進行某些形式的資訊交流，其中包括提供評等給投資人參考。
	*總報酬 = (十二個月目標價-現價)/現價

免責聲明

凱基證券投資顧問股份有限公司（以下簡稱本公司）為凱基金控集團之成員。本報告之內容皆來自本公司認可之資料來源，但不保證其完整性及精確性。報告內容所提及之各項業務、財務等相關檔案資料及所有的意見及預估皆基於本公司於特定日期所做之判斷，故有其時效性限制，邇後若有變更時，本公司將不做預告或更新。本報告內容僅供參考，並不提供或遊說客戶為買賣股票之投資依據。投資人應審慎考量本身之投資風險，並就投資結果自行負責。本公司及所屬集團成員，暨其主管或員工皆有可能持有報告中所提及的證券。本公司及所屬集團成員並可能經常提供投資銀行或其他服務給報告中提及之公司或向其爭取相關業務。本報告之著作權為本公司所有，非經本公司同意，本報告全文或部份內容，不得以任何形式或方式引用、轉載或轉寄。

春源 (2010)　區間操作　報告日期: 20261012

研究員：陳國民/翁國梁

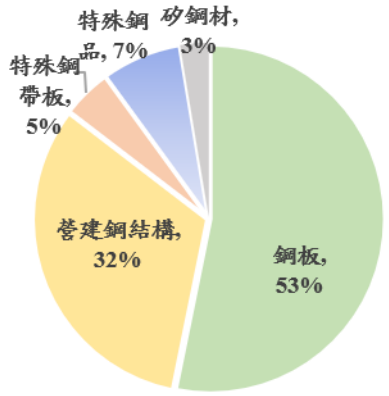
公司基本資料

前日收盤價(元)	22.70
目前股本(億元)	64.77
市值(億元)	147.00
每股淨值(元)	19.36
外資持股比例(%)	8.69
投信持股比例(%)	0.03
董監持股比例(%)	16.52
融資餘額(張)	2,288

投資建議

合庫預估2026/2027年 EPS 2.57/2.61元。受惠於AI建廠需求熱絡及公共支出創高，帶動鋼構訂單能見度已達4Q27。在評價方面，過去5年ROE平均為9.0%，預估2027年PER8.7X，考量高現金股利殖利率，公司具備高防禦屬性，故給予「區間操作」評等，21~26元(2027 PER8~10X)。

1H26 營收比重



公司簡介

春源(2010)成立於1965年，國內設有鋼板、直銷、特殊鋼、特鋼帶、矽鋼、自動倉儲及營建等七大事業部門，廠區分布於汐止、龍潭、台中、台南、高雄等地。海外佈局則採取穩健的自有資本策略，主要據點包括深圳春源(持股86.21%)、深圳紅源(持股87.97%)、上海華騰(持股80.59%)及青島春源(持股95%)，建構完整的中國大陸行銷網絡。

簡易損益表

單位：百萬 NT	2025	2026(F)	2027(F)	3Q25	4Q25	1Q26	2Q26	3Q26(F)	4Q26(F)	1Q27(F)	2Q27(F)
營業收入	24,189	25,620	26,164	6,199	6,030	6,435	6,449	6,221	6,516	5,920	6,776
營業毛利	2,658	2,877	2,964	672	648	683	822	676	696	719	768
營業利益	1,733	1,926	2,002	451	424	464	552	438	472	511	491
稅後淨利	1,505	1,666	1,690	372	380	369	507	376	414	420	417
EPS(NT$)	2.32	2.57	2.61	0.57	0.58	0.57	0.78	0.58	0.64	0.65	0.64
毛利率	11.0%	11.2%	11.3%	10.8%	10.7%	10.6%	12.7%	10.9%	10.7%	12.2%	11.3%
營業利益率	7.2%	7.5%	7.7%	7.3%	7.0%	7.2%	8.6%	7.0%	7.2%	8.6%	7.2%
稅後淨利率	6.2%	6.5%	6.5%	6.0%	6.3%	5.7%	7.9%	6.0%	6.4%	7.1%	6.1%
營收 YoY	0.7%	5.9%	2.1%	0.4%	(6.0%)	15.0%	1.3%	0.3%	8.1%	(8.0%)	5.1%
EPS YoY	40.6%	10.9%	1.5%	21.3%	18.4%	1.8%	30.0%	1.8%	10.2%	13.7%	(17.5%)

營運展望

1. **1H26 因營建鋼構事業帶動獲利雙位數成長：**

1H26 合併營收為 128.8 億元（+7.7% YoY），產品結構以鋼板類占 52.5%、營建鋼結構工程占 31.8%、其他（包含特殊鋼、特鋼帶、矽鋼、自動倉儲等）占 15.7%。累計毛利率為 11.7%（+0.5 ppts YoY），累計營業利益為 10.2 億元（+18.5% YoY）。受惠於 AI 產業熱絡帶動的科技廠辦擴建、商辦需求、都更危老改建及公共工程預算創新高，營建鋼構事業已成為近年最重要的結構性獲利引擎，其營建鋼結構工程占比雖為 31.8%，但獲利貢獻高達 56%。累計 EPS 為 1.35 元（+16.4% YoY），累計 EBITDA 達 13.39 億元（+18.5% YoY）。

2. **受惠鋼市供需調整，2026 年營運樂觀：**

預估 2026 年合併營收為 256 億元（+5.9% YoY）。春源的營收與出貨量主要由鋼板（非工程類）與鋼結構（工程類）兩大主軸構成。公司對第四季鋼市景氣趨於樂觀，因煉鋼成本上漲支撐鋼價，加上中國大陸政策性減產使鋼價不易下跌，第四季基礎建設需求釋放及補庫存啟動，將對亞洲鋼價形成支撐。預估毛利率為 11.2%（+0.2 ppts YoY），高毛利的鋼構工程支撐整體出貨動能，營業利益預估達 19 億元（+11.2% YoY），EPS 預估為 2.57 元（+10.9% YoY）。

3. **科技建廠需求強勁，推升鋼構業務成長：**

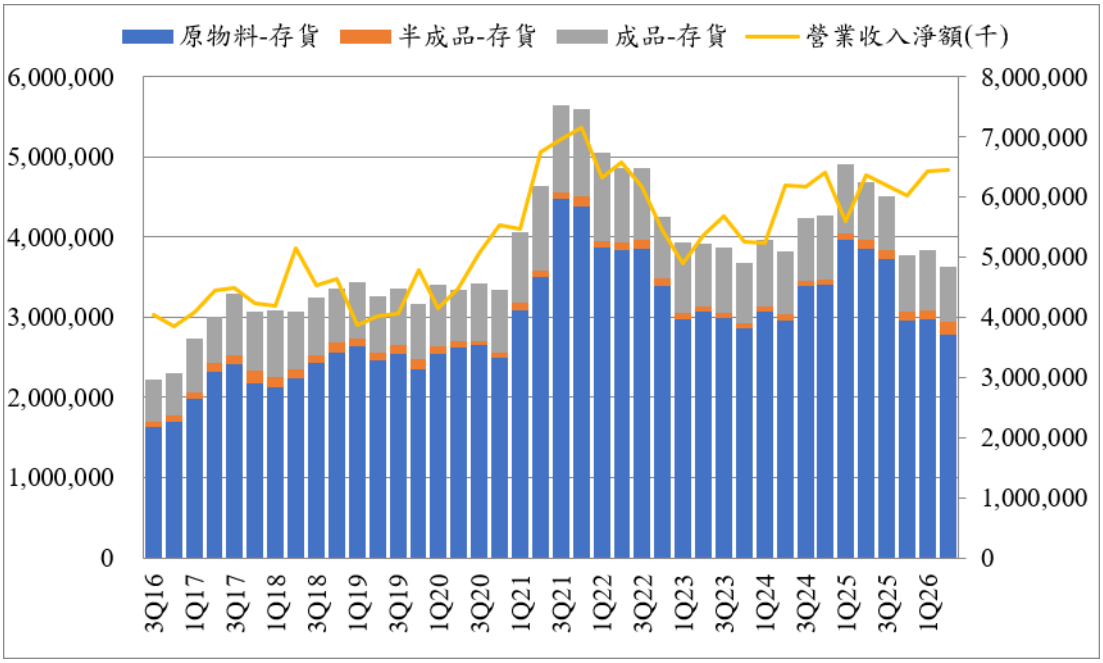
營建事業部門為目前春源主要獲利來源，受惠 2027 年公共預算創十年新高，科技廠辦、商辦、都更危老及公共建設四大動能強勁，推升鋼構需求，且春源具備承攬大型巨蛋等高難度工程的專利工法，享有市場競爭優勢。主要成長動能包含：

(1) **科技建廠需求強勁**：受惠於 AI 產業熱絡帶動的科技廠辦擴建（如半導體廠房、資料中心）及商辦需求，營建鋼構事業已成為近年最重要的結構性獲利引擎。

(2) **訂單能見度已達 2027 年**：受惠國內商辦都更、AI 晶圓廠房擴建及資料中心新建需求爆發，帶動國內原物料與生產活動，未來 3～5 年仍為鋼構業黃金期。目前春源鋼結構工程在手訂單維持在 20 萬噸以上，訂單能見度直達 4Q27，部分高難度特殊鋼構訂單甚至已排至 1H28。

(3) **營建鋼構需求強勁，存貨去化速度快**：公司依鋼市景氣與鋼價波動動態調整安全庫存水位，避免原料過高在鋼價下跌時面臨「存貨跌價損失」風險。由於營建鋼構訂單火熱，生產出的鋼構建材與裁剪鋼板得以迅速向工程端及驗收端去化，不易在倉庫堆積，形成良性的低庫存周轉。

圖：春源鋼構需求強勁，存貨去化速度快(單位：千元)



資料來源：公司財報；合庫投顧整理

4. **鋼板及特殊鋼受益 AI 與汽車外銷雙引擎驅動：**

鋼板及特殊鋼品主要供應汽車、機械及 AI 相關產業，受惠 AI 應用需求擴增，主計總處大幅上修 2026 年經濟成長率，AI 需求已擴散至傳統產業。主要成長動能包含：

(1) **AI 外銷外溢效應：**機械業受惠 AI、半導體、機器人、無人機等需求，出口持續成長。公司正積極開發新特殊鋼品（如微型馬達）以提高附加價值，隨 AI 應用外溢發酵，加上汽車用鋼爭取到外銷增量機會，皆為 2027 年營收帶來支撐。

(2) **汽車鋼板內外銷回暖：**汽車市場因汰舊換新需求及產業政策支持而回暖，預估 2026 年總銷量達 43 萬輛。此外，重要客戶豐田規劃在台建立全球第四個生產基地，國瑞汽車整車回銷日本於 10 月啟航，春源作為其核心零件供應商，中長期車用鋼板與組件出貨動能看俏。

5. **具高股息殖利率，下檔風險有限：**

2Q26 流動比率為 177%，負債比率為 45%，顯示公司短期償債能力良好且財務結構穩健，上半年營業活動現金流入達 18.37 億元。在建工程合約資產、應收帳款及存貨合計達 133 億元，遠高於 107 億元的負債總額。2026 年預估稅後淨利 16.7 億元，擬發放現金股利 13 億元，若股利發放率為 86%，預估現金股利為 2.2 元。以近日收盤價計算，目前現金股利殖利率達 9.77%，下檔風險有限。

表：春源具高股息殖利率(單位：元；%)

	2020	2021	2022	2023	2024	2025	2026E
現金股利(元)	0.60	1.30	0.60	1.00	1.30	2.00	2.22
股票股利(元)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
除息日	2021/7/9	2022/4/21	2023/7/18	2024/7/4	2025/4/17	2026/7/2	
EPS(元)	0.79	1.88	0.95	1.35	1.65	2.32	2.57
現金股利殖利率(%)	1.78	5.27	3.31	4.90	6.93	8.44	9.77
股利發放率(%)	75.9	69.1	63.2	74.1	78.8	86.2	86.2

資料來源：公司財報；合庫投顧整理

6. **ESG（2026 年上市公司治理評鑑排名為 81~100%）：**

春源鋼鐵將永續發展與企業社會責任（CSR）納入核心營運策略。面對鋼鐵產業減碳挑戰，春源透過以下面向落實 ESG：

(1) 減碳與氣候行動:致力於控制溫室氣體排放與節能,設定科學基礎減碳目標,並在「企業減碳溫度計」評比中表現優異。

(2) 環安衛與綠色建築：將「綠色永續」理念融入本業，在廠房營造與結構設計上推動環境友善與循環經濟。

(3) 公司治理：具備穩健財務體質，遵循公司治理準則，並定期編製與公開永續報告書，確保資訊透明度。

PE/PB Band

損益表

單位：百萬 NT	2025	2026(F)	2027(F)	3Q25	4Q25	1Q26	2Q26	3Q26(F)	4Q26(F)	1Q27(F)	2Q27(F)
營業收入	24,189	25,620	26,164	6,199	6,030	6,435	6,449	6,221	6,516	5,920	6,776
營業毛利	2,658	2,877	2,964	672	648	683	822	676	696	719	768
營業費用	926	950	962	221	224	219	270	238	224	208	278
營業利益	1,733	1,926	2,002	451	424	464	552	438	472	511	491
業外收支	133	194	164	35	42	9	90	49	46	13	64
稅前淨利	1,865	2,120	2,165	486	466	473	642	487	518	524	555
所得稅	345	436	456	109	82	99	130	107	100	100	133
稅後淨利	1,505	1,666	1,690	372	380	369	507	376	414	420	417
股本	6,477	6,477	6,477	6,477	6,477	6,477	6,477	6,477	6,477	6,477	6,477
EPS(NT$)	2.32	2.57	2.61	0.57	0.58	0.57	0.78	0.58	0.64	0.65	0.64
營收 QoQ	-	-	-	(2.6%)	(2.7%)	6.7%	0.2%	(3.5%)	4.7%	(9.1%)	14.5%
營收 YoY	0.7%	5.9%	2.1%	0.4%	(6.0%)	15.0%	1.3%	0.3%	8.1%	(8.0%)	5.1%
EPS YoY	40.6%	10.9%	1.5%	21.3%	18.4%	1.8%	30.0%	1.8%	10.2%	13.7%	(17.5%)
毛利率	11.0%	11.2%	11.3%	10.8%	10.7%	10.6%	12.7%	10.9%	10.7%	12.2%	11.3%
營業費用率	3.8%	3.7%	3.7%	3.6%	3.7%	3.4%	4.2%	3.8%	3.4%	3.5%	4.1%
營業利益率	7.2%	7.5%	7.7%	7.3%	7.0%	7.2%	8.6%	7.0%	7.2%	8.6%	7.2%
稅前淨利率	7.7%	8.3%	8.3%	7.8%	7.7%	7.4%	10.0%	7.8%	7.9%	8.9%	8.2%
所得稅率	18.5%	20.6%	21.1%	22.4%	17.7%	20.9%	20.2%	21.9%	19.4%	19.0%	24.0%
稅後淨利率	6.2%	6.5%	6.5%	6.0%	6.3%	5.7%	7.9%	6.0%	6.4%	7.1%	6.1%

資產負債表

(百萬元)	2023	2024	2025	2Q26
資產總計	20,303	22,345	24,224	23,666
流動資產	12,852	14,538	15,483	15,069
現金及約當現金	461	529	609	1,043
應收帳款與票據	4,639	5,073	5,555	5,139
存貨	3,718	4,359	3,835	3,715
採權益法之投資	623	675	739	741
不動產及廠房	5,692	5,977	6,202	6,557
負債總計	8,243	9,698	10,925	10,729
流動負債	6,850	8,108	8,773	8,514
應付帳款	812	1,081	1,290	1,291
應付票據	204	51	64	23
非流動負債	1,392	1,590	2,152	2,216
權益總計	12,061	12,648	13,299	12,937
普通股股本	6,477	6,477	6,477	6,477
保留盈餘	5,313	5,760	6,477	6,058
母公司業主權益	11,692	12,264	12,912	12,537

損益表

(百萬元)	2024	2025	2026(F)	2027(F)
營業收入淨額	24,014	24,189	25,620	26,164
營業成本	21,946	21,531	22,744	23,200
營業毛利淨額	2,069	2,658	2,877	2,964
營業費用	866	926	950	962
營業利益	1,202	1,733	1,926	2,002
業外收支	185	133	194	164
稅前純益	1,387	1,865	2,120	2,165
所得稅	301	345	436	456
稅後純益	1,069	1,505	1,666	1,690
稅後EPS(元)	1.65	2.32	2.57	2.61

註1：稅後淨利係指本期淨利歸屬母公司業主。

註2：稅後EPS以當期股本計算

現金流量表

(百萬元)	2023	2024	2025	2Q26
營業活動現金	2,244	1,037	1,993	1,224
稅前純益	1,121	1,387	1,865	642
折舊	276	294	300	86
攤銷	17	18	22	5
其他營運現金	2,549	1,406	2,701	1,527
投資活動現金	(1,004)	(559)	(1,293)	(46)
其他投資現金	0	0	0	0
籌資活動現金	(1,775)	(450)	(643)	(795)
現金增資	0	0	0	0
發放現金股利	(389)	(648)	(842)	0
其他籌資現金	0	0	0	0
淨現金流量	(578)	68	80	423
期初現金	1,038	461	529	620
期末現金	461	529	609	1,043

比率分析

(%)	2023	2024	2025	2Q26
成長力分析				
營收成長	(13.5)	13.3	0.7	1.3
營利成長	33.2	29.6	44.1	33.3
稅後純益成長	45.1	21.9	40.1	30.0
獲利能力分析				
毛利率	8.3	8.6	11.0	12.74
營利率	4.4	5.0	7.2	8.56
稅後純益率	4.1	4.5	6.2	7.86
償債能力分析				
負債比率	40.6	43.4	45.1	45.34
流動比率	187.6	179.3	176.5	176.99
速動比率	131.3	123.3	131.4	132.5

資料來源：CMoney、合庫投顧

免責聲明:
本報告內容僅供參考，已力求訊息確實與完整，但客戶仍應審慎考量本身之需求與投資風險及承受度，本公司不做任何獲利保證，亦不就投資盈虧負任何法律責任。本刊著作權屬合庫投顧所有，非經本公司同意請勿抄襲、引用或轉載。

本研究報告並不構成要約、招攬、邀請、宣傳、誘使，或任何形式之表示，本報告並非針對客戶於買賣證券、選擇權、期貨或其他證券相關商品提供詢價或推介服務。本研究報告內容係報告撰寫人依所收集之資訊進行合理之判斷，並就其專業能力提供之善意分析，報告撰寫人與相關複核人員的報酬與本篇報告內容表述或觀點無關。出具本報告時，本公司、報告撰寫人與複核人員，均遵守主管機關相關利益衝突防範。

合作金庫金融控股(股)公司旗下子公司暨關係企業從事廣泛金融業務，合作金庫集團對於本報告所涵蓋的標的公司可能有投資或業務往來，合作金庫集團及其所屬員工可能會投資本報告所涵蓋的標的，或以該公司為標的的衍生性金融商品，而其交易的方向與本報告中所提及的投資方向可能不一致。

評等標準說明

評等	定義
買進	預期未來 6 個月潛在報酬率>15%
區間	預期未來 6 個月潛在報酬率 0%~15%
中立	預期未來 6 個月潛在報酬率<0%
未評等	訊息不足或有相關業務往來

大立光(3008)

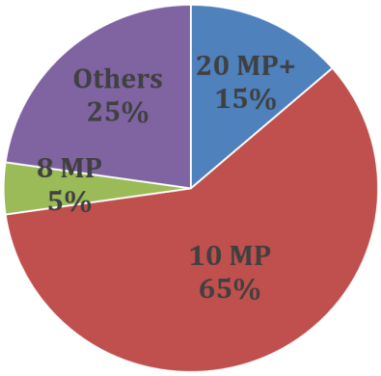
區間操作　　報告日期：20261012

研究員：陳彥翰

公司基本資料

前日收盤價(元)	6,660.00
目前股本(億元)	13.08
市值(億元)	8,711.30
每股淨值(元)	1,464.41
外資持股比例(%)	24.11
投信持股比例(%)	7.93
董監持股比例(%)	19.02
融資餘額(張)	4,804

業務營收比重



投資建議

展望明後年，光纖陣列與稜鏡微透鏡陣列等光學元件新業務增長動能明確，將彌補手機鏡頭缺緩造成的缺口，帶動大立光營運獲利重返成長軌道。合庫預估 2026 / 2027 年獲利分別衰退 1% / 成長 7% ，稅後每股盈餘 158.03 / 169.79 元，建議操作區間 5,435 ~ 7,300 元（2027 PER 32 ~ 43x）。

公司簡介

大立光電股份有限公司創立於 1987 年，專業於各式光學鏡頭模組與光電零組件等產品之研究開發，產品包括智慧型手機鏡頭、3D 結構光鏡頭、ToF、屏下光學指紋、空拍機鏡頭、平板電腦鏡頭、IP 相機鏡頭、智慧型電視鏡頭、AR/VR 鏡頭、IoT 鏡頭、穿戴式裝置鏡頭、虹膜辨識鏡頭、醫療儀器用鏡頭、汽車鏡頭、鏡片等光電零組件之設計、製造與加工。

簡易損益表

單位：百萬	2025	2026(F)	2027(F)	3Q25	4Q25	1Q26	2Q26	3Q26(F)	4Q26(F)	1Q27(F)	2Q27(F)
營業收入	61,148	59,781	66,050	17,677	17,219	15,544	13,665	15,651	14,921	13,776	12,303
營業毛利	30,837	27,993	32,314	8,352	8,260	7,680	6,753	6,607	6,953	6,309	6,071
營業利益	23,558	20,820	24,584	6,265	6,329	5,812	5,148	4,735	5,125	4,551	4,407
稅後純益	21,275	20,694	22,209	7,080	6,720	6,123	4,670	5,182	4,719	4,346	4,122
EPS(NT$)	159.41	158.03	169.79	53.05	50.35	46.63	35.70	39.62	36.08	33.23	31.51
毛利率	50.4%	46.8%	48.9%	47.3%	48.0%	49.4%	49.4%	42.2%	46.6%	45.8%	49.4%
營業利益率	38.5%	34.8%	37.2%	35.4%	36.8%	37.4%	37.7%	30.3%	34.3%	33.0%	35.8%
稅後淨利率	34.8%	34.6%	33.6%	40.1%	39.0%	39.4%	34.2%	33.1%	31.6%	31.5%	33.5%
營收 YoY	2.8%	(2.2%)	10.5%	(6.7%)	(5.4%)	6.6%	17.1%	(11.5%)	(13.3%)	(11.4%)	(10.0%)
EPS YoY	(17.7%)	(0.9%)	7.4%	8.0%	(24.2%)	(5.6%)	347.5%	(25.3%)	(28.3%)	(28.7%)	(11.7%)

營運展望

1. **3Q26 良率與成本明顯衝擊毛利率表現：**受到綠電成本上升以及新機良率較低，造成毛利率低於市場預期。第三季營收 157 億元、+14.5% QoQ、-11.5% YoY，毛利率 42.2%。營業利益 47 億元、-8.0% QoQ、-24.4% YoY，營業利益率 30.3%。歸屬母公司稅後淨利 52 億元，每股盈餘 39.62 元。產品組合分別為 2,000 萬畫素以上鏡頭佔比 10 ~ 20%、1,000 萬畫素鏡頭佔比 60 ~ 70%、800 萬畫素鏡頭佔比小於 10%、低於 500 萬畫素鏡頭佔比 20 ~ 30%。

2. **4Q26 春季新品拉貨動能仍無法彌補旗艦機種下修的缺口：**第四季拉貨動能趨緩，公司預期 10 月拉貨動能與 9 月相當，但 11 月拉貨動能開始下滑。即使春季中階新品開始備貨，仍無法彌補旗艦機種下修的缺口。不過可變光圈良率已經明顯改善，有助毛利率拉升。但綠電成本部分，已成為結構性負面因素，最快需至後年才能漲價彌補差異。

3. **光纖陣列已通過客戶驗證，預期 2027 年下半年產能可建置完成上線：**公司證實光纖陣列（Fiber Array, FA）產品已於九月順利取得終端客戶認證。十月將與客戶正式洽談具體產能規模需求，預估產線最快 2027 年年中能建置完成並開始量產。至於光纖陣列（Fiber Array Unit, FAU）模組現階段尚未進入送樣流程，而是透過產線運作與現場生產數據供客戶評估，本月將與客戶協商確切送樣與認證時程。目前聚焦自製 FA 並外購稜鏡微透鏡陣列（Photonic Module Lens Assembly, PMLA）進行耦合以組裝成 FAU 模組。下一階段目標為爭取載板（carrier）組裝業務，以深化產品附加價值。至於近期購入的七座廠房土地將全數投入建置生產線及無塵室，預計明年開工，2030 年完工。客戶若採用 NPO 方案，只要非採用 VCSEL 技術，其所需元件與公司現行的 FAU 解決方案幾乎一致，技術差異僅存在於封裝形式，公司皆能通用並持續供貨。

4. **手機市場雖趨緩，但光纖業務展望佳將助營運重返成長：**展望明後年，光纖陣列與稜鏡微透鏡陣列等光學元件新業務增長動能明確，將彌補受科技通膨影響的手機鏡頭業務缺口，帶動大立光營運獲利重返成長軌道。合庫預估 2026 年營收 598 億元、-2.2% YoY，毛利率 46.8%。營業利益 208 億元、-11.6% YoY，營業利益率 34.8%。歸屬母公司稅後淨利 207 億元，每股盈餘 158.03 元。

5. **ESG（大立光在 2026 年上市公司治理評鑑結果為 66 ~ 80%）：**大立光 ESG 委員會為跨部門溝通平台，負責規畫與執行及管控各項行動方案，整合與監督公司治理（含經濟）、社會、環境三大面向的執行進度與成效，確保組織橫向與縱向溝通的有效性，具體實踐永續發展。大立光依據六大篩選原則「責任」、「影響力」、「親近度」、「依賴性」、「代表性」、「政策及策略意圖」，在 ESG 委員會會議鑑別出五個主要的利害關係人，包含投資人（股東）、員工、客戶、供應商、政府機關。

PE/PB Band

損益表

單位：百萬	2025	2026(F)	2027(F)	3Q25	4Q25	1Q26	2Q26	3Q26(F)	4Q26(F)	1Q27(F)	2Q27(F)
營業收入	61,148	59,781	66,050	17,677	17,219	15,544	13,665	15,651	14,921	13,776	12,303
營業毛利	30,837	27,993	32,314	8,352	8,260	7,680	6,753	6,607	6,953	6,309	6,071
營業費用	7,279	7,172	7,731	2,088	1,932	1,867	1,605	1,872	1,828	1,758	1,664
營業利益	23,558	20,820	24,584	6,265	6,329	5,812	5,148	4,735	5,125	4,551	4,407
業外收支	2,342	4,561	2,962	1,890	1,907	1,476	951	1,357	777	654	691
稅前純益	25,900	25,382	27,546	8,155	8,236	7,288	6,099	6,092	5,903	5,205	5,098
所得稅	4,340	4,540	5,191	996	1,434	1,112	1,373	909	1,145	828	948
稅後純益	21,275	20,694	22,209	7,080	6,720	6,123	4,670	5,182	4,719	4,346	4,122
股本	1,335	1,308	1,308	1,335	1,335	1,335	1,308	1,308	1,308	1,308	1,308
EPS(NT$)	159.41	158.03	169.79	53.05	50.35	46.63	35.70	39.62	36.08	33.23	31.51
營收 QoQ	-	-	-	51.4%	(2.6%)	(9.7%)	(12.1%)	14.5%	(4.7%)	(7.7%)	(10.7%)
營收 YoY	2.8%	(2.2%)	10.5%	(6.7%)	(5.4%)	6.6%	17.1%	(11.5%)	(13.3%)	(11.4%)	(10.0%)
EPS YoY	(17.7%)	(0.9%)	7.4%	8.0%	(24.2%)	(5.6%)	347.5%	(25.3%)	(28.3%)	(28.7%)	(11.7%)
毛利率	50.4%	46.8%	48.9%	47.3%	48.0%	49.4%	49.4%	42.2%	46.6%	45.8%	49.4%
營業費用率	11.9%	12.0%	11.7%	11.8%	11.2%	12.0%	11.8%	12.0%	12.3%	12.8%	13.5%
營業利益率	38.5%	34.8%	37.2%	35.4%	36.8%	37.4%	37.7%	30.3%	34.3%	33.0%	35.8%
稅前淨利率	42.4%	42.5%	41.7%	46.1%	47.8%	46.9%	44.6%	38.9%	39.6%	37.8%	41.4%
所得稅率	16.8%	17.9%	18.8%	12.2%	17.4%	15.3%	22.5%	14.9%	19.4%	15.9%	18.6%
稅後淨利率	34.8%	34.6%	33.6%	40.1%	39.0%	39.4%	34.2%	33.1%	31.6%	31.5%	33.5%

資產負債表

(百萬元)	2023	2024	2025	2Q26
資產總計	195,138	216,527	220,787	226,131
流動資產	134,321	145,764	154,285	157,493
現金及約當現金	107,490	113,658	115,942	118,482
應收帳款與票據	10,091	10,360	10,462	7,328
存貨	4,591	5,733	6,713	9,088
採權益法之投資	1,369	24	26	28
不動產及廠房	41,135	46,936	51,472	52,530
負債總計	29,627	31,139	29,919	32,642
流動負債	29,517	30,578	29,749	32,442
應付帳款	1,612	1,854	1,724	2,046
應付票據	1	2	1	1
非流動負債	110	561	171	200
權益總計	165,510	185,388	190,868	193,489
普通股股本	1,335	1,335	1,335	1,308
保留盈餘	160,871	175,969	185,818	183,393
母公司業主權益	165,510	183,519	188,867	191,542

損益表

(百萬元)	2024	2025	2026(F)	2027(F)
營業收入淨額	59,458	61,148	59,781	66,050
營業成本	28,248	30,311	31,788	33,736
營業毛利淨額	31,209	30,837	27,993	32,314
營業費用	7,177	7,279	7,172	7,731
營業利益	24,033	23,558	20,820	24,584
業外收支	8,142	2,342	4,561	2,962
稅前純益	32,174	25,900	25,382	27,546
所得稅	5,963	4,340	4,540	5,191
稅後純益	25,915	21,275	20,694	22,209
稅後EPS(元)	194.17	159.41	158.03	169.79

註1：稅後淨利係指本期淨利歸屬母公司業主。

註2：稅後EPS以當期股本計算

現金流量表

(百萬元)	2023	2024	2025	2Q26
營業活動現金	18,199	31,579	26,080	8,968
稅前純益	22,102	32,174	25,900	6,099
折舊	5,299	6,038	7,511	2,119
攤銷	123	192	221	57
其他營運現金	20,323	36,423	28,279	11,688
投資活動現金	(11,151)	(16,066)	(9,236)	(2,399)
其他投資現金	0	0	0	0
籌資活動現金	(9,750)	(10,764)	(13,458)	(6,816)
現金增資	0	0	0	0
發放現金股利	(9,676)	(10,811)	(11,412)	0
其他籌資現金	5	2	3	0
淨現金流量	(2,682)	6,168	2,283	(541)
期初現金	110,172	107,490	113,658	119,023
期末現金	107,490	113,658	115,942	118,482

比率分析

(%)	2023	2024	2025	2Q26
成長力分析				
營收成長	2.5	21.7	2.8	17.1
營利成長	(12.6)	35.0	(2.0)	5.5
稅後純益成長	(20.9)	46.4	(17.7)	347.5
獲利能力分析				
毛利率	48.7	52.5	50.4	49.4
營利率	36.5	40.4	38.5	37.7
稅後純益率	36.7	43.6	34.8	34.2
償債能力分析				
負債比率	15.2	14.4	13.6	14.4
流動比率	455.1	476.7	518.6	485.5
速動比率	439.5	458.0	496.1	457.5

資料來源：CMoney、合庫投顧

免責聲明：
本報告內容僅供參考，已力求訊息確實與完整，但客戶仍應審慎考量本身之需求與投資風險及承受度，本公司不做任何獲利保證，亦不就投資盈虧負任何法律責任。本刊著作權屬合庫投顧所有，非經本公司同意請勿抄襲、引用或轉載。

本研究報告並不構成要約、招攬、邀請、宣傳、誘使，或任何形式之表示，本報告並非針對客戶於買賣證券、選擇權、期貨或其他證券相關商品提供詢價或推介服務。本研究報告內容係報告撰寫人依所收集之資訊進行合理之判斷，並就其專業能力提供之善意分析，報告撰寫人與相關複核人員的報酬與本篇報告內容表述或觀點無關。出具本報告時，本公司、報告撰寫人與複核人員，均遵守主管機關相關利益衝突防範。

合作金庫金融控股(股)公司旗下子公司暨關係企業從事廣泛金融業務，合作金庫集團對於本報告所涵蓋的標的公司可能有投資或業務往來，合作金庫集團及其所屬員工可能會投資本報告所涵蓋的標的，或以該公司為標的的衍生性金融商品，而其交易的方向與本報告中所提及的投資方向可能不一致。

評等標準說明

評等	定義
買進	預期未來 6 個月潛在報酬率>15%
區間	預期未來 6 個月潛在報酬率 0%~15%
中立	預期未來 6 個月潛在報酬率<0%
未評等	訊息不足或有相關業務往來